Calabi–Yau Quantum Groups

Factorization, Hall Algebras, Chiral Centers, Quantum Doubles, and Automorphic Characters

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Abstract

A Calabi–Yau category, its local field observables, its Hall algebra, and a quantum double carry different operations. The comparison problem is to construct maps that preserve those operations in specified coefficient categories and completions. Hochschild operations and shifted symplectic transgression give categorical constructions under their respective hypotheses. A specified elliptic BV theory supplies local observables in its completed coefficient category. Geometric pushforward to a curve retains their factorization maps. A comparison with a separately specified field theory or chiral algebra requires the corresponding action and collision maps. Oriented extension correspondences, under the stated base-change and completion hypotheses, supply Hall multiplication. A paired triangular Hopf datum supplies a completed double and exchange operators on the declared continuous modules.

On smooth projective K3 moduli spaces, selected correspondences define algebras of actual cohomological operators. A specified tensor category and fiber functor instead give a matrix-coefficient bialgebra. Comparing these constructions requires their action maps. For $K3 \times E$, the Borcherds root algebra and its Fock character are independently defined. A geometric primitive comparison requires the Hall coefficients, orientations, brackets, and parity-refined state spaces. The finite Hamiltonian results include the relative Rees comparison and oscillator quotient. Passage to the full jet and mixed open–closed structures uses the specified compatible systems and operadic refinements. The BV constructions describe centers, deformation algebras, and master equations for a supplied complete mixed boundary datum. A physical or gravitational interpretation additionally requires a comparison of fields, boundary actions, propagators, and quantizations.

The comparison problem

Local collision laws and automorphic characters belong to different mathematical constructions. The diagram records the comparison problems between them:

$$\begin{aligned}\text{collision} &\longrightarrow \text{ordered fusion} \longrightarrow \text{center} \longrightarrow \text{Hall extension}\\ &\longrightarrow \text{quantum double} \longrightarrow \text{character}.\end{aligned}$$

Each arrow requires a map in a specified category, together with its grading, completion, and preserved operations.

For a finite-dimensional complex Lie algebra, the relative Rees comparison identifies relative Hochschild cochains with a Chevalley complex up to quasi-isomorphism. The Chevalley differential is scaled by the Rees parameter. Quadratic Hamiltonians generate the even Weyl algebra through the oscillator Casimir quotient. Scalar reduction and an augmentation give continuous stable trace maps of commutative algebras. Monoidal module fusion uses a separately specified coassociative, counital bialgebra in the chosen completed coefficient category.

The Hamiltonian jet limit requires compatible cochain transition maps and limit identifications. Higher mixed open–closed operations require the stated operadic refinements. A Hochschild center and the Drinfeld center of a monoidal module category are separate comparison targets. Hall, quantum-double, and physical bulk identifications require their actual comparison morphisms and the relevant orientation, descent, pairing, and analytic hypotheses. A proper filtration specifies an inverse-limit carrier. Compatibility of the operations with that limit remains part of each comparison theorem.

Notation and normalization

The ground field has characteristic zero. Tensor products on presentable stable coefficient categories are assumed to preserve colimits separately. Small categories of perfect objects are idempotent-complete. Completions are separated inverse limits of finite weight or proper-height quotients.

The Igusa normalization is

$$\Delta_5^{\theta} = 64 D_5, \qquad \chi_{10} = D_5^{\,2},$$

with D_5 monic.

A charge is an additive grading on objects. Gram degree is carried through its polarization cocycle. Bar, factorization, and Hopf coproducts retain distinct notation and distinct geometric meanings.

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Book I

Calabi–Yau moduli and Hall operations

Chapter 1

Two insertions and one extension

An extension distinguishes the subobject from the quotient. This order can remain visible after passing to cohomology. For the quiver with one vertex and one weighted loop, reversing the two inputs changes their product. The same geometry has a nonsplit extension at coincident support. Its invariant lines form a nonreduced scheme of length two.

1.1 The first ordered product

Geometric objects are over $\mathbb{C}$, and cohomology has rational coefficients. Put $R = \mathbb{Q}[\hbar]$ and

$$\mathcal{S}_n = R[x_1, \dots, x_n]^{\mathfrak{S}_n}, \qquad \mathcal{S} = \bigoplus_{n \geq 0} \mathcal{S}_n.$$

The symmetric group permutes the variables. The summand index n is called the *charge*. Write $\mathbf{1}_n$ for the constant polynomial in $\mathcal{S}_n$, including $\mathcal{S}_0 = R$.

Give every x_i and $\hbar$ cohomological degree two. All classes are even. The coefficient ring is polynomial, with no completion.

For $f, g \in \mathcal{S}_1$, define

$$\begin{aligned}(f \star g)(x_1, x_2) = f(x_1)\, g(x_2) \frac{x_2 - x_1 - \hbar}{x_2 - x_1} \\ + f(x_2)\, g(x_1) \frac{x_1 - x_2 - \hbar}{x_1 - x_2}.\end{aligned} \tag{1.1}$$

The two summands correspond to the two choices of a first root. Their residues at $x_1 = x_2$ cancel, so the result is a symmetric polynomial.

Set $a_r = x_1^r \in \mathcal{S}_1$ for $r \geq 0$. Write $[f, g] = f \star g - g \star f$. Direct substitution gives

$$\begin{aligned} a_0 \star a_1 &= x_1 + x_2 - \hbar, \quad & a_1 \star a_0 &= x_1 + x_2 + \hbar, \\ a_0 \star a_0 &= 2\mathbf{1}_2, & [a_1, a_0] &= 2\hbar \mathbf{1}_2. \end{aligned} \tag{1.2}$$

In particular, $[a_1, a_0] = \hbar(a_0 \star a_0)$. The constant $\mathbf{1}_2$ has charge two. The algebra unit will be $\mathbf{1}_0$, and $a_0 = \mathbf{1}_1$ is a third element.

The two-root formula extends by assigning roots to the two inputs. For a subset I, write x_I for its list of roots. For $f \in \mathcal{S}_n$ and $g \in \mathcal{S}_m$, set

$$(f \star g)(x) = \sum_{\substack{I \subset \{1, \dots, n+m\} \\ |I| = n}} f(x_I)\, g(x_{I^c}) \prod_{i \in I,\ j \in I^c} \frac{x_j - x_i - \hbar}{x_j - x_i}. \tag{1.3}$$

Symmetry makes the order within I and I^c irrelevant. An empty product equals one. Each subset occurs once.

Proposition 1.1 (The weighted-loop algebra)**.** *Formula* (1.3) *defines an associative unital R-algebra structure on $\mathcal{S}$. Multiplication adds charges and has cohomological degree zero. Its unit is* $\mathbf{1}_0$.

Proof. The only possible poles are simple poles on root diagonals. Fix $x_i = x_j$ and pair each summand that places these roots in opposite blocks with the summand that exchanges them. On this diagonal the input polynomials and all other pair factors agree, while the two residues have opposite signs. Thus every residue vanishes.

The common denominator is a product of distinct diagonal equations in the unique factorization domain $R[x_1, \dots, x_{n+m}]$. Cancellation at each one proves polynomiality. Permuting the roots permutes the summands.

For three inputs, either bracketing sums over the same ordered partition of the roots into three blocks. Every pair of roots in distinct blocks contributes its factor exactly once, in the order of those blocks. The two rational expressions agree term by term after this common indexing, and hence agree as polynomials. With an empty input there is one subset and no pair factor, which proves the unit identities. Each pair factor has degree zero. □

At $\hbar = 0$ the pair factors are one. The commutator in (1.2) therefore measures a deformation of the unweighted product. Its geometric origin is the difference between the loop weight and the tangent weight of a flag.

1.2 The Euler factor of an invariant subspace

A representation of the one-loop quiver is a vector space with an endomorphism. In dimension n its moduli stack is

$$\mathfrak{M}_n = [\mathrm{End}(\mathbb{C}^n)/GL_n].$$

Let $T = \mathbb{C}^\times$ act on the endomorphism by the character $\chi_{-1}(s) = s^{-1}$, and let $\hbar$ be the first Chern class of the character $s \mapsto s$. Equivariant contraction of the endomorphism space to zero identifies

$$H^\bullet_T(\mathfrak{M}_n; \mathbb{Q}) = H^\bullet(BGL_n \times BT; \mathbb{Q}) = \mathcal{S}_n.$$

The variables x_i are the Chern roots of the universal vector bundle. They have cohomological degree two, independently of the eigenvalues of an individual endomorphism.

Let $\mathfrak{E}_{n,m}$ classify an endomorphism of an $(n+m)$-dimensional space together with an invariant n-dimensional subspace. It gives the correspondence

$$\mathfrak{M}_n \times \mathfrak{M}_m \xleftarrow{q} \mathfrak{E}_{n,m} \xrightarrow{p} \mathfrak{M}_{n+m}. \tag{1.4}$$

The map q takes the restricted endomorphism and its quotient. The map p forgets the subspace. It is proper, since invariant subspaces form a closed subscheme of a Grassmannian.

On $\mathrm{Gr}(n, n+m)$ write S for the tautological subbundle and Q for the quotient bundle. Invariance sets to zero the endomorphism block from S to Q. Its normal bundle is $\mathrm{Hom}(S, Q) \otimes \chi_{-1}$. The Grassmannian tangent bundle is $\mathrm{Hom}(S, Q)$.

Write e for the top Chern class and $f(S)$, $g(Q)$ for evaluation on the respective Chern roots. The Hall operation in ordinary equivariant cohomology is complex-oriented integration of

$$f(S)\, g(Q)\, e\big(\mathrm{Hom}(S, Q) \otimes \chi_{-1}\big) \tag{1.5}$$

over the Grassmannian. The normal Gysin map adds degree $2nm$ and Grassmannian integration subtracts $2nm$.

At the coordinate subspace indexed by I, these two bundles have roots $x_j - x_i - \hbar$ and $x_j - x_i$, respectively, for $i \in I$, $j \in I^c$. Equivariant integration divides by the tangent Euler class and sums over

these subspaces. It gives exactly (1.3). This is the ordinary quiver Hall construction of [100, §§2.4–2.5], with the loop character retained. Proposition 1.1 proves directly that its formula is polynomial over $\mathbb{Q}[\hbar]$.

For $n = m = 1$, $S = \mathcal{O}(-1)$ and $Q = \mathcal{O}(1)$ on $\mathbb{P}^1$. Consequently

$$\int_{\mathbb{P}^1} c_1(Q) = 1, \qquad \int_{\mathbb{P}^1} c_1(S) = -1, \qquad \int_{\mathbb{P}^1} c_1 \operatorname{Hom}(S, Q) = 2.$$

These equations fix the signs and the charge-two coefficient in (1.2). Geometrically, the coefficient integrates invariant-subspace data. Those data change when support points coincide.

1.3 A nonsplit extension at a collision

To see the support, regard an endomorphism as the action of t on a finite-dimensional $A = \mathbb{C}[t]$-module. For $\lambda \in \mathbb{C}$, put $L_\lambda = A/(t - \lambda)$. An extension

$$0 \longrightarrow L_\lambda \longrightarrow E \longrightarrow L_\mu \longrightarrow 0 \tag{1.6}$$

has, after choosing a lift of the quotient basis, the matrix

$$T_E = \begin{pmatrix} \lambda & c \\ 0 & \mu \end{pmatrix}.$$

Replacing that lift by itself plus b times the submodule basis changes c to $c + b(\lambda - \mu)$. If $\lambda \neq \mu$, taking $b = -c/(\lambda - \mu)$ splits the extension. If $\lambda = \mu$, the parameter c is unchanged.

The free resolution $0 \to A \xrightarrow{t-\mu} A \to L_\mu \to 0$ computes the full derived Hom by the two-term complex

$$\operatorname{RHom}_A(L_\mu, L_\lambda) \simeq [\, \mathbb{C} \xrightarrow{\lambda-\mu} \mathbb{C} \,], \qquad \text{in degrees } 0, 1. \tag{1.7}$$

The degree-one basis is chosen for the displayed differential. Both Hom and Ext^1 vanish at distinct supports. At coincident support they are each one-dimensional. In particular, the splitting off the diagonal is unique as a splitting of (1.6).

At $\lambda = \mu = 0$ and $c = 1$ the middle term is $A/(t^2)$. Its exact sequence is

$$0 \longrightarrow A/(t) \xrightarrow{1 \mapsto t} A/(t^2) \longrightarrow A/(t) \longrightarrow 0. \tag{1.8}$$

It is nonsplit because multiplication by t is nonzero on its middle term and zero on $A/(t) \oplus A/(t)$. The characteristic polynomial of T_E is nevertheless $(s - \lambda)(s - \mu)$ for every c. It records the two support points with multiplicity and forgets the extension class.

The flag correspondence retains still more than the invariant lines as a set. In the affine chart of $\mathbb{P}^1$ whose line is spanned by $(1, w)$, invariance under T_E is the equation

$$(\mu - \lambda)w - cw^2 = 0. \tag{1.9}$$

Indeed, the second coordinate of $T_E(1, w)$ must equal w times the first. This equation holds over every commutative $\mathbb{C}$-algebra, so it describes the scheme, including its infinitesimal points.

Proposition 1.2 (The invariant-line scheme of a Jordan block)**.** *For* $T_E = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$*, the scheme of invariant lines is* $\operatorname{Spec} \mathbb{C}[w]/(w^2)$*. It has length two and one underlying point.*

Proof. The only invariant line over $\mathbb{C}$ is the first coordinate line, so the entire invariant-line scheme lies in the displayed affine chart. Equation (1.9) becomes $w^2 = 0$. The quotient has basis $1, w$ over $\mathbb{C}$. □

Thus the two in (1.2) is also the length of this regular zero scheme of the invariant-line section. Replacing the scheme by its reduced point loses that length. For the scalar endomorphism the section instead vanishes identically, and the invariant-line scheme is all of $\mathbb{P}^1$. The Euler-class operation remains the specified pushforward.

1.4 Relations and ordered words

The first commutator belongs to a complete presentation of the weighted-loop algebra. Let $\mathcal{P}_\hbar$ be the unital associative R-algebra generated by e_r, $r \geq 0$, subject to

$$[e_{r+1}, e_s] - [e_r, e_{s+1}] = \hbar(e_r e_s + e_s e_r), \qquad r, s \geq 0. \tag{1.10}$$

Each e_r has charge one and cohomological degree $2r$.

Theorem 1.3 (An integral ordered basis)**.** *The map $e_r \mapsto a_r$ is an isomorphism $\mathcal{P}_\hbar \simeq \mathcal{S}$ of charge-graded R-algebras. The words $e_{r_1} \cdots e_{r_n}$ with $0 \leq r_1 \leq \cdots \leq r_n$, including the empty word, form an R-basis.*

Proof. For two roots x, y, formula (1.1) gives

$$[a_r, a_s] = 2\hbar \frac{y^r x^s - x^r y^s}{y - x}.$$

Subtracting this formula at $(r, s+1)$ from the one at $(r+1, s)$ gives $2\hbar(x^r y^s + x^s y^r)$. This also equals $\hbar(a_r \star a_s + a_s \star a_r)$, so the assignments respect (1.10).

These relations order the words. For $r > s$, write $C_{r,s} = [e_r, e_s]$. Then

$$C_{r,s} = C_{r-1,s+1} + \hbar(e_{r-1} e_s + e_s e_{r-1}).$$

The index gap decreases until $C_{s,s} = 0$ or $C_{s+1,s} = \hbar e_s^2$. Every correction has smaller sum of generator indices. Induction on that sum, and then on the number of inversions in the word, expresses every word in ordered form.

Filter $\mathcal{S}_n$ by total root degree, treating $\hbar$ as a coefficient. The leading part of an ordered product is

$$\sum_{\sigma \in \mathfrak{S}_n} x_1^{r_{\sigma(1)}} \cdots x_n^{r_{\sigma(n)}}.$$

It is the monomial symmetric polynomial with these exponents multiplied by the factorials of their multiplicities. Every such factorial is invertible in R. Distinct ordered words have distinct leading basis elements, which proves independence over R.

Subtracting the leading term of a symmetric polynomial reduces its root degree and proves spanning. The ordered spanning words in $\mathcal{P}_\hbar$ therefore map bijectively to a basis of $\mathcal{S}$. □

The proof works at $\hbar = 0$ and over the full ring $\mathbb{Q}[\hbar]$. It gives an algebraic deformation with its generators and relations. A field realization requires operations with a different target.

1.5 A coefficient transpose and the field comparison

Let $m_{1,1} : \mathcal{S}_1 \otimes_R \mathcal{S}_1 \to \mathcal{S}_2$ be the binary product (1.1). Let $\epsilon_r : \mathcal{S}_1 \to R$ extract the coefficient of x_1^r, and let $\epsilon^{(2)} : \mathcal{S}_2 \to R$ extract the constant term in both roots. Transposing the binary Hall product on this particular functional gives the finite tensor

$$m_{1,1}^* \epsilon^{(2)} = 2\epsilon_0 \otimes \epsilon_0 + \hbar(\epsilon_1 \otimes \epsilon_0 - \epsilon_0 \otimes \epsilon_1). \tag{1.11}$$

To verify the formula, evaluate on $a_r \otimes a_s$. The polynomial part of the product contributes a constant only for $r = s = 0$. Its part proportional to $\hbar$ contributes only when $r + s = 1$. Equation (1.2) gives the three coefficients.

For an R-module V, put $V((z)) = V[[z]][z^{-1}]$. A binary field operation has type

$$Y : V \otimes_R V \longrightarrow V((z)). \tag{1.12}$$

Thus $Y(v, z)w$ has a finite principal part for each pair v, w. This target differs from the tensor of coefficient functionals in (1.11). The formal variable z describes an insertion difference. The Chern roots in (1.1) describe characteristic classes on a moduli stack.

Problem 1.4 (Realizing extension operations as fields)**.** For specified Calabi–Yau moduli and extension correspondences, construct their oriented coefficient complexes on finite support configurations and their maps across collision strata. Construct a state module, a vacuum, translation, and operations of type (1.12), together with a comparison of their coefficients with the geometric operations. The comparison must preserve the empty-object unit, the chosen local vacuum, nested flag composition, and the attaching maps at collisions. Its domain and its tensor completions are part of the construction.

The one-loop model supplies the Hall multiplication and (1.11). Its coincident extension shows what a comparison based only on support would omit. For Calabi–Yau moduli, the coefficient construction additionally uses the cyclic duality class and compatible orientation data. These enter before the extension correspondences can define the required operations on complexes.

Chapter 2

Calabi–Yau Structures and Moduli of Objects

An extension varies, has automorphisms, and can meet obstructions to further variation. Calabi–Yau duality relates these three features. On a threefold it pairs first-order deformations with their obstruction spaces and pairs automorphisms with degree-three endomorphisms. These pairings determine the shifted tangent geometry and determinant lines that enter extension correspondences. Their construction retains both the underlying complexes and the coherence that makes the pairings vary over a moduli space.

Throughout, k is a field of characteristic zero. Complexes are cohomologically graded, with $V[r]^i = V^{i+r}$ and $d_{V[r]} = (-1)^r d_V$. The dual of a complex is $V^* = \mathrm{RHom}_k(V, k)$. A perfect k-complex has finite-dimensional total cohomology. All tensor products and Hom complexes are derived unless their displayed free models make this unnecessary.

2.1 The deformation spaces of a point

Let $A = k[x_1, x_2, x_3]$ and let $E = A/(x_1, x_2, x_3)$ be the module of the origin. Write $W = ke_1 \oplus ke_2 \oplus ke_3$. The Koszul resolution of E is

$$K^{-p} = A \otimes \bigwedge^p W, \qquad d(e_{i_1} \wedge \cdots \wedge e_{i_p}) = \sum_{a=1}^{p} (-1)^{a-1} x_{i_a} e_{i_1} \wedge \cdots \widehat{e_{i_a}} \cdots \wedge e_{i_p}. \tag{2.1}$$

Here each e_i has cohomological degree -1. The sequence x_1, x_2, x_3 is regular: after quotienting by its first $i-1$ terms, x_i is still a non-zero-divisor. Successively tensoring the corresponding two-term resolutions therefore proves that $K \to E$ is a resolution. After applying $\mathrm{Hom}_A(-, E)$, its differential is zero.

Proposition 2.1. *Let ξ_1, ξ_2, ξ_3 be the basis dual to e_1, e_2, e_3. The Yoneda algebra of the origin is*

$$\mathrm{Ext}^*_A(E, E) = \bigwedge^*(k\xi_1 \oplus k\xi_2 \oplus k\xi_3), \qquad |\xi_i| = 1.$$

The functional $\tau(\xi_1\xi_2\xi_3) = 1$, zero in other degrees, gives a perfect pairing $\mathrm{Ext}^p_A(E, E) \otimes \mathrm{Ext}^{3-p}_A(E, E) \to k$ by multiplication.

Proof. The resolution gives the stated graded vector space. To compute its product, let ι_i be contraction with e_i^* on the exterior factor of K. It is an A-linear endomorphism of degree one. The relations $\iota_i\iota_j + \iota_j\iota_i = 0$ and $d = \sum_i x_i \iota_i$ show that each ι_i is closed in $\mathrm{End}_A(K)$.

Composition with $K \to E$ identifies their products with $(-1)^{p(p-1)/2}$ times the dual exterior basis in degree p of $\mathrm{Hom}_A(K, E)$. This composition is a quasi-isomorphism because K is a bounded free resolution. It computes the Yoneda product and proves the exterior presentation. A basis monomial pairs only with its complementary monomial, with coefficient 1 or -1. □

Thus the four dimensions are 1, 3, 3, 1. The three parameters moving the point occur in degree one. Their three duals occur in degree two. Degree zero and degree three supply the remaining pair. This

computation already distinguishes a three-dimensional space of ordinary points from the derived moduli of their modules. We next identify the categorical structure responsible for its duality.

2.2 Two finiteness conditions

A differential graded category has complexes of morphisms and composition compatible with their differentials. We use small k-linear dg categories, and pass to perfect modules when shifts, cones and direct summands are needed. For a dg algebra A, the category $\mathrm{Perf}(A)$ consists of the direct summands of finite iterated cones of shifts of the free right module A. A *generator* is an object whose shifts, cones and direct summands generate the category in this sense.

These operations make an algebra presentation useful without making it intrinsic. If G generates a dg category and $A = \mathrm{RHom}(G, G)$, the functor $\mathrm{RHom}(G, -)$ identifies its perfect completion with $\mathrm{Perf}(A)$. The identification of morphism complexes holds on G and extends under cones and summands in both variables. Its image is precisely the subcategory generated by A. An equivalence of perfect module categories is called a *Morita equivalence*.

Definition 2.2. A dg category C is *proper* if $\mathrm{RHom}_C(E, F)$ is perfect over k for every E, F. It is *smooth* if its diagonal bimodule $(E, F) \mapsto \mathrm{RHom}_C(E, F)$ is perfect over $C^{\mathrm{op}} \otimes C$.

Properness concerns the size of each space of morphisms. Smoothness concerns a finite resolution of the identity functor. For an algebra A, put $A^e = A \otimes A^{\mathrm{op}}$, with the graded opposite and the Koszul tensor signs. Smoothness of $\mathrm{Perf}(A)$ means that the diagonal A is perfect as an A^e-module. Properness means that A is perfect over k.

The polynomial algebra $A = k[x_1, \dots, x_d]$ is smooth: the Koszul complex on $x_i \otimes 1 - 1 \otimes x_i$ resolves its diagonal over A^e. For $d > 0$ it is not proper, since $\mathrm{Hom}_A(A, A) = A$ is infinite-dimensional. The converse independence also has a small example.

A Serre functor on a proper category is an autoequivalence S with natural isomorphisms $\mathrm{RHom}(E, F)^* \simeq \mathrm{RHom}(F, SE)$.

Proposition 2.3. *For $D = k[\epsilon]/(\epsilon^2)$, the category $\mathrm{Perf}(D)$ is proper and has the identity as its Serre functor, but is not smooth.*

Proof. The functional $\tau_D(a + b\epsilon) = b$ has multiplication pairing $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ in the basis $1, \epsilon$. It identifies D with its linear dual as a bimodule. For bounded complexes of finite projective modules, evaluation gives

$$\mathrm{RHom}_D(P, Q)^* \simeq \mathrm{RHom}_D(Q, P \otimes_D D^*) \simeq \mathrm{RHom}_D(Q, P).$$

The first identification can be checked on free modules and then extended under cones and summands. All these Hom complexes are perfect over k. This proves properness and the asserted Serre duality.

For smoothness, set $B = D^e = k[x, y]/(x^2, y^2)$. The diagonal is the quotient sending x, y to ϵ, and has the free resolution

$$\cdots \xrightarrow{x-y} B \xrightarrow{x+y} B \xrightarrow{x-y} B \longrightarrow D \longrightarrow 0. \tag{2.2}$$

Both consecutive products vanish. In the basis $1, x, y, xy$,

$$\ker(x - y) = \langle x + y, xy \rangle = \mathrm{im}(x + y), \qquad \ker(x + y) = \langle x - y, xy \rangle = \mathrm{im}(x - y).$$

The kernel of $B \to D$ is also $\langle x - y, xy \rangle$, proving exactness.

Tensoring with $B/(x, y) = k$ kills every differential, so $\mathrm{Tor}_i^B(D, k) = k$ for every $i \geq 0$. A perfect diagonal would give zero Tor in all sufficiently high degrees. □

Smoothness and properness together ensure its existence on the perfect category. The proposition shows that smoothness is not necessary for existence. The distinction matters when a Calabi–Yau condition is stated using a Serre functor.

2.3 A VOLUME CLASS AND A TRACE

There are two duals of the diagonal. In an algebra presentation they are

$$A^! = \mathrm{RHom}_{A^e}(A, A^e), \qquad A^* = \mathrm{RHom}_k(A, k).$$

In the first expression the resulting right A^e-module is regarded as a bimodule by exchanging the two factors with the graded symmetry. When both finiteness conditions hold, tensoring with A^* is the Serre functor and tensoring with $A^!$ is its inverse [13, Proposition 2.3 and Example 2.12].

To state the coherent duality datum, let $C(A)$ denote the Hochschild chain complex. We write $C_j(A)$ for its cohomological degree $-j$. Its Hochschild differential b lowers j by one. The Connes operator B raises j by one and satisfies

$$b^2 = B^2 = bB + Bb = 0.$$

A complex with these two operators is a *mixed complex*. With a formal variable u of homological degree -2, define the negative cyclic and cyclic complexes by

$$CC^-(A) = (C(A)[[u]], b + uB), \qquad CC(A) = (C(A)((u))/uC(A)[[u]], b + uB). \tag{2.3}$$

The first uses products in each total degree. In the second, each representative has only finitely many nonpositive powers of u. These complexes model the homotopy fixed points and homotopy orbits, respectively, of the circle action on Hochschild chains. Their maps below use our cohomological shift convention.

Definition 2.4. A *left Calabi–Yau structure of dimension d* on a smooth dg category is a class $\eta : k[d] \to CC^-(C)$ with the following nondegeneracy property. Its underlying Hochschild class induces an equivalence of bimodules

$$C^! \xrightarrow{\sim} C[-d].$$

A *right Calabi–Yau structure of dimension d* on a proper dg category is a map $\tau : CC(C) \to k[-d]$ whose underlying Hochschild functional induces an equivalence

$$C[d] \xrightarrow{\sim} C^*.$$

These are the left and right structures of [13, Definitions 3.2 and 3.5].

The maps have different variances: η is a homology class, whereas τ is a functional. Smoothness identifies Hochschild chains with $\mathrm{RHom}_{A^e}(A^!, A)$, which explains the map induced by η. Dualizing Hochschild chains identifies their functionals with $\mathrm{RHom}_{A^e}(A, A^*)$, which explains the map induced by τ. If one retains only these nondegenerate Hochschild data, the structure is called *weak*. A cyclic lift is part of a left or right structure as defined here.

On a smooth proper category the underlying left and right dualities give the same Serre shift $S \simeq [d]$, since the two dualizing functors are inverse. The coherent structures are related by the circle-equivariant evaluation pairing. An object E is *proper* if $\mathrm{RHom}(F, E)$ is perfect over k for every perfect F. A left

structure induces a right structure on the full subcategory of proper objects [13, Theorem 3.1]. This is the duality used for objects with proper support in a nonproper space.

The negative cyclic condition is visible as a sequence of equations. Writing a degree-d cycle as $\eta = \eta_0 + u\eta_1 + u^2\eta_2 + \cdots$, where $\eta_r \in C_{d+2r}(A)$, gives

$$b\eta_0 = 0, \qquad b\eta_{r+1} = -B\eta_r \quad (r \geq 0). \tag{2.4}$$

Thus the first additional requirement is a primitive for $B\eta_0$.

For example, take a mixed complex with basis e, f in homological degrees 0, 1, with $b = 0$ and $Be = f$. Its class e has no negative cyclic lift. Adjoining g in degree two with $bg = -f$ and $Bg = 0$ gives the lift $e + ug$. This example is a statement about mixed complexes. It makes no assertion that the two complexes are Hochschild complexes of smooth proper categories.

2.4 The polynomial volume class

For $A = k[x_1, \ldots, x_d]$, the left structure can be constructed in coordinates. In normalized Hochschild chains write $a_0[a_1|\cdots|a_n]$. A unit in any bracketed slot gives zero. The degree-d chain

$$\eta_0 = \sum_{\sigma \in S_d} \operatorname{sgn}(\sigma) 1[x_{\sigma(1)}|\cdots|x_{\sigma(d)}] \tag{2.5}$$

is a negative cyclic cycle with no higher u-terms.

Proposition 2.5. *The class* (2.5) *defines a left Calabi–Yau structure of dimension d on* $\operatorname{Perf}(A)$. *Its underlying bimodule equivalence is $A^! \simeq A[-d]$.*

Proof. The adjacent multiplication terms in $b\eta_0$ cancel in pairs by commutativity and antisymmetry. The two end terms cancel as well. Every term of $B\eta_0$ has the initial unit of η_0 in a bracketed slot, and therefore vanishes in the normalized complex. This proves the negative cyclic equations.

Let K_Δ be the free bimodule Koszul resolution on $x_1 - y_1, \ldots, x_d - y_d$. The regular-sequence argument used for (2.1) applies to this sequence too. Its dual $\operatorname{Hom}_{A^e}(K_\Delta, A^e)$ has cohomology only in degree d, where it is A.

The antisymmetrization map from this Koszul resolution to the bar resolution sends its top generator to the sum in (2.5) after tensoring with the diagonal. Indeed, its differential telescopes to the terms $x_i - y_i$. All interior products cancel in adjacent transpositions. The induced map from the dual Koszul complex sends the dual top generator to 1 in $A[-d]$. It is an isomorphism on its only cohomology group, proving nondegeneracy and fixing the normalization. □

Write $\mathrm{HH}^p(A) = H^p \operatorname{RHom}_{A^e}(A, A)$ and $\mathrm{HH}_q(A) = H^{-q}(A \otimes^{\mathbf{L}}_{A^e} A)$. The same resolution computes

$$\mathrm{HH}^p(A) = A \otimes \bigwedge^p \langle \partial_1, \ldots, \partial_d \rangle, \qquad \mathrm{HH}_q(A) = A \otimes \bigwedge^q \langle dx_1, \ldots, dx_d \rangle.$$

Cap product with $dx_1 \wedge \cdots \wedge dx_d$ identifies $\mathrm{HH}^p(A)$ with $\mathrm{HH}_{d-p}(A)$ by contraction. For example, $\iota_{\partial_i}(dx_1 \wedge \cdots \wedge dx_d) = (-1)^{i-1} dx_1 \wedge \cdots \widehat{dx_i} \cdots \wedge dx_d$.

This duality changes how the degrees are displayed. The graded commutator of cochain insertions defines the Hochschild Gerstenhaber bracket, of degree −1. Its value on a derivation and a function is evaluation, so $[\partial_i, x_j] = \delta_{ij}$. Deforming an object instead introduces a shift of its endomorphism complex.

2.5 The tangent complex and its pairing

For a connective commutative dg k-algebra R, a *pseudo-perfect family* over R is a dg functor $C^{\mathrm{op}} \to \mathrm{Perf}(R)$. The derived space of such functors and their equivalences defines the moduli prestack $\mathcal{M}_C(R)$. In an algebra presentation, it is the moduli of right $A \otimes R$-modules whose underlying R-complex is perfect. This condition concerns the family over the base. For a smooth category its fibers are perfect modules over the category as well.

We impose a finite-type hypothesis for the geometric statements below. The perfect category of C is of *finite type* if it is compact in the Morita category of small idempotent-complete dg categories. This means that mapping spaces out of it commute with filtered colimits. Under this hypothesis $\mathcal{M}_C$ is locally a derived Artin stack of finite presentation and has a perfect cotangent complex [14, Remark 3.9]. Its cotangent complex $\mathbb{L}_{\mathcal{M}}$ represents derivations into square-zero extensions. Its dual is the tangent complex $\mathbb{T}_{\mathcal{M}}$.

The affine-three-space example satisfies the finite-type hypothesis. One finite free dg algebra model has generators x_1, x_2, x_3 in degree zero, h_{12}, h_{13}, h_{23} in degree -1, and t in degree -2, with differential

$$\begin{aligned} dx_i &= 0, \qquad dh_{ij} = x_i x_j - x_j x_i, \\ dt &= [x_1, h_{23}] - [x_2, h_{13}] + [x_3, h_{12}]. \end{aligned} \tag{2.6}$$

Here $[x, h] = xh - hx$, and d extends by the graded Leibniz rule. The Jacobi identity gives $d^2 t = 0$. The map to $A = k[x_1, x_2, x_3]$ sends h_{ij}, t to zero and is a quasi-isomorphism.

For the augmentation $x_i \mapsto 0$, the reduced bar construction of A has words in its augmentation ideal shifted by one. Its differential multiplies adjacent entries, and its coproduct cuts words. The exterior coalgebra on three degree--1 primitive elements maps to it by antisymmetrization. The Koszul resolution of k proves that this map induces an isomorphism on Tor, hence is a quasi-isomorphism.

The free dg algebra (2.6) is its cobar: desuspend the reduced coalgebra and extend the reduced coproduct as a differential. Give x_i, h_{ij}, t weights 1, 2, 3, respectively. Cobar preserves this quasi-isomorphism, as the tensor-length filtration is finite in each polynomial weight. Bar–cobar contraction then gives the asserted map to A. There are only seven free generators, attached in the displayed order. Maps out of this finite cell presentation commute with filtered colimits, and so do maps out of its perfect category.

At an object E there is a natural equivalence

$$\mathbb{T}_{\mathcal{M},E} \simeq V_E[1], \qquad V_E = \mathrm{RHom}_C(E, E) \quad [14, \text{Proposition 3.3}]. \tag{2.7}$$

The shift can be seen directly. On a cofibrant module representative, a first-order change of differential is $d_E + \varepsilon a$, with $|a| = 1$ and $\varepsilon^2 = 0$. Squaring gives $d_E a + a d_E = 0$. Conjugation by $1 + \varepsilon h$, with $|h| = 0$, changes a by a boundary. Thus degree-one endomorphisms describe deformations, degree-zero endomorphisms describe their infinitesimal automorphisms, and degree-two endomorphisms receive the obstruction to lifting a first-order deformation. The derived statement (2.7) retains this entire complex.

Suppose that E carries the proper trace induced by a d-dimensional Calabi–Yau structure. It gives a map

$$\tau_E : V_E \longrightarrow k[-d]$$

whose cohomology pairing is perfect and satisfies $\tau_E(xy) = (-1)^{|x||y|}\tau_E(yx)$. When we compute with a cyclic chain model, this identity holds on that model. Put $sx \in V_E[1]$ for the shift of a homogeneous element x. Thus $|sx| = |x| - 1$. The corresponding tangent pairing is

$$\omega_E(sx, sy) = (-1)^{|x|}\tau_E(xy) \;:\; V_E[1] \otimes V_E[1] \longrightarrow k[2 - d]. \tag{2.8}$$

Lemma 2.6. *On cohomology, the pairing* (2.8) *is graded alternating and nondegenerate. In a cyclic chain model it is an alternating map of complexes.*

Proof. Write $p = |x|$ and $q = |y|$. Cyclicity gives

$$-(-1)^{(p-1)(q-1)}\omega_E(sy, sx) = -(-1)^{(p-1)(q-1)+q+pq}\tau_E(xy) = (-1)^p\tau_E(xy).$$

A nonzero value has $p + q = d$, so the two tangent degrees sum to $d - 2$. This proves both the alternating sign and the displayed target shift.

For chain compatibility, use $d(sx) = -s(dx)$ and $\tau_E(d(xy)) = 0$. The chain-map equation becomes $(-1)^p\tau_E(d(xy)) = 0$. The underlying trace pairing is perfect, and shifting preserves its nondegeneracy. □

The closedness of the global form has more content than this linear calculation. To specify it, let

$$\mathrm{DR}(\mathcal{M}) = \mathrm{R}\Gamma\left(\mathcal{M}, \prod_{q\geq 0}(\wedge^q \mathbb{L}_{\mathcal{M}})[-q]\right)$$

be the Hodge-completed derived de Rham complex. On its qth factor the internal differential is $(-1)^q d$. The de Rham differential goes to the next factor and has total degree one. Its filtration F^p retains the terms with $q \geq p$.

A closed p-form of degree n is a point of the derived mapping space $\mathrm{Map}(k[-n], F^p\,\mathrm{DR}(\mathcal{M})[p])$. Its cocycle has compatible components in every Hodge weight at least p. Projection to the first term gives the underlying p-form. A closed two-form is *n-shifted symplectic* if its underlying pairing induces an equivalence $\mathbb{T}_{\mathcal{M}} \simeq \mathbb{L}_{\mathcal{M}}[n]$.

Theorem 2.7 (Brav–Dyckerhoff)**.** *Let C be of finite type and equipped with a left Calabi–Yau structure of dimension d. Its moduli of pseudo-perfect objects carries a $(2-d)$-shifted symplectic form. With the suspension convention above, its underlying pairing at E is* (2.8)*.*

Construction and nondegeneracy. The universal family induces a circle-equivariant map on Hochschild chains. Pass to negative cyclic complexes and take the weight-two component of the derived de Rham realization. This sends η to a closed two-form of degree $2 - d$, by [14, Proposition 5.2]. Its higher components retain the cyclic coherence in η.

The underlying form is the composition of endomorphisms followed by the induced trace [14, Proposition 5.3]. Under (2.7), suspension gives the sign in (2.8). The bimodule nondegeneracy of η makes this trace pairing perfect, hence identifies the tangent complex with the shifted cotangent complex. This is [14, Theorem 5.5(1)]. □

For the origin in affine three-space, the tangent cohomology is therefore

$$H^{-1}(\mathbb{T}_E) = k, \quad H^0(\mathbb{T}_E) = k^3, \quad H^1(\mathbb{T}_E) = \bigwedge^2 k^3, \quad H^2(\mathbb{T}_E) = k.$$

The form has degree -1. It pairs the first and last spaces and the middle two, with the explicit signs above. The ordinary family of points is smooth, but this does not remove the higher tangent groups of the derived object moduli. The same formula gives degree zero for a Calabi–Yau surface and degree two for a zero-dimensional Calabi–Yau category.

2.6 Proper support and compact geometry

Over $A = k[x_1, x_2, x_3]$, the free module has infinite-dimensional endomorphisms and is not a pseudo-perfect k-point. A module with finite-dimensional total cohomology is pseudo-perfect. Such a module is also perfect over A. Tensor a finite diagonal resolution with the module. Each resulting free A-term has a perfect k-complex of coefficients. The point calculation thus belongs to a proper-support subcategory of a smooth nonproper category.

For comparison, let X be a smooth projective variety of dimension d with a chosen trivialization of its canonical line ω_X. Serre duality gives, for perfect complexes E, F,

$$\operatorname{Ext}^i_X(E, F)^* \simeq \operatorname{Ext}^{d-i}_X(F, E). \tag{2.9}$$

Here the usual functor $(-) \otimes \omega_X[d]$ becomes the shift $[d]$ by the chosen trivialization. This trace is a geometric input. When a closed form on moduli is used, its compatible cyclic structure is retained as in Theorem 2.7.

Corollary 2.8. *In a nonzero proper d-Calabi–Yau category with $d > 0$, no nonzero object T has $\operatorname{Ext}^i(T, T) = 0$ for every $i \neq 0$. In particular, there is no tilting generator with this vanishing.*

Proof. The identity makes $\operatorname{Hom}(T, T)$ nonzero. Duality identifies its dual with $\operatorname{Ext}^d(T, T)$, which is therefore nonzero. □

A compact generator can still exist. Its endomorphisms must retain their dg structure, including the class dual to its identity. Local tilting presentations on nonproper spaces have different hypotheses and do not contradict this corollary.

2.7 Determinant lines along extensions

For a perfect complex V, its determinant is $\det V = \bigotimes_i (\det H^i V)^{(-1)^i}$. An exponent -1 denotes the dual line. We use determinant superlines: their parity is $\chi(V) \bmod 2$, and interchanging lines of parities a, b has sign $(-1)^{ab}$. The determinant functor is multiplicative on exact triangles, with these symmetry signs.

On the object moduli of a three-dimensional Calabi–Yau category set $\mathcal{D}_E = \det \operatorname{RHom}(E, E)$. Since $\mathbb{T}_E = V_E[1]$, the determinant of the cotangent complex is $\det \mathbb{L}_E = \det(V_E^*[-1]) = \mathcal{D}_E$. If $\operatorname{Ext}^i(E, E) = 0$ outside $0 \leq i \leq 3$, duality gives

$$\mathcal{D}_E \simeq (\det \operatorname{Ext}^0(E, E) \otimes (\det \operatorname{Ext}^1(E, E))^{-1})^{\otimes 2}. \tag{2.10}$$

Indeed, the degree-two space is dual to degree one and degree three is dual to degree zero. The parity of V_E is even, as the same pairings show.

Extensions impose a further relation. Consider an exact triangle $E_1 \to E \to E_2 \to E_1[1]$ on which all Hom complexes are perfect over k. Put $U = \operatorname{RHom}(E_1, E_2)$. Applying Hom in each variable and the determinant functor gives

$$\mathcal{D}_E \simeq \mathcal{D}_{E_1} \otimes \mathcal{D}_{E_2} \otimes \det U \otimes \det \operatorname{RHom}(E_2, E_1).$$

The factors are ordered as displayed using the determinant symmetry.

Calabi–Yau duality identifies the last complex with $U^*[-3]$. Dualizing and shifting by an odd integer each invert a determinant. Together they leave it unchanged. Consequently

$$\mathcal{D}_E \simeq \mathcal{D}_{E_1} \otimes \mathcal{D}_{E_2} \otimes (\det U)^{\otimes 2}. \tag{2.11}$$

An *orientation datum* for these extension spaces consists of a superline bundle $\mathcal{L}$ with $\mathcal{L}^{\otimes 2} \simeq \mathcal{D}$ and isomorphisms

$$\mathcal{L}_E \simeq \mathcal{L}_{E_1} \otimes \mathcal{L}_{E_2} \otimes \det \mathrm{RHom}(E_1, E_2) \tag{2.12}$$

whose squares are (2.11). These are the square-root and triangle relations of [99, §5.2].

We require the choices to be compatible with base change and with successive extensions. For a flag with three successive quotients, (2.12) has two iterated composites. These must coincide, using the determinant symmetry just fixed. The analogous coherences are retained on longer flags.

Formula (2.10) supplies a square root on a stratum where its cohomology spaces form vector bundles. It does not supply the gluing across dimension jumps or the flag isomorphisms (2.12). Those are part of the orientation datum. Once such a datum is chosen, a Hall construction can use the resulting oriented coefficients on extension correspondences. The pullback, direct image and comparison of these coefficients must still be constructed for the chosen cohomology theory. These maps live on flag spaces and their incidence correspondences.

Chapter 3

Derived flags and Hall coefficients

A flag correspondence can be used with coherent sheaves or with constructible sheaves. Coherent coefficients retain nilpotent functions. Constructible coefficients retain the topology and its supported maps. A degeneration of length-three quotients separates these two calculations. Its complete flags acquire a projective line, while their generic point has scheme-theoretic length six. The multiplicities determine a cycle trace.

The coherent direct image is free of rank six over the parameter ring. Its map to functions on the special fiber has a three-dimensional kernel.

3.1 A noncurvilinear point and its complete flags

Let k be a field of characteristic zero. A length-three quotient of the plane need not admit one algebra generator. This first occurs at

$$A_0 = k[x, y]/(x, y)^2.$$

Every element of A_0 is $c + w$, with $c \in k$ and $w^2 = 0$, so it generates a subalgebra of dimension at most two.

No change of the chosen polynomial coordinate makes its multiplication cyclic on A_0. The pair x, y, together with the unit, nevertheless generates the algebra. It gives a chart in which both the quotient and its ideal flags can be calculated through this point.

Definition 3.1. A local length-three k-algebra with residue field k is called *curvilinear* if it is isomorphic to $k[u]/(u^3)$.

Thus A_0 is noncurvilinear.

Failure of a fixed coordinate to generate is weaker. For instance $k[x, y]/(x, y^2)$ is not generated by x, but is generated by $x + y$. The failure at A_0 persists under every such change of coordinate.

3.1.1 The simultaneous multiplication chart

Consider the open of $\mathrm{Hilb}^3(\mathbb{A}^2_k)$ where $1, x, y$ form the tautological basis. Write its border products as

$$x^2 = a + bx + cy, \qquad xy = d + ex + fy, \qquad y^2 = g + hx + iy.$$

Theorem 3.2. *This open is $\mathbb{A}^6_k$, with coordinates b, c, e, f, h, i. Its other coefficients are*

$$a = c(e - i) + f(f - b), \qquad d = ch - ef, \qquad g = e(e - i) + h(f - b). \tag{3.1}$$

The universal multiplication matrices, in the basis $(1, x, y)$, are

$$X = \begin{pmatrix} 0 & a & d \\ 1 & b & e \\ 0 & c & f \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & d & g \\ 0 & e & h \\ 1 & f & i \end{pmatrix}.$$

These formulas represent the open chart over every commutative k-algebra.

Proof. The coefficients of x, y in $x^2 y = x(xy)$ and $xy^2 = (xy)y$ give (3.1). Their remaining constant coefficients vanish after this substitution. Equivalently, direct multiplication gives $XY = YX$.

Over any base algebra, the matrices therefore define a $k[x, y]$-module structure on the free rank-three module. The vector 1 generates it, since $X1 = x$ and $Y1 = y$. The resulting surjection from the polynomial algebra has an ideal as kernel and gives a rank-three quotient with the prescribed basis. Conversely, any quotient in the open has unique matrices of this form, and commutativity forces the displayed eliminations. The two constructions are inverse and commute with base change. □

Set $c = t$ and $b = e = f = h = i = 0$. The resulting algebra over $k[t]$ is

$$\mathcal{A} = k[t, x, y]/(x^2 - ty, xy, y^2). \tag{3.2}$$

It is free with basis $1, x, y$. Its fiber at zero is A_0, whereas for a nonzero value of t it is $k[x]/(x^3)$, with $y = x^2/t$. This last presentation belongs only to the punctured base.

3.1.2 *The full ordinary flag scheme*

Let $T = \mathbb{A}^1_k$. Define $p : \mathcal{F} \to T$ to parameterize ideal flags

$$0 \subset L \subset M \subset \mathcal{A}, \qquad \operatorname{rank} L = 1, \qquad \operatorname{rank} M = 2,$$

with all successive quotients locally free. The invariance conditions under multiplication are closed in the relative flag variety of $\mathcal{A}$, so p is proper. It is the complete nested Hilbert correspondence of lengths 1, 2, 3 pulled back along (3.2).

The quotient $\mathcal{A}/M$ has a unital character $x \mapsto \lambda$, $y \mapsto \mu$. Its equations are

$$\lambda^2 = t\mu, \qquad \lambda\mu = 0, \qquad \mu^2 = 0. \tag{3.3}$$

Its kernel has basis $x - \lambda$, $y - \mu$.

A line in it is locally generated by $A(x - \lambda) + B(y - \mu)$, with $[A : B] \in \mathbb{P}^1$. Multiplication by x and y gives the respective coefficient columns

$$(-A\lambda - B\mu, At), \qquad (0, -A\lambda - B\mu).$$

Taking determinants with (A, B) shows that the line is an ideal precisely when

$$tA^2 + \lambda AB + \mu B^2 = 0, \qquad \lambda A^2 + \mu AB = 0. \tag{3.4}$$

In particular the quotient character retains nilpotent values in a family. It has not been fixed to the origin.

Theorem 3.3. *On $B \neq 0$, put $v = A/B$. The coordinate ring is*

$$R_B = k[t, v, \lambda]/(F, G), \qquad F = tv^3, \quad G = \lambda^2 + tv\lambda + t^2v^2, \tag{3.5}$$

with $\mu = -tv^2 - v\lambda$. On $A \neq 0$, put $r = B/A$. Its coordinate ring is

$$R_A = k[r, \mu]/(\mu^2), \qquad t = 0, \quad \lambda = -r\mu. \tag{3.6}$$

The overlap is $r = v^{-1}$, $\mu = -v\lambda$.

Proof. On the first chart the first line equation eliminates μ. The second becomes $-F = 0$, and the first character equation becomes $G = 0$. The other character equations reduce to

$$\lambda\mu = -vG + tF, \qquad \mu^2 = v^2G + \lambda F.$$

They impose no further ordinary equations.

On the second chart the second line equation gives $\lambda = -r\mu$. The first then gives $t = 0$. The character equations reduce to $\mu^2 = 0$. On the overlap v is a unit, so $tv^3 = 0$ gives $t = 0$ and the stated transition. Conversely these substitutions satisfy every equation (3.3)–(3.4). □

After inverting t, set $q = tv$. Then

$$R_B[t^{-1}] = k[t, t^{-1}, q, \lambda]/(q^3, \lambda^2 + q\lambda + q^2). \tag{3.7}$$

It is free over $k[t, t^{-1}]$ with basis $1, q, q^2, \lambda, q\lambda, q^2\lambda$. Every nonzero fiber is one nonreduced point of length six. The A-chart is empty there.

At $t = 0$, the two charts are $k[v, \lambda]/\lambda^2$ and $k[r, \mu]/\mu^2$, glued by the same transition. The special fiber is a ribbon with reduced curve $\mathbb{P}^1$. Its points are the lines in the square-zero ideal (x, y).

The total reduced scheme is a horizontal copy of T and a vertical $\mathbb{P}^1$, meeting at one point. Their generic scheme multiplicities are six and two, respectively. Although $\mathcal{A}$ is flat over $k[t]$, its flag scheme is not: $v^3 \neq 0$ in R_B, as its image modulo t shows, while $tv^3 = 0$.

Proposition 3.4. *The exchange $q \leftrightarrow \lambda$ defines an automorphism of (3.7). It has no extension to a $k[t]$-algebra endomorphism of R_B with these prescribed root images.*

Proof. The three successive x-eigenvalues on the flag are $q, -q - \lambda, \lambda$. Their elementary symmetric functions vanish. In the displayed quotient, $\lambda^3 = q^3 = 0$, so interchanging q, λ preserves its defining ideal and is an involution. An extension would satisfy $t\varphi(v) = \lambda$. But λ has nonzero image in $R_B/tR_B = k[v, \lambda]/\lambda^2$, which excludes this equation. □

There is a corresponding global obstruction on $\mathscr{F}$. Let κ_L be the eigenvalue of x on the tautological line L. It is tv on the B-chart and zero on the A-chart, so $\kappa_L \in t\mathcal{O}_{\mathscr{F}}$ locally. Every automorphism over T preserves that ideal sheaf. It therefore cannot pull κ_L back to the quotient eigenvalue λ, which is not in the ideal on the B-chart. This excludes the prescribed root exchange on the whole flag scheme as well.

3.1.3 The incidence enhancement and its specialization

We use the cochain grading, with all polynomial variables in degree zero. Before elimination on the B-chart, the five incidence equations are

$$r_1 = \lambda^2 - t\mu, \qquad r_2 = \lambda\mu, \qquad r_3 = \mu^2,$$
$$r_4 = tv^2 + v\lambda + \mu, \quad r_5 = v^2\lambda + v\mu.$$

Their specified derived zero locus has algebra

$$K_{\mathrm{inc}} = k[t, v, \lambda, \mu] \otimes \Lambda(e_1, \dots, e_5), \qquad |e_j| = -1, \quad de_j = r_j. \tag{3.8}$$

The full list of equations is part of this definition.

Theorem 3.5. *There is a quasi-isomorphism of commutative dg algebras*

$$K_{\mathrm{inc}} \simeq \big(k[t, v, \lambda] \otimes \Lambda(e_F, e_G),\ de_F = F,\ de_G = G\big) \otimes \Lambda(\chi_1, \chi_2), \tag{3.9}$$

where the two χ's have degree -1 and are closed. Its cohomology is $R_B \otimes \Lambda(\chi_1, \chi_2)$.

Proof. Set $\nu = r_4$, so $\mu = \nu - tv^2 - v\lambda$. Change the odd generators by

$$\begin{aligned} &e_\nu = e_4, && e_G = e_1 + te_\nu, && e_F = ve_\nu - e_5,\\ &\chi_1 = e_2 - \lambda e_\nu + ve_G - te_F,\\ &\chi_2 = e_3 - \big(\nu - 2v(tv+\lambda)\big)e_\nu - v^2 e_G - \lambda e_F. \end{aligned}$$

The even change has coefficient one. Ordered as $(e_4, e_1, e_5, e_2, e_3)$, the odd change is triangular with diagonal $(1, 1, -1, 1, 1)$. It is therefore invertible.

Substitution gives

$$\begin{aligned} &r_1 = G - t\nu, && r_5 = v\nu - F,\\ &r_2 = \lambda\nu - vG + tF,\\ &r_3 = \big(\nu - 2v(tv+\lambda)\big)\nu + v^2 G + \lambda F. \end{aligned}$$

Thus the new differentials are ν, G, F, 0, 0.

The factor $k[\nu] \otimes \Lambda(e_\nu)$ retracts to k, with homotopy

$$H(\nu^j) = e_\nu \nu^{j-1} \quad (j \geq 1), \qquad H(1) = H(e_\nu \nu^j) = 0.$$

With the Koszul tensor sign this gives (3.9). Finally F is a nonzero divisor in the polynomial ring, and G is monic in λ over $k[t, v]/(tv^3)$. They form a regular sequence, so their Koszul complex resolves R_B. □

At a nonzero value of t, the dimensions of this cohomology in degrees 0, −1, −2 are 6, 12, 6. At zero the derived fiber is instead

$$K_{\mathrm{inc}} \otimes^{\mathbb{L}}_{k[t]} k \simeq k[v, \lambda]/(\lambda^2) \otimes \Lambda(e_F, \chi_1, \chi_2). \tag{3.10}$$

Indeed the free Koszul model computes derived base change. Its F-differential becomes zero, while $de_G = \lambda^2$ remains regular. The new e_F class records the nonflat flag fiber. It is distinct from the two incidence syzygies.

Fixing the quotient character to the origin gives another base change. In (3.8), setting $\lambda = \mu = 0$ leaves $de_4 = tv^2$ and four closed generators. Its ordinary ring is $k[t, v]/(tv^2)$. Thus prescribing a quotient character changes even the ordinary flag ring, as well as the derived equations.

Over $\mathbb{C}$, the equation count also fixes a constructible normalization. For a polynomial section $s = (s_1, \ldots, s_r)$ on a smooth complex m-fold X, put $W = \sum_j \zeta_j s_j$ on $X \times \mathbb{A}^r$, with projection π. Let $j : Z(s) \hookrightarrow X$. Fiber-linear dimensional reduction [35, Theorem A.1] gives

$$R\pi_! \phi_W \mathbb{Q}_{X \times \mathbb{A}^r}[m + r - 1] \simeq j_* \mathbb{Q}_{Z(s)}[m - r]. \tag{3.11}$$

Here ϕ_W is the unshifted vanishing-cycle functor, extended by zero from the zero locus of W, with $\phi_0[-1] = \mathrm{id}$. The coefficients are pulled back rational constants.

For (F, G) on $\mathbb{A}^3$ the output shift is +1. For the five equations on $\mathbb{A}^4$ it is −1. Eliminating (ν, e_ν) preserves $m - r$.

The invertible odd change in the proof of Theorem 3.5 induces its inverse dual change of cut coordinates. In those coordinates the potential is $\zeta_\nu \nu + \zeta_G G + \zeta_F F$. The two cut coordinates corresponding to χ_1, χ_2 are unused. Deleting them changes $m - r$.

These are different normalized critical coefficients on the same classical space. Their identification with an oriented Hall coefficient requires a comparison of the specified obstruction theories.

3.1.4 Proper pushforward and the Gysin parameter

For constructible coefficients take $k = \mathbb{C}$ and work on complex points. All sheaf morphisms below belong to the bounded derived category of rational constructible sheaves. Let $i : \{0\} \hookrightarrow T = \mathbb{A}^1$. The flag family has a section given by $\lambda = \mu = 0$, $L = \mathbb{C}y$. Write $\mathscr{L}$ for its tautological ideal line. On the reduced vertical fiber it is $\mathcal{O}_{\mathbb{P}^1}(-1) \subset \mathcal{O}_{\mathbb{P}^1} \otimes (\mathbb{C}x \oplus \mathbb{C}y)$.

Theorem 3.6. *The proper coefficient object is*

$$\mathscr{P} := R p_* \mathbb{Q}_{\mathscr{F}} \simeq \mathbb{Q}_T \oplus i_* \mathbb{Q}[-2]. \tag{3.12}$$

The unit and restriction to the horizontal section split the first summand. Complex orientation, normalized by $\int_{\mathbb{P}^1} -c_1(\mathscr{L}) = 1$*, fixes the second.*

Proof. The generic fiber has underlying space a point and the special fiber has underlying space $\mathbb{P}^1$. Proper base change gives $R^0 p_* \mathbb{Q} = \mathbb{Q}_T$, $R^2 p_* \mathbb{Q} = i_* \mathbb{Q}$, and no other cohomology sheaves. The section retracts the unit. Its complement is supported at zero and has only $\mathbb{Q}$ in degree two, so it is $i_* \mathbb{Q}[-2]$. The stated integral identifies its generator with the oriented top class. □

Purity supplies the Gysin morphism

$$\gamma : i_* \mathbb{Q}[-2] = i_* i^! \mathbb{Q}_T \longrightarrow \mathbb{Q}_T.$$

Under adjunction it is the identity of $i^! \mathbb{Q}_T = \mathbb{Q}[-2]$, and is nonzero.

Theorem 3.7. *Every degree-zero derived morphism $\mathscr{P} \to \mathbb{Q}_T$ restricting to the identity off zero has, in the splitting* (3.12)*, the unique form*

$$f_a = (\mathrm{id}, a\gamma), \qquad a \in \mathbb{Q}. \tag{3.13}$$

All these maps induce the same ordinary stalk and global cohomology maps. On compactly supported degree-two cohomology, their matrix is $(1, a)$*.*

Proof. Connectedness gives $\mathrm{Hom}(\mathbb{Q}_T, \mathbb{Q}_T) = \mathbb{Q}$. Purity and adjunction give

$$\mathrm{Hom}(i_* \mathbb{Q}[-2], \mathbb{Q}_T) = \mathrm{Hom}(\mathbb{Q}[-2], \mathbb{Q}[-2]) = \mathbb{Q}.$$

The open restriction fixes the first coefficient to one. The second is the unique multiple of γ.

The ordinary pullback of γ at zero is a map $\mathbb{Q}[-2] \to \mathbb{Q}$, which vanishes in $D(\mathbb{Q})$. The same holds after ordinary global sections. Away from zero its source vanishes. Compact sections instead put both source and target in degree two. The Gysin map sends the oriented point class to the compact orientation class of the line, and is the identity in these bases. This gives $(1, a)$ and distinguishes every a. □

The extraordinary costalk detects the same scalar. Applying $i^!$ to the counit makes $i^! \gamma : \mathbb{Q}[-2] \to \mathbb{Q}[-2]$ the identity, so $i^! f_a$ also has matrix $(1, a)$. The costalk ranks alone do not determine this matrix.

The value $a = 0$ is restriction to the section. The value $a = 1$ adds the oriented exceptional contribution. Both are defined maps. Determining which map an independent Hall correspondence supplies requires its compact pairing. Ordinary stalk tests do not choose it.

3.1.5 *Independent products and higher compatibility*

Take r independent copies of p, over T^r. For $S \subseteq \{1, \dots, r\}$, let $i_S : T^{r-|S|} \hookrightarrow T^r$ set the coordinates indexed by S to zero. Proper Künneth and (3.12) give

$$\mathscr{P}^{\boxtimes r} \simeq \bigoplus_S i_{S,*}\mathbb{Q}_{T^{r-|S|}}[-2|S|].$$

Let γ_S be the product of the Gysin maps in those coordinates and the identities in the others.

Proposition 3.8. *Every degree-zero morphism $\mathscr{P}^{\boxtimes r} \to \mathbb{Q}_{T^r}$ equal to the identity on $(\mathbb{G}_m)^r$ is uniquely specified by coefficients $c_S \in \mathbb{Q}$, with $c_\varnothing = 1$, through*

$$f|_S = c_S \gamma_S.$$

A family of such maps preserves all independent external products, with singleton coefficients $a_1, \dots, a_r$, if and only if

$$c_S = \prod_{j \in S} a_j. \tag{3.14}$$

The first two-coordinate defect is $(c_{\{1,2\}} - a_1 a_2)\gamma_{\{1,2\}}$. It is nonzero precisely when its scalar is nonzero.

Proof. Coordinate purity gives $i_S^! \mathbb{Q}_{T^r} = \mathbb{Q}_{T^{r-|S|}}[-2|S|]$. Adjunction therefore identifies the derived Hom from each summand to the target with $R\Gamma(T^{r-|S|}, \mathbb{Q}) = \mathbb{Q}$ in degree zero. This proves the classification. Tensoring the singleton maps multiplies exactly the coefficients indexed by S, so gives (3.14). Conversely every refinement into singleton factors requires these coefficients.

All shifts are even. Under further regrouping the Gysin map is the same product of coordinate orientation classes, which gives the compatible higher comparisons. Equivalently, the indicated mapping spaces have no higher homotopy groups, so these equal classes have unique homotopies when needed. On H_c^{2r}, every summand contributes one oriented line and γ_S has coefficient one. It detects the stated defect. □

This proposition concerns external products of the explicit flag families. To apply its criterion to an interacting flag correspondence requires a comparison with that external product, preserving the supported coefficient map.

3.2 The cycle trace and coherent algebra of the complete flag

The complete flag family of §3.1 has one topological point over a nonzero parameter, but that point has scheme-theoretic length six. Its exceptional projective line has generic length two. These lengths select a trace once the fundamental cycle is specified. They also enter a different calculation: the direct image of the structure sheaf. We compute both operations, keeping their coefficient categories distinct.

Set $R = k[t]$, with k as above. Retain the proper flag family $p : \mathscr{F} \to T = \operatorname{Spec} R$ of (3.2). Its two charts are

$$R_B = R[v, \lambda]/(tv^3, G), \qquad R_A = k[r, \mu]/(\mu^2), \qquad G = \lambda^2 + tv\lambda + t^2v^2. \tag{3.15}$$

On the overlap, $r = v^{-1}$, $t = 0$, and $\mu = -v\lambda$.

Write H for the reduced horizontal section and $V \simeq \mathbb{P}^1_k$ for the reduced vertical component. The two equations in the first chart form a regular sequence. Indeed, tv^3 is a nonzero divisor in the polynomial ring, and G is monic in λ over its quotient. The second chart is a hypersurface. Thus $\mathscr{F}$ is pure of dimension one.

The generic lengths calculated in Theorem 3.3 give its fundamental cycle

$$[\mathscr{F}] = 6[H] + 2[V]. \tag{3.16}$$

3.2.1 *The multiplicities determine a specified trace*

For the topological calculation take $k = \mathbb{C}$ and rational constructible coefficients. Let $\omega_{\mathscr{F}}$ and ω_T denote the dualizing complexes. For either closed inclusion $u : C \hookrightarrow \mathscr{F}$, with $C = H, V$, restriction, complex orientation and the proper counit give

$$\mathbb{Q}_{\mathscr{F}}[2] \longrightarrow u_*\mathbb{Q}_C[2] = u_*\omega_C \longrightarrow \omega_{\mathscr{F}}.$$

Weight these two maps by six and two. Their sum is the fundamental-class morphism $\mathrm{cl}_{[\mathscr{F}]}$.

Define the cycle trace by

$$\mathrm{Tr}_{[\mathscr{F}]} = \left(R p_* \mathbb{Q}_{\mathscr{F}}[2] \xrightarrow{R p_* \mathrm{cl}_{[\mathscr{F}]}} R p_* \omega_{\mathscr{F}} \longrightarrow \omega_T = \mathbb{Q}_T[2] \right)[-2]. \tag{3.17}$$

This construction uses the orientations of the smooth components. It requires no homology-manifold property of their union.

Use the splitting $\mathscr{P} = R p_* \mathbb{Q}_{\mathscr{F}} \simeq \mathbb{Q}_T \oplus i_* \mathbb{Q}[-2]$ of Theorem 3.6, where $i : \{0\} \hookrightarrow T$. Its exceptional generator is normalized by $\int_V -c_1(\mathscr{L}) = 1$. Let $\gamma : i_*\mathbb{Q}[-2] \to \mathbb{Q}_T$ be the oriented Gysin counit of Theorem 3.7.

Theorem 3.9. *The trace* (3.17) *and its normalization to the identity on* $T \setminus \{0\}$ *are*

$$\mathrm{Tr}_{[\mathscr{F}]} = (6\,\mathrm{id}, 2\gamma), \qquad \widehat{\mathrm{Tr}}_{[\mathscr{F}]} = \tfrac{1}{6}\,\mathrm{Tr}_{[\mathscr{F}]} = (\mathrm{id}, \tfrac{1}{3}\gamma). \tag{3.18}$$

Proof. The map from H is restriction to the section followed by the identity trace of $H \to T$. It is therefore six times the projection onto $\mathbb{Q}_T$.

The map from V factors through $V \to \{0\} \to T$. Integration sends its oriented degree-two generator to one, after which the closed embedding gives γ. Its degree-zero class integrates to zero: the integration map has target $\mathbb{Q}[-2]$, so its restriction from $\mathbb{Q}[0]$ vanishes. The coefficient of the exceptional summand is consequently two. The generic coefficient is six, which gives the normalization in (3.18). □

Thus compactly supported degree-two cohomology gives the row $(1, 1/3)$. The reduced cycle $[H] + [V]$ instead gives $(1, 1)$, and restriction to the section gives $(1, 0)$. The orientation fixes each of these maps once its cycle is chosen.

Corollary 3.10. *For r independent copies of the flag family, the normalized product trace has coefficient $3^{-|S|}$ on the summand indexed by $S \subseteq \{1, \dots, r\}$ in Proposition 3.8. In particular its two-axis coefficient is $1/9$. These coefficients preserve every regrouping of the independent factors.*

Proof. The component choosing V in positions S has cycle multiplicity $6^{r-|S|}2^{|S|}$. External products of orientation maps and proper traces give the product Gysin map γ_S. Division by 6^r gives $3^{-|S|}$. Fubini for the component traces, or the coefficient criterion of Proposition 3.8, proves compatibility with regrouping. □

3.2.2 *The vertical ribbon and the coherent boundary map*

Return to the field k. The horizontal flat closure has coordinate algebra

$$B^{\mathrm{flat}} = R[v, \lambda]/(v^3, G). \tag{3.19}$$

It is free over R, with ordered basis $(1, v, v^2, \lambda, v\lambda, v^2\lambda)$. Indeed, first divide by the monic polynomial G in λ, then by v^3. In particular its defining ideal is t-saturated and agrees with (tv^3, G) after inverting t.

This proves $(tv^3, G) : t^\infty = (v^3, G)$. The flat closure has no part in the A-chart.

Let $V_{\mathrm{rib}} = \mathscr{F} \times_T \{0\}$. Its two square-zero coordinates glue by $\mu = -v\lambda$, so it has a retraction $\rho : V_{\mathrm{rib}} \to V$ with

$$\rho_* \mathcal{O}_{V_{\mathrm{rib}}} = \mathcal{O}_V \oplus \mathcal{O}_V(1), \qquad \mathcal{O}_V(1)^2 = 0.$$

Put $\mathcal{J} = \ker(\mathcal{O}_{\mathscr{F}} \to \mathcal{O}_{\mathscr{F}^{\mathrm{flat}}})$.

Lemma 3.11. *The ideal $\mathcal{J}$ is annihilated by t, and*

$$\mathcal{J} \simeq \rho^* \mathcal{O}_V(-3), \qquad \rho_* \mathcal{J} \simeq \mathcal{O}_V(-3) \oplus \mathcal{O}_V(-2). \tag{3.20}$$

For the reduced embedding $\iota : V \hookrightarrow V_{\mathrm{rib}}$, ordinary restriction instead gives $\iota^* \mathcal{J} \simeq \mathcal{O}_V(-3)$. Consequently $H^0(\mathcal{J}) = 0$ and $H^1(\mathcal{J}) = k^3$.

Proof. On the B-chart, $\mathcal{J} = (v^3)$. The ring R_B is free on $1, \lambda$ over $k[t, v]/(tv^3)$, where the annihilator of v^3 is (t). Thus multiplication by v^3 identifies $R_B/(t)$ with $\mathcal{J}$ on this chart. On the A-chart, $\mathcal{J}$ is the whole structure sheaf.

Its frames are therefore $e_B = v^3$ and $e_A = 1$, with transition $e_B = v^3 e_A$. This is the pullback of $\mathcal{O}_V(-3)$. The projection formula for ρ gives the direct sum in (3.20). Quotienting by the square-zero ideal gives the stated restriction. The cohomology follows from that of the two line bundles on $\mathbb{P}^1$. □

The exact sequence defining $\mathcal{J}$ has connecting map $\partial : B^{\mathrm{flat}} \to H^1(\mathcal{J})$. We fix its signs by taking the B-lift minus the A-lift on the overlap. In the e_B-frame the class of a flat section f is represented by f/v^3, with coefficients reduced modulo t. The Laurent quotient retains exactly

$$\zeta_1 = [v^{-2}], \qquad \zeta_2 = [v^{-1}], \qquad \zeta_3 = [\lambda v^{-1}].$$

The B-chart removes nonnegative powers. The A-chart removes powers at most -3 in the even component and at most -2 in the λ-component. It follows that

$$\partial(v) = \zeta_1, \quad \partial(v^2) = \zeta_2, \quad \partial(v^2\lambda) = \zeta_3, \quad \partial(1) = \partial(\lambda) = \partial(v\lambda) = 0. \tag{3.21}$$

Each ζ_i is killed by t.

Theorem 3.12. *The entire coherent direct image is the degree-zero algebra*

$$R p_* \mathcal{O}_{\mathscr{F}} = \mathcal{B}[0], \qquad \mathcal{B} = R\{1, q, b, \lambda, c, d\}, \tag{3.22}$$

where its embedding into B^{flat} is $q = tv$, $b = tv^2$, $c = v\lambda$, and $d = tv^2\lambda$. There is an exact sequence

$$0 \longrightarrow \mathcal{B} \longrightarrow B^{\mathrm{flat}} \xrightarrow{\partial} (R/(t))^3 \longrightarrow 0. \tag{3.23}$$

The inclusion has determinant t^3 in the displayed bases. Besides the unit and commutativity, its nonzero basis products are

$$q^2 = tb, \quad q\lambda = tc, \quad qc = d, \quad b\lambda = d, \qquad \lambda^2 = -t(b + c), \quad \lambda c = -d. \tag{3.24}$$

Proof. Equation (3.21) is surjective. The long exact sequence of coherent cohomology therefore identifies $p_* \mathcal{O}_{\mathscr{F}}$ with its kernel and makes $R^1 p_* \mathcal{O}_{\mathscr{F}}$ zero. The two affine charts have affine intersection, so their Čech complex has no terms above degree one.

The kernel consists precisely of flat sections whose coefficients of $v, v^2, v^2\lambda$ are divisible by t. This gives (3.22) and the diagonal inclusion matrix $\mathrm{diag}(1, t, t, 1, 1, t)$. Multiplication in (3.19) now gives (3.24). For example, $\lambda c = v\lambda^2 = -tv^2\lambda = -d$. All omitted positive-degree products contain v^3. □

Proposition 3.13. *Ordinary specialization is a surjection*

$$\mathcal{B}/t\mathcal{B} \longrightarrow H^0(V_{\mathrm{rib}}, \mathcal{O}) = k\{1, \lambda, c\}$$

with square-zero kernel $k\{q, b, d\}$. Its source has dimension six and its target dimension three. The source is a local Gorenstein algebra: its socle, the annihilator of its maximal ideal, is the line kd. Its maximal-ideal filtration has Hilbert function $(1, 4, 1)$, and it is not a complete intersection. The missing three classes are retained by derived specialization:

$$0 \longrightarrow H^1(\mathcal{J}) \longrightarrow H^0 R\Gamma(\mathscr{F}, \mathcal{O}_{\mathscr{F}} \otimes^L_R k) \longrightarrow H^0(V_{\mathrm{rib}}, \mathcal{O}) \longrightarrow 0, \tag{3.25}$$

where the middle term is $\mathcal{B}/t\mathcal{B}$.

Proof. The split ribbon has H^0 of dimension three and $H^1 = 0$. On it $t = 0$, so the three generators q, b, d restrict to zero. Formula (3.24) proves the assertion about the kernel. This ideal is not annihilated by every nonconstant generator: $qc = b\lambda = d$.

In $\mathcal{B}/t\mathcal{B}$, the maximal ideal $\mathfrak{m} = (q, b, \lambda, c, d)$ satisfies $\mathfrak{m}^2 = kd$ and $\mathfrak{m}^3 = 0$. Multiplication on $\mathfrak{m}/\mathfrak{m}^2$, in the basis (q, b, λ, c) and with values measured in d, is

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{pmatrix}.$$

Its determinant is one. Hence no nonzero class in $\mathfrak{m}/\mathfrak{m}^2$ belongs to the socle, which is exactly kd.

There are nine independent quadratic relations among four minimal generators, since $\dim \mathrm{Sym}^2(k^4) = 10$ and $\dim \mathfrak{m}^2 = 1$. An Artinian complete intersection of embedding dimension four has four defining relations, so this algebra is not one. In particular it differs from $B^{\mathrm{flat}}/tB^{\mathrm{flat}} = k[v, \lambda]/(v^3, \lambda^2)$, whose Hilbert function is $(1, 2, 2, 1)$.

The resolution $[R \xrightarrow{t} R]$ in degrees $-1, 0$ computes the derived fiber. Its degree-minus-one cohomology sheaf is $\mathrm{Ann}(t) = \mathcal{J}$, by the chart computation in Lemma 3.11. Its degree-zero cohomology sheaf is $\mathcal{O}_{V_{\mathrm{rib}}}$. The truncation triangle starts with $\mathcal{J}[1]$. Since $H^0(\mathcal{J}) = 0$ and both sheaves have no cohomology above degree one, that triangle gives (3.25).

Tensoring the two-term Čech complex with the same free resolution identifies its total complex with $R\Gamma(\mathscr{F}, \mathcal{O}) \otimes^L_R k$. Theorem 3.12 and freeness of $\mathcal{B}$ give the asserted middle term. Freeness of the direct image does not imply flatness of $\mathcal{O}_{\mathscr{F}}$ over R. □

3.2.3 A residue pairing over the whole parameter line

Define the R-linear functional $\varepsilon : \mathcal{B} \to R$ by $\varepsilon(d) = 1$ and by zero on the other five basis elements. This specifies an algebraic residue. It makes no identification with the constructible cycle trace.

Definition 3.14. An R-linear functional $f : \mathcal{B} \to R$ is called a *Frobenius form* if the pairing $(x, y) \mapsto f(xy)$ is perfect over R.

Theorem 3.15. *The pairing $(x, y) \mapsto \varepsilon(xy)$ is perfect over R. The basis dual to $(1, q, b, \lambda, c, d)$ is $(d, b + c, q + \lambda, b, q, 1)$. The resulting coproduct is*

$$\begin{aligned} \Delta(1) &= 1 \otimes d + d \otimes 1 + q \otimes (b + c) + (b + c) \otimes q \\ &\quad + b \otimes \lambda + \lambda \otimes b, \end{aligned} \tag{3.26}$$

$$\Delta(x) = (x \otimes 1)\Delta(1). \tag{3.27}$$

It is coassociative, has counit ε, and is a $\mathcal{B}$-bimodule map. Its handle element is $\operatorname{mult}\Delta(1) = 6d$*, whose square is zero.*

Every R-linear functional defining a perfect Frobenius pairing is uniquely $\varepsilon_u(x) = \varepsilon(ux)$ for a unit $u \in \mathcal{B}^\times$. Its coproduct and handle are

$$\Delta_u(1) = (u^{-1} \otimes 1)\Delta(1), \qquad \operatorname{mult}\Delta_u(1) = 6u^{-1}d.$$

In particular, $u = 1 + d$ gives $\varepsilon_u(1) = 1$ and the same handle $6d$. No such Frobenius coproduct preserves the unit.

Proof. The multiplication table gives the Gram matrix

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Its determinant is -1, and multiplication by the listed dual basis gives the identity pairing matrix.

Under this perfect pairing Δ is adjoint to multiplication. Pairing a proposed coassociativity identity with $x \otimes y \otimes z$ gives the associativity identity $(xy)z = x(yz)$, so proves it. Dual-basis evaluation gives both counit identities. The equality $\varepsilon((ax)y) = \varepsilon(x(ay))$ gives $(a \otimes 1)\Delta(1) = (1 \otimes a)\Delta(1)$, and hence the bimodule identity. Multiplying the six dual basis pairs gives $6d$, and $d^2 = 0$ by (3.24).

The perfect pairing identifies $\operatorname{Hom}_R(\mathcal{B}, R)$ with $\mathcal{B}$ as a $\mathcal{B}$-module. Thus every functional is uniquely $\varepsilon(u\,\cdot)$, and its pairing is perfect exactly when multiplication by u is invertible, equivalently when u is a unit. Replacing each dual basis element by its product with u^{-1} gives the displayed formulas. For $u = 1 + d$, the inverse is $1 - d$, and $d^2 = 0$ proves the normalization and handle assertions. Finally, $\Delta_u(1)$ induces an isomorphism from the dual of the rank-six module $\mathcal{B}$ to $\mathcal{B}$. The tensor $1 \otimes 1$ has image of rank one, so cannot equal $\Delta_u(1)$. □

These operations give the usual commutative Frobenius sewing over R. The closed genus-g value $\varepsilon((6d)^g)$ is 0 for $g = 0$, 6 for $g = 1$, and zero for $g \geq 2$. For the normalized form ε_{1+d}, these values are $1, 6, 0, 0, \ldots$.

Normalization is therefore a choice of Frobenius form. It does not construct a physical vacuum expectation. The coproduct obstruction holds for every such choice. The ordinary trace of multiplication is different: it is six on 1 and zero on q, b, λ, c, d. Indeed these last elements have positive degrees 1, 2, 1, 2, 3 in a grading with $\deg t = 0$, so their multiplication matrices have zero diagonal. That trace pairing is degenerate.

3.2.4 *The enhancement remains part of the operation*

Proposition 3.16. *The intrinsic obstruction theory given by the cotangent complex of $\mathscr{F}$ has virtual class $[\mathscr{F}] = 6[H] + 2[V]$. For $k = \mathbb{C}$, the normalized trace of this cycle is* $(\mathrm{id}, \gamma/3)$*. On the B-chart, the two-equation complete intersection and the five-equation incidence presentation of Theorem 3.5 have virtual dimensions one and minus one, respectively. The usual non-equivariant virtual class of the latter presentation is zero. For $k = \mathbb{C}$, their rational normalized linear-potential critical coefficients instead have shifts $+1$ and -1, as in* (3.11)*.*

Proof. For a regular embedding the normal cone is the normal bundle. Pulling its class back by the zero section gives the fundamental cycle. This applies on both complete intersection charts of $\mathscr{F}$. Their intrinsic cotangent

obstruction theories agree on the overlap. The virtual class of that theory is consequently the cycle of Theorem 3.9, with its scheme multiplicities.

The first presentation has three smooth coordinates and two regular equations. The second has four smooth coordinates and five equations. Eliminating its regular pair leaves the same two-equation Koszul algebra together with two closed generators of degree minus one. The resulting perfect obstruction theory has virtual rank -1, so its virtual class belongs to the zero Chow group A_{-1} of this ordinary chart.

For a rank-r linear potential on a smooth m-dimensional base, fiber-linear dimensional reduction gives output shift $m - r$ in the normalization (3.11). The two values are therefore $3 - 2 = 1$ and $4 - 5 = -1$. Neither resulting constant coefficient complex on the nonempty zero locus is zero. □

The dimensionally reduced critical coefficient is a shifted constant sheaf on the underlying topological zero locus. Its object alone does not record the two generic nilpotent lengths. Choosing the reduced fundamental cycle gives the trace (id, γ). Choosing the scheme cycle gives $(\mathrm{id}, \gamma/3)$ after normalization. The coefficient sheaf alone selects neither morphism.

The cycle map (3.17) is consequently a specified comparison target for an oriented Hall operation. Identifying them requires that operation's actual coefficient functor, obstruction theory, orientation and supported map. Such an identification must also preserve the two unit insertions and substitution of successive flags. These conditions compare geometric coefficient maps. The residue pairing on $\mathcal{B}$ does not supply those maps.

3.3 The multiplication varies over the parameter

The coherent flag algebra $\mathcal{B}$ is free over $k[t]$, and its residue pairing is constant in the displayed basis. Its multiplication nevertheless has a nonzero first derivative as an algebra deformation. We compute this derivative before asking for a connection that preserves multiplication.

Retain the characteristic-zero field k, the ring $R = k[t]$, and the basis $(1, q, b, \lambda, c, d)$ of Theorem 3.12. Put $B_0 = \mathcal{B}/t\mathcal{B}$. The basis identifies the underlying module with $B_0 \otimes_k R$, so (3.24) takes the form

$$\mu_t = \mu_0 + t\mu_1.$$

Here μ_0 is the product of B_0.

For a unital algebra A, the normalized Hochschild cochains are $C^n(A, A) = \mathrm{Hom}_k((A/k1)^{\otimes n}, A)$ for $n > 0$, and $C^0(A, A) = A$. Positive-degree cochains vanish on unit inputs. In the degrees needed here their differential is

$$\begin{aligned}(\delta a)(x) &= xa - ax,\\ (\delta D)(x, y) &= xD(y) - D(xy) + D(x)y,\\ (\delta F)(x, y, z) &= xF(y, z) - F(xy, z) + F(x, yz) - F(x, y)z.\end{aligned}$$

Write $HH^n(A)$ for its cohomology. For a two-cochain F, the insertion bracket satisfies

$$\tfrac{1}{2}[F, F](x, y, z) = F(F(x, y), z) - F(x, F(y, z)).$$

Thus the coefficient of t in associativity is the cocycle equation, and the coefficient of t^2 is the bracket equation.

Proposition 3.17. *The first-order deformation class $[\mu_1] \in HH^2(B_0)$ is nonzero. Its representative has precisely the nonzero values*

$$\mu_1(q, q) = b, \qquad \mu_1(q, \lambda) = \mu_1(\lambda, q) = c, \qquad \mu_1(\lambda, \lambda) = -(b + c). \tag{3.28}$$

It satisfies $\delta\mu_1 = [\mu_1, \mu_1] = 0$. *For the residue pairing* $\langle x, y\rangle = \varepsilon(\mu_t(x, y))$ *of Theorem 3.15, it is cyclic:*

$$\langle \mu_1(x, y), z\rangle = \langle x, \mu_1(y, z)\rangle.$$

Proof. Differentiating (3.24) gives (3.28). Associativity of μ_t gives the two equations. If $\mu_1 = \delta D$, evaluation on (q, q) in B_0 would give

$$b = 2qD(q) \in qB_0 = \operatorname{span}_k\{q, d\},$$

which contradicts the basis of B_0.

None of the varying products has a d component, so the pairing is independent of t. Associativity gives $\langle \mu_t(x, y), z\rangle = \langle x, \mu_t(y, z)\rangle$. Its coefficient of t proves cyclicity. □

A change of basis equal to the identity modulo t changes μ_1 by a Hochschild coboundary. The proposition therefore rules out a trivialization as an algebra even over $k[t]/(t^2)$. Allowing an automorphism on the special fiber does not remove the obstruction: it acts invertibly on $HH^2(B_0)$.

3.3.1 Regular and logarithmic connections

For a derivation $\xi : R \to R$, a connection along ξ is a k-linear map $\nabla : \mathcal{B} \to \mathcal{B}$ such that $\nabla(fx) = \xi(f)x + f\nabla x$. It is multiplicative if $\nabla(xy) = (\nabla x)y + x\nabla y$. The derivation $t\partial_t$ vanishes at the special point; a connection along it can preserve a degeneration there.

Proposition 3.18. *There is no multiplicative connection along* ∂_t. *A multiplicative connection along* $t\partial_t$ *is*

$$\nabla_{\log}(f(t)x) = tf'(t)x + w(x)f(t)x, \qquad (w(1), w(q), w(b), w(\lambda), w(c), w(d)) = (0, 1, 1, 1, 1, 2). \tag{3.29}$$

For this connection the residue pairing obeys

$$\langle \nabla_{\log}x, y\rangle + \langle x, \nabla_{\log}y\rangle = t\partial_t\langle x, y\rangle + 2\langle x, y\rangle.$$

Proof. Apply a putative multiplicative connection along ∂_t to $q^2 = tb$. Modulo t it gives $2q\nabla q = b$, the same impossible equality used in Proposition 3.17.

For the logarithmic connection, give t weight one and the basis the weights in (3.29). Every product in (3.24) is homogeneous. The Euler derivation of this grading is $\nabla_{\log}$, so it obeys the product rule. These weights also define a $\mathbb{G}_m$-action on the algebra over the dilation action on the parameter line.

Since ε takes the coefficient of the weight-two element d,

$$\varepsilon(\nabla_{\log}x) = t\partial_t\varepsilon(x) + 2\varepsilon(x).$$

Apply this identity to a product to obtain the pairing formula. □

3.3.2 All first-order associative deformation classes

The geometric family selects one class in $HH^2(B_0)$. The entire group can be calculated from the nondegenerate product on the maximal ideal of B_0.

Theorem 3.19. *The special fiber satisfies*

$$\dim_k HH^1(B_0) = 11, \qquad \dim_k HH^2(B_0) = 26.$$

The second group is the direct sum of a 16*-dimensional space represented by symmetric cocycles and a* 10*-dimensional space represented by antisymmetric cocycles. The class* $[\mu_1]$ *lies in the symmetric summand.*

Proof. Write $z = d$ and $V = k\{q, b, \lambda, c\}$. Then

$$B_0 = k1 \oplus V \oplus kz, \qquad vw = g(v,w)z, \qquad Vz = z^2 = 0,$$

where the matrix of the nondegenerate form g is

$$G = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{pmatrix}. \tag{3.30}$$

Give V weight one and z weight two. The internal weight of a cochain is its output weight minus the sum of its input weights. The differential preserves it. After extending k, we may choose an orthonormal basis of V. Dimensions descend because extension of fields preserves the ranks of these finite matrices.

A derivation of weight one is an arbitrary map $V \to kz$, giving four dimensions. At weight zero it has the form $D(v) = Lv$, $D(z) = rz$, with

$$g(Lv, w) + g(v, Lw) = r\,g(v,w).$$

These are a skew-adjoint map and a scalar multiple of the identity, giving $6 + 1 = 7$ dimensions.

At weight -1, write $D(v) = a(v)1$ and $D(z) = b \in V$. The product rule on vw gives

$$a(v)w + a(w)v = g(v,w)b.$$

Distinct orthonormal basis vectors force $a = 0$, and taking $v = w$ then gives $b = 0$. At weight -2, only $D(z) = c1$ is possible, and the same product rule gives $c = 0$. These exhaust the weights. Since B_0 is commutative, inner derivations vanish. Thus HH^1 has dimension 11, and the boundary map $C^1 \to C^2$ has rank $30 - 11 = 19$.

At weight zero, a two-cochain is a bilinear form $V \otimes V \to kz$. All sixteen are closed. The boundaries are exactly the symmetric forms:

$$(v,w) \longmapsto g(v, Lw) + g(Lv, w) - r\,g(v,w).$$

Consequently this weight contributes the six alternating forms.

At weight -1, a two-cochain has the form

$$F(v,w) = B(v,w) \in V, \qquad F(v,z) = p(v)z, \qquad F(z,v) = q(v)z.$$

Set $C(v,w,u) = g(B(v,w),u)$. Its cocycle equation is

$$C(w,u,v) - C(v,w,u) = g(v,w)q(u) - g(w,u)p(v). \tag{3.31}$$

Sum over cyclic permutations and take $v = w$ a unit vector orthogonal to u. This forces $p = q$. Equation (3.31) now says that

$$C'(v,w,u) = C(v,w,u) + g(v,w)p(u)$$

is cyclically invariant. Conversely, a cyclic tensor C' and a covector p give a cocycle; triples containing z give no additional equation.

The trace of either nontrivial three-cycle on $V^{*\otimes 3}$ is 4. Averaging over the cyclic group therefore gives $(4^3 + 2 \cdot 4)/3 = 24$ invariant tensors. Together with p, they give 28 cocycles. A one-cochain of this weight is $D(v) = a(v)1$, $D(z) = b$. Its boundary corresponds to

$$p = a + g(b, -),$$
$$C'(v,w,u) = a(w)g(v,u) + a(v)g(w,u) + a(u)g(v,w).$$

The absence of derivations of weight -1 makes this eight-dimensional boundary map injective. The quotient therefore has dimension $28 - 8 = 20$.

At weight -2, write

$$F(v,w) = h(v,w)\mathrm{I}, \quad F(v,z) = Lv, \quad F(z,v) = Mv, \quad F(z,z) = cz.$$

The equations on (v,w,z), (v,z,w) and (z,v,w) give $M = L^{*}$ and $h(v,w) = g(v,Lw) - c\,g(v,w)$. The equation on three vectors is

$$h(w,u)v - g(v,w)Mu + g(w,u)Lv - h(v,w)u = 0.$$

Three distinct basis indices force the off-diagonal entries of h, and then of L, to vanish. The indices (i,i,j) give $L_i + L_j = c$ for every $i \neq j$. With four indices this implies $L = M = a\,\mathrm{id}$, $c = 2a$, $h = -ag$. This is precisely the boundary of $D(z) = a\mathrm{I}$, $D(V) = 0$.

At weight -3, the possible values are $F(v,z) = p(v)\mathrm{I}$, $F(z,v) = q(v)\mathrm{I}$ and $F(z,z) = b \in V$. The equation on three vectors forces $p = q = 0$; the equation on (v,w,z) then forces $b = 0$. At weight -4, only $F(z,z) = c\mathrm{I}$ is possible, and the latter equation again gives $c = 0$. There are no cochains of other weights. We obtain

weight	0	-1	-2	-3	-4
$\dim C^2$	16	72	49	12	1
$\dim Z^2$	16	28	1	0	0
$\dim B^2$	10	8	1	0	0.

Thus $\dim HH^2 = 6 + 20 = 26$.

Every boundary is symmetric because B_0 is commutative. At weight -1, symmetric cocycles correspond to fully symmetric C' and arbitrary p. There are $20 + 4$ such cocycles, leaving 16 classes after quotienting. For an antisymmetric cocycle, $p = q$ forces $p = q = 0$, and cyclicity makes C' alternating. These give four classes. The six alternating forms at weight zero supply the other six. The weight -2 cocycle is a boundary, which proves the claimed decomposition. □

3.4 Ramification of the flag family

Replace the marked parameter t by u^e, where $e \geq 1$ is an integer. The coherent algebra remains free, but its inclusion in the flat closure acquires a larger cokernel. At the same time the vertical cycle multiplicity increases. We calculate both changes and then compare the resulting trace with pullback of the oriented Gysin map.

Put $R_e = k[u]$ and let $p_e : \mathscr{F}_e = \mathscr{F} \times_{\operatorname{Spec} R} \operatorname{Spec} R_e \to \operatorname{Spec} R_e$. The charts of (3.15) become

$$\begin{aligned} S_{B,e} &= k[u,v,\lambda]/(u^e v^3, G_e), \quad G_e = \lambda^2 + u^e v\lambda + u^{2e}v^2, \\ S_{A,e} &= k[u,r,\mu]/(u^e, \mu^2). \end{aligned} \tag{3.32}$$

The gluing is $r = v^{-1}$, $\mu = -v\lambda$. Denote the reduced horizontal and vertical components by H_e and V_e, with $V_e \simeq \mathbb{P}^1_k$, and set

$$\mathcal{B}_e = \mathcal{B} \otimes_R R_e, \qquad B_e^{\mathrm{flat}} = k[u,v,\lambda]/(v^3, G_e).$$

Proposition 3.20. *The fundamental cycle and coherent direct image are*

$$[\mathscr{F}_e] = 6[H_e] + 2e[V_e], \qquad Rp_{e*}\mathcal{O}_{\mathscr{F}_e} = \mathcal{B}_e[0]. \tag{3.33}$$

The inclusion into the flat closure fits into an exact sequence

$$0 \longrightarrow \mathcal{B}_e \longrightarrow B_e^{\mathrm{flat}} \longrightarrow (R_e/(u^e))^3 \longrightarrow 0. \tag{3.34}$$

Its matrix in the bases $(1, q, b, \lambda, c, d)$ *and* $(1, v, v^2, \lambda, v\lambda, v^2\lambda)$ *is* $\operatorname{diag}(1, u^e, u^e, 1, 1, u^e)$, *with determinant* u^{3e}.

Proof. Over $k(u)$, the six monomials $1, v, v^2, \lambda, v\lambda, v^2\lambda$ form a basis, so the horizontal length is six. At the generic point of V_e, invert v. The transverse ring is $k(v)[u,\lambda]/(u^e,\lambda^2)$, of length $2e$. Both charts are complete intersections of dimension one: $u^e v^3$ is a nonzero divisor in the polynomial ring, and G_e is monic in λ over its quotient. They have no embedded associated components. The two generic lengths therefore give the fundamental cycle.

The map $R \to R_e$ is free with basis $1, u, \dots, u^{e-1}$. Tensoring the two-chart Čech complex is consequently exact, and computes $Rp_{e*}\mathcal{O}_{\mathscr{F}_e}$. Tensoring (3.23) gives (3.34). The formulas $q = u^e v$, $b = u^e v^2$, $c = v\lambda$, $d = u^e v^2\lambda$ give the displayed inclusion matrix. □

The cokernel has length $3e$. Tensoring it with $k = R_e/(u)$ gives only a three-dimensional space, so ordinary specialization of the cokernel forgets e. The resolution $[R_e \xrightarrow{u^e} R_e]$ also gives $\operatorname{Tor}_1^{R_e}(R_e/(u^e), k) = k$. Thus specialization of (3.34) has a three-dimensional kernel as well as a three-dimensional cokernel. Its kernel in $\mathcal{B}_e/u\mathcal{B}_e$ is exactly $k\{q, b, d\}$.

3.4.1 *The trace under a change of marking*

Take $k = \mathbb{C}$ and rational constructible coefficients. Let D_t, D_u be small disks and $\rho_e : D_u \to D_t$ the map $t = u^e$. Write i_t, i_u for their origins. Normalize the Gysin maps

$$\gamma_t : i_{t,*}\mathbb{Q}[-2] \to \mathbb{Q}_{D_t}, \qquad \gamma_u : i_{u,*}\mathbb{Q}[-2] \to \mathbb{Q}_{D_u}$$

by the complex orientations.

Lemma 3.21. *Under the natural identification of the pulled-back point sheaf with $i_{u,*}\mathbb{Q}$,*

$$\rho_e^* \gamma_t = e\gamma_u. \tag{3.35}$$

Proof. Purity identifies a Gysin class with the oriented generator of $H^2(D, D \setminus \{0\}; \mathbb{Q})$. The boundary isomorphism from $H^1(D \setminus \{0\}; \mathbb{Q})$ identifies it with the positive link generator. On that link, $u \mapsto u^e$ has degree e, so pullback multiplies the generator by e. Naturality of the relative cohomology sequence proves the identity of Gysin classes, hence of the corresponding maps. □

Proposition 3.22. *Use the horizontal section and the exceptional generator $\int_{V_e} -c_1(\mathscr{L}) = 1$ to split*

$$Rp_{e*}\mathbb{Q}_{\mathscr{F}_e} = \mathbb{Q}_{D_u} \oplus i_{u,*}\mathbb{Q}[-2].$$

The cycle trace and its normalization on the punctured disk are

$$\operatorname{Tr}_{[\mathscr{F}_e]} = (6\,\mathrm{id}, 2e\gamma_u), \qquad \widehat{\operatorname{Tr}}_{[\mathscr{F}_e]} = (\mathrm{id}, \tfrac{e}{3}\gamma_u). \tag{3.36}$$

The normalized trace is the pullback of (3.18). For independent marked axes with indices $e_1, \dots, e_r$, its coefficient on the exceptional summand indexed by S is $\prod_{i\in S}(e_i/3)$.

Proof. The reduced space is still a horizontal section with a vertical projective line attached at the origin. The splitting of Theorem 3.6 therefore applies, with the same orientation of the exceptional curve. Apply the fundamental-class construction (3.17) to the cycle in (3.33). The horizontal component gives 6 id. Integration on V_e, followed by its Gysin map, gives $2e\gamma_u$. Division by six gives the normalized trace.

Proper base change identifies the pulled-back coefficient complex with this splitting: the section and exceptional generator pull back to the specified ones. Lemma 3.21 now identifies the pulled-back trace. External products multiply the singleton coefficients, and Proposition 3.8 gives their compatibility with all regroupings. □

For two successive markings $t = u^e$, $u = w^f$, the exceptional coefficient is $ef/3$, whether pulled back successively or along the composite. Choosing the reduced cycle separately after each change of marking instead gives coefficient one; pulling back the reduced trace in t gives coefficient e. These two prescriptions differ when $e > 1$. Restriction to the horizontal section, whose exceptional coefficient is zero, also commutes with pullback.

3.4.2 *Ramification retains the connection obstruction*

The logarithmic connection on $\mathcal{B}_e$ has parameter part $u\,\partial_u$ and basis weights $(0, e, e, e, e, 2e)$. It satisfies

$$\langle \nabla_{\log} x,\, y\rangle + \langle x,\, \nabla_{\log} y\rangle = u\,\partial_u \langle x,\, y\rangle + 2e\langle x,\, y\rangle.$$

These assertions follow from the same homogeneous relations as Proposition 3.18.

A multiplicative connection along ∂_u would give

$$2q\nabla q = eu^{e-1}b + u^e \nabla b.$$

The b coefficient of the left side is divisible by u^e: multiplication by q produces b only through $q^2 = u^e b$. Comparing coefficients, cancelling u^{e-1}, and reducing modulo u would give $0 = e$. Characteristic zero rules this out. The obstruction persists even when the first derivative of u^e vanishes at zero.

The coherent and constructible calculations impose different conditions on a geometric realization. A comparison of coherent algebras over the parameter must carry the nonzero class $[\mu_1]$. A comparison that also identifies the inclusions into the flat closures must carry their ramified cokernels $(R_e/(u^e))^3$. A comparison of the specified constructible cycle traces must carry the coefficient $e/3$ and its products on independent axes. An oriented critical Hall operation has its own coefficient complex and trace. Identifying those maps with the ones calculated here requires the actual oriented correspondence and its supported comparison maps.

3.5 The subobject comes first

An extension correspondence gives a product only after its coefficient maps have been specified. For a quiver without relations, these maps are ordinary pullback and complex-oriented integration. A Calabi–Yau moduli space is generally singular. Its corresponding operation uses a perverse sheaf of vanishing cycles and an orientation. The one-arrow quiver fixes the order and grading before those coefficients are introduced.

For the Hall constructions below, work over $\mathbb{C}$ and use rational constructible coefficients.

For a finite quiver Q, write Q_0 for its vertices and Q_1 for its arrows. A representation consists of vector spaces V_i and maps $V_a : V_i \to V_j$ for each arrow $a : i \to j$. Its dimension vector is $\alpha = (\dim V_i)_{i \in Q_0}$. Put

$$R_\alpha = \bigoplus_{a : i \to j} \operatorname{Hom}(\mathbb{C}^{\alpha_i}, \mathbb{C}^{\alpha_j}), \qquad G_\alpha = \prod_i GL_{\alpha_i}, \qquad \mathfrak{M}_\alpha = [R_\alpha / G_\alpha].$$

Contraction of R_α to zero identifies $H^\bullet(\mathfrak{M}_\alpha; \mathbb{Q})$ with $H^\bullet(BG_\alpha; \mathbb{Q})$. Its Chern roots have degree two.

An extension is written

$$0 \longrightarrow E \longrightarrow V \longrightarrow F \longrightarrow 0. \tag{3.37}$$

The ordered input is (E, F): subobject, then quotient. Let $\mathfrak{E}_{\alpha,\beta}$ classify (3.37) with dimensions (α, β). There are maps

$$\mathfrak{M}_\alpha \times \mathfrak{M}_\beta \xleftarrow{q} \mathfrak{E}_{\alpha,\beta} \xrightarrow{p} \mathfrak{M}_{\alpha+\beta}, \qquad q(V, E) = (E, V/E), \quad p(V, E) = V. \tag{3.38}$$

Over $\mathfrak{M}_{\alpha+\beta}$, the correspondence is the closed incidence subscheme of the product of Grassmannian bundles $\prod_i \operatorname{Gr}(\alpha_i, \alpha_i + \beta_i)$. Hence p is representable and proper.

Take Q to have one arrow $1 \to 2$, and let S_1, S_2 be its one-dimensional simple representations. Write x, y for the Chern classes at the two vertices, and u_1, u_2 for the constant classes of their moduli stacks. A representation of dimension $(1, 1)$ is a scalar map $z : \mathbb{C} \to \mathbb{C}$.

For subobject S_1 and quotient S_2, invariance forces $z = 0$. This is the zero section of the equivariant line of weight $y - x$. For subobject S_2 and quotient S_1, every scalar z is allowed. Complex-oriented Gysin integration therefore gives

$$u_1 \star u_2 = y - x, \qquad u_2 \star u_1 = 1 \quad \text{in } H^\bullet(\mathfrak{M}_{(1,1)}; \mathbb{Q}). \tag{3.39}$$

The two products have ordinary degrees two and zero. Their common target is the same stack, with different extension conditions.

3.5.1 *Normal and tangent Euler classes*

For arbitrary Q, let S_i and T_i be the tautological subbundle and quotient bundle on the product of Grassmannians above. Preservation of the subspaces sets to zero every block $S_i \to T_j$ along an arrow $i \to j$. The normal bundle of this incidence condition and the tangent bundle of the Grassmannians are respectively

$$N_{\alpha,\beta} = \bigoplus_{a:i\to j} \operatorname{Hom}(S_i, T_j), \qquad T_{\alpha,\beta} = \bigoplus_{i} \operatorname{Hom}(S_i, T_i). \tag{3.40}$$

For classes f and g, the Hall product is the Grassmannian integral of $f(S)\,g(T)\,e(N_{\alpha,\beta})$. Here e is the top Chern class. Thus the construction is polynomial before equivariant localization is used to compute it.

Define the quiver Euler form by

$$\chi_Q(\alpha, \beta) = \sum_i \alpha_i \beta_i - \sum_{a:i\to j} \alpha_i \beta_j. \tag{3.41}$$

The normal Gysin map raises degree by 2 rank $N_{\alpha,\beta}$, and integration lowers it by 2 rank $T_{\alpha,\beta}$. Consequently

$$H^p(\mathfrak{M}_\alpha) \otimes H^q(\mathfrak{M}_\beta) \longrightarrow H^{p+q-2\chi_Q(\alpha,\beta)}(\mathfrak{M}_{\alpha+\beta}). \tag{3.42}$$

This is the ordinary cohomological Hall construction [101, §§2.4–2.5].

For its localized expression, partition the roots at each vertex into $I_i \sqcup J_i$, with $|I_i| = \alpha_i$ and $|J_i| = \beta_i$. Each such partition contributes

$$f(x_I)\,g(x_J)\,\frac{\displaystyle\prod_{a:i\to j}\prod_{r\in I_i,\ s\in J_j}(x_{j,s} - x_{i,r})}{\displaystyle\prod_{i}\prod_{r\in I_i,\ s\in J_i}(x_{i,s} - x_{i,r})}. \tag{3.43}$$

Summing over all partitions gives $f \star g$. The numerator records precisely the forbidden arrow blocks. The denominator records the tangent directions of the flag.

For three inputs, both bracketings sum over the same ordered three-block partitions. Every pair of distinct blocks contributes its factor once. This proves associativity of the displayed formula and agrees with composition of three-step flags.

3.5.2 *Perverse normalization*

For a cochain complex K we use $K[s]^n = K^{n+s}$.

The smooth stack $\mathfrak{M}_\alpha$ has dimension

$$d_\alpha = \dim R_\alpha - \dim G_\alpha = -\chi_Q(\alpha, \alpha).$$

Its constant perverse sheaf is $\mathbb{Q}_{\mathfrak{M}_\alpha}[d_\alpha]$. Set

$$\mathsf{H}^p_\alpha = H^p(\mathfrak{M}_\alpha, \mathbb{Q}[d_\alpha]) = H^{p+d_\alpha}(\mathfrak{M}_\alpha; \mathbb{Q}).$$

Changing to this grading changes the degree in (3.42) to

$$\begin{aligned} \nu_Q(\alpha,\beta) &= -2\chi_Q(\alpha,\beta) + d_\alpha + d_\beta - d_{\alpha+\beta} \\ &= \chi_Q(\beta,\alpha) - \chi_Q(\alpha,\beta). \end{aligned} \tag{3.44}$$

In the one-arrow example, all three stack dimensions are -1. Both constant input classes have perverse degree one. The outputs $y - x$ and 1 have perverse degrees three and one, respectively. Their product degrees are $+1$ and -1. This also fixes the shift when the coefficient sheaf is replaced by vanishing cycles.

3.6 Flags and derived composition

For three objects, composing extensions gives $0 \subset E_1 \subset E_2 \subset E_3$. One bracketing first forms E_2, the other E_3/E_1. Both retain the same three successive quotients. In a derived category the quotient is a cofiber, so its nullhomotopy is part of the flag.

Let $\mathcal{D}$ be a small stable ∞-category. Thus it has a zero object and finite limits and colimits, and a square is a pullback precisely when it is a pushout.

For $n \geq 0$, let T_n be the poset of pairs (i, j) with $0 \leq i \leq j \leq n$, ordered coordinatewise. Let $S_n(\mathcal{D})$ be the space of coherent T_n-diagrams

$$X_{ij}, \qquad 0 \leq i \leq j \leq n,$$

with $X_{ii} = 0$, such that, for $i \leq j \leq k$, the square

$$\begin{array}{ccc} X_{ij} & \longrightarrow & X_{ik} \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & X_{jk} \end{array}$$

is a pushout and a pullback. Equivalences of diagrams give the paths in this space. Restriction to a subset of the indices gives its face maps. Repetition of an index inserts a zero quotient and gives its degeneracy maps. This is the Waldhausen S-construction.

Proposition 3.23 (Gluing flags)**.** *Triangulate the polygon with vertices $0, \dots, n$. Restriction identifies $S_n(\mathcal{D})$ with the homotopy limit of its triangle spaces, glued along common edges. In particular, both extension composites are equivalent to $S_3(\mathcal{D})$. Inserting a zero quotient gives the unit.*

Proof. For the diagonal 02 of the quadrilateral 0123, the two triangles specify $X_{01} \to X_{02} \to X_{03}$ together with the corresponding cofibers. The remaining object is $X_{13} = \operatorname{cofib}(X_{01} \to X_{03})$. Pasting pushout squares gives the required map $X_{12} \to X_{13}$ and its cofiber X_{23}. The space of these completions is contractible by the universal property of a colimit.

For the diagonal 13, the triangles specify $X_{01} \to X_{03} \to X_{13}$ and $X_{12} \to X_{13} \to X_{23}$. Recover X_{02} as the pullback of $X_{03} \to X_{13}$ along $X_{12} \to X_{13}$. Stability makes this square a pushout as well, so it recovers the same flag. Again the completion is unique up to a contractible choice.

Gluing triangles successively and applying these universal properties proves the assertion for a polygon. The coherent version is the unital 2-Segal theorem for the S-construction, [51, Theorem 7.3.3]. A stable category has the exact structure in which all maps are admissible. A zero cofiber makes its map an equivalence, which proves the unit assertion. □

The proposition concerns spaces of diagrams, including their automorphisms and higher homotopies. Applied to stable categories of families over a derived test algebra, it gives equivalences of prestacks. Its fiber products are derived.

For example, put $A = \mathbb{C}[t]$ and let both maps $\operatorname{Spec}\mathbb{C} \to \operatorname{Spec} A$ be the origin. Resolving $\mathbb{C}$ by $[A \xrightarrow{t} A]$ in degrees -1, 0 gives

$$\mathbb{C} \otimes^{\mathrm{L}}_{A} \mathbb{C} \simeq \Lambda_{\mathbb{C}}(\eta), \qquad |\eta| = -1, \qquad d\eta = 0. \tag{3.45}$$

The ordinary fiber product retains only degree zero. The missing class belongs to the composed correspondence itself, before a coefficient theory is applied.

3.7 The Calabi–Yau coefficient map

Let C be a dg category of finite type over $\mathbb{C}$, equipped with a left Calabi–Yau structure of dimension three. Here finite type means compactness in the Morita category, and a Calabi–Yau structure is a nondegenerate negative cyclic class. Write $\mathcal{M}_C$ for its derived moduli stack of pseudo-perfect right modules: an R-family is a dg functor $C^{\mathrm{op}} \to \operatorname{Perf}(R)$. It carries the resulting (-1)-shifted symplectic form.

Choose an open substack $\mathcal{M} \subset \mathcal{M}_C$ closed under direct sums and containing the zero module. Put $\Theta = [\mathbb{A}^1/\mathbb{G}_m]$, where $\mathbb{G}_m$ acts by scaling, and set

$$\operatorname{Filt}(\mathcal{M}) = \operatorname{Map}(\Theta, \mathcal{M}), \qquad \operatorname{Grad}(\mathcal{M}) = \operatorname{Map}(B\mathbb{G}_m, \mathcal{M}).$$

Restriction at 1 gives evaluation, and restriction at zero gives the associated graded object. These definitions include the automorphisms of a filtration.

We impose the following geometric hypotheses:

(i) The classical truncation $\mathcal{M}^{\mathrm{cl}}$ is a quasi-separated Artin stack with affine stabilizers.

(ii) The evaluation map $\operatorname{Filt}(\mathcal{M}^{\mathrm{cl}}) \to \mathcal{M}^{\mathrm{cl}}$ is proper on each connected component of its source.

(iii) The zero-module map $\epsilon : \operatorname{Spec}\mathbb{C} \to \mathcal{M}^{\mathrm{cl}}$ is a closed immersion.

The second condition is called Θ-*reductivity* here.

Put $\Gamma = \pi_0(\mathcal{M})$, with addition induced by direct sum, and write $\mathcal{M}_\alpha$ for a component. Map $\mathcal{M}_\alpha \times \mathcal{M}_\beta$ to $\operatorname{Grad}(\mathcal{M})$ by giving E weight one and F weight zero. Define

$$\mathcal{E}_{\alpha,\beta} = \operatorname{Filt}(\mathcal{M}) \overset{\mathrm{R}}{\underset{\operatorname{Grad}(\mathcal{M})}{\times}} (\mathcal{M}_\alpha \times \mathcal{M}_\beta).$$

This is the correspondence of cofiber sequences $E \to V \to F$ with those weights. For modules in a heart it is the Rees convention for $0 \subset E \subset V$. Projection and evaluation give

$$\mathcal{M}_\alpha \times \mathcal{M}_\beta \xleftarrow{q} \mathcal{E}_{\alpha,\beta} \xrightarrow{p} \mathcal{M}_{\alpha+\beta},$$

with the same subobject-first order as (3.38).

3.7.1 *Strong orientation*

An orientation o consists of a $\mathbb{Z}/2$-graded line L_o on $\mathcal{M}^{\mathrm{red}}$ and a squaring isomorphism

$$L_o^{\otimes 2} \simeq \det(\mathbb{L}_{\mathcal{M}})|_{\mathcal{M}^{\mathrm{red}}},$$

where the determinant has its virtual-rank parity. This is [96, Definition 3.15]. On a critical chart the orientation determines the sign local system needed to glue perverse vanishing cycles. The resulting Donaldson–Thomas sheaf will be denoted $\mathcal{P}_o$. It is a rational perverse sheaf on $\mathcal{M}^{\mathrm{cl}}$. We use the vanishing-cycle normalization for which the zero function on a smooth space has coefficient $\mathbb{Q}[\dim]$.

The direct-sum map $\Phi : \mathcal{M} \times \mathcal{M} \to \mathcal{M}$ has an oriented pullback operation $\Phi^{\star}$ supplied by its filtered Lagrangian correspondence. This operation includes the determinant of the moving tangent directions in addition to ordinary pullback.

Definition 3.24. A *strong orientation datum* consists of o, an isomorphism $\lambda : \Phi^{\star} o \simeq o \boxtimes o$, and a trivialization $\epsilon^{\star} o \simeq o_{\mathrm{triv}}$ at the zero module. The two isomorphisms obtained by applying λ to three summands must agree under the canonical comparison of the filtered Lagrangian correspondences. The left and right unit comparisons must agree with the given trivialization.

This is [96, Definition 8.4]. It specifies the compatibility isomorphisms themselves. Existence of a square root at each object does not give these isomorphisms or their associativity.

3.7.2 The index and the product

At $E \oplus F$, decompose the tangent complex into $\mathbb{G}_m$-weights. Define the index

$$I(\alpha, \beta) = \operatorname{rank}\bigl(\mathbb{T}^{>0}_{\mathcal{M}, E \oplus F}\bigr), \qquad \operatorname{rank} K = \sum_{j} (-1)^{j} \dim H^{j}(K). \tag{3.46}$$

This index is locally constant on the graded component. Equivalently, take the positive-minus-negative weight dimension in H^{0} of the classical tangent complex. Subtract the same difference in H^{-1} [96, Definition 6.6 and (6.4)].

At objects with perfect derived Hom complexes, put $\chi(E, F) = \sum_{j} (-1)^{j} \dim \operatorname{Ext}^{j}(E, F)$. The positive-weight tangent is $\mathrm{RHom}(F, E)[1]$. Three-dimensional Calabi–Yau duality therefore gives

$$I(\alpha, \beta) = -\chi(F, E) = \chi(E, F). \tag{3.47}$$

The shift by one in the tangent is essential to this sign. It also shows that I is additive in each charge whenever the charges are taken in the Grothendieck group.

On connected components it satisfies the identity

$$I(\alpha, \beta) + I(\alpha + \beta, \gamma) = I(\beta, \gamma) + I(\alpha, \beta + \gamma). \tag{3.48}$$

This identity also follows directly by comparing the moving directions of a three-step filtration.

Theorem 3.25 (Kinjo–Park–Safronov)**.** *Under the stated geometric hypotheses and a strong orientation datum, write $\mathcal{P}_{\alpha} = \mathcal{P}_o|_{\mathcal{M}_{\alpha}}$. There is a natural coefficient isomorphism*

$$\zeta_{\alpha, \beta} : (\mathcal{P}_{\alpha} \boxtimes \mathcal{P}_{\beta})[I(\alpha, \beta)] \xrightarrow{\sim} \mathrm{R}q_{*} p^{!} \mathcal{P}_{\alpha + \beta}. \tag{3.49}$$

Its two composites on a three-step filtered correspondence agree, and its zero-object restrictions are the identities. Consequently the spaces $\mathsf{H}^{p}_{\alpha} = H^{p}(\mathcal{M}_{\alpha}, \mathcal{P}_{\alpha})$ have a unital associative Hall multiplication

$$\mathsf{H}^{p}_{\alpha} \otimes \mathsf{H}^{q}_{\beta} \longrightarrow \mathsf{H}^{p+q-I(\alpha,\beta)}_{\alpha+\beta}. \tag{3.50}$$

The coefficient theorem is [96, Theorem 8.6 and Proposition 8.29], with the two graded factors ordered as above. Its construction uses hyperbolic localization of oriented vanishing cycles. The following adjunction calculation specifies how it gives the product and fixes the target shift.

Construction of the product. Write $C_\alpha = \mathrm{R}\Gamma(\mathcal{M}_\alpha, \mathcal{P}_\alpha)$. First take the external product of sections. Then use (3.49):

$$\begin{aligned}(C_\alpha \otimes C_\beta)[I(\alpha,\beta)] &\longrightarrow \mathrm{R}\Gamma(\mathcal{M}_\alpha \times \mathcal{M}_\beta, (\mathcal{P}_\alpha \boxtimes \mathcal{P}_\beta)[I]) \\ &\xrightarrow{\zeta_{\alpha,\beta}} \mathrm{R}\Gamma(\mathcal{E}_{\alpha,\beta}, p^!\mathcal{P}_{\alpha+\beta}).\end{aligned}$$

The canonical external product suffices, without a Künneth isomorphism for an unrestricted infinite stack.

Properness of p identifies $\mathrm{R}p_*$ with $\mathrm{R}p_!$. The counit $\mathrm{R}p_! p^! \mathcal{P} \to \mathcal{P}$ gives

$$(C_\alpha \otimes C_\beta)[I(\alpha,\beta)] \longrightarrow C_{\alpha+\beta}, \quad \text{equivalently} \quad C_\alpha \otimes C_\beta \longrightarrow C_{\alpha+\beta}[-I(\alpha,\beta)]. \tag{3.51}$$

Taking cohomology gives (3.50).

For three inputs, identify both composed correspondences with the three-step filtered correspondence. Proper base change and composition of counits identify the two final integrations. The two coefficient maps agree by the associativity part of the coefficient theorem. Equation (3.48) makes their shifts equal, so the induced cohomological products agree with the usual Koszul signs for shifted complexes. The zero module, its orientation trivialization and the unitality of ζ give the unit, as in [96, Corollary 8.8]. □

The source shift in (3.49) determines the minus sign in (3.51). For a quiver with potential, the comparison with the critical quiver construction is [96, Proposition 8.37]. With subobject first, its index is $I(\alpha,\beta) = \chi_Q(\alpha,\beta) - \chi_Q(\beta,\alpha)$. Thus (3.50) recovers (3.44). Reversing the order of the inputs reverses this antisymmetric index. The one-arrow products (3.39) provide the elementary normalization.

3.8 Coherent coefficients and local flag calculations

The 2-Segal statement identifies spaces of composable flags. The coefficient theorem supplies the maps needed for cohomological Hall multiplication. To obtain an algebra of complexes with all associativity homotopies, retain coefficient maps and their coherences on every higher flag.

Here is a sufficient formulation. Take a class of flag correspondences closed under the derived fiber products used in composition. On Γ-graded complexes define the tensor product

$$(V \odot W)_\delta = \bigoplus_{\alpha+\beta=\delta} (V_\alpha \otimes W_\beta)[I(\alpha,\beta)].$$

Equation (3.48) gives its associator, using the standard signed identifications for shifts and tensor products. Assume a monoidal functor from the flag ∞-category of correspondences to this category of complexes.

Require that it sends the extension span to (3.51) and the zero span to its unit. Such a functor includes coherent base-change maps, composition maps and unit maps. Equality of the induced cohomology maps is weaker data.

Proposition 3.26 (Conditional coherent realization)**.** *A coefficient functor with these properties sends the unital 2-Segal flag object to a unital E_1-algebra in the specified category of charge-graded complexes.*

Proof. Each triangulation gives an iterated multiplication span. Proposition 3.23 identifies it with the span of the full flag. These identifications are equivalences of coherent diagrams, not choices of isomorphism classes.

Applying the assumed functor carries them and their compositions to the associativity coherences. Inserting a zero quotient gives the unit coherences. The cocycle identity (3.48) identifies their shifts. These are the operations and coherences of a unital E_1-algebra. □

On compact threefolds, an orientation compatible with direct sums must still satisfy the strong associativity condition before the coefficient theorem applies. A coherent realization must further retain the required higher coefficient comparisons. To produce holomorphic fields, one must in addition construct the coefficient system on ordered collision strata and compare its operations with the Hall maps. These are distinct constructions.

A local flag family can test those comparisons. Its coherent direct image records functions and derived specialization. Its constructible direct image records supported cohomology maps. The Hall integration map is determined by (3.49) and its proper counit. Neither a scheme-cycle multiplicity nor a Frobenius functional on the coherent algebra identifies that trace without a comparison of these coefficient operations.

Chapter 4

Direct sums and spectral Hall operations

Two operations on representations distinguish the geometry needed for a spectral coproduct. The Hall correspondence remembers a subrepresentation and integrates over its possible positions. The direct-sum inclusion fixes two summands and sets both off-diagonal arrow blocks to zero. Their normal bundles differ.

For one loop and two one-dimensional summands, write x, y for their Chern roots. In the Hall integral, the arrow normal weight $y - x$ cancels the tangent weight of the invariant-line flag. The two coordinate lines therefore give $\mathbf{1}_1 \star \mathbf{1}_1 = 2\mathbf{1}_2$. For the direct-sum inclusion, both off-diagonal entries vanish. Their weights are $y - x$ and $x - y$, so its Gysin map sends $1 \otimes 1$ to $-(y - x)^2$.

Here $\mathbf{1}_n$ denotes the constant class in charge n. Only $\mathbf{1}_0$ is the Hall unit.

The direct-sum Gysin map becomes invertible away from its normal Euler divisor. The Hall coproduct instead weights direct-sum pullback by the Euler class of a quiver Hom complex. Translating one summand makes its moving factors invertible as Laurent series. Compatibility with the Hall product then involves four blocks: each input has a part in each output. We first construct these maps on polynomial coefficients, then give their critical-cohomology form. An explicit pair of shuffle algebras will show which collision information their common exchange ratio fails to determine.

4.1 Direct-sum normal bundles

Work over $\mathbb{C}$ with rational cohomology. Let Q be a finite quiver, with vertex set Q_0 and arrow set Q_1. A dimension vector $d = (d_i)_{i \in Q_0}$ has nonnegative integral entries. Its representation space is

$$V_d = \bigoplus_{a : i \to j} \operatorname{Hom}(\mathbb{C}^{d_i}, \mathbb{C}^{d_j}).$$

Retain labeled roots and put

$$P_d = \mathbb{Q}[x_{i,r} : i \in Q_0,\ 1 \le r \le d_i], \qquad C_d = \operatorname{Spec} P_d, \qquad m(d) = \sum_{a : i \to j} d_i d_j.$$

Thus P_d is the equivariant cohomology of V_d for its coordinate torus. A root has degree two. The space C_d is the spectrum of the underlying ordinary ring, with this grading retained on its coefficient modules.

For chosen blocks of dimensions d, e, the inclusion $V_d \times V_e \hookrightarrow V_{d+e}$ has normal representation

$$N_{d,e} = \bigoplus_{a : i \to j} \Big(\operatorname{Hom}(\mathbb{C}^{d_i}, \mathbb{C}^{e_j}) \oplus \operatorname{Hom}(\mathbb{C}^{e_i}, \mathbb{C}^{d_j}) \Big).$$

Consequently its Euler class is

$$\mathcal{E}_{d,e}(x; y) = \prod_{a : i \to j} \left(\prod_{r \le d_i,\ s \le e_j} (y_{j,s} - x_{i,r}) \right) \left(\prod_{r \le e_i,\ s \le d_j} (x_{j,s} - y_{i,r}) \right). \tag{4.1}$$

Let $\mathcal{F}_d$ be the free rank-one graded sheaf on C_d, where a polynomial F has adjusted degree $\deg F - 2m(d)$. Under $C_{d+e} = C_d \times C_e$ for the chosen blocks, define

$$\nu_{d,e}(F \boxtimes G) = \mathcal{E}_{d,e}(x; y)F(x)G(y). \tag{4.2}$$

Proposition 4.1. *The maps $\nu_{d,e}$ have degree zero, are associative, and commute with block exchange and root relabeling. Charge zero gives their unit. On the open set where all cross-block roots are distinct, they are isomorphisms. For any ordered collection of blocks, every iterated map is*

$$\boxtimes_{r=1}^{k} F_r \longmapsto \left(\prod_{r<s} \mathcal{E}_{d_r,d_s}(x_r; x_s)\right) \prod_{r=1}^{k} F_r(x_r). \tag{4.3}$$

Proof. Complex orientation identifies the Gysin map with multiplication by the Euler class of $N_{d,e}$. Its degree is $2(m(d+e) - m(d) - m(e))$, which the three grading adjustments cancel. For three blocks the normal representation is the direct sum of the representations between each pair of blocks. Hence

$$\mathcal{E}_{d,e}\mathcal{E}_{d+e,f} = \mathcal{E}_{d,e}\mathcal{E}_{d,f}\mathcal{E}_{e,f} = \mathcal{E}_{e,f}\mathcal{E}_{d,e+f}.$$

The same decomposition proves (4.3) for every k. Both directions of each arrow occur, so exchanging the blocks preserves the normal representation and its Euler class. Relabeling roots permutes the factors. On the indicated open set each factor is invertible, and division gives the inverse map. An empty block contributes the empty product, equal to one. □

This is the direct-sum factorization over Chern-root parameters [102, §6.5]. Its extension across the Euler divisor contains more information than its restriction to that open set.

For one vertex with $\ell \geq 1$ loops, take two charge-one blocks and put $P = \mathbb{Q}[x, y]$, $\delta = y - x$. The Euler class is $(-1)^{\ell}\delta^{2\ell}$. Forgetting the internal polynomial grading, the cone of the normal map has the free representative

$$[\, P \xrightarrow{(-1)^{\ell}\delta^{2\ell}} P\,], \qquad \text{in cochain degrees } -1, 0. \tag{4.4}$$

It resolves $P/(\delta^{2\ell})$, since P is a domain.

For $D = P/(\delta)$ its derived restriction is $[D \xrightarrow{0} D]$. Thus the diagonal fiber has a class in each of degrees -1 and 0. Ordinary restriction of the degree-zero quotient alone would omit the first class.

4.2 The completion of an Euler inverse

An external torus weight does not by itself invert the Euler class of a bundle over a stack. Let L be the universal line on $B\mathbb{G}_m$, with $a = c_1(L)$, and let another torus $T = \mathbb{G}_m$ scale its fibers. Write u for the external parameter. The zero section has equivariant Euler class $a + u$. The Gysin sequence for its complement $L^{\times}$ gives

$$H_T^{\bullet}(L^{\times}; \mathbb{Q}) = \mathbb{Q}[a, u]/(a + u). \tag{4.5}$$

There is no odd term because multiplication by $a + u$ is injective.

After tensoring over $\mathbb{Q}[u]$ with $\mathbb{Q}(u)$, the quotient is still $\mathbb{Q}(u)$: the relation sets $a = -u$. Equivalently, $u + a$ is not a unit of $\mathbb{Q}(u)[a]$.

The stabilizer accounts for the surviving class. Present the complement as $[\mathbb{G}_m/H]$, where $H = \mathbb{G}_m$ acts by scaling. In the further quotient by T, the point represented by 1 has automorphisms (h, t) satisfying $ht = 1$. Their projection onto T is an isomorphism. The free action on each geometric fiber therefore does not remove the torus stabilizer of the stack point.

For a commutative $\mathbb{Q}$-algebra R, write $R((u^{-1}))$ for series whose powers of u are bounded above, completed toward decreasing powers. Coefficients may have unbounded degree in R.

Proposition 4.2. *For $R = \mathbb{Q}[a]$, the Euler class $u + a$ is a unit of $R((u^{-1}))$, with*

$$(u + a)^{-1} = u^{-1} \sum_{j \geq 0} (-a/u)^j. \tag{4.6}$$

On the finite approximation $\mathbb{P}^m$ of $B\mathbb{G}_m$, the series stops at $j = m$. More generally, every factor $wu + c$, with $0 \neq w \in \mathbb{Q}$ and $c \in R$, is invertible in this completion.

Proof. Multiplication of the partial sum through $j = m$ by $u + a$ gives $1 - (-a/u)^{m+1}$. The remainder tends to zero in the specified topology. On $\mathbb{P}^m$ it is zero because $a^{m+1} = 0$. Factoring $wu + c = wu(1 + c/(wu))$ proves the last assertion. □

The corresponding flag denominator is equally concrete. Let $T_2 = \mathbb{G}_m^2$ act on $\mathbb{C}^2$, put $R = \mathbb{Q}[x_1, x_2]$, and let h be the first Chern class of the tautological line on $\mathbb{P}^1$. Then

$$H^\bullet_{T_2}(\mathbb{P}^1) = R[h]/((h - x_1)(h - x_2)).$$

Restriction to the two coordinate lines is

$$r(f_0 + f_1 h) = (f_0 + x_1 f_1, f_0 + x_2 f_1).$$

Its matrix in the basis $1, h$ has determinant $\delta = x_2 - x_1$. Its image consists precisely of pairs (a, b) with $b - a \in \delta R$. After inverting δ, its inverse is

$$r^{-1}(a, b) = \frac{x_2 a - x_1 b}{\delta} + \frac{b - a}{\delta} h. \tag{4.7}$$

Inverting x_2 alone cannot give this inverse: at $x_1 = x_2 = 1$ the matrix has rank one.

For a bounded complex of bundles with nonzero external weights, its Euler factor is a finite product of factors $wu + c$ and their inverses. Proposition 4.2 constructs this characteristic class in the Laurent ring, independently of a splitting into Chern roots. A Gysin inverse requires a further identification: on the chosen fundamental class, the Gysin map must act by this Euler factor. Even coefficient extraction depends on the expansion: $u/(u + a)$ has constant term one at infinity and zero at the origin over $\mathbb{Q}(a)$.

4.3 The translated Hall coproduct

Allow a torus T to act on the arrows of Q, and let $R = H^\bullet(BT; \mathbb{Q})$. For an arrow a, denote its weight class by $h_a \in R^2$. A zero weight is allowed. For each dimension vector define the even polynomial coefficient module

$$A_d = R[x_{i,r} : 1 \leq r \leq d_i]^{\prod_i S_{d_i}}, \qquad A = \bigoplus_d A_d.$$

The charge sum is algebraic. Neither roots nor charges are completed.

Write

$$\chi(d, e) = \sum_i d_i e_i - \sum_{a : i \to j} d_i e_j.$$

In the subobject-first convention, the weighted Hall kernel is

$$K_{d,e}(x; y) = \frac{\displaystyle\prod_{a : i \to j} \prod_{r \leq d_i,\, s \leq e_j} (y_{j,s} - x_{i,r} + h_a)}{\displaystyle\prod_i \prod_{r \leq d_i,\, s \leq e_i} (y_{i,s} - x_{i,r})}. \tag{4.8}$$

The product $F \star G$ sums $F(x_I)G(x_J)K_{d,e}(x_I; x_J)$ over root partitions $I_i \sqcup J_i$ of sizes d_i, e_i at each vertex. There is no factorial divisor.

This formula is polynomial. A possible pole along a root diagonal at one vertex is simple. Pair the summands obtained by exchanging the two roots across the blocks. Their remaining factors agree on the diagonal, and their residues have opposite signs. Every possible prime denominator therefore cancels. The result is symmetric.

Both bracketings of three inputs sum over the same ordered three-block partitions, with one kernel factor for every ordered pair of blocks. This proves associativity in every charge and, by the same partition argument, coherence of all iterated products. The empty block gives the unit $1 \in A_0 = R$.

Geometrically, the numerator is the normal Euler class of the invariant-subspace condition. The denominator is the tangent Euler class of the flag [102, §§2.4–2.5]. The product has ordinary cohomological degree $-2\chi(d, e)$.

Define the Euler series, first as a rational function, by

$$E_{d,e}(u; x, y) = \frac{\displaystyle\prod_i \prod_{r \le d_i,\, s \le e_i} (u + y_{i,s} - x_{i,r})}{\displaystyle\prod_{a: i \to j} \prod_{r \le d_i,\, s \le e_j} (u + y_{j,s} - x_{i,r} + h_a)}. \tag{4.9}$$

Its leading term is $u^{\chi(d,e)}$. At $u = 0$ it is the inverse of (4.8) in the ring where the displayed Euler factors are inverted.

Now set

$$\Delta_{d,e}(u)F = E_{d,e}(u; x, y)F(x - u, y), \qquad F \in A_{d+e}, \tag{4.10}$$

where every first-block root is shifted by $-u$. Expanding at infinity gives a map $A_{d+e} \to (A_d \otimes_R A_e)((u^{-1}))$. Each input has finite upper Laurent order, and a fixed total charge has only finitely many splits. With $\deg u = 2$, this component has cohomological degree $2\chi(d, e)$.

Theorem 4.3. *On rational representatives the coproduct satisfies*

$$(\Delta_{d,e}(u) \otimes 1)\Delta_{d+e,f}(v) = (1 \otimes \Delta_{e,f}(v))\Delta_{d,e+f}(u + v). \tag{4.11}$$

The equality also holds in either chosen iterated Laurent chamber, using that chamber for every factor on both sides.

Proof. Splitting a block factors its Euler product into the products for its two constituents. Both sides of (4.11) therefore send F to

$$E_{d,e}(u; x, y)E_{d,f}(u + v; x, z)E_{e,f}(v; y, z)F(x - u - v, y - v, z).$$

This is an identity of rational functions before expansion. For example, one may expand in $(A_d \otimes_R A_e \otimes_R A_f)((u^{-1}))((v^{-1}))$, with v the outer variable and $u + v$ expanded at $v = \infty$. Applying this expansion to the common rational expression proves the asserted formal identity. It does not require extending either binary map to every element of an arbitrary two-variable completion. □

Projection onto A_0 is the counit $\varepsilon : A \to R$. The empty-block formulas are

$$(\varepsilon \otimes 1)\Delta(u) = \mathrm{id}, \qquad (1 \otimes \varepsilon)\Delta(u) = e^{uD}, \qquad D = -\sum_{i,r} \partial_{x_{i,r}}. \tag{4.12}$$

The exponential is finite on each polynomial. The even derivation D lowers cohomological degree by two and satisfies

$$\Delta(u)D = \partial_u\Delta(u) + (1 \otimes D)\Delta(u).$$

Indeed, differentiating the Euler factor with respect to u is the same as differentiating simultaneously in all second-block roots. These terms cancel against their contribution from D.

4.3.1 *Exchange and the four blocks*

Define

$$R_{d,e}(u; x, y) = \frac{E_{d,e}(u; x, y)}{E_{e,d}(-u; y, x)}. \tag{4.13}$$

This Laurent unit obeys $R_{d,e}(u; x, y)R_{e,d}(-u; y, x) = 1$. Its pairwise multipliers on three labeled blocks commute, hence satisfy the spectral Yang–Baxter identity. Splitting a block gives the corresponding fusion identities by factorization of E. The signs from negative arguments are included in (4.13).

Let τ exchange two blocks and let T_u shift all roots of a rational representative by $-u$. Then

$$\Delta_{d,e}(u) = R_{d,e}(u)\mathsf{T}_u\tau\Delta_{e,d}(-u). \tag{4.14}$$

Simultaneous translation preserves every root difference. The right side has polynomial part $F(x - u, y)$ and Euler factor $E_{d,e}(u)$, proving the identity on its stated rational domain.

Theorem 4.4 (The four-block identity)**.** *For $F \in A_\alpha$, $G \in A_\beta$ and $r + s = \alpha + \beta$,*

$$\begin{aligned}\Delta_{r,s}(u)(F \star G) = \sum_{\substack{a+c=\alpha,\ b+d=\beta \\ a+b=r,\ c+d=s}} (\star_{a,b} \otimes \star_{c,d})\Big[R_{b,c}(u; x_b, y_c) \\ \cdot \tau_{23}(\Delta_{a,c}(u)F \otimes \Delta_{b,d}(u)G)\Big].\end{aligned} \tag{4.15}$$

Here τ_{23} takes slots (a, c, b, d) to (a, b, c, d). After that permutation, $R_{b,c}$ multiplies the middle two slots. The two Hall products sum over divisions of the output roots. All operations are formed rationally and then expanded at infinity.

Proof. Write the first-output roots as x_a, x_b and the second-output roots as y_c, y_d. The coproduct on the left contributes $E_{a,c}E_{a,d}E_{b,c}E_{b,d}$ at parameter u. The Hall kernel between its two input blocks contributes

$$K_{a,b}(x_a; x_b), \quad E_{a,d}(u; x_a, y_d)^{-1}, \quad E_{c,b}(-u; y_c, x_b)^{-1}, \quad K_{c,d}(y_c; y_d).$$

The a, d factors cancel. The b, c factors give $E_{b,c}(u)/E_{c,b}(-u) = R_{b,c}(u)$. The remaining Euler factors are those of the two input coproducts. The remaining kernels are those of the two output products. Both constructions enumerate the same four labeled subsets. Thus their summands agree under this correspondence. □

For one vertex and two charge-one inputs, write $K(v)$ for the pair kernel and put $v = u + y - x$. The constant-input case is already

$$\Delta_{1,1}(u)(\mathbf{1}_1 \star \mathbf{1}_1) = \frac{K(v) + K(-v)}{K(v)} = 1 + R_{1,1}(u; x, y).$$

The crossed factor is therefore part of the multiplication comparison itself. The Yang–Baxter equation alone would not determine this comparison. Equations (4.15) and (4.12) use the unshifted even coefficient modules. A different grading must transport their tensor shifts and permutation signs together.

4.4 Critical coefficients

The same translation mechanism applies when ordinary coefficients are replaced by vanishing cycles. Let W be a potential, a finite linear combination of cyclic paths in Q, and assume that the arrow action of T preserves W. The arrow weights belong to the character lattice of T. They need not lie in a positive cone. For $G_d = \prod_i GL_{d_i}$, put

$$\mathfrak{M}_d^T = [V_d/(G_d \times T)], \qquad w_d = \mathrm{Tr}(W), \qquad B_d = H^\bullet(\mathfrak{M}_d^T, \phi_{w_d}\mathbb{Q}[-\chi(d,d)]).$$

Use the perverse normalization of vanishing cycles and the shift convention $K[s]^n = K^{n+s}$. The shift $-\chi(d,d)$ is the dimension of $[V_d/G_d]$, before passage to the external torus quotient. The coefficient ring remains $R = H^\bullet(BT;\mathbb{Q})$. The charge-dependent tensor convention is

$$B_d \otimes^T B_e = (B_d \otimes_R B_e)[\chi(d,e) - \chi(e,d)].$$

Assumption 4.5 (Relative Kunneth)**.** *For every pair d, e, the relative Kunneth map and Thom–Sebastiani comparison identify*

$$B_d \otimes_R B_e \xrightarrow{\sim} H^\bullet\Big(\mathfrak{M}_d^T \times_{BT} \mathfrak{M}_e^T, \phi_{w_d \boxplus w_e}\mathbb{Q}[-\chi(d,d) - \chi(e,e)]\Big). \tag{4.16}$$

This is an assumption over the base BT, imposed component by component. A Kunneth isomorphism for sheaves on BT does not by itself give (4.16) for their cohomology modules. All the critical coproducts below use this hypothesis.

On the product stack let E_i and F_i be the tautological bundles of the first and second representation. If L_a is the line on BT associated with the arrow weight, define the perfect complex

$$\mathcal{V}_{d,e} = \left[\bigoplus_i E_i^\vee \otimes F_i \longrightarrow \bigoplus_{a:i\to j} E_i^\vee \otimes F_j \otimes L_a\right], \qquad \text{in degrees } 0, 1. \tag{4.17}$$

The differential sends a family of maps to its commutators with the arrow maps. This is the derived Hom complex for the underlying quiver representations. Its virtual rank is $\chi(d,e)$. It is not the full self-Ext complex in the associated three-dimensional Calabi–Yau category.

For a virtual bundle V, write

$$\Psi(V,z) = z^{\mathrm{rk}\,V} c(V, z^{-1}), \qquad c(V,t) = \sum_{k\ge 0} c_k(V) t^k. \tag{4.18}$$

In a two-term presentation $V = [V^0 \to V^1]$, this is the ratio $\Psi(V^0,z)/\Psi(V^1,z)$. In Chern roots it is the product of z+root for V^0, divided by the corresponding product for V^1, expanded at infinity. Thus

$$\Psi(V \oplus V', z) = \Psi(V,z)\Psi(V',z), \qquad \Psi(V \otimes L, z) = \Psi(V, z + c_1(L)).$$

Tensoring a representation by a line defines an action of $B\mathbb{G}_m$. On coefficients its coaction is $\mathrm{act}_z^* = e^{zT}$, where z is the first Chern class of the line and T has degree -2. It sends each Chern root x to $x + z$. Write $\oplus^*$ for the critical pullback along direct sum, followed by (4.16). The potential identity $\mathrm{Tr}\, W(E \oplus F) = w_d(E) + w_e(F)$ supplies the Thom–Sebastiani comparison used in this map.

A *coassociative vertex coproduct* here means a Laurent map satisfying parameter-dependent coassociativity (4.11) and the translation identity. It also satisfies the two counit identities (4.12). Use z, T in place of u, D and the specified tensor convention. Its right counit is polynomial on each input. This definition imposes no exchange axiom.

Theorem 4.6 (The critical spectral coproduct)**.** *Under Assumption 4.5, the maps*

$$\Delta^{\mathrm{crit}}_{d,e}(z) = \Psi(\mathcal{V}_{d,e}, -z)\,\mathrm{act}^*_{z,1} \oplus^* \tag{4.19}$$

give a coassociative vertex coproduct on $B = \bigoplus_d B_d$. Its component has target

$$(B_d \otimes_R B_e)[\chi(d,e) - \chi(e,d)]((z^{-1})).$$

The covacuum is projection to $B_0 = R$. Its left counit is the identity and its right counit is e^{zT}.

Proof. The direct-sum pullbacks are coassociative and commute with simultaneous line tensoring. Pulling $\mathcal{V}$ back along a direct sum in either variable splits it into the corresponding two Ext complexes. Tensoring the first variable by the line changes its characteristic argument by $-z$. Consequently the two three-block composites have common factor

$$\Psi(\mathcal{V}_{d,e}, -z)\Psi(\mathcal{V}_{d,f}, -z-w)\Psi(\mathcal{V}_{e,f}, -w).$$

They tensor the first block by the line of class $z+w$ and the second by the line of class w.

This proves the coassociativity identity with parameters $z, w, z+w$, in compatible Laurent expansions. The empty Ext complex gives the two counit identities. Differentiating the line action gives $\Delta^{\mathrm{crit}}(z)T = \partial_z \Delta^{\mathrm{crit}}(z) + (1 \otimes T)\Delta^{\mathrm{crit}}(z)$.

The perverse shifts give direct-sum pullback degree $-\chi(d,e) - \chi(e,d)$. Multiplication by the characteristic factor adds $2\chi(d,e)$. The stated target shift makes the resulting map degree zero. These are the geometric and grading identities of [87, Theorem 2.3.1]. □

The localized operation before line translation is

$$\Delta^{\mathrm{loc}}_{d,e} = e(\mathcal{V}_{d,e})\oplus^*, \qquad e(\mathcal{V}_{d,e}) = \frac{e(\mathcal{V}^0_{d,e})}{e(\mathcal{V}^1_{d,e})}.$$

Inverting the indicated Euler classes makes this expression defined. Applying $\mathrm{act}^*_{z,1}$ and expanding at infinity changes $e(\mathcal{V})$ to $\Psi(\mathcal{V}, -z)$. This identifies the localized and spectral constructions [87, Lemma 3.2.4 and Theorem 3.3.2]. For zero potential, the unshifted polynomial formula (4.10) uses $z = -u$.

4.4.1 Braided multiplication and its signs

Assume now that Q is symmetric: for each pair of distinct vertices the numbers of arrows in the two directions agree. Thus χ is symmetric, although the arrow weights need not be paired by negation. The tensor shift in Theorem 4.6 vanishes. Let H be the tautological ring, the R-subalgebra of $\prod_d H^\bullet(\mathfrak{M}^T_d)$ generated by the Chern characters of the vertex bundles, including their ranks. Direct sum and line tensoring give its coproduct $\delta_H(z) = \mathrm{act}^*_{z,1} \oplus^*$. It acts on B by cup product.

Let σ denote the graded swap, with its Koszul sign on cohomology. The characteristic multiplier

$$\mathscr{R}_{d,e}(z) = \frac{\Psi(\sigma^*\mathcal{V}^\vee_{e,d}, z)}{\Psi(\mathcal{V}_{d,e}, z)} \tag{4.20}$$

has coefficients in $H \otimes_R H$ and leading term one. Its spectral hexagon identities follow from direct-sum factorization and line translation of the two complexes. These identities define the meromorphic exchange $\beta(z) = \sigma \circ (\mathscr{R}(z) \cdot -)$ on tautological modules, using $\delta_H(z)$ for the coefficient action.

In Chern-root coordinates its relation to (4.13) is

$$\mathscr{R}_{e,d}(-u; y, x) = (-1)^{\chi(d,e)} R_{d,e}(u; x, y). \tag{4.21}$$

Indeed, duality gives $\Upsilon(V^\vee, z) = (-1)^{\mathrm{rk}\, V}\Upsilon(V, -z)$. Thus the change $z = -u$ includes both a block reversal and the displayed Euler sign.

The perverse grading contributes another sign. Put

$$p(d) = \chi(d,d) \pmod 2, \qquad \tau(d,e) = p(d)\,p(e) + \chi(d,e) \pmod 2. \tag{4.22}$$

Symmetry of χ makes p linear modulo two and makes τ alternating and bilinear. Use the exchange

$$\beta^{\tau}_{d,e}(z) = (-1)^{\tau(d,e)}\beta_{d,e}(z).$$

The critical Hall product m and the coproduct satisfy

$$\Delta^{\mathrm{crit}}(z)m = (m \otimes m)(1 \otimes \beta^{\tau}(z) \otimes 1)(\Delta^{\mathrm{crit}}(z) \otimes \Delta^{\mathrm{crit}}(z)). \tag{4.23}$$

This is the untwisted-product form of the critical vertex-bialgebra theorem [87, Theorem 4.5.2 and its untwisted form]. Its proof starts with the localized Hall compatibility, translates the first and third blocks, and expands their invertible Euler factors. Cup-product linearity permits translation through the two Hall products. The crossed Euler factor becomes (4.20). The Koszul and perverse signs give (4.22).

One can move this sign into the operations, provided both are transported. Order the vertices, with basis dimension vectors e_i, and choose

$$\psi(d,e) = \sum_{i<j} \tau(e_i, e_j)\, d_i e_j \pmod 2.$$

Then $\psi(d,e) + \psi(e,d) = \tau(d,e)$ and ψ is bilinear.

Proposition 4.7 (Simultaneous sign transport)**.** *Define, on their charge components,*

$$m^{\psi}_{d,e} = (-1)^{\psi(d,e)}m_{d,e}, \qquad \Delta^{\psi}_{d,e}(z) = (-1)^{\psi(d,e)}\Delta^{\mathrm{crit}}_{d,e}(z). \tag{4.24}$$

They remain unital, associative and spectrally coassociative, with the same counits and translation. Their multiplication compatibility uses β in place of β^{τ}.

Proof. Bilinearity gives $\psi(d,e) + \psi(d+e,f) = \psi(e,f) + \psi(d,e+f)$, so the two associativity identities retain their signs. Also $\psi(d,0) = \psi(0,d) = 0$.

For four blocks set $r = a+b$, $s = c+d$, $\alpha = a+c$, $\beta = b+d$. Expanding the bilinear terms gives

$$\psi(r,s) + \psi(\alpha,\beta) + \tau(c,b) = \psi(a,b) + \psi(c,d) + \psi(a,c) + \psi(b,d) \pmod 2.$$

The left side is the sign on the left of (4.23), including its old crossed sign. The right side is the sign of the four transported operations. This proves the assertion. □

Changing only multiplication need not preserve this square. For two isolated vertices, χ is the identity matrix and $\psi(e_1, e_2) = 1$. The straight four-block term $(a,c,b,d) = (e_1, 0, 0, e_2)$ changes sign on the product of the two inputs, while neither product in the two outputs changes sign. The coproduct factors in (4.24) supply the required correction.

For a nonsymmetric quiver, the characteristic ratio (4.20) has leading power $z^{\chi(e,d)-\chi(d,e)}$. The componentwise module formulation of [87, §4.5.4] therefore has a different coefficient recipient from the tautological-ring formulation above.

The displayed symmetric sign prescription also has a precise limit. For one arrow $1 \longrightarrow 2$,

$$\chi(e_1,e_2) = -1, \quad \chi(e_2,e_1) = 0, \quad \tau(e_1,e_2) = 0, \quad \tau(e_2,e_1) = 1.$$

No function ψ can then satisfy $\psi + \psi^{\mathrm{op}} = \tau$. Extending that sign convention requires a compatible choice on the shifted componentwise tensor product. The polynomial identity (4.15), with its explicitly even convention, already applies to this quiver.

4.5 An exchange ratio and two integral lattices

Exchange compares a pair kernel with its reverse. Multiplying the kernel by an even function leaves that ratio unchanged, but can change multiplication on the collision divisor. To measure the change, keep the entire polynomial coefficient module in each charge.

For this section take

$$R = \mathbb{Q}[h_1, h_2], \qquad h_3 = -h_1 - h_2, \qquad a = \sum_{i<j} h_i h_j, \qquad b = h_1 h_2 h_3. \tag{4.25}$$

Thus $a = -(h_1^2 + h_1 h_2 + h_2^2)$ and $b = -h_1 h_2 (h_1 + h_2) \neq 0$ in the domain R. No parameter is inverted.

Define two independently specified polynomial carriers

$$A_n^{(j)} = R[x_1, \dots, x_n]^{S_n}, \qquad \mathcal{A}^{(j)} = \bigoplus_{n \geq 0} A_n^{(j)}, \qquad j = 0, 1,$$

with even parity and algebraic charge sum. Their pair kernels are

$$K_0(v) = \frac{\prod_{i=1}^{3}(v + h_i)}{v} = v^2 + a + \frac{b}{v}, \qquad K_1(v) = v^2 K_0(v). \tag{4.26}$$

Here v is a root difference, not a coefficient parameter.

For $F \in A_d^{(j)}$ and $G \in A_e^{(j)}$, set

$$(F *_j G)(X) = \sum_{\substack{I \sqcup J = [d+e] \\ |I| = d}} F(X_I) G(X_J) \prod_{r \in I,\, s \in J} K_j(x_s - x_r). \tag{4.27}$$

The sum has no factorial divisor and the empty charge is its unit.

The pole-cancellation and ordered-partition arguments of Section 4.3 prove polynomiality and associativity. For K_1, polynomiality also follows term by term. The first kernel is the zero-potential three-loop Hall kernel. Two additional loops of weight zero give the second kernel. This is a calculation of ordinary normal Euler classes, without a critical-cohomology identification.

Both kernels have exchange ratio

$$\frac{K_0(-v)}{K_0(v)} = \frac{K_1(-v)}{K_1(v)} = \frac{v^3 + av - b}{v^3 + av + b} = \frac{\prod_i (v - h_i)}{\prod_i (v + h_i)}. \tag{4.28}$$

The equality cancels the even factor v^2. To decide whether that cancellation comes from an invertible change of polynomial coefficients, compute the whole binary image.

4.5.1 *Binary generation and its cokernel*

In charge two put

$$c = \frac{x + y}{2}, \qquad v = y - x, \qquad t = v^2, \qquad B = R[c, t].$$

Then $A_2^{(j)} = B$ and $A_1^{(j)} \otimes_R A_1^{(j)} = R[c, v] = B \oplus vB$. Pure tensors span this entire source. For $p = A + vC$, with $A, C \in B$, linearized multiplication is therefore

$$\mu_0(A + vC) = 2((t + a)A + bC), \qquad \mu_1(A + vC) = 2t((t + a)A + bC). \tag{4.29}$$

Indeed, the two terms for K_0 are $(v^2 + a + b/v)(A + vC)$ and $(v^2 + a - b/v)(A - vC)$. Their sum gives the first formula. The common even factor v^2 gives the second. Since 2 is invertible, their images are

$$\operatorname{im} \mu_0 = J = (t + a, b)B, \qquad \operatorname{im} \mu_1 = tJ. \tag{4.30}$$

Regard each product as the two-term cochain complex

$$M_j = [B \oplus vB \xrightarrow{\mu_j} B], \qquad \text{in degrees } -1, 0.$$

It records binary generation, rather than a full deformation complex.

Proposition 4.8. *The cohomology of these complexes is*

$$H^{-1}(M_0) = H^{-1}(M_1) = B\,(b, -(t + a)), \tag{4.31}$$

$$H^0(M_0) = R[c]/(b), \tag{4.32}$$

$$H^0(M_1) = R[c] \oplus (R[c]/(b))\,t \quad \text{as } R[c]\text{-modules}. \tag{4.33}$$

On the last module, multiplication by t is $t(A, C) = (0, A - aC)$, with the second coordinate reduced modulo b. Evaluation at $t = 0$ gives a cochain map $M_1 \to R[c][0]$. It detects the nonzero class of $b\mathbf{1}_2$.

Proof. If $(t + a)A + bC = 0$, reduction modulo $t + a$ gives $b\overline{C} = 0$ in the domain $R[c]$. Thus $C = -(t + a)H$ and $A = bH$. Multiplication of the row by the non-zero-divisor t changes no kernel, proving (4.31).

The first cokernel sets $t = -a$ and $b = 0$. In the second, the relations are $t^2 = -at$ and $bt = 0$. Division by the monic polynomial $t^2 + at$ gives a unique expression $A(c) + tC(c)$, with C then taken modulo b. This proves both the module description and its t-action.

Every value of μ_1 is divisible by t, so evaluation at zero annihilates the differential. It sends $b\mathbf{1}_2$ to the nonzero element b. By contrast, $\mu_0(v/2) = b\mathbf{1}_2$. □

An *ambient lattice equivalence* will mean a family of R-linear bijections $U_n : A_n^{(0)} \to A_n^{(1)}$ with $U_0 = \mathrm{id}_R$ and

$$U_{d+e}(F *_0 G) = U_d(F) *_1 U_e(G). \tag{4.34}$$

No preservation of root degree, translation or multiplication by c, t is assumed. In particular, these are not restricted to maps given by scalar multiplication.

Theorem 4.9. *There is no ambient lattice equivalence. The conclusion remains true if every U_n is semilinear for one automorphism of R.*

Proof. The R-annihilators of the binary cokernels are

$$\operatorname{Ann}_R H^0(M_0) = (b), \qquad \operatorname{Ann}_R H^0(M_1) = 0.$$

The first follows from (4.32). The second follows from the quotient $H^0(M_1) \twoheadrightarrow R[c]$, a faithful R-module. The charge-two instance of (4.34) would identify the complexes through $U_1 \otimes U_1$ in degree -1 and U_2 in degree zero. Their cokernels would then have the same annihilator, a contradiction.

Under a parameter automorphism σ, the nonzero annihilator b is carried to the nonzero element $\sigma(b)$. The contradiction persists. No step requires U_2 to preserve the coordinate t. □

The same distinction survives inversion of parameters alone. For a submodule $L \subset B$ define its parameter saturation by

$$L_R^{\mathrm{sat}} = \{F \in B : rF \in L \text{ for some } 0 \neq r \in R\}.$$

Since $bB \subset J$ and $btB \subset tJ$,

$$J_R^{\mathrm{sat}} = B, \qquad (tJ)_R^{\mathrm{sat}} = tB. \tag{4.35}$$

For the reverse inclusion in the second equality, if $rF \in tB$, evaluation at $t = 0$ gives $rF(c, 0) = 0$ in the domain $R[c]$. Thus $F \in tB$. Over $\mathrm{Frac}(R)$, the first binary product is consequently surjective and the second still has image $t\,\mathrm{Frac}(R)[c, t]$. In particular, $\mu_0(v/(2b)) = \mathbf{1}_2$ but $\mu_1(v/(2b)) = t\mathbf{1}_2$.

4.5.2 *The comparison that does exist*

Set $D_0 = D_1 = 1$ and

$$D_n(X) = \prod_{r<s} (x_s - x_r)^2, \qquad \Phi_n(F) = D_n F. \tag{4.36}$$

For each partition $I \sqcup J = [d + e]$,

$$D_{d+e}(X) = D_d(X_I) D_e(X_J) \prod_{r \in I,\, s \in J} (x_s - x_r)^2. \tag{4.37}$$

Thus the even factors omitted by exchange supply an actual map.

Theorem 4.10. *The maps Φ_n define a unital, injective, charge-preserving algebra map $\Phi : \mathcal{A}^{(0)} \to \mathcal{A}^{(1)}$. They commute with simultaneous root translation and satisfy*

$$\Phi_{d+e}(F *_0 G) = \Phi_d(F) *_1 \Phi_e(G). \tag{4.38}$$

They are not surjective on A_n for $n \geq 2$. If $C^{(j)}$ is the subalgebra generated by $A_1^{(j)}$, restriction gives an integral isomorphism

$$\Phi : C^{(0)} \xrightarrow{\sim} C^{(1)}, \qquad C_n^{(1)} = D_n C_n^{(0)}. \tag{4.39}$$

Proof. In every summand on the right of (4.38), the internal discriminants and the cross squares from $K_1 = v^2 K_0$ combine, by (4.37), into D_{d+e}. This factor is independent of the subset and can be taken outside the sum. The same factorization over any ordered partition proves compatibility with every iterated product.

Each D_n is nonzero in a domain, so multiplication by it is injective. For $n \geq 2$ its image cannot contain the constant polynomial, since D_n vanishes at $x_1 = x_2$. The factors $D_0 = D_1 = 1$ give the unit and the identity on charge one. A multiplicative injection fixing those generators therefore maps $C^{(0)}$ exactly onto $C^{(1)}$. Division by D_n on this specified image gives its inverse. Finally $\sum_i \partial_{x_i} D_n = 0$, proving translation compatibility. □

This isomorphism of generated algebras and Theorem 4.9 concern different modules. The former identifies the generated images, whereas the latter excludes its extension to bijections of all polynomial coefficients. For compatible even gradings, give roots and parameters raw degree two and set

$$\deg_0 F = \deg_{\mathrm{raw}} F - 2n(n-1), \qquad \deg_1 F = \deg_{\mathrm{raw}} F - 4n(n-1), \qquad F \in A_n.$$

Both products and Φ then have degree zero. The obstruction requires none of these grading restrictions on the proposed gauges.

Define the spectral maps of the two algebras independently by

$$\Delta^{(j)}_{d,e}(s)F(X,Y) = \prod_{r,k} K_j(s + y_k - x_r)^{-1}\, F(X - s, Y). \tag{4.40}$$

They are rationally coassociative by Theorem 4.3, and expand at infinity with coefficients in $A^{(j)}_d \otimes_R A^{(j)}_e$. Their left counit is the identity and their right counit is translation by $-s$.

Proposition 4.11. *The discriminant maps preserve these coproducts:*

$$\Delta^{(1)}_{d,e}(s)\Phi_{d+e} = (\Phi_d \otimes_R \Phi_e)\Delta^{(0)}_{d,e}(s). \tag{4.41}$$

The products extend to $\mathcal{A}^{(j)}_{\mathrm{reg}} = \bigoplus_n A^{(j)}_n[D_n^{-1}]$. On these carriers Φ is an algebra isomorphism preserving translation and the coproducts with localized output coefficients.

Proof. The translated factorization is

$$D_{d+e}(X - s, Y) = D_d(X)D_e(Y)\prod_{r,k}(s + y_k - x_r)^2.$$

The cross factors cancel the extra squares in the denominators of $\Delta^{(1)}$, proving (4.41) before and after expansion.

Every internal discriminant and every cross-root difference is invertible after the total discriminant is inverted. Each localized shuffle therefore has poles only on that total divisor and lies in the stated localized symmetric ring. Multiplication by D_n now has inverse D_n^{-1}. Translation fixes D_n, and the displayed translated factorization proves the coproduct assertion. In charge two the map of binary complexes (id, t) is an isomorphism after t is inverted. No coefficient parameter such as b needs to be inverted. □

4.5.3 *The remaining collision module*

The classes discarded by the discriminant localization form the coefficient modules

$$Q_n = A^{(1)}_n / D_n A^{(1)}_n, \qquad Q = \bigoplus_n Q_n.$$

Their support lies on the internal root discriminant. In particular, $Q_0 = Q_1 = 0$ and $Q_2 = B/(t) = R[c]$. There is an exact sequence

$$0 \longrightarrow \mathcal{A}^{(0)} \xrightarrow{\Phi} \mathcal{A}^{(1)} \longrightarrow Q \longrightarrow 0 \tag{4.42}$$

of $\mathcal{A}^{(0)}$-bimodules, with quotient actions

$$a \cdot [z] = [\Phi(a) *_1 z], \qquad [z] \cdot a = [z *_1 \Phi(a)]. \tag{4.43}$$

These are well-defined because replacing z by $z + \Phi(c)$ changes the two products by $\Phi(a *_0 c)$ and $\Phi(c *_0 a)$. Associativity proves the bimodule laws. Translation preserves the sequence, since it fixes D_n and is a derivation of both products. The sequence is not an algebra quotient: its proper kernel contains the unit and therefore cannot be an ideal of a nonzero unital quotient.

The charge-two cone is $[B \xrightarrow{t} B]$ in degrees -1, 0, with cokernel $B/(t)$. The same cone occurs for the chain map $(\mathrm{id}, t) : M_0 \to M_1$. Indeed, write $P = B^2$. Its cone has terms P, $P \oplus B$, B in degrees -2, -1, 0, with differentials

$$p \longmapsto (p, -\mu_0 p), \qquad (q, z) \longmapsto t(\mu_0 q + z).$$

The coordinate $w = z + \mu_0 q$ splits off the contractible pair $P \xrightarrow{1} P$, leaving $[B \xrightarrow{t} B]$.

On the ordered root cover this quotient is $R[c, v]/(v^2)$. Its normal direction has length two there, while on the symmetric quotient its coordinate $t = v^2$ gives length one.

4.5.4 *Other even factors*

The same argument applies to a general monic even multiplier. Let R now be a commutative $\mathbb{Q}$-algebra that is a domain, and choose $a, b \in R$ with $b \neq 0$. Put $q(v) = P(v^2)$, where $P(t) \in R[t]$ is monic of positive degree.

Theorem 4.12. *The shuffle products for $K(v) = v^2 + a + b/v$ and $q(v)K(v)$ are polynomial and associative. There is no charge-preserving R-linear equivalence of their full polynomial algebras. Their current-generated subalgebras are nevertheless isomorphic through multiplication in charge n by*

$$\prod_{r<s} q(x_s - x_r). \tag{4.44}$$

This map is injective on the full carriers and becomes an algebra isomorphism after the displayed pair product is inverted. The two exchange ratios agree.

Proof. Pole cancellation over $\mathrm{Frac}(R)$ removes the possible simple diagonal denominators. Division by each monic linear root difference already takes place over R. Successive such divisions remove the full denominator: over the fraction field, distinct differences are relatively prime. Ordered partitions then prove associativity over R.

In charge two the images are $J = (t + a, b)$ and $P(t)J$ in $B = R[c, t]$. The first cokernel is killed by b. The second surjects onto $B/(P(t))$, which is a nonzero free $R[c]$-module with basis $1, t, \ldots, t^{\deg P - 1}$. It is faithful over R. The scalar-annihilator argument therefore excludes every proposed equivalence of full carriers. Their parameter saturations are B and $P(t)B$: use $bP(t)B \subset P(t)J$ and the R-torsion-freeness of $B/P(t)B$.

The pair product (4.44) factors into internal and cross-block products. Substitution in each shuffle summand proves multiplicativity and the isomorphism on generated subalgebras. It is injective over the domain.

For localized inputs, a sufficiently high power of the total pair product clears their internal denominators. The remaining diagonal poles cancel by the polynomiality argument. This proves closure under localized multiplication and gives the stated inverse. It does not require that inverting $q(v)$ invert v. Finally $q(-v) = q(v)$ proves equality of the exchange ratios. □

At $(h_1, h_2, h_3) = (1, 1, -2)$, the original example has $a = -3$, $b = -2$. Its two constant-current products are $2(t - 3)$ and $2t(t - 3)$. The tensor polynomial $-v/4$ multiplies to $\mathbf{1}_2$ for K_0 and to $t\mathbf{1}_2$ for K_1.

At a fiber where $b = 0$, the broad class of linear gauges requires a separate invariant. For example, when $a = b = 0$, the two cokernels are $\mathbb{Q}[c, t]/(t)$ and $\mathbb{Q}[c, t]/(t^2)$. Their t-module structures differ, but an arbitrary $\mathbb{Q}$-linear gauge need not preserve that operator. The integral theorem asserts no such fiberwise classification.

4.6 From coefficient collisions to fields

The coefficient constructions now retain their operations: direct-sum Gysin maps and their cones, Hall multiplication, spectral translation, the four-block square, and the discriminant comparison. The even-factor example shows why a common exchange ratio cannot recover the coefficient module supported at a collision. It also supplies a positive comparison on generated algebras and on the complement of that collision.

Problem 4.13. Construct a geometric coefficient family on a spacetime collision space which realizes these spectral operations. Its comparison with the Chern-root family must preserve Hall multiplication, translation, the empty-charge unit and the chosen local vacuum. Across the collision divisor it must realize the derived coefficient cones and their attachment maps.

For (4.42), construct both bimodule actions (4.43) geometrically and identify the first relative term with $[R[c,t] \xrightarrow{t} R[c,t]]$, including the quotient class detected by $t = 0$. At three or more points, require compatibility on all intersecting collision divisors and under every successive merger.

This problem requires a map between two independently specified families. Internal root differences have so far been coefficient coordinates. Their substitution for spacetime separations needs pullback, support and continuity data. The translated counit of the spectral coproduct also remains part of that comparison. It does not become an ordinary Hopf counit by choosing an expansion.

Chapter 5

The local threefold benchmark

The affine threefold $\mathbb{C}^3$ gives a concrete setting in which to compare Hall operations with fields on a curve. Its category of finite-length sheaves has a critical Hall algebra. A free holomorphic gauge theory on the same threefold has curve defects and their operator product expansions (OPEs). We construct these two inputs with their own coefficients.

The Hall algebra admits an affine-Yangian presentation over its polynomial coefficient ring. The defect has an explicit Gaussian Ward identity. A comparison between them must preserve the operations computed on each side.

5.1 A potential and its critical coefficients

Let Q have one vertex and three loops x, y, z. Its path algebra is $\mathbb{C}\langle x, y, z\rangle$. Identify cyclic rotations in the vector space of words, and take the potential

$$W = zxy - zyx = z[x, y]. \tag{5.1}$$

The cyclic derivative with respect to a generator sums over its occurrences, removing each occurrence and reading the remaining word from the next position. Thus

$$\partial_x W = yz - zy, \qquad \partial_y W = zx - xz, \qquad \partial_z W = xy - yx.$$

The Jacobi algebra, obtained by imposing these three derivatives, is $\mathbb{C}[x, y, z]$. Its finite-dimensional modules are precisely finite-length coherent sheaves on $\mathbb{C}^3$.

Let $T = (\mathbb{C}^*)^2$ act on the loops with additive characters $a, b, -a - b$. Each term of W has weight zero. Put

$$R = H_T^\bullet(\mathrm{pt}, \mathbb{Q}) = \mathbb{Q}[a, b], \qquad |a| = |b| = 2.$$

Charge $n \geq 0$ means representation dimension n. The smooth representation stack and its function are

$$\mathcal{M}_n = [\mathrm{Mat}_n(\mathbb{C})^3 / GL_n], \qquad f_n(X, Y, Z) = \mathrm{Tr}\, Z[X, Y].$$

The classical critical locus parametrizes commuting triples. The coefficient complex also records the singularity of f_n in its smooth ambient stack. It is additional to the underlying set of commuting triples.

Write Φ_f for shifted vanishing cycles, extended from the zero fiber, with convention $\Phi_0 \mathbb{Q} = \mathbb{Q}$. The virtual shift on $\mathcal{M}_n$ is its complex dimension $2n^2$. We use the relative equivariant convention: quotienting by the external torus does not subtract $\dim T$ from that shift. Define

$$\mathcal{H}_n^T = H^\bullet(\mathcal{M}_n / T, \Phi_{f_n} \mathbb{Q}[2n^2]), \qquad \mathcal{H}^T = \bigoplus_{n \geq 0} \mathcal{H}_n^T. \tag{5.2}$$

The extension correspondence, with its critical coefficient maps, gives the Hall product. It adds charge and preserves cohomological degree. Its unit is $1 \in \mathcal{H}_0^T = R$.

At charge one the function is zero, so

$$\mathcal{H}_1^T = R[u][2], \qquad |u^r| = 2r - 2. \tag{5.3}$$

Here $u = c_1(L)$ for the universal one-dimensional representation L. It is independent of the external parameters a, b. We fix the root convention for the Hall product below. In particular, charge one has infinite rank over R.

5.2 The cut and the product sign

The potential is linear in every entry of Z. Forgetting Z therefore leads to the moment equation $[X, Y] = 0$. Let

$$C_n = \{(X, Y) \in \mathrm{Mat}_n(\mathbb{C})^2 : [X, Y] = 0\}, \qquad \mathcal{P}_n^T = H_\bullet^{\mathrm{BM},T}([C_n/GL_n], \mathbb{Q}).$$

We regard Borel–Moore homology as cohomologically graded by reversing its index. Its equivariant normalization has $\mathcal{P}_0^T = R$ and $\mathcal{P}_1^T = R[u][2]$. The product on $\mathcal{P}^T = \bigoplus_n \mathcal{P}_n^T$ is the refined pullback along the moment correspondence followed by the proper map forgetting the invariant subspace.

The refinement already matters for a two-dimensional representation. For an invariant first coordinate, write

$$X = \begin{pmatrix} x_1 & r \\ 0 & x_2 \end{pmatrix}, \qquad Y = \begin{pmatrix} y_1 & v \\ 0 & y_2 \end{pmatrix}.$$

Then

$$[X, Y] = \begin{pmatrix} 0 & \rho \\ 0 & 0 \end{pmatrix}, \qquad \rho = (x_1 - x_2)v - (y_1 - y_2)r. \tag{5.4}$$

Each diagonal block commutes, but the extension must still satisfy $\rho = 0$. On the coincident diagonal $x_1 = x_2$, $y_1 = y_2$, this equation vanishes identically. Its derived zero fiber retains the odd generator of the Koszul complex

$$\mathbb{Q}[x_1, x_2, y_1, y_2, r, v][\eta], \qquad |\eta| = -1, \quad d\eta = \rho.$$

After derived specialization to that diagonal, $d\eta = 0$. Ordinary restriction of the classical zero locus would lose this negative-degree class.

Dimensional reduction provides isomorphisms of graded R-modules $\theta_n : \mathcal{P}_n^T \to \mathcal{H}_n^T$. They arise by integrating every cut-loop entry, so the equations imposed on a flag include (5.4). With complex orientations, the two ordinary Hall products obey

$$m_{\mathrm{crit}}(\theta_n \otimes \theta_m) = (-1)^{nm}\theta_{n+m} m_{\mathrm{red}}. \tag{5.5}$$

This is the product comparison in [128, Appendix, Theorem 4.4] and [37, §3.9].

The sign has a two-variable origin. For the quadratic potential st on $\mathbb{A}^2$, the compact critical class maps to $(1, -1)$ in the two complex-oriented axes of $\{st = 0\}$. Both axes specialize with multiplicity one to the smooth fiber, so their sum map has kernel generated by $(1, -1)$. Restriction and integration along the two axes consequently differ by -1.

The flag comparison has nm such paired normal directions. Thom–Sebastiani multiplies their signs and gives (5.5). The remaining maps are the same proper flag map and the same refined moment pullback. Thus this comparison is defined before Euler classes are inverted.

Proposition 5.1. *The maps*

$$\widehat{\theta}_n = (-1)^{\binom{n}{2}}\theta_n$$

give a unital isomorphism of charge-graded, cohomologically graded R-algebras $\mathcal{P}^T \simeq \mathcal{H}^T$.

Proof. The identity $\binom{n+m}{2} = \binom{n}{2} + \binom{m}{2} + nm$ absorbs exactly the sign in (5.5). At charge zero both reduction and its scalar correction are the identity. Each θ_n is an isomorphism by linear dimensional reduction, so their direct sum has an inverse. All scalars used are rational units. □

This theorem concerns the rational Hall operations. A comparison of full derived flag coefficient categories must also retain the closed odd directions illustrated by (5.4).

5.3 The full algebra and its charge-one generators

The spherical subalgebra of $\mathcal{H}^T$ is the unital R-subalgebra generated by $\mathcal{H}_1^T$. In the present model it is the whole algebra. We impose no nilpotence condition on any of the three matrices.

Put $a_1 = a$, $a_2 = b$, $a_3 = -a - b$ and

$$\sigma_2 = a_1a_2 + a_1a_3 + a_2a_3, \qquad \sigma_3 = a_1a_2a_3 = -ab(a+b).$$

Define Y_R^+ to be the unital associative R-algebra with generators e_r, $r \geq 0$, of charge one and degree $2r-2$. Its relations are

$$\begin{aligned}[e_{r+3}, e_s] - 3[e_{r+2}, e_{s+1}] + 3[e_{r+1}, e_{s+2}] - [e_r, e_{s+3}] \\ + \sigma_2([e_{r+1}, e_s] - [e_r, e_{s+1}]) = -\sigma_3(e_re_s + e_se_r),\end{aligned} \tag{5.6}$$

$$\sum_{\pi\in S_3} [e_{r_{\pi(1)}}, [e_{r_{\pi(2)}}, e_{r_{\pi(3)}+1}]] = 0. \tag{5.7}$$

All indices are nonnegative. This is the strictly positive affine Yangian presentation. No Cartan or negative generators occur in its definition.

Theorem 5.2 (Davison)**.** *The assignment of e_r to the normalized charge-one class u^r extends to an isomorphism*

$$Y_R^+ \xrightarrow{\cong} \mathcal{H}^T.$$

It preserves charge, cohomological degree and the unit. It requires neither Euler-class localization nor completion in charge. Let any torus act diagonally on the three loops and preserve W. Its full critical Hall algebra is generated over its equivariant coefficient ring by charge one.

Proof. The algebra comparison is [37, Theorem 5.11], using the rational dimensional-reduction normalization of §3.9 there. Its generators are the powers of the determinant class in charge one. The defining relations are (40)–(41) of that source, with the root convention made explicit in (5.8) below. The all-torus generation statement is its Theorem 5.6. The hypotheses hold here because each cyclic term of (5.1) has total torus weight zero. □

The coefficient convention is essential for supported variants. At charge one, the cut-loop nilpotent locus is $z = 0$ in $\mathbb{A}^3 \times B\mathbb{G}_m$. Ordinary restriction of the critical coefficient $\mathbb{Q}[2]$ gives $\mathbb{Q}[2]$ there. Extraordinary restriction gives $\mathbb{Q}[0](-1)$, since the normal complex line contributes $[-2](-1)$. These define different coefficient problems. The Tate twist (-1) records the normal line's Hodge weight when that refinement is retained.

Theorem 5.2 concerns the unrestricted stack and its displayed coefficient. A supported or nilpotent sector requires its own restriction functor, shift and generation theorem.

5.4 A class lost by specialization of the shuffle map

The full-rank torus gives a faithful polynomial realization [37, Proposition 3.19]. Its ambient charge-n module is

$$\mathcal{A}_n = R[x_1, \dots, x_n]^{S_n}[2n^2].$$

Thus a polynomial of ordinary cohomological degree d has adjusted degree $d - 2n^2$. Put $\mathcal{A} = \bigoplus_{n\geq 0} \mathcal{A}_n$ and define

$$\zeta(v) = \frac{(v+a)(v+b)(v-a-b)}{v} = v^2 + \sigma_2 + \frac{\sigma_3}{v}.$$

We orient a cross-block difference as second root minus first root. For $f \in \mathcal{A}_n$ and $g \in \mathcal{A}_m$, the product is

$$(f \star g)(x) = \sum_{\substack{I \sqcup J = \{1,\dots,n+m\} \\ |I|=n}} f(x_I)\, g(x_J) \prod_{i\in I,\, j\in J} \zeta(x_j - x_i). \tag{5.8}$$

Residues cancel in pairs across $x_i = x_j$, so this expression is polynomial. The three-block product contains one factor for each ordered cross-block pair, which proves associativity.

The convention with $\zeta(x_i - x_j)$ is carried to (5.8) by the R-linear map $f(x_1, \dots, x_n) \mapsto f(-x_1, \dots, -x_n)$. The same map sends the charge-one monomial x^r to $(-x)^r$. Reversing the root direction therefore requires this simultaneous change of generator coordinates. With (5.8), the normalized Hall map is

$$\Phi_T : \mathcal{H}^T \hookrightarrow \mathcal{A}, \qquad \Phi_T(e_r) = x^r. \tag{5.9}$$

Here e_r also denotes its image in $\mathcal{H}^T$ under Theorem 5.2.

For two roots x, y, put $v = y - x$. Direct multiplication gives

$$\begin{aligned} \Phi_T([e_1, e_0]) &= (x-y)\big(\zeta(v) - \zeta(-v)\big) \\ &= -2\sigma_3 = 2ab(a+b). \end{aligned}$$

The result is a constant polynomial in charge two. Its adjusted degree is $6 - 8 = -2$, which agrees with the degree of the commutator. It is not an element of the charge-zero coefficient ring.

Set $b = 0$. The specialized kernel is $v^2 - a^2$, an even function. Exchanging two blocks in (5.8) now leaves every factor unchanged, so the entire specialized ambient algebra is commutative. The Hall algebra at the same parameter is still noncommutative. An explicit model exhibits the surviving class.

Let s be a coordinate on $\mathbb{C}^*$ and let $D = s\,\mathrm{d}/\mathrm{d}s$. Differential operators satisfy

$$(s^n P(D))(s^m Q(D)) = s^{n+m} P(D+m) Q(D). \tag{5.10}$$

Inside the Lie algebra of such operators with Laurent parameter a, take the free $\mathbb{Q}[a]$-module

$$\mathfrak{w}_a = \bigoplus_{n\geq 1,\, r\geq 0} \mathbb{Q}[a]\, s^n D^r a^{r-1}. \tag{5.11}$$

Its displayed generators form a basis, with charge n and degree $2r-2$. The symbol s^n/a is a basis element of this module. Scalar multiplication by a^{-1} is not allowed.

Formula (5.10) proves closure under commutators. The highest powers of D cancel. Each remaining term has a nonnegative coefficient power of a in the displayed basis.

The one-parameter comparison is

$$\mathcal{H}^T \otimes_R \mathbb{Q}[a] \cong U_{\mathbb{Q}[a]}(\mathfrak{w}_a), \qquad e_r \longmapsto s D^r a^{r-1}, \tag{5.12}$$

where $R \to \mathbb{Q}[a]$ sends b to zero [37, Theorem 5.9]. The Hall base change itself is justified by equivariant purity [37, Proposition 3.6]. In this model

$$[e_1, e_0] \longmapsto [sD, s/a] = s^2/a. \tag{5.13}$$

The right side is a nonzero basis vector of $\mathfrak{w}_a$, and remains nonzero in its enveloping algebra by the Poincaré–Birkhoff–Witt theorem.

Proposition 5.3. *The specialization of* (5.9) *at* $b = 0$ *is not injective. Its kernel contains the nonzero Hall class* $[e_1, e_0]$. *The specialized Hall algebra is nevertheless generated by charge one.*

Proof. Its ambient image $2ab(a+b)$ vanishes at $b = 0$, whereas (5.13) proves that the source class survives. The generation statement is the one-parameter case of Theorem 5.2. □

At $a = 0$ as well, the basis in (5.11) gives symbols $w_{n,r}$ with bracket

$$[w_{n,r}, w_{m,j}] = (rm - jn) w_{n+m,r+j-1}. \tag{5.14}$$

When $r = j = 0$, the bracket is zero. This is the Poisson bracket of the symbols $s^n \xi^r$ with $\{\xi, s\} = s$. The nonequivariant Hall algebra is its enveloping algebra [37, Theorem 5.4 and Corollary 5.5]. In particular, $[w_{1,1}, w_{1,0}] = w_{2,0} \neq 0$. Setting the equivariant parameters to zero in the ambient shuffle formula would discard this bracket.

This symbol Lie algebra is generated by the elements $w_{1,r}$. Indeed, induction on $n \geq 2$ uses

$$[w_{1,r+1}, w_{n-1,0}] = (r+1)(n-1) w_{n,r}.$$

The coefficient is a nonzero rational unit. Thus charge-one generation follows from actual brackets.

To lift it equivariantly, purity supplies free coefficient modules with their nonequivariant specialization. In each charge their cohomological grading is bounded below. Lifting products of charge-one generators and inducting on degree then proves generation over the parameter ring. This is the graded argument used in [37, Theorem 5.6].

5.5 An independent free bulk and defect

We now specify the field-theoretic input. On $M = \mathbb{C}_z \times \mathbb{C}^2_{w_1,w_2}$, with volume form $\Omega = \mathrm{d}z\, \mathrm{d}w_1\, \mathrm{d}w_2$, take a one-dimensional abelian gauge algebra with pairing 1. The Batalin–Vilkovisky (BV) field complex of free holomorphic Chern–Simons (hCS) theory is

$$\mathcal{E} = \Omega^{0,\bullet}(M)[1], \qquad Q = \bar\partial.$$

A tangent component of antiholomorphic form degree r has degree $r-1$. Its coordinate has ghost degree $1-r$. Integration of compactly supported variations against Ω therefore gives a pairing of degree -1. The quadratic action is

$$S_0(\mathsf{A}) = \frac{1}{2} \int_M \Omega\, \mathsf{A} \bar\partial \mathsf{A}. \tag{5.15}$$

The action is a local functional, also defined on compactly supported fields. There is no cubic interaction for this abelian algebra.

Choose the flat Dolbeault gauge and its Gaussian heat regularization. The physical component $A = A_{\bar z}$ has ghost number zero. Its gauge differential, denoted by s, acts on A and its ghost c by

$$sA = \bar\partial c, \qquad sc = 0.$$

At a formal point, the Dolbeault jet contraction leaves holomorphic ghost jets. The antiholomorphic Euler contraction divided by total antiholomorphic polynomial and form degree contracts every positive degree. This defines the contraction coefficient by coefficient on formal series.

For nonnegative integers α_1, α_2, r, the resulting ghost algebra has generators

$$c_{\alpha,r} = \frac{1}{\alpha_1!\alpha_2!}\partial_z^r \partial_{w_1}^{\alpha_1}\partial_{w_2}^{\alpha_2} c, \qquad |c_{\alpha,r}| = 1.$$

They anticommute. Their differential is zero and their OPE is regular. Their normal weight is $|\alpha| = \alpha_1 + \alpha_2$. This is the formal ghost-jet object of the specified free theory.

Place an even $\beta\gamma$ system on the curve $w_1 = w_2 = 0$. Introduce independent parameters $\hbar$, q and κ_{def}, with coefficient ring $\mathbb{C}[\kappa_{\mathrm{def}}][[\hbar, q]]$. All three parameters have ghost degree zero. The covariance is fixed by

$$\beta(z)\gamma(w) \sim \frac{\hbar}{z-w}, \qquad \gamma(z)\beta(w) \sim -\frac{\hbar}{z-w}, \qquad \beta\beta \sim \gamma\gamma \sim 0. \tag{5.16}$$

Equivalently, use the Cauchy inverse for the kinetic operator $-\bar\partial$ with the integration convention

$$\tau(f) = \frac{1}{2\pi i}\int_{\mathbb{C}} dz \wedge d\bar z\, f.$$

The state module is the polynomial vacuum module generated by the fields and their derivatives, completed in the stated parameters. Its vacuum is denoted by $\mathbf{1}$.

Write $\partial = \partial_z$ and use primes for this derivative. Choose the current and its level to be

$$J = \beta + \frac{\kappa_{\mathrm{def}}}{2}\partial\gamma, \qquad k_{\mathrm{def}} = \hbar\kappa_{\mathrm{def}}. \tag{5.17}$$

Differentiating the two different slots in (5.16) gives two positive cross contractions. Each contributes $\hbar\kappa_{\mathrm{def}}/2$, so

$$J(z)J(w) \sim \frac{k_{\mathrm{def}}}{(z-w)^2}. \tag{5.18}$$

The level is determined by this covariance and current.

Couple the zeroth normal jet of A by the source exponential $\exp(q\tau(AJ))$. Sources and ghosts are smooth and compactly supported in z. Higher normal jets may be included formally, with finite tangential derivative order at each fixed normal weight and parameter precision. These are the free hCS current couplings of [34, §6.3.1], with a specified Gaussian defect and normalization. The parameter q is the coefficient in this source exponential. No inverse of $\hbar$ has been taken.

Two such source vertices cannot be joined by a bulk propagator. Write $\epsilon_{ij\ell}$ for the alternating three-tensor and p_ℓ for the holomorphic momentum components. The flat physical one-form covariance has a scalar heat multiplier times $\epsilon_{ij\ell}p_\ell$. It is antisymmetric in its component indices. Its $\bar z$, $\bar z$ component is therefore zero at every cutoff, also after normal differentiation. The remaining connected source calculation uses only the defect covariance (5.18).

5.6 The Ward class and a nonzero regular coupling

Let $C(u) = 1/u$ be the locally integrable Cauchy distribution and define $u^{-2} = -\partial_u C$. Normalize δ^τ by $\tau(\delta^\tau f) = f(0)$. Since $dz \wedge d\bar z = -2i\, dx \wedge dy$, the distribution identities are

$$\bar\partial C = -\delta^\tau, \qquad \bar\partial(u^{-2}) = \partial_u \delta^\tau. \tag{5.19}$$

These identities fix the sign in the Ward calculation.

Proposition 5.4. *For the normalized connected Gaussian source functional*

$$G(A) = \log\langle \exp(q\tau(AJ))\rangle,$$

one has

$$G(A) = \frac{q^2 k_{\mathrm{def}}}{2}\,\tau_z\tau_w\left(\frac{A(z)A(w)}{(z-w)^2}\right), \tag{5.20}$$

$$sG(A) = q^2 k_{\mathrm{def}}\,\tau(c'A). \tag{5.21}$$

If $q^2 k_{\mathrm{def}} \neq 0$, this variation cannot be removed by translation-invariant, finite-order differential-polynomial counterterms in the source alone, with constant coefficients and no inverse antiholomorphic derivative.

Proof. The current J is linear in Gaussian fields. All connected cumulants above order two vanish, and its second cumulant is (5.18). Wick's identity gives (5.20). The double integral is the action of the derivative of the Cauchy distribution on smooth compact joint sources. It involves no product of distributions on the diagonal.

The two variations cancel the factor $1/2$. Integration by parts and (5.19) give

$$\begin{aligned} sG(A) &= q^2 k_{\mathrm{def}}\,\tau_z\tau_w\left(\frac{\bar\partial c(z)\,A(w)}{(z-w)^2}\right) \\ &= -q^2 k_{\mathrm{def}}\,\tau_z\tau_w\big(c(z)A(w)\partial_z\partial^\tau(z-w)\big) \\ &= q^2 k_{\mathrm{def}}\,\tau(c'A). \end{aligned}$$

Compact support removes the outer boundary term.

For the counterterm assertion, grade differential monomials by their total antiholomorphic derivative order. The source gauge differential replaces A by $\bar\partial c$ and raises this order. Holomorphic integration by parts preserves it. Antiholomorphic total derivatives have positive order. The monomial $c'A$ has order zero.

In its two-field component with one holomorphic derivative, the only total-derivative relation is $c'A + cA' = \partial(cA)$. Its class is nonzero, so it cannot be the variation of a counterterm in the stated class. □

A smooth detector is explicit. With an independent odd scalar η, take Schwartz functions

$$c(z) = \eta z e^{-|z|^2}, \qquad A(z) = e^{-|z|^2}.$$

Writing $t = |z|^2$ gives

$$\tau(c'A) = -\eta\int_0^\infty (1-t)e^{-2t}\,\mathrm{d}t = -\frac{\eta}{4}.$$

Compact smooth cutoffs preserve nonvanishing for sufficiently large support, by continuity of the distributional pairing.

The same coefficient can be retained algebraically. For the zeroth normal label write $c_r = \partial^r c$, and let $A_{\mathrm{gh}} = \Lambda(c_0, c_1, \ldots)$, with $Tc_r = c_{r+1}$. Let B be the defect vertex algebra with translation T_B, and let $\mathfrak{m}_{\mathrm{gh}}$ be the ideal generated by the ghosts. Form the positive ghost-density space

$$L_B = \frac{\mathfrak{m}_{\mathrm{gh}} \otimes B}{(T + T_B)(\mathfrak{m}_{\mathrm{gh}} \otimes B)}.$$

It is completed in $\hbar, q$, with finite tangential jet support at each precision. The vertex zero product gives a graded Lie bracket on this quotient. Its differential is zero in the present abelian model. For $\alpha = q\overline{c_0 J}$, the residue calculation gives

$$\frac{1}{2}[\alpha,\alpha] = \frac{q^2 k_{\mathrm{def}}}{2}\overline{c_1 c_0}\mathbf{1}. \tag{5.22}$$

Indeed, the linear Taylor term of $c(z)$ multiplies the double pole in (5.18).

Assume $q^2 k_{\mathrm{def}} \neq 0$. Take the vacuum coefficient in the chosen normal-ordered state basis, polarize the two ghosts on Laurent functions f, g, and take the residue. Total translation is killed by this functional: T_B has zero vacuum coefficient, and the residue of a derivative vanishes. On the right side of (5.22) it gives

$$q^2 k_{\mathrm{def}} \operatorname{Res}_0(f' g \, \mathrm{d}z).$$

The pair $(f, g) = (z, z^{-1})$ gives $q^2 k_{\mathrm{def}}$. The vacuum-coefficient functional used here is linear. It is not a scalar augmentation of the vertex algebra. Since $d = 0$, no higher-order correction to α can cancel its order-q^2 curvature in L_B.

Descent agrees with the physical calculation. The odd derivation $\delta c_r = \partial^r A$, $\delta A = 0$, gives

$$\delta(c_1 c_0) = A' c - c' A, \qquad -\tau\delta(c_1 c_0) = 2\tau(c' A).$$

Applying $-\tau\delta$ to (5.22) gives exactly (5.21). For the current ∂J, the fourth pole is $-6k_{\mathrm{def}}/(z-w)^4$. The Taylor factor $1/3!$ then gives curvature $-q^2 k_{\mathrm{def}} \overline{c_3 c_0}/2$ and Ward functional $-q^2 k_{\mathrm{def}} \tau(c''' A)$.

There are nonzero couplings with regular OPE. Take two independent defect systems and currents

$$J_+ = \beta_+ + \frac{\kappa_{\mathrm{def}}}{2} \partial\gamma_+, \qquad J_- = \beta_- - \frac{\kappa_{\mathrm{def}}}{2} \partial\gamma_-.$$

Their mixed contractions vanish and their levels are opposite. Consequently $J_{\mathrm{tot}} = J_+ + J_-$ has regular self-OPE. It is nonzero because the two β coefficients are independent.

Its connected source logarithm vanishes and its Gaussian Ward identity holds to every order. The original nonzero-level current retains the obstruction of Proposition 5.4.

5.7 The comparison targets

Consider a family of mutually regular even currents indexed by normal labels α. Define the free commutative differential algebra

$$U_0 = \operatorname{Sym}\left(\bigoplus_{\alpha \in \mathbb{N}^2} \mathbb{C}[T] \, j_\alpha \right), \qquad Y(u, z)v = (e^{zT} u)v.$$

It has a unital vertex map to their coefficient algebra. This map sends j_α to the chosen current.

Derivatives and the vertex Wick identity show that all generated currents remain mutually regular. The normal product is therefore commutative and associative, which proves this universal property. Parameter completion is taken only after the algebraic normal-label direct sum is specified. After scalar extension and parameter completion, the current J_{tot} gives such a map by sending $j_{(0,0)}$ to $q J_{\mathrm{tot}}$ and the other generators to zero.

Extend the nonequivariant Hall coefficients to $\mathbb{C}$. The commutator $[e_1, e_0]$ in this algebra is nonzero. The normal product on U_0 is commutative. Hence no isomorphism can carry this Hall multiplication to the normal product on U_0. This is an obstruction for this specified operation comparison with the free abelian current target.

Several constructions remain available with their own operations:

Object	Coefficient and operation	Unit and parameter data
$\mathcal{H}^T \cong Y_R^+$	Critical Hall multiplication on all charges	$R = \mathbb{Q}[a, b]$ and charge-zero unit.
$\mathcal{A}$	Polynomial ambient shuffle product	The same ring and grading, with a map that can lose faithfulness after specialization.
B	The Gaussian defect vertex algebra and all singular products	Local vacuum, $\hbar, q, \kappa_{\mathrm{def}}$ and the fixed covariance.
$U_{\circ}$	Universal mutually regular current fields	Local vacuum, algebraic normal labels and parameter completion.

A representation on Hilbert-scheme cohomology is further data. Already $H_T^{\bullet}(\mathrm{Hilb}^{\mathrm{I}}(\mathbb{C}^2)) = R$, whereas the charge-one Hall coefficient is $R[u][2]$. A full double additionally requires a negative half, Cartan data, pairing and compatible completion. The positive-algebra isomorphism of Theorem 5.2 supplies its stated associative comparison.

Problem 5.5. Construct a family of critical Hall coefficient complexes over ordered configurations on a curve, with its collision and unit maps. Compare this family with an independently specified field theory. Specify the coefficient-ring map, the grading dictionary, and the treatment of any coefficient dual. Construct the maps preserving translation, vacuum and every supported collision operation.

Chern roots in (5.8) are internal equivariant parameters. A formal generating series in those parameters is a candidate input to this problem. A spacetime field requires its state-field map and locality. The surviving class of Proposition 5.3 and the explicit Ward class of Proposition 5.4 are constraints on any proposed comparison using these carriers.

Book II

Local coefficients and reconstruction

Chapter 6

Restricted duals and differential jets

For a polynomial $F(x)$, evaluation of its mth derivative at zero is a coefficient functional. Transposing differentiation sends this functional to evaluation of the next derivative. Thus a locally nilpotent operation on polynomials becomes an infinite translation orbit on their restricted dual. This elementary change of carrier permits a singular field to arise from a spectral Hall cooperation.

We first construct that field for the three-loop quiver with zero potential. Its contraction is the full inverse Euler factor, including an even factor invisible to the exchange ratio. The comparison is an identity on polynomial tests, with explicit factorials, translation sign and Laurent bounds. The critical Hall coefficients of the preceding chapter require a further geometric map.

An infinite current orbit also determines the coefficient category for geometric duality. Finite perfect differential modules admit an involutive dual, and dualizing a filtered presentation retains an arbitrary coefficient as a pro-object. Normal support imposes a further condition: it survives additive convolution but can be lost by an ordinary realized dual. After proving these statements, we transpose the coproduct of a full bar construction. The resulting associative algebra has a transported tensor; its realization by independently defined fields is a separate comparison.

6.1 Derivative evaluation

Let k be a field of characteristic zero. On $k[x]$ define

$$\delta_m(F) = F^{(m)}(0), \qquad m \geq 0.$$

Then $\delta_m(x^n) = m!\,\delta_{mn}$. The vector space spanned by the δ_m is the continuous dual of $k[[x]]$, where k is discrete and $k[[x]]$ has its x-adic topology. It is smaller than the algebraic dual of $k[x]$: a continuous functional depends on only finitely many coefficients. If $D = -\partial_x$ and D° denotes precomposition by D, then

$$D^\circ \delta_m = -\delta_{m+1}. \tag{6.1}$$

Consequently the finite span of $\delta_0, \dots, \delta_M$ is not preserved by $-D^\circ$.

For the Hall example take

$$k = \mathbb{Q}(a, b), \qquad h = a + b, \qquad R_N = k[\epsilon]/(\epsilon^N), \qquad N \geq 1.$$

The parameter ϵ is nilpotent and is never inverted. In charge $d \geq 0$, put

$$A_d = R_N[x_1, \dots, x_d]^{\mathfrak{S}_d}, \qquad A_0 = R_N.$$

Let $A_d[m]$ denote its homogeneous root-degree-m part. Each such part is finite free over R_N. Its degree completion and restricted dual are, respectively,

$$\widehat{A}_d = \prod_{m \geq 0} A_d[m], \qquad A_d^\circ = \bigoplus_{m \geq 0} \operatorname{Hom}_{R_N}(A_d[m], R_N). \tag{6.2}$$

The direct sum is the continuous dual of the displayed product, with the finite-degree filtration and discrete target R_N.

Introduce the polynomial state space

$$W_N = R_N[J_0, J_1, J_2, \dots], \qquad W_{N,d} = \operatorname{span}_{R_N}\{J_{m_1} \cdots J_{m_d} : m_i \geq 0\}.$$

Thus $W_N = \bigoplus_d W_{N,d}$ and $W_{N,0} = R_N 1$. Its ordinary polynomial multiplication concatenates the J factors. Define a derivation T by $TJ_m = J_{m+1}$ and $T1 = 0$. All these states have cochain degree zero. Pair states with Hall polynomials by

$$\langle J_{m_1} \cdots J_{m_d}, F \rangle = \partial_{x_1}^{m_1} \cdots \partial_{x_d}^{m_d} F\big|_{x=0}. \tag{6.3}$$

Symmetry of F makes this independent of the ordering of the m_i.

Proposition 6.1. *The pairing identifies A_d° with $W_{N,d}$. Under this identification,*

$$T = -D^\circ, \qquad D = -\sum_{i=1}^{d} \partial_{x_i}.$$

Proof. For a nondecreasing list $\lambda = (\lambda_1, \dots, \lambda_d)$, let m_λ be the sum of the distinct monomials obtained by permuting the exponents of $x_1^{\lambda_1} \cdots x_d^{\lambda_d}$. These form a basis of the symmetric polynomials. In each degree,

$$\langle J_{\lambda_1} \cdots J_{\lambda_d}, m_\mu \rangle = \begin{cases} \prod_i \lambda_i!, & \lambda = \mu, \\ 0, & \lambda \neq \mu. \end{cases}$$

The factorials are units in R_N, proving perfection in that degree and the restricted-dual identification. Precomposing the pairing by D raises one derivative order at a time, with a minus sign. The derivation rule for T gives the asserted identity. □

Write $\iota_d : A_d^\circ \xrightarrow{\sim} W_{N,d}$ for this identification. The generating series

$$G(u) = \sum_{m \geq 0} \frac{u^m}{m!} J_m = e^{uT} J_0 \tag{6.4}$$

records all finite jets. It is a formal series of states, not an additional algebraic state. Taylor's formula gives the useful identity

$$\left\langle \prod_{i=1}^{d} G(u_i), F \right\rangle = F(u_1, \dots, u_d). \tag{6.5}$$

Every formula below involving these series defines an operation on algebraic states by taking finitely many auxiliary derivatives at zero.

6.2 The full Euler factor

Consider the three-loop representation space $\operatorname{End}(\mathbb{C}^d)^3$, with zero potential and arrow weights $\epsilon a, \epsilon b, -\epsilon h$. Its equivariant cohomology, after this base change, is A_d. This zero-potential coefficient system differs from the vanishing-cycle system for $\operatorname{Tr} Z[X, Y]$.

Set $\ell = \epsilon h$ and define

$$\begin{aligned} K_{\mathrm{o}}(s) &= \frac{(s+\epsilon a)(s+\epsilon b)(s-\ell)}{s}, \quad E_{\mathrm{o}}(s) = K_{\mathrm{o}}(s)^{-1}, \\ E_{\mathrm{pl}}(s) &= \frac{s(s+\ell)}{(s+\epsilon a)(s+\epsilon b)}, \qquad\qquad P(s) = s^2 - \ell^2, \\ r(s) &= \frac{E_{\mathrm{pl}}(s)}{E_{\mathrm{pl}}(-s)}. \end{aligned} \tag{6.6}$$

Fractions with nilpotent parameters are expanded in ϵ after inverting the displayed spectral variable. The identity

$$E_{\mathrm{o}}(s) = \frac{E_{\mathrm{pl}}(s)}{P(s)}, \qquad \frac{E_{\mathrm{o}}(s)}{E_{\mathrm{o}}(-s)} = r(s) \tag{6.7}$$

separates the contraction from its exchange ratio.

For $F \in A_d$ and $H \in A_e$, the Hall product is

$$(F * H)(X) = \sum_{\substack{I \sqcup J = [d+e] \\ |I| = d}} F(X_I) H(X_J) \prod_{i \in I,\, j \in J} K_{\mathrm{o}}(x_j - x_i). \tag{6.8}$$

There is no factorial divisor in this sum: its summands are the coordinate flags of the proper invariant-subspace correspondence. Its spectral cooperation is

$$\Delta^{\circ}_{d,e}(s) F = \prod_{i=1}^{d} \prod_{j=1}^{e} E_{\mathrm{o}}(s + y_j - x_i) F(x - s, y), \qquad F \in A_{d+e}. \tag{6.9}$$

Here $x - s$ means $(x_1 - s, \dots, x_d - s)$, and expansion at infinity takes values in $(A_d \otimes_{R_N} A_e)((s^{-1}))$.

Proposition 6.2. *The Hall product is polynomial and associative. The cooperation is rationally coassociative: for blocks x, y, w the two iterates use parameters $s, t, s+t$ and evaluate the input at $(x - s - t, y - t, w)$. In charge split $(1,1)$,*

$$1 * 1 = 2\big((y-x)^2 + \epsilon^2(ab - h^2)\big), \qquad \Delta^{\circ}_{1,1}(s) 1 = E_{\mathrm{o}}(s + y - x). \tag{6.10}$$

Proof. At a coordinate flag, the three moving arrow spaces contribute $y - x + \epsilon a$, $y - x + \epsilon b$ and $y - x - \ell$; the flag tangent space contributes $y - x$ in the denominator. Their Euler ratio is $K_{\mathrm{o}}(y - x)$. The proper pushforward exists before localization. Algebraically, each possible root-diagonal pole in the shuffle sum is simple, and its residues cancel between the flags interchanged across that diagonal. The result is therefore polynomial.

Either bracketing of a triple product sums over ordered three-block partitions, with one K_{o} for every ordered cross-block pair. For the cooperation, both iterates have the common expression

$$\prod_{i,j} E_{\mathrm{o}}(s + y_j - x_i) \prod_{i,k} E_{\mathrm{o}}(s + t + w_k - x_i) \prod_{j,k} E_{\mathrm{o}}(t + w_k - y_j) F(x - s - t, y - t, w).$$

These equalities precede the choice of Laurent expansion. Finally, $K_{\mathrm{o}}(v) + K_{\mathrm{o}}(-v)$ gives the first formula in (6.10). □

Empty blocks give

$$\Delta^{\circ}_{\mathrm{o},d}(s) F = F, \qquad \Delta^{\circ}_{d,\mathrm{o}}(s) F = F(x - s).$$

Thus one empty-block identity includes translation. The cooperation has not been declared an ordinary Hopf coproduct with two unshifted counit identities. Nor is the product $*$ the ordinary multiplication on W_N: the latter merely forms polynomial states.

There is an obstruction to replacing these identities by ordinary counital multiplicativity on the same Hall algebra. Put

$$\alpha = \epsilon^2(ab - h^2), \qquad \gamma = -\epsilon^3 abh, \qquad e_m = x^m \in A_1.$$

Thus $K_0(v) = v^2 + \alpha + \gamma/v$ and e_0 is the constant in charge one, distinct from the unit in charge zero.

Proposition 6.3. *For $N \geq 4$, the charge-graded Hall algebra $\bigoplus_d A_d$ admits no R_N-linear, charge-preserving, unital algebra map to its ordinary tensor square whose two-sided counit is projection onto charge zero. This conclusion does not require coassociativity.*

Proof. The shuffle formula gives

$$[f, g]_* = 2\gamma \frac{f(x)g(y) - g(x)f(y)}{y - x}.$$

Consequently

$$[e_0, e_3]_* - 3[e_1, e_2]_* + \alpha[e_0, e_1]_* = \gamma(e_0 * e_0). \tag{6.11}$$

Indeed the left side is $2\gamma((y-x)^2 + \alpha)$, and $e_0 * e_0 = 2((y-x)^2 + \alpha)$. Charge preservation and the two counit identities force every e_m to be primitive. Commutators of even primitive elements are primitive in the ordinary tensor product. Applying a multiplicative coproduct to (6.11) and taking its charge-$(1,1)$ component therefore gives $0 = 2\gamma e_0 \otimes e_0$. This is impossible in the free module $A_1 \otimes_{R_N} A_1 = R_N[x, y]$, since $\gamma \neq 0$ for $N \geq 4$. □

At $N \leq 3$ this particular obstruction vanishes. The proposition concerns ordinary tensor multiplication and two unshifted counits; it leaves the spectral cooperation above, with its translated empty-block identity, intact.

Finite precision will be essential to the lower-truncated field target. Before nilpotent base change, specialize the arrow weights to $1, 1, -2$. The exchange is

$$r(s) = \frac{(s+2)(s-1)^2}{(s-2)(s+1)^2}, \qquad \operatorname{Res}_{s=2} r(s)\, ds = \frac{4}{9}.$$

A finite Laurent polynomial in s has no pole at $s = 2$. At precision N, the shifted denominators instead expand into finitely many powers of s^{-1}. Dualizing then changes polynomial differentiation into the nonnilpotent translation of Proposition 6.1.

6.3 Fields on polynomial jets

For lists $u = (u_1, \dots, u_d)$ and $v = (v_1, \dots, v_e)$ write $A_u = \prod_i G(u_i)$ and $B_v = \prod_j G(v_j)$. Define the field on their generating series by

$$Y(A_u, z)B_v = \prod_{i,j} E_0(-z - u_i + v_j) \prod_i G(z + u_i) \prod_j G(v_j). \tag{6.12}$$

Expand first in the auxiliary variables and in the nilpotent parameter. Finite auxiliary differentiation defines $Y(a, z)b$ for every $a, b \in W_N$. Define the exchange operator by the same prescription:

$$\mathcal{S}(z)(A_u \otimes B_v) = R_{u,v}(z) A_u \otimes B_v, \qquad R_{u,v}(z) = \prod_{i,j} r(z + u_i - v_j). \tag{6.13}$$

The notation means coefficient extraction from a generating identity; on individual jet monomials the operator generally has several terms.

Theorem 6.4. *The fields have local vacuum* $1 \in W_{N,0}$ *and translation* T*:*

$$Y(1,z) = \mathrm{id}, \qquad Y(a,z)1 = e^{zT}a, \qquad Y(Ta,z) = \partial_z Y(a,z), \qquad [T, Y(a,z)] = \partial_z Y(a,z).$$

For inputs of charges d, e *and total jet order* M*, their values lie in* $W_{N,d+e}((z))$*, with pole order at most* $2de + N - 1 + M$*. After pairing with a Hall polynomial, triple products and iterates are expansions of one rational function. The exchange is unitary and satisfies the spectral Yang–Baxter equation and the block-fusion identities.*

Proof. The empty products give the vacuum identities. Differentiation in the first auxiliary list agrees with differentiation in z. Differentiation in both lists annihilates every contraction, since it depends on their differences. These two observations prove the translation identities after coefficient extraction.

Expand $E_0(q) = \sum_{j=0}^{N-1} c_j \epsilon^j q^{-2-j}$. In a product of de contractions, the baseline pole order is $2de$. A nonzero coefficient has total ϵ-degree at most $N-1$; each auxiliary derivative raises the possible pole order by at most one. The remaining state factor is a power series in z. At each fixed power of z, finitely many of its terms contribute. This proves both the Laurent target and the asserted bound.

For a third list w, put $C_w = \prod_k G(w_k)$. The product $Y(A_u, z_1)Y(B_v, z_2)C_w$ is the indicated expansion of

$$\prod_{i,j} E_0(-z_1 + z_2 - u_i + v_j) \prod_{i,k} E_0(-z_1 - u_i + w_k) \cdot \prod_{j,k} E_0(-z_2 - v_j + w_k) \prod_i G(z_1 + u_i) \prod_j G(z_2 + v_j) \prod_k G(w_k).$$

The iterate $Y(Y(A_u, z_1 - z_2)B_v, z_2)C_w$ gives the same expression. At fixed auxiliary orders and precision, a finite product of powers of z_1, z_2 and $z_1 - z_2$ clears the denominators. This proves rational weak associativity in the respective expansion regions.

Since $r(q)r(-q) = 1$, the exchange is unitary. On three generating inputs the pairwise exchange factors commute, proving the Yang–Baxter equation. Compatibility with fusion uses the stronger identities

$$R_{u,v\sqcup w}(z) = R_{u,v}(z)R_{u,w}(z), \qquad R_{u\sqcup v,w}(z) = R_{u,w}(z)R_{v,w}(z),$$

together with substitution of the translated auxiliary coordinates in a fused block. They follow by partitioning the cross pairs. Thus fusion compatibility follows directly from the block formula, in addition to the Yang–Baxter identity. □

Our exchange convention can be checked without terminology. If $Y_z(a \otimes b) = Y(a,z)b$, then

$$Y_z \mathcal{S}(z)(A_u \otimes B_v) = e^{zT} Y(B_v, -z)A_u. \tag{6.14}$$

Indeed the contraction on the left is $R_{u,v}(z) \prod E_0(-z - u_i + v_j) = \prod E_0(z + u_i - v_j)$, and the translated states on both sides agree.

Modulo ϵ^4, the first coefficients are

$$\begin{aligned} E_0(-z) &= z^{-2} + (h^2 - ab)\epsilon^2 z^{-4} - abh\epsilon^3 z^{-5}, \\ r(z) &= 1 + 2abh\epsilon^3 z^{-3}. \end{aligned} \tag{6.15}$$

They follow by inverting $K_0(s) = s^2 + \epsilon^2(ab - h^2) - \epsilon^3 abh/s$. The even factor P^{-1} changes the leading contraction from 1 to z^{-2} while leaving the exchange ratio fixed. Exchange data alone therefore do not determine the field.

Theorem 6.5 (The restricted transpose)**.** *For finite-jet functionals $\alpha \in A_d^\circ$ and $\beta \in A_e^\circ$,*

$$Y(\iota_d\alpha, z)\iota_e\beta = \iota_{d+e}\big((\Delta^\circ_{d,e}(-z))^\circ(\alpha \otimes \beta)\big). \tag{6.16}$$

The right side is defined coefficientwise on polynomial tests and belongs to $W_{N,d+e}((z))$. Its exchange is the transpose of the inverse Hall multiplier at the reflected coordinate $-z$.

Proof. Pair (6.12) with $F \in A_{d+e}$. Equation (6.5) gives

$$\prod_{i,j} E_0(-z - u_i + v_j)F(z + u_1, \dots, z + u_d, v_1, \dots, v_e).$$

This is precisely (6.9) at $s = -z$, evaluated on $x = u$, $y = v$. Finite auxiliary differentiation proves the equality. A fixed Laurent coefficient uses finitely many derivatives of F, so it is in the restricted dual; the theorem's lower bound is supplied by Theorem 6.4. Finally the Hall multiplier is $\prod r(s + y_j - x_i)$, whose inverse at $s = -z$ is $\prod r(z + x_i - y_j)$ by unitarity. This is (6.13) under the derivative pairing. □

Give roots and ϵ degree two and a, b degree zero. A root-degree m Hall polynomial in charge d has shifted degree $2m - 2d^2$. Its restricted dual state has degree $2d^2 - 2m$. With deg $z = 2$ and deg $T = -2$, the field comparison is homogeneous. Ordinary polynomial multiplication on W_N does not have degree zero for this charge-dependent shift. This grading is separate from the cochain grading, which remains zero.

The completions also have distinct meanings. For charge-completed states the field target is

$$\prod_{d \geq 0} W_{N,d}((z));$$

at a fixed total charge only finitely many input charge splittings contribute. No uniform pole bound over all charges is imposed. For a formal parameter retain the compatible system $\varprojlim_N W_N((z))$, with a bound at each precision. The cooperation on polynomials has not been extended to every root-adic series: translation by the nonnilpotent variable s does not preserve the root-adic topology. Its coefficientwise restricted transpose is the map proved above.

6.4 Moving the centre of evaluation

Extend coefficients to $R = R_N[a_{\mathrm{fr}}, \beta]$ and put $\lambda = a_{\mathrm{fr}} + \epsilon b$. Here a_{fr} is the moving centre parameter; β scales its derivative. The connection on charge-d polynomials is

$$\nabla_d = \beta\left(\partial_{a_{\mathrm{fr}}} + \sum_i \partial_{x_i}\right). \tag{6.17}$$

It obeys the Leibniz rule over the coefficient derivation $\beta\partial_{a_{\mathrm{fr}}}$. On a tensor product over R, that coefficient derivative occurs once, while every root derivative occurs once.

Proposition 6.6. *The Hall product and cooperation are horizontal for this connection. The finite derivative evaluations at $x_i = \lambda$, indexed by nondecreasing derivative orders, form a horizontal basis of the continuous dual of the completion along that centre. Their identification with $W_{N,d} \otimes_{R_N} R$ makes the field transpose horizontal.*

Proof. Every Euler factor depends on root differences, so the sum of its root derivatives is zero. Translation of one root block by s commutes with simultaneous root differentiation. This proves the first assertion, including the flag sum in the product.

For $\xi_i = x_i - \lambda$, one has $\nabla_d \xi_i = 0$. The completion in the ξ_i is therefore preserved. If

$$v_{\mathbf{m},\lambda}(F) = \partial_{x_1}^{m_1} \cdots \partial_{x_d}^{m_d} F\big|_{x_i=\lambda},$$

the chain rule gives

$$\beta\, \partial_{a_{\mathrm{fr}}} \langle v_{\mathbf{m},\lambda}, F\rangle = \langle v_{\mathbf{m},\lambda}, \nabla_d F\rangle.$$

This is horizontality for the dual connection. The assignment $v_{\mathbf{m},\lambda} \mapsto \prod_i J_{m_i}$ reduces the transpose proof to evaluation of F at $(\lambda + z + u, \lambda + v)$. The contraction factors are unchanged. □

For the test $F(x) = x$, both sides of the displayed chain rule equal β. This test detects omission of the coefficient derivative. The moving-centre identification uses new evaluation functionals. When λ is nonnilpotent, the expression $e^{\lambda T} J_0$ is an infinite series and does not give an algebraic vector in W_N. It cannot be substituted for that change of coefficient completion.

6.5 Why a singular current needs an infinite orbit

The jet states above have independent translates. A pole forces this phenomenon in every translation-equivariant field system over a field of characteristic zero, regardless of its Hall origin.

Lemma 6.7. *Let V be a k-vector space with fields $Y(a,z)b \in V((z))$ satisfying $Y(Ta,z)b = \partial_z Y(a,z)b$. If $Y(a,z)b$ has a nonzero negative-power term, the states $a, Ta, T^2a, \dots$ are linearly independent.*

Proof. Write the lowest term of $Y(a,z)b$ as vz^{-p} with $p > 0$ and $v \neq 0$. Suppose $\sum_{j=0}^{m} c_j T^j a = 0$ with $c_m \neq 0$. Applying its field to b gives a constant-coefficient differential equation. Its lowest power is z^{-p-m}, with coefficient

$$c_m(-p)(-p-1)\cdots(-p-m+1)v \neq 0.$$

The other derivatives have larger lowest powers, a contradiction. □

Over R_N, the same conclusion holds for a flat state module when the pole remains nonzero modulo ϵ. For a finite relation, take the least ϵ-adic order of its coefficients. Flatness identifies the corresponding associated-graded piece with the reduction modulo ϵ, where the lemma applies. The hypothesis on the reduced pole is necessary: over $k[\epsilon]/(\epsilon^2)$ the state ϵa is annihilated by ϵ, even if its field has a nonzero polar term.

Lower truncation must also be interpreted separately for each second input. For the level-one Heisenberg module put

$$V = k[q_1, q_2, \dots], \qquad a_{-n} = q_n, \quad a_n = n\partial_{q_n}\ (n > 0), \quad a_0 = 0, \qquad J = q_1.$$

The field is $Y(J,z) = \sum_{n\in\mathbb{Z}} a_n z^{-n-1}$ and the vacuum is 1. Translation on states is the derivation $Tq_n = nq_{n+1}$. In particular,

$$T^r J = r!\, q_{r+1}, \qquad a_{r+1} T^r J = (r+1)!\, 1. \tag{6.18}$$

The fixed field $Y(J,z)$ therefore has no lower Laurent bound uniform over its inputs. Its correct target is $\mathrm{Hom}_k(V, V((z)))$; writing it as an element of the literal space $\mathrm{End}_k(V)((z))$ would impose a uniform bound. Every polynomial input still has a finite lower bound.

Scalar completion introduces a second failure of uniformity, now for one fixed pair of completed inputs. Extend this Heisenberg module to $V[[\epsilon]]$ and consider the state

$$a(\epsilon) = \sum_{n\geq 0} \frac{(-\epsilon)^n}{(n+1)!} T^n J.$$

Its vacuum-valued contraction with J is

$$[1]Y(a(\epsilon), z)J = \sum_{n\geq 0} \epsilon^n z^{-n-2}, \tag{6.19}$$

where $[1]$ extracts the constant polynomial state. To obtain this formula, differentiate the contraction z^{-2} exactly n times: $\partial_z^n z^{-2} = (-1)^n (n+1)! z^{-n-2}$. The resulting series has no finite lower bound in z, although its reduction modulo every ϵ^r does.

For an ϵ-adically complete module M, the appropriate recipient is

$$\mathcal{L}_\epsilon(M) = \varprojlim_r (M/\epsilon^r M)((z)). \tag{6.20}$$

Equivalently, it consists of series $\sum_{j\in\mathbb{Z}} m_j z^j$ such that for each r all sufficiently negative j satisfy $m_j \in \epsilon^r M$. A compatible family of finite-precision Laurent series determines its coefficients in M and has exactly this property, proving the equivalence. Equation (6.19) belongs to this recipient, whereas it does not belong to $V[[\epsilon]]((z))$. Completion of the states therefore requires completion of the Laurent recipient as well.

6.5.1 A quotient and its dual have opposite directions

A finite Hamiltonian quotient gives another useful test of what a jet truncation preserves. Let $\mathfrak{m} = (x, y) \subset k[[x, y]]$ and give $\mathfrak{g} = \mathfrak{m}^2$ the Poisson bracket $\{x, y\} = 1$. For $d \geq 2$, the subspace $\mathfrak{m}^{d+1}$ is a Lie ideal in $\mathfrak{g}$; hence $\mathfrak{g}_d = \mathfrak{m}^2/\mathfrak{m}^{d+1}$ is a finite Lie algebra. Let ρ_{ab} extract the coefficient of $x^a y^b$, where $a, b \geq 0$ and $a + b \geq 2$. These span the continuous dual $\mathfrak{g}^\circ$.

The coadjoint action is defined by $(f \cdot \rho)(g) = -\rho(\{f, g\})$. It gives

$$x^i y^j \cdot \rho_{ab} = \big(j(a+1) - i(b+1)\big)\rho_{a-i+1,b-j+1}, \tag{6.21}$$

with the result zero when the indicated functional is outside $\mathfrak{g}^\circ$. To verify the coefficient, pair with $x^p y^q$: the only possible indices are $p = a - i + 1$, $q = b - j + 1$, and the coadjoint sign changes $iq - jp$ to the displayed number.

Each shifted cotangent Lie algebra $\mathfrak{g}_d \ltimes \mathfrak{g}_d^\vee[-1]$ is well defined. However, simultaneous projection of both factors by total degree need not give its transition maps. Already

$$x^3 \cdot \rho_{21} = -6\rho_{02}. \tag{6.22}$$

Projection to degree at most two kills both inputs and preserves the nonzero output. It is therefore not a Lie map on the cotangent algebra. The quotient $\mathfrak{g} \to \mathfrak{g}_2$ instead dualizes to an inclusion $\mathfrak{g}_2^\vee \to \mathfrak{g}^\circ$. A compatible family of cotangent theories must supply maps with their actual variance; it cannot use the simultaneous projection.

6.6 Perfect differential modules and their pro-duals

Let $X = \mathbb{A}^1_k$ with coordinate u and differential-operator algebra

$$D = k\langle u, \partial\rangle/(\partial u - u\partial - 1).$$

A state space (V, T) defines a right differential module $\omega_X \otimes_k V$ by multiplication by functions and

$$(f\,du \otimes v)\partial = -f'\,du \otimes v - f\,du \otimes Tv. \tag{6.23}$$

This formula satisfies the Weyl relation. If $k[T]a$ is a free translation orbit, the right module $\omega_X \otimes k[T]a$ is free of rank one over D: the generator $du \otimes a$ sends ∂^m to $(-1)^m du \otimes T^m a$. The formula with polynomial coefficients is triangular in the derivative order, so it gives an isomorphism. Its characteristic variety is the full two-dimensional cotangent bundle. In particular it is not holonomic, which on a curve would require characteristic dimension one.

The finite object appropriate to such a coefficient is therefore a finite complex of free differential modules, rather than a finite span of current modes. Let X now be a smooth affine coordinate chart of dimension n, and let $D_X = \Gamma(X, \mathcal{D}_X)$. A perfect right D_X-module is a retract of a finite complex of finite free right modules. Denote their stable category by $\mathrm{Perf}_r(D_X)$. All module categories in this section are derived categories with their mapping spaces, so homotopies are retained.

A coordinate volume form converts left differential modules to right ones. The formal transpose is the anti-involution

$$t(f) = f, \qquad t(\partial_i) = -\partial_i, \qquad t(PQ) = t(Q)t(P).$$

For $P \in \mathrm{Perf}_r(D_X)$ define

$$P^\dagger = \big(\mathrm{RHom}_{D_X}(P, D_X)\big)^t[n]. \tag{6.24}$$

The superscript converts the left module into a right module; the shift is the dimension normalization. Intrinsically the canonical line supplies the left–right conversion.

Lemma 6.8. *The functor $(-)^\dagger$ is an involutive anti-equivalence on $\mathrm{Perf}_r(D_X)$. It commutes with open restriction and with external products on products of smooth charts.*

Proof. Evaluation is an isomorphism from a finite projective module to its double D_X-dual. Finite complexes and retracts preserve it. Duality reverses a cochain shift, so the two shifts by n cancel in the double dual. Localization gives the open-restriction identity. For an external product, evaluation gives the comparison on free modules over the two operator algebras, and finite cones and retracts extend it to perfect modules. The dimensions add. Canonical-line conversion makes these comparisons agree on intersections of charts. □

For an associative ring, compact objects in its derived module category are exactly the perfect modules, and every module is a filtered colimit of perfect modules [114, Proposition 7.2.4.2]. The perfect left–right duality is also expressed in [114, Proposition 7.2.4.7]. No holonomicity condition occurs in these statements. Thus write

$$M \simeq \varinjlim_\alpha P_\alpha.$$

The maps in this presentation need not be inclusions of submodules. Dualize the presentation as a diagram:

$$M^\sharp = \{P_\alpha^\dagger\}_\alpha \in \mathrm{Pro}\,\mathrm{Perf}_r(D_X). \tag{6.25}$$

For a pro-perfect diagram $Q = \{Q_\beta\}$ set $\mathcal{U}Q = \varinjlim_\beta Q_\beta^\dagger$.

Theorem 6.9 (Full coefficient duality)**.** *These constructions give inverse anti-equivalences*

$$\mathrm{Mod}_r(D_X)^{\mathrm{op}} \simeq \mathrm{Pro}\,\mathrm{Perf}_r(D_X). \tag{6.26}$$

In particular $M^\sharp$ is independent of the chosen perfect presentation.

Proof. Let $M = \varinjlim_\alpha P_\alpha$ and $N = \varinjlim_\beta Q_\beta$. The pro-category mapping formula, perfect biduality and compactness give

$$\begin{aligned}\mathrm{Map}_{\mathrm{Pro}}(N^\sharp, M^\sharp) &\simeq \varprojlim_\alpha \varinjlim_\beta \mathrm{Map}(Q_\beta^\dagger, P_\alpha^\dagger)\\ &\simeq \varprojlim_\alpha \varinjlim_\beta \mathrm{Map}(P_\alpha, Q_\beta)\\ &\simeq \mathrm{Map}(M, N).\end{aligned}$$

The equivalence respects composition. It proves full faithfulness and makes the presentation immaterial. Conversely, the same formula identifies any $\{Q_\beta\}$ with $(\varinjlim_\beta Q_\beta^\dagger)^\sharp$. Evaluation supplies both inverse comparisons and their naturality. □

6.6.1 The diagram cannot be replaced by its limit

On the affine line, the localized free right module has the perfect presentation

$$D[u^{-1}] = \varinjlim\bigl(D \xrightarrow{L_u} D \xrightarrow{L_u} D \longrightarrow \cdots\bigr), \tag{6.27}$$

where L_u is left multiplication by u. The stage-m map sends a to $u^{-m}a$. Normal form in powers of ∂ shows that these images exhaust the localization. Its dual is

$$D[u^{-1}]^\sharp = \{\cdots \xrightarrow{L_u} D[1] \xrightarrow{L_u} D[1]\}. \tag{6.28}$$

Indeed precomposition by L_u sends a left-module value b to bu, and transpose converts this to L_u on a right module. Every transition to a fixed stage is a nonzero power of L_u. The pro-object is therefore nonzero.

Its underived inverse limit is zero. The first component of a compatible sequence belongs to $\bigcap_m u^m D = 0$, since each coefficient in its finite normal form would be divisible by every power of u. The same argument applies at every stage. This calculation does not assert vanishing of a derived inverse limit.

The vacuum coefficient has a different dual. The right-module resolution

$$0 \longrightarrow D \xrightarrow{L_\partial} D \longrightarrow \omega_X \longrightarrow 0$$

and (6.24) give $\omega_X^\dagger \simeq \omega_X$, whereas $D^\dagger = D[1]$. The same category therefore contains the vacuum and the free current, with different resolutions and shifts.

6.6.2 Finite perfect pieces of the polynomial current

Return to $P = k[J_0, J_1, \ldots]$ with $TJ_m = J_{m+1}$, and give J_m weight $m+1$. Each weight space is finite-dimensional. The derivation T is injective on the positive-weight part P_+. To see this, take a nonconstant polynomial whose highest occurring variable is J_M. The coefficient of J_{M+1} in its derivative is $\partial F/\partial J_M \neq 0$.

Choose a complement Q_w to $T(P_+ \cap P_{w-1})$ in P_w, and put $Q = \bigoplus_{w>0} Q_w$. Induction on weight, using injectivity, gives

$$P = k1 \oplus k[T] \otimes_k Q. \tag{6.29}$$

For example the Hilbert series of Q is

$$\sum_{w>0} \dim(Q_w) t^w = (1-t)\left(\prod_{n\geq 1}(1-t^n)^{-1} - 1\right).$$

Consequently the current module of P is a filtered union of ω_X and finite sums of free copies of D, one for each chosen finite set of homogeneous basis elements of Q. Every such piece retains the entire translation orbit of each generator. The finite presentation is by differential modules, not by a cutoff on the energy or derivative order.

6.7 Derived normal support

The pro-dual retains the full coefficient. A collision also singles out the coefficients supported where a normal coordinate vanishes. This support condition must be checked after derived scalar reduction: ordinary reduction can omit torsion cohomology. We examine it on one normal line, where both reconstruction and convolution are explicit.

In this section put $R = k[[\epsilon]]$, $R_r = R/(\epsilon^r)$, and

$$D_R = R\langle u, \partial\rangle/(\partial u - u\partial - 1).$$

The formal scalar ϵ has cochain degree zero and is independent of the normal coordinate u. For a complex M with an R-action write

$$M_r = M \otimes^{\mathbb{L}}_R R_r, \qquad \Lambda M = R\lim_{\longleftarrow r} M_r.$$

Call M complete when $M \to \Lambda M$ is an equivalence. Restriction $M[u^{-1}]$ uses flat localization of the underlying $R[u]$-module and its induced differential action. It does not treat u as central in the Weyl algebra.

Let $\mathcal{G}_\epsilon$ be the derived category of complete scalar R-complexes. Let $\mathcal{C}_\epsilon$ be the full category of complete right D_R-complexes satisfying

$$M_r[u^{-1}] \simeq 0 \qquad \text{for every } r \geq 1. \tag{6.30}$$

This is the derived adic normal support condition. It imposes no boundedness or scalar-flatness assumption. The condition at $r = 1$ already suffices: it says that multiplication by ϵ is an equivalence on $M[u^{-1}]$, and hence that all its derived reductions vanish. For an R-flat module the condition becomes

$$\text{for every } m \in M \text{ and } r \geq 1 \text{ there is } n \text{ with } mu^n \in \epsilon^r M.$$

Without flatness, this elementwise condition controls only the ordinary quotients and is not the definition in (6.30).

The basic supported right module is the principal-parts module

$$P = R[u, u^{-1}]/R[u] = \bigoplus_{m\geq 0} R\beta_m, \qquad \beta_m = [u^{-m-1}]. \tag{6.31}$$

Right differentiation is minus ordinary differentiation. Thus

$$\beta_0 u = 0, \qquad \beta_m u = \beta_{m-1}\ (m > 0), \qquad \beta_m \partial = (m+1)\beta_{m+1}.$$

The map $1 \mapsto \beta_0$ identifies $P = D_R/uD_R$ and gives the resolution

$$0 \longrightarrow D_R \xrightarrow{L_u} D_R \longrightarrow P \longrightarrow 0. \tag{6.32}$$

Ordered Weyl monomials prove injectivity of L_u.

Theorem 6.10 (Normal reconstruction)**.** *The functors*

$$I(E) = \Lambda(P \otimes^{\mathbb{L}}_R E), \qquad K(M) = \mathrm{RHom}_{D_R}(P, M)$$

give inverse equivalences $I : \mathcal{G}_\epsilon \simeq \mathcal{C}_\epsilon : K$. For a right differential module complex,

$$K(M) = [M \xrightarrow{u} M]_{[0,1]}. \tag{6.33}$$

The last display is a complex of scalar modules; right multiplication by u need not be a differential-module map.

Proof. First work over any commutative k-algebra B, with the analogous Weyl algebra D_B and $P_B = B[u, u^{-1}]/B[u]$. If a right D_B-module V is supported at $u = 0$, multiplication by u is locally nilpotent. For $v \in V$ the finite sum

$$\pi(v) = \sum_{j \geq 0} \frac{(-1)^j}{j!} (vu^j)\partial^j$$

belongs to $\ker u$. Indeed $\partial^j u = u\partial^j + j\partial^{j-1}$ makes consecutive terms of $\pi(v)u$ cancel. The same binomial cancellation gives

$$v = \sum_{m \geq 0} \frac{\pi(vu^m)\partial^m}{m!}.$$

Consequently

$$P_B \otimes_B \ker u \xrightarrow{\sim} V, \qquad \beta_m \otimes e \longmapsto e\partial^m/m!. \tag{6.34}$$

Surjectivity follows from the preceding identity. In a finite relation, applying the largest occurring power of u extracts the highest coefficient, proving injectivity. This also proves that $u : V \to V$ is surjective.

Resolution (6.32) gives (6.33). For a complex with supported cohomology, its long exact sequence therefore identifies $H^q K(V)$ with $\ker(u : H^q V \to H^q V)$. Since P_B is B-free, (6.34) proves that the adjunction counit $P_B \otimes^{\mathbb{L}}_B K(V) \to V$ is an equivalence in every degree. The unit is an equivalence because $u : P_B \to P_B$ is a split surjection with kernel $B\beta_0$. No truncation of the complex was used.

Derived completion preserves every derived reduction: $(\Lambda X)_r \simeq X_r$. To check this, tensor the defining limit with the perfect R-complex R_r. For $n \geq r$ the resulting tower has a constant copy of X_r and a shifted copy whose transitions multiply by ϵ. Any r successive transitions on the latter copy vanish, so its homotopy limit is zero. Completion is the reflection onto complete complexes: its kernel consists of complexes on which ϵ is invertible, and maps from such a complex to a complete one vanish.

The finite free resolution of P shows that K commutes with reduction and completion. Tensor–Hom adjunction and the reflection just described therefore give $I \dashv K$ on complete objects. Its unit and counit reduce to the equivalences over $B = R_r$ proved above. Complete objects are limits of their reductions, so these maps are equivalences over R as well. □

6.7.1 Additive convolution

There is a geometric tensor preserving this support. For addition $a : \mathbb{A}^1_x \times \mathbb{A}^1_y \to \mathbb{A}^1_t$, $t = x + y$, let a_+ denote right differential-module direct image, computed here by the relative right de Rham complex. Define

$$M *_+ N = \Lambda a_+(M \boxtimes^{\mathbb{L}}_R N), \qquad E \widehat{\otimes}_R F = \Lambda(E \otimes^{\mathbb{L}}_R F).$$

Proposition 6.11. *There are natural coherent equivalences*

$$I(E) *_+ I(F) \simeq I(E \widehat{\otimes}_R F).$$

They make normal reconstruction symmetric monoidal, with convolution unit $\widehat{P} = I(R)$. Relative convolution over an algebra $I(B)$ is given by

$$I(E) *^{\mathbb{L}}_{I(B)} I(F) \simeq I\big(\Lambda(E \otimes^{\mathbb{L}}_B F)\big) \tag{6.35}$$

for right and left B-modules E, F in complete scalar complexes.

Proof. At precision r, change coordinates to $t = x + y$, $s = y$. Since $\partial_x = \partial_t$ and $\partial_y = \partial_t + \partial_s$, divided derivatives give

$$\beta_i(x)\beta_j(y) \longmapsto \sum_{q=0}^{j} \binom{i+q}{q} \beta_{i+q}(t)\beta_{j-q}(s).$$

The relative de Rham complex $[P_s \xrightarrow{\partial_s} P_s]_{[-1,0]}$ has injective differential and cokernel $R_r\beta_0$. Thus direct image retains the $q = j$ term:

$$\beta_i\beta_j \longmapsto \binom{i+j}{i} \beta_{i+j}. \tag{6.36}$$

There is no cohomological shift. This split computation remains valid after tensoring with any scalar complexes. Associativity on three modes is

$$\binom{i+j}{i}\binom{i+j+k}{i+j} = \frac{(i+j+k)!}{i!\,j!\,k!} = \binom{j+k}{j}\binom{i+j+k}{i}.$$

The coherent comparisons come from iterated addition and relative de Rham direct image. Passing through all derived reductions proves the completed assertion. The two-sided bar for relative convolution has terms $E \widehat{\otimes} B^{\widehat{\otimes} n} \widehat{\otimes} F$; its realization gives (6.35). □

This convolution retains the normal coordinate. Balanced tensor over the Weyl algebra gives a different output. With formal transpose $u \mapsto u$, $\partial \mapsto -\partial$, the same resolution gives

$$\mathrm{RHom}_{D_R}(P, P) \simeq R, \qquad P \otimes^{\mathbb{L}}_{D_R} P^t \simeq R[1].$$

The lowering operator is surjective with kernel $R\beta_0$. Its Hom complex has degrees 0, 1, while its tensor complex has degrees -1, 0. These calculations explain both the shift and the scalar output of the balanced tensor. Identifying additive convolution with a collision convolution requires a comparison of their actual correspondences and units.

Derived support is stable under the relative operation even when scalar flatness is lost. For example, let $B = \widehat{R[t]}$ be the ϵ-adic completion, and give $E_0 = R$ and $E_\epsilon = R$ the actions $t = 0$ and $t = \epsilon$. The exact free resolution of E_0 by multiplication by t gives

$$E_0 \otimes^{\mathbb{L}}_B E_\epsilon = [R \xrightarrow{\epsilon} R]_{[-1,0]} \simeq k.$$

Both inputs are scalar-flat; the output is complete and nonflat. Its derived reduction has a copy of k in each of degrees -1, 0. Applying I retains normal support in both degrees.

Scalar perfection alone is also not stable under arbitrary relative tensor products. For $B = R[\eta]/(\eta^2)$ and the augmentation module R, the resolution with every negative differential equal to multiplication by η gives

$$H^{-n}(R \otimes_B^{\mathbb{L}} R) = R \qquad (n \geq 0).$$

Both scalar inputs are perfect, but their relative tensor is not. Perfection of the right module over B, together with perfection of the left module over R, is a sufficient stronger hypothesis: a finite B-cell presentation then tensors to a finite R-cell presentation.

6.7.2 Limits and ordinary realized duals

Normal reconstruction describes the supported limits intrinsically:

$$\varprojlim_{C_\epsilon} M_i \simeq I\left(\varprojlim_{i} K(M_i)\right).$$

They need not agree with limits in all complete differential modules. Indeed

$$\widehat{P} = \left\{ \sum_{m \geq 0} a_m \beta_m : a_m \in R,\ a_m \longrightarrow 0 \text{ adically} \right\}$$

is flat, complete and derived-supported. Its ambient countable product $W = \prod_{i \geq 0} \widehat{P}$ is flat and complete, with $W_1 = \prod_i P_1$. The element

$$e = (\beta_0, \beta_1, \beta_2, \ldots) \in W_1 \tag{6.37}$$

is killed by no power of u: the nth coordinate of eu^n is $\beta_0 \neq 0$. Hence $W_1[u^{-1}] \neq 0$ and $W \notin C_\epsilon$.

In matrix coordinates $a_{i,m}$, the supported product $I(\prod_i R)$ consists of those elements of W satisfying

$$\text{for each } r \text{ there is } M_r \text{ such that } a_{i,m} \in \epsilon^r R \quad \text{for every } i \text{ and } m \geq M_r.$$

The ambient product imposes the bound separately in each row. The diagonal element (6.37) distinguishes the two. This is a failure of normal support, even though the tower of finite products has split surjective transitions.

The same example obstructs an ordinary realized coefficient dual. Put $\widehat{D}_R = \Lambda D_R$ and define

$$\mathbb{D}_\epsilon(M) = \mathrm{RHom}_{D_R}(M, \widehat{D}_R)^t[1].$$

This takes values in ordinary complete differential-module complexes. Dualizing (6.32) yields an injective right-multiplication map R_u on $\widehat{D}_R$, now regarded as a left module. Ordered monomials $\partial^j u^i$ identify its cokernel with the completed left delta module. Thus

$$\mathrm{RHom}_{D_R}(P, \widehat{D}_R) \simeq \widehat{P}^t[-1].$$

For the flat, complete, supported module

$$M = \Lambda\left(\bigoplus_{i \geq 0} P\right)$$

the reflection property of completion and mapping out of a direct sum now give

$$\mathbb{D}_\epsilon(M) \simeq \prod_{i\geq 0} \widehat{P} = W \notin C_\epsilon. \tag{6.38}$$

For a perfect scalar ground E, finite evaluation instead gives

$$\mathbb{D}_\epsilon(I(E)) \simeq I(\mathrm{RHom}_R(E, R)).$$

This is a sufficient domain for ordinary supported duality. The relative-tensor example above shows why scalar perfection alone does not make that domain stable under every relative fusion. The perfect-presentation argument of Theorem 6.9 also applies to D_R. It retains a diagram. The failure in (6.38) concerns the ordinarily realized dual.

6.8 Transposing the full bar coalgebra

Coefficient duality reverses maps. It becomes a comparison of operations only after the tensor and geometric functors on the dual coefficients have been specified. On a smooth scheme use the affine construction above and open descent to define $\mathcal{D}^\sharp(X)$. The canonical-line identifications agree on overlaps, and taking opposite categories preserves the descent limit. Thus the derived category $\mathcal{D}(X)$ of right differential modules satisfies $\mathcal{D}(X)^{\mathrm{op}} \simeq \mathcal{D}^\sharp(X)$.

For a given functor $F : \mathcal{D}(X) \to \mathcal{D}(Y)$ define

$$F^\sharp = \sharp_Y \circ F \circ \mathcal{U}_X. \tag{6.39}$$

In particular, a collision pullback $\delta^!$ gives $\delta^\dagger = \sharp\delta^!\mathcal{U}$, and an open direct image j_* gives $j_{!,\mathrm{pro}} = \sharp j_*\mathcal{U}$. The latter takes values in pro-perfect coefficients; it has not been identified with an ordinary $j_!$ on all differential modules. The localized module in (6.28) explains why retaining the pro-diagram matters.

Suppose now that C is a stable category of coherent coefficient systems on smooth collision spaces X_I, indexed by finite label sets I. For an ordered decomposition $I = A \sqcup B$, let $U_{A|B} \subset X_I$ be the open locus where points in different blocks are distinct, and let $j_{A|B}$ be its inclusion. Collisions within one block remain allowed. Assume that external products and the collision maps define an associative convolution

$$(M \star N)_I = \bigoplus_{I=A\sqcup B} j_{A|B,*}(M_A \boxtimes N_B)|_{U_{A|B}}$$

with unit $K_\varnothing = k$, $K_I = 0$ for $I \neq \varnothing$. The collision equivalences and their higher compatibilities are part of a coefficient system. All ordered subset decompositions occur in this convolution. Keeping only consecutive cuts of an ordered word would define another tensor.

Componentwise duality gives a category $C^\sharp$ and an anti-equivalence $C^{\mathrm{op}} \simeq C^\sharp$. Define

$$E \star_\sharp F = (\mathcal{U}E \star \mathcal{U}F)^\sharp. \tag{6.40}$$

Equivalently its component at I is

$$\bigoplus_{I=A\sqcup B} j_{A|B,!,\mathrm{pro}}(E_A \boxtimes^\sharp F_B)|_{U_{A|B}},$$

where $\boxtimes^\sharp$ is the two-index external product of perfect diagrams. The sum is finite. Associativity, its coherent homotopies and the unit $K^\sharp = K$ are transported from $\star$.

Proposition 6.12. *For finite lists of coefficient objects there is a natural equivalence*

$$\mathrm{Map}_C\left(\underset{i}{\star} M_i, \underset{j}{\star} N_j\right) \simeq \mathrm{Map}_{C^\sharp}\left(\underset{j}{\star_\sharp} N_j^\sharp, \underset{i}{\star_\sharp} M_i^\sharp\right).$$

It reverses composition and preserves identities and all coherent homotopies. Collision maps are compared by (6.39)*.*

Proof. Apply coefficient anti-equivalence to the two tensor objects. Equation (6.40) identifies their images. Naturality of the mapping equivalence gives composition and every collision diagram, including their specified homotopies. □

Let A be an augmented associative algebra for $\star$, with $A_\varnothing = k$. Its augmentation ideal $\bar{A}$ has zero empty coefficient. A normalized bar word of length r on I uses an ordered decomposition into r nonempty blocks. Hence $r \le |I|$. This finiteness is in configuration support and makes no assertion about the dimension of a current coefficient.

Write $B_\star A$ for its full normalized bar coalgebra, with suspensions of $\bar{A}$, deconcatenation coproduct and complete bar differential. For a coherent associative presentation, that differential includes all higher products. Define

$$\mathcal{K}_\sharp A = (B_\star A)^\sharp. \tag{6.41}$$

Its multiplication is the explicit transpose

$$\mathcal{K}_\sharp A \star_\sharp \mathcal{K}_\sharp A \simeq (B_\star A \star B_\star A)^\sharp \xrightarrow{\Delta_B^\sharp} (B_\star A)^\sharp.$$

The bar counit becomes the unit; the coaugmentation becomes the augmentation. This constructs a second associative algebra, with the tensor in (6.40).

Theorem 6.13. *Assume that normalized associative bar and cobar are inverse equivalences between the augmented algebras with positive augmentation ideal in $(C, \star)$ and their connected conilpotent coalgebras. Assume their normalized words have the finite configuration bound above. Then $\mathcal{K}_\sharp$ is an anti-equivalence to the corresponding augmented associative algebras for $\star_\sharp$. Its inverse is*

$$E \longmapsto \mathcal{U} B_{\star_\sharp} E.$$

Proof. Bar gives a coalgebra equivalence by hypothesis, and Proposition 6.12 transposes coalgebras to algebras for the transported tensor. To identify the inverse, let C be a connected conilpotent $\star$-coalgebra. On normalized words,

$$\mathcal{U} B_{\star_\sharp}(C^\sharp) \simeq \Omega_\star C.$$

A bar suspension dualizes to a cobar desuspension. Multiplication dualizes to the coproduct of C, while deconcatenation dualizes to concatenation. The full differentials are transposed with these maps. At each I there are only finitely many positive words, so exact duality commutes with their finite extensions. For $C = B_\star A$, the comparison back to A is therefore

$$\mathcal{U} B_{\star_\sharp}(\mathcal{K}_\sharp A) \simeq \Omega_\star B_\star A \xrightarrow{\epsilon_A} A,$$

the stipulated bar–cobar counit. The other comparison is the coalgebra-side unit. Their triangle identities and coefficient biduality prove the inverse-functor assertion. □

The theorem isolates coefficient transposition from the geometric bar–cobar construction: its hypotheses must be proved for the chosen collision convolution. They do not follow from Lie–commutative chiral duality. Once supplied, the full bar coalgebra yields the second associative algebra above, even for nonholonomic currents. No quadratic approximation of the bar has been used.

At finite parameter precision the same argument uses $D_X \otimes_k R_N$. Perfect modules remain the compact modules over this associative ring, although R_N is not regular. Coefficient reduction is transported by (6.39), and the compatible inverse system is part of the dual datum. The finite word bound does not exchange an ordinary pro-limit, a parameter-adic limit and a Laurent expansion.

6.9 The further field comparison

Theorem 6.5 identifies the zero-potential Hall cooperation with a field product on restricted jets. Theorem 6.13 instead transposes a full coalgebra to an algebra for a transported geometric tensor. Their carriers and operations specify two different comparison problems. Neither identifies a Hall charge split with a level-changing field coproduct: a map sending a charge-one state to two charge-one factors has output charges $(1, 1)$, whereas a Hall split has output charges adding to one.

The empty-configuration unit K is also different from the local vacuum coefficient ω_X. An augmentation is a specified algebra map to the tensor unit, and evaluation in a prepared state is additional linear data. The coefficient duality constructs none of these identifications by changing notation.

6.9.1 The full bar contact marking

Suppose a native field system has already supplied a connected full collision coalgebra C in the category above. Its contact maps retain

$$r(u^m; a, b) = \mathrm{pp}_u\big(u^m Y(a, u)b\big), \qquad m \in \mathbb{Z},$$

where pp_u takes the finite principal part at the stated precision. The coalgebra also retains the higher collision maps, spectators and normal derivatives. Form $U_*(C) = \Omega_\star C$. Under the bar–cobar hypotheses of Theorem 6.13, the unit is the identity at empty support and has the following formula on $c \in \bar{C}$:

$$\eta_C : C \xrightarrow{\sim} B_\star\Omega_\star C, \qquad \eta_C(c) = \sum_{r\geq 1} (s\tau)^{\star r}\bar{\Delta}^{r-1}c. \tag{6.42}$$

Here s has degree -1, t has degree 1, and $\tau(c) = tc$ is the universal twisting cochain into the cobar. At a configuration I the sum has $r \leq |I|$. With bar signs $b_1(sa) = -s(da)$ and $b_2(sa, sb) = (-1)^{|a|}s(ab)$, cobar splitting cancels adjacent bar multiplication. The remaining terms give the chain identity for η_C; cutting the outer word gives its coalgebra identity. The stipulated bar–cobar theorem supplies its inverse equivalence.

This marking retains every native binary mode:

$$a_{(n)}b = [u^{-1}]r(u^n; a, b), \qquad n \in \mathbb{Z}. \tag{6.43}$$

Indeed multiplication by u^n moves the coefficient of u^{-n-1} to u^{-1}. Negative multipliers are essential for recovering regular modes. Transposition gives the operation-preserving coefficient comparison

$$(B_\star\Omega_\star C)^\sharp \xrightarrow{\eta_C^\sharp} C^\sharp.$$

These operations are marked on the bar of the algebra realization. They have not been identified with its bare singleton multiplication. The higher native collision system was used to construct C; equation (6.43) alone does not construct it.

6.9.2 *Equal fields can carry different jet markings*

A further ambiguity remains even when all unmarked fields agree. Take $R_2 = k[\hbar]/(\hbar^2)$, with $\hbar$ of weight and cochain degree zero, and the charge-graded algebra

$$A = R_2[e_0, e_1, \dots]/(e_0, e_1, \dots)^3.$$

Every e_n has charge one, weight $n+1$ and cochain degree zero; the differential is zero. There is no cutoff on n. Define derivations by

$$T e_n = (n+1)e_{n+1}, \qquad \partial_i e_n = e_{n-i+1} \quad (i \geq 1),$$

where $e_j = 0$ for $j < 0$. Then ∂_1 is the charge derivation, the ∂_i commute, and $[\partial_i, T] = (i-1)\partial_{i-1}$, with $\partial_0 = 0$. For each $\kappa \in k$, change the marking to

$$J_1^\kappa = \partial_1, \quad J_2^\kappa = \partial_2, \quad J_i^\kappa = \partial_i + \hbar\kappa \binom{i-1}{2} \partial_{i-2} \quad (i \geq 3). \tag{6.44}$$

Every operator is charge preserving and bounded for the weight filtration. Only finitely many act nontrivially on any fixed state. They commute with the differential and satisfy the same jet and translation relations. The latter assertion follows from

$$\binom{i-1}{2}(i-3) = (i-1)\binom{i-2}{2}.$$

For these markings define the contraction and fields

$$\begin{aligned} C_1^\kappa(u) &= \sum_{p,q \geq 0} (-1)^{p+1} \binom{p+q}{p} u^{-p-q-1} J_{p+1}^\kappa \otimes J_{q+1}^\kappa, \\ Y^\kappa(a,u)b &= \mu(e^{uT} \otimes 1)(1 - \hbar C_1^\kappa(u))(a \otimes b), \\ \mathcal{S}^\kappa(u) &= 1 + 2\hbar C_1^\kappa(u). \end{aligned}$$

The contraction is finite on each pair of algebraic inputs. The commuting derivations and their translation identities give the field identities at this precision: in a three-input product only the sum of the three pair contractions survives. This also gives exchange and the block-fusion identities.

Proposition 6.14. *All the displayed unmarked fields and exchanges are identical. Distinct values of κ remain inequivalent if R_2-linear morphisms preserve the markings and weak equivalences are state quasi-isomorphisms through nonpositive dg algebras whose marked operators are chain maps.*

Proof. The difference $J_i^\kappa - \partial_i$ is divisible by $\hbar$. Since the contraction already has an external factor $\hbar$, its change is zero in R_2. Thus $Y^\kappa = Y^0$ and $\mathcal{S}^\kappa = \mathcal{S}^0$. In particular all iterates and Laurent-multiplier residues agree.

On charge one the standard operators satisfy $\partial_3 = \partial_2^2$. The modified marking has

$$J_3^\kappa - (J_2^\kappa)^2 = \hbar\kappa \, \mathrm{id}\,. \tag{6.45}$$

Chain compatibility makes this an operator on cohomology. A zigzag of marked quasi-isomorphisms induces an invertible intertwiner f on charge-one cohomology. Between parameters κ, λ, equation (6.45) would force $\hbar(\kappa - \lambda)f = 0$. Composing with f^{-1} contradicts freeness of the charge-one R_2-module unless $\kappa = \lambda$. □

The assumption that the marked operators are chain maps is necessary for this cohomological argument. Exact weight would also change the admitted class: the correction in (6.44) has weight $3 - i$, whereas δ_i has weight $1 - i$. Filtered boundedness admits both. On the bottom state the difference is already visible as $J_2 e_0 = 0$ and $J_3^\kappa e_0 = \hbar\kappa e_0$, but the translation orbit remains infinite.

This ambiguity does not destroy a specified current realization. For example, in charge n let $p_m = \sum_{a=1}^{n} x_a^m$ and $p_0 = n$. The corresponding marked moments and their compensation are

$$P_m^\kappa = p_m + \hbar\kappa \binom{m}{2} p_{m-2}, \qquad \widehat{P}_m = P_m^\kappa - \hbar\kappa \binom{m}{2} P_{m-2}^\kappa = p_m,$$

with the correction zero for $m < 2$. Thus compensated moments give the original action of $k[x_1, \dots, x_n]^{\mathfrak{S}_n}$. Any current presentation with this specified action retains its localization at $\prod_{a<b}(x_a - x_b)^2$ and its rational pair multipliers. Raw moments cannot replace that action: in charge one, $P_2^\kappa - (P_1^\kappa)^2 = \hbar\kappa$, while the polynomial identity is $p_2 = p_1^2$.

The bar contact comparison and the specified current markings are therefore positive additional data. Recovering those markings intrinsically from an unmarked field is a stronger problem. The first-order equality above makes no assertion of compatible equality at every higher parameter precision.

Problem 6.15. For an independently defined family of local fields, construct its coherent collision coefficient system and compare its operation maps with the transported full-bar algebra. Specify the coefficient class, tensor, unit, translation and Laurent completion, and retain the maps supported on collision diagonals. For critical Hall input, construct the oriented coefficient map before applying the restricted transpose; the zero-potential polynomial calculation alone does not supply it.

The next geometric input is the coefficient at a collision. The derived relation directions can survive where their ordinary equations vanish. Retaining those directions is necessary before testing a comparison with the fields constructed here.

Chapter 7

Critical coefficients at a collision

An extension of two commuting representations must itself satisfy the moment equation. When their eigenvalues meet, that equation can lose rank. Its derived zero fiber then retains classes that disappear if one first passes to ordinary polynomial functions. We begin by computing those classes on a two-dimensional flag chart.

A second phenomenon appears when three positions meet. A local system can extend cleanly across each punctured pair-collision divisor and still have cohomology at their common intersection. An explicit cubic cover realizes this situation. Its critical direct image contains a point constituent that its intermediate extension does not contain. Both examples explain why a comparison of Hall coefficients with local fields must specify the coefficient complex and its extension across collisions.

7.1 The moment equation on a flag

Let $V = \mathbb{C}^2$, with its first coordinate line. Consider the matrices and the framing maps $i : \mathbb{C} \to V$, $j : V \to \mathbb{C}$ given by

$$X = \begin{pmatrix} x_1 & B_0 \\ 0 & x_2 \end{pmatrix}, \qquad Y = \begin{pmatrix} y_1 & D_0 \\ 0 & y_2 \end{pmatrix}, \qquad i = \begin{pmatrix} i_1 \\ i_2 \end{pmatrix}, \qquad j = (1, 0).$$

The equation is $[X, Y] + ij = 0$. The framing is *cocyclic* when no nonzero subspace invariant under both matrices lies in $\ker j$. On this slice that condition is $(B_0, D_0) \neq (0, 0)$: the only possible such subspace is the second coordinate line.

Direct multiplication gives the four entries of the moment map:

$$i_1, \qquad (x_1 - x_2)D_0 - (y_1 - y_2)B_0, \qquad i_2, \qquad 0. \tag{7.1}$$

The last entry is a zero equation. It still contributes a generator to the specified derived zero fiber.

We use cohomological grading: the differential has degree 1, and $H^q(E[s]) = H^{q+s}(E)$. For functions $f_1, \dots, f_r$ on an affine scheme with ring A, their Koszul algebra is $A \otimes \Lambda(\xi_1, \dots, \xi_r)$, with $|\xi_i| = -1$ and $d\xi_i = f_i$. This algebra specifies the derived zero fiber of the given list of functions. Replacing the list by equations for the same ordinary zero set can change that fiber.

The pairs for i_1 and i_2 in (7.1) are contractible. Indeed $A[i] \otimes \Lambda(\xi)$, with $d\xi = i$, retracts onto A: the projection sets $i = \xi = 0$, and the degree--1 homotopy is

$$H(i^m) = \xi i^{m-1} \quad (m \geq 1), \qquad H(1) = H(\xi i^m) = 0.$$

Tensoring these contractions with the other factors leaves

$$d\eta = (x_1 - x_2)D_0 - (y_1 - y_2)B_0, \qquad d\chi = 0, \qquad |\eta| = |\chi| = -1.$$

On $B_0 \neq 0$, the remaining diagonal change of basis fixes $B_0 = 1$, with the corresponding invertible rescaling of the equation generators. Take the derived restriction to the curve $y_1 = y_2 = 0$ in the plane. The Koszul complex is termwise free over the eigenvalue base, so derived restriction is computed by substitution in that complex. The polynomial model is defined over $\mathbb{Q}$. After base change to a nonzero commutative $\mathbb{Q}$-algebra R, it is

$$K = R[x_1, x_2, D_0] \otimes \Lambda(\eta, \chi), \qquad d\eta = (x_1 - x_2)D_0, \qquad d\chi = 0. \tag{7.2}$$

The three polynomial variables have cochain degree zero. We retain only this grading in the following calculations.

The variables x_i, y_i are plane eigenvalues. They are distinct from the Chern roots used in an equivariant Hall formula. An identification of the moment complex with a Hall direct-image complex requires the actual support maps, orientation and shift.

Proposition 7.1. *Put $u = x_1 - x_2$ and $w = x_2$. Then*

$$H^*(K) = R[u, w, D_0]/(uD_0) \otimes \Lambda(\chi). \tag{7.3}$$

After inverting u, the complex retracts onto $R[u, u^{-1}, w] \otimes \Lambda(\chi)$ with zero differential. Its derived fiber over $u = 0$ is instead

$$K \otimes^L_{R[u]} R = R[w, D_0] \otimes \Lambda(\eta, \chi), \qquad d = 0. \tag{7.4}$$

In particular the collision fiber has a nonzero class $\eta\chi$ in degree -2.

Proof. Multiplication by uD_0 is injective on $R[u, w, D_0]$: it sends distinct polynomial monomials to distinct monomials without changing their coefficients. This remains true if R has nilpotents. Thus the two-term complex with generator η has only its degree-zero cokernel. Tensoring with the closed exterior factor $\Lambda(\chi)$ proves (7.3).

After localization, let p set $D_0 = \eta = 0$ and let i be the inclusion. Define

$$H(D_0^m \chi^e) = u^{-1}\eta D_0^{m-1}\chi^e \quad (m \geq 1,\ e = 0, 1), \qquad H(\chi^e) = H(\eta D_0^m \chi^e) = 0.$$

The identity $dH + Hd = 1 - ip$ follows on each monomial. It retains the closed generator χ.

Finally K is bounded and termwise free over $R[u]$. It therefore computes its derived tensor product with R. Setting $u = 0$ makes the differential vanish and proves (7.4). □

Ordinary restriction of the algebra in (7.3) would miss η and $\eta\chi$. They are the additional derived-fiber classes. The closed class χ is present even before this restriction; it comes from the zero entry in the original moment map.

7.2 What polynomial tests forget

Complete the polynomial variables in K at (x_1, x_2, D_0). Its continuous R-linear dual, with R discrete, consists of functionals depending on finitely many Taylor coefficients. Denote it by K°, with differential

$$(d^\circ\phi)(k) = (-1)^{|\phi|+1}\phi(dk).$$

It lies in degrees 0, 1, 2. For $m, n, l \geq 0$, write

$$v_{mnl}(F) = \partial_{x_1}^m \partial_{x_2}^n \partial_{D_0}^l F\Big|_0.$$

The notation $\eta^\vee v$, $\chi^\vee v$ or $(\eta\chi)^\vee v$ means extraction of the indicated exterior coefficient followed by v. These are linear dual coordinates; no multiplication of dual odd generators is being asserted.

Give $P_2 = R[x_1, x_2]^{\mathfrak{S}_2}$ the topology induced by the (x_1, x_2)-adic topology. Its continuous dual is identified with the free R-module W_2 on the commutative symbols $J_m J_n = J_n J_m$, by

$$\langle J_m J_n, F\rangle = \partial_{x_1}^m \partial_{x_2}^n F\big|_0.$$

To see that the pairing is nondegenerate, use the symmetric monomial basis. The diagonal entries are $m!n!$, including when $m = n$, and these are units in R. Thus restriction along the inclusion $P_2 \hookrightarrow K$ defines a cochain map

$$\Theta : K^\circ \longrightarrow W_2[0].$$

Proposition 7.2. *The map Θ is surjective and has the formula*

$$\Theta(v_{mnl}) = \begin{cases} J_m J_n, & l = 0, \\ 0, & l > 0, \end{cases} \qquad \Theta(\eta^\vee v) = \Theta(\chi^\vee v) = \Theta((\eta\chi)^\vee v) = 0. \tag{7.5}$$

It kills the nonzero classes

$$[v_{001}] \in H^0(K^\circ), \qquad [\chi^\vee v_{000}] \in H^1(K^\circ). \tag{7.6}$$

It intertwines the transpose of $\partial_{x_1} + \partial_{x_2}$ with $T(J_m J_n) = J_{m+1} J_n + J_m J_{n+1}$. In particular translation compatibility does not make Θ a quasi-isomorphism.

Proof. Restriction to symmetric polynomials gives (7.5), and the images span W_2. The transpose of multiplication by $(x_1 - x_2)D_0$ is

$$L v_{mnl} = ml\, v_{m-1,n,l-1} - nl\, v_{m,n-1,l-1};$$

a term with zero integer coefficient is zero. The dual differential is consequently

$$d^\circ v = -\eta^\vee L v, \qquad d^\circ(\eta^\vee v) = 0,$$
$$d^\circ(\chi^\vee v) = (\eta\chi)^\vee L v, \qquad d^\circ((\eta\chi)^\vee v) = 0.$$

Both $L v_{001}$ and $L v_{000}$ vanish. The first class in (7.6) has no incoming boundary and evaluates to 1 on D_0. Every boundary of degree one has only an $\eta^\vee$ component, so the second class, which evaluates to 1 on χ, is not a boundary either.

The simultaneous derivative commutes with d, since it kills $x_1 - x_2$. Its transpose sends v_{mnl} to $v_{m+1,n,l} + v_{m,n+1,l}$. Formula (7.5) now gives the asserted translation identity, in every cochain degree. □

The first lost class differentiates in the moment coordinate. The second is dual to a relation direction. The argument persists under any scalar specialization of R: both detecting evaluations are equal to 1. A degree-preserving map to a complex concentrated in degree zero cannot retain the second class. A geometric direct image may change the complex and shift; restriction of polynomial tests does neither.

7.3 The separated coefficient and its homotopy

To compare with a spectral coefficient, specialize the rational moment model to

$$R_4 = \mathbb{Q}(a, b)[\epsilon]/(\epsilon^4), \qquad h = a + b, \qquad B = R_4[x, y]((s^{-1})), \qquad Z = s + y - x.$$

The coordinate Z is invertible in B. Expand fractions first in ϵ and then at $s = \infty$. The scalar

$$E(Z) = \frac{Z(Z+\epsilon h)}{(Z+\epsilon a)(Z+\epsilon b)} = 1 - \frac{ab\epsilon^2}{Z^2} + \frac{abh\epsilon^3}{Z^3} \tag{7.7}$$

is exact at this precision. Define

$$K_s = B[D_0] \otimes \Lambda(\eta, \chi), \qquad d\eta = -ZD_0, \qquad d\chi = 0.$$

Substitution $x_1 = x - s$, $x_2 = y$ defines a cochain map $\sigma_s : K \to K_s$, with K now taken over R_4. Multiplying its output by $E(Z)$ gives the coefficient map

$$\mathcal{L}_s = E(Z)\sigma_s. \tag{7.8}$$

This map is linear. Its value on 1 is $E(Z)$, so it is not asserted to be a unital algebra homomorphism.

Let $p : K_s \to B \otimes \Lambda(\chi)$ set $D_0 = \eta = 0$, and let i be the inclusion. With the exterior factors ordered as $\eta\chi$, put

$$H(D_0^m \chi^e) = -\frac{\eta D_0^{m-1}\chi^e}{Z} \quad (m \geq 1,\ e = 0, 1), \qquad H(\chi^e) = H(\eta D_0^m \chi^e) = 0. \tag{7.9}$$

Proposition 7.3. *The map $\mathcal{L}_s$ preserves the two odd coefficient directions and commutes with the simultaneous derivative in the two roots. The homotopy satisfies*

$$dH + Hd = 1 - ip, \qquad d(EH) + (EH)d = E(Z)(1 - ip). \tag{7.10}$$

For a symmetric polynomial F,

$$p\mathcal{L}_s(F) = E(s + y - x)F(x - s, y).$$

The first corrections to the homotopy are

$$(EH)(D_0) = -\frac{\eta}{Z} + \frac{ab\epsilon^2\eta}{Z^3} - \frac{abh\epsilon^3\eta}{Z^4}. \tag{7.11}$$

Proof. Substitution takes $(x_1 - x_2)D_0$ to $-ZD_0$, and multiplication by the closed scalar $E(Z)$ commutes with the differential. Both maps preserve multiplication by χ. The simultaneous derivative kills Z, so it commutes with $\mathcal{L}_s$.

For $m \geq 1$, dH is the identity on $D_0^m\chi^e$ and Hd vanishes there. On $\eta D_0^m \chi^e$, the two roles are reversed. Both vanish on $1, \chi$. These monomials prove the first identity in (7.10). Multiplication by E proves the second. Projection gives the polynomial formula, and (7.7) gives (7.11). □

Thus the scalar coefficient also determines a correction to the moment homotopy. Retaining E in the coefficient map while omitting it from the homotopy would leave the defect

$$\big(dH + Hd - E(Z)(1 - ip)\big)(D_0) = \frac{ab\epsilon^2 D_0}{Z^2} \pmod{\epsilon^3}. \tag{7.12}$$

At $a = b = Z = D_0 = 1$, its ϵ^2 coefficient is 1.

7.4 Principal parts on the collision

Return to the coefficient ring R of (7.2). The localized calculation divides by $u = x_1 - x_2$. Coefficients supported at $u = 0$ have a different action of u. To compute it, define the principal-parts module

$$P_u = R[u, u^{-1}]/R[u] = \bigoplus_{p \geq 1} R\, u^{-p}.$$

Multiplication by u kills u^{-1} and takes u^{-p} to $u^{-(p-1)}$ for $p > 1$. Define P_v in the same way, and set

$$\mathcal{T} = R[w] \otimes_R P_u \otimes_R P_v, \qquad \mathcal{T}_1 = \ker(u : \mathcal{T} \to \mathcal{T}).$$

Here v is a second normal coordinate and w is a common position coordinate. These are algebraic direct sums of principal parts. In particular every element has finite pole order in each normal coordinate. We make no assertion about an ambient product or an adic completion of this module.

Tensor the moment complex with $\mathcal{T}$ over its coefficient ring in u, w. The result is

$$M = \mathcal{T}[D_0] \otimes \Lambda(\eta, \chi), \qquad d\eta = uD_0, \qquad d\chi = 0. \tag{7.13}$$

This is a complex of modules; principal parts are not being given a quotient-ring multiplication.

Proposition 7.4. *The cohomology of* (7.13) *is*

$$\begin{aligned} H^0(M) &= \mathcal{T}, \\ H^{-1}(M) &= \eta\mathcal{T}_1[D_0] \oplus \chi\mathcal{T}, \\ H^{-2}(M) &= \eta\chi\mathcal{T}_1[D_0]. \end{aligned} \tag{7.14}$$

The copies of $\mathcal{T}$ in degrees $0, -1$ *are represented at* $D_0 = 0$. *In particular*

$$[\eta u^{-1} v^{-1}], \qquad [\eta\chi u^{-1} v^{-1}]$$

are nonzero. There is no deformation retraction of M onto $\mathcal{T} \otimes \Lambda(\chi)$ with zero differential.

Proof. Multiplication by u is surjective on $\mathcal{T}$. Therefore the image of uD_0 on $\mathcal{T}[D_0]$ is $D_0\mathcal{T}[D_0]$, and its kernel is $\mathcal{T}_1[D_0]$. These give the cokernel and kernel of the two-term η complex. Tensoring with the closed factor $\Lambda(\chi)$ proves (7.14).

Explicitly, $\eta u^{-p-1} v^{-r} D_0^{m-1}$ is a primitive of $u^{-p} v^{-r} D_0^m$ for $m \geq 1$. By contrast $\eta u^{-1} v^{-r} D_0^m$ is closed. A boundary in degree -1 has only a χ component, and there are no boundaries in degree -2. This proves the two stated nonvanishing claims. The proposed retract has no degree--2 cohomology, whereas M does. A deformation retraction is therefore impossible. □

The coefficient derivative ∂_u alone is not a connection on this moment complex. Holding D_0 fixed gives

$$[\partial_u, d]\eta = D_0.$$

A horizontal comparison must supply the additional action on the moment factors or the connection after geometric direct image. The failure of the localized contraction proves a statement about that contraction and recipient. It leaves open a comparison retaining the full supported complex.

7.5 The link of a triple collision

We now use bounded constructible complexes of finite-dimensional k-vector spaces on complex algebraic varieties with their analytic topology. The field k has characteristic zero. All sheaf operations are derived. For an open inclusion j, write $j_!$ for extension by zero and Rj_* for right direct image. For a closed point inclusion i, i^* is its ordinary stalk and $i^!$ its costalk, or extraordinary restriction. These two restrictions detect different supported maps.

For three positions put $u = z_1 - z_2$, $v = z_2 - z_3$, and remove the common-position coordinate. The normal surface and the separated open are

$$X = \mathbb{C}^2, \qquad D = \{uv(u+v) = 0\}, \qquad U = X \setminus D. \tag{7.15}$$

Write $j : U \hookrightarrow X$ and $i : \{0\} \hookrightarrow X$. The line $u + v = 0$ away from 0 is the collision $z_1 = z_3$ with z_2 distinct. It is one of the three branches, and its coefficient must be included before studying their common intersection.

The coordinates $(u, v) = (t, tr)$ give

$$U \simeq \mathbb{C}^* \times \big(\mathbb{C} \setminus \{0, -1\}\big).$$

The second factor retracts onto a bouquet of two circles. Let c be the positive loop in the first factor, and let a, b be based positive meridians around 0, -1 in the second. Then $\pi_1(U) = \mathbb{Z} \times F(a, b)$, where $F(a, b)$ is the free group on a, b. With compatible base paths, the remaining branch meridian is conjugate to $c(ab)^{-1}$.

A finite-dimensional representation M of this group has invertible actions C, A, B, with $CA = AC$ and $CB = BC$. The operators A, B need not commute. Let $\mathcal{L}_M$ be the corresponding local system. Define

$$K(M) = \left[M \xrightarrow{d^0} M^3 \xrightarrow{d^1} M^2 \right] \quad \text{in degrees } 0, 1, 2, \tag{7.16}$$

where

$$d^0 m = ((C-1)m, (A-1)m, (B-1)m),$$
$$d^1(p, q, r) = ((C-1)q - (A-1)p,\ (C-1)r - (B-1)p).$$

The commutation with C gives $d^1 d^0 = 0$.

Proposition 7.5. *There are natural identifications*

$$R\Gamma(U, \mathcal{L}_M) \simeq K(M) \simeq i^* R j_* \mathcal{L}_M.$$

If $C - 1$ is invertible, this complex is contractible. The same formulas apply to bounded complexes of representations after totalization with the internal differential and its Koszul signs.

Proof. The universal cover of the angular graph is a tree. Its cellular resolution has boundary $(a - 1, b - 1)$. The radial circle has boundary $c - 1$. Tensoring these two free resolutions and applying $\mathrm{Hom}(-, M)$ gives (7.16), including its minus signs. The product of the circle and graph is a classifying space for $\mathbb{Z} \times F(a, b)$, so this complex computes the stated cohomology. Positive real dilation identifies the complement in a sufficiently small ball, up to homotopy, with the global complement. This proves the stalk assertion.

When $C - 1$ is invertible, the maps

$$h^1(p, q, r) = (C-1)^{-1} p, \qquad h^2(y, z) = (0, (C-1)^{-1} y, (C-1)^{-1} z)$$

satisfy $dh + hd = 1$. Commutation with A, B proves the identity in degree one; the other two degrees follow directly. □

For the constant local system both differentials vanish, giving Betti numbers (1, 3, 2). The complement of two independent coordinate axes instead has Betti numbers (1, 2, 1). Thus the third collision branch changes the link even with constant coefficients.

Noncommuting monodromy already occurs in dimension two. Take

$$C = I, \qquad A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

The common invariant space is zero and both differentials have rank two. The cohomology dimensions are (0, 2, 2). This example has invariant vectors around individual branches; it will be useful to compare it with coefficients having no such invariants.

7.6 Clean branches and a surviving point

We say that a local system extends *cleanly* across a stratum if the canonical map $j_!\mathcal{L} \to Rj_*\mathcal{L}$ is an isomorphism near that stratum. At a smooth branch of D, the normal link is a circle. For scalar monodromy λ its cochain complex is $[k \xrightarrow{\lambda-1} k]$. Thus $\lambda \neq 1$ gives clean extension along the punctured branch.

Take a rank-one coefficient with $C = 1$, $A = 2$ and $B = 3$. Its three branch monodromies are 2, 3, $1/6$, so each punctured branch is clean. At their intersection the link is

$$d^0 m = (0, m, 2m), \qquad d^1(p, q, r) = (-p, -2p).$$

It has $H^0 = 0$ and one-dimensional groups $V_1 = H^1K, V_2 = H^2K$. Representatives are $(0, 0, 1)$ in degree one and $(0, 1)$ modulo $k(1, 2)$ in degree two.

Proposition 7.6. *For this coefficient there is a distinguished triangle*

$$j_!\mathcal{L} \longrightarrow Rj_*\mathcal{L} \longrightarrow i_*K \longrightarrow j_!\mathcal{L}[1]. \tag{7.17}$$

Moreover $i^! j_!\mathcal{L} \simeq K[-1]$, with one-dimensional cohomology in degrees 2, 3. The same conclusions hold for $C = 1$, $A = a$, $B = b$ whenever $a, b, ab \neq 1$.

Proof. The cone of $j_!\mathcal{L} \to Rj_*\mathcal{L}$ vanishes on U and on every punctured branch. It is supported at 0, where its stalk is K. This identifies the cone with i_*K. The localization adjunction gives $i^! Rj_*\mathcal{L} = 0$; applying $i^!$ to the triangle gives the shift $K[-1]$. For the scalar family in the statement, d^0 and d^1 both have rank one. The branch monodromies are a, b, $(ab)^{-1}$, all different from 1. □

Consequently clean pairwise collisions need not imply clean extension at a triple collision. Among rank-one coefficients, the locus with $H^1K \neq 0$ is exactly $C = 1$. Away from that locus the radial contraction applies. On it the angular graph has nonzero first cohomology. In branch coordinates, the resonance equation is that the product of the three monodromies equals 1.

The same phenomenon allows noncommuting monodromy. For

$$C = I, \qquad A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix}, \tag{7.18}$$

the first two branch matrices have eigenvalues 2, 3, while $(AB)^{-1}$ has eigenvalues $1/9, 1/4$. Every branch is clean. The invertibility of $A - I$ makes both link differentials have rank two, again giving dimensions (0, 2, 2). Here $AB \neq BA$. Thus the point cohomology is not an artifact of requiring scalar monodromy.

7.7 A cubic cover and its intermediate extension

There is an algebraic realization of the scalar resonance. Adjoin a primitive cube root of unity ζ to $\mathbb{Q}$ and consider the finite map

$$p : Z = \{w^3 = uv(u+v)\} \longrightarrow X. \tag{7.19}$$

Over U it is an etale cover with deck group μ_3. A nontrivial character summand of its constant direct image has $C = 1$, $A = B = \zeta$, after choosing the character. Indeed, with $(u, v) = (t, tr)$, the variable $y = w/t$ satisfies $y^3 = r(r+1)$. A radial loop fixes the character, and each of the three branch loops has monodromy ζ.

The projective normalization of $y^3 = r(r+1)$ has genus one: its degree-three map to $\mathbb{P}^1$ is fully ramified at 0, -1, ∞, so Riemann–Hurwitz gives $2g - 2 = -6 + 3(3-1) = 0$. Over $\mathbb{Q}$, the two nontrivial characters together have matrices

$$C = I, \qquad A = B = J, \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}.$$

Since $J^3 = I$ and $J - I$ is invertible, the rational link has dimensions $(0, 2, 2)$. Each character has dimensions $(0, 1, 1)$ after adjoining ζ.

To describe the extension across 0, recall the perverse normalization. For a stratum S of complex dimension s, a constructible complex P is perverse if its restriction to S has no cohomology above $-s$ and its extraordinary restriction has none below $-s$. A local system on a smooth complex d-fold becomes perverse after shift by d. The intermediate extension $j_{!*}\mathcal{L}[d]$ is the perverse extension with no subobject or quotient supported on the complement. Equivalently, it is the image of $j_!\mathcal{L}[d] \to Rj_*\mathcal{L}[d]$ in the perverse heart, with perverse truncations when needed. We use the shifted convention of [69, §15.6].

Retain a rank-one coefficient with $C = 1$ and all three branch monodromies nontrivial. Let K be its link and $V_i = H^i K$ for $i = 1, 2$. Define

$$F = \operatorname{fib}\big(Rj_*\mathcal{L} \longrightarrow i_* V_2[-2]\big), \qquad \mathcal{I} = F[2]. \tag{7.20}$$

The arrow is point restriction followed by the canonical top-cohomology quotient $K \to H^2(K)[-2]$.

Theorem 7.7. *The complex $\mathcal{I}$ is $j_{!*}\mathcal{L}[2]$, and*

$$i^*\mathcal{I} = V_1[1], \qquad i^!\mathcal{I} = V_2[-1]. \tag{7.21}$$

It also has the following proper geometric realization. Let $\pi : \widetilde{X} = \mathrm{Bl}_0 X \to X$ and $q : \widetilde{X} \to \mathbb{P}^1$ be the line-bundle projection. The local system descends to ℓ on $Y = \mathbb{P}^1 \setminus \{0, -1, \infty\}$. For $j_Y : Y \hookrightarrow \mathbb{P}^1$, set $\bar{\ell} = j_{Y,!}\ell = Rj_{Y,*}\ell$. Then

$$R\pi_* q^* \bar{\ell}[2] \simeq \mathcal{I}, \tag{7.22}$$

with the isomorphism restricting to the identity on U.

Proof. Ordinary restriction of (7.20) is $\tau_{\le 1}K = V_1[-1]$. Extraordinary restriction is $V_2[-3]$, because $i^! Rj_*\mathcal{L} = 0$. The shift by 2 gives (7.21). These restrictions satisfy the perverse bounds at the point, and both have zero degree-zero cohomology. The punctured branches are clean. Thus $\mathcal{I}$ is perverse, with no subobject or quotient supported on the complement of U. This proves the intermediate-extension assertion.

The trivial radial monodromy gives the descent to Y; the three nontrivial puncture monodromies make $\bar{\ell}$ clean. Proper base change identifies the stalk of the left side of (7.22) with $R\Gamma(Y, \ell)[2] = H^1(Y, \ell)[1] = V_1[1]$. Proper Verdier duality computes its costalk from the same calculation for $\ell^\vee$. Poincare duality on the punctured direction

curve identifies $H^1(Y,\ell^\vee)^\vee$ with $H^1_c(Y,\ell)$. Clean extension identifies compactly supported and ordinary cohomology here. Tensoring with the positive radial circle class identifies $H^1(Y,\ell)$ with V_2. The costalk is therefore $V_2[-1]$. Away from 0, π is an isomorphism. The same intermediate-extension characterization proves (7.22) and its uniqueness over U. □

The blowup separates the branches. Its exceptional direction curve nevertheless retains their joint cohomology. This is the geometric reason that replacing the three-line arrangement by independent axes would change the coefficient.

7.8 The critical direct image and its point constituent

The cubic cover also defines a critical coefficient. Put

$$B=\mathbb{A}^3_{u,v,w},\qquad W=\lambda\big(w^3-uv(u+v)\big),\qquad \pi_B: B\times\mathbb{A}^1_\lambda\longrightarrow B,$$

and let $a: Z\hookrightarrow B$ be the zero locus of the factor multiplying λ. Write ϕ_W for ordinary vanishing cycles, and $\varphi_W=\phi_W[-1]$ for their perverse normalization. All coefficients in the next formula are rational, with Tate data forgotten.

Theorem 7.8. *There is a deck-equivariant isomorphism*

$$R\pi_{B,!}\varphi_W\mathbb{Q}_{B\times\mathbb{A}^1}[4]\simeq a_*\mathbb{Q}_Z[2]. \tag{7.23}$$

After finite pushforward along $p: Z\to X$, scalar extension to $\mathbb{Q}(\zeta)$, and projection to a nontrivial deck character, the coefficient is $\mathcal{P}=j_!\mathcal{L}[2]$. It is perverse and fits into the nonsplit exact sequence

$$0\longrightarrow i_*V_1\longrightarrow\mathcal{P}\longrightarrow\mathcal{I}\longrightarrow 0,\qquad \dim V_1=1. \tag{7.24}$$

Thus this critical direct image and the intermediate extension have the same restriction to U and different extensions across its complement.

Proof. The potential is linear in λ. Dimensional reduction for one linear fiber coordinate gives

$$R\pi_{B,!}\phi_W\mathbb{Q}[-1]\simeq a_*\mathbb{Q}_Z[-2]$$

[35, Theorem A.1]. Shifting by the ambient complex dimension 4 gives (7.23). The deck action preserves W, so naturality of the comparison makes it deck-equivariant. In particular its character summands are preserved.

Over a point of D, the fiber of p is the single topological point $w=0$, fixed by the deck group. Finite proper base change therefore gives zero ordinary boundary stalks for any nontrivial character summand of $Rp_*\mathbb{Q}_Z$. On U that summand is $\mathcal{L}$. The extension-by-zero adjunction is an isomorphism on every stalk, identifying the summand with $j_!\mathcal{L}$.

The point stalk of $\mathcal{P}$ is zero. By (7.17), its costalk is $K[1]$, with cohomology in degrees 0, 1. The punctured branches are clean. These restrictions prove perversity. The two truncation triangles of K give a triangle

$$j_!\mathcal{L}\longrightarrow F\longrightarrow i_*V_1[-1]\longrightarrow .$$

After shifting by 2 and rotating, perverse cohomology gives (7.24). If the sequence split, its middle term would have point stalk $V_1\oplus V_1[1]$, whereas that stalk is zero. The sequence is therefore nonsplit. □

The point constituent can be seen directly on Z. Let $i_Z:\{0\}\hookrightarrow Z$ and $j_Z: Z\setminus\{0\}\hookrightarrow Z$. The surface is the affine cone over the projective cubic

$$E=\{w^3=uv(u+v)\}\subset\mathbb{P}^2.$$

The cubic E is smooth: the partial derivatives of $w^3-uv(u+v)$ have no common projective zero. Let $\mathrm{IC}_Z=j_{Z,!*}\mathbb{Q}[2]$ denote the intersection-cohomology complex.

Proposition 7.9. *There is an exact sequence of perverse sheaves*

$$0 \longrightarrow i_{Z,*}H^1(E,\mathbb{Q}) \longrightarrow \mathbb{Q}_Z[2] \longrightarrow \mathrm{IC}_Z \longrightarrow 0. \tag{7.25}$$

Each nontrivial deck character of $H^1(E)$ has dimension one after adjoining ζ. Finite pushforward and character projection take (7.25) *to* (7.24).

Proof. The link of the cone is the unit circle bundle of $\mathcal{O}_E(-1)$, whose Euler number is -3. Its Gysin sequence gives $H^1(\mathrm{Link}(Z),\mathbb{Q}) = H^1(E,\mathbb{Q})$. The cone localization triangle computes the costalk of $\mathbb{Q}_Z$ as the fiber of $\mathbb{Q} \to R\Gamma(\mathrm{Link}(Z),\mathbb{Q})$. After shift by 2 this costalk has no negative cohomology, and its degree-zero cohomology is $H^1(E,\mathbb{Q})$. The ordinary stalk is $\mathbb{Q}[2]$.

These are the perverse bounds at the vertex; away from the vertex the surface is smooth. There is no point quotient, since the ordinary stalk has no degree-zero cohomology. The maximal point subobject is the degree-zero costalk, namely $i_{Z,*}H^1(E,\mathbb{Q})$. Taking its quotient removes the point subobject without introducing a point quotient, so the quotient is IC_Z.

The deck quotient of E is $\mathbb{P}^1$. Averaging over the finite group identifies invariant cohomology with the cohomology of the quotient, hence $H^1(E)$ has no invariant part. It has rational dimension two, and the two conjugate nontrivial characters each occur once. Finite pushforward is perverse exact and commutes with intermediate extension. This proves the last assertion. □

The inclusion $i_*V_1 \to \mathcal{P}$ in (7.24) is a nonzero supported morphism whose map on the ordinary point stalk is zero. Indeed adjunction identifies it with the inclusion of $H^0(i^!\mathcal{P}) = V_1$ into $i^!\mathcal{P} = K[1]$. It is therefore detected by the costalk. This is why ordinary stalk maps alone cannot determine a critical operation at the collision.

7.9 A coefficient concentrated at the deepest collision

The cubic construction extends to any number of insertions. For $n \geq 3$, remove common translation by setting $z_1 = 0$, and define

$$\begin{aligned} X_n &= \mathbb{A}^{n-1}, \qquad N = \binom{n}{2}, \qquad Q_n = \prod_{1\leq i<j\leq n} (z_j - z_i), \\ U_n &= X_n \setminus V(Q_n), \qquad j : U_n \hookrightarrow X_n, \qquad i : \{0\} \hookrightarrow X_n. \end{aligned} \tag{7.26}$$

The origin is the deepest collision, where all positions meet. For $q \in k^\times$, let $L_n(q)$ be the pullback by $Q_n : U_n \to \mathbb{C}^*$ of the rank-one local system with positive meridian monodromy q. Every pair-collision hyperplane consequently has the same monodromy q.

When $q^N = 1$, this coefficient is a character summand of the finite cover

$$p_n : Z_n = \{w^N = Q_n\} \longrightarrow X_n. \tag{7.27}$$

The coefficient field is assumed to contain the required roots of unity. The character is faithful when q has order N, but faithfulness will not be necessary.

Lemma 7.10. *Suppose*

$$q^{\binom{r}{2}} \neq 1 \qquad (2 \leq r < n). \tag{7.28}$$

Then the normal coefficient link vanishes at every nontrivial proper collision stratum. Hence $j_!L_n(q) \to Rj_*L_n(q)$ is an isomorphism on $X_n \setminus \{0\}$.

Proof. A collision stratum is specified by a partition into clusters of equal positions. At a point of that stratum, differences between distinct clusters are nonzero. Its normal arrangement is the product of the arrangements internal to the clusters. A cluster of size r has an internal Vandermonde of degree $\binom{r}{2}$. Its positive radial circle therefore acts by $q^{\binom{r}{2}}$.

The radial circle complex contracts by the inverse of $q^{\binom{r}{2}} - 1$. Tensoring its contraction with the remaining normal factors contracts the entire normal link. Every nontrivial proper stratum has a cluster of size $2 \le r < n$, so (7.28) applies. The cone of $j_! \to R\,j_*$ vanishes on these strata, proving the assertion. □

For a primitive Nth root, condition (7.28) holds because $0 < \binom{r}{2} < N$. In particular the calculation includes nonconsecutive clusters and simultaneous disjoint collisions. At four insertions, for example, it includes $13 \mid 24$. Only the coefficient link vanishes; the stratum itself remains part of the configuration geometry.

7.9.1 The direction space

On U_n, set $t = z_2$ and $x_i = z_i/t$ for $i \ge 3$. Then

$$U_n \simeq \mathbb{C}^* \times Y_n, \qquad Y_n = \{x_i \ne 0, 1,\ x_i \ne x_j\}. \tag{7.29}$$

The marked points $0, 1, x_3, \dots, x_n, \infty$ identify Y_n with the moduli space $\mathcal{M}_{0,n+1}$ of ordered distinct points on a projective line, modulo projective automorphisms. Its complex dimension is $m = n - 2$. Since $Q_n = t^N q_n(x)$, the radial monodromy is q^N. For $q^N = 1$, the coefficient in (7.29) is the external product of the constant radial coefficient and a rank-one system $\ell_n(q)$ on Y_n.

Theorem 7.11. *Assume $q^N = 1$ and (7.28). The direction cohomology is concentrated in degree $m = n - 2$:*

$$H^a(Y_n, \ell_n(q)) = 0 \quad (a \ne m), \qquad \dim_k H^m(Y_n, \ell_n(q)) = m!. \tag{7.30}$$

The deepest link $K_n = i^* R\,j_* L_n(q)$ has precisely

$$H^{n-2}(K_n) = V_n, \qquad H^{n-1}(K_n) = W_n, \qquad \dim V_n = \dim W_n = (n-2)!. \tag{7.31}$$

Proof. The smooth compactification $\overline{\mathcal{M}}_{0,n+1}$ has normal-crossing boundary, whose divisors are indexed by stable bipartitions of the marked points [18, §2.6]. Choose the side not containing ∞. It is a cluster of r finite marks, with $2 \le r \le n-1$. The Vandermonde winding around its collision divisor is $\binom{r}{2}$. Renormalizing the coordinates to $z_1 = 0$, $z_2 = 1$ changes this winding by a multiple of N, since simultaneous rescaling multiplies Q_n by an Nth power. Thus the boundary monodromy is $q^{\binom{r}{2}}$.

At an intersection of boundary divisors the local normal link is a product of circles, with these nontrivial monodromies. Its coefficient complex contracts. Extension by zero to the compactification equals right direct image, and properness gives

$$H^a_c(Y_n, \ell_n(q)) \xrightarrow{\sim} H^a(Y_n, \ell_n(q)). \tag{7.32}$$

Successively forgetting a moving point expresses Y_n as a tower of locally trivial fibrations by punctured projective lines. Local trivializations follow by moving the punctures in disjoint disks. Each fiber has the homotopy type of a finite graph. The local-coefficient spectral sequence, applied inductively also to the finite-rank systems given by fiber cohomology, proves vanishing above complex dimension m. The same bound applies to the dual coefficient. Poincare duality on the oriented real $2m$-manifold then gives compact-support vanishing below m. Together with (7.32), these bounds prove concentration in degree m.

The fiber Euler characteristics in the tower are $-1, -2, \dots, -m$, so $\chi(Y_n) = (-1)^m m!$. Euler characteristic with a rank-one local system is the same as for the constant system, as seen from a finite cellular complex. Concentration proves the dimension in (7.30). Finally (7.29) tensors this group with $H^*(\mathbb{C}^*, k)$ in degrees $0, 1$. Positive real scaling identifies the complement with its local link at the origin, proving (7.31). □

The first ranks are $1, 2, 6, 24$ for $n = 3, 4, 5, 6$. The cubic cover is the case $n = 3$.

7.9.2 The critical extension in every arity

Put $d = n - 1$, $B_n = X_n \times \mathbb{A}^1_w$, and define

$$\mathcal{W}_n = \lambda(w^N - Q_n), \qquad \pi : B_n \times \mathbb{A}^1_\lambda \longrightarrow B_n, \qquad a : Z_n \hookrightarrow B_n.$$

We continue to use $\varphi = \phi[-1]$ for perverse vanishing cycles and omit Tate data.

Theorem 7.12. *Under the hypotheses of Theorem 7.11, dimensional reduction is deck-equivariant and gives*

$$R\pi_! \varphi_{\mathcal{W}_n} k_{B_n \times \mathbb{A}^1}[n+1] \simeq a_* k_{Z_n}[n-1]. \tag{7.33}$$

After pushforward by p_n and projection to the character with monodromy q, the coefficient is the perverse sheaf $\mathcal{P}_n = j_! L_n(q)[d]$. It has a nonsplit exact sequence

$$0 \longrightarrow i_* V_n \longrightarrow \mathcal{P}_n \longrightarrow j_{!*} L_n(q)[d] \longrightarrow 0, \qquad \dim V_n = (n-2)!. \tag{7.34}$$

Proof. The one-coordinate dimensional-reduction identity is $R\pi_! \phi_{\mathcal{W}_n} k[-1] \simeq a_* k_{Z_n}[-2]$. Shifting by the ambient dimension $n+1$ gives (7.33); naturality gives deck equivariance. Every fiber of p_n over $Q_n = 0$ is the single topological point $w = 0$. Since the selected character is nontrivial, its ordinary boundary stalks vanish. Its direct image is therefore $j_! L_n(q)$.

By Lemma 7.10, the difference between the two open extensions is supported only at 0:

$$j_! L_n(q) \longrightarrow R j_* L_n(q) \longrightarrow i_* K_n \longrightarrow .$$

Define

$$F_n = \mathrm{fib}\big(R j_* L_n(q) \longrightarrow i_* W_n[-d]\big), \qquad \mathcal{I}_n = F_n[d].$$

The arrow is the top-cohomology quotient on the point link. The two restrictions are $i^* \mathcal{I}_n = V_n[1]$ and $i^! \mathcal{I}_n = W_n[-1]$. Thus $\mathcal{I}_n$ is the intermediate extension. The point stalk of $\mathcal{P}_n$ is zero and its costalk is $K_n[d-1]$, whose cohomology is in degrees 0, 1. The extension is clean at every other boundary stratum, so $\mathcal{P}_n$ is perverse.

The truncation triangles of K_n now give $i_* V_n \to \mathcal{P}_n \to \mathcal{I}_n \to i_* V_n[1]$, which is (7.34) in the perverse heart. A splitting would give the nonzero point stalk $V_n \oplus V_n[1]$ to $\mathcal{P}_n$. This contradicts its zero stalk. □

7.9.3 Keeping the monodromy fixed

The preceding construction can choose a primitive $\binom{n}{2}$th root for each n. That choice changes the coefficient with the number of insertions. For a fixed nontrivial q of finite order ℓ, set

$$n(\ell) = \min\left\{ r \geq 2 : \ell \mid \binom{r}{2} \right\}. \tag{7.35}$$

The minimum exists, since $\ell \mid \binom{2\ell}{2}$.

Corollary 7.13. *Among $n \geq 3$, the coefficient $L_n(q)$ has vanishing proper-collision links and a nonzero deepest link exactly at $n = n(\ell)$. For $\ell = 2, 3, 4, 5, 6, 7, 8$, these arities are respectively 4, 3, 8, 5, 4, 7, 16.*

Proof. Before the first such arity the total radial monodromy $q^{\binom{n}{2}}$ differs from 1, so the deepest link contracts. At the first arity, all proper cluster monodromies are nontrivial and Theorem 7.11 applies. At any larger arity choose a cluster of size $n(\ell)$ and keep the other points distinct. Its normal link is the nonzero link just computed, so proper-collision vanishing fails. Testing the intervening triangular numbers gives the listed values. □

For example $q = -1$ gives the deepest-only phenomenon at four insertions, although it is not a faithful character of the sixfold cover. No fixed nontrivial finite-order scalar character gives that phenomenon in every arity. Compatibility between different arities requires actual coefficient maps along substitutions. The family of finite covers alone does not supply those maps.

7.10 The geometric comparison still required

The moment chart and the finite covers answer different coefficient questions. The chart computes relation directions in a derived flag fiber. The covers determine constructible extensions and their point constituents. Neither calculation identifies its coefficient with the oriented direct image of a Hall correspondence.

Problem 7.14. Construct the oriented critical Hall or Hecke coefficient map on the flag chart (7.2), with its support maps and cohomological shift. On the separated chamber, compare its coefficient operation with $\mathcal{L}_s$ in (7.8), including the homotopy (7.11). At the collision retain both odd directions through the full supported complex (7.13) before applying the geometric direct image.

For three and more positions, determine the extension of that same geometric coefficient across every collision stratum. Compare its ordinary stalks, costalks and attaching maps with the extensions calculated here. In particular determine whether a critical point constituent occurs and how the actual operation acts on it. The comparison must include the nonconsecutive collision $13 \mid 2$ and its common intersection with the other pair collisions.

A framing and its moving position connection must be part of this map when framed coefficients are used. The charge-zero scalar unit, the local vacuum under the diagonal tensor and a functional defining a state remain distinct data. Their behavior cannot be inferred from a dimension or an Euler ratio. The examples in this chapter specify which classes and supported maps the geometric comparison must account for.

Chapter 8

Ordered collisions and cut reconstruction

Consider three positions in the order (x, y, x), where $x \neq y$. Neither cut

$$(x) \mid (y, x), \qquad (x, y) \mid (x)$$

separates disjoint sets of positions. The outer collision therefore survives a construction that permits only consecutive cuts. Collapsing the two occurrences of x gives two distinct positions, which do admit a cut. This discrepancy determines both a useful reconstruction theorem and a limitation of its comparison with full collision geometry.

We begin with the cut product and its bar and cobar constructions. Strong cut coproducts identify a finite localization complex; its support contains every configuration whose equalities obstruct all cuts. We then construct the ordered collision atlas and its product over arbitrary ordered subsets. It has its own bar–cobar equivalence. For the dualizing coalgebra, the map from its cobar to the cut cobar has a kernel with explicit principal-part classes, including negative-degree classes in arbitrarily large arity.

The last comparison concerns factorization. Invertibility of each coproduct component differs from invertibility of their joint map. Full exchange data identify the atlas with the corresponding word systems, while a local field realization must also preserve the actual collision operations. Throughout, these statements concern associative algebras and coassociative coalgebras on specified geometric carriers.

8.1 Coefficients on ordered words

Let k be a field of characteristic zero and let X be a smooth separated curve over k. Write $\mathrm{DMod}(Y)$ for the enhanced stable category of unbounded right differential-module complexes on a smooth scheme Y, with quasi-coherent underlying modules. External products, direct images and extraordinary inverse images are derived. Our shift convention is $M[r]^i = M^{i+r}$, with differential $(-1)^r d_M$.

An ordered word coefficient system consists of a complex F_0 of k-vector spaces and objects $F_n \in \mathrm{DMod}(X^n)$ for $n > 0$. For every monotone surjection $\alpha : [n] \twoheadrightarrow [m]$, it includes an equivalence

$$d_\alpha^! F_n \simeq F_m, \qquad d_\alpha(y)_i = y_{\alpha(i)},$$

compatible with identities, composites and their higher coherences. Here $[n] = \{1, \dots, n\}$ in its usual order. Let $D(k)$ be the derived category of k-vector spaces, and let $\mathsf{OrdSurj}_+$ have the nonempty words $[n]$ as objects and monotone surjections as arrows. The category of coefficient systems is

$$\mathcal{W}_X = D(k) \times \varprojlim_{\mathsf{OrdSurj}_+} \mathrm{DMod}(X^n).$$

The first factor is isolated: there are no collision maps joining it to a nonempty word.

For $1 \leq p < n$, define

$$U_p^{(n)} = \{x_i \neq x_j \text{ whenever } i \leq p < j\}, \qquad j_p : U_p^{(n)} \hookrightarrow X^n. \tag{8.1}$$

Every pair crossing the cut occurs in this condition, including pairs of nonadjacent labels. Set $U_0^{(n)} = U_n^{(n)} = X^n$. Define

$$(F \star_{\mathrm{cut}} G)_n = \bigoplus_{p=0}^{n} j_{p,*} j_p^!(F_p \boxtimes G_{n-p}). \tag{8.2}$$

At an empty end, the external product means tensoring by the scalar complex F_0 or G_0. The candidate unit is

$$K_0 = k, \qquad K_n = 0 \quad (n > 0).$$

This empty-support unit is distinct from a vacuum coefficient on X.

We will repeatedly use open base change. If $j : U \hookrightarrow Y$ is open, $f : Y' \to Y$, and $j' : U' \hookrightarrow Y'$ is its pullback, then

$$f^! j_* \simeq j'_* f_U^!. \tag{8.3}$$

To see this, let i, i' be the closed complements. Recollement gives $i^! j_* = 0$. The composite $f i'$ factors through i, so $i'^! f^! j_* = 0$. The localization triangle identifies $f^! j_* M$ with the direct image of its restriction to U', which is $f_U^! M$. All comparisons restrict to transitivity of extraordinary inverse image on U'. Full faithfulness of j'_* therefore supplies their composition coherences as well.

A consecutive partition divides a word into nonempty intervals, listed in their original order.

Proposition 8.1. *Formula* (8.2) *makes $\mathcal{W}_X$ a monoidal category with unit K. Its iterated products are finite sums over consecutive partitions, localized where distinct blocks have disjoint positions.*

Proof. A monotone collision kills a cut inside one of its fibres: that fibre would have to meet both sides at distinct positions. Every other cut is the inverse image of exactly one quotient cut. Equation (8.3) identifies the surviving summands with the quotient product, compatibly with further collisions.

Each parenthesized product imposes the same final consecutive partition and the same cross-block inequalities. Successive open direct images identify its coefficient with the external product on that common open. These identifications give associativity and all its coherences. The two empty-end summands give the unit laws. □

Call a coefficient system *positive* if its empty coefficient vanishes. More generally, let $\mathcal{W}_{X,\geq a}$ consist of systems vanishing in arities below a. A nonzero summand of a product requires the labels used by each factor, so

$$\mathcal{W}_{X,\geq a} \star_{\mathrm{cut}} \mathcal{W}_{X,\geq b} \subseteq \mathcal{W}_{X,\geq a+b}. \tag{8.4}$$

In particular, r positive factors have zero coefficient at $n < r$. Every reduced word construction below is finite at each configuration. This finiteness uses arity, with no cohomological boundedness assumption.

8.2 The two associative constructions

Here E_1 means homotopy-coherent associative. Let A be an augmented E_1 algebra for $\star_{\mathrm{cut}}$, with $A_0 = k$ and augmentation ideal $a = \mathrm{fib}(A \to K)$. Its unit splits $A \simeq K \oplus a$, and a is positive. Write $s = [1]$, so $|sx| = |x| - 1$. The normalized bar has underlying word object

$$B_{\mathrm{cut}} A = K \oplus \bigoplus_{r \geq 1} (sa)^{\star_{\mathrm{cut}} r}. \tag{8.5}$$

Its coproduct deconcatenates words, retaining both empty ends. On a strict differential graded representative, its degree-one coderivation is determined by

$$b_1(sx) = -s(dx), \qquad b_2(sx, sy) = (-1)^{|x|} s(xy). \tag{8.6}$$

Insertion after a suspended prefix contributes the Koszul sign of that prefix. Thus for closed degree-zero inputs,

$$b[x|y] = [xy], \qquad b[x|y|z] = [xy|z] - [x|yz].$$

The second differential on the last expression is zero by associativity. These words denote geometric coefficient summands; they need not be global pure tensors. A coherent E_1 model retains the higher faces of its normalized bar diagram.

Let C instead be a coaugmented counital E_1 coalgebra with $C_0 = k$, and put $c = \operatorname{cofib}(K \to C)$. Its reduced coproduct is $\bar{\Delta} : c \to c \star_{\mathrm{cut}} c$. With $t = [-1]$, define

$$\Omega_{\mathrm{cut}} C = K \oplus \bigoplus_{r \geq 1} (tc)^{\star_{\mathrm{cut}} r}. \tag{8.7}$$

Multiplication concatenates consecutive block lists. On a strict model, the differential is the derivation

$$d(tc) = -t(d_C c) + \sum_{\bar{\Delta} c = \sum c' \otimes c''} (-1)^{|c'|+1} (tc')(tc''). \tag{8.8}$$

The notation for the reduced coproduct abbreviates its localized coefficient map. Internal differentials cancel the mixed terms in d^2; disjoint splits cancel by the odd derivation rule; the two refinements of one block into three cancel by coassociativity.

Theorem 8.2. *Bar and cobar are inverse equivalences between augmented E_1 cut algebras with empty coefficient k and coaugmented counital E_1 cut coalgebras with empty coefficient k. The equivalence includes their derived mapping spaces and higher structure maps.*

Proof. Use the enhanced bar and cobar, with normalized expressions (8.5)–(8.7). Their natural comparisons have strict-model forms

$$\Omega_{\mathrm{cut}} B_{\mathrm{cut}} A \longrightarrow A, \qquad C \longrightarrow B_{\mathrm{cut}} \Omega_{\mathrm{cut}} C.$$

The first uses the universal twisting map: a one-factor bar word desuspends to its generator, and longer generators map to zero. The second uses all iterated reduced coproducts. Positivity makes every component of these formulas finite.

Flatten a word in $\Omega_{\mathrm{cut}} B_{\mathrm{cut}} A$ into N original factors of a. Filter by N. Multiplication of original factors lowers this number. In the associated graded, each of its $N - 1$ gaps has two states: it is inside a bar block, or between cobar factors. Changing the state is the identity on the same final localized coefficient, in adjacent degrees. The gap complex is the signed tensor product of these two-term identity complexes. For $N > 1$, contracting the first gap contracts the whole complex. At $N = 0, 1$, the comparison is the identity on K and the original factor.

For $B_{\mathrm{cut}} \Omega_{\mathrm{cut}} C$, use the decreasing filtration by the number of original factors of c. Splitting a factor raises that number. Its associated graded has the same identity gap complexes, with the two states interchanged. The comparison is again an equivalence in lengths zero and one and has contractible higher pieces.

At arity n, both filtrations have length at most n by (8.4). Thus their associated-graded contractions prove both comparisons to be equivalences. In coherent models, higher products lower the first filtration and higher coproducts raise the second; the associated-graded gap diagrams are unchanged. Normalized realization and totalization retain their higher faces. Their natural comparisons and triangle coherences therefore give inverse functors on the enhanced categories, including mapping spaces. □

The bar is a coalgebra and the cobar is an algebra. An exchanged associative algebra requires a coefficient duality as well. For a contravariant equivalence $\sharp$ on an admissible coefficient subcategory, the coefficient transposition of Chapter 6 takes the following form for the cut product:

$$A^{!}_{\sharp} = (B_{\mathrm{cut}} A)^{\sharp}, \qquad F^{\sharp} \star_{\sharp} G^{\sharp} = (F \star_{\mathrm{cut}} G)^{\sharp}.$$

The transposed deconcatenation multiplies $A^!_\sharp$ for this transported tensor. Its carrier and admissibility hypotheses remain part of the construction. No Lie–commutative duality is used to identify it with an algebra of fields.

8.3 Which collisions the cut cobar retains

Write $\Delta_{ij} \subset X^n$ for the equality locus $x_i = x_j$, $\Delta_n : X \hookrightarrow X^n$ for the small diagonal, and $\Delta = \Delta_2$. Define the reduced closed support

$$Z_1 = X, \qquad Z_n = X^n \setminus \bigcup_{p=1}^{n-1} U_p^{(n)} = \bigcap_{p=1}^{n-1} \bigcup_{i \le p < j} \Delta_{ij}. \tag{8.9}$$

A tuple lies in Z_n precisely when every cut has an equality crossing it. The first examples are

$$Z_2 = \Delta_{12}, \qquad Z_3 = \Delta_{13}, \qquad Z_4 = \Delta_{14} \cup (\Delta_{13} \cap \Delta_{24}). \tag{8.10}$$

For the four-point formula, equality of the outer positions blocks every cut. If those positions differ, failure of the first and last cuts leaves the possibilities 12, 34 or 13, 24. The first still admits the middle cut unless the outer positions become equal. The second blocks all three cuts. These are identities of reduced supports, without a claim about scheme multiplicities.

A coaugmented cut coalgebra is *strong* if each of its supplied coproduct maps

$$C_n|_{U_p^{(n)}} \longrightarrow (C_p \boxtimes C_{n-p})|_{U_p^{(n)}}$$

is an equivalence. For a reduced closed support Z with inclusion i, write $R\Gamma_Z M = i_* i^! M$. Recollement expresses it as the fibre of restriction followed by direct image from the complementary open.

Theorem 8.3. *For a strong cut coalgebra, the reduced cobar has a natural equivalence*

$$\overline{\Omega_{\mathrm{cut}} C}_n \simeq R\Gamma_{Z_n} C_n[-1]. \tag{8.11}$$

It commutes with monotone collisions. Its map to $C_n[-1]$ is the length-one projection of the cobar.

Proof. A subset $S \subseteq [n-1]$ of gaps specifies a consecutive partition P_S. Let $U_S = \bigcap_{p \in S} U_p^{(n)}$, with inclusion j_S, and put $L_S = j_{S,*} j_S^!$; the empty subset gives the identity. The iterated strong coproduct identifies the entire refinement diagram of cobar coefficients with $S \mapsto L_S C_n$. Coassociativity identifies its faces and higher homotopies, not merely its vertices.

The nonempty subsets form the finite Čech diagram of the union of cut opens. Adding the empty vertex takes its total fibre and therefore gives $R\Gamma_{Z_n} C_n$. A partition with r blocks has $r-1$ cuts. Its cobar term has r desuspensions, producing the additional shift $[-1]$ in (8.11).

For the signs, consolidate the block shifts by

$$(tc_1) \otimes \cdots \otimes (tc_r) \longmapsto (-1)^{\sum_{j=1}^{r} (r-j)|c_j|} (c_1 \otimes \cdots \otimes c_r)[-r]. \tag{8.12}$$

Splitting block i into degrees a, b changes this exponent by $\sum_{j<i} |c_j| + a$. Adding the cobar splitting exponent gives i modulo two. Thus the face has sign $(-1)^i$, the negative of the corresponding Čech insertion sign; the internal differential has sign $(-1)^r$. These are precisely the shifted-cube signs.

A monotone collision deletes cuts inside its fibres and identifies the other cuts with all quotient cuts. Open base change identifies both augmented cubes. In particular $d_\alpha^{-1} Z_n = Z_m$. This proves compatibility, naturality and the assertion about the empty vertex of the cube. □

Write $L_p = L_{\{p\}}$ and $L_{12} = L_{\{1,2\}}$. At two positions, the resolution is

$$\mathrm{Tot}[C_2[-1] \xrightarrow{-\mathrm{loc}} L_1 C_2[-2]].$$

At three positions it is

$$\mathrm{Tot}\left[C_3[-1] \xrightarrow{(-\mathrm{loc},-\mathrm{loc})} (L_1C_3 \oplus L_2C_3)[-2] \xrightarrow{(u,v)\mapsto u|_{12}-v|_{12}} L_{12}C_3[-3]\right]. \tag{8.13}$$

The two composites cancel. For a smooth Y, let $p_Y : Y \to \mathrm{Spec}\, k$ be its structure map, ω_Y its canonical right differential module, and $J_Y = p_Y^! k$. The dualizing word system is

$$J_n = p_{X^n}^! k = \omega_{X^n}[n], \qquad J_0 = k, \tag{8.14}$$

the coproduct is localization and is strong. Transitivity of !-pullback gives its collision identifications. If $i_{13}(x, y) = (x, y, x)$, then

$$\overline{\Omega_{\mathrm{cut}} J}_2 \simeq \Delta_{12,*} J_X[-1], \qquad \overline{\Omega_{\mathrm{cut}} J}_3 \simeq i_{13,*} J_{X^2}[-1]. \tag{8.15}$$

The latter coefficient remains nonzero at $x_1 = x_3 \neq x_2$.

8.3.1 The product on the localization complex

The support equivalence retains an algebra structure through the specified strong coproduct. Its inverse on $U_p^{(p+q)}$ is a map

$$m_p : (C_p \boxtimes C_q)|_{U_p} \longrightarrow C_{p+q}|_{U_p}.$$

Write the finite resolution as

$$\mathcal{K}(C)_n = \mathrm{Tot} \bigoplus_{S \subseteq [n-1]} L_S C_n[-|S| - 1].$$

The internal differential on its S term is $(-1)^{|S|+1} d$; adding a gap p has sign $(-1)^{1+\#\{a \in S : a < p\}}$. For $S \subseteq [p-1]$ and $T \subseteq [q-1]$, put

$$r = |S| + 1, \quad s = |T| + 1, \quad G = S \cup \{p\} \cup (p + T).$$

Their product is defined on U_G by the restriction of m_p and then extended by $j_{G,*}$. On unshifted homogeneous coefficients,

$$u_S v_T = (-1)^{s|u|} \big(m_p(u \boxtimes v)\big)_G. \tag{8.16}$$

The total degree of u_S is $|u| + r$.

Proposition 8.4. *These operations make $K \oplus \mathcal{K}(C)$ an augmented E_1 cut algebra, and the support comparison identifies it with $\Omega_{\mathrm{cut}} C$ as an algebra.*

Proof. For a third term with block length h, the two multiplication signs have exponents $s|u| + h(|u| + |v|)$ and $h|v| + (s + h)|u|$, which agree. Both coefficient maps are the inverse factorization map on the same final open. Its coherences give the higher associativity data.

An added cut in the first word retains its insertion position. An added cut in the second word changes its insertion position by r. The Leibniz sign and changed product sign contribute $|u| + r + (s + 1)|u| = r + s|u|$ modulo two, as required. An internal differential on the first coefficient gives the equal exponents $s|u| + r + s$ and $r + s(|u| + 1)$; the second coefficient is identical after adding $|u|$ to both. Thus the differential is a derivation. Finally, the consolidation exponent for concatenated block lists differs from the two separate exponents by $s|u|$. Equation (8.12) therefore intertwines this product with cobar concatenation, including the unit and coherent faces. □

Call an augmented cut algebra *Z-supported* when its augmentation ideal in arity n is supported on Z_n.

Theorem 8.5. *Bar and cobar restrict to inverse equivalences between Z-supported augmented cut algebras and strong coaugmented cut coalgebras, with empty coefficient k on both sides.*

Proof. Theorem 8.3 gives one direction. Conversely, restrict the bar of $A = K \oplus a$ to a cut open U_p. A factor whose interval crosses p has an open internal cut there, so its Z-supported coefficient vanishes. Each surviving bar word has a separator at p. Deconcatenation identifies it with a word on the prefix and a word on the suffix, with the same coefficient. Every multiplication face crossing that separator has zero target; the other faces and all higher coherences belong to the two factors. The supplied coproduct component is consequently an equivalence. The ambient inverse functors of Theorem 8.2 restrict to these full subcategories and their mapping spaces. □

An equality partition is *cut-indecomposable* if each gap is crossed by one of its blocks. Its exact equality stratum S_π then lies in Z_n; write $i_\pi : S_\pi \hookrightarrow X^n$ for its inclusion. The smaller support consisting only of the small diagonal requires an additional condition.

Corollary 8.6. *For a strong cut coalgebra C, its reduced cobar is supported on the small diagonal in every arity if and only if*

$$i_\pi^! C_n = 0$$

for each cut-indecomposable partition π with more than one equality block. In that case the proper diagonal counit gives

$$\Delta_{n,*}(C_1[-1]) \simeq \overline{\Omega_{\mathrm{cut}} C}_n.$$

Proof. On a stratum in Z_n, extraordinary restriction of $R\Gamma_{Z_n} C_n$ is that of C_n; outside Z_n it vanishes. The finite filtration by equality strata gives the stated criterion. Kashiwara equivalence and the monotone collapse to the singleton identify the remaining coefficient and its counit. □

8.3.2 A coefficient beyond binary data

Let X have two distinct geometric points and let J be (8.14). Define

$$P_n = \mathrm{cofib}\big(R\Gamma_{\Delta_n} J_n \longrightarrow R\Gamma_{Z_n} J_n\big)[-1], \qquad P_0 = 0. \tag{8.17}$$

Both supports pull back correctly under monotone collisions, so P is a word system. It is Z-supported. If $v : X^2 \setminus \Delta \hookrightarrow X^2$, then

$$P_1 = P_2 = 0, \qquad P_3 \simeq i_{13,*} v_* J_{X^2 \setminus \Delta}[-1].$$

Set $A^0 = \Omega_{\mathrm{cut}} J$ and give P its A^0-bimodule structure through the augmentation. The square-zero extension $A' = A^0 \ltimes P$ has zero product on $P \star_{\mathrm{cut}} P$.

Proposition 8.7. *The strong cut coalgebras J and $B_{\mathrm{cut}} A'$ have identical empty, unary and binary data and are not equivalent.*

Proof. Theorem 8.5 makes $B_{\mathrm{cut}} A'$ strong. A positive bar word containing P needs at least three labels, so projection to $B_{\mathrm{cut}} A^0 \simeq J$ is an equivalence through arity two. At arity three the only additional word is $P_3[1]$. Hence

$$(B_{\mathrm{cut}} A')_3 \simeq J_3 \oplus P_3[1].$$

On $x_1 = x_3 \neq x_2$ its extraordinary restriction has two copies of the dualizing coefficient, while J_3 has one. Their ranks differ. □

This example uses derived coefficients. On the affine line, put $M = i_{13,*}v_*\omega_{X^2\setminus\Delta}$. Then

$$(B_{\mathrm{cut}}A')_3[-3] \simeq \omega_{X^3} \oplus M[-1], \qquad H^1((B_{\mathrm{cut}}A')_3[-3]) = M \neq 0.$$

It therefore fails the condition that the arity-n coefficient shifted by $[-n]$ be an ordinary differential module. A reconstruction theorem on that smaller class requires its own hypotheses and comparisons.

8.4 A COLLISION THAT DESTROYS CUTS

The preceding support formula is compatible with monotone collisions. To examine another collision, order the blocks of a set partition by their last positions. A surjection $\alpha : [n] \twoheadrightarrow [m]$ with this *root order* satisfies

$$\max \alpha^{-1}(1) < \cdots < \max \alpha^{-1}(m).$$

Its retained quotient gaps and source positions are

$$T_\alpha = \{q \in [m-1] : \alpha^{-1}([q]) = [p] \text{ for some } p\}, \qquad p_\alpha(q) = p. \tag{8.18}$$

For these surjections, too, let $d_\alpha : X^m \to X^n$ be the ordinary diagonal $d_\alpha(y)_i = y_{\alpha(i)}$. Composites retain root order: the last position of a composite fibre belongs to its last intermediate fibre. The map p_α is strictly increasing on its domain.

Proposition 8.8. *A source cut crossed by an α-fibre has empty inverse image. Every other cut is $p_\alpha(q)$ for a unique $q \in T_\alpha$, and its inverse image is $U_q^{(m)}$. For root-ordered collisions $\alpha : [n] \twoheadrightarrow [m]$ and $\beta : [m] \twoheadrightarrow [\ell]$,*

$$T_{\beta\alpha} = \{q \in T_\beta : p_\beta(q) \in T_\alpha\}, \qquad p_{\beta\alpha}(q) = p_\alpha(p_\beta(q)). \tag{8.19}$$

All quotient gaps survive precisely when α is monotone.

Proof. A crossing fibre imposes an equality forbidden by the cut open. Otherwise each fibre lies on one side. The left fibres have all their last positions before those of the right fibres, so root order makes their quotient labels a prefix. This gives the asserted inverse image.

If $\alpha^{-1}(E)$ is a prefix, the maxima of the fibres indexed by E precede all other maxima; hence E is itself a prefix. Apply this observation to $E = \beta^{-1}([q])$. It proves (8.19) in both directions. If every quotient prefix lifts to a prefix, successive differences make all fibres consecutive intervals in their given order. This is monotonicity. □

For $T \subseteq [m-1]$, put

$$Z_T^{(m)} = \bigcap_{q \in T} (X^m \setminus U_q^{(m)}), \qquad Z_\varnothing^{(m)} = X^m.$$

Taking complements in the proposition gives

$$d_\alpha^{-1} Z_n = Z_{T_\alpha}^{(m)}, \qquad Z_m \subseteq Z_{T_\alpha}^{(m)}.$$

Let $A = \Omega_{\mathrm{cut}}J$, and write $\bar{A}$ for its augmentation ideal. The support inclusion defines a map

$$\Phi_\alpha : \bar{A}_m \longrightarrow d_\alpha^! \bar{A}_n, \qquad d_\alpha^! \bar{A}_n \simeq R\Gamma_{Z_{T_\alpha}^{(m)}} J_m[-1]. \tag{8.20}$$

In the finite localization cubes it keeps exactly the subsets of retained gaps. Equation (8.19) and coherent open base change prove

$$\Phi_{\beta\alpha} \simeq d_\beta^! \Phi_\alpha \circ \Phi_\beta.$$

The map preserves the length-one projection to $J_m[-1]$. It respects multiplication at separators retained by the collision: both composites restrict the same inverse factorization map to the common open, with the signs of (8.16). No monoidal structure on arbitrary nonmonotone pullback is assumed.

Theorem 8.9. *Let $O_m = \bigcup_{q=1}^{m-1} U_q^{(m)}$ and $j_m : O_m \hookrightarrow X^m$. The cone of Φ_α is*

$$E_\alpha \simeq j_{m,*} R\Gamma_{Z_{T_\alpha}^{(m)} \cap O_m}(j_m^! J_m)[-1]. \tag{8.21}$$

If X has two distinct geometric points, Φ_α is an equivalence exactly for monotone α.

Proof. Apply the localization triangle for Z_m to $R\Gamma_{Z_{T_\alpha}^{(m)}} J_m$. Its supported term is $R\Gamma_{Z_m} J_m$, giving (8.21).

For a missing quotient gap q, choose distinct x, y over an algebraically closed extension κ/k, and take $t_q = (x, \dots, x, y, \dots, y)$, with q occurrences of x. Only the cut q is open there. The retained-cut cube has just its empty vertex after extraordinary restriction, giving $\kappa[-1]$. The full cube is acyclic, because its q direction is an identity. Consequently

$$t_q^! E_{\alpha,\kappa} \simeq \kappa[-1] \neq 0.$$

For a monotone collision every gap is retained, so the two cubes and their comparison agree. □

For the outer merger $\alpha = (2, 1, 2)$, no quotient gap survives. Equation (8.20) becomes

$$R\Gamma_\Delta J_2[-1] \longrightarrow J_2[-1], \qquad E_\alpha = j_* j^! J_2[-1], \tag{8.22}$$

where j includes the distinct pairs. The two endpoint coefficients already differ on that open. Changing representatives of the map cannot make them equivalent.

Even a surviving cut retains its full localization map. Take $X = \mathbb{A}^1$. For $\alpha = (2, 1, 2, 3)$, the last source cut has defining equation $(z_1 - z_4)(z_2 - z_4)(z_3 - z_4)$. With $x = y_1 - y_3$, $y = y_2 - y_3$, its pullback is xy^2, whereas the quotient cut is defined by xy. For a polynomial numerator g and an integer $a \geq 0$, the inverse presentations are

$$\frac{g}{(xy^2)^a} \longmapsto \frac{x^a g}{(xy)^{2a}}, \qquad \frac{g}{(xy)^a} \longmapsto \frac{y^a g}{(xy^2)^a}. \tag{8.23}$$

They preserve the rational section and commute with differentiation. The next collision $x = y = d$ sends $1/(xy^2)$ to d^{-3} by either route. The numerator in the first map is essential: omitting it would instead give d^{-4}.

8.5 The ordered collision atlas

A word system records monotone identifications. To retain an order within every collision, start with one copy X_σ^I of X^I for each total order σ on a finite nonempty set I. For two orders put

$$V_{\sigma\rho} = X^I \setminus \bigcup_{\substack{\{a,b\} \subset I \\ \sigma,\rho \text{ disagree on } a,b}} \Delta_{ab}. \tag{8.24}$$

Glue the two charts by the identity on this open. On the affine line, this is the ravioli configuration construction of [2, §4.1]. The definition also makes sense for every separated finite-type scheme, since it uses closed equality loci. Here X remains smooth of dimension one.

Proposition 8.10. *The charts glue to a smooth quasi-compact quasi-separated scheme $\mathscr{R}_I$ with a locally isomorphic map $q_I : \mathscr{R}_I \to X^I$. Over a geometric tuple with equality blocks B, its fibre has $\prod_B |B|!$ points.*

Proof. If two orders disagree on a pair, a third order disagrees with at least one of them. Hence $V_{\sigma\rho} \cap V_{\sigma\eta} = V_{\sigma\rho} \cap V_{\rho\eta}$, the gluing condition. There are finitely many noetherian charts and quasi-compact overlaps. Smoothness is local on them. Two charts represent the same point over the tuple exactly when their orders agree within each equality block. Independent total orders on these blocks extend to an order on I, proving the fibre count. □

At two labels the scheme has two diagonal branches and one common distinct-pair open. At a total triple collision it has six branches. The chosen branch records the order inside each equality block.

A surjection $\alpha : I \twoheadrightarrow J$ with an order on each fibre expands a total order σ of J into an order $e_\alpha(\sigma)$ of I. Diagonal substitution on those charts defines

$$\delta_\alpha : \mathscr{R}_J \longrightarrow \mathscr{R}_I, \qquad x_i = y_{\alpha(i)}. \tag{8.25}$$

Indeed, $d_\alpha^{-1} V_{e_\alpha(\sigma), e_\alpha(\rho)} = V_{\sigma\rho}$: there is no disagreement inside a fibre, and disagreements between fibres are exactly the quotient disagreements. Thus the chart maps glue. Successive expansion concatenates the same ordered lists, giving $\delta_\alpha \delta_\beta = \delta_{\beta\alpha}$.

These maps need not be closed immersions. For the ordered fibre $1 < 3$ and singleton fibre 2, the quotient order $2|13$ expands to chart 213. Where the quotient roots differ, this chart agrees with chart 123 along $x_1 = x_3 \neq x_2$. At the triple collision, the image instead lies on branches where 1, 3 are consecutive. Thus its intersection with a fixed chart need not be the entire ordinary diagonal of that chart.

Let OSurj_+ denote finite nonempty sets and surjections with ordered fibres. Define

$$C_X = D(k) \times \varprojlim_{\mathsf{OSurj}_+} \mathrm{DMod}(\mathscr{R}_I), \tag{8.26}$$

with transitions $\delta_\alpha^!$. On a nonseparated atlas, differential modules are obtained by descent from its chart cover. An object of C_X includes all the collision equivalences and their coherences.

For $I = A \sqcup B$, the locus $U_{A,B} \subset \mathscr{R}_I$ excluding every cross equality is canonically the disjoint locus in $\mathscr{R}_A \times \mathscr{R}_B$. Orders on each block remain, and all their shuffles agree because positions in different blocks are distinct. With $j_{A,B}$ its inclusion, put

$$(F \star G)_I = \bigoplus_{I = A \sqcup B} j_{A,B,*}(F_A \boxtimes G_B)|_{U_{A,B}}. \tag{8.27}$$

The sum includes all ordered decompositions, including empty blocks; an empty block contributes its scalar coefficient.

Theorem 8.11. *The product $\star$ is symmetric monoidal on C_X, with unit K supported at the empty configuration. Augmented E_1 algebras and coaugmented counital E_1 coalgebras with empty coefficient k are exchanged by inverse bar and cobar equivalences for this product.*

Proof. A decomposition splitting a collision fibre has empty inverse-image disjoint locus. Every other decomposition comes from a unique quotient decomposition. Open base change proves compatibility with the coefficient systems. All parenthesizations of an iterated product have one summand for each final ordered partition and the same coefficient on its cross-disjoint open. This supplies the associativity coherences. Permuting named blocks gives the symmetry, with the usual Koszul signs; the empty summands give the unit.

Each positive factor consumes a label. The bar and cobar therefore have at most $|I|$ reduced factors at I. The proof of Theorem 8.2 applies with arbitrary ordered partitions in place of consecutive partitions: each gap still has the same localized coefficient on both sides of its identity map. The finite filtrations and coherent comparisons give the asserted inverse equivalences. □

Equation (8.22) now has a geometric consequence. No Cartesian atlas system can have chart coefficients $\overline{\Omega_{\mathrm{cut}}J}_2$ and $\overline{\Omega_{\mathrm{cut}}J}_3$ on the increasing charts in these two arities. On the distinct quotient roots of the outer merger, the latter pulls back to $J_2[-1]$, whereas the former vanishes. The actual atlas map agrees with the ordinary outer diagonal on this open, so a collision equivalence would identify zero with a nonzero object. An atlas cobar has its own chart coefficients; Theorem 8.11 retains that independent construction.

8.6 The dualizing coalgebra and its singular product

On the atlas let $J_I = p^!_{\mathscr{R}_I} k = \omega_{\mathscr{R}_I}[|I|]$ and $J_\emptyset = k$. Its coproduct for an ordered decomposition is localization to its disjoint locus. Successive localizations impose the same inequalities, giving coherent coassociativity. Coaugmentation and counit are the identity at the empty component and zero elsewhere. The two decompositions with an empty end give the counit identities. Define

$$\mathscr{U} = \Omega_\star J.$$

Its bar is equivalent to J by Theorem 8.11. We compute the operation retained by its first coefficients.

Fix a reference order on the labels and local coordinates on X. For $B = \{b_1 < \cdots < b_m\}$ let $c_B = dz_{b_1} \wedge \cdots \wedge dz_{b_m}$. It has degree $-m$ in J_B; set $\xi_B = t c_B$, of degree $1 - m$. If $B = A \sqcup C$, define $\epsilon(A, C)$ by $c_A \wedge c_C = \epsilon(A, C) c_B$. For an ordered partition $P = (B_1|\cdots|B_r)$, let $j_P : U_P \hookrightarrow \mathscr{R}_I$ include the locus where different blocks have disjoint positions. The cobar coefficient at $n = |I|$ is

$$\mathscr{U}_I = \bigoplus_{r=1}^{n} \bigoplus_{I = B_1 \sqcup \cdots \sqcup B_r} j_{P,*}\omega_{U_P}[n - r], \qquad P = (B_1|\cdots|B_r), \quad B_i \neq \varnothing. \tag{8.28}$$

Its multiplication concatenates block lists on the corresponding disjoint open. Splitting block $B_i = A \sqcup C$ has coefficient

$$(-1)^{\sum_{j<i}(1-|B_j|)+|A|+1}\epsilon(A, C). \tag{8.29}$$

This includes both the suspension and the density permutation. Thus, writing $a_i = \xi_{\{i\}}$,

$$d\xi_{12} = a_1 a_2 - a_2 a_1, \tag{8.30}$$

$$d\xi_{123} = a_1\xi_{23} - a_2\xi_{13} + a_3\xi_{12} - \xi_{12}a_3 + \xi_{13}a_2 - \xi_{23}a_1. \tag{8.31}$$

Every three-factor word occurs twice with opposite signs after the second differential. In general, splits in different blocks cancel by the odd derivation rule. For three subblocks of one block, the two refinements have equal total density sign and opposite suspension sign. This proves $d^2 = 0$ in every arity.

On the affine-line chart put $R = k[z_1, z_2]$, $u = z_1 - z_2$ and $L = R[u^{-1}]$. Suppress the common density while retaining its signs. Put $\xi = \xi_{12}$, $e_{12} = a_1 a_2$ and $e_{21} = a_2 a_1$. The binary coefficient is

$$R\xi \xrightarrow{d} Le_{12} \oplus Le_{21}, \qquad d\xi = e_{12} - e_{21}, \qquad |\xi| = -1. \tag{8.32}$$

The map is injective. Its degree-zero cohomology has the explicit isomorphism

$$[(f, g)] \longmapsto (f + g, [f]) \in L \oplus L/R, \qquad (h, [p]) \longmapsto [(p, h - p)]. \tag{8.33}$$

Changing a lift of $[p]$ changes the pair by $(r, -r)$ for $r \in R$, which is a boundary. This proves both well-definedness and inversion. For a constant right density ν, $(f\nu)\partial_i = -(\partial_i f)\nu$. All displayed maps commute with this action.

The regular commutator is a boundary, but

$$\beta = \left[\frac{e_{12} - e_{21}}{u}\right] \longleftrightarrow (0, [u^{-1}]) \tag{8.34}$$

is nonzero. A primitive would require $u^{-1}\xi$, which is not an allowed coefficient in degree -1. Normal differentiation generates all higher principal parts. Restoring densities, the second summand in (8.33) is the diagonal direct image $\Delta_{2,*}\omega_X$.

The coefficient interchange $E(f, g) = (g, f)$ acts in these splitting coordinates by

$$E(h, [p]) = (h, [h] - [p]).$$

The half-difference coordinate $[q] = [(f - g)/2]$ instead gives $(h, [q]) \mapsto (h, -[q])$. A geometric label permutation also changes coordinates and density: exchanging z_1, z_2 sends u to $-u$ and $dz_1 \wedge dz_2$ to its negative.

The full bar retains the homotopy and its multiplication together. Its reduced binary complex is

$$R \oplus L \oplus L \longrightarrow L \oplus L, \qquad (r, f, g) \longmapsto (-r + f, r + g), \tag{8.35}$$

in degrees $-2, -1$. It is surjective with kernel $(r, r, -r)$. The corresponding cycle is

$$[\xi] + [a_1|a_2] - [a_2|a_1].$$

Its three differential terms cancel by (8.30). Restoring the density gives $H^{-2}(B_\star \mathscr{U})_2 = \omega_{X^2}$ and zero cohomology in other degrees, hence the chart coefficient is equivalent to $J_2 = \omega_{X^2}[2]$. The suspended class of β is killed by the allowed two-factor bar word $u^{-1}([a_1|a_2] - [a_2|a_1])$. Thus the bar comparison retains the singular multiplication and the differential resolving it.

8.6.1 Derived coefficients on ordinary position space

The map $q_I : \mathscr{R}_I \to X^I$ permits a second coefficient calculation. Choose an auxiliary ordering of its chart set $\Sigma_I = \mathrm{Ord}(I)$. For a nonempty subset S of charts, let V_S remove the diagonals whose pair order varies in S, and let $j_S : V_S \hookrightarrow X^I$. Define the following complex of left differential modules:

$$\check{C}_I^p = \bigoplus_{\sigma_0 < \cdots < \sigma_p} j_{\{\sigma_0,\ldots,\sigma_p\},*}\mathcal{O}_{V_{\{\sigma_0,\ldots,\sigma_p\}}}, \qquad d = \sum_a (-1)^a \operatorname{res}_a . \tag{8.36}$$

Different subsets remain different summands even when their opens coincide.

Write $q_{I,+}$ for the derived direct image of right differential modules along q_I.

Proposition 8.12. *There are equivalences of right differential modules*

$$q_{I,+}(B_\star \mathscr{U})_I \simeq q_{I,+}J_I \simeq (\omega_{X^I} \otimes_{\mathcal{O}_{X^I}} \check{C}_I)[|I|].$$

Proof. The first equivalence is the direct image of $J \simeq B_\star \Omega_\star J$. Since q_I is locally an isomorphism, its transfer module contributes neither a relative de Rham differential nor a dimension shift. Its remaining sheaf direct image is computed by the finite chart cover. Locally on the base, every intersection is localization at a product of diagonal equations and has affine open inclusion. The augmented Čech resolution is exact, contracted locally by inserting a chart containing the point. Thus its direct image is (8.36). The canonical line pulls back from X^I; tensoring by it converts the left connection into a right differential module and gives the stated shift. □

For two labels,

$$\check{C}_2 = [R \oplus R \xrightarrow{(f,g)\mapsto g-f} L], \qquad H^0(\check{C}_2) = R, \quad H^1(\check{C}_2) = L/R.$$

The chart coefficient J_2 and its derived direct image therefore have different cohomological descriptions.

A commutative dg model retains the products of the derived functions. On an affine coordinate open $V \subset X^I$, let $R_S = \Gamma(V \cap V_S, \mathcal{O})$. Polynomial forms on the simplex with vertices S form the differential graded algebra

$$\Omega_{\mathrm{poly}}(\Delta^S) = k[t_\sigma, dt_\sigma : \sigma \in S]/(\textstyle\sum_\sigma t_\sigma - 1, \sum_\sigma dt_\sigma),$$

where t_σ has degree zero and dt_σ degree one. Define $\mathrm{TW}_I(V)$ to consist of families

$$\eta_S \in R_S \otimes_k \Omega_{\mathrm{poly}}(\Delta^S), \qquad S \neq \varnothing,$$

whose face restrictions agree with the localization maps for every inclusion of subsets. Multiplication and differential are componentwise. Integration over oriented faces gives a connection-preserving quasi-isomorphism of complexes $\mathrm{TW}_I(V) \to \Gamma(V, \check{C}_I)$. These local models glue by restriction. One proof of the equivalence filters by simplex skeleta: polynomial forms relative to the boundary of a p-simplex have a single cohomology class in degree p, identified with k by integration; the attaching maps are the alternating restrictions. The filtration is finite. This is the all-face model of [2, Theorem 5].

Integration need not preserve a chosen cochain product strictly. In the two-endpoint path model, a degree-zero path $f(x) = x$ and the one-form $g = u^{-1}dx$ satisfy $\int_0^1 f\,g = 1/(2u)$, whereas $f(0)\int_0^1 g = 0$. The commutative product is retained in TW_I; its comparison with an endpoint cochain product includes higher maps.

8.7 The maps between the two bar–cobar constructions

Restrict an atlas system to its increasing chart and monotone collisions:

$$r : C_X \longrightarrow \mathcal{W}_X.$$

Inclusion and projection of consecutive decompositions give natural maps

$$rF \star_{\mathrm{cut}} rG \xrightarrow{\iota} r(F \star G) \xrightarrow{\pi} rF \star_{\mathrm{cut}} rG, \qquad \pi\iota = 1. \tag{8.37}$$

Every iterated inclusion retains the same consecutive decomposition; every iterated projection selects it. The collision comparisons of the products show that ι is lax monoidal and π is colax monoidal. Thus r transports algebras by ι and coalgebras by π; write the resulting functors as r_{alg} and r_{co}.

These products have different coefficients. For n positive inputs with rank-one singleton coefficients at n distinct positions, the all-subset product has $n!$ summands and the cut product has one. The retraction in (8.37) preserves this difference.

Proposition 8.13. *For an augmented atlas algebra A and a coaugmented atlas coalgebra C, with empty coefficient k, there are natural maps*

$$B_{\mathrm{cut}} r_{\mathrm{alg}} A \longrightarrow r_{\mathrm{co}} B_\star A, \tag{8.38}$$

$$r_{\mathrm{alg}} \Omega_\star C \longrightarrow \Omega_{\mathrm{cut}} r_{\mathrm{co}} C. \tag{8.39}$$

The first is a coalgebra map including consecutive words; the second is an algebra map projecting onto them. On strict normalized models, the second is a degreewise surjection with a dg two-sided ideal as kernel.

Proof. Adjacent merges of consecutive blocks remain consecutive. Inclusion therefore commutes with bar faces, internal differentials and higher faces. Deconcatenation followed by the cut projection retains exactly the cuts of that consecutive word, proving the coalgebra assertion.

For cobar, a nonconsecutive partition cannot refine to a consecutive one: merging two adjacent intervals in such a refinement would make the original partition consecutive. Hence discarded summands have only discarded refinements. On retained summands the retained refinements are exactly the cut coproduct, with the same signs and localizations. Projection is therefore a chain map. Concatenation of a prefix partition and a suffix partition is consecutive exactly when both are, proving multiplicativity and the ideal assertion. Empty terms are unchanged.

A monotone collision kills a partition splitting one of its fibres. A surviving partition descends to a quotient partition and is consecutive precisely when that quotient is. Thus both maps respect collisions. These constructions commute with the normalized finite diagrams, including their coherent faces, so they also give the stated enhanced maps. The strict-model description does not assert strictification on unchanged coefficient objects. □

Projection in the reverse direction for the bar generally fails to commute with the differential. A discarded reversed word $[a_2|a_1]$ can have differential $[a_2a_1]$, which lies in the retained one-block term. In a free atlas algebra that product is nonzero. Thus the bar map in (8.38) has an essential direction.

For the dualizing coalgebra, the binary cobar map is especially simple:

$$\begin{array}{ccc} R & \xrightarrow{(1,-1)} & L \oplus L \\ {\scriptstyle 1}\downarrow & & \downarrow{\scriptstyle (f,g)\mapsto f} \\ R & \xrightarrow{1} & L. \end{array} \tag{8.40}$$

The rows have degrees $-1, 0$; density bases fix the displayed signs. Its kernel is L in degree zero. On cohomology it gives

$$0 \longrightarrow L \xrightarrow{h\mapsto[(0,h)]} (L \oplus L)/\{(a,-a) : a \in R\} \xrightarrow{[(f,g)]\mapsto[f]} L/R \longrightarrow 0.$$

Indeed, if $f \in R$, subtracting $(f,-f)$ leaves the unique representative $(0, f+g)$ of its kernel class. After inverting u, the full cobar has one nonzero degree-zero coefficient and the cut cobar is acyclic. This comparison is not an equivalence.

8.7.1 The ternary kernel

For the affine line put $R_n = k[z_1, \dots, z_n]$, $z_{ij} = z_i - z_j$, and

$$L_n = R_n\left[\left(\prod_{i<j} z_{ij}\right)^{-1}\right], \qquad M_P = R_n[z_{ij}^{-1} : i, j \text{ lie in different blocks of } P]. \tag{8.41}$$

All modules are tensored with the common right density, suppressed in the coordinate matrices. Equal rings with different partition labels remain separate summands. Let F_n be the degreewise kernel of (8.39) on this normalized model for J, and put $e_P = \xi_{B_1} \cdots \xi_{B_r}$. Then

$$F_n^{r-n} = \bigoplus_{\substack{P \text{ has } r \text{ blocks} \\ P \text{ not consecutive}}} M_P e_P. \tag{8.42}$$

The differential has the signs of (8.29).

At three positions, order the degree-minus-one terms by 13|2, 3|12, 23|1, 2|13, and the degree-zero terms by 132, 312, 321, 231, 213. The matrix is

$$D = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix} : \bigoplus_P M_P \longrightarrow L_3^5. \tag{8.43}$$

An entry ± 1 denotes a signed localization inclusion. Replace the target coordinates $(a_0, \dots, a_4)$ by

$$\left(\sum_i a_i,\ a_0,\ a_0 + a_1,\ a_0 + a_1 + a_2,\ a_0 + a_1 + a_2 + a_3 \right).$$

Successive differences invert this integral basis change. It turns D into a zero row followed by diagonal entries $1, 1, -1, -1$. Consequently

$$H^{-1}(F_3) = 0, \qquad H^0(F_3) \simeq \omega_{\mathbb{A}^3} \otimes_{R_3} \left(L_3 \oplus \bigoplus_P L_3/M_P \right). \tag{8.44}$$

In particular, the class of z_{13}^{-1} in $L_3/M_{13|2}$ is nonzero: at the generic point of Δ_{13}, the factors z_{12}, z_{23} are units and every element of $M_{13|2}$ is regular, whereas z_{13}^{-1} has a pole. Fully localizing every coefficient before taking cohomology would lose this class.

8.8 Outer principal parts in every arity

The discarded complex has negative cohomology beginning at four positions. Its first class has an explicit representative in every arity. In the fully localized complex, let T_n be the part of $d\xi_{[n]}$ keeping the first and last labels in the same block:

$$T_n = \sum_{\substack{A \sqcup C = [n],\ A, C \neq \varnothing \\ 1, n \text{ in the same part}}} (-1)^{|A|+1} \epsilon(A, C) \xi_A \xi_C, \qquad b_n = dT_n. \tag{8.45}$$

All its terms are discarded. Its degree is $2 - n$, and b_n has degree $3 - n$. Write $u = z_{1n}$.

Theorem 8.14. *For $n \geq 3$ there is an injection of right differential modules*

$$\omega_{\mathbb{A}^n} \otimes_{R_n} (R_n[u^{-1}]/R_n) \hookrightarrow H^{3-n}(F_n), \qquad [f] \longmapsto [f b_n]. \tag{8.46}$$

Thus these negative cohomology groups have no lower bound independent of arity.

Proof. The permutahedron is the convex hull of the permutations of $(1, \dots, n)$ in $\mathbb{R}^n$. After extending coefficients to L_n, ordered partitions form its cellular chain complex with degrees negated. The face of a partition with r blocks has dimension $n - r$; the density orientations give exactly (8.29). Convexity makes this cellular complex acyclic below its degree-zero class. The kernel on its facets is the rank-one L_n-module spanned by the boundary of the unique top cell. That boundary has coefficient ± 1 on every two-block partition. The discarded facets omit all $n - 1$ consecutive two-block partitions, so the differential on their span is injective. The same holds on their actual localization submodules.

If a three-block partition still keeps $1, n$ together, both its facet coarsenings occur in T_n and cancel in dT_n. Every nonzero coordinate of b_n therefore separates $1, n$ and inverts u. Thus $f b_n$ is an allowed cochain for every $f \in R_n[u^{-1}]$, and it is closed. For $f \in R_n$ its primitive $f T_n$ is allowed.

Conversely, suppose $f b_n = dv$ for an allowed discarded two-block cochain v. Injectivity of the fully localized facet differential forces $v = f T_n$. A facet of T_n does not invert u. Unique factorization gives $M_P \cap R_n[u^{-1}] = R_n$, so its coordinate $\pm f$ can belong to M_P only when $f \in R_n$. This proves injectivity. Such a facet exists for $n \geq 3$, and the same injectivity proves $b_n \neq 0$. Its constant coefficients make differentiation of $f b_n$ equal to differentiation of f followed by the map. Restoring the density proves differential-module linearity. □

For four positions,

$$T_4 = -\xi_{124}\xi_3 + \xi_{134}\xi_2 - \xi_{14}\xi_{23} - \xi_2\xi_{134} - \xi_{23}\xi_{14} + \xi_3\xi_{124}. \tag{8.47}$$

The cycle $z_{14}^{-1} dT_4$ is allowed, but its fully localized primitive $z_{14}^{-1} T_4$ is not. In particular, $F_4 \to H^0(F_4)[0]$ is not a quasi-isomorphism.

This class also fixes the sign of the connecting map from the cut cobar. Let

$$U_n = d_{\mathrm{cut}}\xi_{[n]} = \sum_{p=1}^{n-1} (-1)^{p+1} \xi_{[1,p]}\xi_{[p+1,n]}.$$

Every term inverts u, so $f U_n$ is a cut cocycle for a principal part f. Split the full top boundary as $U_n + T_n + V_n$, where V_n consists of discarded facets separating the outer labels. The exact-sequence connecting map $H^{2-n}((\Omega_{\mathrm{cut}} J)_n) \to H^{3-n}(F_n)$ then satisfies

$$[f U_n] \longmapsto -[f b_n]. \tag{8.48}$$

Indeed, differentiating its full-complex lift gives $d(f U_n) = -f b_n - d(f V_n)$, and $f V_n$ is an allowed discarded cochain. The obstruction belongs to this comparison of cobars; it leaves both ambient bar–cobar equivalences intact.

8.8.1 *The complete negative group at four positions*

The four-point complex has eleven discarded facets, thirty-three discarded edges and twenty-three vertices:

$$\bigoplus_{11} M_P \xrightarrow{A} \bigoplus_{33} M_Q \xrightarrow{B} L_4^{23}, \tag{8.49}$$

in degrees $-2, -1, 0$. After full localization, the ranks of A, B are 11, 22. The first rank follows from the facet argument above. For the second, the permutahedron graph with vertex 1234 removed is connected: increasing inversion number connects every remaining permutation to 4321 without visiting 1234.

Put $R = R_4$, $L = L_4$, and for a pair $h = \{i, j\}$ set

$$L_h = R[z_{ab}^{-1} : \{a, b\} \neq h], \qquad W_h = L/L_h.$$

Let I_h index discarded facets keeping h together, and let J_h index three-block partitions keeping h together. The submatrix $K_h = A_{J_h, I_h}$ has the following dimensions and ranks:

h	12	13	14	23	24	34
$\lvert I_h \rvert$	4	5	6	4	5	4
$\lvert J_h \rvert$	5	6	6	5	6	5
$\operatorname{rank} K_h$	4	5	5	4	5	4.

These are constant matrices defined by (8.29). For $h = 14$, ordering the columns as in (8.47) gives

$$K_{14} = \begin{pmatrix} 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & -1 \end{pmatrix}.$$

Its kernel is generated by $(-1, 1, -1, -1, -1, 1)$, the coefficients of T_4. For the other pairs, lexicographic ordering of block lists gives the following row equations:

$$\begin{array}{c|l} 12 & -v_1,\ -v_2,\ v_2 + v_3,\ v_1 - v_4,\ -v_3 + v_4 \\ 13 & -v_1,\ v_1 - v_2,\ -v_3,\ v_3 + v_4,\ v_2 + v_5,\ -v_4 - v_5 \\ 23 & v_1,\ -v_2,\ v_2 - v_3,\ -v_1 + v_4,\ v_3 - v_4 \\ 24 & v_1,\ v_2,\ -v_1 - v_4,\ v_3 + v_4,\ -v_2 + v_5,\ -v_3 - v_5 \\ 34 & v_1,\ v_2,\ -v_2 + v_3,\ -v_1 - v_4,\ -v_3 + v_4. \end{array}$$

Setting each row list to zero successively kills every coordinate. These eliminations use only ± 1, so the same kernel descriptions hold with coefficients in W_h.

Theorem 8.15. *Globally on the standard four-position chart,*

$$H^{-2}(F_4) = 0, \qquad H^{-1}(F_4) \simeq \omega_{\mathbb{A}^4} \otimes_R (R[z_{14}^{-1}]/R). \tag{8.50}$$

The second map is $[f] \mapsto [f\, dT_4]$.

Proof. The facet differential is injective, proving the first assertion. For a degree-minus-one cocycle w, full localization gives a unique $v \in L^{11}$ with $Av = w$. A three-block term keeping h together has coefficient L_h. Thus w belongs to its actual coefficient modules precisely when

$$K_h(v_i \bmod L_h)_{i \in I_h} = 0 \quad \text{for every } h.$$

For $h \neq 14$, the displayed eliminations force the corresponding coordinates of v to be regular along h.

Unique factorization gives $M_P = \bigcap_{h \text{ internal to } P} L_h$ inside L. Writing $u = z_{14}$, it follows that $v_P \in M_P[u^{-1}]$. The coordinates of facets separating 1, 4 already lie in M_P and can be subtracted as legal boundaries. Modulo those boundaries, the other six principal parts are proportional to the coefficients of T_4. The remaining module is therefore the intersection, in W_{14}, of the six images $M_P[u^{-1}]/M_P$ for $P \in I_{14}$.

Use coordinates $u = z_{14}$, $a = z_{12}$, $b = z_{23}$, $y = z_4$, and write $c = a + b = z_{13}$, $D = k[y, a, b]$. Finite negative Laurent parts at $u = 0$ identify

$$W_{14} \simeq \bigoplus_{j \geq 1} D[(abc)^{-1}]u^{-j}.$$

The factors $u - a, u - c$ need only finitely many Taylor coefficients for any fixed pole order. Vanishing of the negative part means divisibility by the required power of u, hence membership in L_{14}. For 124|3, 134|2 and 14|23, the images are respectively

$$\bigoplus_{j \geq 1} D[(bc)^{-1}]u^{-j}, \qquad \bigoplus_{j \geq 1} D[(ab)^{-1}]u^{-j}, \qquad \bigoplus_{j \geq 1} D[(ac)^{-1}]u^{-j}.$$

Reversed facets repeat these three modules. Since $a, b, c = a + b$ are pairwise nonassociated prime factors in D, the intersection of their coefficient rings is D. The desired intersection is therefore $\bigoplus_{j \geq 1} Du^{-j} = R[u^{-1}]/R$. This identifies the class with $[f\, dT_4]$, and the density argument in Theorem 8.14 proves differential-module linearity. □

The degree-zero cohomology also has a global finite presentation. Let G be the permutahedron graph on the orders of $[4]$ other than 1234, fix $\sigma_0 = 4321$, and label each edge by its swapped pair h. Set

$$\mathcal{G}_R = \frac{\bigoplus_{\sigma \in V(G)\setminus\{\sigma_0\}} L e_\sigma}{\sum_{\{\sigma,\tau\}\in E(G)} L_{h(\sigma,\tau)}(e_\sigma - e_\tau)}, \qquad e_{\sigma_0} = 0. \tag{8.51}$$

Then

$$H^0(F_4) \simeq \omega_{\mathbb{A}^4} \otimes_R (L \oplus \mathcal{G}_R). \tag{8.52}$$

To prove this, sum all vertex coordinates. Every incidence column has sum zero, and the σ_0 coordinate splits this sum. In its kernel use $e_\sigma - e_{\sigma_0}$ as basis. The edge relations are precisely (8.51), with twenty-two generator modules and thirty-three relation modules. Constant basis changes preserve the differential-module action. This presentation retains the extensions across intersections of the six diagonals; a decomposition into indecomposable modules is additional information.

8.8.2 The crossed collision

On the open set where the blocks $\{1, 3\}$ and $\{2, 4\}$ are separated, write

$$u = z_{13}, \quad v = z_{24}, \quad A = R[(z_{12}z_{14}z_{23}z_{34})^{-1}],$$
$$A_u = A[u^{-1}], \quad A_v = A[v^{-1}], \quad L = A[(uv)^{-1}].$$

Here u, v are independent normal coordinates. Suppressing the common density, the coefficient types of F_4 are

$$A^2 \oplus A_u^3 \oplus A_v^3 \oplus L^3 \longrightarrow A_u^6 \oplus A_v^6 \oplus L^{21} \longrightarrow L^{23}. \tag{8.53}$$

A scalar pivot between two *identical* coefficient modules can be cancelled by a chain homotopy. For a nonzero scalar pivot $a = D_{ij}$ in row i, column j of a differential matrix D, the remaining matrix is the Schur complement

$$D'_{rc} = D_{rc} - D_{rj}a^{-1}D_{ic}. \tag{8.54}$$

Triangular row and column operations isolate the two-term isomorphism, whose contracting homotopy is a^{-1}. All other entries remain composites of allowed localization maps. The equality of the pivot modules is necessary: a localization inclusion is not an isomorphism merely because its displayed scalar is 1.

Order partitions lexicographically by their increasing block lists. Choose the first nonzero pivot between identical modules, first by lower degree, then row and column; retain the order after cancellation. The nine pivots from degree -2 have types

$$A_u, L, A_v, A_v, L, A_u, A_u, L, A_v$$

and scalars 1, 1, -1, -1, 1, 1, 1, 1, 1. The next eighteen pivots have type L and scalar 1. Formula (8.54) leaves

$$A^2 \xrightarrow{M_0} A_u^3 \oplus A_v^3 \xrightarrow{M_1} L^5,$$

with retained middle order $(A_u, A_v, A_v, A_u, A_u, A_v)$ and matrices

$$M_0 = \begin{pmatrix} 1 & 0 \\ -1 & 0 \\ 0 & 1 \\ 0 & -1 \\ -1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad M_1 = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & -1 \end{pmatrix}. \tag{8.55}$$

For completeness, the required final basis changes are

$$S = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 \end{pmatrix}, \qquad W = \begin{pmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 1 & 0 & -1 & 0 \\ -1 & 1 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 \end{pmatrix}.$$

Their determinants are units. The first three columns of S have type A_u and the last three type A_v, so it respects the coefficient modules. Direct multiplication gives

$$S^{-1}M_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ -1 & 0 \\ 0 & -1 \\ 0 & 0 \end{pmatrix}, \qquad W^{-1}M_1S = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \tag{8.56}$$

Theorem 8.16. *On this open set, F_4 is chain-homotopy equivalent by differential-module maps to*

$$L[0] \oplus [A_u \to L] \oplus [A_v \to L] \oplus [A \to A_u \oplus A_v \to L]^{\oplus 2}. \tag{8.57}$$

The two-term complexes have degrees $-1, 0$ and the three-term complexes degrees $-2, -1, 0$. Hence

$$H^{<0}(F_4) = 0, \qquad H^0(F_4) \simeq \omega \otimes_A \left(L \oplus L/A_u \oplus L/A_v \oplus (L/(A_u + A_v))^{\oplus 2}\right).$$

Proof. The two paired columns in (8.56) give complexes with maps $a \mapsto (a, -a)$ and $(f, g) \mapsto f + g$. The third and sixth columns give the two single localization maps; the last target row gives $L[0]$. The preceding cancellations and constant basis changes prove (8.57).

The localization maps are injective. In either three-term complex, the middle kernel consists of $(f, -f)$ with $f \in A_u \cap A_v$. Unique factorization and independence of u, v give $A_u \cap A_v = A$. Only the displayed cokernels therefore remain. All transformations are scalar matrices or localization maps, so they commute with the right differential operators. □

Both codimension-two summands are nonzero. Near $u = v = 0$, the double residue of $1/(uv)$ is 1, while it vanishes on sections having poles in only one of these variables. The cross-block denominators are units there. Thus $1/(uv)$ survives modulo $A_u + A_v$. On this open z_{14} is invertible, so the negative outer module of (8.50) vanishes, consistently with the local calculation.

8.9 Exchange and the two factorization conditions

To recover an atlas system from a word system, one must specify how its charts agree on their full overlaps. For a word system F, let F_I^σ be its transport to the chart with order σ. An *exchange datum* consists of equivalences

$$g_{\sigma\rho} : F_I^\sigma|_{V_{\sigma\rho}} \xrightarrow{\sim} F_I^\rho|_{V_{\sigma\rho}},$$

with coherent Čech cocycles and relabeling compatibility. It also requires collision compatibility: expanding an ordered-fibre surjection into charts must intertwine these maps with the monotone collision equivalences, coherently under every composite. The overlaps in (8.24) may contain partial collisions.

Proposition 8.17. *Restriction identifies C_X with word systems equipped with this exchange datum. The corresponding tensor is the all-subset product glued chartwise. Componentwise strong atlas coalgebras correspond to strong cut coalgebras with factorization-compatible exchange.*

Proof. Differential modules satisfy descent for the finite open chart cover. The g maps and their cocycles therefore glue F_I^σ to a coefficient on $\mathscr{R}_I$, and descent of morphisms gives the same statement on mapping spaces. Collision compatibility glues the chartwise !-pullback equivalences and their higher coherences. Restriction and gluing are inverse.

For an arbitrary ordered decomposition, choose a chart in which its blocks are consecutive. Its coefficient product is then the cut product on that chart. The exchange compatibilities glue these products to (8.27). The same choice identifies each supplied factorization component with a cut component, proving the last assertion. □

Write $\mathrm{Conf}_2(X) = X^2 \setminus \Delta$ for distinct pairs. Such data need not exist for an underlying word system. Suppose an augmented cut algebra has augmentation ideal $a_n = \Delta_{n,*}M$, and $(M \boxtimes M)|_{\mathrm{Conf}_2(X)} \neq 0$. On distinct pairs its cut bar has coefficient $(M \boxtimes M)[2]$. On the stratum $x_1 = x_3 \neq x_2$ its ternary coefficient has zero extraordinary restriction: a one-block word requires the small diagonal, a two-block word requires 12 or 23 to coincide, and a three-block word requires all positions to be distinct. An outer-merger collision equivalence in the atlas would identify this zero coefficient with the nonzero binary one. Thus the cut bar has no extension with unchanged chart coefficients.

8.9.1 Joint factorization

For the atlas let D_I be the union of the small-diagonal branches $\delta_\sigma : X \hookrightarrow \mathscr{R}_I$. A coefficient $N \in \mathrm{DMod}(X)$ has the diagonal insertion

$$(i_{\mathrm{ord}}N)_I = \bigoplus_\sigma \delta_{\sigma,*}N, \qquad (i_{\mathrm{ord}}N)_0 = 0.$$

An ordered-fibre collision pulls back exactly the branches whose orders are expanded quotient orders. Proper base change gives the collision structure; Kashiwara equivalence identifies every positive system supported on D_I with i_{ord} of its singleton.

For an atlas coalgebra C put $N = C_1$ and let U_I be its fully distinct locus. Its iterated coproducts define a joint map

$$\Theta_I : C_I|_{U_I} \longrightarrow \bigoplus_{\sigma \in \mathrm{Ord}(I)} (N_{\sigma(1)} \boxtimes \cdots \boxtimes N_{\sigma(n)})|_{U_I}. \tag{8.58}$$

Call C *jointly factorizing* if these supplied maps are equivalences in every arity.

Theorem 8.18. *Atlas bar and cobar restrict to inverse equivalences between augmented algebras with positive coefficients supported on D_I and jointly factorizing coalgebras. For such a coalgebra,*

$$i_{\mathrm{ord}}(C_1[-1]) \simeq \overline{\Omega_\star C}.$$

Proof. On distinct positions, a diagonal-supported bar factor occupying more than one label vanishes. Its surviving terms are the singleton words, one per total order, and every multiplication face has zero target. Their deconcatenation is exactly (8.58).

Conversely, projection to singletons followed by all iterated coproducts maps the separated coefficient system of C to the tensor coalgebra on its singleton. Its component is Θ_I, so joint factorization makes it an equivalence. The gap contraction of Theorem 8.2 contracts its cobar at more than one distinct position, leaving $C_1[-1]$ at one position. Every equality stratum with at least two roots is the distinct locus of an ordered-fibre collision. Collision compatibility therefore kills its extraordinary restriction. Finite-stratum recollement puts the reduced cobar on D_I, and the proper branch counits give the displayed equivalence. The ambient bar–cobar maps restrict to these full subcategories and their mapping spaces. □

Now assume that *each* binary coproduct component is an equivalence on distinct pairs. Using one of them identifies the joint map with

$$(1, R) : V \longrightarrow V \oplus V, \qquad V = (N \boxtimes N)|_{\mathrm{Conf}_2(X)},$$

where R is an equivalence. Its cofiber is V, even for an arbitrary chain endomorphism R. Explicitly, cone coordinates $(x, y, z) \in V^q \oplus V^q \oplus V^{q+1}$ have differential

$$D(x, y, z) = (dx + z, dy + Rz, -dz).$$

The maps $q(x, y, z) = y - Rx$, $s(v) = (0, v, 0)$ and $H(x, y, z) = (0, 0, x)$ satisfy $qs = 1$ and $DH + HD = 1 - sq$. Thus the ordinary graph kernel is zero but its derived fibre is $V[-1]$, and the separated binary cobar is $V[-2]$.

Corollary 8.19. *A componentwise strong atlas coalgebra C is jointly factorizing if and only if its singleton satisfies*

$$(C_1 \boxtimes C_1)|_{\mathrm{Conf}_2(X)} = 0.$$

In this case its positive coefficients are diagonal-supported and its reduced coproduct vanishes.

Proof. The graph cofiber proves necessity. Conversely, the assumed vanishing kills every separated singleton tensor of length at least two. Successive strong components identify the coalgebra with these tensors on distinct loci. Ordered-fibre collisions and finite-stratum recollement put its positive coefficients on the small-diagonal branches. Their two-factor convolution is zero by the same separated square condition, so the reduced coproduct vanishes. For $|I| > 1$ both sides of (8.58) vanish; for $|I| = 1$ the map is the identity. □

For an atlas algebra with diagonal-supported augmentation ideal, one can also see the component coproduct directly in its bar. On a disjoint decomposition $I = P \sqcup Q$, every surviving block has one of these two colors. A multiplication face between colors has zero target, so words split into subcomplexes indexed by their maximal consecutive runs of colors. The coproduct for $P|Q$ projects onto the sector with all P blocks before all Q blocks. That projection has a section as a coefficient complex. Reverse and alternating sectors remain in its kernel.

Finally, derived primitives have a precise associative description. For a positive object V, let $\mathrm{Prim}_0(V) = K \oplus V$ have zero reduced coproduct and trivial higher reduced cooperations. In either carrier, bar–cobar duality and the free-algebra property give

$$\begin{aligned} \mathrm{Map}_{\mathrm{Coalg}_{E_1}}(\mathrm{Prim}_0(V), C) &\simeq \mathrm{Map}_{\mathrm{Alg}_{E_1}}(T(V[-1]), \Omega C) \\ &\simeq \mathrm{Map}(V, \overline{\Omega C}[1]). \end{aligned}$$

The tensor algebra here uses the chosen geometric product, with finite words at each configuration. The right adjoint thus retains derived fibres; an ordinary kernel of the reduced coproduct would lose them.

8.10 Operations on the small diagonals

The support of a coefficient determines which associativity homotopies it can carry. In this section let X be a smooth separated scheme of pure dimension $d \geq 1$. The word category and cut product have the same definitions. For $M \in \mathrm{DMod}(X)$, let

$$(iM)_n = \Delta_{n,*}M \quad (n > 0), \qquad (iM)_0 = 0,$$

where Δ_n is the small diagonal. Kashiwara equivalence and monotone collapse identify these coefficients coherently. Let $j_n : U_n \hookrightarrow X^n$ include the pairwise-distinct locus. The derived complex of n-input chiral operations is

$$P_n(M_1, \dots, M_n; P) = R\,\mathrm{Hom}_{\mathcal{D}_{X^n}}\big(j_{n,*}j_n^*(M_1 \boxtimes \cdots \boxtimes M_n), \Delta_{n,*}P\big). \tag{8.59}$$

It retains all degrees of the operation, not just ordinary module maps.

Proposition 8.20. *For every word system G, restriction to the singleton-block summand at n positions gives*

$$R\,\mathrm{Hom}_{\mathcal{W}_X}(iM_1 \star_{\mathrm{cut}} \cdots \star_{\mathrm{cut}} iM_n, G) \simeq R\,\mathrm{Hom}_{\mathcal{D}_{X^n}}\big(j_{n,*}j_n^*(\boxtimes_i M_i), G_n\big).$$

For $G = iP$, this identifies the mapping complex with (8.59)*, compatibly with composition along monotone partitions.*

Proof. Put $Q = j_{n,*}j_n^*(\boxtimes_i M_i)$. At m positions the cut product is the direct sum of $d_{\alpha,*}Q$ over monotone surjections $\alpha : [m] \twoheadrightarrow [n]$. Adjunction and the structure maps of G identify each component mapping complex with $Y = R\,\mathrm{Hom}(Q, G_n)$. Under another monotone collision, a summand survives precisely when its partition descends to the quotient; otherwise its extraordinary restriction meets a forbidden diagonal of Q and vanishes. The mapping diagram is thus the product of copies of Y indexed by surjections to $[n]$.

The category of these surjections has final object $\mathrm{id}_{[n]}$, so its nerve is contractible. The limit of the mapping diagram is Y, detected on that singleton summand. For composition, an inner operation localizes within each consecutive block and an outer operation localizes between the blocks. Open base change identifies the common source with the fully localized input. The resulting composite is the external product of the inner maps followed by the direct image of the outer map, with the Koszul signs for tensoring maps. This is exactly composition of the displayed chiral operations. □

Now let every M_i and P be an ordinary right differential module. The object $j_{n,*}j_n^*(\boxtimes_i M_i)$ has cohomology in degrees 0 through $(n-1)(d-1)$. To prove the upper bound, add the points one at a time. Over the distinct first $n-1$ points, the forbidden graphs for the last point are disjoint smooth loci of codimension d. Locally their complement is covered by d affine opens, giving direct-image amplitude at most $d-1$. Induction gives the stated bound, also after any smooth auxiliary factor.

Set $r = (n-1)(d-1)$. Shifting all inputs and the output by $[d-1]$ changes the operation complex to

$$R\,\mathrm{Hom}(j_{n,*}j_n^*(\boxtimes_i M_i), \Delta_{n,*}P)[-r].$$

It has no negative cohomology, and its degree-zero part is $\mathrm{Hom}(H^r(j_{n,*}j_n^*(\boxtimes_i M_i)), \Delta_{n,*}P)$. The mapping spaces of operations on $i(M[d-1])$ are therefore discrete. An associative structure on it is determined by a binary degree-zero operation whose two ternary composites are equal.

Theorem 8.21. *Every such associative structure on $i(M[d-1])$ has zero ternary composites. More generally, a degree-zero ternary operation on shifted modules that factors through both channels* 12|3 *and* 1|23 *is zero, even when the intermediate modules differ.*

Proof. First let $d = 1$ and use an étale coordinate near the small diagonal. Put $s = z_1 - z_2$, $v = z_2 - z_3$, $h = s + v = z_1 - z_3$. The intermediate module for the first channel has locally nilpotent s and invertible h; for the second it has locally nilpotent v and invertible h. Their first maps are linear over $O[h^{-1}]$, and their last maps to the small-diagonal target are O-linear.

For a fixed input e, choose exponents a, b killing its images by s^a, v^b in the two intermediate modules. For $N = a + b - 1$,

$$e = h^{-N}(s+v)^N e = \sum_{j=0}^{N} \binom{N}{j} s^j v^{N-j} h^{-N} e.$$

If $j \geq a$, its common output vanishes by the first factorization; otherwise $N - j \geq b$ and it vanishes by the second. The inverse of h is used before the final map. This proves the curve case without a uniform nilpotence or pole bound.

For $d \geq 2$, put $F = M_1 \boxtimes M_2 \boxtimes M_3$ and $E = H^{2d-2}(j_{3,*}j_3^* F)$. The degree bound identifies a ternary operation with a module map $E \to \Delta_{3,*}P$. If Z is the union of the three pair diagonals, recollement gives $E = H_Z^{2d-1}(F)$. Each pair diagonal has local-cohomology amplitude at most d, and their common small diagonal has amplitude at most $2d$.

For pair diagonals A, B, let $S_{AB} \subset E$ be the image of $H^{2d-1}_{A\cup B}(F)$. Mayer–Vietoris gives $H^{2d-1}_{A\cup B}(F) \simeq H^{2d}_{\Delta}(F)$ and $H^{2d}_{A\cup B}(F) = 0$. With C the third pair diagonal, the next Mayer–Vietoris sequence is

$$0 \longrightarrow S_{AB} \longrightarrow E \longrightarrow H^{2d}_{\Delta}(F) \longrightarrow 0.$$

The first map is injective because the preceding map from $H^{2d-1}_{\Delta}(F)$ factors through $H^{2d-1}_{A}(F) = 0$. Naturality for $A \subset A \cup B$ identifies the image of S_{AC} in the quotient with the connecting isomorphism for A, C. Hence $S_{AB} + S_{AC} = E$.

An inner operation through $12|3$ is a map $j_* j^*(M_1 \boxtimes M_2) \to \Delta_* N[1-d]$. Its restriction to $M_1 \boxtimes M_2$ is a negative Ext class between modules and is zero. The composition formula of Proposition 8.20 therefore makes the ternary operation vanish on S_{AB} for $A = \Delta_{13}$ and $B = \Delta_{23}$. A factorization through $1|23$ makes it vanish on S_{AC} for $C = \Delta_{12}$. Their sum is E, proving the assertion. Associativity identifies these two factorizations. Every binary tree with at least three leaves contains two adjacent internal vertices, so all longer composites vanish as well. □

The hypothesis is a shifted ordinary module on the small diagonals. The cut-support complexes and atlas cobars above have larger supports and additional cochain degrees. Their multiplication homotopies and outer classes occupy precisely the information that this hypothesis excludes. The finite positive-word bound places no two-step-nilpotence condition on those larger algebras.

Problem 8.22. Construct collision-compatible maps from the oriented Hall correspondence coefficients to the operation complexes of an independently specified field theory. The maps must preserve the coefficient shifts, translations, supported operations and all substitutions, including the nonconsecutive collisions. If a word model is used, supply its full-overlap exchange datum and compare it with the atlas through (8.38) and (8.39), retaining the action of their kernel classes. Specify separately the empty-support unit, the local vacuum and any prepared state.

An order in the collision atlas records branches over coincident positions. A comparison with real-time ordered observables additionally requires the spacetime operation maps and their support conditions. The two geometric dualities, their support theorems and their explicit comparison maps provide coefficient constructions for that problem.

Chapter 9

Coherent comparison, descent and formal limits

Two fields can have the same product and different collision extensions. Let $B = \mathbb{C}1 \oplus \mathbb{C}\epsilon$, with $\epsilon^2 = 0$. The constant fields

$$Y(a,u)b = ab, \qquad T = 0,$$

retain every ordered iterated product in B. They do not specify how a particle in position three passes a particle in position two before positions one and three coincide. Giving this passage either sign produces the same binary coefficient and different ternary coefficients. The distinction survives regular holonomicity, concentration in the ordinary differential-module heart, and all local vacuum insertions.

We first construct these coefficients and identify the exchange datum that makes their reconstruction fully faithful. We then turn to current cochains, where a quadratic differential records multiplication in the state algebra. Rational current fields require a jointly localized tensor product, and their reconstruction must retain a current framing. The final descent argument keeps the nonlinear comparison maps before passing to finite cochains or to a formal limit.

9.1 A collision invisible to the binary fields

Work over $k = \mathbb{C}$ on $X = \mathbb{A}^1$. Coefficients are enhanced derived right algebraic differential modules with quasi-coherent underlying modules. For a finite-type scheme Z, write $J_Z = a_Z^! k$, where $a_Z : Z \to \operatorname{Spec} k$ is its structure map. If Z is smooth of dimension n, then $J_Z = \omega_Z[n]$. We use cohomological shifts, so that $H^i(M[r]) = H^{i+r}(M)$.

Definition 9.1. An ordered coefficient system with local vacua assigns F_I to each finite ordered set I, with $F_\varnothing = k$. A monotone map $f : S \to I$ determines

$$d_f : X^I \longrightarrow X^S, \qquad (x_i) \longmapsto (x_{f(s)}), \qquad \nu_f : d_f^! F_S \longrightarrow F_I.$$

These maps compose coherently and are equivalences when f is surjective. On the open set where every coordinate in I_1 differs from every coordinate in I_2, a consecutive decomposition $I = I_1 \sqcup I_2$ has a specified isomorphism

$$F_{I_1} \boxtimes F_{I_2} \xrightarrow{\sim} F_I.$$

It is compatible with refinements, monotone collisions and the maps ν_f.

For an active subset $S \subset I$, the vacuum-marked coefficient is $p_{I,S}^! F_S$, where $p_{I,S} : X^I \to X^S$ is projection. Its evaluation is $\nu_{S \hookrightarrow I}$.

The system satisfies the *arity heart condition* when $F_I[-|I|]$ is an ordinary right differential module. A morphism is a degree-zero family preserving these operations and the identity of the empty coefficient.

Deleting an inactive position identifies two pullbacks of the same active coefficient. It does not require F_I itself to be pulled back from a smaller configuration. This distinction permits a local vacuum without

imposing a scalar augmentation on its state algebra. The shift in the heart condition is the canonical normalization of [6, §2.1.1].

Let W be finite-dimensional and concentrated in degree zero. Choose an involutive Yang–Baxter operator $c : W \otimes W \to W \otimes W$:

$$c^2 = 1, \qquad c_{12}c_{23}c_{12} = c_{23}c_{12}c_{23}.$$

Here the subscripts specify the two tensor positions on which c acts. Adjacent copies of c define an action of the symmetric group S_r on $W^{\otimes r}$. Put $B = k1 \oplus W$, with $W^2 = 0$, and extend c to $R_c : B^{\otimes 2} \to B^{\otimes 2}$ by

$$R_c(1 \otimes b) = b \otimes 1, \qquad R_c(b \otimes 1) = 1 \otimes b, \qquad R_c|_{W^{\otimes 2}} = c.$$

For $0 \le r \le n$, consider the reduced incidence scheme

$$Z_{n,r} = \left\{ (x,t) \in X^n \times X^r : \prod_{i=1}^{n} (t_a - x_i) = 0 \ (1 \le a \le r), \quad t_a \ne t_b \ (a \ne b) \right\}_{\mathrm{red}}. \tag{9.1}$$

The selected roots $t_1, \dots, t_r$ mark the occupied positions. Unselected positions carry the vacuum. Write $p_{n,r} : Z_{n,r} \to X^n$ for projection, and set $Z_{n,0} = X^n$. The geometric permutation of selected roots and the c-action on their states give an S_r-action. Define

$$F_n(W,c) = \bigoplus_{r=0}^{n} \big((p_{n,r})_* J_{Z_{n,r}} \otimes W^{\otimes r} \big)^{S_r}. \tag{9.2}$$

Theorem 9.2. *The coefficients* (9.2) *form a translation-equivariant ordered system with local vacua. Its consecutive factorization maps are isomorphisms. Every coefficient is regular holonomic and satisfies the arity heart condition. Its singleton is $J_X \otimes B$.*

Proof. At a point of $Z_{n,r}$, let $I_a = \{ i : x_i = t_a \}$. The sets I_a are nonempty and disjoint. In a neighborhood of this point the equations, after removing invertible factors, are

$$\prod_{i \in I_a} (x_i - t_a) = 0 \qquad (1 \le a \le r).$$

Use $x_i - t_a$, the t_a, and the remaining x_i as coordinates. The equations are square-free monomials in disjoint groups of variables. They define a reduced local complete intersection of pure dimension n. Thus $J_{Z_{n,r}}[-n]$ lies in the right-module heart.

Before the root inequalities, projection is finite because each root satisfies a monic equation. Inverting the product of the root differences makes $p_{n,r}$ affine and quasi-finite. Its direct image is exact on that heart. These are the local complete-intersection and direct-image statements of [6, §§2.1.2,2.1.4]. Finite sums and invariants are exact. Regular holonomicity follows from preservation under the corresponding algebraic direct images [83, Theorem 6.1.5].

For a monotone surjection $f : S \twoheadrightarrow I$, reduced base change of the incidence scheme replaces repeated factors in its root polynomial by single factors. It is $Z_{|I|,r}$. Extraordinary base change and pullback of the dualizing object therefore give $d_f^! F_S \simeq F_I$. Differential modules are unchanged by the discarded nilpotent thickenings.

For an active subset $S \subset I$, let $Y_{I,S,r}$ be the closed reduced locus in $Z_{|I|,r}$ where every selected root occurs among the coordinates of S. It is the reduced pullback of $Z_{|S|,r}$. The closed embedding $h : Y_{I,S,r} \hookrightarrow Z_{|I|,r}$ has the proper counit

$$h_* J_{Y_{I,S,r}} = h_* h^! J_{Z_{|I|,r}} \longrightarrow J_{Z_{|I|,r}}.$$

Direct image and invariants give the vacuum evaluation.

Factor a general monotone map through its image to obtain ν_f. Inverse images of active-root loci are the corresponding active-root loci. Nested active subsets give nested closed embeddings. Base change and transitivity of their proper counits prove composition, including the empty active subset. The latter retains only $r = 0$.

On a consecutive cross-disjoint open, each selected root belongs to one block. Split the incidence scheme by that choice. Its pieces are external products of the two incidence schemes, with $r = r_1 + r_2$. An invariant section on a free orbit of root orderings is determined by its value at the ordering in which the first block comes first. This identifies the S_r-invariants with the appropriate $S_{r_1} \times S_{r_2}$-invariants, with coefficient one. It gives the factorization isomorphism.

For several blocks, every parenthesization uses the same concatenated root ordering. Alternative reorderings agree by the symmetric-group relations. Splitting roots commutes with selecting active roots, so the same proper base change proves compatibility with vacua and collisions. These canonical enhanced-functor identities supply the coherences for chains of maps. Simultaneous translation of all roots and positions preserves every scheme and map used above. The empty and singleton assertions follow directly from the formula. □

The normal extension makes the dependence on c explicit. For a surjection $q : V \to Q$ of finite-dimensional vector spaces, put

$$\mathcal{P} = k[u, u^{-1}]/k[u], \qquad M(q) = \ker\bigl(V[u, u^{-1}] \xrightarrow{q\otimes \mathrm{pp}} Q \otimes \mathcal{P}\bigr).$$

Here pp takes the principal part. In the left-module normal coordinate,

$$M(q) = V[u] + (\ker q) \otimes u^{-1}k[u^{-1}], \qquad \frac{v}{u} \in M(q) \iff q(v) = 0. \tag{9.3}$$

Multiplication by u and differentiation preserve this module. The right-module version is obtained by tensoring with the density. Its supported quotient retains all normal derivatives, normalized by $\mathrm{Res}(du/u) = 1$ and $u^{-r-1} = (-1)^r \partial_u^r(u^{-1})/r!$. Completion replaces $k[u]$ by $k[[u]]$ and keeps finite Laurent poles.

Lemma 9.3. *The binary normal module of $F_2(W, c)[-2]$ is $M(m_B)$. Near $x_1 = x_3 \neq x_2$, with the output blocks ordered as $(13), 2$, the ternary normal module is $M(q_c)$, where*

$$q_c = (m_B \otimes 1)(1 \otimes R_c) : B^{\otimes 3} \longrightarrow B^{\otimes 2}. \tag{9.4}$$

These descriptions include the collision quotient and the local vacua.

Proof. The union of the two sections $t = x_1$ and $t = x_2$ has, on the normal line $u = x_1 - x_2$, one additional principal-part quotient. Each section has multiplicity one, and the quotient is $(a(u), b(u)) \mapsto \mathrm{pp}(a(u) + b(u))$. Its extra allowed pole is $(1, -1)/u$. One can also obtain this description by dualizing the exact sequence of constant sheaves on the union. The de Rham comparison preserves that sequence and its maps [83, Theorem 7.1.1].

For two positions, the zero-particle summand is regular. The one-particle summand is the preceding union of sections tensored with W. The two-particle summand is fully localized, since two selected roots cannot coincide. The quotient is consequently

$$1 \otimes 1 \mapsto 1, \quad w \otimes 1 \mapsto w, \quad 1 \otimes w \mapsto w, \quad w \otimes w' \mapsto 0.$$

This is m_B, independent of c. A vacuum includes one of the two sections and is independent of c as well.

At the outer collision, the middle root remains separate. The two branches of a particle approaching the outer root have opposite orders relative to a particle at the middle root. Expressing both in the closed ordering $(13), 2$ applies R_c to positions two and three. Multiplication then joins the outer positions. This is q_c.

Terms with two particles at the outer positions remain fully localized, and the other summands have the same two-section description. The vacuum maps are the corresponding proper counits. □

Binary residue testing sends $f(u)a \otimes b$ to $\mathrm{pp}(f(u)ab)$. Tests with negative powers recover the regular coefficient ab as well as every singular coefficient. Thus the full binary fields are $Y(a,u)b = ab$, with $T = 0$, for every c. All planar iterates are products in B.

Proposition 9.4. *For $W = k\epsilon$, the systems with $c = 1$ and $c = -1$ have identical binary coefficients, fields and local vacua. They are not isomorphic. The identity of their binary fields does not lift to a coefficient map.*

Proof. Set $v = \epsilon \otimes \epsilon \otimes 1 - 1 \otimes \epsilon \otimes \epsilon$. Then $q_+(v) = 0$, whereas $q_-(v) = 2\epsilon \otimes \epsilon$. Hence $v/(x_1 - x_3)$ extends in the first system and not in the second. Factorization would make an identity unary lift the identity on the generic ternary tensor, contradicting this extension test.

More generally a unital isomorphism of the state algebras sends ϵ to $\lambda\epsilon$, with $\lambda \neq 0$. Preserving the outer quotient and its vacuum inclusions would require $\lambda^2 = -\lambda^2$. This is impossible in characteristic zero. □

9.2 Reconstruction with an exchange

The particle example suggests retaining the operator used to move a state past a spectator. Multiplication can then be nonzero on two non-vacuum states. No scalar augmentation is needed.

Definition 9.5. An algebra with involutive exchange is a nonzero finite-dimensional unital associative algebra B, in degree zero, together with $R : B^{\otimes 2} \to B^{\otimes 2}$ satisfying

$$R^2 = 1, \qquad R_{12}R_{23}R_{12} = R_{23}R_{12}R_{23},$$

$$R(1 \otimes a) = a \otimes 1, \qquad R(a \otimes 1) = 1 \otimes a, \qquad mR = m, \tag{9.5}$$

$$R(m \otimes 1) = (1 \otimes m)R_{12}R_{23}, \qquad R(1 \otimes m) = (m \otimes 1)R_{23}R_{12}. \tag{9.6}$$

Morphisms are unital algebra maps intertwining R.

These are identities on state tensors. A tensor category of representations, with its own exchange, remains additional structure. For a function $f : I \to J$ of finite ordered sets, sort the inputs into increasing f-fibers using adjacent copies of R. Multiply inside each fiber and insert 1 in empty fibers. Denote this map by

$$Q_f : B^{\otimes I} \longrightarrow B^{\otimes J}.$$

Lemma 9.6. *The maps Q_f satisfy $Q_gQ_f = Q_{gf}$ and $Q_{\mathrm{id}} = 1$. They preserve consecutive external products. If f is surjective, Q_f has the section Q_s for any set-theoretic section s of f.*

Proof. Involution, Yang–Baxter and commutation in disjoint slots give a symmetric-group action. An exchange inside a single fiber disappears after multiplication, by associativity and $mR = m$.

In Q_gQ_f, use (9.6) to move each multiplication through the exchanges above it. Each move replaces a crossing of products by crossings of their separate inputs. The process terminates at individual inputs.

The unit equations remove empty fibers, leaving the permutation that sorts by gf and the products in its fibers. Associativity removes parentheses, and $mR = m$ removes permutations internal to a final fiber. This is Q_{gf}. Juxtaposition gives the external-product identity, and $fs = \mathrm{id}$ gives $Q_fQ_s = 1$. Longer composites have the same map Q, so the same argument supplies their coherences. □

Every unital associative algebra admits an example:

$$R_B(a \otimes b) = ab \otimes 1 + 1 \otimes ab - a \otimes b. \tag{9.7}$$

Indeed, for $p(a \otimes b) = ab \otimes 1$ and $q(a \otimes b) = 1 \otimes ab$, one has $p^2 = p$, $q^2 = q$, $pq = p$ and $qp = q$. Hence $(p + q - 1)^2 = 1$. Both braid words on $a \otimes b \otimes c$ give

$$abc \otimes 1 \otimes 1 - ab \otimes c \otimes 1 + 1 \otimes 1 \otimes abc + 1 \otimes ab \otimes c$$
$$+ a \otimes bc \otimes 1 - 1 \otimes a \otimes bc - a \otimes b \otimes c.$$

The first fusion identity gives $abc \otimes 1 + 1 \otimes abc - ab \otimes c$ on either side. The second gives $abc \otimes 1 + 1 \otimes abc - a \otimes bc$. The remaining identities follow from the unit and associativity.

We now specify the collision extension independently of its fields. Let X be a connected smooth separated complex variety of positive dimension. An equality partition π of $[n]$ specifies the stratum $S_\pi \subset X^n$ on which coordinates agree exactly within its blocks. Order those blocks by first occurrence, let $f_\pi : [n] \to [|\pi|]$ be the quotient, and put $q_\pi = Q_{f_\pi}$. For a coarsening ρ of π, fusion on their block sets gives

$$q_\rho = q_{\pi\rho} q_\pi, \qquad q_{\rho\sigma} q_{\pi\rho} = q_{\pi\sigma}.$$

Write $j_n : \mathrm{Conf}_n(X) \hookrightarrow X^n$ for the fully distinct locus and set

$$E_n = (B^*)^{\otimes n}, \qquad E_\pi = \mathrm{im}(q_\pi^*) \subset E_n.$$

Coarsening makes the subspace smaller.

Define $K_R(B)_n$ as the subsheaf of the ordinary analytic sheaf $(j_n)_* \underline{E}_n$ whose germ on S_π belongs to E_π. Near such a point choose disjoint coordinate neighborhoods of the distinct positions. The fully distinct locus in their product is connected, since complex diagonals have real codimension at least two. Its generic germ is a single vector.

Every vector in E_π extends constantly nearby: the nearby partitions refine π and their subspaces contain E_π. Thus the stalk is exactly E_π, and its generizations are the specified inclusions. This is an algebraically constructible sheaf.

Normalize the regular Riemann–Hilbert equivalence by $\mathrm{DR}(\omega_Y) = k_{Y^{\mathrm{an}}}[\dim Y]$ on a smooth variety. Let $\mathbb{D}$ denote Verdier duality, and define

$$F_R(B)_n = \mathrm{DR}^{-1}(\mathbb{D} K_R(B)_n), \qquad F_R(B)_0 = k. \tag{9.8}$$

Duality and the six operations are compatible with this equivalence [83, Theorems 7.1.1,7.2.2].

Theorem 9.7. *Formula (9.8) gives a fully faithful functor from finite-dimensional algebras with involutive exchange to ordered coefficient systems with local vacua on X. Its coefficients are bounded regular-holonomic complexes, and its consecutive factorization maps are isomorphisms. For $\iota_\pi : S_\pi \hookrightarrow X^n$,*

$$\iota_\pi^! F_R(B)_n \simeq J_{S_\pi} \otimes B^{\otimes |\pi|}.$$

On the affine line it is translation-equivariant, has unary translation zero, and satisfies the arity heart condition.

Proof. The constructible sheaf has finite-dimensional stalks, so its Verdier dual corresponds to a bounded regular-holonomic complex. Duality turns stalk restriction into extraordinary restriction. Since q_π is split surjective, the dual of E_π is canonically $B^{\otimes |\pi|}$, proving the displayed formula.

On a monotone collision diagonal, equality partitions are the inverse images of partitions of the smaller ordered set. Their first-occurrence orders agree. The sheaf stalks and generizations therefore identify the ordinary pullback with the smaller sheaf. Duality gives the collision equivalence.

For an active subset, use the injection of its equality-block set into the full block set. The map Q for this injection moves the active factors into the required order and inserts units elsewhere. Its dual gives $K_R(B)_I \to p_{I,S}^* K_R(B)_S$. After duality this is the local vacuum evaluation. Fusion and insertion commute because both composites are Q of the same function.

On a consecutive cross-disjoint open, every equality block lies on one side of the cut. The quotient and its dual image are external products of the two block constructions. This gives strong factorization.

Refinements, collisions and vacuum insertions again reduce to equalities of Q-maps. Enhanced derived functors carry these sheaf identities to coherent structure maps. A morphism of exchange algebras intertwines all Q_f. Its dual tensor powers and then duality give the claimed covariant functor.

For heart concentration on a curve, filter B by $L_0 B = k1$, $L_r B = B$ for $r \geq 1$, and tensor powers by total filtration. Each Q_f preserves the filtration. For surjective f its filtered section proves strictness. The associated graded is $k1 \oplus W$, $W = B/k1$, with $W^2 = 0$. Its exchange has the form R_c of the particle construction.

The decreasing dual filtration on E_n induces on each E_π the dual output filtration. Strictness identifies its associated graded inclusion with $(\operatorname{gr} q_\pi)^*$.

The graded sheaf is the Verdier dual of (9.2). In particle degree r, its stalk selects r distinct equality blocks and one W^* factor at each. Under generization these blocks select finer labels, reordered by c^*. These are exactly the components of the graded inclusion. The compactly supported direct image from the selected-root cover has finite fibers, so this identification has no higher stalk terms. Evaluation at one ordering gives coefficient one.

The particle theorem puts every dual graded piece in the shifted right heart. There are at most $n + 1$ filtration steps. Closure of the heart under extensions proves concentration for $F_R(B)_n[-n]$.

It remains to identify morphisms. The singleton is constant, and X is connected, so a unary map is a constant linear map $f : B \to B'$. The vacuum makes it unital, and factorization makes its generic arity-n map $f^{\otimes n}$. Binary generization imposes $f m = m'(f \otimes f)$.

At $13 \mid 2$ the quotient is

$$q_R = (m \otimes 1)(1 \otimes R), \qquad q_R(a \otimes b \otimes 1) = a \otimes b, \quad q_R(1 \otimes a \otimes b) = R(a \otimes b).$$

If h is the closed coefficient map, naturality gives $h q_R = q_{R'} f^{\otimes 3}$. The first vacuum insertion forces $h = f \otimes f$, and the second forces $(f \otimes f) R = R'(f \otimes f)$.

Conversely, such an f has the structured lift already constructed. After duality, the difference between two lifts would have zero generic restriction and image in a sheaf embedded in its generic direct image. It is zero.

The sheaves and the domains of their structure maps, formed by ordinary pullbacks and external products, are in degree zero. Their mapping spaces are discrete, and limits imposing compatibility remain discrete. This proves full faithfulness with its coherent comparisons.

Simultaneous translation on the line preserves every equality stratum and subspace. The singleton has constant connection, giving $T = 0$. □

This theorem recovers an exchange from a vacuum-marked outer attachment. It does not assert that the constant binary fields determine that attachment. Matrix states make the difference visible without a square-zero multiplication.

Example 9.8. In $B = \mathrm{Mat}_2(k)$, let

$$X_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Z_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad e_{a,b} = X_0^a Z_0^b,$$

where $(a, b) \in (\mathbb{Z}/2)^2$. Multiplication is $e_{a,b} e_{c,d} = (-1)^{bc} e_{a+c,b+d}$. Besides (9.7), define

$$R_{\mathrm{col}}(e_{a,b} \otimes e_{c,d}) = (-1)^{bc+ad} e_{c,d} \otimes e_{a,b}.$$

The exponent is symmetric and bilinear. It proves involution, both fusion identities and Yang–Baxter. The latter has on either side the sum of the three pairwise exponents. Multiplication after this exchange has sign $(-1)^{bc+ad+da} = (-1)^{bc}$, as required. All states remain in cohomological degree zero.

Both extensions have binary quotient m and fields $Y(a,u)b = ab$. Let $s(a \otimes b) = a \otimes b \otimes 1$ and put $v = 1 \otimes X_0 \otimes Z_0 - sR_B(X_0 \otimes Z_0)$. The outer quotient for R_B kills v, while that for R_{col} gives

$$-Z_0 \otimes X_0 - X_0 Z_0 \otimes 1 - 1 \otimes X_0 Z_0 + X_0 \otimes Z_0.$$

Its four terms are distinct basis tensors. Hence the two normal modules admit different poles $v/(x_1 - x_3)$.

They are not related by a change of unary framing either: $\operatorname{tr} R_B = 4 + 4 - 16 = -8$, whereas $\operatorname{tr} R_{\mathrm{col}} = 4$. An isomorphism would conjugate one exchange to the other by $f \otimes f$.

For $P = E_{12}$ and $Q = E_{21}$, both exchanges retain the ordered outer product $PQP = P$. An ordinary flip would give $P^2 Q = 0$. There is also no unital algebra map $B \to k$: it would kill P, Q by nilpotence and contradict $PQ + QP = 1$. Thus the local vacua of the coefficient system cannot be replaced by a scalar counit on B.

9.3 Multiplication recorded by a current differential

The preceding reconstruction retains exchange as part of the input. A different construction records a commutative multiplication inside a quadratic differential. Its scalar object already explains why the empty field unit and the coefficient of constant states must be kept separate.

Let $\mathcal{A}$ consist of unital commutative dg algebras in nonpositive degrees, with a degree-zero derivation T_A commuting with d_A. Require

$$A = \bigoplus_{q \geq 0} A_q, \qquad A_p A_q \subset A_{p+q}, \quad d_A A_q \subset A_q, \quad T_A A_q \subset A_{q+1},$$

where every A_q is a bounded finite-dimensional complex and $1 \in A_0$. Maps preserve the unit, differential and translation. They need not preserve the auxiliary weight decomposition. The direct product $A \times A'$ has componentwise multiplication and underlying complex $A \oplus A'$. We admit the zero algebra as its monoidal unit.

Fix $\mathfrak{g} = \mathfrak{sl}_2(k)$, with basis e, f, h and $[h,e] = 2e$, $[h,f] = -2f$, $[e,f] = h$. Define

$$L_A = \mathfrak{g} \otimes A, \quad [x \otimes a, y \otimes b] = [x,y] \otimes ab, \quad d_L = 1 \otimes d_A, \quad T_L = 1 \otimes T_A. \tag{9.9}$$

Its Chevalley cochain algebra is

$$\mathcal{B}_A = \widehat{\operatorname{Sym}}(W_A), \qquad W_A = (L_A[1])^\vee. \tag{9.10}$$

On a discrete graded vector space, $\vee$ means its full graded dual with the product topology. On a linearly compact space it means the continuous dual. The completion is the product over weights in each cohomological degree. A generator dual to L_A^i has degree $1 - i > 0$, so only finitely many symmetric degrees occur in each cohomological degree.

We fix signs before using the differential. For a homogeneous basis v_i of L_A, put $a_i = |v_i|$ and write

$$d_L v_i = \sum_k D_i^k v_k, \qquad [v_i, v_j] = \sum_k f_{ij}^k v_k.$$

The dual generator c^i has degree $1 - a_i$. The cochain differential is the continuous derivation

$$dc^k = \sum_i (-1)^{a_i} D_i^k c^i - \frac{1}{2} \sum_{i,j} (-1)^{a_i(1-a_j)} f_{ij}^k c^i c^j. \tag{9.11}$$

Its linear and quadratic parts are denoted d_1, d_2. The dg Lie identities give $d_1^2 = d_1 d_2 + d_2 d_1 = d_2^2 = 0$. For degree-zero currents this says $(d\lambda)(x, y) = -\lambda([x, y])$.

Set $T_{\mathcal{B}} = -T_L^\vee$ on generators and extend it as a derivation. The regular fields are

$$Y_{\mathcal{B}}(b, u)c = (e^{uT_{\mathcal{B}}} b)c. \tag{9.12}$$

The completed recipient is explicit: an output through dual weight $-q$ and Taylor degree $N - 1$ needs inputs only through weight $q + N - 1$. Equation (9.12) denotes the compatible operations on these finite quotients. It does not impose a uniform bound on the input weights. Since T_L is a dg Lie derivation, $T_{\mathcal{B}}$ is a product derivation commuting with d. Associativity, graded commutativity and the exponential identity give the dg vertex-superalgebra identities. For example, weak associativity is

$$(e^{(u+v)T_{\mathcal{B}}} b)(e^{vT_{\mathcal{B}}} c)a = e^{vT_{\mathcal{B}}}\big((e^{uT_{\mathcal{B}}} b)c\big)a.$$

Constant currents give $\rho_A : \mathcal{B}_A \to \mathcal{B}_k$. The latter algebra is

$$\mathcal{B}_k = \Lambda(\xi_e, \xi_f, \xi_h), \qquad d\xi_e = 2\xi_e\xi_h, \quad d\xi_f = -2\xi_f\xi_h, \quad d\xi_h = -\xi_e\xi_f, \qquad T_{\mathcal{B}_k} = 0.$$

The differential identifies degrees one and two and vanishes in degrees two and three. Thus $H^*(\mathcal{B}_k)$ is k in degrees zero and three. It is different from the field unit $k = \mathcal{B}_0$. An augmentation $A \to k$ commuting with translation gives a section of ρ_A.

Definition 9.9. A quadratic current presentation is a complete regular dg field algebra $\mathcal{B} = \widehat{\mathrm{Sym}}(W)$ with a specified split symmetric-degree grading and a map $\rho : \mathcal{B} \to \mathcal{B}_k$. Its generators have the form $W = (L[1])^\vee$, with L nonpositive and with bounded finite nonnegative weight pieces. The differential has only linear and quadratic parts, with the signs (9.11). Translation preserves W and has the stated weight behavior. The fields are (9.12).

The transpose of ρ is a dg Lie map $j : \mathfrak{g} \to L$, with $T_L j = 0$. Require evaluation to be an isomorphism

$$\mathfrak{g} \otimes \mathrm{Hom}_{\mathfrak{g}}(\mathfrak{g}, L) \xrightarrow{\sim} L, \qquad x \cdot v = [j(x), v]. \tag{9.13}$$

The recovered multiplicity space has the weight properties of $\mathcal{A}$. The map ρ preserves weight and symmetric degree. Morphisms are continuous dg field maps over $\mathcal{B}_k$ that preserve the split symmetric-degree grading. Denote this category by $\mathcal{P}_{\mathfrak{g}}$.

The split grading is supplied data. It makes morphisms linear on generators even when a generator has cohomological degree greater than one. No assertion that every dg field algebra has such a canonical grading is needed.

Theorem 9.10. *There is an equivalence*

$$\mathcal{A}^{\mathrm{op}} \xrightarrow{\sim} \mathcal{P}_{\mathfrak{g}}, \qquad A \longmapsto (\mathcal{B}_A, \rho_A).$$

Its inverse recovers

$$A_{\mathcal{B}} = \mathrm{Hom}_{\mathfrak{g}}(\mathfrak{g}, W^\vee[-1]).$$

The linear differential recovers d_A, the quadratic differential recovers multiplication, and $-T_{\mathcal{B}}^\vee$ recovers T_A. Direct products correspond to completed tensor products, whose scalar map uses multiplication in $\mathcal{B}_k$.

Proof. The adjoint representation of $\mathfrak{sl}_2$ satisfies

$$\mathrm{End}_{\mathfrak{g}}(\mathfrak{g}) = k, \qquad \mathrm{Hom}_{\mathfrak{g}}(\mathfrak{g} \otimes \mathfrak{g}, \mathfrak{g}) = k[\,,\,].$$

The first identity follows from the weight spaces $2, 0, -2$ and the actions of e, f between them. For the second, $\Lambda^2\mathfrak{g} \simeq \mathfrak{g}$, whereas $\mathrm{Sym}^2 \mathfrak{g}$ is the five-dimensional irreducible representation generated by $e \otimes e$ plus the invariant line generated by $2(e \otimes f + f \otimes e) + h \otimes h$.

Put $M = \mathrm{Hom}_{\mathfrak{g}}(\mathfrak{g}, L)$. The bracket on $\mathfrak{g} \otimes M$ therefore has the form $[x, y] \otimes m(a, b)$. Graded antisymmetry makes m graded commutative. Jacobi on $e \otimes a, e \otimes b, f \otimes c$ gives $a(bc) = (-1)^{|a||b|} b(ac)$. Consequently

$$(ab)c = (-1)^{(|a|+|b|)|c|} c(ab) = (-1)^{|b||c|} a(cb) = a(bc).$$

The prescribed adjoint action makes $j \in M^0$ the unit.

Both d_L and T_L commute with that action. The first adjoint identity makes them $1 \otimes d_M$ and $1 \otimes T_M$. Their Lie derivation equations are precisely the product derivation equations on M.

The symmetric-degree components of $d^2 = 0$ recover the differential, derivation and Jacobi equations of L. A morphism of presentations transposes on generators to a dg Lie map preserving j and translation. It is $1 \otimes f$, and its bracket equation says exactly that f preserves multiplication. Continuous duality and the free completed presentation recover the whole field map from this linear map. This proves the inverse comparisons and full faithfulness.

Finally $L_{A \times A'} = L_A \oplus L_{A'}$ gives

$$\mathcal{B}_{A \times A'} \simeq \mathcal{B}_A \widehat{\otimes} \mathcal{B}_{A'}, \qquad \mathcal{B}_0 = k.$$

The diagonal constant-current inclusion transposes to multiplication of the two scalar maps, as asserted. □

9.4 The full coefficient comparison

Recovering the state multiplication is only the first step. We must also recover the normal derivatives and compare all ways of reaching a multiple collision. The elementary three-point identity fixes the recipient of that comparison.

Put $s = z_1 - z_3$, $t = z_2 - z_3$ and $r = s - t$. For a vector space P, let

$$\mathcal{G}_3(P) = P[[s, t]][(st(s - t))^{-1}], \qquad E_{s,t}(P) = \bigoplus_{a,b \geq 1} P\, s^{-a} t^{-b}.$$

The latter is the full normal contact module, with multiplication by coordinates followed by principal part and with normal differentiation. Change from contact coordinates (r, t) to (s, t) by

$$J(r^{-a} t^{-b}) = \sum_{j=0}^{b-1} \binom{a + j - 1}{j} s^{-a-j} t^{-b+j}. \tag{9.14}$$

This map sends the simple contact to $s^{-1}t^{-1}$, carries ∂_r to ∂_s, and carries the old ∂_t to $\partial_s + \partial_t$. Applying these derivatives to the simple contact proves the formula for every a, b.

Lemma 9.11. *For $F \in \mathcal{G}_3(P)$, define*

$$\begin{aligned} L(F) &= J\,\mathrm{pp}_t \mathrm{pp}_r F(r + t, t) && (r \ll t), \\ R(F) &= \mathrm{pp}_s \mathrm{pp}_t F(s, t) && (t \ll s), \\ N(F) &= \mathrm{pp}_t \mathrm{pp}_s F(s, t) && (s \ll t). \end{aligned}$$

The rightmost principal part is applied first. Then $L(F) - R(F) + N(F) = 0$ in $E_{s,t}(P)$.

Proof. Write $F = G/(s^a t^b (s-t)^c)$. A monomial surviving two principal parts has total degree at most -2. The denominator is homogeneous, so numerator terms above degree $a+b+c-2$ contribute to none of the three maps. We may take G polynomial. Over $k[t,t^{-1}]$, partial fractions in the coprime factors s, $s-t$ reduce to polynomials in s, terms $a_j(t)s^{-j}$, and terms $b_j(t)(s-t)^{-j}$, with Laurent coefficients in t.

A polynomial in s contributes nothing. For $a_j(t)s^{-j}$, the r-expansion is regular, while both other channels give $s^{-j}\operatorname{pp}_t(a_j)$.

For $b_j(t)(s-t)^{-j}$, the expansion for $s \ll t$ is regular in s. In the other domain,

$$(s-t)^{-j} = \sum_{n\geq 0} \binom{j+n-1}{n} s^{-j-n} t^n.$$

A coefficient $t^{-\beta}$ contributes only for $\beta \geq 1$, and gives $\sum_{n<\beta} \binom{j+n-1}{n} s^{-j-n}t^{n-\beta}$. This is exactly $J(r^{-j}t^{-\beta})$. The identity follows term by term, with the coefficients of P unchanged. □

For $A \in \mathcal{A}$, put $\mathscr{M}_A = \omega_X \otimes A$, with its internal differential and right action

$$(f\,dz \otimes a)\partial_z = -f'\,dz \otimes a - f\,dz \otimes T_A a. \tag{9.15}$$

On an ordered chart, a partition π of I has its blocks in root order. Let i_π be its diagonal inclusion and j_π the open inclusion of distinct block positions. Define the partition coefficient complex

$$(C_A)_I = \bigoplus_\pi i_{\pi,*} j_{\pi,*} j_\pi^! (\mathscr{M}_A^{\boxtimes|\pi|})[|\pi|], \qquad (C_A)_\varnothing = k. \tag{9.16}$$

All direct images and pullbacks are derived. In particular the closed direct image includes every normal derivative.

Regard each block state as suspended, with $|sa| = |a| - 1$ and $d(sa) = -s(d_A a)$. A merger $i < j$ moves sa_i beside sa_j through the intervening suspended factors, then applies

$$sa \otimes sb \longmapsto (-1)^{|a|} s \operatorname{pp}_u\big(f(u)(e^{uT_A}a)b\big), \qquad u = z_i - z_j, \tag{9.17}$$

retaining the root of j. The sign of tensoring this degree-one map past the preceding factors is included. Equivalently, the total sign is

$$(-1)^{\sum_{k<i}(|a_k|-1)+|a_i|(1+\sum_{i<k<j}(|a_k|-1))}.$$

It is $(-1)^{i-1}$ for degree-zero states. The differential of C_A is the internal differential plus these mergers.

Use the full ordered collision carrier of Chapter 8: all ordered-subset cuts are retained, including $13 \mid 2$. Denote its all-subset convolution by $\star$ and its empty unit by K. Write $\sharp$ for the full coefficient duality of Theorem 6.9, and $\star_\sharp$ for the tensor transported by that duality. These are differential-module duals. They are distinct from the continuous vector-space dual in (9.10).

Proposition 9.12. *The complexes C_A form a functor to full ordered collision coalgebras with local vacua. It sends quasi-isomorphisms in $\mathcal{A}$ to equivalences. There are coherent isomorphisms*

$$C_{A\times A'} \simeq C_A \star C_{A'}, \qquad C_0 = K. \tag{9.18}$$

Under this comparison the diagonal $A \to A \times A$ induces the full cut coproduct of C_A.

Proof. The regular fields of A are $Y_A(a,u)b = (e^{uT_A}a)b$. For three states their iterates are expansions of the same graded-commutative product $(e^{sT_A}a)(e^{tT_A}b)c$. Multiplication by an allowed rational test puts them in the common recipient of Lemma 9.11. That lemma, with the other positions as parameters, gives the three-channel residue identity.

Two disjoint mergers cancel in d^2 because their suspended maps have degree one. The three mergers on a fixed triple cancel by the displayed identity and the same suspended permutations. These exhaust pairs of mergers. The internal differential anticommutes with them since d_A is a product derivation commuting with T_A. Thus the proposed differential squares to zero.

Cuts, diagonal restriction and exchange of ordered charts retain these same residue maps and their full normal-transfer modules. The unit identities follow from $Y_A(1,u)a = a$ and $Y_A(a,u)1 = e^{uT_A}a$. Associativity and the three-channel identity give compatible compositions on successive collisions. Ordinary graded exchanges give the chart coherences. This constructs the full coefficient coalgebra and its local vacua.

Filter a coefficient with n labels by its number of blocks. The internal differential preserves this number and a merger lowers it. There are only n nonempty steps.

A quasi-isomorphism $A \to A'$ gives an equivalence of the singleton differential modules. Finite external products and derived geometric functors preserve equivalences, giving the same conclusion on every associated graded term. Finite filtered comparison gives the assertion on every configuration. The natural collision, cut and vacuum maps are unchanged by this argument.

For (9.18), expand every block state of $A \oplus A'$ in its two colors. The unions I_0, I_1 of the blocks of each color specify exactly an all-subset cut. The two internal distinct-block opens, together with every cross-color diagonal inverted by convolution, give the original open. The closed direct images and normal modules agree.

Group suspended factors by color with their Koszul permutation sign. Opposite colors have zero product. Same-color mergers and internal differentials are those of the corresponding factor. Moving a degree-one merger through the other color gives precisely the tensor differential sign. Thus grouping is a chain isomorphism.

Collisions either identify same-color blocks or kill a cross-color localized term. Cuts keep the same colored blocks. Grouping three colors in either order crosses each pair once, proving associativity, symmetry and the unit law.

Finally the diagonal expands each block in both colors. A given cut keeps the term when every block stays on one side and gives zero otherwise. These are the full coproduct maps. □

Define $E_A = C^\sharp_A$ and, for a quadratic current presentation,

$$\mathfrak{F}(\mathcal{B}) = C^\sharp_{A_{\mathcal{B}}}.$$

Recognition reverses arrows and full coefficient duality reverses them again.

Theorem 9.13. *The functor $\mathfrak{F}$ is covariant and strong symmetric monoidal. There is a natural isomorphism $\mathfrak{F}(\mathcal{B}_A) \simeq E_A$ preserving every collision, all normal derivatives, arbitrary spectators and every ordered-subset cut. It carries field multiplication to $\Delta^\sharp_A$ in both orders. The field unit maps to K, while*

$$\mathfrak{F}(\mathcal{B}_k) = C^\sharp_k, \qquad \mathfrak{F}(\rho_A) : E_A \longrightarrow C^\sharp_k.$$

Any scalar section induced by an augmentation of A is also preserved.

Proof. Recognition identifies the recovered state algebra with A, including its differential and translation. Apply that identification to each partition block and each normal derivative in (9.16). It is a coefficient chain isomorphism before duality, compatible with all structure maps. The tensor comparison is the transpose of (9.18), together with the cochain tensor comparison. Their coherence identities were proved above.

On generators, multiplication of cochains transposes to the diagonal $L_A \to L_A \oplus L_A$. Its recovered state map is $A \to A \times A$, so the last assertion of the proposition identifies the resulting product with $\Delta^\sharp_A$. Color interchange

proves the opposite-order assertion. The zero algebra and the maps $k \to A \to k$, when the latter is supplied, give the stated empty and local scalar maps.

The translation sign is fixed: $e^{uT_{\mathcal{B}}}$ transposes to e^{-uT_A}. Its coefficient map is $C(e^{-uT_A})^\sharp$ at every finite Taylor precision. For a normal test u^{-m}, the complete residue is the finite expression

$$r(u^{-m}; a, b) = \sum_{j=0}^{m-1} \frac{(T_A^j a)b}{j!} u^{j-m}. \tag{9.19}$$

This compares every normal derivative, including at an outer collision with spectators. The singleton is $(\mathscr{M}_A[1])^\sharp$. It is obtained from the recovered linear generators, not from one uniform shift of the entire Chevalley cochain algebra. □

9.5 Formal equivalence and the linearized states

The current differential recovers the state algebra before cohomology is taken. To make reconstruction invariant under a change of dg presentation, its weak equivalences must retain the linearized states. Nilpotent test algebras detect exactly this requirement.

Let $\mathfrak{a}$ be the augmentation ideal of a quadratic current presentation, and put

$$F^r\mathcal{B} = \overline{\mathfrak{a}^r}, \qquad Q\mathcal{B} = \mathfrak{a}/\overline{\mathfrak{a}^2}.$$

The differential on $Q\mathcal{B} = W$ is d_1.

Let R be a finite-dimensional local commutative dg algebra in nonpositive degrees, augmented to k, with nilpotent augmentation ideal $\mathfrak{m}$. The polynomial de Rham algebra of the n-simplex is

$$\Omega_n = k[t_0, \dots, t_n, dt_0, \dots, dt_n]/(\textstyle\sum t_i - 1, \sum dt_i), \qquad |t_i| = 0, \quad |dt_i| = 1.$$

Define a simplicial set of formal points by

$$\mathcal{X}_{\mathcal{B}}(R)_n = \left\{ \phi : \mathcal{B} \longrightarrow R \otimes \Omega_n : \begin{array}{l} \phi \text{ is a continuous dg algebra map,} \\ \phi(\mathfrak{a}) \subset \mathfrak{m} \otimes \Omega_n \end{array} \right\}. \tag{9.20}$$

The target is discrete. Faces and degeneracies are induced by the simplex maps. The scalar map gives $\mathcal{X}_{\mathcal{B}_k} \to \mathcal{X}_{\mathcal{B}}$.

A morphism $f : \mathcal{B} \to \mathcal{B}'$ in $\mathcal{P}_{\mathfrak{g}}$ is a *formal equivalence* when $\mathcal{X}_{\mathcal{B}'}(R) \to \mathcal{X}_{\mathcal{B}}(R)$ is a weak equivalence for every such R. This definition uses the field algebra and its augmentation. It does not use the coefficient functor.

Theorem 9.14. *For a morphism of quadratic current presentations, the following conditions are equivalent:*

(i) *it is a formal equivalence.*

(ii) *its map on $Q\mathcal{B}$ is a quasi-isomorphism.*

(iii) *its recovered map of state algebras is a quasi-isomorphism.*

Each condition implies an ordinary quasi-isomorphism of field algebras.

Proof. The images of the generators under a map in (9.20) determine a total degree-one element $\alpha \in L \otimes \mathfrak{m} \otimes \Omega_n$. The signs in (9.11) identify the chain-map equation with $d\alpha + \frac{1}{2}[\alpha, \alpha] = 0$.

Continuity means finite weight support on each simplex. Only finitely many generator degrees occur, since $R \otimes \Omega_n$ has bounded degree range. Conversely, a Maurer–Cartan element determines a continuous map on the generators. Nilpotence of $\mathfrak{m}$ makes its extension to the completed symmetric algebra well defined. Thus

$$\mathcal{X}_{\mathcal{B}}(R) = \mathrm{MC}_\bullet(L \otimes \mathfrak{m}).$$

The dg Lie algebra on the right is nilpotent and nonpositive. A quasi-isomorphism of L therefore induces a weak equivalence of these simplicial sets by [66, Theorem 4.8].

Conversely take $R = k[\varepsilon]/(\varepsilon^2)$, in degree zero. The bracket on $L\varepsilon$ vanishes. Polynomial closed forms give the Dold–Kan realization of the shifted complex, so

$$\pi_n \mathcal{X}_{\mathcal{B}}(R) = H^{1-n}(L)\varepsilon \quad (n \geq 1), \qquad \pi_0 = *. \tag{9.21}$$

See [80, §9.2] for this polynomial-forms comparison. Every degree of L is nonpositive and hence occurs in (9.21). A formal equivalence induces a quasi-isomorphism of L.

Continuous duality between discrete and linearly compact vector spaces is exact. Since $L = (Q\mathcal{B})^\vee[-1]$, this is equivalent to (ii). The current recognition theorem identifies the recovered map of L with $1_{\mathfrak{g}} \otimes f_A$, proving (iii).

Finally filter cochains by F^r. The associated graded map is the completed symmetric algebra on the map of linearized complexes. Finite tensor products over k, finite symmetric coinvariants in characteristic zero, and continuous duality preserve the relevant quasi-isomorphisms. Every generator has positive cohomological degree, so $F^{m+1}\mathcal{B}^m = 0$. The filtration is bounded in each degree, and filtered comparison proves the last assertion. □

Example 9.15. Ordinary field quasi-isomorphisms need not be formal equivalences. Take $A = k[t]$, with $T_A = 0$ and auxiliary weight $\mathrm{wt}(t) = 1$. Constant-current inclusion and evaluation at $t = 0$ give split field maps

$$\mathcal{B}_{k[t]} \rightleftarrows \mathcal{B}_k.$$

Both are ordinary quasi-isomorphisms, including on every finite weight quotient. To see this, fix chain weight q and choose $N > q$. The weight-q Chevalley complex of $\mathfrak{g}[t]$ agrees with that of $\mathfrak{g}[t]/t^N\mathfrak{g}[t]$: no bracket of total weight at most q meets the truncation.

For $\mathfrak{g} = \mathfrak{sl}_2$, the strong Macdonald theorem gives exterior cohomology generators of degree three and cochain weights

$$0, -(N+1), \dots, -(2N-1)$$

[60, Theorem A and Remark 1.2(iii)]. Thus every positive chain-weight component is acyclic, and weight zero is exactly $\mathcal{B}_k$. Products of complexes of vector spaces are exact, so the assertion also holds for the complete cochains. Translation is zero, and the two maps preserve the scalar map and the fields.

On the dual-number test, however, the first homotopy groups are $\mathfrak{g}[t]\varepsilon$ and $\mathfrak{g}\varepsilon$. Evaluation kills every positive polynomial weight. The coefficient maps also fail to be equivalences: their singleton maps are inclusion and evaluation between $\omega_X[1]$ and $(\omega_X \otimes k[t])[1]$. Full coefficient duality reflects this failure. Hence $\mathfrak{F}$ does not descend through ordinary field quasi-isomorphisms, even with the scalar map retained.

Let $\mathcal{P}_{\mathfrak{g}}^{\mathrm{form}}$ denote localization at formal equivalences, retaining all mapping spaces. Recognition is an equivalence of relative categories, and therefore gives

$$\mathcal{P}_{\mathfrak{g}}^{\mathrm{form}} \simeq \left(\mathcal{A}[\mathrm{qis}^{-1}]\right)^{\mathrm{op}}.$$

Direct products preserve quasi-isomorphisms, so the tensor structure descends. These mapping spaces concern the specified quadratic presentations with their split grading.

Corollary 9.16. *The coefficient realization descends to a covariant strong symmetric monoidal functor on $\mathcal{P}_{\mathfrak{g}}^{\mathrm{form}}$. It preserves both products, all supported collision comparisons, every finite normal derivative, arbitrary spectators and every cut. The field unit maps to K, the constant-current object maps to $C_k^\sharp$, and the scalar maps and translation are preserved. Full coefficient biduality recovers C_A with these operations.*

Proof. The formal equivalence theorem and the finite block filtration in the proof of (9.18) show that $\mathfrak{F}$ inverts the designated maps. Its structure comparisons are natural before localization, so they descend with their coherences and mapping spaces. Theorem 9.13 identifies the products, units and translation. The full retained coefficient duality of Theorem 6.9 returns every coefficient map on applying the inverse duality. □

Example 9.17. Formal equivalence permits changes of presentation with all translation jets. Set

$$J = k[x_0, x_1, \dots] \otimes \Lambda(\eta_0, \eta_1, \dots), \quad |x_j| = 0, \quad |\eta_j| = -1, \quad d\eta_j = x_j, \quad dx_j = 0,$$

and define $Tx_j = x_{j+1}$, $T\eta_j = \eta_{j+1}$. Both generators with index j have weight $j+1$. Every weight is a bounded finite-dimensional complex, so J and $A \otimes J$ belong to $\mathcal{A}$.

Let N count total monomial degree, including the odd generators. On positive monomial degree put

$$h = \frac{1}{N} \sum_{j \geq 0} \eta_j \frac{\partial}{\partial x_j}, \qquad h(1) = 0.$$

Every value is a finite sum. The supercommutator of $d = \sum_j x_j \partial_{\eta_j}$ with $\sum_j \eta_j \partial_{x_j}$ is N. Both preserve monomial degree, so $dh + hd = 1 - \iota\epsilon$, where $\epsilon : J \to k$ is augmentation.

Translation commutes with N and with both sums. Hence $Th = hT$. The tensor homotopy is $a \otimes v \mapsto (-1)^{|a|} a \otimes hv$.

Inclusion and augmentation between A and $A \otimes J$ are therefore quasi-isomorphisms commuting with translation. They give formal equivalences of current fields and equivalences of the full coefficient duals, with all scalar maps, vacua, cuts and collisions preserved. The positive polynomial states in Example 9.15 admit no such contraction.

9.6 Interacting currents and their rational coefficient space

A quadratic current differential can split one current into two. If the current carries an internal pole, that pole must survive the split. We therefore keep the constituent jets distinct from their grouping into currents. The differential changes the grouping and leaves the constituent variables in place.

Work compatibly at every precision $\mathsf{R}_r = k[\hbar]/(\hbar^r)$. Let R be a finite-dimensional commutative unital algebra, and let E be a bounded nonpositive complex of finite-dimensional R-modules. Put

$$F_R(E) = \mathrm{Sym}_R(E \otimes k[t]).$$

The symmetric algebra uses the cohomological signs. A generator $e_n = et^n$ has charge one and weight $n+1$. R has charge and weight zero.

Define even derivations, zero on R, by

$$T(e_n) = (n+1)e_{n+1}, \qquad \delta_i(e_n) = e_{n-i+1}, \qquad e_j = 0 \quad (j < 0). \tag{9.22}$$

They commute with the differential and obey

$$[\delta_i, \delta_j] = 0, \qquad [\delta_i, T] = (i-1)\delta_{i-1}, \qquad \delta_0 = 0.$$

In particular, δ_1 acts by charge.

Definition 9.18. A charged jet quotient is an algebra $A = F_R(E)/\mathfrak{a}$, where $\mathfrak{a}$ is a dg ideal of positive charge. It is homogeneous in charge and weight and stable under T and every ∂_i. Require each weight piece to be a bounded finite-dimensional complex.

Maps preserve the unit, differential, charge and jet actions. The chosen presentation is not part of the data. Denote this category, with the zero algebra admitted, by $\mathcal{A}_{\mathrm{cj}}$. At precision r use scalar extensions and R_r-linear maps preserving the same data.

The algebra has a canonical presentation. On its charge-one part A_1, put $D = \partial_2$ and $V_A = \ker D$. Since $[D, T] = 1$ and D is locally nilpotent, the finite-input formulas

$$\pi = \sum_{j \geq 0} \frac{(-1)^j}{j!} T^j D^j, \qquad a = \sum_{n \geq 0} \frac{T^n}{n!} \pi(D^n a) \tag{9.23}$$

give $A_1 = V_A \otimes k[t]$. Adjacent binomial terms cancel to give $D\pi = 0$ and the second identity. Successive powers of D prove uniqueness.

Thus V_A is a finite quotient complex of E, and multiplication gives a canonical surjection $F_R(V_A) \to A$. Its kernel is the exact relation ideal of this presentation.

Let C_A be the finite-weight polynomial core of the Chevalley cochains of $\mathfrak{g} \otimes A$. On its generators put

$$T_B = -(1 \otimes T)^\vee, \qquad D_i = -(1 \otimes \partial_i)^\vee,$$

and extend as derivations. These restricted transposes commute with the current differential. Write $C_A^{(n)}$ for total charge n. On it $D_1 = -n$.

Define

$$\Lambda_n = k[x_1, \dots, x_n]^{S_n}, \qquad p_m = \sum_{a=1}^{n} x_a^m, \qquad d_n = \prod_{a<b} (x_a - x_b)^2. \tag{9.24}$$

Set $\Lambda_0 = k$ and $d_0 = d_1 = 1$. The x_a are constituent dual-jet variables, independent of the positions on the curve.

Lemma 9.19. *On $C_A^{(n)}$, the assignments $p_m \mapsto -D_{m+1}$ define a Λ_n-action. The current differential is linear over this ring, multiplication by d_n is injective, and*

$$[T_B, p_m] = -m\, p_{m-1}, \qquad T_B(d_n) = 0.$$

Proof. First take $A = F_R(E)$. Let K_n be the multidegree $(1, \dots, 1)$ current-cochain complex on independent copies $E_1, \dots, E_n$ of E. It is finite. A current uses a subset of the copies, and a product of currents partitions them among its factors. Constant-current ghosts are retained. Polarization gives

$$(K_n \otimes k[x_1, \dots, x_n])^{S_n} \xrightarrow{\sim} C^{(n)}_{F_R(E)}. \tag{9.25}$$

Pair x^j with et^j by one, replace the copy indices of each current by those jet indices, and use the graded symmetric pairing. Repeated even indices contribute their factorials. Repeated identical odd indices vanish. Each permutation orbit therefore maps bijectively to its corresponding monomial in every finite weight. The R-balancing relations concern the coefficient factors alone.

For example, if $R = E = k$, a cochain dual to the current $a \otimes e_1 \cdots e_n$ maps to $n!$ times the cochain dual to $a \otimes e_0^n$. Multiplication under (9.25) is concatenation followed by the normalized projector $P_{p+q} = ((p+q)!)^{-1} \sum_{\sigma \in S_{p+q}} \sigma$. There is no further binomial factor.

The quadratic differential splits a current's copy set into two subsets. It changes no x_a. The linear differential acts on the copies of E. In these coordinates

$$T_B = -\sum_a \partial_{x_a}, \qquad D_{m+1} = -\sum_a x_a^m.$$

The Newton identities, linearity and commutator formula follow. The nonzero polynomial d_n is injective before and after taking invariants, and simultaneous translation annihilates it.

For a quotient, the state surjection induces an injection of current cores. This follows on each finite weight and then on its symmetric cochains. It preserves the differential and all jet actions. The preceding identities and injectivity restrict to this subcomplex. The moment formula makes the construction independent of the chosen presentation. □

The meromorphic state complex is the intrinsic localization

$$\mathcal{B}_A^{\mathrm{mer}} = \bigoplus_{n\ge 0} C_A^{(n)} \otimes_{\Lambda_n} \Lambda_n[d_n^{-1}]. \tag{9.26}$$

Its differential extends uniquely. In a free presentation this is $(K_n \otimes k[x_1,\dots,x_n,\Delta_n^{-1}])^{S_n}$ in charge n, where $\Delta_n = \prod_{a<b}(x_a - x_b)$: localizing at $d_n = \Delta_n^2$ commutes with the averaging projector. An internal pole therefore remains in the same coefficient space when its current splits.

Choose

$$w(u) \in 1 + \hbar k[u,u^{-1}][[\hbar]]$$

and reduce at each precision. For groups of charges p, q, set

$$\Omega_{p,q}(v) = \prod_{a=1}^{p}\prod_{b=1}^{q} w(v + x_a - y_b), \qquad R_{p,q} = \prod_{a,b}(x_a - y_b).$$

These expressions are symmetric within the two groups.

Their joint tensor is

$$(C_A^{(p)} \otimes C_{A'}^{(q)})[(d_p d_q R_{p,q}^2)^{-1}]. \tag{9.27}$$

Indeed the union of the two sets of variables sends d_{p+q} exactly to $d_p d_q R_{p,q}^2$.

For field parameters also invert all translated cross differences $x_a - y_b + mv$, $m \in \mathbb{Z}$. With several parameters use their integral translates. The resulting rational spaces are preserved by block translations and permutations. Every element has finitely many denominator factors at fixed precision.

The ordinary current product μ_0 is linear over Λ_{p+q}, since each D_i is primitive for multiplication. It extends to (9.27). Define the native product and fields by

$$\mu_w = \mu_0 \Omega_{p,q}(0), \qquad Y_w(b,v)c = \mu_w(e^{vT_B} b \otimes c). \tag{9.28}$$

Proposition 9.20. *The product μ_w is unital, associative and differential graded. Translation is locally nilpotent at each precision. The displayed fields are regular dg nonlocal fields. On the joint rational space their unpermuted exchange is*

$$S_w(v) = \frac{\Omega_{q,p}(-v)^{21}}{\Omega_{p,q}(v)}. \tag{9.29}$$

They satisfy both vacuum identities, translation, exchange, unitarity, Yang–Baxter and the field hexagon. The construction is contravariant in $\mathcal{A}_{\mathrm{cj}}$.

Proof. In a free presentation, the product is the normalized permutation average of the ordinary product multiplied by $\Omega_{p,q}(0)$. For three groups the final projector absorbs either intermediate projector, and both associations have the factor $\Omega_{p,q}(0)\Omega_{p,s}(0)\Omega_{q,s}(0)$. The empty group gives the unit. The differential preserves the constituent variables, so it commutes with every cross multiplier. The ordinary dg product rule proves the new one.

For a quotient, the localized core embeds in the free localized core. A two-group multiplier has numerator symmetric within each group and denominator dividing a power of the total discriminant. The numerator preserves the quotient-current factors through their jet actions. Localized Λ_{p+q}-linear multiplication lands in the quotient-current object, so all identities restrict to it.

Translation lowers polynomial weight and annihilates the internal discriminant. It is locally nilpotent on every localized state. Total translation annihilates the cross differences and is a derivation of μ_w. Associativity and this derivation give the regular field identities, including both vacuum identities.

Keep the two groups unsymmetrized to check exchange. Translating the first group by $-v$ changes $\Omega_{p,q}(v)$ to $\Omega_{p,q}(0)$. The ratio in (9.29) changes this factor to the reversed product factor, while the final projector supplies the Koszul permutation. Reversing the pair inverts the ratio.

The pair multipliers commute, proving Yang–Baxter. Splitting their pair sets proves the hexagon at independent output separations. Maps preserving the core jet actions preserve all these formulas and extend uniquely through localization. □

At finite Taylor order, a translated denominator is inverted by its finite nilpotent expansion. The compatible state tower has fields in $\varprojlim_r \mathcal{B}^{\mathrm{mer}}_{A,r}[v]$. The exchange space includes cross-group internal poles. Replacing it by the ordinary Laurent tensor square would change the theorem.

9.6.1 *Recognition and the tensor of meromorphic currents*

A charged meromorphic current presentation retains its polynomial quadratic current core, charge grading, scalar map, translation and jet actions. Its completed core satisfies Definition 9.9. On the recovered states, charge zero is a finite algebra R in degree zero. Charge one has the decomposition (9.23) and generates the algebra over R. The relations satisfy Definition 9.18.

Its extension and native operations are exactly (9.26) and (9.28). The core is marked data. For polar w it need not be a subalgebra for μ_w.

Morphisms preserve core generators, differential, charge, scalar map and jet actions, and extend by localization. Write this category as $\mathcal{G}^{\mathrm{cj}}_w$. At finite precision use scalar extensions and maps over R_r.

Proposition 9.21. *There is an equivalence $\mathcal{A}^{\mathrm{op}}_{\mathrm{cj}} \simeq \mathcal{G}^{\mathrm{cj}}_w$. It recovers the algebra, differential, unit, translation, every jet action and the canonical relation ideal.*

Proof. Current recognition recovers $A_B = \mathrm{Hom}_{\mathfrak{g}}(\mathfrak{g}, W^\vee[-1])$. A core derivation commuting with its differential and killing scalar currents transposes to a current Lie derivation. Adjoint-isotypic evaluation makes it $1 \otimes \partial$ for a state derivation. Thus negative transposition recovers all prescribed jet actions and their commutators.

Charge zero recovers R, and (9.23) recovers $V_B = \ker(\delta_2|(A_B)_1)$. Generation gives the surjection $F_R(V_B) \to A_B$ and its kernel. The symmetric-ring action and localization are intrinsic, so they recover the entire meromorphic object.

A morphism commuting with δ_2, T extends its map on V_B constantly along the jets. Its quadratic differential makes it preserve multiplication and relations. Conversely every such state map induces the core map and its unique localization. These constructions are inverse on objects and morphisms. □

The tensor is also specified. On the joint localization write $F_{B,C} = \Omega_{B,C}(0)$ and put

$$\sigma_w = \tau(F^{21}_{C,B})^{-1}F_{B,C}, \qquad \mu_{B\otimes_w C} = (\mu_B \otimes \mu_C)(1 \otimes \sigma_w \otimes 1),$$

where τ is the Koszul permutation. If U removes the native cross factor, transport the polynomial core through

$$U_2 : UB \otimes UC \longrightarrow U(B \otimes_w C), \qquad U_2 = F_{B,C}^{-1}. \tag{9.30}$$

Lemma 9.22. *This is a symmetric monoidal tensor corresponding to $A \times A'$. Native multiplication is a morphism $B \otimes_w B \to B$, and $U(\mu_w)U_2 = \mu_0$. The tensor unit is $\mathsf{R}_r = \mathcal{B}^{\mathrm{mer}}_{0,r}$.*

Proof. The ordinary current core of $A \times A'$ is the tensor of their cores. Its total discriminant gives exactly the joint localization. For three groups,

$$F_{B,C\otimes D} = F_{B,C}F_{B,D}, \qquad F_{B\otimes C,D} = F_{B,D}F_{C,D}.$$

These equalities of commuting factors prove the associativity and unit diagrams for U_2. Moving τ through the factors proves the symmetry diagram and $\mu_w\sigma_w = \mu_w$.

For completeness, on four groups the braided product contributes $w_{13}w_{24}w_{23}w_{32}^{-1}$. The factor on its two outputs contributes $w_{12}w_{14}w_{32}w_{34}$. Their product is $\prod_{i<j} w_{ij}$, the factor obtained by twisting within each input and then between the inputs. Here w_{ij} includes all constituent pairs between groups i, j. This proves the algebra tensor identity on the joint localization.

Finally $\mu_w F^{-1} = \mu_0$. The tensor constraint is the identity on core generators, so the transported multiplication is the recognized map corresponding to $A \to A \times A$. □

9.6.2 The state fields and the full product comparison

There is an independent field operation on the state algebra. Put $\ell(u) = \log w(u)$ and define

$$\begin{aligned} C_\ell(u) &= \sum_{i,j\ge 1} \frac{(-1)^{j-1}\ell^{(i+j-2)}(u)}{(i-1)!(j-1)!}\partial_i \otimes \partial_j, \\ \mathcal{W}_A(u) &= e^{C_\ell(u)}, \qquad \mathcal{E}_A(u) = \mathcal{W}_A(u)^{-1}. \end{aligned} \tag{9.31}$$

At precision r the exponential has finitely many terms. On each fixed state input only finitely many jet actions in each term survive. The operators therefore have a finite pole bound on that input.

For $A \neq 0$ set

$$Y_A^w(a,u)b = \mu_A(e^{uT} \otimes 1)\mathcal{E}_A(u)(a \otimes b), \qquad \mathcal{S}_A^w(u) = \mathcal{E}_A^{21}(-u)\mathcal{E}_A(u)^{-1}. \tag{9.32}$$

Taylor translation is applied after contraction. It introduces nonnegative powers and preserves the finite lower bound.

Lemma 9.23. *These formulas define dg quantum vertex fields with vacuum 1_A and rational exchange at every finite precision. They are functorial in $\mathcal{A}_{\mathrm{cj}}$.*

Proof. The jet derivations commute. Their commutators with T give

$$[T \otimes 1, C_\ell(u)] = -\partial_u C_\ell(u), \qquad [T \otimes 1 + 1 \otimes T, C_\ell(u)] = 0.$$

This follows directly by shifting i and j in (9.31). The factorials cancel their commutator coefficients. Every derivation annihilates the unit, so a contraction with either vacuum is the identity.

The derivation rule splits a contraction with a product into its two constituent contractions. Moving Taylor translation through it shifts its separation by that parameter. Both three-field associations have the same commuting pair factors before expansion. Their reversed ratio is $\mathcal{S}_A^w$, proving associativity, exchange and the field hexagon. Commutativity of pair factors proves Yang–Baxter, and reversing a pair proves unitarity.

Each identity is finite at fixed input, precision and Taylor coefficient. The relation ideal is preserved by the derivations and multiplication, so all operations descend and commute with the differential and maps. □

The native current multiplier determines this contraction. Undoing translation in its first group changes $\Omega_{p,q}(0)$ to $\Omega_{p,q}(u)$. With u invertible its expansion is

$$\log \Omega_{p,q}(u) = \sum_{a,b\ge 0} \frac{(-1)^b \ell^{(a+b)}(u)}{a!b!} p_a(x)\, p_b(y). \tag{9.33}$$

The charge terms $p_0(x) = p$, $p_0(y) = q$ are included.

Transposition of the moment actions recovers the two jet derivations. Their two minus signs cancel. The finite-input transpose of (9.33) is exactly $C_\ell(u)$.

More explicitly, multiplication by x_i transposes to lowering the corresponding constituent jet index. If N_{ij} is the difference of the two lowerings, a factor $w(u + x_i - y_j)$ transposes to

$$w(u + N_{ij})a = \sum_{m\ge 0} \frac{w^{(m)}(u)}{m!} N_{ij}^m a.$$

The sum is finite on each jet input. Multiplying the factors gives $\mathcal{W}_A(u)$, whose symmetric-ring expression descends through the relations.

The normal-frame map retains density and derivatives:

$$[f(v)\,dv] \otimes a \longmapsto \sum_m [f^{(m)}(u)\,du] \otimes \frac{N_{ij}^m a}{m!}.$$

It intertwines differentiation and lets v act as $u + N_{ij}$. No constituent variable is identified with a geometric position.

Let C_A° be the regular state collision complex and C_A^w the collision complex of (9.32). Both use the partition coefficients, suspended signs and full normal modules of the preceding sections. In a block order ω, put

$$\mathcal{E}_A^\omega = \prod_{a <_\omega b} \mathcal{E}_{A,ab}(z_a - z_b).$$

For ordered groups G, H, let $\mathcal{E}_{G,H}$ be the product over all pairs with one input in each group, and put $\mathcal{W}_{G,H} = \mathcal{E}_{G,H}^{-1}$.

Lemma 9.24. *The ordered operators glue to an isomorphism $\mathscr{E}_A : C_A^w \xrightarrow{\sim} C_A^\circ$. For every ordered-subset decomposition $G \sqcup H$,*

$$(\mathscr{E}_G \boxtimes \mathscr{E}_H)\Delta_{G,H}^w \mathscr{E}_A^{-1} = \mathcal{W}_{G,H}\Delta_{G,H}^\circ. \tag{9.34}$$

The cross operators commute with internal mergers, satisfy the three-group coherence identities and preserve both local vacuum actions, all normal derivatives and arbitrary spectators.

Proof. The translation commutators make each pair operator horizontal for $\partial_{z_a} + T_a$, so it extends through the full closed direct image. At an adjacent merger $z_a = z_b + u$, the derivation rule gives

$$\mu_{ab} e^{uT_a} \mathcal{E}_{ac}(z_b + u - z_c)\mathcal{E}_{bc}(z_b - z_c) = \mathcal{E}_{(ab)c}(z_b - z_c)\mu_{ab} e^{uT_a}.$$

The corresponding identity holds in the other tensor place. The internal pair factor is the contraction in (9.32). Thus $r_{ab}^\circ \mathcal{E}^\omega = \mathcal{E}^{\omega'} r_{ab}^w$, before and after full principal parts. No square-zero condition on positive-charge products is used.

Changing neighboring orders contributes $\mathcal{E}_{ab}\mathcal{E}_{ba}^{-1}$, which is exactly the quantum passage and cancels the change of ordered contractions. Bring any nonconsecutive pair together to obtain the same merger identity. The operators are even and preserve the suspended signs.

For an arbitrary subset, move G before H using these passages. The order-change comparison cancels the passage, and the total contraction factors as $(\mathscr{E}_G \boxtimes \mathscr{E}_H)\mathcal{E}_{G,H}$. This proves (9.34). Splitting pair sets among three groups proves coherence, and the merger identity proves compatibility with each internal differential. Horizontality retains the density and every normal differential action. All contractions with a vacuum are identities. □

For the outer merger 13, write $u = z_1 - z_3$, $t = z_2 - z_3$ and $X_{12}(u-t) = \mathcal{E}_{12}(u-t)\mathcal{E}_{21}(t-u)^{-1}$. The identity includes the middle state explicitly:

$$r^{\circ}_{13}\mathcal{E}_{12}(u-t)\mathcal{E}_{13}(u)\mathcal{E}_{23}(t) = \mathcal{E}_{2,(13)}(t)\,\mathrm{pp}_{u=0}\,\mu_{13}e^{uT_1} \cdot \mathcal{E}_{13}(u)X_{12}(u-t). \tag{9.35}$$

It remains valid after multiplication by u^{-m} for every $m \geq 1$. Further spectators supply the same pair factors on both sides.

Recover A_B from the marked current core and set $G(B) = (C^{\circ}_{A_B})^{\sharp}$. Let J° be the transpose of the colored-partition comparison combined with (9.30). On the $G \sqcup H$ component define

$$J^w = J^{\circ}(\mathcal{W}_{G,H})^{\sharp}. \tag{9.36}$$

The cross factor is fixed by the native kernel calculation (9.33).

Theorem 9.25. *The functor G with tensorator J^w is covariant and strong monoidal. It has natural isomorphisms of full coefficient algebras*

$$\mathscr{E}^{\sharp}_A : \mathfrak{F}_w(\mathcal{B}^{\mathrm{mer}}_A, \mu_w) \xrightarrow{\sim} (C^w_A, \Delta^w)^{\sharp}.$$

On every ordered-subset component the native product satisfies

$$G(\mu_w)J^w = (\Delta^{\circ}_{G,H})^{\sharp}(\mathcal{W}_{G,H})^{\sharp}. \tag{9.37}$$

Both orders, all collisions and coherent cut compositions are retained.

The singleton is $(\mathscr{M}_A[1])^{\sharp}$. The field unit maps to K, the constant-current object maps to $C^{\sharp}_k$, and ρ_A maps to the local scalar projection. A chosen augmentation $R \to k$, extended by zero on positive charge, gives its section. Both local vacuum actions and their full normal modules are preserved.

Proof. Recognition and full coefficient duality each reverse arrows, making G covariant. The colored-partition comparison gives J°, with the source tensor of Lemma 9.22. For three colors both composites of the new tensorator contain $\mathcal{W}_{G,H}\mathcal{W}_{G,J}\mathcal{W}_{H,J}$. Empty groups contribute the identity. Lemma 9.24 makes these natural maps on every full open direct image, proving the strong monoidal identities.

Untwisting native multiplication gives μ_0 on the core. Its generator transpose is the state diagonal $A \to A \times A$, whose colored-partition map is Δ°. Hence $G(\mu_w)J^{\circ} = (\Delta^{\circ})^{\sharp}$. Inserting the cross factor proves (9.37). Transpose (9.34) to obtain the algebra isomorphism.

The merger identity, including (9.35), retains the full normal-transfer differential. Full coefficient duality preserves the compositions and their coherent homotopies.

There is no pair contraction on a singleton. The zero state algebra gives the empty unit, and $k \to A$ and the specified augmentations preserve all jet actions. Their two transpositions give the stated scalar maps. Contractions involving either scalar factor vanish, which proves the vacuum assertions. □

Native fields are also actual morphisms after transporting their jet marking. At finite Taylor precision give $B\{v\}$ the actions

$$D_i^{[v]} = e^{-vT_B}D_i e^{vT_B} = \sum_{j=0}^{i-1}\binom{i-1}{j}v^j D_{i-j}.$$

Then $T_v = e^{vT_B} : B\{v\} \to B$ preserves the localized presentation, and $Y_w(v) = \mu_w(T_v \otimes 1)$. State transposition recovers $a \mapsto e^{-vT}a$.

With $\Theta_v = C_A^\circ(e^{-vT})^\sharp$ and $m^w = G(\mu_w)J^w$, monoidal naturality reads

$$\mathfrak{F}_w(Y_w(v))J^w_{B\{v\},B} = m^w(\Theta_v \star_\sharp 1).$$

The same conjugation changes u to $u + v$ in the native kernel. v remains separate from all spatial coordinates.

The transported symmetry is

$$(J^w_{C,B})^{-1}G(\sigma_w)J^w_{B,C} = ((\mathcal{W}_{H,G})^\sharp)^{-1}\tau_\sharp(\mathcal{W}_{G,H})^\sharp.$$

Removing the permutation and inverting recovers the field convention (9.29). Reflection $w(u) \mapsto w(-u)$ gives opposite multiplication. Inversion $w \mapsto w^{-1}$ gives inverse contraction. Full coefficient biduality returns C^w_A with its density shifts, collisions and vacua. This is biduality of the retained coefficient diagram.

9.6.3 *An interaction visible in cohomology*

Take

$$A_2 = F_k(k)/(\text{charge at least three}), \qquad w(u) = 1 - \hbar/u,$$

and write $a = e_0$. Its positive-charge ideal has nonzero square and zero cube. For example,

$$Y^w_{A_2}(a, u)a = \frac{(e^{uT}a)a}{1 - \hbar/u};$$

the coefficient of $\hbar u^{-1}$ is a^2. The polarized current complex retains partitions whose current factors have at most two constituent indices. Let c^e_i, c^h_i, c^f_i be its odd charge-one current ghosts and put

$$\kappa_{12} = c^e_1 c^f_2 + c^f_1 c^e_2 + 2c^h_1 c^h_2, \qquad \beta = \frac{\kappa_{12}}{x_1 - x_2}, \qquad \gamma = (x_1 + x_2)\beta.$$

The ghost permutation sign and the odd denominator make these degree-two states S_2-invariant.

Proposition 9.26. *Over* R_2*, the states* β, γ *are closed and* $[\beta] *_w [\gamma] \neq [\gamma] *_w [\beta]$ *in cohomology. Thus this dg algebra is not quasi-isomorphic to a commutative dg algebra over* R_2*. The same holds for the free state algebra* $F_k(k)$ *at this precision.*

Proof. The invariant trace form gives $d\kappa_{12} = 0$: its constant-current action cancels, a charge-one current has no positive-charge splitting, and the state differential vanishes. The differential does not act on constituent variables, proving closedness of both states.

On charge-four degree-four cochains, let L extract the coefficient of $c^h_1 c^h_2 c^h_3 c^h_4$ and evaluate the variables at $(0, 1, 2, 3)$. All differences are invertible there. The functional kills every boundary. A constant-current differential term retains a constant ghost. A positive-current split could give the selected coefficient only from a bracket of two Cartan currents, which is zero.

Set $f(x, y) = 1/(x - y)$ and $g(x, y) = (x + y)/(x - y)$. The normalized permutation average gives

$$\begin{aligned} L(\beta *_w \gamma - \gamma *_w \beta) = &-\frac{8\hbar}{4!}\sum_{\sigma\in S_4}\operatorname{sgn}(\sigma)f(x_{\sigma 1}, x_{\sigma 2})\,g(x_{\sigma 3}, x_{\sigma 4}) \\ &\cdot\left(\sum_{a=1}^{2}\sum_{b=3}^{4}\frac{1}{x_{\sigma a} - x_{\sigma b}}\right)\Bigg|_{(x_1,x_2,x_3,x_4)=(0,1,2,3)} = \frac{104}{9}\hbar. \end{aligned} \tag{9.38}$$

The two Cartan coefficients contribute four, the difference of first-order cross factors contributes minus two, and the projector contributes $1/4!$. The nonzero value cannot be a boundary. A commutative dg algebra has graded-commutative cohomology, so it cannot be quasi-isomorphic to this algebra. The same detector and boundary argument use only charge-one ghosts and remain valid for the free state algebra. □

The formal criterion also extends to these marked currents. Complete the polynomial core in weight, remove its native cross factor, and test continuous maps to $\mathsf{R}_r \otimes S \otimes \Omega_n$ carrying the core augmentation ideal into $\mathsf{R}_r \otimes \mathfrak{m}_S \otimes \Omega_n$. Here S ranges over the nonpositive nilpotent test algebras used in (9.20). Call a map a formal equivalence when it induces weak equivalences on these tests for every r, S.

Corollary 9.27. *A morphism of charged meromorphic current presentations is a formal equivalence precisely when its recovered state map is a quasi-isomorphism at every precision. The tensor, coefficient realization and native field comparisons descend to localization at these maps, retaining their mapping spaces and coherences.*

Proof. The coordinate calculation in the formal equivalence theorem identifies the tests with $\mathrm{MC}_\bullet((\mathfrak{g} \otimes A_r) \otimes \mathfrak{m}_S)$. The argument is over k, with the finite algebra R_r retained in its coefficients. Quasi-isomorphisms give equivalences by nilpotence. Dual numbers have homotopy groups $H^{1-n}(\mathfrak{g} \otimes A_r)\varepsilon$ for every $n \geq 1$, so they detect the converse.

The linear dual in this coordinate identification is exact. The pairing $(a, b) \mapsto [\hbar^{r-1}](ab)$ identifies R_r with its k-dual. In the power basis its matrix is antidiagonal with entries one. Thus R_r-linear duality agrees with k-linear duality, weight by weight. The scalar-extended finite weight pieces give the required continuous biduals.

The finite collision-block filtration sends a state quasi-isomorphism to an equivalence of C°_A, using derived external products and geometric functors. Naturality of $\mathscr{E}_A$ gives the same for C^{w}_A. Direct products preserve state quasi-isomorphisms. Full duality and localization therefore carry the tensor and the native field and product comparisons to the localized categories. The universal property carries their coherent compositions as well. □

9.7 Current framings and the information they retain

The meromorphic construction retains its polynomial core, charge and jet actions. Full coefficient duality preserves the geometric current module, but a geometric current module need not select its constant state sections. This distinction affects morphisms even when every collision and vacuum map has been retained.

Before suspension the singleton is $\mathscr{M}_A = \omega_X \otimes A$, with right action

$$(f\, dz \otimes a)\partial_z = -f'\, dz \otimes a - f\, dz \otimes Ta.$$

An $\mathcal{O}_X$-linear map $F(z)$ is differential-linear exactly when

$$\partial_z F(z) + [T, F(z)] = 0. \tag{9.39}$$

A *current framing* retains the constant state sections in this presentation. A map induced by a translation-preserving state map is constant in that framing. Equation (9.39) alone allows position-dependent maps.

Proposition 9.28. *On the essential image of the preceding state-field construction, retain a current framing and restrict arrows to actual constant singleton maps. The binary boundary recovers the input fields by*

$$a_{(n)}b = [u^{-1}]r_Y(u^n; a, b), \qquad Y(a, u)b = \sum_{n\in\mathbb{Z}} (a_{(n)}b)u^{-n-1}. \tag{9.40}$$

This recovery preserves translation, both vacuum identities and the full collision and cut operations.

Suppose further that a comparison is supplied by an actual isomorphism f of recovered charged dg state algebras. Require its singleton to be constant and horizontal and its state map to preserve charge and scalar maps. It lifts to a marked meromorphic current isomorphism if and only if it preserves δ_2.

Proof. Finiteness of each principal part gives the lower bound on field powers. Function-linearity gives

$$[u^{-q-1}]r_Y(u^n;a,b) = [u^{-1}]r_Y(u^{n+q};a,b) \qquad (q \geq 0).$$

Thus decoding and re-encoding are inverse on every Laurent input. Differential-linearity gives both translation identities with the stated right-action sign. The two specified vacuum boundary equations recover the two vacuum identities. A constant singleton commutes with each coefficient extraction.

Full inverse duality retains every higher coefficient and normal module. On an essential-image object, decoding returns its original input field, so its full coefficient construction returns those same collisions and cuts. The arrows here are induced by the stated actual singleton maps, and their coefficient constructions are natural. This recovers the original input fields. It does not discard the marked core needed for the meromorphic realization.

Necessity of preserving δ_2 follows from recognition. For sufficiency, constant horizontality gives $[T,f]=0$. On charge one, $\delta_i = \delta_2^{\,i-1}$, by (9.23). Hence f preserves every jet there. The jet actions are derivations, vanish on the scalar algebra, and charge one generates the algebra. The same identities hold on all products and through the relation ideal.

The current transpose of f preserves the split core and its differential. It preserves the moment actions, so localization extends it uniquely to each internal and joint rational space. The native product, fields and tensorator then commute with it term by term. Applying the construction to f^{-1} gives the inverse comparisons. This is a criterion for the supplied actual state isomorphism, not a replacement for a derived lift by a test on cohomology. □

Example 9.29. Even a constant field automorphism can change the jet marking. For the interacting algebra A_2 above, let $c \in \hbar\mathsf{R}_r$ and put $\psi_c = e^{cT}$. It is a finite exponential, preserves charge, commutes with translation and has inverse ψ_{-c}. Total translation commutes with every pair contraction. Thus ψ_c preserves the input fields and all their coefficient operations.

Nevertheless

$$[\delta_2,\psi_c] = c\psi_c\delta_1, \qquad [\delta_2,\psi_c]e_0 = c\psi_c e_0 \neq 0 \quad (c\neq 0).$$

It is not an automorphism of the fixed marked presentation.

Transporting the marking gives

$$\delta_i^{(c)} = e^{cT}\delta_i e^{-cT} = \sum_{j=0}^{i-1}\binom{i-1}{j}(-c)^j\delta_{i-j}.$$

With that transported marking ψ_c is an isomorphism. The assertion distinguishes a fixed marking from its isomorphic transport. It does not distinguish the underlying unmarked field algebra from itself.

Proposition 9.30. *For A_2 and every admissible w, the polynomial map*

$$F_c(z) = \exp\bigl(c(z\delta_1+\delta_2)\bigr), \qquad c\in\hbar\mathsf{R}_r,$$

induces a full coefficient automorphism, with inverse $F_{-c}(z)$. It preserves the scalar maps, empty unit and both local vacua. For $c\neq 0$ it is not induced by a map of the fixed constant-state fields preserving translation. Its fiber $f_c = e^{c\delta_2}$ at $z=0$ satisfies

$$f_cTf_c^{-1} = T + c\delta_1, \qquad f_cY^w(f_c^{-1}a,u)f_c^{-1}b = e^{pcu}Y^w(a,u)b \tag{9.41}$$

for an input a of charge p.

Proof. The derivation $Z_2(z) = z\partial_1 + \partial_2$ satisfies $\partial_z Z_2 + [T, Z_2] = 0$. Its finite exponential is horizontal, multiplicative and invertible. Explicitly,

$$F_c(z)e_n = e^{cz} \sum_{j=0}^{n} \frac{c^j}{j!} e_{n-j}.$$

Every scalar exponential is finite in $\mathsf{R}_r[z]$. The conjugation identity $e^{uT}\partial_2 e^{-uT} = \partial_2 - u\partial_1$ gives

$$e^{uT} F_c(y+u) = F_c(y) e^{uT}.$$

Each F_c commutes with the jet derivations in both factors of every contraction. Multiplicativity therefore gives the geometric field identity

$$F_c(y) Y^w(a,u)b = Y^w(F_c(y+u)a, u) F_c(y) b. \tag{9.42}$$

It also intertwines exchange.

Place $F_c(z_B)$ at each block of each partition coefficient. Horizontal differential-linearity extends it through open and closed direct images and every normal generator. Equation (9.42) intertwines adjacent residues on all Laurent multipliers. Exchange compatibility gives the same identity for every nonconsecutive merger with spectators.

Cuts are external products of these same maps. All derivations kill 1 and preserve the augmentation ideal, proving the scalar and vacuum assertions. The empty coefficient is fixed separately. The negative exponential constructs the inverse. Reductions in r commute with these maps, giving a compatible formal tower.

A constant state map has position-independent singleton. But $F_c(z)e_0 = e^{cz}e_0$ has nonzero linear coefficient ce_0. Its fiber also fails to commute with T: $[f_c, T] = cf_c\partial_1$, since $[\partial_2, T] = \partial_1$ and the next commutator vanishes. Conjugating the state-field formula replaces T by $T + c\partial_1$ and leaves its contraction unchanged. This proves (9.41).

At $r = 2$, $c = \hbar$, the creation defect has coefficient $\hbar e_0$ at u^1, and the interacting pair (e_0, e_0) has coefficient $\hbar e_0^2$. Both are nonzero in the degree-zero state complex. □

Thus a reconstruction equivariant for all coefficient automorphisms cannot recover these fixed constant-state fields with the supplied framing. Full inverse duality still recovers the geometric current contact system. The transported field with translation $T + c\partial_1$ is isomorphic to the original through f_c. It is a different translation-equipped presentation of that input.

The meromorphic current side detects the same distinction using its own state complex. Full duality reverses compositions, so identify a coalgebra automorphism Ψ with the algebra automorphism $(\Psi^{-1})^\sharp$. The inverse-transpose fiber of F_c on current generators is

$$g_c = (1 \otimes f_c^{-1})^\vee = e^{cD_2}.$$

It extends multiplicatively and through all joint localizations. It preserves the native product, differential and scalar maps, since D_2 commutes with the moment actions and is a product derivation. But

$$g_c T_B g_c^{-1} = T_B + cD_1, \qquad g_c Y_w(g_c^{-1}b, v)\, g_c^{-1} d = e^{-qcv} Y_w(b,v) d$$

for b of charge q.

For the closed states β, γ of the preceding section, $D_2\beta = -\gamma, T_B\beta = 0$, and $T_B\gamma = -2\beta$. At precision two this gives $g_\hbar\beta = \beta - \hbar\gamma$ and $[g_\hbar, T_B]\beta = -2\hbar\beta$. The creation difference is $-2\hbar v\beta$. Extracting $c_1^h c_2^h$ and evaluating $(x_1, x_2) = (0, 1)$ kills every boundary: a differential term either contains a constant ghost or has Cartan coefficient $[h, h] = 0$. The functional takes β to -2, and the coefficient of v in this defect to $4\hbar \neq 0$. The failure to preserve the fixed native field survives cohomology.

Example 9.31. The joint localization is also visible after a supported collision. At precision two take $w(u) = 1 - \hbar/u$, $s = z_1 - z_3$ and $t = z_2 - z_3$, with t invertible. On the ordered cut $G = \{1, 3\}$, $H = \{2\}$, the merged state is e_0^2 at z_3 and the spectator is e_0 at z_2. Their separation is $-t$. All higher jet actions vanish on these zero-jet states, so

$$\mathcal{W}_{G,H}(e_0^2 \otimes e_0) = w(-t)^2(e_0^2 \otimes e_0) = (1 + 2\hbar/t)(e_0^2 \otimes e_0).$$

The supported term lost by replacing this cross factor by the identity is

$$\frac{2\hbar}{t}[ds/s] \otimes (e_0^2 \otimes e_0). \tag{9.43}$$

The two states lie in different tensor factors. Their tensor is not the vanishing product e_0^3.

For $\eta_m = [ds/s^{m+1}]$, $m \geq 0$, the full normal module has

$$\eta_m \partial_s = (m+1)\eta_{m+1}, \qquad \eta_m s = \begin{cases} \eta_{m-1} & m > 0, \\ 0 & m = 0. \end{cases}$$

These actions give $(\eta_m \partial_s)s - (\eta_m s)\partial_s = \eta_m$. In particular η_0 and all its normal derivatives are retained. The Laurent coefficient ring and the state modules are free over R_2, so (9.43) is nonzero.

Before the merger its cross factor is $w(s-t)w(-t) = 1 - \hbar/(s-t) + \hbar/t$. On the normal test s^{-m} its difference from the identity has principal part

$$\hbar\left(\frac{2}{t}s^{-m} + \sum_{j=1}^{m-1} \frac{s^{j-m}}{t^{j+1}}\right).$$

Thus the comparison retains every normal order.

Already on two charge-one cores, the native tensor constraint is $w(x-y)^{-1} = 1 + \hbar/(x-y)$. Their individual discriminants are one, so the ordinary external tensor has no inverse of $x - y$. The joint tensor inverts $(x-y)^2$ and contains the required term. This domain difference precedes the supported coefficient difference in (9.43).

9.8 Coherent descent and passage to a formal limit

The preceding comparisons give actual maps on full collision diagrams. A global Hall-to-field comparison needs maps of this kind on overlaps, with their product coherences and orientations. A linear comparison of local complexes is only one part of that datum. The cover must see every finite configuration. Its totalization must retain the nonlinear equation governing the comparison.

A cover is a *Weiss cover* when every finite subset is contained in one of its members. An ordinary cover by two proper opens U, V need not have this property: choose $p \in U \setminus V$ and $q \in V \setminus U$. Their pair is contained in neither open.

Starting with a sufficiently fine polydisc basis subordinate to an ordinary cover, take all finite disjoint unions of basis polydiscs. For each finite subset choose pairwise disjoint small polydiscs around its distinct points. Their union contains the subset, proving the Weiss property. Such a union need not lie in a single member of the original cover.

The Ran space records finite nonempty subsets, with its presentation indexed by all nonempty finite sets and surjections. A surjection $I \twoheadrightarrow J$ gives the diagonal $X^J \to X^I$. There is no canonical map

$\mathrm{Sym}^n X \to \mathrm{Sym}^{n+1} X$ obtained by repeating a first entry: the representatives (p, q) and (q, p) give different multiplicities. The finite-set diagram retains the intended identifications.

Its configuration arity is independent of the dimension of a Čech simplex. One insertion lying in three cover opens already gives a two-simplex. A finite descent model therefore specifies its normalized simplicial diagram and its coefficient complexes separately from any arity or charge bound.

9.8.1 The degree and product in the comparison equation

Let C be a coaugmented conilpotent dg coalgebra and H a dg associative algebra over a field of characteristic zero. Write $\overline{C} = \mathrm{coker}(k \to C)$ and let $\overline{\Delta}_C$ be its reduced coproduct. On $\mathrm{Hom}(\overline{C}, H)$ use

$$Df = d_H f - (-1)^{|f|} f d_C, \qquad f * g = \mu_H(f \otimes g)\overline{\Delta}_C.$$

The tensor of maps uses the Koszul sign. The commutator of this convolution product has degree zero.

Lemma 9.32. *A degree-one map $t : \overline{C} \to H$ satisfying $Dt + t * t = 0$ determines a dg algebra map $\Omega C \to H$. These assignments are inverse. For $C = BA$, this encodes a comparison from the associative bar construction of A, with its higher components.*

Proof. Write $\Omega C = T(\overline{C}[-1])$. A generator $s^{-1}c$ has degree $|c| + 1$, so its image is $t(c)$. Take the cobar differential on generators to be

$$d(s^{-1}c) = -s^{-1}(d_C c) - \sum_{\overline{\Delta}c} (-1)^{|c_1|}(s^{-1}c_1)(s^{-1}c_2).$$

The coalgebra identities make its square zero.

The chain-map equation on this generator is

$$d_H t(c) + t(d_C c) + \sum_{\overline{\Delta}c} (-1)^{|c_1|} t(c_1)t(c_2) = 0,$$

which is exactly $Dt + t * t = 0$. The tensor algebra then extends the map uniquely, and checking its generators suffices. Conilpotence makes each reduced coproduct calculation finite. □

For a degree-zero map Θ in this same convolution algebra, $[\Theta, \Theta] = 0$. Its Maurer–Cartan expression cannot impose multiplicativity. For instance the unital chain map on $k[\epsilon]/(\epsilon^2)$ given by $f(\epsilon) = 1$ is not an algebra map. A shifted convolution convention is possible, but requires the corresponding bracket, differential and map degrees.

On a finite semicosimplicial diagram of dg Lie algebras, retain compatible polynomial forms on every simplex, as in the all-face model of Chapter 8. Differential and bracket are componentwise. A compatible family of relative Maurer–Cartan elements is then a Maurer–Cartan element of this total algebra. Both the equation and its face restrictions are componentwise equalities. Integration is a comparison of complexes. Transporting the nonlinear equation to ordinary simplex cochains needs the corresponding higher comparison maps.

Example 9.33. Linear integration need not preserve a Maurer–Cartan equation with the Alexander–Whitney product. Let V have an ordered basis of degrees $(1, 0, 0, 0)$ and zero differential. In $\mathrm{End}(V)$ put $A = E_{01}$ of degree one and $N = E_{12} + E_{23}$ of degree zero. Then $N^3 = 0$, $NA = 0$. Define

$$a(t) = A(1 + tN + t^2N^2/2), \qquad \alpha = a(t) + N\,dt.$$

The tensor differential gives $d\alpha = -a'(t)dt$, and $\alpha^2 = (a(t)N - Na(t))dt$. Since $a' = aN$ and $Na = 0$, $d\alpha + \alpha^2 = 0$.

Integration gives endpoint cochains $a_0 = A$, $a_1 = A + AN + AN^2/2$ and edge cochain N. The interval cochains have vertex idempotents v_0, v_1 and an edge e of degree one, with

$$dv_0 = -e, \quad dv_1 = e, \qquad v_0 e = e = e v_1, \quad v_1 e = e v_0 = e^2 = 0.$$

These define its Alexander–Whitney product.

In the signed tensor product with the matrix algebra, the edge curvature is

$$-(a_1 - a_0) + a_0 N - N a_1 = -AN^2/2 = -E_{03}/2 \neq 0.$$

Here the sign before the endpoint difference comes from the degree-one matrix coefficient.

Replacing the edge by $N + N^2/2 = e^N - 1$ makes the curvature zero and recovers $a_1 = a_0 e^N$. The missing quadratic term is a nonlinear comparison term. This calculation excludes the raw-integration inference. It leaves the compatible-form construction intact.

Parameter dependence also affects the differential of a local critical model. On $\mathbb{A}^1_x \times \mathbb{A}^1_t$ with $W = tx$, the operator

$$D = \hbar d_x + d_x W \wedge + d_t$$

satisfies $D^2(1) = -dx \wedge dt$. After inverting $\hbar$, the operator

$$D' = \hbar d_x + d_t + d_x W \wedge + \hbar^{-1} d_t W \wedge$$

has square zero: the connection form is closed for $\hbar d_x + d_t$, and its wedge square vanishes. The additional term is $\hbar^{-1} x\, dt$, so it need not preserve a prescribed $k[[\hbar]]$ lattice. A parameter-independent potential, a compatible flat connection, or a specified larger model is therefore part of the relative critical datum.

9.8.2 Finite comparisons and inverse limits

Proposition 9.34. *Let $f : C^\bullet \to D^\bullet$ be a map of semicosimplicial complexes, with $C^p = D^p = 0$ for $p > d$. If each f^p is a quasi-isomorphism, its finite totalization is a quasi-isomorphism. If a sequence of such comparisons forms an actual compatible tower $f_r : C_r \to D_r$, then it induces an equivalence of derived inverse limits. When the tower maps are degreewise surjective on both sides, ordinary inverse limits compute these derived limits.*

Proof. Filter the cone of the totalized map by semicosimplicial degree. Its associated graded pieces are shifts of $\mathrm{Cone}(f^p)$. There are finitely many, and each is acyclic, so the total cone is acyclic.

For a tower of complexes C_r, a model for its derived limit is

$$\mathrm{Cone}\Big(1 - s : \prod_r C_r \longrightarrow \prod_r C_r\Big)[-1],$$

where s uses the transition maps. Products of vector-space complexes preserve quasi-isomorphisms, so the tower of comparisons induces a quasi-isomorphism of these cones. If every transition map is degreewise surjective, then $1 - s$ is degreewise surjective: solve its components successively, choosing a lift at each step. Its kernel is the ordinary inverse limit. The short exact sequence identifies that kernel with the displayed derived-limit model. □

For an operation-preserving comparison the operations must also be compatible across this tower. In particular the tensor products, normal-support completions and continuous maps on both sides must be specified before passing to the limit. The proposition does not identify a limit of ordinary tensors

with a completed tensor. Independent finite comparisons without compatible transition maps do not define the tower to which it applies.

For a Hall-to-field application the finite inputs include oriented critical complexes on extension correspondences and their actual products. They also include local field complexes and comparison maps on a common configuration diagram. Local chain equations, product equations and higher overlap equations are separate conditions on those maps.

A cyclic contraction of finite models consists of maps

$$(M, d_M) \underset{p}{\overset{i}{\rightleftarrows}} (V, d), \qquad h : V \longrightarrow V[-1].$$

Its underlying contraction identities are

$$pi = 1, \qquad di = id_M, \qquad pd = d_M p, \qquad ip + dh + hd = 1.$$

Its pairing compatibility and its compatibility with the extension blocks are additional equations.

Finite bases turn supplied equations into exact matrix equalities. They do not construct the contractions or identify their coefficient spaces with vanishing cycles.

Problem 9.35. Construct a comparison from independently oriented framed critical Hall correspondences to an independently specified field recipient, on a common full collision diagram. Require charge, the empty and local units, all flag substitutions, both product orders, and the supported outer coefficients to be preserved. Construct its coherent overlap maps and compatible precision tower, with the stated completion and normal transfer. Determine when the resulting comparison is an equivalence and when a relative operation class obstructs it under the admitted changes of orientation and comparison.

The retained-exchange and marked-current constructions prove reconstruction on their declared classes. The failures of raw integration, an external tensor substitute, or an unframed field recipient delimit those constructions. They do not decide the question for every admissible field realization or every geometric Hall comparison.

Book III

Spectral forms and quantum symmetries

Chapter 10

Spectral forms and level-changing tensor products

10.1 Addition of levels

Let k be a field of characteristic zero. Let J be a Heisenberg current with singular product $J(z)J(w) \sim K(z-w)^{-2}$. In the ordinary tensor product of two such algebras, the current $J \otimes \mathbf{1} + \mathbf{1} \otimes J$ has level $2K$. The two self-contractions add, and the mixed contractions vanish. Thus this formula cannot define a vertex-algebra coproduct at a fixed nonzero level. It instead gives a map from level $K_1 + K_2$ to the tensor product of levels K_1 and K_2.

There is a further effect when translation changes the contraction. Over $k[\lambda, c_1, c_2]/(c_1^2, c_2^2)$, consider

$$P_c(z) = \frac{\lambda}{z-c} - \frac{\lambda}{z},$$

where a shifted denominator means its finite nilpotent expansion. Each separate field has $P_{c_i}(z) = \lambda c_i z^{-2}$, whereas

$$P_{c_1+c_2}(z) - P_{c_1}(z) - P_{c_2}(z) = \frac{2\lambda c_1 c_2}{z^3}. \tag{10.1}$$

The right side is nonzero in this coefficient ring. A tensor product receiving a coproduct from the summed level must retain it. We shall construct the extra contraction and prove its compatibility with fields, iterated coproducts and module actions.

The construction starts with an ordinary Hopf algebra, a translation derivation and a bilinear form depending on the separation of two points. The form determines both exchange and contraction. A Hopf automorphism preserving it specifies a level. The coproduct then composes levels in their given order, which need not be commutative. Formal translation levels recover additive parameters and the rational quantum affine example. Their comparison with Hall charge splitting requires an additional map between the two coefficient systems.

10.2 Rational fields and a spectral form

Throughout the algebraic construction, k has characteristic zero, R is a commutative k-algebra, and tensor products are over R. States lie in cohomological degree zero. For an R-module V, put

$$\mathcal{G}_r(V) = V[[z_1, \dots, z_r]][z_1^{-1}, \dots, z_r^{-1}, (z_i - z_j)^{-1} \mid 1 \le i < j \le r]. \tag{10.2}$$

Each element has a finite common denominator. The bound may depend on the input states. Products and iterates of fields are compared as expansions of the same element of this module.

Definition 10.1. A *quantum vertex algebra with rational exchange* consists of a flat module V, an operator T, and bilinear fields $Y(a,z)b \in V((z))$. Its vacuum $\mathbf{1}$ spans a free direct summand. They obey

$$Y(\mathbf{1},z) = \mathrm{id}, \qquad Y(a,z)\mathbf{1} = e^{zT}a, \qquad Y(Ta,z) = \partial_z Y(a,z) = [T, Y(a,z)].$$

They are weakly associative: a product and its consecutive iterate agree in their common expansion domain after clearing a finite power of the collision coordinate.

The exchange is an invertible $R[z, z^{-1}]$-linear map $\mathcal{S}(z)$ on $(V \otimes V)[z, z^{-1}]$, fixing both vacuum tensors. It satisfies

$$[T \otimes 1, \mathcal{S}(z)] = -\partial_z \mathcal{S}(z), \tag{10.3}$$

$$\mathcal{S}_{12}(u)\mathcal{S}_{13}(u+v)\mathcal{S}_{23}(v) = \mathcal{S}_{23}(v)\mathcal{S}_{13}(u+v)\mathcal{S}_{12}(u), \tag{10.4}$$

$$\mathcal{S}_{21}(-z)\mathcal{S}(z) = 1. \tag{10.5}$$

Here $\mathcal{S}_{21}$ conjugates $\mathcal{S}$ by the flip. Writing $Y(z)(a \otimes b) = Y(a, z)b$, the field identities are

$$\begin{aligned}&(z-w)^N Y(b,w)Y(a,z)c\\ &\quad = (z-w)^N Y(z)(1 \otimes Y(w))\big(\mathcal{S}(z-w)(a \otimes b) \otimes c\big),\end{aligned} \tag{10.6}$$

$$\mathcal{S}(z)(Y(w) \otimes 1) = (Y(w) \otimes 1)\mathcal{S}_{23}(z)\mathcal{S}_{13}(z+w). \tag{10.7}$$

The exponent N in the first identity depends on a, b, not on c. In the second identity $\mathcal{S}_{13}(z+w)$ is expanded at $w = 0$.

This is the rational, algebraic version of the exchange and hexagon formalism of Etingof and Kazhdan [57, §§1.3–1.4]. The formal version below uses completed tensors and these identities at each finite precision. The definition itself supplies no coproduct.

Let H be a flat Hopf algebra over R, with coproduct Δ, counit ε and bijective antipode s. Write $\Delta a = a_1 \otimes a_2$, with the finite sum understood. Repeated subscripts denote iterated coproducts. Let T be a derivation and coderivation with $\varepsilon T = 0$. It commutes with s. Write $a(z) = e^{zT}a$.

Definition 10.2. A *unitary spectral coquasitriangular form* is an R-bilinear map $\rho_z : H \otimes H \to R[z, z^{-1}]$ satisfying

$$\rho_z(1, a) = \varepsilon(a) = \rho_z(a, 1), \tag{10.8}$$

$$\rho_z(ab, c) = \rho_z(a, c_1)\rho_z(b, c_2), \tag{10.9}$$

$$\rho_z(a, bc) = \rho_z(a_1, c)\rho_z(a_2, b), \tag{10.10}$$

$$\rho_z(a_1, b_1)a_2(z)b_2 = b_1 a_1(z)\rho_z(a_2, b_2), \tag{10.11}$$

$$\rho_z(Ta, b) = \partial_z \rho_z(a, b) = -\rho_z(a, Tb), \tag{10.12}$$

$$\rho_z(a_1, b_1)\rho_{-z}(b_2, a_2) = \varepsilon(a)\varepsilon(b). \tag{10.13}$$

The commutation identity takes values in $H[[z]][z^{-1}]$. The reversed order in (10.10) fixes our convention.

The two multiplication rules alone do not impose exchange or unitarity. When $T = 0$ and the form is constant, the first four rules are the coquasitriangular identities and the last is cotriangularity. For two forms, use convolution $(f * g)(a, b) = f(a_1, b_1)g(a_2, b_2)$, whose unit is $\varepsilon \otimes \varepsilon$.

Lemma 10.3. *The convolution inverse of ρ_z is*

$$\bar{\rho}_z(a, b) = \rho_{-z}(b, a) = \rho_z(\mathsf{s}a, b) = \rho_z(a, \mathsf{s}^{-1}b). \tag{10.14}$$

Simultaneous application of the antipode preserves the form, and

$$\begin{aligned}\rho_u(a_1,b_1)\rho_{u+v}(a_2,c_1)\rho_v(b_2,c_2)\\ =\rho_v(b_1,c_1)\rho_{u+v}(a_1,c_2)\rho_u(a_2,b_2).\end{aligned}\tag{10.15}$$

The latter is an identity in $R[u,v,u^{-1},v^{-1},(u+v)^{-1}]$.

Proof. Apply the first multiplication rule to $\mathsf{s}(a_1)a_2$ and $a_1\mathsf{s}(a_2)$. The antipode identities make $\rho_z(\mathsf{s}-,-)$ a two-sided inverse. The second multiplication rule, applied to $b_2\mathsf{s}^{-1}(b_1)$ and $\mathsf{s}^{-1}(b_2)b_1$, gives the other inverse. Uniqueness and (10.13) give (10.14). Substituting $\mathsf{s}b$ then gives $\rho_z(\mathsf{s}a,\mathsf{s}b)=\rho_z(a,b)$.

Taylor expansion gives $\rho_v(a(u),c)=\rho_{u+v}(a,c)$. Pair the two sides of (10.11), at u, with c using ρ_v. Expanding their products gives exactly the two sides of (10.15), expanded at $u=0$. The expansion map from the displayed localization is injective, so the rational identity follows. □

Translation puts the commutation identity at two positions:

$$\rho_{z-w}(a_1,b_1)a_2(z)b_2(w)=b_1(w)a_1(z)\rho_{z-w}(a_2,b_2).\tag{10.16}$$

Before introducing contractions, this already gives a regular example.

Proposition 10.4. *On* H*, the fields* $Y^{\mathrm{reg}}(a,z)b=a(z)b$ *and exchange*

$$\mathcal{S}^{\mathrm{reg}}(z)(a\otimes b)=\rho_z(a_1,b_1)a_2\otimes b_2\bar\rho_z(a_3,b_3)$$

form a quantum vertex algebra. Its Hopf coproduct and counit preserve fields and exchange, using ordinary tensor fields on the target.

Proof. The derivation rule proves the vacuum, translation and associativity identities. Convolving (10.16) on the right with $\bar\rho$ proves locality. The finitely many scalar coefficients on a,b give its bound independently of the terminal state.

For a convolution-invertible form f, put $\Phi_f(a\otimes b)=f(a_1,b_1)a_2\otimes b_2f^{-1}(a_3,b_3)$. Coassociativity gives $\Phi_f\Phi_g=\Phi_{g*f}$. Equations (10.15) and (10.14) give Yang–Baxter and unitarity. Differentiating the scalar factors gives (10.3). The first multiplication rule and its reversed inverse rule $\bar\rho_z(ab,c)=\bar\rho_z(b,c_1)\bar\rho_z(a,c_2)$ give the hexagon, with the translated argument producing $z+w$.

The Hopf maps preserve multiplication and translation, hence fields. In the target exchange $\mathcal{S}^{\mathrm{reg}}_{13}\mathcal{S}^{\mathrm{reg}}_{24}$, the middle factors $\bar\rho_z(a_3,b_3)\rho_z(a_4,b_4)$ cancel. The result is $(\Delta\otimes\Delta)\mathcal{S}^{\mathrm{reg}}$. Applying both counits cancels all scalar factors. □

10.3 Levels, coactions and contractions

Definition 10.5. A *spectral level* is a Hopf automorphism η of H commuting with T and satisfying $\rho_z(\eta a,\eta b)=\rho_z(a,b)$. Products of levels mean composition: $\eta_1\eta_2=\eta_1\circ\eta_2$.

The levels form a group. It contains s^2. For a nilpotent $c\in R$, it also contains e^{cT}: this exponential is finite, and simultaneous translation leaves the form unchanged. No commutativity of levels is assumed.

Fix a level η. On the underlying module H, define

$$\delta_\eta(b)=b_2\otimes b_3\mathsf{s}^{-1}(\eta b_1).\tag{10.17}$$

For a right coaction we use parentheses, $\delta(b)=b_{(0)}\otimes b_{(1)}$, to distinguish its terms from Hopf coproduct components.

Lemma 10.6. *The map δ_η is a counital right H-coaction, sends 1 to $1 \otimes 1$, and satisfies*

$$\delta_\eta T = (T \otimes 1 + 1 \otimes T)\delta_\eta, \tag{10.18}$$

$$\delta_\eta(xb) = x_2 b_{(0)} \otimes x_3 b_{(1)} \mathsf{s}^{-1}(\eta x_1). \tag{10.19}$$

Proof. The two coaction composites are both

$$b_3 \otimes b_4 \mathsf{s}^{-1}(\eta b_2) \otimes b_5 \mathsf{s}^{-1}(\eta b_1).$$

Applying the last counit gives b. The coderivation rule and commutation of T with η, s give (10.18). Expanding $\delta_\eta(xb)$ gives the second formula because s^{-1} reverses products. □

For any right comodule M, set

$$A_z^M(x)m = m_{(0)} \rho_{-z}(m_{(1)}, x). \tag{10.20}$$

It is a finite Laurent polynomial for fixed x, m. Write A_z for $M = (H, \delta_\eta)$, and let $L(x)$ denote left multiplication by x.

Lemma 10.7. *On every right comodule,*

$$A_z(x)A_z(y) = A_z(yx), \tag{10.21}$$

$$\rho_{w-z}(y_1, x_1)A_z(x_2)A_w(y_2) = A_w(y_1)A_z(x_1)\rho_{w-z}(y_2, x_2). \tag{10.22}$$

For δ_η one further has

$$A_z(x)L(b(w)) = \rho_{w-z}(b_3, x_1)\rho_{w-z}(\mathsf{s}^{-1}\eta b_1, x_3)L(b_2(w))A_z(x_2), \tag{10.23}$$

$$A_z(x)1 = \varepsilon(x)1. \tag{10.24}$$

Proof. Coassociativity writes the first composite as $m_{(0)}\rho_{-z}(m_{(1)}, x)\rho_{-z}(m_{(2)}, y)$. The second multiplication rule combines the scalar factors with the product yx. Pairing the components of m, x, y in (10.15), at positions $0, z, w$, gives the second identity.

For the mixed identity, use (10.19) on $b(w)m$. Its last factor is $b_3(w)m_{(1)}\mathsf{s}^{-1}\eta b_1(w)$. The first multiplication rule separates these three factors. Translation changes the first and last parameters to $w - z$, leaving $A_z(x_2)$ in the middle. The vacuum identity follows from $\delta_\eta(1) = 1 \otimes 1$. □

Creation and annihilation now define the fields:

$$\begin{aligned} Y_\eta(a, z)b &= L(a_1(z))A_z(a_2)b \\ &= a_1(z)b_{(0)}\rho_{-z}(b_{(1)}, a_2) \\ &= a_1(z)b_2\rho_{-z}(b_3, a_2)\rho_{-z}(\mathsf{s}^{-1}\eta b_1, a_3). \end{aligned} \tag{10.25}$$

The finite coproducts give an element of $H((z))$. Denote this state module with its fields by V_η.

Theorem 10.8. *The fields* (10.25)*, vacuum* 1*, translation T and exchange*

$$\mathcal{S}_\eta(z)(a \otimes b) = a_{(0)} \otimes b_{(0)}\rho_z(a_{(1)}, b_{(1)}) \tag{10.26}$$

make V_η a quantum vertex algebra with rational exchange. Every finite ordered field product on a fixed state has a common germ in (10.2)*. Moreover,*

$$\delta_\eta(Y_\eta(a, z)b) = Y_\eta(a_{(0)}, z)b_{(0)} \otimes b_{(1)}a_{(1)}(z), \tag{10.27}$$

$$Y_\eta(a_{(0)}, z)b_{(0)}\rho_z(a_{(1)}, b_{(1)}) = e^{zT}Y_\eta(b, -z)a. \tag{10.28}$$

Proof. The unit rule for ρ gives the vacuum and creation identities. The Hopf counit splits off the vacuum summand as an R-module. Using (10.18) gives

$$[T, A_z(x)]b = -b_{(0)}\rho_{-z}(Tb_{(1)}, x) = \partial_z(A_z(x)b) = A_z(Tx)b.$$

Together with the coderivation rule for Δ, this proves both translation identities for Y_η.

Move annihilation past creation using (10.23). The product becomes

$$\begin{aligned} Y_\eta(a,z)Y_\eta(b,w) = {} & \rho_{w-z}(b_3, a_2)\rho_{w-z}(\mathsf{s}^{-1}\eta b_1, a_4) \\ & \cdot L(a_1(z)b_2(w))A_z(a_3)A_w(b_4). \end{aligned} \tag{10.29}$$

Expanding the inner field in the iterate instead gives

$$\begin{aligned} Y_\eta(Y_\eta(a,u)b,w) = {} & \rho_{-u}(b_4, a_3)\rho_{-u}(\mathsf{s}^{-1}\eta b_1, a_4) \\ & \cdot L(a_1(u+w)b_2(w))A_w(b_3)A_{w+u}(a_2). \end{aligned} \tag{10.30}$$

The annihilation exchange identifies these expressions at $u = z - w$.

On a fixed terminal state the remaining annihilations are finite Laurent polynomials. The other poles lie on $z = w$, so both expressions come from one element of $\mathcal{G}_2(H)$. Repeatedly moving annihilations to the right proves the same assertion for any finite number of fields.

For covariance, expand the coaction of (10.25). The middle factors are interchanged by

$$a_3(z)b_{(1)}\rho_{-z}(b_{(2)}, a_4) = \rho_{-z}(b_{(1)}, a_3)b_{(2)}a_4(z),$$

which is (10.16) at 0, z. The outer factors agree directly, proving (10.27).

For skew-symmetry, expand the two coactions on its left, separate products inside each form, and cancel inverse pairings. Invariance under η leaves

$$a_3(z)b_2\rho_z(a_2, b_1)\rho_z(\mathsf{s}^{-1}\eta a_1, b_3).$$

Direct expansion of the right side gives $b_1a_2(z)\rho_z(a_3, b_2)\rho_z(\mathsf{s}^{-1}\eta a_1, b_3)$. The commutation identity for ρ equates them.

Associativity and skew-symmetry identify the two orders of a pair of fields in their collision expansion. The scalar coefficients in (10.29) have a pole bound depending only on a, b. The exchanged side has finitely many analogous coefficients. Their maximum clears the collision denominators independently of the terminal state, proving (10.6).

Coassociativity of δ_η turns (10.15) and (10.13) into Yang–Baxter and unitarity for $\mathcal{S}_\eta$. Differentiation of the paired coaction components gives (10.3). Finally, the tail of a field in (10.27) is $b_{(1)}a_{(1)}(w)$. Pairing it with $c_{(1)}$ gives $\rho_z(b_{(1)}, c_{(1)})\rho_{z+w}(a_{(1)}, c_{(2)})$, the two factors of the hexagon. The unit rule fixes its vacuum tensors. □

10.4 The tensor product and its cross contraction

An *H-covariant field algebra* is a quantum vertex algebra with rational exchange and a right coaction preserving its vacuum and total translation, satisfying (10.27). Its exchange is given by (10.26), and every finite ordered product on a fixed state has a common rational germ. Morphisms preserve fields, vacuum, translation and coaction. The algebras V_η and the scalar algebra R with trivial coaction are examples.

For any such algebra V, covariance gives

$$A^V_z(x)Y_V(a,w) = \rho_{w-z}(a_{(1)}, x_2)Y_V(a_{(0)}, w)A^V_z(x_1). \tag{10.31}$$

Indeed, its left side pairs $v_{(1)}a_{(1)}(w)$ with x. The first multiplication rule separates precisely the two factors on the right.

On $V_1 \otimes V_2$ define

$$Y_{12}(a \otimes b, z)(c \otimes d) = Y_1(a, z)c_{(0)} \otimes Y_2(b_{(0)}, z)d\ \rho_{-z}(c_{(1)}, b_{(1)}), \tag{10.32}$$

$$\delta_{12}(a \otimes b) = a_{(0)} \otimes b_{(0)} \otimes b_{(1)}a_{(1)}. \tag{10.33}$$

The vacuum is $\mathbf{1} \otimes \mathbf{1}$ and translation is $T_1 + T_2$. Pairing the last factors of the coaction defines its exchange. Write $V_1 \boxtimes_\rho V_2$ for this field algebra on the ordinary tensor module. The symbol records its cross contraction at one point.

Theorem 10.9. *These formulas define a monoidal product on H-covariant field algebras, with unit R and the usual associativity map of tensor modules.*

Proof. The reversed product of tails is a coaction because Δ is multiplicative. The coderivation identity gives its translation rule. The two counits give the vacuum and creation identities. Differentiating the pairing proves translation for the fields.

In operator notation the tensor field is

$$Y_{12}(a \otimes b, z) = Y_1(a, z)A_z^{V_1}(b_{(1)}) \otimes Y_2(b_{(0)}, z).$$

Using (10.31), a product of two such fields is

$$\begin{aligned}\rho_{w-z}(c_{(1)}, b_{(2)})Y_1(a, z)Y_1(c_{(0)}, w)A_z^{V_1}(b_{(1)})A_w^{V_1}(d_{(1)}) \\ \otimes Y_2(b_{(0)}, z)Y_2(d_{(0)}, w).\end{aligned} \tag{10.34}$$

For the iterate, covariance in V_2 instead gives the annihilation $A_w^{V_1}(d_{(1)}b_{(1)}(z-w))$. The anti-action and translation rules turn this into $A_z^{V_1}(b_{(1)})A_w^{V_1}(d_{(1)})$. Associativity in the two factors therefore gives (10.34) again.

Finite coactions and the common-germ property in each factor bound all denominators. Repeating the calculation gives the common germ at every arity.

For covariance, the middle tails are related by

$$b_{(1)}(z)c_{(1)}\rho_{-z}(c_{(2)}, b_{(2)}) = \rho_{-z}(c_{(1)}, b_{(1)})c_{(2)}b_{(2)}(z).$$

This is (10.16). Multiplication on the left by $d_{(1)}$ and on the right by $a_{(1)}(z)$ gives the desired reversed tensor coaction.

To prove skew-symmetry, expand the exchange pairing of $b_{(1)}a_{(1)}$ with $d_{(1)}c_{(1)}$. Alongside the two factor fields its scalar coefficient is

$$\begin{aligned}\rho_{-z}(c_{(1)}, b_{(1)})\rho_z(b_{(2)}, c_{(2)})\rho_z(b_{(3)}, d_{(1)}) \\ \cdot \rho_z(a_{(1)}, c_{(3)})\rho_z(a_{(2)}, d_{(2)}).\end{aligned}$$

The first two factors cancel. Skew-symmetry in each factor leaves

$$e^{z(T_1+T_2)}\big(Y_1(c, -z)a_{(0)} \otimes Y_2(d_{(0)}, -z)b\big)\rho_z(a_{(1)}, d_{(1)}),$$

the reversed tensor field. The proof of Theorem 10.8 now gives locality and all exchange identities. Its pole bound uses the two factor bounds and their finitely many cross pairings.

For associativity of the tensor itself, the reversed tails give

$$A_z^{V_1\boxtimes_\rho V_2}(x) = A_z^{V_1}(x_2) \otimes A_z^{V_2}(x_1). \tag{10.35}$$

Both triple parenthesizations consequently have the field

$$\begin{aligned}Y_1(a, z)A_z^{V_1}(c_{(2)}b_{(1)}) \otimes Y_2(b_{(0)}, z)A_z^{V_2}(c_{(1)}) \\ \otimes Y_3(c_{(0)}, z),\end{aligned} \tag{10.36}$$

and their coaction ends in $c_{(1)}b_{(1)}a_{(1)}$. Thus the ordinary associator preserves the fields and coaction. Its pentagon is the tensor-module pentagon. A trivial coaction makes every cross factor a counit, proving the unit assertion. Colinear field maps preserve the formulas. □

The tensor is also an instance of Li and Sun's twisted tensor construction [110, Definition 2.2 and Theorem 2.4]. In their convention the twisting map is

$$\mathscr{T}(x)(b\otimes c)=c_{(0)}\otimes b_{(0)}\rho_x(c_{(1)},b_{(1)}).$$

The two nontrivial twisting identities follow by expanding the tails $d_{(1)}c_{(1)}(y)$ and $d_{(1)}b_{(1)}(y)$ in the covariance rule. They read

$$\mathscr{T}(x)(1\otimes Y_1(y))=(Y_1(y)\otimes 1)\mathscr{T}^{23}(x)\mathscr{T}^{12}(x+y),$$
$$\mathscr{T}(x)(Y_2(y)\otimes 1)=(1\otimes Y_2(y))\mathscr{T}^{12}(x-y)\mathscr{T}^{23}(x).$$

The unit identities are the two counits. Their field formula $(Y_1(z)\otimes Y_2(z))\mathscr{T}^{23}(-z)$ is exactly (10.32).

10.5 Ordered coproducts and covariant modules

The tensor target has now been constructed independently of a level. The coproduct into it is a shifted Hopf coproduct.

Theorem 10.10. *For any two levels, the map*

$$D_{\eta_1,\eta_2}=(\eta_2\otimes 1)\Delta : V_{\eta_1\eta_2}\longrightarrow V_{\eta_1}\boxtimes_\rho V_{\eta_2} \tag{10.37}$$

preserves fields, vacuum, translation, coaction and exchange. It is coassociative as a family:

$$(D_{\eta_1,\eta_2}\boxtimes_\rho 1)D_{\eta_1\eta_2,\eta_3}=(1\boxtimes_\rho D_{\eta_2,\eta_3})D_{\eta_1,\eta_2\eta_3}. \tag{10.38}$$

The Hopf counit is a covariant field map $\varepsilon : V_{\mathrm{id}}\to R$, and

$$(1\boxtimes_\rho \varepsilon)D_{\eta,\mathrm{id}}=1,\qquad (\varepsilon\boxtimes_\rho 1)D_{\mathrm{id},\eta}=1.$$

Proof. Let $D=(\eta_2\otimes 1)\Delta$ and $W=V_{\eta_1}\boxtimes_\rho V_{\eta_2}$. Expanding its target coaction gives

$$\delta_W D(a)=\eta_2 a_2\otimes a_5\otimes a_6\mathsf{s}^{-1}(\eta_2 a_4)\eta_2 a_3\mathsf{s}^{-1}(\eta_1\eta_2 a_1).$$

The middle inverse-antipode pair cancels, leaving $\eta_2 a_2\otimes a_3\otimes a_4\mathsf{s}^{-1}(\eta_1\eta_2 a_1)$. This is $(D\otimes 1)\delta_{\eta_1,\eta_2}(a)$. The order of the levels has not changed.

Colinearity intertwines annihilations. Using (10.35) and the multiplicativity of D for the original Hopf product gives

$$DY_{\eta_1\eta_2}(a,z)=\big(L(\eta_2 a_1(z))A^1_z(a_4)\otimes L(a_2(z))A^2_z(a_3)\big)D. \tag{10.39}$$

Here A^1, A^2 denote the two target annihilations.

Expanding $Y_W(Da,z)$ gives in its first slot the factors

$$A^1_z(\eta_2 a_2)A^1_z(a_6\mathsf{s}^{-1}(\eta_2 a_3))=A^1_z(a_6\mathsf{s}^{-1}(\eta_2 a_3)\eta_2 a_2).$$

Canceling the inverse-antipode pair gives the parenthesis in (10.39). Hence D preserves fields. It preserves the vacuum and translation by construction, and colinearity also preserves the paired exchange.

The two iterates in (10.38) both send a to

$$\eta_2\eta_3 a_1\otimes\eta_3 a_2\otimes a_3. \tag{10.40}$$

Theorem 10.9 identifies their field targets. At the identity level, $(\varepsilon\otimes 1)\delta_{\mathrm{id}}(b)=\varepsilon(b)1$. Applying ε to the field formula therefore gives $\varepsilon(Y_{\mathrm{id}}(a,z)b)=\varepsilon(a)\varepsilon(b)$. The same coaction identity preserves exchange. The family counits follow from the Hopf counits and $\varepsilon\eta=\varepsilon$. □

Proposition 10.11. *The Hopf counit $(H, \delta_\eta) \to R$, with trivial coaction on R, is colinear if and only if $\eta = \mathrm{id}$. Thus it is a covariant field counit precisely at the identity level.*

Proof. Colinearity is the identity $b_2 \mathsf{s}^{-1}(\eta b_1) = \varepsilon(b)1$. On $\mathrm{Hom}_R(H, H)$ use opposite convolution $(f \star_{\mathrm{op}} g)(b) = f(b_2)g(b_1)$. The displayed equation says that $\mathsf{s}^{-1}\eta$ is an inverse of id, whose inverse is already s^{-1}. Uniqueness gives $\eta = \mathrm{id}$. The converse and preservation of fields are part of Theorem 10.10. □

Preservation of fields alone has a different answer. Take $H = k[x]$ with x primitive, $T = 0$ and $\rho_z = \varepsilon \otimes \varepsilon$. The level $\eta x = 2x$ gives ordinary multiplication as its field. The polynomial augmentation preserves that field, but $\delta_\eta(x) = x \otimes 1 - 1 \otimes x$, so it is not colinear. Thus a scalar field character and a covariant counit are distinct.

For translation levels $\eta_i = e^{c_i T}$,

$$D_{c_1,c_2}(a) = a_{(1)}(c_2) \otimes a_{(2)}.$$

Its source parameter is $c_1 + c_2$, and its counit belongs to parameter zero. The first target factor is translated by the second parameter.

Definition 10.12. A *covariant module* for an H-covariant field algebra V is a flat right H-comodule M equipped with fields $Y_M(a, z)m \in M((z))$. These satisfy the vacuum and input-translation identities. Every finite ordered product on a fixed m has a common germ in $\mathcal{G}_r(M)$. Every consecutive field substitution or terminal module substitution expands that germ.

The terminal module is placed at 0. Covariance means

$$\delta_M(Y_M(a, z)m) = Y_M(a_{(0)}, z)m_{(0)} \otimes m_{(1)} a_{(1)}(z). \tag{10.41}$$

Morphisms preserve the fields and coaction. Write $\mathcal{M}_\eta$ for these modules over V_η.

No translation operator on the module itself is required here. The identity $Y_M(Ta, z) = \partial_z Y_M(a, z)$ concerns its state input. Its coaction supplies the annihilation operators (10.20) and the identity (10.31).

Theorem 10.13. *For covariant modules M, N over V_1, V_2, respectively, the formula*

$$\begin{aligned} &Y_{M \boxtimes_\rho N}(a \otimes b, z)(m \otimes n) \\ &\qquad = Y_M(a, z)m_{(0)} \otimes Y_N(b_{(0)}, z)n \, \rho_{-z}(m_{(1)}, b_{(1)}) \end{aligned} \tag{10.42}$$

defines a covariant $V_1 \boxtimes_\rho V_2$-module. Its coaction has tail $n_{(1)} m_{(1)}$. Pullback along (10.37) gives

$$\mathcal{M}_{\eta_1} \times \mathcal{M}_{\eta_2} \longrightarrow \mathcal{M}_{\eta_1 \eta_2}.$$

The disjoint union over levels is monoidal, with the ordinary associativity map and unit $R \in \mathcal{M}_{\mathrm{id}}$ acting through ε.

Proof. The action operator is $Y_M(a, z)A^M_z(b_{(1)}) \otimes Y_N(b_{(0)}, z)$. Moving annihilation past the next action produces (10.34), now acting on $M \otimes N$. Module associativity in each factor identifies it with the action of the tensor field. Repetition proves all consecutive substitutions. Every pairing on fixed inputs is finite, so the same calculation supplies the common denominator. The unit rule gives the vacuum identity, and $\partial_z A^M_z(x) = A^M_z(Tx)$ gives input translation.

Applying (10.41) in both slots leaves the same middle interchange (10.16) as in the tensor-algebra proof. It gives the stated reversed tail. For three modules both actions are (10.36) with Y_M, Y_N, Y_P in the respective slots. Their pullbacks coincide by (10.40).

Consequently the usual associator is colinear and intertwines actions. Its pentagon, the two family counits and the counital coactions give all monoidal coherences. The formulas preserve module morphisms. □

The theorem uses the specified coaction on each module. Without it, the annihilation A_z^M in the tensor formula has no definition. Likewise, the exchange on states does not by itself give an intertwiner exchanging arbitrary modules.

10.6 Polynomial fields and additive parameters

Let U be a free R-module and $H = \mathrm{Sym}_R(U)$, with primitive generators and $T(U) \subset U$. A bilinear kernel G_z on U extends uniquely by the unit and multiplication rules to

$$\rho_z(u_1 \cdots u_r, v_1 \cdots v_r) = \sum_{\sigma \in S_r} \prod_{i=1}^{r} G_z(u_i, v_{\sigma(i)}). \tag{10.43}$$

The pairing of unequal polynomial degrees is zero. Commutativity and cocommutativity give the form commutation rule. Unitarity is equivalent to $G_z(u,v) + G_{-z}(v,u) = 0$, and the translation rule is its restriction to U.

For a level preserving U, put

$$P_z^\eta(u,v) = \rho_{-z}(v - \eta v, u). \tag{10.44}$$

Since $\delta_\eta(v) = v \otimes 1 + 1 \otimes (v - \eta v)$, the primitive field is $Y_\eta(u,z)v = u(z)v + P_z^\eta(u,v)$.

For a basis (u_i) and the corresponding polynomial derivatives, all fields are given by Wick contraction:

$$Y_\eta(a,z)b = \mu(e^{zT} \otimes 1) \exp\left(\sum_{i,j} P_z^\eta(u_i, u_j)\, \partial_i \otimes \partial_j \right)(a \otimes b). \tag{10.45}$$

Each input pair uses finitely many derivatives and finitely many contractions. Selecting r contracted factors in the coproducts and pairing them gives (10.43). The exponential's $1/r!$ cancels the ordering of the same contractions. This proves the formula in every polynomial degree.

Take $H = R[x_1, x_2, \ldots]$, with $T x_n = n x_{n+1}$. An odd Laurent polynomial $g(z)$ defines the jet kernel

$$\rho_z(x_p, x_q) = \frac{(-1)^{q-1}}{(p-1)!(q-1)!} \partial_z^{p+q-2} g(z). \tag{10.46}$$

Indeed, multiplication by p in the $(p+1, q)$ entry is its z-derivative, and multiplication by q in the $(p, q+1)$ entry is its negative derivative. Oddness gives the unitary condition.

For nilpotent c and $\eta = e^{cT}$, the current $J = x_1$ has

$$P_c(z) = g(z-c) - g(z), \tag{10.47}$$

$$\mathcal{S}_c(z)(J \otimes J) = J \otimes J + \big(2g(z) - g(z+c) - g(z-c)\big) 1 \otimes 1. \tag{10.48}$$

The coproduct current is $J^{(1)}(c_2) + J^{(2)}$. Its two self-contractions are P_{c_1} and P_{c_2}, and its mixed contraction is

$$Q_{c_1,c_2}(z) = g(z - c_1 - c_2) - g(z - c_1) - g(z - c_2) + g(z). \tag{10.49}$$

Their sum is $P_{c_1+c_2}$. With $g(z) = \lambda/z$ and $c_1^2 = c_2^2 = 0$, this is precisely (10.1).

10.6.1 An independent simple pole

The double pole and simple pole of a current can vary independently. Use $R = k[\kappa, \lambda]$ and two families of primitive jets x_n, y_n, translated by $Tx_n = nx_{n+1}$ and $Ty_n = ny_{n+1}$. At the first jets the kernel and level are

$$G_z = \begin{pmatrix} -\lambda/(2z) & \kappa/z^2 \\ -\kappa/z^2 & 0 \end{pmatrix}, \qquad \eta x_n = -x_n + y_n, \quad \eta y_n = -y_n. \tag{10.50}$$

Rows and columns of G index its first and second inputs. Extend each entry by the jet derivative formula and then by (10.43).

Proposition 10.14. *The data* (10.50) *define a spectral form and a level. The polynomial submodule generated by the x_n is a quantum vertex subalgebra, with primitive contraction $\kappa z^{-2} + \lambda z^{-1}$ and exchange coefficient $-2\lambda z^{-1}$ on $x_1 \otimes x_1$.*

Proof. The kernel satisfies $G_z + G^{t}_{-z} = 0$. The matrix $E = \left(\begin{smallmatrix} -1 & 0 \\ 1 & -1 \end{smallmatrix}\right)$ of the level satisfies $E^{t} G_z E = G_z$ and commutes with jet translation. Thus it preserves the extended form.

Since $x_1 - \eta x_1 = 2x_1 - y_1$,

$$\rho_{-z}(2x_1 - y_1, x_1) = \frac{\kappa}{z^2} + \frac{\lambda}{z}, \qquad \rho_z(2x_1 - y_1, 2x_1 - y_1) = -\frac{2\lambda}{z}.$$

Differentiation gives all jet coefficients. Wick contraction leaves only x-jets in the output, and the same holds for exchange. This proves closure on all polynomial states. □

The full two-current algebra has the level-changing coproduct. Its one-current subalgebra need not be preserved by the target levels: already $\eta x_1 = -x_1 + y_1$ introduces the second current. The larger state space is therefore part of this coproduct construction. There is no vacuum-preserving R-linear field map from this singular member to the scalar algebra. Such a map would send the nonzero singular coefficients $\kappa 1$ and $\lambda 1$ to themselves, whereas scalar fields have no pole.

10.6.2 A noncommutative constant example

The order of the tails can also be seen without a spectral parameter. Over $k[\alpha]$, let H have basis 1, g, x, gx with

$$g^2 = 1, \quad x^2 = 0, \quad xg = -gx, \qquad \Delta g = g \otimes g, \quad \Delta x = x \otimes 1 + g \otimes x.$$

Its counit sends g to 1 and x to 0. Its antipode is $\mathsf{s}(g) = g$, $\mathsf{s}(x) = -gx$. Take $T = 0$ and, in the displayed basis, the constant form

$$(\rho(e_i, e_j)) = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & \alpha & \alpha \\ 0 & 0 & -\alpha & \alpha \end{pmatrix}. \tag{10.51}$$

The multiplication and coproduct tables give both multiplicativity rules, the commutation rule and unitarity.

The levels id and s^2 have

$$\delta_{\mathrm{id}}(x) = x \otimes g + (1 - g) \otimes gx,$$
$$\delta_{\mathsf{s}^2}(x) = x \otimes g - (1 + g) \otimes gx.$$

Substitution in (10.25), writing $*_\eta$ for the constant field product, gives

$$x *_{\mathrm{id}} x = \alpha(1-g), \qquad x *_{s^2} x = \alpha(1+g).$$

The Hopf counit sends the second product to 2α. Thus it fails to preserve fields at that level, while the identity-level counit remains multiplicative.

10.7 An integral formal family

Let $\hbar$ be formal. A completed field on V takes values in

$$V((z))^\wedge = \varprojlim_{N\geq 1} (V/\hbar^N V)((z)).$$

Each precision has a finite pole bound on fixed inputs. The inverse limit need not have one bound valid at every precision. All tensors in this section are $\hbar$-adically completed.

On $k[K][x_1, x_2, \dots][[\hbar]]$, with the same jet translation, consider the Wick propagator

$$P_K(z) = \frac{1}{\hbar}\Big(\frac{1}{z-K\hbar} - \frac{1}{z}\Big) = \frac{K}{z^2} + \frac{K^2\hbar}{z^3} + \frac{K^3\hbar^2}{z^4} + \cdots . \tag{10.52}$$

Every fraction is expanded in $\hbar$.

Proposition 10.15. *These fields define compatible quantum vertex algebras at all finite precisions, with Heisenberg classical limit of level K. There are coassociative field maps from level $K_1 + K_2$ given by*

$$x_n \longmapsto e^{K_2\hbar T} x_n \otimes 1 + 1 \otimes x_n. \tag{10.53}$$

The target has diagonal propagators P_{K_1}, P_{K_2}. Its contraction from the first current to the second is zero. The contraction from the second to the first is

$$Q^{21}_{K_1,K_2}(z) = \frac{1}{\hbar}\left(-\frac{1}{z} + \frac{1}{z+K_2\hbar} + \frac{1}{z-K_1\hbar} - \frac{1}{z+(K_2-K_1)\hbar}\right). \tag{10.54}$$

The family counit is polynomial augmentation at $K = 0$.

Proof. The numerator in (10.52) has zero constant term. In (10.54), both constant and linear terms cancel, so $Q^{21} = O(\hbar)$. The displayed fields are therefore integral.

At finite precision every propagator is a Laurent polynomial. The contraction operators in the Wick formula commute. A product and its iterate contract exactly the same pairs, giving weak associativity on a common germ.

Exchanging the two inputs replaces the contraction matrix by its transposed value at $-z$. The exponential of the difference gives exchange, with unitarity, Yang–Baxter and the hexagon following from addition of these commuting pair contractions. Jet differentiation gives translation. Reduction commutes with all these finite formulas. At $\hbar = 0$ the propagator is K/z^2 and the exchange is the identity.

The contraction of the image current is

$$P_{K_1}(z) + P_{K_2}(z) + Q^{21}_{K_1,K_2}(z - K_2\hbar) = P_{K_1+K_2}(z).$$

The four fractions cancel to give this identity. The state map preserves translation and polynomial multiplication, so Wick contraction proves preservation of every field and exchange.

Writing $Q_{A,B} = Q^{21}_{A,B}$, the same fractions give

$$Q_{K_1+K_2,K_3}(z) = Q_{K_1,K_3}(z - K_2\hbar) + Q_{K_2,K_3}(z),$$
$$Q_{K_1,K_2+K_3}(z) = Q_{K_1,K_2}(z + K_3\hbar) + Q_{K_1,K_3}(z).$$

Thus both triple parenthesizations have the same pairwise kernels. Both iterated state maps shift the first factor by $(K_2 + K_3)\hbar$, the second by $K_3\hbar$, and the third by zero. This proves coassociativity at every precision and in the limit. At $K = 0$ the fields are regular polynomial fields, whose augmentation gives the counit. □

Formally, the primitive spectral form behind this example is $1/(\hbar z)$, which is not integral. The difference between a state and its level translate supplies the power of $\hbar$ canceled in the field formula. Thus integrality of fields and integrality of the underlying form are separate assertions.

10.8 Divided states and the rational quantum affine algebra

The same distinction occurs for matrix currents. Write R for an auxiliary matrix, reserving R for the coefficient ring. If $L = 1 + \hbar\ell$ and $\rho_z(L_1(u), L_2(v)) = \mathsf{R}(z + u - v)$, the unit rule forces

$$\rho_z(\ell_1(u), \ell_2(v)) = \hbar^{-2}\big(\mathsf{R}(z + u - v) - 1\big). \tag{10.55}$$

Thus $\mathsf{R} = 1 + \hbar r + O(\hbar^2)$ with $r \neq 0$ gives a term $\hbar^{-1} r$ on divided generators. The formula follows simply by subtracting the identity in both inputs. An integral field construction must explain how its actual coaction avoids this nonintegral pairing.

Work now over $\mathbb{C}[[\hbar]]$ in rank two. Let P flip $\mathbb{C}^2 \otimes \mathbb{C}^2$, and normalize

$$\begin{aligned} \mathsf{R}(w) &= f(\hbar/w)(1 - \hbar P/w), \qquad f(0) = 1, \\ f\left(\frac{\hbar}{w + 2\hbar}\right) &= f(\hbar/w)(1 - \hbar^2/w^2). \end{aligned} \tag{10.56}$$

This determines the scalar series, whose first terms are

$$f(t) = 1 + \frac{1}{2}t + \frac{5}{8}t^2 + \frac{11}{16}t^3 + \frac{83}{128}t^4 + O(t^5).$$

The matrix is unitary and has crossing identity

$$\big((\mathsf{R}(w)^{t_1})^{-1}\big)^{t_1} = \mathsf{R}(w - 2\hbar)^{-1}, \tag{10.57}$$

where t_1 transposes the first auxiliary factor. The scalar equation and $(P^{t_1})^2 = 2P^{t_1}$ prove these identities.

Let H be the $\hbar$-adic dual Yangian generated by

$$L(u) = 1 + \hbar\ell(u), \qquad \ell(u) = \sum_{r \geq 1} \ell^{(r)} u^{r-1},$$

with RTT relation

$$\mathsf{R}(u - v) L_1(u) L_2(v) = L_2(v) L_1(u) \mathsf{R}(u - v). \tag{10.58}$$

Its generator relations are obtained after canceling $\hbar^2$. We use the torsion-free algebra with its topological PBW basis of ordered monomials in the $\ell_{ij}^{(r)}$.

The coproduct is matrix multiplication, $\Delta L = L^{[1]} L^{[2]}$, the counit sends L to 1, and $TL(u) = \partial_u L(u)$. The internal variables u, v are expanded at zero. The normalized matrix defines the spectral form on polynomial states after inverting $\hbar$. The multiplication rules determine every pairing.

The independent comparison object is the double Yangian at level c. It has a creation matrix $T^+(u) = 1 - \hbar \sum_{r \geq 1} t^{(-r)} u^{r-1}$ and an annihilation matrix $M_c(s) = 1 + \hbar \sum_{r \geq 1} m^{(r)} s^{-r}$. Each satisfies RTT, and their mixed relation is

$$\mathsf{R}(s - v + c\hbar/2) M_{c,1}(s) T_2^+(v) = T_2^+(v) M_{c,1}(s) \mathsf{R}(s - v - c\hbar/2). \tag{10.59}$$

It is expanded in s^{-1} and v and divided by $\hbar^2$ at the level of generator relations.

The vacuum module $\mathcal{V}_c$ is the completed quotient by the left ideal generated by the $m_{ij}^{(r)}$, so $M_c(s)\mathbf{1} = 1$. The PBW theorem identifies its states with the completed dual Yangian [89, Theorem 2.2 and §4.1]. The full scalar-matrix direction is retained.

Lemma 10.16. *For $\eta_c = e^{-c\hbar T}$, the fields and exchange of V_{η_c} are integral on the divided states H. The cross contractions in $V_{\eta_{c_1}} \boxtimes_\rho V_{\eta_{c_2}}$ are integral as well. They extend continuously to the $\hbar$-adic state completions.*

Proof. Let F_pH be PBW degree at most p before completion. The completed subalgebra B generated by entries of L consists, in PBW coordinates, of series $\sum_{r\ge 0}\hbar^r b_r$ with $b_r \in F_rH$. The divided RTT relations give $[\ell_{ij}^{(r)}, \ell_{kl}^{(s)}] \in F_1H + \hbar F_2H$. Consequently $[\ell_{ij}^{(r)}, B] \subset B$. Both antipodes, the coproduct and translation preserve B, and $(\eta_c - 1)B \subset \hbar B$.

The pairing has the estimate

$$\rho_z(F_pH, F_qH) \subset \hbar^{-\min(p,q)}\mathbb{C}((z))[[\hbar]]. \tag{10.60}$$

To see it, pair two products of L's and subtract the identity in every input. Each surviving product of factors $\mathsf{R} - 1$ must meet every input generator. At least $\max(p, q)$ such factors are needed. Dividing by $\hbar^{p+q}$ gives the estimate.

It makes pairings of B with polynomial states integral. For an input of degree q, the term $\hbar^r b_r$ contributes at order at least $r - q$, proving convergence.

The coaction has its second component in B. On a divided generator the inverse-antipode identity gives

$$\begin{aligned}\delta_{\eta_c}(\ell_{ij}(u)) = {} & \sum_{a,b} \ell_{ab}(u) \otimes L_{bj}(u)\mathsf{s}^{-1}\eta_c L_{ia}(u) \\ & + 1 \otimes \hbar^{-1}\sum_a L_{aj}(u)\mathsf{s}^{-1}(\eta_c - 1)L_{ia}(u).\end{aligned} \tag{10.61}$$

Both tails lie in B. For products, (10.19) introduces only commutators $[\ell_{ij}^{(r)}, b]$ and the divided translation difference in the second line, multiplied by elements of B. Induction gives $\delta_{\eta_c}(H) \subset H\widehat{\otimes}B$.

Annihilation therefore pairs B with a polynomial state. Exchange and cross contraction pair two tails in B. The estimate proves integrality and convergence for each operation. Its continuous $\hbar$-linear extension gives the completed fields. At finite precision a fixed tuple of states still has a finite Laurent pole bound. □

Theorem 10.17. *The map*

$$\Phi_c(\ell_{i_1j_1}^{(r_1)} \cdots \ell_{i_mj_m}^{(r_m)}) = (-1)^m t_{i_1j_1}^{(-r_1)} \cdots t_{i_mj_m}^{(-r_m)}\mathbf{1} \tag{10.62}$$

identifies $V_{e^{-c\hbar T}}$ with the rational quantum affine vertex algebra on $\mathcal{V}_c$. It preserves the vacuum, translation, all fields and exchange. On products of generating matrices the fields are

$$\begin{aligned}Y_c(L_1(u_1)\cdots L_m(u_m)\mathbf{1}, z) = {} & L_1(z + u_1)\cdots L_m(z + u_m) \\ & \cdot \big(M_{c,1}(z + u_1 + c\hbar/2)\cdots M_{c,m}(z + u_m + c\hbar/2)\big)^{-1}.\end{aligned} \tag{10.63}$$

Proof. PBW gives the continuous inverse replacing $t_{ij}^{(-r)}$ by $-\ell_{ij}^{(r)}$. Differentiation of generating series proves translation compatibility. Put $K_0(u, z) = A_z(L_0(u))$ and $w = z + u - v$. The antipode pairing rule and crossing identity give $\rho_w(\mathsf{s}^{-1}L_1, L_2) = \mathsf{R}(w - 2\hbar)^{-1}$. Substitution in (10.23) yields

$$K_0(u, z)\mathsf{R}_{01}(w + c\hbar)^{-1}L_1(v) = L_1(v)\mathsf{R}_{01}(w)^{-1}K_0(u, z), \qquad K_0(u, z)\mathbf{1} = 1. \tag{10.64}$$

Inverting (10.59) shows that $M_c(z + u + c\hbar/2)^{-1}$ satisfies this same recurrence.

Its value on the vacuum determines its action on every ordered product of creation matrices. Subtracting their constant terms and dividing by powers of $\hbar$ gives equality on polynomial PBW states. Lemma 10.16 extends it to the completion. The anti-action reverses annihilation order, giving (10.63).

The exchange uses four contractions of the two coaction tails. For an auxiliary matrix $F = \sum F' \otimes F''$, write $F^{\mathrm{lr}}(X \otimes Y) = \sum F' X \otimes Y F''$. The superscripts ll, rl, rr indicate the other left or right placements.

Expanding (10.26), using the two multiplicativity rules and the inverse-antipode pairing, gives the operator

$$\big(\mathsf{R}(w - c\hbar)^{\mathrm{lr}}\big)^{-1}\mathsf{R}(w)^{\mathrm{ll}}\mathsf{R}(w)^{\mathrm{rr}}\big(\mathsf{R}(w + c\hbar)^{-1}\big)^{\mathrm{rl}}. \tag{10.65}$$

The first inverse is an inverse of an operator. The last factor places an inverse matrix. These two operations differ.

Equivalently,

$$\begin{aligned}\mathcal{S}_c(z)\big(\mathsf{R}(w)^{-1}L_2(v)\mathsf{R}(w - c\hbar)L_1(u)\big)\\ = L_1(u)\mathsf{R}(w + c\hbar)^{-1}L_2(v)\mathsf{R}(w).\end{aligned} \tag{10.66}$$

The state factors remain separate throughout the auxiliary products. This is the exchange on the independently defined vacuum module [89, Theorem 4.1, equations (4.17)–(4.19)].

For n matrices in the first state and m in the second, replace each $\mathsf{R}(w)$ by $\prod_{i=1}^{n}\prod_{j=m}^{1}\mathsf{R}_{i,n+j}(z + u_i - v_j)$, in those orders. Appending one creation matrix inserts its four contractions by the multiplicativity rules. Yang–Baxter moves them past the existing factors, and crossing cancels the inverse pairs. Induction proves the same exchange identity for all polynomial states. Continuity proves it for completed states. □

The first divided coefficients make the level visible. Put $t = \hbar/w$ and let $A = P^{\mathrm{lr}}$, $B = P^{\mathrm{ll}}$, $C = P^{\mathrm{rr}}$, $D = P^{\mathrm{rl}}$. Then $A^2 = 2A$, $D^2 = 2D$ and $B^2 = C^2 = 1$. Writing $a = c + 2$, expansion of (10.65) is

$$\begin{aligned}Q_c(t) = {}& (1 + t^2 - c(c+2)t^3 + t^4)\\ &\cdot (1 + (t + at^2 + a^2t^3 + a^3t^4)A)\\ &\cdot (1 - t(B + C) + t^2BC)\\ &\cdot (1 + (t - ct^2 + c^2t^3 - c^3t^4)D) \pmod{t^5}.\end{aligned} \tag{10.67}$$

The scalar factor comes from

$$\frac{f(t)^2}{f(t/(1-ct))f(t/(1+ct))(1 - t^2/(1+ct)^2)}.$$

Let Q_j be the coefficient of t^j. For $X = \ell_1(u) + \ell_2(v)$ and $Z = \ell_1(u)\ell_2(v)$, subtracting the vacuum and one-state terms of L_1L_2 gives

$$\begin{aligned}\mathcal{S}_c(z)Z = Z + \hbar\left(\frac{Q_1Z}{w} + \frac{Q_2X}{w^2} + \frac{Q_3(1)}{w^3}\right)\\ + \hbar^2\left(\frac{Q_2Z}{w^2} + \frac{Q_3X}{w^3} + \frac{Q_4(1)}{w^4}\right) + O(\hbar^3).\end{aligned} \tag{10.68}$$

Direct multiplication gives $Q_1(1) = Q_2(1) = Q_1X = 0$ and, with $\Omega = P - \frac{1}{2}I$,

$$Q_3(1) = 2c^2\Omega, \qquad Q_4(1) = 4c^2\Omega. \tag{10.69}$$

For the 11, 11 input, applying both Hopf counits gives $c^2\hbar w^{-3} + 2c^2\hbar^2w^{-4} + O(\hbar^3)$. Replacing the level translate by the identity sets $c = 0$ and loses this coefficient already at first order on divided states.

The fields retain the classical affine level as well. For $J_{ij}(z) = Y_c(-\ell^{(1)}_{ij}, z)$, expansion of the mixed relation through $\hbar^2$ gives, modulo $\hbar$,

$$J_{ij}(z)J_{kl}(w) \sim \frac{c(\delta_{jk}\delta_{il} - \frac{1}{2}\delta_{ij}\delta_{kl})}{(z-w)^2} + \frac{\delta_{jk}J_{il}(w) - \delta_{il}J_{kj}(w)}{z-w}. \tag{10.70}$$

The difference of the two shifted R-matrices supplies the double pole, while their linear matrix terms supply the Lie bracket. The invariant form is $\operatorname{tr}(XY) - \frac{1}{2}\operatorname{tr}(X)\operatorname{tr}(Y)$. Its scalar-matrix direction has zero level. For $c \neq 0$, the 12, 21 double pole excludes a vacuum-preserving field map to the scalar vertex algebra.

10.9 The two rational coproducts

Under (10.62), the spectral coproduct is

$$\begin{aligned} \mathscr{D}_{c_1,c_2}L(u) &= L^{[1]}(u - c_2\hbar)L^{[2]}(u), \\ \mathscr{D}_{c_1,c_2}\ell(u) &= \ell^{[1]}(u - c_2\hbar) + \ell^{[2]}(u) + \hbar\ell^{[1]}(u - c_2\hbar)\ell^{[2]}(u). \end{aligned} \tag{10.71}$$

Its source is $\mathcal{V}_{c_1+c_2}$ and its target is $\mathcal{V}_{c_1} \boxtimes_\rho \mathcal{V}_{c_2}$. Theorems 10.10 and 10.17 prove that it preserves all fields and exchange. Both triple iterates give

$$L^{[1]}(u - (c_2 + c_3)\hbar)L^{[2]}(u - c_3\hbar)L^{[3]}(u).$$

The counit belongs to level zero.

The field of its generating state is

$$\begin{aligned} Y_{12}(\mathscr{D}L(u), z) = {} & L^{[1]}(u + z - c_2\hbar)L^{[2]}(u + z) \\ & \cdot M^{[2]}_{c_2}(u + z + c_2\hbar/2)^{-1}M^{[1]}_{c_1}(u + z + c_1\hbar/2)^{-1}. \end{aligned} \tag{10.72}$$

This follows by inserting the annihilation recurrence into (10.39). The first creation matrix is shifted by $-c_2\hbar$, while its annihilation argument has no such shift. The cross contraction accounts for the difference.

The ordinary double Yangian has a different pair of half-coproducts. With central generator C, put $C_1 = C \otimes 1$, $C_2 = 1 \otimes C$. In the normalization of (10.59), they are

$$\Delta_{\mathrm{DY}}L(u) = L^{[1]}(u - \hbar C_2/4)L^{[2]}(u + \hbar C_1/4), \tag{10.73}$$

$$\Delta_{\mathrm{DY}}M(s) = M^{[1]}(s + \hbar C_2/4)M^{[2]}(s - \hbar C_1/4), \tag{10.74}$$

$$\Delta_{\mathrm{DY}}C = C_1 + C_2.$$

These opposite quarter shifts are part of the Hopf structure [89, §4.1].

Scalar recentering of each level cannot identify the families. If $L(u)$ is replaced by $L(u + b(c)\hbar)$ at level c, the two spectral shifts become $b(c_1) - b(c_1 + c_2) - c_2$ and $b(c_2) - b(c_1 + c_2)$. Equality with the quarter shifts, first at $c_1 = 0$ and then at $c_2 = 0$, would require simultaneously

$$b(c) - b(0) = -3c/4, \qquad b(c) - b(0) = -c/4.$$

A tensor-dependent translation does give an exact comparison.

Definition 10.18. Adjoin to the double Yangian an element d with

$$[d, C] = 0, \qquad [d, L(u)] = \partial_u L(u), \qquad [d, M(s)] = \partial_s M(s),$$

and make d primitive, with counit zero and antipode $-d$. This is the *translation extension* of the double Yangian.

Simultaneous differentiation leaves every spectral difference in the defining relations unchanged, so it defines a derivation. Differentiating the two half-coproducts proves the coderivation rule. The extension therefore exists as a completed Hopf algebra. On $\mathcal{V}_c$, C acts by c and d by the state translation T.

Theorem 10.19. *In the translation extension,*

$$F = \exp\left(\frac{\hbar}{4}(3d \otimes C + C \otimes d)\right) \tag{10.75}$$

is an invertible normalized Hopf two-cocycle. The conjugated coproduct $\Delta_{\mathrm{sp}} = F^{-1}\Delta_{\mathrm{DY}}F$ satisfies

$$\Delta_{\mathrm{sp}} L(u) = L^{[1]}(u - \hbar C_2) L^{[2]}(u), \tag{10.76}$$

$$\Delta_{\mathrm{sp}} M(s) = M^{[1]}(s - \hbar C_2/2) M^{[2]}(s - \hbar C_1/2). \tag{10.77}$$

It restricts to the unextended double. On vacuum state spaces,

$$F_{c_1,c_2} = e^{\hbar(3c_2 T_1 + c_1 T_2)/4}, \qquad \Delta^{\mathrm{vac}}_{\mathrm{DY},c_1,c_2} = F_{c_1,c_2}\mathscr{D}_{c_1,c_2}. \tag{10.78}$$

This is a continuous invertible comparison preserving the vacuum and total translation.

Proof. A normalized two-cocycle satisfies

$$F_{12}(\Delta \otimes 1)F = F_{23}(1 \otimes \Delta)F, \qquad (\varepsilon \otimes 1)F = (1 \otimes \varepsilon)F = 1.$$

Here C, d commute and are primitive. For $f = (3d \otimes C + C \otimes d)/4$, all placements commute and $f_{12} + (\Delta \otimes 1)f = f_{23} + (1 \otimes \Delta)f$. Exponentiation proves the two-cocycle equation and counitality. The same argument applies to F^{-1}.

Conjugation by F^{-1} shifts first-factor arguments by $-3\hbar C_2/4$ and second-factor arguments by $-\hbar C_1/4$. Applied to the two opposite quarter shifts, it gives (10.76)–(10.77). Their coefficients lie in the unextended tensor square. The twisted antipode is conjugated by $m(1 \otimes \mathsf{s}_{\mathrm{DY}})(F^{-1}) = e^{\hbar C d}$. Its action translates both matrix arguments by $\hbar C$, so it also preserves the unextended algebra.

Both half-coproducts make the annihilation matrices act as the identity on $\mathbf{1} \otimes \mathbf{1}$ and therefore descend to the vacuum quotients of source level $c_1 + c_2$. There F acts as the exponential in (10.78). Conjugation proves the state identity on every creation generator, then on their products and their PBW completion. □

The comparison transports the field tensor as well. Put $p = 3\hbar c_2/4$, $q = \hbar c_1/4$ and $\tau_a = e^{aT}$. On the same completed tensor module, set

$$Y^F(a \otimes b, z)(x \otimes y) = Y_{c_1}(a, z)x_{(0)} \otimes Y_{c_2}(b_{(0)}, z)\, y\, \rho_{-z-p+q}(x_{(1)}, b_{(1)}), \tag{10.79}$$

$$\delta^F(a \otimes b) = a_{(0)} \otimes b_{(0)} \otimes \tau_{-q}(b_{(1)})\tau_{-p}(a_{(1)}). \tag{10.80}$$

The shifts are $\hbar$-adic Taylor series. The pairings use the integral tails of Lemma 10.16.

Proposition 10.20. *The map $F_{c_1,c_2} = \tau_p \otimes \tau_q$ is an isomorphism from $\mathcal{V}_{c_1} \boxtimes_\rho \mathcal{V}_{c_2}$ to the field tensor (10.79)–(10.80). It preserves fields, vacuum, translation, coaction and exchange. In this target, the ordinary double-Yangian vacuum coproduct is a field homomorphism. The comparison is coherent on all iterated tensors.*

Proof. Translation is a field automorphism, and $\delta\tau_a = (\tau_a \otimes \tau_a)\delta$. The form satisfies $\rho_w(\tau_a x, \tau_b y) = \rho_{w+a-b}(x, y)$. Inserting the translated inputs in (10.79) therefore changes its scalar factor to $\rho_{-z-p+q+p-q} = \rho_{-z}$. All other factors are the translates of the original fields. Thus $Y^F(Fv, z)Fw = FY_{12}(v, z)w$. The same substitution gives $\delta^F F = (F \otimes 1)\delta_{12}$, and hence preservation of the paired exchange.

These identities hold at every precision, so also on the completion.

Composing with $\mathscr{D}$ and using (10.78) gives the ordinary vacuum coproduct as a field map. On three factors both comparison composites are

$$\exp\left(\frac{\hbar}{4}\big(3(c_2 + c_3)T_1 + (c_1 + 3c_3)T_2 + (c_1 + c_2)T_3\big)\right).$$

This is the cocycle identity on state spaces. It gives the associativity coherence, and iteration gives all higher coherences. The scalar unit has level and translation zero, so both unit comparisons are the identity. □

10.9.1 Obstructions without translation

The translation extension changes the admissible comparison maps. This can be detected by a central series in the unextended double. Define the quantum determinant

$$Q(u) = L_{11}(u)L_{22}(u - \hbar) - L_{21}(u)L_{12}(u - \hbar). \tag{10.81}$$

Its coefficients are central [89, Proposition 2.8]. For commuting RTT matrices it is multiplicative: the identity $A^{(2)}L_1(u)L_2(u - \hbar) = A^{(2)}Q(u)$, with $A^{(2)} = (1 - P)/2$, applied successively to the two matrices proves it.

Proposition 10.21. *At a nonzero level pair (c_1, c_2), no invertible element of the unextended double-Yangian tensor square conjugates the spectral coproduct to the ordinary double-Yangian coproduct. The obstruction is already visible modulo $\hbar^3$.*

Proof. The images of the quantum determinant are

$$\Delta_{\mathrm{sp}}Q(u) = Q^{[1]}(u - c_2\hbar)Q^{[2]}(u),$$
$$\Delta_{\mathrm{DY}}Q(u) = Q^{[1]}(u - c_2\hbar/4)Q^{[2]}(u + c_1\hbar/4).$$

Both are central in the target and hence fixed by conjugation. Since $L = 1 + \hbar\ell$, $\Delta_{\mathrm{DY}}Q - \Delta_{\mathrm{sp}}Q$ begins with

$$\frac{\hbar^2}{4}\Big(3c_2\,\partial_u \operatorname{tr}\ell^{[1]}(u) + c_1\,\partial_u \operatorname{tr}\ell^{[2]}(u)\Big). \tag{10.82}$$

PBW makes the two trace derivatives independent, so this coefficient is nonzero for every nonzero level pair.

For the scalar test $c_1 = 0$, $c_2 = 1$, take the creation-algebra characters $L^{[1]}(u) = (1 + \hbar u)I_2$, $L^{[2]}(u) = I_2$. The coefficient in (10.82) is $3/2$. Only the creation subalgebras are evaluated in this test. The characters need not extend to the annihilation matrices. □

At $(c_1, c_2) = (0, 0)$ the state coproducts agree. The determinant-one quotient removes this particular central coefficient. In the translation extension, $[d, Q(u)] = \partial_u Q(u)$, so this coefficient is no longer central and the explicit twist applies.

A different restriction concerns the completion. Let

$$A_c = U(\mathfrak{sl}_2[t, t^{-1}] \oplus \mathbb{C}K)/(K - c),$$

where $[X_m, Y_n] = [X, Y]_{m+n} + m\delta_{m,-n}\,\mathrm{tr}(XY)K$. An element of A_c is a finite sum of PBW monomials. No completion in the mode index is included. Take $H = \mathrm{diag}(1, -1)$, $E = E_{12}$ and $F = E_{21}$.

Lemma 10.22. *For $(\alpha, \beta) \neq (0, 0)$, no $a \in A_{c_1} \otimes A_{c_2}$ satisfies, for every $r \geq 1$,*

$$[a, H_{-r} \otimes 1 + 1 \otimes H_{-r}] = r(\alpha H_{-r-1} \otimes 1 + \beta 1 \otimes H_{-r-1}).$$

Proof. Use PBW symmetrization to express a as a finite polynomial in loop generators in each slot. Choose r larger than the absolute value of every mode index in it. Then its commutator with H_{-r} has no central term. The adjoint action sends an E_n or F_n to a root vector of mode $n - r$ and sends H_n to zero. Symmetrization intertwines this action, so it preserves PBW bidegree.

Only bidegrees $(1, 0)$ and $(0, 1)$ could produce the right side. Their brackets are root vectors, while that side is a nonzero combination of independent Cartan vectors. This is impossible. □

Consequently, suppose a flat determinant-one quantization has classical reduction A_c and translation $[d, H_{-r}] = rH_{-r-1}$. A twist $1 + \hbar a + O(\hbar^2)$ in its purely $\hbar$-adic tensor completion would give the forbidden identity with $\alpha = 3c_2/4$, $\beta = c_1/4$. This consequence assumes the specified classical reduction. A completion allowing infinitely many modes in a has a different admissible class.

Such mode completions occur in the double-Yangian cocycle of Enriquez and Felder [54, §1, Theorem 3.1 and Proposition 5.1]. Their algebra includes a derivation and central element, and their cocycle compares Drinfeld's coproduct with the Yangian coproduct. Those two named structures require their own current and completion comparison with (10.73)–(10.77). The finite-mode lemma does not exclude their construction.

10.9.2 *The field tensor remains nontrivial*

Even when the two state coproducts agree at level zero, their field target retains a mixed exchange. Let $j_1(a) = a \otimes \mathbf{1}$, $j_2(b) = \mathbf{1} \otimes b$ in $W = \mathcal{V}_0 \boxtimes_\rho \mathcal{V}_0$. Write its exchange as $1 + \hbar\sigma_W(z) + O(\hbar^2)$. Use the ordered PBW decomposition to project to degree one in each state factor, denoting this projection by $\mathrm{pr}_{1,1}$.

Proposition 10.23. *There is no exchange-preserving isomorphism $W \to \mathcal{V}_0 \otimes \mathcal{V}_0$ equal to the identity modulo $\hbar$, when the target has ordinary tensor fields and exchange.*

Proof. The four first-order contractions in the mixed exchange give

$$\mathrm{pr}_{1,1}\,\sigma_W(z)(\ell_{11}^{(1)} \otimes \ell_{12}^{(1)}) = z^{-1}\ell_{12}^{(1)} \otimes (\ell_{22}^{(1)} - \ell_{11}^{(1)}). \tag{10.83}$$

The inputs denote $j_1(\ell_{11}^{(1)})$ and $j_2(\ell_{12}^{(1)})$ in the two copies of W. In auxiliary notation their coefficient is $P^{\mathrm{lr}}Z - PZ - ZP + P^{\mathrm{rl}}Z$, where $Z = \ell_1\ell_2$. Its 11, 12 entry is the displayed tensor.

Equivalently, the adjoint Casimir calculation is

$$-\sum_{i,j}[E_{ij}, E_{11}] \otimes [E_{ji}, E_{12}] = E_{12} \otimes (E_{22} - E_{11}).$$

PBW makes (10.83) nonzero. The ordinary tensor exchange is the identity on these mixed inputs.

If $G = 1 + \hbar g + O(\hbar^2)$ intertwined the exchanges, the coefficient of $\hbar$ in $\mathcal{S}_{\mathrm{ord}}(G \otimes G) = (G \otimes G)\mathcal{S}_W$ would give $\sigma_{\mathrm{ord}} = \sigma_W$. The nonzero mixed coefficient contradicts this equality. □

Thus the translation twist compares two presentations of the same level family with their specified field tensors. It does not identify that tensor with the ordinary field tensor. The obstruction just proved fixes the identity classical state map. It makes no assertion about every possible intrinsic comparison.

10.10 Differentials and the spectral tensor of a line

A spectral tensor can also change a differential. For a dg algebra (A, Q), an element $r(z)$ of degree one with $Q_{\otimes} r + r^{2} = 0$ defines the differential $Q_{\otimes} + [r(z), -]$ on the completed tensor square containing r. Here $Q_{\otimes} = Q \otimes 1 + 1 \otimes Q$, with the Koszul sign in the second summand. This construction has a different degree from the degree-zero exchange operators considered above. The gauge-theory example makes the relation between the two degrees explicit [46, §§6.5, 7.2].

Let $\mathfrak{g}$ be a finite-dimensional complex Lie algebra, with basis t_a and structure constants $[t_a, t_b] = f_{ab}^{\ c} t_c$. Fix an invariant symmetric form κ and a scalar $k \in \mathbb{C}$. In this section k denotes the level. The ground field is $\mathbb{C}$. Repeated Lie algebra indices are summed.

Write

$$h_{\mathfrak{g}}(x, y) = \tfrac{1}{2} \operatorname{tr}(\operatorname{ad} x \ \operatorname{ad} y).$$

For a simple algebra with long roots of square length two, $h_{\mathfrak{g}} = h^{\vee} \kappa$.

Definition 10.24. The graded algebra $\mathcal{Y}_k$ has even generators B_n^a of cohomological degree zero and odd generators $c_{a,n}$ of degree one, for $n \geq 0$, with relations

$$[B_n^a, B_m^b] = f_{ab}^{\ c} B_{n+m}^c, \qquad [B_n^a, c_{b,m}] = -f_{ac}^{\ b} c_{c,n+m}, \qquad c_{a,n} c_{b,m} = -c_{b,m} c_{a,n}. \tag{10.84}$$

Define a degree-one derivation on the generators by

$$Q c_{a,n} = \frac{1}{2} f_{bc}^{\ a} \sum_{m+p=n-1} c_{b,m} c_{c,p}, \tag{10.85}$$

$$Q B_n^a = nk\kappa_{ab} c_{b,n-1} + \frac{1}{2} f_{ab}^{\ c} \sum_{m+p=n-1} (c_{b,m} B_p^c + B_p^c c_{b,m}). \tag{10.86}$$

Every index in a sum is nonnegative, so the sum is empty for $n = 0$. The weights $\operatorname{wt}(B_n) = n$ and $\operatorname{wt}(c_n) = n + 1$ make Q weight preserving. The algebra consists of finite polynomial expressions in these modes.

The symmetric ordering in (10.86) determines the level seen by a cochain model. Moving every ghost to the left gives

$$Q B_n^a = n(k\kappa + h_{\mathfrak{g}})_{ab} c_{b,n-1} + f_{ab}^{\ c} \sum_{m+p=n-1} c_{b,m} B_p^c. \tag{10.87}$$

Indeed, each of the n commutators contributes $-\frac{1}{2} f_{ab}^{\ c} f_{cd}^{\ b} c_{d,n-1} = (h_{\mathfrak{g}})_{ad} c_{d,n-1}$.

Proposition 10.25. *The derivation Q respects* (10.84) *and satisfies $Q^{2} = 0$.*

Proof. Put $\mathfrak{g}_{+} = \mathfrak{g}[u]$ and $\mathfrak{g}_{-} = u^{-1}\mathfrak{g}[u^{-1}]$. For an invariant form b, let U_b be the enveloping algebra of the central extension

$$[x u^{p}, y u^{q}] = [x, y] u^{p+q} + p \delta_{p+q,0} b(x, y) \mathbf{1},$$

with the central element specialized to 1. Take $b = k\kappa + h_{\mathfrak{g}}$ and $M_b = U_b \otimes_{U(\mathfrak{g}_{-})} \mathbb{C}$. PBW identifies M_b with $U(\mathfrak{g}_{+})$.

Use restricted Chevalley cochains for $\mathfrak{g}_-$ on this module. The ghost dual to $t_a u^{-n-1}$ is $c_{a,n}$. Each cochain is a finite sum of homogeneous energy components. The energies of B_n and c_n are n and $n+1$, respectively. Write $\mathfrak{g}_-^\circ$ for this restricted graded dual.

Choose the ghost sign for which $dc^x = \frac{1}{2}c^x([y,z])c^y c^z$ and the action term is $-\sum_x c^x \rho(x)$. Here x, y, z range over a homogeneous basis of $\mathfrak{g}_-$, c^x is the dual ghost, and ρ is its action on M_b. On $B = t_a u^n$, the identity

$$[t_b u^{-m-1}, B] = f_{ba}^{\ c} t_c u^{n-m-1} - (m+1)\delta_{m,n-1} b_{ba}$$

gives exactly (10.87) after acting on the inducing vector. The ghost differential gives (10.85).

This generator calculation extends to products as follows. Transport left multiplication by B to $a_B \otimes 1 + 1 \otimes L_B$ on $\Lambda(\mathfrak{g}_-^\circ) \otimes U(\mathfrak{g}_+)$, where a_B is the coadjoint derivation for the projected bracket $[B,-]_-$. The projected Jacobi identity gives

$$[d_{\mathrm{CE}}, a_B] = -\sum_x \varepsilon(c^x) a_{[x,B]_+}.$$

Here $\varepsilon(c^x)$ is exterior multiplication by c^x. Together with the central term in the displayed bracket, this says $[d, L_B] = L_{QB}$. For ghost multiplication the same equation is the exterior Leibniz rule.

Induction on word length now identifies the transported differential with Q on every PBW monomial. The multiplication operators already satisfy the relations, so Q respects them. The Chevalley identity $d^2 = 0$ gives $Q^2 = 0$. □

Let $TB_n = nB_{n-1}$ and $Tc_n = nc_{n-1}$. This degree-zero derivation commutes with Q and is locally nilpotent on the polynomial algebra. Put $\tau_z = e^{zT}$ and let ϵ kill every generator. For each generator a, define

$$\Delta_z(a) = \tau_z(a) \otimes 1 + 1 \otimes a, \qquad r(z) = \sum_{n,m \geq 0} (-1)^n \binom{n+m}{n} \frac{c_{a,n} \otimes B_m^a - B_n^a \otimes c_{a,m}}{z^{n+m+1}}. \tag{10.88}$$

The coproduct extends multiplicatively.

The tensor r lies in $(\mathcal{Y}_k \otimes \mathcal{Y}_k)((z^{-1}))$. This Laurent completion consists of series bounded above in powers of z, with algebraic coefficients. It is larger than algebraic scalar extension by $\mathbb{C}((z^{-1}))$.

Theorem 10.26. *The spectral data satisfy*

$$Q_\otimes r + r^2 = 0, \qquad \Delta_z Q = (Q_\otimes + [r,-])\Delta_z, \tag{10.89}$$

$$(\epsilon \otimes 1)\Delta_z = 1, \qquad (1 \otimes \epsilon)\Delta_z = \tau_z, \tag{10.90}$$

$$(T \otimes 1 + 1 \otimes T)r = 0, \qquad r(-z) = Pr(z). \tag{10.91}$$

In the first counit identity, 1 denotes the identity map. The flip P uses the cohomological sign. Coassociativity takes the form

$$(\Delta_w \otimes 1)\Delta_z = (1 \otimes \Delta_z)\Delta_{z+w}.$$

The two iterated twists both equal $r_{12}(w) + r_{13}(z+w) + r_{23}(z)$.

The shifted kernels are first evaluated on $0 < |w| < |z| < |z+w|$, with sufficiently small auxiliary jet variables. Their fixed-mode coefficients then continue rationally to $zw(z+w) \neq 0$. The assertion concerns these kernels and their finite products. It does not extend translation to every formal Laurent series.

Proof. Translation, the two counits and coassociativity follow on generators from $\tau_z \tau_w = \tau_{z+w}$. The binomial coefficients in (10.88) give

$$(\Delta_w \otimes 1)r(z) = r_{13}(z+w) + r_{23}(z), \qquad (1 \otimes \Delta_z)r(z+w) = r_{12}(w) + r_{13}(z+w).$$

They also give (10.91). For fixed output modes, the first shifted sum converges normally when $|w| < |z|$, and the second when $|z| < |z + w|$. Indeed, their tails are polynomial in the summation index times the corresponding geometric ratio. The stated common domain permits both substitutions and their finite products before rational continuation.

For the two differential identities, write $B^a(s) = \sum_{n\geq 0} B^a_n s^{-n-1}$ and similarly for $c_a(s)$. Then $r(z)$ is the coefficient of s^{-1} in $c_a(s) \otimes B^a(s+z) - B^a(s) \otimes c_a(s+z)$, expanded in s/z. The terms of $Q_\otimes r$ containing $k\kappa + h_{\mathfrak{g}}$ are the residue of a total s-derivative and vanish.

Its remaining terms are the residue of

$$f_{ab}{}^{c}\big(\tfrac{1}{2}c_a c_b \otimes B^c - c_a \otimes c_b B^c - c_b B^c \otimes c_a - \tfrac{1}{2}B^c \otimes c_a c_b\big).$$

The first and second factors are evaluated at s and $s+z$. The commutators (10.84) in r^2 give the negatives of these four terms. This is the residue calculation of [46, (7.23)–(7.24)].

Both sides of the second identity in (10.89) are derivations along Δ_z after subtracting $Q_\otimes \Delta_z$. On $c_{a,n}$ their common value is $f_{bc}{}^{a} \sum_{m+p=n-1} \tau_z(c_{b,m}) \otimes c_{c,p}$. On B^a_n it is

$$f_{ab}{}^{c} \sum_{m+p=n-1} \big(\tau_z(c_{b,m}) \otimes B^c_p + \tau_z(B^c_p) \otimes c_{b,m}\big).$$

For $\Delta_z Q - Q_\otimes \Delta_z$, these are the cross terms of the quadratic differential. For $[r, \Delta_z(-)]$, inserting the commutator relations and summing the binomial expansion gives the same finite sums [46, §7.2.3]. The identity therefore holds on every polynomial. □

For dg left modules M, N, the tensor space $(M \otimes N)((z^{-1}))$ consequently has differential $D_z = d_\otimes + \rho(r(z))$ and $\mathcal{Y}_k$-action through Δ_z. Here ρ denotes the tensor product representation. Its iterated products use the common triple twist just displayed. This is the spectral tensor product of the dg line modules.

10.10.1 *A degree-zero kernel from a translation homotopy*

Assume $k \neq 0$ and that κ is nondegenerate. Let κ^{ab} be the inverse matrix. Define a derivation L of degree -1 by

$$L c_{a,n} = k^{-1}\kappa^{ab} B^b_n, \qquad L B^a_n = 0.$$

Invariance of κ makes these assignments respect all relations. Put $\Omega = \kappa^{ab} t_a \otimes t_b$.

Proposition 10.27. *The derivation L satisfies $[Q, L] = T$. The degree-zero tensor $X = (L \otimes 1)r$ is*

$$X(z) = \frac{1}{k} \sum_{n,m\geq 0} (-1)^n \binom{n+m}{n} \frac{\kappa^{ab} B^a_n \otimes B^b_m}{z^{n+m+1}} = \frac{\Omega}{k(z+u-v)}. \tag{10.92}$$

In the last expression, B^a_n is read as $t_a u^n$ in the first factor and $t_a v^n$ in the second. On the tensor product of dg modules, $\nabla_z = \partial_z + \rho(X)$ commutes with D_z.

Proof. On B_n, the level term of LQB_n is nB_{n-1}. The remaining terms vanish because invariance makes their Lie coefficient antisymmetric, while the symmetric ordering and the sum over (m, p) make their even product symmetric. On c_n, the level term of QLc_n is nc_{n-1}, and the two quadratic contributions in $QLc_n + LQc_n$ cancel by the same invariance. These checks prove $[Q, L] = T$ on the full algebra.

Only the first summand of r survives $L \otimes 1$. Expanding $(z+u-v)^{-1}$ proves (10.92). Apply $L \otimes 1$ to the Maurer–Cartan equation. Since $(L \otimes 1)(r^2) = Xr - rX$ and $(T \otimes 1)r = \partial_z r$, the result is

$$\partial_z r = Q_\otimes X + [r, X].$$

This is exactly $[\partial_z + \rho(X), D_z] = 0$. □

The inverse in L is the inverse of $k\kappa$. For a normalized simple algebra, replacing it by $(k+h^\vee)^{-1}\kappa^{-1}$ gives $[Q,L]B_1 = k(k+h^\vee)^{-1}B_0$, which differs from $TB_1 = B_0$ when $h^\vee \neq 0$. Thus the normal-ordering shift in (10.87) is not a replacement for the level in the translation homotopy.

A Wilson module is obtained from a $\mathfrak{g}$-representation by letting the ghosts and all B_n, $n>0$, act by zero. On two such modules r acts by zero, while the connection becomes $\partial_z + \rho(\Omega)/(kz)$. For two defining $\mathfrak{sl}_2$ modules with $\kappa = \mathrm{tr}_{\mathbb{C}^2}$,

$$\Omega = P - \tfrac{1}{2}\mathrm{id}.$$

Its connection residues are $1/(2k)$ and $-3/(2k)$ on the symmetric and antisymmetric summands. With affine level $k' = -k - h^\vee$, these are the residues of the Knizhnik–Zamolodchikov connection in the convention $d - \Omega\, d\log z/(k'+h^\vee)$.

The ideal generated by the ghosts is stable under Q, and its quotient is $U(\mathfrak{g}[u])$ with zero differential. The tensor r maps to zero in this quotient, whereas X survives. The operation $r \mapsto (L\otimes 1)r$ therefore identifies a rational kernel. It does not identify the dg line algebra with a degree-zero quantum group. It also requires $k \neq 0$, although the dg spectral structure itself exists at $k=0$.

10.11 The comparisons still to be constructed

An algebra isomorphism on states does not determine a field tensor. Proposition 10.23 proves this for the rational rank-two family, even at level zero. The dg example adds a second distinction: a differential twist and its degree-zero connection kernel have different types. A comparison must specify how these types are related.

Current Ward identities provide a useful intermediate target for maps from line modules to affine modules. For smooth affine modules X_1, X_2, X_3, such an identity has the form

$$xt^m I_z(v_1,v_2) = \sum_{j\geq 0}\binom{m}{j} z^{m-j} I_z(xt^j v_1, v_2) + I_z(v_1, xt^m v_2).$$

Smoothness means that sufficiently positive modes kill each vector.

Suppose their energy gradings are bounded below with finite-dimensional energy spaces. Take I_z with values in $\prod_e (X_{3,e}\otimes \mathbb{C}((z^{-1})))$ and require

$$\mathrm{pr}_e\, I_z(X_{1,e_1}\otimes X_{2,e_2}) \subset X_{3,e} z^{e-e_1-e_2}.$$

Then its restricted transpose converts the Ward equation into linearity for the mode coproduct

$$\delta_z(xt^m) = \sum_{j\geq 0}\binom{-m}{j} z^{-m-j}\, xt^{-j}\otimes 1 + 1\otimes xt^m.$$

The transpose uses the anti-involution $xt^m \mapsto -xt^{-m}$ and Laurent completion after the algebraic tensor product. This is a comparison of current actions. The full intertwining Jacobi identity imposes further equations.

Example 10.28. Let H_λ be the Heisenberg vacuum algebra with $[J_m, J_n] = \lambda m \delta_{m+n,0}$, where $\lambda \neq 0$. For $\alpha \in \mathbb{C}$, set $A_\alpha(z) = J(z) + \alpha z^{-1}\mathrm{id}$ and define

$$I_\alpha(J_{-n_1}\cdots J_{-n_r}\mathbf{1}, z) = :\prod_{i=1}^{r} \frac{\partial_z^{n_i-1}A_\alpha(z)}{(n_i-1)!}:.$$

The empty product is the identity. Normal ordering places the negative current modes first. The scalar summand commutes with every current, so

$$[J_m, \partial_z^{n-1} A_\alpha(z)/(n-1)!] = \lambda n \binom{m}{n} z^{m-n}\mathrm{id}.$$

The product rule gives the current Ward identity on every PBW state. The relation $[L_{-1}, J_{-n}] = nJ_{-n-1}$ also gives $I_\alpha(L_{-1}v, z) = \partial_z I_\alpha(v, z)$.

The full intertwining Jacobi identity, applied with first state $\mathbf{1}$, would instead require $I_\alpha(J_{-1}\mathbf{1}, z) = :J(z)I_\alpha(\mathbf{1}, z): = J(z)$. It fails when $\alpha \neq 0$. Thus the current and derivative equations alone do not construct an affine intertwining operator.

Problem 10.29. Start from a coassociative comparison through an affine bimodule resolution of the line-module functor. Choose a derived affine fusion category and coefficient models on which the comparison computes derived images and admits restricted transposition. Does its transpose satisfy the full intertwining Jacobi identity and the required grading and completion conditions? If so, determine when the resulting comparison from affine fusion to the image of the spectral line tensor is invertible. Retain the unit and normalization on the resolution. Agreement of Wilson connection residues determines neither this fusion morphism nor its inverse.

10.11.1 A rational Hall cooperation

A Hall cooperation can be multiplicative after both its coefficients and its tensor multiplication are specified. Set $R_H = \mathbb{Q}[h_1, h_2, h_3]/(h_1 + h_2 + h_3)$ and

$$N(v) = \prod_{i=1}^{3}(v + h_i), \qquad K_0(v) = N(v)/v, \qquad K_1(v) = vN(v).$$

For $j = 0, 1$, give $\mathcal{P}_d = R_H[x_1, \dots, x_d]^{\mathfrak{S}_d}$ the shuffle product

$$(F *_j G)(Z) = \sum_{Z = X \sqcup Y} F(X)G(Y) \prod_{x \in X,\, y \in Y} K_j(y - x),$$

where the ordered partitions have the prescribed input sizes.

The first kernel has only simple root-diagonal poles, whose paired residues cancel in the sum. The second is polynomial. Thus both products are polynomial. Summing over ordered three-block partitions proves associativity.

For ordered lists X, Y of sizes d, e, put

$$Q^{(j)}_{d,e}(X, Y) = \prod_{x \in X,\, y \in Y} K_j(y - x), \qquad \Delta^{\mathrm{rat}}_{d,e} F = F(X, Y)/Q^{(j)}_{d,e}(X, Y).$$

The recipient is the algebra of block-symmetric polynomials localized at all pair differences and their translates by $\pm h_i$. No parameter h_i is inverted. Write $r_H(v) = K_j(-v)/K_j(v)$. The ratio is the same for both j.

For bidegrees (d, e) and (d', e'), define a product by summing over $X = X_1 \sqcup X_2$, $Y = Y_1 \sqcup Y_2$, with these sizes and summand

$$\begin{aligned} F(X_1, Y_1)G(X_2, Y_2) \prod_{x \in X_1,\, x' \in X_2} K_j(x' - x) \prod_{y \in Y_1,\, y' \in Y_2} K_j(y' - y) \\ \cdot \prod_{y \in Y_1,\, x' \in X_2} r_H(y - x'). \end{aligned} \tag{10.93}$$

The final factors specify the interchange of the middle two blocks. They are absent from ordinary tensor multiplication.

Proposition 10.30. *The product* (10.93) *is associative and unital. The sum of the maps* $\Delta^{\mathrm{rat}}_{d,e}$ *is multiplicative into this recipient, rationally coassociative and counital for both charge-zero projections.*

Proof. Fix four ordered groups X_1, Y_1, X_2, Y_2 in a source product. Dividing by the kernel between $X_1 \sqcup X_2$ and $Y_1 \sqcup Y_2$ cancels the factors from X_1 to Y_2. The other mixed factors leave

$$K_j(x' - y)/K_j(y - x') = r_H(y - x'), \qquad y \in Y_1,\ x' \in X_2.$$

The remaining denominators are those of $\Delta^{\mathrm{rat}} F$ and $\Delta^{\mathrm{rat}} G$. The factors within the two output lists are exactly those in (10.93). Summing proves multiplicativity.

For three input pairs, either bracketing has the same factors within each output list. It also has the same factor from every earlier right block to every later left block. Hence the target product is associative. On three output lists, either iterated cooperation divides by $Q^{(j)}_{12} Q^{(j)}_{13} Q^{(j)}_{23}$. The empty lists give the unit and both counits. All charge sums on a fixed input are finite. □

The construction leaves the polynomial carrier:

$$\Delta^{\mathrm{rat}}_{1,1} 1 = \frac{v}{N(v)} \quad (j = 0), \qquad \Delta^{\mathrm{rat}}_{1,1} 1 = \frac{1}{vN(v)} \quad (j = 1).$$

It also removes the closed loci at the inverted denominators. It therefore supplies neither a supported comparison across those loci nor an ordinary-tensor coproduct on the original polynomial Hall algebra. The obstruction in Proposition 6.3 retains its scope.

Hall charge now introduces another comparison index. For a charge monoid Γ, a Hall cooperation has components $\Delta_{\gamma_1,\gamma_2}$ from charge $\gamma_1 + \gamma_2$ to a completed tensor of charges γ_1, γ_2. The field coproduct D_{η_1,η_2} instead changes the level from $\eta_1\eta_2$ to (η_1, η_2) and takes values in the covariant field tensor of this chapter. Even when both structures have a Yangian presentation, these indices and their coefficient completions remain to be compared.

Problem 10.31. For a specified oriented Hall system and a specified spectral field family, construct the coefficient and state comparison maps. Choose a map from Hall charges to levels and prove its compatibility with their ordered composition. Construct the tensor comparison that identifies each Hall cooperation with D_{η_1,η_2}, including the cross contractions, vacuum maps and expansions at every collision. Determine the coefficient localization, integral form and completion on which it is defined and invertible.

There is also a question internal to the level family. The formulas above put all $\mathcal{V}_c$ on a common state module, but their classical current products have level-dependent central terms. The translation twist compares two coproduct presentations. It does not remove that dependence.

One may instead seek an intrinsic category or a category of coefficient objects in which the different levels admit comparison functors preserving the relevant tensor. The first required construction is such a functor together with its map on the central operation. The ordinary-tensor obstruction fixes one classical state identification and leaves this broader question open.

Chapter II

Reconstruction from modules and fusion

II.1 Recovering the charge of a Heisenberg module

A tensor product of representations can determine a Hopf algebra even when the underlying fields have no coproduct at a fixed level. The Heisenberg algebra gives an explicit example. Its oscillator representation is fixed, while the charge varies. Removing the oscillators recovers the charge space, and tensoring the charge spaces defines fusion at the original level. The finite-dimensional charge spaces also retain the logarithmic exchange of the fields.

Work over $\mathbb{C}$ and fix a nonzero current level κ_{H}. The modes satisfy

$$[J_m, J_n] = m\kappa_{\mathrm{H}}\delta_{m+n,0}, \qquad m, n \in \mathbb{Z}. \tag{II.1}$$

Let $\mathscr{F}_{\kappa_{\mathrm{H}}}$ consist of modules with locally nilpotent positive modes. Require also $J_n v = 0$ for all sufficiently large n, depending on v. No semisimplicity or local finiteness is imposed on J_0. Morphisms commute with every mode.

Put $A = \mathbb{C}[q]$ and $P = \mathbb{C}[x_1, x_2, \dots]$, with finite polynomial states. For an A-module M, write Q_M for the action of q and set

$$F(M) = P \otimes M, \qquad J_{-n} = x_n, \qquad J_n = n\kappa_{\mathrm{H}}\partial_{x_n} \quad (n > 0), \qquad J_0 = Q_M. \tag{II.2}$$

Conversely, the highest space of a module V is

$$\Omega_+(V) = \bigcap_{n>0} \ker J_n,$$

with charge J_0. The word highest refers only to the positive oscillator modes. The charge can have arbitrary Jordan blocks.

Theorem II.1. *The functors F and Ω_+ are inverse exact equivalences between A-Mod and $\mathscr{F}_{\kappa_{\mathrm{H}}}$. The equivalence preserves all fields obtained from polynomial Heisenberg states by derivatives and normal products.*

Proof. On V, write $X_n = J_{-n}$ and $D_n = J_n/(n\kappa_{\mathrm{H}})$, so $[D_n, X_m] = \delta_{nm}$. Define the projection

$$E_n = \sum_{r \ge 0} \frac{(-1)^r}{r!} X_n^r D_n^r, \qquad E = \prod_{n>0} E_n. \tag{II.3}$$

Each sum is finite on a vector. Only finitely many factors of the product act nontrivially, because different oscillator pairs commute and the large positive modes annihilate that vector. The Weyl relation gives $D_n E_n = 0$, $E_n X_n = 0$, and $E_n|_{\ker D_n} = 1$. Thus E projects onto $\Omega_+(V)$.

For a finitely supported multi-index α, put $X^\alpha = \prod_n X_n^{\alpha_n}$, $D^\alpha = \prod_n D_n^{\alpha_n}$ and $\alpha! = \prod_n \alpha_n!$. The inverse to multiplication is

$$V \longrightarrow P \otimes \Omega_+(V), \qquad v \longmapsto \sum_{\alpha} \frac{x^\alpha}{\alpha!} \otimes E D^\alpha v. \tag{II.4}$$

Commuting locally nilpotent derivatives make this sum finite. The one-variable identity

$$\sum_{r\geq 0} X^r E_n D_n^r / r! = 1$$

proves that one composite is the identity. For $w \in \Omega_+(V)$, the equality $ED^\alpha X^\beta w = \delta_{\alpha\beta}\alpha! w$ proves the other. These maps commute with J_0 as well as the oscillators.

The equivalence is additive, hence exact between the two abelian categories. A mode-intertwining map intertwines derivatives and normal products of the current. Such products define every polynomial Heisenberg field, proving the last assertion. □

11.1.1 *The action monad and its tensor product*

Forget the zero mode for a moment. The same proof identifies the oscillator category $\mathcal{E}_{\kappa_H}$ with Vect. Its tensor is the balanced tensor over P:

$$(P \otimes M) \otimes_P (P \otimes N) = P \otimes (M \otimes N).$$

Positive modes act by the sum of the two derivatives, which descends to the balanced tensor, and negative modes multiply the common polynomial factor. The unit is P. This construction uses a single oscillator system at level κ_H.

Adjoining an unconstrained zero mode is free. The left adjoint to forgetting that mode sends $P \otimes M$ to $P \otimes A \otimes M$. It defines the monad

$$\mathbb{T}(P \otimes M) = P \otimes A \otimes M,$$

whose multiplication and unit come from multiplication and unit in A. A module over this monad is exactly an action of q on M. The primitive Hopf structure

$$\Delta q = q \otimes 1 + 1 \otimes q, \qquad \epsilon(q) = 0, \qquad S(q) = -q \tag{11.5}$$

makes $\mathbb{T}$ a comonoidal monad.

For a comonoidal monad, the left and right fusion maps combine its comonoidal map with multiplication on the appropriate factor of $\mathbb{T}^2$. It is called a *Hopf monad* when both fusion maps are invertible. Here the two copies of A can be written as $\mathbb{C}[u, v]$. The fusion maps and their inverses are

	fusion map on $f(u,v)$	inverse on $f(u,v)$
left	$f(u+v, v)$	$f(u-v, v)$
right	$f(u+v, u)$	$f(v, u-v)$.

(11.6)

Both assertions follow by substitution, over the polynomial ring itself.

Proposition 11.2. *The monad $\mathbb{T}$ is a Hopf monad. Its category of modules has tensor and unit*

$$F(M) \star F(N) = F(M \otimes N), \qquad Q_{M\otimes N} = Q_M \otimes 1 + 1 \otimes Q_N, \qquad \mathbf{1}_\star = F(\mathbb{C}_0). \tag{11.7}$$

When M is finite-dimensional, the dual object is $F(M^*)$, with charge $-Q_M^{\mathrm{t}}$.

Proof. Equation (11.6) verifies the two Hopf conditions. The tensor of monad modules is induced by (11.5), giving the displayed charge sum. Evaluation and coevaluation for M are charge-intertwining precisely with the indicated dual action. Their usual triangular identities remain identities after adjoining the common oscillator factor. □

The finite-charge category, where $\dim M < \infty$, is closed under this tensor and these duals. The free-action monad does not preserve it. In fact the forgetful functor from finite-dimensional A-modules to finite-dimensional vector spaces has no left adjoint. If its value L on $\mathbb{C}$ existed, adjunction would give $\mathrm{Hom}_A(L, \mathbb{C}_\lambda) = \mathbb{C}$ for every $\lambda \in \mathbb{C}$. A finite-dimensional L has finite charge spectrum, so this Hom space vanishes for all other λ. Finite-charge reconstruction therefore requires a different algebraic construction.

II.2 Matrix coefficients and the finite dual

An action algebra describes operators on representations. Matrix coefficients describe the representations as comodules. Their universal quotient imposes every morphism in the category, so it retains information that a chosen exchange matrix alone may omit.

Let k be a field and $\mathcal{C}$ an essentially small k-linear abelian category. Let $\omega : \mathcal{C} \to \mathrm{Vect}_{\mathrm{fd}}(k)$ be faithful and exact. Choose a small skeleton. Its matrix-coefficient space is the quotient

$$H_\omega = \left(\bigoplus_{X \in \mathcal{C}} \omega(X)^* \otimes \omega(X) \right) \Big/ \left\langle [\lambda\omega(f) \otimes v]_X - [\lambda \otimes \omega(f)v]_Y \right\rangle, \tag{II.8}$$

where $f : X \to Y$, $v \in \omega(X)$ and $\lambda \in \omega(Y)^*$. This is the coend $\int^X \omega(X)^* \otimes \omega(X)$ in vector spaces. Its elements are finite sums of matrix coefficients.

Proposition II.3. *For a basis (e_i) of $\omega(X)$ and its dual (e^i), the formulas*

$$\Delta[\lambda \otimes v]_X = \sum_i [\lambda \otimes e_i]_X \otimes [e^i \otimes v]_X, \tag{II.9}$$

$$\epsilon[\lambda \otimes v]_X = \lambda(v), \qquad \delta_X(v) = \sum_i e_i \otimes [e^i \otimes v]_X \tag{II.10}$$

define a coalgebra and natural right coactions on $\omega(X)$.

Proof. The tensor $\sum_i e_i \otimes e^i$ is the identity endomorphism, so the formulas are basis-independent. Inserting it twice gives the same triple coefficient for either iterated coproduct. Evaluation proves the counit and coaction identities. For $f : X \to Y$, the relation in (II.8) is exactly the assertion that $\omega(f)$ intertwines the coactions. Inserting the identity before or after $\omega(f)$ shows that the same relation is preserved by Δ and ϵ. □

Theorem II.4 (Reconstruction from an exact fiber functor)**.** *The canonical functor*

$$\mathcal{C} \longrightarrow \mathrm{Comod}_{\mathrm{fd}}(H_\omega)$$

is an equivalence. If $\mathcal{C}$ is rigid monoidal and ω is strong monoidal, H_ω is a Hopf algebra and the equivalence is monoidal. If $\mathcal{C}$ is braided, its braiding determines a coquasitriangular form on H_ω and the equivalence is braided.

Proof. The coalgebra reconstruction theorem for an exact faithful functor to finite-dimensional vector spaces gives the first equivalence [55, Theorem I.10.1]. Its universal coalgebra is the matrix-coefficient quotient (II.8). The hypotheses imply finite-dimensional Hom spaces and finite length. Faithfulness embeds each Hom space into a finite matrix space. Exactness sends every strict subobject chain to a strict chain of subspaces. No separate comonadic hypothesis is needed.

Write $J_{X,Y} : \omega(X) \otimes \omega(Y) \to \omega(X \otimes Y)$ for the tensor constraint. Multiplication is

$$[\lambda \otimes v]_X [\mu \otimes w]_Y = [(\lambda \otimes \mu) J_{X,Y}^{-1} \otimes J_{X,Y}(v \otimes w)]_{X \otimes Y}. \tag{II.11}$$

Naturality of J makes it well defined. The associativity and unit constraints make it an associative unital product, and (11.9) makes it a coalgebra map. Dual evaluation and coevaluation give the antipode identities. This is the rigid reconstruction of [55, Theorem 5.4.1].

For a braiding $c_{X,Y}$, the form is specified by

$$\rho([\lambda \otimes v]_X, [\mu \otimes w]_Y) = (\mu \otimes \lambda) J_{Y,X}^{-1} \omega(c_{X,Y}) J_{X,Y}(v \otimes w). \tag{11.12}$$

Naturality descends it through the coefficient relations. The two hexagons give its two multiplicativity identities, and invertibility of c gives its convolution inverse. On right comodules it recovers $v \otimes w \mapsto \sum w_{(0)} \otimes v_{(0)} \rho(v_{(1)}, w_{(1)})$, hence the original braiding. □

11.2.1 *The completion of natural endomorphisms*

The vector-space dual of a coend is an end. Accordingly,

$$H_\omega^* = \mathrm{End}(\omega)$$

as algebras, with natural endomorphisms multiplied by composition. For infinitely many matrix coefficients, this dual generally needs a completed tensor product to carry its coproduct.

Example 11.5. Let $C = \mathrm{Rep}_{\mathrm{alg}}(\mathbb{G}_m)$ and let ω forget the action. Every representation is a finite sum of integral weight spaces. Naturality forces an endomorphism of ω to be scalar on each weight, with independently chosen scalars. Thus

$$\mathrm{End}(\omega) = \prod_{n \in \mathbb{Z}} k, \qquad (\Delta f)(m, n) = f(m + n).$$

For $f = \delta_0$, restriction of this matrix to $-N \le m, n \le N$ has rank $2N + 1$. An element of the ordinary tensor product of two function spaces is a finite sum $\sum_{i=1}^{r} a_i(m) b_i(n)$, whose every finite submatrix has rank at most r. Therefore $\Delta\delta_0$ is not in that ordinary tensor product.

The product topology instead gives the continuous coproduct into $k^{\mathbb{Z}\times\mathbb{Z}}$, with $\epsilon(f) = f(0)$ and $S(f)(n) = f(-n)$. The ordinary coefficient Hopf algebra is

$$H_\omega = k[t, t^{-1}], \qquad \Delta t = t \otimes t, \quad \epsilon(t) = 1, \quad S(t) = t^{-1}.$$

Its finite-dimensional comodules recover the original category.

Likewise, an exchange equation constrains a presentation without necessarily presenting the coend. For the one-dimensional object $X = k$ in $\mathrm{Vect}_{\mathrm{fd}}$, its self-exchange equation alone imposes no relation on its single matrix coefficient t. Adjoining a dual makes t invertible and gives $k[t, t^{-1}]$. The actual morphism $k = \mathbf{1} \to X$ imposes $t = 1$ in (11.8), and the full coend is k. An RTT presentation therefore also needs the relations expressing the morphisms and tensor constraints of the specified category.

11.2.2 *Finite charge and distributions*

Return to $A = \mathbb{C}[q]$. Its finite dual A° consists of linear functionals annihilating an ideal of finite codimension. Every such ideal is generated by a nonzero polynomial. Factoring that polynomial and using the Chinese remainder theorem gives

$$A^\circ = \bigoplus_{\lambda \in \mathbb{C}} \bigoplus_{n \ge 0} \mathbb{C} d_{\lambda,n}, \qquad d_{\lambda,n}(p) = \frac{p^{(n)}(\lambda)}{n!}. \tag{11.13}$$

Each element has finite support in both indices.

Proposition 11.6. *The finite dual is an ordinary Hopf algebra with*

$$\Delta d_{\lambda,n} = \sum_{i+j=n} d_{\lambda,i} \otimes d_{\lambda,j}, \qquad \epsilon(d_{\lambda,n}) = \delta_{n0}, \tag{11.14}$$

$$d_{\lambda,m} d_{\mu,n} = \binom{m+n}{m} d_{\lambda+\mu,m+n}, \qquad S(d_{\lambda,n}) = (-1)^n d_{-\lambda,n}. \tag{11.15}$$

Its unit is $d_{0,0}$. Its finite-dimensional comodules are the finite-dimensional A-modules, and all its comodules are the locally finite A-modules.

Proof. The Leibniz rule gives the coproduct. Differentiating $p(u+v)$ at (λ, μ) gives the binomial product, and substitution $q \mapsto -q$ gives the antipode. These are the duals of the multiplication, coproduct and antipode of A, and their values remain in the displayed direct sums.

On a generalized eigenspace M_λ, write $Q_M = \lambda + N_\lambda$, with N_λ nilpotent. The coaction is

$$v \longmapsto \sum_{n \geq 0} N_\lambda^n v \otimes d_{\lambda,n}. \tag{11.16}$$

It is finite on v and recovers the action by evaluation against $p(q)$. Conversely, a coalgebra coaction places each vector in a finite-dimensional subcomodule, and evaluation gives a locally finite A-action. These two constructions are inverse and preserve the primitive charge tensor. □

The same Hopf algebra is the coend for the forgetful functor on finite-dimensional A-modules. Indeed their matrix coefficients are exactly the functionals in (11.13). Forgetting an A°-coaction has cofree right adjoint $V \mapsto V \otimes A^\circ$. This is distinct from the free left adjoint $V \mapsto A \otimes V$ on arbitrary A-modules.

11.3 Logarithmic exchange and changes of presentation

The primitive charge tensor has more than one possible braiding on finite-dimensional modules. The endpoint operation determines the relevant one. Give x_n oscillator weight n and take the product completion in output oscillator weight. Inputs remain polynomial. In the endpoint constructions, M, N, L are finite-dimensional charge modules. Write

$$J(z) = \sum_{m \in \mathbb{Z}} J_m z^{-m-1} = J^-(z) + J^+(z),$$

where J^- uses $m < 0$ and J^+ uses $m \geq 0$.

For a ground vector $v \in M$, define

$$K_{M,N}(z)(v \otimes f(x) w) = \exp\left(\frac{Q_M}{\kappa_{\mathrm{H}}} \sum_{r>0} \frac{x_r z^r}{r} \right) f(x_r - Q_M z^{-r})(v \otimes w). \tag{11.17}$$

In every output weight the exponential has finitely many terms. Extend to the first input by

$$K(J_{-n} a, z) = \frac{\partial_z^{n-1} J^-(z)}{(n-1)!} K(a,z) + K(a,z) \frac{\partial_z^{n-1} J^+(z)}{(n-1)!}. \tag{11.18}$$

The currents act on the output and on the second input, respectively. Positive-mode sums are finite on an input, and negative-mode sums are finite at each output weight.

Proposition 11.7. *The operation K satisfies the Ward identities*

$$J_m K(a,z) - K(a,z) J_m = \sum_{j \geq 0} \binom{m}{j} z^{m-j} K(J_j a, z). \tag{11.19}$$

Every holomorphic operation with these identities and (11.18)*, valued in the weight completion of $F(L)$, is uniquely $F(\theta(z))K_{M,N}$. Here $\theta(z) : M \otimes N \to L$ is a holomorphic charge-intertwining map on the chosen open subset of $\mathbb{C}^*$.*

Proof. Commuting a positive mode through the exponential differentiates it and gives $z^m Q_M$. Commuting a negative mode through the shift gives $z^m Q_M$ as well. The zero-mode equation is charge additivity. These are (11.19) for a highest first input. Normal ordering gives the descendant equation and extends the Ward identity by induction on its oscillator degree.

Conversely, remove the exponential and shift from a highest-input operation. The remaining map commutes with every oscillator multiplication and derivative, so it is the extension of its ground map. The zero-mode identity makes that map charge-intertwining. The descendant equation then determines all remaining values. □

The translation on $F(M)$ is

$$T_M = \sum_{n>0} n x_{n+1} \partial_{x_n} + \frac{x_1 Q_M}{\kappa_{\mathrm{H}}}. \tag{11.20}$$

It satisfies $[T_M, J_m] = -m J_{m-1}$, including $m = 1$, where the last term supplies $-Q_M$. Put $t_{M,N} = Q_M \otimes Q_N / \kappa_{\mathrm{H}}$. Direct differentiation of (11.17) gives

$$T_{M\otimes N} K - K T_N - \partial_z K = z^{-1} t_{M,N} K. \tag{11.21}$$

The oscillator terms cancel consecutively, leaving $Q_M Q_N / (\kappa_{\mathrm{H}} z)$.

Thus the connection on representing states and the connection on operations have opposite signs:

$$\nabla^{\mathrm{state}} = d + t_{M,N}\, d\log z, \qquad \nabla^{\mathrm{op}} \theta = d\theta - \theta\, t_{M,N}\, d\log z. \tag{11.22}$$

Their horizontal factors are $z^{-t_{M,N}}$ and $z^{t_{M,N}}$. For finite charge spaces these are ordinary matrix exponentials after a branch of $\log z$ is chosen. Denote the normalized horizontal endpoint operation by $\mathcal{E}_{M,N} = F(z^{t_{M,N}}) K_{M,N}$.

Proposition 11.8. *The normalized horizontal endpoint operations have braiding*

$$c^{+}_{M,N} = \tau \exp\!\left(\frac{\pi i}{\kappa_{\mathrm{H}}} Q_M \otimes Q_N \right), \tag{11.23}$$

where the half-turn of the separation is counterclockwise and τ exchanges the two charge factors. Their associator is the usual identification of the threefold charge tensor.

Proof. For three charges put $t_{ij} = Q_i Q_j / \kappa_{\mathrm{H}}$. For inputs $u_i \in F(M_i)$, the ordered and nested operations are

$$\Phi_{23} = \mathcal{E}_{M_1, M_2 \otimes M_3}(u_1, z_1) \mathcal{E}_{M_2, M_3}(u_2, z_2) u_3,$$
$$\Phi_{12} = \mathcal{E}_{M_1 \otimes M_2, M_3}(\mathcal{E}_{M_1, M_2}(u_1, z_1 - z_2) u_2, z_2) u_3.$$

They are expanded in $|z_1| > |z_2| > 0$ and $|z_2| > |z_1 - z_2| > 0$, respectively.

All pair residues commute. Reordering the creation and annihilation factors in the ordered channel gives

$$z_1^{t_{12}+t_{13}}(1 - z_2/z_1)^{t_{12}} z_2^{t_{23}}.$$

The nested channel gives

$$(1 + (z_1 - z_2)/z_2)^{t_{13}} z_2^{t_{13}+t_{23}} (z_1 - z_2)^{t_{12}}.$$

With consistent logarithms these are both $z_1^{t_{13}} z_2^{t_{23}} (z_1 - z_2)^{t_{12}}$. Their series converge on $|z_1| > |z_2| > |z_1 - z_2| > 0$. The Ward and descendant recursions reduce every further coefficient to this ground expression, by finite operations in each output weight. Their rational coefficients have poles only at the three collisions. Hence the equality extends to every polynomial input.

The exchanged channel replaces $z_1 - z_2$ by $z_2 - z_1$. A counterclockwise half-turn contributes $\exp(\pi i t_{12})$. Charge additivity and the commuting residues prove both hexagons for (11.23). The zero charge proves its two unit identities. This compares the separated and nested operations on their logarithmic domains. □

Under Proposition 11.6, this braiding is represented by the scalar-valued form

$$\rho^+(d_{\lambda,m}, d_{\mu,n}) = \frac{1}{m!n!} \partial_\lambda^m \partial_\mu^n \exp(\pi i \lambda \mu / \kappa_H). \tag{11.24}$$

Taylor expansion on each pair of generalized eigenspaces gives (11.23). The two exponential addition identities give the coquasitriangular identities, and changing the sign of the exponent gives the convolution inverse. The scalar exponential therefore presents the full finite-charge category, including nonsemisimple charges.

Proposition 11.9. *On all $\mathbb{C}[q]$-modules with the primitive charge tensor, the only natural braiding satisfying the unit identities is the flip.*

Proof. Remove the flip from a proposed braiding. On the pair of free modules (A, A) its value on $1 \otimes 1$ is a polynomial $r(u, v)$. Naturality with respect to the module maps $A \to M, 1 \mapsto m$, shows that the transformation on every pair is the action of this polynomial. The inverse transformation likewise comes from a polynomial, so r is a unit in $\mathbb{C}[u, v]$. It is constant. The unit identities set that constant equal to one. □

11.3.1 *The full turn and a nonsplit extension*

A monoidal change of presentation with tensorator $J_{M,N}$ changes the braiding and its full turn by

$$c'_{M,N} = J_{N,M}^{-1} c_{M,N} J_{M,N}, \qquad D'_{M,N} = J_{M,N}^{-1} D_{M,N} J_{M,N}, \tag{11.25}$$

where

$$D_{M,N} = c^+_{N,M} c^+_{M,N} = \exp(2\pi i Q_M \otimes Q_N / \kappa_H).$$

Consequently a nonidentity full turn cannot become the identity through an equivalence preserving this exchange.

For a scalar example take $\kappa_H = 1$ and charges $\lambda = \mu = 1/\sqrt{2}$. Then $D = -1$.

There is also an example with only zero eigenvalues. On $M = \mathbb{C}e_0 \oplus \mathbb{C}e_1$, let $Ne_1 = e_0$ and $Ne_0 = 0$. The charge module with $Q_M = N$ is the nonsplit self-extension

$$0 \longrightarrow \mathbb{C}_0 \longrightarrow M \longrightarrow \mathbb{C}_0 \longrightarrow 0.$$

It cannot split because a charge-zero complement would lie in $\ker N = \mathbb{C}e_0$.

On its tensor square,

$$D_{M,M} = 1 + \frac{2\pi i}{\kappa_H} N \otimes N \neq 1, \qquad z^{N\otimes N/\kappa_H} = 1 + \frac{\log z}{\kappa_H} N \otimes N. \tag{11.26}$$

The nonzero rank-one operator $N \otimes N$ detects both the logarithm and its monodromy. Replacing a charge by its eigenvalues would lose this extension and its exchange.

Proposition 11.10. *Let $\mathcal{D}$ be a full replete subcategory of finite-dimensional charge modules, containing the unit and closed under finite sums, subquotients, tensor products and duals. The braiding* (11.23) *is symmetric on $\mathcal{D}$ exactly under the following conditions. Every charge is semisimple, and the simple charges form an additive subgroup $\Gamma \subset \mathbb{C}$ satisfying*

$$\lambda\mu/\kappa_H \in \mathbb{Z} \qquad (\lambda, \mu \in \Gamma). \tag{11.27}$$

Every nonzero such subgroup is $\Gamma = \gamma\mathbb{Z}$, with $\gamma^2/\kappa_H \in \mathbb{Z} \setminus \{0\}$.

Proof. Suppose a charge has a nontrivial Jordan block at λ. A two-dimensional subquotient has charge $\lambda + N$, with $N^2 = 0$ and $N \neq 0$. If $\lambda \neq 0$, its full turn with the simple subquotient $\mathbb{C}_\lambda$ contains the nonzero nilpotent term $2\pi i \lambda N/\kappa_H$ after its scalar factor. It cannot be the identity. If $\lambda = 0$, (11.26) gives the contradiction. Thus all charges are semisimple.

Closure under tensor and dual makes the simple charges an additive subgroup. On two simples, symmetry is exactly (11.27). Conversely, that condition makes the full turn the identity on every sum of these simples.

Choose $\lambda_0 \neq 0$ in Γ. The injective map $\lambda \mapsto \lambda\lambda_0/\kappa_H$ embeds Γ as a subgroup of $\mathbb{Z}$. Its image is $d\mathbb{Z}$ for some positive integer d, so $\Gamma = \gamma\mathbb{Z}$. Applying (11.27) to γ proves the last assertion. □

These semisimple subcategories need not be closed under extensions in the ambient charge category. The preceding neutral extension already prevents that closure. For $\Gamma = \gamma\mathbb{Z}$, the coefficient Hopf algebra is $\mathbb{C}[t, t^{-1}]$ and its form is $\rho(t^m, t^n) = (-1)^{Nmn}$, where $N = \gamma^2/\kappa_H$. This describes a symmetric charge subcategory. It does not identify an arbitrary category of fields with a spectral presentation.

11.4 The affine resolution of a dg line algebra

The charge equivalence separates a fixed oscillator representation from its multiplicity space. For a nonabelian current algebra the corresponding construction uses a resolution, and its endomorphisms form a dg algebra. The opposite algebra and the shifted affine level are both forced by the action on that resolution.

Use the polynomial dg algebra $\mathcal{Y}_k$ of Definition 10.24, for a finite-dimensional complex Lie algebra $\mathfrak{g}$ with invariant symmetric form κ. The scalar k is its level parameter. Put

$$h_{\mathfrak{g}}(x, y) = \tfrac{1}{2}\operatorname{tr}(\operatorname{ad} x \operatorname{ad} y), \qquad \ell = -k\kappa - h_{\mathfrak{g}}.$$

Let A_ℓ be the affine enveloping algebra with central element acting as one, so that

$$[xt^r, yt^s] = [x, y]t^{r+s} + r\delta_{r+s,0}\ell(x, y)1. \tag{11.28}$$

Write $\mathfrak{n} = t\mathfrak{g}[t]$ and $\mathfrak{l} = \mathfrak{g}[t^{-1}]$. The induced module

$$\mathbb{W}_\ell = A_\ell \otimes_{U(\mathfrak{n})} \mathbb{C}$$

has underlying vector space $U(\mathfrak{l})$ by PBW. Its cyclic vector is killed by $\mathfrak{n}$, while its horizontal zero modes act freely.

11.4.1 *A resolution with a right action*

The induced Chevalley resolution has

$$\mathcal{P}^{-p} = A_\ell \otimes \Lambda^p \mathfrak{n}, \qquad p \geq 0, \tag{11.29}$$

with augmentation $\pi : \mathcal{P} \to \mathbb{W}_\ell$ and differential

$$\begin{aligned} d_{\mathcal{P}}(u \otimes y_1 \wedge \cdots \wedge y_p) = {} & \sum_i (-1)^{i-1} u y_i \otimes y_1 \wedge \cdots \widehat{y_i} \cdots \wedge y_p \\ & + \sum_{i<j} (-1)^{i+j} u \otimes [y_i, y_j] \wedge y_1 \wedge \cdots \widehat{y_i} \cdots \widehat{y_j} \cdots \wedge y_p. \end{aligned} \tag{11.30}$$

The surviving exterior factors retain their order. PBW makes A_ℓ free as a right $U(\mathfrak{n})$-module, so inducing the free Chevalley resolution proves that π is a quasi-isomorphism. Every chain is an algebraic finite sum.

Give xt^r energy $-r$, including when it is an exterior generator. The differential preserves energy, and $\mathbb{W}_\ell$ has nonnegative energy. Define

$$\mathcal{E} = \bigoplus_{w \in \mathbb{Z}} \operatorname{End}^{\bullet, w}_{A_\ell}(\mathcal{P}), \qquad d_{\mathcal{E}}(F) = d_{\mathcal{P}} F - (-1)^{|F|} F d_{\mathcal{P}}. \tag{11.31}$$

Here w is the internal energy shift. Each Hom complex uses the usual product totalization over the cohomological terms of $\mathcal{P}$, while the outer sum allows only finitely many energy shifts. The graded opposite product is $F \cdot_{\mathrm{op}} G = (-1)^{|F||G|} GF$.

For $x \in \mathfrak{l}$, let R_x multiply the affine factor on the right, and let a_x be the exterior derivation induced by $y \mapsto \mathrm{pr}_{\mathfrak{n}}[x, y]$. Projection is along $\mathfrak{l}$ and the central direction. For the basis (t_a) of $\mathfrak{g}$, let $\iota_{a,s}$ contract with the functional dual to $t_a t^s$, for $s \geq 1$. Set

$$\beta_{a,n} = R_{t_a t^{-n}} - a_{t_a t^{-n}}, \qquad \gamma_{a,m} = \iota_{a,m+1}, \qquad n, m \geq 0. \tag{11.32}$$

Their cohomological degrees are zero and one, and their energies are n and $m + 1$.

Lemma 11.11. *With ordinary operator composition, the lifts satisfy*

$$[\beta_{a,n}, \beta_{b,m}] = -f_{ab}^{\ c} \beta_{c,n+m}, \qquad [\beta_{a,n}, \gamma_{b,m}] = f_{ac}^{\ b} \gamma_{c,n+m}, \tag{11.33}$$

$$[d_{\mathcal{P}}, \gamma_{a,n}] = \tfrac{1}{2} f_{bc}^{\ a} \sum_{m+p=n-1} \gamma_{b,m} \gamma_{c,p}, \tag{11.34}$$

$$[d_{\mathcal{P}}, \beta_{a,n}] = -n \ell_{ab} \gamma_{b,n-1} + f_{ab}^{\ c} \sum_{m+p=n-1} \beta_{c,p} \gamma_{b,m}. \tag{11.35}$$

The contractions anticommute, and negative indices are absent.

Proof. Right multiplications obey $[R_x, R_y] = -R_{[x,y]}$. The projected exterior action is a Lie action and commutes with right multiplication. These facts give the first bracket. The identity $[a_x, \iota_\lambda] = -\iota_{\lambda \circ a_x}$ gives the second.

Writing ε_y for exterior multiplication, the two pieces of the differential are

$$d_1 = \sum_i R_{y_i} \iota_i, \qquad d_2 = \tfrac{1}{2} \sum_{i,j} \varepsilon_{[y_i, y_j]} \iota_i \iota_j.$$

Only contractions from the input exterior monomial contribute. The anticommutator with γ comes from d_2 and gives (11.34). In $[d_1, \beta_{a,n}]$ the positive-mode terms cancel the projected exterior action. The remaining current and central terms, for $1 \leq s \leq n$, are $f_{ab}^{\ c} t_c t^{s-n} - n \delta_{s,n} \ell_{ab}$.

Jacobi, projected to $\mathfrak{n}$, gives

$$[d_2, \beta_x] = -\sum_i a_{\mathrm{pr}_1[x,y_i]}\iota_i.$$

It supplies the exterior correction to those remaining currents. Putting $s = m + 1$ proves (11.35). □

Theorem 11.12. *The map*

$$\rho : \mathcal{Y}_k \longrightarrow \mathcal{E}^{\mathrm{op}}, \qquad B^a_n \longmapsto \beta_{a,n}, \qquad c_{a,n} \longmapsto \gamma_{a,n} \tag{11.36}$$

is a unital energy-preserving quasi-isomorphism of dg algebras. In particular,

$$\mathcal{Y}_k \simeq R\,\mathrm{Hom}^{\mathrm{wt}}_{A_\ell}(\mathbb{W}_\ell, \mathbb{W}_\ell)^{\mathrm{op}}, \qquad \ell = -k\kappa - h_{\mathfrak{g}},$$

where the superscript denotes the direct sum of homogeneous energy shifts. No nondegeneracy of κ is required.

Proof. The graded opposite changes the two brackets of (11.33) into the line relations and leaves the product of two odd contractions unchanged. For the even differential, invariance gives $-\frac{1}{2}f^{\,c}_{ab}f^{\,b}_{cd} = (h_{\mathfrak{g}})_{ad}$. There are n pairs with $m + p = n - 1$, so symmetric reordering gives

$$[d_{\mathcal{P}}, \beta_{a,n}] = -n(\ell + h_{\mathfrak{g}})_{ab}\gamma_{b,n-1} + \tfrac{1}{2}f^{\,c}_{ab}\sum_{m+p=n-1}(\beta_{c,p}\gamma_{b,m} + \gamma_{b,m}\beta_{c,p}).$$

Since $-(\ell + h_{\mathfrak{g}}) = k\kappa$, this is the image of the line differential. Thus ρ is a dg algebra map.

Compose ρ with augmentation of its output:

$$\mathcal{Y}_k \longrightarrow \mathcal{E}^{\mathrm{op}} \xrightarrow{\pi\circ -} \bigoplus_w \mathrm{Hom}^{\bullet,w}_{A_\ell}(\mathcal{P}, \mathbb{W}_\ell). \tag{11.37}$$

The last target is used only as a complex. Order the basis (y_i) of $\mathfrak{n}$ and write its dual ghosts c_i. On the ordered wedge $y_{i_1} \wedge \cdots \wedge y_{i_p}$, the ghost-left PBW monomial $c_{i_1}\cdots c_{i_p}u$, with $u \in U(\mathfrak{l})$, evaluates to $(-1)^{p(p-1)/2}u|0\rangle$ and vanishes on other basis wedges of length p. The reversed composition applies the contractions first, after which all exterior corrections vanish.

Fixing the energy shift w makes this a basis of the entire Hom complex. A wedge whose positive mode indices sum to s has energy $-s$, and its output must have energy $w - s \geq 0$. There are only finitely many such wedges. Their outputs remain algebraic elements of $U(\mathfrak{l})$, including its possibly infinite-dimensional energy-zero part. Hence the composite in (11.37) is an isomorphism of complexes.

The bounded-above graded-free resolution $\mathcal{P}$ is K-projective in internally graded modules. Applying each homogeneous Hom complex to π makes the second arrow a quasi-isomorphism. Direct sums over shifts preserve it, so ρ is a quasi-isomorphism as well. □

For $\mathfrak{sl}_2$, with $\kappa(E,F) = 1$ and $\kappa(H,H) = 2$, one has $h_{\mathfrak{g}} = 2\kappa$ and affine level $-k - 2$. On the single exterior generator Et, $[d_{\mathcal{P}}, \beta_{F,1}]$ gives $-H + k + 2$. Using affine level $-k$ would instead lose the constant two and fail to intertwine the unchanged line differential.

The quasi-isomorphism also has a useful limit on adjoining a formal differential generator. Let

$$\mathcal{Y}^q_k = \mathcal{Y}_k\langle q\rangle/(q^2,\ qa - (-1)^{|a|}aq - Qa), \qquad |q| = 1,$$

with differential $[q, -]$ and energy zero for q. The map extends by $q \mapsto -d_{\mathcal{P}}$. However, if N multiplies $\mathcal{P}^i$ by i, then $d_{\mathcal{E}}N = -d_{\mathcal{P}}$. The charge image is exact, whereas the energy-zero summand $U(\mathfrak{g})q$ has zero differential and nonzero class $[q]$ in the source. The extended map is therefore not a quasi-isomorphism.

11.4.2 *The derived functor and the spectral comparison*

The opposite algebra in (11.36) defines the right action

$$p \cdot a = (-1)^{|p||a|}\rho(a)\,p$$

on the left affine module $\mathcal{P}$. It gives a derived functor

$$\Phi_k(M) = \mathcal{P} \otimes^{\mathbb{L}}_{\mathcal{Y}_k} M \quad \text{in } D(A_\ell\text{-Mod}). \tag{11.38}$$

It sends the regular line module to $\mathbb{W}_\ell$. Suppose the internally graded complex underlying M has energy bounded below. Every cohomology module of $\Phi_k(M)$ retains that lower bound and is smooth for the affine action.

To prove the energy assertion, use the bar resolution over $\mathbb{C}$. The line algebra has nonnegative energy, so each free coefficient complex retains the lower bound of M. After tensoring with $\mathcal{P}$, its free associated piece can be replaced by $\mathbb{W}_\ell$, also of nonnegative energy. Cones and the homotopy colimit of the finite bar stages preserve vanishing below the bound. A sufficiently positive mode sends any fixed-energy cohomology class below that bound, proving smoothness.

When $\mathfrak{g}$ is abelian and $k\kappa$ is nondegenerate, the positive-energy pairs in $\mathcal{Y}_k$ contract, so $U(\mathfrak{g}) \hookrightarrow \mathcal{Y}_k$ is a quasi-isomorphism. The zero-mode exterior correction vanishes and π respects the right $U(\mathfrak{g})$-action. Consequently

$$\Phi_k(M) \simeq \mathbb{W}_{-k\kappa} \otimes_{U(\mathfrak{g})} M|_{U(\mathfrak{g})}.$$

PBW makes the right module free, so this ordinary tensor computes the derived tensor. In rank one it is the Fock induction already constructed above.

For the spectral comparison take $k \neq 0$ and put $K_z = \mathbb{C}((z^{-1}))$. The notation $V((z^{-1}))$ means Laurent series bounded above in powers of z, with algebraic coefficients in V. It is generally larger than $V \otimes K_z$.

Define the affine mode map

$$\delta_z(xt^m) = \sum_{j\geq 0} \binom{-m}{j} z^{-m-j}\, xt^{-j} \otimes 1 + 1 \otimes xt^m. \tag{11.39}$$

Its target is $(U(\mathfrak{l}) \otimes A_\ell)((z^{-1}))$. The first component expands $x(z + t^{-1})^{-m}$. It has no affine cocycle, because the residue of $f\,dg$ vanishes for series in t^{-1} with nonnegative powers of t^{-1}. The second component retains the central term. Thus (11.39) extends to an algebra map.

On algebraic mode inputs it satisfies

$$(\delta_w \otimes 1)\delta_z = (1 \otimes \delta_z)\delta_{z+w}, \qquad (\epsilon \otimes 1)\delta_z = 1. \tag{11.40}$$

Both iterates send xt^m to the sum of $x(z + w + t^{-1})^{-m}$, $x(z + t^{-1})^{-m}$ and xt^m in the three respective factors. The counit here is the augmentation of $U(\mathfrak{l})$ in the first factor.

Use the line coproduct Δ_z and degree-one twist $r(z)$ of (10.88). On $(\mathcal{P} \otimes \mathcal{P})((z^{-1}))$ put

$$D(v) = (d_{\mathcal{P}} \otimes 1 + 1 \otimes d_{\mathcal{P}})v - (-1)^{|v|} v \cdot r(z). \tag{11.41}$$

The right tensor action is $(p_1 \otimes p_2)(a_1 \otimes a_2) = (-1)^{|p_2||a_1|}\, p_1a_1 \otimes p_2a_2$. The Maurer–Cartan identity for r gives $D^2 = 0$.

Theorem 11.13. *The map*

$$\Delta_{\mathcal{P}}(u \otimes \omega) = \partial_z(u)\big(1 \otimes (1 \otimes \omega)\big) \tag{11.42}$$

is a chain map from $\mathcal{P}$ to the twisted complex (11.41). It intertwines the left affine action along ∂_z and the right line action along Δ_z. It is counital on the first factor and coassociative on the common coefficient domain of the iterated spectral maps.

Proof. All exterior factors are placed in the second resolution. The discrepancy between its differential and the image of (11.30) consists of the first components of $\partial_z(y_i)$, one for each exterior contraction. The terms $B \otimes c$ in r give precisely these components with the sign in (11.41). The terms $c \otimes B$ contract the first vacuum and vanish. This proves $D\Delta_{\mathcal{P}} = \Delta_{\mathcal{P}} d_{\mathcal{P}}$.

For a right ghost the first translated contraction likewise annihilates the first vacuum, leaving the contraction of ω in the second factor. For a right even mode, the two translated right multiplications are the two components of $\partial_z(t_a t^{-n})$, and its exterior correction remains in the second factor. These identities prove right linearity on generators, hence on the entire polynomial line algebra. Left linearity is built into the definition.

Both triple maps put the exterior factors in the last copy. Equation (11.40) identifies their affine factors, and Theorem 10.26 identifies their differential twists. For shifted kernels this equality is read first on $0 < |w| < |z| < |z + w|$ with the stated small auxiliary jet variables, then on their rationally continued mode coefficients. It asserts no extension of translation to arbitrary iterated Laurent completions. The first-factor augmentation proves the counit identity. □

For left dg line modules M, N, let $M \otimes_z N$ be $(M \otimes N)((z^{-1}))$ with differential $d + \rho_{M,N}(r)$ and line action through Δ_z. Let $X_1 \circledast_z X_2 = (X_1 \otimes X_2)((z^{-1}))$ with affine action through ∂_z. These specify two different completed tensor constructions.

Corollary 11.14. *There is a natural chain map*

$$\theta_{M,N} : \mathcal{P} \otimes_{\mathcal{Y}_k} (M \otimes_z N) \longrightarrow (\mathcal{P} \otimes_{\mathcal{Y}_k} M) \circledast_z (\mathcal{P} \otimes_{\mathcal{Y}_k} N), \qquad p \otimes v \longmapsto \Delta_{\mathcal{P}}(p) \otimes v. \tag{11.43}$$

It is affine-linear and inherits the counit and coassociativity identities on the coefficient domains of Theorem 11.13. Suppose M, N are homotopically flat over $\mathcal{Y}_k$, and $M \otimes_z N$ is homotopically flat over $\mathcal{Y}_k \otimes K_z$. Then the map computes the comparison of derived images, with algebraic scalar extension to K_z on the source.

Proof. Right linearity of $\Delta_{\mathcal{P}}$ makes the map balanced. The right and left twists cancel under this balancing, leaving the ordinary tensor differential on the target. Multiplication of Laurent coefficients is defined because each series has an upper bound, so each output coefficient is a finite sum. The remaining identities follow from the preceding theorem. Under the flatness assumptions, the indicated tensors compute derived tensors. Laurent completion of complexes of vector spaces is exact, being a filtered union of products, and therefore preserves the target quasi-isomorphisms. □

This target carries an affine action, but its Laurent completion need not be smooth as an algebraic affine module. Nor does the construction identify $\circledast_z$ with a separately defined affine fusion product. Those are additional questions about the category and the comparison map.

11.4.3 Translation and its homotopy

The coordinate change $t^{-1} \mapsto t^{-1} + z$ is generated by $L_1 = -t^2\partial_t$. Its action on the resolution is

$$T_{\mathcal{P}} = \operatorname{ad} L_1 \otimes 1 + 1 \otimes L_1.$$

It commutes with $d_{\mathcal{P}}$ because L_1 preserves the bracket and its central cocycle. It is a derivation along the affine action and the line translation, and

$$\partial_z \Delta_{\mathcal{P}} = (T_{\mathcal{P}} \otimes 1)\Delta_{\mathcal{P}}. \tag{11.44}$$

Thus $\partial_z - T_{\mathcal{P}} \otimes 1$ commutes with D.

Assume now that κ is nondegenerate. An infinite Sugawara sum requires a completion. Put $\mathfrak{n}_N = t^N \mathfrak{g}[t]$ and let $F_N\mathcal{P}$ be spanned by chains whose affine factor lies in $A_\ell \mathfrak{n}_N$ or whose exterior factor contains $\mathfrak{n}_N$. Define

$$\widehat{\mathcal{P}} = \varprojlim_N \mathcal{P}/F_N\mathcal{P}$$

degreewise.

PBW identifies each quotient with $U(\mathfrak{l})$ tensored with the Chevalley resolution for $\mathfrak{n}/\mathfrak{n}_N$. Its cohomology is $\mathbb{W}_\ell$ in degree zero, and the transition maps induce the identity there. The chain towers are surjective, so the inverse-limit exact sequence proves $\mathcal{P} \simeq \widehat{\mathcal{P}}$.

Right multiplication by xt^{-m} and its exterior correction send F_N into F_{N-m} for $N > m$. Fixed contractions are continuous as well. The bimodule action therefore extends. On this completion define

$$L_1^{\mathrm{Sug}} = -\frac{1}{k} \sum_{m \geq 0} \kappa^{ab} t_a t^{-m} t_b t^{m+1}, \tag{II.45}$$

$$L_{\mathcal{P}} = \frac{1}{k} \sum_{m \geq 0} \kappa^{ab} \big(R_{t_b t^{-m}} - \tfrac{1}{2} a_{t_b t^{-m}} \big) \varepsilon_{t_a t^{m+1}}. \tag{II.46}$$

The first acts by left multiplication and the second has cohomological degree -1. Their inverse level form is $(\ell + h_{\mathfrak{g}})^{-1} = -k^{-1}\kappa^{-1}$.

Proposition II.15. *The sums converge continuously on $\widehat{\mathcal{P}}$ and satisfy*

$$[L_{\mathcal{P}}, \beta_{a,n}] = 0, \qquad [L_{\mathcal{P}}, \gamma_{a,n}]_+ = k^{-1}\kappa^{ab}\beta_{b,n}, \tag{II.47}$$

$$d_{\mathcal{P}} L_{\mathcal{P}} + L_{\mathcal{P}} d_{\mathcal{P}} = T_{\mathcal{P}} - L_1^{\mathrm{Sug}}. \tag{II.48}$$

Proof. The term of index m in $L_{\mathcal{P}}$ lies in F_{m+1}. The only exterior correction that could remove its new high factor is proportional to $\kappa^{ab}[t_b, t_a]t = 0$. Modulo F_d only $m < d$ remains, and those finitely many operators send F_{2d} into F_d. This proves continuity. The same estimate after commuting a high positive mode past a fixed algebraic input proves convergence of L_1^{Sug}.

Invariance of κ^{-1} cancels the commutator with each β. For a contraction, use $[\varepsilon_y, \iota_\lambda]_+ = \lambda(y)$ and $[a_x, \iota_\lambda] = -\iota_{\lambda \circ a_x}$. Contracting the new exterior factor gives $k^{-1}\kappa^{ab}(R_{t_b t^{-n}} - \frac{1}{2} a_{t_b t^{-n}})$. The remaining contraction through the exterior correction gives the other $-\frac{1}{2}k^{-1}\kappa^{ab} a_{t_b t^{-n}}$. This proves (II.47).

Let Z be the left side of (II.48) minus its right side. The derivation identities make Z left affine-linear and right line-linear, and it commutes with the differential. On the cyclic generator, $d_{\mathcal{P}} L_{\mathcal{P}}(1) = -L_1^{\mathrm{Sug}} 1$, so $Z(1) = 0$. Induction on exterior degree, commuting Z with every contraction, shows that $Z(1 \otimes \omega)$ is killed by all contractions. A positive-exterior-degree element with that property is zero, also in the indicated completion. Thus $Z = 0$. □

Let L be the degree -1 line derivation with $L(B_n^a) = 0$ and $L(c_{a,n}) = k^{-1}\kappa^{ab} B_n^b$. Then $[Q, L] = T$, and $X(z) = (L \otimes 1)r(z)$ is the degree-zero connection kernel of Proposition 10.27. On the twisted resolution tensor, set

$$\mathfrak{D} = \partial_z - T_{\mathcal{P}} \otimes 1 + [D, L_{\mathcal{P}} \otimes 1].$$

Equations (II.47) and (II.48) give

$$\mathfrak{D} = \partial_z - L_1^{\mathrm{Sug}} \otimes 1 - R_{X(z)}. \tag{II.49}$$

When balanced with the left-module connection $\nabla = \partial_z + \rho_{M,N}(X)$, the last terms cancel. The induced tensor connection is $\partial_z - L_1^{\mathrm{Sug}} \otimes 1$.

Let p be a cycle of $\mathcal{P}$ and σ a cycle of $M \otimes_z N$. Suppose $\nabla\sigma$ is a boundary. Then the derivative of $\widehat{\theta}(p \otimes \sigma)$ under this affine connection is a boundary. Indeed it is

$$[D, L_{\mathcal{P}} \otimes 1](\Delta_{\mathcal{P}} p) \otimes \sigma + \Delta_{\mathcal{P}} p \otimes \nabla\sigma.$$

The first term is the boundary of the homotopy applied to $\Delta_{\mathcal{P}} p$, and the second is a boundary by assumption.

Here $\widehat{\theta}$ is obtained by including the two target resolutions in $\widehat{\mathcal{P}}$. Tensor products with M, N remain algebraic and Laurent completion is performed last. With homotopically flat modules the inclusion of resolutions preserves the derived images.

This supplies the comparison and the translation homotopy required at the start of Problem 10.29. Restricted transposition can then give current Ward identities on specified energy-graded models. The full intertwining Jacobi identity and invertibility for an independently defined affine fusion remain additional requirements.

11.5 Two line bundles on the conifold

An endomorphism algebra reconstructs a derived category when its object detects every complex. The exceptional locus makes that condition visible. On the small resolution of the conifold, the structure sheaf misses a nonzero sheaf on the exceptional curve. One more line bundle detects it and completes the generating object. This is the basic conifold example of Van den Bergh's reconstruction [134, Introduction].

Set

$$C = \mathbb{C}[X, Y, U, V]/(XY - UV), \qquad I = (X, U) \subset C. \tag{11.50}$$

The gradient of the equation vanishes only at the origin. The ring is a domain: substituting $U = qX$, $Y = qV$ embeds it in $\mathbb{C}[q, X, V]$. To check injectivity, reduce monomials by $XY = UV$ until X and Y do not occur together. The image of $X^b Y^d U^e V^f$, with $bd = 0$, has exponents $(d + e, b + e, d + f)$ in (q, X, V), which determine the original exponents.

The ideal I has rank one and is locally free away from the origin. At the origin its fiber $I/\mathfrak{m}I$ has independent classes X, U, so it is not locally free there. Record their ratio by the incidence variety

$$\mathcal{Y} = \{(X, Y, U, V, [s : t]) : Xt = Us,\ Vt = Ys\} \subset \operatorname{Spec} C \times \mathbb{P}^1. \tag{11.51}$$

Let ρ forget $[s : t]$ and π retain it. The two charts have coordinates (q, X, V) and (q', U, Y), with transitions

$$q' = q^{-1}, \qquad U = qX, \qquad Y = qV. \tag{11.52}$$

Thus $\mathcal{Y}$ is the total space of $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$ and is smooth.

Away from the origin, a nonzero row of $\left(\begin{smallmatrix} X & U \\ V & Y \end{smallmatrix}\right)$ determines $[s : t]$. Over the origin the fiber is $\mathbb{P}^1$. The projection is projective, and the exceptional locus has codimension two, so this is a small resolution. The Jacobian of (11.52) has determinant -1. Consequently $dq \wedge dX \wedge dV$ and $-dq' \wedge dU \wedge dY$ define a nowhere-zero global three-form.

The image of $\rho^* I$ in $\mathcal{O}_{\mathcal{Y}}$ is a line bundle $\mathcal{L}$. Its frames are the nonzero divisors X and U on the two charts, with transition $\ell_1 = q\ell_0$. Hence

$$\mathcal{L} = \pi^* \mathcal{O}_{\mathbb{P}^1}(1), \qquad \mathcal{T} = \mathcal{O}_{\mathcal{Y}} \oplus \mathcal{L}.$$

The vanishing of a frame as a function on the exceptional curve does not make its principal ideal a nonfree module.

11.5.1 *The full composition algebra*

A monomial $q^a X^b V^c$ extends across the second chart exactly when $0 \le a \le b + c$. These global functions generate C. For sections of $\mathcal{L}$ the upper bound becomes $b + c + 1$, and multiplication by its frame X identifies the sections with I. For $\mathcal{L}^{-1}$ it becomes $b + c - 1$, and division by X identifies the sections with $C + (V/X)C$.

Every map $I \to C$ is multiplication by a fraction after passing to the fraction field. If $rX, rU \in C$, write $a = rX$. Modulo X, the relation $aU = 0$ lies in $\mathbb{C}[Y, U, V]/(UV)$, where the annihilator of U is (V). It follows that $r = b + (V/X)c$. Both summands really map I to C, so

$$I^\vee = \operatorname{Hom}_C(I, C) = C + (V/X)C.$$

If the fraction also preserves I, then $Vc \in I$. The quotient $C/I = \mathbb{C}[Y, V]$ has no V-torsion, so $c = Xd + Ue$ and $r = b + Vd + Ye \in C$. Therefore $\operatorname{End}_C(I) = C$ and

$$A_{\mathrm{con}} = \operatorname{End}_{\mathcal{Y}}(\mathcal{T}) = \begin{pmatrix} C & I^\vee \\ I & C \end{pmatrix} = \operatorname{End}_C(C \oplus I). \tag{11.53}$$

The multiplication is matrix composition with the indicated fractional entries, each of which is an actual module map.

For example, put

$$Z = \begin{pmatrix} 0 & 1 \\ X & 0 \end{pmatrix}, \qquad P_0 = \begin{pmatrix} 0 & V/X \\ U & 0 \end{pmatrix}, \qquad G = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{11.54}$$

Then

$$Z^2 = X1, \quad P_0^2 = Y1, \quad ZP_0 = \operatorname{diag}(U, V), \quad P_0 Z = \operatorname{diag}(V, U), \quad [P_0, Z] = (V - U)G.$$

These maps remain regular on $\mathcal{Y}$, including its exceptional curve. A central element of A_{con} commutes with the two projections, hence is diagonal. Commuting with $I \to C$ makes its diagonal entries equal. Thus $Z(A_{\mathrm{con}}) = C$.

The same algebra has a potential presentation. Let $a, c : 1 \to 0$ and $b, d : 0 \to 1$ be four arrows, with paths composed from right to left, and put $\Phi = abcd - adcb$. The Jacobi algebra is the path algebra modulo the four cyclic derivatives of this potential.

Proposition 11.16. *The Jacobi algebra of* (Q, Φ) *is* A_{con}. *The arrows are the four off-diagonal parts of* Z, P_0 *in* (11.54)*, and its central element*

$$(cb - ad) + (bc - da) \tag{11.55}$$

maps to $\lambda = V - U$.

Proof. Consider

$$\mathcal{B} = \mathbb{C}[\lambda]\langle z, p, g\rangle/(g^2 - 1,\ gz + zg,\ gp + pg,\ pz - zp - \lambda g),$$

with central λ. Its ordered basis is $z^i p^j g^\epsilon$, $i, j \geq 0$, $\epsilon \in \{0, 1\}$. The mixed ambiguity gpz reduces in both ways to $zpg + \lambda$, and the ambiguities with g^2 agree as well. With $e_0 = (1 + g)/2$, $e_1 = (1 - g)/2$, the arrows are $e_0 z e_1, e_1 z e_0, e_0 p e_1, e_1 p e_0$.

The cyclic derivatives are $bcd - dcb, cda - adc, dab - bad$ and $abc - cba$. They make (11.55) central. The maps $z \mapsto a + b$, $p \mapsto c + d$ and $g \mapsto e_0 - e_1$ then give inverse maps between the Jacobi algebra and $\mathcal{B}$.

The matrices define a surjection $\mathcal{B} \to A_{\mathrm{con}}$: their arrows generate $I, I^\vee$ and the central diagonal ring. In $\mathcal{B}$, the elements

$$X = z^2, \quad Y = p^2, \quad W = zp + \lambda g/2, \quad U = W - \lambda/2, \quad V = W + \lambda/2$$

are central and obey $XY = UV$.

Filter $\mathcal{B}$ by the number of z, p factors. Then $\operatorname{gr} \mathcal{B} = \mathbb{C}[\lambda, z, p] \rtimes \mu_2$. Its central subring

$$\mathbb{C}[\lambda, X, Y, W]/(XY - W^2)$$

embeds, so its nonzero elements act injectively. The filtered algebra is therefore torsion-free over C. Over $\operatorname{Frac}(C)$, the elements $1, z, g, zg$ are a basis with independent matrix images. The kernel of the surjection is consequently both torsion and torsion-free, hence zero. □

11.5.2 *Generation and the inverse functor*

A compact object is one whose maps commute with arbitrary direct sums. A generator detects zero objects by its derived Hom complex. A compact generating bundle is *tilting* when its positive self-extensions vanish. These are properties of the entire derived category, including objects supported on the exceptional curve.

Theorem 11.17. *The bundle $\mathcal{T}$ is a compact tilting generator of $D_{\mathrm{qc}}(\mathcal{Y})$. The functors*

$$F \longmapsto R\operatorname{Hom}_{\mathcal{Y}}(\mathcal{T}, F), \qquad N \longmapsto N \otimes^{\mathbb{L}}_{A_{\mathrm{con}}} \mathcal{T} \tag{11.56}$$

are inverse C-linear equivalences with $D(\text{Mod-}A_{\mathrm{con}})$. They restrict to the categories of perfect objects.

Proof. The diagonal equation on $\mathbb{P}^1 \times \mathbb{P}^1$ gives the resolution

$$0 \to \mathcal{O}(-1) \boxtimes \mathcal{O}(-1) \to \mathcal{O} \boxtimes \mathcal{O} \to \mathcal{O}_\Delta \to 0.$$

For any complex F on $\mathbb{P}^1$, its integral transform gives the triangle

$$\mathcal{O}(-1) \otimes R\Gamma(F(-1)) \longrightarrow \mathcal{O} \otimes R\Gamma(F) \longrightarrow F.$$

Hence $\mathcal{O} \oplus \mathcal{O}(1)$ generates. The affine map π has exact pushforward on quasi-coherent sheaves. It is called *conservative* because it detects zero objects. Adjunction then proves generation by $\mathcal{T}$ on $\mathcal{Y}$.

Moreover, $\pi_* \mathcal{O}_{\mathcal{Y}} = \bigoplus_{n \geq 0} \mathcal{O}(n)^{\oplus(n+1)}$. The self-extension groups reduce to $H^i(\mathbb{P}^1, \mathcal{O}(n + j))$ for $n \geq 0$ and $j = -1, 0, 1$. In the two-chart calculation,

$$H^1(\mathbb{P}^1, \mathcal{O}(d)) = \mathbb{C}[q, q^{-1}]/(\mathbb{C}[q] + q^d \mathbb{C}[q^{-1}]),$$

which vanishes for $d \geq -1$. Higher cohomology vanishes because the affine cover has two terms. The same cover and finite local freeness make derived Hom from $\mathcal{T}$ commute with direct sums. This proves tilting and compactness, also for unbounded complexes.

The right algebra action on a map $f : \mathcal{T} \to F$ is $f \cdot a = f \circ a$, fixing the variance of (11.56). Resolve a right module by successively adjoining free generators. The tensor–Hom adjunction unit is an equivalence on the free module A_{con}, by the self-extension calculation. Both functors preserve cones, direct sums and the homotopy

colimits of these free stages, so the unit is an equivalence on every module. The image of the counit under derived Hom is then an equivalence. Its cone vanishes by generation, proving the other inverse identity.

Perfect complexes are the compact objects on this quasi-compact separated scheme [131, Tag 09M1], and perfect modules are the compact objects on the algebra side [92, Corollary 3.7]. Thus the equivalence restricts to perfect objects. Every construction retains the central C-action. The argument is the compact-generator form of derived Morita reconstruction [92, Theorem 3.8]. □

For the exceptional inclusion $i : \mathbb{P}^1 \hookrightarrow \mathcal{Y}$, the nonzero sheaf $F = i_*\mathcal{O}(-1)$ has

$$R\operatorname{Hom}(\mathcal{O}, F) = 0, \qquad H^1 R\operatorname{Hom}(\mathcal{L}, F) = H^1(\mathbb{P}^1, \mathcal{O}(-2)) = \mathbb{C}. \tag{11.57}$$

The latter class is represented by q^{-1} in the Laurent quotient. Thus the second summand is essential to the equivalence. The ordinary center C alone does not record this generating test.

Corollary 11.18. *For every $c \in \mathbb{C}$, the restriction $\mathcal{T}_c$ to $\mathcal{Y}_c = \{V - U = c\}$ is a compact tilting generator, with endomorphism algebra $A_{\mathrm{con}}/(\lambda - c)A_{\mathrm{con}}$. The equivalence (11.56) holds on each fiber with this quotient algebra.*

Proof. Multiplication by $\lambda - c$ is injective on $\mathcal{E}nd(\mathcal{T})$, since it is a coordinate on each affine chart. Its restriction sequence and the preceding cohomology vanishing compute the stated endomorphism algebra and zero positive self-extensions. Closed pushforward from the fiber is conservative, and its adjunction with $\mathcal{T}$ proves generation. The same affine cover proves compactness. The tensor–Hom proof therefore applies on every fiber. □

The fiber charts have transition $q' = q^{-1}$, $Y = q^2X + cq$. They are smooth for every c. At $c = 0$ the fiber is $T^*\mathbb{P}^1$. For $c \neq 0$, it is the smooth affine surface $XY = U(U + c)$, which avoids the conifold origin. This parameter describes surfaces inside the fixed threefold $XY = UV$. It is distinct from a smoothing parameter in an equation $XY - UV = c$.

11.6 Intrinsic Hall fibers and ordinary cooperation

The derived equivalence of the conifold retains modules and their extensions. A cooperation on the Hall coefficients asks for more: it must preserve the extension product in a specified tensor category. Specializing a coefficient map to polynomials can obscure that question. Its kernel can grow even when the coefficient module is free over the parameter ring. The conifold gives an obstruction supported on precisely these surviving classes.

Let the torus $(\mathbb{C}^*)^2$ act on the four arrows (a, b, c, d) with weights $(\alpha, \beta, -\alpha, -\beta)$. Put

$$R = \mathbb{C}[\alpha, \beta], \qquad P = \alpha^2, \quad Q = \beta^2, \quad \delta = P - Q,$$

with $|\alpha| = |\beta| = 2$. These equivariant parameters are independent of the central function $\lambda = V - U$ above.

For charge $\mathbf{d} = (n, m)$ use the representation stack $\mathfrak{M}_{n,m}$ of the four-arrow quiver, the function $f_{n,m} = \operatorname{Tr}\Phi$, and the critical complex

$$H_{n,m} = R\Gamma_T\big(\mathfrak{M}_{n,m}, \phi^p_{f_{n,m}}\mathbb{C}[4nm - n^2 - m^2]\big). \tag{11.58}$$

Here $\phi^p_f = \phi_f[-1]$, so $\phi^p_0\mathbb{C} = \mathbb{C}$. Use the critical-chart orientation with ordered arrow and vertex determinants, and twist the product of charges (n, m), (r, s) by $(-1)^{mr}$. All sums in charge and degree are direct sums.

The critical transformations on invariant flags give an augmented unital associative algebra on these complexes. Their hypotheses hold because all representations are allowed, the potential is homogeneous with every critical value zero, and the Euler form

$$\chi((n,m),(r,s)) = nr + ms - 2ns - 2mr$$

is symmetric. The product therefore has degree zero [35, §3.2]. Chains of flag refinements give its coherent higher identifications.

Choose a charge-graded K-flat dg algebra B representing it, with $B_{0,0} = R$. Such a model can be obtained by bar and cobar from a strictly unital semifree higher-algebra representative. Reduced word length in charge (n,m) is at most $n+m$, so this requires no charge completion and assumes no formality.

11.6.1 *Integral coefficient modules*

Deleting the arrow d writes the potential as $\mathrm{Tr}(d(abc-cba))$. Dimensional reduction identifies the critical coefficients with shifted equivariant Borel–Moore homology of

$$Z_{n,m} = \{abc - cba = 0\}$$

in the three-arrow representation space. The shift is $2nm + \chi((n,m),(n,m))$ [35, Theorem A.1 and Corollary A.9].

Closed pushforward gives a coefficient map

$$\theta_{n,m} : H^*(B_{n,m}) \longrightarrow R[x_1,\dots,x_n,y_1,\dots,y_m]^{\mathfrak{S}_n\times\mathfrak{S}_m}[-\chi((n,m),(n,m))]. \tag{11.59}$$

The roots and torus parameters have raw weight one and cohomological degree two. An output root polynomial of raw weight w has Hall degree $2w + \chi((n,m),(n,m))$. Position coordinates will be introduced separately.

In charges $(1,0)$, $(0,1)$ and $(1,1)$ the maps identify

$$H^*(B_{1,0}) = R[x][-1], \quad H^*(B_{0,1}) = R[y][-1], \quad H^*(B_{1,1}) = R[x,u][2], \qquad u = y - x. \tag{11.60}$$

In the mixed charge all matrices are scalar and the potential vanishes. If $a_r = x^r$ and $b_s = y^s$ are the simple-charge states, the two flags give

$$a_r b_s = x^r y^s (u^2 - Q), \qquad b_s a_r = -x^r y^s (u^2 - P). \tag{11.61}$$

The first flag sets $b = d = 0$ and the second sets $a = c = 0$. Their normal Euler classes are the two quadratics, with the second sign supplied by the chosen twist. Thus their products span the proper ideal $(u^2 - P, \delta)$, while the full mixed coefficient contains the degree -2 class $1_{(1,1)}$.

For charge $(2,1)$ put

$$L = x_1 + x_2 - 2y, \qquad H = (x_1 - y)(x_2 - y) + P.$$

For $(1,2)$ use $L' = y_1 + y_2 - 2x$, $H' = (y_1 - x)(y_2 - x) + P$. Then the coefficient maps are injective with images

$$I_{21} = (L, H, \delta)R[y, L, H], \qquad I_{12} = (L', H', \delta)R[x, L', H'], \tag{11.62}$$

both with Hall shift $[3]$. In charge $(2,1)$ the arrows a, c are columns, and the cut equation is

$$\det[a\ c]\ (b_2, -b_1)^{\mathsf{t}} = 0.$$

Its complement has $[a\ c]$ invertible and $b \neq 0$.

Gauge the two columns to a basis. The remaining punctured vector (ba, bc) has weights $\beta + \alpha$, $\beta - \alpha$ and Euler class $-\delta$. Its equivariant cohomology is $R[y]/(\delta)$ and is even. Restriction from the ambient ring is surjective, setting $L = H = 0$. The closed–open Borel–Moore sequence is therefore injective on the closed locus, with kernel ideal I_{21}. The row version of the same argument proves the formula for I_{12}.

The next output charge requires one more coefficient. Put

$$\begin{aligned} \tau &= y_1 + y_2, \qquad \xi = x_1 + x_2 - y_1 - y_2, \\ A_0 &= (x_1 - x_2)^2, \quad C_0 = (y_1 - y_2)^2, \quad \zeta = A_0 - C_0, \\ h &= \frac{\xi^2 - A_0 - C_0}{2} + P + Q, \qquad g = \delta\xi, \\ k_0 &= \frac{\zeta^2 - \xi^4 + (12P - 4Q)\xi^2 - 16PQ}{16}. \end{aligned} \tag{11.63}$$

The invariant coefficient ring is $\mathscr{B} = R[\tau, \xi, h, \zeta]$.

Lemma 11.19. *The map $\theta_{2,2}$ is injective with image $I_{22} = (h, k_0, g) \subset \mathscr{B}$ and Hall shift* [8]. *The sequence h, k_0, g is regular.*

Proof. For 2×2 matrices, expansion in the entries of d gives

$$\operatorname{Tr}(abcd - adcb) = \det[\operatorname{vec}(a), \operatorname{vec}(c), \operatorname{vec}(\operatorname{adj} b), \operatorname{vec}(\operatorname{adj} d)].$$

The cut locus is the rank-at-most-two locus of the first three columns, a generic 4×3 matrix. Its complement is the bundle of injections from a rank-three bundle U_3 into a rank-four bundle E_4. Here the roots of E_4 are $x_i - y_j$, and those of U_3 are $-\alpha$, α, $\xi - \beta$. Division of their Chern polynomials gives

$$\begin{aligned} c_2(E_4 - U_3) &= h, \\ c_3(E_4 - U_3) &= \beta h + g, \\ c_4(E_4 - U_3) &= k_0 + \tfrac{1}{4}\xi^2 h + (P - \beta\xi + Q)h - (\xi - \beta)g. \end{aligned}$$

The universal injection bundle has reductive stabilizer $GL_3 \times GL_1$. Its characteristic-class restriction sends E_4 to $U_3 \oplus L$ and has kernel generated by c_2, c_3, c_4 of the difference bundle. Pulling back its polynomial cochain model gives the Koszul complex on these three classes. The classifying bases are simply connected, so this computes the equivariant cohomology degreewise.

In the present ring, h is a coordinate and $16k_0$ is monic in ζ of degree two. Modulo these two elements the ring is free on $1, \zeta$ over $R[\tau, \xi]$. Multiplication by $g = \delta\xi$ is injective on that free module, proving regularity. Hence the complement has even cohomology $\mathscr{B}/I_{22}$, and restriction is surjective. The closed–open sequence makes the coefficient map injective with image I_{22}. Equation (11.59) gives the shift. □

The critical flag product is integral and intertwines these coefficient maps with ambient flag induction. This follows from the general-cut comparison, which retains the full off-diagonal derivative $abc - cba$ in its Cartesian flag square [140, Appendix A, Theorem A.2]. In the ambient composite, the Euler class of the diagonal inclusion cancels its inverse in the refined pullback formula. The remaining Euler classes are exactly the four forbidden arrow blocks. Proper parabolic induction contributes the same-color Grassmannian denominators.

Explicitly, for a splitting $(n,m)+(r,s)$, let L_0, L_1 be subsets of the output roots of sizes n, m, and let M_0, M_1 be their complements. The first factor is the subobject. The product of coefficient polynomials is

$$(-1)^{mr}\sum_{L_0,L_1}\frac{f(L_0,L_1)g(M_0,M_1)\prod_{u\in L_0,v\in M_1}((v-u)^2-Q)\prod_{u\in L_1,v\in M_0}((v-u)^2-P)}{\prod_{u\in L_0,v\in M_0}(v-u)\prod_{u\in L_1,v\in M_1}(v-u)}. \tag{11.64}$$

Each term is one fixed flag, so there is no factorial. Proper induction makes the sum polynomial. Equality with the critical product follows first after localization, then integrally because the ambient polynomial ring embeds in that localization. In each injective output charge above, this determines the actual critical class, including products of full coefficients outside the simple-current subalgebra.

11.6.2 *Classes that survive specialization*

Let $S=R/(\delta)$ and take the intrinsic derived coefficient fiber $B_S=B\otimes^{\mathbb{L}}_R S$. The base ring has the resolution

$$K_\delta=R\otimes\Lambda(\epsilon),\qquad |\epsilon|=3,\quad d\epsilon=\delta.$$

Its odd generator has internal degree four and resolution degree -1. The K-flat model $B\otimes_R S$ represents the same fiber as an S-algebra. This is coefficient base change. Identifying it with critical cohomology for a changed geometric equivariance would require a separate theorem.

The ideals in (11.62) are free over R, as the decompositions

$$I_{21}=(L,H)R[y,L,H]\oplus\delta R[y],\qquad I_{12}=(L',H')R[x,L',H']\oplus\delta R[x] \tag{11.65}$$

show. Denote the unique classes with coefficient δ by Γ_D,Γ_E. Both have Hall degree one. Their images D,E in $H^*(B_S)$ are nonzero, although polynomial evaluation kills them. Projection to the constant term of their second summands gives S-linear coefficient detectors $D\mapsto 1$, $E\mapsto 1$.

Likewise monic division first by h and then by $16k_0$ gives the free R-module decomposition

$$I_{22}=h\mathscr{B}\oplus k_0R[\tau,\xi,\zeta]\oplus\delta\xi R[\tau,\xi]\{1,\zeta\}. \tag{11.66}$$

Let Γ_g be the unique Hall class with coefficient $g=\delta\xi$. Its Hall degree is -2, and its intrinsic image G also survives. These free decompositions imply that the relevant derived tensors over S are the ordinary tensors of specialized cohomology modules. Indeed graded-free cohomology admits a choice of lifted basis cycles giving a quasi-isomorphism from its zero-differential complex. Base change preserves that comparison.

In the simple and first mixed charges define marked states

$$a=1_{(1,0)},\quad b=1_{(0,1)},\quad q=1_{(1,1)},\quad q_x=x_{(1,1)},\quad r_0=u_{(1,1)},\quad r_1=(xu)_{(1,1)}.$$

Their degrees are $1,1,-2,0,0,2$, respectively. The subscript specifies the charge. In particular q is not the Hall unit, which belongs to charge zero.

Proposition 11.20. *The actual critical Hall classes over R satisfy*

$$[a,r_0]=-\Gamma_D,\quad [b,r_0]=\Gamma_E,\quad [q,q_x]=-2\Gamma_g,\quad [r_0,r_1]+\Gamma_E a+2P\Gamma_g=0. \tag{11.67}$$

They retain these identities under intrinsic base change to S and to the homogeneous origin $\alpha=\beta=0$.

Proof. All displayed commutators have an even input. The two flags in charge $(2,1)$ give

$$\theta_{2,1}([a,f]) = \partial \frac{f(x_2,y) - f(x_1,y)}{x_2 - x_1}.$$

For $f = y - x$ this is $-\partial$. The reflected divided difference is $+1$, giving $+\partial$ in charge $(1,2)$.

For two mixed inputs denote the output roots by $X, U; Y, V$, and put

$$A_1 = Y - X, \quad B_1 = V - U, \quad C_1 = V - X, \quad D_1 = Y - U, \qquad \xi = X + U - Y - V.$$

Pairing complementary flags uses

$$(D_1^2 - Q)(C_1^2 - P) - (C_1^2 - Q)(D_1^2 - P) = \partial(C_1^2 - D_1^2).$$

The two pairs in the commutator $[q, q_x]$ then give $-2\partial\xi$, proving the third relation by injectivity.

For $[r_0, r_1]$ the remaining coefficient is

$$\frac{\partial}{V - Y}\big(A_1B_1(C_1^2 - D_1^2) + C_1D_1(B_1^2 - A_1^2)\big) = -\partial\xi K,$$

where $K = X^2 + U^2 - (X+U)(Y+V) + 2YV$. The two flags for $\Gamma_E a$ instead give $\partial(F(U) - F(X))/(U - X) = \partial\xi(K - 2P)$, where $F(t) = ((t-Y)^2 - P)((t-V)^2 - P)$. Adding $2P\partial\xi$ gives zero. Injectivity before specialization proves the relation on Hall classes. The free models above then transport it to the intrinsic fibers, including the classes whose polynomial images vanish. □

Theorem II.21. *There is no degree-zero, charge-preserving unital map*

$$B_S \longrightarrow B_S \otimes_S^{\mathbb{L}} B_S$$

preserving binary multiplication in $D(S)$, with the ordinary Koszul interchange and charge-zero projection as a two-sided counit. The same conclusion holds at the intrinsic homogeneous origin. Coassociativity is not needed for this obstruction.

Proof. Write $\overline{\Delta}(v) = \Delta(v) - v \otimes 1 - 1 \otimes v$. The counits fix the charge-zero summands. The simple charges have no proper positive splits, so a, b are primitive. The only proper splits of charge $(1,1)$ have minimum degree two, whereas q, q_x, r_0 have degrees $-2, 0, 0$. They too are primitive.

In the ordinary graded tensor product, a supercommutator of primitives is primitive. The first three relations in (II.67) therefore force D, E, G to be primitive.

The full possible correction to the degree-two state r_1 is

$$\overline{\Delta}(r_1) = \lambda a \otimes b + \mu b \otimes a, \qquad \lambda, \mu \in \mathbb{C}.$$

Since r_0 is even and a, E are odd, the last relation then has reduced coproduct

$$\lambda D \otimes b + \mu b \otimes D - (\lambda + 1)a \otimes E + (1 - \mu)E \otimes a. \tag{II.68}$$

The four summands have different charge splits. Their coefficient detectors take value one, by (II.65) and the constant simple-charge coefficients. The first forces $\lambda = 0$, while the third forces $\lambda = -1$. The same free decompositions and detectors survive at the origin. The contradiction occurs in cohomology of the actual derived tensor components, so no choice of chain homotopy removes it. □

The four equations can be expressed by the two-term complex

$$S^2 \xrightarrow{\Psi} S^4, \qquad \Psi = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \omega = (0,0,1,-1)^{\mathsf{t}}. \tag{II.69}$$

The map $J(v_1, v_2, v_3, v_4) = (v_1 + v_3, v_2 + v_4)$ has kernel $\operatorname{im} \Upsilon$ and a section $(v, w) \mapsto (0, 0, v, w)$. Hence the obstruction class maps to $J\omega = (1, -1)$ in its cokernel S^2. Even allowing arbitrary S-valued corrections cannot kill it. This is a quotient of the necessary multiplication equations, without an identification with the full bialgebra deformation complex.

Deleting the two terms containing D would give the apparent solution $(\lambda, \mu) = (-1, 1)$. Its residual on the intrinsic fiber is $-D \otimes b + b \otimes D \neq 0$. Thus passing to the specialized polynomial image changes the coefficient object on which the obstruction is tested.

11.6.3 *The regular collision tensor*

The coefficient tensor in the theorem is forced by a specific regular collision construction. Let the position curve be $X = \mathbb{A}^1$ with coordinate z, independent of all Chern roots.

For a partition π of $[N]$, a run is a maximal consecutive interval with the same block label, and $\ell(\pi)$ is the number of runs. In particular $13|2$ has three runs. Let i_π repeat block coordinates and j_π include their pairwise-distinct locus. With trivial coefficient connection put

$$M_\pi(B) = i_{\pi,*} j_{\pi,*} j_\pi^! (\omega_{X^{|\pi|}} \otimes B^{\otimes_R \ell(\pi)}).$$

The canonical module here is unshifted.

A merger multiplies exactly the adjacent runs that acquire the same label. It leaves all three factors on $13|2$ and multiplies them in order on the subsequent total collision. Order blocks by their largest elements and use the full normal transfer $v^{-r-1} dv \leftrightarrow \delta_v \partial_v^r / r!$. The run complex is

$$C_0 = R, \qquad C_N = \bigoplus_{\pi \vdash [N]} M_\pi(B)[|\pi|], \qquad d|_\pi = (-1)^{|\pi|} d_B + \sum_{i<j} (-1)^{i-1} r_{ij}. \tag{11.70}$$

Here d_B means the tensor differential on all run factors. The principal-part map r_{ij} includes both the normal transfer and the ordered multiplication of the newly adjacent factors.

For disjoint mergers the state maps agree and the signs cancel. For three overlapping blocks the state map is the same ordered product on each route. The scalar identity is $L - R + N = 0$ for the three full principal-part transfers. In coordinates $s = z_1 - z_3$, $t = z_2 - z_3$, its change of normal coordinates is

$$(s - t)^{-a} t^{-b} \longmapsto \sum_{j=0}^{b-1} \binom{a + j - 1}{j} s^{-a-j} t^{-b+j}.$$

Partial fractions at $s = 0$ and $s = t$ prove the identity, and right differential linearity retains all normal derivatives. Thus (11.70) has square-zero differential.

On a cut open, every block crossing that cut restricts to zero. The remaining partitions split uniquely, and their runs concatenate. This gives a strong cut coalgebra. Extraordinary pullback along a monotone surjection retains exactly the partitions pulled back from its target and deletes repeated adjacent run labels. These identifications compose and preserve the residues.

The cut bar–cobar equivalence of Theorem 8.2 then gives its cut cobar A_Φ. The gap contraction retains exactly partitions crossing every cut, with coefficient $M_\pi(B)[|\pi| - 1]$. The finite perturbation $\sum_{a=0}^{N-1} (-\partial_{\mathrm{col}} h_0)^a$, where h_0 contracts the first admissible gap, supplies the comparison and its higher operations.

The scalar unit of B supplies the local vacuum. Its Nth coefficient is

$$\omega_{X^N} \otimes R, \qquad \omega_Y = \omega_Y[\dim_{\mathbb{C}} Y].$$

It is distinct from the empty cut unit, which has only arity zero. Under coefficient base change, the finite partition sums, residue maps and gap contractions commute with derived scalar extension. Thus the same construction represents B_S.

The decisive tensor calculation is already binary. On a normal complex disk D, let $j : D^* \hookrightarrow D$ and $i : \{0\} \hookrightarrow D$. For a map of coefficient complexes $\mu : V \to W$, define the regular attachment

$$G(\mu) = \mathrm{Fib}\Big(R j_* \omega_{D^*} \otimes V \xrightarrow{\mathrm{Res} \otimes \mu} i_* W[1] \Big). \tag{11.71}$$

Its open coefficient is V and its costalk is W. After removing the common tangent density, the binary run coefficient is $G(\mu_B : B \otimes^{\mathbb{L}}_R B \to B)$.

Lemma 11.22. *For perfect coefficient arrows, and for their Ind-extension with K-flat coefficients,*

$$G(\mu) \otimes^! G(\nu) \simeq G(\mu \otimes^{\mathbb{L}} \nu).$$

A map of attachments inducing open and closed maps a, b obeys $b\mu = \nu a$ in the derived category.

Proof. For perfect coefficients, Verdier duality identifies $G(\mu)$ with the dual of a constructible diagram. Its closed stalk is $W^\vee$, its punctured stalk is $V^\vee$, its monodromy is the identity and its generization is $\mu^\vee$. Ordinary tensor tensors its stalks and generization, and duality gives the exceptional tensor formula. Every coefficient arrow is a filtered homotopy colimit of perfect arrows. The fixed two-stratum kernels, their localization triangles and derived tensors commute with these colimits, so the formula extends without dualizing nondualizable coefficients. Naturality of the localization boundary then gives $b\mu = \nu a$ on its normal-circle residue component. □

Transport the exceptional tensor through cut bar and cobar:

$$A \diamond A' = \Omega_{\mathrm{cut}}(B_{\mathrm{cut}} A \otimes^! B_{\mathrm{cut}} A').$$

Suppose a unital cooperation $A_\Phi \to A_\Phi \diamond A_\Phi$ preserves cuts, collisions and the local vacuum, with a two-sided counit to the transported vacuum unit. Its unary coefficient is a map $\Delta_B : B \to B \otimes^{\mathbb{L}}_R B$. The cut comparison forces its open binary map to be two copies of Δ_B, with the middle factors interchanged. Its closed map is Δ_B by the common-root comparison.

Lemma 11.22 therefore forces

$$\Delta_B \mu_B = (\mu_B \otimes \mu_B)(1 \otimes \tau \otimes 1)(\Delta_B \otimes \Delta_B)$$

with the ordinary Koszul swap τ. The geometric counit induces the charge-zero projection. Theorem 11.21 excludes this cooperation on the intrinsic fiber S and at the homogeneous origin.

Over R itself there is an earlier obstruction. The mixed products (11.61) satisfy

$$\{a_{r+2}, b_s\} - 2\{a_{r+1}, b_{s+1}\} + \{a_r, b_{s+2}\} - P a_r b_s - Q b_s a_r = 0.$$

Both simple charges are odd and primitive in any such cooperation. Projection of the proposed coproduct onto charge split $(1,0)|(0,1)$ gives $-\delta a_r \otimes b_s \neq 0$. This class remains nonzero after inverting δ. Its vanishing on S is the reason the surviving intrinsic classes were necessary for the stronger fiber calculation.

A tensor with meromorphic interchange or different collision attachment changes these hypotheses. The rational Hall tensor of the preceding chapter is an example of a construction with a specified changed interchange. Comparing it with the global critical coefficients requires a map retaining the intrinsic classes, the vacuum, the counit and the triple-collision identities.

11.7 Derived interfaces at a node

There is an associative tensor available without a coproduct on the coefficient algebra: composition of bimodules by derived tensor over that algebra. At a singular point its higher Tor classes are essential. The node gives both an explicit composition and a test of how much of the center one branch can detect.

Let k have characteristic zero and put

$$A = k[u,m]/(um), \qquad P = A/(m), \qquad Q = A/(u), \qquad A^e = A \otimes_k A^{\mathrm{op}}.$$

Regard the two branch modules as diagonal A-bimodules. In $D(A^e)$, interface composition is $M \otimes^{\mathbb{L}}_A N$, with unit A. The two-sided bar complex gives its associator: both parenthesizations realize the same multibar complex, and its successive face maps give the coherence identities. All tensors here are algebraic.

11.7.1 *The derived intersection retains infinitely many classes*

A semifree resolution of P as an A-algebra is

$$R_P = A[e,t], \qquad |e| = -1, \quad |t| = -2, \qquad de = m, \quad dt = ue. \tag{11.72}$$

The odd variable is exterior. Use divided powers $t^{[j]} = t^j/j!$ for its even variable. The underlying free complex has one cell in each nonpositive degree and differential alternately $m, u, m, u, \ldots$. Since $\operatorname{ann}(m) = uA$ and $\operatorname{ann}(u) = mA$, it is exact away from degree zero, where its cohomology is P. The analogous resolution of Q interchanges u and m.

Proposition 11.23. *There are quasi-isomorphisms of derived A-bimodules*

$$P \otimes^{\mathbb{L}}_A Q \simeq k[t], \qquad |t| = -2, \tag{11.73}$$

$$P \otimes^{\mathbb{L}}_A Q \otimes^{\mathbb{L}}_A P \simeq k[t_1,t_2] \otimes \Lambda(\epsilon), \qquad |\epsilon| = -1. \tag{11.74}$$

The targets have zero differential, and both A-actions factor through $A/(u,m) = k$.

Proof. Tensor R_P with Q. This gives $k[m,e,t]$ with $de = m$, $dt = 0$. Projection killing m, e is a bimodule map to $k[t]$. The regular Koszul pair contracts by $H(m^j) = em^{j-1}$ for $j > 0$, with H zero on the constant and odd terms. The homotopy is k-linear, which suffices to prove that the bimodule projection is a quasi-isomorphism.

For the triple tensor, resolve both outer copies of P. The common complex is

$$k[m,e_1,e_2,t_1,t_2], \qquad de_1 = de_2 = m, \quad dt_1 = dt_2 = 0.$$

Put $\epsilon = e_2 - e_1$ and contract (m,e_1). The remaining variables give (11.74). Flatness of the outer resolutions makes this one complex compute both parenthesizations. □

The ordinary intersection $P \otimes_A Q = k$ discards all positive powers of t. Replacing the first derived intersection by that point changes its next composition to $k \otimes^{\mathbb{L}}_A P \simeq k[t] \otimes \Lambda(e)$. Its degree -2 has dimension one, whereas that degree in (11.74) has dimension two. The distinction is present before any completion.

The bounded coherent category therefore does not contain the composite, even though it contains both branches. Nor is $\mathrm{Perf}(A^e)$ closed under this composition:

$$A^e \otimes_A A^e \simeq A \otimes_k A \otimes_k A$$

is an infinite-rank free module over the two outer factors A^e. Its degree-zero cohomology is not finitely generated over the Noetherian ring A^e, so it cannot be perfect.

There is a different closed class. Bimodules perfect as left A-modules and as right A-modules are closed under relative derived tensor and contain the unit. For left perfection, express the second factor by finite cones and retracts of left free modules. Tensoring turns these into finite cones and retracts of the first factor. The right-module argument is the reverse.

The branch modules do not belong to this class, as their periodic self-extensions below show. Their composition is instead defined in the full $D(A^e)$.

11.7.2 *Cross-branch extensions and their composition*

Let $P_\bullet, Q_\bullet$ be the alternating free resolutions just constructed. Applying Hom to them gives

$$\begin{aligned} &\operatorname{Ext}_A^0(P,P) = P, && \operatorname{Ext}_A^{2j}(P,P) = k \quad (j \geq 1), && \operatorname{Ext}_A^{2j+1}(P,P) = 0, \\ &\operatorname{Ext}_A^{2j+1}(P,Q) = k \quad (j \geq 0), && \operatorname{Ext}_A^{2j}(P,Q) = 0 \quad (j \geq 0). \end{aligned} \tag{11.75}$$

For the first row the cochain differentials alternate between zero and multiplication by u on $k[u]$. For the second they alternate between multiplication by m and zero on $k[m]$. The formulas with the branches exchanged follow by symmetry.

The degree-one cross classes have nonzero Yoneda composite. Define $F : P_\bullet \to Q_\bullet$ by sending the cell in degree $-n$, for $n \geq 1$, to $(-1)^{n-1}$ times the cell in degree $-n+1$, and killing degree zero. Define $G : Q_\bullet \to P_\bullet$ by the same coefficients. Adjacent coefficients cancel, so $dF + Fd = dG + Gd = 0$. Their composites equal minus the two-step shift, which represents the nonzero class in $P/uP = \operatorname{Ext}_A^2(P,P)$.

The periodic matrices make the sign visible:

$$d_P = \begin{pmatrix} 0 & m \\ u & 0 \end{pmatrix}, \qquad d_Q = \begin{pmatrix} 0 & u \\ m & 0 \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{11.76}$$

The two differentials square to um, while $d_Q F + F d_P = 0$ and $F^2 = -1$. Thus these are matrix factorizations of the nodal equation, with the same nonzero cross-branch composition after periodization.

11.7.3 *The Hochschild class detected by one branch*

The ordinary center of the commutative node is A. Its derived center also sees the singular point through a periodic degree-two class. To calculate it, use left and right coordinates on A^e and the diagonal Tate resolution

$$\mathcal{T}_A = A^e \otimes \Lambda(e_u, e_m) \otimes \Gamma(h), \quad |e_u| = |e_m| = -1, \quad |h| = -2, \tag{11.77}$$

with

$$de_u = u_L - u_R, \qquad de_m = m_L - m_R, \qquad dh = m_L e_u + u_R e_m.$$

The last differential squares to $u_L m_L - u_R m_R = 0$. The even variable has divided powers and $dh^{[r]} = (dh)h^{[r-1]}$.

To verify exactness, put $a = u_L - u_R, b = m_L - m_R, x = u_R, y = m_R$. Then $A^e = A[a,b]/(ay + bx + ab)$. The defining polynomial is a nonzero divisor, with leading monomial ab and coefficient one. The Koszul complex on a, b therefore has, after this quotient, homology A in degrees zero and -1. The latter generator is $m_L e_u + u_R e_m$.

Filtering (11.77) by divided-power degree pairs each positive divided power with this generator and contracts them. The filtration is finite in each chain degree, so the augmentation to A is a free diagonal resolution.

Applying $\mathrm{Hom}_{A^e}(-, A)$ gives the additive cochain model

$$A[\tau] \otimes \Lambda(\xi_u, \xi_m), \qquad |\tau| = 2, \quad |\xi_u| = |\xi_m| = 1, \qquad d\xi_u = m\tau, \quad d\xi_m = u\tau. \tag{11.78}$$

The exterior notation specifies the basis and differential. Products will be obtained from maps of the resolution, without identifying this exterior product with a strict Hochschild cup product.

Proposition 11.24. *With $E_u = u\xi_u$ and $E_m = m\xi_m$, one has*

$$\begin{aligned} \mathrm{HH}^0(A) &= A, \\ \mathrm{HH}^1(A) &= uk[u]\xi_u \oplus mk[m]\xi_m, \\ \mathrm{HH}^{2r}(A) &= k\tau^r \quad (r \geq 1), \\ \mathrm{HH}^{2r+1}(A) &= kE_u\tau^r = kE_m\tau^r \quad (r \geq 1). \end{aligned}$$

Multiplication by τ is periodic in degrees at least two. The Gerstenhaber brackets are

$$[E_u, \tau] = [E_m, \tau] = -\tau. \tag{11.79}$$

The products $E_u\tau = E_m\tau$ are nonzero, while all products of two degree-one classes vanish.

Proof. A degree-one cycle has coefficients A_1, A_2 with $mA_1 + uA_2 = 0$. Since $(u) \cap (m) = 0$, they lie in (u) and (m), respectively. In positive even degree the scalar τ coefficient is reduced modulo (u, m), and no nonzero coefficient of $\xi_u\xi_m$ is annihilated by both u and m. In odd degrees at least three, the boundary $(m\xi_m - u\xi_u)\tau$ identifies the Euler terms and kills their higher polynomial multiples. This computes the groups.

The chain map lowering the divided power of h by one and killing $h^{[0]}$ represents τ. It commutes with the divided-power differential, including its first step. Precomposition with it gives cup multiplication by τ, proving periodicity and the asserted nonzero products.

Scale u, m independently by a two-dimensional torus. The resolution generator h has weight $(1, 1)$, so τ has weight $(-1, -1)$. Bracket with an Euler derivation is this weight action, proving (11.79). Degree-one classes have weights $(r, 0)$ or $(0, s)$ with $r, s \geq 0$, whereas degree two has only weight $(-1, -1)$. Weight preservation forces all degree-one pair products to vanish. □

A Hochschild class acts on a module by tensoring the corresponding diagonal endomorphism with that module. For the branch P, this is the characteristic map

$$\chi_P : \mathrm{HH}^*(A) \longrightarrow \mathrm{Ext}^*_A(P, P).$$

Let β denote the positive two-step shift of $P_\bullet$. Then

$$\chi_P(\tau) = \beta \neq 0, \qquad \chi_P(E_u) = 0. \tag{11.80}$$

Indeed, after tensoring the diagonal resolution on the right with P, put $h' = h + e_u e_m$. The differential becomes $dh' = u_L e_m$. The regular pair $(u_L - u_R, e_u)$ contracts, leaving exactly $A[e_m, h']$ with $de_m = m_L$, $dh' = u_L e_m$. Lowering the divided power of h becomes lowering that of h', which is the branch periodic shift. The degree-one image vanishes by (11.75).

Consequently no E_2-algebra map to an E_2 refinement of $R\,\mathrm{Hom}_A(P, P)$ can induce this characteristic action. Such a map would preserve the Gerstenhaber bracket and give

$$-\beta = \chi_P([E_u, \tau]) = [\chi_P(E_u), \chi_P(\tau)] = [0, \beta] = 0.$$

The ordinary derived characteristic action remains defined. This obstruction concerns its specified values and that proposed higher structure on its target.

11.8 Adding constrained systems

Relative interface composition uses one fixed algebra A. Adding the coordinates of two independent constrained systems asks instead for a coproduct compatible with their equations. For the node this produces a cross term that is not a boundary in the ordinary tensor of the two systems.

Consider the commutative dg algebra

$$K = k[u, m] \otimes \Lambda(\eta, \chi), \qquad |\eta| = |\chi| = -1, \quad d\eta = um, \quad d\chi = 0. \tag{11.81}$$

Its cohomology is $A \otimes \Lambda(\chi)$. The proposed sums $u \mapsto u_1 + u_2$ and $m \mapsto m_1 + m_2$ send the relation to

$$b_{12} = u_1 m_2 + u_2 m_1 \mod (u_1 m_1, u_2 m_2).$$

This is nonzero, as setting $m_1 = u_2 = 0$ shows. Thus no dg algebra map $K \to K \otimes K$ can induce those coordinate sums. Higher components of an A_∞ map cannot change the induced cohomology algebra relation either.

11.8.1 *A common refinement and a noninvertible comparison*

Adjoin an odd generator θ_{12} with $d\theta_{12} = b_{12}$. The coupled algebra T_2 then receives the map

$$u \mapsto u_1 + u_2, \quad m \mapsto m_1 + m_2, \quad \eta \mapsto \eta_1 + \eta_2 + \theta_{12}, \quad \chi \mapsto \chi_1 + \chi_2.$$

The added generator makes it a chain map. It does not yet make successive binary constructions equivalent.

For three leaves put $b_{ij} = u_i m_j + u_j m_i$. The left parenthesization has additional generators with differentials b_{12} and $b_{13} + b_{23}$. The right parenthesization uses b_{23} and $b_{12} + b_{13}$. Their degree-zero cohomology algebras are the quotients of the fixed six-coordinate polynomial ring by

$$\begin{aligned} I_L &= (u_i m_i \ (1 \le i \le 3),\ b_{12},\ b_{13} + b_{23}), \\ I_R &= (u_i m_i \ (1 \le i \le 3),\ b_{23},\ b_{12} + b_{13}). \end{aligned} \tag{11.82}$$

At $u = (1, -1, 0)$ and $m = (0, 0, 1)$, every generator of I_L vanishes but $b_{23} = -1$. Thus the ideals differ. An equivalence over the fixed leaf-coordinate ring would have to identify their kernels in degree-zero cohomology, which is impossible.

There is nevertheless a coherent common refinement. For r leaves let

$$T_r = k[u_i, m_i] \otimes \Lambda(\eta_i, \chi_i, \theta_{ij} : i < j), \quad d\eta_i = u_i m_i, \quad d\chi_i = 0, \quad d\theta_{ij} = b_{ij}.$$

For a partition into ordered blocks, the refinement map sends

$$\begin{aligned} u_B &\mapsto \sum_{i \in B} u_i, & m_B &\mapsto \sum_{i \in B} m_i, \\ \eta_B &\mapsto \sum_{i \in B} \eta_i + \sum_{i < j \in B} \theta_{ij}, & \chi_B &\mapsto \sum_{i \in B} \chi_i, \\ \theta_{B,C} &\mapsto \sum_{i \in B,\, j \in C} \theta_{\{i,j\}}. \end{aligned} \tag{11.83}$$

Expansion proves compatibility with differentials. Every pair occurs exactly once under successive refinement, so these maps compose strictly. Both binary parenthesizations map to T_3. Their maps to that common target are not thereby invertible associators.

11.8.2 *Retaining the equation value gives a Hopf algebra*

Instead of imposing $um = 0$, keep its value as a coordinate:

$$\widetilde{K} = k[u, m, c] \otimes \Lambda(\eta, \chi), \qquad d\eta = um - c, \quad d\chi = 0.$$

The coordinate $w = c - um$ makes the differential linear, $d\eta = -w$. Declare u, m, w, η, χ primitive in the free graded-commutative algebra on these generators.

Proposition 11.25. *This makes $\widetilde{K}$ a dg Hopf algebra, with*

$$\Delta c = c \otimes 1 + 1 \otimes c + u \otimes m + m \otimes u, \qquad S(c) = -c + 2um. \tag{11.84}$$

Its counit kills every displayed generator. The quotient $c = 0$ is not a Hopf quotient, and $H^*(\widetilde{K}) = k[u, m] \otimes \Lambda(\chi)$.

Proof. In the variables u, m, w, η, χ, the algebra is the free graded-commutative Hopf algebra on a complex of primitives. The linear differential commutes with all its Hopf maps. Substitution $c = w + um$ gives both formulas in (11.84). Modulo c in both factors, the cross term remains nonzero, so (c) is not a coideal. Contracting the pair (w, η) computes the cohomology. □

The enlargement provides a coherent tensor by retaining the equation value, but changes the cohomology of the constrained system. Relative interface composition, the common-refinement maps, and this enlarged Hopf tensor therefore solve distinct problems.

Problem 11.26. Construct a category of constrained nodal field theories with a specified tensor product, local vacuum and collision operations. Determine whether it receives the associative interface composition over $\mathcal{A}$, or the all-pair refinement maps (11.83), through functors preserving their derived coefficients. If an enlargement by equation values is used, construct the comparison with the original constrained cohomology and identify which operations it preserves.

Chapter 12

Positive halves, doubles, and Fock spaces

Adding a point to a surface gives a creation operator. Its commutators can generate a positive quantum group, and its successive applications can generate a Fock space. Neither assertion specifies the annihilation operators, the pairing between two halves, or a state–field correspondence. These additional structures can be constructed, but their coefficient rings and tensor products matter. A lowering operator may require an equivariant parameter in its denominator even when every creation operator is polynomial. A tensor product may retain relative oscillators after its total charge has been fixed.

The Hilbert scheme of two points provides a small model in which these distinctions can be calculated. Its nested correspondence is a ramified double cover. The trace of this cover fixes the normalization of the first nontrivial current. The resulting rational half-current has a nonzero regular boundary; projecting that boundary away changes the operation. We first retain it, extend the creation action to all charges, and construct polynomially scaled lowering and Cartan modes.

We then construct doubles in two settings. A perfect residue pairing gives an oscillator Hopf algebra with an actual braiding on continuous modules. A formal coordinate Hopf algebra gives divided states and a cotangent Lie algebra; the determinant-one rational matrix algebra realizes this construction as a quantum affine vertex algebra. Its induced modules retain both the coordinate coaction and the creation action. Finally, total and relative Fock coordinates explain why extracting a charge line is generally only a lax tensor operation.

12.1 The first two charges

Let $T = (\mathbb{C}^\times)^2$ act on $\mathbb{C}^2$ with tangent weights α, β. Put

$$R_0 = \mathbb{Q}[\alpha, \beta], \qquad R = R_0[a], \qquad h = \alpha + \beta, \quad p = \alpha\beta.$$

Here a is a framing weight; h is an equivariant parameter, not the formal quantization parameter $\hbar$ used later. Write $H_n = \mathrm{Hilb}^n(\mathbb{C}^2)$, and let

$$H_n \xleftarrow{q_n} H_{n,n+1} \xrightarrow{p_n} H_{n+1}$$

be the correspondence of inclusions $I_{n+1} \subset I_n$. The map p_n is proper. The tautological bundles fit into

$$0 \longrightarrow \mathcal{L}_n \longrightarrow p_n^* \mathcal{V}_{n+1} \longrightarrow q_n^* \mathcal{V}_n \longrightarrow 0, \qquad \mathcal{L}_n|_{(I_n, I_{n+1})} = I_n / I_{n+1}.$$

Use the negative Chern classes

$$D_n = -c_1^T(\mathcal{V}_n), \qquad \ell_n = -c_1^T(\mathcal{L}_n).$$

On $\mathcal{H} = \bigoplus_{n \geq 0} H_T^*(H_n; \mathbb{Q}) \otimes_{R_0} R$ define

$$E_r^{(n)}(v) = (p_n)_*((a + \ell_n)^r q_n^* v), \qquad r \geq 0. \tag{12.1}$$

The direct sum is algebraic, and the charge is n.

Proposition 12.1. *There are identifications*

$$H_{1,2} \simeq \mathbb{C}^2_c \times \operatorname{Tot} \mathcal{O}_{\mathbb{P}^1}(-1), \qquad H_2 \simeq \mathbb{C}^2_c \times \operatorname{Tot} \mathcal{O}_{\mathbb{P}^1}(-2).$$

On the chart with difference vector $v = (s, sr)$, the maps are

$$p_1(c, s, r) = (c, t = s^2, r), \qquad q_1(c, s, r) = c - \tfrac{1}{2}(s, sr).$$

Thus p_1 is finite flat of degree two, with involution $s \mapsto -s$.

Proof. A pair of distinct points has center c and unordered difference $\{v, -v\}$. At coincidence its limiting direction is a point of $\mathbb{P}^1$. The resolution of $\mathbb{C}^2/\{\pm 1\}$ is the total space of $\mathcal{O}(-2)$; choosing which point is retained replaces the squared difference by v, giving $\mathcal{O}(-1)$. The two chart transition functions are those of these line bundles. In either chart the coordinate map is $t = s^2$, whose coordinate algebra is free on $1, s$ over the target. This also proves the assertion along the zero section. □

Let $E \subset H_2$ be the exceptional divisor and $j : H_2 \setminus E \hookrightarrow H_2$ its complement. On the complement, the anti-invariant summand of the double cover is the rank-one local system $\mathscr{L}_-$ with normal monodromy -1. In the constructible derived category,

$$R(p_1)_* \mathbb{Q} \simeq \mathbb{Q}_{H_2} \oplus j_! \mathscr{L}_-, \qquad j_! \mathscr{L}_- = R j_* \mathscr{L}_-, \qquad \operatorname{Tr}_{p_1} = (2, 0). \tag{12.2}$$

Indeed the idempotents $(1 \pm \iota)/2$ split the direct image. The punctured normal disk has zero cohomology with coefficients in $\mathscr{L}_-$, since its cellular differential is $(-1) - 1 = -2$. Both extensions therefore have zero stalk on E. The trace is two on the invariant summand over the dense unramified locus and zero on the anti-invariant summand. Proper trace extends these formulas across E.

The trace-free line in $\mathcal{V}_2$ has negative Chern class ℓ, with restrictions α, β at the two fixed directions. Equivariant retraction to $\mathbb{P}^1$ gives

$$H^*_T(H_2; \mathbb{Q}) \otimes_{R_0} R = R[\ell]/(\ell^2 - h\ell + p). \tag{12.3}$$

Restriction to the fixed points is injective, with image

$$\{(F_\alpha, F_\beta) \in R^2 : F_\alpha - F_\beta \in (\alpha - \beta)\}.$$

To see the image, write a class as $A + B\ell$; conversely the displayed divisibility recovers B and then A. This description retains the lattice at $\alpha = \beta$. It does not replace it by two independent fixed-point values.

The class ℓ pulls back to ℓ_1, so the projection formula and (12.2) give, for every r,

$$E_r^{(1)}(1) = 2(a + \ell)^r.$$

Writing $Z = z - a$, the current expanded at infinity is

$$E(z)1_{H_1} = \frac{2}{Z - \ell} = \frac{2(Z - h + \ell)}{(Z - \alpha)(Z - \beta)}. \tag{12.4}$$

The equality follows by multiplying by $Z - \ell$ and using $\ell^2 - h\ell + p = 0$. At $\alpha = \beta = 0$ it becomes $2Z^{-1} + 2\ell Z^{-2}$. The nilpotent class remains present.

12.2 The polynomial Fock action

Let

$$\mathcal{F}_R = R[p_1, p_2, \ldots], \qquad \operatorname{charge}(p_j) = j, \qquad \mathbf{1} = 1, \qquad \partial_j = \partial/\partial p_j.$$

The coefficient $p = \alpha\beta$ is distinct from every variable p_j. For a partition λ, put $p_\lambda = \prod_i p_{\lambda_i}$, and let $\ell(\lambda)$ be its number of parts. Define the charge-preserving operator

$$\mathsf{D} = \frac{1}{2}\sum_{i,j\geq 1} ij\,p_{i+j}\partial_i\partial_j - \frac{p}{2}\sum_{i,j\geq 1}(i+j)\,p_i p_j \partial_{i+j} + \frac{h}{2}\sum_{j\geq 1} j(j-1)\,p_j\partial_j. \tag{12.5}$$

Every sum is finite on each polynomial. If M_f denotes multiplication by f, set

$$e_0 = M_{p_1}, \qquad e_r = (\operatorname{ad}_{\mathsf{D}} + a)^r e_0, \qquad \mathsf{E}(z) = \sum_{r\geq 0} e_r z^{-r-1}. \tag{12.6}$$

Each mode raises charge by one.

Theorem 12.2. *The Nakajima correspondence map is an isomorphism*

$$\Psi_n : R_0\{p_\lambda : |\lambda| = n\} \xrightarrow{\sim} H_T^*(H_n;\mathbb{Q})$$

in every charge. After adjoining a, it intertwines D with D_n, and e_r with E_r. The identities and the inverse map persist under every commutative coefficient change from R.

Proof. The Hilbert scheme has affine strata of complex dimensions $n + \ell(\lambda)$, hence Betti polynomial $\sum_{\lambda\vdash n} t^{2(n-\ell(\lambda))}$ [105, Theorem 6]. The equivariant spectral sequence degenerates by parity. Homogeneous lifts of ordinary cohomology give a free R_0-basis with degrees $2(n-\ell(\lambda))$. The correspondence class of p_λ has precisely this degree. The geometric Fock theorem makes Ψ_n invertible over $\operatorname{Frac}(R_0)$ [119, 120].

The determinant of a degree-zero map between these two graded free modules has degree zero: each nonzero determinant monomial has degree equal to the difference of the two sums of basis degrees. Since α, β have positive degree, this determinant is a rational scalar. Generic invertibility makes it a nonzero scalar, so the adjugate matrix gives an inverse over R_0.

The equivariant tautological divisor formula gives (12.5); its splitting and joining terms are the classical terms of the Hilbert-scheme divisor operator [124, Section 2.2]. Our use of $-c_1(\mathcal{V}_n)$ fixes the sign of the operator. The tautological exact sequence gives $[D, E_r] = E_{r+1} - aE_r$ by the projection formula. The equality at $r = 0$ therefore identifies all modes. All identities are matrices over R, so extending coefficients preserves them and the inverse of Ψ_n. □

The two smallest divisor matrices are

$$D_2 = \begin{pmatrix} 0 & -p \\ 1 & h \end{pmatrix}, \qquad D_3 = \begin{pmatrix} 0 & -p & 0 \\ 3 & h & -3p \\ 0 & 2 & 3h \end{pmatrix}, \tag{12.7}$$

in the bases (p_1^2, p_2) and (p_1^3, p_1p_2, p_3). In charge two, $\Psi(p_1^2) = 2$ and $\Psi(p_2) = 2\ell$; these formulas identify the resolvent of D_2 with (12.4). They also identify the sign convention with a tautological c_1-convention: if its modes are $E_r^{\mathrm{taut}}(a)$, then

$$e_r(a) = (-1)^r E_r^{\mathrm{taut}}(-a).$$

Thus spectral reflection, including the framing shift, must accompany this change of divisor.

Put $\sigma_2 = p - h^2$ and $\sigma_3 = -ph$. The positive affine Yangian in this convention has generators e_r with quadratic relations

$$\begin{aligned}[e_{i+3}, e_j] - 3[e_{i+2}, e_{j+1}] + 3[e_{i+1}, e_{j+2}] - [e_i, e_{j+3}] \\ + \sigma_2([e_{i+1}, e_j] - [e_i, e_{j+1}]) = \sigma_3(e_i e_j + e_j e_i)\end{aligned} \tag{12.8}$$

and cubic Serre relations

$$\sum_{\pi \in S_3} [e_{i_{\pi(1)}}, [e_{i_{\pi(2)}}, e_{i_{\pi(3)}+1}]] = 0. \tag{12.9}$$

The operators (12.6) satisfy both families over R. Indeed the localized geometric action satisfies them [133, Theorem 2.2], with parameters $(\alpha, \beta, -\alpha - \beta)$ after the displayed reflection. Every matrix entry of either relation is a polynomial in α, β, a. A polynomial matrix vanishing over their fraction field vanishes over R. This proves every mode relation, not only a bounded charge check.

The same argument descends the action of the current-generated shuffle algebra. Over the fraction field its identification with the positive Yangian is [133, Theorem 7.6]. Multiplication or division by $\prod_{i<j}(x_i - x_j)^2$ changes the two shuffle kernels as in Chapter 4. For any relation in the kernel of the free algebra on the currents, its action on each polynomial is zero after localization and hence before localization. This proves the factorization of the action through the integral current-generated algebra. No commutation between an endomorphism ring and base change is required.

The module is cyclic on every nonzero coefficient fibre. In fact

$$S = e_1 - a e_0 = \sum_{j \geq 1} j\, p_{j+1} \partial_j, \qquad [S, M_{p_n}] = n M_{p_{n+1}},$$

so

$$M_{p_n} = \frac{1}{(n-1)!} (\mathrm{ad}_S)^{n-1} e_0. \tag{12.10}$$

Products of these operators generate every monomial from 1. For example $[e_1, e_0] = M_{p_2}$ survives at $\alpha = \beta = a = 0$. This survival concerns the intrinsic current action even when an ambient specialized shuffle image loses the corresponding commutator. The character is the partition product $\prod_{n \geq 1}(1 - q^n)^{-1}$. A plane-partition character belongs to a different representation.

Example 12.3. Let $l = -c_1^T(\mathcal{V}_3)$. Then

$$H_T^*(H_3; \mathbb{Q}) = R_0[l]/((l - h)(l^2 - 3hl + 9p)).$$

Indeed $\Upsilon(p_1^3) = 6$, since the three distinct points admit six ordered flags. In the charge-three oscillator basis, the columns for $1, l, l^2$ are

$$\begin{pmatrix} 1/6 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1/2 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -p/2 \\ h/2 \\ 1 \end{pmatrix}.$$

Their determinant is $1/12$, a unit. The characteristic polynomial of D_3 supplies the displayed relation; comparison of the two free rank-three modules proves there are no others. At the parameter origin the ring is $\mathbb{Q}[l]/(l^3)$.

There is also an exact current matrix from charge two to three. Put

$$Q(Z) = (Z^2 - hZ + p)(Z^2 - 2hZ + 4p),$$

and

$$N(Z)=\begin{pmatrix} Z^3-3hZ^2+(2h^2+3p)Z-2ph & 2p^2 \\ 2Z^2-4hZ+2p & Z^3-3hZ^2+(2h^2+p)Z-4ph \\ 2Z & 2Z^2-2hZ+4p \end{pmatrix}. \tag{12.11}$$

Then $\mathsf{E}^{(2)}(z)=N(Z)/Q(Z)$, with $Z=z-a$. To verify the formula, let $E_0=\left(\begin{smallmatrix}1&0\\0&1\\0&0\end{smallmatrix}\right)$. Multiplication gives $ZN-D_3N+ND_2=QE_0$. The current recursion has the unique solution $ZE-D_3E+ED_2=E_0$ with leading term $Z^{-1}E_0$. Since Q is monic, every mode remains polynomial. At $h=p=0$ the matrix becomes

$$\begin{pmatrix} Z^{-1} & 0 \\ 2Z^{-2} & Z^{-1} \\ 2Z^{-3} & 2Z^{-2} \end{pmatrix}.$$

Its higher poles retain the nilpotent divisor action.

12.3 Regular terms, lowering, and Cartan modes

Already on the vacuum the complete rational half-currents have no scalar exchange. With $\Delta(Z)=(Z-\alpha)(Z-\beta)$, the first two charges give

$$\mathsf{E}(z)1=\frac{p_1}{Z},\qquad \mathsf{E}(z)p_1=\frac{(Z-h)p_1^2+p_2}{\Delta(Z)}.$$

The coefficient columns of $\mathsf{E}(z)\mathsf{E}(w)1$ and $\mathsf{E}(w)\mathsf{E}(z)1$, with $W=w-a$, have determinant

$$\det\begin{pmatrix} (Z-h)/(W\Delta(Z)) & (W-h)/(Z\Delta(W)) \\ 1/(W\Delta(Z)) & 1/(Z\Delta(W)) \end{pmatrix}=\frac{Z-W}{ZW\Delta(Z)\Delta(W)}\neq 0. \tag{12.12}$$

Consequently no scalar rational function can exchange these unmodified operators.

This obstruction is consistent with the mode relations. Let π_z^- retain strictly negative powers of z in $A[[z^{-1},w^{-1}]][z,w]$, for any associative R-algebra A; define π_w^- in the same way. Put $P_\pm(s)=s^3+\sigma_2 s\pm\sigma_3$ and

$$\begin{aligned} B(z,w)&=P_-(z-w)\mathsf{E}(z)\mathsf{E}(w)-P_+(z-w)\mathsf{E}(w)\mathsf{E}(z),\\ C(z,w)&=(z^2-3zw+3w^2+\sigma_2)e_0+(z-3w)e_1+e_2. \end{aligned} \tag{12.13}$$

Proposition 12.4. *Relation* (12.8) *is equivalent to the exact identity*

$$B(z,w)=[C(z,w),\mathsf{E}(w)]+[C(w,z),\mathsf{E}(z)]-6[e_0,e_1]. \tag{12.14}$$

Proof. The coefficient of $z^{-i-1}w^{-j-1}$ in B is the difference of the two sides of (12.8). Thus that relation says $\pi_z^-\pi_w^-B=0$. Set $q_z=1-\pi_z^-$. The formula

$$q_z(z^r\mathsf{E}(z))=\sum_{i=0}^{r-1}e_i z^{r-1-i}$$

gives $q_zB=[C(z,w),\mathsf{E}(w)]$ and its transposed counterpart. Their common polynomial part is $3[e_0,e_1]-3[e_1,e_0]=6[e_0,e_1]$. The identity $1-\pi_z^-\pi_w^-=q_z+q_w-q_zq_w$ proves the formula. Conversely its right side has no term strictly negative in both variables. □

At the parameter origin the boundary is visibly nonzero:

$$B(z,w)_1 = -\frac{(z-w)^4}{z^2w^2}p_2 = \left(-\frac{z^2}{w^2} + \frac{4z}{w} - 6 + \frac{4w}{z} - \frac{w^2}{z^2}\right)p_2.$$

Here regularity refers to spectral powers. Turning these terms into spacetime distributions or supported operations requires their own coefficient maps.

Their higher overlaps are also operator-valued. Write $X_i = \mathsf{E}(z_i)$, $r_{ij} = P_+(z_i - z_j)/P_-(z_i - z_j)$, and $c_{ij} = B(z_i, z_j)/P_-(z_i - z_j)$, over the generic spectral field. Then $X_iX_j = r_{ij}X_jX_i + c_{ij}$. Expanding the two three-swap paths gives

$$\begin{aligned} 0 = {} & c_{12}X_3 + r_{12}X_2c_{13} + r_{12}r_{13}c_{23}X_1 \\ & - X_1c_{23} - r_{23}c_{13}X_2 - r_{23}r_{13}X_3c_{12}. \end{aligned} \tag{12.15}$$

Two disjoint swaps give

$$X_1X_2X_3X_4 = r_{12}r_{34}X_2X_1X_4X_3 + r_{12}X_2X_1c_{34} + r_{34}c_{12}X_4X_3 + c_{12}c_{34}.$$

Both identities follow by substitution in the indicated order. They retain the noncentral corrections and their product. They alone do not construct homotopies on a critical flag complex.

The generic geometric pairing has normalization

$$\langle p_\lambda, p_\mu\rangle = \delta_{\lambda\mu}z_\lambda\frac{(-1)^{n-\ell(\lambda)}}{p^{\ell(\lambda)}}, \qquad z_\lambda = \prod_{j\ge1} j^{m_j}m_j!, \quad |\lambda| = n, \tag{12.16}$$

where m_j is the multiplicity of j. It is defined after inverting p; Nakajima's signed form differs by $(-1)^n$ in charge n [120, Sections 2(i), 3(iii)]. Thus

$$M_{p_j}^\dagger = (-1)^{j-1}jp^{-1}\partial_j, \qquad \mathsf{D}^\dagger = \mathsf{D}.$$

Polynomial lowering operators are obtained by retaining the scale:

$$\widetilde{f}_r = (-\operatorname{ad}_{\mathsf{D}} + a)^r\partial_1, \qquad \widetilde{\psi}_r = [e_r, \partial_1]. \tag{12.17}$$

They have charge shifts -1 and 0.

For two mode families x_i, y_j, let

$$\mathcal{D}_{i,j}(x,y) = [x_{i+3}, y_j] - 3[x_{i+2}, y_{j+1}] + 3[x_{i+1}, y_{j+2}] - [x_i, y_{j+3}] + \sigma_2([x_{i+1}, y_j] - [x_i, y_{j+1}]).$$

With $\{x, y\} = xy + yx$, the full polynomial relations are

$$\begin{aligned} [e_i, \widetilde{f}_j] &= \widetilde{\psi}_{i+j}, & [\widetilde{\psi}_i, \widetilde{\psi}_j] &= 0, \\ \mathcal{D}_{i,j}(e,e) &= \sigma_3\{e_i, e_j\}, & \mathcal{D}_{i,j}(\widetilde{f},\widetilde{f}) &= -\sigma_3\{\widetilde{f}_i, \widetilde{f}_j\}, \\ \mathcal{D}_{i,j}(\widetilde{\psi},e) &= \sigma_3\{\widetilde{\psi}_i, e_j\}, & \mathcal{D}_{i,j}(\widetilde{\psi},\widetilde{f}) &= -\sigma_3\{\widetilde{\psi}_i, \widetilde{f}_j\}. \end{aligned} \tag{12.18}$$

Both e and $\widetilde{f}$ obey (12.9). The initial Cartan modes are

$$\begin{aligned} &\widetilde{\psi}_0 = -1, \qquad \widetilde{\psi}_1 = -a, \qquad \widetilde{\psi}_2 = 2p\mathsf{N} - a^2, \\ &\widetilde{\psi}_3 = 6p\mathsf{D} + (2ph + 6ap)\mathsf{N} - a^3, \qquad \mathsf{N} = \sum_{j\ge1} jp_j\partial_j. \end{aligned} \tag{12.19}$$

In particular their commutators with e_j are $0, 0, 2pe_j$ for indices $0, 1, 2$; with $\widetilde{f}_j$ the last sign is negative.

To prove these relations in all charges, work first over the fraction field. Adjunction identifies the geometric lowering modes with $p^{-1}\widetilde{f}_r$, and the Cartan modes with $p^{-1}\widetilde{\psi}_r$. The geometric Yangian relations, multiplied by the required powers of p, give (12.18). Their matrices are polynomial, so the same injectivity argument as above descends every relation and every Serre identity. The low formulas follow directly from (12.5). This proves an integral representation with a scaled negative half; it does not define a nonproper geometric pushforward without localization.

For $\widetilde{F}(w) = \sum_j \widetilde{f}_j w^{-j-1}$ and $\widetilde{\Psi}(z) = \sum_m \widetilde{\psi}_m z^{-m-1}$, the mixed relation has the exact form

$$[\mathsf{E}(z), \widetilde{F}(w)] = \frac{\widetilde{\Psi}(z) - \widetilde{\Psi}(w)}{w - z}. \tag{12.20}$$

The coefficient identity is $(z^{-m-1} - w^{-m-1})/(w - z) = \sum_{i+j=m} z^{-i-1} w^{-j-1}$; the coincident limit is $-\widetilde{\Psi}'(z)$.

At the origin the positive Lie algebra remains large. Define

$$W_{b,r} = \sum_{i_1,\dots,i_r \ge 1} i_1 \cdots i_r \, p_{b+i_1+\cdots+i_r} \, \partial_{i_1} \cdots \partial_{i_r}, \qquad b \ge 1,\ r \ge 0,$$

where $W_{b,0} = M_{p_b}$. Then

$$e_r = W_{1,r}, \qquad [W_{b,r}, W_{c,s}] = (rc - sb) W_{b+c,r+s-1}. \tag{12.21}$$

When $r = s = 0$, the right side means zero. In a composite at most one derivative can hit the other operator's linear coefficient. Terms with no such hit cancel. Symmetrizing the remaining indices cancels their sums and leaves the constants $rc - sb$, proving the bracket. The same calculation with $W_{0,2}/2 = \mathsf{D}$ proves the formula for e_r. Charge distinguishes b, and the coefficient of ∂_1^r is p_{b+r}; successive highest differential orders prove linear independence. Thus $x^b y^r \mapsto W_{b,r}$ is a faithful Lie action of $x\mathbb{Q}[x, y]$ with bracket $\{f, g\} = x(f_y g_x - f_x g_y)$. The identity $[W_{1,0}, W_{b-1,r+1}] = -(r+1) W_{b,r}$ shows that its image is generated by the positive modes. By contrast, $\widetilde{f}_0 = \partial_1$ and $\widetilde{f}_r = 0$ for $r > 0$ on this fibre.

12.4 What a double must add

The half-current obstruction also prevents a simple change of coordinate from supplying the creation axiom. Suppose a vector space V has a distinguished vector $\mathbf{1}$ and operators $E(z) = \sum_{r \ge 0} e_r z^{-r-1}$ with $E(z)\mathbf{1} = v/(z - a)$, where $v \neq 0$. Let $q(s) \in s\,k[[s]]$ be an invertible formal coordinate and $g(s) \in k((s))$. If a state–field map satisfied

$$Y(v, s) = g(s) E(1/q(s)), \qquad Y(w, s)\mathbf{1} = e^{sT} w, \qquad Y(Tw, s) = \partial_s Y(w, s),$$

then all the e_r would lie in a single scalar line of $\mathrm{End}(V)$. Indeed creation says

$$e^{sT} v = \frac{g(s) q(s)}{1 - a q(s)} v.$$

Its linear coefficient gives $Tv = cv$. Translation then forces $\partial_s Y(v, s) = c Y(v, s)$. The Laurent coefficient recurrence in characteristic zero gives $Y(v, s) = e^{cs} A$ for one operator A. Dividing by the nonzero scalar g and reversing the coordinate change proves the claim. For the Hecke action e_0 and e_1 are independent, so this prescription fails.

There is nevertheless a direct way to realize these operators as current modes on a larger space. Let $\mathfrak{l}$ be their Lie algebra inside $\mathrm{End}(V)$, acting on V. Regard V as a $\mathfrak{l}[\zeta]$-module by evaluation at $\zeta = 0$. The induced module is

$$\widehat{V} = U(\mathfrak{l}[\zeta, \zeta^{-1}]) \otimes_{U(\mathfrak{l}[\zeta])} V \simeq U(\zeta^{-1}\mathfrak{l}[\zeta^{-1}]) \otimes V.$$

It is restricted: a sufficiently high positive mode kills any fixed finite word, after commuting past its negative modes. Consequently

$$J_b(s) = \sum_{n \in \mathbb{Z}} (b\zeta^n) s^{-n-1}$$

is a field on $\widehat{V}$, and its zero mode on the inducing top is b. The loop bracket gives the current vertex-algebra module identities at level zero. This construction adds states and negative modes; it does not turn the original rational half-current on V into the field of its vacuum output.

Integral doubles impose a further constraint. Let B be a separated q-adic algebra with $[B, B] \subset qB$. If an ordinary Yangian maps to B, its relations $[\,\psi_2, e_j\,] = 2e_j$ and $[\,\psi_2, f_j\,] = -2f_j$ force every image of e_j, f_j into qB, then into q^2B, and inductively into every q^nB. Separation makes them zero; the mixed commutators then kill all ψ_j. Thus such a map factors through the scalar augmentation. This argument concerns the full ordinary presentation. It does not obstruct the positive action above, or a presentation whose Cartan relations have been scaled.

The pairing in a triangular double must likewise be normalized before denominators can be cancelled. For example, suppose

$$\Delta e = e \otimes 1 + k \otimes e, \qquad \Delta f = 1 \otimes f + f \otimes l$$

with k, l grouplike. Let a skew Hopf pairing satisfy $\langle e, f\rangle = b$, $\langle k, f\rangle = \langle e, l\rangle = 0$, and the unit–counit normalization. In the convention

$$\sum a_{(1)} c_{(1)} \langle a_{(2)}, c_{(2)}\rangle = \sum \langle a_{(1)}, c_{(1)}\rangle c_{(2)} a_{(2)},$$

substitution of $a = e, c = f$ gives

$$[e, f] = b(l - k). \tag{12.22}$$

If $l - k = \sigma_3\theta$, the normalization $b = \sigma_3^{-1}$ produces $[e, f] = \theta$ over the generic field. An integral double requires lattices on which this product, both coproducts and the full pairing are defined. The cancellation in one commutator does not by itself provide them.

12.5 An oscillator Hopf algebra

Let k have characteristic zero, Q be a discrete vector space, and $E = Q^*_{\mathrm{cont}}$ its linearly compact dual. Fix the perfect evaluation pairing $\beta : E \times Q \to k$. The residue example is

$$Q = k((t))/k[[t]], \qquad E = k[[t]]\,dt, \qquad \beta(\omega, f) = \mathrm{Res}_{t=0}(f\omega).$$

Write $q(v)$, $p(e)$ for copies of $v \in Q, e \in E$. The Heisenberg bracket is $[\,p(e), q(v)] = \beta(e, v)c$, with c central. On $\mathrm{Sym}(Q)$ it has the Fock action

$$q(v) = M_v, \qquad p(e) = \kappa D_e, \qquad c = \kappa, \qquad D_e(v) = \beta(e, v).$$

Each polynomial is killed by an open subspace of E. For $\kappa \neq 0$, the quotient algebra $c = \kappa$ has no scalar character: applying one to $[\,p(e), q(v)] = \kappa$ with $\beta(e, v) = 1$ gives a contradiction. Its vacuum vector therefore cannot serve as a Hopf counit.

The canonical tensor $t = \sum_i q(v_i) \otimes p(e_i)$ is interpreted through finite continuous evaluations. It obeys

$$[t_{12}, t_{13}] = [t_{13}, t_{23}] = 0, \qquad [t_{12}, t_{23}] = \Theta := \sum_i q(v_i) \otimes c \otimes p(e_i) \neq 0.$$

The coefficient of one dual pair detects the last inequality. In particular t does not solve the constant classical Yang–Baxter equation.

There is also a polar obstruction. Let M_2, M_3 be tensor coefficient spaces for the square and cube of the Heisenberg enveloping algebra, in which the displayed contractions and brackets are defined and a dual-pair coefficient detects Θ. A strict spectral solution

$$S(z) = 1 + \hbar(t/z + a(z)) + O(\hbar^2), \qquad a(z) \in M_2[[z]],$$

would have classical Yang–Baxter coefficient $\Theta/(uv)$ in the expansion at $u = 0$, followed by $v = 0$. Every term involving the regular correction belongs to

$$M_3((v))[[u]] + M_3[[v]]((u)) \subset M_3((v))((u)).$$

The double residue kills this sum and sends $\Theta/(uv)$ to Θ. Second-order coefficients of S cancel between the two sides of Yang–Baxter. Hence no such correction can give a strict solution while retaining this polar coefficient and the central generator.

Adjoin a weight derivation d with $[d, q(v)] = -q(v)$, $[d, p(e)] = p(e)$. For $r_\gamma = \gamma d \otimes c + t$, direct commutation gives

$$\mathrm{CYB}(r_\gamma) = (1 - \gamma)\Theta.$$

The coefficient $\gamma = 1$ supplies a constant classical solution. It has a quantization with a controlled tensor topology. Put $R_\hbar = k[[\hbar]]$ and $g_\hbar(c) = (e^{\hbar c} - 1)/\hbar$. Define $\mathcal{U}_\hbar$ by the relations

$$[p(e), q(v)] = \beta(e, v)\, g_\hbar(c), \qquad [d, q(v)] = -q(v), \qquad [d, p(e)] = p(e), \tag{12.23}$$

with all other displayed brackets zero. Its vectors are algebraic in Q, E, c, d, completed only in $\hbar$.

Theorem 12.5. *The algebra $\mathcal{U}_\hbar$ is topologically free, with ordered PBW module*

$$\big(\mathrm{Sym}(Q) \otimes k[c, d] \otimes \mathrm{Sym}(E)\big)[[\hbar]].$$

It is a Hopf algebra with primitive $q(v), c, d$, and

$$\Delta p(e) = p(e) \otimes 1 + e^{\hbar c} \otimes p(e), \qquad S(p(e)) = -e^{-\hbar c} p(e). \tag{12.24}$$

The counit kills all generators, and the antipode negates the primitive ones.

Proof. Order q's before c, d, and p's last. The mixed commutator is central. The only overlap involving three different types reduces to

$$[d, [p, q]] = 0 = [[d, p], q] + [p, [d, q]].$$

Commuting differential Ore extensions, followed by the weight derivation, therefore give the indicated basis at each precision. Every expression uses finitely many vectors of Q, E, so this argument has no finite-dimensional restriction.

The coproduct preserves the mixed bracket because

$$g_\hbar(c_1 + c_2) = g_\hbar(c_1) + e^{\hbar c_1} g_\hbar(c_2).$$

The weight relations use primitivity of d. Since $e^{\hbar c}$ is grouplike, both iterated coproducts of p are

$$p \otimes 1 \otimes 1 + e^{\hbar c} \otimes p \otimes 1 + e^{\hbar c} \otimes e^{\hbar c} \otimes p.$$

The counit and both antipode identities follow on the generators; the antipode respects the relations in reverse order. All maps take values in the $\hbar$-completed algebraic tensor. □

A smooth module is a compatible system of discrete $\mathcal{U}_\hbar/(\hbar^N)$-modules M_N, with

$$M_{N+1} \otimes_{k[\hbar]/(\hbar^{N+1})} k[\hbar]/(\hbar^N) \simeq M_N,$$

such that $p(-)m : E \to M_N$ is continuous for every m. Equivalently an open finite-codimension subspace of E kills m. Morphisms are compatible module maps. The coproduct defines their tensor product stage by stage.

Theorem 12.6. *On smooth modules the operator*

$$\mathcal{R} = \exp(\hbar d \otimes c)\, \exp\!\left(\hbar \sum_i q(v_i) \otimes p(e_i)\right) \tag{12.25}$$

is continuous, invertible and independent of the dual coordinates. It satisfies

$$\mathcal{R}\Delta(x)\mathcal{R}^{-1} = \Delta^{\mathrm{op}}(x), \qquad (\Delta \otimes 1)\mathcal{R} = \mathcal{R}_{13}\mathcal{R}_{23}, \qquad (1 \otimes \Delta)\mathcal{R} = \mathcal{R}_{13}\mathcal{R}_{12}.$$

Thus $\tau\mathcal{R}$ is a braiding, with scalar unit at level zero.

Proof. For $n \in N_N$, choose an open annihilator $U \subset E$. The contraction in the second exponential then uses only the finite-dimensional pair E/U and $U^\perp \subset Q$. The $p(e)$'s commute, so this same annihilator suffices for all their products on n. Modulo $\hbar^N$ both exponentials are finite sums. Their definitions are canonical contractions and commute with reduction. The intersection of annihilators proves smoothness of the tensor product.

Write $A = \hbar d_1 c_2$, $B = \hbar \sum_i q_i^{(1)} p_i^{(2)}$. Conjugation first by e^B, then by e^A, sends

$$\Delta q = q_1 + q_2 \;\longmapsto\; q_1 e^{\hbar c_2} + q_2 \;\longmapsto\; q_1 + q_2,$$

and

$$\Delta p = p_1 + e^{\hbar c_1} p_2 \;\longmapsto\; p_1 + p_2 \;\longmapsto\; e^{\hbar c_2} p_1 + p_2.$$

Both exponentials commute with Δc, Δd. This proves the intertwining identity. Primitivity gives the first hexagon. For the second, the coproduct of p gives

$$(1 \otimes \Delta)\mathcal{R} = e^{\hbar d_1 (c_2 + c_3)} e^{\hbar \sum_i q_i^{(1)} p_i^{(2)}} e^{\hbar e^{\hbar c_2} \sum_i q_i^{(1)} p_i^{(3)}}.$$

Conjugation by $e^{-\hbar d_1 c_2}$ multiplies $q_i^{(1)}$ by $e^{\hbar c_2}$, so moving the middle factors of $\mathcal{R}_{13}\mathcal{R}_{12}$ gives exactly this expression. These calculations use finite contractions on every input at every precision. They prove the identities on the complete module systems. The hexagons and intertwining give Yang–Baxter; the associator is the ordinary tensor associator. □

On $F_{\kappa,\sigma} = \mathrm{Sym}(Q)[[\hbar]]$, put

$$c = \kappa, \qquad q(v) = M_v, \qquad p(e) = g_\hbar(\kappa) D_e, \qquad d = \sigma - \mathsf{E},$$

where E is polynomial degree. At $\sigma = 0$, on two such modules of levels κ, λ,

$$\mathcal{R} f(x, y) = f(bx, y + (1 - b)x), \qquad b = e^{-\hbar\lambda}. \tag{12.26}$$

The right exponential translates y by $(e^{\hbar\lambda}-1)x$; the left then rescales x. This proves the formula on every polynomial coefficient. Levels add, so a fixed nonzero-level sector is not closed under this tensor product.

The categorical braiding is not the vacuum-normalized exchange of a field algebra on these Fock states:

$$\mathcal{R}(x\otimes 1)=e^{-\hbar\lambda}x\otimes 1,\qquad \mathcal{R}(1\otimes x)=1\otimes x+(1-e^{-\hbar\lambda})x\otimes 1.$$

For $\lambda\neq 0$, the first defect begins with $-\hbar\lambda x\otimes 1$. An invertible comparison preserving the vacuum insertions cannot kill this defect. The scalar monoidal unit and the vacuum in a nonzero-level module have different central actions.

Nor can the algebra be enlarged by every formal normal-ordered symbol without checking the antipode. In the ordinary Heisenberg algebra let $E_N=\sum_{i<N}q_ip_i$, with $[p_i,q_i]=c$. Then $S(E_N)=E_N+Nc$. If a separated linear completion contained $\lim_N E_N$ and retained a continuous scalar detector with $\ell(q_ip_i)=0$, $\ell(c)=1$, continuity of the primitive antipode would force the sequence N to converge in discrete k. It does not. The smooth-module interpretation of (12.25) avoids asserting this larger symbol completion.

12.6 Dividing the augmentation ideal

We now return to the spectral forms of Chapter 10. Work over $R_\hbar=k[[\hbar]]$, with k of characteristic zero. A formal spectral datum consists of a topologically free Hopf algebra H, a bijective antipode S, a continuous derivation and coderivation T with $\varepsilon T=0$, and a form

$$\rho_z : H\widehat{\otimes}H\longrightarrow k[z,z^{-1}][[\hbar]].$$

The form satisfies the unit, two product, commutation, translation and unitarity identities of Definition 10.2 at every $\hbar$-adic precision. Assume $H/\hbar H$ commutative and $\rho_z\equiv\varepsilon\otimes\varepsilon \pmod\hbar$. Levels are continuous Hopf automorphisms η satisfying Definition 10.5, with $(\eta-1)H\subset\hbar H$. The target of the form permits its Laurent degree to grow with the $\hbar$-adic precision.

Write

$$J=\ker\varepsilon,\qquad I=J+\hbar H,\qquad H^\times=\bigcup_{n\geq 0}\hbar^{-n}I^n\subset H[\hbar^{-1}],\qquad H^\vee=\varprojlim_N H^\times/\hbar^N H^\times.$$

This is the divided augmentation construction [64]. The superscript $\vee$ here denotes this algebra, not a linear dual. The original coordinate algebra H and the divided state algebra $H^\vee$ have different roles.

Lemma 12.7. *For $m,n\geq 0$, with closures understood at each finite precision,*

$$\begin{aligned}\Delta(I^n)&\subset\sum_{r+s=n}I^r\widehat{\otimes}I^s,\\ \delta_\eta(I^n)&\subset I^n\widehat{\otimes}H,\qquad \delta_\eta(x)=x_2\otimes x_3S^{-1}(\eta x_1),\\ \rho_z(I^m,I^n)&\subset\hbar^{\max(m,n)}k[z,z^{-1}][[\hbar]].\end{aligned}\tag{12.27}$$

The Hopf maps and T extend to $H^\vee$, and the coaction extends to $H^\vee\to H^\vee\widehat{\otimes}H$. The form extends integrally on either mixed carrier $H\widehat{\otimes}H^\vee$ or $H^\vee\widehat{\otimes}H$.

Proof. Since $H=R_\hbar 1\oplus J$, $\Delta(J)\subset J\otimes H+H\otimes J$. Including $\hbar$ and taking products gives the first estimate. Both antipodes, T and the levels preserve I^n; also $\varepsilon(I^n)\subset\hbar^nR_\hbar$. Division therefore extends these maps.

For $x \in I$, applying ε to the first factor of $\delta_\eta(x)$ gives $x_2 S^{-1}(\eta x_1) \in \hbar H$. Here $\eta \equiv 1 \pmod{\hbar}$ and $x_2 S^{-1}(x_1) = \varepsilon(x)$. The splitting by J proves the second estimate for $n = 1$. For $b \in I^{n-1}$, the coaction product formula reads

$$\delta_\eta(xb) = x_{(0)} b_{(0)} \otimes b_{(1)} x_{(1)} + x_2 b_{(0)} \otimes [x_3, b_{(1)}] S^{-1}(\eta x_1).$$

The first term has its first factor in I^n. Commutativity modulo $\hbar$ places the second in $\hbar I^{n-1} \otimes H \subset I^n \otimes H$. This proves the induction.

For $x \in J$, the normalization of ρ gives $\rho_z(H, x) \subset \hbar k[z, z^{-1}][[\hbar]]$; the same holds for the scalar generator $\hbar$ of I. The second product rule gives a factor $\hbar^n$ on I^n in the second argument. The first product rule gives $\hbar^m$ in the first. Both estimates hold simultaneously. Dividing in just one argument is therefore integral. All extended identities follow at finite precision and pass to the separated inverse limit. □

Dividing both arguments is different: on $\hbar^{-m} I^m \otimes \hbar^{-n} I^n$, the estimate allows a pole of order $\min(m, n)$ in $\hbar$. The field construction uses the mixed pairing precisely where integrality is available. Writing $\delta_\eta(b) = b_{(0)} \otimes b_{(1)}$, its formulas are

$$\begin{aligned} A_z(a)b &= b_{(0)} \rho_{-z}(b_{(1)}, a), \\ Y_\eta(a, z)b &= a_1(z) b_{(0)} \rho_{-z}(b_{(1)}, a_2), \\ \mathcal{S}_\eta(z)(a \otimes b) &= a_{(0)} \otimes b_{(0)} \rho_z(a_{(1)}, b_{(1)}), \qquad a(z) = e^{zT} a. \end{aligned} \tag{12.28}$$

In the second line a_i are divided states and $b_{(1)}$ is a coordinate. In the third line both paired factors are coordinates. Lemma 12.7 therefore makes all formulas integral. The unit, translation, common-germ substitutions and exchange identities of Theorem 10.8 extend by division to algebraic words and then by continuity. The level-changing coproduct $(\eta_2 \otimes 1)\Delta$ extends in the same way, with the crossed field tensor of Theorem 10.10. The coaction is still by H.

12.6.1 *The full cotangent space*

Let $A = H/\hbar H$ and $\mathfrak{m} = \ker(\varepsilon : A \to k)$. Its Poisson bracket is $\{f, g\} = \hbar^{-1}[\widetilde{f}, \widetilde{g}] \bmod \hbar$. The cotangent Lie algebra is

$$Q = \mathfrak{m}/\mathfrak{m}^2 = J/(J^2 + \hbar J), \qquad [\langle x \rangle, \langle y \rangle] = \langle \hbar^{-1}[x, y] \rangle.$$

Define $U_t(Q)$ as the graded $k[t]$-algebra generated by Q in degree one, with t central of degree one and relations $xy - yx = t[x, y]_Q$.

Theorem 12.8. *There are canonical isomorphisms*

$$\operatorname{gr}_I H \simeq U_t(Q), \qquad H^\vee/\hbar H^\vee \simeq U(Q). \tag{12.29}$$

The second is a Hopf isomorphism taking $\langle x \rangle$ to $x/\hbar \bmod \hbar$. Ordered monomials in lifts of Q, divided by $\hbar$, form a topological basis of $H^\vee$.

Proof. A commutative Hopf algebra over a characteristic-zero field is formally smooth, including without finite generation [52, Theorem 1.2(1), Section 3.1]. Choose an ordered basis v_b of Q and lifts $x_b \in J$. The local coordinate lemma for the augmentation filtration [64, Lemma 4.1, Section 4.8] identifies $\widehat{H}_I$ with ordered series

$$\sum_{r,\gamma} c_{r,\gamma} \hbar^r x^\gamma$$

having finitely many terms below each total degree $r + |\gamma|$. Reduction modulo $\hbar$ gives the formal coordinates at the counit of A; successive lifting gives existence and uniqueness of the expansion. Reordering preserves total degree

because $[x_b, x_c] \in \hbar J$. At any fixed degree only finitely many coordinates occur. In particular multiplication by $\hbar^s$ raises the filtration by exactly s. Pulling back to H gives

$$(I^n : \hbar^s) = I^{n-s} \quad (0 \le s \le n). \tag{12.30}$$

Elements in the kernel of I-adic completion belong to every I^n and cause no exception to this formula.

The symbols of $\hbar$, x_b generate $\operatorname{gr}_I H$. Their commutator in degree two is $t[x_b, x_c]_Q$. This defines a surjection from $U_t(Q)$. The ordered coordinate basis identifies its PBW basis with the associated-graded basis, proving the first isomorphism.

The divided images generate $H^\times/\hbar H^\times$. If $x/\hbar \in \hbar H^\times$, then $\hbar^{n-2}x \in I^n$ for some $n \ge 2$; (12.30) gives $x \in I^2$. Intersecting with J gives $J \cap I^2 = J^2 + \hbar J$. Thus no additional cotangent class vanishes. Applying the same ordered-coordinate argument to a finite linear combination of divided PBW monomials proves their independence modulo $\hbar$. They span by the definition of $H^\times$, so this quotient is $U(Q)$. Successive lifting of its basis gives the topological basis of $H^\vee$. The reduced coproduct makes each cotangent generator primitive, proving the Hopf assertion. □

The characteristic-zero hypothesis has content. In characteristic three the free rank-three Hopf algebra

$$H = k[[\hbar]][y]/(y^3 - \hbar y), \qquad \Delta y = y \otimes 1 + 1 \otimes y$$

has one-dimensional cotangent space. But $X = y/\hbar$ satisfies $X = \hbar X^3$, hence $X = \hbar^{(3^r-1)/2} X^{3^r}$ for every r. Its separated divided completion is only $k[[\hbar]]$. The failure is already $\hbar y = y^3 \in I^3$ with $y \notin I^2$. Disconnected components can also disappear without contradicting (12.29): for $H = k[C_2][[\hbar]]$, the nontrivial augmentation idempotent belongs to every I^n; the cotangent space is zero and the divided completion is scalar.

The coproduct of A also induces a Lie cobracket

$$\delta_Q(\pi f) = (\pi \otimes \pi)(\Delta_A f - \Delta_A^{\mathrm{op}} f), \qquad \pi f = \langle f - \varepsilon(f) \rangle.$$

A product of augmentation elements contributes symmetrically modulo $\mathfrak{m}^2$ in each factor, so this is well defined. Coassociativity and the Poisson coproduct identity prove coJacobi and compatibility with the bracket. It is the co-Poisson limit of $H^\vee$.

For instance a finite free coordinate comodule M with matrix (a_{ij}) has character $\chi = \sum_i a_{ii}$, and

$$\frac{\chi - \operatorname{rank}(M)}{\hbar} \in H^\vee$$

has cocommutative coproduct: $\Delta\chi = \sum_{i,j} a_{ij} \otimes a_{ji}$. A grouplike q with counit one gives the primitive $\hbar^{-1} \log q$, since $q - 1 \in \hbar H^\vee$. These are positive lifting constructions. For a general $a \in U(Q)$, the first obstruction is

$$\omega_1(a) = \hbar^{-1}(\Delta - \Delta^{\mathrm{op}})\widetilde{a} \bmod \hbar.$$

It is independent of the lift: replacing $\widetilde{a}$ by $\widetilde{a} + \hbar b$ changes it by a coproduct defect which vanishes modulo $\hbar$. A cocommutative lift requires $\omega_1(a) = 0$; on Q this is precisely δ_Q.

12.7 The cotangent double and current fields

Give Q the discrete topology and $\mathfrak{g} = \operatorname{Hom}_k(Q, k)$ the topology of pointwise convergence on finite-dimensional subspaces. The continuous dual of $\mathfrak{g}$ is Q. The bracket dual to δ_Q is convolution:

$$[\xi, \zeta](f) = \xi(f_1)\zeta(f_2) - \zeta(f_1)\xi(f_2).$$

Elements of $\mathfrak{g}$ are derivations of A at the counit. For the coadjoint actions use

$$\langle \zeta, \operatorname{ad}^*_\xi q\rangle = -\langle [\xi,\zeta], q\rangle, \qquad (\operatorname{ad}^*_q \xi)(q') = -\xi([q,q']).$$

The first action takes values in Q, since $\delta_Q(q)$ is an algebraic tensor. The continuous double $\mathfrak{D} = Q \oplus \mathfrak{g}$ has mixed bracket

$$[\xi, q] = \operatorname{ad}^*_\xi q - \operatorname{ad}^*_q \xi$$

and invariant form $B(q+\xi, q'+\zeta) = \xi(q') + \zeta(q)$. Mixed Jacobi, evaluated against either summand, is exactly Lie-bialgebra compatibility. Thus the construction applies to this infinite-dimensional dual pair with its specified topology.

Let $\theta = \hbar^{-1}(\eta - 1) \bmod \hbar$ and $D\langle f\rangle = \langle \theta f\rangle$. Expansion of the Hopf identities makes θ a derivation and coderivation; expansion of the commutator identity to second order shows that it preserves the Poisson bracket. Extend D to $\mathfrak{D}$ by $D\xi = -\xi \circ D$. Then

$$c_\eta(u,v) = B(Du, v) \tag{12.31}$$

is a continuous Lie two-cocycle. Indeed D is a derivation skew for B. For $\varphi(u,v,w) = B([u,v],w)$, invariance and skewness give

$$\varphi(Du,v,w) + \varphi(u,Dv,w) + \varphi(u,v,Dw) = 0;$$

the cocycle identity is twice this equality. In particular $c_\eta(\xi, q) = -\xi(Dq)$.

On $V = H^\vee/\hbar H^\vee = U(Q)$, the reduced coaction defines

$$\mathcal{A}^\xi b = b_{(0)}\xi(b_{(1)}), \qquad L(q)b = qb.$$

These operators represent the central extension $\widehat{\mathfrak{D}}_\eta = \mathfrak{D} \oplus k\mathbf{c}$, with $\mathbf{c}$ acting by one. Coassociativity proves $[\,\mathcal{A}^\xi, \mathcal{A}^\zeta\,] = \mathcal{A}^{[\xi,\zeta]}$. Expanding the divided coaction product gives the mixed relation

$$[\mathcal{A}^\xi, L(q)] = L(\operatorname{ad}^*_\xi q) - \mathcal{A}^{\operatorname{ad}^*_q \xi} - \xi(Dq)\,\mathrm{id}\,. \tag{12.32}$$

Its three terms come respectively from the adjoint coaction of A, its Poisson bracket and the first-order level. For example the first term, evaluated by ζ, is $(\zeta * \xi - \xi * \zeta)(f)$ for $q = \langle f\rangle$; this fixes its sign. The vacuum is killed by $\mathfrak{g}$. PBW consequently identifies

$$U(\widehat{\mathfrak{D}}_\eta) \otimes_{U(\mathfrak{g}\oplus k\mathbf{c})} k_1 \xrightarrow{\sim} U(Q), \tag{12.33}$$

where $\mathfrak{g}$ kills k_1 and $\mathbf{c}$ acts as one. The source is algebraic $U(Q)$, and its action on the vacuum is the identity under Theorem 12.8.

The spectral form supplies fields as well as these Lie actions. Put

$$r_z(f,g) = \hbar^{-1}(\rho_z(f,g) - \varepsilon(f)\varepsilon(g)) \bmod \hbar.$$

It is a derivation at the counit in each argument. For $q = \langle x\rangle$, define $\xi_q(z)(\langle f\rangle) = r_{-z}(f,x)$. The filtration estimate makes this independent of the lifts. Its carrier is

$$\operatorname{Hom}_k(Q, k[z,z^{-1}]) = \varprojlim_{\substack{F\subset Q\\ \dim F<\infty}} F^* \otimes k[z,z^{-1}],$$

with a Laurent bound on each finite-dimensional F, rather than a uniform bound on Q. Reduction of (12.28) gives

$$Y(q,z) = L(e^{zT}q) + \mathcal{A}^{\xi_q(z)}. \tag{12.34}$$

The coaction of a fixed state has finitely many coordinate factors, so the annihilation term is Laurent on that state. The quantum exchange is the identity modulo $\hbar$; the common-germ field identities therefore reduce to the ordinary vertex-algebra identities on V.

Its singular products preserve $k\mathbf{1} \oplus Q$. In fact (12.32) applied to the vacuum shows that the singular part of $Y(q,z)q'$ belongs to $Q \oplus k\mathbf{1}$. With the inherited translation and $[q_\lambda q'] = \sum_{n\geq 0} \lambda^n q_{(n)}q'/n!$, this is a Lie conformal algebra R_η. The distinguished central $\mathbf{1}$ is retained. Its pointed vertex envelope is canonically V.

Here is a PBW proof that does not assume R_η free over $k[T]$. Its coefficient Lie algebra is

$$\mathcal{L}(R_\eta) = \frac{R_\eta \otimes k[t,t^{-1}]}{(T\otimes 1 + 1\otimes \partial_t)(R_\eta \otimes k[t,t^{-1}])}, \qquad [u(m),v(n)] = \sum_{j\geq 0} \binom{m}{j} (u_{(j)}v)(m+n-j).$$

Its negative part identifies with R_η by $u(-r) \leftrightarrow T^{r-1}u/(r-1)!$. Inducing from the nonnegative part and imposing $\mathbf{1}(-1) = 1$ gives associated graded $\mathrm{Sym}(Q)$. The coefficient fields are local and create the generators; their iterated normal products construct the universal vertex algebra. On $U(Q)$, relation (12.32) shows that every $\mathcal{A}^\xi$ preserves PBW degree. Thus the (-1)-mode of q is $L(q)$ plus a degree-preserving operator. The map from the envelope has associated graded the identity on $\mathrm{Sym}(Q)$. Induction on the exhaustive nonnegative filtration proves it an isomorphism, preserving all fields and translation.

12.8 The determinant-one rational double

The rational matrix example makes both halves independent before their action on the vacuum is considered. Fix $N \geq 2$, $R_\hbar = \mathbb{C}[[\hbar]]$, and let P flip $\mathbb{C}^N \otimes \mathbb{C}^N$. Normalize

$$\mathsf{R}(w) = f(w)(I - \hbar P/w), \qquad f(w+N\hbar) = f(w)(1-\hbar^2/w^2), \qquad f(w) = 1 + O(\hbar/w). \tag{12.35}$$

The scalar recurrence determines f coefficientwise. It gives unitarity and the crossing identity

$$\big((\mathsf{R}(w)^{t_1})^{-1}\big)^{t_1} = \mathsf{R}(w-N\hbar)^{-1};$$

the first term is $I - \hbar(P - N^{-1}I)/w$. These are the determinant-normalized conventions of [56, Sections 1.1–1.6].

Use independent matrices

$$L(v) = I + \hbar \sum_{r\geq 1} \ell^{[r-1]} v^{r-1}, \qquad C(p) = I + \hbar \sum_{r\geq 1} m^{(r)} p^{-r}.$$

For either matrix X, define the descending-column determinant

$$q_X(u) = \sum_{\sigma\in S_N} (-1)^\sigma X_{\sigma(1),1}(u) X_{\sigma(2),2}(u-\hbar)\cdots X_{\sigma(N),N}(u-(N-1)\hbar).$$

The creation algebra U and annihilation algebra U_+ are generated by the divided coefficients of L, C, respectively. Each satisfies RTT and $(q_X - 1)/\hbar = 0$. For $K \in \mathbb{C}$, impose in addition

$$\mathsf{R}_{12}(p - v + K\hbar/2) C_1(p) L_2(v) = L_2(v) C_1(p) \mathsf{R}_{12}(p - v - K\hbar/2). \tag{12.36}$$

All RTT and mixed defects are divided by $\hbar^2$ before their coefficients are imposed. Clear auxiliary denominators in the creation relation; expand the mixed relation in $p^{-1}, v, \hbar$. The closed ideal of these divided relations defines D_K. Initially every $\hbar$-coefficient is a finite polynomial in modes. Flatness is not imposed as an extra quotient.

Theorem 12.9. *The algebras U, U_+, D_K are topologically free over $R_\hbar$, and multiplication is an isomorphism of topological modules*

$$U \widehat{\otimes}_{R_\hbar} U_+ \xrightarrow{\sim} D_K. \tag{12.37}$$

In each half, ordered monomials in the off-diagonal generators and $N-1$ independent diagonal generators give a topological basis. Moreover

$$D_K/\hbar D_K \simeq U(\mathfrak{sl}_N[t,t^{-1}] \oplus \mathbb{C}\mathbf{c})/(\mathbf{c}-K),$$

where

$$[xt^a, yt^b] = [x,y]t^{a+b} + a\,\partial_{a+b,0}\,\mathrm{tr}(xy)\mathbf{c}.$$

The creation and annihilation halves reduce respectively to $U(t^{-1}\mathfrak{sl}_N[t^{-1}])$ and $U(\mathfrak{sl}_N[t])$.

Proof. Retain the full diagonal families and a variable central $\mathbf{c}$ first. The general linear double has the PBW basis of [89, Theorem 2.2, Corollary 2.3, Section 4.1]. Their matrices are $T = C, T^+ = L$, with $t^{(-r)} = -\ell^{[r-1]}$, $t^{(r)} = m^{(r)}$. The filtration

$$\deg t^{(r)} = r-1, \qquad \deg t^{(-r)} = -r, \qquad \deg \mathbf{c} = 0$$

produces the present divided relations in its Rees algebra, after spectral rescaling. Negative filtration degrees cause no difficulty before completion: expressions in the localization are finite sums. The Rees PBW basis and then $\hbar$-completion give a topologically free general linear algebra, free also over the central parameter. Specializing $\mathbf{c} = K$ retains this basis.

The divided determinant coefficients $d_L = (q_L - 1)/\hbar$, $d_C = (q_C - 1)/\hbar$ are central. For the mixed commutation, the antisymmetrizer A_N obeys

$$A_N \mathsf{R}_{0N}(w + (N-1)\hbar) \cdots \mathsf{R}_{01}(w) = A_N.$$

Without the scalar factors the product is $A_N(1 - \hbar/w)$, while $\prod_{j=0}^{N-1} f(w + j\hbar) = (1 - \hbar/w)^{-1}$. Moving C through the fused determinant of L therefore cancels both numerical products, including their central shifts. The argument with the halves interchanged is the same. Within either half this is the RTT determinant identity. Torsion-freeness permits division of the commutator by $\hbar$.

At each mode the leading term of a divided determinant coefficient is the trace of the divided matrix. Replace one diagonal generator by that central coefficient. Modulo $\hbar$ this is an invertible change of basis in $\mathfrak{gl}_N$. PBW and successive lifting prove that the new ordered monomials remain a topological basis: subtract a lift at each precision and divide by $\hbar$; the first nonzero coefficient rules out a kernel. Put the central coefficients last. The closed ideal they generate is precisely the span of monomials containing one of them, by centrality and factorization of that coefficient. The remaining monomials form a free complementary summand. This proves the determinant-one quotient is saturated, its factorization, and its displayed classical limit. □

Formal positive modes require a different completion. Let $I_{d,M}$ be the left ideal generated by $\hbar^d$ and the $m_{ij}^{(r)}$ with $r > M$, and put

$$D_K^{\mathrm{sm}} = \varprojlim_{d,M} D_K/I_{d,M}.$$

These are module quotients, not asserted to be algebra quotients. The corresponding classical carrier is

$$t^{-1}\mathfrak{sl}_N[t^{-1}] \oplus \mathfrak{sl}_N[[t]] \oplus \mathbb{C}\mathbf{c},$$

with the positive half t-adic and the negative half discrete. The same left-ideal construction defines its completed enveloping algebra at level K.

PBW extends to this completion. At precision $\hbar^d$, commuting a high positive mode past a fixed creation word lowers its index by a bounded amount, as follows from the expansion in (12.36). A determinant correction of order $\hbar^j$ contains at most $j+1$ mode factors, whose total positive index is at least $r-j$ if it replaces a trace of index r. Following a marked high mode during ordering therefore leaves some factor of index at least $(r-d)/(d+1)$. If $B_{d,M}$ is the span of $\hbar^d D_K$ and ordered PBW monomials containing a positive index greater than M, then

$$B_{d,M} \subset I_{d,M}, \qquad I_{d,L} \subset B_{d,M} \quad \text{for } L > (d+1)M + d.$$

The two systems are cofinal. Thus the completion consists of PBW series with finite support at each precision after bounding all positive indices. This proves separation, $\hbar$-torsion-freeness and the stated completed classical limit. Left multiplication preserves the left-ideal neighborhoods; right multiplication by a fixed word is continuous by the index bound. Approximating one factor at a time extends multiplication to Cauchy series, and density proves associativity.

For a finite free continuous U_+^{sm}-module E, the separated completed balanced tensor consequently satisfies

$$D_K^{\mathrm{sm}} \widehat{\otimes}^{\mathrm{cont}}_{U_+^{\mathrm{sm}}} E \simeq U \widehat{\otimes} E.$$

At each precision all sufficiently high positive modes kill E. PBW ordering therefore evaluates a completed element on the top to a finite creation polynomial at that precision. The inverse is the inclusion of creation tensors. Both composites are identities on finite words and hence on their completions. No infinite sum of creation modes has been added to the state space.

12.8.1 *The independent field comparison*

Let F denote the determinant-one coordinate Hopf algebra: its matrix $L(v)$ is not divided by $\hbar$. Its commutative reduction is the coordinate algebra of $SL_N[[t]]$; its map to the divided algebra sends L to $I + \hbar\ell$. The form on F is $\rho_z(L_1(u), L_2(v)) = \mathsf{R}(z+u-v)$. Take $\eta_K = \tau_{-K\hbar}$, where $\tau_s L(v) = L(v+s)$. The mixed pairing and coaction of Lemma 12.7 give the fields on U.

On the independently induced vacuum module of D_K, the matrix C satisfies $C(p)\mathbf{1} = I$. The mixed relation recursively determines its value on every creation word. The coaction annihilation operator obeys the same recursion, with

$$C_0(z + u + K\hbar/2) = A_z(S^{-1}L_0(u)) = A_z(L_0(u))^{-1}. \tag{12.38}$$

This is an inverse of matrices of operators on all states. It is not obtained by silently replacing a partial inverse of a numerical tensor with an ordinary inverse. The recursion follows by pairing the coaction product against $L_0(u)$; the inverse-antipode pairing is evaluated with the crossing identity following (12.35). Both sides equal I on the vacuum. PBW induction and the integral divided formulas prove the equality on every state.

For a word of creation matrices this gives

$$\begin{aligned} Y_K(L_1(u_1)\cdots L_r(u_r), z) = {} & L_1(u_1+z)\cdots L_r(u_r+z) \\ & \cdot \big(C_1(u_1+z+K\hbar/2)\cdots C_r(u_r+z+K\hbar/2)\big)^{-1}. \end{aligned} \tag{12.39}$$

The reversed order inside the inverse is the annihilation anti-action. This is the quantum affine field formula of [57, Section 2.1, equation (2.10)]. Subtract I in each input matrix and divide by $\hbar^r$ to obtain the equality on every divided monomial. One can first work to r extra powers of $\hbar$; integrality then permits division, and continuity gives the complete state space. The PBW identity is therefore an isomorphism preserving the vacuum, translation and all fields, for every rank and level.

For completeness, the exchange comparison retains the full auxiliary block order. Put

$$A = L_1(u_1)\cdots L_r(u_r), \qquad B = L_{r+1}(v_1)\cdots L_{r+s}(v_s),$$

with entries in the first and second state factors, and

$$r_0 = \prod_{i=1}^{r}\prod_{j=s}^{1} \mathsf{R}_{i,r+j}(z+u_i-v_j), \qquad r_\pm = r_0(z\pm c), \qquad c = -K\hbar.$$

Then the coordinate-coaction exchange satisfies

$$\mathcal{S}_K(z)(r_0^{-1}Br_+A) = Ar_-^{-1}Br_0. \tag{12.40}$$

To check the order, let $b_+ = ((r_+^{t_A})^{-1})^{t_A}$, where t_A transposes the first auxiliary block. The coefficient of $A_{ab}\otimes B_{cd}$ in $\mathcal{S}_K(z)(A_{ij}\otimes B_{kl})$ is

$$\sum_{p,q,r',s'} (r_-^{-1})_{bs',pc}(r_0)_{pd,jr'}(r_0)_{qk,as'}(b_+)_{ir',ql}.$$

These are the four pairings of the ordinary and inverse-antipode tails. Substitution into the left side of (12.40) first contracts

$$\sum_{e,d}(r_+)_{ad,el}(b_+)_{er',qd} = \delta_{aq}\delta_{r'l},$$

which is $b_+^{t_A}r_+^{t_A} = I$, and then contracts $r_0^{-1}r_0$. The remaining expression is the right side. No entries in one state factor are commuted. The numerical input transformation is the identity modulo $\hbar$, hence invertible. It determines the exchange on coordinate words; divided finite differences and continuity extend it to every pair of states. This is equation (2.11) of [57], with its displayed level shifts.

The reduction is $V^K(\mathfrak{sl}_N)$. The cotangent double is the Laurent current algebra, paired by $\operatorname{Res}\operatorname{tr}(ab)\,dt$, and (12.31) becomes

$$K\operatorname{Res}_{t=0}\operatorname{tr}(\partial_t a(t)b(t))\,dt.$$

Indeed the negative and positive halves are isotropic and complementary, so invariance determines the mixed bracket. On the negative half $T = -\partial_t$ and $\theta = -KT = K\partial_t$; its skew extension is also $K\partial_t$ by integration by parts. The coaction derivation associated to at^m, $m\ge 0$, is the current mode $a(m)$ on every PBW state. This follows first on current generators from the rational kernel, then on products from the mixed relation and the vacuum. Sufficiently high modes kill a fixed state, so the positive formal series acts by a pointwise finite sum.

The first exchange coefficient is consequently

$$s_K(z) = -\sum_{m,n\ge 0}(-1)^m\binom{m+n}{m}z^{-m-n-1}\sum_\gamma e_\gamma(m)\otimes e^\gamma(n),$$

for trace-dual bases e_γ, e^γ. The sign is the negative rational kernel in (12.35). For two current states a, b, only modes zero and one remain:

$$s_K(z)(a\otimes b) = -\frac{\sum_\gamma [e_\gamma, a]\otimes [e^\gamma, b]}{z} + \frac{K(1\otimes [a,b] + [a,b]\otimes 1)}{z^2} + \frac{2K^2\operatorname{tr}(ab)}{z^3} 1\otimes 1. \tag{12.41}$$

The critical specialization retains a scalar pole. Set $s = z + u - v$ and $\omega_{ij,kl} = \delta_{il}\delta_{kj} - N^{-1}\delta_{ij}\delta_{kl}$. The scalar component, extracted by the state counit, is

$$\varepsilon(Y_K(\ell_{ij}(u), z)\ell_{kl}(v)) = \omega_{ij,kl}\left(\frac{K}{s^2} - \hbar\frac{K(K+N)}{s^3}\right) + O(\hbar^2). \tag{12.42}$$

At $K = -N$, its exact value is

$$-\frac{N\omega_{ij,kl}}{s^2 - \hbar^2}. \tag{12.43}$$

To derive both formulas, put $w = -s$, $c = -K\hbar$, $\varphi = f(w)/f(w+c)$ and $g = \hbar/(w + c - N\hbar)$. The four contractions give

$$\frac{\varphi - 1}{\hbar^2}\delta_{ij}\delta_{kl} + \frac{\varphi}{\hbar^2}\left(g - \frac{\hbar}{w} - \frac{N\hbar g}{w}\right)\delta_{il}\delta_{kj}.$$

For $c = \lambda\hbar$, the scalar recurrence gives

$$\varphi = 1 + \frac{\lambda\hbar^2}{Nw^2} - \frac{\lambda(\lambda - N)\hbar^3}{Nw^3} + O(\hbar^4).$$

Also the parenthesis is $-\hbar c/[w(w + c - N\hbar)]$. These yield (12.42). At $K = -N$, $c = N\hbar$ and $\varphi = (1 - \hbar^2/w^2)^{-1}$, giving (12.43). No construction above divides by $K + N$. The identity $S^2 = \tau_{N\hbar}$ gives $\eta_K = S^2\tau_{-(K+N)\hbar}$; at critical level the extra translation disappears, while the central contraction remains nonzero.

12.9 Induced modules and the level twist

A coordinate comodule alone has an annihilation action through the form. Covariant fields also need creation. This can be seen before constructing the induced module. Let E be the vector comodule with coefficient matrix $L(0)$, and try to use only

$$A^E_z(a)e = e_{(0)}\rho_{-z}(e_{(1)}, a).$$

Write $G(v)$ for the universal classical loop. For the identity level, the first covariance defect on coefficients $L^\circ_{ij}, L^\circ_{kl}$ is

$$\frac{\delta_{jk}G_{il}(0) - G_{il}(w)(G(w)^{-1}G(0))_{kj}}{w}. \tag{12.44}$$

It follows by inserting the first-order kernel $-\big(P - N^{-1}I\big)/w$ in the two coaction tails; the scalar terms cancel. In rank two, evaluate at $g(v) = I + vE_{21}$ and $(i, j, k, l) = (2, 1, 1, 1)$. The result is -1.

Changing the level cannot repair these prescribed operators and this coaction. If covariance held at a level η, its zeroth-order identity would imply

$$G_{ij}(0)\big(\varepsilon(a) - \tau_w(a_2 S_0^{-1}(\eta_0 a_1))\big) = 0$$

for every a in the classical coordinate algebra. Invertibility of $G(0)$ gives $a_2 S_0^{-1}(\eta_0 a_1) = \varepsilon(a)$. Uniqueness of the convolution inverse for the opposite coproduct then gives $\eta_0 = 1$. At first order the change of level modifies (12.44) by $-G_{ij}(0) q^{\eta}_{kl}(w)$ for a scalar coordinate function q^{η}_{kl}. The same test loop has $g_{21}(0) = 0$; its defect remains -1. This rules out the annihilation-only prescription, while leaving modules with additional states and fields available.

Let now E be any finite free $R_\hbar$-module with a continuous right F-coaction $e \mapsto e_0 \otimes e_1$. No translation on E is assumed. Set $W_K(E) = U\widehat{\otimes}E$, with left creation action $\lambda(a)(b \otimes e) = ab \otimes e$.

Theorem 12.10. *There is a unique continuous integral coaction on $W_K(E)$ whose restriction to the top is $(1 \otimes e) \mapsto (1 \otimes e_0) \otimes e_1$ and whose compatibility is*

$$\delta_K(\lambda(a)w) = \lambda(a_2) w_0 \otimes a_3 w_1 S^{-1}\eta_K(a_1). \tag{12.45}$$

The fields

$$A_z(a)w = w_0 \beta_{-z}(w_1, a), \qquad Y_W(a,z)w = \lambda(a_1(z)) A_z(a_2) w \tag{12.46}$$

are integral, with values in $\varprojlim_d (W_K(E)/\hbar^d)((z))$. They satisfy the module field identities and covariance

$$\delta_K(Y_W(a,z)w) = Y_W(a_{(0)},z)w_0 \otimes w_1 a_{(1)}(z).$$

Here $\beta : F\widehat{\otimes}U \to \mathbb{C}[z,z^{-1}][[\hbar]]$ is the mixed pairing.

Proof. The generic formula is

$$\delta_K(a \otimes e) = (a_2 \otimes e_0) \otimes a_3 e_1 S^{-1}\eta_K(a_1).$$

Its integral meaning is fixed by a recursion. Put $B_{ia}(v) = S^{-1}L_{ia}(v)$. The opposite-antipode identity is $\sum_a L_{aj}(v) B_{ia}(v) = \delta_{ij}$. For $f \in F$ define

$$d^v_{ij}(f) = \hbar^{-1}\Big(\sum_a L_{aj}(v) f B_{ia}(v - K\hbar) - \delta_{ij} f \Big).$$

Its numerator is the sum of $\sum_a L_{aj}[f, B_{ia}]$ and $\sum_a L_{aj} f (B_{ia}(v-K\hbar) - B_{ia}(v))$. Both are divisible by $\hbar$, using commutativity of $F/\hbar F$ and $\eta_K \equiv 1 \pmod{\hbar}$. For a divided generator this gives

$$\begin{aligned}\delta_K(\ell_{ij}(v)w) = {} & \sum_{a,b} \lambda(\ell_{ab}(v)) w_0 \otimes L_{bj}(v) w_1 B_{ia}(v - K\hbar) \\ & + w_0 \otimes d^v_{ij}(w_1).\end{aligned}$$

Starting at the top defines integral values on all divided words. The generic formula kills the RTT relations; the integral torsion-free target and the saturated PBW presentation prove descent to U. For determinant descent, the coordinate determinant q is central and grouplike. With $d = (q-1)/\hbar$ and $g = q(\eta_K q)^{-1}$, its coaction is

$$(d \otimes e_0) \otimes g e_1 + (1 \otimes e_0) \otimes \frac{g-1}{\hbar} e_1.$$

Imposing $d = 0$ in U and $q = 1$ in F kills it.

The two coaction iterates are

$$(a_3 \otimes e_0) \otimes a_4 e_1 S^{-1}\eta_K(a_2) \otimes a_5 e_2 S^{-1}\eta_K(a_1).$$

This proves coassociativity; the counit identity is immediate. The recursion proves uniqueness. The mixed pairing is integral by Lemma 12.7, so (12.46) is integral. The two pairing product rules and (12.45) give covariance by cancelling consecutive antipode factors. The same cancellations in products of fields give the consecutive-substitution common germ, exactly as in Theorem 10.8. The unit and derivative identities follow from the counit and $a(z) = e^{zT}a$. All calculations hold on divided words at finite precision; continuity gives the stated field carrier. □

This is the integral version of dimodule induction [56, Proposition 3.2]. When the top is realized as a continuous annihilation-module, PBW identifies it with induction from U_+ to D_K. For example the evaluation top is defined by the spectral pairing; the mixed relation (12.36) recursively determines the action on $U\widehat{\otimes}E$. The equality (12.38) on words identifies that action with (12.46). This comparison preserves creation, coaction and every field.

Let $\mathscr{D}_K$ be the category of such continuous structured modules: a U-action λ, an F-coaction δ and the compatibility (12.45). Morphisms preserve both maps. For $M \in \mathscr{D}_K$, $N \in \mathscr{D}_L$, define

$$\begin{aligned} \delta_{M\odot N}(m\otimes n) &= m_{(0)}\otimes n_{(0)}\otimes n_{(1)}m_{(1)},\\ \lambda_{M\odot N}(a) &= (\lambda_M\eta_L\otimes\lambda_N)\Delta a. \end{aligned} \tag{12.47}$$

The reverse coordinate tail is essential. Expansion of the two compatibility identities cancels the adjacent factors $S^{-1}\eta_L(a_4)\eta_L(a_3)$, leaving (12.45) at level $K+L$. Thus $M\odot N\in\mathscr{D}_{K+L}$. Its fields are the crossed tensor fields pulled back along $(\eta_L\otimes 1)\Delta$. Both descriptions reduce to the same normal expression

$$\lambda_M(\eta_L a_{(1)}(z))A^M_z(a_4)m\otimes\lambda_N(a_{(2)}(z))A^N_z(a_3)n.$$

On a divided matrix generator the coproduct is explicitly

$$\ell_{ij}(u-L\hbar)\otimes 1+1\otimes\ell_{ij}(u)+\hbar\sum_a \ell_{ia}(u-L\hbar)\otimes\ell_{aj}(u).$$

The triple creation product is $L^{(1)}(u-(L+J)\hbar)L^{(2)}(u-J\hbar)L^{(3)}(u)$ for a third level J, and the triple tail is $p_{(1)}n_{(1)}m_{(1)}$. These prove associativity with the ordinary associator. The scalar counit module is the unit at level zero. Two induced modules retain two creation factors $(U\widehat{\otimes}E)\widehat{\otimes}(U\widehat{\otimes}E')$; the tensor is not identified with $U\widehat{\otimes}(E\otimes E')$.

For a scalar A, define the level twist on $N\in\mathscr{D}_L$ by

$$\lambda_{\mathrm{Tw}_A N}(a)=\lambda_N(\eta_A^{-1}a),\qquad \delta_{\mathrm{Tw}_A N}(n)=n_{(0)}\otimes\eta_A(n_{(1)}).$$

It remains at level L. There are integral field intertwiners

$$c_{M,N}:M\odot N\longrightarrow(\mathrm{Tw}_K N)\odot M,\qquad c_{M,N}(m\otimes n)=n_{(0)}\otimes\lambda_M(n_{(1)})m. \tag{12.48}$$

The action of $n_{(1)}\in F$ uses the map $F\to U$. The inverse, using the untwisted coaction of N, is

$$n\otimes m\longmapsto\lambda_M(Sn_{(1)})m\otimes n_{(0)}.$$

The antipode identities check both composites. For creation intertwining, the source expression reduces to $\lambda_N(a_{(1)})n_{(0)}\otimes\lambda_M(a_{(2)}n_{(1)})m$. On the target the twist cancels the η_K in its tensor action and gives the same expression. For colinearity the inverse-antipode pair cancels to leave $n_{(0)}\otimes\lambda_M(n_{(1)})m_{(0)}\otimes n_{(2)}m_{(1)}$. A map preserving creation and coaction preserves (12.46), proving all-field intertwining.

The two hexagons have respective values

$$n_{(0)}\otimes p_{(0)}\otimes\lambda_M(p_{(1)}n_{(1)})m,\qquad p_{(0)}\otimes\lambda_M(\eta_L p_{(1)})m\otimes\lambda_N(p_{(2)})n.$$

Thus they hold with their specified twisted targets; naturality, the unit and compatibility with every twist follow from the same formulas. At level zero this is an ordinary braiding. At general levels erasing the twist changes the creation action and loses the intertwining equation.

12.10 Coefficients at finite precision

The rational matrix construction has a representation-theoretic counterpart. Its coefficient algebra must retain extensions and the scalar part of the universal matrix. Both affect the integral lattice. We construct it over $R = \mathbb{C}[[\hbar]]$, with $R_N = R/\hbar^N R$.

Let $\mathfrak{g}$ be a complex simple Lie algebra with a fixed nondegenerate invariant form and inverse tensor Ω. Let A be its formal Yangian, completed at the current origin. A finite-free R-module is continuous if sufficiently high current modes act by zero modulo each $\hbar^N$. Write $M_N = M/\hbar^N M$ for its reduction. Use the conventions

$$\Delta_A J(x) = J(x) \otimes 1 + 1 \otimes J(x) + \frac{\hbar}{2}[x \otimes 1, \Omega], \qquad \tau_s J(x) = J(x) + sx.$$

The universal formal matrix has expansion $\mathsf{R}(z) = 1 + \hbar\Omega/z + O(z^{-2})$. Its translation, cabling, unitarity and shifted intertwining identities hold with its full scalar factor [63, Theorem 7.4].

Let C have all finite-free continuous A-modules as objects and all intertwiners as morphisms. Exactness means exactness on the underlying R-modules. For $m \in M \in C$ and $\phi \in M^\vee$, put

$$c_{\phi,m}(a) = \phi(am), \qquad H^\circ = \operatorname{span}_R\{c_{\phi,m}\} \subset A^\vee_{\mathrm{cont}}.$$

A finite sum of coefficients is a coefficient of a direct sum.

Proposition 12.11. *The category C is exact and rigid. Evaluation identifies its coend with H°:*

$$\left(\bigoplus_{M \in C} M^\vee \otimes_R M\right) \Big/ \langle [\psi f, m]_M - [\psi, fm]_{M'} \rangle \xrightarrow{\sim} H^\circ.$$

Here $f : M \to M'$ ranges over all intertwiners. Write $Q(C)$ for the displayed quotient. The coefficient module is saturated in the continuous dual and is a flat Hopf algebra.

Proof. The continuous coproduct and both antipodes preserve the indicated class under tensor products and duals. Stable finite-free quotients inherit continuity. For a stable submodule $L \subset M$, Smith normal form gives $d \geq 0$ such that $L \cap \hbar^{n+d} M \subset \hbar^n L$ for every n. Continuity on M at precision $n + d$ therefore implies continuity on L at precision n. Such submodules are finite free because R is a discrete valuation ring. These facts also give the exact structure and its idempotent splittings.

Represent a coend class by $[\phi, m]_M$, and take the stable lattice $L = Am$. Its inclusion gives $[\phi, m]_M = [\phi|_L, m]_L$. If the coefficient vanishes, $\phi|_L = 0$; hence evaluation is injective. If instead its values lie in $\hbar^r R$, the functional $\psi = \hbar^{-r}\phi|_L$ is integral and continuous, and

$$c_{\phi,m} = \hbar^r c^L_{\psi,m}, \qquad [\phi, m]_M = \hbar^r [\psi, m]_L.$$

This proves saturation. Torsion-freeness proves flatness.

For a basis (e_i) of M, the Hopf maps are

$$(f\,g)(a) = (f \otimes g)(\Delta_A a), \qquad \Delta_H c_{\phi,m} = \sum_i c_{\phi,e_i} \otimes c_{e^i,m}, \qquad \varepsilon_H(f) = f(1), \quad S_H(f) = f \circ S_A.$$

The unit is ε_A. Tensor products and duals keep these maps inside H°; evaluation verifies their identities. □

This argument also gives finite coalgebras on which reduction loses no coefficient relations. For the coefficient space C_U of a finite direct sum $U \in C$, define

$$B_U = (\mathbb{C}((\hbar))C_U) \cap A^\vee_{\mathrm{cont}}.$$

Finitely many evaluations separate the finite-dimensional fraction space and embed B_U into some R^d. Thus B_U is finite free. Saturation puts it inside H° with flat quotient. Consequently $B_U \otimes B_U$ is saturated in $H^\circ \otimes H^\circ$. Since its fraction space contains $\Delta_H(B_U)$, it is a subcoalgebra.

There is a stronger evaluation statement. Saturation embeds $B_U/\hbar B_U$ into $(A/\hbar A)^\vee$. Choose evaluations giving a square matrix invertible modulo $\hbar$; its determinant is a unit. They identify B_U with $R^{\operatorname{rank} B_U}$ and separate tensors at every precision. The regular action $(a \cdot b)(y) = b(ya)$ is continuous in these coordinates. Hence $B_U \in \mathcal{C}$. The maps

$$M \longrightarrow B_U, \quad m \longmapsto c_{\phi,m}, \qquad \varepsilon_H(c_{\phi,m}) = \phi(m),$$

are intertwiners whenever the coefficients of M lie in B_U. The B_U form a directed family exhausting H°.

Set $H_N = H^\circ/\hbar^N H^\circ$, viewed by saturation inside $(A/\hbar^N A)^\vee$. Define $\mathcal{C}_N$ to have the finite-free continuous $A/\hbar^N A$-modules whose coefficients belong to this specified H_N. A module in $\mathcal{C}_N$ need not lift to R.

Theorem 12.12. *The algebras H_N are flat over R_N, commute with reduction, and satisfy*

$$\mathcal{C}_N \simeq \operatorname{Comod}_{\mathrm{ff}}(H_N), \qquad Q(\mathcal{C}_N) \cong H_N \cong Q(\mathcal{C})/\hbar^N Q(\mathcal{C}).$$

The complete Hopf algebra $H = \varprojlim_N H_N$ is topologically free, with $H/\hbar^N H = H_N$, and

$$\mathcal{C} \simeq \operatorname{Comod}^{\mathrm{cont}}_{\mathrm{ff}}(H), \qquad H \widehat{\otimes} H = \varprojlim_N (H_N \otimes_{R_N} H_N).$$

Proof. Flatness and reduction follow from the flatness of H°. The finite-free coalgebras $B_U/\hbar^N B_U$ exhaust H_N. On a basis of $M \in \mathcal{C}_N$, define $\delta(m) = \sum_i e_i \otimes c_{e^i,m}$. The action identities give the coaction identities because the finite evaluation coordinates just constructed separate their tensors. Conversely, evaluate a coaction by $am = \sum m_0\langle a, m_1\rangle$. These constructions are inverse on objects and morphisms and preserve exact sequences, tensor products and both duals.

Every coend class reduces to a regular coefficient object by $[\phi, m]_M = [\varepsilon_H, c_{\phi,m}]_{B_U}$. A lifted class with zero coefficient modulo $\hbar^N$ is $\hbar^N$ times an actual coefficient class by the cyclic-lattice argument. Additional intertwiners at finite precision have zero coefficient defect, so introduce no further kernel. This proves the two coend identifications without lifting those intertwiners.

Choose compatible lifts of a complex basis of H_1. Flatness identifies each successive $\hbar$-quotient of H_N with H_1, so the lifts give compatible isomorphisms $H_N \cong R_N^{(I)}$. Their limit is $H_1[[\hbar]]$. Evaluation of a continuous finite-free H-coaction gives a continuous A-action at every precision, and its coefficient coaction is the original one. This proves the final equivalence. □

The recognition theorem concerns the actual Yangian coefficient category just defined. It does not characterize an arbitrary category of local fields by a unitary spectral presentation. Its ordinary tensor product has tail $m_1 n_1$. For the reverse-tail convention of (12.47), reverse the tensor category and then take opposite multiplication on its raw coend. The associated spectral interchange is $P\mathsf{R}(z)^{-1}$; ordinary Yangian commutativity uses $P\mathsf{R}(z)$.

12.11 The full scalar and the pole bound

A finite pole bound at each precision is enough for formal fields. A single bound independent of precision is a stronger condition, and fails even on a fixed pair of vector coefficients. We first prove the finite-precision bound, including the abelian factor of the universal matrix.

For nonzero $M \in C$, its reduction factors through a finite current Lie algebra $\mathfrak{g}[t]/t^J\mathfrak{g}[t]$. The positive-current ideal kills every simple constituent: its solvable radical acts there by a character, and the commutator with the constant semisimple algebra forces this character to vanish. Choose a filtration of length $e_M \le \operatorname{rank}_R M$ which positive currents lower by one, and put $\lambda_M = 2e_M - 1$. Take $e_M = 1$ if positive currents vanish.

Write $x^{\pm}_{i,r}$ and $\xi_{i,r}$ for the Drinfeld root and Cartan modes. Define their logarithmic Cartan modes by

$$\hbar \sum_{r\ge 0} t_{i,r} u^{-r-1} = \log\left(1 + \hbar \sum_{r\ge 0} \xi_{i,r} u^{-r-1}\right).$$

If (a_{ij}) is the Cartan matrix and d_i its symmetrizers, then

$$t_{i,1} = \xi_{i,1} - \tfrac{\hbar}{2}\xi_{i,0}^2, \qquad [t_{i,1}, x^{\pm}_{j,r}] = \pm d_i a_{ij} x^{\pm}_{j,r+1}$$

[63, §2.4]. For the matrix $B_i = t_{i,1}|_M$, we have $B_i^{e_M} \in \hbar \operatorname{End}(M)$. Every term of $(\operatorname{ad} B_i)^{2e_M-1}$ therefore contains a factor divisible by $\hbar$. Iterating the displayed commutator with $i = j$, and using $[x^+_{i,r}, x^-_{i,0}] = \xi_{i,r}$, proves

$$v_\hbar(x^{\pm}_{i,r}|_M), \quad v_\hbar(\xi_{i,r}|_M), \quad v_\hbar(t_{i,r}|_M) \ \ge\ \lfloor r/\lambda_M \rfloor. \tag{12.49}$$

For the last inequality, a logarithmic term has the form $\hbar^{k-1}\xi_{i,r_1}\cdots\xi_{i,r_k}$, where $\sum r_j = r + 1 - k$. If its lower valuation bound is $v = k - 1 + \sum\lfloor r_j/\lambda_M\rfloor$, division of each index by λ_M gives $r \le \lambda_M(v+1) - 1$. This proves the claim for all three families.

Theorem 12.13. *For nonzero $M, W \in C$, put $D = \lambda_M + \lambda_W - 1 = 2(e_M + e_W) - 3$. Then*

$$\mathsf{R}_{M,W}(z) \bmod \hbar^N \in \operatorname{End}_{R_N}(M_N \otimes W_N)[z^{-1}], \qquad \operatorname{pole}_0 \mathsf{R}_{M,W} \le D(N-1).$$

The bound includes the full scalar normalization.

Proof. It suffices to show that the coefficient of z^{-d} has valuation at least $\lceil d/D\rceil$. This region is preserved by products and inversion of a series congruent to one modulo $\hbar$. Use the Gauss factors $\mathsf{R}^+\mathsf{R}^0\mathsf{R}^-$. For a regular element h_0 of the positive Cartan chamber, let $T_A(h_0)$ be the corresponding linear combination of logarithmic first modes, and set $Q = T_A(h_0)|_M \otimes 1 + 1 \otimes T_A(h_0)|_W$. The tensor filtration gives $Q^{e_M+e_W-1} \in \hbar \operatorname{End}(M \otimes W)$, hence $(\operatorname{ad} Q)^D$ is divisible by $\hbar$.

For a positive root-lattice element β, the triangular recursion is

$$\mathsf{R}^-_\beta(z) = \hbar \sum_{r\ge 0} \frac{(\operatorname{ad} Q)^r}{(z\beta(h_0))^{r+1}} \sum_{\alpha>0} \alpha(h_0)\mathsf{R}^-_{\beta-\alpha}(z)(x^-_\alpha|_M \otimes x^+_\alpha|_W)$$

[63, §4.2]. A path of q steps with indices $r_1, \dots, r_q$ has valuation at least $q + \sum\lfloor r_j/D\rfloor$ and pole at most $\sum(r_j+1)$. Since $r + 1 \le D(1 + \lfloor r/D\rfloor)$, it lies in the required region. At precision N, only $q < N$ and bounded indices contribute; there are finitely many roots. The inverse $\mathsf{R}^+(z) = \mathsf{R}^-_{21}(-z)^{-1}$ has the same bound.

The logarithm of the abelian factor is a sum of terms

$$C\hbar^a t_{i,r}|_M \otimes t_{j,s}|_W \, z^{-(a+r+s)}, \qquad a \ge 1.$$

This follows directly from the explicit logarithm in [63, equation (6.7)]: its two series $\hbar\sum t_{i,r}u^r/r!$ cancel the prefactor $\hbar^{-2}$, while its remaining scalar series begins with $\hbar/z$. By (12.49), a lower valuation is $v = a + \lfloor r/\lambda_M\rfloor + \lfloor s/\lambda_W\rfloor$. The sum of the two remainders is at most $D - 1$, so $a + r + s \le Dv$. Exponentiation preserves the region. All three factors obey the stated bound. □

In type A_{n-1}, use the opposite spectral sign from Section 12.8. Write

$$R_+(z) = f_+(z)(I + \hbar P/z), \qquad \prod_{r=0}^{n-1} f_+(z + r\hbar) = \frac{z + (n-1)\hbar}{z + n\hbar}.$$

Let F_+ be the corresponding determinant-one coordinate Hopf algebra. The vector representation $V(u)$ has $J(x) = ux$. Its parameter jets define a Hopf map

$$F_+ \longrightarrow H, \qquad L_{ij}^{[r]} \longmapsto \big(a \longmapsto [u^r] e_i^* \pi_{V(u)}(a) e_j\big). \tag{12.50}$$

The map is an isomorphism preserving translation and the full spectral pairing. Here is the determinant and scalar check. The alternating line in $V(u + (n-1)\hbar) \otimes \cdots \otimes V(u)$ is trivial: on its alternating vector w, $[x_i, P_{ij}]w = (x_j - x_i)w$, and the coefficient of each $x_i w$ in the iterated $J(x)$ is $u + (n - i)\hbar + \hbar(2i - n - 1)/2 = u + (n-1)\hbar/2$. The sum $\sum_i x_i w$ vanishes. Its coefficient is therefore one. The bare cabled matrix on this line tensored with V is

$$\prod_{i=1}^{n} \left(1 + \frac{\hbar P_{i,n+1}}{z + (n-i)\hbar}\right) = \frac{z + n\hbar}{z + (n-1)\hbar}\, 1.$$

Compression of a product of k distinct permutations to the alternating line is $(-1)^{k-1}/n$; expansion reduces the formula to a telescoping product. Trivial-line cabling forces the displayed equation for f_+, whose constant term one determines it uniquely, order by order. The forms thus agree on vector coefficients and, by their product identities, on the whole algebra.

Modulo $\hbar$, every coefficient representation integrates to a finite jet group $\mathrm{SL}_n(\mathbb{C}[t]/t^J)$: its constant semisimple action integrates, and the nilpotent positive-current action exponentiates polynomially. Differentiation of functions on this connected smooth group is injective. Conversely the parameter jets of $V(u)$ give every jet matrix coordinate. These generate $\mathbb{C}[\mathrm{SL}_n[[t]]]$, proving that (12.50) is an isomorphism modulo $\hbar$. Both sides are topologically free, by the rational coordinate flatness theorem [56, §1.5] and Theorem 12.12. Successive lifting proves surjectivity; successive divisibility and separatedness prove injectivity. The change of spectral sign identifies the source level K with $c = -K$. This coordinate algebra remains distinct from the divided state algebra U, whose pairing has domain $F \widehat{\otimes} U$.

The scalar cannot in general be discarded or replaced by a Laurent polynomial. For $\mathfrak{sl}_2$, normalize $\Omega = P - \frac{1}{2}I$, and set $M = V(\hbar) \otimes V(0)$. If $s = e_+ \otimes e_- - e_- \otimes e_+$, $t = e_+ \otimes e_- + e_- \otimes e_+$, and $H_0 = \mathrm{diag}(1, -1)$, then $J(H_0)t = 2\hbar s$. The trivial line Rs has no stable triplet complement, so this module is a nonsplit extension.

Put $x = \hbar/z$, and write the vector matrix as $F(x)(I + xP)$. Cabling with Rs gives

$$F\left(\frac{x}{1+x}\right) F(x) = \frac{1+x}{1+2x}, \qquad F(x) = 1 - \frac{x}{2} + \frac{5x^2}{8} - \frac{11x^3}{16} + \frac{83x^4}{128} - \frac{143x^5}{256} + O(x^6).$$

The matrix entry from 010 to 001 on $M \otimes V(0)$ is

$$\frac{x(1+x)}{1+2x} = x - x^2 + 2x^3 - 4x^4 + \cdots.$$

Both reductions have zero positive currents, so $D = 1$. This entry attains pole order $N - 1$ at every precision $N \geq 2$. Moreover, if F were polynomial, the left side of its equation could have a finite pole only at -1, whereas the right side has one at $-1/2$. Thus F is not polynomial. On two fixed opposite off-diagonal coefficients the pairing equals $xF(x)$. Every reduction has finite pole order, but the complete value does not belong to $R[z, z^{-1}]$.

12.12 Total and relative Fock oscillators

The tensor of two induced modules retains two creation factors. In the abelian case we can identify the extra factor explicitly. Let $\Gamma \subset \mathbb{C}$ be an additive charge group, and put

$$H_\Gamma = (\mathbb{C}[b_1, b_2, \dots] \otimes \mathbb{C}[\Gamma])[[\hbar]].$$

The b_n are primitive, the charge elements g_μ are group-like, and

$$T b_n = n b_{n+1}, \qquad T g_\mu = -\mu b_2 g_\mu.$$

The spectral pairing is determined by the Hopf product identities, translation, and

$$\rho_z(b_1, b_1) = \frac{\hbar}{z}, \qquad \rho_z(g_\mu, b_1) = \rho_z(b_1, g_\mu) = -\frac{\hbar\mu}{z}, \qquad \rho_z(g_\mu, g_\nu) = e^{\hbar\mu\nu/z}. \tag{12.51}$$

For example, applying translation to the last formula agrees with $T g_\mu = -\mu b_2 g_\mu$; this fixes its sign. All exponentials are $\hbar$-adic. In this section the level translation is $\eta_K = e^{K\hbar T}$, the opposite spectral-sign convention from $\eta_K = \tau_{-K\hbar}$ in the rational matrix example.

For $a \in \mathbb{C}$, define

$$A_a(T) = \sum_{r \ge 0} \frac{a^{r+1}\hbar^r T^r}{(r+1)!}, \qquad \beta_{K,n} = -T A_K(T) b_n = \frac{b_n - e^{K\hbar T} b_n}{\hbar}. \tag{12.52}$$

The notation $(e^{a\hbar T} - 1)/(\hbar T)$ for $A_a(T)$ is a power-series abbreviation, requiring no inverse of T or $\hbar$. The field-state algebra is the divided primitive polynomial algebra $V_K = \mathbb{C}[x_1, x_2, \dots][[\hbar]]$, with $x_n = b_n/\hbar$ inside the divided algebra. The group elements remain coaction coefficients. For level K and charge μ, set

$$F_{K,\mu} = \mathbb{C}[x_1, x_2, \dots][[\hbar]] v_\mu, \qquad T x_n = n x_{n+1},$$

with coaction

$$\delta(f(x) v_\mu) = f(x \otimes 1 + 1 \otimes \beta_K)(v_\mu \otimes g_\mu). \tag{12.53}$$

The $\beta_{K,n}$ are primitive, so this is coassociative.

Define the rational formal series and annihilation derivation

$$P_K(z) = \frac{1}{\hbar}\Big(\frac{1}{z - K\hbar} - \frac{1}{z}\Big), \qquad P_{K,n}(z) = \frac{(-\partial_z)^{n-1}}{(n-1)!} P_K(z), \qquad D_K(z) = \sum_{n \ge 1} P_{K,n}(z)\partial_{x_n}.$$

At each precision the derivative sum is finite on each polynomial. Writing $C(z) = \sum_{n \ge 1} x_n z^{n-1}$, the current is

$$J^{K,\mu}(z) = C(z) + \frac{\mu}{z} + D_K(z). \tag{12.54}$$

It represents the state $x_1 = b_1/\hbar$. For every monomial of V_K, the field is

$$Y(x_{n_1} \cdots x_{n_r}, z) =: \prod_{i=1}^{r} \frac{\partial_z^{n_i - 1} J^{K,\mu}(z)}{(n_i - 1)!} : .$$

Normal ordering puts the multiplication operators on the left and the scalar and differentiation operators on the right. The annihilation map of b_1 is $\hbar(\mu/z + D_K(z))$, as follows by pairing (12.53) with (12.51). The creation factors and this annihilation map give the field construction of Section 12.9.

For the level-additive product $F_{K,\mu} \star F_{L,\nu}$, the first creation current is shifted by $L\hbar$. The charge term also shifts, but the crossed pairing supplies

$$\rho_{-z}(g_\mu, \beta_{L,1}) = \mu\Big(\frac{1}{z} - \frac{1}{z + L\hbar}\Big).$$

It cancels the shifted charge pole. Consequently the total current is

$$C_1(z + L\hbar) + C_2(z) + \frac{\mu + \nu}{z} + D_K^{(1)}(z) + D_L^{(2)}(z). \tag{12.55}$$

The substitution $x_n \mapsto q_n = e^{L\hbar T} x_n^{(1)} + x_n^{(2)}$, $v_{\mu+\nu} \mapsto v_\mu \otimes v_\nu$, defines a continuous injective field-module and comodule map

$$I_{12} : F_{K+L,\mu+\nu} \longrightarrow F_{K,\mu} \star F_{L,\nu}.$$

Injectivity follows already by setting the first oscillator family to zero. This construction holds at zero and cancelling levels. It identifies a cyclic submodule, not the whole tensor.

Theorem 12.14. *Suppose $K + L \neq 0$, and set*

$$y_n = A_L(T) x_n^{(1)} - A_K(T) x_n^{(2)}.$$

There is an isomorphism of continuous covariant field modules

$$F_{K,\mu} \star F_{L,\nu} \cong F_{K+L,\mu+\nu} \widehat{\otimes}_R \mathbb{C}[y_1, y_2, \ldots][[\hbar]]. \tag{12.56}$$

The current and coaction act on the first factor. The second is exactly the positive-mode ground-space multiplicity, and I_{12} is its constant-multiplicity inclusion. Multiplication by y_1 and differentiation ∂_{y_1} are covariant field endomorphisms with $[\partial_{y_1}, y_1] = 1$.

Proof. The identity

$$e^{L\hbar T} A_K(T) + A_L(T) = A_{K+L}(T)$$

provides the inverse coordinate change

$$\begin{aligned} x_n^{(1)} &= A_{K+L}(T)^{-1}(A_K(T) q_n + y_n), \\ x_n^{(2)} &= A_{K+L}(T)^{-1}(A_L(T) q_n - e^{L\hbar T} y_n). \end{aligned}$$

The constant term of $A_{K+L}(T)$ is the nonzero scalar $K + L$. Every coefficient uses finitely many higher oscillator indices, so these changes are inverse on polynomial modules modulo each $\hbar^N$, and on their inverse limits.

On a linear oscillator, $D_K(z) T^r x_n = (-\partial_z)^r P_{K,n}(z)$, while $P_K(z) = A_K(-\partial_z) z^{-2}$. Commutativity of the two A operators gives

$$(D_K^{(1)} + D_L^{(2)}) y_n = 0, \qquad (D_K^{(1)} + D_L^{(2)}) q_n = P_{K+L,n}(z).$$

The second equality uses $P_K(z - L\hbar) + P_L(z) = P_{K+L}(z)$. Thus (12.55) acts only on the q variables, and equals the total-level current. Its normal-ordered derivatives give the assertion for every field.

The primitive coaction of y_n is $A_L(T)\beta_{K,n} - A_K(T)\beta_{L,n} = 0$, and that of q_n is $\beta_{K+L,n}$. The highest-vector tail is $g_\mu g_\nu = g_{\mu+\nu}$. This proves the coaction assertion. Write $J(z) = \sum_{n\in\mathbb{Z}} J_n z^{-n-1}$. Modulo $\hbar$, its positive modes on the first factor are $n(K + L)\partial_{q_n}$. Their common kernel contains no q variables. Removing the constant coefficient and repeating at each precision proves that the exact common kernel is precisely the completed polynomial space in the y's. Conversely the exact annihilation derivations kill all these polynomials. Multiplication and differentiation in y_1 commute with both structures, and their Weyl commutator proves the last assertion. □

For $K = 2$, $L = 3$, for instance,

$$y_1 = 3x_1^{(1)} - 2x_1^{(2)} + \hbar\Big(\frac{9}{2}x_2^{(1)} - 2x_2^{(2)}\Big) + \hbar^2\Big(9x_3^{(1)} - \frac{8}{3}x_3^{(2)}\Big) + O(\hbar^3).$$

Its constant term is already a ground vector distinct from the tensor of highest vectors. At cancelling levels the inverse coordinate argument fails, and no such decomposition is asserted.

Let $\mathscr{C}_F$ be the full continuous R-linear category generated by $F_{K,\mu}$, for $K \in \mathbb{Z}_{>0}$ and $\mu \in \Gamma$, their finite $\star$-products, finite direct sums at each level, and the scalar unit at level zero. Morphisms are colinear field-module maps. Positive levels ensure that each nonempty total level is invertible over $\mathbb{C}$. Let b generate the charge algebra $R[b]$, with primitive coproduct. Let $\mathscr{W}$ consist of charge modules $P[[\hbar]] \otimes W_\mu$, and finite sums of these, where P is a complex vector space and $W_\mu = R$ has $b = \mu$. Morphisms preserve charge, and tensor product adds charges.

Proposition 12.15. *The functor*

$$\mathsf{G}(M) = \bigcap_{n>0} \ker J_n, \qquad b|_{\mathsf{G}(M)} = J_0,$$

from $\mathscr{C}_F$ to $\mathscr{W}$ is lax monoidal, with tensor comparison the natural inclusion of ground spaces. It sends $F_{K,\mu}$ to W_μ. There is no R-linear strong monoidal functor on this full category having these values on every generating Fock module.

Proof. Iterate Theorem 12.14 to express a tensor word as its total-level Fock module with a trivial polynomial multiplicity space. Morphisms preserve positive-mode kernels and J_0, so restriction defines the functor. The tensor of two ground vectors is ground for the crossed total current, by the charge-pole cancellation in (12.55). Its charge is the sum. This gives $\mathsf{G}(M)\widehat{\otimes}\mathsf{G}(N) \to \mathsf{G}(M \star N)$. Both three-factor composites are the same inclusion of underlying ground tensors, and the scalar unit map is the identity. These observations prove the lax coherence identities.

For two generators the inclusion is $R \hookrightarrow \mathbb{C}[y_1, y_2, \dots][[\hbar]]$. Under a strong functor with the stated values, their product would have rank-one image $W_{\mu+\nu}$, whose endomorphism algebra is R. The relative endomorphisms y_1, ∂_{y_1} would map to commuting scalars. Linearity, composition and the identity would then imply $0 = [\mathsf{G}'(\partial_{y_1}), \mathsf{G}'(y_1)] = \mathsf{G}'(1) = 1$. This contradiction already holds modulo $\hbar$. □

Through the abelian charge reconstruction of Chapter 11, $\mathscr{W}$ also has a Wilson-line interpretation at a fixed nonzero physical level κ. That fixed-level fusion is a different source operation from the level-additive product $\star$. On ground vectors the pairing is $e^{\hbar\mu\nu/z}$. Its logarithmic residue agrees with the Wilson coefficient $\mu\nu/\kappa$ after the parameter relation $\hbar\kappa = 1$ in a ring where this relation is meaningful; it is not a specialization of the formal parameter to an invertible complex number inside $\mathbb{C}[[\hbar]]$.

Even the ground coefficient does not provide a field intertwiner on all descendants. At zero charge the spectral pairing fixes a tensor with at most one oscillator. If $\mathcal{E}(u)$ denotes its exchange and τ flips factors, then

$$\tau\mathcal{E}(u)I_{12}(x_1v_0) - I_{21}(x_1v_0) = 1 \otimes (e^{L\hbar T} - 1)x_1 + (1 - e^{K\hbar T})x_1 \otimes 1 = \hbar(L\,1 \otimes x_2 - K x_2 \otimes 1) + O(\hbar^2).$$

The right side is nonzero. A comparison retaining this highest coefficient requires a translation-corrected intertwiner acting also on the relative multiplicity space. It would still leave the strong-functor obstruction intact. A physical line–boundary comparison must separately specify its junction complex, fixed-level tensor operation, vacuum and residue maps.

12.13 Stable lattices and meromorphic families

An equivariant formula with denominators can have a good integral lattice even if its displayed basis has poles. This is a finite linear-algebra question before it is a question about fields. Let $\mathcal{O}$ be a discrete valuation ring with fraction field Q, and let $T \in \mathrm{End}_Q(V)$, with $\dim_Q V = d < \infty$. Then

$$T \text{ preserves an } \mathcal{O}\text{-lattice} \quad \Longleftrightarrow \quad \det(t - T) \in \mathcal{O}[t]. \tag{12.57}$$

Necessity follows by writing T in an integral basis. For sufficiency choose any lattice L_0 and set $L = \sum_{i=0}^{d-1} T^i L_0$. This is a lattice, and Cayley–Hamilton gives $TL \subset L$. If T is invertible, the stronger equality $TL = L$ can be arranged exactly when its characteristic polynomial is integral and its determinant is a unit. The constant term then expresses T^{-1} integrally in powers of T. A unit determinant alone is insufficient: $\mathrm{diag}(t, t^{-1})$ over $\mathbb{C}((t))$ has determinant one but nonintegral trace.

A second distinction appears in the energy completion. Suppose $E = \prod_{n\ge 0} E_n$, with every E_n finite-dimensional, and $T_n(u)$ is a meromorphic endomorphism of E_n. Here a meromorphic family with values in $\prod_n \mathrm{End}(E_n)$ means a family which becomes coordinatewise holomorphic near each point after multiplication by one scalar holomorphic function not identically zero. Then $(T_n(u))_n$ has this property precisely when the union of its actual poles is locally finite and the pole orders at each point are bounded independently of n. Necessity follows because a common scalar denominator has isolated zeros of finite order. For sufficiency, choose a disk containing at most its central pole and multiply by a sufficiently high power of the local coordinate. Product holomorphy is coordinatewise holomorphy. Thus the criterion is exact. Blockwise rationality alone is insufficient: the scalar families u^{-n} have unbounded orders, while $(u - 1/n)^{-1}$ have accumulating poles. Formal $\hbar$-adic fields instead permit pole bounds depending on precision, as in Theorem 12.13.

These distinctions persist for transfer operators. To state a conditional construction precisely, let $A = \bigoplus_{r\ge 0} A_r$ be an auxiliary energy-graded space with finite-dimensional pieces, and let $d_A|_{A_r} = r$. Let V be a quantum space, and suppose a monodromy $M_A(u)$ and an invertible auxiliary matrix $R_{AB}(u - v)$ satisfy

$$R_{AB}(u - v)M_A(u)M_B(v) = M_B(v)M_A(u)R_{AB}(u - v), \qquad [R_{AB}, d_A + d_B] = 0.$$

Assume all restrictions and sums below are defined at each finite total auxiliary energy. Define the formal transfer series $\mathsf{T}(u; q) = \mathrm{tr}_A(q^{d_A} M_A(u))$. For the same parameter q in two auxiliary copies, $[\mathsf{T}(u; q), \mathsf{T}(v; q)] = 0$. Indeed, at each total energy, cyclicity of the finite partial trace moves the numerical auxiliary matrix R_{AB} through the product. It commutes with $q^{d_A+d_B}$, and its inverse cancels. The monodromy relation then reverses the two factors. If the energy-zero term is a nonzero scalar, normalize it to one; coefficients of the formal logarithm belong to the same commuting algebra. Convergence, analytic continuation, and a geometric realization of the monodromy are additional assertions.

The positive Hilbert-scheme action, its integral scaled negative modes, the paired doubles, and their Fock modules are therefore separate constructions with explicit links. To identify a compact geometric Hall theory with one of these field theories still requires a map preserving the actual coefficient system, annihilation pairing, local vacuum and collision operations. A character identity, a generic fixed-point action or a transfer trace alone does not construct that map.

Chapter 13

Local centres and boundary generators

A representation of a boundary algebra becomes a reconstruction of the bulk only after its operations determine the bulk action. The tensor used by the action matters: convolution combines separated positions, whereas the diagonal tensor places an interior input above a boundary input at the same position. Each tensor has its own representing problem. We begin with a local geometric model in which the boundary data and their maps can be recovered completely.

On a disk, restriction to the puncture and to the closed point misses the homotopy attaching them. The exceptional component of the trace (3.18) is such a homotopy. Nontrivial monodromy supplies a second difficulty: every finite collection of finite-rank probes misses some rank-one local system. The regular monodromy module solves this problem in an explicitly enlarged category. Its derived centre then remembers the attaching point through logarithmic derivatives.

This calculation leads to three further constructions: the categorical centre of a tensor category, the internal centre for convolution, and the ambient endomorphism coalgebra for the diagonal tensor. The last exists without a factorization condition. Requiring factorizing actors changes the answer: a nonzero vertex derivation produces a parallel pair that no receiving factorizing monoid action can identify. The proof retains the action paths and multiplication coherences.

Let D be a complex disk, $U = D \setminus \{0\}$, and let $j : U \hookrightarrow D$ and $i : \{0\} \hookrightarrow D$ be the inclusions. Fix a field k of characteristic zero. Work in the presentable stable category $C(D)$ of sheaves of k-complexes whose restriction to U is locally constant. Fibers and complexes are unrestricted in rank and degree.

All tensor products and mapping complexes below are derived unless a chain model is explicitly given. Derived categories retain their mapping complexes, not only their homotopy categories. Cycle traces are extended from $\mathbb{Q}$ to k.

Choose the positively oriented generator s of $\pi_1(U)$, and write $\Lambda = k[s, s^{-1}]$. A local system of complexes is determined by a left Λ-module complex M. Denote it by $\mathcal{L}_M$. The link functor is

$$K(M) = R\operatorname{Hom}_\Lambda(k, M) \simeq [M \xrightarrow{s-1} M]. \tag{13.1}$$

The link degrees are zero and one, totalized with the internal differential of M. The free resolution $0 \to \Lambda \xrightarrow{s-1} \Lambda \to k \to 0$ both proves this formula and supplies a functorial chain model. In particular $i^* R j_* \mathcal{L}_M = K(M)$.

13.1 Objects and maps in the attaching diagram

Proposition 13.1. *An object of $C(D)$ is equivalently a diagram*

$$(M, V, \alpha : V \longrightarrow K(M)), \tag{13.2}$$

with $M \in D(\Lambda)$ and $V \in D(k)$. The reconstructed sheaf and its costalk are

$$F = \mathrm{Fib}\Big(R j_* \mathcal{L}_M \oplus i_* V \xrightarrow{\mathrm{res} - i_* \alpha} i_* K(M) \Big), \tag{13.3}$$

$$i^! F = \mathrm{Fib}\big(V \xrightarrow{\alpha} K(M) \big). \tag{13.4}$$

For two such diagrams, the mapping complex is

$$\mathrm{Fib}\Big(R\,\mathrm{Hom}_\Lambda(M, M') \oplus R\,\mathrm{Hom}_k(V, V') \xrightarrow{\Phi} R\,\mathrm{Hom}_k(V, K(M')) \Big), \tag{13.5}$$

where $\Phi(a, b) = K(a)\alpha - \alpha' b$.

Proof. Restricting (13.3) to U gives $\mathcal{L}_M$. At the point, the fiber of $K(M) \oplus V \xrightarrow{(1, -\alpha)} K(M)$ is V, with the prescribed attaching map α. Conversely, the adjunction maps from any F to $R j_* j^* F$ and to $i_* i^* F$ define a map to this fiber. Its restrictions to both strata are equivalences. The open–closed localization triangle shows it is an equivalence. Applying that triangle at the closed stratum gives (13.4).

A map of the reconstructed diagrams consists of maps on the two vertices and a homotopy commuting the attaching square. The homotopy limit of these mapping complexes is exactly (13.5). This computes maps in the derived diagram category, including their higher homotopies. The reconstruction equivalence transfers it to $C(D)$. □

Choose compatible resolutions so that α, α' and Φ are actual chain maps. In the fiber convention used here, a degree-n element is (a, b, h), where a, b have degree n and h degree $n - 1$, with

$$d(a, b, h) = (da, db, K(a)\alpha - \alpha' b - dh). \tag{13.6}$$

Thus a degree-zero map with fixed closed endpoints exists precisely when $K(a)\alpha - \alpha' b$ is a boundary. Its homotopy component is an actual primitive in that mapping complex. The supported extension problem below uses the same homotopy-fiber construction.

Proposition 13.2. *In (13.5), the oriented Gysin morphism $\gamma : i_* k[-2] \to k_D$ has representative $(0, 0, h_\gamma)$, where h_γ sends the degree-two point generator to the degree-one link generator. This class spans the degree-zero cohomology of the mapping complex. With these conventions the normalized cycle trace of Theorem 3.9 has open component $a = \mathrm{id}$, closed component $b = (\mathrm{id}, 0) : k \oplus k[-2] \to k$, and homotopy component $\frac{1}{3} h_\gamma$ on the second summand.*

Proof. The constant sheaf has diagram (k, k, α), where $K(k) = [k \xrightarrow{0} k]$ and α is inclusion into link degree zero. The point source has diagram $(0, k[-2], 0)$.

Its mapping complex has a b-component in degree -2. The h-components from link degrees zero and one have degrees -1 and zero, respectively. Formula (13.6) makes the first differential $-\mathrm{id}$. The second is zero. Exactly the asserted class survives.

The costalk fiber differential is $d(v, u) = (dv, \alpha v - du)$. A closed triple (a, b, h) induces the costalk map $(v, u) \mapsto (bv, K(a)u - hv)$.

Let η be the positive link class with integral one. On a small closed subdisk D_ϵ, relative integration is $\int_{D_\epsilon} v - \int_{\partial D_\epsilon} u$. Thus $(0, -\eta)$ is the positive degree-two orientation class in the relative cone.

Consequently $h_\gamma(1) = \eta$ induces the identity on the oriented costalk $k[-2]$. This is precisely the Gysin counit, also normalized by $H^2_c(\gamma) = 1$. Adding the identity on the constant summand and multiplying this class by $1/3$ gives the last assertion. □

This calculation locates the lost datum. The exceptional ordinary stalk map is zero, but its attaching homotopy is nonzero. Extraordinary costalks and compact support already detect its coefficient, as in Theorem 3.7.

13.2 Probes and monodromy

Definition 13.3. For $P, F \in C(D)$, the *probe value* of F at P is $R\operatorname{Hom}(P, F)$. A family of probes is *conservative* if their values vanish simultaneously only on the zero object.

The preceding Gysin calculation uses constant monodromy. Nontrivial monodromy can make both constant-probe values vanish, even for a rank-one local system.

For $\lambda \in k^\times$, let k_λ be the one-dimensional Λ-module on which s acts by λ, and write $\mathcal{L}_\lambda$ for its local system.

Proposition 13.4. *If $\lambda \neq 1$, the nonzero object $F_\lambda = j_!\mathcal{L}_\lambda$ satisfies*

$$F_\lambda \simeq R\,j_*\mathcal{L}_\lambda, \qquad R\operatorname{Hom}(k_D, F_\lambda) = 0, \qquad R\operatorname{Hom}(i_*k, F_\lambda) = 0, \qquad R\Gamma_c(D, F_\lambda) = 0.$$

More generally, a finite family of bounded finite-rank constructible objects cannot be conservative as probes $R\operatorname{Hom}(P, -)$ on all bounded finite-rank objects of $C(D)$.

Proof. The link complex of k_λ has differential $\lambda - 1$, hence is acyclic. The localization triangle identifies extension by zero with derived direct image, and (13.4) makes its costalk zero. Ordinary sections are the same acyclic circle complex. Compact sections of $U \simeq S^1 \times (0,1)$ are that complex tensored with the compact cochains of the open interval, so vanish as well.

For finitely many bounded probes, choose $\lambda \neq 0, 1$ outside the spectra of all their finite-dimensional open cohomology modules. There are only finitely many excluded values, and k is infinite. If W is a finite-dimensional monodromy module with operator T_W, it has free resolution

$$0 \longrightarrow \Lambda \otimes_k W \xrightarrow{s\otimes 1 - 1\otimes T_W} \Lambda \otimes_k W \longrightarrow W \longrightarrow 0.$$

Injectivity follows by comparing the highest Laurent power. The cokernel imposes exactly the given monodromy relation. Applying $\operatorname{Hom}_\Lambda(-, k_\lambda)$ gives an invertible differential when λ is outside the spectrum. Truncation triangles give the same vanishing for each bounded open complex.

In (13.5), the target has zero closed term and acyclic link, so these open vanishings prove $R\operatorname{Hom}(P, F_\lambda) = 0$ for every chosen probe. Its restriction to U is a nonzero rank-one local system. □

13.2.1 The ramified double cover

The monodromy -1 occurs in a branched correspondence. Let $\pi : \widetilde{D} \to D$ be the squaring map $z = u^2$ between unit disks, and let $\sigma(u) = -u$.

Example 13.5. The coefficient object and its complex-oriented trace are

$$R\pi_* k_{\widetilde{D}} = k_D \oplus j_!\mathcal{L}_{-1}, \qquad \operatorname{Tr}_\pi = (2, 0), \qquad j_!\mathcal{L}_{-1} = R\,j_*\mathcal{L}_{-1}.$$

Indeed, the idempotents $(1 \pm \sigma)/2$ split the permutation representation on the two sheets. At zero there is one invariant stalk. The anti-invariant link has differential -2, so its stalk and costalk vanish. The finite map has no higher direct images. On the unramified open the oriented trace sums the two sheets. It extends as multiplication by two on the constant summand, including the ramification point.

This is the normal form of the first nested Hilbert correspondence on $\mathbb{C}^2$. After centering a length-two quotient, a chart with basis $1, X$ has equations

$$Y - rX = 0, \qquad X^2 = z/4.$$

Here r fixes its tangent direction. Choosing a retained length-one quotient means choosing the root $X = -u/2$, so the forgetful map is $z = u^2$. The center and r are smooth parameters. Thus the normal anti-invariant coefficient is exactly $\mathcal{L}_{-1}$.

This coefficient is missed by the two constant probes. A generator for all monodromies must contain more information.

13.3 The regular monodromy generator

The regular module Λ supplies all monodromies at once. Its local system has infinite rank. No nonzero Laurent polynomial lies in a finite-dimensional s-stable subspace: its translates by increasing powers of s are linearly independent. Thus $\mathcal{L}_\Lambda$ is not a filtered union of finite-rank local systems. The unrestricted category specified above is essential.

Set

$$A = j_!\mathcal{L}_\Lambda, \qquad B = i_*k, \qquad \mathcal{G} = A \oplus B. \tag{13.7}$$

For a sheaf with diagram (M, V, α), adjunction gives

$$R\operatorname{Hom}(A, F) = M, \qquad R\operatorname{Hom}(B, F) = i^!F. \tag{13.8}$$

Definition 13.6. An object P of $\mathcal{C}(D)$ is *compact* if $R\operatorname{Hom}(P, -)$ commutes with arbitrary sums. A family *generates* if the smallest stable subcategory containing it and closed under arbitrary sums is $\mathcal{C}(D)$.

Theorem 13.7. *The object $\mathcal{G}$ is a compact generator of $\mathcal{C}(D)$. Its derived endomorphism algebra has the triangular model*

$$\mathcal{E} = \begin{pmatrix} \Lambda & k[-2] \\ 0 & k \end{pmatrix}, \qquad d = 0, \tag{13.9}$$

where the left action of Λ on the off-diagonal entry is evaluation at $s = 1$. Consequently $\mathcal{C}(D)$ is equivalent to the derived category of right $\mathcal{E}$-modules, retaining their full derived mapping complexes.

Proof. The first functor in (13.8) commutes with arbitrary sums. The second does too: by (13.4) it is a fiber of stalk and link functors, and (13.1) is a finite free resolution. Hence A, B are compact.

If both probe values vanish, the open restriction is zero, so localization makes $F = i_*V$. Its costalk is then V, which is zero. More explicitly, Λ generates $D(\Lambda)$ and k generates $D(k)$. The triangle $j_!j^*F \to F \to i_*i^*F$ puts every F in the localizing subcategory generated by A, B.

Use their diagrams $(\Lambda, 0, 0)$ and $(0, k, 0)$ in (13.5). The four blocks are

$$R\operatorname{Hom}(A, A) = \Lambda, \qquad R\operatorname{Hom}(A, B) = 0,$$
$$R\operatorname{Hom}(B, B) = k, \qquad R\operatorname{Hom}(B, A) = K(\Lambda)[-1].$$

Let Q be the complex with $Q^1 = Q^2 = \Lambda$ and differential $s - 1$. It identifies with $K(\Lambda)[-1]$ by multiplying degree one by -1. Composition of diagrams therefore gives a strict model

$$\widetilde{\mathcal{E}} = \begin{pmatrix} \Lambda & Q \\ 0 & k \end{pmatrix}.$$

Evaluation at one in degree two and zero in degree one give a quasi-isomorphism of bimodules $Q \to k[-2]$. It extends to a dg algebra quasi-isomorphism $\widetilde{\mathcal{E}} \to \mathcal{E}$: two off-diagonal entries cannot compose, and all remaining products are the indicated left and right module actions.

The functor $R\operatorname{Hom}(\mathcal{G}, -)$ takes values in right $\widetilde{\mathcal{E}}$-modules by precomposition. Its left adjoint is $N \mapsto N \otimes^L_{\widetilde{\mathcal{E}}} \mathcal{G}$. The unit and counit are equivalences on the two idempotent free modules and the corresponding summands of $\mathcal{G}$. Both functors preserve sums and triangles, so the equivalences extend to their generated categories, which are the entire categories just specified.

Finally transport modules along $\widetilde{\mathcal{E}} \to \mathcal{E}$. If an explicit inverse over the zero-differential algebra is desired, put $\mathcal{G}_{\mathcal{E}} = \mathcal{E} \otimes^L_{\widetilde{\mathcal{E}}} \mathcal{G}$. The inverse is $N \mapsto N \otimes^L_{\mathcal{E}} \mathcal{G}_{\mathcal{E}}$. Derived transport defines the left action without factoring the original chain action strictly through the quotient. □

Example 13.8. Full derived module maps carry more than their underlying graded cohomology maps. Fix $\lambda \neq 0, 1$. The sequence

$$0 \longrightarrow k_\lambda \longrightarrow \Lambda/(s-\lambda)^2 \longrightarrow k_\lambda \longrightarrow 0$$

does not split as Λ-modules. Indeed, any lift of the quotient generator is sent by $s - \lambda$ to the nonzero submodule generator. Its connecting morphism is a nonzero element of $\mathrm{Ext}^1_\Lambda(k_\lambda, k_\lambda)$.

Since $j_!$ is fully faithful, it gives a nonzero map $F_\lambda \to F_\lambda[1]$. The underlying map of A-probe complexes is zero in $D(k)$, because the sequence splits as vector spaces. Both B-probe complexes are zero. The derived module map in Theorem 13.7 nevertheless remains nonzero. Conservativity on objects cannot be replaced by faithfulness of the underlying graded vector-space maps.

13.4 Independent normal directions

Give D^r the coordinate stratification, and allow arbitrary local systems of complexes on each stratum. For each $I \subseteq \{1, \dots, r\}$, take the external product which uses A in coordinates in I and B in the others.

Proposition 13.9. *These 2^r external products form a compact generating family. The derived endomorphism algebra of their sum is quasi-isomorphic to the graded tensor product $\mathcal{E}^{\otimes r}$, with its Koszul signs. It reconstructs all objects and maps in this stratified category. In particular it detects a nonzero discrepancy from the two-axis coefficient $1/9$ of Corollary 3.10.*

Proof. Apply open–closed localization one coordinate at a time. On a stratum with open coordinates I, the regular local system is the free module over $k[s_i^{\pm 1} : i \in I]$. It generates that stratum's local systems. The localization triangles give generation on the whole polydisk. Compactness follows from the finite tensor product of the two-term link resolutions.

The same resolutions compute each Hom block between the displayed generators. A factor with source A and target B gives zero. The other factors give Λ, k, or Q, as in the preceding proof. The product cellular complex is the tensor product of these finite free resolutions. Composition is tensor composition with the cohomological signs. Thus its full dg endomorphism algebra is $\widetilde{\mathcal{E}}^{\otimes r}$.

Tensoring the block quasi-isomorphisms over the field k gives $\mathcal{E}^{\otimes r}$, and the same adjunction argument reconstructs the category. The exceptional two-axis Gysin class is nonzero by coordinate purity, so a nonzero scalar multiple remains a nonzero derived module map under this equivalence. □

For independent flags this model tests the whole normalized trace, including its attaching homotopies. An interacting flag requires its own stratification and link functors. The reconstruction theorem applies once those coefficient objects and their attaching maps have been supplied.

13.5 Derived centres and change of generator

A compact generator reconstructs objects and maps from one algebra. The operations that act naturally on every module are its derived centre. This requires the bimodule structure of the algebra itself.

For a unital differential graded algebra A over k, put $\mathsf{A}^e = \mathsf{A} \otimes_k \mathsf{A}^{\mathrm{op}}$ and define

$$Z_{\mathrm{dg}}(\mathsf{A}) = R\,\mathrm{Hom}_{\mathsf{A}^e}(\mathsf{A}, \mathsf{A}), \qquad HH^n(\mathsf{A}) = H^n Z_{\mathrm{dg}}(\mathsf{A}).$$

The ordinary centre is HH^0 when A is concentrated in degree zero. The whole complex has a product and coherent operations refining its commutativity.

Write $s = [1]$ for suspension, so $|sa| = |a| - 1$. The normalized two-sided bar resolution gives the product complex

$$C(\mathsf{A}) = \prod_{q \geq 0} \operatorname{Hom}_k\big((s\bar{\mathsf{A}})^{\otimes q}, \mathsf{A}\big), \qquad \bar{\mathsf{A}} = \mathsf{A}/k\mathbf{1}. \tag{13.10}$$

Derived representatives are understood. An arity-q cochain vanishes if any of its inputs is the unit. The product over arities is part of the totalization; it is not replaced by a direct sum when infinitely many arities contribute in one degree.

The signs can be specified on suspended inputs. For a homogeneous cochain f, set $F = sf$, with $|F| = |f| - 1$. The brace $F\{G_1, \dots, G_r\}$ sums over ordered disjoint insertions of the G_j into F. On suspended inputs x_i, an insertion beginning at positions ℓ_j has sign

$$(-1)^{\sum_j |G_j| \sum_{i<\ell_j} |x_i|}. \tag{13.11}$$

An arity-zero insertion consumes no input. The inner sum includes inputs consumed by earlier inserted maps. Projection to $s\bar{\mathsf{A}}$ precedes every insertion into a normalized input.

Let $b = b_1 + b_2$ be the bar coderivation on $T^c(s\mathsf{A})$, with

$$b_1(sa) = -s(da), \qquad b_2(sa, sb) = (-1)^{|a|} s(ab).$$

For suspended maps put $[F, G] = F\{G\} - (-1)^{|F||G|} G\{F\}$. The cochain differential, cup product and Gerstenhaber bracket are

$$Df = -s^{-1}[b, F], \qquad f \smile g = (-1)^{|f|} s^{-1} b_2\{F, G\}, \qquad [f, g]_{\mathrm{G}} = s^{-1}[F, G]. \tag{13.12}$$

The bracket has degree -1. An ordinary arity-q multilinear map is put in this convention by precomposing with $(s^{-1})^{\otimes q}$, including its tensor Koszul sign.

Proposition 13.10. *The complex* (13.10) *computes* $Z_{\mathrm{dg}}(\mathsf{A})$. *Its cup product represents composition of derived bimodule endomorphisms. Its cup and braces give the Hochschild* E_2 *structure. On cohomology,* (13.12) *defines a Gerstenhaber algebra.*

Proof. The two-sided bar is augmented to A by multiplication. Inserting the unit contracts the augmented resolution. Applying bimodule Hom produces (13.10). Its differential has the internal differential, the adjacent products and the two outer actions, exactly the terms of $-s^{-1}[b, F]$.

The Alexander–Whitney diagonal cuts a bar string at every position. Applying g to its final segment and then f to the preceding segment gives $f \smile g$, with the sign of moving the maps through the suspended inputs. The diagonal commutes with the bar differential: products across a cut cancel against its outer actions. Its coassociativity identifies cup with composition of the resulting chain endomorphisms.

A second family of brace insertions either lies inside an existing insertion or occupies a remaining slot. These choices partition the same ordered substitutions, with the same Koszul signs. This proves the brace identities; $b^2 = 0$ gives their differential identities. The operad generated by cup, unit and these braces is a chain model of the little two-cubes operad [117, Theorem 1.1]. Its cohomology operations are the asserted product and bracket. □

Example 13.11. Let $\mathsf{A} = k[\epsilon]/(\epsilon^2)$. The periodic bimodule resolution has successive differentials alternating between multiplication by $x - y$ and $x + y$ in $\mathsf{A}^e = k[x, y]/(x^2, y^2)$, beginning with $x - y$. The kernels are the images of the other multiplication: this follows by evaluating on the basis $1, x, y, xy$. After applying bimodule Hom, the differentials alternate between zero and multiplication by 2ϵ. Hence

$$HH^0(\mathsf{A}) = \mathsf{A}, \qquad HH^{2r+1}(\mathsf{A}) = k\epsilon \quad (r \geq 0), \qquad HH^{2r}(\mathsf{A}) = k \quad (r \geq 1).$$

In contrast, $\operatorname{End}_k(\mathsf{A})$ is four-dimensional and concentrated in degree zero. Linear endomorphisms do not compute the derived centre.

Theorem 13.12. *A k-linear equivalence of presentable derived module categories, preserving colimits, identifies their derived centres as E_2 algebras and intertwines their actions on modules. In particular, changing a compact generator preserves the full Hochschild structure.*

Proof. A colimit-preserving k-linear functor from left A-modules to left B-modules is tensoring with a B–A bimodule P. Indeed, take P to be its value on A; the right action comes from the right action on this free module. The assertion holds on free modules by linearity and on every module by its free bar resolution. The same construction identifies natural transformations and their higher homotopies.

Thus endofunctors of the module category correspond to bimodules, composition corresponds to balanced tensor, and the identity corresponds to A. Its endomorphism complex is $Z_{\mathrm{dg}}(\mathsf{A})$. Vertical composition cuts bar strings; horizontal composition substitutes their components. These are cup and braces. Conjugating endofunctors by the given equivalence and a chosen inverse preserves both compositions and their interchange. It therefore identifies the E_2 algebras and their evaluations. This is also the derived invariance of Hochschild higher operations [94, Sections 3.2 and 4.6]. □

For $\mathsf{B} = M_n(\mathsf{A})$ the comparison is visible entry by entry. With $e = E_{11}$, the bimodules $P = \mathsf{B}e$ and $Q = e\mathsf{B}$ satisfy

$$Q \otimes^{\mathbb{L}}_{\mathsf{B}} P \simeq \mathsf{A}, \qquad P \otimes^{\mathbb{L}}_{\mathsf{A}} Q \simeq \mathsf{B}.$$

For an unsuspended arity-q cochain,

$$(\Phi_n f)(X_1, \dots, X_q)_{i_0 i_q} = \sum_{i_1, \dots, i_{q-1}} f\big((X_1)_{i_0 i_1}, \dots, (X_q)_{i_{q-1} i_q}\big). \tag{13.13}$$

At arity zero it is the diagonal matrix with entry f. Concatenating index paths proves compatibility with cup; substituting paths proves compatibility with every brace. The relative bar over $M_n(k)$ computes the ordinary bar Hom because this algebra is separable: inserting $\sum_i E_{i1} \otimes E_{1i}$ contracts the extra balancing directions. Its relative cochains identify with $C(\mathsf{A})$ by (13.13). Thus the displayed map is a quasi-isomorphism with all these operations.

13.6 The centre retains the attaching point

Return to the disk generator $A \oplus B$ of (13.7). Replace it by $A \oplus B[-2]$. The off-diagonal derived Hom is now in degree zero, so its endomorphism algebra is quasi-isomorphic to

$$T_x = \begin{pmatrix} \Lambda & k \\ 0 & k \end{pmatrix}, \qquad \Lambda = k[x, (1+x)^{-1}], \quad x = s - 1. \tag{13.14}$$

The arrow a in the upper-right entry satisfies $xa = 0$. The shift changes the compact generator; it does not discard the degree-two attachment. More explicitly, a dg algebra with cohomology only in degree zero has the multiplicative truncation zigzag $E \leftarrow \tau_{\le 0} E \rightarrow H^0(E)$. Theorem 13.12 therefore identifies the derived centre of the disk category with that of T_x.

To vary the attachment, let $F \in k[x]$ be monic and nonconstant, with $F(-1) \neq 0$, and put

$$M_F = \Lambda/(F), \qquad T_F = \begin{pmatrix} \Lambda & M_F \\ 0 & k \end{pmatrix}. \tag{13.15}$$

Its diagonal idempotents are e_1, e_2; the class of 1 in M_F is $a = e_1 a e_2$, and $F(x)a = 0$. The algebra T_{x^r} for $r > 1$ defines a different attaching category. A geometric realization of that category requires additional data.

Theorem 13.13. *The algebra $T = T_F$ has the projective bimodule resolution*

$$0 \longrightarrow P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{\mu} T \longrightarrow 0,$$

where

$$\begin{aligned} P_0 &= (Te_1 \otimes e_1T)[e_1] \oplus (Te_2 \otimes e_2T)[e_2],\\ P_1 &= (Te_1 \otimes e_1T)[x] \oplus (Te_1 \otimes e_2T)[a],\\ P_2 &= (Te_1 \otimes e_2T)[R]. \end{aligned}$$

Here all tensors are over k, and brackets name generators. Writing $F(x) = \sum_j f_j x^j$, the differentials are

$$\begin{aligned} d_1[x] &= x[e_1] - [e_1]x, & d_1[a] &= a[e_2] - [e_1]a,\\ d_2[R] &= \sum_{j\geq 1} f_j \sum_{b=0}^{j-1} x^b[x]x^{j-1-b}a + F(x)[a]. \end{aligned} \tag{13.16}$$

Proof. Every summand is an idempotent summand of a free T^e-module. The composite d_1d_2 telescopes to $F(x)a[e_2] - [e_1]F(x)a = 0$. Check exactness in the four idempotent blocks. The $(2,1)$ block is zero, the $(2,2)$ augmentation is the identity, and the $(1,1)$ block is the diagonal resolution of Λ, whose kernel is generated by the nonzero divisor $x - y$ in $\Lambda \otimes_k \Lambda$.

For the $(1,2)$ block set

$$L = \Lambda_x \otimes_k M_{F,y}, \qquad D_F(x,y) = \frac{F(x) - F(y)}{x - y}.$$

The augmented complex is

$$0 \longrightarrow \Lambda_x \longrightarrow L \oplus \Lambda_x \longrightarrow L \oplus M_F \longrightarrow M_F \longrightarrow 0$$

with maps

$$W \longmapsto (D_FW, FW), \quad (U,V) \longmapsto ((x-y)U - V, \overline{V}), \quad (Z,m) \longmapsto Z(y,y) + m.$$

The first is injective because F is nonzero in the domain Λ_x. Multiplication by $x - y$ on L is injective: it is monic in x before localization, even when M_F is nonreduced. If $(x-y)U = V$ and $\overline{V} = 0$, write $V = FW$; then $U = D_FW$. If $Z(y,y) + m = 0$, choose a lift $V \in \Lambda$ of m. The element $Z + V$ is divisible by $x - y$, and its quotient gives a preimage of (Z,m). The final map is surjective. These arguments prove exactness. □

Let $S = ke_1 \oplus ke_2$. The relative normalized bar cochains

$$C^n_S(T_F, T_F) = \mathrm{Hom}_{S\text{-}S}((T_F/S)^{\otimes_S n}, T_F)$$

retain cup and braces and compute the derived centre, since S is separable. The finite resolution maps to this bar by sending $[x]$ and $[a]$ to their one-bar classes and

$$[R] \longmapsto \sum_{j\geq 1} f_j \sum_{b=0}^{j-1} x^b[x \mid x^{j-1-b}a]. \tag{13.17}$$

Its bar boundary telescopes to (13.16); the relation $F(x)a = 0$ supplies the constant term. The short complex below computes cohomology. Its length does not define strict brace operations on it.

Theorem 13.14. *The underlying derived-centre complex is computed by*

$$\Lambda \oplus k \xrightarrow{d^0} \Lambda \oplus M_F \xrightarrow{d^1} M_F, \quad d^0(f,c) = (0, c - \overline{f}), \qquad d^1(v,w) = \overline{F'v}. \tag{13.18}$$

These terms are in degrees zero, one and two. If $g = \gcd(F, F')$ is monic, then

$$\begin{aligned} HH^0(T_F) &= k + F\Lambda, \qquad HH^1(T_F) = (F/g)\Lambda\,\partial_x, \\ HH^2(T_F) &= \Lambda/(F, F'), \quad HH^n(T_F) = 0 \quad (n \geq 3). \end{aligned} \tag{13.19}$$

The function $f = c + Fh$ represents the central matrix $\operatorname{diag}(f, c)$.

Proof. Bimodule Hom sends the vertex generators to (f, c) and the loop and arrow generators to (v, w). Evaluating their coboundaries on (13.16) gives d^0 and $F'v + Fw = F'v$ modulo F. The first differential surjects onto the arrow component. The remaining kernel is $\{v : F \mid F'v\} = (F/g)\Lambda$, and the cokernel is $\Lambda/(F, F')$. The resolution excludes higher groups. Represent a degree-one class by the derivation $\delta_v(x) = v$, $\delta_v(a) = \delta_v(e_i) = 0$. The divisibility condition is exactly the condition preserving $F(x)a = 0$. □

For the simple attachment this gives

$$HH^0(T_x) = \Lambda, \qquad HH^1(T_x) = x\Lambda\partial_x, \qquad HH^{\geq 2}(T_x) = 0. \tag{13.20}$$

The contraction of (13.18) onto these groups is explicit:

$$\begin{aligned} i^0(f) &= (f, f(0)), \quad p^0(f,c) = f, \\ i^1(v) &= (v, 0), \qquad\quad p^1(v,w) = v - v(0), \end{aligned} \qquad h^1(v,w) = (0,w), \quad h^2(z) = (z,0).$$

It satisfies $dh + hd = 1 - ip$ and $pi = 1$. The cup action is multiplication of functions on vector fields. The derivation representatives give

$$[v\partial_x, f] = vf', \qquad [v\partial_x, w\partial_x] = (vw' - wv')\partial_x, \qquad v, w \in x\Lambda. \tag{13.21}$$

Products of two positive-degree classes vanish by (13.20).

Corollary 13.15. *There is no Gerstenhaber-algebra map $HH^\bullet(\Lambda) \to HH^\bullet(T_x)$ that is the identity on degree-zero functions. Consequently there is no unital E_2 map of derived centres with that property.*

Proof. The image of ∂_x would be $v\partial_x$ for $v \in x\Lambda$. Preservation of $[\partial_x, x] = 1$ would force $v = 1$, whereas x is not invertible in Λ. No section property is needed for this contradiction. □

The coordinate $x = s - 1$ measures monodromy relative to the attaching point. It is not a spacetime insertion coordinate. Restriction to the open object does give a higher Hochschild map in the opposite direction, by restriction of bar cochains to a full subcategory [94, Section 4.3]. On cohomology it is the identity on functions and the inclusion $x\Lambda\partial_x \subset \Lambda\partial_x$.

13.6.1 *Several attachments and flat deformations*

For d independent simple attachments, tensor the finite bimodule resolutions. Their terms are finitely generated projectives, so Hom commutes with the tensors in this calculation. Künneth gives

$$HH^\bullet(T_x^{\otimes d}) = k[x_i, (1+x_i)^{-1} : 1 \le i \le d] \otimes \Lambda(\theta_1, \dots, \theta_d), \qquad \theta_i = x_i \partial_{x_i}, \quad |\theta_i| = 1. \tag{13.22}$$

The brackets obey $[\theta_i, x_j] = \delta_{ij} x_i$ and $[\theta_i, \theta_j] = 0$, extended by the graded Leibniz rule. In two directions, $\theta_1 \smile \theta_2 \neq 0$.

Let D_i be the derivation of $T_x^{\otimes 2}$ which acts by $x_i \partial_{x_i}$ on its open coordinate and kills both idempotents and both arrows. It preserves the arrow relations and the localized functions. The commuting operators D_1, D_2 define

$$a \star_\hbar b = \sum_{n \ge 0} \frac{\hbar^n}{n!} D_1^n(a) D_2^n(b). \tag{13.23}$$

This is an associative unital product on $T_x^{\otimes 2}[[\hbar]]$. Both triple products equal ordinary triple multiplication after

$$\exp\bigl(\hbar(D_1 \otimes D_2 \otimes 1 + D_1 \otimes 1 \otimes D_2 + 1 \otimes D_1 \otimes D_2)\bigr).$$

The three operators commute, proving associativity in every formal degree. The derivations kill the unit. The first-order class is $\theta_1 \smile \theta_2$, and

$$x_1 \star_\hbar x_2 = e^{\hbar} x_1 x_2, \qquad x_2 \star_\hbar x_1 = x_1 x_2.$$

The antisymmetrization of a Hochschild coboundary on the central pair (x_1, x_2) is zero, whereas that of the displayed first-order term is $x_1 x_2$. This also detects its nonzero class.

For a thick attachment $F = x^r$, $r \ge 2$, the groups are

$$HH^0(T_{x^r}) = k + x^r \Lambda, \quad HH^1(T_{x^r}) = x\Lambda\partial_x, \quad HH^2(T_{x^r}) = k[x]/(x^{r-1}).$$

The degree-two directions distinguish it from the simple attachment even up to derived Morita equivalence. For $v = xw$ their Gerstenhaber action is

$$[xw\partial_x, g] = xwg' - rwg \pmod{x^{r-1}}. \tag{13.24}$$

Indeed, $x \mapsto x + \varepsilon xw$ carries x^r to $(1 + \varepsilon r w)x^r$. Apply this change to $x^r - \tau g$ and divide the relation by that unit. Its new deformation term is $g + \varepsilon(xwg' - rwg)$, proving the formula. In particular $[x\partial_x, x^j] = (j - r)x^j$ in degree two.

All these directions integrate together. Put

$$R = k[[t_0, \dots, t_{r-2}]], \qquad F_{\mathbf{t}}(x) = x^r - \sum_{j=0}^{r-2} t_j x^j,$$

and complete in the maximal ideal of R. Then

$$\mathcal{T}_{\mathbf{t}} = \begin{pmatrix} \Lambda \widehat{\otimes}_k R & R[x]/(F_{\mathbf{t}}) \\ 0 & R \end{pmatrix} \tag{13.25}$$

is a formally flat algebra. The arrow module is free over R on $1, x, \dots, x^{r-1}$, and $1 + x$ is invertible on it because $F_{\mathbf{t}}(-1)$ is a unit. Each Artinian reduction is free over its parameter ring. The t_j tangent class

is $+x^j$: in the companion presentation, the first product variation sends $(x, x^{r-1}a)$ to $x^j a$, and (13.17) identifies this with the positive relation class.

For $r = 2$, multiplication by $1 + x$ on the arrow module is

$$S_\tau = \begin{pmatrix} 1 & \tau \\ 1 & 1 \end{pmatrix}, \qquad (S_\tau - I)^2 = \tau I, \qquad \det S_\tau = 1 - \tau.$$

The rank and trace are constant while the attachment deforms. The algebraic families (13.23) and (13.25) construct operations on ordinary probe algebras. A Hall coproduct and factorization sewing require their own maps.

13.7 Half-braidings and central functors

The derived centre above is an algebra of operations. A tensor category has a different centre, itself a braided category. Its objects remember how one object exchanges with every other object of the category.

Let $(\mathscr{M}, \otimes, \mathbf{1})$ be a monoidal category. An object of its *Drinfeld centre* $\mathcal{Z}(\mathscr{M})$ is a pair (V, β), where $V \in \mathscr{M}$ and

$$\beta_W : V \otimes W \xrightarrow{\sim} W \otimes V$$

is natural in W, equals the unit constraint at $\mathbf{1}$, and obeys the half-braiding identity

$$\beta_{W \otimes W'} = (1_W \otimes \beta_{W'})(\beta_W \otimes 1_{W'}).$$

Associators are understood in this formula. Morphisms commute with every β_W. Tensoring the half-braidings defines the monoidal product, and $\beta_{V'}$ gives its braiding.

A central structure on a strong monoidal functor $F : \mathscr{B} \to \mathscr{M}$ consists of half-braidings on every $F(B)$, compatible with the tensor constraints. If $\mathscr{B}$ is braided and the lift is to be braided, its braiding must also map to the resulting half-braiding between objects in the image. These structures are additional to the monoidal constraints of F.

Example 13.16. For a finite group G, let Vec_G be the category of finite-dimensional G-graded vector spaces, with $(V \otimes W)_g = \bigoplus_{ab=g} V_a \otimes W_b$. Write δ_g for the one-dimensional object of degree g. The symmetric category Vec_{C_2} maps strongly monoidally to Vec_{S_3} by sending the nontrivial degree to $s = (12)$. This functor has no central lift. Indeed, a half-braiding of δ_s with δ_h, $h = (123)$, would identify objects of degrees sh and hs. These degrees differ, so every graded map between them is zero.

Thus the centre construction does not provide a right adjoint from arbitrary monoidal categories to braided categories with all strong monoidal functors as morphisms. There is a genuine adjunction for the forgetful functor from a fixed centre. The finite-group case computes it completely.

Proposition 13.17. *The category $\mathcal{Z}(\mathrm{Vec}_G)$ consists of G-graded vector spaces V with a G-action ρ such that*

$$\rho(h) V_g \subset V_{hgh^{-1}}.$$

The associated half-braiding on homogeneous elements is

$$\beta(v \otimes w_h) = w_h \otimes \rho(h^{-1})v. \tag{13.26}$$

The forgetful functor U has right adjoint

$$I(X) = \bigoplus_{a \in G} (a, X), \qquad \deg(a, x_g) = a g a^{-1}, \quad \rho(h)(a, x) = (ha, x). \tag{13.27}$$

Proof. The grading condition makes (13.26) degree-preserving. Exchanges first with degree h and then with degree k apply $\rho(k^{-1})\rho(h^{-1}) = \rho((hk)^{-1})$; this is the half-braiding identity. Conversely, the exchanges with all δ_h determine these operators. The unit and tensor identities give their group law, and degree preservation gives the conjugation condition. Naturality extends the formula from graded lines to all objects.

The object (13.27) has the required grading and action. Given a graded map $f : UV \to X$, define

$$\widetilde{f}(v) = \sum_{a \in G} (a, f(\rho(a^{-1})v)).$$

Each summand has the degree of v. Reindexing $a = hb$ proves equivariance, so $\widetilde{f}$ is a central morphism. Projection to the identity-labelled summand recovers f. Conversely, equivariance determines every summand of a central map from that projection. The two constructions are inverse and natural, proving $\mathrm{Hom}_{\mathcal{Z}}(V, I(X)) = \mathrm{Hom}_{\mathrm{Vec}_G}(UV, X)$. □

For $G = S_3$ and $X = \delta_{(12)}$, the object $I(X)$ has dimension six. Each of the three conjugate transposition degrees occurs twice. The adjoint adds these conjugate components; it does not place a half-braiding on the original one-dimensional object.

Proposition 13.18. *A strong monoidal equivalence $F : \mathscr{M} \xrightarrow{\sim} \mathscr{N}$ induces a braided equivalence $\mathcal{Z}(\mathscr{M}) \xrightarrow{\sim} \mathcal{Z}(\mathscr{N})$.*

Proof. Choose a monoidal inverse G and counit $\varepsilon : FG \xrightarrow{\sim} \mathrm{Id}_{\mathscr{N}}$. If ϕ denotes the tensor constraint of F, transport a half-braiding by the composite

$$\begin{aligned} F(V) \otimes W &\xrightarrow{1 \otimes \varepsilon_W^{-1}} F(V) \otimes F(GW) \xrightarrow{\phi} F(V \otimes GW) \\ &\xrightarrow{F(\beta_{GW})} F(GW \otimes V) \xrightarrow{\phi^{-1}} F(GW) \otimes F(V) \xrightarrow{\varepsilon_W \otimes 1} W \otimes F(V). \end{aligned}$$

The monoidal coherence equations and the original half-braiding identity prove the new one. The same construction with G supplies the inverse; the unit and counit give its central equivalences. Applying the formula to two central objects shows preservation of their braiding. □

A derived equivalence of geometric categories therefore transports their ordinary derived centres by Theorem 13.12. To use Proposition 13.18, it must additionally preserve the specified monoidal products. An equivalence of underlying categories supplies no Hall tensor constraint.

13.8 The internal centre for convolution

The coefficient category of the ordered collision schemes carries a symmetric disjointness convolution. We now apply the derived-centre construction internally, retaining the coefficient object on every configuration.

Let $(\mathscr{C}_R, \star, K_R)$ be the presentable stable coefficient category of Chapter 8, extended over a commutative k-algebra R. Its convolution preserves colimits in both variables, and K_R is supported at the empty configuration. For an associative algebra A in this category, define $Z^{\mathrm{int}}_{\star}(A)$ by

$$\mathrm{Map}_{\mathscr{C}_R}(V, Z^{\mathrm{int}}_{\star}(A)) \simeq \mathrm{Map}_{A\text{-}A}(V \star A, A). \tag{13.28}$$

The right side uses the two actions of A on itself. Its representability follows from presentability and the colimit-preserving coefficient action.

An action on A through convolution is a left-module structure over an E_2 algebra F in the category of associative algebras of $\mathscr{C}_R$. Thus $F \star A \to A$ is an algebra map with unit, multiplication and action coherences. Morphisms of actions preserve all these data.

Theorem 13.19. *The internal centre is an E_2 algebra, and its evaluation is final among convolution actions on A. Composition with evaluation identifies maps of E_2 algebras $F \to Z^{\mathrm{int}}_\star(A)$ with such actions of F. Evaluation at the unit of A gives an algebra map $Z^{\mathrm{int}}_\star(A) \to A$.*

Proof. The centre theorem for a presentable symmetric monoidal category whose tensor preserves colimits constructs the final action [114, Corollary 5.3.1.15]. Its centralizer of the identity is the morphism object of A as a bimodule [114, Proposition 5.3.1.8 and Theorem 5.3.1.30]. That morphism object is precisely (13.28). The two compatible compositions make it an associative algebra in associative algebras, hence an E_2 algebra. Finally, compose evaluation with $Z^{\mathrm{int}}_\star(A) \star K_R \to Z^{\mathrm{int}}_\star(A) \star A$. Both arrows are algebra maps. □

For a left A-module M, tensoring evaluation over A gives its characteristic action

$$Z^{\mathrm{int}}_\star(A) \star M \simeq (Z^{\mathrm{int}}_\star(A) \star A) \star^{\mathbb{L}}_A M \longrightarrow A \star^{\mathbb{L}}_A M \simeq M. \tag{13.29}$$

It is natural in M, including all homotopies. A $\mathscr{C}_R$-linear equivalence of module categories intertwines these actions and identifies their internal centres as E_2 algebras. The proof of Theorem 13.12 applies with $\star$ in place of tensor over k: the coefficient-linear endofunctors are bimodules and the identity is A. The coefficient linearity is necessary to retain the internal object.

For $A = K_R \oplus a$ with positive augmentation ideal a, let $[V, W]_\star$ denote the internal Hom for convolution. The two-sided bar gives

$$Z^{\mathrm{int}}_\star(A) \simeq \left(\prod_{q \geq 0} [(sa)^{\star q}, A]_\star, \partial + [\tau, -] \right), \tag{13.30}$$

where $\tau : BA \to A$ is the universal twisting cochain, $\partial f = d_A f - (-1)^{|f|} f d_{BA}$, and $f \smile g = \mu_A(f \star g)\Delta_{BA}$. The adjacent bar products contribute to ∂; the two outer module actions give the commutator with τ. The Maurer–Cartan equation for τ makes this differential square zero. Cup and braces are those of (13.11)–(13.12), using evaluation and the symmetry of convolution.

The product in (13.30) can be infinite even on one fixed output configuration: internal Hom from a high convolution power can have a nonzero empty coefficient. Define the scalar centre by

$$Z^{\mathrm{sc}}_\star(A) = R\,\mathrm{Hom}_{\mathscr{C}_R}(K_R, Z^{\mathrm{int}}_\star(A)).$$

It retains only this scalar coefficient of the internal object.

For the square-zero algebra $K \oplus a$, the cochains with values in a have only their internal differential. The positive-to-empty mapping objects vanish, so

$$Z^{\mathrm{int}}_\star(K \oplus a) \simeq K \oplus \prod_{q \geq 0} [a^{\star q}, a]_\star[-q].$$

Their cup products vanish, but their braces need not. If δ is the identity of a, regarded as a degree-one Hochschild class, then any a-valued arity-q cochain obeys

$$[\delta, f]_{\mathrm{G}} = (1 - q)f. \tag{13.31}$$

The output insertion gives f and the q input insertions give qf; $s\delta$ has degree zero, so no insertion sign occurs.

For $a = i_{\mathrm{ord}}\omega_{\mathbb{A}^1}$, every binary reduced word gives the operation complex $k \oplus k[-1]$. Indeed, applying derived Hom into the diagonal module to the localization triangle gives a triangle with other terms $k[-1]$

and $k[1]$. The map between them is zero over k; the remaining complex is $k \oplus k[-1]$, with the residue in degree zero. The middle-term calculation uses the one-variable Spencer resolution: derivatives act freely on the delta module, leaving one degree-one cokernel. The Hochschild shift by two therefore gives scalar-centre classes μ_w, ν_w of degrees two and three, with

$$[\partial, \mu_w]_{\mathrm{G}} = -\mu_w, \qquad [\partial, \nu_w]_{\mathrm{G}} = -\nu_w.$$

Thus even this square-zero example requires the brace operations.

13.9 The ambient transverse endomorphism coalgebra

Convolution permits disjoint position sets. A transverse bulk input may share every position of its boundary input. The corresponding tensor is the diagonal tensor $\otimes^{!}$. Let Coalg_{+} be the category of coaugmented counital convolution coalgebras whose empty coefficient is k. Its diagonal unit is the dualizing coalgebra $\mathbb{J}$, with a vacuum at every configuration. Its reduced part is positive. The empty convolution unit remains K.

A *transverse monoid* is an associative algebra object of $(\mathsf{Coalg}_{+}, \otimes^{!}, \mathbb{J})$. Its product, unit and action on a coalgebra C have the forms

$$F \otimes^{!} F \longrightarrow F, \qquad \mathbb{J} \longrightarrow F, \qquad F \otimes^{!} C \longrightarrow C.$$

They are coalgebra maps with coherent monoid or module laws. In particular the product intertwines the coproduct through the common-decomposition interchange of the two tensors.

Theorem 13.20. *The category* Coalg_{+} *is closed for* $\otimes^{!}$. *Write* $[[C, D]]$ *for its internal mapping coalgebra. Then*

$$\mathcal{E}(C) = [[C, C]]$$

is a transverse monoid, and evaluation classifies every transverse monoid action on C*:*

$$\mathrm{Map}_{\mathrm{Mon}}(F, \mathcal{E}(C)) \simeq \{\textit{coherent transverse actions of } F \textit{ on } C\}. \tag{13.32}$$

At one position,

$$\mathcal{E}(C)_{\{1\}} \simeq \underline{\mathrm{Hom}}^{!}_{X}(C_{\{1\}}, C_{\{1\}}).$$

No factorization condition on $\mathcal{E}(C)$ *is asserted.*

Proof. Associative bar–cobar identifies Coalg_{+} with augmented convolution algebras having positive augmentation ideal. The latter form a presentable category. Limits of nonunital algebras in the opposite coefficient category are computed on underlying objects, so coalgebra colimits are computed on reduced parts. The diagonal tensor preserves these colimits componentwise. The adjoint functor theorem therefore gives the mapping coalgebra.

An internal endomorphism object has composition and an identity. The enriched endomorphism universal property identifies its algebra maps with module structures, including their coherences [114, Corollary 4.7.1.40]. This gives (13.32).

For the last statement, test the pointwise Hom on objects supported on the small-diagonal branches. Local cohomology on those branches is $i_{\mathrm{ord}}(C_{\{1\}})$, and the projection formula reduces the adjunction to the one-position mapping object. Bar words of length at least two vanish at one position. The suspension in bar cancels the desuspension in cobar, leaving the displayed endomorphism object. □

There is also an explicit cochain description. For reduced parts c of C and a of an augmented convolution algebra A, put

$$\mathcal{H}(C, A) = \Omega[[C, BA]].$$

Its augmentation ideal is $\underline{\mathrm{Hom}}^{!}(c, a)$, and on strict derived representatives its product is determined by

$$\mathrm{ev}(\cup \otimes^{!} 1) = \mu_a(\mathrm{ev} \star \mathrm{ev})\zeta(1 \otimes^{!} \overline{\Delta}_C). \tag{13.33}$$

Here ζ projects onto common ordered decompositions. The differential is the mapping differential, without a twist by a distinguished cochain.

To verify the assertion, test on a positive coefficient V. The bar of the free tensor algebra $T_\star V$ is equivalent to the primitive coalgebra $K \oplus sV$. Its diagonal product with C is primitive on $sV \otimes^{!} c$, so free-algebra adjunction gives

$$\mathrm{Map}(V, \overline{\mathcal{H}(C, A)}) \simeq \mathrm{Map}(V \otimes^{!} c, a).$$

The product (13.33) is associative by coassociativity, associativity and the coherent interchange. Compose the bar twisting cochain of this cup algebra with evaluation. The resulting twisting cochain defines a map from its bar tensored with C to BA; its adjoint induces the identity on the displayed augmentation ideal. It is therefore an equivalence of algebras. Taking $A = \Omega C$ gives

$$\Omega\mathcal{E}(C) \simeq \mathcal{H}(C, \Omega C).$$

The identity of C specifies its transverse unit.

The two action tensors cannot be interchanged in these statements. For a positive coefficient V,

$$(K \star V)_{\{1\}} = V_{\{1\}}, \qquad (K \otimes^{!} V)_{\{1\}} = 0.$$

The common-decomposition interchange has a section, but its other composite is a projection, not an equivalence. The convolution centre has the twisted differential in (13.30); the transverse endomorphism coalgebra has the untwisted cochains above. A comparison must specify its tensor and its evaluation map.

13.10 A Heisenberg monoid and its augmentation obstruction

The ambient construction does not require a chosen augmentation to $\mathbb{J}$. A concrete factorizing monoid acting on the Heisenberg boundary shows why such an augmentation can fail even when its singleton cohomology is an augmented algebra.

Fix $\kappa \in k^{\times}$, and let V_κ be the Heisenberg vertex algebra generated by J, with $[J_\lambda J] = \kappa\lambda$. Adjoin an odd field c of degree one, with regular brackets, and set

$$\mathcal{Z}_\kappa = V_\kappa \otimes \Lambda[c, Tc, T^2c, \ldots], \qquad dJ = \kappa Tc, \quad dc = 0.$$

The differential is the zero mode of $:cJ:$ and is a vertex derivation. It squares to zero. All states are algebraic polynomials in finitely many jets.

Let E be the Weyl–Clifford vertex algebra on even fields t, p of degree zero and odd fields η, ι of degrees 1, -1, with

$$[p_\lambda t] = 1, \qquad [\iota_\lambda \eta] = 1, \qquad Qt = \eta, \quad Q\iota = p, \quad Q\eta = Qp = 0. \tag{13.34}$$

The other generator brackets are zero or determined by skewsymmetry. On $W = V_\kappa \otimes E$, the maps

$$i(J) = J + p, \qquad j(J) = J - \kappa T t, \qquad j(c) = -\eta \tag{13.35}$$

define vertex maps $i : V_\kappa \to W$ and $j : \mathcal{Z}_\kappa \to W$ with commuting images. Indeed,

$$[(J + p)_\lambda (J - \kappa T t)] = \kappa\lambda - \kappa\lambda = 0, \qquad Q(J - \kappa T t) = -\kappa T \eta.$$

Both image currents retain level κ.

The quartet is contractible away from its scalar vacuum. On polynomial mode coordinates its differential pairs each t mode with its η mode and each ι mode with its p mode. The reverse pairing, divided by the total positive quartet degree, is a contraction. Thus i is a quasi-isomorphism. The commutant of $J + p$ has coordinates

$$v_n = x_n - \kappa n \tau_n, \quad \pi_n \quad (n \geq 1), \qquad \bar\eta_m \quad (m \geq 0), \quad \bar\iota_n \quad (n \geq 1),$$

Here $x_n = T^{n-1} J/(n-1)!$ and $\pi_n = T^{n-1} p/(n-1)!$ for $n \geq 1$, whereas $\tau_m = T^m t/m!$ and $\bar\eta_m = T^m \eta/m!$ for $m \geq 0$. Finally $\bar\iota_n = T^{n-1}\iota/(n-1)!$. These are polynomial jet coordinates. Their differential is $Q v_n = -\kappa n \bar\eta_n$ and $Q\bar\iota_n = \pi_n$. Contracting the latter pairs shows that j is a quasi-isomorphism onto the commutant.

For b quartets put $W_b = V_\kappa \otimes E^{\otimes b}$, $K_b = J + \sum_{l=1}^{b} p_l$ and $\mathcal{Z}^{(b)} = \mathrm{Com}(K_b, W_b)$. These commutants have generators $J - \kappa T t_1$, $t_j - t_1$ for $j > 1$, and all p_j, η_j, ι_j. For an order-preserving surjection $\phi : [b'] \twoheadrightarrow [b]$, define a merge $m_\phi : W_b \to W_{b'}$ by

$$p_j \mapsto \sum_{\phi(l)=j} p_l, \quad \iota_j \mapsto \sum_{\phi(l)=j} \iota_l, \quad t_j \mapsto t_{\min \phi^{-1}(j)}, \quad \eta_j \mapsto \eta_{\min \phi^{-1}(j)}.$$

It fixes J and carries K_b to $K_{b'}$. The generator brackets prove that it is a dg vertex map. The same polynomial contractions show that it and its restriction to commutants are quasi-isomorphisms. Taking the least element of a fibre commutes with composition, so these merges form a diagram.

The shift $r_a : W_b \to W_{a+b}$ sends J to K_a and the jth quartet to the $(a + j)$th. The products

$$\mathcal{Z}^{(a)} \otimes W_b \longrightarrow W_{a+b}, \qquad z \otimes w \longmapsto {:}z\, r_a(w){:}$$

are vertex maps because the two images commute. They restrict to the commutants, are strictly associative and commute with merges. Iteration sends the ith cochain factor to

$$J^{(i)} \longmapsto J + \sum_{l<i} p_l - \kappa T t_i, \qquad c^{(i)} \longmapsto -\eta_i, \tag{13.36}$$

and the boundary current to K_b.

For a dg vertex algebra B on $X = \mathbb{A}^1$, let

$$L_B = i_{\mathrm{ord}}(\omega_X \otimes B)$$

be its main-diagonal Lie object for $\star$. Here i_{ord} is the diagonal insertion of Section 8.9; the right differential-module structure is induced by translation, and the bracket by the vertex products. Write

$$C_B = \mathrm{CE}_\star(L_B) = K \oplus \bigoplus_{r \geq 1} \mathrm{Sym}^r_\star(L_B[1])$$

for its Chevalley–Cousin factorization coalgebra. The differential combines the internal differential with pairwise bracket contractions, with the suspended Koszul signs; the coproduct is the unshuffle coproduct. Each coefficient has finitely many positive factors. Write $U_\star L_B$ for the associative enveloping algebra of L_B.

Proposition 13.21. *The coalgebra $F_\kappa = C_{\mathcal{Z}_\kappa}$ has a coherent unital transverse monoid structure and acts on C_{V_κ}. Its products are the roofs*

$$C_{\mathcal{Z}_\kappa}^{\otimes^! b} \longrightarrow C_{\mathcal{Z}^{(b)}} \xleftarrow{\sim} C_{\mathcal{Z}^{(1)}} \xleftarrow{\sim} C_{\mathcal{Z}_\kappa}.$$

Its action is induced by the commuting maps in (13.35) *and the inverse of C_i.*

Proof. For these free fields the factorization lattices are K-flat: their PBW filtration has graded pieces equal to symmetric powers of free principal-part modules, with a linear differential. Tensoring lattices therefore computes the derived diagonal tensor. The finite Chevalley–Cousin comparison gives factorization and carries quasi-isomorphisms to equivalences. It also carries polynomial homotopies to homotopies. Hence $B \mapsto C_B$ preserves the tensor, merges and strict products above.

The category of nonempty ordered finite sets and surjections has a final singleton. Its nerve, and the nerves of its finite products, are contractible. Localizing the diagram at its merge equivalences therefore descends the products and action to the singleton. The same argument for the two-coloured algebra/module operad retains all their coherences: localization of the associated cocartesian fibration localizes its fibres [81, Proposition 2.1.4].

It remains to verify the unit. In two quartets, the merge $m\,j$ and shift $r_1 j$ are joined by the following two polynomial homotopies, with parameter u and forms on the left:

	J	c
1	$J - \kappa T((1-u)t_1 + ut_2)$	$-(1-u)\eta_1 - u\eta_2 - du(t_2 - t_1)$
2	$J + up_1 + du\,\iota_1 - \kappa T t_2$	$-\eta_2$.

The first starts at $m\,j$; the second ends at $r_1 j$. They preserve brackets and differential and commute with K_2. Right multiplication by the vacuum is the inclusion, which agrees with m on j; left multiplication is the shift. Thus the vacuum is a quasi-unit. On the module its action is an equivalence, since $r_1 i_1 = i_2$. Unitalization of quasi-unital algebras and modules supplies the coherent units [114, Theorem 5.4.3.5 and Proposition 5.4.3.14]. Transport along C_j and C_i gives the stated monoid and action. □

13.10.1 *The supported binary class*

The specified homotopies also determine a supported operation. Put $Z = \mathcal{Z}_\kappa$ and $Z' = \mathcal{Z}^{(2)}$. Its two embeddings are

$$\jmath_1(J) = U_1 = J - \kappa T t_1, \quad \jmath_1(c) = -\eta_1, \qquad \jmath_2(J) = U_1 + p_1 - \kappa T(t_2 - t_1), \quad \jmath_2(c) = -\eta_2.$$

Integrating the form components of the two homotopies gives h of degree -1, with

$$Qh + hd = \jmath_2 - \jmath_1, \qquad h(J) = \iota_1, \quad h(c) = -(t_2 - t_1).$$

On a binary chart the unshifted coefficient is the fibre of the chiral bracket $\mu : j_* j^*(Z^r \boxtimes Z^r) \to \Delta_* Z^r$, where $Z^r = \omega_X \otimes Z$. Its differential is $d(\sigma, \zeta) = (d\sigma, \mu\sigma - d\zeta)$. The two products of vacuum insertions are $(\jmath_1 \boxtimes \jmath_2, 0)$ and $(\jmath_2 \boxtimes \jmath_1, 0)$.

The external homotopy

$$H(a \boxtimes b) = -h(a) \boxtimes \jmath_2(b) + (-1)^{|a|}\jmath_2(a) \boxtimes h(b)$$

has differential their difference in the first component. Subtracting its boundary in the fibre leaves $(0, -\mu H)$. Its Fourier expression is therefore

$$\beta(a,b) = [h(a)_\lambda \jmath_2(b)] - (-1)^{|a|}[\jmath_2(a)_\lambda h(b)]. \tag{13.37}$$

This calculation specifies the overlap nullhomotopy as well as the supported map.

Contract the extra quartet coordinates by the polynomial contraction, obtaining $R : Z' \to Z$ with $R\jmath_1 = 1$. The sign-adjusted pairing $\{a_\lambda b\} = (-1)^{|a|} R\beta(a,b)$ has degree -1 and generator values

$$\{J_\lambda J\} = \{c_\lambda c\} = 0, \qquad \{c_\lambda J\} = 1, \quad \{J_\lambda c\} = -1.$$

For example $[\jmath_2(J)_\lambda(t_2 - t_1)] = -1$ and $[\iota_{1\lambda}\eta_2] = 0$, giving $\beta(J,c) = -1$.

The underlying polynomial complex has coordinates $x_n = J_{-n}\mathbf{1}$ for $n \geq 1$ and $\xi_n = T^n c/n!$ for $n \geq 0$, with

$$dx_n = \kappa n \xi_n, \quad d\xi_n = 0, \quad Tx_n = nx_{n+1}, \quad T\xi_n = (n+1)\xi_{n+1}.$$

Its cohomology is $\Lambda(\xi_0)$. The derivation $K = \kappa^{-1}\sum_{n\geq 0} x_{n+1}\partial_{\xi_n}$ satisfies $dK + Kd = T$. Thus the secondary value of the specified pairing is

$$-\{K\xi_{0\lambda}\xi_0\}\big|_{\lambda=0} = \kappa^{-1}. \tag{13.38}$$

The homotopy K refers to the polynomial complex; its existence does not identify all its vertex operations with those of a commutative vertex algebra.

To locate this value geometrically, record a moving coefficient complex as a module over the de Rham algebra, with differential $d + dz(\partial_z + T)$. The operator $1 - dzK$, with inverse $1 + dzK$, changes this differential to $d + dz\partial_z$. Indeed, $[d + dz\partial_z, dzK] = -dzT$. Apply this change on the two inputs and the diagonal output of the bracket. The coefficient of dz_1 at Fourier parameter zero is $(-1)^{|a|}\{Ka_\lambda b\}|_{\lambda=0}$. For $a = b = \xi_0$ it is κ^{-1}. The de Rham generator of the diagonal direct image is $\varepsilon = [(dz_1 - dz_2)\otimes 1]$, the residue image of dz/z, where $z = z_1 - z_2$. Thus the supported lift has class

$$\kappa^{-1}(\xi_0 \otimes \xi_0)^\vee \otimes \varepsilon. \tag{13.39}$$

13.10.2 *Why the singleton augmentation does not extend*

Let $Y = \mathbb{A}^2$, $U = Y \setminus \Delta$ and $\mathbb{J}_Y = \omega_Y[2]$. Put $H_F = \Lambda(\xi_0)$ and $P = (H_F \otimes H_F)\otimes \mathbb{J}_Y$. The relative mapping group is

$$\frac{H^{-1}R\,\mathrm{Hom}_U(P|_U, \mathbb{J}_U)}{\mathrm{im}\, H^{-1}R\,\mathrm{Hom}_Y(P, \mathbb{J}_Y)} \cong k(\xi_0\otimes\xi_0)^\vee[dz/z]. \tag{13.40}$$

This follows from $H^*_{\mathrm{dR}}(Y) = k$ and $H^*_{\mathrm{dR}}(U) = k \oplus k[dz/z]$. The connecting map is residue. If N is the external nullhomotopy used above and L a chart nullhomotopy after applying a map f to $\mathbb{J}_Y$, the fibre convention gives

$$\Delta_*\epsilon_1\,\beta = -\partial[\, j^!fN - j^!L\,].$$

Here $\epsilon_1 : H_F \to k$ is the singleton augmentation. Indeed, the iterated fibre with coordinates (σ, ζ, N) maps to the supported complex by $\zeta - \mu N$. Hence a pure overlap loop contributes minus its residue. By (13.39), the loop has residue $-\kappa^{-1}$ for every such L.

Theorem 13.22. *The transverse monoid F_κ has no augmentation to $\mathbb{J}$. Every transverse monoid receiving a monoid map from F_κ has the same obstruction, including $\mathcal{E}(C_{V_\kappa})$. These assertions persist under constant extension to any nonzero commutative k-algebra concentrated in degree zero.*

Proof. Suppose an augmentation exists. Use the roof model Z', and let f be its two-position coefficient. Its coproduct comparison supplies a homotopy H from the localized f to the external product of singleton augmentations. Its multiplicativity comparison supplies a chart homotopy M, and compatibility of that homotopy with coproduct supplies a degree--2 homotopy Γ on U. All three belong to the coherent augmentation data.

Denote by μ the two-factor product in the roof, by v', v'' the two vacuum insertions, and by s their symmetry. The difference

$$L = M(v' \otimes^{!} v'') - M(v'' \otimes^{!} v')s$$

is a chart nullhomotopy of the augmented product difference: the two products in the commutative unit cancel. Expand the coproduct-compatibility equation for M on these two insertions. Its terms involving H applied to the product difference cancel against the same terms in $f N$. The remaining loop is a chart restriction, a Γ boundary, and products of an insertion of H with a closed chart map.

Each latter term is again a chart restriction plus a boundary. On a single vacuum insertion, the possible residue of H belongs to $\mathrm{Hom}^{-2}(H_F, k)[dz/z] = 0$. The degree-zero restriction on its boundary is injective, so that boundary has a chart primitive. Subtracting it leaves a closed insertion of H whose only possible non-chart component is the vanishing residue just described. Multiplication by the other closed factor preserves exactness. Thus the loop of this particular L is zero in (13.40). This contradicts the value $-\kappa^{-1}$ computed above.

An augmentation of a receiving monoid would restrict to one of F_κ, proving the inheritance assertion. Its action gives the map to $\mathcal{E}(C_{V_\kappa})$ by Theorem 13.20. Over a degree-zero coefficient algebra $R \neq 0$, the same finite free mapping calculation replaces H_F by $R \oplus R\xi_0$ and the residue by $-\kappa^{-1}1_R \neq 0$. The vanishing unary residue and the cancellation argument remain valid. □

A module whose underlying coalgebra is $\mathbb{J}$ is precisely an augmentation: the endomorphism object of the monoidal unit is the unit. Therefore neither left nor right unit modules exist for these receiving monoids. The relative bar with both endpoints equal to this fixed unit cannot be formed. Other endpoint modules remain possible. Already $M_2(k)$ has no one-dimensional unit module, but its row and column modules have a well-defined relative tensor product. Likewise a vacuum family varying through collisions requires its own module and comparison data.

13.11 Factorizing monoid actions

We now impose factorization on the acting coalgebra, while retaining the transverse tensor and every action coherence. The resulting universal problem can fail even for an ordinary vertex algebra. The obstruction comes from a submonoid whose action is trivial although its multiplication with another submonoid leaves a nonzero binary operation.

For this section work over $\mathbb{C}$ on $X = \mathbb{A}^1$. Let Q be a unital vertex algebra in cohomological degree zero, with algebraic state space and translation T. A *vertex derivation* is a linear map $D : Q \to Q$ such that

$$D\mathbf{1} = 0, \qquad DT = TD, \qquad D(x_{(n)}y) = (Dx)_{(n)}y + x_{(n)}Dy \quad (n \in \mathbb{Z}).$$

Write C_Q for its Chevalley–Cousin factorization coalgebra on the full ordered collision carrier, and put $\mathfrak{g} = L_Q$, $\ell = L_{\mathbb{C}}$ and $\mathbb{J} = C_{\mathbb{C}}$. The singleton coefficient of $\mathfrak{g}$ is the right differential module $\omega_X \otimes Q$, in degree zero.

The category of *native monoid actions on* C_Q has connected, coaugmented, counital factorizing coalgebras as actors. Each actor has a coherent unital monoid structure for $\otimes^{!}$ and a coherent action on C_Q. Morphisms preserve the monoid and action structures, including their comparison paths. Neither an augmentation to $\mathbb{J}$ nor a propagation condition is imposed. Receiving actors may have unbounded derived coefficients. A native monoid centre would be a final object of this category.

Theorem 13.23. *If Q admits a nonzero vertex derivation, its category of native monoid actions has no final object. There are two parallel morphisms of monoid actions which no receiving factorizing monoid action can identify up to homotopy.*

We construct the actors first. This also explains why a calculation of formal coordinate maps alone would miss the obstruction.

13.11.1 Exterior actors and marked forms

For a finite-dimensional vector space W in degree zero, set

$$E_W = \Lambda(W[-1]), \qquad C_W = C_{E_W}.$$

Thus the exterior generators have degree one; their differential and translation are zero. The augmentation and unit of E_W give maps under and over $\mathbb{J}$. A natural operation between these actors means a compatible family formed from the ordinary chiral bracket, products in the exterior coefficients, and the induced Chevalley and bar maps. Only finite linear combinations are allowed in each source weight and cochain degree. Completing over source weights is harmless on a configuration of m positions: its exterior actor weight is at most $m \dim W$.

A natural map $C_W \to C_V$ has a marked matrix

$$A(t) \in \widehat{\mathrm{Sym}}(W^*) \otimes W^* \otimes V, \qquad \Phi_A(t) = A(t)t.$$

The completion is in polynomial degree. One source root is marked; exponential polarization supplies the factorials for the remaining roots. The contraction Φ_A forgets that mark. For example, $t\,ds - s\,dt$ is a nonzero marked form with zero contraction.

Lemma 13.24. *Components of the natural mapping space under and over $\mathbb{J}$ are the marked matrices $A(t)$, and each component is contractible. Composition and exterior tensor product are*

$$A_{gf}(t) = A_g(\Phi_f(t))A_f(t), \qquad A_{f\otimes g} = A_f \oplus A_g. \tag{13.41}$$

Different marked matrices remain different under geometric homotopy, although this statement does not classify all geometric maps.

Proof. Here is the natural mapping complex and its contraction. Let $\mathfrak{q}_W$ be the Lie cobar of the reduced dual coalgebra $(E_W^+)^*$, and adjoin the ordinary-input generator z of degree one, with $dz = \frac{1}{2}[z, z]$. Before taking its positive source-weight and positive target-exterior summand, the natural convolution algebra is the completed envelope of

$$\big(\mathrm{Lie}(z) \amalg \mathfrak{q}_W\big) \otimes E_V.$$

The coproduct in E_W^* gives the cobar differential. This description follows by resolving the input coalgebra: connected rooted bracket expressions give the free Lie generators, their ordered forests give its envelope, and splitting an exterior input gives the cobar differential. The relation $dz = z^2$ in the envelope accounts for the ordinary bar input. Taking a positive target-exterior factor imposes the relative condition under and over $\mathbb{J}$.

The ordinary generator can be removed without changing cohomology. In the augmentation ideal of $(\mathbb{C}\langle z\rangle, dz = z^2)$, use

$$h(z^{2j}) = z^{2j-1}, \qquad h(z^{2j-1}) = 0 \quad (j \geq 1).$$

The graded Leibniz rule gives $dh + hd = 1$. In the free product with $U(\mathfrak{q}_W)$, apply this contraction to the first ordinary factor, with the sign of its prefix. PBW symmetrization then proves the same assertion on the Lie algebra, after tensoring with E_V, and after taking its envelope.

The projection $\mathfrak{q}_W \to W^*$ kills brackets and nonsingleton exterior labels and has abelian target. To check it, write $E_W = \Lambda(\alpha_1, \dots, \alpha_r)$ and resolve the augmentation module by

$$E_W \otimes \mathbb{C}[v_1, \dots, v_r], \qquad |v_i| = 0, \quad dv_i = \alpha_i.$$

At positive combined weight N, the map $N^{-1}\sum_i v_i\partial_{\alpha_i}$ contracts this complex. Its dual derived Hom is concentrated in degree zero and is $\mathrm{Sym}(W^*)$. The bar resolution computes the same Hom; dualizing each finite weight identifies it with $U(\mathfrak{q}_W)$. In degree zero, its boundaries are exactly the commutators of the singleton generators. PBW gives the asserted Lie quasi-isomorphism.

Consequently the natural convolution algebra projects surjectively, with weightwise acyclic kernel, to

$$\mathcal{B}(W,V) = \Big[\widehat{\mathrm{Sym}}(W^*\otimes E_V)\Big]_{E_V^+}.$$

The subscript denotes the ideal generated by $W^*\otimes E_V^+$. It has zero differential and no terms in degrees at most zero. Its degree-one part is precisely the space of marked matrices.

Lift the Maurer–Cartan equation one source weight at a time. Its first curvature is a closed degree-two kernel element, removed by a degree-one primitive. The first difference of two lifts is removed by a degree-zero primitive. Tensoring the same acyclic kernel with polynomial forms relative to the boundary of a simplex supplies all parameterized lifts. Thus every fibre is contractible. In $\mathcal{B}$ a degree-one Maurer–Cartan simplex is constant: its square is zero by graded commutativity, and its parameter differential vanishes.

To compute composition, keep independent ordinary root variables and project their brackets to zero. The marked source variable has one continuation through a composite, giving matrix multiplication; the unmarked variables substitute by Φ_f. This gives (13.41). Disjoint coefficient blocks give the tensor formula. The same top-root coefficient is unchanged by a geometric homotopy: after the ordinary variables become independent, its possible cobar boundary has zero top coefficient. It therefore distinguishes unequal matrices without asserting that every geometric map is natural. □

The preceding contractions involve no infinite sum of states or normal derivatives. At fixed source weight and cochain degree the number of forest slots is bounded; at a fixed configuration only finitely many source weights occur. These are the finiteness conditions used below.

13.11.2 A monoid with a trivially acting direction

Let $A = \mathbb{C}a \oplus \mathbb{C}b$ be the auxiliary commutative algebra

$$a^2 = a, \qquad ab = ba = b, \qquad b^2 = 0.$$

Its unit a is distinct from the vacuum of the exterior vertex algebra. Put

$$F = C_A, \qquad G = C_{\mathbb{C}a}, \qquad H = C_{\mathbb{C}b}.$$

For $x \in A$, let L_x be multiplication by x, and set

$$Q(x) = e^{L_x}, \qquad M(x) = \sum_{n\ge 0} \frac{L_x^n}{(n+1)!}.$$

The inverse of M exists as a formal series. Define the multiplication of F by the two marked blocks

$$\big(M(x+y)^{-1}M(x) \mid M(x+y)^{-1}Q(x)M(y)\big). \tag{13.42}$$

Lemma 13.25. *This is a coherent unital transverse monoid under and over $\mathbb{J}$. The inclusions $i : G \to F$ and $j : H \to F$ are monoid maps. The multiplication of H is additive, and the coordinate contraction of the multiplication of F is $x + y$.*

Proof. In the affine group with law

$$(x,U)(y,V) = (x+y, U + Q(x)V),$$

associativity follows from $Q(x+y)=Q(x)Q(y)$. The change of fibre coordinates $U=M(x)u$ gives (13.42), so its three marked associativity identities and its unit identities hold. In the algebra A,

$$M(x)x=e^{x}-a, \qquad e^{x+y}-a=(e^{x}-a)+e^{x}(e^{y}-a).$$

Hence the diagonal $u=x$ is a homomorphism, which proves the coordinate contraction. Both lines are stable under these operators. On the b-line, $L_{sb}b=0$, so both multiplication blocks are the identity. Lemma 13.24 realizes the identities and all their coherences. Its contractions preserve the auxiliary grading $\deg a=0$, $\deg b=1$, as does the multiplication. □

Lemma 13.26. *For a vertex derivation D of Q, the monoid F has a coherent action on C_Q whose restriction to G is the trivial monoid action and whose restriction to H is the single-derivation action. Both inclusions preserve these actions with their specified comparisons.*

Proof. First construct the operations with one formal derivation. For r copies of F, let t_ν, s_ν be the coordinates dual to a, b. Give D auxiliary weight one, each s_ν weight minus one and each t_ν weight zero. Prescribe the entire restriction with no b-input to be trivial.

The scalar-bar and Koszul contractions in the preceding proof apply also to differential polynomials in D. They give the commutative relative model

$$[(s_1,\dots,s_r)\mathbb{C}[[t_1,s_1,\dots,t_r,s_r]]\otimes\Lambda(\mathbb{C}D[-1])]_{\text{weight }0}. \tag{13.43}$$

The odd generator in this formula is the suspended derivation, not the degree-zero endomorphism of Q. The differential is zero. There are no nonpositive degrees: a scalar monomial with a positive s-count has negative auxiliary weight. The degree-one part is exactly

$$\sum_{\nu=1}^{r} s_\nu f_\nu(t_1,\dots,t_r)D.$$

The projections to this model have the same weightwise acyclic kernels as before.

Deleting an actor or deleting all its a-generators commutes with the projection deleting the b-generators. Their inclusions split the projections. Inclusion–exclusion therefore extends compatible data on the union of these faces; the remaining kernel is a sum of multigraded acyclic kernels. The lifting argument works relative to the unit faces, the trivial G-restriction and a prescribed H-action.

Choose the class sD for one copy. Pullback along the monoid product is $(s_1+s_2)D$, since its coordinate contraction is addition. Composition has the same class: the formal operator identity $e^{s_1D}e^{s_2D}=e^{(s_1+s_2)D}$ involves one derivation. Relative contractibility supplies the comparison path. Every higher module associahedron has its boundary in the component with class $(s_1+\cdots+s_r)D$. The same relative lifting fills it, including all parameterized comparisons.

To evaluate these operations on the whole collision carrier, the functor $\mathcal{T}_Q(B)=B\otimes^{!}C_Q$ is strong symmetric monoidal for $\star$. On each ordered disjoint decomposition its comparison is the factorization isomorphism of C_Q followed by open projection. Iteration gives the same ordered partition and coproduct, and hence preserves collisions and normal transfers. On the main diagonal it sends ℓ to $\mathfrak{g}$ with its bracket. In particular

$$C_W\otimes^{!}C_Q\simeq \mathrm{CE}_\star(\mathfrak{g}\otimes E_W).$$

These comparisons commute with envelopes and bars. The vertex automorphism $1+uD$ over $\mathbb{C}[u]/(u^2)$ carries D to $D\otimes 1$ through them, so the differential polynomials and their homotopies evaluate to the required action. Exterior degree bounds the total actor weight; derivative order equals its b-count. Every evaluated operation is finite on its specified input. □

The requirement that the whole G-action be trivial is used here. The zero auxiliary degree of a alone would not remove scalar homotopies.

13.11.3 A commutator with a nonzero marked form

Let $S = C_{\mathbb{C}e_1 \oplus \mathbb{C}e_2}$. Map its exterior generators into four copies of F by

$$e_1 \longmapsto a^{(1)} - a^{(3)}, \qquad e_2 \longmapsto b^{(2)} - b^{(4)},$$

and multiply in the displayed order. Denote the resulting pointed coalgebra map by $w : S \to F$. This map need not preserve a monoid structure on S.

Lemma 13.27. *There is a natural pointed map $f : S \to H$ with $w \simeq jf$. Its marked matrix is*

$$A_f(t,s) = \big(-s\chi(t) \mid t\chi(t)\big), \qquad \chi(t) = \sum_{n \geq 0} \frac{t^n}{(n+1)!}. \tag{13.44}$$

Its coordinate contraction and singleton components are zero, but its first marked form is $t\,ds - s\,dt$.

Proof. Compute the fourfold product in the affine coordinates of Lemma 13.25, with base points ta, sb, $-ta$, $-sb$. Their sum is zero, so the final change of frame is the identity. If dt, ds denote the marked inputs, the fibre coordinate is

$$\begin{aligned} &M(ta)a\,dt + Q(ta)M(sb)b\,ds \\ &\quad - Q(ta+sb)M(-ta)a\,dt - Q(sb)M(-sb)b\,ds. \end{aligned}$$

The two a-components cancel. Using $b^2 = 0$, the remaining component is $b(-s\chi(t)\,dt + (e^t - 1)\,ds)$. This is (13.44) in the b-line. The natural classification supplies the map and the homotopy. The other assertions follow by contraction and by taking the lowest polynomial degree. □

13.11.4 Detection in an arbitrary receiving actor

Fix $D \neq 0$. Let $h : H \to Z$ be a receiving pointed map of actions, and put $B = \Omega Z$. Write $\iota : \ell \to B$ for its vacuum pointing and h_0 for the pure singleton component of h.

Lemma 13.28. *On every nonempty affine open $Y \subset X$,*

$$[h_0]|_Y \neq 0 \quad \text{in} \quad H^1 R\,\mathrm{Hom}_Y(\ell|_Y, B_{\{1\}}|_Y).$$

No boundedness or holonomicity assumption on B is needed.

Proof. We first examine a plain degree-zero coefficient class $a : \ell|_Y \to B_{\{1\}}|_Y$. Its action gives an endomorphism M of $\mathfrak{g}|_Y$. We claim

$$M\mu_{\mathfrak{g}} = \mu_{\mathfrak{g}}(1 \star M) = \mu_{\mathfrak{g}}(M \star 1). \tag{13.45}$$

Both ordered cobar products of B are equivalences on distinct roots. Indeed, the quotient by one product summand is the cone of the other factorization coproduct. Subtract the open exchange part of $[\iota, a]$, with its nullhomotopy. Its image in $U_\star\mathfrak{g}$ is zero there, because the commutator in this envelope is the supported chiral bracket.

The remaining map factors through

$$\mathrm{cofib}\big(j_! j^*(\ell \boxtimes \ell) \longrightarrow j_* j^*(\ell \boxtimes \ell)\big),$$

where j is the distinct-root inclusion. The source is regular holonomic, so this $j_!$ is defined. On either diagonal branch the cofiber has a two-step filtration with successive terms $\ell[1]$ and ℓ. Tensoring by C_Q replaces them by $\mathfrak{g}[1]$ and $\mathfrak{g}$. The PBW envelope is in the right-module heart and its extraordinary diagonal restriction is $\mathfrak{g}$. Negative Ext kills the first term. The remaining action has the form $N\mu_{\mathfrak{g}}$ and also $\mu_{\mathfrak{g}}(1 \star M)$. Inserting the vacuum in the first slot gives $N = M$: the possible kernel of the principal-part vacuum precomposition is again a negative Ext between heart objects. Skewsymmetry gives the other equality in (13.45).

Suppose now that $[h_0]|_Y = 0$. A relative gauge in the resolved pointed mapping complex removes h_0, fixing $\mathbb{J}$. Its first actor-weight term is then a degree-zero cocycle on one ordinary input and one actor input. Choose $q \in Y$, put the actor at q, and vary the ordinary input on $Y' = Y \setminus \{q\}$.

The finite Spencer resolution of ℓ commutes with tensoring by an arbitrary coefficient complex. Algebraic Künneth therefore expresses this operation as a finite sum of bidegrees $(r, -r)$. After the action, the second factor lies in $H^{-r} R\,\mathrm{Hom}_{\mathbb{C}}(Q, Q)$, which vanishes unless $r = 0$. The action path gives

$$D \otimes 1_Q = \sum_{\nu} M_\nu \otimes N_\nu,$$

where every M_ν satisfies (13.45). This equality holds for the two-output classes: the possible zero-column boundary vanishes by open adjunction, $R\,\mathrm{Hom}(\mathfrak{g}, \mathrm{Sym}^2_\star \mathfrak{g}) = 0$.

Choose a linear functional on the finite span of $1, N_1, \dots, N_m$ taking 1 to 1. It expresses $D|_{Y'}$ as a linear combination of the M_ν. Thus D is both a module map and a derivation. Subtracting the derivation identity from (13.45) and inserting the vacuum in the second slot gives $D = 0$. Localization is faithful on the free algebraic state module $O_X \otimes Q$. This contradicts the choice of D. □

Lemma 13.29. *The composite $hf : S \to Z$ is not homotopic under $\mathbb{J}$ to the map through $\mathbb{J}$.*

Proof. On the distinct-root slice $(X \setminus \{q\}) \times \{q\}$, its first binary class is

$$[\iota]|_{X\setminus\{q\}} \otimes [h_0]_q - [h_0]|_{X\setminus\{q\}} \otimes [\iota]_q. \tag{13.46}$$

The normalized fibre at q is $i_q^! B[1]$. The two terms have Künneth degrees $(0, 1)$ and $(1, 0)$. The second is nonzero by the preceding lemma and the unit identity: $[\iota]_q$ acts as 1_Q. Changing the cobar product order adds a cobar boundary, so centrality of ι is not being assumed.

We verify invariance under an arbitrary pointed geometric homotopy. Represent it in the full coefficient resolution by $\tau(u) + c(u)\,du$, where τ and c have degrees one and zero. Both endpoint maps have zero pure singleton components. On those inputs the interval equation gives

$$\tau_{1,0} = \partial E, \qquad \tau_{0,1} = \partial F, \qquad E(u) = \int_0^u c_{1,0}(v)\,dv, \quad F(u) = \int_0^u c_{0,1}(v)\,dv.$$

Here $E(0) = F(0) = 0$ and $\partial E(1) = \partial F(1) = 0$. Restriction to distinct roots kills the supported source bracket. The binary equation becomes

$$\tau'_{1,1} = \partial c_{1,1} + [\partial E, F'] + [\partial F, E'].$$

The brackets include the labelled external products and the target multiplication. Since E, F have degree zero, the graded Leibniz rule gives

$$[\partial E, F'] + [\partial F, E'] = \partial[E, F'] - \frac{d}{du}[E, \partial F].$$

Integration makes the difference of the endpoint binary components a boundary; the endpoint term $[E, \partial F]|_0^1$ is zero. Their classes are therefore equal under any such homotopy. The map through $\mathbb{J}$ has zero binary class, whereas (13.46) is nonzero. □

Proof of Theorem 13.23. Let $i_0 : G \to \mathbb{J} \to F$ be augmentation followed by the unit. The relative action construction gives two morphisms

$$(G, \mathrm{triv}) \rightrightarrows (F, \alpha), \qquad i,\ i_0.$$

If a receiving monoid-action map $k : F \to Z$ identified ki and ki_0, its multiplication coherences would send the four-factor commutator w through $\mathbb{J}$: the first and third factors become units, and the remaining H-factors cancel. But $w \simeq jf$, and the map $h = kj$ is a receiving pointed action map. Lemma 13.29 forbids this homotopy. A final object would identify the parallel maps after their map into it, so no final object exists. □

For the Heisenberg vertex algebra, translation itself is a nonzero vertex derivation, since $TJ \neq 0$. Thus the theorem applies to its ordinary factorization coalgebra. The acting monoid of Proposition 13.21 and the ambient universal monoid of Theorem 13.20 remain valid constructions. The latter is not required to factorize. The parallel-pair obstruction also differs from a claim that two objects have no common recipient; no such claim is used or deduced here.

A physical bulk reconstruction must consequently specify its actor class, action tensor, local vacuum and morphisms independently. Restricting that class may change the universal problem, but then the restriction and the comparison to physical observables are additional mathematical data.

13.12 Supported comparison equations

The same open–closed decomposition decides whether a map on the separated locus extends through a collision. Let $j : U \hookrightarrow X$ be an open embedding, $i : Z \hookrightarrow X$ its closed complement, and work in a stable coefficient category with recollement

$$i_* i^! Q \longrightarrow Q \longrightarrow R j_* j^* Q \longrightarrow i_* i^! Q[1].$$

For a source P, a map $f_U : j^* P \to j^* Q$ has a connecting class

$$\mathrm{ob}(f_U) \in H^1 R \operatorname{Hom}_X(P, i_* i^! Q).$$

Proposition 13.30. *The map f_U extends to $P \to Q$, with a specified homotopy on restriction, exactly when* $\mathrm{ob}(f_U) = 0$. *After choosing one extension, the space of extensions is identified with* $\operatorname{Map}_X(P, i_* i^! Q)$.

Proof. Apply $R \operatorname{Hom}_X(P, -)$ to the recollement triangle. Open adjunction identifies the third term with $R \operatorname{Hom}_U(j^* P, j^* Q)$. Exactness gives the obstruction and the lift. Translation by a chosen lift identifies the homotopy fiber over f_U with the fiber over zero. This retains the restriction homotopy and all higher homotopies of an extension. □

Extension is only the first equation for a comparison preserving composition. To specify the higher equations, let A, B be augmented homotopy associative algebras over k, with augmentation ideals $\overline{\mathsf{A}}$, $\overline{\mathsf{B}}$.

Their bar coderivations b^{A}, b^{B} on the tensor coalgebras of the suspended ideals have degree one and square zero. A coaugmentation-preserving coalgebra map F of degree zero has Taylor components

$$f_n : (\overline{\mathsf{A}}[1])^{\otimes n} \longrightarrow \overline{\mathsf{B}}[1], \qquad n \geq 1.$$

All tensor substitutions use the Koszul signs on these suspended complexes. Suppose the allowed derived Hom complexes are closed under these substitutions, and let ∂ be their differential induced by b^{A}_1 and b^{B}_1. Projection of $b^{\mathsf{B}} F = F b^{\mathsf{A}}$ to one output gives

$$\begin{aligned} \partial f_n = &\sum_{\substack{r+s+t=n \\ s \geq 2}} f_{r+1+t}(1^{\otimes r} \otimes b^{\mathsf{A}}_s \otimes 1^{\otimes t}) \\ &- \sum_{\substack{\ell \geq 2 \\ i_1 + \cdots + i_\ell = n}} b^{\mathsf{B}}_\ell (f_{i_1} \otimes \cdots \otimes f_{i_\ell}). \end{aligned} \tag{13.47}$$

The first sum inserts one source operation. The second runs over every ordered partition of the inputs into blocks. In particular, at four inputs it contains $-b^{\mathsf{B}}_2(f_2 \otimes f_2)$, even when the binary defect has a primitive.

For an oriented Hall-to-field comparison, the endpoints, orientations, supported mapping complexes and both unit markings must first be specified. The regular generator then reconstructs their maps. It neither chooses those operations nor supplies the primitives required by (13.47).

Chapter 14

Quantization and supported transfer

A classical contraction identifies a field complex with a smaller complex. Quantization asks for more: the interaction, the measure, and the collision operations must descend with their identities. Three explicit constructions will distinguish these requirements. On $K3 \times E$, harmonic forms give a strict model of the classical holomorphic Chern–Simons Lie algebra, while its full residual action retains two cubic terms. In a nodal field algebra, the BRST differential retains both the coordinate relation and its momentum fields. In a rational current algebra, a completed product and its curved differential can be constructed through specified powers of the deformation parameter. Each construction has its own carrier and its own comparison problem.

The final example concerns support. Dividing by a normal coordinate on principal parts has a right inverse with a nonzero contact term. That term survives in cohomology and prevents a particular division homotopy from becoming module-linear. Its explicit representative makes it possible to state the remaining relative comparison equation, including the normal derivatives and the vacuum.

14.1 The full harmonic Chern–Simons model

Let $X = S \times E$, where S is a compact K3 surface and E is an elliptic curve. Fix a product Ricci-flat Kähler metric, a nowhere vanishing holomorphic three-form Ω, and a finite-dimensional complex Lie algebra $\mathfrak{g}$ with a nondegenerate invariant symmetric form κ. We work about the trivial gauge background. The differential graded Lie algebra and its cyclic pairing are

$$L = \mathfrak{g} \otimes \Omega^{0,\bullet}(X), \qquad d = \bar{\partial}, \qquad \pi(x\alpha, y\beta) = \int_X \Omega \wedge \alpha \wedge \beta\, \kappa(x, y). \tag{14.1}$$

The pairing has degree -3. The shifted space of BV fields is $L[1]$; in particular it includes all four Dolbeault degrees. Its odd symplectic pairing is

$$\sigma(sx, sy) = (-1)^{|x|}\pi(x, y), \qquad |s| = -1.$$

The restriction to Dolbeault degree one is the classical physical field space. The full BV extension also contains ghosts and antifields [32, §1.5].

Choose parallel harmonic representatives ϵ and u of the elliptic $(0,1)$-class and the K3 $(0,2)$-class. Normalize them so that $\int_X \Omega \wedge \epsilon u = 1$. Their algebra is

$$H = \mathbb{C}[u]/(u^2) \otimes \Lambda(\epsilon), \qquad |u| = 2, \quad |\epsilon| = 1. \tag{14.2}$$

Write $\tau(\epsilon u) = 1$, with τ zero in the other degrees.

Proposition 14.1. *The inclusion $\mathfrak{g} \otimes H \longrightarrow L$ is a cyclic dg Lie quasi-isomorphism. Homotopy transfer along the product Hodge contraction has no operations of arity at least three on $\mathfrak{g} \otimes H$. Its binary bracket is*

$$[x \otimes v, y \otimes w] = [x, y] \otimes vw.$$

Proof. The Hodge dimensions of S and E, together with Künneth, give the four classes $1, \epsilon, u, \epsilon u$. Their chosen representatives are parallel and closed under multiplication, so their inclusion is a multiplicative quasi-isomorphism. The trace normalization identifies its pairing with $\kappa \otimes \tau$.

The Hodge homotopy kills harmonic forms. Every rooted binary tree with at least three leaves has an internal edge immediately above a vertex whose two inputs are harmonic. Their bracket is harmonic, and the homotopy on that edge kills it. Thus every higher transferred tree is zero. The binary tree has no such internal edge and gives the displayed bracket. □

The strictness of this Lie model does not remove its cubic Hamiltonian. Write the universal unshifted field, with its coordinate coefficients on the right, as

$$\omega = 1c + \epsilon a + ub + \epsilon u d.$$

The coordinate degrees are $|c| = 1$, $|a| = 0$, $|b| = -1$, and $|d| = -2$. Consequently c, b are odd and a, d are even. The residual action is the degree-zero function

$$S_H = \frac{1}{6}\tau\kappa(\omega, [\omega, \omega]) = \frac{1}{2}\kappa(d, [c, c]) - \kappa(c, [a, b]). \tag{14.3}$$

To compute it, a nonzero trace must have form degree three. The possible patterns are $(0, 0, 3)$, $(0, 1, 2)$, and $(1, 1, 1)$. The last vanishes because $\epsilon^2 = 0$. Moving the right-hand coordinate coefficients past the forms in the first two patterns gives exactly the two signs in (14.3).

For example, take $\mathfrak{g} = \mathfrak{sl}_2$ with $e = E_{12}$, $f = E_{21}$, $h = \operatorname{diag}(1, -1)$ and $\kappa(x, y) = \operatorname{tr}(xy)$. Since $[e, f] = h$, $\kappa(h, h) = 2$, and $\kappa(e, f) = 1$, the coefficients of $c_h a_e b_f$ and $c_e c_f d_h$ in S_H are respectively -2 and 2. The action vanishes on the restriction $c = 0$, but it is nonzero on the full residual field space.

Proposition 14.2. *Use left graded derivatives and the constant-density BV Laplacian*

$$\Delta = \sum_{i,j} \kappa^{ij} \left(\frac{\partial}{\partial c_i} \frac{\partial}{\partial d_j} + \frac{\partial}{\partial a_i} \frac{\partial}{\partial b_j} \right).$$

For the bracket $\{-, -\}$ generated by Δ, one has $\{S_H, S_H\} = 0$ and $\Delta S_H = 0$. The Hamiltonian differential is

$$Qc = \tfrac{1}{2}[c, c], \qquad Qa = [c, a], \qquad Qb = [c, b], \qquad Qd = [c, d] - [a, b]. \tag{14.4}$$

Thus S_H solves the finite algebraic quantum master equation.

Proof. Expand $d_{\mathrm{tot}}\omega = \frac{1}{2}[\omega, \omega]$, where $d_{\mathrm{tot}} = 1_H \otimes Q$ uses the tensor sign. Its four coefficients give (14.4). Invariance of κ identifies this differential with the Hamiltonian derivation of S_H. The graded Jacobi identity gives $Q^2 = 0$ and the classical master equation. A metric Lie algebra in characteristic zero is unimodular: each adjoint endomorphism is skew with respect to its nondegenerate symmetric form, hence has trace zero. The contractions in ΔS_H are precisely these adjoint traces, so they vanish. □

This finite cyclic construction is the algebraic residual mechanism of [26, §3.1]. Its differential on H is zero, so the quadratic kinetic term restricts to zero. The nondegenerate odd symplectic form supplies the BV bracket, not a nondegenerate quadratic action. An effective quantum action for the analytic theory further requires a regulated integration over nonzero modes, its local counterterms, and a choice of residual BV integration. Neither the finite master equation nor $\chi(S \times E) = 0$ computes those data.

14.2 A product propagator with one time variable

Let A and B be closed Hermitian manifolds. On their Dolbeault complexes write $d_A = \bar\partial_A$, $q_A = d_A^*$, $\Delta_A = d_A q_A + q_A d_A$, and $\Gamma_A = (-1)^{\deg}$, and use analogous notation for B. All adjoints refer to the specified metrics. On the product complex,

$$\begin{aligned} d &= d_A \otimes 1 + \Gamma_A \otimes d_B, \\ q &= q_A \otimes 1 + \Gamma_A \otimes q_B, \\ \Delta &= \Delta_A \otimes 1 + 1 \otimes \Delta_B. \end{aligned} \tag{14.5}$$

The last identity follows because Γ_A anticommutes with both odd operators. If $K_A(t) = e^{-t\Delta_A}$, then $K(t) = K_A(t) \otimes K_B(t)$.

Proposition 14.3. *For $0 < \varepsilon < L$, the product Hodge propagator is*

$$P_{[\varepsilon,L]} = \int_\varepsilon^L \big(q_A K_A(t) \otimes K_B(t) + \Gamma_A K_A(t) \otimes q_B K_B(t)\big)\, dt. \tag{14.6}$$

It satisfies

$$dP_{[\varepsilon,L]} + P_{[\varepsilon,L]} d = K(\varepsilon) - K(L).$$

Let H_A, H_B be the orthogonal harmonic projections, put $H = H_A \otimes H_B$, and let G be the inverse of Δ on $H^\perp$, extended by zero on H. Then $P = qG$ satisfies $dP + Pd = 1 - H$. If $\lambda_1 > 0$ is the first positive eigenvalue of Δ, then

$$\|P - P_{[\varepsilon,L]}\|_{L^2 \to L^2} \le \sqrt{\varepsilon} + \frac{e^{-L\lambda_1}}{\sqrt{\lambda_1}}. \tag{14.7}$$

Proof. The integrand is $qK(t)$. Since $dq + qd = \Delta$, its anticommutator with d is $-\partial_t K(t)$, proving the finite-cutoff identity. On a positive λ-eigenspace, $\|q\| \le \sqrt{\lambda}$. The omitted short-time integral is bounded there by $(1 - e^{-\varepsilon\lambda})/\sqrt{\lambda} \le \sqrt{\varepsilon}$, using $1 - e^{-x} \le \min(x, 1)$. The long-time tail is bounded by $e^{-L\lambda}/\sqrt{\lambda}$. Taking the supremum proves (14.7). On H, $qH = 0$, so there is no term growing linearly with L. The spectral theorem gives the limit qG, and elliptic regularity gives its action on smooth forms. □

There is only one time integral in (14.6). On a joint eigenspace with eigenvalues $\lambda, \mu > 0$, its Green denominator is $\lambda + \mu$. Multiplying two separate Green operators would instead give $\lambda\mu$. The mixed harmonic sectors also remain:

$$P|_{\operatorname{im} H_A \otimes \ker H_B} = \Gamma_A H_A \otimes P_B, \qquad P|_{\ker H_A \otimes \operatorname{im} H_B} = P_A \otimes H_B.$$

The sector harmonic in both factors contributes zero. If h_A is an even harmonic vector and $d_B c = bd$ with $b \neq 0$, then $P(h_A \otimes d) = b^{-1} h_A \otimes c$. The sequential contraction $P_A \otimes 1 + \Gamma_A H_A \otimes P_B$ is another valid contraction; it is a different choice from the product-metric Hodge contraction.

On $E = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$, write $z = u + \tau v$, $\tau = a + ib$, $b > 0$, and give the metric $g = (V/b)|dz|^2$ area V. In the convention $\Delta_{\bar\partial} = \Delta_{\text{real}}/2$, the characters and eigenvalues are

$$e_{mn}(u, v) = e^{2\pi i(mu+nv)}, \qquad \lambda_{mn} = \frac{2\pi^2 |m\tau - n|^2}{Vb}.$$

The scalar heat kernel is therefore

$$K_E(t; z, z') = \frac{1}{V} \sum_{m,n \in \mathbb{Z}} e^{-t\lambda_{mn}} e_{mn}(z) \overline{e_{mn}(z')}. \tag{14.8}$$

Every summand is periodic in both variables. Parallel orthonormal form frames give the degree-one formula. These conventions agree with the Dolbeault Hodge identities in [44, Ch. VI, §5].

For fixed positive cutoffs the propagator kernel is smooth even on the diagonal. Its limiting singularity has a different order from the scalar Green kernel. In real dimension six the positive flat Laplacian has

$$G(r) = \frac{1}{4\pi^3 r^4}, \qquad G'(r) = -\frac{1}{\pi^3 r^5}.$$

The first-order adjoint thus produces leading coefficients of order r^{-5} in fixed orthonormal frames. Directly,

$$\int_0^\infty r t^{-4} e^{-r^2/(4t)}\, dt = \frac{128}{r^5}, \qquad \int_\varepsilon^L r t^{-4} e^{-r^2/(4t)}\, dt \le \frac{r}{3}(\varepsilon^{-3} - L^{-3}).$$

The first expression is locally integrable in six dimensions; the second has no short-distance pole. These linear identities underlie the heat-kernel construction of [31, §§7.2.3–7.2.5]. An interacting master equation requires the corresponding local functional calculation.

14.3 Cyclic transfer and a nonformal minimal algebra

Let (A, d, m_2) be a dg algebra with a cyclic pairing, and let $(H, 0) \underset{p}{\overset{i}{\rightleftarrows}} (A, d)$ be a contraction with a homotopy h of degree -1:

$$pi = 1, \qquad dh + hd = 1 - ip, \qquad h^2 = hi = ph = 0.$$

A cyclic transfer also requires p, i, h to be compatible with the pairing in the shifted sign convention. The transferred operations are the signed planar-tree compositions of i, m_2, h, p: leaves carry i, vertices m_2, internal edges h, and the root p. This gives $|m_n| = 2 - n$. It preserves cyclicity when the contraction is cyclic [83]. A linear contraction without these pairing conditions does not by itself supply cyclic transfer.

We use the following unsuspended convention throughout this section:

$$\sum_{r+s+t=n} (-1)^{r+st+(2-s)(|x_1|+\cdots+|x_r|)} m_{r+1+t}(x_1, \dots, x_r, m_s(x_{r+1}, \dots, x_{r+s}), x_{r+s+1}, \dots, x_n) = 0. \tag{14.9}$$

For a degree--3 pairing, cyclicity means

$$\begin{aligned} \langle m_n(x_1, \dots, x_n), x_{n+1} \rangle = (-1)^{n+|x_1|(|x_2|+\cdots+|x_{n+1}|)} \\ \cdot \langle m_n(x_2, \dots, x_{n+1}), x_1 \rangle. \end{aligned} \tag{14.10}$$

Both sides place the operation in the first pairing slot. The shift, the argument order, and the sign must be transported together when using another convention [27, §2].

Proposition 14.4. *Over a characteristic-zero field, let A_λ have basis $1, a, b, v$ in degrees $0, 1, 2, 3$. Give it a strict unit 1, the nonzero products*

$$m_2(a, b) = m_2(b, a) = v, \qquad m_3(a, a, a) = \lambda b,$$

and no other products on nonunit inputs. Give it the pairing

$$\langle 1, v \rangle = \langle v, 1 \rangle = \langle a, b \rangle = \langle b, a \rangle = 1,$$

zero on the other basis pairs. This is a minimal cyclic A_∞-algebra of dimension three. If $\lambda \neq 0$, it is nonformal and has both a nonzero ternary product and a nonzero binary product on its augmentation ideal.

Proof. The binary product is associative: three elements in the augmentation ideal always multiply to zero, and the remaining cases use the unit. The pairing is the Frobenius pairing $\tau(xy)$ with $\tau(v) = 1$. It is nondegenerate, graded symmetric, and of degree -3. The only nonzero ternary cyclic tensor is the value on four copies of a. Its rotation sign in (14.10) is $(-1)^{3+3} = 1$.

At arity four the only nonunit input that contributes is (a, a, a, a). Equation (14.9) becomes

$$-m_2(m_3(a,a,a),a) + m_2(a, m_3(a,a,a)) = -\lambda v + \lambda v = 0.$$

The inner binary terms vanish because $a^2 = 0$. With one unit the nonzero terms cancel in pairs; with more units there is no contribution. At arity five every composite of two ternary products vanishes, because the inner output is b. No higher arity can occur in a composite of the available operations. This proves all the identities.

If f_2 has degree -1, then $f_2(a, a)$ is a scalar multiple of a. Every term of its Hochschild coboundary on (a, a, a) is consequently zero: the inner or outer binary product contains a^2. Thus the value $m_3(a, a, a) = \lambda b$ cannot be removed by a quadratic change of coordinates. An invertible graded linear change only rescales the one-dimensional a and b spaces. This proves nonformality. □

For the degree-one element $x = ta$, the cyclic potential is

$$\Phi(x) = \sum_{n \geq 2} \frac{1}{n+1} \langle m_n(x, \dots, x), x \rangle = \frac{\lambda}{4} t^4.$$

The quartic term records the ternary operation. It does not force the binary product to vanish: cyclicity with the unit gives $\langle m_2(a, b), 1\rangle = \langle a, b\rangle = 1$. This finite example makes no assertion of homological smoothness or of a geometric realization by a compact Calabi–Yau category. Those are additional properties of a specified model.

14.3.1 *Compact conifold objects and their ternary product*

The conifold gives a geometric example in which the cubic relations determine a nonzero ternary Ext product. Put

$$C = \mathbb{C}[X, Y, U, V]/(XY - UV), \qquad I = (X, U), \qquad A = \operatorname{End}_C(C \oplus I).$$

Use arrows $a, c : 1 \to 0$ and $b, d : 0 \to 1$, with composition written right to left, and the potential $\Phi = abcd - adcb$. The arrow matrices on $C \oplus I$ are

$$a = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 0 & 0 \\ X & 0 \end{pmatrix}, \quad c = \begin{pmatrix} 0 & V/X \\ 0 & 0 \end{pmatrix}, \quad d = \begin{pmatrix} 0 & 0 \\ U & 0 \end{pmatrix}.$$

Here V/X is a regular homomorphism $I \to C$, since it sends X to V and U to Y. These matrices identify A with the Jacobi algebra of Φ. One way to verify the presentation is to write $z = a + b$, $p = c + d$, $g = e_0 - e_1$, with e_i the vertex idempotents, and $\lambda = (cb - ad) + (bc - da)$. The cyclic derivatives say that λ is central and give

$$g^2 = 1, \quad gz = -zg, \quad gp = -pg, \quad pz - zp = \lambda g.$$

The normal monomials $z^i p^j g^\varepsilon$ form a $\mathbb{C}[\lambda]$-basis: the only mixed ambiguity gpz has the same reduction $zpg + \lambda$ in both orders. The central elements $X = z^2$, $Y = p^2$, $W = zp + \lambda g/2$, $U = W - \lambda/2$, $V = W + \lambda/2$ satisfy $XY = UV$. The matrix map is surjective and becomes an isomorphism over $\operatorname{Frac}(C)$ on the basis $1, z, g, zg$. Filter the source by the number of z, p factors, giving λ weight zero. Its associated graded is $\mathbb{C}[\lambda, z, p] \rtimes \mu_2$; the central graded ring $\mathbb{C}[\lambda, X, Y, W]/(XY - W^2)$ embeds by

$X = z^2, Y = p^2, W = zp$ and acts without torsion. The filtered source is therefore torsion-free over C, so the kernel is zero.

Let S_i be the right vertex simples and $P_i = e_i A$. Give the arrows weight one; in $P_i(-w)$ the free generator has internal weight w. The compact projective resolutions are

$$\begin{aligned} 0 \to P_0(-4) &\xrightarrow{\binom{b}{d}} P_1(-3)^2 \xrightarrow{\left(\begin{smallmatrix} -dc & bc \\ da & -ba \end{smallmatrix}\right)} P_1(-1)^2 \xrightarrow{(a\ c)} P_0 \to S_0 \to 0, \\ 0 \to P_1(-4) &\xrightarrow{\binom{a}{c}} P_0(-3)^2 \xrightarrow{\left(\begin{smallmatrix} cd & -ad \\ -cb & ab \end{smallmatrix}\right)} P_0(-1)^2 \xrightarrow{(b\ d)} P_1 \to S_1 \to 0. \end{aligned} \tag{14.11}$$

The composites vanish by the four cyclic derivatives. For exactness, the first middle matrix in the fractional-ideal model is $\left(\begin{smallmatrix} -Y & V \\ U & -X \end{smallmatrix}\right)$. Its kernel is (Xh, Uh), with h in C or $I^\vee$ according to the relevant component. At the next place the required colon identities are

$$\{n \in C : Vn \in XC\} = I, \qquad \{n \in I : Vn \in XI\} = I^2.$$

The first follows by reducing modulo X, where $C/(X) = \mathbb{C}[Y, U, V]/(UV)$. For the second, write $n = X\alpha + U\beta$ and reduce the condition $V\alpha + Y\beta \in I$ modulo I. In $\mathbb{C}[Y, V]$ it gives $\bar{\alpha} = Yh$, $\bar{\beta} = -Vh$, whence $n \in I^2$ by $XY = UV$. These identities prove exactness of the first resolution; exchange the vertices for the second. All entries have positive weight, so the resolutions are minimal. They give self-Ext dimensions $(1, 0, 0, 1)$ and off-diagonal dimensions $(0, 2, 2, 0)$.

Proposition 14.5. *A minimal model of* $\mathrm{RHom}_A(S_0 \oplus S_1, S_0 \oplus S_1)$ *has only* m_2, m_3 *as nonunit operations. Its ternary product is nonzero and cannot be removed by a change of minimal model.*

Proof. Adjoin opposite arrows α^* of degree -1 and loops τ_i of degree -2 to the path algebra, with

$$d\alpha^* = \partial_\alpha \Phi, \qquad d\tau_i = e_i \sum_\alpha [\alpha, \alpha^*] e_i.$$

The cyclic derivative identity proves $d^2 = 0$. The generating weights are 1, 3, 4. The one-sided twisted complexes of the dual generating coalgebra are precisely (14.11). The universal twisted complex over its cobar resolves the vertex ring; the positive internal weights make the bar contraction and its perturbation finite at each weight. Tensoring it with A gives the exact complexes above. If the cone of the cobar map to A had cohomology, its least internal weight would survive tensoring with the vertex ring, a contradiction. Thus the cobar map is a quasi-isomorphism.

The Ext weights are 0, 1, 3, 4 in cohomological degrees 0, 1, 2, 3. Preservation of weight and the rule $|m_n| = 2 - n$ permit only the binary products on degree pairs $(1, 2)$, $(2, 1)$ and the ternary product on $(1, 1, 1)$. In suspended dual coordinates the relation $d(d^*) = abc - cba$ gives

$$b_3(sa^\vee, sb^\vee, sc^\vee) = s(d^*)^\vee, \qquad b_3(sc^\vee, sb^\vee, sa^\vee) = -s(d^*)^\vee.$$

One can see the first value directly on the resolutions. Choose closed degree-one lifts A_1, B_1, C_1 of the three arrow classes, with components from homological degree n to $n - 1$ as follows:

	$n = 1$	$n = 2$	$n = 3$
A_1	$(1, 0)$	$\left(\begin{smallmatrix} 0 & -c \\ c & 0 \end{smallmatrix}\right)$	$(1, 0)^t$
B_1	$(1, 0)$	$\left(\begin{smallmatrix} 0 & d \\ -d & 0 \end{smallmatrix}\right)$	$(1, 0)^t$
C_1	$(0, 1)$	$\left(\begin{smallmatrix} 0 & a \\ -a & 0 \end{smallmatrix}\right)$	$(0, 1)^t$

Take homotopies whose only nonzero components are $H_{AB,2} = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and $H_{BC,2} = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. For $\delta f = d_{\text{target}} f - (-1)^{|f|} f d_{\text{source}}$, matrix multiplication gives $\delta H_{AB} = A_1 B_1$ and $\delta H_{BC} = B_1 C_1$. The closed representative $A_1 H_{BC} + H_{AB} C_1$ projects to the row $(0, 1)$ in degree two. All products of two degree-one Ext classes vanish by weight, so the triple Massey value has zero indeterminacy. It cannot be erased. □

Under the conifold tilting equivalence these objects are $\mathcal{O}_E$ and $\mathcal{O}_E(-1)[1]$, where E is the exceptional curve. Their degree-three Serre pairing pairs arrows with relations and units with top classes. Its compact support is essential: it supplies no finite trace on all of A. The ternary calculation retains that pairing and the quartic potential without identifying them with a quantized field action.

14.4 Deforming a specified action

The equations for a formal action deformation belong to a convolution complex over a characteristic-zero field k. Fix an augmented dg algebra A, a complex V, and a unital dg action $\rho : A \to E = \operatorname{End}(V)$. Put $\bar{A} = \ker(A \to k)$ and give $s\bar{A}$ the shift $|sa| = |a| - 1$. On the reduced tensor coalgebra $B(A) = \bigoplus_{r\geq 1}(s\bar{A})^{\otimes r}$, use the bar components

$$b_1(sa) = -s(da), \qquad b_2(sa \otimes sb) = (-1)^{|a|}s(ab).$$

Its coproduct is deconcatenation. The completed convolution algebra is

$$\mathcal{L} = \prod_{r\geq 1} \operatorname{Hom}((s\bar{A})^{\otimes r}, E).$$

For homogeneous f, its differential is $d_{\mathcal{L}}f = d_E f - (-1)^{|f|}fb$; its product is composition in E after the coproduct, with the tensor Koszul sign. The commutator makes $\mathcal{L}$ a dg Lie algebra.

The degree-one map $\tau(sa) = \rho(a)$, zero on longer tensors, satisfies $d_{\mathcal{L}}\tau + \tau \star \tau = 0$. At tensor length two the two terms are $(-1)^{|a|}\rho(ab)$ and $(-1)^{|a|-1}\rho(a)\rho(b)$, so multiplicativity gives their cancellation. Set

$$D = d_{\mathcal{L}} + [\tau, -].$$

A deformation of this action, allowing higher action components, is a degree-one series $a(t) = \sum_{n\geq 1} t^n a_n \in t\mathcal{L}[[t]]$ satisfying

$$Da(t) + \tfrac{1}{2}[a(t), a(t)] = 0. \tag{14.12}$$

The t-adic completion makes every coefficient a finite sum; each a_n can itself have arbitrary tensor arity.

Proposition 14.6. *Suppose* (14.12) *holds through order $n-1$. Put*

$$R_n = -\frac{1}{2}\sum_{i=1}^{n-1}[a_i, a_{n-i}].$$

Then $DR_n = 0$. An extension to order n exists exactly when $[R_n] = 0$ in $H^2(\mathcal{L}, D)$. Its choices form an affine space over $Z^1(\mathcal{L}, D)$, and modulo order-n gauge transformations they form an affine space over $H^1(\mathcal{L}, D)$.

Proof. The coefficient of t^n in the curvature of $\tau + a(t)$ is $Da_n - R_n$. The Bianchi identity says that its twisted differential is zero. Since the lower curvature coefficients vanish, the coefficient of t^n in that identity is $DR_n = 0$. Solving $Da_n = R_n$ gives the obstruction statement. Differences of solutions are closed degree-one elements. A gauge parameter $t^n\xi$, $|\xi| = 0$, changes a_n by $D\xi$ without changing the lower coefficients, proving the last claim. □

Suppose the twisted complex has a contraction onto its cohomology, $Dh + hD = 1 - ip$, with $|h| = -1$. If $pR_n = 0$, the choice $a_n = hR_n$ solves the order-n equation. Through order six this gives

$$\begin{aligned} a_2 &= -\tfrac{1}{2}h[a_1, a_1], \\ a_3 &= -h[a_1, a_2], \\ a_4 &= -h\big([a_1, a_3] + \tfrac{1}{2}[a_2, a_2]\big), \\ a_5 &= -h\big([a_1, a_4] + [a_2, a_3]\big), \\ a_6 &= -h\big([a_1, a_5] + [a_2, a_4] + \tfrac{1}{2}[a_3, a_3]\big). \end{aligned} \tag{14.13}$$

Every line assumes the preceding obstruction projections vanish. The factor $1/2$ on the diagonal terms comes from the symmetry of the bracket on degree-one elements; off-diagonal terms occur twice.

If $a_1 = u\beta$ and the contraction is linear over $k[u]$, this recursion gives $a_n \in u^n \mathcal{L}$. The first terms are

$$a_2 = -\tfrac{u^2}{2}h[\beta, \beta], \qquad a_3 = \tfrac{u^3}{2}h[\beta, h[\beta, \beta]].$$

The divisibility is a consequence of the recursion. It does not specify the tensor arity of a_n, and it need not be the exact order of vanishing if a coefficient is zero.

Two degree-zero algebras separate extension from obstruction. For $A = k[x]$ acting on k by augmentation, the strict deformed action $x \mapsto tu$ gives

$$a_n(sx^r) = \begin{cases} u^n & r = n, \\ 0 & r \neq n, \end{cases}$$

with no higher tensor-length component. All orders exist because the polynomial algebra has no relation to violate. For $A = k[\epsilon]/(\epsilon^2)$, the same first-order value $a_1(s\epsilon) = u$ has a nonzero order-two obstruction: the required strict relation would give $t^2u^2 = 0$. In the convolution complex the relevant differential is zero and the residual value on $s\epsilon \otimes s\epsilon$ is nonzero. No higher action component with values in $\mathrm{End}(k)$ can remove it. Thus an integer sequence attached to another generating problem cannot count the extensions without an actual identification of complexes.

14.5 Compact supports and holomorphic moments

The support condition changes the coefficient space already in the linear Dolbeault problem. On $\mathbb{C}^3$, write $z_j = x_j + iy_j$, $dV = \prod_j (dx_j \wedge dy_j)$, and $\Omega = dz_1 \wedge dz_2 \wedge dz_3$. Then

$$\Omega \wedge d\bar{z}_1 \wedge d\bar{z}_2 \wedge d\bar{z}_3 = -8i\, dV.$$

Choose a smooth radial bump $\rho(|z|^2)$ supported inside the unit disk with $\int_{\mathbb{C}} \rho\, dA = 1$. For a multi-index $\beta \in \mathbb{N}^3$, define compactly supported forms

$$\alpha_0 = \frac{1}{-8i} \prod_j \rho(|z_j|^2)\, d\bar{z}_1 \wedge d\bar{z}_2 \wedge d\bar{z}_3, \qquad \alpha_\beta = \frac{(-1)^{|\beta|}}{\beta!}\partial_z^\beta \alpha_0. \tag{14.14}$$

They are $\bar{\partial}$-closed because they have top antiholomorphic degree.

Proposition 14.7. *For every entire holomorphic function f,*

$$\int_{\mathbb{C}^3} f\,\Omega \wedge \alpha_\beta = \frac{\partial_z^\beta f(0)}{\beta!}.$$

The classes $[\alpha_\beta]$ *are linearly independent in* $H_c^{0,3}(\mathbb{C}^3)$. *For every* N, *the moment map to the dual of polynomials of total degree at most* N *is split surjective and has rank* $\binom{N+3}{3}$.

Proof. Angular averaging in each coordinate gives the formula for $\beta = 0$. Repeated integration by parts gives the general formula; compact support kills every boundary term. If $\alpha = \bar\partial\eta$ with η compactly supported, Stokes gives $\int f\,\Omega \wedge \alpha = 0$. Pairing a finite relation among the $[\alpha_\beta]$ with z^γ therefore extracts its γ-coefficient. The same monomial pairing gives the asserted finite-dimensional splitting. □

There is a useful one-dimensional quotient, but the cohomology itself is infinite-dimensional. To identify that quotient, put $F(t) = -\int_t^1 \rho(s)\,ds$, extended by zero for $t \geq 1$. The function $F(|z|^2)$ is smooth and compactly supported, and $\partial_{\bar z} F(|z|^2) = z\rho(|z|^2)$. Replacing the jth bump factor in α_0 by F, omitting $d\bar z_j$, and using the sign $(-1)^{j-1}$ gives a compactly supported primitive of $z_j\alpha_0$. Differentiation and the Leibniz rule consequently give

$$z_j[\alpha_\beta] = \begin{cases} [\alpha_{\beta-e_j}] & \beta_j > 0, \\ 0 & \beta_j = 0, \end{cases} \qquad \partial_{z_j}[\alpha_\beta] = -(\beta_j+1)[\alpha_{\beta+e_j}]. \tag{14.15}$$

These operators satisfy $[\partial_{z_j}, z_k] = \delta_{jk}$. The span of the moment classes is the polynomial delta module for the Weyl algebra. Its cyclic generator $[\alpha_0]$ is killed by every z_j. Its quotient by $\sum_j \partial_{z_j}$ of that span is the single line generated by $[\alpha_0]$. This quotient retains only the zeroth moment; it is a different coefficient object from $H_c^{0,3}(\mathbb{C}^3)$. The compact-support moment mechanism is also visible in the Cauchy formula of [44, Ch. I, §3].

Gauge coefficients can introduce further dimensions. If $\mathfrak{g} = \bigoplus_{i=1}^r \mathfrak{g}_i$ is complex semisimple with simple summands, every invariant symmetric form has zero cross terms: for $x = [a, b] \in \mathfrak{g}_i$ and $y \in \mathfrak{g}_j$, $i \neq j$, invariance gives $B(x, y) = B(a, [b, y]) = 0$. Each simple summand contributes one line of forms. A finite-dimensional local deformation claim must therefore specify both its gauge restriction and its support or translation quotient, together with the map from the field-theoretic deformation complex.

14.6 A nodal BRST field algebra

Let u, m be target coordinates on $\mathbb{A}^2$, and let z denote the independent field-insertion coordinate. Start with even fields u, m, β_u, β_m and odd fields $\eta, \chi, b_\eta, b_\chi$. Their elementary singular products are

$$\beta_u(z)u(w) \sim \frac{1}{z-w}, \quad \beta_m(z)m(w) \sim \frac{1}{z-w}, \quad b_\eta(z)\eta(w) \sim \frac{1}{z-w}, \quad b_\chi(z)\chi(w) \sim \frac{1}{z-w}. \tag{14.16}$$

Reversing an even pair changes the sign; reversing an odd pair keeps it. All other elementary products are regular.

The coordinates have conformal weight zero, their momenta weight one. The cochain degrees of η, χ are -1, those of b_η, b_χ are 1, and all even fields have degree zero. For each coordinate and its momentum use oscillator expansions $x(z) = \sum_n x[n]z^{-n}$ and $\beta_x(z) = \sum_n \beta_x[n]z^{-n-1}$. The vacuum representation has zero-mode space $\mathbb{C}[u, m] \otimes \Lambda(\eta, \chi)$; coordinate zero modes multiply and momentum zero modes differentiate. Positive modes kill this space. Negative modes freely generate polynomial or exterior factors according to parity. Normal ordering and Wick contraction define the vertex superalgebra V, including every translated state [115, §§2.1, 3.2].

Proposition 14.8. *The charge field and its residue are*

$$q(z) = :u(z)m(z)b_\eta(z):, \qquad Q = \operatorname{Res}_{z=0} q(z).$$

The operator Q is a square-zero, weight-preserving vertex derivation of degree one. It kills the vacuum, commutes with translation, and acts by

$$Q\eta = um, \qquad Q\beta_u = -:mb_\eta:, \qquad Q\beta_m = -:ub_\eta:,$$

with zero value on $u, m, \chi, b_\eta, b_\chi$. Its weight-zero cohomology is $\mathbb{C}[u,m]/(um) \otimes \Lambda(\chi)$, and its vacuum is nonzero in the full cohomology.

Proof. A zero mode is a derivation of every vertex product, kills the vacuum, and commutes with translation. The charge has no self-contractions, so its self-OPE is regular and $2Q^2 = [Q,Q] = 0$. Its weight is one, making its residue weight zero. Wick contraction gives the displayed values, with both minus signs coming from the reversed even pairs. In weight zero the complex is $(\mathbb{C}[u,m] \otimes \Lambda(\eta,\chi), um\partial_\eta)$. Multiplication by um is injective in the polynomial ring, so its cohomology is the stated quotient and the unit survives. Weight preservation prevents a higher-weight primitive from killing it. □

Every translated relation is retained:

$$Q(T^n\eta) = \sum_{a=0}^{n} \binom{n}{a} :(T^a u)(T^{n-a} m):.$$

The momentum zero mode $\beta_{u,(0)}$ is not closed. The construction therefore makes normal differentiation part of a dg operator action; it does not turn it into a closed normal connection on the unchanged nodal quotient.

Define closed currents

$$J_u = :u\beta_u: + :\eta b_\eta:, \qquad J_m = :m\beta_m: + :\eta b_\eta:, \qquad J_\chi = :\chi b_\chi:.$$

Their pairing matrix, in this order, is

$$J_i(z)J_j(w) \sim \frac{B_{ij}}{(z-w)^2}, \qquad B = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{14.17}$$

Indeed the two terms of QJ_u, and of QJ_m, cancel. A bosonic number current has double contraction $-(z-w)^{-2}$ and an odd number current has $+(z-w)^{-2}$; the single contractions cancel. The two mixed currents share only the odd pair. This proves the formula. It embeds the rank-three Heisenberg vacuum algebra in $H(V,Q)$: the nondegenerate pairing lets positive modes differentiate a highest negative-mode monomial to a nonzero scalar, so the Heisenberg vacuum module is simple. Its map to cohomology preserves the nonzero vacuum and is therefore injective.

14.6.1 A rational dressing with a finite pole bound

Work over $R_N = \mathbb{C}[a,b,\varepsilon]/(\varepsilon^N)$ and expand first in ε. Set

$$K(s) = \frac{s(s+\varepsilon(a+b))}{(s+\varepsilon a)(s+\varepsilon b)}, \qquad r(s) = K(s)/K(-s).$$

Write $J_u(z) = \sum_n J_{u,n} z^{-n-1}$. For $g(z) \in 1 + \varepsilon R_N[z^{-1}]$, define

$$A_g(z) = \sum_{n \geq 0} \frac{(-1)^n}{n!} \partial_z^n \log g(z)\, J_{u,n}, \qquad U_g(z) = e^{A_g(z)}. \tag{14.18}$$

The logarithm and exponential terminate in ε. On a state of weight L, the current modes with $n > L$ vanish, and the other nonnegative modes do not increase weight. Thus the formula defines an operator on every state.

The modes $J_{u,n}$ commute with one another and with Q. Their action $[J_{u,n}, Y(u,w)] = w^n Y(u,w)$ gives

$$U_g(z) Y(u,w) U_g(z)^{-1} = g(z-w) Y(u,w), \qquad |z| > |w|.$$

The coefficient relation $a'_n = -(n+1)a_{n+1}$ in (14.18), together with $[T, J_{u,n}] = -n J_{u,n-1}$, gives $[T, U_g] = \partial_z U_g$. Every U_g fixes the vacuum. Consequently $C(z) = Y(u,z) U_K(z)$ and $H(p) = U_r(p)$ obey

$$\begin{aligned} C(z)C(w) &= r(z-w) C(w) C(z), \\ H(p) C(z) H(p)^{-1} &= r(p-z) C(z), \\ [Q, C(z)] = [Q, H(p)] &= 0, \qquad C(z)\mathbf{1} = e^{zT} u. \end{aligned} \tag{14.19}$$

The equalities compare rational continuations and their specified Laurent expansions. For any finite number of insertions,

$$C(z_1) \cdots C(z_r) = \prod_{i<j} K(z_i - z_j)\, Y(u, z_1) \cdots Y(u, z_r) \prod_i U_K(z_i).$$

At order ε^j the pair-kernel product has total diagonal pole order at most j; derivatives of total order M add at most M. A fixed oscillator state contributes its own finite pole bound. These bounds give compatible operators for all N.

For example, modulo ε^4,

$$K(s) = 1 - \frac{ab\varepsilon^2}{s^2} + \frac{ab(a+b)\varepsilon^3}{s^3}, \qquad r(s) = 1 + \frac{2ab(a+b)\varepsilon^3}{s^3}.$$

This constructs the displayed operator family. A new state-field map on every state with these assigned operators has not been defined by the dressing alone. Its comparison with an oriented Hall operation requires the corresponding morphism of coefficients and residues.

14.6.2 Induction from differential modules

Put $A_0 = \mathbb{C}[u, m]$ and let $\mathcal{D}_{A_0}$ be its polynomial differential-operator algebra. The super differential-operator algebra of $P = A_0 \otimes \Lambda(\eta, \chi)$ has differential $[um\partial_\eta, -]$. In particular,

$$[um\partial_\eta, \partial_u] = -m\partial_\eta, \qquad [um\partial_\eta, \partial_m] = -u\partial_\eta.$$

For a complex (M, d_M) of left $\mathcal{D}_{A_0}$-modules, set

$$L(M) = M \otimes \Lambda(\eta, \chi), \qquad D_M = d_M + um\partial_\eta$$

with the graded tensor sign. Its square is zero because d_M is $\mathcal{D}_{A_0}$-linear. Let $\widehat{\mathfrak{o}}$ be the oscillator Lie superalgebra of (14.16), with central element acting by one. Give $L(M)$ its indicated zero-mode action and let strictly positive modes act by zero. Oscillator induction gives

$$\mathbb{M}(M) = U(\widehat{\mathfrak{o}}_{<0}) \otimes L(M), \qquad d_{\mathbb{M}} = d_M + q_{(0)}. \tag{14.20}$$

PBW supplies polynomial factors for even negative modes and exterior factors for odd ones. Normal-ordered state fields act with finitely many terms on each vector. On the top space, $q_{(0)}$ is $um\partial_\eta$; on the full induced module its square vanishes by the same regular self-OPE. Thus (14.20) defines a dg V-module.

A $\mathcal{D}_{A_0}$-linear chain map extends by the identity on all oscillator factors. It commutes with every state-field mode, with the dressings, and with principal-part residues and their finite normal derivatives. Induction preserves short exact sequences of underlying complexes because these factors are free vector spaces. Positive oscillator modes act locally nilpotently by differentiating finite monomials. A spectator connection commuting with the ambient action and d_M and annihilating um extends through the same formulas.

14.6.3 *The constant critical coefficient*

For $f = um$, let

$$N = A_0[f^{-1}]/A_0, \quad N_u = A_0[u^{-1}]/A_0, \quad N_m = A_0[m^{-1}]/A_0.$$

Their quotient sequence

$$0 \longrightarrow N_u \oplus N_m \longrightarrow N \longrightarrow \delta_0 \longrightarrow 0$$

has δ_0 spanned by $u^{-r}m^{-s}$, $r, s \geq 1$. It is nonsplit even as an A_0-module sequence. Indeed, $\ker(u : N \to N)$ is spanned by $u^{-1}m^b$, $b \geq 0$, while $\ker(m : N \to N)$ is spanned by $u^a m^{-1}$, $a \geq 0$. Their intersection is zero; the nonzero class $u^{-1}m^{-1}$ in δ_0 is killed by both. Induction carries the sequence and its actual maps to an exact sequence of dg field modules. This exactness makes no assertion that every derived extension remains nonsplit.

The constant critical coefficient is the holonomic dual $P_Z = \mathbb{D}_{\mathcal{D}} N$. With the normalization $\mathrm{DR}(\mathcal{O}_{\mathbb{A}^2}) = \mathbb{C}_{\mathbb{A}^2}[2]$, regular-holonomic duality gives $\mathrm{DR}(P_Z) = i_{Z,*}\mathbb{C}_Z[1]$ for $Z = \{um = 0\}$. There is an explicit presentation:

$$P_Z = \ker\Big(J = \mathcal{D}_{A_0}/(\mathcal{D}_{A_0} u\partial_u + \mathcal{D}_{A_0} m\partial_m) \xrightarrow{g \mapsto 1} A_0\Big). \tag{14.21}$$

Here g is the class of one in J. To see the presentation, $A_0[f^{-1}]$ is cyclic on $1/(um)$ with commuting relations $u\partial_u + 1$ and $m\partial_m + 1$. Their principal symbols form a regular sequence. Formal transposition in its Koszul resolution sends them to $-u\partial_u$ and $-m\partial_m$. Dualizing the localization sequence proves (14.21).

A basis of its kernel is

$$A_{r,b} = \partial_u^r m^b g \ (r \geq 1, b \geq 0), \quad B_{a,s} = u^a \partial_m^s g \ (a \geq 0, s \geq 1), \quad C_{r,s} = \partial_u^r \partial_m^s g \ (r, s \geq 1).$$

The polynomial–polynomial sector maps isomorphically to A_0; these other three sectors give the kernel. The equality $\partial_m A_{r,0} = C_{r,1}$ retains the attachment of a branch to the crossing. Commuting the coordinates through the derivatives gives

$$\begin{aligned} f A_{r,b} &= -(r-1) A_{r-1,b+1}, \\ f B_{a,s} &= -(s-1) B_{a+1,s-1}, \\ f C_{r,s} &= (r-1)(s-1) C_{r-1,s-1}, \end{aligned} \tag{14.22}$$

with zero whenever a coefficient vanishes. The cokernel has basis $A_{r,0}$, $B_{0,s}$, and the kernel has basis $A_{1,b}$, $B_{a,1}$, $C_{1,s}$, $C_{r,1}$, counting $C_{1,1}$ once. Every nonzero image in each sector is a distinct basis vector, which proves these assertions. Hence

$$H(L(P_Z)) = \big(\operatorname{coker} f \oplus \eta \ker f\big) \otimes \Lambda(\chi).$$

The first two summands have degrees zero and -1. On its degree-zero top cohomology, $(J_{u,0}, J_{m,0})$ has weights $(-r, 0)$ on $A_{r,0}$ and $(0, -s)$ on $B_{0,s}$. This is a concrete coefficient module, distinct from the vacuum representation.

For comparison, $f : N \to N$ is surjective with kernel $A_0/(f)$, via $a \mapsto [a/f]$. Thus $H^{-1}(L(N)) = A_0/(f)$, $H^{-2}(L(N)) = \chi A_0/(f)$, and $H^0(L(N)) = 0$. The field-module construction preserves the actual dg differential-operator action, including its nonclosed normal momenta, rather than identifying these two cohomologies.

Finally a moving spectator coefficient can be added with generator δ_λ satisfying

$$(w - \lambda)\delta_\lambda = 0, \qquad (\partial_w + \partial_\lambda)\delta_\lambda = 0.$$

The two operators commute. Tensoring it with the nodal coefficient retains this common connection under induction, since um is independent of w, λ. If $\lambda = (a_{\mathrm{fr}} + \varepsilon y)/\beta$ with β invertible, the common derivative is $\partial_w + \beta\partial_{a_{\mathrm{fr}}}$. Its identification with a framed Hall direct image still requires the oriented correspondence map.

14.7 The interacting curved dual of the Miura algebra

The next example has noncentral ghost brackets. Work over $k = \mathbb{C}(\alpha)$ on $X = \mathbb{A}^1_k$, with translation $T = \partial_z$. Let four even Heisenberg fields satisfy

$$\sum_{i=1}^{4} a_i = 0, \qquad a_i(z)a_j(w) \sim \frac{\delta_{ij} - 1/4}{(z-w)^2}.$$

Define U_s by the right-nested, normally ordered Miura operator

$$:(\alpha T + a_1)(\alpha T + a_2)(\alpha T + a_3)(\alpha T + a_4): = \sum_{s=0}^{4} U_s(\alpha T)^{4-s}, \qquad U_0 = \mathbf{1}, \quad U_1 = 0, \tag{14.23}$$

moving every T to the right. Let V be the generated vertex subalgebra. Its stress field is $L = -U_2$ and its central charge is $c = 3 - 60\alpha^2$. Relative to the negative Heisenberg contraction in [142], the convention change is $a_i^Z = ia_i$, $\alpha_Z = i\alpha$, $U_s^Z = i^s U_s$.

All coefficients can be computed with one Wick rule:

$$(T^r a_i)(z)(T^s a_j)(w) \sim (\delta_{ij} - 1/4)(-1)^r (r+s+1)!(z-w)^{-r-s-2}. \tag{14.24}$$

Sum over partial cross-matchings and Taylor-expand each uncontracted left factor. This includes regular terms that become poles after multiplication by a coefficient. For instance, if PP $Y(U_4, u)U_4 = \sum_{p=1}^{8} A_p u^{-p}$, then

$$A_8 = \frac{c(5 - 180\alpha^2 + 2016\alpha^4)}{32}\mathbf{1}, \qquad A_6 = -\frac{5}{4}(1 - 12\alpha^2)^2 U_2.$$

The coefficient A_2 contains $\frac{3}{4}(U_3)_{(-1)}U_3$. It cannot be discarded: the $a_1^4 a_2^2$ coefficient detects it. The odd coefficients are fixed by self-skewsymmetry; for example, $A_5 = \frac{1}{2}TA_6$ and $A_3 = \frac{1}{2}TA_4 - \frac{1}{24}T^3A_6$.

14.7.1 *The complete finite relation datum*

Put $I_4 = \{0, 2, 3, 4\}$ and let N° have even frame e_i, with $e_i \mapsto U_i$. The index 0 is a formal scalar generator. The right $\mathcal{D}$ generators are $N^\circ \otimes \omega$. Their shifted dual is

$$\check{N}^\circ = s^{-1}(N^\circ)^\vee \otimes \omega^{-1}, \qquad \langle -, - \rangle : (N^\circ \boxtimes N^\circ) \otimes (\check{N}^\circ \boxtimes \check{N}^\circ) \longrightarrow s^{-2}\omega_{X^2}^{-1}.$$

Here s^{-1} raises degree by one. Write its odd frame as $\Gamma_0 = S, \Gamma_s = \gamma_s$ for $s = 2, 3, 4$; all four generators have cochain degree one and weight $1 - s$. In the affine coordinate their density types are $s^{-1}\omega^{s-1}$, including $s^{-1}\omega^{-1}$ at $s = 0$.

The following finite matrix specifies both relation lattices. For $u = z_1 - z_2$, define

$$t_{ij} = \Gamma_i \boxtimes \Gamma_j + \sum_{k,l} E_{ij}^{kl}\, u^{-(k+l-i-j)} \Gamma_k \boxtimes \Gamma_l, \qquad t = D(\Gamma \boxtimes \Gamma). \tag{14.25}$$

The independent nonzero coefficients are in Table 14.1. Set all others to zero and use $E_{ji}^{lk} = (-1)^{k+l-i-j} E_{ij}^{kl}$ for reversed rows. Let $M = D^{-\mathrm{t}}$ and $r = M(e \boxtimes e)$. The $\mathcal{O}_{X^2}$-span P° of the rows r is the primal quadratic–linear–scalar relation lattice; the span of t is its annihilator for the displayed density-valued pairing.

Table 14.1: The annihilator matrix; $m = k + l - i - j$.

ij	kl	m	E_{ij}^{kl}
00	22	4	$\frac{3}{2} - 30\alpha^2$
00	23	5	$-6\alpha(20\alpha^2 - 1)$
00	24	6	$-9\alpha^2(20\alpha^2 - 1)$
00	32	5	$6\alpha(20\alpha^2 - 1)$
00	33	6	$(36\alpha^2 - 1)(20\alpha^2 - 1)$
00	34	7	$3\alpha(20\alpha^2 - 1)(24\alpha^2 - 1)$
00	42	6	$-9\alpha^2(20\alpha^2 - 1)$
00	43	7	$-3\alpha(20\alpha^2 - 1)(24\alpha^2 - 1)$
00	44	8	$(3 - 60\alpha^2)(2016\alpha^4 - 180\alpha^2 + 5)/32$
02	22	2	-1
02	23	3	-4α
02	24	4	$(1 - 36\alpha^2)/4$
02	33	4	$12\alpha^2 - 1$
02	34	5	$4\alpha(12\alpha^2 - 1)$
02	43	5	$-\alpha(12\alpha^2 - 1)$
02	44	6	$-5(12\alpha^2 - 1)^2/8$
03	23	2	-2
03	24	3	-3α
03	32	2	-1
03	33	3	2α
03	34	4	$12\alpha^2 - 1$
03	43	4	$-1/2$
03	44	5	$-3\alpha(6\alpha^2 - 1)$

ij	kl	m	E_{ij}^{kl}
04	24	2	-3
04	33	2	-2
04	34	3	6α
04	42	2	-1
04	43	3	2α
04	44	4	$(3-6\alpha)(2\alpha+1)/2$
22	33	2	1
22	34	3	α
22	43	3	$-\alpha$
22	44	4	$5/8-9\alpha^2/2$
23	34	2	$1/2$
23	44	3	$-3\alpha/2$
24	44	2	-1
33	44	2	$3/4$

Proposition 14.9. *The lattice P° is free of rank sixteen, agrees with the full generator tensor space away from the diagonal, and satisfies the unit intersection condition. Its unital vertex quotient is V. The rows of* (14.25) *are exactly its annihilator, and the nine nonunit rows give the annihilator of its projected quadratic lattice.*

Proof. Order pairs by $i+j$. The matrix D is unipotent triangular; each correction raises the pair sum by at least two. Its inverse transpose has the opposite triangularity. With unit rows first it has block form

$$M=\begin{pmatrix} 1 & 0 \\ L_0 & A_0 \end{pmatrix}, \qquad D=\begin{pmatrix} 1 & -L_0^{\mathrm{t}}A_0^{-\mathrm{t}} \\ 0 & A_0^{-\mathrm{t}} \end{pmatrix}.$$

Both diagonal blocks are unipotent. A linear combination of primal rows whose nonunit projection vanishes therefore has zero coefficients on all nine nonunit rows. Its seven remaining coefficients are regular. This is the unit intersection condition.

Put $F_{ij}(u)=Y(U_i,u)U_j$. Applying the Wick rule (14.24) gives $\mathrm{PP}(MF)=0$. This is an exact finite polynomial calculation: each contraction lowers the Heisenberg degree, and only the Taylor terms that can reach a negative power of u are needed. In particular the scalar coefficient of r_{44} includes the poles of lower full products; it is not the isolated coefficient A_8 of the self-OPE.

Give U_s filtration s and T filtration zero. Every relation expresses a singular product in smaller filtration, so ordered normal monomials in the jets T^rU_s span the quotient. In the Heisenberg realization the leading symbols are the jets of the elementary symmetric polynomials e_2,e_3,e_4 on $\sum a_i=0$. Their Jacobian is nonzero at distinct roots. At every finite jet order its prolongation is block triangular with these same invertible diagonal blocks. The jets are algebraically independent, so the spanning monomials remain independent in V. This proves the presentation.

Finally $MD^{\mathrm{t}}=1$. Pairing regularly against every primal row is equivalent to having regular coordinates in the dual row frame. This proves equality with the annihilator, including its density factor. The block formula proves the projected assertion. □

14.7.2 A finite recursion and the full PBW algebra

For every $n\ge 0$, impose

$$(\Gamma_i)_{(n)}\Gamma_j=-\sum_{kl}E_{ij}^{kl}(\Gamma_k)_{(n-m)}\Gamma_l, \qquad m=k+l-i-j. \tag{14.26}$$

For $r \ge 0$ use $(\Gamma_k)_{(-r-1)}\Gamma_l = (T^r\Gamma_k)_{(-1)}\Gamma_l/r!$; for remaining nonnegative indices apply the rule again. The positive filtration

$$\delta(S) = 9, \quad \delta(\gamma_2) = 5, \quad \delta(\gamma_3) = 3, \quad \delta(\gamma_4) = 1, \quad \delta(T) = 0$$

drops by $2m \ge 4$ at each correction. Thus the recursion terminates.

Proposition 14.10. *The rule* (14.26) *defines a vertex superalgebra B° with the ordered odd-jet PBW basis. The subalgebra B generated by $\gamma_2, \gamma_3, \gamma_4$ embeds in it. Every nonnegative product $S_{(n)}B$ and $S_{(n)}S$ lies in B.*

Proof. We specify the finite generator verification and its extension to all states. Reduce words by the supercommutator and iterate formulas

$$[x_{(m)}, y_{(n)}\} = \sum_{r\ge 0}\binom{m}{r}(x_{(r)}y)_{(m+n-r)}, \tag{14.27}$$

$$(x_{(-p)}v)_{(n)}b = \sum_{r\ge 0}\binom{p+r-1}{r}\Big(x_{(-p-r)}v_{(n+r)}b - (-1)^{p+|v|}v_{(n-p-r)}x_{(r)}b\Big), \tag{14.28}$$

where x is odd and $p \ge 1$. Exchange wrong-ordered creation pairs with the odd sign and their commutator; replace an equal pair by half its commutator. Each correction has lower filtration. For a differentiated generator,

$$(\Gamma_{i,(-p)}\mathbf{1})_{(n)} = (-1)^{p-1}\binom{n}{p-1}(\Gamma_i)_{(n-p+1)}.$$

Binomial coefficients with negative upper entry have their polynomial meaning. Translation replaces a creation index p by $p+1$ and multiplies by p.

A degree-d normal word has weight at least the sum of the d smallest entries in $\{1 - i + r : i \in I_4, r \ge 0\}$. For degrees two and three these bounds are -5 and -7. Thus the generator skew identities have only $0 \le n \le 6 - i - j$ to consider, and generator Jacobi only $m, n \ge 0$, $m + n \le 8 - i - j - k$. Substitution of Table 14.1 into

$$\begin{aligned} K_{ij,n} &= (\Gamma_i)_{(n)}\Gamma_j - \sum_{r\ge 0}\frac{(-1)^{n+r}}{r!}T^r(\Gamma_j)_{(n+r)}\Gamma_i, \\ J_{ijk;mn} &= (\Gamma_i)_{(m)}((\Gamma_j)_{(n)}\Gamma_k) + (\Gamma_j)_{(n)}((\Gamma_i)_{(m)}\Gamma_k) \\ &\quad - \sum_{r=0}^{m}\binom{m}{r}((\Gamma_i)_{(r)}\Gamma_j)_{(m+n-r)}\Gamma_k \end{aligned}$$

reduces each coefficient to zero over $\mathbb{Q}[\alpha]$ by (14.27)–(14.28). There are respectively 41 and 438 identities. The weight bounds exclude every other generator case; reductions use only the relation module in lower filtration, so confluence has not been assumed.

Choose tensor-algebra lifts of the resulting brackets and replace $P_{ij}(\lambda)$ by $\frac{1}{2}(P_{ij}(\lambda) + P_{ji}(-\lambda - T))$. The difference is already a lower-filtration relation. Induction on the positive filtration preserves both the relation module and the Jacobi congruences. These exactly skew lifts meet the nonlinear PBW hypotheses of [42, Theorem 3.9]. That theorem gives the vertex algebra and its ordered odd-jet basis. All corrections in the table use indices 2, 3, 4, so the three-generator algebra closes, its PBW words remain independent, and all asserted scalar-dual products lie in it. The commutator and iterate formulas extend this preservation to every state. □

The algebra B is noncommutative. If $\chi = \gamma_4$, the recursion gives

$$(\gamma_2)_{(0)}\chi = -\chi_{(-1)}\chi_{(-2)}\mathbf{1}, \qquad (\gamma_3)_{(0)}\gamma_3 = \tfrac{3}{4}\chi_{(-1)}\chi_{(-2)}\mathbf{1}.$$

The common ordered word is nonzero by PBW.

14.7.3 Curvature and the exact coupling

Set

$$d = S_{(0)}|_B, \qquad w = \tfrac{1}{2} S_{(0)} S, \qquad q = \sum_{s=2}^{4} U_s \otimes \gamma_s .$$

The zero-mode commutator gives $d^2 = w_{(0)}$ on every state. Put $A_S = S_{(0)} S$. Jacobi gives $2 S_{(0)} A_S = (A_S)_{(0)} S$; super-skewsymmetry then gives the explicit Bianchi primitive

$$dw = Tv, \qquad v = \frac{1}{6} \sum_{j=1}^{8} \frac{(-1)^{j+1}}{j!} T^{j-1} (S_{(j)} A_S). \tag{14.29}$$

The upper limit follows from the degree-three minimum weight -7. The action of the curvature is nonzero. For example,

$$\begin{aligned} w_{(0)}\chi = {} & \frac{c}{2} \gamma_{2,(-4)} \chi_{(-1)} \chi_{(-2)} \mathbf{1} - 2c\alpha \gamma_{3,(-5)} \chi_{(-1)} \chi_{(-2)} \mathbf{1} \\ & + 3c\alpha^2 \chi_{(-1)} \chi_{(-2)} \chi_{(-6)} \mathbf{1}. \end{aligned} \tag{14.30}$$

The three PBW words are distinct and $c \neq 0$ in k.

Theorem 14.11. *With $e = S + q$, one has $e_{(n)} e = 0$ for every $n \geq 0$. Consequently $Q = d + q_{(0)}$ is a square-zero differential on $V \otimes B$. For any unital vertex superalgebra C concentrated in cohomological degree zero, homomorphisms $V \to C$ are naturally equivalent to tuples of even states (x_2, x_3, x_4) such that*

$$(S + \textstyle\sum_s x_s \otimes \gamma_s)_{(n)} (S + \sum_s x_s \otimes \gamma_s) = 0 \quad (n \geq 0).$$

The assertion concerns this generator-linear form of coupling.

Proof. Let $F_{ij} = Y(U_i, u) U_j$ and $G_{ij} = Y(\Gamma_i, u) \Gamma_j$. The primal generators are even, so the tensor OPE has no extra cross sign. The dual-frame identity gives

$$Y(e, u) e = F^{\mathrm{t}} G = (MF)^{\mathrm{t}} (M^{-\mathrm{t}} G).$$

Both factors on the right are regular by their defining lattices. This proves the full chiral identity. Its zero mode gives $2e_{(0)}^2 = (e_{(0)} e)_{(0)} = 0$, and its restriction to $V \otimes B$ is Q.

For the converse, put $x_0 = \mathbf{1}$, $R = M(Y(x_i, u) x_j)_{ij}$, and $H = M^{-\mathrm{t}} G$. The seven unit components of R are regular and $R_{ji}(u) = e^{uT_C} R_{ij}(-u)$. The leading dual-filtration symbol of H_{ij} is $(e^{uT} \xi_i) \xi_j$ for independent odd differential generators ξ_i.

We use their jet independence. If PP $\sum R_{ij}(u) (e^{uT} \xi_i) \xi_j = 0$, the coefficient of $(T\xi_i)\xi_j$ shows that every R_{ij} has at most a simple pole. For $i < j$ the coefficient of $\xi_i \xi_j$ and the transported skew identity give twice its residue; for $i = j$ the same skew identity makes its residue equal its negative. All residues vanish in characteristic zero. This reasoning also applies to a fixed collection of unordered pairs.

Apply it to the highest filtration of the coupling equation. First R_{22} is regular. Its entire contribution may then be removed from principal parts because H_{22} is regular. Repeat at pair sums 5, 6, 7, 8. Distinct unordered ghost pairs are independent, so all nine nonunit components become regular. The tuple therefore satisfies the complete primal relation lattice and extends uniquely to $V \to C$. A homomorphism conversely transports the coupling identity. No conformal grading on C is required: its trial coupling must satisfy all nonnegative modes, with locality giving a finite principal part for each tuple. □

This argument does not identify the dual lattice with the entire kernel of its chiral operation. In fact $u^{-1} \chi \boxtimes \chi$ is in that kernel, since $\chi_{(n)} \chi = 0$ for $n \geq 0$ and $\chi_{(-1)} \chi = 0$. Its coefficient in the 44 relation row is u^{-1}, so it is outside the lattice. The recognition theorem uses independent odd jets, which retain more information than this kernel comparison.

14.7.4 *Primary coordinates and a translation homotopy*

Set $\Delta_4 = 22 + 5c = 37 - 300\alpha^2$ and

$$d_0 = \frac{480\alpha^4 - 124\alpha^2 + 3}{8(300\alpha^2 - 37)}, \qquad e_0 = -\frac{108\alpha^2 - 5}{4(300\alpha^2 - 37)}.$$

The triangular primary generators are

$$W = U_3 - \alpha T U_2, \qquad H = U_4 - \tfrac{\alpha}{2} T U_3 + d_0 T^2 U_2 + e_0 {:}U_2U_2{:}.$$

The Wick rule gives $L_{(0)}W = TW$, $L_{(1)}W = 3W$, and $L_{(n)}W = 0$ for $n \geq 2$, with the analogous identities of weight four for H. The linear system for the three corrections to U_4 has a minor $-24(300\alpha^2 - 37)$, proving uniqueness of this ansatz. The change is invertible over $\mathbb{C}[\alpha, \Delta_4^{-1}]$; the Miura lattices themselves had coefficients in $\mathbb{Q}[\alpha]$. The inverse gives the exact coupling

$$\begin{aligned} q = {} & -L\gamma_2 + W\gamma_3 - \alpha(TL)\gamma_3 + H\gamma_4 + \tfrac{\alpha}{2}(TW)\gamma_4 \\ & + (d_0 - \tfrac{\alpha^2}{2})(T^2L)\gamma_4 - e_0 {:}LL{:}\gamma_4. \end{aligned} \tag{14.31}$$

Products between primal and ghost factors denote tensor products. Integrating derivatives by parts changes this state by an explicit total derivative. Such a change preserves its zero mode but need not preserve its full chiral coupling. The nonlinear term in (14.31) is also part of the comparison.

There is nevertheless an exact homotopy for translation. Define a degree--1 derivation on B by

$$h\gamma_2 = -\mathbf{1}, \qquad h\gamma_3 = h\gamma_4 = 0.$$

It exists because $\gamma_2 \mapsto \gamma_2 - \varepsilon\mathbf{1}$, with $|\varepsilon| = 1$, $\varepsilon^2 = 0$ and $T\varepsilon = 0$, preserves every generator singular product: their right sides involve only jets of γ_3, γ_4. Taking the coefficient of that constant in the resulting automorphism gives h. It commutes with T and has $h^2 = 0$. In $d\gamma_i$ the only term with undifferentiated γ_2 is $-\gamma_{2,(-1)}T\gamma_i$. The recursion therefore gives $[d, h]\gamma_i = T\gamma_i$ and hence $[d, h] = T_B$ on every state. Extend h by zero on V. The exact coupling gives $hq = -U_2 = L$, so

$$[Q, h] = T_V + T_B. \tag{14.32}$$

For Q-cocycles x, y and $n \neq -1$, sesquilinearity gives

$$x_{(n)}y = Q\Big(-\frac{1}{n+1}(hx)_{(n+1)}y\Big).$$

Thus cohomology has only its (-1)-product, which is unital, graded-commutative, and associative by the vertex identities. This proves an algebraic translation homotopy. A realization by a theory on $\mathbb{R} \times X$ must additionally compare the full coupling and the operations on collision strata. A cooperation on the curved field dual and its comparison with the completed ordered bar dual likewise require maps preserving all those operations, the nonempty vacuum, and the bar unit.

14.8 Completed rational ghost modes

Return to rational exchange and put $R_N = \mathbb{C}[\hbar]/(\hbar^N)$. Let $\mathfrak{g} = \mathfrak{gl}_2$, with

$$B(x, y) = \operatorname{tr}(xy) - \tfrac{1}{2}\operatorname{tr}(x)\operatorname{tr}(y), \qquad P = \sum_{i,j} E_{ij} \otimes E_{ji}.$$

The identity matrix is retained; it is central and lies in the radical of B. Fix a basis E_a and write $[E_a, E_b] = f_{ab}^{c} E_c$, summing repeated colour indices. The notation $\sum_P x \otimes y$ means contraction with the trace tensor P.

Let A be the exterior algebra on c_n^a, $n \in \mathbb{Z}$, with cochain degree one and weight $-n-1$. Give $\hbar$ degree zero and weight -1. For every integer L let I_L be the exterior ideal generated by modes $n \geq L$, and set

$$G = \varprojlim_{L\to\infty} A/I_L, \qquad \widehat{I}_L = \ker(G \to A/I_L).$$

Each quotient is an ordinary polynomial exterior algebra. Thus only finitely many monomials of a completed element survive any fixed upper mode bound. This topology is different from a completion by the total number of generators.

The continuous even loop derivations and translation are

$$L_{x,j} c_n^a = -f_{xb}^{a} c_{n+j}^{b}, \qquad [L_{x,i}, L_{y,j}] = L_{[x,y],i+j}, \qquad \partial c_n^a = -n c_{n-1}^a.$$

Define the first-order product on $G \otimes R_2$ by

$$f * g = fg + \hbar \mu_1(f,g), \qquad \mu_1(f,g) = -\frac{1}{2} \sum_{r\geq 0} \sum_P (L_{x,-r-1} f)(L_{y,r} g). \tag{14.33}$$

Proposition 14.12. *This is a continuous associative unital product, free over R_2. Every $\widehat{I}_L \otimes R_2$ is a closed left ideal. The smooth left modules $F_L = (G \otimes R_2)/(\widehat{I}_L \otimes R_2)$ are jointly faithful; their weight pieces are finite-dimensional and their weights are bounded below.*

Proof. Modulo $\widehat{I}_L$, the right input is a finite exterior polynomial. For sufficiently large r, $L_{y,r}$ kills it in that quotient. After this finite range is fixed, an enlarged bound on the left input suffices. This proves convergence and continuity. The nonnegative right shifts preserve $\widehat{I}_L$, proving the left ideal assertion. For even derivations U, V, the map $(f,g) \mapsto (Uf)(Vg)$ is a Hochschild two-cocycle by the two Leibniz rules. Thus μ_1 is a cocycle and the product is associative modulo $\hbar^2$. Derivations kill 1, which remains its unit; the underlying module is unchanged.

Write ρ_L for the left action and 1_L for the quotient of 1. Then $\rho_L(f)1_L = f \bmod \widehat{I}_L$. The intersection of these kernels is zero. For any fixed polynomial vector only finitely many right shifts occur, so sufficiently large left generator modes kill it as well. There are finitely many nonpositive-weight generators in each quotient and four generators in each sufficiently large positive weight. Exterior monomials give the claimed weight bounds. □

These quotients are modules, not quotient algebras for $*$. For example, in the basis $H = E_{11} - E_{22}$, $E = E_{12}$, $F = E_{21}$, $I = E_{11} + E_{22}$,

$$c_1^E * c_{-1}^H = -\hbar(c_{-1}^H c_0^E + c_0^H c_{-1}^E) \pmod{\widehat{I}_1},$$

although $c_1^E \in I_1$. This one-sided feature is essential when taking transposes.

14.8.1 Fields and the first curved identities

Separate jet states from modes. Let $B_0 = \Lambda(\xi_j^a : j \geq 0)$ with $T\xi_j^a = (j+1)\xi_{j+1}^a$ and $c^a = \xi_0^a$. Each ξ_j^a has cochain degree one and weight j. Its classical fields are $Y_0(b,z)c = (e^{zT}b)c$. The coadjoint operators $\ell_{x,m}\xi_j^a = -f_{xb}^{a}\xi_{j-m}^b$, with negative jets zero, extend by the Leibniz rule. Set

$$s_B(z) = \sum_{m,r\geq 0} (-1)^m \binom{m+r}{m} z^{-m-r-1} \sum_P \ell_{x,m} \otimes \ell_{y,r}. \tag{14.34}$$

This sum is finite on each state tensor. The state fields and exchange to first order are

$$Y_B = Y_0(1 - \tfrac{\hbar}{2} s_B), \qquad S_B = 1 + \hbar s_B.$$

They define a quantum vertex superalgebra with the unchanged vacuum and translation. Indeed translation of the kernel $P/(z + t_1 - t_2)$ gives $[T \otimes 1, s_B] = -\partial_z s_B$, and its Leibniz rule on a product gives the two hexagon identities. Both iterated fields have the same correction

$$1 - \tfrac{\hbar}{2}\big(s_{12}(z_1 - z_2) + s_{13}(z_1) + s_{23}(z_2)\big),$$

so classical associativity gives weak associativity. Reversing the first two inputs changes the sign of s_{12} and gives the specified exchange.

For a state b, substitute the full mode series $c^a(z) = \sum_n c^a_n z^{-n-1}$ and $T = \partial_z$ in its classical exterior expression, obtaining coefficients $b_{b,n} \in G$. At a fixed coefficient, the sum of indices is fixed; an upper bound on all indices bounds them below as well. Hence the sum converges in G. Its field on F_L is

$$Y_L(b, z) = \sum_n \rho_L(b_{b,n}) z^{-n-1}.$$

The covariance identity

$$L_{x,j} b_b(z) = \sum_{m \geq 0} \binom{j}{m} z^{j-m} b_{\ell_{x,m} b}(z)$$

follows first on jets, then on products. At $j = -r - 1$ it turns (14.33) into precisely the half-twisted field formula, proving the module identities for every state. The exchange is nonidentity: $(S_B - 1)(c^E \otimes c^E) = 2\hbar z^{-1} c^E \otimes c^E$.

Fix a level $K \in \mathbb{C}$. On the classical exterior algebra define odd derivations

$$\begin{aligned} D_0 c^c_n &= -\tfrac{1}{2} f^c_{ab} \sum_{p+q=n-1} c^a_p c^b_q, \\ D_1 &= -\tfrac{K}{2} \sum_{r \geq 0, b} (r+1)\big(c^b_{-r-2} L_{E_b, r} + c^b_r L_{E_b, -r-2}\big), \end{aligned} \tag{14.35}$$

and, writing $B_{ab} = B(E_a, E_b)$, curvature coefficients

$$\chi_0 = -K \sum_{r \geq 0, a, b} B_{ab}(r+1) c^a_{-r-2} c^b_r, \qquad \chi_1 = K^2 \sum_{r \geq 0, a, b} B_{ab} \binom{r+2}{2} c^b_{-r-3} c^a_r. \tag{14.36}$$

For a fixed output bound L, the first two maps need only input bounds $2L$ and $2L + 1$ respectively. The curvature sums converge because their right factors have index r. Thus all formulas define continuous maps in the chosen completion.

Theorem 14.13. *The operator $D = D_0 + \hbar D_1$ is an odd derivation of $*$ and, with $\chi = \chi_0 + \hbar \chi_1$, satisfies*

$$D^2 = [\chi, -]_*, \qquad D\chi = 0.$$

For $K \neq 0$, one has $D^2 \neq 0$.

Proof. Pair c_n^a with $E_a t^{-n-1}$. Then D_0 is the loop Chevalley–Eilenberg differential, so $D_0^2 = 0$ and $[D_0, L_{x,j}] = 0$. It differentiates each term of $*$; $\hbar D_1$ is a classical derivation at its order. For the contraction $\iota_{x,j} c_n^a = x^a \delta_{n,-j-1}$, Cartan's identity gives $[D_0, \iota_{x,j}] = L_{x,j}$. The cochain χ_0 evaluates to $-K\, j B(x,y)\delta_{j+\ell,0}$ on (xt^j, yt^ℓ). Invariance of B makes it a cocycle. Thus, with $x_n^\flat = B(x, E_a)c_n^a$,

$$D_0\chi_0 = 0, \qquad \iota_{x,j}\chi_0 = -K\, j x_{j-1}^\flat, \qquad L_{x,j}\chi_0 = -K\, j D_0 x_{j-1}^\flat.$$

Substitution in the antisymmetrization of $\mu_1(\chi_0, -)$ gives $[D_0, D_1]$. The contraction of B with P is the traceless projection; the missing scalar acts by the zero adjoint operator, so no trace current is removed. This proves the curvature commutator. Reindexing the same formula gives $D_1\chi_0 = -D_0\chi_1$, proving the Bianchi identity. All rearrangements are finite after an output cutoff. For nonvanishing, the coefficient of $c_{-6}^E c_1^H c_1^F$ in $D^2 c_{-1}^H$ is $-\hbar K$. □

The classical zero mode of $w_K = \frac{K}{2} B_{ab} c^a T c^b$ is χ_0. The additional term χ_1 is a convergent mode operator. For $K \neq 0$ it is not the zero mode of a finite quadratic local state: on the fixed mode-sum line its antisymmetrized coefficient is proportional to $(q+1)(q+2)\operatorname{sgn}(q+\frac{3}{2})/2$, whereas a finite quadratic differential polynomial has a polynomial coefficient in q. The two half-lines cannot be values of that same polynomial. The common completion permits the operator without turning it into such a local state.

14.8.2 The affine central term from an auxiliary generator

Adjoin an odd generator τ of weight zero with

$$[\tau, g]_s = D_0 g, \qquad \tau^2 = \chi_0.$$

Its normal form is $G \oplus G\tau$: the overlapping reductions agree by $D_0^2 = 0$, $D_0\chi_0 = 0$, and classical centrality of χ_0. Extend the loop derivations by

$$\widetilde{L}_{x,j}\tau = \alpha_{x,j} := -K\, j x_{j-1}^\flat.$$

The identities above preserve both relations, and invariance gives $L_{x,i}\alpha_{y,j} - L_{y,j}\alpha_{x,i} = \alpha_{[x,y],i+j}$. Thus these are actual loop derivations. Twisting their product by the same formula as (14.33) gives

$$\tau \star c + c \star \tau = (D_0 + \hbar D_1)c, \qquad \tau \star \tau = \chi_0 + \hbar\chi_1.$$

The auxiliary generator records the curved data; it is not an extra polarized ghost state.

Let z now denote the central homogenizing generator dual to τ in this quadratic presentation, rather than an insertion coordinate. Its dual relations are

$$[J_x(m), J_y(n)] = z J_{[x,y]}(m+n) + K m B(x,y)\delta_{m+n,0} z^2, \qquad [z, J_x(m)] = 0.$$

The negative transpose of $\widetilde{L}$ is

$$\delta_{x,j} J_y(n) = J_{[x,y]}(j+n) + K\, j B(x,y)\delta_{j+n,0} z, \qquad \delta_{x,j} z = 0.$$

The central summand is the pairing with $\widetilde{L}_{x,j}\tau$. After $z = 1$, this is the affine inner derivation $[J_x(j), -]$. For $j \geq 0$ it descends to the vacuum module and gives

$$\delta_{x,1}(J_y(-1)\mathbf{1}) = K B(x,y)\mathbf{1}. \tag{14.37}$$

Omitting the auxiliary image would lose this vacuum component. The finite quadratic transpose consequently retains the level and scalar affine relation before taking the vacuum quotient.

14.9 The second-order product and its exact range

The second coefficient of the product must correct associativity and remain compatible with translation. Write

$$\mathcal{K} = \sum_{r\ge 0}\sum_{P} L_{x,-r-1}\otimes L_{y,r}, \qquad \kappa_2 = \tfrac{1}{8}\mu_0\mathcal{K}^2,$$

where μ_0 is the classical exterior product. The defect left after κ_2 is a triderivation. On generators it is

$$-\frac{1}{8}\sum_{\substack{i+j+k=-2\\ i<0,\ k\ge 0}} \theta^{xyz}(L_{x,i}u)(L_{y,j}v)(L_{z,k}w), \qquad \theta = [P_{12},P_{23}].$$

An explicit correction will cancel it by its three-mode shifts.

For $\xi\in\mathfrak{g}^*$ write $\xi_n = \xi(E_a)c^a_n$. Let π be the projection onto the annihilator of I, and let Ω_{ab} be the trace Casimir acting coadjointly in factors a,b. Equivalently $\pi = \frac{1}{4}\sum_P x\,y$ in the coadjoint representation. Put $\Sigma_{ab} = \pi_a\pi_b + \Omega_{ab}/2$ and $\{A,B\} = AB + BA$. Define six colour operators

$$\begin{aligned}
X_{12} &= \tfrac{1}{6}\Sigma_{23}\Omega_{12}, & X_{13} &= \tfrac{1}{6}\Sigma_{23}\Omega_{13},\\
X_1 &= \tfrac{1}{6}\pi_1\Omega_{23} + \tfrac{1}{24}\big(\{\Omega_{12}+\Omega_{13},\Omega_{23}\} - \{\Omega_{12},\Omega_{13}\}\big), & &\\
Y_1 &= \tfrac{1}{6}\Sigma_{12}\Omega_{23}, & Y_2 &= \tfrac{1}{6}\Sigma_{12}\Omega_{13},\\
Y_{12} &= \tfrac{1}{6}\pi_3\Omega_{12} + \tfrac{1}{24}\big(\{\Omega_{13}+\Omega_{23},\Omega_{12}\} - \{\Omega_{13},\Omega_{23}\}\big). & &
\end{aligned} \tag{14.38}$$

For shifts $e_1+e_2+e_3=-2$, let $J(e)=\{i : e_i<0\}$. It is a nonempty proper subset of $\{1,2,3\}$. Set $X(e)=X_{J(e)}$ when $1\in J(e)$ and zero otherwise; set $Y(e)=Y_{J(e)}$ when $3\notin J(e)$ and zero otherwise. For a kernel Z, define

$$\Psi_Z((\xi_1)_{n_1},(\xi_2)_{n_2},(\xi_3)_{n_3}) = \sum_{e_1+e_2+e_3=-2}\big(Z(e)(\xi_1\otimes\xi_2\otimes\xi_3)\big)_{n+e},$$

where $(\eta_1\otimes\eta_2\otimes\eta_3)_q = (\eta_1)_{q_1}(\eta_2)_{q_2}(\eta_3)_{q_3}$. Only finitely many terms survive each upper bound.

Define $\beta(u;v\wedge w)=\Psi_X(u,v,w)$ and $\beta(u\wedge v;w)=\Psi_Y(u,v,w)$. To extend it to ordered exterior monomials, choose each subset of one or two factors once. For factorizations $f=\varepsilon u f'$, $g=\varepsilon' v w g'$ add $\varepsilon\varepsilon'\beta(u;v\wedge w)f'g'$. For $f=\varepsilon u v f'$, $g=\varepsilon' w g'$ add $\varepsilon\varepsilon'(-1)^{|f'|}\beta(u\wedge v;w)f'g'$. Here ε,ε' are the signs of moving the selected factors to the front. The colour symmetries make each selected pair alternating, so this defines an exterior operation.

For an equivariant colour operator $A=\sum_i X^i\otimes Y^i$, let $X^i_j\xi_n = (X^i\xi)_{n+j}$ be its extension as an even derivation, and put

$$\pi_A(f,g) = \sum_{r\ge 0,i}(r+1)(X^i_{-r-2}f)(Y^i_r g).$$

On generators this is the expansion of $A/(z-w)^2$. If τ_{12} denotes exchange of the two colour factors, set

$$A_* = \tfrac{1}{2}(1-\tau_{12})(\pi\otimes\pi) - \tfrac{1}{4}\Omega, \qquad A_2 = \tfrac{1}{2}(1+\tau_{12})(\pi\otimes\pi), \qquad A_K = A_* + (K+2)A_2. \tag{14.39}$$

The subscript on A_2 here names this colour operator, not a precision quotient. Define

$$f *_K g = fg + \hbar\mu_1(f,g) + \hbar^2\mu_2^{(K)}(f,g), \qquad \mu_2^{(K)} = \kappa_2 + \beta + \pi_{A_K}. \tag{14.40}$$

Theorem 14.14. *Over R_3, the product (14.40) is continuous, associative, and unital. Translation ∂ and the constant coadjoint operators are derivations. Each $\widehat{I}_L \otimes R_3$ is a closed left ideal, and its smooth quotient modules are jointly faithful. The polarized quotient $B_3 = (G \otimes R_3)/(\widehat{I}_0 \otimes R_3)$ has a quantum vertex superalgebra structure with vacuum 1_0, translation ∂, and exchange $\exp(\hbar s_B)$. On every quotient F_L, each composite state acts by its classical exterior coefficients through the product $*_K$.*

Proof. In the colour representation,

$$[\Omega_{12}, \Omega_{23}] = [\Omega_{23}, \Omega_{13}] = [\Omega_{13}, \Omega_{12}].$$

The operator Σ_{ab} has image symmetric in factors a, b: on their antisymmetric traceless summand the Casimir is -2, so $\pi_a \pi_b + \Omega_{ab}/2$ vanishes. A shift kernel constant on one half-plane, with image symmetric in the two remaining factors, acts by zero after exterior multiplication. Indeed the finite set of surviving mode pairs is stable under exchange and its terms cancel in pairs, including zero at a fixed pair.

The colour identities give

$$X_1 - Y_1 + \tfrac{1}{8}\theta = X_{12} - Y_{12} + \tfrac{1}{8}\theta = X_{13} - Y_2 = \tfrac{1}{12}(\Omega_{23} - \Omega_{12})\Omega_{13}.$$

On the six sign regions these turn the remaining associator into two of the vanishing half-plane kernels just described. The extension rule for β makes its Hochschild coboundary a triderivation: terms selecting factors from only two inputs cancel in pairs, leaving one selected factor from each input. The generator computation therefore proves associativity on all monomials. The additional π_{A_K} is a biderivation and hence a Hochschild cocycle.

The same half-plane cancellation proves $\beta(G, \widehat{I}_L) \subset \widehat{I}_L$. For a right-input polynomial with least mode $p_0 < L$, the bound $M = 3L - 2p_0 + 2$ gives $\beta(\widehat{I}_M, g) \subset \widehat{I}_L$. For the one-plus-two part this follows from the sum of its three output modes; for the two-plus-one part the possible low output lies in a symmetric half-plane and cancels. The other product terms satisfy the same bound because their right shifts are nonnegative. These estimates prove continuity, the left ideals, smoothness, and the extension of associativity to the completion. Evaluation at 1_L again separates coefficients.

For translation, a kernel depending only on $J(e)$ has defect $\sum_i (e_i' + 1) Z(e' + \delta_i)$ on $\sum e_i' = -3$. All terms with nonzero coefficient use the same sign region, and their coefficients sum to zero. This proves translation for β. For the double-pole kernel the boundary term is $(a-1)(a-2)[a=2]A_K = 0$. Thus π_{A_K} also commutes with translation. Equivariance is built into the colour operators.

Finally $\tau_{12} A_K \tau_{12} = A_K$. The two expansions of its double pole therefore agree after clearing the diagonal, giving the generator exchange. Reconstruction from the creating generating fields gives the state algebra. For a fixed module vector with least mode p_0, use the same bound $M = 3L - 2p_0 + 2$ before taking an iterate. A word of r fields with total derivative order d has no power below $-rM - d$, so one power of the associativity denominator suffices independently of output weight. Taylor expansion identifies the iterate with the field of the composite state. This proves the all-state module assertion and extends exchange from generators to all states. □

14.9.1 The derivation and curvature at precision three

Let $\delta : \mathfrak{g}^* \to \mathfrak{g}^* \otimes \mathfrak{g}^*$ be $\delta c^c = -\frac{1}{2} f^c_{ab} c^a \otimes c^b$. Put

$$\Delta_K = [D_0, \mu_2^{(K)}] + [D_1, \mu_1].$$

These brackets denote the failure to be a derivation, with the odd Leibniz sign. Write $\delta(z-w) = \sum_{n \in \mathbb{Z}} z^{-n-1} w^n$ for the formal delta function, and let a superscript + mean the sum of a bilinear expression and its input reversal on odd generators. The diagonal symmetrization vanishes:

$$\Delta_K^+ = -(1 + K/2)\partial(z-w) S_1 + \tfrac{K+2}{2} \partial(z-w) S_1 = 0, \tag{14.41}$$

where $S_1(\xi,\eta) = (\partial_1(\pi\xi\otimes\partial\eta + \pi\eta\otimes\partial\xi))(w,w,w)$. The first term comes from the internal collisions in β and the level term D_1; the second is $[D_0, \pi_{(K+2)A_2}]^+$. The identity follows by replacing the difference of the two double-pole expansions by $\partial_w\delta(z-w)$ and taking the first diagonal Taylor coefficient.

Order generators by mode and then colour. There is a unique continuous odd map vanishing on generators that corrects this defect:

$$D_2(u_1\cdots u_k) = -\sum_{i=1}^{k-1}(-1)^{i-1}u_1\cdots u_{i-1}\Delta_K(u_i, u_{i+1}\cdots u_k). \tag{14.42}$$

To verify well-definedness, exchange two neighbouring odd generators. The odd Hochschild cocycle identity for Δ_K reduces the difference to their symmetrization, which is zero by (14.41). The same argument handles a repeated generator. The first term of the recursion then proves the derivation identity on a generator and a monomial, and the cocycle identity extends it to every pair. The upper mode estimates extend it continuously. The difference of two such maps would be a continuous classical derivation zero on all generators, proving uniqueness.

Define the classical derivation E_2 and the next curvature by

$$\begin{aligned} E_2\xi_n &= -\frac{3K+2}{2}\sum_{p+q=n-3}(p+1)(p+2)(\partial\xi)_{p,q},\\ \chi_2 &= \frac{K(K^2+9K+6)}{2}\sum_{r\ge 0,a,b}\binom{r+3}{3}B_{ab}c^a_{-r-4}c^b_r,\\ D &= D_0+\hbar D_1+\hbar^2(D_2+E_2), \qquad \chi = \chi_0+\hbar\chi_1+\hbar^2\chi_2. \end{aligned} \tag{14.43}$$

The map D is a derivation of $*_K$. The curved identities will follow from a current pairing in every weight, rather than from a finite weight window.

Let $V = V^{KB}(\mathfrak{gl}_2)$, with currents J_x satisfying $Y(J_x,z)J_y \sim KB(x,y)z^{-2}\mathbf{1} + z^{-1}J_{[x,y]}$. For trace-dual bases e_a, e^a of $\mathfrak{sl}_2$, put

$$\begin{aligned} s_A(z) &= \sum_{m,n\ge 0}(-1)^m\binom{m+n}{m}z^{-m-n-1}\sum_a e_a(m)\otimes e^a(n),\\ M^p_b &= \sum_{a,n\ge 0} e_a(-n-1)[e^a,e_b](n+p),\\ a_0(z) &= \sum_{p,q\ge 0}(-1)^p\binom{p+q}{p}z^{-p-q-1}\sum_b M^p_b\otimes e^b(q), \qquad \check{a}_0(z) = \tau_{12}a_0(-z)\tau_{12}. \end{aligned}$$

Each sum is finite on a fixed pair of states. The polar fields at this precision are

$$Y^P = Y\Big(1+\tfrac{\hbar}{2}s_A+\hbar^2\big(s_A^2/8+(a_0+\check{a}_0)/24\big)\Big), \qquad S_A = \exp(-\hbar s_A). \tag{14.44}$$

The diagonal completion a_0 compensates the second-order associator of the half twist. The loop commutator, its primitive coproduct, and the rational Casimir identity give the corresponding associativity and exchange identities coefficientwise. The scalar current is an unchanged commutative differential-polynomial factor: all its nonnegative modes vanish on V.

Define equivariant weight-three states for $x\in\mathfrak{sl}_2$ by

$$\Psi_1(x) = \sum_a [e_a,x](-2)e^a, \qquad \Psi_2(x) = \sum_a \operatorname{sym}(e_a(-1)e^a(-1)x),$$

where sym averages the six orders. Set

$$\theta_K(x) = -\frac{3K^2+33K+22}{72}T^2x + \frac{3}{8}\Psi_1(x) - \frac{1}{6}\Psi_2(x), \quad \theta_K(I) = 0, \quad \widehat{J}_x = J_x + \hbar^2\theta_K(x). \tag{14.45}$$

This is an invertible change on the same polar state space.

Theorem 14.15. *For every level K, smooth polar current module M, and polarized ghost module F_L with $L \geq 0$, put*

$$Q_{M,L} = \sum_{a,j} \widehat{J}_a(j) \otimes \rho_L(c^a_{-j-1}).$$

Then, modulo $\hbar^3$,

$$Q_{M,L}^2 + \mathcal{D}Q_{M,L} + 1 \otimes \rho_L(\chi) = 0, \qquad D^2 = [\chi, -]_{*_K}, \qquad D\chi = 0. \tag{14.46}$$

Here $\mathcal{D}$ differentiates the completed ghost coefficients before their action on F_L; no descent of D to an arbitrary quotient is assumed.

Proof. We give the residue and exchange terms in the tensor state space $W = V^P \otimes B_3$. Let T be its total translation and put $d_{\mathrm{CE}}c^c = -\frac{1}{2}f^c_{ab}c^a c^b$. Define $\tau = \sum_a J_a \otimes c^a$, $\Gamma = (1 \otimes d_{\mathrm{CE}})\tau$, $\widehat{\tau} = \sum_a \widehat{J}_a \otimes c^a$, and let T_A act on the current factor. Define

$$e(c^c) = \tfrac{3K+2}{4} f^c_{ab}(T^2c^a)c^b, \qquad \zeta = \sum_c J_c \otimes e(c^c),$$

$$v_K = -\tfrac{K(K^2+9K+6)}{24} B_{ab}c^a T^3 c^b, \quad w_K = \tfrac{K}{2} B_{ab}c^a T c^b.$$

The classical modes of e are E_2 and the zero modes of w_K, v_K are χ_0, χ_2. Direct residue expansion of (14.44) and the ghost generator field gives

$$\tfrac{1}{2}\widehat{\tau}_{(0)}\widehat{\tau} + (1 \otimes d_{\mathrm{CE}})\widehat{\tau} + w_K = \tfrac{\hbar K}{2} T(\Gamma + w_K) + \hbar^2\Big(-\zeta - v_K + \tfrac{3K+2}{4} T T_A \Gamma\Big). \tag{14.47}$$

This is a finite state calculation. More explicitly, write $[\hbar^i]Y^P(J_a, z)J_b = \sum_\ell z^\ell P^{ab}_{i,\ell}$ and $C_0 = 1$, $C_1 = -\Omega/2$, $C_2 = \Omega^2/8 + A_K$. The order-r residue of $\tau_{(0)}\tau$ is

$$\sum_{i+j=r} \sum_{a,b,u,v} \sum_{\ell=-i-2}^{j-1} (C_j)^{uv}_{ab} P^{ab}_{i,\ell} \otimes \xi^u_{j-\ell-1}\xi^v_0.$$

The sixteen current pairs and these finite intervals, together with the two linear insertions of θ_K, give respectively the coefficients 0, $\frac{K}{2}T(\Gamma + w_K)$, and the last coefficient in (14.47). The cubic and quadratic current terms cancel. Thus the local residue in the desired operator identity is a total translation and has zero zero mode.

There are also exchange contributions. Put $R(x) = \sum_a J_{[x,E_a]} \otimes c^a$, $\vartheta(x) = 1 \otimes x^\flat$, and

$$\mathcal{R} = \sum_P (R(x) \otimes \vartheta(y) - \vartheta(x) \otimes R(y)), \qquad \mathcal{H} = \sum_P \vartheta(x) \otimes \vartheta(y).$$

The tensor exchange is $S_W = \exp(-\hbar s_W)$ for $s_W = s_A - s_B$. Invariance of the canonical tensor gives

$$s_W(z)\tau^{\otimes 2} = Kz^{-2}\mathcal{R} - 2K^2 z^{-3}\mathcal{H}, \qquad s_W(z)^2\tau^{\otimes 2} = 2Kz^{-3}\mathcal{R} - 8K^2 z^{-4}\mathcal{H}.$$

The second equation uses the ghost Casimir eigenvalues $-2, -4$ on the adjoint and invariant colour summands. For a rational tensor term $z^{-p}u \otimes v$, the correction to a mode commutator is computed by

$$\mathfrak{B}_{m,n}(z^{-p}; u, v) = (-1)^p \sum_{r \geq 0} \binom{p+r-1}{r} v_{(n-p-r)}u_{(m+r)},$$

with an additional $(-1)^{|u||v|}$ for the tensor term. The rightmost mode makes the sum finite on each smooth vector. Expanding S_W^{-1} gives both $\hbar\mathfrak{B}(s_W)$ and $\hbar^2\mathfrak{B}(s_W^2)/2$. In the first term the reversed product must itself be retained through first order.

Its classical part cancels D_1 and χ_1. Its first-order part has coefficient

$$-\hbar^2\left(\frac{K}{2}\sum_{c,j} J_c(j)\otimes U^c_{-j-1} + 2K^2\,1\otimes N_3\right),$$

where

$$U^c_n = \sum_{a,b,r\geq 0}\binom{r+2}{2} f^c_{ba}(c^b_{-r-3}c^a_{n+r} - c^a_{n-r-3}c^b_r),$$

$$N_3 = \sum_{a,b,r\geq 0}\binom{r+3}{3} B_{ab}c^b_{-r-4}c^a_r.$$

The second exchange term is its negative. The coefficients follow from the same two Casimir eigenvalues and $\sum_{r+t=u}(r+1) = \binom{u+2}{2}$, $\sum_{r+t=u}\binom{r+2}{2} = \binom{u+3}{3}$. This proves the first identity in (14.46). Every rearrangement is finite after the ghost upper bound and the smooth current bound on a fixed vector.

For the curved identities let $\mathcal{E} = D^2 - [\chi,-] = \hbar^2\mathcal{E}_2$ and $D\chi = \hbar^2 C_2$. The first is a continuous classical derivation. Apply the operator identity to the affine vacuum module and all 1_L; their separation gives an identity of completed coefficients. Its Bianchi identity is

$$\sum_{a,j}\widehat{J}_a(j)\otimes\mathcal{E}(c^a_{-j-1}) + 1\otimes D\chi = 0.$$

The trace ghost is killed by D and is supercentral for $*_K$: every pair or triple correction has a coadjoint action or traceless projection on that input. Fixing a cubic ghost monomial of mode sum s, the remaining coefficient is $J_x(j) + \lambda 1 = 0$ on the classical affine vacuum module, with $x \in \mathfrak{sl}_2$ and $j = -s-5$. For $j < 0$ the vacuum and PBW give $x = 0$. For $j \geq 0$ the vacuum first gives $\lambda = 0$; applying to $J_y(-j-1)\mathbf{1}$ gives $J_{[x,y]}(-1)\mathbf{1} = 0$. The central term is absent because the indices sum to -1. PBW and the zero centre of $\mathfrak{sl}_2$ imply $x = 0$ at every level. Thus $\mathcal{E}_2$ vanishes on all generators and $C_2 = 0$. Continuity proves the two full identities. □

Transposing the quadratic product, differential, and curvature on finite perfect coefficient blocks now gives a current algebra at each precision $1 \leq N \leq 3$. Homogenize the linear and scalar terms with the generator dual to τ, set that generator to one, and take the quotient killing nonnegative current modes on the vacuum. The map to the corrected polar currents satisfies every relation by (14.46). It is an isomorphism on the singleton vacuum module. Indeed the classical affine relations reorder every word, with an error divisible by $\hbar$; iterating this process terminates because $\hbar^N = 0$. Thus ordered PBW words span the source. Their images are the PBW basis of the flat polar module, which proves independence without assuming source flatness. The transposed singleton isomorphism preserves translation, the outgoing vacuum, and every ordered-word coefficient, including the trace direction. Native binary germs and their supported residue maps require the additional comparison described below.

14.9.2 The integral current comparison retains the trace

The polar current algebra also has a field comparison with the integral quantum affine algebra; its trace correction differs from the change (14.45) used in the ghost pairing. Let $\boldsymbol{\mathcal{U}}_K$ be the full matrix quantum affine vacuum algebra with creation matrix $L(v) = I - \hbar j(v)$ and normalized rational matrix

$$R(w) = f(\hbar/w)(I - \hbar P/w), \qquad f(t) = 1 + \tfrac{1}{2}t + \tfrac{5}{8}t^2 + \tfrac{11}{16}t^3 + \tfrac{83}{128}t^4 + O(t^5).$$

The scalar normalization and the integral creation lattice are those of the rational construction in Section 12.8; here the full trace-current sector is retained. Write $j = j^{[0]}$, $Tj^{[n]} = (n+1)j^{[n+1]}$, and

interpret products of entries of $j^{[n]}$ as products in the creation algebra acting on the vacuum. They are not being identified with quantum (-1)-products.

Put $u = J_I$ and $C = \sum_a e_a(-1)e^a$. For the polar completion $a_\mu = a_0 + \mu\partial_z s_A$ define

$$\begin{aligned}\Theta_\mu(x) &= \left(-\frac{K^2}{12} - \frac{K}{24} + \frac{13}{72} + \frac{\mu K}{24}\right)T^2x + \left(\frac{K+1}{8} - \frac{\mu}{24}\right)\Psi_1(x) + \frac{1}{6}\Psi_2(x)\\ &\quad + \frac{1}{2}u(-1)Tx + \frac{1}{4}u(-1)u(-1)x, \qquad x \in \mathfrak{sl}_2,\\ \Theta_\mu(I) &= \frac{1}{2}u(-1)C.\end{aligned}$$

The map on currents

$$J_{E_{ij}} \longmapsto j_{ij} + \frac{\hbar}{2}\sum_a j_{ia}j_{aj} + \hbar^2\widetilde{\Theta_\mu(E_{ij})} \tag{14.48}$$

extends to a graded, equivariant isomorphism of quantum vertex algebras $V_\mu^P \simeq \mathcal{U}_K/\hbar^3\mathcal{U}_K$. The tilde is any lift of the classical PBW state; multiplication by $\hbar^2$ makes it independent of that choice. This comparison fixes the vacuum, commutes with translation, and preserves all fields and exchanges without localizing the level.

Here is how the second coefficient is detected. For a current change θ, its singular-product coboundary is

$$q_\theta(x, y; z) = \operatorname{Sing}_z\big(Y(\theta(x), z)y + Y(x, z)\theta(y)\big) - z^{-1}\theta([x, y]).$$

The residual of the first logarithmic change $j \mapsto j + \hbar j^2/2$ on all sixteen matrix-current pairs is $-q_{\Theta_\mu}$. It follows by expanding the mixed RTT relation through the required scalar order, straightening the creation words, and using the affine commutator for the polar coefficients. Both recursions terminate: the classical commutator shortens a word, and its quadratic correction raises the power of $\hbar$. The exchange residual after the first coordinate change is $s_A^2/2$. Thus (14.48) matches the complete generating products and exchange. Translation-compatible PBW extension and weak associativity then give the equality on every state. Flatness and the identity classical reduction give the inverse by a finite $\hbar$-adic recursion.

In particular, for $y^\circ = y - \operatorname{tr}(y)I/2$, the uncorrected trace residual is

$$q(I, y; z) = -(K+2)\left(\frac{u(-1)y^\circ}{z^2} + \frac{T(u(-1)y^\circ)}{z}\right), \qquad q(y, I; z) = -(K+2)\frac{u(-1)y^\circ}{z^2}.$$

The Casimir identity $\operatorname{Sing}_z Y(C, z)y^\circ = 2(K+2)(z^{-2}y^\circ + z^{-1}Ty^\circ)$ shows that $I \mapsto u(-1)C/2$ removes exactly these terms. It is essential away from $K = -2$. A change only on traceless currents would leave the trace field comparison false.

14.9.3 The next associative order

The specified precision-three product has a genuine obstruction at noncritical level. We state its admissible class before the conclusion. Let ν be a continuous even biderivation of the classical exterior product, of degree zero and weight two, equivariant for the constant coadjoint action. Require its symmetrization on two generating fields to be supported on their diagonal. Allow a continuous odd first-order correction D' of degree one and weight one with $[D_0, D'] = [D_0, D_1]$.

Proposition 14.16. *Suppose $D_0 + \hbar D' + \hbar^2 D_2'$ is a derivation of $\mu_0 + \hbar\mu_1 + \hbar^2(\kappa_2 + \beta + \pi_{A_*} + \nu)$ modulo $\hbar^3$. If this product has any parity-preserving bilinear associative correction at order $\hbar^3$, then $K = -2$. In particular the fixed product $*_K$ has no associative extension modulo $\hbar^4$ when $K \neq -2$.*

Proof. Pass to the quotient by the closed ideal of trace ghosts. Equivariance and diagonal support express the relevant symmetric colour coefficient as $Y = y_1P_1 + y_3P_3 + y_5P_5$, where P_i project onto the dimensions 1, 3, 5 of the tensor square of the adjoint representation. The correction $F = D' - D_1$ is a D_0-closed odd derivation. Its finitely many coefficients used here may be averaged over $SU(2)$, so it may be taken equivariant. Its remaining component has the form $F(\xi_n) = \sum_{p+q=n-2} a(p,q)(\delta\xi)_{p,q}$ modulo trace ghosts, with $a(p,q) = a(q,p)$.

Evaluate the second-order derivation defect on $u = c_{-1}^H$ twice and the third-order associator on u three times. Their unknown correction terms vanish: $u^2 = 0$, while $D_2'u$ and $\mu_3(u,u)$ are even and commute classically with u. Write h, e, f for the three traceless ghost colours and take the coefficients of $m_2 = e_{-2}f_{-2}h_{-1}$, $m_3 = h_{-3}e_{-1}f_{-1}$, and $m_4 = f_{-4}h_{-1}e_{-1}$. The $[D_0, F](h_{-2})$ coefficient at $h_{-2}e_{-2}f_{-1}$ is $2(a(-3,-1) - a(-2,-1))$. The m_2, m_3 coefficients of $[D_0, \nu](u,u)$ are $(-y_1 + 3y_3 + y_5)/3$ and $-(y_1 + 2y_5)/3$; the corresponding $[F, \mu_1]$ coefficients are zero and $2(a(-2,-1) - a(-3,-1))$. The third-order m_4 coefficient is $(2y_1 - 3y_3 + y_5)/12$. Together with the base level contribution $K + 2$ at m_3, these yield

$$\begin{aligned} a(-3,-1) &= a(-2,-1),\\ -y_1 + 3y_3 + y_5 &= 0,\\ K + 2 - (y_1 + 2y_5)/3 &= 0,\\ 2y_1 - 3y_3 + y_5 &= 0. \end{aligned}$$

They can be computed from the double-pole expansion and the three Casimir eigenvalues $-4, -2, 2$; the first equality removes the possible contribution of F to the third line. Adding the second and fourth lines gives $y_1 + 2y_5 = 0$, and the third gives $K = -2$. For $*_K$, take $\nu = (K+2)\pi_{A_2}$ and $D' = D_1$. An arbitrary bilinear extension would have a parity-preserving component solving the same linear extension equation, so the fixed-product conclusion excludes it as well. □

At $K = -2$ there is a positive associative extension. If a_3 is the third-order associator of $*_{-2}$, its total symmetrization on odd generators is zero. Substitution of the colour operators into the rational three-point terms proves their cancellation; the differences of Laurent expansions are derivatives of diagonal delta functions, and their contact terms cancel by the same odd exchange and Casimir identities. This is an all-mode rational identity.

Here is an explicit continuous primitive of that cocycle. Order generators by mode and colour. For $v > x$ put $\mathcal{S}_{vx}(F) = a_3(F, v, x) + a_3(F, x, v)$ and put $\mathcal{S}_{xx}(F) = a_3(F, x, x)$. For $g_1 < \cdots < g_k$ set

$$\rho_x(g_1 \cdots g_k) = -\sum_{j : g_j \geq x} (-1)^{k-j} \mathcal{S}_{g_j x}(g_1 \cdots g_{j-1}) g_{j+1} \cdots g_k,$$

and define

$$\begin{aligned} \mu_3(f, g_1 \cdots g_k) = \sum_{j=1}^{k} \big(& a_3(f, g_1 \cdots g_{j-1}, g_j) - f\rho_{g_j}(g_1 \cdots g_{j-1}) \\ & + \rho_{g_j}(f g_1 \cdots g_{j-1}) \big) g_{j+1} \cdots g_k. \end{aligned} \tag{14.49}$$

The Hochschild cocycle identity and $\operatorname{Sym} a_3 = 0$ show that exchanging adjacent generators leaves this formula consistent with the exterior signs. Induction on the second word, then the third input of its coboundary, gives $\delta_0\mu_3 = a_3$. The right ideal and enlarged first-input bounds for a_3 give the same bounds for these recursions, so μ_3 extends continuously and preserves every right-input upper mode ideal. Therefore $*_{-2} + \hbar^3\mu_3$ is associative modulo $\hbar^4$. The ordering used in this primitive is not preserved by translation. A translation-compatible field extension and the next curved differential remain further requirements. The obstruction above also retains its stated admissible class; it does not exclude every possible presentation of a dual field theory.

14.10 Finite adjunction before stabilization

The singleton comparison does not determine a binary field operation. That operation uses a relative germ, its quadratic transpose, and the adjoint in one specified input. We first construct these maps classically and then retain the loop cutoff during the first-order transpose.

Let $V = V^{KB}(\mathfrak{gl}_2)$ and $V_{\le a} = \bigoplus_{0\le d\le a} V_d$. The classical ghost germ sends

$$\xi_j^{\alpha} \longmapsto \frac{1}{j!}\partial_u^j \sum_n c_n^{\alpha} u^{-n-1}.$$

Its finite first-input transpose is

$$A_x^a(m;u) = \sum_{j=0}^{a} \binom{m}{j} u^{m-j} J_x(j)|_{V_{\le a}}. \tag{14.50}$$

Negative m occur too: their coefficients pair with the full annihilation germ before transposition. On $\mathscr{H}_a = \mathrm{Hom}(V_{\le a}, V((u)))$, finite tensor–Hom adjunction gives

$$R_x^a(m)F = J_x(m)F - FA_x^a(m;u). \tag{14.51}$$

The minus sign is the antipode action on the dual input. Let $\epsilon_a(v) = e^{uT}v$ be the creation kernel. Write U_K for the enveloping algebra of affine mode words with central generator acting by K.

Proposition 14.17. *The cyclic module $\mathscr{K}_a = U_K\epsilon_a$ is canonically isomorphic to V, by a map Ψ_a with $\Psi_a(\mathbf{1}) = \epsilon_a$. Its inverse is evaluation on the first vacuum. These maps commute with restriction of the first-input bound, and they retain the affine linear and scalar relations.*

Proof. Vandermonde's identity gives $[A_x^a(m), A_y^a(n)] = A_{[x,y]}^a(m+n)$. The indices in these first-input operators are nonnegative, so their central cocycle is zero. Negative precomposition is a representation, because precomposition reverses order. Hence (14.51) satisfies the affine relation, with its central term occurring once in the output. For $m \ge 0$,

$$J_x(m)e^{uT}v = e^{uT}\sum_{j=0}^{m}\binom{m}{j}u^{m-j}J_x(j)v,$$

so $R_x^a(m)\epsilon_a = 0$. The vacuum quotient supplies Ψ_a. Every $A_x^a(m)$ kills the first vacuum, so evaluation there intertwines $R_x^a(m)$ with $J_x(m)$ and returns $\mathbf{1}$ from ϵ_a. Its composite with Ψ_a is the identity, proving injectivity; surjectivity holds onto the cyclic module by definition. Restriction preserves the first-input matrices and the creation kernel. Smoothness follows from the resulting identification with V. □

For output projection $\pi_c : V \to V_{\le c}$, the resulting coefficient block is

$$\Gamma_{a,b,c} : V_{\le b} \longrightarrow \mathrm{Hom}\big(V_{\le a}, V_{\le c}\otimes\langle u^{-a-b},\dots,u^{c}\rangle\big), \qquad \Gamma_{a,b,c}(w)(v) = \pi_c\Psi_a(w)(v).$$

The range is finite: if the inputs have weights i, j and the output has weight k, the Laurent power is $k-i-j$. The recursion in negative creation words uses finitely many precomposition indices, whose total is at most i, and finitely many terms of e^{uT} after output projection. For at most d additional translations or derivatives, put $s_0 = a+b+c+d+2$. All required ghost generators lie in $[-4s_0-1, 4s_0-1]$; the quadratic rows can be read with upper bound $4s_0+1$ and weight interval $[-8s_0-2, 8s_0+2]$. These bounds follow by straightening the finite current words and then transposing their linear terms. They provide a common finite source for each matrix, rather than assuming that a fixed cutoff is preserved by the differential.

The compatible operation $\mu(v,u)w = \Upsilon(w)(v)$ satisfies

$$\mu(J_x(-1)\mathbf{1},u)J_y(-1)\mathbf{1} = J_y(-1)e^{uT}J_x(-1)\mathbf{1} + u^{-1}e^{uT}J_{[x,y]}(-1)\mathbf{1} + Ku^{-2}B(x,y)\mathbf{1}.$$

The same recursion on all creation words identifies it with the classical affine field, including $\mu(\mathbf{1},u)w = w$, $\mu(v,u)\mathbf{1} = e^{uT}v$, and its reversed operation. Consequently the two ordered coproduct transposes give separate commuting product comparisons. The actual residue identity is equality of $\mathrm{pp}_u(f\mu(v,u)w)$ with the corresponding field residue. For pole order p in f, only output weights below $a+b+p$ contribute. Spectators and any fixed finite number of normal derivatives are therefore covered by a common finite enlargement. This also gives the exchanged 13 | 2 residue and compatibility with finite perfect biduality at the classical precision.

14.10.1 *The two finite terms that cancel*

Let H be the enveloping Hopf algebra of the centrally extended loop algebra, with primitive generators $x[m]$ and central generator $\mathbf{c}$. Its bracket is $[x[m],y[n]] = [x,y][m+n] + mB(x,y)\delta_{m+n,0}\mathbf{c}$. On ghosts $\mathbf{c}$ acts by zero, and on currents by K. Write S for the antipode of H. The Hopf counit here belongs to H; no character of the fixed-level affine algebra is required. For a loop cutoff M put

$$r_M = \sum_{0\le r<M}\sum_P x[-r-1]\otimes y[r], \qquad \sigma_M^{\pm} = 1 \pm \tfrac{\hbar}{2}r_M.$$

These are counital twists modulo $\hbar^2$, since their linear twist equations are the sums of the three placements of primitive tensors. Use $\Delta_\sigma = \sigma\Delta\sigma^{-1}$ and $m_\sigma = m\sigma^{-1}$.

Twist the auxiliary relative ghost algebra, including its shared τ, by σ_M^+. Its product is the finite sum in (14.33); commutator with τ and its square give the finite D and χ. The generator-linear relative germ remains an equivariant quadratic map. Finite inverse transposition changes the twist to σ_M^-, and after setting the generator dual to τ to one its relative current map is

$$\mathsf{J}_x(m) \longmapsto \mathbb{A}_x^a(m;u)\otimes 1 + 1\otimes J_x(m)$$

into the twist of $(U(\mathfrak{g}[t]/t^{a+1})\otimes U_K)[u,u^{-1}]$. Here $\mathbb{A}$ is (14.50) with formal truncated-loop generators in place of their operators. The whole relative algebra is twisted, including its mixed rows. The first factor has central level zero and the second has level K.

Take the right dual of the finite first-input module $U = V_{\le a}\otimes R_2[u,u^{-1}]$, with evaluation in the order $U\otimes U^{\mathrm{r}}\to R_2[u,u^{-1}]$. Its action uses the inverse antipode. For $\sigma = \sigma_M^-$,

$$Q_M = m_H(1\otimes S)(\sigma) = 1 + \tfrac{\hbar}{2}\sum_{r<M,P} x[-r-1]y[r], \qquad S_\sigma^{-1}(h) = S^{-1}(Q_M)S^{-1}(h)S^{-1}(Q_M)^{-1}.$$

The right-dual comparison contributes precomposition in the order $A_y^a(r)A_x^a(-r-1)$. The tensor comparison contributes negative precomposition on the first factor and the output current on the second. Their sum is the actual adjoint comparison

$$\begin{aligned}\Theta_M(F) &= F + \tfrac{\hbar}{2}\sum_{r<M,P} FA_y^a(r)A_x^a(-r-1)\\ &\quad - \tfrac{\hbar}{2}\sum_{r<M,P} J_y(r)FA_x^a(-r-1) = F - \hbar C_M^a F,\\ C_M^a F &= \tfrac{1}{2}\sum_{r<M,P}(R_y^a(r)F)A_x^a(-r-1).\end{aligned} \tag{14.52}$$

The first sum is fixed by right-dual evaluation, not a choice of normalization. Its inverse is $1 + \hbar C^a_M$.

Theorem 14.18. *At every level K, the first-order native quadratic germ, with its full differential and curvature, has a finite adjunction on each coefficient block. Its stabilized cyclic kernel is*

$$W_u(w)(v) = \Upsilon(w)(v) + \frac{\hbar}{2} \sum_{m,n\ge 0,P} (-1)^m \binom{m+n}{m} u^{-m-n-1} \Upsilon(J_y(n)w)(J_x(m)v). \tag{14.53}$$

It preserves both cyclic vacua and all native quadratic, linear, and scalar relations. Its finite coefficient blocks are compatible with restriction, inclusion, projection, and finite translation and derivative enlargements.

Proof. At cutoff M, twisting the classical equivariant adjunction leaves its underlying map Υ_a unchanged. Applying the actual right adjoint comparison (14.52) gives $W_{a,M} = (1 + \hbar C^a_M)\Upsilon_a$. The finite twisting functor preserves the module law, so this map satisfies the full transposed relative relations, including those containing the differential and curvature.

On $w \in V_{\le b}$, $R^a_y(r)\Upsilon_a(w) = \Upsilon_a(J_y(r)w)$ vanishes for $r > b$. Thus $W_{a,M}(w)$ stabilizes as soon as $M > b$, before output projection. Expand $A^a_x(-r-1)$ using $\binom{-r-1}{m} = (-1)^m\binom{m+r}{m}$ to obtain (14.53). Only $m \le a$, $n \le b$ contribute. Every $A^a_x(m)$ kills the first vacuum, and $R^a_y(r)\Upsilon_a(\mathbf{1}) = 0$ for $r \ge 0$; hence

$$W_u(w)(\mathbf{1}) = w, \qquad W_u(\mathbf{1})(v) = e^{uT}v.$$

Evaluation is the inverse onto the cyclic image, as in the classical case. Trace states remain, although their correction operators are zero.

A word of current indices $m_1, \dots, m_d$ acting on weight at most b has intermediate weight at most $b + 1 + \sum_i \max(0, -m_i)$ at this precision. Choose M above this bound, and enlarge the ghost input block for its finite quadratic rows. Then the native action, its adjoint, and all cyclic comparisons for the word stabilize together. On input weights i, j and output weight k, the first-order Laurent exponent is $k - i - j - 1$. Thus the finite projected range is $[-a - b - 1, c]$. These bounds prove compatibility on common enlargements and with every finite collection of derivatives. Stabilization is asserted on the cyclic image; no action on the entire Hom space is needed. □

The cancellation is visible on $v_E = J_E(-1)\mathbf{1}$ at the second vacuum. Each of the two sums in (14.52) has u^{-1} coefficient $2Mv_E$ before subtraction: the zero-mode trace Casimir is 4 on the adjoint representation. Their difference is zero, as required by $W_u(\mathbf{1})(v_E) = e^{uT}v_E$. Dropping the right-dual term would leave a coefficient growing with the cutoff. For $F = E_{21}$ and $H = E_{11} - E_{22}$, the first-order root coefficients are

$$[u^{-3}\mathbf{1}](W_u - \Upsilon)(v_F)(v_E) = -\hbar K(K+2), \qquad [u^{-2}v_H](W_u - \Upsilon)(v_F)(v_E) = -\hbar(K+1).$$

Both the scalar curvature contribution and the right-dual contribution are needed; no denominator excludes zero or critical level.

The translation identities are

$$TW_u(w) - W_u(w)T = W_u(Tw), \qquad \partial_u W_u(w)(v) = W_u(w)(Tv).$$

They identify the singleton coefficient diagrams as follows. For finite PBW subsets $U \subset W$ with $T\langle U\rangle \subset \langle W\rangle$, let ι_{UW} be inclusion and T_{WU} the finite translation matrix. The perfect coefficient transpose of the right-$\mathcal{D}$ Spencer complex gives the pro diagram

$$\left\{ \left[\mathcal{D}^{W^*}_{R_N} \xrightarrow{\partial\iota^{\mathrm{t}}_{UW} - T^{\mathrm{t}}_{WU}} \mathcal{D}^{U^*}_{R_N} \right] \right\}_{U,W}, \qquad N = 1, 2, \tag{14.54}$$

in degrees zero and one. The right module convention is $(f\,dz \otimes v)\partial = -f'\,dz \otimes v - f\,dz \otimes Tv$. Formal transposition sends ∂ to $-\partial$, and the singleton suspension supplies the displayed degrees. Each map is made before passing to the pro diagram; an ordinary inverse limit of these complexes is not substituted for it. The outgoing vacuum map to $\omega[-1]$ is preserved. It remains distinct from the empty convolution unit.

The classical all-state comparison gave both ordered residue diagrams. The first-order theorem above gives the native finite adjunction and its singleton diagrams. To identify the two quantum ordered products one must still compare their separately typed tensor constraints, both vacuum actions, arbitrary spectators, normal derivatives, and the exchanged $13 \mid 2$ residue. At precision three the native adjunction must also incorporate (14.40), (14.43), and (14.45), reducing to this first-order map. The all-mode singleton theorem does not replace these binary requirements.

14.11 The contact term of supported division

Let $a = z_2$, $w = z_1 - z_2$, and $\kappa \neq 0$. Let $\widehat{A}_\kappa$ be the degreewise energy completion of the commutative dg algebra on x_n of cochain degree zero and ξ_n of cochain degree one, for $n \geq 0$, with

$$dx_n = \kappa(n+1)\xi_{n+1}, \qquad d\xi_n = 0, \qquad Tx_n = (n+1)x_{n+1}, \quad T\xi_n = (n+1)\xi_{n+1}.$$

Their energies are $n+1$ and n. Extend scalars to $\mathbb{C}[a]$ before completing, so each energy coefficient remains polynomial in a. A first subscript i denotes the copy at z_i. On the disjoint locus, put $\theta = dw/w$ and place differential forms before field coefficients. In the total de Rham complex the charge generators $y_i = \xi_{i,0} + \kappa^{-1} dz_i\, x_{i,0}$ are closed. Let $\pi(y_i) = \epsilon_i$, where ϵ_i are the odd generators of the exterior charge algebra. The charge comparison with $F_2(\epsilon_1, \epsilon_2) = 0$ and $F_2(\epsilon_2, \epsilon_1) = -\kappa^{-1}\theta$ differs from the binary roof representative

$$C_2(y_1, y_2) = \frac{da}{\kappa w}, \qquad C_2(y_2, y_1) = -\frac{da + dw}{\kappa w}$$

by opposite values $\beta, -\beta$, where $\beta = da/(\kappa w)$. Their sum vanishes on commuting charge insertions, but

$$d\beta = -\frac{dw \wedge da}{\kappa w^2}.$$

These two values do not define a homotopy on the whole bar construction. Extending their division by w to a supported coefficient requires its actual polynomial module action.

Fix a finite-dimensional space V in cochain degree zero and commuting endomorphisms A, B. Set $\Sigma = A + B$ and $P_\Sigma = (\widehat{A}_\kappa \otimes V, d + \xi_0\Sigma)$. For $e_n = [w^{-n}]$, $n \geq 1$, and $e_0 = 0$, the supported carrier is

$$\mathcal{M}_{A,B} = \bigcup_{r \geq 1} \bigoplus_{n=1}^{r} e_n P_\Sigma, \qquad we_n = e_{n-1}. \tag{14.55}$$

The field energies are completed at each fixed pole bound. Each vector has one finite pole bound independent of its energy; the union and the energy product are not interchanged.

In its original normal frame, the differential and connections are

$$D(w) = d + c(w)A + \xi_0 B, \qquad c(w) = \sum_{n \geq 0} w^n \xi_n,$$

$$\nabla_w = \partial_w + \kappa^{-1} j(w) A, \qquad j(w) = \sum_{n \geq 0} w^n x_n,$$

$$\nabla_a = \partial_a + T + \kappa^{-1}(j(w)A + x_0 B).$$

Only finitely many Taylor coefficients act at each pole bound. The normal gauge

$$\Gamma_A(w) = \exp\left(\frac{A}{\kappa}\sum_{n\geq 0}\frac{w^{n+1}x_n}{n+1}\right) = \sum_{n\geq 0} w^n\gamma_n \tag{14.56}$$

has $\gamma_0 = 1$ and energy n in γ_n. Multiplication by it intertwines these operators with $d + \xi_0\Sigma$, ∂_w, and $\partial_a + T + \kappa^{-1}x_0\Sigma$.

In the normal frame let $S(e_n m) = e_{n+1}m$ and define

$$S_A = \Gamma_A^{-1}S\Gamma_A, \qquad R_A(s) = [e_1](\Gamma_A s), \qquad \iota(m) = e_1 m.$$

The bracket $[e_1]$ denotes coefficient extraction, equivalently the normalized covariant residue of $dw \wedge s$. The shift uses the unique negative-power representative of a principal part. It is not multiplication by a Laurent function on the quotient.

Proposition 14.19. *The exact formulas and identities are*

$$\begin{gathered} S_A(e_n m) = e_{n+1}m - e_1\gamma_n m, \qquad R_A(e_n m) = \gamma_{n-1}m, \\ wS_A = 1, \qquad S_A w = 1 - \iota R_A, \qquad [w, S_A] = \iota R_A, \qquad R_A\iota = 1, \\ [D(w), S_A] = 0, \qquad [\nabla_a, S_A] = 0, \qquad [\nabla_w, S_A] = -S_A^2. \end{gathered} \tag{14.57}$$

The contact operator ιR_A is an idempotent cochain map. The shift is continuous at each pole bound through all energies, raises that bound by one, and does not lower energy.

Proof. On $e_n m$, the gauge retains precisely its first n coefficients. Therefore $S\Gamma_A(e_n m) = \Gamma_A(e_{n+1}m) - e_1\gamma_n m$. Since $\Gamma_A^{-1}(e_1 v) = e_1 v$, this proves both explicit formulas. At pole bound r there are only r gauge coefficients; at a fixed output energy each uses finitely many input energies. This proves continuity in the stated order of completion.

In the normal frame $wS = 1$ and $Sw = 1 - e_1[e_1]$. Conjugate by Γ_A, which commutes with w, to obtain the contact identities and idempotence. The normal differential and tangential connection have no w-dependence, so they commute with S. Finally $\partial_w e_n = -ne_{n+1}$ gives $[\partial_w, S]e_n = -e_{n+2} = -S^2 e_n$. Conjugation proves all the connection identities and cochain compatibility. □

For instance, $\gamma_1 = \kappa^{-1}Ax_0$ and $\gamma_2 = \frac{1}{2}(\kappa^{-1}Ax_1 + \kappa^{-2}A^2x_0^2)$. Thus the differential-compatible shift already has a field correction on $e_1 v$; its commutator with w there is still $e_1 v$.

Theorem 14.20. *If $\ker(A + B) \neq 0$, then ιR_A is not null-homotopic for the internal differential, even after forgetting the module actions. Consequently the cochain-homotopy class of S_A contains no $\mathbb{C}[a, w]$-linear cochain map and cannot be the linear term of an A_∞ endomorphism of that module. The obstruction persists on a quasi-isomorphic resolution with the corresponding module action and maps. It applies to $B = -A$ without a nilpotence assumption on A.*

Proof. For $0 \neq v \in \ker(A + B)$, the vector $e_1 v$ is closed: positive powers of w kill it and the remaining differential is $e_1\xi_0(A + B)v = 0$. The complex has no negative cochain degrees, so this vector is not a boundary. The contact operator fixes it. Hence its cohomology action is nonzero.

If $K - S_A = Dh + hD$, then $[D, w] = 0$ gives $[w, K] - [w, S_A] = D[w, h] + [w, h]D$. A module-linear K has $[w, K] = 0$, contradicting the nonzero cohomology action of $[w, S_A]$. For an A_∞ module map, the identity with one algebra input w similarly says that the commutator of its linear term with w is the differential of its next component. The same contradiction applies. Compatible quasi-isomorphisms conjugate the cohomology maps and cannot remove this nonzero action. □

This is an obstruction to the specified class of supported division. The two Laurent values $\beta, -\beta$ do not force an arbitrary relative comparison homotopy into that class. The remaining binary problem is to solve

$$d_{\mathrm{rel}} H = C - F\pi$$

in the complex of continuous module cochains retaining finite Taylor action and residue, or to prove that the actual right-hand side has a nonzero class there. Its other bar components and polynomial representatives must be allowed before asserting a universal obstruction. A solution must then be compatible with every arity, both ordered charts, spectators, all finite normal derivatives, vacuum insertion, and the normal gauge. The transverse arity and the carrier's pole bound remain independent parameters throughout.

Chapter 15

Compact Hall Geometry on a K3 Product

Translation along an elliptic curve removes a common position while preserving the positions of objects relative to one another. An extension depends on these relative positions: sheaves in different fibres have no extensions, whereas sheaves in the same fibre can have both deformations and obstructions. A Hall construction on the translation quotient must therefore retain one elliptic torsor throughout its flag correspondences.

We develop this geometry for $X = S \times E$, where S is a smooth projective complex K3 surface and E is an elliptic curve. The disjoint point locus has explicit self-dual coefficients and finite trace maps. Compact rational-curve pairs give an oriented critical model with computable cohomology. At a polystable collection, cyclic deformation theory gives one formal model for all finite flags. These constructions also expose the data needed at collisions: scalar trace complexes, paired orbit directions, square-root sewing and supported coefficient maps. Their compatibility on the full compact extension stack is the global comparison problem.

Fix nonzero holomorphic forms $\omega_S \in H^0(S, K_S)$ and $\omega_E \in H^0(E, K_E)$, and use their product as the volume form on X. Coherent complexes are over $\mathbb{C}$; constructible complexes have rational coefficients. All shifts are cohomological, so $V[a]^j = V^{j+a}$. Derived pullbacks and tensor products retain this convention. Put $W = T_0 E$.

15.1 One translation torsor for a whole flag

Let $\mathcal{M}_\alpha$ be an E-invariant moduli stack of sheaves of charge α, and let $\mathcal{Q}_\alpha = [\mathcal{M}_\alpha / E]$. The morphism $\mathcal{Q}_\alpha \to BE$ remembers the principal E-bundle in the definition of the quotient stack. An object of $\mathcal{Q}_\alpha$ over a base B is thus an E-torsor $P \to B$ together with an equivariant family on P.

Proposition 15.1. *There is a natural equivalence*

$$\mathcal{Q}_\alpha \times_{BE} \mathcal{Q}_\beta \simeq [(\mathcal{M}_\alpha \times \mathcal{M}_\beta)/E_{\mathrm{diag}}].$$

Exact sequences on the common torsor give a correspondence

$$\mathcal{Q}_\alpha \times_{BE} \mathcal{Q}_\beta \xleftarrow{a} \mathcal{H}_{\alpha,\beta} \xrightarrow{p} \mathcal{Q}_{\alpha+\beta}. \tag{15.1}$$

Finite filtrations identify all successive compositions of these correspondences, compatibly with refinement. The zero object gives the relative unit BE.

Proof. An object of the fibre product consists of two torsors, their identification, and the two equivariant families. Transport along the identification makes this one torsor with a pair of families. This is precisely the diagonal quotient, including morphisms and descent over B. An exact sequence in that family defines (15.1).

For three successive quotients, either composite correspondence classifies a filtration $0 \subset F_1 \subset F_2 \subset F_3$ on one torsor. A further refinement inserts further subobjects into that same filtration. Forgetting specified levels is independent of the order in which the other levels are forgotten. This proves the assertions for every finite flag. The zero sheaf is fixed by E, so its quotient is BE; inserting it makes no additional choice. □

The relative product is already necessary for a single point. Let $\mathfrak{M}_1$ denote the stack of flat length-one sheaves on X. Then

$$\mathfrak{M}_1 = X \times B\mathbb{G}_m, \qquad [\mathfrak{M}_1/E] \times_{BE} [\mathfrak{M}_1/E] \simeq S^2 \times E \times B\mathbb{G}_m^2. \tag{15.2}$$

To see the first statement in families, the support of a flat length-one sheaf is finite over the base. Its pushforward is a line bundle L. The support algebra acts faithfully on L, and its action identifies it with the scalar algebra of the base. The support is therefore the graph of a map to X, and the sheaf is that graph's pushforward of L. This construction respects isomorphisms and descent. For the second statement, write the two elliptic coordinates as (e_1, e_2) and quotient by their common translation. The surviving coordinate is $e_2 - e_1$.

15.2 Disjoint point coefficients

Let $U_n \subset X^n$ be the variety of ordered pairwise distinct points. The torus $T_n = (\mathbb{G}_m)^n$ acts trivially and records the automorphisms of the coefficient lines. The symmetric group $\mathfrak{S}_n$ permutes points and torus factors. Put

$$\mathfrak{Q}_n = [U_n/(E \times (T_n \rtimes \mathfrak{S}_n))], \qquad \mathfrak{Q}_0 = BE.$$

After trivializing the elliptic torsor, the morphism $q_n : \mathfrak{Q}_n \to BE$ becomes the smooth stack $[U_n/(T_n \rtimes \mathfrak{S}_n)]$, of complex dimension $3n - n = 2n$. Complex orientation consequently gives

$$\omega_{q_n} = \mathbb{Q}_{\mathfrak{Q}_n}[4n], \qquad C_n = \mathbb{Q}_{\mathfrak{Q}_n}[2n], \qquad C_n \otimes C_n \simeq \omega_{q_n}. \tag{15.3}$$

Here ω_{q_n} is the relative dualizing complex. Thus C_n is self-dual for relative Verdier duality and $C_0 = \mathbb{Q}_{BE}$. These formulas specify smooth coefficients on the disjoint locus.

Let $D_{m,n} \subset \mathfrak{Q}_m \times_{BE} \mathfrak{Q}_n$ be the open locus of disjoint supports, and restrict the extension correspondence to

$$D_{m,n} \xleftarrow{a} \mathfrak{F}_{m,n} \xrightarrow{p} \mathfrak{Q}_{m+n}.$$

Proposition 15.2. *The map a is an equivalence, and p is representable finite étale of degree $\binom{m+n}{m}$. There is a canonical identification*

$$a^*(C_m \boxtimes_{BE} C_n) = p^* C_{m+n}.$$

The adjunction and finite trace maps have composite

$$C_{m+n} \longrightarrow p_* p^* C_{m+n} \xrightarrow{\mathrm{Tr}_p} C_{m+n}, \qquad \binom{m+n}{m} \mathrm{id}\,.$$

They are adjoint for the relative pairings, and their refinement identities hold for every finite flag, including empty blocks.

Proof. Derived Hom vanishes between sheaves with disjoint support. Every extension therefore splits uniquely. A subobject of length m in a fixed middle sheaf chooses an m-element subset of its support. After an étale ordering of that support, p is the constant cover by those subsets. This proves both geometric assertions. The coefficient shifts agree because $2m + 2n = 2(m + n)$.

On that cover, the first map repeats a section on every sheet, and trace sums its values. Their composite and adjunction follow from summing $\binom{m+n}{m}$ identical values and from $\sum_I u v_I = u \sum_I v_I$. A refined flag is an ordered partition of the support. Every parenthesization sums over the same partitions, which proves all refinements at once. An empty block contributes one choice and the coefficient C_0. □

The scalar automorphisms also enter the determinant line. After fixing a basis of W, its pullback to U_n is $L_n = K_{U_n} \otimes \det \mathbb{C}^n$. A permutation acts by sgn^3 on the support volume and by sgn on the torus determinant. Their product is one. If λ_n places the support factors before the scalar factors, binary sewing has sign $(-1)^{mn}$. The generators

$$\widetilde{\lambda}_n = (-1)^{n(n-1)/2}\lambda_n \tag{15.4}$$

sew with coefficient one, since $\binom{m+n}{2} = \binom{m}{2} + \binom{n}{2} + mn$. The same identity gives every associative regrouping.

15.2.1 A two-sheet coefficient cannot be replaced by a scalar

Choose a nonzero two-torsion point $\tau \in E$. Translation by τ preserves the sheaf supported at (s, e) and $(s, e + \tau)$, and interchanges its two length-one subobjects. On that fibre the flag coefficient is $A = \mathbb{Q}^2$, with coordinatewise product and the transposition action. Its trace is $\operatorname{Tr}(a, b) = a + b$.

Proposition 15.3. *There is no equivariant unital algebra map $A \to \mathbb{Q}$ when the target has the trivial action. The complex $[A \xrightarrow{\operatorname{Tr}} \mathbb{Q}]$ in degrees zero and one has only one cohomology group: the sign representation in degree zero.*

Proof. A unital algebra map sends the two orthogonal idempotents $(1, 0)$, $(0, 1)$ to $0, 1$ in some order. It therefore cannot be invariant under their transposition. Trace is surjective and has kernel $\mathbb{Q}(1, -1)$, on which the transposition acts by -1. □

This kernel is not an ideal: $(1, -1)^2 = (1, 1)$ has trace two. Nor does adding a homologically scalar dg algebra avoid the multiplicity. Indeed, let B be a unital dg algebra with $H^\circ(B) = \mathbb{C}$, and let a finite group G act transitively on a nonempty finite set Ω. Let G act on $M_m(B)$ by conjugation through a representation on $\mathbb{C}^m$, trivially on B. A G-equivariant unital dg algebra map $\mathbb{C}^\Omega \to M_m(B)$ induces orthogonal projectors in $M_m(\mathbb{C})$. Transitivity makes their ranks equal. Since they sum to the identity, $m = d|\Omega|$ for an integer $d \geq 1$. The permutation representation and its diagonal projectors attain $m = |\Omega|$.

15.3 Trace complexes on split and nonsplit flags

The structure-sheaf cohomology of the product has the finite Dolbeault model

$$H = \mathbb{C}[\sigma]/(\sigma^2) \otimes \Lambda(\eta), \qquad |\sigma| = 2, \quad |\eta| = 1, \quad \int_X \eta\sigma = 1. \tag{15.5}$$

Parallel representatives for a product Ricci-flat metric give the multiplicative inclusion into the Dolbeault complex used in Chapter 14. There is one cohomology class in each degree zero through three, and multiplication followed by the displayed integral is a perfect pairing of degree -3.

For a vector bundle A write $C_A = R\Gamma(X, R\mathcal{H}om(A, A))$ and let C_A° be its trace-free summand. Positive rank is needed for the scalar splitting $h \mapsto h\,\mathrm{id}_A/\mathrm{rk}\,A$. Write $C(A, B) = R\Gamma(X, R\mathcal{H}om(A, B))$.

Proposition 15.4. *For bundles of positive ranks n, m there is a decomposition*

$$C^\circ_{A\oplus B} \simeq C^\circ_A \oplus C^\circ_B \oplus C(A, B) \oplus C(B, A) \oplus H. \tag{15.6}$$

The last summand is $h \mapsto \operatorname{diag}(h/n, -h/m)$, where the two entries multiply the respective identities. Its induced pairing is

$$(h,k) \longmapsto \frac{n+m}{nm}\int_X hk.$$

The construction applies to relative endomorphism complexes in families of bundles, with the relative structure-sheaf complex.

Proof. Split the endomorphism matrix into its four blocks. The two diagonal traces must sum to zero, leaving one copy of the scalar complex. Composition of their scalar matrices has trace $(1/n + 1/m)hk$. All decompositions and maps are defined before taking cohomology, so they also give the relative claim. □

For r bundles of ranks $n_i > 0$, put

$$U_r = \{(a_1, \dots, a_r) \in \mathbb{C}^r : \textstyle\sum_i a_i = 0\}.$$

The scalar complex is $H \otimes U_r$. In its trace coordinates $h_i \in H$, its pairing is

$$\langle h, k\rangle = \int_X \sum_i \frac{h_i k_i}{n_i}.$$

The scalar matrix in the ith block is h_i/n_i. For a partition into nonempty groups $I_1, \dots, I_s$, set $n_{I_a} = \sum_{i\in I_a} n_i$. The maps

$$G(h)_a = \sum_{i\in I_a} h_i, \qquad J(k)_i = \frac{n_i}{n_{I_a}} k_a \quad (i \in I_a) \tag{15.7}$$

give, on scalar coordinates with form $\sum_i a_i b_i/n_i$, the orthogonal splitting

$$0 \longrightarrow \bigoplus_a U_{I_a} \longrightarrow U_r \xrightarrow{G} U_s \longrightarrow 0.$$

Tensoring with H gives the scalar flag-complex splitting. Indeed $GJ = 1$, and a within-group trace-zero vector pairs trivially with $J(k)$. The induced pairing on the grouped coordinates has denominator n_{I_a}. Successive sums compose, and the section coefficients multiply as $(n_i/n_{I_a})(n_{I_a}/n_J) = n_i/n_J$. Thus the splittings agree under every refinement.

15.3.1 *A family with torsion in its flag cohomology*

Let $B = \mathbb{C}[t]$ and consider the rank-two Dolbeault operator

$$\bar\partial_t = \bar\partial + t\eta e_{12}, \qquad e_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

It squares to zero because $\bar\partial\eta = \eta^2 = 0$. It defines extensions $0 \to \mathcal{O}_X \to F_t \to \mathcal{O}_X \to 0$. They are algebraic in t: a Čech representative of the elliptic extension class gives transition matrices $1 + t\eta_{ij}e_{12}$, whose products are finite.

The endomorphism model is $M_2(H) \otimes B$ with differential $[t\eta e_{12}, -]$. For trace-free endomorphisms preserving the first summand, put $e = e_{12}$ and $z = \operatorname{diag}(1, -1)$. Their complex is

$$\mathfrak{b}_t^{\circ} = (Hz \oplus He) \otimes B, \qquad dz = -2t\eta e, \quad de = 0. \tag{15.8}$$

The full endomorphism model retains associative composition; the trace-free subcomplexes retain its Lie commutator. To justify the model after twisting, filter matrices by the diagonal weight for which $[e_{12}, -]$ strictly raises weight. The filtration is finite. On its associated graded, the inclusion of (15.5) into Dolbeault forms is a quasi-isomorphism. It follows that the twisted inclusion is also a quasi-isomorphism. The same argument works for upper-triangular endomorphisms and for any fixed finite strictly upper-triangular matrix.

Directly from (15.8),

j	$H^j(\mathfrak{b}_t^\circ)$
0	Be
1	$B\eta z \oplus (B/(t))\eta e$
2	$B\sigma e$
3	$B\eta\sigma z \oplus (B/(t))\eta\sigma e.$

(15.9)

The only nonzero differentials are multiplication by $-2t$ from z to ηe and from σz to $\eta\sigma e$. Every nonzero fibre has dimension one in each degree. The derived zero fibre has dimension two in each degree, since the differential of the free model becomes zero there.

Translations act trivially on $H^1(E, \mathcal{O}_E)$, so for $t \neq 0$ the extension on E is invariant under translation up to isomorphism. It has no translation linearization. An E-equivariant vector bundle for the free translation action on E descends to a vector space on a point and is therefore trivial. The nontrivial extension has $h^0 = 1$: its connecting map $H^0(\mathcal{O}_E) \to H^1(\mathcal{O}_E)$ is the nonzero extension class. A trivial rank-two bundle has $h^0 = 2$.

More generally, let V be a finite-dimensional matrix subspace stable under $L = \operatorname{ad} N$, and put $\ell = \operatorname{rank} L$. The complex $H \otimes V \otimes B$ with differential $t\eta L$ is the direct sum of the two-term complex $[V_B \xrightarrow{tL} V_B]$ and its shift by $[-2]$. Choosing bases for the image of L gives

$$H^0 = (\ker L) \otimes B, \qquad H^1 \simeq (\operatorname{coker} L) \otimes B \oplus (B/(t))^\ell, \qquad H^{j+2} = \sigma H^j.$$

The displayed splitting is a choice; its torsion submodule and multiplicity are intrinsic. Derived specialization retains V in each of the four degrees.

15.3.2 *The endpoint traces in the flag pairing*

Let N be strictly upper triangular, and use differential $[\eta N, -]$ on $\mathfrak{g}_r^\circ = \mathfrak{sl}_r \otimes H$ and on its upper-triangular subcomplex $\mathfrak{b}_r^\circ$. Put $V_r = H \otimes U_r$ for rank-one diagonal blocks, with zero differential. Diagonal projection on upper-triangular matrices is a chain map. The map

$$j : \mathfrak{b}_r^\circ \longrightarrow \mathfrak{g}_r^\circ \oplus V_r, \qquad x \longmapsto (x, \operatorname{diag} x)$$

is isotropic for

$$Q((x,h),(y,k)) = \int_X \operatorname{tr}(xy) - \int_X \sum_i h_i k_i.$$

For upper-triangular x, y, the diagonal of xy is the componentwise product of their diagonals, proving isotropy. The pairing on the target is perfect. In each coefficient degree the matrix dimensions satisfy $2(r(r+1)/2 - 1) = (r^2 - 1) + (r - 1)$. Pairing complementary upper and lower matrix units then shows that the annihilator is exactly the image of j. Hence

$$0 \longrightarrow \mathfrak{b}_r^\circ \xrightarrow{j} \mathfrak{g}_r^\circ \oplus V_r \longrightarrow (\mathfrak{b}_r^\circ)^\vee[-3] \longrightarrow 0 \tag{15.10}$$

is an exact sequence of complexes, compatible with base change. For $r = 2$, omitting V_r leaves $\operatorname{tr}(z^2) = 2$; the flag subspace would no longer be isotropic. The endpoint trace term supplies the cancellation.

15.4 Paired orbit directions and finite centres

To preserve Calabi–Yau duality, removing an orbit tangent direction also requires its paired obstruction direction. The following complex makes sense even when the orbit map changes rank. Let R be a commutative $\mathbb{Q}$-algebra, let C be a finite-projective complex in degrees zero through three, and equip it with a perfect graded-symmetric pairing $C \otimes_R C \to R[-3]$ compatible with its differential. Let W_R be finite projective and let $a : W_R \to C^1$ satisfy $d_1 a = 0$. Define $a^\dagger : C^2 \to W_R^*$ by $(a^\dagger v)(w) = \langle aw, v\rangle$.

Proposition 15.5. *The complex*

$$C_{\mathrm{orb}} = [C^0 \oplus W_R \xrightarrow{(d_0,a)} C^1 \xrightarrow{d_1} C^2 \xrightarrow{(d_2,-a^\dagger)} C^3 \oplus W_R^*] \tag{15.11}$$

is perfect and self-dual with pairing of degree -3. It commutes with derived base change, and

$$\det C_{\mathrm{orb}} = \det C \otimes (\det W_R)^2.$$

A square root P_C therefore gives the square root $P_C \otimes \det W_R$.

Proof. Pairing compatibility and $d_1 a = 0$ imply $a^\dagger d_1 = 0$, so consecutive differentials compose to zero. Add the evaluation pairing between W_R in degree zero and W_R^* in degree three. The two new terms in its differential are $\langle aw, v\rangle - \langle w, a^\dagger v\rangle = 0$. The resulting pairing is perfect. Finite projectivity gives derived base change and identifies both new determinant contributions with $\det W_R$. □

Over a field, the outer terms record the kernel of the induced orbit map $[a] : W_R \to H^1(C)$ and its dual. When $[a]$ is injective, its image in $H^1(C)$ and its dual quotient in $H^2(C)$ disappear. A cochain-level centre $p : C^1 \to W_R$ with $pa = 1$ and $pd_0 = 0$ gives an actual complement: its middle terms are $\ker p$ and $\ker a^\dagger$. The projections ap and $p^* a^\dagger$ split off two contractible pairs in (15.11). A splitting only in the derived category gives the same conclusion by splitting its homotopy idempotent in perfect complexes.

The rank-jumping example is already nonconstant over $R = \mathbb{C}[t]$:

$$[R \xrightarrow{t} R \xrightarrow{0} R \xrightarrow{-t} R]. \tag{15.12}$$

Its cohomology is $R/(t)$ in degrees one and three and zero elsewhere. Its generic fibre is acyclic; its derived zero fibre has one class in each degree zero through three. A fixed-rank deletion cannot recover these fibres.

For r disjoint point sheaves, the original Ext dimensions are $(r, 3r, 3r, r)$. Removing the simultaneous orbit pair gives $(r, 3r-1, 3r-1, r)$. Removing an orbit pair at every endpoint would give $(r, 2r, 2r, r)$ and lose the relative contribution

$$U[-1] \oplus U^*[-2], \qquad U = W^r/W_{\mathrm{diag}}, \qquad \det\big(U[-1] \oplus U^*[-2]\big) = (\det U)^{-2}. \tag{15.13}$$

All regroupings give coordinates on the same quotient U. At the zero object, however, (15.11) is $W[0] \oplus W^*[-3]$. After the tangent shift it is $W[1] \oplus W^*[-2]$, the tangent of $T^*[-1]BE$ along its zero section. A comparison with the relative unit BE must therefore specify what happens to its cotangent partner.

15.4.1 *A centre map retains a finite group*

Proposition 15.6. *Let M be an algebraic stack with an E-action and a map $s : M \to E$ satisfying $s(t \cdot m) = s(m) + Nt$ for an integer $N \neq 0$. Then, with $M_0 = s^{-1}(0)$,*

$$[M/E] \simeq [M_0/E[N]]. \tag{15.14}$$

This equivalence preserves stabilizers.

Proof. The multiplication map $[N] : E \to E$ is finite étale and surjective. After that base change a point of M admits a translation into M_0. The choices differ exactly by $E[N]$. Thus $E \times^{E[N]} M_0 \to M$ is an isomorphism after a faithfully flat base change and hence an isomorphism of stacks. Quotienting by E proves (15.14), with the same isomorphism groupoids on each side. □

For length-N point sheaves, the sum of the elliptic support cycle is such a map. On a bounded sector of fixed Hilbert polynomial whose sheaves have finite image in E, choose an ample line bundle A on S and a sufficiently large integer m. The determinant of the perfect complex $R\pi_{E,*}(F \otimes \pi_S^* A^m)$ is a line bundle on E. Its divisor degree is the sum of the positive Hilbert polynomials of the fibre-supported pieces. This gives a positive integer N and, after identifying $\mathrm{Pic}^N(E)$ with E, a centre map of the stated weight. The choice of m is made on that bounded sector.

Centred factors still have relative centres. For weights $N_1, \dots, N_r$, the condition for a centred total object is $\sum_i N_i e_i = 0$. Its solution is the full kernel of $E^r \to E$, including finite components. Replacing this kernel by a point would again erase extension parameters. There is no positive-weight isogeny construction at $N = 0$.

15.5 A compact rational-curve coefficient

Assume now that S contains a smooth rational curve C. Adjunction gives $C^2 = -2$. A stable pair on X consists of a pure one-dimensional sheaf F and a section $\mathcal{O}_X \to F$ with zero-dimensional cokernel. Let $P_n(X, ([C], 0))$ be its projective moduli scheme with $\chi(F) = n$ and curve class $([C], 0)$.

Theorem 15.7. *For $r \geq 0$, the universal effective divisor gives an isomorphism*

$$E \times \mathrm{Sym}^r C \xrightarrow{\sim} P_{r+1}(X, ([C], 0)), \qquad (e, D) \longmapsto (\mathcal{O}_X \longrightarrow i_{e,*}\mathcal{O}_C(D)).$$

Here $i_e : C \hookrightarrow X$ is inclusion in the fibre over e. Translation is free and its quotient is $Y_r = \mathrm{Sym}^r C \simeq \mathbb{P}^r$. The moduli scheme is empty for $n \leq 0$.

Proof. An effective divisor in the class of C must contain C, since otherwise its intersection with C would be nonnegative. The remainder has zero class and hence zero intersection with an ample divisor, so it vanishes. Thus C is the unique effective divisor in its class.

The degree against a positive divisor from E is zero. Every component of the sheaf's support cycle is consequently constant in E. Projection to S and the uniqueness just proved show that the cycle is $C \times \{e\}$ with multiplicity one. Its reduced ideal acts on F with generically zero image. That image is zero-dimensional, so purity makes it zero. Hence $F = i_{e,*}L$ for a torsion-free rank-one sheaf L on the smooth curve C. It is a line bundle, and its nonzero section gives $L = \mathcal{O}_C(D)$ for an effective divisor D. Then $\chi(F) = \deg D + 1$.

This classification gives the stated morphism in families. To verify its scheme structure, use $N_{C \times \{e\}/X} = \mathcal{O}_C(-2) \oplus \mathcal{O}_C$. The local-to-global Ext sequence gives

$$0 \to H^1(\mathcal{O}_C) \to \mathrm{Ext}^1_X(F, F) \to H^0(\mathcal{O}_C(-2) \oplus \mathcal{O}_C) \to H^2(\mathcal{O}_C),$$

so $\operatorname{Ext}^1_X(F,F)=\mathbb{C}$. Also $\operatorname{Hom}(F,F)=\mathbb{C}$, $h^0(F)=r+1$ and $H^1(F)=0$. The tangent sequence for deforming a sheaf with its section is

$$0\to \operatorname{Hom}(F,F)\to H^0(F)\to T_{(F,s)}P\to \operatorname{Ext}^1(F,F)\to H^1(F).$$

The first map is injective because the section is nonzero. Thus every tangent space has dimension $r+1$.

The morphism from $E\times\mathbb{P}^r$ is proper and bijective on closed points, hence finite. It surjects onto an irreducible target of dimension $r+1$. At each closed point the local dimension equals the tangent dimension, so the target is regular and normal. The finite map is birational in characteristic zero and therefore an isomorphism. Translation acts on the first factor. □

We can now specify an oriented critical enhancement on this smooth quotient. Give Y_r the zero function and take $\mathcal{Y}_r=T^*[-1]Y_r$. Its virtual canonical line is $K_{Y_r}^{\otimes 2}$, with chosen square root

$$L_r=K_{Y_r}=\mathcal{O}_{\mathbb{P}^r}(-r-1). \tag{15.15}$$

This is a specified enhancement of the classical quotient. An equivalence with a reduced derived stable-pair stack requires a comparison of their derived structures.

Use the perverse vanishing-cycle normalization $\phi^p_f=\phi_f[-1]$. For the zero function, the nearby-cycle complex is zero. The defining triangle gives $\phi_0F=F[1]$, and hence the oriented coefficient of (15.15) is

$$\varphi_r=\mathbb{Q}_{Y_r}[r],\qquad H_r=R\Gamma_c(Y_r,\varphi_r)\simeq\bigoplus_{k=0}^{r}\mathbb{Q}v_{r,k},\qquad |v_{r,k}|=2k-r. \tag{15.16}$$

The differential is zero. The basis comes from powers of the hyperplane class, since projective space has one cell in each even real dimension. Exterior products of the canonical lines give coherent orientations on ordered products of these charts. Determinants of Jacobians multiply under composition, so these product comparisons agree under regrouping, with the usual Koszul signs.

There are two different functionals here. Integration is

$$\epsilon_r: H_r\longrightarrow\mathbb{Q}[-r],\qquad \epsilon_r(v_{r,r})=1,\quad \epsilon_r(v_{r,k})=0\quad (k<r).$$

Its target has its nonzero term in degree r, the degree of the top class. The categorical trace instead has domain $\operatorname{End}^\bullet(H_r)$:

$$\tau_r(T)=\sum_j(-1)^j\operatorname{Tr}(T|_{H_r^j})\quad(|T|=0),\qquad \tau_r(T)=0\quad(|T|\neq 0).$$

For a formal degree-zero variable y, equip $\prod_{r\geq 0}\operatorname{End}^\bullet(H_r)$ with the tail topology and componentwise composition. Define $\mathcal{T}((T_r)_r)=\sum_{r\geq 0}y^{r+1}\tau_r(T_r)$.

Proposition 15.8. *The maps τ_r and $\mathcal{T}$ are continuous cyclic chain maps, and*

$$\tau_r(\mathrm{id})=(-1)^r(r+1),\qquad \mathcal{T}((\mathrm{id})_r)=\frac{y}{(1+y)^2}. \tag{15.17}$$

The first number is also the Behrend-weighted Euler characteristic of the quotient Y_r.

Proof. For homogeneous composable endomorphisms, pairing the two matrix entries in opposite order gives $\tau_r(AB)=(-1)^{|A||B|}\tau_r(BA)$. In particular $\tau_r([d,T])=0$. This is the evaluation trace of a perfect complex and is invariant under a change of perfect model. All $r+1$ summands in (15.16) have parity r. Summing their signs and then the geometric-series derivative gives (15.17). Each coefficient uses one finite complex, proving continuity. Finally the Behrend function of the smooth scheme $\mathbb{P}^r$ is $(-1)^r$, and its Euler characteristic is $r+1$. □

Oberdieck's quotient-incidence comparison identifies this number with the reduced stable-pair incidence invariant for the nonzero K3 class $[C]$ [121]. Upstairs, the smooth zero-function coefficient is $\mathbb{Q}_E[1] \boxtimes \varphi_r$. Its restriction to the slice over 0, shifted by $[-1]$, is φ_r. Its compact Euler characteristic vanishes because $\chi(E) = 0$, while the first quotient value is $+1$. The shifted restriction and the quotient virtual-class comparison specify how the nonzero invariant is obtained.

15.6 Relative Ext and square roots along a curve

The unframed sheaves $G_e = i_*\mathcal{O}_C \boxtimes \mathcal{O}_e$, with $i : C \hookrightarrow S$, test extension coefficients. On S, the divisor resolution gives

$$R\operatorname{Hom}_S(i_*\mathcal{O}_C, i_*\mathcal{O}_C) \simeq \mathbb{C} \oplus \mathbb{C}[-2]$$

as a complex of vector spaces: its two contributions are $H^0(\mathcal{O}_C)$ and $H^1(\mathcal{O}_C(-2))$, the latter shifted by one. The point resolution on E then gives Ext dimensions $(1, 1, 1, 1)$ when the two parameters coincide. For distinct parameters the supports are disjoint and derived Hom vanishes.

For $m \geq 1$, let $B_m = E^m / E_{\mathrm{diag}} \simeq E^{m-1}$, using the first coordinate as origin. For $i \neq j$ let $\delta_{ij} : B_m \to E$ be the difference $e_j - e_i$, let $D_{ij} = \delta_{ij}^{-1}(0)$, and denote its canonical divisor section by s_{ij}.

Proposition 15.9. *After choosing the displayed splitting of the surface Hom complex, the relative Hom from G_{e_i} to G_{e_j} is represented by*

$$\mathcal{K}_{ij} = (\mathbb{C} \oplus \mathbb{C}[-2]) \otimes [\mathcal{O}_{B_m} \xrightarrow{s_{ij}} \mathcal{O}_{B_m}(D_{ij})]_{0,1}. \tag{15.18}$$

It computes derived restriction to every equality stratum, including their nontransverse intersections, and all further derived base changes.

Proof. On $E_z \times E_\delta$, resolve the fixed-point sheaf by $[\mathcal{O}(-\{z = 0\}) \to \mathcal{O}]$. Apply sheaf Hom into the graph sheaf $\mathcal{O}_{z=\delta}$. Projection of that graph to E_δ is an isomorphism, and the resulting complex is $[\mathcal{O}_E \xrightarrow{s_0} \mathcal{O}_E(0)]_{0,1}$. The section has a simple zero at $\delta = 0$. Translation and pullback by δ_{ij} give the elliptic factor in (15.18). Tensoring with the surface factor proves the formula. Its terms are vector bundles, so it computes derived pullback along every map. □

On a formal normal disk, the elliptic factor is $[\mathbb{C}[[t]] \xrightarrow{t} \mathbb{C}[[t]]]_{0,1}$. It is quasi-isomorphic to its degree-one cokernel, but the derived fibre of that cokernel has a degree-zero Tor term. Both terms, and their shifts by $[-2]$, are needed for the four collision groups. A coefficient pulled back from the independent orbit quotient, which is a point, would have constant derived fibres and cannot give this complex.

For a perfect complex use $\det V^\bullet = \bigotimes_j (\det V^j)^{(-1)^j}$, with parity its virtual rank modulo two. The two line terms of (15.18) occur in degrees one and three; consequently $\det \mathcal{K}_{ij} = \mathcal{O}(-2D_{ij})$ and its parity is even. Choose constant determinant trivializations of the diagonal self-Ext factors using the volume forms. Taking the endomorphism determinant relatively over B_m gives

$$\det R\operatorname{Hom}_X\left(\bigoplus_i G_{e_i}, \bigoplus_i G_{e_i}\right) = \mathcal{O}_{B_m}\left(-4\sum_{i<j} D_{ij}\right), \qquad \mathcal{L}_m = \mathcal{O}_{B_m}\left(-2\sum_{i<j} D_{ij}\right) \tag{15.19}$$

is a specified square root.

For ordered blocks A, B, multiplication of divisor lines gives

$$\mathcal{L}_{A\sqcup B} \simeq r_A^*\mathcal{L}_A \otimes r_B^*\mathcal{L}_B \otimes \det\Bigg(\bigoplus_{i\in A,\ j\in B} \mathcal{K}_{ij}\Bigg), \tag{15.20}$$

where r_A, r_B retain the internal differences. Every unordered pair of leaves is separated at exactly one vertex of a binary tree, its least common ancestor. Each parenthesization therefore counts the same divisor with exponent -2. Multiplication of line bundles is associative, and all cross determinants are even. Thus the maps themselves agree under every ordered regrouping. They commute with derived restriction, and $\mathcal{L}_\emptyset = \mathcal{O}$ supplies the empty-block identity.

These roots are defined on labelled split parameter spaces. At coincident parameters the direct sum acquires more automorphisms. Descent under those automorphisms, extension to nonsplit sheaves and comparison with a reduced critical structure are further requirements. The divisor formula does not specify a zero-function coefficient on B_m.

15.7 Determinant sewing and orientation coherence

For point sheaves there is a complementary global calculation. Over X^2, let $\mathcal{C}$ be the relative derived Hom between the two universal graph sheaves. It is perfect and acyclic off the diagonal. Its determinant has a canonical acyclic trivialization there. Since X^2 is smooth and its diagonal has codimension three, this trivialization extends uniquely: a rational section of a line bundle on a normal variety with no zeros or poles in codimension one is regular and invertible. Uniqueness makes it invariant under common translation. The coefficient-line automorphisms act on its determinant with exponent the Euler characteristic, which is zero. It therefore descends to the common-torsor pair stack.

On the diagonal, the complex has cohomology $\Lambda^* T_x X$ and dimensions $(1, 3, 3, 1)$. Its determinant is trivial because the exponent of $\det T_x X$ is $-1 + 2 - 1 = 0$. The full complex still has all four groups. For r coincident copies its Yoneda algebra is $M_r(\mathbb{C}) \otimes \Lambda^* T_x X$. For independent tangent vectors v_1, v_2 and $r = 2$, the first-order deformation $E_{12}v_1 + E_{21}v_2$ has quadratic obstruction $\operatorname{diag}(1, -1)v_1v_2 \neq 0$ although its scalar trace is zero.

Let $0 = F_0 \subset F_1 \subset F_2 \subset F_3 = F$ be a family over a locally noetherian complex scheme B, with flat length-one quotients A_i. For projection $\pi : X \times B \to B$, put

$$C_{ij} = R\pi_* R\mathcal{H}om(A_i, A_j), \qquad L_{ij} = \det C_{ij}, \qquad D(F) = \det R\pi_* R\mathcal{H}om(F, F).$$

Proposition 15.10. *There is a natural isomorphism*

$$D(F) \simeq \bigotimes_{i=1}^{3} D(A_i) \otimes \bigotimes_{i<j} L_{ij}^{\otimes 2}. \tag{15.21}$$

It is compatible with derived base change and common-torsor descent. Both parenthesizations give the same determinant map at nonsplit flags and collisions.

Proof. The graph sheaves are perfect, and the extensions make the F_i perfect. Locally choose bounded locally free resolutions of the three quotients. Lifting their successive extensions gives an upper-triangular differential on their graded direct sum. Filtering the endomorphism complex by matrix-block weight has associated graded the nine Hom complexes C_{ij}. This statement uses the filtered resolution, so it does not split the original extensions.

For a short exact sequence, the determinant isomorphism concatenates a basis of the subobject with lifts of a basis of the quotient. Changing the lifts leaves the top wedge unchanged. Concatenating three quotient bases

is independent of parentheses. Degreewise application, using determinant duals in odd degrees, gives the same refinement rule for perfect complexes. Acyclic factors cancel canonically, so filtered resolution comparisons preserve it.

Apply this rule to the nine blocks in the order C_{31}, C_{21}, C_{32}, C_{11}, C_{22}, C_{33}, C_{12}, C_{23}, C_{13}. Serre duality gives $C_{ji} \simeq C_{ij}^{\vee}[-3]$ and therefore $\det C_{ji} = L_{ij}$. Every block has virtual rank zero. The reorderings are thus even and give (15.21). Both parenthesizations refine to this same ordered filtration. Perfect base change and the translation-invariant Serre pairing give the remaining compatibilities. □

An orientation must lift this determinant identity to chosen square roots. A line D has a square root precisely when its étale Kummer class $\delta(D) \in H^2(-, \mu_2)$ vanishes. After square roots have been chosen, two binary sewing isomorphisms may differ by a sign. On a three-step flag, the ratio of the two parenthesized sewing maps is a μ_2-valued associator. Its pentagon is a cocycle equation on the full flag and overlap diagram. One must trivialize that cocycle coherently. This is the extra associativity condition in strong orientation data from Definition 3.24.

Ordinary orientation data on compact Calabi–Yau threefold moduli are available [90]. Their existence does not identify the square roots after orbit reduction or supply the required reduced flag maps. Even a trivial determinant permits an incoherent choice: multiply only the binary map for masses $(2, 1)$ by -1, and leave the others unchanged. All squares remain the same, but the two maps on masses $(1, 1, 1)$ differ by -1.

15.7.1 *Two sign defects that can be solved explicitly*

Suppose the associator depends only on masses. Write it as $a : \mathbb{N}^3 \to \mu_2$, normalized to one if any argument is zero, and assume its pentagon

$$a(n, r, s)a(m, n + r, s)a(m, n, r) = a(m + n, r, s)a(m, n, r + s).$$

Define $\varepsilon(m, 0) = \varepsilon(m, 1) = 1$ and

$$\varepsilon(m, n) = \prod_{j=1}^{n-1} a(m, j, 1) \quad (n \geq 2). \tag{15.22}$$

Multiplying binary sewing by ε changes the associator to

$$a'(m, n, r) = a(m, n, r)\frac{\varepsilon(m, n)\varepsilon(m + n, r)}{\varepsilon(n, r)\varepsilon(m, n + r)}.$$

The recursion gives $a'(m, n, 1) = 1$. Its pentagon with last argument one gives $a'(m, n, r + 1) = a'(m, n, r)$. Induction from $r = 0$ proves $a' = 1$. The correction preserves units and squares.

There is also an explicit finite-group calculation. If G has odd order and $\alpha : G^2 \to \mu_2$ is a normalized two-cocycle with trivial coefficient action, put $\beta(g) = \prod_{k \in G} \alpha(g, k)$. Multiplying the cocycle identity over k gives

$$\alpha(g, h)^{|G|}\beta(gh) = \beta(g)\beta(h).$$

Odd order makes $\alpha = \delta\beta$. For $G = (\mathbb{Z}/2)^2$, however, $\alpha(g, h) = (-1)^{g_1 h_2}$ is a cocycle and is not a coboundary: coboundaries on an abelian group are symmetric, and this one is not. Thus $E[N]$ has no constant sign obstruction of this kind for odd N, while even torsion can retain one. Neither calculation trivializes a gerbe that varies over the sheaf moduli.

15.8 A polystable spherical collection

An Euler matrix alone does not produce a polystable object. We construct three spherical sheaves with one polarization and one Hilbert polynomial, so that every multiplicity vector belongs to an actual polystable locus.

Theorem 15.11. *There is a projective K3 surface with an ample class H, $H^2 = 8$, and line bundles $L_i = \mathcal{O}_S(D_i)$, $1 \le i \le 3$, such that*

$$D_i^2 = -8, \qquad D_i D_j = -4 \ (i \neq j), \qquad H D_i = 0.$$

They are stable with common Hilbert polynomial $4m^2 - 2$, and their Ext dimensions are

$$\dim \operatorname{Ext}^\bullet_S(L_i, L_j) = \begin{cases} (1, 0, 1), & i = j, \\ (0, 2, 0), & i \neq j. \end{cases}$$

Thus each L_i is spherical, their Euler matrix is

$$G = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}, \tag{15.23}$$

and every nonzero $F_{\mathbf{n}} = \bigoplus_i L_i \otimes \mathbb{C}^{n_i}$, with $n_i \ge 0$, is polystable.

Proof. In three hyperbolic planes in the K3 lattice choose $e_i^2 = f_i^2 = 0$, $e_i f_i = 1$, with different planes orthogonal. Set

$$D_1 = e_1 - 4f_1, \qquad D_2 = -4f_1 + e_2 - 4f_2,$$
$$D_3 = -4f_1 - 4f_2 + e_3 - 4f_3, \qquad V = f_1 + f_2 + 2f_3.$$

Their lattice N has Gram matrix

$$\begin{pmatrix} -8 & -4 & -4 & 1 \\ -4 & -8 & -4 & 1 \\ -4 & -4 & -8 & 2 \\ 1 & 1 & 2 & 0 \end{pmatrix}, \qquad \det N = -128.$$

The minor in coordinates e_1, e_2, e_3, f_1 is one, so the embedding is primitive. The first three generators span a negative-definite subspace and their Schur complement is positive; the signature is $(1, 3)$.

Put $H = D_3 + 4V = e_3 + 4f_3$. Then $H^2 = 8$, $H D_i = 0$ and $HV = 2$. The integral orthogonal complement of H in N is spanned by the D_i, with

$$\left(\sum_i a_i D_i\right)^2 = -4\left(\sum_i a_i^2 + \left(\sum_i a_i\right)^2\right).$$

The bracket is even, so this complement has no vector of square -2. The period and lattice embedding theorems realize N as the Néron–Severi lattice of a projective K3 surface. Since H lies on no root wall, the Weyl chamber description permits a marking in which it is ample [84, Chapters 8 and 14].

For $i \neq j$, the class $D_j - D_i$ is nonzero, has square -8 and is orthogonal to H. Neither sign is effective, so its line bundle has $H^0 = H^2 = 0$. Riemann–Roch gives $\chi = 2 - 8/2 = -2$, proving the cross-Ext calculation. Self-Ext is $H^*(S, \mathcal{O}_S)$. Finally $\chi(L_i(mH)) = 2 + (D_i + mH)^2/2 = 4m^2 - 2$. Rank-one torsion-free sheaves are Gieseker stable: a proper same-rank subsheaf has a nonzero torsion quotient and a strictly smaller reduced Hilbert polynomial. The common polynomial makes the direct sums polystable. □

The Mukai vectors are $v_i = (1, D_i, -3)$ and satisfy $-\langle v_i, v_j\rangle = G_{ij}$. They are independent, so this gives an additive isometry from their charge lattice with the negative Mukai form to the lattice with matrix (15.23). Budur–Zhang formality applies to every $F_{\mathbf{n}}$ [21]. We will also need compatibility with the geometric trace and with the whole collection.

15.8.1 One cyclic model for all multiplicities

Let $\mathcal{D}$ be the three-object Dolbeault dg category with morphisms $\mathcal{A}^{0,*}(S, L_i^\vee \otimes L_j)$ and differential $\bar\partial$. Composition is wedge product and bundle composition. Its degree--2 trace pairing is

$$\langle a, b\rangle = \int_S \omega_S \wedge \operatorname{tr}(ba).$$

Put $A(i,j) = \operatorname{Ext}^*_S(L_i, L_j)$ and $\ell = \bigoplus_i \mathbb{C}1_i$. A cyclic comparison preserves the associated Calabi–Yau class; on a cyclic minimal model that class is represented by this perfect pairing.

Proposition 15.12. *There is a strictly unital cyclic minimal model A of $\mathcal{D}$ with only its Yoneda product m_2 nonzero. One cyclic quasi-equivalence $F = (F_n) : A \to \mathcal{D}$ can be chosen for the entire three-object collection.*

Proof. The collection is simple-minded: its negative Ext groups vanish and $A^0 = \ell$ consists of the vertex identities. The cyclic minimal-model and strict-unit theorem for such a 2-Calabi–Yau collection applies, since Serre duality is perfect of degree -2 [36, Lemmas 4.4–4.5 and Proposition 4.8]. It supplies a cyclic model with strict units over the semisimple algebra ℓ and the given geometric trace.

The higher products then vanish for a direct reason. For $n \geq 3$, an input in $A^0 = \ell$ makes m_n zero by strict unitality. If all inputs have positive degree and one has degree two, the output has degree at least three, since $|m_n| = 2 - n$. Only inputs all in degree one remain; their output is in A^2. Pair that output with each vertex unit. Cyclicity moves the unit into an input of m_n, giving zero. The pairing between A^2 and A^0 is perfect, so the output is zero. □

This is a formal statement in the completed tensor algebra over ℓ. The formal correction of a cyclic form can be seen weight by weight. In the noncommutative de Rham complex, give every coordinate and differential weight one. For a closed error ω_w of weight $w > 2$ beyond the constant two-form, Cartan's identity gives $\omega_w = d_{\mathrm{dR}}(w^{-1}\iota_{\mathcal{E}}\omega_w)$, where $\mathcal{E}$ is the Euler vector field. The constant nondegenerate form identifies that primitive with a vector field whose formal flow removes the error. Increasing weights converge in tensor length. Analytic convergence is a separate condition.

For multiplicity spaces V_i, amplify the morphisms to $A(i,j) \otimes \operatorname{Hom}(V_i, V_j)$ and define the Taylor coefficients of F^{mat} by summing over composable matrix paths:

$$(F_n^{\mathrm{mat}}(a_n, \dots, a_1))_{i_n i_0} = \sum_{i_1, \dots, i_{n-1}} F_n((a_n)_{i_n i_{n-1}}, \dots, (a_1)_{i_1 i_0}). \tag{15.24}$$

The formula includes multiplication of the matrix coefficients. Use the suspended bar convention on $A[1]$: the bar differential has degree one and F has degree zero. Its tensor products determine the Koszul signs.

A finite twisted flag is an ordered direct sum with a strictly upper-triangular degree-one matrix δ satisfying its Maurer–Cartan equation, here $\delta^2 = 0$. Twisting inserts copies of δ into the operations; upper-triangular morphisms preserve the filtration. The cyclic class is carried by the full matrix morphism complexes; its restriction to flag-preserving morphisms need not be nondegenerate.

Theorem 15.13. *The fixed functor F extends to a cyclic quasi-equivalence on finite twisted objects, retaining their flag-preserving subcomplexes. For morphisms a_i between flags with twisting matrices δ_i, it is given by*

$$F_*(\delta) = \sum_{n\geq 1} F_n^{\mathrm{mat}}(\delta,\dots,\delta), \tag{15.25}$$

$$F_k^{\mathrm{tw}}(a_k,\dots,a_1) = \sum_{j_0,\dots,j_k\geq 0} F^{\mathrm{mat}}_{k+j_0+\cdots+j_k}(\delta_k^{j_k},a_k,\dots,a_1,\delta_0^{j_0}). \tag{15.26}$$

A power denotes repeated inputs. These maps are equivariant under changes of multiplicity bases and coherent under every ordered regrouping. They also apply over nilpotent Artinian bases and in their inverse-limit formal completion.

Proof. Each upper-triangular input advances along a finite flag. The object formula terminates before its length, and the morphism formula terminates between each pair of its distinguished inputs. With Artinian parameters, combine flag distance with powers of the maximal ideal. Every formula is then finite at each precision.

Evaluate the bar identity $b_{\mathcal{D}}F = Fb_{\mathcal{A}}$ on repeated copies of δ. It gives the Maurer–Cartan equation for $F_*(\delta)$. Marking $a_1,\dots,a_k$ in the same identity gives all twisted functor equations. A finite filtration by matrix distance has associated graded a direct sum of the original quasi-isomorphisms F_1. Thus the twisted functor is quasi-fully faithful.

For essential surjectivity, induct on the flag length. The last extension class lifts through quasi-full-faithfulness; a change by a boundary gives an isomorphic twisted object. Taking its cone extends the lifted flag. Over a square-zero Artinian extension with kernel J, a tentative lift has curvature in degree-two cohomology of the twisted endomorphism complex tensored with J. The existing target lift makes its image zero; the quasi-isomorphism already proved on these complexes removes it. Surjectivity on degree-one cohomology matches the extension class and a degree-zero gauge change matches representatives. Induction proves the Artinian claim.

The amplified trace is the original cyclic trace followed by matrix trace. Its terms are closed matrix paths, whose rotation has exactly the original Koszul sign. This also transports the cyclic class after twisting. Basis changes conjugate the path sums. Regrouping partitions the same paths into consecutive subpaths, leaving their sum unchanged. All subdivisions therefore give the required coherences. The compatible finite-precision formulas pass to the formal inverse limit. □

For three blocks the first nonlinear term is already visible: $(F_*\delta)_{13} = F_1(\delta_{13}) + F_2(\delta_{12},\delta_{23})$, with the composition order of (15.24). Separate choices for the two binary extensions need not retain that term.

15.9 The cubic critical model and its gauge directions

Choose a quiver Q with vertices 1, 2, 3 and two arrows $i\to j$ for each $i<j$. Write $t(a), h(a)$ for the tail and head of an arrow. With $\dim V_i = n_i$, its doubled representation space has maps $X_a : V_{t(a)} \to V_{h(a)}$ and $Y_a : V_{h(a)} \to V_{t(a)}$. Put $G_{\mathbf{n}} = \prod_i GL(V_i)$ and $\mathfrak{g}_{\mathbf{n}} = \bigoplus_i \mathrm{End}(V_i)$. The degree-one Yoneda products, using Serre-dual arrow bases, give the moment map

$$\mu_i = \sum_{h(a)=i} X_aY_a - \sum_{t(a)=i} Y_aX_a. \tag{15.27}$$

The equation $x^2 = 0$ in the surface Ext algebra is $\mu = 0$. Its pointed formal gauge group over an Artinian base is $\exp(\mathfrak{g}_{\mathbf{n}}\otimes\mathfrak{m})$, where $\mathfrak{m}$ is the maximal ideal. The full affine quotient by $G_{\mathbf{n}}$ is the quiver moduli model.

For the embedding $S\times\{e\}\hookrightarrow X$, a normal parameter normalized by ω_E gives the exterior Koszul factor $\Lambda(\theta)$, with $|\theta| = 1$ and $\mathrm{tr}(\theta) = 1$. Tensoring the formal cyclic collection with this factor gives a formal cyclic threefold comparison. A degree-one element is $x + z\theta$, with x in the surface degree one and z in degree zero. Its square is $x^2 + (xz - zx)\theta$.

On $\mathcal{V}_{\mathbf{n}} = \operatorname{Rep}(\overline{Q}, \mathbf{n}) \oplus \mathfrak{g}_{\mathbf{n}}$ these equations are the critical equations of

$$\mathcal{W}_{\mathbf{n}}(X, Y, Z) = \sum_i \operatorname{Tr}(Z_i \mu_i) = \sum_a \operatorname{Tr}\big(Y_a (Z_{h(a)} X_a - X_a Z_{t(a)})\big). \tag{15.28}$$

Differentiation with respect to Z, X, Y gives, respectively,

$$\mu_i = 0, \qquad Y_a Z_{h(a)} = Z_{t(a)} Y_a, \qquad Z_{h(a)} X_a = X_a Z_{t(a)}.$$

Thus the classical critical quotient parametrizes preprojective representations with a commuting normal operator.

The derived critical quotient also keeps the gauge duals. For a smooth affine V with a smooth affine group G acting and an invariant function f, write $\rho : \mathfrak{g} \to \Gamma(V, T_V)$ for its action. In étale coordinates x_i, write $\rho(a) = \sum_i \rho_a^i \partial_{x_i}$ for a basis element $a \in \mathfrak{g}$. Introduce odd variables ξ_i of degree -1 and variables b_a of degree -2, and set

$$K_f = \operatorname{Sym}_{\mathcal{O}(V)}(T_V[1] \oplus (\mathfrak{g} \otimes \mathcal{O}(V))[2]), \qquad d\xi_i = \partial_i f, \quad db_a = -\sum_i \rho_a^i \xi_i, \quad dx_i = 0. \tag{15.29}$$

The algebra is graded commutative. The action on all generators is the induced G-action.

Proposition 15.14. *The stack* $[\operatorname{Spec} K_f / G]$ *is the derived critical locus of* f *on* $[V/G]$. *At a critical point, its tangent complex shifted to endomorphism degrees is*

$$[\mathfrak{g} \xrightarrow{\rho} T_V \xrightarrow{\operatorname{Hess} f} T_V^* \xrightarrow{-\rho^*} \mathfrak{g}^*] \tag{15.30}$$

in degrees zero through three.

Proof. The tangent of $[V/G]$ pulled back to V is $[\mathfrak{g} \to T_V]$ in degrees -1, 0. The Koszul model for the intersection of its zero section and df in the cotangent stack has generators $T_V[1]$ and $\mathfrak{g}[2]$, with the differential (15.29) [8, Section 4]. It squares to zero because $d^2 b_a = -\rho(a) f = 0$. Linearization and the quotient tangent give (15.30). Differentiating invariance at a critical point gives $\operatorname{Hess}(f)\rho = 0$ and its dual identity. This tangent construction makes the coordinate models compatible on overlaps. □

For $N = \sum_i n_i > 0$, set $c = N^{-1} \sum_i \operatorname{Tr} Z_i$ and $Z_i^\circ = Z_i - cI$. Let $\mathcal{V}_{\mathbf{n}}^\circ$ be the hyperplane $\sum_i \operatorname{Tr} Z_i = 0$ and $\mathcal{W}_{\mathbf{n}}^\circ$ the restricted potential. Since $\sum_i \operatorname{Tr} \mu_i = 0$, the potential is independent of c and

$$\operatorname{Crit}_{[\mathcal{V}_{\mathbf{n}}/G_{\mathbf{n}}]}(\mathcal{W}_{\mathbf{n}}) \simeq T^*[-1]\mathbb{A}^1 \times \operatorname{Crit}_{[\mathcal{V}_{\mathbf{n}}^\circ/G_{\mathbf{n}}]}(\mathcal{W}_{\mathbf{n}}^\circ). \tag{15.31}$$

At the algebra level the first factor is $\mathbb{C}[c] \otimes \Lambda(\xi_c)$ with $d\xi_c = 0$. The centred factor retains all degree--2 gauge generators. This removes the orbit pair, whereas quotienting only the translation of c leaves ξ_c.

The local orientation is the canonical line of the smooth quotient,

$$K_{[\mathcal{V}/G]} = \det(\mathcal{V}^*) \otimes \det \mathfrak{g}.$$

Every arrow space pairs with its dual reverse-arrow space, and the loop space is the adjoint representation. Their determinant characters cancel. An ordering of the paired spaces gives a trivialization. Removing the centre changes this line by $\det W$, in agreement with Proposition 15.5.

15.9.1 *Dimensional reduction and the point chart*

For a function $f(x,z)=\sum_{a=1}^{d} z_a f_a(x)$ linear in the fibre coordinates, dimensional reduction identifies compactly supported vanishing-cycle cohomology with cohomology of $(f_a=0)$, shifted by $[-2d]$ and Tate-twisted by $(-d)$ [35, Appendix A]. With $\varphi_f=\phi_f[-1]$, the constructed potential gives

$$R\Gamma_c(\mathcal{V}_{\mathbf{n}},\varphi_{\mathcal{W}}\mathbb{Q})\simeq R\Gamma_c(\mu^{-1}(0),\mathbb{Q})[-2\dim\mathfrak{g}_{\mathbf{n}}](-\dim\mathfrak{g}_{\mathbf{n}}). \tag{15.32}$$

The equivariant version uses finite-dimensional approximations. Centring replaces the loop dimension by $\dim\mathfrak{g}_{\mathbf{n}}-1$; the zero locus stays the same because μ already has total trace zero. Equation (15.32) is a direct-image comparison, with the shift from compact affine fibres.

For a point sheaf there is an analytic cyclic model as well. The volume germ identifies its cyclic minimal algebra with $\Lambda^* T_x X$, its exterior product and its top-degree trace [135]. With multiplicity n, matrix amplification gives

$$f_n(X,Y,Z)=\operatorname{Tr}(Z[X,Y]). \tag{15.33}$$

For an oriented tangent basis, if $a=Xe_1+Ye_2+Ze_3$, then $f_n=\operatorname{vol}_x\operatorname{Tr}(a^3)/3$. This fixes the normalization. Its critical equations are the three commutator equations. In the formal neighbourhood of the origin, or in a sufficiently small analytic neighbourhood, they describe the corresponding finite-length local-coordinate modules.

Writing the dual odd matrices as X^*, Y^*, Z^* and the degree--2 gauge matrix as b, the full differential is

$$dX^*=[Y,Z],\quad dY^*=[Z,X],\quad dZ^*=[X,Y],\qquad db=-[X,X^*]-[Y,Y^*]-[Z,Z^*].$$

The Jacobi identity gives $d^2b=0$. A single point retains endomorphism dimensions $(1,3,3,1)$; after the paired centre removal they are $(1,2,2,1)$. The ambient orientation line is $\det(\operatorname{End}\mathbb{C}^n)^{-2}$, trivialized by the evaluation determinant. The local model retains the scalar stabilizer as well as its dual.

15.10 The first two-colour collision

For two one-dimensional vertex spaces, remove their common normal centre and retain the relative coordinate s. With r arrows in each dual direction the potential is

$$f=sQ,\qquad Q(x,y)=\sum_{i=1}^{r}x_i y_i,\qquad (s,x,y)\in\mathbb{A}^{2r+1}.$$

The effective relative group $\mathbb{G}_m$ has weights $1,-1,0$ on x, y, s.

Proposition 15.15. *The critical ideal is*

$$J=(Q,sx_1,\dots,sx_r,sy_1,\dots,sy_r)=(x,y)\cap(s,Q). \tag{15.34}$$

For $r=2$, the affine quotient of $\{s=Q=0\}$ is $\{u^2+vw=0\}$. Its quotient on $x\neq 0$ is $T^*\mathbb{P}^1$.

Proof. The derivatives give the first ideal. With $\mathfrak{m}=(x,y)$, multiplication by s is injective modulo $\mathfrak{m}$. Since $Q\in\mathfrak{m}$, an element $sa+Qb$ belongs to $\mathfrak{m}$ exactly when $a\in\mathfrak{m}$, proving the intersection formula. For $r=2$, the invariant matrix $(x_i y_j)$ has rank at most one and trace zero. Write it as $\left(\begin{smallmatrix} u & v\\ w & -u\end{smallmatrix}\right)$; its determinant is $-u^2-vw$. These entries generate the invariant ring. On $x\neq 0$, the vector x defines a line and $y(x)=0$ defines a cotangent vector there. The reciprocal rescaling is precisely the cotangent bundle quotient. □

For $r \geq 2$ the two components are the spectral axis and the irreducible collision quadric. The relative gauge generator remains in the derived model, with

$$db = -\sum_i (x_i \xi_{x_i} - y_i \xi_{y_i}), \qquad d^2 b = -\sum_i (x_i s y_i - y_i s x_i) = 0.$$

Set $\Phi_f = \phi_f[-1]\mathbb{Q}[2r+1]$ on the smooth atlas. Its stalks detect more than the ordinary critical locus.

Proposition 15.16. *For $r \geq 2$, the stalk at the origin has one copy of $\mathbb{Q}$ in degrees $1-2r, -1, 0$ and no other cohomology. At a nonzero point of the spectral axis only degree -1 survives. At a smooth nonzero point of the collision quadric only degree $1-2r$ survives.*

Proof. The map $(s, x, y) \mapsto (s, x, sy)$ identifies $f^{-1}(1)$ with $\mathbb{C}^* \times Q^{-1}(1)$. Projection of $Q^{-1}(1)$ to $x \neq 0$ is an affine bundle with fibre $\mathbb{C}^{r-1}$, so it has the homotopy type of S^{2r-1}. Homogeneity identifies a small local Milnor fibre with a sufficiently large ball sublevel in this global fibre. Those sublevels stabilize: the squared norm is proper and semialgebraic on the smooth affine fibre, its critical set has finitely many connected components, and its value is constant on each. Above these critical values its gradient gives deformation retractions. The Milnor fibre at the origin therefore has type $S^1 \times S^{2r-1}$.

Its reduced cohomology lies in degrees $1, 2r-1, 2r$. The normalization gives $H^j(\Phi_f)_0 = \widetilde{H}^{j+2r}(F_0, \mathbb{Q})$, proving the first assertion. Away from the origin on the axis the transverse function is the nondegenerate quadratic form Q. Away from the origin on the quadric use Q as a coordinate; the transverse function is the node sQ. Their Milnor cohomology degrees are $2r-1$ and one, respectively, giving the remaining shifts. □

For $r = 1$, the two degree-one classes of the Milnor fibre coincide in degree, giving a rank-two stalk there. These are atlas coefficients. Passing to the quotient requires their equivariant structures and retains the classifying-stack directions. Inverting s cannot decide a class supported at the collision: the complex $[\mathbb{C}[s]e \xrightarrow{d} \mathbb{C}[s]]$ with $|e| = -1$ and $de = s$ becomes contractible there, but before localization has $H^0 = \mathbb{C}[s]/(s)$.

15.10.1 *Real-root convolution relations*

In the constructible Hall algebra of nilpotent representations of the preprojective algebra of Q, use Euler-characteristic convolution. Let e_i be the characteristic function of the vertex simple. The divided power $e_i^{(b)} = e_i^{*b}/b!$ is the characteristic function of its b-fold direct sum, since a complete flag in $\mathbb{C}^b$ has Euler characteristic $b!$.

Proposition 15.17. *For distinct vertices i, j, $(\operatorname{ad} e_i)^3(e_j) = 0$. Consequently the real positive Kac–Moody algebra with matrix (15.23) maps to this constructible Hall algebra as a Lie algebra.*

Proof. Evaluate the relation on dimension vector $(3, 1)$ at i, j. Combine the forward arrows into $X : \mathbb{C}^3 \to \mathbb{C}^2$ and the reverse arrows into $Y : \mathbb{C}^2 \to \mathbb{C}^3$. The preprojective relation $YX = 0$ implies $\operatorname{rank} X + \operatorname{rank} Y \leq 2$. Put $I = \operatorname{im} Y$ and $K = \ker X$. A flag counted by $e_i^{(b)} * e_j * e_i^{(3-b)}$ is a b-plane with $I \subset U \subset K$. If $I \not\subset K$, all terms are empty. Otherwise write $r_0 = \dim I$, $s_0 = \dim K$. The flag Euler characteristic is $\binom{s_0 - r_0}{b - r_0}$, and $s_0 - r_0 = 3 - \operatorname{rank} X - \operatorname{rank} Y \geq 1$. Thus

$$\sum_{b=0}^{3} (-1)^b \binom{s_0 - r_0}{b - r_0} = (-1)^{r_0}(1-1)^{s_0 - r_0} = 0.$$

Multiplying by $3!$ gives the cubic Serre relation, up to a common sign. Interchanging the vertices proves the other relation. □

This constructs an actual homomorphism and its enveloping-algebra map. It does not determine its kernel or every imaginary primitive. Those require the root-graded operators and their relations, beyond the Euler matrix and the real-root Serre identities.

15.11 An oriented normal coefficient map

A local comparison can be made on the actual normal Gysin map. Let U be a smooth complex variety, let m be a regular function, and put

$$p : Y = U \times \mathbb{A}^1_{\tau_g} \to U, \qquad f = m\tau_g.$$

Let $j : U \hookrightarrow Y$ be the zero section, $i : Z = (m = 0) \hookrightarrow U$, and $k : Z \times \mathbb{A}^1 \hookrightarrow Y$. The geometric coordinate τ_g has degree zero. Write $\Phi_f = \phi_f[-1]$, extended by zero to Y; in particular Φ_0 is the identity.

For $F \in D^b_c(U;\mathbb{Q})$, normalize the complex Thom map $j_*F \to p^*F[2]$ to have compact trace $+1$. Applying vanishing cycles gives

$$G_F : j_*F \longrightarrow \Phi_f p^*F[2]. \tag{15.35}$$

The source identification follows from proper base change and $f j = 0$. Rank-one dimensional reduction, constructed from the coefficient unit $p^*F \to k_*k^*p^*F$, gives

$$\theta_F : R p_! \Phi_f p^*F[2] \xrightarrow{\sim} i_* i^* F.$$

It applies to these pulled-back coefficients without a smoothness assumption on Z.

Lemma 15.18. *With this orientation,*

$$\theta_F R p_! G_F = (F \longrightarrow i_* i^* F). \tag{15.36}$$

The equality is natural in F and in the normal frame.

Proof. Pull the Thom map to $Z \times \mathbb{A}^1$. The Cartesian zero-section square identifies its composite with restriction to Z followed by the same normal Thom map over Z. Its compact trace is the identity. Naturality of the coefficient unit before and after Φ_f proves (15.36). A complex frame change preserves the real orientation and its trace normalization. □

We now specify the coefficient change. Pull the algebraic structure sheaf to the analytic site, and write it as $\mathscr{O}_U$. Define the dg algebra

$$K_m = \mathscr{O}_U \otimes \Lambda(\eta, \chi), \qquad |\eta| = |\chi| = -1, \quad d\eta = m, \quad d\chi = 0. \tag{15.37}$$

It is the Koszul algebra of the two specified equations $(m, 0)$. The covariant exact realization is

$$\mathcal{R}_m(A) = K_m \otimes^{\mathbb{L}}_{\mathbb{Q}_U} R p_! A, \qquad D^b_c(Y;\mathbb{Q}) \longrightarrow D(K_m). \tag{15.38}$$

The target consists of sheaves of left dg modules; its objects need not be coherent. This construction includes extension of scalars by the specified Koszul algebra. In unrestricted derived categories the right adjoint forgets the module structure and then applies $p^!$.

The algebraic normal resolution uses a separate variable τ_a of degree two. Set $\mathcal{E} = \mathscr{O}_U[\tau_a] \otimes \Lambda(e)$, with $|e| = 1$. Let λ multiply by e and let η contract with e. Thus $\lambda\eta + \eta\lambda = 1$, and

$$D_{\mathcal{E}} = m\lambda + \tau_a \eta, \qquad D^2_{\mathcal{E}} = m\tau_a.$$

Put

$$\mathscr{A} = \operatorname{End}_{\mathscr{O}_U[\tau_{\mathrm{a}}]}(\mathcal{E}) \otimes \Lambda(\chi), \qquad \mathscr{S} = (\mathscr{A}[\tau_{\mathrm{a}}^{-1}]/\mathscr{A})[-1].$$

The differential of $\mathscr{A}$ is $[D_{\mathcal{E}}, -]$; on the shifted quotient it has the opposite sign. Only finite negative Laurent sums are admitted. Order η, χ before λ, and use the shifted left action $a \cdot [x] = (-1)^{|a|}[ax]$.

For homogeneous $\xi \in K_m$ and $n \geq 1$, define

$$r(\xi\lambda\tau_{\mathrm{a}}^{-n}) = \begin{cases} (-1)^{|\xi|}\xi, & n = 1, \\ 0, & n > 1, \end{cases} \qquad (15.39)$$

$$r(\xi\tau_{\mathrm{a}}^{-n}) = 0, \qquad s(\xi) = (-1)^{|\xi|}\xi\lambda\tau_{\mathrm{a}}^{-1}.$$

They are degree-zero K_m-linear chain maps with $rs = 1$. The degree--1 homotopy

$$h(\xi\tau_{\mathrm{a}}^{-n}) = (-1)^{|\xi|+1}\xi\lambda\tau_{\mathrm{a}}^{-n-1}, \qquad h(\xi\lambda\tau_{\mathrm{a}}^{-n}) = 0$$

satisfies $dh + hd = 1 - sr$ and $h^2 = rh = hs = 0$. Indeed $[D_{\mathcal{E}}, \eta] = m$ and $[D_{\mathcal{E}}, \lambda] = \tau_{\mathrm{a}}$. Substitution cancels the successive pole terms; at pole one the regular term is discarded and the remainder is exactly $1 - sr$.

Proposition 15.19. *The coefficient restriction $q_F : K_m \otimes F \to K_m \otimes i_*i^*F$ is a quasi-isomorphism. If $\alpha_F = s \otimes 1_F$ and $\gamma_F = q_F^{-1}(1 \otimes \theta_F)$ in the derived category, then*

$$(r \otimes 1_F)\alpha_F = \gamma_F \mathcal{R}_m(G_F). \qquad (15.40)$$

For constant coefficients, both realized objects have the model K_m and the realized arrow is the identity. The oriented normal class is represented by $\lambda/\tau_{\mathrm{a}}$, with residue 1.

Proof. On Z, the coefficient restriction is the identity on stalks. Off Z, multiplication by η/m contracts the Koszul complex. Tensoring over the field $\mathbb{Q}$ preserves these assertions, so q_F is a quasi-isomorphism. Lemma 15.18 identifies the composite of $\mathcal{R}_m(G_F)$ with $1 \otimes \theta_F$ as q_F. Now $rs = 1$ proves (15.40). The resulting arrow is objectwise equivalent to the identity of $K_m \otimes F$, giving the stated models and zero cochain homotopy for their square. □

The geometric coordinate has been integrated out. The algebraic coordinate has degree two, with the Thom shift $[2]$ and the resolution shift $[-1]$ making $\lambda/\tau_{\mathrm{a}}$ degree zero. These are distinct coordinate algebras. External products of independent normal factors are compatible by compact Künneth, the product orientation and the Koszul tensor rule. The construction also glues for a normal line bundle with a section of its dual.

15.11.1 *The character after fixing the framed slice*

Consider the framed matrices

$$A = \begin{pmatrix} w + u & B_0 \\ 0 & w \end{pmatrix}, \quad B = \begin{pmatrix} y + v & D_0 \\ 0 & y \end{pmatrix}, \quad \iota = \begin{pmatrix} \iota_1 \\ \iota_2 \end{pmatrix}, \quad j_{\mathrm{fr}} = (1, 0).$$

Their moment matrix is

$$[A, B] + \iota j_{\mathrm{fr}} = \begin{pmatrix} \iota_1 & uD_0 - vB_0 \\ \iota_2 & 0 \end{pmatrix}.$$

Eliminate the two regular framing equations by their Koszul contractions and, on $B_0 \neq 0$, use diagonal gauge to set $B_0 = 1$. The remaining equations are $m = uD_0 - v$ and zero, with generators η, χ. For an

auxiliary matrix L, the trace potential is $L_{11}\iota_1 + L_{21}(uD_0 - \nu B_0) + L_{12}\iota_2$. Thus $\tau_g = L_{21}$ is the normal coordinate.

Let $(s,t) \in (\mathbb{C}^*)^2$ scale A, B by s, t, and let h_1, h_2 denote its additive characters. The compensating gauge is $g = \mathrm{diag}(s^{-1}, 1)$ with framing change $h = s^{-1}$. The transformations

$$A' = g(sA)g^{-1}, \quad B' = g(tB)g^{-1}, \quad \iota' = st\, g\iota h^{-1}, \quad j'_{\mathrm{fr}} = h\, j_{\mathrm{fr}} g^{-1}$$

preserve both slice conditions. They give

variables	w, u	D_0	ν, η	χ	τ_g, τ_a, λ
character	h_1	$h_2 - h_1$	h_2	$h_1 + h_2$	$-h_2$.

(15.41)

In fact $D'_0 = (t/s)D_0$, $u' = su$, $\nu' = t\nu$, and the lower-left entry of $g((st)^{-1}L)g^{-1}$ is $t^{-1}\tau_g$. The zero $(2,2)$ moment component retains character $h_1 + h_2$. Hence the residue frame is invariant. Under any degree-zero frame unit a,

$$\tau'_a = a\tau_a, \quad \lambda' = a\lambda, \quad m' = a^{-1}m, \quad \eta' = a^{-1}\eta,$$

and the residue and section agree in the two frames. Writing h_i also for the equivariant first Chern classes, the normal Euler class is $-h_2$; compact Thom normalization requires no inversion of that class.

15.11.2 *Supported specialization keeps the incidence term*

Set $A_0 = \mathbb{C}[w,u]$, $z = w + u$ and $v = t_0 - w$. Use proper rational representatives at $t_0 = \infty$ in the principal-parts modules

$$\begin{aligned} P_z &= A_0[t_0, (t_0 - z)^{-1}]/A_0[t_0], \\ P_w &= A_0[t_0, (t_0 - w)^{-1}]/A_0[t_0], \\ P_{zw} &= A_0[t_0, ((t_0 - z)(t_0 - w))^{-1}]/A_0[t_0]. \end{aligned}$$

Let $C = [P_z \oplus P_w \to P_{zw}]$ in degrees $-1, 0$, with differential $(a,b) \mapsto a - b$. The target of the Cauchy comparison is

$$\mathcal{T} = \bigoplus_{p,r \geq 1} \mathbb{C}[w] u^{-p} v^{-r}.$$

Multiplication discards terms once either pole exponent becomes nonnegative. This is a direct sum of finite principal parts, with no completion in pole order.

The map $\Theta : C \to \mathcal{T}$ is zero in degree -1; in degree zero take first the v-principal part at zero and then the u-principal part. Explicitly,

$$\Theta\left(\frac{1}{v^a (v-u)^b}\right) = (-1)^b \sum_{j=0}^{a-1} \binom{b+j-1}{j} u^{-b-j} v^{j-a}, \qquad a, b \geq 1. \tag{15.42}$$

It is a quasi-isomorphism. The single-pole modules intersect trivially by partial fractions after inverting u and torsion-freeness before inversion. At each total pole order, the binomial expansion in (15.42) is triangular with unit diagonal, proving surjectivity. If a proper rational function has zero image, its w-principal part has no u-pole and lies in P_w. Subtract it; the remainder has only a z-pole and lies in P_z. Monic division by $t_0 - z$ keeps its coefficients in A_0, proving the kernel assertion.

Take $m = uD_0 - v$ and put

$$M_v = \mathcal{T}[D_0, v] \otimes \Lambda(\eta, \chi), \qquad d\eta = uD_0 - v, \quad d\chi = 0.$$

The residue contraction tensored with C and the Cauchy map give quasi-isomorphisms

$$\mathscr{S} \otimes_{A_0} C \xrightarrow{r\otimes 1} K_m \otimes_{A_0} C \xrightarrow{1\otimes\Theta} M_v. \tag{15.43}$$

They commute with derived specialization at $v = 0$: the Koszul terms are bounded free modules and the principal-parts terms are free over $\mathbb{C}[v]$. On the constructible side the vanishing-cycle specialization need not be an isomorphism before normal integration. After $Rp_!$, dimensional reduction identifies it with the restriction of the coefficient unit. The closed inclusion, that unit and the relative compact trace commute with base change, giving the same specialization after extension to K_m.

Set

$$F_1 = \frac{1}{(z - t_0)(t_0 - w)}, \qquad c_1 = ((t_0 - z)^{-1}, (t_0 - w)^{-1}).$$

Then $dc_1 = -uF_1$ and $\Theta(F_1) = u^{-1}v^{-1}$. Before specialization the identities are

$$\begin{aligned} d(\eta F_1 + D_0 c_1) &= -vF_1, \\ d(\eta\chi F_1 - \chi D_0 c_1) &= -v\chi F_1. \end{aligned} \tag{15.44}$$

At $v = 0$, the signed section lifts them to the cycles

$$\begin{aligned} B_\eta &= -\eta\lambda\tau_a^{-1}F_1 + D_0\lambda\tau_a^{-1}c_1, \\ B_{\eta\chi} &= \eta\chi\lambda\tau_a^{-1}F_1 + \chi D_0\lambda\tau_a^{-1}c_1. \end{aligned}$$

Together with $\lambda\tau_a^{-1}F_1$, their images are

$$[u^{-1}v^{-1}], \quad [\eta u^{-1}v^{-1}], \quad [\eta\chi u^{-1}v^{-1}] \tag{15.45}$$

in degrees $0, -1, -2$, and all are nonzero. For the first class, evaluate a prospective boundary at $D_0 = 0$. A degree--1 boundary has only a χ component, so cannot give the second class. There is no incoming differential in degree -2. Multiplication by u kills the displayed principal part, proving closedness. Before projection, omitting $D_0 c_1$ leaves $d(\eta F_1) = uD_0/[v(u - v)] \neq 0$.

The specialization can also be read as Tor. On $V = \mathcal{T}[D_0]$ let $N = uD_0$. The two-term complex $[V[v] \xrightarrow{N-v} V[v]]$ has degree-zero cohomology V, with v acting as N, and no degree--1 cohomology. Injectivity follows by taking the highest v coefficient; polynomial division gives the cokernel. Its derived zero fibre therefore retains

$$\operatorname{Tor}_1^{\mathbb{C}[v]}(V, \mathbb{C}) = \ker N = \bigoplus_{r\geq 1} \mathbb{C}[w, D_0]u^{-1}v^{-r}.$$

The cokernel of N is $\mathcal{T}$, since u acts surjectively on $\mathcal{T}$. Adjoining χ retains both copies with their shifts. On a generic fibre, however, $(N - v)^{-1} = -\sum_{j\geq 0} N^j v^{-j-1}$; the sum is finite on each principal part. The generic complex is contractible, while its derived special fibre has the classes (15.45).

15.11.3 *The kernel test for a Hall comparison*

The functor (15.38) has a concrete nonzero kernel. Let $a : U \times \mathbb{C}^* \hookrightarrow Y$ and put $L = Ra_*\mathbb{Q}_{U\times\mathbb{C}^*}$. On each normal line the localization triangle is

$$\mathbb{Q}_{\{0\}}[-2] \xrightarrow{\text{Thom}} \mathbb{Q}_{\mathbb{C}} \longrightarrow Ra_*\mathbb{Q}_{\mathbb{C}^*} \longrightarrow .$$

Compact pushforward takes its first arrow to the identity of $\mathbb{Q}[-2]$. Hence $Rp_!L = 0$, although $L \neq 0$. Its restriction to the zero section is $\mathbb{Q}_U \oplus \mathbb{Q}_U[-1]$. For the ordinary internal tensor,

$$\mathcal{R}_m(L) = 0, \qquad \mathcal{R}_m(L \otimes j_*\mathbb{Q}_U) = K_m \oplus K_m[-1]. \tag{15.46}$$

When Z is nonempty the second object is nonzero. Thus this realization is not strong monoidal for that tensor. Its external-product comparison remains valid.

There is a separate proper-image test on the algebraic models. For $f : \mathbb{P}^1 \to \mathrm{pt}$, take $m = 0$ and $K_0 = \mathbb{C} \otimes \Lambda(\eta, \chi)$. Constructible proper pushforward followed by realization gives $K_0 \oplus K_0[-2]$, since $H^*(\mathbb{P}^1, \mathbb{Q})$ has degrees zero and two. Ordinary coherent pushforward of the model $\mathcal{O}_{\mathbb{P}^1} \otimes K_0$ gives K_0. Indeed the degree-one Čech quotient is

$$\mathbb{C}[z, z^{-1}]/(\mathbb{C}[z] + \mathbb{C}[z^{-1}]) = 0.$$

The two complexes have cohomological widths four and two; a uniform shift cannot identify them. This excludes that proposed use of ordinary coherent direct image. It is not a statement about every functor on the sheaf-module target.

For an actual critical Hall convolution $\star_{\mathrm{H}}$ and specified cut realizations $\mathcal{R}_d$, descent through their kernels $\mathcal{N}_d$ requires

$$\mathcal{R}_{d+e}(A \star_{\mathrm{H}} B) = 0 \quad \text{if } A \in \mathcal{N}_d \text{ or } B \in \mathcal{N}_e. \tag{15.47}$$

The ordinary tensor test does not decide this different product. The normal square constructs its stated coefficient arrow; a Hall lift must still prove (15.47) and compare the actual proper traces and supported incidence maps.

15.12 Vertical, wrapped and mixed support

The projection $\pi_E : X \to E$ gives a useful support decomposition. Call a coherent sheaf *vertical* if its support has finite image in E, and let $\mathcal{T}$ be their full subcategory. Let $\mathcal{F}$ consist of coherent sheaves flat over E.

Proposition 15.20. *The pair $(\mathcal{T}, \mathcal{F})$ is a torsion pair in* $\mathrm{Coh}(X)$. *Every sheaf has a functorial exact sequence*

$$0 \longrightarrow t_E(F) \longrightarrow F \longrightarrow F/t_E(F) \longrightarrow 0 \tag{15.48}$$

with the first term vertical and the last flat over E, and $\mathrm{Hom}(\mathcal{T}, \mathcal{F}) = 0$.

Proof. The sum of all vertical subsheaves stabilizes because F is noetherian; its support has finite image in E. This defines $t_E(F)$. The quotient has no torsion over the local discrete valuation rings of the smooth curve E, so it is flat over them. A map from a vertical sheaf to an E-flat sheaf has torsion image and is zero. These characterizations also prove functoriality. □

Sheaves pulled back from S are examples of wrapped support, and belong to $\mathcal{F}$. The torsion-pair sequence does not yield a product decomposition of families. On E, let F_t be the family of extensions

$$0 \to \mathcal{O}_E(-p) \to F_t \to \mathcal{O}_p \to 0$$

with extension parameter t. For $t \neq 0$, $F_t \simeq \mathcal{O}_E$ has no torsion; at zero it is $\mathcal{O}_E(-p) \oplus \mathcal{O}_p$ and its torsion has length one. The total family is flat in t, but its torsion parts are not. Pushing it into $\{s\} \times E$ gives the same example on X. Hall identities using (15.48) must therefore use the corresponding strata of the extension stack.

The mixed Ext groups retain the normal direction. For $i : S \times \{e\} \hookrightarrow X$ and perfect A, B on S,

$$\begin{aligned} R\operatorname{Hom}_X(\pi_S^* B, i_* A) &\simeq R\operatorname{Hom}_S(B, A), \\ R\operatorname{Hom}_X(i_* A, \pi_S^* B) &\simeq R\operatorname{Hom}_S(A, B) \otimes W[-1], \\ R\operatorname{Hom}_X(i_* A, i_* B) &\simeq R\operatorname{Hom}_S(A, B) \otimes (\mathbb{C} \oplus W[-1]), \\ R\operatorname{Hom}_X(\pi_S^* A, \pi_S^* B) &\simeq R\operatorname{Hom}_S(A, B) \otimes R\Gamma(E, \mathcal{O}_E). \end{aligned} \tag{15.49}$$

The first two follow from the point resolution on E: $R\operatorname{Hom}_E(\mathcal{O}_E, \mathcal{O}_e) = \mathbb{C}$ and $R\operatorname{Hom}_E(\mathcal{O}_e, \mathcal{O}_E) = W[-1]$. The other two follow from its self-Ext and from Künneth. Thus the two directions of a mixed extension carry different shifts, and both are needed in a cyclic comparison.

15.13 Reduced invariants and the remaining coefficient map

A reduced virtual-class theory exists for stable pairs on $S \times E$. Let $\beta \neq 0$ be an effective K3 curve class, $d \geq 0$, and put $P = P_n(X, (\beta, d))$. Its standard perfect obstruction theory $\mathbb{E} \to L_P$ has amplitude $[-1, 0]$. The semiregularity morphism is

$$\operatorname{sr} : \mathbb{E}^\vee \longrightarrow H^{1,3}(X) \otimes \mathcal{O}_P[-1].$$

It is surjective on obstruction sheaves. Its dual cone gives a reduced obstruction theory of virtual dimension one. The existence of the map from that cone to L_P uses the twistor deformation of the K3 surface; it does not follow merely by deleting a vector from the obstruction space.

The elliptic orbit vector and semiregularity are Serre dual. One way to see their relationship is to express both through the universal Atiyah class: contraction with the invariant vector field gives the orbit direction, while composition and trace give semiregularity. This duality allows a second cone, removing the orbit direction, to carry a symmetric obstruction theory on an actual centre fibre. Determinant and Fourier–Mukai constructions give a centre map of nonzero weight, so that this fibre has a finite-group quotient as in Proposition 15.6.

For the incidence convention, let $\mathcal{I}$ be the universal stable-pair complex on $X \times P$, with projections π_X, π_P. For $\gamma \in H^4(X, \mathbb{Q})$ put

$$\tau_0(\gamma) = -\pi_{P,*}\big(\operatorname{ch}_2(\mathcal{I}) \cup \pi_X^* \gamma\big).$$

This is a degree-two class on P.

Theorem 15.21 (Reduced stable-pair comparison)**.** *For these nonzero K3 classes, the quotient P/E has a zero-dimensional virtual class whose pullback along $\pi : P \to P/E$ is*

$$\pi^*[P/E]^{\mathrm{vir}} = [P]^{\mathrm{red}}. \tag{15.50}$$

Its degree is the Behrend-weighted Euler characteristic of the quotient, with finite stabilizers counted orbifoldly. It equals the reduced incidence invariant obtained by inserting $\tau_0(\beta^\vee \boxtimes [0])$, where $\langle \beta, \beta^\vee \rangle = 1$.

This is the quotient and incidence theorem of Oberdieck [121, Theorems 1–2]. Concretely, if K is a centre fibre with finite group G, its class on the quotient is $|G|^{-1}q_*[K]^{\mathrm{vir}}$. The symmetric obstruction theory computes its weighted Euler characteristic. The virtual-class comparison uses the common Fulton class and Segre-class expression; the normalized incidence intersects the elliptic orbit with degree one. This construction is independent of the chosen centre fibration.

The original standard virtual class is zero in this sector. Likewise, a surjective cosection has empty degeneracy locus for the cosection-localized standard class [95, Theorem 1.1]. The reduced class above uses the separate reduced obstruction theory. For $\beta \neq 0$, the reduced Donaldson–Thomas and stable-pair series also agree [121, Theorem 3(i)]. That comparison uses the local Hall identity and Behrend integration for the quotient; the vanishing of $\chi(S \times E)$ alone does not construct it.

Equivariant Hall integration gives another precise numerical construction. For the elliptic translation action on the smooth projective Calabi–Yau threefold, the regular equivariant Hall algebra has a semiclassical Poisson quotient. Its integration map takes values in the charge algebra with dual numbers $\mathbb{Q}[\varepsilon]/(\varepsilon^2)$ [123, Theorem 5]. The two terms integrate the fixed locus and the moving quotient, respectively, with the sign $(-1)^{\dim E}$ on the latter. With χ the sheaf Euler pairing, the target basis c_α has product $c_\alpha c_\beta = (-1)^{\chi(\alpha,\beta)}c_{\alpha+\beta}$ and Poisson bracket $\{c_\alpha, c_\beta\} = (-1)^{\chi(\alpha,\beta)}\chi(\alpha,\beta)c_{\alpha+\beta}$.

Write the integration as $I_0 + \varepsilon I_1$. Then I_0 is unital and multiplicative, and

$$I_1(ab) = I_0(a)I_1(b) + I_1(a)I_0(b), \qquad I_1(1) = 0. \tag{15.51}$$

Thus the moving contribution is an I_0-derivation. It supplies reduced numbers, with a different domain and unit law from a state-valued Hall action.

The preceding constructions fix substantial parts of a global coefficient comparison: the common torsor and its relative unit; the disjoint coefficient and trace; the full scalar and orbit complexes; explicit compact coefficients and split roots; one formal cyclic comparison on finite flags; and an oriented normal Gysin map with its supported specialization. Their global assembly has a definite remaining equation. For a proposed reduced critical enhancement of the common-torsor extension stack, write $\mathrm{Filt}_{\alpha,\beta}$ for its filtered objects, let gr take its graded pair and let ev forget the filtration. For specified virtual index $\iota_{\alpha,\beta}$, the desired critical coefficient map is

$$\mathrm{gr}^*(\varphi_\alpha \boxtimes_{BE} \varphi_\beta)[\iota_{\alpha,\beta}] \longrightarrow \mathrm{ev}^!\,\varphi_{\alpha+\beta}. \tag{15.52}$$

Problem 15.22. Construct the compact reduced critical coefficient system φ with its relative unit, cyclic pairing and strong orientation, and the maps (15.52) on the full extension stack. Prove their compatibility with every flag refinement, finite-isogeny descent, collision specialization and the support conditions needed for the chosen pushforward. Compare their trace and orbit reductions with (15.10) and (15.11), and their normal restriction with (15.43).

The ordinary critical Hall theorem of Chapter 3.5 supplies its coefficient map under the stated Calabi–Yau, geometric and strong orientation hypotheses. Applying it to this reduced relative stack requires those data on that stack. A further comparison is then needed with the independently defined chiral states: it must preserve normal oscillators, mixed wrapped collisions, the local vacuum and cycle joining. The numerical trace (15.17) and integration (15.51) remain positive consequences in their own coefficient categories.

Chapter 16

Primitive operators and geometric states

An automorphic product can prescribe the difference between even and odd root multiplicities without prescribing either multiplicity. The first interior root of the Igusa half-denominator is a useful example. Its root space has dimensions $29 \mid 93$. An independently defined elliptic coefficient on an Enriques surface has dimensions $1 \mid 65$. Both have superdimension -64. A differential could remove twenty-eight pairs, but it would also have to respect the real-root operators. Their Serre relations detect an obstruction in height six.

The geometric operators carry further information. Incidence correspondences on Hilbert schemes give creation and annihilation; the surface diagonal gives joining and splitting. Elliptic characteristic classes transport these operations to new lattices. Their degeneration at $y = 1$ has parameter torsion, higher products and nontrivial derived fibres. Chiral de Rham sheaves provide states with fields before an index is taken. Their normal-coordinate systems make the remaining diagonal and Hall comparisons concrete.

We work over $\mathbb{Q}$ for ordinary cohomology and extend to $\mathbb{C}$ for holomorphic fields. A complex is cohomologically graded, with $C[r]^j = C^{j+r}$. For a complex of super vector spaces, the sign in tensor interchange is the sum of cochain degree and intrinsic parity. We specify a different grading whenever a parameter is used to form a geometric cochain model.

16.1 The isotropic face on an Enriques surface

Let T be a smooth projective complex Enriques surface. Its rational cohomology is concentrated in degrees 0, 2, 4, with dimensions 1, 10, 1. Choose an ample class h, put $d = \int_T h^2$, and normalize $\omega \in H^4(T;\mathbb{Q})$ by $\int_T \omega = 1$. Then

$$H^*(T;\mathbb{Q}) = \mathbb{Q}1 \oplus \mathbb{Q}h \oplus P \oplus \mathbb{Q}\omega, \qquad P = h^{\perp} \subset H^2(T;\mathbb{Q}), \qquad \dim P = 9.$$

The intersection pairing is nondegenerate and $d \neq 0$. These facts also follow from the canonical K3 cover [47, § 3].

Proposition 16.1. *Make P central and define brackets on the remaining three directions by*

$$[h, \omega] = -2\omega, \qquad [h, 1] = 2 \cdot 1, \qquad [\omega, 1] = \frac{2}{d}h. \tag{16.1}$$

This gives an even Lie algebra $\mathfrak{l}_T \cong \mathfrak{sl}_2 \oplus P$ on $H^*(T;\mathbb{Q})[2]$. With

$$E_+ = \frac{d}{2}\omega, \qquad H_0 = -h, \qquad F_- = -1,$$

the adjoint action of E_+ is cup product by h. The Lie subalgebra

$$\mathfrak{b}_T = \mathbb{Q}F_- \oplus t\mathfrak{l}_T[t] \subset \mathfrak{l}_T[t]$$

has root-space denominator

$$(1-y^{-1})\prod_{n\geq 1}(1-p^n y)(1-p^n)^{10}(1-p^n y^{-1}), \tag{16.2}$$

where F_- has degree $(0,-1)$ and t^nE_+, t^nH_0, t^nP, t^nF_- have degrees $(n,1)$, $(n,0)$, $(n,0)$, $(n,-1)$, respectively.

Proof. The three displayed elements satisfy

$$[H_0,E_+]=2E_+,\qquad [H_0,F_-]=-2F_-,\qquad [E_+,F_-]=H_0.$$

Together with central P, these relations prove Jacobi. Moreover

$$[E_+,1]=h,\qquad [E_+,h]=d\omega,\qquad [E_+,\omega]=[E_+,P]=0,$$

which is precisely $h\cup -$. Bracket adds the t-degree, so $\mathfrak{b}_T$ is closed. The two elements F_-, tE_+ first generate tH_0, tF_- and then all three $\mathfrak{sl}_2$ directions in every positive degree. Each space t^nP is an additional central summand. There are therefore ten directions at $(n,0)$ and one at each adjacent degree, giving (16.2). □

Both cubic relations

$$(\operatorname{ad}F_-)^3(tE_+)=0,\qquad (\operatorname{ad}(tE_+))^3F_-=0$$

hold by the three-dimensional $\mathfrak{sl}_2$ calculation. The nine central directions at each positive isotropic multiple agree with the simple isotropic multiplicity in the Gritsenko–Nikulin presentation [74, Theorem 2.3]. This is an explicitly defined Lefschetz Lie structure. The commutator of the cup product itself is zero, and the same imaginary roots need not remain central after the third real generator is included.

There are actual current fields for this algebra. Induce the trivial module of $\mathfrak{b}_T\otimes\mathbb{Q}[z]$ to the loop Lie algebra $\mathfrak{b}_T\otimes\mathbb{Q}[z,z^{-1}]$. Its vacuum module has PBW space

$$U\big(\mathfrak{b}_T\otimes z^{-1}\mathbb{Q}[z^{-1}]\big),$$

and the current of $a\in\mathfrak{b}_T$ is $\sum_r (az^r)w^{-r-1}$. The loop bracket gives $[a_\lambda b]=[a,b]$; translation sends az^{-m} to maz^{-m-1}. These are the universal level-zero current fields of the face algebra. Their vacuum and their products are defined algebraically; a compact Hall realization requires an action comparison.

16.2 Real Hall generators and primitive dimensions

The spherical collection of Theorem 15.11 has Euler matrix

$$G=\begin{pmatrix}2&-2&-2\\-2&2&-2\\-2&-2&2\end{pmatrix}=4I-2\mathbf{1}\mathbf{1}^{\mathrm{t}}. \tag{16.3}$$

Let Q be the quiver with two arrows $i\to j$ for each $i<j$. Its doubled quiver has the preprojective algebra Π_Q, obtained by imposing the moment relations $\sum_{a:i\to j}a^*a-\sum_{a:j\to i}aa^*=0$ at each vertex, with the rightmost map acting first. We use finite-dimensional nilpotent Π_Q-modules: sufficiently long paths act by zero.

For invariant constructible functions, define convolution by Euler integration over submodules. The *composition algebra* $\mathcal{C}_Q$ is the subalgebra generated by the characteristic functions e_i of the vertex simples. Its unit is the characteristic function of the zero module. Every iterated product is the Euler integral over

the corresponding module flags, so forgetting intermediate subobjects proves associativity in every finite arity.

Proposition 15.17 proves the cubic relations directly from those flags. In dimension $(3,1)$ on two vertices, the arrows give $X : \mathbb{C}^3 \to \mathbb{C}^2$ and $Y : \mathbb{C}^2 \to \mathbb{C}^3$ with $YX = 0$. A contributing b-plane lies between $\operatorname{im} Y$ and $\ker X$. If this inclusion is impossible the flag variety is empty; otherwise its Euler characteristic is $\binom{s-r}{b-r}$, with $r = \operatorname{rk} Y$ and $s = 3 - \operatorname{rk} X$. The relation $YX = 0$ implies $\operatorname{rk} X + \operatorname{rk} Y \le 2$, and hence

$$\sum_{b=0}^{3} (-1)^b \binom{s-r}{b-r} = (-1)^r (1-1)^{s-r} = 0.$$

After the divided-power factors are restored this is $(\operatorname{ad} e_i)^3 e_j = 0$.

Lusztig's semicanonical realization theorem gives the stronger identification

$$U(\mathfrak{n}_G^+) \xrightarrow{\sim} C_Q, \qquad e_i \longmapsto \mathbf{1}_{S_i}, \tag{16.4}$$

where $\mathfrak{n}_G^+$ is the positive Kac–Moody algebra with matrix (16.3). The precise nilpotent constructible-function algebra and its flag basis are described in [65, §§ 1.1, 2.1]. This theorem identifies the real composition algebra. It does not identify all imaginary primitives of a cohomological Hall algebra or transport their operations to compact K3 moduli.

The difference is visible in two exact quiver counts. Let $A_{Q,\mathbf{d}}(v)$ count absolutely indecomposable Q-representations of dimension $\mathbf{d}$ over a field with v elements.

Proposition 16.2. *For the quiver Q,*

$$A_{Q,(1,1,0)}(v) = v + 1, \qquad A_{Q,(1,1,1)}(v) = (v+1)^2(v^2+2) = v^4 + 2v^3 + 3v^2 + 4v + 2.$$

Proof. At two vertices the two arrows form a nonzero vector in $\mathbb{F}_v^2$, modulo scalar multiplication. This gives $(v^2-1)/(v-1) = v+1$ classes. At three vertices each edge of the triangle carries a two-vector. The representation is indecomposable exactly when the nonzero edges connect the vertices. Connectedness forces every endomorphism to be a common scalar, over every field extension, so these objects are absolutely indecomposable. Put $a = v^2 - 1$. There are $3a^2 + a^3$ connected assignments, and the effective vertex torus acts freely with order $(v-1)^2$. Dividing gives the asserted polynomial. □

For comparison with BPS spaces, retain their coefficient ring. Let $\mathbb{T}$ be the arrow-scaling torus preserving the canonical tripled-quiver potential, in the equivariant theory of [88, § 1.4]. For its full BPS space $\mathfrak{g}_{\mathbf{d}}^{\mathbb{T}}$, the Kac–BPS identity has convention

$$\operatorname{grdim} \mathfrak{g}_{\mathbf{d}}^{\mathbb{T}} = (1-t)^{-\dim \mathbb{T}} A_{Q,\mathbf{d}}(t^{-1}), \qquad \operatorname{grdim} V = \sum_j \dim V^j \, t^{j/2}. \tag{16.5}$$

The equivariant module is free. Reducing it along the augmentation of $\mathbb{Q}[\operatorname{Lie} \mathbb{T}]$ therefore gives total dimensions 2 and 12 in the two displayed charges [88, § 1.4, after (19)]. The current variable in Hall PBW is a separate factor. Since $|\operatorname{sdim} V| \le \dim V$, these finite spaces cannot be the Igusa root spaces of signed dimensions 10 and -64. The conclusion concerns the finite parts in (16.5); restoring equivariant variables, current modes or new geometric coefficients changes the comparison.

16.3 The first interior root

Let $\alpha_1, \alpha_2, \alpha_3$ have Gram matrix (16.3). The eigenvalues are $4, 4, -2$ and the determinant is -32. Set

$$\delta_{ij} = \alpha_i + \alpha_j, \qquad \gamma = \alpha_1 + \alpha_2 + \alpha_3.$$

The pair roots are isotropic and $(\gamma, \gamma) = -6$. The Weyl vector $\rho = (\alpha_1 + \alpha_2 + \alpha_3)/2$ satisfies $(\rho, \alpha_i) = -1$ and $(\rho, \rho) = -3/2$.

The weak Jacobi form needed for these multiplicities is

$$\phi_{0,1}(q, y) = 4 \sum_{j=2}^{4} \frac{\vartheta_j(q, y)^2}{\vartheta_j(q, 1)^2},$$

where

$$\vartheta_2 = \sum_{n \in \mathbb{Z}} q^{(n+1/2)^2/2} y^{n+1/2}, \quad \vartheta_3 = \sum_{n \in \mathbb{Z}} q^{n^2/2} y^n, \quad \vartheta_4 = \sum_{n \in \mathbb{Z}} (-1)^n q^{n^2/2} y^n.$$

The fractional powers cancel in the sum, giving

$$\phi_{0,1} = y + 10 + y^{-1} + q(10y^2 - 64y + 108 - 64y^{-1} + 10y^{-2}) + O(q^2). \tag{16.6}$$

For example, the q coefficient of the first ratio multiplied by four is $2y^2 - 4 + 2y^{-2}$. The other two together contribute $8((y + y^{-1})^2 - 8(y + y^{-1}) + 12)$.

Use the minimal Gritsenko–Nikulin presentation associated to the Igusa half-denominator [74, §§ 2–3]. "Minimal" here fixes the parity prescription: positive simple supermultiplicities give even simple generators, negative ones give odd simple generators, and no cancelling pairs are added. There are three even real generators e_i and nine even imaginary simple generators at each δ_{ij}. Write B_{ij} for the latter space. The full root exponent at δ_{ij} is ten, including $[e_i, e_j]$; the exponent at γ is -64.

Proposition 16.3. *The presentation has 93 odd simple generators c^u at γ. Its root space at γ has parity dimensions*

$$\dim \mathfrak{g}_\gamma = (29 \mid 93). \tag{16.7}$$

Proof. Put $x_i = e^{-\alpha_i}$. Up to squarefree degree $x_1x_2x_3$, the denominator after removing its Weyl monomial is

$$\prod_i (1 - x_i) \prod_{i<j} (1 - x_ix_j)^{10} (1 - x_1x_2x_3)^{-64}.$$

Its coefficient of $x_1x_2x_3$ is $-1 + 30 + 64 = 93$. No nonidentity term of the Weyl alternating sum contributes there. A reduced word of length two already displaces the Weyl vector by $3\alpha_i + \alpha_j$, of height four; longer reduced words add positive roots. A simple reflection has displacement α_i. It fixes a null pair containing i, so the resulting monomial repeats α_i; it adds $4\alpha_i$ to the opposite pair. Two imaginary corrections have height at least four. The coefficient 93 is therefore the negative of the simple supermultiplicity at γ. The minimal prescription makes these 93 generators odd.

Real Serre relations start in height four. The multilinear part of the free Lie algebra on e_1, e_2, e_3 has dimension two. For each pair $\{i, j\}$, the nine brackets $[e_k, B_{ij}]$, with k the remaining index, give independent even vectors. The relations $[e_i, B_{ij}] = [e_j, B_{ij}] = 0$ occur in other root degrees, and no further defining relation occurs here. Thus the even dimension is $2 + 3 \cdot 9 = 29$. The odd part consists of the 93 simple generators. □

This calculation distinguishes simple multiplicities from root multiplicities. It also distinguishes either parity dimension from the signed exponent $29 - 93 = -64$. A finite Lie algebra retaining the calculation is obtained from the generators e_i, B_{ij}, c^u by imposing the incident commutators above and

killing all root components of height at least four. Call it $L_{\le 3}$. Its height-one dimension is 3, its height-two dimension is 30, and its height-three dimensions are $35 \mid 93$: eight real Lie words and twenty-seven mixed even brackets. Hence $\dim L_{\le 3} = 161$. It is a Lie quotient, so Jacobi holds without a choice of bracketing table.

Height truncation of a Lie algebra and height truncation of its enveloping algebra are different operations. Even for one primitive element x, the quotient $\mathbb{C}[x]/(x^{H+1})$ is not a Hopf quotient in characteristic zero: the coproduct of x^{H+1} has nonzero terms $\binom{H+1}{j} x^j \otimes x^{H+1-j}$ with $0 < j < H+1$. By contrast, the enveloping algebra of a finite-height Lie quotient is a Hopf algebra, generally infinite-dimensional.

16.4 The canonical cover and an elliptic coefficient

The canonical cover explains the half-index geometrically, but its two characters must be retained when morphisms are composed. Fix $K_T^{\otimes 2} \cong \mathcal{O}_T$ and write

$$\pi : S = \operatorname{Spec}_T(\mathcal{O}_T \oplus K_T) \longrightarrow T.$$

This is the étale K3 double cover. For perfect complexes F, G on T, put

$$C^\epsilon(F,G) = R\operatorname{Hom}_T(F, G \otimes K_T^\epsilon), \qquad \epsilon \in \mathbb{Z}/2.$$

Compose f of character ϵ and g of character η by $(g \otimes 1_{K_T^\epsilon})f$, using the chosen square trivialization. The character is independent of cochain parity.

Proposition 16.4. *There is a composition-compatible equivalence*

$$R\operatorname{Hom}_S(\pi^*F, \pi^*G) \simeq C^0(F,G) \oplus C^1(F,G).$$

The K3 Serre trace pairs opposite characters. The invariant character alone is not a 2-Calabi–Yau category.

Proof. Finite étale adjunction and the projection formula give $\pi_*\pi^*G = G \otimes (\mathcal{O}_T \oplus K_T)$. The algebra multiplication on this direct sum gives precisely the stated composition. The deck involution acts by -1 on the K3 volume form, so the trace is nonzero on the anti-invariant character; two paired morphisms must have opposite characters. For $F = G = \mathcal{O}_T$, the invariant endomorphisms have cohomology $(1,0,0)$, while the K_T character has cohomology $(0,0,1)$. Removing the latter removes the Serre partner of the identity. □

There is a related geometric caution. The free involution $(\iota, -1)$ on $S \times E$ preserves its holomorphic three-form, so the quotient has trivial canonical bundle. Translation by $a \in E$ descends to this quotient precisely when $2a = 0$, since it must commute with $z \mapsto -z$. This quotient does not retain the full translation action used in the common-torsor construction of Section 15.1.

For a smooth projective variety Y of even complex dimension d, define the elliptic tensor class

$$\mathcal{E}_{q,y}(Y) = y^{-d/2} \bigotimes_{r \ge 1} \Big(\Lambda_{-yq^{r-1}} T_Y^* \otimes \Lambda_{-y^{-1}q^r} T_Y \otimes \operatorname{Sym}_{q^r} T_Y^* \otimes \operatorname{Sym}_{q^r} T_Y \Big). \tag{16.8}$$

An exterior power of degree a has intrinsic parity a; its coefficient retains a positive monomial weight. Thus the minus signs in (16.8) record supercharacters rather than negative vector spaces. Every $(q^m y^\ell)$ coefficient is a finite superbundle. Its derived global sections form a perfect supercomplex $V_{m,\ell}(Y)$, with index

$$e_Y(m,\ell) = \sum_{j,\epsilon} (-1)^{j+\epsilon} \dim H^j(V_{m,\ell}(Y))_\epsilon.$$

Proposition 16.5. *For an Enriques surface, the $q\,y$ coefficient of* (16.8) *is internally even and equals*

$$B_{11} = \operatorname{Sym}^2 \Omega^1_T \oplus K_T \oplus \Omega^1_T \oplus T_T. \tag{16.9}$$

Its cohomology dimensions in degrees 0, 1, 2 *are* (0, 65, 1)*, so its total parity dimensions are* 1 | 65*. The $q\,y^2$ coefficient is ΠT_T and has ten even cohomology classes. On a K3 surface the $q\,y$ coefficient has cohomology dimensions* (1, 130, 1).

Proof. Extracting energy one in the tensor product gives

$$\Omega^1_T \otimes \Omega^1_T \oplus K_T \otimes \Omega^1_T \oplus K_T \otimes T_T.$$

Use $\Omega^1_T \cong T_T \otimes K_T$ and $K_T^{\otimes 2} \cong \mathcal{O}_T$ to obtain (16.9). Every summand is even. At Laurent weight two the only summand is the odd bundle $K_T \otimes \Omega^1_T \cong T_T$.

A K3 surface has no positive symmetric holomorphic tangent or cotangent tensors, by the Bochner and holonomy argument [98, Corollary 8]. Pullback to the canonical cover gives the same vanishing for the corresponding Enriques bundles and their canonical twists. Serre duality therefore concentrates the cohomology of $\operatorname{Sym}^2 \Omega^1_T$, Ω^1_T, T_T in degree one. Rationally $c_1(T_T) = 0$, $\int_T c_2 = 12$, and

$$\operatorname{ch}(\operatorname{Sym}^2 \Omega^1_T) = 3 - 4c_2, \qquad \operatorname{td}(T_T) = 1 + c_2/12.$$

Riemann–Roch gives their degree-one dimensions 45, 10, 10. The canonical line has only $H^2(K_T) = \mathbb{C}$. This proves (0, 65, 1); the odd parity of ΠT_T changes its ten degree-one classes to total even parity. For K3 the three degree-one dimensions are 90, 20, 20, and $K_S = \mathcal{O}_S$ contributes in degrees zero and two. □

The coefficient bundle is the symbol of a field sheaf, whose filtration need not split. Section 16.10 will recover the same dimensions for the actual chiral coefficient by a field-theoretic projector. Neither calculation identifies it with the root space (16.7).

16.4.1 Cyclotomic coefficient states

The coefficient complexes themselves give another state construction. On the energy-r coefficient let μ_n act by ζ^r, for a primitive nth root ζ. The averaging projector $n^{-1} \sum_{j=0}^{n-1} \zeta^{jr}$ retains exactly the energies divisible by n. Regrade $r = nm$ to energy m and assign length n. This gives the complex

$$\mathcal{P}_Y = \bigoplus_{n>0, m\geq 0, \ell\in\mathbb{Z}} p^n q^m y^\ell V_{nm,\ell}(Y), \qquad \mathcal{F}_Y^{\mathrm{cyc}} = \widehat{\operatorname{Sym}}_{\mathrm{super}}(\mathcal{P}_Y).$$

The differential extends as a derivation; the scalar at length zero is the vacuum. Use total cochain and intrinsic parity in the symmetry, and complete coefficientwise in nonnegative length and energy. At fixed $p^N q^M$, only $n \leq N, m \leq M$ occur, and each tensor coefficient has finitely many Laurent weights. Thus this completion is defined before cancellation in the index.

For a homogeneous vector v and a coefficientwise continuous dual vector λ, creation by multiplication and annihilation by the corresponding derivation satisfy $[D_\lambda, M_v]_{\mathrm{super}} = \lambda(v)$. The index is

$$\operatorname{sch} \mathcal{F}_Y^{\mathrm{cyc}} = \prod_{n>0, m\geq 0, \ell} (1 - p^n q^m y^\ell)^{-e_Y(nm,\ell)}. \tag{16.10}$$

Indeed each finite complex splits over the ground field into cohomology and contractible pairs. A contractible even–odd pair contributes one to the symmetric-power index; an even class of weight x contributes $(1-x)^{-1}$ and an odd class contributes $1 - x$. The finite coefficientwise product proves the formula. This constructs complexes and state operators from the elliptic tensor coefficients. A geometric twisted-sector sheaf is defined separately below. Equal indices do not identify these complexes or their geometric operations.

16.5 A differential tested by the Serre relations

Let W_γ be the even part of the root space, and let R_γ be its subspace of Lie words in the real generators. There is a direct sum decomposition

$$W_\gamma = R_\gamma \oplus [e_3, B_{12}] \oplus [e_2, B_{13}] \oplus [e_1, B_{23}], \qquad \dim R_\gamma = 2.$$

Choose independent $w_1, \dots, w_{28} \in W_\gamma$. Put the even generators in cochain degree zero and the odd generators $c^1, \dots, c^{93}$ in degree -1. Their former super parity is now the parity of cochain degree, with no additional intrinsic grading. In $L_{\le 3}$ the assignments

$$dc^u = \begin{cases} w_u, & u \le 28, \\ 0, & u > 28, \end{cases} \qquad de_i = dB_{ij} = 0 \tag{16.11}$$

define a dg Lie algebra. Its degree-γ complex has cohomology $\mathbb{C}^{65}$ in degree -1 and $\mathbb{C}$ in degree zero. After choices, it is quasi-isomorphic to $R\Gamma(T, B_{II})[2]$. This is a comparison of complexes in one root degree.

The full positive presentation has more relations. Let e_i, f_i, h_i be the Chevalley triple for the ith real root in the full root algebra, and write $E_i = \operatorname{ad} e_i$, $F_i = \operatorname{ad} f_i$ and $H_i = \operatorname{ad} h_i$. On W_γ,

$$[E_i, F_i] = H_i, \qquad H_i w = -2w.$$

The real–imaginary Serre relation is $E_i^3 c^u = 0$, since $1 - (\alpha_i, \gamma) = 3$.

Lemma 16.6. *The map $w \mapsto (F_1 w, F_2 w, F_3 w)$ is injective on W_γ, and $F_i^2 W_\gamma = 0$ for every i.*

Proof. Use $u = [e_1, [e_2, e_3]]$ and $v = [e_2, [e_1, e_3]]$ as a basis of R_γ. Jacobi and $[f_i, e_i] = -h_i$ give

	u	v
F_1	$4[e_2, e_3]$	$2[e_2, e_3]$
F_2	$2[e_1, e_3]$	$4[e_1, e_3]$
F_3	$-2[e_1, e_2]$	$2[e_1, e_2]$.

For $b \in B_{ij}$ and the remaining index k,

$$F_k[e_k, b] = 4b, \qquad F_i[e_k, b] = F_j[e_k, b] = 0.$$

The B_{ij} directions are independent of the real pair brackets. The three lowering maps first detect the twenty-seven mixed coefficients; their first two rows on R_γ have determinant 12. This gives an injective 30×29 matrix. Its ith component lies in the pair space on the other two indices, which F_i kills. □

Lemma 16.7. *If $[E, F] = H$, $[H, E] = 2E$, $[H, F] = -2F$, $Hw = -2w$ and $F^2 w = 0$, then*

$$F^4 E^3 w = 144 F w. \tag{16.12}$$

Proof. The commutation relations imply

$$F E^a F^b w = E^a F^{b+1} w - a(-2 - 2b + a - 1) E^{a-1} F^b w.$$

Four applications to $E^3 w$ give successively $E^3 F w$, $6E^2 F w$, $36 E F w$ and $144 F w$. □

Theorem 16.8. *Every root-preserving odd derivation of the unchanged positive Borcherds presentation which satisfies $de_i = 0$ and $dc^u \in W_\gamma$ has $dc^u = 0$ for all u. The rank-28 differential* (16.11) *extends to the positive Lie quotients of height at most five, but it does not descend at height six.*

Proof. Differentiate $E_i^3 c^u = 0$ to obtain $E_i^3 dc^u = 0$. Equation (16.12) gives $F_i dc^u = 0$ for all i, and Lemma 16.6 gives $dc^u = 0$. The negative generators only detect a relation in the positive algebra; the derivation need not act on them. Square-zero was not used in this obstruction.

For the finite quotients, extend (16.11) by zero on every other simple generator. The first odd simple roots have height three. A relation containing c^u is either a real Serre relation, first appearing at height six, or a commutator for orthogonal simple roots. But $(\gamma, \alpha_i) = -2$ and $(\gamma, \delta_{ij}) = -4$, so no such orthogonality relation has height at most five. The remaining relations are closed or discarded by height. The differential preserves height and squares to zero on generators. At height six the relations just tested prevent descent. □

There is already a loss of real-root information in degree γ. Every 28-dimensional subspace $W' \subset W_\gamma$ meets R_γ in dimension at least $28 + 2 - 29 = 1$. If W' consists of boundaries and the real generators are closed, a nonzero real Lie word becomes zero in cohomology. The real composition algebra (16.4) cannot then act faithfully through that cohomology with its original generators.

16.5.1 A free attachment changes the presentation

One can construct an all-height dg Lie algebra with the desired boundaries by changing the relations. Let $\mathfrak{a}$ be the ordinary Lie algebra generated by the real and null simple generators with their prescribed relations, in degree zero. Form the graded free product

$$L = \mathfrak{a} * \mathrm{Lie}(c^1, \dots, c^{93}), \qquad |c^u| = -1, \tag{16.13}$$

with root γ for each c^u, and use (16.11). No real–imaginary Serre relation is imposed on the new generators.

The free-product universal property extends the differential uniquely. It squares to zero on generators and hence everywhere. Its degree-zero cycles are $\mathfrak{a}$, and its degree-zero boundaries form exactly the Lie ideal generated by the w_u. Consequently

$$H^0(L) = \mathfrak{a}/\langle w_1, \dots, w_{28} \rangle_{\mathrm{Lie}}.$$

Positive-height completion preserves this construction: every fixed root component is finite and the differential preserves it. A homotopy-invariant derived quotient can instead be formed from a semifree resolution of $\mathfrak{a}$ and chosen closed lifts of the w_u. Both constructions retain the changed quotient. Merely adjoining a generator with $dz = E_i^3 c^u$ would not repair the old presentation, since $d^2 z = E_i^3 w_u$ can be nonzero.

Thus a geometric primitive comparison has two different possibilities: retain the ordinary $29 \mid 93$ root space, or construct a larger derived presentation together with its geometric differential, brackets and syzygies. Equality of the signed coefficient does not choose between them.

16.6 Incidence and the surface diagonal

Let Y be a K3 or Enriques surface, put $X_n = Y^{[n]}$, and write $A = H^*(Y; \mathbb{Q})$. The vacuum space is $H^*(X_0) = \mathbb{Q}$. Three coefficient theories occur naturally on the same incidence geometry: constructible functions, ordinary cohomology and coherent sheaves. Their pushforwards and adjoints have different normalizations.

First consider zero-dimensional coherent sheaves on Y. For constructible functions on their isomorphism classes, the Euler Hall product is

$$(h_1 * h_2)(Q) = \int_{U \subset Q} h_1(U) h_2(Q/U)\, d\chi.$$

Lengths are fixed in each integral; constructibility is understood on finite-type strata. A cyclic quotient is a surjection $\mathcal{O}_Y \twoheadrightarrow Q$, and the length-$n$ cyclic quotients form X_n.

Proposition 16.9. *Constructible functions on the cyclic quotients carry the unital Hall action*

$$(h \cdot v)(\mathcal{O}_Y \twoheadrightarrow Q) = \int_{U \subset Q} h(U) v(\mathcal{O}_Y \twoheadrightarrow Q/U)\, d\chi.$$

All finite iterates are compatible with forgetting stages of the extension flag.

Proof. The quotient Q/U inherits a surjective framing. Subobjects of a fixed length are parameterized by a projective Quot variety, so the integral is defined. Both orders of applying h_1, h_2 integrate $h_1(U) h_2(W/U) v(\mathcal{O}_Y \twoheadrightarrow Q/W)$ over $0 \subset U \subset W \subset Q$. Functoriality of Euler pushforward proves the action identity and its all-flag version. The length-zero subobject gives the unit. □

In geometric notation, $Z \subset Z'$ gives the sequence

$$0 \longrightarrow I_Z/I_{Z'} \longrightarrow \mathcal{O}_{Z'} \longrightarrow \mathcal{O}_Z \longrightarrow 0.$$

The framing belongs to the middle object. This orientation is necessary for the quotient to remain cyclic.

For $r > 0$ let $\mathfrak{Q}_{n,r}$ be the incidence cycle in $X_{n+r} \times Y \times X_n$ consisting of (Z', x, Z) with $Z \subset Z'$ and $I_Z/I_{Z'}$ of length r supported at x. Its fundamental cycle is the closure of the generic punctual incidence locus. With projections π_+, s, π_-, define

$$C_r(a)v = (\pi_+)_*\big([\mathfrak{Q}_{n,r}] \cup s^* a \cup \pi_-^* v\big).$$

Here cap product and Poincaré duality are implicit; all pushforwards are proper. Write $\dagger$ for the adjoint under ordinary Poincaré integration and set $A_r(a) = (-1)^{r-1} C_r(a)^\dagger$. The geometric Heisenberg theorem gives

$$[A_r(a), C_s(b)] = r\delta_{rs} \int_Y ab, \qquad [C_r(a), C_s(b)] = [A_r(a), A_s(b)] = 0. \tag{16.14}$$

It identifies

$$\operatorname{Sym}\left(\bigoplus_{r \geq 1} p^r A\right) \xrightarrow{\sim} \bigoplus_{n \geq 0} H^*(X_n; \mathbb{Q}), \qquad p_r(a) \longmapsto C_r(a)|0\rangle, \tag{16.15}$$

with products of generators interpreted as successive creations [107, 119]. The degree of $C_r(a)$ is $\deg(a) + 2(r-1)$. Over distinct points the iterated length-one correspondence orders the points in $n!$ ways. Since both sides have degree zero, this proves

$$\frac{1}{n!} C_1(1)^n |0\rangle = 1_{X_n}. \tag{16.16}$$

There are also coherent kernels. On the nested scheme $Z_{n,n+1}$ let $\mathcal{L}$ be its universal length-one quotient line. For $F \in \operatorname{Perf}(Y)$ and $k \in \mathbb{Z}$, the kernel

$$R(\pi_+, \pi_-)_*(\mathcal{O}_{Z_{n,n+1}} \otimes \mathcal{L}^k \otimes s^* F) \in \operatorname{Perf}(X_{n+1} \times X_n)$$

is bounded coherent by properness and perfect by smoothness of the product. Composition uses derived convolution. The right adjoint kernel is its dual, tensored by the canonical bundle of the source and shifted by its dimension. For $n = k = 0$ and $F = \mathcal{O}_Y$, creation is $\mathbb{C} \mapsto \mathcal{O}_Y$ and its right adjoint is $R\Gamma(Y, -)$. Their composite is $\mathbb{C}$ for Enriques and $\mathbb{C} \oplus \mathbb{C}[-2]$ for K3. Thus the passage to cohomological adjoints includes the Todd class; it cannot be made by identifying the two transposes.

16.6.1 *Frobenius multiplication and graph genus*

Choose a basis u_i of $H^2(Y)$, put $Q_{ij} = \int_Y u_i u_j$, write $(Q^{ij}) = Q^{-1}$, and normalize $\int_Y \omega = 1$. Then $u_i u_j = Q_{ij}\omega$, and positive-degree classes kill ω. Write m for cup multiplication. The diagonal Gysin map, adjoint to the cup product for $\int_Y$, is

$$\begin{aligned} \Delta 1 &= 1 \otimes \omega + \omega \otimes 1 + \sum_{i,j} Q^{ij} u_i \otimes u_j, \\ \Delta u_i &= u_i \otimes \omega + \omega \otimes u_i, \qquad \Delta\omega = \omega \otimes \omega. \end{aligned} \tag{16.17}$$

Testing against basis pairs proves the formulas. In particular

$$e_A = m\Delta 1 = (b_2(Y) + 2)\omega = c_2(T_Y), \qquad m\Delta(a) = e_A a. \tag{16.18}$$

A split followed by a join therefore inserts the Euler class, with coefficient 12 or 24.

For $\sigma \in \mathfrak{S}_N$, label each cycle of σ by a factor of A and form $\mathscr{A}_N = \bigoplus_\sigma A^{\otimes c(\sigma)}\sigma$. Here $c(\sigma)$ counts cycles. On a joint orbit O, write $c_\sigma(O)$ for the number of cycles of its restriction. To multiply a σ-labelled element by a τ-labelled element, consider each joint orbit O of $\langle \sigma, \tau \rangle$. Set

$$g_O = \frac{|O| + 2 - c_\sigma(O) - c_\tau(O) - c_{\sigma\tau}(O)}{2}.$$

The three permutations $\sigma, \tau, (\sigma\tau)^{-1}$ define a connected branched cover of the sphere on O. Riemann–Hurwitz identifies g_O as its genus. Let a_O, b_O be the products of the two sets of input labels on O, and let Δ_k be the iterated diagonal, with $\Delta_1 = \mathrm{id}$. Multiply these labels, insert $e_A^{g_O}$, and apply the iterated diagonal to obtain the labels on the $c_{\sigma\tau}(O)$ output cycles. Explicitly, the factor of the product on O is

$$\Delta_{c_{\sigma\tau}(O)}(a_O b_O e_A^{g_O}). \tag{16.19}$$

The Frobenius identities make this multiplication associative: cutting a connected covering surface into cylinders, pairs of pants and handles gives multiplication, diagonal and $m\Delta$, respectively. Associativity, coassociativity and the Frobenius projection formula identify different cuts; (16.18) accounts for every handle. This is the Frobenius permutation construction and its orbifold realization [107, §2] [59, § 3]. For $s = (12) \in \mathfrak{S}_2$ and $r = (123) \in \mathfrak{S}_3$,

$$(as)(bs) = \Delta(ab)\,\mathrm{id}, \qquad (ar)(br) = abe_A r^2, \qquad (ar)(br^{-1}) = \Delta_3(ab)\,\mathrm{id}.$$

The middle formula has genus one. Genus at least two gives zero because $e_A^2 = 0$.

To obtain the Hilbert-scheme product, multiply each product by

$$\varepsilon(\sigma, \tau) = (-1)^{(|\sigma|+|\tau|-|\sigma\tau|)/2}, \qquad |\sigma| = N - c(\sigma). \tag{16.20}$$

The exponent is integral by permutation parity. The cocycle identity follows by cancelling the middle lengths in a triple product, so associativity is preserved. The invariant algebra is $H^*(X_N; \mathbb{Q})$ when the canonical class is numerically trivial, including both surfaces used here [107, Theorem 3.2]. The sum over conjugates, with repetitions, corresponds to the product of incidence creators. Its identity sector is therefore $N!$ times the unit, as in (16.16). Over $\mathbb{C}$ the sector rescaling by $i^{|\sigma|}$ removes the sign twist.

16.6.2 *Joining has order two*

In a homogeneous basis a_α of A, write $a_\alpha a_\beta = m^\gamma_{\alpha\beta} a_\gamma$ and $\Delta a_\gamma = \Delta_\gamma^{\alpha\beta} a_\alpha \otimes a_\beta$. Let $\partial_{i,\alpha}$ differentiate the Fock variable $p_i(a_\alpha)$. Define

$$\begin{aligned} J &= \frac{1}{2} \sum_{i,j\geq 1} ij\, m^\gamma_{\alpha\beta} p_{i+j}(a_\gamma)\, \partial_{i,\alpha}\partial_{j,\beta}, \\ S &= \frac{1}{2} \sum_{i,j\geq 1} (i+j) \Delta_\gamma^{\alpha\beta}\, p_i(a_\alpha)\, p_j(a_\beta)\, \partial_{i+j,\gamma}, \qquad D_H = J - S. \end{aligned} \tag{16.21}$$

Each fixed length sees finitely many terms. There are ij choices of endpoints for joining cycles of lengths i, j. A cycle of length $i+j$ has $i+j$ choices for a specified split, with the ordered sum counting each split twice. These counts give $J+S$ for the orbifold transposition sum and $J-S$ for the Hilbert sign. Under the incidence isomorphism, D_H is multiplication by $-c_1(\mathcal{O}_Y^{[N]})$; the canonical-class term in Lehn's formula vanishes [106, Theorem 3.10].

Proposition 16.10. *If $M_i(a)$ is multiplication by $p_i(a)$, then*

$$[[D_H, M_i(a)], M_j(b)] = ij M_{i+j}(ab), \qquad [[[D_H, M_i(a)], M_j(b)], M_k(c)] = 0.$$

The operator preserves length, raises cohomological degree by two and normalized Hodge charge by one. It satisfies

$$D_H^2 p_2(a) = -p_2(e_A a), \qquad D_H^{2N+1}|_{H^*(X_N)} = 0.$$

Proof. Only the second-order term J contributes to the double commutator. Its two derivative slots give the displayed product. A further commutator is zero. A join raises age by one, contributing cohomological degree two; a split adds diagonal degree four and lowers age by one. Both have degree two and Hodge bidegree $(1,1)$. On a cycle of length i, a class in $H^{r,s}(Y)$ has normalized charge $r+i-1-i = r-1$, so the charge increases by one. Splitting $p_2(a)$ and then joining yields $-p_2(m\Delta a)$, proving the square. Finally the degrees of $H^*(X_N)$ range from zero to $4N$. □

For $a = 1$ the square is nonzero. In particular D_H is even and second order; it is neither a derivation of the Fock product nor a square-zero operator.

16.7 Elliptic transport and its integral image

Put $R = \mathbb{Q}(y)[[q]]$ and

$$\epsilon_n(q,y) = \mathrm{td}(T_{X_n})\, \mathrm{ch}(\mathcal{E}_{q,y}(X_n)).$$

Its integral is the elliptic genus. Its cohomological degree-zero part r_n has constant q coefficient $y^{-n}(1-y)^{2n}$ and is a unit of R. The remaining part $\nu_n = \epsilon_n - r_n$ has positive degree and $\nu_n^{2n+1} = 0$. Hence

$$\epsilon_n^{-1} = r_n^{-1} \sum_{j=0}^{2n} (-r_n^{-1}\nu_n)^j. \tag{16.22}$$

Multiplication by ϵ_n is an invertible, generally inhomogeneous map U_n of ordinary cohomology over R.

A cohomology kernel K from X_m to X_n acts by pullback, cup product and integration along the source. Write p_n, p_m for the two projections of $X_n \times X_m$. Define its transport by

$$\widetilde{K} = p_n^* \epsilon_n \cup K \cup p_m^* \epsilon_m^{-1}. \tag{16.23}$$

Proposition 16.11. *Transport preserves identity kernels and all finite convolutions. On operators it is* $T_{\widetilde{K}} = U_n T_K U_m^{-1}$.

Proof. In a convolution the two classes on every middle factor multiply to $\epsilon_j^{-1}\epsilon_j = 1$. The projection formula moves the two outer classes through the remaining pushforward. This proves the operator and composition identities, in every finite arity. On the diagonal the two classes cancel, giving the identity kernel. □

Thus the transported creators and annihilators satisfy (16.14). Cup multiplication by the tautological first Chern class commutes with U_n, so $\widetilde{D}_H = D_H$. Let $\langle \mathcal{I}|$ integrate each length component. Completing in length, the vacuum normalization gives

$$\frac{1}{n!}\widetilde{C}_1(1)^n|0\rangle = \epsilon_n, \qquad \left\langle \mathcal{I}\middle| e^{p\widetilde{C}_1(1)}\middle|0\right\rangle = \sum_{n\geq 0} p^n \operatorname{Ell}(X_n; q, y). \tag{16.24}$$

This constructs a specified matrix coefficient. The parameter q here belongs to the coefficient ring; the inverse in (16.22) is an inverse in cohomology. It is not a tensor inverse of the elliptic coefficient complex.

There is considerable freedom in this construction. Any even invertible classes α_n with $\alpha_0 = 1$ transport kernels in the same way. Taking $\alpha_n = 1 + c_n[\mathrm{pt}]$ for $n > 0$ gives any chosen coefficients c_n after integration. A scalar series therefore does not characterize a critical Hall action on an independently defined state object.

For integral behaviour near $y = 1$, use $z = \log y$, $\mathcal{R} = \mathbb{Q}[[q, z]]$, and $H_n = H^*(X_n; \mathbb{Q}) \otimes \mathcal{R}$. The class $\epsilon_n(q, e^z)$ is regular and becomes invertible after inverting z. Its multiplication map is injective because H_n is free and the localized map is invertible. Define the image lattice

$$\mathcal{M}_n = \epsilon_n H_n \subset H_n.$$

Multiplication identifies H_n with $\mathcal{M}_n$ as abstract free modules. A transported kernel preserves these lattices exactly when the original kernel preserves H_m, H_n, since

$$\widetilde{K}(\epsilon_m v) = \epsilon_n K(v). \tag{16.25}$$

All rational geometric kernels above are therefore regular on the image lattices. The hypothesis is necessary: z^{-1} times an identity kernel does not preserve any nonzero free $\mathcal{R}$-lattice of this form.

16.7.1 The Euler factor at every energy

For a smooth projective d-fold X, with formal tangent Chern roots x_i, put

$$e_z(TX) = \prod_{i=1}^{d}(x_i - z) = \sum_{k=0}^{d}(-z)^{d-k}c_k(TX), \qquad s(w) = \frac{\sinh(w/2)}{w/2}.$$

Here z is an even coefficient parameter, $c_0 = 1$ and $s(0) = 1$. Use the elliptic tensor formula with $y^{-d/2} = e^{-dz/2}$ also when d is odd, and put $\epsilon_X = \mathrm{td}(TX)\,\mathrm{ch}(\mathcal{E}_{q,e^z}(X))$. It has the factorization

$$\epsilon_X(q, e^z) = e_z(TX)U_X(q, z), \qquad U_X \in (H^*(X) \otimes \mathcal{R})^\times, \qquad U_X(q, 0) = 1, \tag{16.26}$$

where

$$U_X = \prod_i \left[\frac{s(x_i - z)}{s(x_i)} \prod_{r\geq 1} \frac{(1 - e^z q^r e^{-x_i})(1 - e^{-z} q^r e^{x_i})}{(1 - q^r e^{-x_i})(1 - q^r e^{x_i})}\right].$$

Indeed the one-root, energy-zero factor is

$$e^{-z/2}\frac{x(1-e^{z-x})}{1-e^{-x}} = (x-z)\frac{s(x-z)}{s(x)}.$$

The positive-energy factors are the second product. Every fixed q coefficient uses finitely many factors; positive cohomological degrees are nilpotent. All remaining factors are units and become one at $z=0$, proving the identity by the splitting principle.

Theorem 16.12. *Let $d>0$, let $\pi:\mathbb{P}(TX)\to X$ be the bundle of tangent lines, and put $\lambda=c_1(\mathcal{O}(-1))$. Then*

$$\frac{H^*(X;\mathbb{Q})\otimes\mathcal{R}}{(\epsilon_X)} \cong H^*(\mathbb{P}(TX);\mathbb{Q})[[q]], \qquad z\longmapsto\lambda. \tag{16.27}$$

The image ideals of ϵ_X and $e_z(TX)$ coincide. Their two-term multiplication complexes in resolution degrees $-1,0$ are isomorphic by U_X on the source and the identity on the target.

Proof. The unit factorization proves the ideal and complex assertions. The tautological inclusion $\mathcal{O}(-1)\subset\pi^*TX$ is a nowhere-zero section of $\pi^*TX\otimes\mathcal{O}(1)$, so $\sum_k(-\lambda)^{d-k}\pi^*c_k(TX)=0$. The classes $1,\lambda,\dots,\lambda^{d-1}$ restrict to a basis on each projective fibre. Leray–Hirsch identifies the quotient by this monic relation with the projective-bundle cohomology. The class λ is nilpotent, so completion in z leaves the quotient unchanged; adjoining q gives the result. □

Write $D=\dim H^*(X;\mathbb{Q})$. Over the discrete valuation ring $\mathbb{Q}[[z]]$, multiplication by $e_z(TX)$ has determinant $(-z)^{dD}$: in a basis ordered by degree, each positive Chern class is strictly triangular. It is therefore injective. Its positive Smith exponents are the Jordan lengths of $\lambda\cup-$ on $H^*(\mathbb{P}(TX))$, and their sum is dD. The assertion extends to $\mathcal{R}$ by the unit factorization, without invoking Smith normal form over a two-variable ring. If X is connected and $\chi(X)\neq0$, multiplication at $z=0$ is by $c_d(TX)=\chi(X)[\mathrm{pt}]$ and has rank one. There is one unit Smith factor and $D-1$ positive ones. The derived fibre of the cokernel has rank $D-1$ in each resolution degree $-1,0$. At length zero, $\epsilon_{X_0}=1$ and this cokernel is zero.

For a K3 or Enriques surface, $e_z(TY)=z^2+\chi(Y)\omega$. On $\langle1,\omega\rangle$ its matrix is $\left(\begin{smallmatrix} z^2 & 0\\ \chi(Y) & z^2\end{smallmatrix}\right)$, whose elementary divisors are $1,z^4$. Each of the $b_2(Y)$ middle directions contributes z^2. Hence

$$\operatorname{coker}\epsilon_Y \cong \mathcal{R}/(z^4)\oplus\big(\mathcal{R}/(z^2)\big)^{b_2(Y)}. \tag{16.28}$$

The derived fibre has dimensions 11, 11 for Enriques and 23, 23 for K3, over $\mathbb{Q}[[q]]$. The abstract image lattice has rank $b_2(Y)+2$, while its image in ambient cohomology at $z=0$ has rank one. The difference is accounted for by the Tor term in the specialization of $0\to M_1\to H_1\to\operatorname{coker}\epsilon_Y\to0$.

There is an explicit pole before passing to the image lattice. At $q=0$, set $t=y-1$, $s=y+y^{-1}-2=t^2/(1+t)$ and $a=1+s/12$. The surface class is

$$\epsilon_Y(0,y)=s+a\,c_2(TY), \qquad [q]\epsilon_Y=-2s^2+\Big(\frac{5}{6}s^2-2s\Big)c_2(TY).$$

These follow by expanding the opposite Chern roots to degree four. On $\langle1,\omega\rangle$ the inverse matrix has lower-left entry $-a\chi(Y)s^{-2}$. Thus $\widetilde{A}_1(1)(1)=\int_Y\epsilon_Y^{-1}$ has a fourth-order pole in t, although the elliptic genus $\int_Y\epsilon_Y=a\chi(Y)$ is regular.

16.7.2 The Hilbert square

For $X = S^{[2]}$ with S K3, the Fock decomposition gives even Betti numbers (1, 23, 276, 23, 1) and no odd cohomology. Its Chern numbers and Chern pairing, for $\alpha, \beta \in H^2(X)$, are

$$\int_X c_4 = 324, \qquad \int_X c_2\alpha\beta = 30 q_X(\alpha,\beta), \qquad \int_X c_2^2 = 828, \tag{16.29}$$

where q_X has its standard integral Beauville–Bogomolov normalization. The first two are the Chern–Fujiki identities [24, Theorem 4.2]; the last also follows from $\chi(\mathcal{O}_X) = 3$ and $720\chi(\mathcal{O}_X) = 3\int c_2^2 - \int c_4$. In particular $c_2 : H^2(X) \to H^6(X)$ is an isomorphism.

Proposition 16.13. *The Smith exponents of Euler multiplication on $H^*(X)[[z]]$ are*

$$0^{1}, \qquad 2^{23}, \qquad 4^{276}, \qquad 6^{23}, \qquad 8^{1}.$$

The same elementary divisors describe ϵ_X at every elliptic energy. Its cokernel has derived fibre of rank 323 in each resolution degree $-1, 0$.

Proof. Symplecticity gives $e_z(TX) = z^4 + z^2 c_2 + c_4$. Normalize $\int_X \omega = 1$ and choose a basis a_i of $H^2(X)$. On $\langle 1, c_2, \omega\rangle$ its matrix is

$$\begin{pmatrix} z^4 & 0 & 0 \\ z^2 & z^4 & 0 \\ 324 & 828 z^2 & z^4 \end{pmatrix}.$$

There is a unit entry. The two-by-two minors generate (z^4), since all are divisible by z^4 and the lower left minor is $504 z^4$. The determinant is z^{12}, giving exponents 0, 4, 8. Each of the 23 pairs $\langle a_i, c_2 a_i\rangle$ in degrees two and six has matrix $\left(\begin{smallmatrix} z^4 & 0 \\ z^2 & z^4 \end{smallmatrix}\right)$ and exponents 2, 6. The kernel of $\int c_2(-)$ in H^4 has dimension 275 and is complementary to $\mathbb{Q}c_2$; every such direction contributes 4. There are 323 positive exponents, summing to $1296 = 4 \cdot 324$. The unit factorization and the two-term free resolutions prove the remaining assertions. □

These lengths constrain operators on the degeneration. For $L_0 = \mathbb{Q}[[z]]^D$ and $M = \operatorname{diag}(z^{\ell_i}) L_0$ with $\ell_i \geq 0$, an ambient endomorphism preserves M precisely when

$$T_{ij} \in z^{\max(0, \ell_i - \ell_j)} \mathbb{Q}[[z]]. \tag{16.30}$$

This follows by conjugating with the diagonal matrix. The inequality $\max(0, a-b) + \max(0, b-c) \geq \max(0, a-c)$ proves closure under composition. For exponents $(0, m)$, $m > 0$, the order has basis $e_{11}, e_{22}, e_{12}, z^m e_{21}$. Its abstract fibre is four-dimensional, with two opposite arrows whose composites vanish. Its image on $L_0 / z L_0$ is three-dimensional because the lower arrow acts by zero. Thus even the algebra of regular operators has distinct abstract and ambient fibres.

There cannot be a regular Heisenberg action on the unchanged ambient modules with the transported creators and the same empty vacuum. Indeed $\widetilde{C}_1(\omega)|0\rangle = z^2 u(q,z)\omega$ with u a unit. Every regular annihilator sends this vector into $z^2\mathcal{R}$, whereas (16.14) requires its commutator with $\widetilde{A}_1(1)$ on the vacuum to be 1. The reversed product is zero since there are no negative lengths. Equation (16.25) nevertheless supplies the regular action on the image lattices themselves.

16.8 Multiplicative and parameter fibres

Two gradings describe different aspects of the Euler degeneration. The Smith calculation placed z in resolution degree zero. A geometric cochain model instead gives z degree two. Its derived fibre is a sphere bundle and can have nonzero Massey products. The unit U_X in (16.26) is generally inhomogeneous in ordinary cohomology, so the equality of completed ideals does not identify these two graded constructions.

Let X be a compact smooth manifold and let $E \to X$ be a complex bundle of rank $r \geq 1$. Use rational polynomial forms $A_X = A_{\mathrm{PL}}(X)$, with chosen closed Chern representatives, and put

$$\mathcal{P}_E = A_X[z] \otimes \Lambda(\eta), \quad |z| = 2, \quad |\eta| = 2r - 1, \quad Dz = 0, \quad D\eta = e_z(E). \tag{16.31}$$

This is a commutative cochain model of $\mathbb{P}(E)$. To construct the comparison, send z to a representative of $c_1(\mathcal{O}(-1))$ and η to a primitive of the top Chern relation. Its induced cohomology map is an isomorphism: multiplication by the monic polynomial $e_z(E)$ on $H^*(X)[z]$ is injective, and the projective bundle theorem identifies the resulting quotient. Over real de Rham forms, that primitive has the explicit Chern–Simons expression

$$\Theta = r \int_0^1 P_r(\nabla_1 - \nabla_0, F_t, \dots, F_t)\, dt.$$

Here $P_r(F, \dots, F) = c_r(\nabla)$, ∇_1 is the tensor connection on $\pi^*E \otimes L^*$, and ∇_0 is trivial on its tautological nowhere-zero line; $L = \mathcal{O}(-1)$, $\nabla_t = (1-t)\nabla_0 + t\nabla_1$ and F_t is its curvature. Write $\lambda = c_1(\nabla L)$. The Bianchi identity gives $d\Theta = c_r(\nabla_1) - c_r(\nabla_0) = e_\lambda(E)$. Changing a Chern representative by $D\sigma$ changes the relation generator by $\eta \mapsto \eta + \sigma$.

Proposition 16.14. *The multiplicative derived fibre at $z = 0$ has model*

$$\mathcal{S}_E = A_X \otimes \Lambda(\eta'), \qquad |\eta'| = 2r - 1, \quad D\eta' = c_r(E).$$

It is the rational cochain algebra of the unit sphere bundle $S(E)$. Equivalently,

$$C^*(\mathbb{P}(E); \mathbb{Q}) \otimes^{\mathbb{L}}_{\mathbb{Q}[z]} \mathbb{Q} \simeq C^*(S(E); \mathbb{Q}), \qquad |z| = 2.$$

Proof. Resolve $\mathbb{Q}$ over $\mathbb{Q}[z]$ by a generator ν of degree one with $D\nu = z$. For $Q_E(z) = (e_z(E) - c_r(E))/z$, set $\eta' = \eta - \nu Q_E(z)$. Then $D\eta' = c_r(E)$. The remaining factor $\mathbb{Q}[z] \otimes \Lambda(\nu)$ contracts to its constants. Geometrically, the unit circle bundle of the tautological line is $S(E)$, the homotopy fibre of its classifying map to $BU(1)$. Its odd-sphere relative model adjoins the oriented fibre generator with differential $c_r(E)$, giving the asserted model. □

16.8.1 A nonzero ternary multiplication

Suppose X is connected and smooth projective, of complex dimension $d > 0$, with no odd rational cohomology and $\chi = \chi(X) \neq 0$. Set $A = H^*(X; \mathbb{Q})$, $\int_X \omega = 1$ and $\tau(a) = \int_X a$, extended by zero outside top degree. Kähler formality allows the tangent-sphere model to be written

$$B = A \otimes \Lambda(\eta), \qquad |\eta| = 2d - 1, \quad d\eta = \chi\omega$$

[43, § 6]. Let $A' = \bigoplus_{0<j<2d} A^j$. Representatives for its cohomology are

$$H = \mathbb{Q}1 \oplus A' \oplus \eta A^{>0}.$$

The projection p kills η, ω, inclusion i is the displayed representative inclusion, and the homotopy is $h(\omega) = \eta/\chi$, zero elsewhere. Thus $dh + hd = 1 - ip$ and $h^2 = hi = ph = 0$.

Proposition 16.15. *There is a strictly unital minimal A_∞ model of B on H. Its binary multiplication is $p(i(-)i(-))$, and its only possible nonzero higher multiplication is*

$$m_3(a,b,c) = \frac{\tau(bc)a - \tau(ab)c}{\chi}\,\eta, \qquad a,b,c \in A'. \tag{16.32}$$

It vanishes on other homogeneous input patterns, and $m_n = 0$ for $n \geq 4$.

Proof. Fix suspension s of degree -1 and bar signs $b_1(sa) = -s(da)$, $b_2(sa,sb) = (-1)^{|a|}s(ab)$. The binary product loses exactly $\tau(ab)\omega$. Its primitive is $h(ab) = \tau(ab)\eta/\chi$. The two placements of this primitive in arity three give $p(a\,h(bc) - h(ab)c)$, which is the displayed formula. The comparison to B has first term i, suspended second term $I_2(sa,sb) = s(\tau(ab)\eta/\chi)$ on A' and all later terms zero. The binary bar equation is $-s\,d(\tau(ab)\eta/\chi) + s(ab) = s(ab - \tau(ab)\omega)$.

Every nonzero internal h edge in a transfer tree has output proportional to η. A second h above it vanishes; two such outputs multiply to zero. Every tree with at least four leaves has one of these forms. The same argument kills higher comparison terms and all terms with an η-marked input. Normalization of the contraction gives strict units. These identities prove the complete bar equation, with I_1 inducing the identity on cohomology. □

If $\dim H^2(X) \geq 2$, choose independent $a, c \in A^2$ and $b \in A^{2d-2}$ with $\tau(ab) = 1$, $\tau(bc) = 0$. Then the triple Massey product is

$$\langle [a],[b],[c]\rangle = -[c\eta]/\chi \neq 0. \tag{16.33}$$

Its indeterminacy is zero because $H^{2d-1}(B) = 0$. This proves nonformality of the tangent-sphere cochains. It applies to every positive-length $Y^{[n]}$: the incidence decomposition gives no odd cohomology, retains the surface degree-two classes, and gives positive Euler characteristic from $\prod_{m\geq 1}(1-p^m)^{-\chi(Y)}$. For a surface the cohomology degrees are $0, 2, 5, 7$, with dimensions $1, b_2, b_2, 1$, and

$$m_3(u_i,u_j,u_k) = \frac{Q_{jk}u_i - Q_{ij}u_k}{\chi(Y)}\eta.$$

For a hyperbolic pair $u^2 = v^2 = 0, uv = \omega$, the choice $a = u + v, c = u - v$ gives $\langle a,a,c\rangle = -2c\eta/\chi(Y)$.

16.8.2 Jordan lengths as higher module operations

Return to resolution grading: let $R_0 = \mathbb{Q}[[z]]$ with z of degree zero, and let C be a finite torsion R_0-module. Write N for multiplication by z. For $E = \Lambda(\xi) = \operatorname{Ext}^*_{R_0}(\mathbb{Q},\mathbb{Q})$, $|\xi| = 1$, a dg model of $R\operatorname{Hom}_{R_0}(\mathbb{Q}, C)$ is

$$M_C = C \otimes \Lambda(\xi), \qquad d(a) = \xi N a, \quad d(\xi a) = 0. \tag{16.34}$$

This uses the free resolution with differential $-z$. An additional internal degree two for z is compatible with internal degree -2 for ξ.

For $C_m = R_0/(z^m)$, choose $e_0, \dots, e_{m-1}$ with $Ne_j = e_{j+1}$ and $Ne_{m-1} = 0$. Cohomology has u in resolution degree zero and v in degree one. Representatives and homotopy are

$$i(u) = e_{m-1}, \quad i(v) = (-1)^{m-1}\xi e_0, \quad h(\xi e_j) = e_{j-1}\ (j>0),$$

with $h = 0$ elsewhere. The projection sends e_{m-1} to u and ξe_0 to $(-1)^{m-1}v$, and kills the other basis vectors. The unique nonzero reduced minimal-module operation is

$$m_{m+1}(\xi,\dots,\xi,u) = v. \tag{16.35}$$

Indeed the term with j algebra inputs is $p\xi(-h\xi)^{j-1}i$. Projection first survives at $j=m$; the chosen sign of v makes its coefficient one. Every operation on v vanishes. The comparison components on the reduced bar comodule are

$$I_j((s\xi)^{\otimes j},u)=(-1)^j e_{m-1-j}\quad (0\le j<m),\qquad I_0(v)=(-1)^{m-1}\xi e_0.$$

The equation $dI_j+\xi I_{j-1}=\sum_{k=1}^{j} I_{j-k}m_{k+1}$ cancels for $j<m$, equals $I_0(v)$ for $j=m$, and is zero above it. The minimal coderivation squares to zero since its output is v. This verifies every arity, not just the first nonzero one.

Applied to $C=H^*(\mathbb{P}(TX))$ with $N=\lambda\cup -$, the multiplicities of these operations are

X	m_3	m_5	m_7	m_9
Enriques surface	10	1	0	0
K3 surface	22	1	0	0
K3 Hilbert square	23	276	23	1.

These are higher module actions in resolution grading. The ternary multiplication (16.32) belongs to the separate multiplicative sphere model. Deleting the module action (16.35) loses the relation z^m even though the two cohomology dimensions remain one.

Maps commuting with N induce strict maps of (16.34). After transfer, write $F(T)=P'TI$ for bar projection, the strict map and bar inclusion. If the bar homotopy H satisfies $IP-1=-dH-Hd$, then

$$F(T_2)F(T_1)-F(T_2T_1)=dK+Kd,\qquad K=-P_3T_2H_2T_1I_1.$$

The perturbation sums are finite at fixed bar length. For two blocks $T_m=R_0/(z^m)$ and $T_n=R_0/(z^n)$, their free resolutions also give

$$\operatorname{Hom}_{R_0}(T_m,T_n)=z^{\max(n-m,0)}R_0/(z^n),\qquad \operatorname{Ext}^1_{R_0}(T_m,T_n)=R_0/(z^{\min(m,n)}),$$

and no higher Ext. Both nonzero groups have dimension $\min(m,n)$. A correspondence that does not commute with N does not define such a parameter-module map.

16.8.3 Gysin maps and maps of cones

Some geometric maps act on the Euler cochain models without a transfer. Let E, F be bundles of ranks $r\ge 1$, $s\ge 0$ on the same base, with compatible split Chern representatives. For $j:\mathbb{P}(E)\hookrightarrow\mathbb{P}(E\oplus F)$, write $e_E=e_z(E)$, $e_F=e_z(F)$ and use relation generators η_E,η_{EF}. The normal bundle is $L^*\otimes F$. Pullback and Gysin have representatives

$$\begin{aligned} j^*(a)&=a,\qquad & j^*(\eta_{EF}b)&=\eta_E e_F b,\\ j_!(a)&=e_F a,\qquad & j_!(\eta_E b)&=\eta_{EF}b,\qquad a,b\in A_X[z]. \end{aligned}\tag{16.36}$$

Pullback is a dg algebra map. Gysin has degree $2s$ and is a cochain map: on the odd generator both differentials give $e_Ee_Fb-\eta_{EF}Db$. Multiplication verifies

$$j_!(j^*v\cdot w)=v\cdot j_!(w),\qquad j^*j_!(w)=e_Fw.$$

Successive split inclusions multiply the appropriate summand by $e_Fe_G\cdots$, so every finite composition is independent of regrouping.

These maps represent the geometric operations. The normal Euler class gives $j_!(1) = e_F(\lambda)$, and the projective bundle is generated over the base by λ, so the projection formula determines the cohomology map. At the cochain level, take the universal pair over $BU(r) \times BU(s)$. Its rational Chern-class model is even. A degree-$2s$ map out of the Euler resolution must send η_E to the odd degree $2(r+s)-1$, which is zero in the target quotient. Its value on 1 therefore determines it, and lifting the universal Gysin map gives (16.36). Derived base change along the bundle classifying map gives the assertion on X. This argument concerns split vector bundles. A compact extension deformation complex also contains cross derived-Hom terms; its comparison with these bundles and its orientation are additional requirements.

For general degeneration maps, a compatible square includes its homotopy. If $f : A \to B$ and $g : C \to D$ are cochain maps, set

$$\operatorname{Cone}(f) = B \oplus A[1], \qquad d_f = \begin{pmatrix} d_B & f \\ 0 & -d_A \end{pmatrix}.$$

A filtered degree-r map of cones has form $T = \left(\begin{smallmatrix} b & h \\ 0 & (-1)^r a \end{smallmatrix}\right)$, with $|a| = |b| = r$, $|h| = r-1$. The differential $\partial T = d_g T - (-1)^r T d_f$ and matrix composition make these maps a dg category. In degree zero, closedness is precisely the cochain condition on a, b together with

$$d_D h + h d_A = bf - ga. \tag{16.37}$$

The composite homotopy is $b_2 h_1 + h_2 a_1$. These assertions follow by multiplying the two matrices; $d_f^2 = d_g^2 = 0$ gives $\partial^2 = 0$ and the graded Leibniz identity.

At length zero take $f_0 = 1 : R_0 \to R_0$, and at length one take Euler multiplication $f_1 = e_z(TY) : H^*(Y)[[z]] \to H^*(Y)[[z]]$ in resolution grading. The ordinary creator $b(1) = 1_Y$ admits no closed square: it would require $1_Y = e_z(TY)a(1)$, contrary to the nonzero unit in $H^*(\mathbb{P}(TY))$. The transported creator $b(1) = e_z(TY)$ admits $a(1) = 1_Y$, $h = 0$. Its cone map is nevertheless zero in the derived category because $\operatorname{Cone}(f_0)$ is contractible. Regularity on an image lattice therefore does not supply a nonzero vacuum action on its cone.

16.9 Normal coordinates and derived morphisms

A compact sheaf on a formal fibre has more information than a sheaf with zero normal operator. For a smooth projective surface Y, coherent sheaves on $Y \times \operatorname{Spf} \mathbb{C}[[t]]$ killed by some t^M are equivalent to pairs (F, N_F) with N_F nilpotent. Restriction of scalars and the rule $t \mapsto N_F$ are inverse. They preserve kernels, cokernels, extensions and finite flags. On flat families with a common nilpotence bound they commute with base change.

Choose a dg enhancement $C(F,G)$ of the surface Hom complexes with associative composition and strict closed nilpotent representatives for N_F. Put $\delta_{GF}(a) = N_G a - a N_F$ and adjoin $|\theta| = 1$. A model of the formal-fibre Hom is

$$\begin{aligned} \mathcal{K}_N(F,G) &= C(F,G) \otimes \Lambda(\theta), \\ D(a\theta^\epsilon) &= d_C a\, \theta^\epsilon + (-1)^{|a|} \delta_{GF}(a) \theta^{\epsilon+1}, \\ (a\theta^\epsilon)(b\theta^\eta) &= (-1)^{\epsilon|b|} ab\, \theta^{\epsilon+\eta}. \end{aligned} \tag{16.38}$$

The resolution $[\mathcal{O}_Y[t] \otimes F \xrightarrow{t - N_F} \mathcal{O}_Y[t] \otimes F]$ together with a surface resolution gives the two copies of $C(F,G)$ and their commutator map. The exterior diagonal gives composition. Equivalently, this is

the one-dimensional Lie algebra cochain model for the action δ. Since d_C and δ commute and satisfy the Leibniz rule, $D^2 = 0$ and the displayed composition is a dg product. No formality of $C(F, G)$ is required.

For K3 with a chosen holomorphic two-form, write $\tau_Y : C(F, F) \to \mathbb{C}[-2]$ for its Serre trace. The threefold trace is its θ coefficient, normalized by the normal one-form. In particular the pairing of $a\theta^\epsilon$ and $b\theta^\eta$ is $(-1)^{\epsilon|b|}\tau_Y(ab)$ if $\epsilon + \eta = 1$, and is zero otherwise. Cyclicity and the vanishing of a commutator trace give compatibility with the differential.

Let $f(t) = \sum_{r\geq 1} a_r t^r$ with $a_1 \neq 0$ be a new formal coordinate. For composable morphisms $b_i : F_{i-1} \to F_i$, define

$$D_n^f(b_n, \dots, b_1) = \sum_{r\geq n} a_r \sum_{j_0+\cdots+j_n=r-n} N_{F_n}^{j_n} b_n N_{F_{n-1}}^{j_{n-1}} \cdots b_1 N_{F_0}^{j_0}. \tag{16.39}$$

At fixed arity, nilpotence makes this finite. The first difference satisfies $D_1^f(N_G b - bN_F) = f(N_G)b - bf(N_F)$. Thus the linear map is the identity on unmarked morphisms and D_1^f on their θ components. Its product defect is generally nonzero.

Theorem 16.16. *Use suspension s of degree -1 and $b_1(sx) = -s(Dx)$, $b_2(sx, sy) = (-1)^{|x|}s(xy)$. For $n \geq 2$, set*

$$(\mathcal{F}_f)_n(s(b_n\theta), \dots, s(b_1\theta)) = s(D_n^f(b_n, \dots, b_1)\theta),$$

and set this component to zero on every other input pattern. Together with the linear map just described, these give a strictly unital A_∞ functor $\mathcal{K}_N \to \mathcal{K}_{f(N)}$. For invertible coordinates f, g, there is an exact bar identity

$$\mathcal{F}_g\mathcal{F}_f = \mathcal{F}_{g\circ f},$$

where the second coordinate is evaluated at $f(N)$. In particular $\mathcal{F}_{f^{-1}}$ is an inverse.

Proof. The d_C terms cancel by the graded Leibniz rule, because every N_F is closed of degree zero. Among the remaining bar terms, all-marked inputs have zero adjacent product since $\theta^2 = 0$. Two or more unmarked inputs kill every higher term, leaving only the ordinary binary identity. With one unmarked input, the commutator inserts $N_{F_i}b_i - b_i N_{F_{i-1}}$ in a divided difference. For a monomial t^r the powers at successive positions telescope; the endpoints are the adjacent products. For example

$$D_1^f(ab) - aD_1^f(b) = D_2^f(N_H a - aN_G, b),$$
$$D_1^f(a)b - D_1^f(ab) = D_2^f(a, N_G b - bN_F).$$

At an interior position the same cancellation has common suspended sign $(-1)^{1+\Sigma_{i=k}^n |b_i|}$ on its three terms. This proves the graded functor equation in every arity. Linearity and fixed-arity finiteness extend it to formal series.

For composition, put the endomorphisms N_{F_i} on the diagonal of an upper-triangular block matrix and the b_i on its first superdiagonal. The outer block of f of this matrix is D_n^f. The paths through $g(f(-))$ run over all ordered partitions of the inputs, exactly as in bar composition. Functional calculus gives the asserted identity. Mixed-input higher terms vanish on both sides. An unmarked unit kills every higher component, proving strict unitality. □

For $f(t) = t + at^2$ the correction terminates at $D_2^f(x, y) = axy$, with $D_1^f(x) = x + a(N_G x + xN_F)$. A cubic coordinate has a third component. The inverse coordinate can have arbitrarily many components, each finite on a fixed string of nilpotent objects.

16.9.1 The coordinate contribution to the determinant

For a cyclic trace and a formal series h, cyclic permutation of the r terms of $D_1^{t^r}$ gives

$$\tau_Y\big(h(N)f'(N)^{-1}D_1^f(b)\big) = \tau_Y(h(N)b). \tag{16.40}$$

The inverse exists since $f'(0) \neq 0$ and N is nilpotent. This transports the normal volume form $h(t)\,dt$ and its trace pairing on cohomology. It does not choose negative-cyclic representatives.

On a finite perfect equivariant chart over a local Artin $\mathbb{C}$-algebra R, put

$$\chi_Y(F,G) = \sum_j (-1)^j \operatorname{rk}_R C^j(F,G).$$

The determinant contribution is

$$d_f(F,G) = \prod_j \det(D_1^f|_{C^j(F,G)})^{(-1)^{j+1}}. \tag{16.41}$$

The extra sign comes from the degree of θ. At a complex point $D_1^f = a_1\,\mathrm{id} + U$ with U nilpotent, so $d_f(F,G) = a_1^{-\chi_Y(F,G)}$. Over a nonreduced base the nilpotent determinant need not vanish. For K3, take $R = \mathbb{C}[\epsilon]/(\epsilon^2)$, $F = \mathcal{O}_Y \otimes R$, $N = \epsilon\,\mathrm{id}$, and $f(t) = t + at^2$. Then

$$D_1^f = 1 + 2a\epsilon, \qquad d_f(F,F) = 1 - 4a\epsilon, \qquad d_f(F,F)^{1/2} = 1 - 2a\epsilon.$$

On K3 self-Hom, Serre duality makes $\chi_Y(F,F)$ even: degrees $j, 2-j$ pair, and the middle pairing is skew and nondegenerate. Put $\rho_f(F) = a_1^{\chi_Y(F,F)}\, d_f(F,F)$. Its reduction is one, and the square root with specified reduction is uniquely

$$j_f(F) = a_1^{-\chi_Y(F,F)/2} \exp\big(\tfrac{1}{2}\log \rho_f(F)\big). \tag{16.42}$$

Logarithm and exponential are finite on the Artin maximal ideal. The linear part of the coordinate functor proves the determinant cocycle; uniqueness proves the square-root cocycle. On a strict upper-triangular extension chart $0 \to F \to E \to G \to 0$ whose Hom filtration is preserved by the endomorphisms, the two cross determinants agree by Serre adjunction. Consequently

$$d_f(E,E) = d_f(F,F)\,d_f(G,G)\,d_f(F,G)^2, \qquad j_f(E) = j_f(F)\,j_f(G)\,d_f(F,G).$$

The second identity follows from the first and the specified reductions. All finite flag refinements use the same diagonal and unordered cross blocks; compatible inverse limits of Artin charts preserve the identities.

This supplies the normal-coordinate contribution to square-root sewing on these charts. A global orientation and stabilizer characters are separate data. On an elliptic factor, a fixed invariant one-form permits coordinates with translation transitions. After recentering at a fibre the transition is the identity, so $D_1 = 1$, $D_{n>1} = 0$ and all these determinant factors are one. Descent of critical Hall coefficient functors still needs its supported operation maps.

16.10 States with chiral fields

The elliptic tensor class has a field-bearing realization. On a coordinate chart of a smooth projective complex d-fold Y, the chiral de Rham sheaf Ω^{ch}_Y has even fields γ^i, β_i and odd fields c^i, b_i, with singular products

$$\beta_i(z)\gamma^j(w) \sim \frac{\delta_i^j}{z-w}, \qquad b_i(z)c^j(w) \sim \frac{\delta_i^j}{z-w}. \tag{16.43}$$

Their weights are 0, 1, 0, 1; the charges of c, b are 1, -1 and those of γ, β are zero. Functions of γ, translation and normal products generate the local vertex superalgebra. Write ∂ for vertex translation and $a_{(n)}b$ for its products; for $n \geq 0$ these are the coefficients of $(z-w)^{-n-1}$ in the singular OPE. Let L_0, J_0 denote conformal weight and charge. For $\widetilde{\gamma} = f(\gamma)$ with inverse g, put $f_i^a = \partial_i f^a$ and $g_a^i = \partial_a g^i$. The transformations are

$$\begin{aligned} \widetilde{c}^a &= f_i^a(\gamma)c^i, \qquad \widetilde{b}_a =: g_a^i(f(\gamma))b_i :, \\ \widetilde{\beta}_a &=: \beta_i g_a^i(f(\gamma)) : + : \partial_k(g_a^i \circ f)(\gamma)c^k b_i : . \end{aligned} \tag{16.44}$$

The order of the normal product in the last line is part of the formula. The coordinate theorem proves preservation of OPEs and composition on overlaps [116, § 3]. For $\widetilde{\gamma} = \gamma^2$ on a punctured one-dimensional chart, the last expression is $: \beta(2\gamma)^{-1} : -(2\gamma^2)^{-1} : cb :$; discarding the fermionic term would change the coordinate map.

Choose a finite affine cover with affine intersections, or a Stein cover acyclic for each coherent graded piece. Let $V^\bullet$ be its cosimplicial section object. The Thom–Whitney totalization

$$\mathbb{V}_Y = \mathrm{TW}(V^\bullet)$$

uses compatible polynomial simplex forms tensored with local sections. A form-field tensor has total parity $\deg\omega + p(v)$, and its products are

$$(\omega \otimes v)_{(n)}(\eta \otimes w) = (-1)^{p(v)\deg\eta}\omega\eta \otimes v_{(n)}w. \tag{16.45}$$

The simplex differential is a derivation for these signs. Translation acts on fields and the compatible constant unit is the vacuum. Thus $\mathbb{V}_Y$ is a dg vertex superalgebra. Integration of simplex forms compares it with the Čech complex; the polynomial Poincaré contraction proves that this comparison is a quasi-isomorphism. At fixed weight, the coordinate construction has a finite filtration by coherent tensor bundles. Acyclicity on the intersections and induction on that filtration show that its underlying complex computes $R\Gamma(Y, \Omega^{\mathrm{ch}}_Y)$ weight by weight. Only the sheaf-resolution differential is included.

The associated-graded supercharacter is (16.8). The filtration need not split as a field sheaf. Nevertheless its index is preserved, giving

$$\sum_i (-1)^i \operatorname{str}\bigl(q^{L_0}y^{J_0-d/2}; H^i(Y, \Omega^{\mathrm{ch}}_Y)\bigr) = \mathrm{Ell}(Y; q, y).$$

16.10.1 *A projector onto the canonical descendant*

On the Enriques surface T, choose local square-root volume forms for a fixed trivialization of K_T^2, and volume coordinates for them. Their transition determinants are constant ± 1. The transformation of $J = \sum_i : c^i b_i :$ adds the vertex derivative of the logarithm of that determinant, hence adds zero. Thus J is global. Its zero mode is the charge. Set $\mathscr{V}_{1,2} = (\Omega^{\mathrm{ch}}_T)_{L_0=1, J_0=2}$; this is the normalized qy coefficient. Weight zero and charge two are K_T.

Proposition 16.17. *As sheaves of complex vector spaces,*

$$\mathscr{V}_{1,2} = \partial K_T \oplus \ker J_{(1)}, \qquad P = \tfrac{1}{2}\partial J_{(1)}, \qquad P^2 = P.$$

The field coefficient has cohomology dimensions $(0, 65, 1)$ *in degrees zero, one and two. The kernel has only degree-one cohomology, of dimension* 65*. For K3 the corresponding field coefficient has dimensions* $(1, 130, 1)$.

Proof. The translation relation is $[J_{(1)}, \partial] = J_{(0)}$. For $a \in K_T$, weight gives $J_{(1)}a = 0$ and charge gives $J_{(0)}a = 2a$. Hence $J_{(1)}\partial a = 2a$, proving the split injection and projector. These maps need not be $\mathcal{O}_T$-linear.

The finite symbol filtration has factors $\operatorname{Sym}^2 \Omega^1_T$, K_T, Ω^1_T, T_T. Proposition 16.5 therefore bounds the field cohomology by $h^0 = 0$, $h^1 \le 65$, $h^2 \le 1$, and fixes its Euler characteristic at -64. The split canonical summand preserves $H^2(K_T) = \mathbb{C}$, so $h^2 = 1$ and $h^1 = 65$. Removing that summand gives the kernel assertion. On K3 the canonical summand preserves one class in each of degrees zero and two; the remaining symbol dimensions are 90, 20, 20 in degree one, giving the stated result. □

This proves the actual total parity dimensions $1 \mid 65$, independently of a splitting of the symbol. It supplies no cancellation differential on the $29 \mid 93$ Borcherds root space.

There is a natural additional differential, but it has a different effect. Locally put $Q = \sum_i : \beta_i c^i :$ and $G = \sum_i : b_i \partial\gamma^i :$. Their global operators satisfy

$$[Q_{(0)}, G_{(1)}]_+ = L_0. \tag{16.46}$$

Coordinate changes of Q differ by total vertex derivatives, so its zero mode is global [116, § 4]. Adding $Q_{(0)}$ to the resolution differential contracts every positive-weight subcomplex, with homotopy $G_{(1)}/m$ at weight $m > 0$. The total cochain signs make this homotopy anticommute with the resolution differential. Moreover $Q_{(0)}$ raises charge by one. The state complex used here therefore retains derived sections without this extra differential.

16.10.2 Permutation sectors and their operations

The twisted chiral de Rham construction gives a sheaf $\Omega^{\mathrm{ch},g}_{Y^N}$ on the fixed locus of each $g \in \mathfrak{S}_N$, with a compatible centralizer action [62]. Write $C(g)$ for the centralizer. The age is the sum of the fractions $\theta \in (0,1)$ for its normal eigenvalues $e^{2\pi i\theta}$, counted with multiplicity. Define

$$\mathcal{H}_N(Y) = \bigoplus_{[g]\subset\mathfrak{S}_N} R\Gamma((Y^N)^g, \Omega^{\mathrm{ch},g}_{Y^N})^{C(g)}. \tag{16.47}$$

Use only the sheaf differential, twisted conformal weight and charge $J_0 - Nd/2$. Oscillator parity is independent of the age contribution to J_0. For an n-cycle, let

$$\mathcal{K}_n(Y) = R\Gamma(Y, \Omega^{\mathrm{ch},(1\cdots n)}_{Y^n})^{\mu_n}.$$

Its oscillator indices lie in $n^{-1}\mathbb{Z}$. For a surface, age $n-1$ makes the normalized ground Laurent degree -1, as for the seed.

Proposition 16.18. *Coefficientwise in length and conformal weight, there is a natural equivalence of supercomplexes*

$$\prod_{N\ge 0} p^N \mathcal{H}_N(Y) \simeq \widehat{\operatorname{Sym}}_{\mathrm{super}}\left(\bigoplus_{n\ge 1} p^n \mathcal{K}_n(Y)\right). \tag{16.48}$$

Proof. A permutation with a_n cycles of length n has fixed locus $Y^{\sum a_n}$ and centralizer $\prod_n(\mu_n^{a_n} \rtimes \mathfrak{S}_{a_n})$. The independent coordinate sets give external products of the single-cycle twisted sheaves. Their finite coherent filtrations justify coefficientwise Künneth. Finite-group invariants are exact over $\mathbb{C}$. Taking cyclic invariants first and symmetric invariants second gives the super symmetric powers, with cochain and intrinsic parities combined as in (16.45). There are finitely many partitions at each length, which proves the completed statement. □

On the symmetric model, concatenation of cycles and assignment of each cycle to one of two groups give a commutative, cocommutative dg super Hopf algebra. In coinvariant coordinates,

$$\Delta(v_1 \cdots v_r) = \sum_{I \subset \{1,\dots,r\}} (-1)^{\epsilon(I)} v_I \otimes v_{I^c}.$$

Order is retained in each subword and $\epsilon(I)$ is the total Koszul sign. In invariant coordinates the multiplication is the averaged shuffle, fixing the factorials. The empty cycle word is the unit, single cycles are primitive and their antipode is negation. Splitting a word into three groups proves coassociativity directly. A continuous homogeneous linear functional λ on the single-cycle space gives a derivation with $[D_\lambda, M_v]_{\mathrm{super}} = \lambda(v)$. All operations are finite at fixed length.

There is also an action of singular field modes. For $V_Y = H^0(Y, \Omega_Y^{\mathrm{ch}})$, its mode Lie superalgebra is

$$\mathfrak{m}(V_Y) = \frac{V_Y \otimes \mathbb{C}[t, t^{-1}]}{\mathrm{im}(\partial \otimes 1 + 1 \otimes \partial_t)}, \qquad [a[m], b[n]] = \sum_{j \geq 0} \binom{m}{j} (a_{(j)} b)[m + n - j].$$

Here $a[m]$ is the class of $a \otimes t^m$. The invariant field $a^\Delta = \sum_{i=1}^N a^{(i)}$ acts on every twisted sector by integer modes. Write $\rho(a[m]) = (a^\Delta)_{(m)}$ for this action. For $j \geq 0$, different factors have regular products, so $(a^\Delta)_{(j)} b^\Delta = (a_{(j)} b)^\Delta$. Translation is preserved, and the central vacuum mode $1[-1]$ acts by N. Under (16.48) the action is the sum on the individual cycles; therefore

$$[\rho(A), M_v]_{\mathrm{super}} = M_{\rho(A)v}. \tag{16.49}$$

The diagonal field assignment is not a unital vertex map: its vacuum is multiplied by N, and regular products have cross terms.

If $\mathrm{Ell}(Y; q, y) = \sum c(m, \ell) q^m y^\ell$, the twisted trace theorem and the symmetric-product elliptic genus formula give

$$\sum_{N \geq 0} p^N \chi_s \mathcal{H}_N(Y) = \prod_{n>0, m \geq 0, \ell} (1 - p^n q^m y^\ell)^{-c(nm,\ell)} \tag{16.50}$$

[62, Theorem 6.1] [12, Corollary 6.7]. Here χ_s includes cochain and oscillator signs. The same scalar series is the Hilbert-scheme elliptic genus. Its equality does not construct an equivalence of chiral coefficient complexes. The cycle product has zero primitive commutator, while the seed mode action preserves length; neither has yet been identified with the compact extension action.

16.11 An unshifted comparison detected by a field mode

Let S be a smooth projective K3 surface and set $C_{H,N} = R\Gamma(S^{[N]}, \Omega_{S^{[N]}}^{\mathrm{ch}})$ and $C_{O,N} = \mathcal{H}_N(S)$. Retain sheaf degree k, weight w, charge J and intrinsic parity ϵ. Total parity is $k + \epsilon$, and the differential has degree $(1, 0, 0, 0)$. A degree-zero comparison preserving total parity, weight and charge also preserves intrinsic parity at each sheaf degree.

Every untwisted local monomial satisfies $\epsilon \equiv J \pmod 2$: each c or b contributes one to parity and respectively 1 or -1 to charge. Coordinates, momenta, coefficient functions and translation preserve this

relation. The coordinate maps do also. Thus the charge-even, intrinsically odd part of $C_{H,N}$ is a zero sheaf complex, and its cohomology is zero in every homogeneous model.

The field $G = \sum : b_i \partial\gamma^i :$ is global on every smooth complex manifold. The antighosts transform by the inverse transpose Jacobian and coordinate derivatives by the Jacobian. All the factors in these two transformations have regular mutual OPEs, so the Jacobians cancel without an extra contraction. The field is odd, of weight two and charge -1.

In the transposition sector of S^2, let τ be the constant ground section. Its fixed locus is S, and the natural centralizer action fixes τ. Its gradings are $(0, 0, 1, 0)$: age contributes one to charge but no oscillator parity. The global invariant descendant

$$s = (G^{(1)} + G^{(2)})_{(0)}\tau \tag{16.51}$$

has gradings $(0, 1, 0, 1)$.

Proposition 16.19. *The class of s is nonzero in $H^0(C_{O,2})$. Every degree-zero map $C_{H,2} \to C_{O,2}$ preserving weight, charge and total parity has a nonzero degree-zero comparison-cone class in this coefficient. A reverse map has a nonzero cone class in degree -1. There is no zigzag of quasi-isomorphisms with these gradings.*

Proof. Choose dual sum and difference coordinates locally. Use oscillator expansions $\gamma(z) = \sum_t \gamma_t z^{-t}$ and $\beta(z) = \sum_r \beta_r z^{-r-1}$, with the same exponents for c and b, respectively. The two normal coordinates have half-integral modes with $[\beta^-_{a,r}, \gamma^{b,-}_t] = \{b^-_{a,r}, c^{b,-}_t\} = \delta^b_a \delta_{r+t,0}$. Positive normal modes annihilate τ. The normal part of the G descendant is

$$s = \frac{1}{2}\sum_{a=1}^{2} \gamma^{a,-}_{-1/2} b^-_{a,-1/2}\tau.$$

The derivative supplies the factor $1/2$; the integral tangential terms kill the constant ground section. The actual mode operation

$$2\beta^-_{1,1/2} c^{1,-}_{1/2} s = \tau \tag{16.52}$$

proves nonvanishing. Globality was established before the local coordinates were chosen. The sheaf resolution has no negative degree, so a nonzero global section gives a nonzero H^0 class.

For a forward map use $\mathrm{Cone}(F)^k = C^k_{O,2} \oplus C^{k+1}_{H,2}$. Its $(w, J, \epsilon) = (1, 0, 1)$ component is precisely the orbifold component, since the Hilbert component vanishes. It retains $[s]$. The reverse cone has this component shifted by one, giving degree -1. The same cohomology mismatch excludes a homogeneous zigzag. Resolution changes, contractible additions and coordinate changes preserve the nonzero class; coefficientwise completion does not change its fixed length and weight. □

The detector extends to every cycle length $N \geq 2$. Let τ_N be the constant ground section for $(1 \cdots N)$ and put $s_N = (\sum_i G^{(i)})_{(0)}\tau_N$. In dual Fourier bases put $\theta_r = r/N$ for $1 \leq r < N$. The coordinate and c modes have indices in $-\theta_r + \mathbb{Z}$, while their momentum and antighost partners have indices in $\theta_r + \mathbb{Z}$. The global state is

$$s_N = \sum_{a=1}^{2}\sum_{r=1}^{N-1} \theta_r \gamma^{a,r}_{-\theta_r} b_{a,r;-(1-\theta_r)}\tau_N,$$

and each selected summand is detected by

$$\theta_r^{-1}\beta_{a,r;\theta_r} c^{a,r}_{1-\theta_r} s_N = \tau_N.$$

The paired centralizer eigenvalues cancel. Age $N-1$ gives gradings $(0, 1, N-2, 1)$ and normalized charge -2. For every even N the same untwisted parity mismatch excludes the homogeneous comparison. For odd N this test gives no obstruction.

A different recipient can shift both gradings. For a single m-cycle replace

$$\mathcal{K}_m(S) \quad \text{by} \quad \Pi^{m-1}\mathcal{K}_m(S)[-(m-1)]. \tag{16.53}$$

This adds age to sheaf degree and intrinsic parity, where Π reverses intrinsic parity and the shifted differential has sign $(-1)^{m-1}$. It preserves total parity and the elliptic supertrace. It removes the mismatch above: at length two s moves to $(1, 1, 0, 0)$. The shifted complex is not degree-zero quasi-isomorphic to the original one, whose ground class occurs in degree zero. For a general permutation use age $N - c(g)$ in each summand. Comparing this new recipient with $C_{H,N}$ remains an open problem requiring field, creation and joining maps with their supported compatibilities. The grading change alone supplies none of those maps.

16.12 Resolving the target diagonal by normal fields

Ordinary joining uses a diagonal Gysin map. A map on fields has to retain the conjugate normal variables as well. In one normal coordinate, write the even fields as v, β and the odd fields as c, b, with $\beta_{(0)}v = b_{(0)}c = 1$. A unital vertex homomorphism killing v would give $1 = \phi(\beta)_{(0)}0 = 0$. Restriction of functions therefore does not extend to such a map on the full field system.

Instead define

$$\delta = (:vb:)_{(0)}, \qquad \delta c = v, \quad \delta\beta = -b, \quad \delta v = \delta b = 0. \tag{16.54}$$

Use normal cochain degrees $|c| = -1$, $|b| = 1$, $|v| = |\beta| = 0$. In this grading the normal fermion parity is carried by cochain degree, and is not counted a second time as intrinsic parity. Tangential fermion parity remains separate. The defining current has regular self-OPE, so $\delta^2 = 0$. Its zero mode is a derivation of every vertex product and commutes with translation. The minus sign is forced by $\delta(\beta_{(0)}c) = -b_{(0)}c + \beta_{(0)}v = 0$.

For a formal chart, complete the coefficient functions only in the undifferentiated even normal coordinates at each fixed conformal weight, and take the direct sum over weights. Every state then has bounded conformal weight; positive-weight oscillators remain polynomial. For a pair of states the singular products differentiate their coefficients only finitely many times, so vertex products are continuous on this carrier.

Proposition 16.20. *On the polynomial carrier and on this formal carrier, the inclusion of a tangential chiral de Rham algebra into its tensor product with finitely many normal quartets is a quasi-isomorphism of dg vertex superalgebras.*

Proof. In normalized derivative PBW variables,

$$\delta = \sum_j (v_j\partial_{c_j} - b_j\partial_{\beta_j}), \qquad h_0 = \sum_j (c_j\partial_{v_j} - \beta_j\partial_{b_j}).$$

Use left exterior derivatives. Their anticommutator is the total normal-number operator $\mathcal{N}$, since it is a derivation acting identically on each generator. Sum over all normal directions. On positive normal number put $h = h_0/\mathcal{N}$, and put $h = 0$ on the normal vacuum. If i includes the vacuum and π extracts its coefficient,

$$\delta h + h\delta = 1 - i\pi.$$

This is coefficientwise valid on formal functions; differentiation lowers their adic order by at most one. Tensor with the tangential factor. Its inclusion preserves all products, translation and vacuum, proving the asserted quasi-isomorphism. □

The projection π is a complex map, not a vertex map; $\beta_{(0)}v = 1$ excludes the latter. The homotopy is also only linear. Bounded weight is essential: completing simultaneously to admit $\sum_{N\geq 0} : \beta^N :$ and $\sum_{N\geq 0} : v^N :$ would admit complete contractions $N!(z-w)^{-N}$ of unbounded pole order. The differential preserves weight and lowers the original fermionic charge by one. It is therefore not a differential on each Laurent coefficient separately. Unlike the extra de Rham differential (16.46), it leaves tangential positive-weight fields.

For r affine copies, put

$$P_r = I_r - \frac{1}{r}\mathbf{1}\mathbf{1}^{\mathrm{t}}, \qquad \delta_r = \left(: \sum_{i,j} (P_r)_{ij}\gamma_i b_j : \right)_{(0)}.$$

The tangential fields are $\bar\gamma = r^{-1}\sum \gamma_i$, $\bar c = r^{-1}\sum c_i$, $\bar\beta = \sum \beta_i$ and $\bar b = \sum b_i$; these have unit pairings. Their complement has $r-1$ normal quartets. For a partition into blocks let Q be the orthogonal projector onto block-constant vectors, with entries $1/|B|$ within a block B. Then

$$P_r = (I_r - Q) + \left(Q - \frac{1}{r}\mathbf{1}\mathbf{1}^{\mathrm{t}}\right). \tag{16.55}$$

The summands are orthogonal projectors onto within-block differences and differences of block means. Their differentials add to δ_r. Further refinement gives the same full projector. Changing normal basis sends v, c by its matrix and β, b by its inverse transpose, preserving OPEs and $\sum : vb :$. Matrix composition proves compatibility of every affine regrouping. These are target diagonals in Y^r. Insertion diagonals for supported chiral collisions have different ambient spaces and coefficient functors; a comparison must relate their operations.

16.12.1 The global absolute normal bundle

Let $E \to Y$ be a holomorphic vector bundle and consider the chiral de Rham sheaf on $\mathrm{Tot}(E)$, with fibrewise polynomial states or the same fixed-weight formal completion. In a local frame write u^i, v^a for base and fibre coordinates and c_u^i, c_v^a for their odd partners. The fibre Euler vector field $\mathsf{E} = v^a\,\partial_{v^a}$ is intrinsic. Its chiral contraction is intrinsic under (16.44), so $\delta = (\iota_{\mathsf{E}})_{(0)}$ is global.

The global grading now uses $-J$ for every fermion, including the base ones. Indeed a frame change $v' = g(u)v$ gives $c'_v = g(u)c_v + (dg)_u(v)c_u$; a normal-only grading would not respect this transition. Intrinsic fermion parity is absorbed into this cochain grading. Locally the differential is the normal quartet differential and kills the tangential fields.

For the vertical frame vector field $W = A^b_a(u)v^a\,\partial_{v^b}$, the current

$$H_A =: A^b_a c^a_v \beta_b : - : \partial_i A^b_a\, c^i_u c^a_v b_b :$$

has $[\delta, (H_A)_{(0)}] = \mathcal{L}_W$, where $\mathcal{L}_W$ is the infinitesimal chiral de Rham action of W. Apply the odd derivation to the two terms: the result is $: A^b_a v^a \beta_b : + : A^b_a c^a_v b_b : + : \partial_i A^b_a\, v^a c^i_u b_b :$, the full chiral Lie derivative. The sign of the last term uses $\delta(c^i_u c^a_v b_b) = -c^i_u v^a b_b$.

Theorem 16.21. *The cohomology vertex sheaf of the absolute Euler complex is canonically Ω^{ch}_Y, with zero differential and cochain grading $-J$. The assertion holds on both the polynomial and fixed-weight formal carriers.*

Proof. The local normal contraction leaves precisely the tangential states. Locally any varying frame matrix is a constant matrix times $\exp A(u)$. The constant factor fixes the normal vacuum and tangential states. For the varying factor, let F_s be the frame action of $\exp(sA)$. It commutes with δ, and the current identity gives

$$F_1 - 1 = \delta K + K\delta, \qquad K = \int_0^1 (H_A)_{(0)} F_s \, ds.$$

Hence vertical frame changes induce the identity on local cohomology. Base coordinate changes induce exactly the base chiral de Rham maps. These identifications compose on triple overlaps and preserve the vertex products. □

No global strict inclusion of tangential vertex fields is asserted. For the full diagonal in Y^r, its linear normal bundle is $T_Y \otimes (\mathbb{C}^r/\mathbb{C}_{\mathrm{diag}})$; the theorem applies to this bundle and its linear regroupings.

16.12.2 The actual formal diagonal

The normal bundle and the formal neighbourhood are different geometric objects. In coordinates $(u, u+v)$, a change $u' = f(u)$ gives

$$v' = f(u+v) - f(u) = df_u(v) + \tfrac{1}{2} d^2 f_u(v,v) + O(v^3).$$

A holomorphic formal linearization over the first projection, inducing the identity on the normal bundle, would have quadratic term $\mathrm{Exp}_u(v) = u + v - \tfrac{1}{2}\Gamma_u(v,v) + O(v^3)$. Here Γ_u is symmetric in its two inputs. Comparing coordinates gives

$$\Gamma_{f(u)}(df_u v, df_u v) = df_u \Gamma_u(v,v) - d^2 f_u(v,v).$$

This is the transformation law of a torsion-free holomorphic connection on T_Y. On a complex surface its curvature would have type $(2,0)$, so the degree-four Chern–Weil form for c_2 would have type $(4,0)$ and vanish. This contradicts $\int_Y c_2 = 12$ or 24 for Enriques or K3. It rules out this holomorphic linearization. The perfect diagonal kernel $\mathcal{O}_\Delta$ still exists globally, with local Koszul resolutions by the coordinate differences.

For Kähler Y, a smooth formal exponential instead identifies the Dolbeault algebra of the formal diagonal with

$$\left(\mathcal{A}_Y^{0,*}(\widehat{\mathrm{Sym}} T_Y^*), D_B = \bar\partial + R\right), \quad R = \sum_{n\ge 2} R_n, \quad \bar\partial R + \tfrac{1}{2}[R,R] = 0. \tag{16.56}$$

Here R_n is a $(0,1)$-form valued in homogeneous vertical vector fields of degree n. We use the actual Taylor coefficients, including their factorials; R_2 represents the Atiyah class [141, Theorem 3.2].

Let $\mathcal{V}_{\mathrm{rel}}$ be the relative formal bc–$\beta\gamma$ system of T_Y. Its coefficients $\mathcal{A}_Y^{0,*}$ are central vertex elements with translation zero; there are no base momentum fields. Use total cochain degree Dolbeault degree minus J. A vertical vector field $V = V^i(v)\partial_{v^i}$ has the lift

$$\mathcal{L}_V = \left(: V^i(v)\beta_i : + : \partial_j V^i(v) c^j b_i :\right)_{(0)}.$$

The bosonic and fermionic anomalies cancel, giving $[\mathcal{L}_V, \mathcal{L}_W] = \mathcal{L}_{[V,W]}$. Consequently

$$D_{\mathrm{form}} = \bar\partial + \mathcal{L}_R \tag{16.57}$$

is a global square-zero vertex derivation, preserving weight and charge and restricting to D_B on functions. Indeed its square is $\mathcal{L}_{\bar\partial R + [R,R]/2}$. The full chiral coordinate transformation intertwines it under a change

of formal exponential: the base-dependent derivative of that transformation gives the same conjugation term as for $\bar\partial + R$. The coordinate composition law proves the exact cocycle identity. Fixed weight bounds the number of contractions that lower the degree of a coefficient, so these operations are continuous in the specified completion.

The Euler contraction does not commute with this differential. Write $\iota_V =: V^i b_i :$ for chiral contraction, extending to form-valued vector fields with the tensor signs. The Cartan identity and $[V_n, \mathsf{E}] = -(n-1)V_n$ give

$$[D_{\mathrm{form}}, \partial] = -\sum_{n\geq 2}(n-1)(\iota_{R_n})_{(0)}. \tag{16.58}$$

Thus $D_{\mathrm{form}} + \partial$ need not square to zero. The curved formal neighbourhood requires more than adding the linear contraction.

16.12.3 A conjugated resolution of the central base

A different construction resolves the central base while retaining explicit curvature terms. For a homogeneous degree-n vector field set

$$H_{V_n} = \frac{1}{n} : \partial_j V_n^i c^j \beta_i :, \qquad \delta H_{V_n} = K_{V_n} =: V_n^i \beta_i : + \frac{1}{n} : \partial_j V_n^i c^j b_i : .$$

Euler's identity gives the first term; the odd derivation sign gives the second. Its coefficient $1/n$ distinguishes this lift from $\mathcal{L}_{V_n}$. Let $\mathscr{S} = \sum_n H_{R_n}$ and $S_0 = \mathscr{S}_{(0)}$. Form degree one and ghost degree -1 make S_0 even of total degree zero. Put

$$d_0 = \bar\partial + \partial, \qquad D_K = e^{S_0} d_0 e^{-S_0}. \tag{16.59}$$

On a complex d-fold the exponentials are finite since $S_0^{d+1} = 0$ by form degree.

Proposition 16.22. *The operator D_K is a global square-zero vertex derivation. The central inclusion $(\mathcal{A}_Y^{0,*}, \bar\partial) \to (\mathcal{V}_{\mathrm{rel}}, D_K)$ is a quasi-isomorphism. For two choices S_0, S_0', the intertwiner $e^{S_0'} e^{-S_0}$ composes exactly with the corresponding intertwiners.*

Proof. The currents H_{R_n} transform tensorially under relative linear frames with central coefficients. Their zero modes are global derivations, and the finite exponentials are vertex automorphisms fixing the base. Conjugation proves square-zero. The normal-number homotopy is equivariant under these frames. With its tensor sign on forms, it anticommutes with $\bar\partial$, so $d_0 h + h d_0 = 1 - i\pi$ globally. Transport it to

$$h_K = e^{S_0} h e^{-S_0}, \qquad \pi_K = \pi e^{-S_0}, \qquad D_K h_K + h_K D_K = 1 - i\pi_K.$$

This proves the cohomology assertion. The intertwiner identities follow by cancelling adjacent exponentials. Neither projection nor homotopy is claimed to preserve vertex products. □

Conjugation gives a square-zero differential for any such homogeneous tensors. The Maurer–Cartan equation is needed when comparing with the actual formal diagonal. In particular D_K is not identified with D_{form}. It resolves only the central base Dolbeault algebra, whereas the absolute Euler complex retains the base chiral de Rham fields.

On a surface all corrections are visible. For $K_1 = \sum_n K_{R_n}$, the total sign gives $d_0 \mathscr{S} = \bar\partial \mathscr{S} - K_1$, hence

$$D_K = d_0 + (K_1)_{(0)} - (\bar\partial \mathscr{S})_{(0)} + \tfrac{1}{2}[S_0, (K_1)_{(0)}]. \tag{16.60}$$

Terms of form degree at least three vanish. The first-form action on functions is R; the second-form term can contain negative ghost fields, so functions need not be a subcomplex for D_K.

For an explicit calculation, take fibre coordinates v_1, v_2 and exterior form variables η_1, η_2, ordered before the odd fibre variables. Let $V = v_1^2 \partial_{v_2}$, $W = v_2^2 \partial_{v_1}$ and $S_0 = \eta_1 (: v_1 c_1 \beta_2 :)_{(0)} + \eta_2 (: v_2 c_2 \beta_1 :)_{(0)}$. Suppress $\bar\partial$, write $A_1 = [S_0, \delta]$, $A_2 = [S_0, A_1]/2$, and compute by single contractions:

$$\begin{aligned} A_1 v_1 &= \eta_2 v_2^2, \qquad A_1 v_2 = \eta_1 v_1^2, \\ A_2 v_1 &= -\eta_1\eta_2 \left(\tfrac{3}{2} v_1 v_2 c_1 + \tfrac{1}{2} v_1^2 c_2\right), \\ A_2 v_2 &= \eta_1\eta_2 \left(\tfrac{1}{2} v_2^2 c_1 + \tfrac{3}{2} v_1 v_2 c_2\right). \end{aligned}$$

For example $S_0 v_1 = \eta_2 v_2 c_2$, $S_0 A_1 v_1 = -2\eta_1\eta_2 v_1 v_2 c_1$, and $A_1 S_0 v_1 = \eta_1\eta_2 (v_1 v_2 c_1 + v_1^2 c_2)$. Their half-difference gives the displayed second correction. These are polynomial tensors, not the curvature of a specified K3 metric.

16.12.4 Charge restoration retains torsion

To offset the charge decrease of δ, introduce an even parameter u of charge $+1$ and cochain degree zero. Then $u\delta$ preserves combined charge. Let K be a finite normal PBW subcomplex stable under δ, h, with only its scalar vacuum in cohomology. Write $B^k = \operatorname{im}(\delta : K^{k-1} \to K^k)$. Since u acts injectively on $K[u]$, its cycles are $(\ker \delta)[u]$ and its boundaries are $uB[u]$. Splitting the vacuum from the boundaries gives

$$H^k(K[u], u\delta) \cong B^k \otimes \mathbb{C}[u]/(u) \oplus \begin{cases} \mathbb{C}[u]|0\rangle, & k = 0, \\ 0, & k \neq 0. \end{cases} \tag{16.61}$$

The complex is termwise free, so its derived fibre at $u = 0$ is $(K, 0)$. The assertion extends coefficientwise to the fixed-weight formal normal system. For two directions, no derivative jets and total normal number at most three, its dimension is $1 + 8 + 32 + 88 = 129$, from $(1 + t)^4/(1 - t)^4$. There are 64 acyclic pairs and one vacuum, hence 64 torsion generators after rescaling.

For the curved construction, charge restoration starts over $\mathbb{C}[u, u^{-1}]$:

$$e^{S_0/u}(\bar\partial + u\delta)e^{-S_0/u} = \bar\partial + u\delta + (K_1)_{(0)} + u^{-1}K_2 + \cdots .$$

On a surface the last term is $K_2 = -(\bar\partial \mathscr{S})_{(0)} + [S_0, (K_1)_{(0)}]/2$. A form-degree-$r$ correction has charge $r - 1$ and coefficient u^{1-r}. Regular extension on the unchanged carrier requires the negative-power coefficients to vanish, or to gain the corresponding divisibility in a parameter-dependent family. The polynomial example shows they can be nonzero. Their value for a specified K3 curvature requires a separate calculation.

16.13 The compact comparison

The preceding constructions provide concrete state objects and operations: incidence creators, Frobenius joining, elliptic image lattices, parameter actions, twisted fields and normal resolutions. The compact critical Hall theory of Chapter 15 has a different coefficient system. Its common elliptic torsor, relative orbit pair, zero object over BE, orientation and square-root flag data remain part of that system when a state is extracted.

One useful recipient for its reduced state is an extension module. Let H, R be unital associative $\mathbb{C}$-algebras, let $I_0 : H \to R$ be a unital algebra map, and let $D : H \to R$ be a linear map. If $D(ab) = I_0(a)D(b) + D(a)I_0(b)$, then

$$\rho(a) = \begin{pmatrix} I_0(a) & D(a) \\ 0 & I_0(a) \end{pmatrix} \tag{16.62}$$

acts on the right R-module $R \oplus R$. Matrix multiplication is the derivation identity, and $D(1) = 0$ gives the unit. The extension splits as an (H, R)-bimodule exactly when $D(a) = bI_0(a) - I_0(a)b$ for some $b \in R$: a right-linear section is $q \mapsto (bq, q)$. For a finite product,

$$D(a_1 \cdots a_n) = \sum_j I_0(a_1) \cdots I_0(a_{j-1}) D(a_j) I_0(a_{j+1}) \cdots I_0(a_n).$$

The order is retained even when R is noncommutative. Degree-zero chain maps give the dg version; a strict splitting also requires $db = 0$. An intertwiner with this extension and its off-diagonal coefficient is additional to the elliptic vacuum coefficient whose constant term is one.

Problem 16.23. Construct a charge dictionary and an action of the reduced compact critical Hall coefficients on independently defined geometric state complexes. Specify the pullbacks, oriented pushforwards and base-change maps on extension flags, over the common elliptic torsor. Retain the relative unit, marked vacuum and the extension-module coefficient (16.62). For a chiral action, construct the supported collision maps, including failed cuts and their higher compatibilities. For a target diagonal transfer, retain its curved normal fields and derived specialization. For an age-shifted chiral McKay comparison, use the explicit degrees in (16.53).

Fix supported bar objects with coderivations b_A, b_B and comparison components $F_{<n}$ satisfying all identities below arity n. The obstruction in arity n is the closed residual

$$R_n = (b_B F_{<n} - F_{<n} b_A)_n, \qquad \partial_{\mathrm{Hom}} F_n = -R_n.$$

A contraction onto cohomology, with projection P, $\partial h + h\partial = 1 - P$ gives the particular solution $F_n = -hR_n$ precisely when $PR_n = 0$. The normal-coordinate functor solves this equation for its own Hom complexes. The normal quartet supplies another explicit contraction. Neither computes the compact supported residuals. The proposed rank-28 differential on the unchanged root presentation is excluded by Theorem 16.8; an enlarged derived presentation remains a distinct possibility. Likewise the normal-coordinate determinant does not identify a global geometric line with an automorphic line. Such a comparison needs its base map, character and boundary normalization, to which we turn next.

Chapter 17

Indices, denominators and elliptic coefficient lines

Taking an Euler characteristic turns a complex of geometric coefficients into a number. For Hilbert schemes of a K3 surface these numbers assemble into the reciprocal of an Igusa product, after a boundary factor is removed. An Enriques surface gives the corresponding half-index. The root algebra supplies another source of the same product, with a Lie bracket and a Chevalley differential. Comparing the two constructions requires the coefficient complexes and their operations.

The boundary factor is already significant at length zero. The elliptic coefficient line is significant at a collision: its section can have a double zero, and the resulting quotient retains a normal nilpotent. For three interacting colours, the simultaneous incidence has a nonzero higher multiplication. We will compute this complex, its deformation over the spectral elliptic curve, and the cover on which its coefficient line has a square root. These constructions retain data that their Euler characteristics forget.

17.1 The elliptic index and its coefficient complexes

Let Y be a smooth projective complex variety of dimension d. For a vector bundle V, write

$$\Lambda_z V = \sum_{j\ge 0} z^j \Lambda^j V, \qquad \operatorname{Sym}_z V = \sum_{j\ge 0} z^j \operatorname{Sym}^j V.$$

The elliptic class is the formal virtual bundle

$$\begin{aligned}\mathcal{E}_{q,y}(Y) = y^{-d/2} \bigotimes_{r\ge 1} \big(&\Lambda_{-yq^{r-1}} T_Y^* \otimes \Lambda_{-y^{-1}q^r} T_Y \\ &\otimes \operatorname{Sym}_{q^r} T_Y^* \otimes \operatorname{Sym}_{q^r} T_Y\big) \in K^0(Y)[y^{\pm 1/2}][[q]].\end{aligned} \tag{17.1}$$

At a fixed power of q, only finitely many positive-energy factors contribute. The remaining exterior factor has finite rank, so the Laurent support in y is finite as well. Define

$$\mathrm{Ell}(Y; q, y) = \chi(Y, \mathcal{E}_{q,y}(Y)), \qquad \chi(Y, V) = \sum_j (-1)^j \dim H^j(Y, V),$$

coefficient by coefficient.

There is a complex before this Euler characteristic. Give each exterior power in (17.1) its exterior degree modulo two, keep the symmetric powers even, and retain the displayed q, y weights. This defines a formal superbundle $\mathscr{E}_{q,y}(Y)$ whose super-K-class is $\mathcal{E}_{q,y}(Y)$. Its coefficient $\mathscr{E}_{m,\ell}(Y)$ is a finite sum of bundles with specified internal parity. The complex

$$\mathscr{V}_{m,\ell}(Y) = R\Gamma(Y, \mathscr{E}_{m,\ell}(Y)) \tag{17.2}$$

is perfect over $\mathbb{C}$. Its Euler superdimension uses $(-1)^{j+\epsilon}$, where j is cohomological degree and ϵ is internal parity.

Fix a K3 surface S and an Enriques surface T. Write $\widetilde{T} \to T$ for the canonical étale double cover. The K3 elliptic genus is

$$\mathrm{Ell}(S; q, y) = 2\phi_{0,1}(q, y), \qquad \phi_{0,1}(\tau, z) = 4 \sum_{a=2}^{4} \frac{\vartheta_a(\tau, z)^2}{\vartheta_a(\tau, 0)^2}, \tag{17.3}$$

with $q = e^{2\pi i \tau}$, $y = e^{2\pi i z}$. The theta normalization is that of [53]. Write

$$\phi_{0,1}(q, y) = \sum_{m \geq 0, \ell \in \mathbb{Z}} f(4m - \ell^2) q^m y^\ell.$$

The elliptic transformation law identifies the coefficients at (m, ℓ) and $(m + a\ell + a^2, \ell + 2a)$. Reducing ℓ to zero or one proves that $f(D) = 0$ for $D < -1$. The constant row is

$$[q^0]\phi_{0,1} = y^{-1} + 10 + y.$$

Indeed the K3 Hodge numbers give $y^{-1}(\chi(\mathcal{O}_S) - y\chi(\Omega^1_S) + y^2\chi(\Omega^2_S)) = 2y^{-1} + 20 + 2y$. In particular $f(-1) = 1$, $f(0) = 10$, whereas $\phi_{0,1}(\tau, 0) = 12$.

Proposition 17.1. *The Enriques elliptic genus is $\phi_{0,1}$. At $y = 1$, the elliptic genus of any smooth projective Y is its topological Euler characteristic, independent of q.*

Proof. The tangent bundle pulls back along an étale map. The Hirzebruch–Riemann–Roch characteristic class defining the elliptic genus therefore pulls back, and its integral over the double cover is twice its integral over T. Apply (17.3) to $\widetilde{T}$.

At $y = 1$, use $\Lambda_{-z} V \ \mathrm{Sym}_z V = 1$. All positive-$q$ factors cancel, leaving $\Lambda_{-1} T^*_Y$. Its holomorphic Euler characteristic is $\sum_{p,j} (-1)^{p+j} h^{p,j}(Y) = e(Y)$. □

17.1.1 Length and the boundary factor

For a surface Y, let $Y^{[N]}$ be its Hilbert scheme of length-N subschemes, and put

$$F_Y(p, q, y) = \sum_{N \geq 0} p^N \, \mathrm{Ell}(Y^{[N]}; q, y).$$

Thus p records length, q elliptic energy, and y the Laurent weight. Each coefficient has the complex (17.2) on $Y^{[N]}$; the empty Hilbert scheme contributes 1.

For triples (a, b, ℓ), the notation $(a, b, \ell) > 0$ means $a, b \geq 0$, $a + b > 0$, or $a = b = 0$, $\ell < 0$. Choose square roots of the formal monomials and define

$$D_5 = (pqy)^{1/2} \prod_{(a,b,\ell) > 0} (1 - p^a q^b y^\ell)^{f(4ab - \ell^2)}, \qquad \Xi = D_5^2. \tag{17.4}$$

The length-zero factor, including its Weyl monomial, is

$$\begin{aligned} b(q, y) &= q^{1/2}(y^{1/2} - y^{-1/2}) \prod_{r \geq 1} (1 - q^r)^{10} (1 - yq^r)(1 - y^{-1}q^r), \\ \mathcal{B}(q, y) &= b(q, y)^2 \\ &= q(y + y^{-1} - 2) \prod_{r \geq 1} (1 - q^r)^{20} (1 - yq^r)^2 (1 - y^{-1}q^r)^2. \end{aligned} \tag{17.5}$$

Here the symbol $b(q, y)$ is a function; the integer b in a root triple is its energy coordinate.

Theorem 17.2. *In* $\mathbb{Z}[y, y^{-1}][[p, q]]$,

$$F_T = \prod_{a>0, b\geq 0, \ell\in\mathbb{Z}} (1 - p^a q^b y^\ell)^{-f(4ab-\ell^2)} = \frac{p^{1/2} b(q, y)}{D_5}, \tag{17.6}$$

$$F_S = \prod_{a>0, b\geq 0, \ell\in\mathbb{Z}} (1 - p^a q^b y^\ell)^{-2f(4ab-\ell^2)} = \frac{p\mathcal{B}(q, y)}{\Xi} = F_T^2. \tag{17.7}$$

The quotients are expanded after canceling their length-zero factors.

Proof. The smooth-surface Hilbert-scheme formula of Borisov–Libgober identifies the elliptic genus of the Hilbert–Chow resolution with the symmetric-product orbifold genus [12, Corollary 6.7]. It applies to both surfaces; triviality of the canonical bundle is not a hypothesis of that formula. Their seed coefficients are f and $2f$, respectively. At fixed $p^N q^M$, only $a \leq N$, $b \leq M$ contribute, and $4ab - \ell^2 \geq -1$ bounds ℓ. This proves the formal finiteness of the products. Extracting the factors with $a = 0$ from (17.4) gives (17.5). Squaring doubles the positive-length exponents. □

The first coefficients make the normalization visible:

$$F_T(p, q, 1) = 1 + 12p + 90p^2 + 520p^3 + O(p^4), \tag{17.8}$$

$$F_S(p, q, 1) = 1 + 24p + 324p^2 + 3200p^3 + O(p^4). \tag{17.9}$$

They are the products $\prod_{a\geq 1}(1 - p^a)^{-12}$ and $\prod_{a\geq 1}(1 - p^a)^{-24}$. For a more refined coefficient, put $A(y) = y + 10 + y^{-1}$. The length-one modes contribute $\frac{1}{2}(A(y)^2 + A(y^2))$ at length two, and the length-two modes contribute $A(y)$. Hence

$$[p^2 q^0] F_T = y^2 + 11y + 66 + 11y^{-1} + y^{-2}, \tag{17.10}$$

$$[p^2 q^0] F_S = 3y^2 + 42y + 234 + 42y^{-1} + 3y^{-2}. \tag{17.11}$$

Both $\mathcal{B}$ and Ξ vanish at $y = 1$. Their canceled quotient has the regular specialization (17.9); their separate reciprocals do not.

The complexes $\mathscr{V}_{m,\ell}(T^{[N]})$ provide geometric coefficients for F_T. Formula (17.7) compares their Euler supercharacters. It supplies neither a tensor-square equivalence with the K3 complexes nor extension pull–push maps on them. A perfect complex can have zero Euler characteristic and nonzero cohomology, so taking the index cannot recover such maps.

17.2 The positive-length root ideal

Use the lattice $L = \mathbb{Z}^3$ with coordinates (a, b, ℓ) and pairing

$$((a, b, \ell), (a', b', \ell')) = 2\ell\ell' - 4(ab' + ba').$$

Its real simple roots and Weyl vector are

$$\delta_1 = (0, 1, 1), \quad \delta_2 = (1, 0, 1), \quad \delta_3 = (0, 0, -1), \qquad \rho = (1/2, 1/2, 1/2). \tag{17.12}$$

Their Gram matrix is 2 on the diagonal and -2 off the diagonal, and $(\rho, \delta_i) = -1$. Fix

$$e^{-\delta_1} = qy, \qquad e^{-\delta_2} = py, \qquad e^{-\delta_3} = y^{-1}, \qquad e^{-\rho} = (pqy)^{1/2}.$$

These are root-torus coordinates; the analytic doubling in the Gritsenko–Nikulin convention will be specified below.

Let $\mathfrak{n}_+$ be the positive part of the Gritsenko–Nikulin Lie superalgebra, with the full labelled imaginary simple-root spaces and parities of its presentation [74, Sections 3 and 6]. The real generators are even. The presentation supplies both parity spaces and the bracket. The denominator theorem gives their difference:

$$\operatorname{sdim}(\mathfrak{n}_+)_{a,b,\ell} = f(4ab - \ell^2).$$

Here $\operatorname{sdim} V = \dim V_{\bar{0}} - \dim V_{\bar{1}}$. Actual root spaces are finite-dimensional in each parity.

Length a defines

$$\mathfrak{u} = \bigoplus_{a>0,b,\ell} (\mathfrak{n}_+)_{a,b,\ell}, \qquad \mathfrak{n}_0 = \bigoplus_{b,\ell} (\mathfrak{n}_+)_{0,b,\ell}.$$

For Chevalley chains, internal parity and chain degree are separate. Write

$$\chi_s C_*(\mathfrak{u}) = \sum_{j\ge 0} (-1)^j \operatorname{sch} C_j(\mathfrak{u}).$$

All characters are formed degree by degree in the positive root cone.

Proposition 17.3. *There is a semidirect decomposition* $\mathfrak{n}_+ = \mathfrak{u} \rtimes \mathfrak{n}_0$, *and*

$$\operatorname{sch} U(\mathfrak{u}) = F_T, \qquad \chi_s C_*(\mathfrak{u}) = F_T^{-1}, \qquad D_5 = p^{1/2} b(q,y)\, \chi_s C_*(\mathfrak{u}). \tag{17.13}$$

Proof. The bracket adds the nonnegative length. Its positive part is therefore an ideal, and its zero part a subalgebra. Their vector-space sum is direct.

In super PBW, an even root vector of monomial weight x contributes $(1-x)^{-1}$, and an odd root vector contributes $1-x$. In Chevalley chains the contributions are inverse. For an odd vector, symmetric powers have equal internal and chain parities, whose signs cancel; for an even vector, the chain sign gives its exterior factor $1-x$. The root supermultiplicities and (17.6) prove the first two identities. Extracting the length-zero part gives the third. □

The full enveloping character is $(pqy)^{1/2}/D_5$; the character of $U(\mathfrak{n}_0)$ is $(qy)^{1/2}/b(q,y)$. Their quotient is F_T. The ideal still carries the adjoint action of $\mathfrak{n}_0$. Removing a scalar factor from a character does not discard that action from the algebra.

17.2.1 *Completion by the total height of a monomial*

Every root charge has the expression

$$(a,b,\ell) = b\delta_1 + a\delta_2 + (a+b-\ell)\delta_3, \qquad h(a,b,\ell) = 2(a+b) - \ell. \tag{17.14}$$

On the positive cone this is a positive integral height. For an active product factor with $a+b>0$, the support bound implies $\ell \le a+b$, hence $h \ge a+b$. Indeed $\ell \ge a+b+1$ would give $\ell^2 - 4ab \ge (a-b)^2 + 2(a+b) + 1 > 1$. The only active factor with $a=b=0$ has $\ell=-1$. Bounded height therefore contains finitely many root degrees.

For a height-graded vector space $V = \oplus_{r\ge 0} V_r$, set

$$\widehat{V} = \prod_{r\ge 0} V_r, \qquad \widehat{V} \mathbin{\widehat{\otimes}} \widehat{W} = \prod_{r\ge 0} \bigoplus_{i+j=r} V_i \otimes W_j.$$

The neighborhoods of zero are the products over $r \ge H$. The height of a monomial in $U(\mathfrak{n}_+)$ is the sum of the heights of all its generators, including repetitions. The reduced bar and Chevalley complexes inherit this grading.

Theorem 17.4. *Antisymmetrization gives a quasi-isomorphism*

$$\prod_{r\ge 0} C_*(\mathfrak{n}_+)_r \longrightarrow \prod_{r\ge 0} B_*(U(\mathfrak{n}_+))_r. \tag{17.15}$$

At each total height, the terms are finite-dimensional and only finitely many chain degrees occur.

Proof. A monomial of height r contains at most r positive-height generators. Only finitely many root spaces can occur, and each is finite-dimensional in both parities. This proves both finiteness assertions, including the bar length bound.

Use the bar differential whose first multiplication term is positive and the Chevalley differential sending $x \wedge y$ to $[x, y]$. Antisymmetrization sums the bar words with the permutation sign and the Koszul sign of the internal parities. Adjacent multiplication terms cancel in pairs except for the brackets, so it is a chain map. Its PBW associated graded is the Koszul comparison for a free supercommutative algebra: exterior chains for even generators, symmetric chains for odd generators. The filtration is finite at each total height. Thus the comparison is a quasi-isomorphism there. Products of complexes of complex vector spaces are exact, so the product of these comparisons is again a quasi-isomorphism. □

Truncating the Lie algebra alone gives a different topology. For $\mathfrak{a} = \mathbb{C}x$, with x even and $h(x) = 1$,

$$\lim_{\longleftarrow H} U\Bigg(\mathfrak{a}/\bigoplus_{r>H}\mathfrak{a}_r\Bigg) = \mathbb{C}[x], \qquad \widehat{U(\mathfrak{a})} = \mathbb{C}[[x]]. \tag{17.16}$$

The Lie quotient is constant for $H \ge 1$. Total-height completion also permits infinite powers of this one fixed generator.

For the level-zero current envelope, the same distinction applies to products of negative modes. Total root height defines a pro-vertex algebra by its quotients; at a fixed height its states remain algebraic in the mode variables. The coefficient of a fixed height and conformal energy is finite-dimensional. Uniform locality for arbitrary infinite sums is a further requirement if one seeks a vertex algebra on the completed vector space. The ordinary comparison (17.15) does not construct that locality or the supported maps on collision configurations.

17.3 The automorphic section and its roots

Let

$$\mathfrak{H}_2 = \left\{ Z = \begin{pmatrix} \tau & z \\ z & \sigma \end{pmatrix} : \operatorname{Im} Z > 0 \right\}.$$

Use $q = e^{2\pi i\tau}$, $y = e^{2\pi iz}$, $p = e^{2\pi i\sigma}$. For even characteristics $a, b \in \{0,1\}^2$, meaning $a^{\mathsf{T}}b$ is even, fix the Gritsenko–Nikulin phase convention

$$\vartheta^{\mathrm{GN}}_{a,b}(Z) = \sum_{m\in\mathbb{Z}^2} \exp\!\big(\pi i\big[(m + a/2)^{\mathsf{T}} Z (m + a/2) + b^{\mathsf{T}} m\big]\big). \tag{17.17}$$

There are ten such characteristics. This formula fixes the phases as well as the characteristics.

Theorem 17.5. *The product D_5 of (17.4) converges on a cusp chamber and extends to a holomorphic Siegel modular form of weight 5, with a character ν_5 of order two. In the convention (17.17),*

$$\Delta_5 = \prod_{a^{\mathsf{T}}b\equiv 0\ (2)} \vartheta^{\mathrm{GN}}_{a,b} = 64 D_5, \qquad \Xi = D_5^2 = 2^{-12}\Delta_5^2. \tag{17.18}$$

The divisor of D_5 in $\mathfrak{H}_2$ is the $\mathrm{Sp}_4(\mathbb{Z})$-orbit of $z = 0$, with multiplicity one.

Proof. The product, weight, character and divisor are the Borcherds–Gritsenko–Nikulin theorem [74, Proposition 3.1 and Theorem 4.1]. The weight is $f(0)/2 = 5$; the negative-discriminant coefficient $f(-1) = 1$ supplies the diagonal divisor.

The phase convention is [74, equation (1.2)]. The four characteristics with $a = 0$ have leading coefficient 1. The two with $a = (1, 0)$ contribute $2q^{1/8}$ each, and those with $a = (0, 1)$ contribute $2p^{1/8}$ each. For $a = (1, 1)$, their lowest q, p terms are

$$2q^{1/8}p^{1/8}(y^{1/4} + y^{-1/4}), \qquad 2q^{1/8}p^{1/8}(y^{1/4} - y^{-1/4}).$$

Their product gives the leading Laurent factor $64q^{1/2}p^{1/2}(y^{1/2} - y^{-1/2})$. This is 64 times the corresponding factor of (17.4), fixing the scalar. Squaring removes the quadratic character. □

The formal root variables in (17.12) give D_5 directly. In the Lorentzian analytic coordinates of the same Gritsenko–Nikulin construction, the root denominator is $D_5(2Z)$. The map $Z \mapsto Z/2$ relates these coordinate conventions. It does not alter the root multiplicities.

The first Fourier–Jacobi coefficient is $b(q, y)$. For the odd theta function normalized by

$$\vartheta(\tau, z) = q^{1/8}(y^{1/2} - y^{-1/2}) \prod_{r \geq 1} (1 - q^r)(1 - yq^r)(1 - y^{-1}q^r),$$

it is $b = \eta(\tau)^9 \vartheta(\tau, z)$. Its weight is 5 and its index is $1/2$. The weight-zero, index-one form $\phi_{0,1}$ is the multiplicative input; this first coefficient is the additive-lift input. Their roles and weights differ.

17.3.1 *A line, a section and a trace*

On the moduli stack $\mathcal{A}_2 = [\mathfrak{H}_2/\mathrm{Sp}_4(\mathbb{Z})]$, let λ be the determinant of the Hodge bundle, with the convention that weight-k forms are sections of λ^k. The character ν_5 also denotes its flat line. Thus

$$D_5 \in H^0(\mathcal{A}_2, \lambda^5 \otimes \nu_5), \qquad \Xi \in H^0(\mathcal{A}_2, \lambda^{10}). \tag{17.19}$$

The square relation identifies $(\lambda^5 \otimes \nu_5)^{\otimes 2}$ with λ^{10}. The full line retains its Hodge factor; its character alone is only the flat factor.

Roots of functions and roots of lines have different descent data. If a nowhere-zero function f has local n-th roots s_i, their ratios s_i/s_j form a μ_n-valued one-cocycle. It defines the torsor of roots in $H^1(-, \mu_n)$. For a line with transition functions g_{ij}, choose local n-th roots h_{ij}. The products $h_{ij}h_{jk}h_{ki}$ form a μ_n-valued two-cocycle, the Kummer obstruction in $H^2(-, \mu_n)$ to an n-th root line. The local functions s_i are generally $\mathbb{G}_m$-valued, so they do not by themselves trivialize the first cocycle in the sheaf μ_n.

Near a smooth point of the diagonal divisor, choose a transverse coordinate u and trivialize the line. Then $D_5 = u\,v(u)$, with $v(0) \neq 0$, and $\Xi = u^2 v(u)^2$. A branch of $\Xi^{1/n}$ on the punctured disc has monodromy $e^{4\pi i/n}$, of order $n/\gcd(n, 2)$. In particular an eighth root of Ξ has local order four, while an eighth root of D_5 has local order eight. These local branching orders do not specify an orientation line on compact sheaf moduli.

Proposition 17.6. *Let V be a bounded finite-dimensional complex and $s : V \to V$ a chain involution. Its supertrace is an integer. A homotopy equivalence intertwining two such endomorphisms up to chain homotopy preserves their supertraces. An arbitrary intertwining map need not do so.*

Proof. The supertrace is the alternating trace on cohomology. The eigenvalues of an involution in characteristic zero are 1 and -1, proving integrality. A homotopy equivalence induces an isomorphism on cohomology, and the intertwining homotopy makes the induced endomorphisms conjugate. Their traces agree. The inclusion $\mathbb{C} \hookrightarrow \mathbb{C}^2$ intertwines the identity endomorphisms, whose traces are 1 and 2. □

Consequently a half-integral automorphic weight cannot be the ordinary supertrace of an involution on a perfect complex over $\mathbb{C}$. An infinite graded trace, a regularization or another coefficient-valued trace requires its own definition. A duality from one object to a different object also requires an identification before it becomes an endomorphism to which a trace applies.

17.4 Products at higher level

At full level the weight is half the constant coefficient of one Jacobi form. At a congruence level every cusp can contribute. This distinction explains the weights of the forms with a simple diagonal divisor.

For positive integers t, N, set

$$\Gamma_t(N) = \mathrm{Sp}_4(\mathbb{Q}) \cap \begin{pmatrix} \mathbb{Z} & t\mathbb{Z} & \mathbb{Z} & \mathbb{Z} \\ \mathbb{Z} & \mathbb{Z} & \mathbb{Z} & t^{-1}\mathbb{Z} \\ N\mathbb{Z} & Nt\mathbb{Z} & \mathbb{Z} & \mathbb{Z} \\ Nt\mathbb{Z} & Nt\mathbb{Z} & t\mathbb{Z} & \mathbb{Z} \end{pmatrix}.$$

Its extension $\Gamma_t(N)^+$ adjoins the involution $(\tau, z, \sigma) \mapsto (t\sigma, z, \tau/t)$. For $t = 1$, the first group is the Siegel congruence subgroup with lower-left block divisible by N.

Let ϕ be a nearly holomorphic Jacobi form of weight zero and index t on $\Gamma_0(N)$. Choose a representative f/e of each cusp, with $e \mid N$ and $(f, e) = 1$, and a matrix taking ∞ to that cusp. Write its transformed expansion as

$$\phi_{f/e}(\tau, z) = \sum_{n,\ell} c_{f/e}(n, \ell) q^n y^\ell, \qquad h_e = \frac{N}{\gcd(e^2, N)}, \qquad N_e = \frac{N}{e}.$$

The exponents n belong to $h_e^{-1}\mathbb{Z}$. Nearly holomorphic means that each such expansion has only finitely many negative powers of q.

Theorem 17.7 (Cléry–Gritsenko). *Suppose the principal coefficients satisfy $(h_e/N_e)c_{f/e}(n, \ell) \in \mathbb{Z}$ whenever $4tn - \ell^2 \le 0$, at every cusp. Then*

$$q^A y^B p^C \prod_{f/e} \prod_{(n,m,\ell)>0} \left(1 - (q^n y^\ell p^{tm})^{N_e}\right)^{(h_e/N_e)c_{f/e}(nm,\ell)} \tag{17.20}$$

is a meromorphic modular form for $\Gamma_t(N)^+$, with a character or multiplier and weight

$$k = \frac{1}{2} \sum_{f/e} \frac{h_e}{N_e} c_{f/e}(0, 0). \tag{17.21}$$

Here $n, m, \ell \in \mathbb{Z}$, positivity means $m > 0$, or $m = 0, n > 0$, or $m = n = 0, \ell < 0$, and

$$A = \frac{1}{24} \sum_{f/e,\ell} h_e c_{f/e}(0, \ell), \quad B = \frac{1}{2} \sum_{f/e,\ell>0} h_e \ell c_{f/e}(0, \ell), \quad C = \frac{1}{4} \sum_{f/e,\ell} h_e \ell^2 c_{f/e}(0, \ell).$$

The product initially converges in a Weyl chamber. Its zeros and poles lie on the rational quadratic divisors prescribed by the negative-discriminant data. Nonnegative such data give a holomorphic form.

This is [28, Theorem 3.1]; the cusp widths in the integrality hypothesis are part of the statement. At level two the seed $4\vartheta_2(\tau,z)^2/\vartheta_2(\tau,0)^2$ has constant Laurent row $y^{-1}+2+y$ at infinity and constant row 4 at zero. The two ratios h_e/N_e equal one, so its weight is $(2+4)/2=3$. The constant term at infinity alone would give a different weight.

A form with a simple diagonal divisor vanishes precisely on the translates of $z=0$, with multiplicity one. The classification in [28, Theorem 1.4] gives, up to nonzero scalar, the following eight possibilities. The final column records the order of the character or multiplier on $\Gamma_t(N)$.

t	N	form	weight	order
1	1	Δ_5	5	2
2	1	Δ_2	2	4
3	1	Δ_1	1	6
4	1	$\Delta_{1/2}$	1/2	8
1	2	∇_3	3	2
1	3	∇_2	2	2
1	4	$\nabla_{3/2}$	3/2	4
2	2	Q_1	1	4

The proof first bounds the possibilities by an index calculation, then constructs these forms by additive and multiplicative lifts. Its remaining numerical candidate $(t,N,k)=(2,4,1/2)$ is excluded by its first Fourier–Jacobi coefficient. Thus a numerical weight condition alone does not produce an additional form.

17.4.1 *Divisor-indexed inputs and geometric quotients*

The Mathieu construction has a different input. For an element g of M_{24}, let ϕ_g denote its twisted elliptic genus, of weight zero and index one. Write the cycle shape of its permutation on twenty-four points as $\prod_{r\ge1} r^{b_r}$, so $\sum rb_r=24$. Its level N_g is the congruence level of the twisted genus; it need not equal the order of the permutation. The associated product Φ_g uses the whole family

$$(\phi_{g^d})_{d\mid N_g}.$$

The divisor-indexed Jacobi-to-Weil correspondence and Borcherds lift give [137, Theorem 1.3]

$$\operatorname{wt}(\Phi_g)=\frac{1}{2}\sum_r b_r-2, \qquad L_g=U(N_g)\oplus U\oplus A_1. \tag{17.22}$$

Here U is the hyperbolic plane, $U(N_g)$ has its pairing multiplied by N_g, and A_1 has a generator of square two. The orthogonal domain of this lattice of signature $(3,2)$ is the genus-two Siegel domain. The products are single-valued meromorphic forms there.

For example, the cycle shapes $1^{24}, 1^8 2^8, 1^6 3^6, 1^4 2^2 4^4, 1^2 2^2 3^2 6^2$ give the classes $1A, 2A, 3A, 4B, 6A$ and weights $10, 6, 4, 3, 2$, respectively. In the full classification the holomorphic cases are exactly $1A, 2A, 2B, 3A, 4B$. A meromorphic square root exists exactly for

$$1A, 2A, 2B, 3A, 3B, 4A, 4B, 4C, 5A, 6A, 6B, 10A, 11A.$$

The additive lift from the first Fourier–Jacobi coefficient equals Φ_g for

$$1A, 2A, 2B, 3A, 4A, 4B, 5A, 6A, 7AB, 8A.$$

For $3B$ and $4C$, an additional weight-zero Borcherds factor occurs [137, Theorem 1.4]. These assertions concern the stated automorphic forms. They do not identify every twisted genus with a geometric automorphism of a chosen K3 surface.

There is a geometric quotient construction when a K3 surface S carries a symplectic automorphism g of order N. Choose an elliptic curve E and a point $a \in E$ of exact order N. Then

$$X = (S \times E)/\langle (g, t_a) \rangle \tag{17.23}$$

is smooth: every nonidentity power translates E by a nonzero point and has no fixed point. The product volume form descends, so $K_X \cong \mathcal{O}_X$. The elliptic one-form also descends, giving $h^{1,0}(X) = 1$. In particular the construction uses a torsion translation, whose order is unrestricted by the possible orders of linear automorphisms of E. Symplectic K3 automorphisms of order five supply valid examples. These quotients and their reduced curve-counting conjectures are studied in [20]. The general primitive counting formula there is a conjecture; proved base cases retain that narrower scope.

For the untwisted product $S \times E$, the primitive Igusa formula is a theorem. To specify its invariants, take a primitive class β_h on S with $\beta_h^2 = 2h - 2$, choose $\beta_h^\vee$ pairing to one with it, and let $\mathrm{pt} \in H^2(E)$. Let $N_{g,h,d}$ be the reduced, disconnected Gromov–Witten invariant in class (β_h, d), with no contracted connected components and with the one-point insertion $\pi_S^* \beta_h^\vee \cup \pi_E^* \mathrm{pt}$. In the monic normalization Ξ of (17.4), Oberdieck–Pixton prove

$$\sum_{g,h,d \geq 0} N_{g,h,d} u^{2g-2} q^{h-1} p^{d-1} = -\frac{1}{\Xi(p, q, e^{iu})}. \tag{17.24}$$

The right side is expanded as a Laurent series at $u = 0$ [122, Theorem 1 and Section 4]. Their proof shows that Jacobi-form constraints and two holomorphic-anomaly relations determine the series from its coefficient $N_{0,0,0} = 1$. The stable-pair variable is $-e^{iu}$; this sign matters when comparing the two counting conventions. Equation (17.24) is a reduced numerical identity. A Hall product on the coefficient complexes still requires its correspondences and compatibility with the reduction.

17.5 A second root system: the fake monster

The fake-monster denominator illustrates the information carried by a root cone and its imaginary simple roots. Let Λ be the positive-definite Leech lattice and write

$$I I_{25,1} = \Lambda \oplus U, \qquad r = (\lambda, m, n), \qquad r^2 = \lambda^2 - 2mn, \qquad \rho = (0, 0, 1).$$

Its positive roots consist of the vectors of square at most two with $(r, \rho) < 0$, together with $n\rho$, $n > 0$. Define $p_{24}(j)$ and $\tau(j)$ by

$$\eta(\tau)^{-24} = q^{-1} \sum_{j \geq 0} p_{24}(j) q^j, \qquad \eta(\tau)^{24} = \sum_{j \geq 1} \tau(j) q^j.$$

Set $p_{24}(j) = 0$ for $j < 0$.

The physical-state construction from the lattice vertex algebra, followed by the quotient by the radical of its invariant form, gives a generalized Kac–Moody algebra. For every nonzero root r, its multiplicity is $p_{24}(1 - r^2/2)$. The real simple roots have $r^2 = 2$, $(r, \rho) = -1$, and each $n\rho$ is an imaginary simple root with multiplicity twenty-four [9, Theorems 1 and 3]. In the completed positive cone its denominator is

$$\mathbf{e}^\rho \prod_{r > 0} (1 - \mathbf{e}^r)^{p_{24}(1 - r^2/2)} = \sum_{w \in W} \det(w) \sum_{n > 0} \tau(n) \mathbf{e}^{w(n\rho)}. \tag{17.25}$$

The group W is generated by the real-root reflections. The imaginary simple roots are mutually orthogonal; their correction term is the eta product on the right. Thus its coefficients are those of η^{24}, while root multiplicities come from η^{-24}.

The oscillator character also differs from the Leech lattice-vertex character:

$$\eta^{-24} = q^{-1} + 24 + 324q + 3200q^2 + \cdots, \qquad \frac{\Theta_\Lambda}{\eta^{24}} = j - 720 = q^{-1} + 24 + 196884q + \cdots.$$

Here $\Theta_\Lambda = 1 + 196560q^2 + \cdots$. The coefficient $196884 = 196560 + 324$ displays the lattice contribution explicitly. Neither character by itself specifies the Lie bracket used in (17.25).

A comparison with the weight-twelve orthogonal Borcherds form Φ_{12} must also preserve automorphic lines. An ordinary restriction retains weight twelve, so a nonzero identity with D_5^k in the same Hodge-line convention would require $12 = 5k$. There is no integral k satisfying this. A quasi-pullback is different: one first divides by the linear equations of the root divisors containing the subdomain. If there are M such factors, and the resulting restriction is defined and nonzero, its weight is $12 + M$. Its divisor, character and normalization must then be computed. The weight obstruction to a raw restriction does not decide this latter construction.

17.6 A convergent determinant model

The denominator admits an operator realization after a summability estimate. This construction also makes the limits of a scalar product identity precise. For every active positive root $\alpha = (a, b, \ell)$, put $c_\alpha = f(4ab - \ell^2)$. Define a minimal super vector space V_α of dimension $(c_\alpha, 0)$ if $c_\alpha \geq 0$, and $(0, -c_\alpha)$ otherwise. This realizes the supermultiplicity with total dimension $|c_\alpha|$. It need not be the actual root space of $\mathfrak{n}_+$, which may have both parities.

Lemma 17.8. *For $n \geq 1$, every coefficient of $q^n y^\ell$ in $\phi_{0,1}$ has absolute value at most $12\exp(4\pi\sqrt{2n/3})$. Consequently*

$$|c_\alpha| \leq 12 e^{\kappa h(\alpha)}, \qquad \kappa = 2\pi\sqrt{2/3}. \tag{17.26}$$

Proof. Put $r = |q|^{1/2}$. Apply the Jacobi triple products to the three theta ratios in (17.3). The sum of absolute Laurent coefficients of their expansions is bounded, at $0 < r < 1$, by

$$12 \prod_{j \geq 1} \frac{(1 + r^j)^4}{(1 - r^j)^4} \leq 12 \prod_{j \geq 1} (1 - r^j)^{-8}.$$

For ϑ_2 the factors use the even integers j and its initial Laurent polynomial has absolute coefficient sum one; the other two ratios use the odd integers. Enlarging each to all j gives the bound. For $r = e^{-t}$,

$$\log \prod_{j \geq 1} (1 - e^{-tj})^{-8} = 8 \sum_{k \geq 1} \frac{1}{k(e^{tk} - 1)} \leq \frac{4\pi^2}{3t}.$$

Cauchy's coefficient bound at $q^n = r^{2n}$ therefore gives $12\exp(2nt + 4\pi^2/(3t))$. Minimize at $t = \pi\sqrt{2/(3n)}$. For $n = ab$, the inequality $2\sqrt{ab} \leq h(\alpha)$ proves (17.26). The constant row and the root $(0, 0, -1)$ satisfy it directly. □

Give each V_α an orthonormal homogeneous basis and form the Hilbert direct sum $\mathcal{H}$. Let T_Z act on its α-summand by $x_\alpha = p^a q^b y^\ell$. For a real $c > \kappa$ consider the open chamber in which $|x_\alpha| < e^{-ch(\alpha)}$ for every positive root. It is enough to impose these strict inequalities on the three real simple roots: every positive charge is their nonnegative integral combination. It is nonempty: take $p = q = e^{-4\pi t}$, $y = e^{2\pi t}$ with $2\pi t > c$. The corresponding imaginary Siegel matrix is $t\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right) > 0$.

Proposition 17.9. *On this chamber T_Z is trace class, locally uniformly in Z, and*

$$\det_s(1 - T_Z) = \frac{\det(1 - T_Z|_{\mathcal{H}_0})}{\det(1 - T_Z|_{\mathcal{H}_1})} = \prod_{\alpha>0}(1 - x_\alpha)^{c_\alpha} = (pqy)^{-1/2} D_5. \tag{17.27}$$

Proof. At fixed height h, there are at most $(h+1)^2$ root triples: choose $a, b \le h$, after which $\ell = 2(a+b) - h$ is fixed. Thus

$$\|T_Z\|_1 = \sum_\alpha |c_\alpha|\,|x_\alpha| \le 12 \sum_{h\ge 1} (h+1)^2 e^{-(c-\kappa)h} < \infty.$$

On compact subsets the exponent has a positive uniform margin. The trace-class determinant is therefore holomorphic and equals the absolutely convergent product of eigenvalue factors. All these factors are nonzero in the chamber. Taking their even-to-odd ratio proves (17.27). □

There is a relative Pfaffian version. In each parity, set $D = 1 - T_Z$ and use the skew operator on $\mathcal{H} \oplus \mathcal{H}^*$

$$A_Z = \begin{pmatrix} 0 & D \\ -D^{\mathsf{T}} & 0 \end{pmatrix}.$$

Choose the relative orientation for which the Pfaffian ratio to A_0, where $D = 1$, is $\det D$. The finite-dimensional identity follows by pairing each basis vector with its dual; the common sign cancels in the ratio. Trace-norm convergence of finite-rank truncations gives the displayed determinant in the limit. Taking the ratio between the two parity sectors recovers (17.27).

This is a bounded identity-plus-trace-class family. Its skew doubling is not a first-order differential operator on a manifold. To realize D_5 by such an operator requires a geometric family, its symbol, boundary conditions, determinant line and normalization. Similarly, the modular continuation of the scalar in (17.27) does not supply intertwiners between the Hilbert spaces in different cusp charts. Using the actual root superspaces would require a bound on their total dimensions, in addition to the bound on their superdimensions proved above.

17.7 Elliptic coefficient lines on collision flags

We now retain the sheaf from which an elliptic coefficient map starts. Let E be a smooth complex elliptic curve, $\Theta = \mathcal{O}_E(0)$, and let ϑ be its divisor section. This E parametrizes spectral variables. It is separate from the elliptic factor of a threefold and from a curve on which fields are inserted. On E^2, put $u = z_2 - z_1$ and $\Delta = \{z_1 = z_2\}$. For two distinct colours joined by two arrows, the elliptic arrow factor at zero equivariant translations is $\vartheta(u)^2$. In the convention that a factor is a section of the dual kernel line, it gives

$$\mathcal{L} = u^*\Theta^{-2} = \mathcal{O}(-2\Delta) \xrightarrow{\vartheta(u)^2} \mathcal{O}_{E^2}. \tag{17.28}$$

There is no same-colour factor for this dimension vector. These are the arrow lines in the elliptic shuffle construction of [139, Section 2.1].

Proposition 17.10. *The line $\mathcal{L}$ is nontrivial. The coefficient map has image $\mathcal{O}(-2\Delta)$ and cokernel $\mathcal{O}_{2\Delta}$. The exact sequence*

$$0 \longrightarrow \mathcal{O}_\Delta(-\Delta) \longrightarrow \mathcal{O}_{2\Delta} \longrightarrow \mathcal{O}_\Delta \longrightarrow 0 \tag{17.29}$$

does not split as a module over the ambient local ring.

Proof. Restrict $\mathcal{L}$ to $\{z_1\} \times E$: its degree is -2. A formal logarithm at the origin gives a normal parameter, again denoted u, in which $\vartheta(u) = uv(u)$ with v a unit. Change the source frame by this unit. The map becomes $u^2 : \mathbb{C}[[u]] \to \mathbb{C}[[u]]$. Its quotient is $\mathbb{C}[[u]]/(u^2)$, and the submodule $(u)/(u^2)$ gives the first term of (17.29). A module splitting would lift 1 to an element with nonzero constant coefficient killed by u, which is impossible. □

The doubled zero remains even when one untwists the tensor product by changing its coefficient objects. To see this at every arity, fix a finite simple graph G on a set of labels. An edge records an interacting pair of multiplicity two. For disjoint sets A, B, let

$$D_{A,B} = \sum_{\substack{i\in A,\, j\in B \\ \{i,j\}\in E(G)}} \Delta_{ij}, \qquad \mathcal{L}_{A,B} = \mathcal{O}(-2D_{A,B}), \qquad \mathcal{B}_A = \mathcal{O}\left(-2 \sum_{\substack{i<j,\, i,j\in A \\ \{i,j\}\in E(G)}} \Delta_{ij}\right).$$

All lines are pulled back to the common product of copies of E. Each has its indicated divisor inclusion.

Proposition 17.11. *There are canonical, section-preserving isomorphisms*

$$\mathcal{L}_{A,B} \otimes \mathcal{L}_{A\cup B,C} \cong \mathcal{L}_{B,C} \otimes \mathcal{L}_{A,B\cup C}, \tag{17.30}$$

$$\mathcal{L}_{A,B} \cong \mathcal{B}_{A\cup B} \otimes \mathcal{B}_A^{-1} \otimes \mathcal{B}_B^{-1}. \tag{17.31}$$

They satisfy every finite regrouping identity. The functors $F \mapsto F \otimes \mathcal{B}_A^{-1}$ untwist the labelled tensor product $(F \boxtimes G) \otimes \mathcal{L}_{A,B}$. Replacing every divisor coefficient -2 by -1 gives compatible square roots of these line identities.

Proof. In (17.30), each side contains exactly once every edge between the three blocks. Every longer parenthesization counts an edge at the unique step joining the blocks containing its endpoints. It therefore gives the same divisor and the same section. Separating internal edges from cross-edges gives (17.31). Tensoring with $\mathcal{B}_{A\cup B}^{-1}$ cancels its first factor and proves the untwisting assertion. The empty set has $\mathcal{B}_\varnothing = \mathcal{O}$, so units are preserved. All the divisor equalities remain true after halving their coefficients. □

For two labels the transformed coefficient map is $\mathcal{O} \to \mathcal{O}(2\Delta)$, with the same doubled zero. Equivalently, multiplication between the image lines is an isomorphism

$$\mathcal{B}_A \otimes \mathcal{B}_B \otimes \mathcal{L}_{A,B} \xrightarrow{\sim} \mathcal{B}_{A\cup B};$$

the cokernel occurs in the final inclusion $\mathcal{B}_{A\cup B} \hookrightarrow \mathcal{O}$. These are labelled coefficient lines. Shuffle symmetrization, same-colour factors and critical coefficient maps require the corresponding extra data.

17.7.1 Three different derived constructions

For a commutative ring R and elements $f_1, \dots, f_r$, write

$$K_R(f_1, \dots, f_r) = (R \otimes \Lambda(\epsilon_1, \dots, \epsilon_r), d), \qquad |\epsilon_i| = -1, \qquad d\epsilon_i = f_i.$$

The differential has degree one and is a graded derivation. For $R = \mathbb{C}[[u]]$, the algebra $K_R(u^2)$ has cohomology $R/(u^2)$ in degree zero. By contrast, subtracting the two odd generators in $K_R(u, u)$ and resolving the remaining regular equation gives

$$K_R(u, u) \simeq \mathbb{C} \otimes \Lambda(\epsilon), \qquad d\epsilon = 0.$$

The first construction has a normal nilpotent; the second has an excess class in degree minus one.

For a triangle, use formal differences $x = z_2 - z_1$, $y = z_3 - z_2$. The cone of the total coefficient product is

$$K_{\mathbb{C}[[x,y]]}\big(x^2y^2(x+y)^2\big).$$

It has only degree-zero cohomology, supported on three doubled lines. The simultaneous derived intersection of the three pair cokernels is instead

$$K_{\mathbb{C}[[x,y]]}(x^2, y^2, (x+y)^2). \tag{17.32}$$

It is supported at the origin and has negative cohomology. It is an incidence object built from the pair maps; it is not an associativity defect of their tensor product. Even the ordinary product cone has a filtration, rather than a tensor-product identity: for nonzero f, g in a domain,

$$0 \longrightarrow R/(f) \xrightarrow{g} R/(fg) \longrightarrow R/(g) \longrightarrow 0.$$

The global version retains the source line twists.

17.7.2 A compact coefficient algebra and its trace

Suppose G is connected, with $n \geq 1$ vertices and m edges. Its spectral translation quotient is $B_n = E^n/E_{\mathrm{diag}} \cong E^{n-1}$. For an edge e, let $D_e \subset B_n$ be its equality divisor, let $L_e = \mathcal{O}(2D_e)$, and let s_e be its divisor section. Define

$$\mathscr{K}_G = \bigotimes_{e\in E(G)} [L_e^{-1} \xrightarrow{s_e} \mathcal{O}_{B_n}], \qquad C_G = R\Gamma(B_n, \mathscr{K}_G). \tag{17.33}$$

The left terms have degree -1. Thus $\mathscr{K}_G$ is a perfect commutative differential graded sheaf. Its cohomology is supported at the origin: away from the simultaneous zero locus, an invertible section contracts its Koszul factor, and connectedness makes that locus the diagonal. The cohomology sheaves have finite length and no higher cohomology. Completion at the origin therefore computes C_G, including its multiplication.

Adding a constraint gives the exact triangle

$$\mathscr{K}_G \otimes L_e^{-1} \xrightarrow{s_e} \mathscr{K}_G \longrightarrow \mathscr{K}_{G+e} \longrightarrow \mathscr{K}_G \otimes L_e^{-1}[1]. \tag{17.34}$$

Locally write an element as $a + \epsilon_e b$, placing the new generator first. The last map takes it to b. The identity $d(\epsilon_e b) = s_e b - \epsilon_e db$ gives its shift sign. Inclusion of the even summand is a map of differential graded algebras. Permuting edge factors gives the exterior signs and all subset-cube coherences, including disconnected intermediate graphs.

Choose an oriented spanning tree T. Its $k = n - 1$ edge differences $u_1, \dots, u_k$ are formal coordinates, and a chord c has difference $\ell_c = \sum_a \sigma_{ca} u_a$, with $\sigma_{ca} \in \{-1, 0, 1\}$. Put $\beta = m - n + 1$.

Theorem 17.12. *After choosing a formal logarithm and theta frames, C_G has the finite model*

$$\frac{\mathbb{C}[u_1, \dots, u_k]}{(u_1^2, \dots, u_k^2)} \otimes \Lambda(\epsilon_c : c \notin T), \qquad d\epsilon_c = 2\sum_{a<b} \sigma_{ca}\sigma_{cb} u_a u_b. \tag{17.35}$$

Its total dimension is 2^m, and its degrees lie in $[-\beta, 0]$. Different trees give quasi-isomorphic algebras through the full edge Koszul complex.

Proof. The local theta units can be removed by changing the odd generators. The tree equations form the regular sequence $u_1^2, \dots, u_k^2$. Its Koszul algebra maps multiplicatively to its quotient. Tensoring with the free chord factors preserves the quasi-isomorphism. In the quotient, ℓ_c^2 is the differential in (17.35). Squarefree tree monomials and exterior chord monomials give $2^k 2^\beta = 2^m$ basis elements. Each tree reduction starts from the same completed full complex, proving the comparison without requiring a strict identification of the finite presentations. □

Fix orders of both sets of generators. The functional

$$\tau(a) = [u_1 \cdots u_k \epsilon_1 \cdots \epsilon_\beta] a \tag{17.36}$$

gives a perfect pairing $C_G \otimes C_G \to \mathbb{C}[\beta]$ by multiplication. Indeed each basis monomial pairs with its complementary monomial, with the exterior permutation sign. The differential lowers exterior length, so $\tau(da) = 0$. Applying this to $d(ab)$ proves the cochain-pairing identity. Hence

$$\dim H^{-j}(C_G) = \dim H^{-(\beta - j)}(C_G).$$

We use $C[r]^i = C^{i+r}$. A graph symmetry can act on this normalized trace by a character, which must be retained before passing to invariant coefficients. For one vertex the algebra is $\mathbb{C}$. The empty configuration instead has translation quotient BE; it lies outside this nonempty spectral family.

17.8 Displacing the elliptic coefficient

The doubled factor splits when an equivariant displacement is retained. Give the two arrows the weight pairs $(2, 0)$ and $(0, 2)$. Their compatibility condition is $2t_1 = 2t_2$ in the elliptic group law. On the formal chart at the origin it forces $t_1 = t_2 = t$. We keep this specialization and put $h = 2t$. The forward coefficient and its line are

$$s_h(u) = \vartheta(u)\vartheta(u + h), \qquad \mathcal{S} = u^*\Theta \otimes (u + h)^*\Theta, \qquad \mathcal{S}^{-1} \xrightarrow{s_h} \mathcal{O} \tag{17.37}$$

on $E_u \times E_h$. These are the compatible weights of [139, Assumption 2.1 and Remark 2.2]. They do not produce two independent formal parameters.

The reverse ordering has factor $\vartheta(u)\vartheta(u - h)$ in an odd-theta frame. Write $D_a = \{u = a\}$. Its image and the forward image are respectively $\mathcal{I}_- = \mathcal{O}(-D_0 - D_h)$ and $\mathcal{I}_+ = \mathcal{O}(-D_0 - D_{-h})$.

Proposition 17.13. *On a fixed parameter fibre their sum is*

$$\mathcal{J}_h = \mathcal{I}_+ + \mathcal{I}_- = \begin{cases} \mathcal{O}(-D_0), & 2h \neq 0, \\ \mathcal{O}(-D_0 - D_h), & 2h = 0,\ h \neq 0, \\ \mathcal{O}(-2D_0), & h = 0. \end{cases}$$

In the completed local ring at $(u, h) = (0, 0)$, the relative image ideal is

$$J = (u(u + h), u(u - h)) = (u^2, uh). \tag{17.38}$$

The quotient has a nonzero class $[u]$ killed by h. Formation of the image does not commute with base change to $h = 0$.

Proof. The sum of divisor ideals on a smooth curve takes the minimum of their vanishing orders. Comparing the two divisors gives the three cases. Formal logarithms and theta frames give the two generators in (17.38); their sum and difference give u^2, uh, since two is invertible. The element u is absent from J, but hu belongs to it. Moreover $hu \neq 0$ in J/hJ: an equality $hu = hj$, $j \in J$, would imply $u = j$. Its image in the specialized ambient ring is zero. Thus $J/hJ \to \mathbb{C}[[u]]$ is not injective, whereas the image of the specialized maps is the ideal (u^2). □

When $2h \neq 0$, the quotient $\mathcal{J}_h/\mathcal{I}_+$ is supported at $-h$, and the reverse quotient at h. When $2h = 0$, the two image divisors agree. These statements identify divisor lines and their inclusions. A full operation also retains its scalar normalization and equivariant structure.

17.8.1 The triangle over the formal parameter disc

Take three ordered colours with arrows from lower to higher colour. After removing common translation, use $x = z_2 - z_1$, $y = z_3 - z_2$. Over $R = \mathbb{C}[[h]]$, the three formal equations are

$$x(x+h), \qquad y(y+h), \qquad (x+y)(x+y+h). \tag{17.39}$$

Their simultaneous incidence is the Koszul algebra.

Theorem 17.14. *It is multiplicatively quasi-isomorphic to*

$$B_h = R[x,y]/(x^2+hx, y^2+hy), \qquad A_h = B_h \otimes \Lambda(\eta), \quad |\eta| = -1, \quad d\eta = xy. \tag{17.40}$$

Both terms are free of rank four over R. Its cohomology is free, with bases

$$H^0 = R\{1, x, y\}, \qquad x^2 = -hx, \quad y^2 = -hy, \quad xy = 0, \tag{17.41}$$

$$H^{-1} = R\{k_1, k_2, k_3\}, \qquad k_1 = [(x+h)\eta], \quad k_2 = [(y+h)\eta], \quad k_3 = [(xy-h^2)\eta]. \tag{17.42}$$

The action on negative classes is

	k_1	k_2	k_3
x	0	$k_3 + hk_1$	$-hk_3 - h^2k_1$
y	$k_3 + hk_2$	0	$-hk_3 - h^2k_2$.

(17.43)

Products of two negative classes vanish.

Proof. The first two equations are monic in separate variables and form a regular sequence. Their Koszul resolution maps to B_h. In that quotient the third equation is $2xy$; divide its odd generator by two. This gives (17.40) and the multiplicative quasi-isomorphism. Completion gives the same finite R-algebra.

In the basis $1, x, y, xy$, multiplication by xy has only its xy-row nonzero, equal to $(1, -h, -h, h^2)$. The cokernel is free on $1, x, y$. Its kernel has basis $x+h$, $y+h$, $xy-h^2$: these elements are killed by xy, and subtracting their multiples leaves a scalar killed by the unit entry of the displayed row. This proves the cohomology assertions. Multiplication of the representatives proves (17.43); $\eta^2 = 0$ proves the last assertion. □

The contraction onto cohomology is useful for retaining the higher operations. Let $i : H \to A_h$ use the representatives above. Let $p : A_h \to H$ fix $1, x, y$, kill xy, η, and send $x\eta, y\eta, xy\eta$ to k_1, k_2, k_3. Define a degree-minus-one map by

$$\mathsf{H}(xy) = \eta, \qquad \mathsf{H} = 0 \text{ on the other basis elements.} \tag{17.44}$$

Then $pi = 1$, $d\mathsf{H} + \mathsf{H}d = 1 - ip$, and $\mathsf{H}^2 = \mathsf{H}i = p\mathsf{H} = 0$. For example, $\mathsf{H}d(x\eta) = -h\eta = x\eta - i(k_1)$, so the parameter term is required.

The functional $\tau : A_h \to R[1]$ extracting the coefficient of $xy\eta$ gives a perfect cyclic cochain pairing by multiplication. Indeed its matrix between the two copies of B_h is

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & -h \\ 0 & 1 & 0 & -h \\ 1 & -h & -h & h^2 \end{pmatrix}, \qquad \det = 1. \tag{17.45}$$

The trace kills boundaries, and the derivation rule gives the cochain identity. Graded commutativity gives cyclicity. On the displayed cohomology bases the matrix is

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & -h \\ 1 & 0 & -h \end{pmatrix}, \qquad \det = -1. \tag{17.46}$$

Thus the induced pairing is perfect over the parameter ring, including its central fibre.

17.8.2 A ternary class killed by the displacement

Put $V = Rx \oplus Ry \subset H^0$, and define $q : V \otimes V \to R$ by $q(x,y) = q(y,x) = 1, q(x,x) = q(y,y) = 0$. Let $\psi(x) = k_1, \psi(y) = k_2$. We specify the signs of the transferred operations. Augment both A_h and H by sending every displayed nonunit basis element to zero, and write $\overline{A}$ for the augmentation ideal. On $T^c(s\overline{A})$, use $|sa| = |a| - 1$ and

$$b_1(sa) = -s(da), \qquad b_2(sa,sb) = (-1)^{|a|}s(ab), \qquad b_3(sa,sb,sc) = (-1)^{2|a|+|b|}s\,m_3(a,b,c).$$

Each component extends as a degree-one coderivation, with the Koszul sign from its preceding suspended inputs.

Theorem 17.15. *The cohomology algebra extends to a strictly unital minimal A_∞ algebra by*

$$m_3(a,b,c) = q(b,c)\psi(a) - q(a,b)\psi(c) \quad (a,b,c \in V). \tag{17.47}$$

It is zero on every other basis pattern, and $m_n = 0$ for $n \geq 4$. A strictly unital quasi-isomorphism to A_h has suspended components

$$I_1(sa) = s\,i(a), \qquad I_2(sa,sb) = s\mathsf{H}(i(a)i(b)), \qquad I_{n\geq 3} = 0. \tag{17.48}$$

Proof. The binary transfer is $p(i(a)i(b))$. The arity-two comparison equation is exactly $d\mathsf{H}(i(a)i(b)) = i(a)i(b) - i\,m_2(a,b)$. For three degree-zero inputs, the two binary trees give

$$p\big(i(a)\mathsf{H}(i(b)i(c)) - \mathsf{H}(i(a)i(b))i(c)\big).$$

The map H extracts the xy coefficient, and $p(x\eta) = k_1$, $p(y\eta) = k_2$; this is (17.47). A negative input makes both terms zero, and unit inputs vanish by the contraction identities.

In a larger transfer tree, two internal homotopies either lie on one branch or on different branches. On one branch the second homotopy encounters η and vanishes. On different branches the two outputs multiply to zero by $\eta^2 = 0$. Thus all operations of arity at least four vanish. The final homotopy in the comparison map also kills an expression containing η, so $I_{n\geq 3} = 0$.

In arity three the bar comparison is the displayed two-tree identity and $d\mathsf{H} + \mathsf{H}d = 1 - ip$. In higher arity its terms contain either two I_2 outputs or a homotopy on a negative input, and vanish. Hence the induced coalgebra map satisfies $b_A I = I b_H$. Its highest tensor-length component on length n is $i^{\otimes n}$, which is injective. It follows from $b_A^2 = 0$ that $b_H^2 = 0$. Finally I_1 induces the identity on cohomology, proving the quasi-isomorphism and all arity identities. □

Nonzero formulas can become boundaries after a change of minimal coordinates. Here the invariant is more precise. In the normalized Hochschild complex of (H, m_2), an arity-two cochain of internal degree -1 is supported on $V \otimes V$, with values in H^{-1}. Define

$$G_2(x,x) = 0, \quad G_2(x,y) = G_2(y,x) = k_1, \quad G_2(y,y) = k_2. \tag{17.49}$$

On degree-zero inputs its coboundary is

$$(\partial G_2)(a,b,c) = aG_2(b,c) - G_2(ab,c) + G_2(a,bc) - G_2(a,b)c.$$

Proposition 17.16. *The arity-three, internal-degree-minus-one Hochschild class of m_3 is nonzero and has annihilator exactly (h). A primitive for its multiple is*

$$\partial G_2 = h\, m_3. \tag{17.50}$$

Proof. The arity-four identity makes m_3 a Hochschild cocycle. Order the triples as

$$xxx,\ xxy,\ xyx,\ xyy,\ yxx,\ yxy,\ yyx,\ yyy.$$

Substitution of (17.43) gives

$$\partial G_2 = h(0, k_1, 0, -k_2, -k_1, 0, k_2, 0) = h\, m_3.$$

Every other normalized pattern vanishes by degree. Modulo h, the coboundary of any normalized two-cochain G at (x, y, y) belongs to $\mathbb{C}k_3$: the products xy, y^2 vanish, and both xH^{-1} and $H^{-1}y$ lie in that line. But $m_3(x, y, y) = -k_2$. Hence the class is nonzero after reduction and therefore over R. Every scalar outside (h) is a unit; together with (17.50), this proves the exact annihilator. □

At $h = 0$, the same calculation gives the triple Massey set

$$\langle [x], [y], [y] \rangle = -k_2 + \mathbb{C}k_3. \tag{17.51}$$

The representative uses $xy = d\eta$, $y^2 = 0$, and its indeterminacy is $xH^{-1} + H^{-1}y$. It excludes zero, so the central algebra is nonformal. The cohomology modules themselves are free and commute with base change. The torsion in Proposition 17.16 belongs to the Hochschild class of multiplication.

17.8.3 The punctured family and a quadratic comparison

Over $K = \mathbb{C}((h))$, the tree algebra splits into the four points $x, y \in \{0, -h\}$. The idempotents are

$$\begin{aligned} e_{00} &= (x+h)(y+h)/h^2, & e_{10} &= -x(y+h)/h^2, \\ e_{01} &= -(x+h)\,y/h^2, & e_{11} &= xy/h^2. \end{aligned} \tag{17.52}$$

They are orthogonal and sum to one. On the first three factors $xy = 0$; on the last $xy = h^2$. Projection therefore gives a multiplicative quasi-isomorphism

$$A_h \otimes_R K \longrightarrow \prod_{(i,j)\neq(1,1)} K \otimes \Lambda(\eta_{ij}), \qquad d = 0. \tag{17.53}$$

Multiplication by $h^{-2}e_{11}\eta$ contracts its kernel, since its differential is e_{11}. The poles of the idempotents prevent this decomposition from extending across the origin.

An Euler connection is explicit on the punctured model:

$$\nabla_{\partial_h} = \partial_h + h^{-1}(x\partial_x + y\partial_y + 2\eta\partial_\eta). \tag{17.54}$$

It sends $x^2 + hx$ to $2h^{-1}(x^2 + hx)$, and similarly for y, so it descends to the quotient. It commutes with $d\eta = xy$. The elements x/h, y/h, η/h^2 are horizontal. The residue weights on $1, x, y, k_1, k_2, k_3$

are 0, 1, 1, 3, 3, 4, and analytic continuation around the puncture has trivial monodromy. This specifies an algebraic connection; a comparison with geometric Gauss–Manin transport is an additional construction.

Independent quadratic parameters give a different deformation. Recentering by $X = x + h/2$, $Y = y + h/2$ gives

$$x^2 + hx = X^2 - h^2/4, \qquad y^2 + hy = Y^2 - h^2/4, \qquad xy = XY - \frac{h}{2}(X + Y) + \frac{h^2}{4}.$$

The linear term in the last expression cannot be dropped. In fact the obstruction is independent of this choice of coordinates.

Proposition 17.17. *No homomorphism $\mathbb{C}[a, b, c] \to R$ makes the derived pullback of*

$$\frac{\mathbb{C}[a, b, c][X, Y]}{(X^2 - a, Y^2 - b)} \otimes \Lambda(\zeta), \qquad |\zeta| = -1, \qquad d\zeta = XY - c,$$

quasi-isomorphic to A_h as a differential graded R-algebra.

Proof. The displayed family is termwise free over the parameter ring, so ordinary tensor product computes its derived pullback. Over K, multiplication by $XY - c$ has matrix

$$F = \begin{pmatrix} -c & 0 & 0 & ab \\ 0 & -c & b & 0 \\ 0 & a & -c & 0 \\ 1 & 0 & 0 & -c \end{pmatrix} \tag{17.55}$$

in the basis $1, X, Y, XY$. To have three-dimensional kernel and cokernel it must have rank one. Its two-by-two minors include a, b, c, up to signs, so all three must vanish in K, hence in R. The pullback would be the constant central algebra. Its Massey set $-k_2 + Kk_3$ still excludes zero. But (17.53) makes the generic fibre of A_h formal. This contradiction proves the assertion for every base map. □

17.9 The whole spectral parameter curve

The formal family comes from a global perfect complex. On $E_x \times E_y \times E_h$, set $u_1 = x, u_2 = y, u_3 = x + y$, $\mathcal{S}_i = u_i^*\Theta \otimes (u_i + h)^*\Theta$, and $s_i = \vartheta(u_i)\vartheta(u_i + h)$. For the projection π to E_h, define

$$\mathscr{K} = \bigotimes_{i=1}^{3} [\mathcal{S}_i^{-1} \xrightarrow{s_i} \mathcal{O}], \qquad \mathscr{C} = R\pi_*\mathscr{K}. \tag{17.56}$$

Write $E[n] = \ker([n] : E \to E)$.

Theorem 17.18. *The complex $\mathscr{C}$ has two locally free terms of rank four on E_h, in degrees -1, 0. Its derived fibres have dimensions*

h	$\dim H^0$	$\dim H^{-1}$
0	3	3
$h \neq 0,\ 2h \neq 0$	3	3
$h \neq 0,\ 2h = 0$	4	4.

Near a nonzero two-torsion point, three summands have zero differential. The fourth is $[\mathcal{O} \xrightarrow{v} \mathcal{O}]$, where v is a local parameter times a unit. It contributes a skyscraper to $H^0(\mathscr{C})$; the additional negative class of its derived fibre is a Tor_1 *class.*

Proof. The first two sections cut out

$$Z = \{x = 0 \text{ or } -h\} \times_{E_h} \{y = 0 \text{ or } -h\},$$

with their Cartier multiplicities. Each factor is finite flat of degree two: it is the sum of two sections, with a monic quadratic equation near their meeting point and no vertical component. Their product is finite flat of degree four, and the first two equations form a regular sequence. The complex therefore reduces to the finite pushforward of $[\mathcal{S}_3^{-1}|_Z \xrightarrow{s_3} \mathcal{O}_Z]$. An invertible sheaf on this finite flat scheme has locally free pushforward of rank four. The resulting two-term complex computes every derived fibre.

For $h \neq 0$, the four points of Z_h are distinct. The third section vanishes at $(0,0)$, $(0,-h)$, $(-h,0)$. At $(-h,-h)$ its value is, in the relevant line,

$$\vartheta(-2h)\vartheta(-h).$$

This is nonzero unless $2h = 0$. At a nonzero two-torsion point the second factor is nonzero, and the first has a simple zero because multiplication by two is étale. Local frames give the stated fourth complex. Its kernel over the base is zero and its cokernel is $\mathcal{O}/(v)$. Derived specialization supplies the additional degree-minus-one class. At $h = 0$, formal logarithms reduce to (17.40), whose cohomology has ranks three and three. □

Every nonzero fibre is formal: each surviving reduced tree point carries one exterior generator with zero differential, and a nonsurviving point is contractible. At the three nonzero points of $E[2]$, four exterior factors survive. This is a local description and does not choose global frames for the persistent summands. On the original parameter cover $h = 2t$, the extra fibres occur at the twelve points $E[4] \setminus E[2]$; the nonformal doubled fibre occurs at the four points of $E[2]$.

17.9.1 Square roots on the halving cover

Rigidify the symmetric line Θ at zero. The coefficient line $\mathcal{S} = u^*\Theta \otimes (u+h)^*\Theta$ has degree two on each u-fibre, and hence four degree-one square roots on that fibre. These roots do not vary continuously over the original base.

Theorem 17.19. *There is no line $\mathcal{R}$ on $E_u \times E_h$ with $\mathcal{R}^2 \cong \mathcal{S}$, nor a continuous relative choice of degree-one root, even after a base line is tensored with $\mathcal{S}$. On the cover $h = 2t$, the cube theorem gives*

$$u^*\Theta \otimes (u+2t)^*\Theta \cong (u+t)^*\Theta^2 \otimes t^*\Theta^2. \tag{17.57}$$

Thus

$$\mathcal{R}_t = (u+t)^*\Theta \otimes t^*\Theta \tag{17.58}$$

is a root line. A nonzero deck shift $t \mapsto t + \xi$, $\xi \in E[2]$, changes its restriction to a u-fibre by the corresponding nontrivial two-torsion line. For $2t \neq 0$, the section $\vartheta(u)\vartheta(u+2t)$ is not the square of a regular section of $\mathcal{R}_t$.

Proof. Restrict $\mathcal{S}$ to $\{u_0\} \times E_h$. Its degree is one, whereas the square of a line has even degree. This rules out an absolute root. Under $\mathrm{Pic}^1(E) \cong E$, a root on a fixed fibre is a point m with $2m = -h$. A continuous choice would induce $2m_* = -\mathrm{id}$ on $\pi_1(E) = \mathbb{Z}^2$, which is impossible. Tensoring by a base line does not change these relative Picard classes.

For a symmetric rigidified line the cube theorem implies

$$(f+g)^*\Theta \otimes (f-g)^*\Theta \cong (f^*\Theta)^2 \otimes (g^*\Theta)^2.$$

The rigidification removes the constant fibre factor; see [132]. Apply this identity to $f = u + t$, $g = t$, proving (17.57) and its specified root. The ratio after a deck shift restricts to a translate of $T_\xi^*\Theta \otimes \Theta^{-1}$. The degree-one polarization identifies E with $\mathrm{Pic}^0(E)$, so this is nontrivial of order two for $\xi \neq 0$. Finally the two zeros $0, -2t$ of the section are distinct and simple. A square of a regular section has even vanishing order at every point. □

Equation (17.57) is an identity of lines. It is not the scalar identity $\vartheta(u)\vartheta(u+2t) = (\vartheta(u+t)\vartheta(t))^2$, whose right side vanishes identically at $t = 0$. For an ordered flag, tensor the root (17.58) over its interacting cross-pairs, using the fixed square isomorphism for each labelled pair. Every parenthesization uses each pair once. Associativity of tensor products therefore gives all flag coherences, and squaring gives the coefficient-line identifications. Negative kernel lines use the inverse roots. Same-colour and dynamical factors require their own line constructions.

17.10 Shifted graph coefficients and integral torsion

Let G be connected on ordered vertices $0, \dots, n-1$, and orient every edge from lower to higher vertex. Set $z_0 = 0$ and assign the edge ij the equation $(z_j - z_i)(z_j - z_i + h)$. Choose an oriented spanning tree, with coordinates $u_1, \dots, u_k$, $k = n-1$, and retain the signed chord paths $\ell_c = \sum_a \sigma_{ca} u_a$. Let $\beta = |E(G)| - n + 1$.

Theorem 17.20. *The full edge Koszul algebra is multiplicatively quasi-isomorphic to*

$$C_{G,h} = \frac{R[u_1, \dots, u_k]}{(u_a(u_a + h) : 1 \le a \le k)} \otimes \Lambda(\epsilon_c : c \notin T), \tag{17.59}$$

where $|\epsilon_c| = -1$ *and*

$$d\epsilon_c = 2\sum_{a<b} \sigma_{ca}\sigma_{cb} u_a u_b + h \sum_a (\sigma_{ca} - \sigma_{ca}^2) u_a. \tag{17.60}$$

Its degree $-j$ *term is free of rank* $2^k \binom{\beta}{j}$*, and its total rank is* $2^{|E(G)|}$*. Define*

$$\mathcal{A}_G = \{s \in \{0,1\}^k : \textstyle\sum_a \sigma_{ca} s_a \in \{0,1\} \text{ for every chord } c\}.$$

After inverting h*, projection gives a multiplicative quasi-isomorphism*

$$C_{G,h} \otimes_R K \longrightarrow \prod_{s \in \mathcal{A}_G} \Lambda_K(\epsilon_1, \dots, \epsilon_\beta), \qquad d = 0. \tag{17.61}$$

Proof. Tree differences are integral unimodular coordinates on the translation quotient. Their monic quadratics form a regular sequence; quotienting by them gives the multiplicative reduction as in Theorem 17.12. Expanding $\ell_c^2 + h\ell_c$ and using $u_a^2 = -hu_a$ gives (17.60). In particular a negatively traversed edge contributes $-2hu_a$. Squarefree tree monomials and exterior subsets give the ranks.

Over K, the tree algebra splits into points $u_a = -hs_a$. On the s-factor a chord differential becomes $h^2 L_c(s)(L_c(s) - 1)$, where $L_c(s) = \sum_a \sigma_{ca} s_a$. If one is nonzero, it is invertible and multiplication by its reciprocal times ϵ_c contracts that factor. On every remaining factor all differentials are zero, proving the projection formula. □

The central and generic dimensions do not determine the order of integral torsion. For the following graphs it can be computed from their finite matrices. Write ij for the edge $\{i, j\}$; dimensions are listed in degrees $0, -1, \dots$.

Proposition 17.21. *The shifted graph algebras have the following cohomology.*

G	*central dimensions*	*generic dimensions*	R*-torsion*
K_3	$(3, 3)$	$(3, 3)$	0
01, 12, 23, 03	$(6, 6)$	$(4, 4)$	$(R/(h))^2$ *in* H^0
$K_4 \setminus \{01\}$	$(5, 10, 5)$	$(5, 10, 5)$	0
K_4	$(4, 14, 14, 4)$	$(4, 12, 12, 4)$	$(R/(h))^2$ *in* H^{-1}

For the third graph one may give the vertices colours (1, 1, 2, 3) *and include only intercolour factors. The last graph has four distinct colours.*

Proof. Use the ordered trees and chord paths

G	T	$(\sigma_c)_c$
K_3	(01, 12)	(1, 1)
01, 12, 23, 03	(01, 03, 23)	(−1, 1, −1)
$K_4 \setminus \{01\}$	(02, 12, 13)	(1, −1, 1), (0, −1, 1)
K_4	(01, 12, 23)	(1, 1, 0), (1, 1, 1), (0, 1, 1).

There are respectively 3, 4, 5, 4 admissible bit assignments. Theorem 17.20 gives the generic dimensions.

For the integral calculation, order squarefree monomials by bit mask, with u_1 the least significant bit, and exterior subsets lexicographically. Multiplication by a tree coordinate has the matrix $\left(\begin{smallmatrix} 0 & 0 \\ 1 & -h \end{smallmatrix}\right)$ on that coordinate and the identity on the others. Equation (17.60) then specifies every matrix. Removing the r-th exterior factor has sign $(-1)^{r-1}$.

Cancel a nonzero constant entry $w = D_{rc}$ by replacing the remaining entries with $D_{ij} - D_{ic}w^{-1}D_{rj}$. Apply the inverse basis changes to adjacent differentials. Their components in the cancelled summand vanish because $d^2 = 0$, so this splits off $[R \xrightarrow{1} R]$. Reading from lowest cochain degree upwards and then by rows, the first available constant entries give respectively 1, 2, 6, 14 cancellations. For the first and third graphs the remaining differential is zero, with ranks (3, 3) and (5, 10, 5).

For the four-cycle the remaining map is 6×6. After factoring out h, its nonzero rows are

$$\begin{pmatrix} -2h & 2h^2 & 4h & -4h^2 & 2h^2 & -2h^3 \\ -2 & 2h & -2 & 2h & 2h & -2h^2 \\ -2 & 2h & 4 & -4h & 2h & -2h^2 \end{pmatrix}.$$

The first row is h times the third; the last two have a constant invertible minor in columns one and three. Every other column is in their R-span. The two nonzero Smith factors of the original residual map are therefore h, h.

For K_4, the remaining ranks are (4, 14, 14, 4). Only the map from degree −2 to degree −1 is nonzero. Dividing it by h and removing zero rows and columns gives

$$\begin{pmatrix} -2 & 2h & 2 & 0 & -2h & -2h & 0 \\ 2 & -2h & 0 & -2 & 2h & 0 & 2h \\ 0 & 0 & -2 & 2 & 0 & 2h & -2h \end{pmatrix}.$$

The rows sum to zero. The first two have an invertible constant minor in columns one and three, and all remaining columns are in their span. Its nonzero Smith factors are again h, h. Thus the two torsion summands now lie in degree −1. Every residual differential vanishes modulo h, which gives the central dimensions and excludes higher torsion orders. □

The repeated-colour example concerns only its indicated intercolour factors; a Hall coefficient must also include the same-colour contribution. The number of admissible bit assignments is a count of branches of this spectral incidence algebra. Identifying it with a compact BPS multiplicity or an Igusa coefficient requires a map from the corresponding compact theory.

17.11 The compact comparison and the weight-five line

The constructions in this chapter retain several different kinds of coefficients. The Hilbert-scheme elliptic complexes yield F_T and F_S after taking Euler supercharacters. The root Lie algebra has a bracket, and its Chevalley differential gives the inverse enveloping character. The spectral incidence algebra has its cyclic trace, normal jets and the relative ternary class $[m_3]$. The automorphic form is a section of $\lambda^5 \otimes \nu_5$.

For a K3 surface S and an elliptic curve E_{geom}, a reduced oriented critical Hall theory on $S \times E_{\mathrm{geom}}$ must be defined on its compact moduli, with extension correspondences and their common translation torsors. The empty sector lies over BE_{geom}. Its translation reduction differs from the quotient of spectral coordinates used in (17.33). The dual-number-valued reduced integration of [123] and the local trace (17.45) have distinct domains and coefficient rings.

Problem 17.22. Construct a comparison from independently defined reduced oriented compact critical coefficients to a collision recipient carrying (17.56) over the whole spectral parameter curve. Determine whether a multiplicative comparison identifies the two ordered coefficient lines, their exceptional fibres, and the halving cover with its $E[2]$-action on root choices. A proposed equivalence of coefficient algebras must identify the minimal-model class $[m_3]$ and its integral relation $\partial G_2 = h\, m_3$.

Replacing G_2 by G_2/h changes the coefficient ring and cannot give a regular comparison over the whole disc. A representation on a state complex has another target: its equations hold in the endomorphism algebra of that state and need not preserve the auxiliary incidence multiplication.

Problem 17.23. Construct a geometric family over genus-two moduli whose determinant or Pfaffian line is $\lambda^5 \otimes \nu_5$, and identify its section with the normalized D_5. Relate its degeneration and sewing maps to the compact critical coefficients and their elliptic collision lines.

The line (17.58) on a fixed spectral elliptic curve supplies a square root with explicit deck action. It does not by itself give a family over genus-two moduli. The determinant model (17.27) supplies a convergent scalar in a cusp chamber. A solution of the last problem must additionally construct the geometric family, its transition maps and the operation-preserving comparisons.

Book IV

Geometry, operations, and representations

Chapter 18

Extensions of geometric objects

A sheaf retains more information than its support or its cohomology dimension. Let $R = \mathbb{C}[x]$ and $k = R/(x)$. The sequence

$$0 \longrightarrow k \xrightarrow{1 \mapsto x} R/(x^2) \longrightarrow k \longrightarrow 0$$

is a nonsplit extension. Indeed, x acts nontrivially on its middle term and acts trivially on $k \oplus k$. The free resolution $[R \xrightarrow{x} R]$ of k gives $\operatorname{Ext}^1_R(k,k) = k$. Thus one extension parameter distinguishes objects with the same two composition factors.

For a moduli stack $\mathcal{M}_\gamma$ of objects of charge γ, extensions form a correspondence

$$\mathcal{M}_\alpha \times \mathcal{M}_\beta \xleftarrow{p} \mathcal{E}_{\alpha,\beta} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

A coefficient theory with admissible pullback, pushforward, and base change gives the Hall product $q_* p^*$. Critical coefficients additionally require orientation isomorphisms compatible on triple flags. A framed moduli problem gives a representation when its action correspondences satisfy the corresponding flag identities. These geometric operations precede a trace on the representation.

A Calabi–Yau structure supplies a nondegenerate cyclic duality class. At an object E, the tangent complex of its moduli problem is $R\operatorname{Hom}(E,E)[1]$. Composition and trace pair tangent vectors with shift $2-d$ for a d-Calabi–Yau category. Closedness uses the negative-cyclic lift. Under the hypotheses of shifted-symplectic transgression, this defines the moduli form. The ordinary Hochschild cochains have their E_2 operations; an action of another operad or a holomorphic field realization requires its specified chain-level construction.

Chapter 19

Central actions and monoidal transport

For an associative algebra object A in a specified derived coefficient category, its Hochschild centre is the derived endomorphism object of its diagonal bimodule. For a monoidal category $\mathcal{M}$, its categorical centre $Z(\mathcal{M})$ consists instead of pairs (X, β), where

$$\beta_Y : X \otimes Y \xrightarrow{\sim} Y \otimes X$$

is natural in Y, compatible with the unit, and satisfies the half-braiding hexagon. In strict notation that identity is $\beta_{Y\otimes Z} = (\mathrm{id}_Y \otimes \beta_Z)(\beta_Y \otimes \mathrm{id}_Z)$. Associators are inserted in a general monoidal category. The centre has braiding $c_{(X,\beta),(Y,\gamma)} = \beta_Y$.

Theorem 19.1 (Transport of categorical centres)**.** *Let $F : \mathcal{M} \to \mathcal{N}$ be a strong monoidal equivalence. Write $J_{X,Y} : FX \otimes FY \to F(X \otimes Y)$ for its tensor constraint. Choose a monoidal quasi-inverse H and a monoidal natural isomorphism $\epsilon : FH \to \mathrm{id}_{\mathcal{N}}$. Then F induces a braided equivalence $Z(F) : Z(\mathcal{M}) \to Z(\mathcal{N})$. For (X, β) its half-braiding at $Y \in \mathcal{N}$ is*

$$\begin{aligned} FX \otimes Y &\xrightarrow{\mathrm{id}\otimes \epsilon_Y^{-1}} FX \otimes FHY \xrightarrow{J_{X,HY}} F(X \otimes HY) \\ &\xrightarrow{F\beta_{HY}} F(HY \otimes X) \xrightarrow{J^{-1}_{HY,X}} FHY \otimes FX \xrightarrow{\epsilon_Y \otimes \mathrm{id}} Y \otimes FX. \end{aligned}$$

Proof. Naturality follows by applying naturality of J, ϵ, and β to each arrow. For $Y \otimes Z$, monoidality of ϵ replaces its inverse by the two inverses and the tensor constraint of FH. The pentagon for J then identifies the transported hexagon with F applied to the hexagon for β. The same argument with the unit constraints proves the unit identity. The constraints J are central morphisms by this formula. For two central objects their braiding is the half-braiding evaluated on the second object, so the construction preserves braiding. Applying the same construction to H gives an inverse up to the monoidal unit and counit isomorphisms. □

A Morita equivalence of associative algebras supplies an equivalence of module categories. To use this theorem, those categories must have specified tensor products and the equivalence must have coherent tensor and unit constraints. A realization of the Hochschild centre by a categorical centre is a further comparison with its own source and target.

Chapter 20

Field actions and amplitudes

Fix a spacetime, fields, action, gauge symmetry, boundary condition, coefficient category, and approximation regime. A quantization must satisfy the quantum master equation on its stated domain before it supplies quantum observables. A coherent action on the boundary has target the independently defined boundary centre:

$$\beta : \mathrm{Obs}_{\mathrm{bulk}}\big|_{\partial} \longrightarrow Z_{\mathrm{ord}}(\mathcal{A}_{\partial}).$$

An equivalence of this map on a specified sector is a bulk reconstruction statement. A supplied formal factorization envelope is a different object until its realization map to the spacetime observables is constructed.

Hall convolution produces a positive algebra. A compatible coproduct gives tensor products of representations. A paired negative half and Cartan part permit a completed double when the triangular decomposition and canonical tensors converge. Exchange requires the corresponding intertwiners and their hexagon identities. Theorem 19.1 then applies to actual monoidal comparisons of the representation categories.

A trace closes an action into an amplitude only on its declared domain. Its cyclicity, grading, value line, and normalization are additional data. A Borcherds character computes signed graded dimensions of its root algebra. Its identification with a geometric Hall amplitude requires root states, orientation maps, and a trace comparison on those states.

Book entrance: Ordered and Associative Chiral Geometry

Collision on a complex curve, order on a transverse line, exchange in a plane, and the three corresponding bar constructions.

Book V

The invariant object

Chapter 21

Two directions of locality

21.1 The local experiment

Let X be a smooth complex algebraic curve and let $I \subset \mathbb{R}$ be an oriented interval. Place two protected observables at (z_1, t_1) and (z_2, t_2) in $X \times I$. The limits $z_1 \to z_2$ and $t_1 \to t_2$ carry independent information. The first limit produces a singular distribution along the diagonal of X^2. The second composes operators in the order selected by the orientation of I.

Principle 21.1 (Two-direction principle). Holomorphic collision belongs to the factorization coefficient on X. Ordered fusion belongs to an internal E_1 multiplication. Exchange belongs to an additional two-dimensional transport structure.

This separation gives five geometric carriers:

carrier	operation
diagonal in X^2	singular chiral operation
oriented interval	ordered fusion and bar construction
oriented plane	braid transport
circle	Hochschild trace
stable curve	modular sewing.

Each later structure is introduced by the datum carried by its row.

21.2 Carrier and operation

For the diagonal embedding $\Delta : X \hookrightarrow X^2$ and its complement $j : X^2 \setminus \Delta(X) \hookrightarrow X^2$, a Beilinson–Drinfeld chiral Lie operation is a morphism

$$\mu : j_* j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}$$

in the chiral pseudo-tensor category of right D-modules [6]. The D-module is the carrier. The morphism μ is the operation. A factorization object packages carriers on all powers X^I together with their disjoint-locus structure maps. The complete operator product is the pair consisting of that system and its multiplication maps.

In a formal coordinate z near the diagonal, the pole filtration of $\mathcal{O}_{X^2}(*\Delta)$ has

$$\operatorname{gr}^F_m \mathcal{O}_{X^2}(*\Delta) \cong \mathcal{O}_\Delta(m\Delta) \cong N_{\Delta/X^2}^{\otimes m}, \qquad m \geq 1.$$

A principal part of order m therefore has an intrinsic normal symbol. Coordinate changes act triangularly on the filtered object. Residues recover the first symbol; normal jets recover the higher symbols.

Chapter 22

The Ran coefficient category

22.1 The fixed six-functor setting

Fix a presentable stable category $\mathcal{D}_X$ of right D-modules on $\mathrm{Ran}(X)$ with the following operations:

(i) the pointwise tensor product $\otimes^!$;

(ii) chiral convolution $\otimes^{\mathrm{ch}}$ along disjoint union;

(iii) base change and projection formulae for the disjoint-union correspondences;

(iv) geometric realizations and totalizations preserved by the tensor products used below.

The standard renormalized D-module setting of chiral Koszul duality has these operations [61]. Every construction below is formed in this fixed setting.

Let

$$\mathrm{Ran}(X)\times\mathrm{Ran}(X) \xleftarrow{\ j\ } (\mathrm{Ran}(X)\times\mathrm{Ran}(X))_{\mathrm{disj}} \xrightarrow{\ u\ } \mathrm{Ran}(X)$$

be the disjoint-union correspondence. Define

$$F\otimes^{\mathrm{ch}} G = u_! j^!(F\boxtimes G).$$

The pointwise product is induced by the diagonal of $\mathrm{Ran}(X)$.

Definition 22.1 (Aligned-decomposition projector)**.** For a finite set I, let $\mathrm{Dec}(I)$ be the finite groupoid of ordered decompositions $I = A\sqcup B$ and let

$$a_I : \mathrm{Dec}(I)\longrightarrow \mathrm{Dec}(I)\times\mathrm{Dec}(I), \qquad d\longmapsto (d,d)$$

be the diagonal. In the linear species model define

$$e_I = a_{I,!}a_I^!.$$

Thus e_I projects a pair of decompositions onto the summands in which the two decompositions agree. In the six-functor Ran model the same formula denotes extraordinary pullback to the diagonal correspondence followed by the available renormalized pushforward.

Theorem 22.2 (Aligned sector in the finite decomposition model)**.** *Let $\mathcal{S}_I$ be a stable category indexed by ordered decompositions of a finite set I, and assume the direct-sum decomposition*

$$\mathcal{M}_I \simeq \bigoplus_{d_1,d_2\in\mathrm{Dec}(I)} \mathcal{M}_{d_1,d_2}.$$

The projection

$$e_I : \mathcal{M}_I\longrightarrow\mathcal{M}_I, \qquad e_I\big|_{\mathcal{M}_{d_1,d_2}} = \begin{cases}\mathrm{id}, & d_1 = d_2,\\ 0, & d_1\neq d_2,\end{cases}$$

is idempotent. Its image is the aligned sector

$$\mathcal{M}_I^{\mathrm{al}} = \bigoplus_{d \in \mathrm{Dec}(I)} \mathcal{M}_{d,d}.$$

The family e_I is natural under bijections and compatible with iterated disjoint union. A mixed distributive law in this model is a natural transformation defined on $\mathcal{M}_I^{\mathrm{al}}$ whose two restrictions to the triple diagonal $d_1 = d_2 = d_3$ agree.

Proof. Idempotence is checked on each direct summand. Naturality follows because a bijection carries an ordered decomposition to an ordered decomposition and preserves equality of the two indices. For three products, both iterated alignments project

$$\bigoplus_{d_1,d_2,d_3} \mathcal{M}_{d_1,d_2,d_3} \quad \text{onto} \quad \bigoplus_{d} \mathcal{M}_{d,d,d}.$$

The two projectors are therefore equal. This equality is precisely the triple-alignment coherence. □

Theorem 22.3 (Descent of an aligned projector)**.** *Suppose the common-decomposition inclusions are clopen on the chosen charts. Suppose every transition correspondence gives Cartesian squares with these inclusions and the corresponding Beck–Chevalley maps are isomorphisms. Then the functors*

$$e_I = a_{I,!} a_I^!$$

assemble to a natural idempotent e_{al} on the mixed correspondence object. Its image is the common-decomposition correspondence. The two iterated projectors on three decomposition indices coincide, and the assumed base-change maps transport this equality through the transitions.

Proof. Restriction and extension by zero along a clopen inclusion form a projector. The Cartesian-square hypothesis and base change identify these composites across transitions. For three indices, both iterated projectors select the same locus $d_1 = d_2 = d_3$, so the identification respects triple alignment. □

The Cartesian-square hypothesis must be checked on the actual correspondences. Preserving equal indices alone does not suffice: a map can also send distinct indices to equal ones, as the map $\{0, 1\} \to \{*\}$ does. On a common disjoint-union correspondence, the projection formula gives the aligned comparison. Its inclusion and retraction are inverse on the full object exactly when every off-aligned summand vanishes. This criterion concerns that canonical comparison. It does not exclude every noninvertible distributive law.

22.2 Factorization coalgebras

A chiral commutative coalgebra is a commutative coalgebra object for $\otimes^{\mathrm{ch}}$. It is factorizing when every coproduct becomes an equivalence on the relevant disjoint locus. Let

$$\mathrm{Fact}_D(X) \subset \mathrm{CoCAlg}_{\mathrm{ch}}(\mathcal{D}_X)$$

be the full subcategory of factorizing objects. Chiral Koszul duality identifies the corresponding Lie-theoretic and factorization descriptions on the completed Ran category [61, 79].

Proposition 22.4 (Pointwise closure)**.** *The componentwise tensor product of two factorization objects is a factorization object. Thus* $\mathrm{Fact}_D(X)$ *is symmetric monoidal under $\otimes^!$ whenever the fixed Ran enhancement is closed under that tensor product.*

Proof. For a decomposition $I = I_1 \sqcup I_2$, restrict to its disjointness locus. The factorization equivalences of F and G give

$$\begin{aligned} j^!(F_I \otimes^! G_I) &\simeq j^! F_I \otimes^! j^! G_I \\ &\simeq (F_{I_1} \boxtimes F_{I_2}) \otimes^! (G_{I_1} \boxtimes G_{I_2}) \\ &\simeq (F_{I_1} \otimes^! G_{I_1}) \boxtimes (F_{I_2} \otimes^! G_{I_2}). \end{aligned}$$

Compatibility under refinement and permutation follows componentwise from the corresponding compatibilities for F and G. □

Chapter 23

The two noncommutative chiral objects

23.1 Transverse order

Two operators at the same chiral support can compose in an independent oriented real direction. Their multiplication is internal to the pointwise coefficient category. This is the structure carried by a boundary or line defect in a holomorphic–topological theory on $X \times \mathbb{R}$.

Definition 23.1 (Transversely ordered chiral algebra)**.** A transversely ordered chiral algebra is an augmented object

$$A^{\perp} \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathrm{Fact}_D(X), \otimes^{!}).$$

The superscript $\perp$ records the transverse oriented direction. The shorter phrase *ordered chiral algebra* will always mean this object.

23.2 Associativity inside chiral convolution

Chiral convolution combines disjoint finite subsets before allowing them to collide. Associativity for this product is intrinsic to the curve and has a different source and a different bar construction.

Definition 23.2 (Associative chiral algebra)**.** An associative chiral algebra is an augmented algebra

$$A^{\mathrm{ch}}_{\mathsf{Ass}} \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathcal{D}_X, \star^{\mathrm{ch}}).$$

A chiral Lie algebra is a Lie algebra object

$$\mathfrak{g}^{\mathrm{ch}} \in \mathrm{Alg}_{\mathsf{Lie}}(\mathcal{D}_X, \star^{\mathrm{ch}}).$$

The commutator of an associative chiral algebra is a chiral Lie algebra.

The operation $m^{\perp} : A \otimes^{!} A \to A$ composes transverse intervals. The operation $m^{\mathrm{ch}} : A \star^{\mathrm{ch}} A \to A$ composes chiral configurations. They have different domains, different units, and different deformation complexes.

Theorem 23.3 (Independence of the two associative structures)**.** *Let M be an object for which the split square-zero extensions below exist. A pointwise associative product $\mu_{\perp} : M \otimes^{!} M \to M$ defines a transversely ordered algebra on $\mathbf{1}_{\perp} \oplus M$, while the reduced chiral-convolution product may independently be zero. A chiral-convolution product $\mu_{\mathrm{ch}} : M \star^{\mathrm{ch}} M \to M$ defines an associative chiral algebra on $\mathbf{1}_{\mathrm{ch}} \oplus M$, while the reduced pointwise product may independently be zero. Hence neither associative structure determines the other.*

A common carrier equipped with both products is a mixed object after a mixed distributive datum has been supplied and its unit, binary, and triple coherences have been verified.

Proof. Adjoining the unit to an associative product on the augmentation ideal gives a unital split extension. The zero product is associative in either monoidal category. These two constructions vary independently. On a common carrier, compatibility is additional structure because the mixed composites have different a priori factorizations through the two monoidal products. □

Definition 23.4 (Mixed associative chiral datum)**.** A mixed associative chiral datum is a factorization object A equipped with

(i) a pointwise multiplication $m^{\perp} : A \otimes^{!} A \to A$ and a unit map $\eta_{\perp} : \mathbf{1}_{\perp} \to A$;

(ii) a chiral-convolution multiplication $m^{\mathrm{ch}} : A \star^{\mathrm{ch}} A \to A$ and a unit map $\eta_{\mathrm{ch}} : \mathbf{1}_{\mathrm{ch}} \to A$;

(iii) an aligned mixed distributive law identifying the two composites on $\mathrm{im}(e)$.

The coherence diagrams are the two color-3 faces of the mixed forest complex. A bare symbol $\mathbf{1}$ below denotes the unit of the monoidal category in which the displayed construction is formed.

23.3 Formal-disc realization

Fix a formal coordinate on a disc. Translation-equivariance turns a chiral-convolution product into a state–field operation $Y(a, z)b$ with coefficients in Laurent series. The following theorem packages the exact relationship with the vertex-algebra language.

Theorem 23.5 (Formal-disc realizations)**.** *Let V be a translation-equivariant state–field system with vacuum and injective state–field map.*

(i) *If its fields are weakly associative in the sense of Li, then V is a nonlocal vertex algebra.*

(ii) *If, in addition, V is topologically free over $\mathbb{k}[[\hbar]]$, carries a unitary rational quantum Yang–Baxter operator $S(z)$, and satisfies S-locality and the hexagon compatibility, then V is a quantum vertex algebra.*

(iii) *A translation-equivariant associative chiral-convolution algebra produces such a state–field system whenever its formal-disc restriction satisfies the reconstruction, lower-truncation, and weak-associativity hypotheses. Conversely, a nonlocal or quantum vertex algebra produces the corresponding formal-disc factorization system by its products of fields.*

Proof. The first assertion is Li's reconstruction theorem for quasi-compatible fields [108]. The second is the quantum-vertex reconstruction theorem: unitarity gives inverse exchange, the quantum Yang–Baxter equation compares the two three-field exchange paths, and the hexagon relates exchange to the vertex operation [41, 109]. For the third assertion, the formal-disc chiral product supplies the iterated field products; the stated analytic-formal hypotheses are exactly those used by the reconstruction theorems. The converse assigns to a configuration of formal points the iterated field product and uses weak associativity, or braided associativity, for factorization under refinement. □

23.4 Three separation laws

The complete principal-part filtration, transverse order, and planar exchange form a strict hierarchy of supplied data. The mth normal symbol of a pole is recovered by the mth normal-jet map. An oriented line supplies contractible order chambers and hence E_1 fusion. A plane supplies braid paths and hence exchange. These positive constructions locate each operation in the first geometry that carries it.

Chapter 24

Transversely ordered chiral algebras

Definition 24.1 (Ordered chiral algebra)**.** An ordered chiral algebra on X is an augmented E_1-algebra object

$$A \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}\big(\mathrm{Fact}_D(X), \otimes^!\big).$$

Equivalently, A is a factorization object equipped with multiplication maps

$$m_I : A_I \otimes^! A_I \longrightarrow A_I$$

and a unit, compatible with all diagonal transition maps and all disjoint-locus factorization equivalences.

For every decomposition $I = I_1 \sqcup I_2$, locality is the componentwise identity

$$\Delta_{I_1,I_2} \circ m_I = (m_{I_1} \boxtimes m_{I_2}) \circ \sigma \circ (\Delta_{I_1,I_2} \otimes^! \Delta_{I_1,I_2}),$$

where σ reorders the middle two factors. This equation says that ordered fusion commutes with separation of chiral supports.

Theorem 24.2 (Locality recognition)**.** *To give an ordered chiral algebra is equivalent to giving a factorization family A_I, an associative unital multiplication m_I on every component, and the displayed locality identities, compatibly with refinements, permutations, and diagonal transitions.*

Proof. An E_1-algebra in the symmetric monoidal category $\mathrm{Fact}_D(X)$ is, by definition, an associative multiplication and unit whose structure maps are morphisms in $\mathrm{Fact}_D(X)$. A morphism in $\mathrm{Fact}_D(X)$ is exactly a componentwise morphism compatible with the transition and factorization maps. Expanding this compatibility for a binary decomposition gives the displayed identity; the higher decompositions follow by iteration. □

24.1 Product field theories

Let $\mathrm{Fact}_{\mathrm{hol,lc}}(X^{\mathrm{an}} \times \mathbb{R})$ denote factorization algebras that are holomorphic along X and locally constant along $\mathbb{R}$. Product additivity gives

$$\mathrm{Fact}_{\mathrm{hol,lc}}(X^{\mathrm{an}} \times \mathbb{R}) \simeq \mathrm{Alg}_{\mathsf{E}_1}\big(\mathrm{HolFact}(X^{\mathrm{an}})\big)$$

for product-generated objects. Algebraization by regular holonomic Riemann–Hilbert then gives ordered chiral algebras in $\mathrm{Fact}_D(X)$ whenever the coefficient system is regular holonomic and the comparison preserves the required colimits.

Theorem 24.3 (Holomorphic–topological recognition)**.** *Let $\mathbb{F}$ be a product-generated factorization algebra on $X^{\mathrm{an}} \times \mathbb{R}$ that is holomorphic on X and locally constant on $\mathbb{R}$. Then the restriction of $\mathbb{F}$ to a transverse point is an E_1-algebra in holomorphic factorization algebras on X. A regular-holonomic algebraization produces an ordered chiral algebra on X.*

Proof. An ordered family of disjoint intervals in $\mathbb{R}$ acts on the restriction by factorization. Contractibility of each ordered configuration chamber gives the coherent E_1 structure. Factorization in the X direction is preserved because product embeddings commute. Riemann–Hilbert transports the structure in the regular-holonomic subcategory. $\square$

Chapter 25

Block-ordered operations

The preceding definition is intrinsic. A geometric model arises after one chooses an infinity-operad Ch_X representing the selected chiral factorization theory. The Boardman–Vogt tensor product

$$\mathsf{OrdCh}_X = \mathsf{E}_1 \otimes \mathsf{Ch}_X$$

then represents compatible ordered and chiral operations.

Theorem 25.1 (Operadic exponential law)**.** *Let P and Q be infinity-operads whose Boardman–Vogt tensor product exists, and let $\mathcal{V}$ be a compatible presentable symmetric monoidal infinity-category. Then*

$$\mathrm{Alg}_{P\otimes Q}(\mathcal{V}) \simeq \mathrm{Alg}_P\big(\mathrm{Alg}_Q(\mathcal{V})\big).$$

In particular, after a chiral operad Ch_X has been constructed, the tensor product $\mathsf{E}_1 \otimes \mathsf{Ch}_X$ classifies E_1-algebra objects in Ch_X-algebras.

Proof. The tensor product of infinity-operads is characterized by the displayed universal property [114]. The second statement is the specialization $P = \mathsf{E}_1$ and $Q = \mathsf{Ch}_X$. This formal equivalence applies to each geometric model of Ch_X after that model and its operadic tensor product have been constructed. □

A profile is an ordered list $(I_1, \dots, I_r)$ of finite chiral label sets. Its geometric carrier is $X^{I_1} \times \cdots \times X^{I_r}$. Chiral morphisms refine and collide labels inside one block. Transverse morphisms merge adjacent blocks. The tensor-product operad imposes their mixed squares.

25.1 Mixed forests

Fix a cofibrant dg-operad $Q_X \to \mathsf{Ch}_X$. Apply the functorial W-construction to $\mathsf{E}_1 \otimes Q_X$. A cell is represented by a levelled forest with transverse and chiral internal edges. Put

$$\mathrm{or}(F) = \det(E_\perp(F)) \otimes \det(E_{\mathrm{ch}}(F)).$$

The differential is the sum of the screen differential and one-edge contractions.

Theorem 25.2 (Two-color forest differential)**.** *Let C_* be the chain complex freely generated by oriented two-color forests, with one determinant-line generator for every transverse edge and one for every chiral edge. Suppose contraction of an edge is functorial on the coefficient system and every codimension-two contraction square commutes before orientations are inserted. With the product orientation, the transverse and chiral contraction operators satisfy*

$$d_\perp^2 = 0, \qquad d_{\mathrm{ch}}^2 = 0, \qquad d_\perp d_{\mathrm{ch}} + d_{\mathrm{ch}} d_\perp = 0.$$

Thus $d = d_{\mathrm{int}} + d_\perp + d_{\mathrm{ch}}$ squares to zero whenever the internal differential anticommutes with both contraction operators.

Proof. Every term of $d_{\perp}^{2}$ contracts the same pair of transverse edges in two orders. The underlying coefficient maps agree by functoriality, while exchanging the two transverse determinant generators reverses the sign. The terms cancel in pairs. The same argument proves $d_{\mathrm{ch}}^{2} = 0$. For one edge of each color the coefficient maps again form a commuting square and exchanging the two determinant generators contributes the Koszul sign -1, proving the mixed anticommutator identity. The internal terms vanish by the chain-map assumption. □

This construction supplies a geometric model whenever a concrete Q_X and its screen coefficients are available. The intrinsic definition of ordered chiral algebra remains independent of that choice.

Book VI

Bar, cobar, and the boundary state

Chapter 26

Three bar constructions

The phrase *the chiral bar* denotes three functors once its monoidal product and algebraic operad have been named.

Definition 26.1 (Transverse, associative-chiral, and Lie-chiral bars)**.** For augmented objects of the indicated types define

$$\mathrm{Bar}^{\perp}(A^{\perp}) := \mathbf{1} \otimes^{\mathbb{L},!}_{A^{\perp}} \mathbf{1},$$

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A^{\mathrm{ch}}_{\mathsf{Ass}}) := \mathbf{1} \bigstar^{\mathrm{ch},\mathbb{L}}_{A^{\mathrm{ch}}_{\mathsf{Ass}}} \mathbf{1},$$

and

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(\mathfrak{g}^{\mathrm{ch}}) := C^{\mathrm{ch}}_{*}(\mathfrak{g}^{\mathrm{ch}}) = \mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g}^{\mathrm{ch}}[1])$$

with its chiral Chevalley differential. The first bar is formed with $\otimes^{!}$, the second with $\bigstar^{\mathrm{ch}}$, and the third is the Lie-to-commutative-coalgebra Koszul functor.

Theorem 26.2 (Typed bar theorem)**.** *The three bars have the following chain models.*

(i) $\mathrm{Bar}^{\perp}(A^{\perp})$ *is the tensor coalgebra* $T^{c}_{\otimes^{!}}(s\bar{A})$ *with coderivation induced by* $m^{\perp}$*.*

(ii) $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A^{\mathrm{ch}}_{\mathsf{Ass}})$ *is the tensor coalgebra* $T^{c}_{\bigstar^{\mathrm{ch}}}(s\bar{A})$ *with coderivation induced by* m^{ch}*.*

(iii) $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(\mathfrak{g}^{\mathrm{ch}})$ *is the chiral symmetric coalgebra with differential induced by the chiral Lie bracket.*

Each differential squares to zero by the defining relation of its own operad. The outputs live in different coalgebra categories and are compared by a displayed monoidal or PBW morphism.

Proof. Normalize the two simplicial two-sided bars in their respective monoidal categories. Degeneracies remove units, leaving the two tensor coalgebras, and faces contract adjacent factors using the relevant multiplication. The Lie formula is the ordinary operadic bar of the Lie operad internal to chiral convolution. Associativity and Jacobi are exactly the quadratic identities that make the corresponding coderivations square to zero. □

Definition 26.3 (PBW-admissible chiral Lie algebra)**.** A chiral Lie algebra $\mathfrak{g}^{\mathrm{ch}}$ is PBW-admissible when its chiral enveloping algebra $U^{\mathrm{ch}}(\mathfrak{g})$ has a complete separated PBW filtration, the canonical map

$$\mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g}) \longrightarrow \mathrm{gr}_{\mathrm{PBW}}\, U^{\mathrm{ch}}(\mathfrak{g})$$

is an equivalence, and the bar filtration is strongly convergent in the chosen nilcomplete Ran category.

Theorem 26.4 (Filtered chiral PBW comparison)**.** *Let* $\mathfrak{g}$ *be a chiral Lie algebra in a complete filtered chiral category. Assume:*

(i) *its chiral enveloping algebra* $U^{\mathrm{ch}}(\mathfrak{g})$ *exists;*

(ii) *the PBW filtration is exhaustive, complete, and separated;*

(iii) *the chiral PBW morphism*

$$\mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g}) \longrightarrow \mathrm{gr}\, U^{\mathrm{ch}}(\mathfrak{g})$$

is an isomorphism;

(iv) *the induced filtrations on the two bar complexes are strongly convergent.*

Then antisymmetrization induces a filtered quasi-isomorphism

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}\big(U^{\mathrm{ch}}(\mathfrak{g})\big) \simeq \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(\mathfrak{g}).$$

Proof. The associated graded of the associative chiral bar is the Koszul bar of $\mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g})$. The usual Koszul antisymmetrization identifies this coalgebra with the chiral Chevalley coalgebra. Hypotheses (2)–(4) give a complete, separated, strongly convergent comparison spectral sequence. Since its associated-graded map is a quasi-isomorphism, the filtered map is a quasi-isomorphism. This is the PBW lane of chiral Koszul duality [6, 61]. □

Definition 26.5 (Bar-admissible mixed datum)**.** A mixed associative chiral datum is bar-admissible when its aligned distributive law acts on the full bisimplicial two-sided bar and both geometric realizations commute with the tensor products occurring in that bisimplicial object.

Theorem 26.6 (Mixed Fubini theorem)**.** *Let A be a mixed algebra in a presentable stable category. Assume the mixed distributive datum extends to the bisimplicial two-sided bar, and that both monoidal products preserve its geometric realizations separately. Then the two iterated realizations and the diagonal realization are canonically equivalent:*

$$\Big|\, p \mapsto \big|q \mapsto B_{p,q}(A)\big|_{\mathrm{ch}}\Big|_{\perp} \simeq \big|\mathrm{diag}\, B_{\bullet,\bullet}(A)\big| \simeq \big|\, q \mapsto \big|p \mapsto B_{p,q}(A)\big|_{\perp}\big|_{\mathrm{ch}}.$$

Proof. The distributive datum makes the two simplicial directions commute, so $B_{\bullet,\bullet}(A)$ is a genuine bisimplicial object. Geometric realization is a colimit. Since the two tensor products preserve the relevant colimits, Fubini for colimits identifies both iterated realizations with the realization of the diagonal simplicial object. □

The notation used in the remainder of this part is

$$\mathrm{Bar}^{\mathrm{ord}}_{X} = \mathrm{Bar}^{\perp}.$$

It always refers to transverse order.

26.1 Realization and factorization

Factorization compares the coefficient at a disjoint configuration with an external tensor product of its coefficients at the separate configurations. This condition is nonlinear. For example, suppose those coefficients are k at one point and at two distinct points, where k is a field. The componentwise sum of two copies has coefficient k^2 at both configurations. Factorization would instead require $k^2 \otimes_k k^2 \cong k^4$ at two points. A componentwise shift also fails: its two-point coefficient is $k[1]$, whereas the tensor product of its singleton coefficients is $k[2]$. Thus an ambient stable coefficient category does not make its factorization objects closed under componentwise sums or shifts.

Geometric realization has a different colimit property. A small indexing ∞-category J is *sifted* when it is nonempty and its diagonal $J \to J \times J$ is cofinal. Here cofinal means that, for every diagram D on $J \times J$, the canonical map $\mathrm{colim}_J(D \circ \Delta) \to \mathrm{colim}_{J\times J} D$ is an equivalence whenever these colimits exist. Iteration gives the same property for $J \to J^m$, for every finite $m \geq 1$. The simplex category with arrows reversed is sifted, so simplicial geometric realization is a sifted colimit.

Proposition 26.7. *Fix coefficient categories on configuration charts, with specified restriction and collision functors and their coherent identifications. Suppose the coefficient categories admit J-indexed colimits and these functors preserve them. Suppose external tensor products preserve colimits separately in each variable. Let $(F_j)_{j\in J}$ be a diagram of factorization objects and morphisms preserving their factorization equivalences. If J is sifted, the componentwise colimit is a factorization object.*

Proof. Write $F_I = \operatorname{colim}_{j\in J}(F_j)_I$ on a chart indexed by I. The collision maps and their coherences pass to the colimit by hypothesis. For a decomposition $I = I_1 \sqcup \cdots \sqcup I_m$, let u denote restriction to its disjointness locus. Restriction, factorization, cofinality, and separate preservation give

$$\begin{aligned} uF_I &\simeq \operatorname*{colim}_{j\in J} u(F_j)_I \\ &\simeq \operatorname*{colim}_{j\in J} \boxtimes_{a=1}^{m} (F_j)_{I_a} \\ &\simeq \operatorname*{colim}_{(j_1,\dots,j_m)\in J^m} \boxtimes_{a=1}^{m} (F_{j_a})_{I_a} \simeq \boxtimes_{a=1}^{m} F_{I_a}. \end{aligned}$$

Each comparison is natural in the diagram and the decomposition. Refinements and permutations therefore preserve the given coherence diagrams. If an empty-configuration unit is specified, its constant diagram also has that unit as colimit, since a sifted category has contractible groupoid completion. □

For a simplicial two-sided bar, this proposition applies when every simplicial term and every face and degeneracy are already in the specified factorization category. It also requires the actual restriction and collision functors to preserve realization. Normalization may then be computed in an ambient stable category. Its individual shifted summands need not be factorization objects. A coproduct on the normalized complex requires its own compatibility with the factorization equivalences.

Chapter 27

The internal ordered bar

Let A be an augmented ordered chiral algebra. Assume the restriction and collision functors preserve geometric realizations, and the external and pointwise tensor products preserve them separately. The two-sided simplicial bar has

$$B_r(\mathbf{1}, A, \mathbf{1}) = \mathbf{1} \otimes^! A^{\otimes^! r} \otimes^! \mathbf{1}.$$

Every face and degeneracy is a factorization morphism by the algebra and augmentation axioms. The sifted-colimit proposition therefore constructs its realization $\mathrm{Bar}^{\mathrm{ord}}_X(A)$ as a factorization object.

Theorem 27.1 (Normalized bar). *Choose a symmetric monoidal dg model for the ambient coefficients in which the displayed tensors compute derived tensors and direct-sum normalization computes realization. Write $\bar{A}$ for the augmentation ideal in that model. The underlying bar complex is*

$$\mathrm{Bar}^{\mathrm{ord}}_X(A) \simeq \big(T^c(s\bar{A}), d_{\mathrm{int}} + b_m\big).$$

Here s decreases cohomological degree by one. The direct sum and its individual shifted summands are formed in the ambient coefficient category. Deconcatenation makes this complex a conilpotent coassociative coalgebra there:

$$\Delta_{\mathrm{dec}}[a_1|\cdots|a_n] = \sum_{i=0}^{n} [a_1|\cdots|a_i] \otimes [a_{i+1}|\cdots|a_n].$$

Proof. Normalization removes words containing a unit and leaves the shifted augmentation ideal in every slot. The faces give the internal differential and adjacent multiplication. The simplicial identities give a square-zero differential. Two successive cuts enumerate the same three subwords in either order. An internal differential or multiplication lies on one side of any surviving cut, with the tensor sign of its prefix. Thus the differential is a coderivation. Every element has finite word support, so sufficiently many reduced cuts give zero. □

27.1 The bar comparison on separated configurations

The normalized word decomposition is not itself a decomposition in $\mathrm{Fact}_D(X)$. Its compatibility with factorization is expressed by a map between whole bar complexes. For augmented dg algebras P, Q in the chosen model, define

$$\mathrm{sh} : BP \otimes BQ \longrightarrow B(P \otimes Q)$$

by shuffling the letters $p_i \otimes 1$ and $1 \otimes q_j$, keeping their respective orders. The sign is the Koszul sign for the permutation of the suspended letters. For degree-zero letters,

$$\mathrm{sh}([p] \otimes [q]) = [p \otimes 1|1 \otimes q] - [1 \otimes q|p \otimes 1].$$

Proposition 27.2 (Shuffle and factorization). *The shuffle is a natural quasi-isomorphism and a coalgebra map, where its source has the tensor-product coalgebra structure. For the geometric coefficient models above,*

suppose these maps commute with the specified restriction and collision functors. Then they give factorization equivalences for the whole bar and make deconcatenation a morphism of factorization objects in the derived category.

Proof. In the bar differential of a shuffle, a merge of letters from different factors occurs twice with opposite suspended signs. The equality $(p \otimes 1)(1 \otimes q) = (-1)^{|p||q|}(1 \otimes q)(p \otimes 1)$ cancels these terms. Merges within one factor give the differential of that factor. Internal differentials commute with the same signed permutation. Thus the shuffle is a cochain map.

A cut of a shuffled word determines one cut in each input word, a shuffle of their prefixes, and a shuffle of their suffixes. Conversely these four choices determine exactly that cut shuffle. The signs on both sides come from the same permutation of suspended letters. This proves the coalgebra identity with the tensor-product coproduct.

To verify the quasi-isomorphism, form the bisimplicial object $B_p(k, P, k) \otimes B_q(k, Q, k)$, with k denoting the tensor unit. Its iterated realization is $BP \otimes BQ$. Its diagonal is the simplicial bar of $P \otimes Q$. Cofinality of the simplicial diagonal identifies these realizations. On normalized chains the map from the iterated realization to the diagonal is the shuffle just defined: the monotone paths from $(0, 0)$ to (p, q) triangulate the product of its two simplices. Each path inserts precisely the degeneracies of its shuffle. Adjacent paths have opposite orientations on a shared face, so their signed sum has exactly the product boundary. This is the realization comparison, hence an equivalence.

For $I = I_1 \sqcup I_2$ on the disjointness locus, the given algebra factorization identifies A_I with $A_{I_1} \boxtimes A_{I_2}$. The shuffle therefore gives the comparison $BA_{I_1} \boxtimes BA_{I_2} \to BA_I$. Its inverse in the derived category is the bar factorization map. The coalgebra identity just proved gives compatibility with deconcatenation. For several blocks, iterated shuffles enumerate the same ordered multishuffles, with the same Koszul signs. This proves refinement coherence. Naturality and the stated functor compatibility give the collision coherence. □

The restriction and collision compatibility in this proposition is a hypothesis on the chosen geometric model. It does not follow from naming an ambient stable category. Conilpotence here refers to finite normalized words in that model. No word-length completion is implicit.

27.2 Three coproducts

The theorem system uses three coproducts with distinct geometric meanings:

$$\Delta_{\text{fact}} : \text{separation of chiral support},$$
$$\Delta_{\text{dec}} : \text{cutting an ordered bar word},$$
$$\Delta_{\text{Hopf}} : \text{fusion of representations}.$$

The first two are present on the transverse bar. The third is additional bialgebra data on an acting algebra.

Proposition 27.3 (Coproduct separation)**.** *Factorization separation is carried by every ordered coefficient. Deconcatenation is generated functorially by the bar construction. A Hopf coproduct is supplied by an independent bialgebra structure. A coalgebra comparison morphism relates the Hopf coproduct to either geometric coproduct whenever a theory identifies their carriers.*

Proof. Factorization separation is part of the coefficient object, and deconcatenation is the canonical cut map on tensor words. A Hopf coproduct is a map from the algebra to a completed tensor square satisfying algebraic coassociativity. Its source, target, and defining universal property differ from the first two, so a relation among them is exactly the data of a comparison morphism. □

27.3 Interval factorization homology

For a right A-module R and a left A-module L, factorization homology of an interval with endpoint labels gives

$$\int_{([0,1],\{0,1\})} (A; R, L) \simeq R \otimes^{\mathbb{L}}_{A} L.$$

This statement holds internally in every presentable monoidal category supporting factorization homology [4, 114]. Taking $R = L = \mathbf{1}$ identifies the interval amplitude with the ordered bar.

Chapter 28

Cobar and twisting cochains

Let C be a coaugmented conilpotent differential graded coalgebra in the ambient coefficient model. The ordinary, direct-sum tensor cobar algebra is

$$\mathrm{Cobar}_X^{\mathrm{ord}}(C) = \big(T(s^{-1}\bar{C}),\, d_{\mathrm{int}} + d_{\Delta}\big).$$

A twisting cochain $\tau : C \to A$ has degree 1, vanishes on the coaugmentation, and has zero composite with the augmentation of A. It satisfies

$$d\tau + \tau d + m(\tau \otimes \tau)\bar{\Delta} = 0.$$

The universal twisting cochains are

$$\tau_A : \mathrm{Bar}_X^{\mathrm{ord}}(A) \to A, \qquad \tau_C : C \to \mathrm{Cobar}_X^{\mathrm{ord}}(C).$$

Theorem 28.1 (Bar–cobar adjunction)**.** *For ordinary augmented dg algebras and coaugmented conilpotent dg coalgebras in the specified coefficient model, tensor cobar is left adjoint to tensor bar. Their mapping set is the set of twisting cochains. An internal adjunction on factorization objects additionally requires both constructions to produce factorization objects and their twisting maps to preserve the geometric data.*

Proof. An algebra map from the tensor algebra $T(s^{-1}\bar{C})$ is determined by its restriction to $s^{-1}\bar{C}$. Compatibility with the cobar differential is the Maurer–Cartan equation above. The same equation describes coalgebra maps from C to the cofree conilpotent coalgebra $T^c(s\bar{A})$. □

Chapter 29

Complete bar–cobar reconstruction

Ordinary tensor cobar uses finite words. Completion changes that carrier. Let k be a commutative ring and $A_{\mathrm{alg}} = \bigoplus_{w \geq 0} A_w$ an augmented algebra in degree zero, with $A_0 = k$, augmentation given by projection onto A_0, and multiplication adding weights. Put $D = \Omega B(A_{\mathrm{alg}})$ using ordinary direct-sum tensors, and let D_w be its weight-w subcomplex. The complete algebra and its complete resolution are

$$\widehat{A} = \prod_{w \geq 0} A_w, \qquad \widehat{D} = \prod_{w \geq 0} D_w.$$

Products of complexes are taken degreewise, with the weight topology.

Theorem 29.1 (Complete reconstruction). *The componentwise counit $\widehat{D} \to \widehat{A}$ is a quasi-isomorphism of complete differential graded algebras.*

Proof. A weight-w block word has at most w positive-weight entries. Filter by their number. Multiplication lowers this number and splitting preserves it. At a fixed list of $N \geq 2$ entries, the splitting complex is the tensor product of the coefficient module with $N - 1$ copies of $[k \xrightarrow{1} k]$ in degrees $-1, 0$. Changing the first outer cut to an inner gap contracts it. The other differential terms cancel by their prefix signs. This identity survives tensoring with any coefficient module. Subtracting a primitive at the largest entry number terminates, so negative cohomology vanishes. In degree zero, the boundaries impose $c(a)c(b) = c(ab)$, leaving exactly A_w. Taking the degreewise product preserves kernels and images of these differentials, and therefore preserves the computed cohomology. □

For $k[x]$ with weight one on x, the ordinary counit recovers $k[x]$, whereas this weight completion recovers $k[[x]]$. A completed geometric bar–cobar equivalence needs a specified completed tensor product, coalgebra category, and comparison maps compatible with collision and inverse limits. An operadic nilcompletion or a boundary double centralizer is not identified with the ordinary tensor counit by this weight calculation. The coalgebra-unit and geometric duality statements require their own convergence and comparison arguments.

29.1 Quadratic recognition

Let $A = T(V)/(R)$ with $R \subset V \otimes V$. Its quadratic dual coalgebra $A^{\mathrm{q!}}$ is cogenerated by sV with corelations $s^2 R$. The canonical map $A^{\mathrm{q!}} \to \mathrm{Bar}(A)$ gives the Koszul complex.

Theorem 29.2 (Quadratic Koszul criterion). *Assume positive weight and degreewise finite duals. The following conditions are equivalent:*

(i) $A^{\mathrm{q!}} \to \mathrm{Bar}(A)$ *is a quasi-isomorphism;*

(ii) $\mathrm{Cobar}(A^{\mathrm{q!}}) \to A$ *is a quasi-isomorphism;*

(iii) *the left and right Koszul complexes have homology concentrated in weight zero;*

(iv) $\mathrm{Tor}_i^A(\mathbf{1}, \mathbf{1})$ *is concentrated in internal weight i.*

Proof. The universal twisting morphism $A^{\mathrm{q}!} \to A$ is Koszul exactly when the associated twisted tensor products resolve the unit. The fundamental theorem of twisting morphisms identifies this condition with the bar and cobar quasi-isomorphisms. Weight purity is the homological form of the same statement [111, 126]. □

Chapter 30

Modules, comodules, and Morita theory

The ordinary tensor counit reconstructs an algebra. Reconstruction of its modules requires the unit and counit of the twisted module functors as well. The following weight conditions make these maps finite in every component.

Let k be a field and let $A_{\mathrm{alg}} = \bigoplus_{w \geq 0} A_w$ be an associative algebra in cohomological degree zero. Assume $A_0 = k$, multiplication adds weights, and augmentation projects onto A_0. Put $C_{\mathrm{alg}} = B(A_{\mathrm{alg}})$, with its bar weight grading, and set

$$A = \prod_{w \geq 0} A_w, \qquad C = \prod_{w \geq 0} (C_{\mathrm{alg}})_w.$$

Products are degreewise in complexes. The universal twisting map $\tau : C \to A$ has degree one, sends $[a]$ to a, and vanishes on the empty word and all longer words. The bar differential has the earlier convention $b_2[a|b] = [ab]$.

A complete weight-graded left A-module here means a product $M = \prod_{w \geq w_0} M_w$ of complexes, for some integer w_0, with differential preserving weight and actions $A_u \otimes M_v \to M_{u+v}$. A complete weight-graded left C-comodule means such a product N with a counital coassociative coaction $\rho : N \to C \widehat{\otimes} N$ preserving weight and differential. The completed tensor product is defined componentwise by

$$(C \widehat{\otimes} N)_w = \bigoplus_{u+v=w} (C_{\mathrm{alg}})_u \otimes N_v.$$

This sum is finite since $u \geq 0$ and $v \geq w_0$. Each reduced coaction lowers the weight of its final N-component. It therefore vanishes after more than $w - w_0$ iterations on N_w. These comodules need no chosen coaugmentation.

Write $D_{\mathrm{wt}}(A)$ and $D_{\mathrm{wt}}(C)$ for the categories obtained by inverting maps that are quasi-isomorphisms in every weight, respectively for these modules and comodules. This specifies the weak equivalences; no coderived localization is implicit. Define

$$F(M) = C \widehat{\otimes}_\tau M, \qquad G(N) = A \widehat{\otimes}_\tau N.$$

The first has the coaction induced by Δ_C and the second the left A-action. Writing $\Delta(c) = \sum c_{(1)} \otimes c_{(2)}$ and $\rho(n) = \sum n_{(-1)} \otimes n_{(0)}$, their differentials are

$$d_F(c \otimes m) = d_C c \otimes m + (-1)^{|c|} c \otimes d_M m + \sum (-1)^{|c_{(1)}|} c_{(1)} \otimes \tau(c_{(2)}) m,$$
$$d_G(a \otimes n) = a \otimes d_N n - \sum a \tau(n_{(-1)}) \otimes n_{(0)}.$$

All sums are finite in each weight. The twisting equation, associativity, and coassociativity make both differentials square to zero.

Theorem 30.1 (Koszul module equivalence in positive weights)**.** *These functors induce inverse equivalences*

$$D_{\mathrm{wt}}(A) \simeq D_{\mathrm{wt}}(C).$$

Their counit and unit are

$$\epsilon_M(a \otimes c \otimes m) = a\epsilon_C(c)m, \qquad \eta_N(n) = \sum n_{(-1)} \otimes 1 \otimes n_{(0)}.$$

Proof. The differential formulas show that these are cochain maps, respectively A-linear and C-colinear. The twisting terms cancel in the unit equation; the counit equation uses the module action. Coassociativity and the counit identities give the two triangle identities.

The complex $GF(M)$ is the two-sided normalized bar resolution $A\widehat{\otimes}B(A_{\mathrm{alg}})\widehat{\otimes}M$. For its contraction, use the simplicial boundary convention $\partial = \sum_i (-1)^i d_i$. Multiplying a bar word of length r by $(-1)^r$ translates this convention to the displayed twisting differentials. Writing $\bar{a} = a - \epsilon_A(a)1$, the extra degeneracy is

$$h\big(a_0[a_1|\cdots|a_r]m\big) = 1[\bar{a}_0|a_1|\cdots|a_r]m.$$

The first face returns the original term minus its scalar first factor. The other faces cancel against applying h after the boundary, including the module-action face. The internal differential of M cancels by the suspension sign. Thus $\partial h + h\partial = 1 - i\epsilon_M$, with $i(m) = 1[\,]m$. This is a linear contraction; i need not be A-linear. It preserves total weight, so the counit is a quasi-isomorphism in every weight.

At fixed total weight, filter $FG(N)$ by the weight of its final N-factor. This filtration is finite. The reduced-coaction term lowers that weight and disappears in the associated graded. The associated graded unit is therefore the inclusion into $B(k, A_{\mathrm{alg}}, A_{\mathrm{alg}}) \otimes N_v$. The rightmost extra degeneracy contracts this augmented bar onto k:

$$h\big([a_1|\cdots|a_r]a\big) = (-1)^{r+1}[a_1|\cdots|a_r|\bar{a}]1$$

in the same simplicial convention. The last face gives $1 - i\epsilon$; all other faces cancel in pairs. Consequently the unit is a quasi-isomorphism on each associated graded component, and the finite filtration proves the same for each weight of N.

Both functors preserve weightwise quasi-isomorphisms. Filter by the weight of the final module or comodule component. The twisting terms strictly change that weight in one direction. On associated graded components, the functors tensor complexes of k-vector spaces, which preserves quasi-isomorphisms. The filtration is finite at fixed total weight. Hence F, G descend to the stated localizations, where their unit and counit are isomorphisms. □

This theorem uses an explicit weight decomposition and a lower weight bound on every module and comodule. Derived I-adic completion, with $I = \ker(A \to k)$, instead uses $M \to \varprojlim_p (A/I^p \otimes^{\mathbb{L}}_A M)$. The two completions require a comparison of their filtrations. A coderived category also has a separately specified class of coacyclic complexes. Neither identification follows from the algebra counit alone. For right and left modules, the two-sided bar computes their relative derived tensor product in the same finite weight components. An identification with a geometric derived cotensor product additionally requires the two-sided twisting comparison and its compatibility with the chosen completion and collision maps.

Let C be a presentable factorization category and let b be a compact factorization generator. The internal endomorphism object

$$A_b = \mathrm{RHom}_C(b, b)$$

has ordered multiplication by composition.

Theorem 30.2 (Factorization Morita theorem)**.** *Assume b generates each local category and the local Morita equivalences commute with factorization descent. Then*

$$C \simeq \mathrm{Mod}_{A_b}(\mathrm{Fact}_D(X)).$$

Proof. The compact-generator Morita theorem applies on every factorization chart. The descent hypothesis identifies the equivalences on intersections and hence on the Ran colimit. □

Chapter 31

Derived centre and bulk action

The ordered Hochschild cochains are

$$Z_{\mathrm{ord}}(A) = \mathrm{RHom}_{A \otimes A^{\mathrm{op}}}(A, A).$$

Theorem 31.1 (Higher Deligne centre)**.** *Let the ambient coefficient category be presentable, stable, symmetric monoidal, closed, and let tensor product preserve geometric realizations separately. Then $Z_{\mathrm{ord}}(A)$ carries a canonical E_2-algebra structure. It is the universal E_2 algebra acting centrally on A.*

Proof. Hochschild cochains carry the brace operations, and the brace operad is equivalent to chains on little two-disks. This is the Deligne theorem [104, 117]. Universality is the identification of Hochschild cochains with endomorphisms of the identity A-bimodule. □

A boundary condition in a field theory gives a morphism

$$\beta : \mathrm{Obs}_{\mathrm{bulk}}|_{\partial} \longrightarrow Z_{\mathrm{ord}}(A_{\partial}).$$

Locality makes the action central. A boundary condition is open–closed complete on a protected sector when β is an equivalence on that sector.

For a Calabi–Yau category, Hochschild cochains carry the framed E_2 refinement of the cyclic Deligne theorem [16].

Chapter 32

Cyclic closure and modular sewing

A cyclic trace becomes a surface theory after dualizability and nondegeneracy are supplied.

Definition 32.1 (Perfect Calabi–Yau ordered algebra)**.** A perfect Calabi–Yau ordered algebra is a dualizable ordered algebra A together with a trace

$$\mathrm{tr} : \mathrm{HH}_0(A) \longrightarrow \mathbf{1}[-d]$$

whose induced pairing $A \otimes A \to \mathbf{1}[-d]$ is nondegenerate.

Theorem 32.2 (Ribbon-graph realization of a Calabi–Yau algebra)**.** *Let A be a homologically finite, smooth and proper cyclic A_∞ algebra with a nondegenerate pairing of degree d. Then the cyclic bar construction of A defines an open topological conformal field theory. Its operations are indexed by chains on moduli spaces of bordered surfaces, equivalently by the corresponding ribbon-graph complexes. The value on the circle is Hochschild homology $\mathrm{HH}_*(A)$.*

Proof. The nondegenerate cyclic pairing identifies outgoing and incoming boundary labels and contracts the two flags of every internal ribbon edge. The A_∞ identities are exactly the boundary identities for metric ribbon graphs. Smoothness and properness give the dualizability and finite pairing required for all contractions. Costello's construction then produces the open TCFT and identifies the closed state complex with Hochschild chains [29]. □

Remark 32.3. The trace functional supplies the annular operation. Smoothness, properness, and the nondegenerate Calabi–Yau pairing extend it to the complete ribbon-surface theory of the theorem.

Stable-curve sewing is supplied by a modular operad action. Its contractions form the modular structure carried by the chiral coefficient. The two sewing geometries coexist when their contraction maps supercommute.

Book VII

Exact noncommutative models

Chapter 33

Free and square-zero pairs

Work in a presentable stable linear closed monoidal coefficient category C whose tensor preserves coproducts separately and is exact. For $M \in C$, put

$$T_X(M) = \mathbf{1} \oplus M \oplus M^{\otimes 2} \oplus \cdots$$

This is the free augmented algebra in C. All sums and shifts are formed in C. Componentwise freeness alone supplies no separated-locus equivalences or collision maps for a factorization realization.

Theorem 33.1 (Free–square-zero duality)**.** *Assume M is dualizable and use the continuous dual for word weight. Let* Bar *in this statement denote the ordinary augmented bar in C, and put $sM = M[1]$. Then*

$$\mathrm{Bar}(T_X(M)) \simeq \mathbf{1} \oplus sM, \qquad \mathrm{RHom}_{T_X(M)}(\mathbf{1}, \mathbf{1}) \simeq \mathbf{1} \oplus M^\vee[-1]$$

with square-zero multiplication on the reduced summand. Conversely, for $A = \mathbf{1} \oplus M$ with $M^2 = 0$,

$$\mathrm{Bar}(A) = T^c(sM), \qquad A^! \simeq \widehat{T}(M^\vee[-1]).$$

Proof. In weight r, bar terms are copies of $M^{\otimes r}$ indexed by cuts of a word of length r. For $r \geq 2$, insertion of the first possible cut contracts the cut complex. The suspension signs give the total complex homotopy. In weight one the primitive class sM survives.

After the suspension signs are identified, the cut complex is the exterior algebra on the $r - 1$ possible cuts, with differential ι_λ, where $\lambda(e_i) = 1$. Exterior multiplication by e_1 is a contraction because $\iota_\lambda(e_1 \wedge -) + e_1 \wedge \iota_\lambda = 1$. Dualizing this primitive coalgebra gives square-zero multiplication. For the square-zero algebra all multiplication faces vanish, leaving the cofree tensor coalgebra. Its continuous weight dual is the completed tensor algebra. □

Chapter 34

The quantum plane

Let

$$A_q = \Bbbk\langle x, y\rangle/(yx - qxy), \qquad q \in \Bbbk^\times.$$

Give x and y weight one.

Theorem 34.1 (PBW basis of the quantum plane)**.** *The monomials $x^a y^b$, $a, b \geq 0$, form a basis of A_q.*

Proof. Use the rewriting rule $yx \mapsto qxy$. Every reduction decreases the inversion number of a word. The leading word yx has distinct initial and terminal letters, so every overlap decomposes into independent reductions. Bergman's diamond argument therefore gives the ordered monomials as normal forms. □

The quadratic dual is

$$A_q^! = \Bbbk\langle \xi, \eta\rangle/(\xi^2, \eta^2, \xi\eta + q\eta\xi).$$

Theorem 34.2 (Koszulness)**.** *A_q is Koszul, and its Koszul complex is*

$$0 \to A_q \otimes \Bbbk(\xi\eta) \xrightarrow{d_2} A_q \otimes (\Bbbk\xi \oplus \Bbbk\eta) \xrightarrow{d_1} A_q \to \Bbbk \to 0,$$

with

$$d_1(a \otimes \xi + b \otimes \eta) = ax + by, \quad d_2(a \otimes \xi\eta) = ay \otimes \xi - qax \otimes \eta.$$

Proof. Direct substitution gives $d_1 d_2 = ayx - qaxy = 0$. The image of d_1 is the augmentation ideal (x, y), so the complex is exact at A_q and has cokernel $\Bbbk$. We now compute $\ker d_1$.

Write

$$u = \sum_{i,j\geq 0} u_{ij} x^i y^j, \qquad v = \sum_{i,j\geq 0} v_{ij} x^i y^j.$$

The relation $y^j x = q^j x y^j$ gives

$$ux + vy = \sum_{a,b\geq 0} \left(q^b u_{a-1,b} + v_{a,b-1}\right) x^a y^b,$$

where coefficients with a negative index are zero. Thus $ux + vy = 0$ implies $u_{i,0} = 0$, $v_{0,j} = 0$, and

$$v_{a,b} = -q^{b+1} u_{a-1,b+1}.$$

Define $c = \sum_{i,j\geq 0} u_{i,j+1} x^i y^j$. Then $u = cy$ and, using $x^i y^j x = q^j x^{i+1} y^j$,

$$v = -qcx.$$

Hence $(u, v) = d_2(c)$. Finally, $d_2(c) = 0$ implies $cy = 0$; the PBW basis and $q \neq 0$ show that A_q is a domain, so $c = 0$. The complex is exact, and the quadratic Koszul criterion proves that A_q is Koszul. □

Theorem 34.3 (Yoneda algebra of the quantum plane)**.** *Use the convention that a Yoneda product $\alpha\beta$ means first β and then α. The augmentation module has*

$$\operatorname{Ext}^{*}_{A_q}(\Bbbk,\Bbbk) \cong \Bbbk\langle \xi, \eta\rangle/(\xi^2, \eta^2, \xi\eta + q\eta\xi) = A_q^{!}$$

as a bigraded algebra, where $|\xi| = |\eta| = 1$ in cohomological degree and each generator retains its internal weight one.

Proof. Apply $\operatorname{Hom}_{A_q}(-,\Bbbk)$ to the minimal Koszul resolution in the preceding theorem. Every matrix entry of d_1 and d_2 lies in the augmentation ideal, so the dual differential vanishes. The cohomology has basis

$$1, \quad \xi, \eta, \quad \xi\eta$$

in cohomological degrees 0, 1, 1, 2. Splicing the two degree-one extensions gives the degree-two class. The syzygy $y \otimes \xi - qx \otimes \eta$ identifies the two splicings by $\xi\eta = -q\eta\xi$, while the absence of degree-two x^2 and y^2 syzygies gives $\xi^2 = \eta^2 = 0$. These relations account for the complete four-dimensional cohomology, and therefore present the Yoneda algebra. □

The bar, quadratic dual and derived augmentation boundary of A_q are ordinary algebraic constructions. Tensoring with a coefficient object gives an ordered algebra in the ambient pointwise coefficient category when that tensor is defined. A factorization realization additionally requires its separated-locus equivalences and collision maps. These do not follow from componentwise tensoring with A_q.

Chapter 35

Quiver path algebras

Let Q be a finite quiver over a field $\Bbbk$ and $S = \bigoplus_{i \in Q_0} \Bbbk e_i$. Assign a vector space M_a to each arrow $a : i \to j$ and set

$$M = \bigoplus_{a \in Q_1} e_j M_a e_i, \qquad T_S(M) = S \oplus M \oplus M \otimes_S M \oplus \cdots .$$

Words are composable paths. These are ordinary S-bimodules. Placing the construction in geometric coefficients requires a functor preserving the relative tensors and bar realization, together with the intended separated-locus equivalences and collision maps.

For the quiver $1 \xrightarrow{a} 2$, the path algebra has basis e_1, e_2, a and relation $a = e_2 a e_1$. The simple modules S_1, S_2 satisfy

$$\operatorname{Ext}^1(S_1, S_2) \cong \Bbbk, \qquad \operatorname{Ext}^1(S_i, S_i) = 0, \qquad \operatorname{Ext}^1(S_2, S_1) = 0.$$

The bar differential records concatenation of paths, and the quadratic dual reverses the arrow with cohomological shift.

Theorem 35.1 (Relative path-algebra bar)**.** *For a tensor path algebra $T_S(M)$ over semisimple S,*

$$S \otimes^{\mathbb{L}}_{T_S(M)} S \simeq S \oplus sM$$

as an S-bicomodule. A quadratic quotient $T_S(M)/(R)$ is Koszul precisely when its relative quadratic complex is exact away from weight zero.

Proof. The last-arrow contraction is S-bilinear and gives the same extra degeneracy as in the free case. The quotient statement follows from the relative twisting-morphism criterion. □

Chapter 36

Weyl and Clifford deformations

Let V be finite-dimensional and let $\omega : V \otimes V \to \Bbbk$ be alternating. The Weyl algebra is

$$W_\hbar(V, \omega) = T(V)[[\hbar]]/\langle uv - vu - \hbar\omega(u, v)\rangle.$$

For a purely odd space and symmetric ω, the same formula with the supercommutator $uv + vu$ gives a Clifford algebra.

Theorem 36.1 (Filtered PBW theorem)**.** *For alternating ω in the even Weyl case, or symmetric ω in the purely odd Clifford case, the associated graded algebra is* $\mathrm{Sym}(V)$ *in the Weyl case and* $\bigwedge(V)$ *in the Clifford case.*

Proof. Filter by word length. The constant term lowers filtration by two, so the associated graded relation is the commutative or skew-commutative quadratic relation. The cubic ambiguities reduce to the cocycle identity for a central scalar form, which holds because scalars commute. The ordered monomials give the PBW basis. □

The quadratic-linear-constant presentation has a curved Koszul dual. The constant term becomes the curvature functional

$$m_0 : s^2 R \longrightarrow \Bbbk[2].$$

This construction is the algebraic origin of the curvature; geometric anomaly curvature enters through a separate comparison morphism.

Chapter 37

Associative chiral algebras on the formal disc

Let V be a complete topological vector space with vacuum $\mathbf{1}$ and translation T. A state–field map

$$Y(-,z) : V \longrightarrow \mathrm{End}(V)((z))$$

is a coordinate presentation of an associative chiral-convolution product when its iterated products are expansions of one factorization kernel in the nested regions $|z_1| > |z_2|$ and $|z_2| > |z_1|$.

Proposition 37.1 (Associative convolution from weak associativity)**.** *Suppose $Y(\mathbf{1},z) = \mathrm{id}$, $Y(a,z)\mathbf{1} \in a + zV[[z]]$, $[T, Y(a,z)] = \partial_z Y(a,z)$, and for every a, b, c there is N such that*

$$(z_1 - z_2)^N Y(a,z_1)Y(b,z_2)c = (z_1 - z_2)^N Y(Y(a,z_1 - z_2)b, z_2)c$$

in the common formal expansion ring. Then the fields define an associative chiral algebra on the formal disc. The converse holds for a vacuum-generated translation-equivariant associative chiral algebra admitting state–field reconstruction.

Proof. The displayed identity is the equality of the two nested restrictions of the three-point chiral product. Vacuum and translation give the unit and coordinate covariance. Factorization on higher configuration spaces is obtained by iterating the equality; coherence follows from associativity of the chiral-convolution product. □

Example 37.2 (Braided current fields)**.** Let $R(z)$ be an invertible solution of the spectral Yang–Baxter equation, normalized by $R^{21}(-z)R(z) = 1$. Generating fields $a_i(z)$ with exchange

$$a_i(z_1)a_j(z_2) = \sum_{k,l} R_{ij}^{kl}(z_1 - z_2)\, a_l(z_2)a_k(z_1)$$

form a quantum vertex algebra whenever the vacuum, translation, and weak associativity relations hold. The R-matrix supplies the comparison between the two formal expansion regions and records planar exchange in addition to transverse order.

Chapter 38

Zamolodchikov–Faddeev exchange

Let V be finite-dimensional and let $R(u) \in \operatorname{End}(V \otimes V) \otimes \Bbbk(u)$ be invertible. Introduce generators $Z_i(u)$ with exchange relation

$$Z_1(u) Z_2(v) = R_{12}(u - v) Z_2(v) Z_1(u).$$

Theorem 38.1 (Yang–Baxter confluence)**.** *The exchange algebra has a consistent normal ordering for three generators precisely when*

$$R_{12}(u-v) R_{13}(u-w) R_{23}(v-w) = R_{23}(v-w) R_{13}(u-w) R_{12}(u-v).$$

Proof. Reverse the word $Z_1(u) Z_2(v) Z_3(w)$. Exchanging positions in the order (12), (13), (23) gives the left side. Exchanging in the order (23), (13), (12) gives the right side. These are the two reductions of the unique length-three overlap. Equality is therefore the diamond condition. □

For the rational permutation solution

$$R(u) = 1 - \frac{\hbar P}{u},$$

the Yang–Baxter equation follows from $P_{ij}^2 = 1$ and $P_{12} P_{13} P_{23} = P_{23} P_{13} P_{12}$.

Chapter 39

The RTT Yangian of $\mathfrak{gl}_2$

Let

$$T(u) = 1 + \sum_{r \geq 1} T^{(r)} u^{-r}, \qquad R(u) = 1 - \frac{\hbar P}{u}.$$

The RTT Yangian $Y_\hbar(\mathfrak{gl}_2)$ is generated by the coefficients $t_{ij}^{(r)}$ subject to

$$R(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R(u-v).$$

Equivalently,

$$(u-v)[t_{ij}(u), t_{kl}(v)] = \hbar\big(t_{kj}(u)t_{il}(v) - t_{kj}(v)t_{il}(u)\big).$$

Proposition 39.1 (Coefficient relations)**.** *Put $t_{ij}^{(0)} = \delta_{ij}$. The RTT identity is equivalent to*

$$[t_{ij}^{(r+1)}, t_{kl}^{(s)}] - [t_{ij}^{(r)}, t_{kl}^{(s+1)}] = \hbar\big(t_{kj}^{(r)} t_{il}^{(s)} - t_{kj}^{(s)} t_{il}^{(r)}\big), \qquad r, s \geq 0.$$

In particular,

$$[t_{ij}^{(1)}, t_{kl}^{(1)}] = \hbar(\delta_{kj} t_{il}^{(1)} - \delta_{il} t_{kj}^{(1)}),$$

so the first modes form the enveloping algebra of the copy $\hbar\mathfrak{gl}_2$ inside the Yangian.

Proof. Multiply the generating-series relation by $u - v$ and compare the coefficient of $u^{-r-1}v^{-s-1}$. The two terms contributed by multiplication by $u - v$ produce the two commutators on the left, and the two ordered products on the right are the corresponding coefficients of the RTT numerator. Setting $r = 0$ gives the displayed first-mode relation. □

Theorem 39.2 (Hopf structure)**.** *The formulas*

$$\Delta(T(u)) = T(u) \dot{\otimes} T(u), \qquad \varepsilon(T(u)) = 1, \qquad S(T(u)) = T(u)^{-1}$$

define a Hopf algebra structure in the u^{-1}-adic completion. Coefficientwise,

$$\Delta(t_{ij}^{(r)}) = \sum_{a=1}^{2} \sum_{p=0}^{r} t_{ia}^{(p)} \otimes t_{aj}^{(r-p)}.$$

Proof. Apply the RTT relation in each tensor factor. Matrix multiplication in the auxiliary space gives the RTT relation for $\Delta(T)$. Coassociativity is associativity of matrix multiplication. Expanding the matrix product gives the coefficient formula. The inverse series exists uniquely because $T(u) = 1 + O(u^{-1})$; its defining identities give the antipode axioms. □

The evaluation map

$$\mathrm{ev}_a : T(u) \longmapsto 1 + \frac{\hbar E}{u - a}, \qquad E = (E_{ij}),$$

lands in $U(\mathfrak{gl}_2)[[u^{-1}]]$ and gives the standard evaluation modules. The PBW theorem identifies the associated graded with $U(\mathfrak{gl}_2[t])$ [49, 118].

This example contains three structures: ordered multiplication in the Yangian, coproduct on tensor products of modules, and exchange by the rational R-matrix.

Chapter 40

The Hall algebra of the A_2 quiver

Let $Q : 1 \to 2$ over $\mathbb{F}_q$. Write S_1, S_2 for the simples and M for the unique non-split extension

$$0 \to S_2 \to M \to S_1 \to 0.$$

In the twisted Hall algebra, multiplication counts subobjects with Euler-form normalization. Direct counting gives

$$[S_1] * [S_2] = v^{-1}[S_1 \oplus S_2] + v^{-1}[M], \qquad [S_2] * [S_1] = [S_1 \oplus S_2], \qquad v^2 = q.$$

Hence the root vector

$$E_{12} = E_1E_2 - v^{-1}E_2E_1$$

is represented by the indecomposable extension.

Theorem 40.1 (Quantum Serre relation)**.** *The Hall generators satisfy*

$$E_1^2E_2 - (v + v^{-1})E_1E_2E_1 + E_2E_1^2 = 0,$$

and the analogous relation with 1 *and* 2 *exchanged.*

Proof. Every representation of dimension vector $(2, 1)$ is a linear map $f : \mathbb{F}_q^2 \to \mathbb{F}_q$ of rank zero or one. Read a word from the top quotient to the bottom quotient. The complete-flag counts are

$\operatorname{rk} f$	112	121	211
0	$q+1$	$q+1$	$q+1$
1	$q+1$	1	0.

For rank zero every one-dimensional subspace of $\mathbb{F}_q^2$ is admissible, giving $q + 1$ flags in each word. For rank one, the word 121 forces the unique kernel line, the word 211 is incompatible with the final subrepresentation, and the word 112 allows every line at the remaining stage.

The total Euler-form exponents of the three words are respectively $-1, 0, 1$. Their twisted coefficients are therefore $v^{-1}, 1, v$. On the rank-zero stratum the Serre combination has coefficient

$$(q + 1)\big(v^{-1} - (v + v^{-1}) + v\big) = 0.$$

On the rank-one stratum it has coefficient

$$v^{-1}(q + 1) - (v + v^{-1}) = v^{-1}(v^2 + 1) - (v + v^{-1}) = 0.$$

The two isomorphism classes exhaust the dimension vector $(2, 1)$, proving the relation. This is Ringel's realization of $U_v^+(\mathfrak{sl}_3)$ [73, 129]. □

Chapter 41

Matrix traces and the necklace bracket

Let $F = \Bbbk\langle x, y\rangle$ and $F_{\mathrm{cyc}} = F/[F, F]$. Cyclic derivatives are defined by

$$\partial_x^{\mathrm{cyc}}[a_1 \cdots a_n] = \sum_{a_i = x} a_{i+1} \cdots a_n a_1 \cdots a_{i-1}$$

and similarly for y. Define

$$\{f, g\}_{\mathrm{neck}} = \left[\partial_x^{\mathrm{cyc}} f \, \partial_y^{\mathrm{cyc}} g - \partial_y^{\mathrm{cyc}} f \, \partial_x^{\mathrm{cyc}} g\right]_{\mathrm{cyc}}.$$

Theorem 41.1 (Trace realization)**.** *For matrices $X, Y \in \mathfrak{gl}_N$ and the canonical symplectic bracket on matrix entries,*

$$\{\operatorname{Tr} f(X,Y), \operatorname{Tr} g(X,Y)\} = \operatorname{Tr}\{f, g\}_{\mathrm{neck}}(X,Y).$$

Proof. Differentiate the trace by moving the varied matrix entry cyclically to the first position. This gives the cyclic derivative. Contract the X derivative of one trace with the Y derivative of the other and subtract the reversed contraction. The matrix indices close into the trace of the cyclic product. □

In the stable commuting-matrix quotient, let

$$p_{ab} = \operatorname{Tr}(X^a Y^b).$$

Then

$$\{p_{ab}, p_{cd}\} = (ad - bc)\, p_{a+c-1, b+d-1}.$$

For two eigenvalues, the relative invariants

$$A = (x_1 - x_2)^2, \quad B = (x_1 - x_2)(y_1 - y_2), \quad C = (y_1 - y_2)^2$$

satisfy $B^2 = AC$ and carry the standard $\mathfrak{sl}_2$ Poisson bracket after linear rescaling.

Book **VIII**

The nonlocal ladder

Chapter 42

Spectral exchange as an associative chiral operation

42.1 The coefficient field and the exchange operator

Let V be a finite-dimensional vector space and let K be a field of meromorphic functions in spectral differences. Fix an invertible operator

$$R(u) \in \operatorname{End}(V \otimes V) \otimes K$$

that satisfies the spectral Yang–Baxter equation

$$R_{12}(u-v)R_{13}(u-w)R_{23}(v-w) = R_{23}(v-w)R_{13}(u-w)R_{12}(u-v). \tag{42.1}$$

The equation states that two routes through the chamber complex of three labelled points give the same transport.

Definition 42.1 (Zamolodchikov–Faddeev algebra)**.** The exchange algebra $\mathcal{Z}_R$ is generated by the components of a current $A(u) \in V \otimes \mathcal{Z}_R$ with relation

$$A_1(u)A_2(v) = R_{12}(u-v)A_2(v)A_1(u). \tag{42.2}$$

A normal word has spectral labels in a fixed chamber $u_1 \succ \cdots \succ u_n$.

Theorem 42.2 (Confluence is Yang–Baxter)**.** *The quadratic rewriting system defined by* (42.2) *is confluent on words with pairwise distinct spectral labels precisely when* (42.1) *holds. In that case the normal ordered words form a K-basis of $\mathcal{Z}_R$.*

Proof. Every two-letter inversion is removed by one application of (42.2). The minimal overlap occurs in a word of length three. Moving the leftmost label through the other two gives the coefficient $R_{12}R_{13}R_{23}$, while moving the rightmost label in the opposite sequence gives $R_{23}R_{13}R_{12}$. Equality of the two coefficients is (42.1). Bergman's diamond lemma then gives the normal-word basis. □

42.2 A diagonal exchange model

Choose a basis $e_1, \dots, e_N$ of V and functions $q_{ij}(u)$ satisfying

$$q_{ij}(u)q_{ji}(-u) = 1, \qquad q_{ii}(u) = 1.$$

The diagonal operator

$$R(u)(e_i \otimes e_j) = q_{ij}(u)e_i \otimes e_j$$

satisfies the Yang–Baxter equation because all three factors are scalars. The generators $a_i(u)$ obey

$$a_i(u)a_j(v) = q_{ij}(u-v)a_j(v)a_i(u).$$

Every ordered monomial is a basis vector. The quantum plane is the constant two-generator specialization $q_{12} = q$, $q_{21} = q^{-1}$.

Calculation 42.3 (Three-particle coefficient)**.** For $u \succ v \succ w$,

$$a_i(w)a_j(v)a_k(u) = q_{ij}(w-v)q_{ik}(w-u)q_{jk}(v-u)a_k(u)a_j(v)a_i(w).$$

The coefficient is independent of the sequence of adjacent exchanges. This scalar identity is the diagonal Yang–Baxter equation.

Proof. Each inversion of two adjacent letters contributes its scalar exchange factor. Both reduced words for the longest permutation invert the same three pairs. Scalar commutativity makes the products identical. □

Chapter 43

RTT and reflection algebras

43.1 The RTT bialgebra

Let $T(u) = (t_{ij}(u))$ be a matrix of formal currents. The RTT algebra $\mathcal{T}_R$ is defined by

$$R_{12}(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R_{12}(u-v). \tag{43.1}$$

Theorem 43.1 (RTT coproduct)**.** *The formula*

$$\Delta(T(u)) = T_{13}(u)T_{12}(u), \qquad \epsilon(T(u)) = I,$$

defines a coassociative bialgebra structure on $\mathcal{T}_R$.

Proof. Apply (43.1) in each tensor factor. The Yang–Baxter equation moves the three R-operators through the three matrix factors. The resulting identity is the RTT relation for $\Delta(T)$. Matrix multiplication is associative, hence the two iterated coproducts both equal $T_{14}T_{13}T_{12}$. □

For the Yang matrix $R(u) = 1 - \hbar P/u$, expansion at $u = \infty$ gives the RTT Yangian. The coefficient of $u^{-1}v^{-1}$ is

$$[t_{ij}^{(1)}, t_{kl}^{(1)}] = \hbar\big(\delta_{kj}t_{il}^{(1)} - \delta_{il}t_{kj}^{(1)}\big),$$

so the first current layer is the Rees enveloping algebra of $\mathfrak{gl}_N$.

43.2 Boundary exchange

A reflecting boundary reverses the spectral path. For a constant invertible $R \in \mathrm{End}(V^{\otimes 2})$ satisfying Yang–Baxter, define the reflection algebra $\mathcal{K}_R$ by

$$R_{21}K_1R_{12}K_2 = K_2R_{21}K_1R_{12}. \tag{43.2}$$

Theorem 43.2 (Boundary confluence)**.** *The relation* (43.2) *gives a coherent exchange law for three boundary insertions. The two reductions of every length-three overlap agree by the Yang–Baxter equation and its inverse.*

Proof. A boundary exchange consists of an incoming crossing, reflection, and outgoing crossing. The two reductions of three insertions sweep the same braid in the doubled half-plane. Algebraically, one inserts (43.2) three times and uses Yang–Baxter to pass the middle R-matrix through the outer pair. The remaining factors cancel in inverse pairs. □

The RTT algebra governs bulk line exchange. The reflection algebra governs boundary line exchange. The ZF algebra governs charged fields in a fixed representation. Their common coefficient is the same R-matrix, while their products live in three different endomorphism objects.

Chapter 44

Quantum vertex realization

44.1 Two expansion regions

Let V be a complete topological vector space with vacuum $\mathbf{1}$, translation T, state-field map Y, and a unitary rational operator

$$S(z) : V \otimes V \longrightarrow V \otimes V((z)).$$

The two formal regions $|z_1| > |z_2|$ and $|z_2| > |z_1|$ are related by $S(z_1 - z_2)$.

Definition 44.1 (Rational quantum vertex datum)**.** A rational quantum vertex datum satisfies vacuum and translation covariance together with

$$(z_1 - z_2)^N Y(a, z_1) Y(b, z_2) = (z_1 - z_2)^N Y^{21}(S(z_1 - z_2)(a \otimes b); z_2, z_1) \tag{44.1}$$

for a sufficiently large N, and weak associativity in every triple of states.

Theorem 44.2 (Exchange coherence)**.** *Suppose S is unitary,*

$$S^{21}(-z) S(z) = 1,$$

and satisfies the spectral Yang–Baxter equation. Then the pairwise continuation maps obtained from (44.1) *define a braid-group action on every n-point field product. Weak associativity makes the action compatible with iterated operator products.*

Proof. Unitarity identifies a crossing followed by its reverse with the identity. Yang–Baxter identifies the two continuations around every codimension-two triple-collision stratum. These are the defining relations of the braid group. Weak associativity identifies the collision limit in each chamber with the corresponding iterated field. □

When S is the graded flip, the datum is an ordinary local vertex algebra. A nontrivial S retains the ordered chiral product and its exchange monodromy. This is the formal-disc realization of an associative chiral algebra with braiding data in the sense of quantum vertex algebra theory [108, 109].

44.2 The Yangian field

Let $T(u)$ act on a family of evaluation modules. The RTT relation becomes S-locality for the matrix coefficients of $T(u)$. The normalized Yang operator

$$S(u) = f(u)\left(1 - \frac{\hbar P}{u}\right), \qquad f(u) f(-u) = \left(1 - \frac{\hbar^2}{u^2}\right)^{-1},$$

is unitary. Its expansion at infinity gives the quantum vertex exchange of Yangian currents. The scalar factor changes normalization and preserves the Yang–Baxter equation.

Chapter 45

Shuffle realization and the positive current algebra

45.1 Pairwise kernel

Let $K(z)$ be a rational function and put

$$\mathrm{Sh}_n = \mathbb{C}(z_1, \dots, z_n)^{\mathfrak{S}_n}.$$

Write Sym for the sum over ordered two-block partitions, with no factorial divisor. For $f \in \mathrm{Sh}_m$ and $g \in \mathrm{Sh}_n$ define

$$(f * g)(z_1, \dots, z_{m+n}) = \mathrm{Sym}\left(f(z_1, \dots, z_m)\, g(z_{m+1}, \dots, z_{m+n}) \prod_{i \le m < j} K(z_i - z_j) \right). \tag{45.1}$$

Theorem 45.1 (Associativity of the shuffle product). *The product* (45.1) *is associative on every subspace closed under the displayed symmetrization and pole conditions.*

Proof. In a triple product, every ordered pair of variables belonging to different blocks contributes exactly one factor $K(z_i - z_j)$. The product of all pairwise factors is independent of the parenthesization. Final symmetrization gives the same function. □

A meromorphic three-parameter kernel is

$$K(z) = \prod_{a=1}^{3} \frac{z + \epsilon_a}{z}, \qquad \epsilon_1 + \epsilon_2 + \epsilon_3 = 0.$$

A critical Hall or Yangian realization additionally specifies its coefficient lattice, wheel conditions and comparison map [130, 133]. The displayed meromorphic kernel alone does not identify that realization. The polynomial Hall kernel of Section 6.2 has its own coefficient lattice and a different power of the root difference.

Calculation 45.2 (Degree one against degree one). For $f(z_1) = z_1^r$ and $g(z_1) = z_1^s$,

$$f * g = z_1^r z_2^s K(z_1 - z_2) + z_2^r z_1^s K(z_2 - z_1).$$

Put $u = z_1 - z_2$ and $\sigma_3 = \epsilon_1\epsilon_2\epsilon_3$. The antisymmetric part is

$$\frac{2\sigma_3}{u^3}(z_1^r z_2^s - z_1^s z_2^r).$$

For generic nonzero parameters, $u = -\epsilon_a$ is a zero of $K(u)$ and contributes no residue of this commutator. For $(r, s) = (2, 0)$, the residue at the true diagonal $u = 0$, with z_2 fixed, is $2\sigma_3$.

Proof. The two ordered partitions give the displayed product. Subtract the reversed product and use $K(u) - K(-u) = 2\sigma_3/u^3$. For $(r,s) = (2,0)$, substitute $z_1 = z_2 + u$. The result is $4\sigma_3 z_2/u^2 + 2\sigma_3/u$. For distinct nonzero parameters, the reciprocal has instead

$$\operatorname{Res}_{u=-\epsilon_a} K(u)^{-1}\,du = \frac{(-\epsilon_a)^3}{\prod_{b\neq a}(\epsilon_b - \epsilon_a)}.$$

A geometric operation must specify which meromorphic coefficient it uses. □

45.2 Coproduct and double

A chamber expansion fixes a coefficient completion, but it does not prove compatibility of a proposed coproduct with multiplication. Proposition 6.3 gives a charge-two obstruction for the ordinary tensor on the polynomial Hall carriers. Proposition 10.30 constructs a positive rational cooperation with a specified changed middle interchange. A Drinfeld double additionally requires compatible positive and negative Hopf data, a continuous pairing and a completion in which its canonical tensor exists. Under those hypotheses the double construction gives the quasitriangular structure.

Chapter 46

The ladder in one diagram

The nonlocal structures form the chain

$$\text{ordered coefficient} \longrightarrow \mathcal{Z}_R \longrightarrow \mathcal{T}_R \longrightarrow (V, Y, S),$$

$$(V, Y, S) \longrightarrow \mathrm{Sh}^{+} \longrightarrow D\big(\mathrm{Sh}^{+}\big).$$

The arrows retain different data:

Object	Defining operation	Coherence
$\mathcal{Z}_R$	exchange of charged fields	Yang–Baxter equation
$\mathcal{T}_R$	matrix-current product and coproduct	RTT and coassociativity
$\mathcal{K}_R$	boundary exchange	reflection equation
(V, Y, S)	formal field product	S-locality and weak associativity
Sh^{+}	extension shuffle	pairwise-kernel associativity
$D(\mathrm{Sh}^{+})$	positive–negative pairing	quasitriangularity

This ladder gives a concrete generator-and-relation theory of doubly noncommutative chiral observables. The first noncommutativity is ordered composition. The second is exchange between distinct order chambers. Chiral singularity remains the diagonal coefficient throughout.

Book IX

Field theory and master equations

Chapter 47

A finite cotangent BF model

Let L be a finite-dimensional dg Lie algebra. Its cotangent BV space is

$$\mathcal{E}_L = L[1] \oplus L^*[-2],$$

with the canonical degree -1 symplectic pairing. Write $A \in L[1]$ and $B \in L^*[-2]$. The cotangent action is

$$S_L(A, B) = \langle B, d_L A + \tfrac{1}{2}[A, A] \rangle.$$

Theorem 47.1 (Cotangent master equation)**.** *The Hamiltonian vector field of S_L is square-zero, and consequently $\{S_L, S_L\} = 0$.*

Proof. The A component is the Maurer–Cartan vector field $d_L A + \frac{1}{2}[A, A]$; the B component is its cotangent lift. The square of the first component is the sum of d_L^2, the derivation identity for d_L, and the Jacobi identity. The cotangent lift preserves commutators of vector fields, so the full Hamiltonian vector field is square-zero. □

Proposition 47.2 (Coordinate form of the modular term)**.** *Let L be an ordinary finite-dimensional Lie algebra with basis e_a and structure constants $[e_a, e_b] = f_{ab}^{c} e_c$. Let c^a be the odd coordinates on $L[1]$ and b_a their BV-conjugate coordinates. With*

$$S = \frac{1}{2} b_c f_{ab}^{c} c^a c^b, \qquad \Delta = -\sum_a \frac{\partial}{\partial c^a} \frac{\partial}{\partial b_a},$$

one has

$$\Delta S = \sum_b \operatorname{Tr}(\operatorname{ad}_{e_b}) c^b.$$

Thus the one-loop class of the finite cotangent theory is exactly the modular character.

Proof. Differentiation first in b_a gives $\frac{1}{2} f_{ij}^{a} c^i c^j$. Odd differentiation in c^a gives $f_{aj}^{a} c^j$, and the chosen sign in Δ converts this to $f_{ja}^{a} c^j = \operatorname{Tr}(\operatorname{ad}_{e_j}) c^j$. This is the displayed formula. □

A finite-dimensional dg Lie algebra is unimodular when the supertrace of every adjoint endomorphism vanishes. Every finite-dimensional nilpotent Lie algebra in degree zero is unimodular because each ad_x is nilpotent.

Theorem 47.3 (Finite-dimensional quantum master equation)**.** *Let L be unimodular and let Δ be the BV Laplacian determined by the translation-invariant Berezinian on $\mathcal{E}_L$. Then*

$$\Delta S_L = 0, \qquad \frac{1}{2}\{S_L, S_L\} + \hbar \Delta S_L = 0,$$

and equivalently

$$\hbar \Delta\left(e^{S_L/\hbar}\right) = 0.$$

The quantum observable differential

$$\delta_{S_L} = \{S_L, -\} + \hbar\Delta$$

satisfies $\delta_{S_L}^2 = 0$.

Proof. The classical master equation gives $\{S_L, S_L\} = 0$. The scalar ΔS_L is the divergence of the Chevalley vector field on $L[1]$. Its linear coefficient is the modular character $x \mapsto \mathrm{str}(\mathrm{ad}_x)$, which vanishes by unimodularity. The BV exponential identity

$$\hbar^2 e^{-S_L/\hbar}\Delta(e^{S_L/\hbar}) = \frac{1}{2}\{S_L, S_L\} + \hbar\Delta S_L$$

proves the exponential form. The standard BV commutator calculation then gives

$$\delta_{S_L}^2 = \left\{\frac{1}{2}\{S_L, S_L\} + \hbar\Delta S_L, -\right\} = 0.$$

□

Now take the local dg Lie algebra

$$\mathcal{L}_X = \Omega^{0,*}(X) \mathbin{\widehat{\otimes}} \Omega^*(\mathbb{R}) \otimes \mathfrak{l}$$

for a finite-dimensional nilpotent $\mathfrak{l}$, and form its shifted cotangent theory with compactly supported density dual. The preceding finite theorem governs every pairing-preserving finite-dimensional deformation retract. The local classical observables are Chevalley cochains and are holomorphic on X and locally constant on $\mathbb{R}$.

Choose the standard interval polarization in which the $L[1]$ variables are fixed at one end and the cotangent variables at the other. Canonical quantization of the boundary linear Poisson algebra gives the associative zero-mode algebra $U(\mathfrak{l})$. Gluing the two augmentation boundary states across an interval gives the derived tensor product

$$\mathbf{1} \otimes^{\mathbb{L}}_{U(\mathfrak{l})} \mathbf{1} \simeq \mathrm{Bar}(U(\mathfrak{l})) \simeq C_*(\mathfrak{l}),$$

the Chevalley–Eilenberg chains. A solution of the renormalized quantum master equation in the local BV deformation complex produces the corresponding infinite-dimensional quantum factorization algebra.

Chapter 48

BRST and ordered reduction

Let (A, Q) be a dg ordered algebra, with Q a derivation. Extend Q to the bar factorwise. Write b for the multiplication bar differential.

Proposition 48.1 (BRST–bar bicomplex)**.** *With the suspension sign convention,*

$$Q_{\mathrm{Bar}} b + b Q_{\mathrm{Bar}} = 0.$$

Hence $D = Q_{\mathrm{Bar}} + b$ satisfies $D^2 = 0$.

Proof. Apply Q to a bar face that multiplies adjacent entries. The derivation formula splits the result into the two terms obtained by applying Q before that face. The totalization sign makes their sum the anticommutator. □

The filtration by bar length gives

$$E_1 \cong \mathrm{Bar}(H^*(A, Q)) \Longrightarrow H^*(\mathrm{Bar}(A), D)$$

under boundedness and completeness. Quantum Drinfeld–Sokolov reduction is a current-algebra realization of this bicomplex.

Chapter 49

Stokes and algebraic identities

Let $C_*(K)$ be the oriented cellular chains of a regular manifold with corners. Suppose each codimension-one face F is assigned a degree-one operator ρ_F on a coefficient complex E, and the assignments are compatible with restrictions to corners.

Theorem 49.1 (Stokes-to-master)**.** *Assume that each ρ_F is a chain map of total degree one, that its restriction to a corner depends on the induced contraction, and that disjoint contractions act with the Koszul sign prescribed by the oriented normal bundle. Then the component of*

$$D^2, \qquad D = d_E + d_K + \sum_F \rho_F,$$

supported on a codimension-two corner S is the oriented sum

$$\sum_{F \supset S} \epsilon(F, S)\, \rho_{F/S} \rho_F.$$

Thus D is a differential precisely when these corner sums vanish.

Proof. Expand D^2 and group terms by support strata. The chain-map condition cancels the mixed internal terms. A term supported on S is obtained by crossing two incident faces. The two orders carry the boundary orientations of the two sides of the oriented normal square. Their signed sum is the displayed expression. □

Associahedra give Stasheff identities. Fulton–MacPherson collision strata give chiral and graph identities after a coefficient map is supplied. Stable-graph boundaries give BV and modular master equations.

Chapter 50

Two edge colors and two loop operators

Let $\mathsf{G}^{\perp,\mathrm{ch}}$ be the completed vector space spanned by stable graphs with transverse and chiral edges. Contracting an edge that joins distinct vertices gives brackets $[-,-]_{\perp}$ and $[-,-]_{\mathrm{ch}}$. Contracting a loop gives operators $\Delta_{\perp}$ and Δ_{ch}. Product orientation gives

$$\Delta_{\perp}\Delta_{\mathrm{ch}} + \Delta_{\mathrm{ch}}\Delta_{\perp} = 0.$$

Theorem 50.1 (Bicolored master equation). *Let $\mathsf{P}^{\perp,\mathrm{ch}}$ be a two-colored modular operad whose edge contractions carry the product determinant orientation, and let $\mathfrak{g}_{\mathsf{P}}$ be the completed convolution Lie algebra of its Feynman transform with an endomorphism modular operad. Algebra structures on the transform are Maurer–Cartan elements $\Theta \in \mathfrak{g}_{\mathsf{P}}^{1}$ satisfying*

$$d\Theta + \frac{1}{2}[\Theta,\Theta]_{\perp} + \frac{1}{2}[\Theta,\Theta]_{\mathrm{ch}} + \hbar_{\perp}\Delta_{\perp}\Theta + \hbar_{\mathrm{ch}}\Delta_{\mathrm{ch}}\Theta = 0.$$

Proof. The Feynman-transform differential contracts one edge. Separating edges give the two brackets and loop edges give the two BV operators. The product orientation makes mixed contractions anticommute. The condition that the structure map commute with the transform differential is exactly the displayed Maurer–Cartan equation [67]. □

Chapter 51

Typed anomalies

An anomaly is a cohomology class in the deformation complex that governs the attempted construction.

construction	complex	class
central extension	local Lie algebra cochains	cocycle
BRST quantization	BRST deformation complex	nilpotence obstruction
BV quantization	local BV cochains	quantum master anomaly
conformal-block family	Atiyah algebra of its determinant line	projective curvature
curved Koszul duality	curved convolution Lie algebra	curvature m_0

A comparison theorem consists of a morphism between two complexes and an equality of the transported classes. This formulation keeps the geometric origin of every scalar visible.

Book X

Calabi–Yau and automorphic interfaces

Chapter 52

Calabi–Yau categories

Let C be a smooth linear category with a d-Calabi–Yau structure. Two universal consequences are available.

Theorem 52.1 (Cyclic and shifted outputs)**.** *The Hochschild cochains $CC^*(C)$ form a framed E_2 algebra. The derived moduli stack of objects carries a $(2-d)$-shifted symplectic structure whenever the representability hypotheses hold.*

Proof. The first assertion is the cyclic Deligne theorem [16]. The second is the PTVV–Brav–Dyckerhoff construction: the Calabi–Yau trace pairs the tangent complex $\mathrm{RHom}(E,E)[1]$ with itself in degree $2-d$ [15, 125]. □

For a geometric BV theory or an open factorization category, an ordered chiral realization requires a coefficient family with the coherent multiplication and locality data of Theorem 24.2. Constructing this family and its comparison with the geometric theory is additional to the Calabi–Yau structure.

Chapter 53

Hall and BPS algebras

A Hall multiplication starts from the extension correspondence

$$\mathcal{M}_\alpha \times \mathcal{M}_\beta \xleftarrow{p} \mathcal{M}_{\alpha,\beta}^{\mathrm{ext}} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

With a coefficient theory for which p^* and q_* are defined,

$$a * b = q_* p^*(a \boxtimes b).$$

Associativity follows from the stack of two-step filtrations.

For 2-Calabi–Yau categories with good moduli spaces, the BPS algebra is the positive enveloping algebra of a generalized Kac–Moody Lie algebra [39]. For quivers and surface resolutions, preprojective and surface CoHAs recover positive Yangian halves in established cases [45, 130].

The ordered chiral structure enters when Hall multiplication is promoted to a factorization coefficient over a curve or defect parameter. A coproduct, pairing, and completion then produce the corresponding double.

Chapter 54

The Igusa denominator as a bar character

Let $\mathfrak{g}_\Delta$ be the Gritsenko–Nikulin Borcherds–Kac–Moody superalgebra and let $\mathfrak{n}_+$ be its positive nilpotent part. Root spaces are finite-dimensional and the positive cone is locally finite. Fix a proper integral height h that is positive on the effective root cone, and set

$$F^{>H}\mathfrak{n}_+ = \bigoplus_{h(\alpha)>H} \mathfrak{g}_{\Delta,\alpha}, \qquad \mathfrak{n}_+^{(H)} = \mathfrak{n}_+/F^{>H}\mathfrak{n}_+.$$

Height additivity makes $F^{>H}\mathfrak{n}_+$ an ideal; properness makes $\mathfrak{n}_+^{(H)}$ finite-dimensional and nilpotent. Every bar, Chevalley, enveloping, and character formula in this chapter is interpreted coefficientwise in the separated height completion.

Theorem 54.1 (Bar–Chevalley identification)**.** *For each height H there is a natural quasi-isomorphism of conilpotent coalgebras*

$$\mathrm{Bar}\big(U(\mathfrak{n}_+^{(H)})\big) \simeq C_*(\mathfrak{n}_+^{(H)}).$$

The quotient maps are compatible with these quasi-isomorphisms, and therefore induce a quasi-isomorphism in the separated height-complete category

$$\mathrm{Bar}_h^\wedge\big(U(\mathfrak{n}_+)\big) := \varprojlim_H \mathrm{Bar}\big(U(\mathfrak{n}_+^{(H)})\big) \simeq C_{*,h}^\wedge(\mathfrak{n}_+) := \varprojlim_H C_*(\mathfrak{n}_+^{(H)}).$$

The coefficientwise Euler supercharacter of the completed Chevalley complex is

$$\chi\, C_{*,h}^\wedge(\mathfrak{n}_+) = \prod_{\alpha>0} (1-e^{-\alpha})^{\operatorname{sdim}\mathfrak{g}_{\Delta,\alpha}}.$$

Proof. For every H, the PBW filtration on $U(\mathfrak{n}_+^{(H)})$ identifies the associated graded with $\mathrm{Sym}(\mathfrak{n}_+^{(H)})$. The bar complex of the enveloping algebra and the Chevalley complex have the same associated-graded Koszul resolution. The standard antisymmetrization map is a filtered quasi-isomorphism. Naturality for the quotient maps gives the inverse-limit comparison; proper height makes every homogeneous coefficient stabilize. For an even root vector the exterior Chevalley factor contributes $1-e^{-\alpha}$; for an odd root vector it contributes its inverse. Multiplication over the root spaces gives the product. □

With the positive-root convention $e^{-\rho} = q^{1/2}r^{1/2}s^{1/2}$, the denominator theorem reads

$$e^{-\rho}\,\chi\, C_{*,h}^\wedge(\mathfrak{n}_+) = D_5(2\Omega) = 2^{-6}\Delta_5(2\Omega)$$

[74, 76]. This identity is the exact meeting point of ordered bar theory and automorphic products.

Chapter 55

Finite-height currents and an exact physical model

Retain the finite-height quotients $\mathfrak{n}_+^{(H)}$ fixed in the preceding chapter.

The current Lie conformal superalgebra is

$$\operatorname{Cur}(\mathfrak{n}_+^{(H)}) = \Bbbk[\partial] \otimes \mathfrak{n}_+^{(H)}, \qquad [a_\lambda b] = [a, b].$$

Its universal enveloping vertex superalgebra is V_H. The current generators carry the cocommutative vertex-bialgebra coproduct

$$\Delta\big(a(-1)\mathbf{1}\big) = a(-1)\mathbf{1} \otimes \mathbf{1} + \mathbf{1} \otimes a(-1)\mathbf{1}, \qquad a \in \mathfrak{n}_+^{(H)},$$

extended uniquely as a vertex-algebra morphism.

Theorem 55.1 (Height-complete current realization). *Let $\mathfrak{n} = \bigoplus_{\alpha>0} \mathfrak{n}_\alpha$ be a root-graded Lie superalgebra with finite-dimensional root spaces and a proper additive height function. Put*

$$\mathfrak{n}^{(H)} = \mathfrak{n} \Big/ \bigoplus_{h(\alpha)>H} \mathfrak{n}_\alpha.$$

Then every $\mathfrak{n}^{(H)}$ is finite-dimensional and nilpotent, the quotient maps form an inverse system, and

$$\widehat{U}_h(\mathfrak{n}) = \varprojlim_H U(\mathfrak{n}^{(H)}), \quad \widehat{C}_{*,h}(\mathfrak{n}) = \varprojlim_H C_*(\mathfrak{n}^{(H)}), \quad \widehat{V}_h(\mathfrak{n}) = \varprojlim_H U^{\mathrm{ch}}(\operatorname{Cur}_X \mathfrak{n}^{(H)})$$

are separated complete algebras, coalgebras, and current algebras. Their associated graded objects are the corresponding completed symmetric objects, and the bar–Chevalley comparison is compatible with the inverse system.

Proof. Properness gives finitely many roots below a fixed height. Additivity implies that the roots of height $> H$ form a Lie ideal and that the quotient is nilpotent. PBW and Chevalley antisymmetrization are natural under quotient maps. Taking inverse limits in each finite total height therefore gives the displayed complete objects and preserves the coefficientwise quasi-isomorphism. □

Apply the finite BF model of Chapter 47 to every $\mathfrak{n}_+^{(H)}$. The inverse system gives a height-complete holomorphic–topological classical field theory with completed boundary algebra and interval complex

$$U_h^\wedge(\mathfrak{n}_+) := \varprojlim_H U(\mathfrak{n}_+^{(H)}), \qquad C_{*,h}^\wedge(\mathfrak{n}_+) := \varprojlim_H C_*(\mathfrak{n}_+^{(H)}).$$

This gives an exact algebraic realization, a local classical holomorphic–topological realization, and an exact quantum realization of the finite-dimensional zero-mode sectors of the positive enveloping algebra and its denominator character. The compact $K3 \times E$ identification is the geometric comparison studied in the companion volumes.

Chapter 56

Character, bracket, parity, and orientation

A character is an Euler invariant. Faithful reconstruction combines it with the bracket, parity decomposition, and orientation line carried by the intended object.

Theorem 56.1 (Three reconstruction fibres)**.** *The following fibres contain distinct structured objects with the same scalar image.*

(i) *The PBW-character fibre of a graded vector space with basis x, y, z in degrees a, b, $a + b$ contains both the abelian Lie algebra and the Heisenberg Lie algebra $[x, y] = z$.*

(ii) *The superdimension-zero fibre contains both the zero super vector space and $\Bbbk^{1|1}$.*

(iii) *The squaring fibre of a line-valued section contains s and $s \otimes \chi$ for every sign local system χ.*

Consequently a faithful reconstruction datum consists respectively of a bracket, a parity-refined dimension, and an oriented square-root line.

Proof. PBW identifies the graded vector-space character of an enveloping algebra with the symmetric-superalgebra character of its underlying graded Lie space, so the two displayed brackets have the same PBW character. The two super vector spaces in the second item both have superdimension zero and have parity dimensions $(0, 0)$ and $(1, 1)$. Finally $\chi^{\otimes 2} \cong \mathbf{1}$, hence

$$(s \otimes \chi)^{\otimes 2} \cong s^{\otimes 2}.$$

The extra structures listed in the conclusion distinguish each pair. □

Recognition Theorem 56.2 (Minimal PBW recognition)**.** *Let $\phi : \mathfrak{g} \twoheadrightarrow \mathfrak{h}$ be a surjective morphism of positively graded Lie superalgebras with finite-dimensional homogeneous pieces. If the parity-refined Hilbert series of $U(\mathfrak{g})$ and $U(\mathfrak{h})$ agree, then ϕ is an isomorphism. Equality of signed PBW characters suffices when the parity of every primitive degree is prescribed in advance.*

Proof. If the kernel is nonzero, choose its first positive degree. In that degree all decomposable PBW contributions agree by minimality, while the primitive even or odd dimension in $U(\mathfrak{h})$ is strictly smaller. This contradicts equality of parity-refined Hilbert series. With prescribed parity, the formal logarithm of the PBW product recovers the signed primitive dimension inductively in height, and prescribed parity recovers the ordinary dimension. □

Chapter 57

Exact dependency theorem

Theorem 57.1 (Ordered-chiral closure)**.** *Fix:*

(i) *a six-functor Ran category and its factorization subcategory;*

(ii) *a transversely ordered algebra, an associative chiral algebra, or a chiral Lie algebra, as appropriate;*

(iii) *every mixed distributive, completion, duality, trace, or analytic datum named by the intended construction.*

Then the three bars, their complete cobars, their module/comodule Koszul transforms, the Hochschild centre, the formal-disc realization, the ribbon theory, and the finite-height current realization are defined functorially and satisfy the identities proved in this volume. Every passage between the pointwise and chiral-convolution monoidal products is carried by the displayed mixed distributive datum.

Proof. Each output is one of the functors constructed in the preceding chapters. The source and target monoidal categories are displayed in every definition. The comparison theorems invoke the explicitly supplied distributive, PBW, completion, duality, or trace datum. Composition of these functors gives the asserted functorial system. □

Chapter 58

Synthesis

The invariant language has three algebraic axes. Transverse order is an E_1 structure for $\otimes^!$, associative chirality an E_1 structure for $\star^{\mathrm{ch}}$, and chiral Lie theory a Lie structure for $\star^{\mathrm{ch}}$. Their bars are $\mathrm{Bar}^{\perp}$, $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}$ and $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}$. The aligned projector describes the canonical common-decomposition comparison under its stated descent hypotheses. A mixed distributive law and comparisons among the three bars retain their own maps and coherence equations.

The constructed operations form the two functorial rows

$$\text{HT boundary theory} \xrightarrow{\mathrm{Obs}_{\partial}} \mathrm{Alg}_{\mathsf{E}_1}(\mathrm{Fact}_D(X)) \xrightarrow{\mathrm{Bar}^{\mathrm{ord}}} \mathrm{Coalg}_{\mathrm{conil,cpl}}(\mathrm{Fact}_D(X)),$$

$$\mathrm{Alg}_{\mathsf{E}_1}(\mathrm{Fact}_D(X)) \xrightarrow{Z_{\mathrm{ord}}} \mathrm{Alg}_{\mathsf{E}_2}(\mathrm{Fact}_D(X)).$$

A primitive Lie algebra enters when a separate Chevalley or Hall comparison identifies the bar coalgebra with the Chevalley chains of that Lie algebra. Hall geometry supplies this comparison through extension correspondences. Quantum groups require an additional coproduct and exchange structure. Automorphic forms arise from graded Euler characters after a grading and completion have been fixed. Each arrow therefore preserves the geometric operation named at its source.

Book **XI**

Technical appendices to the preceding book

Chapter 59

Sign calculation for the length-three bar

For homogeneous $a, b, c \in \bar{A}$, write $[a|b|c] = sa \otimes sb \otimes sc$. With the standard suspension convention,

$$b_m[a|b|c] = (-1)^{|a|}[ab|c] + (-1)^{|a|+|b|+1}[a|bc].$$

Applying b_m again gives

$$b_m^2[a|b|c] = (-1)^{|b|}\big[(ab)c - a(bc)\big],$$

which vanishes by associativity.

Chapter 60

Rational R-matrix verification

Let P_{ij} permute tensor factors i and j. Expanding

$$R_{12}(u)R_{13}(u+v)R_{23}(v)$$

and its reversed product shows equality coefficientwise. The cubic coefficient uses

$$P_{12}P_{13}P_{23} = P_{23}P_{13}P_{12},$$

the quadratic coefficients use the symmetric-group relations, and the linear coefficients agree termwise.

Chapter 61

Carriers and comparison maps

The five-carrier geometry, transverse bar construction, mixed-forest signs, and noncommutative examples use four kinds of categorical data:

(i) a factorization carrier is paired explicitly with its operation;

(ii) the Ran category is fixed before its tensor products are used;

(iii) the block-ordered operad is introduced after a representing chiral operad is chosen;

(iv) ribbon and stable-curve sewing are derived from nondegenerate Calabi–Yau and modular data.

A comparison between these constructions therefore specifies its source and target categories and the operations preserved by its map.

Chapter 62

Enveloping bars and planar path sequences

The dimension of a bar chain space and the dimension of its homology are different invariants. For an enveloping algebra, the Chevalley comparison gives a Lie-theoretic model for the homology. This makes it possible to compare Motzkin and Riordan path counts with the ordered bar on a specified carrier.

Bar length and Chevalley degree are homological degrees: H_n denotes degree-n homology. The cohomology of the dual Chevalley complex is written H^n. For finite-dimensional weight pieces their dimensions agree.

62.1 Chevalley comparison

Theorem 62.1 (Bar–Chevalley comparison). *Let k be a field of characteristic zero and let $\mathfrak{g}$ be an ordinary Lie algebra over k. Give $U(\mathfrak{g})$ its augmentation killing $\mathfrak{g}$. There is a natural quasi-isomorphism of ordinary conilpotent differential graded coalgebras*

$$\kappa : C_*(\mathfrak{g}) \longrightarrow \mathrm{Bar}^{\perp}\big(U(\mathfrak{g})\big),$$

where both constructions use direct sums and ordinary tensors over k. It preserves the PBW filtration. If $\mathfrak{g} = \bigoplus_{w \geq 1} \mathfrak{g}_w$ and $[\mathfrak{g}_u, \mathfrak{g}_v] \subseteq \mathfrak{g}_{u+v}$, the map is a quasi-isomorphism in each total weight. It extends to the degreewise product over weights, using completed tensor products by total weight. The completed coalgebras are conilpotent in each weight; no global conilpotence is asserted for infinite sums.

Proof. Put $C_n(\mathfrak{g}) = \Lambda^n \mathfrak{g}$ in homological degree n, with differential

$$d(x_1 \wedge \cdots \wedge x_n) = \sum_{i<j} (-1)^{i+j+1} [x_i, x_j] \wedge x_1 \wedge \cdots \widehat{x_i} \cdots \widehat{x_j} \cdots \wedge x_n.$$

Its coproduct splits the exterior factors between two ordered sublists, with the sign of the corresponding permutation. The ordinary normalized bar differential joins positions $i, i+1$ with sign $(-1)^{i-1}$. Define

$$\kappa(x_1 \wedge \cdots \wedge x_n) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) [x_{\sigma(1)} | \cdots | x_{\sigma(n)}].$$

At degree zero this map preserves the unit. At degree two its differential is $[x_1 x_2 - x_2 x_1] = [[x_1, x_2]]$. In every degree, pair permutations whose merged adjacent entries occur in opposite orders. Their difference is the bracket of those entries, with the displayed exterior sign. This proves the chain-map equation. Cutting an antisymmetrized word amounts to choosing an ordered subset and antisymmetrizing its two subwords. The two procedures give the same permutations and signs, proving the coalgebra identity.

Filter $U(\mathfrak{g})$ by PBW degree, the bar by total PBW degree, and $\Lambda^n \mathfrak{g}$ by degree n. PBW identifies the associated graded algebra with $S(\mathfrak{g})$. The associated graded comparison is the antisymmetrization $\Lambda(\mathfrak{g}) \to B(S(\mathfrak{g}))$, with zero differential on its source. To verify that it is a quasi-isomorphism, consider the free Koszul resolution $S(V) \otimes \Lambda(V)$ for $V = \mathfrak{g}$. For a basis (x_i) of V, its differential is $\partial = \sum_i x_i \iota_i$, where ι_i contracts the exterior factor x_i. On polynomial degree p and exterior degree q, with $p + q > 0$, the map

$$h = \frac{1}{p+q} \sum_i (x_i \wedge -) \, \frac{\partial}{\partial x_i}$$

satisfies $\partial h + h\partial = 1$. Only finitely many summands act on each element, even when V is infinite-dimensional. The identity follows by cancellation of cross terms and the Euler identity in the two degrees. Thus this free resolution has augmentation homology k only. The free normalized bar resolution has the same homology by its extra-degeneracy contraction. Antisymmetrization defines an augmentation-preserving comparison between these free resolutions. The simplicial bar sign convention is related to the displayed bar convention by multiplication by $(-1)^n$ in degree n. Free bases permit a reverse comparison and homotopies to be lifted inductively, since every positive-degree cycle in a resolution is a boundary. Tensoring these homotopies with k proves the asserted quasi-isomorphism on associated graded complexes.

Every cycle in the cone of κ has finite PBW filtration degree. A primitive for its highest associated graded part lowers that degree; iteration terminates because the filtration is bounded below. The cone is acyclic, proving the ordinary assertion. For a positive weight grading, PBW degree is at most total weight. The same argument is therefore finite within each weight. Finally, kernels and images of a degreewise product differential are products of kernels and images, so the componentwise quasi-isomorphisms extend to the stated completion. □

The ordinary comparison does not itself construct a morphism in $\mathrm{FactD}(X)$. For geometric coefficient systems, one must give compatible enveloping and Chevalley constructions on every chart and coherent factorization equivalences for the whole Chevalley complex. On each disjointness locus, with restriction denoted by $j^!$, the required equation is

$$f_{B,I}\, j^! \kappa_I = (\kappa_{I_1} \boxtimes \kappa_{I_2})\, f_{C,I}, \qquad I = I_1 \sqcup I_2.$$

Here $f_{C,I}$ and $f_{B,I}$ are the specified Chevalley and bar factorization maps. The bar map may use the signed shuffle comparison under its geometric functor hypotheses. The Chevalley factorization maps, collision compatibility, refinement coherence, and continuity of the completed maps remain additional geometric data to construct or verify. Internal tensor formulas and PBW alone do not provide them.

Calculation 62.2 (Finite-dimensional simple case)**.** For a simple Lie algebra of rank r with exponents $m_1, \dots, m_r$, the Chevalley cohomology is the exterior algebra on primitive classes of degrees $2m_i + 1$. The corresponding homology Poincaré polynomials are

$$\mathrm{Bar}^{\perp} U(\mathfrak{sl}_2) : 1 + t^3, \qquad \mathrm{Bar}^{\perp} U(\mathfrak{sl}_3) : 1 + t^3 + t^5 + t^8, \qquad \mathrm{Bar}^{\perp} U(\mathfrak{g}_2) : 1 + t^3 + t^{11} + t^{14}.$$

The corresponding Betti sequences for $\mathfrak{sl}_2$ and $\mathfrak{sl}_3$ are $1, 0, 0, 1$ and $1, 0, 0, 1, 0, 1, 0, 0, 1$. For a finite-dimensional Lie algebra, the Chevalley chain dimensions are $\binom{\dim \mathfrak{g}}{n}$.

For an ordinary augmented algebra A over $\mathbb{C}$, let $I = \ker(A \to \mathbb{C})$. Its normalized two-sided bar with trivial coefficients has homological chain spaces

$$B_0(A) = \mathbb{C}, \qquad B_n(A) = I^{\otimes n} \quad (n \geq 1).$$

Here n is bar length, independently of any internal weight or PBW degree. If I has finite dimension d, then $\dim B_n(A) = d^n$; in particular, an augmentation ideal of dimension three gives the chain counts 3^n. In a cohomological convention that places B_n in degree $-n$, its homology is denoted H^{-n} instead of H_n.

For $A = U(\mathfrak{sl}_2)$, PBW gives the basis $e^a h^b f^c$, with $a, b, c \geq 0$, where $[h, e] = 2e$, $[h, f] = -2f$, and $[e, f] = h$. The augmentation ideal has all these basis elements except 1. Consequently $B_n(A)$ is countably infinite-dimensional for every $n \geq 1$. Its total PBW filtration has finite-dimensional associated graded pieces, with generating function

$$\sum_{d \geq 0} \dim \mathrm{gr}^{\mathrm{PBW}}_d B_n(A)\, z^d = \left((1 - z)^{-3} - 1\right)^n.$$

In particular, 3^n is the dimension in the least possible total PBW degree $d = n$. These least-degree dimensions are 3, 9, 27, 81, 243 for $n = 1, 2, 3, 4, 5$; the full bar chain spaces are infinite-dimensional. The quotient of indecomposables is zero: the perfectness of $\mathfrak{sl}_2$ implies $I/I^2 = 0$.

The Chevalley chain model is different: its degree-n space is $\Lambda^n \mathfrak{sl}_2$, of dimensions 1, 3, 3, 1 in degrees 0, 1, 2, 3. With the convention $d(x \wedge y) = [x, y]$, the map $d : \Lambda^2 \mathfrak{sl}_2 \to \mathfrak{sl}_2$ sends

$$e \wedge h \longmapsto -2e, \qquad e \wedge f \longmapsto h, \qquad h \wedge f \longmapsto -2f,$$

and is an isomorphism. The next differential vanishes, since

$$d(e \wedge h \wedge f) = -2e \wedge f - h \wedge h - 2f \wedge e = 0.$$

Together with $d : \mathfrak{sl}_2 \to \mathbb{C}$ equal to zero, this gives Chevalley homology dimensions 1, 0, 0, 1. The ordinary bar–Chevalley comparison identifies these homology groups with those of $U(\mathfrak{sl}_2)$; it does not identify the two chain spaces.

62.2 The Virasoro row

The transverse presentation of the Virasoro family is the enveloping algebra of the positive Witt algebra

$$L_1 = \bigoplus_{n \geq 1} \mathbb{C}\, e_n, \qquad [e_i, e_j] = (j - i) e_{i+j},$$

to which Theorem 62.1 applies: L_1 is positively graded with one-dimensional weight spaces, so every weight has finitely many decompositions.

Theorem 62.3 (Pentagonal rigidity)**.** *The bar homology of $U(L_1)$ satisfies*

$$\dim H_n\big(\mathrm{Bar}^{\perp} U(L_1)\big) = 2 \qquad \text{for every } n \geq 1,$$

one dimension in each of the weights $(3n^2 - n)/2$ and $(3n^2 + n)/2$. Its graded Euler character is

$$\chi_q\big(\mathrm{Bar}^{\perp} U(L_1)\big) = \prod_{n \geq 1} (1 - q^n), \tag{62.1}$$

and (62.1) is accounted for degree by degree with no residual freedom.

Proof. By Theorem 62.1 the bar of $U(L_1)$ is quasi-isomorphic to $C_*(L_1)$. Each weight subcomplex is finite-dimensional, so its homology has the same dimensions as the weightwise dual cohomology. Goncharova's calculation gives two dimensions in each degree $n \geq 1$, in the weights $(3n^2 \mp n)/2$. Identity (62.1) is the denominator product applied to L_1, where every root space is one dimensional and even, so $\mathrm{smult}(L_1)_n = 1$. Euler's pentagonal number theorem expands the product as $\sum_{m \in \mathbb{Z}} (-1)^m q^{m(3m-1)/2}$, whose support is exactly $\{(3n^2 \pm n)/2\}$ with coefficient $(-1)^n$ at each of $(3n^2 - n)/2$ and $(3n^2 + n)/2$. Comparing with $\sum_n (-1)^n \dim H_{n,w}$, each weight of the Euler character is saturated by a single class in the degree Goncharova predicts, so no further classes and no cancellation are available. $\square$

For $n = 1, 2, 3, 4$, the nonzero homology groups occupy the weight pairs $(1, 2)$, $(5, 7)$, $(12, 15)$, $(22, 26)$, one dimension in each. The corresponding nonzero coefficients of $\prod_{n \geq 1}(1 - q^n)$ occur at 0, 1, 2, 5, 7, 12, 15, 22, 26 with signs $+, -, -, +, +, -, -, +, +$.

62.3 Planar path sequences

A Motzkin path is a lattice path from $(0,0)$ to $(n,0)$, staying at nonnegative height, with steps $(1,1)$, $(1,0)$, and $(1,-1)$. Let $M(n)$ count all such paths of length n, and let $R(n)$ count those with no horizontal step at height zero. Both counts include the empty path: $M(0) = R(0) = 1$, while $M(1) = 1$ and $R(1) = 0$. For $n \geq 1$ consider the two sequences

$$a_n = M(n+1) - M(n) = 1, 2, 5, 12, 30, 76, \ldots, \qquad b_n = R(n+3) = 3, 6, 15, 36, 91, \ldots.$$

Proposition 62.4 (Motzkin and Riordan identities)**.** *For every $n \geq 0$,*

$$M(n) = R(n) + R(n+1), \qquad M(n+1) - M(n) = R(n+2) - R(n).$$

The ordinary generating functions of the two path counts, and hence those of a_n and b_n, belong to $\mathbb{Q}(z, \sqrt{1-2z-3z^2})$.

Proof. Work in the formal power-series ring $\mathbb{Q}[[z]]$, and write $M(z) = \sum_{n\geq 0} M(n)z^n$ and $R(z) = \sum_{n\geq 0} R(n)z^n$. A nonempty Motzkin path either starts with a horizontal step followed by an arbitrary Motzkin path, or has a unique first-return decomposition $UPDQ$, where U and D are the up and down steps, P and Q are arbitrary Motzkin paths, and P is drawn one unit above the horizontal axis. Thus

$$M = 1 + zM + z^2M^2.$$

A nonempty path counted by R has no initial horizontal step. Its first return similarly has the form $UPDQ$, with P arbitrary and Q counted by R. Consequently

$$R = 1 + z^2MR, \qquad R = (1 - z^2M)^{-1}.$$

The inverses here exist because their denominators have constant term one. Using $z^2M^2 = M - 1 - zM$, compute

$$(1 + zM)(1 - z^2M) = 1 + zM - z^2M - z^3M^2 = 1 + z.$$

It follows that $(1+z)R = 1 + zM$. Comparing coefficients of z^{n+1} gives $R(n+1) + R(n) = M(n)$ for all $n \geq 0$. Subtracting consecutive identities gives the difference formula.

Solving the quadratic equation for the formal series with constant term one gives

$$M(z) = \frac{1 - z - \sqrt{1-2z-3z^2}}{2z^2}, \qquad R(z) = \frac{1 + zM(z)}{1+z},$$

where the square root has constant term one. The apparent pole in the formula for M is removable as a formal series. Finite index shifts and differences preserve the indicated quadratic function field, proving the last assertion. These identities are consequences solely of the path decompositions. □

Remark 62.5 (Low-degree homology). Theorem 62.3 gives homology dimensions 2, 2, 2 in degrees 1, 2, 3, whereas a_1, a_2, a_3 are 1, 2, 5. These triples agree only in degree two. The degree-one discrepancy already excludes equality of the full homology dimension sequences.

Theorem 62.6 (Enveloping bars and planar path counts)**.** *Let $B_\bullet(A)$ be the ordinary normalized bar over $\mathbb{C}$, with trivial coefficients and homological degree given by bar length. With $a_n = M(n+1) - M(n)$ and $b_n = R(n+3)$ indexed by $n \geq 1$ as above,*

$$\dim H_1B_\bullet(U(\mathfrak{sl}_2)) = 0 \neq b_1 = 3, \qquad \dim H_1B_\bullet(U(L_1)) = 2 \neq a_1 = 1,$$

where $L_1 = \bigoplus_{m\geq 1} \mathbb{C}e_m$ and $[e_i, e_j] = (j-i)e_{i+j}$. Hence neither sequence is the bar homology dimension sequence of the corresponding enveloping algebra. Neither is its full bar chain dimension sequence: in both cases B_n is countably infinite-dimensional for every $n \geq 1$.

Proof. For any augmented algebra A with augmentation ideal I, the bar differential out of $B_1 = I$ is zero and the image from $B_2 = I \otimes I$ is I^2. Therefore

$$H_1 B_\bullet(A) = I/I^2.$$

For an ordinary Lie algebra $\mathfrak{g}$, the inclusion of generators induces an isomorphism

$$\mathfrak{g}/[\mathfrak{g},\mathfrak{g}] \xrightarrow{\sim} I/I^2, \qquad I = \ker(U(\mathfrak{g}) \to \mathbb{C}).$$

Indeed the generators span I/I^2, and $[x, y] = xy - yx$ vanishes there. For injectivity, equip $\mathbb{C} \oplus \mathfrak{g}/[\mathfrak{g},\mathfrak{g}]$ with the square-zero product on its second summand. The quotient map from $\mathfrak{g}$ to that summand is a Lie map into the commutator Lie algebra of this associative algebra. The universal property of $U(\mathfrak{g})$ extends it to an augmented algebra map, which kills I^2 and provides an inverse on the displayed quotient.

The brackets $[h, e] = 2e$, $[h, f] = -2f$, and $[e, f] = h$ show that $\mathfrak{sl}_2$ equals its commutator algebra. For L_1, every nonzero bracket has weight at least three. Conversely, for $m \geq 3$,

$$e_m = \frac{1}{m-2}[e_1, e_{m-1}].$$

Thus $[L_1, L_1] = \bigoplus_{m\geq 3} \mathbb{C}e_m$, and its abelianization has basis the classes of e_1, e_2, in weights one and two. This proves both degree-one assertions without using any higher-degree calculation.

PBW gives a countably infinite basis of the augmentation ideal of either enveloping algebra. In the L_1 case the basis consists of nonempty ordered monomials in the countable set $e_1, e_2, \ldots$, each monomial having finite length. Finite tensor powers of either augmentation ideal are therefore countably infinite-dimensional. The path counts are finite in every degree, so they cannot count these full chain spaces. □

In particular, a complex whose homology dimensions are b_n cannot be quasi-isomorphic to $B_\bullet(U(\mathfrak{sl}_2))$, and one whose homology dimensions are a_n cannot be quasi-isomorphic to $B_\bullet(U(L_1))$, with the stated degree convention. A discrepancy of chain dimensions alone would not imply this conclusion; the degree-one homology discrepancy does.

The weight grading gives a different, finite comparison. An ordered PBW monomial of weight w specifies a partition of w, so for $n \geq 1$,

$$\sum_{w\geq 0} \dim B_n(U(L_1))_w q^w = \left(\prod_{m\geq 1}(1-q^m)^{-1} - 1\right)^n.$$

Each coefficient is finite. Thus a weightwise dimension, a dimension after a specified weight cutoff, and an ungraded dimension are distinct invariants. In a weight completion, dimensions must likewise be interpreted weightwise; the countable algebraic dimensions above refer to the ordinary direct-sum enveloping algebras.

Remark 62.7 (Chiral coefficients and transverse multiplication). Independently of these scalar calculations, a Virasoro chiral coefficient does not determine the transverse bar: a fixed coefficient may be combined with free, square-zero, enveloping, or quantum-plane transverse algebras whose bars differ, while the coefficient and the central charge are unchanged. A chain-level comparison with an ordered bar therefore also requires a specified transverse multiplication and a quasi-isomorphism from the proposed complex to that bar.

Chapter 63

The obstruction constitution

Each statement below names what is *not* determined, and therefore what must be supplied. Without them the preceding theorems admit false corollaries.

Proposition 63.1 (Invertibility of the aligned comparison)**.** *In the finite decomposition model, the canonical aligned inclusion and retraction are inverse on the full object exactly when its off-aligned summands vanish. Descent to geometric coefficients additionally requires compatibility with the actual transition correspondences.*

Proof. The canonical maps satisfy $\pi\iota = \mathrm{id}$ and $\iota\pi = e_{\mathrm{al}}$. The latter is the identity exactly when all off-aligned summands vanish. On geometric charts the required base-change squares must identify these projectors across transitions. The conditional descent theorem states those hypotheses. □

Failure of this canonical comparison to be invertible does not exclude every noninvertible distributive law. Such a law must be given as a natural transformation satisfying its own unit and composition diagrams. For Zamolodchikov–Faddeev algebras, the spectral variable and transverse word supply the two inputs to this separate construction problem.

Theorem 63.2 (No quantum group from dimension loss)**.** *The dimensions of the kernels of $V^{\otimes n} \to \mathrm{Sym}^n V$ do not determine a multiplication or a Hopf structure. Such a determination requires operations and their compatibility maps in addition to dimensions.*

Proof. The abelian and Heisenberg Lie algebras with generator weights 1, 1, 2 have enveloping algebras with the same PBW character $((1-t)^2(1-t^2))^{-1}$. The first algebra is commutative and the second is not. Their graded vector spaces, and hence the stated kernel dimensions, are nevertheless isomorphic.

Even fixing the multiplication and augmentation is insufficient. On $\Bbbk[x, y, z]$, make x, y primitive and put

$$\Delta_\eta z = z \otimes 1 + 1 \otimes z + \eta x \otimes y, \qquad \eta = 0, 1.$$

Both maps are coassociative: their extra triple terms are $\eta(x \otimes y \otimes 1 + x \otimes 1 \otimes y + 1 \otimes x \otimes y)$. Polynomial augmentation is the counit, and the antipode sends (x, y, z) to $(-x, -y, -z + \eta xy)$. The first coproduct is cocommutative and the second is not, so the two Hopf algebras are not isomorphic. □

Theorem 63.3 (No modular object from topology alone)**.** *The identity $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$ assigns no coefficient operations to boundary strata. A modular master element exists only after cyclic contractions and clutching-compatible coefficient maps have been supplied.*

Theorem 63.4 (No bulk from a boundary centre)**.** *The derived centre is terminal among algebraic central actions on the boundary algebra. It does not detect a decoupled bulk sector, so a physical bulk is not determined by the boundary chiral algebra unless a bulk-to-centre comparison is supplied and proved to be an equivalence.*

Proof. Terminality is the universal property of the centre. It is a statement about the boundary category and is insensitive to any sector acting trivially on it, so a bulk with trivial boundary action is invisible. □

Theorem 63.5 (No classification by central charge)**.** *Central charge determines the Virasoro anomaly coefficient. It does not determine the chiral operation, the representation category, the braid monodromy, or the higher-genus conformal blocks. Distinct holomorphic theories at equal central charge exist.*

Theorem 63.6 (No identification of anomalies of different types)**.** *A current-algebra level, a Virasoro central charge, a determinant-line curvature, a BRST ghost-number anomaly, and a Borcherds-product weight are invariants in different theories, living in different deformation complexes. They may be related by a theorem in a particular family. They are not one universal scalar, and no identity among them follows from numerical agreement.*

Proof. Each is a class in a distinct complex: an Atiyah-algebra extension class, a Virasoro cocycle, the curvature of a determinant line, a class in H^1 or H^2 of a BRST convolution complex, and a weight in the character lattice of an automorphic form. A map between two of these complexes is additional data; in its absence, equality of numbers is a coincidence of values, not an identification of classes. □

Theorem 63.7 (No unnamed comparison of bars)**.** *An asserted equality* $\mathrm{Bar}^{\perp}(A) = \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A)$ *has no meaning without a functor carrying the pointwise multiplication problem to the chiral-convolution multiplication problem. The two have different underlying gradings, counting interval subdivisions and collision trees respectively. Agreement of Euler characteristics is not a comparison map.*

Theorem 63.8 (A scalar sequence is not homology data)**.** *A sequence of integers becomes bar homology only after a chain complex, a differential, a grading convention, and a rank computation or quasi-isomorphism are specified. Matching initial values, discriminants, or growth rates constructs no such map.*

Proof. Chain-space dimensions and homology dimensions differ by the ranks of the differentials, which are not functions of the chain dimensions. Calculation 62.2 exhibits an algebra whose positive bar chain spaces are countably infinite-dimensional and whose bar homology dimensions are 1, 0, 0, 1. The numbers 3^n count only the least-PBW-weight piece in bar length n. Proposition 62.4 exhibits two sequences with the same discriminant and growth rate that are transforms of one another and of neither homology. □

Theorem 63.6 together with the central-charge statement retracts any programme treating a tuple of scalars drawn from these sources as a single classifying invariant, and any trace identity asserted among them without a named comparison morphism.

Book entrance: Mixed Holomorphic–Topological Deligne Theory

Hamiltonian jets, shifted cotangent fields, universal branes, Duflo quantization, and the all-arity open–closed map.

Book XII

Finite Hamiltonian geometry

Chapter 64

The formal symplectic plane

64.1 Hamiltonians and the Poisson bracket

Let $V = \mathbb{C}^2$ with coordinates (z_1, z_2) and symplectic form

$$\omega = \mathrm{d}z_1 \wedge \mathrm{d}z_2.$$

Let $\mathfrak{m} = (z_1, z_2)$. The pointed formal Hamiltonian Lie algebra is

$$\mathfrak{h} = \mathfrak{m}^2 \subset \mathbb{C}[\![z_1, z_2]\!], \qquad \{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g.$$

It consists of Hamiltonians whose vector fields fix the chosen Darboux point. The shifted order filtration $F^p\mathfrak{h} = \mathfrak{m}^{p+2}$ is complete and obeys

$$\{F^p\mathfrak{h}, F^q\mathfrak{h}\} \subset F^{p+q}\mathfrak{h}.$$

The affine Hamiltonian algebra $\mathfrak{h}^{\mathrm{aff}} = \mathbb{C}[\![z_1, z_2]\!]/\mathbb{C}$ also contains translations. Its linear sector is adjoined separately when a moving brane or center-of-mass coordinate is required.

Write $R_r = \mathrm{Sym}^r(V^\vee)$ for the homogeneous polynomials of degree r. The bracket obeys

$$\{R_r, R_s\} \subset R_{r+s-2}.$$

For $d \geq 2$ define the finite jet quotient

$$\mathfrak{h}_{\leq d} = \bigoplus_{r=2}^{d} R_r,$$

where every bracket component of degree greater than d is set equal to zero.

Theorem 64.1 (Finite Hamiltonian jet algebra)**.** *The truncated bracket makes $\mathfrak{h}_{\leq d}$ a finite-dimensional Lie algebra. Its dimension is*

$$\dim \mathfrak{h}_{\leq d} = \frac{d(d+3)}{2} - 2.$$

The quadratic part is

$$R_2 \cong \mathfrak{sl}_2,$$

and each R_r is the irreducible $\mathfrak{sl}_2$-module $\mathrm{Sym}^r(\mathbb{C}^2)$.

Proof. The Jacobiator of the polynomial Poisson bracket vanishes before truncation. Projection to degrees at most d preserves this identity because the discarded terms form a Lie ideal. The dimension is

$$\sum_{r=2}^{d} (r+1) = \frac{d(d+3)}{2} - 2.$$

Quadratic Hamiltonians act linearly on V and preserve ω; this identifies R_2 with $\mathfrak{sp}(V) = \mathfrak{sl}_2$. The standard action on homogeneous polynomials gives the final statement. □

64.2 Unimodularity

Theorem 64.2 (Unimodularity of every finite jet)**.** *For every $d \geq 2$ and every $x \in \mathfrak{h}_{\leq d}$,*

$$\operatorname{Tr}_{\mathfrak{h}_{\leq d}}(\operatorname{ad}_x) = 0.$$

Proof. A homogeneous Hamiltonian of degree greater than two raises polynomial degree. Its adjoint action is strictly upper triangular in the decomposition $\bigoplus_{r=2}^{d} R_r$, hence has trace zero. A quadratic Hamiltonian belongs to $\mathfrak{sl}_2$. Its action on each finite-dimensional $\mathfrak{sl}_2$-module R_r has trace zero. Linearity gives the result. □

Corollary 64.3 (Canonical finite BV measure)**.** *The linear odd symplectic space $T^*[-1]\mathfrak{h}_{\leq d}[1]$ has a translation-invariant Berezinian for which the cotangent BF Hamiltonian satisfies the quantum master equation.*

Proof. Choose dual linear coordinates on the shifted cotangent space and use their constant Berezinian. The classical term vanishes by Jacobi, while the BV divergence is the adjoint trace. Unimodularity makes that trace zero. □

Chapter 65

The shifted cotangent theory

65.1 The Lie algebra and its functions

Put

$$\mathfrak{g}_d = T^*[-1]\mathfrak{h}_{\le d} = \mathfrak{h}_{\le d} \ltimes \mathfrak{h}_{\le d}^\vee[-1],$$

where $\mathfrak{h}_{\le d}$ acts on the dual by the coadjoint action and the shifted dual is abelian.

Proposition 65.1 (Chevalley–Poisson identification)**.** *There is a canonical isomorphism of commutative dg algebras*

$$C^*_{\mathrm{CE}}(\mathfrak{g}_d) \cong C^*\big(\mathfrak{h}_{\le d}, \mathrm{Sym}(\mathfrak{h}_{\le d})\big).$$

The differential on the right is the Chevalley differential for the adjoint action on $\mathrm{Sym}(\mathfrak{h}_{\le d})$*. Under the identification* $\mathrm{Sym}(\mathfrak{h}_{\le d}) = \mathcal{O}(\mathfrak{h}^*_{\le d})$*, it is the linear Poisson differential.*

Proof. The dual of $\mathfrak{h}^\vee_{\le d}[-1]$ shifted by $[-1]$ is $\mathfrak{h}_{\le d}$ in degree zero. Chevalley cochains therefore contain polynomial functions on $\mathfrak{h}^*_{\le d}$ and exterior ghosts for $\mathfrak{h}_{\le d}$. The semidirect bracket produces the coadjoint action, which is the Hamiltonian action of the linear Poisson tensor. □

65.2 The BV action

Choose dual bases e_a and e^a. Let c^a be the ghost coordinates and b_a the cotangent coordinates. The finite action is

$$S_d = \frac{1}{2} b_c f^c{}_{ab} c^a c^b.$$

The canonical BV bracket satisfies

$$\frac{1}{2}\{S_d, S_d\} = 0$$

by Jacobi.

Theorem 65.2 (Exact finite quantum master equation)**.** *With the translation-invariant BV Laplacian,*

$$\Delta S_d = \sum_b \mathrm{Tr}(\mathrm{ad}\, e_b) c^b = 0.$$

Consequently

$$\frac{1}{2}\{S_d, S_d\} + \hbar \Delta S_d = 0$$

for every d.

Proof. Contracting one field and one antifield in the cubic Hamiltonian gives the divergence of the adjoint vector field. The preceding unimodularity theorem makes every coefficient vanish. □

Chapter 66

The universal boundary brane

66.1 Polarization

Place the cotangent theory on a half-plane. The standard Lagrangian boundary condition fixes the cotangent field and leaves the $\mathfrak{h}_{\leq d}$ field free. Boundary observables form the Rees enveloping algebra

$$U_\hbar(\mathfrak{h}_{\leq d}) = T(\mathfrak{h}_{\leq d})[\![\hbar]\!]/\langle x\,y - y\,x - \hbar[x,y]\rangle.$$

The Poincaré–Birkhoff–Witt symmetrization gives a filtered vector-space isomorphism

$$\mathrm{sym}_\hbar : \mathrm{Sym}(\mathfrak{h}_{\leq d})[\![\hbar]\!] \xrightarrow{\sim} U_\hbar(\mathfrak{h}_{\leq d}).$$

Theorem 66.1 (Boundary quantization)**.** *The factorization algebra of boundary observables of the finite cotangent theory is locally constant and its value on an interval is $U_\hbar(\mathfrak{h}_{\leq d})$. Fusion of adjacent intervals is multiplication in the enveloping algebra.*

Proof. Canonical quantization turns the linear boundary phase space $\mathfrak{h}^*_{\leq d}$ into the Gutt star algebra. For a linear Poisson bracket the Gutt product is transported by PBW to $U_\hbar(\mathfrak{h}_{\leq d})$. Excision identifies ordered interval fusion with multiplication. □

Book XIII

Relative Hochschild comparison

Chapter 67

Hochschild cochains of the universal brane

67.1 The coefficient ring and the Chevalley differential

The Rees relation scales both the Lie bracket and its adjoint action. Its Chevalley differential must retain the same parameter. For example, take the quadratic jet $\mathfrak{l} = \mathfrak{sl}_2$ with $[h, e] = 2e$. Modulo $\hbar$, the enveloping algebra is commutative, so the Hochschild differential of e vanishes. The unscaled Chevalley differential instead satisfies $(d_{\mathrm{CE}} e)(h) = 2e$.

Let $\mathfrak{l}$ be a finite-dimensional complex Lie algebra. For $N \geq 0$, put

$$R_N = \mathbb{C}[\hbar]/(\hbar^{N+1}), \qquad \mathfrak{l}_N = R_N \otimes_{\mathbb{C}} \mathfrak{l}, \qquad [x, y]_N = \hbar[x, y],$$

and

$$A_N = U_{R_N}(\mathfrak{l}_N) = T_{R_N}(\mathfrak{l}_N)/(xy - yx - \hbar[x, y]).$$

Here N denotes the order in $\hbar$, independently of a Chan–Paton rank. Write $\bar{A}_N$ for the augmentation ideal and $S = \mathrm{Sym}_{\mathbb{C}} \mathfrak{l}$. PBW symmetrization is the R_N-module isomorphism

$$\mathrm{sym}_N : S \otimes R_N \longrightarrow A_N, \qquad x_1 \cdots x_r \longmapsto \frac{1}{r!} \sum_{\sigma \in S_r} x_{\sigma(1)} \cdots x_{\sigma(r)}.$$

The source has its ordinary symmetric product. Use normalized relative Hochschild cochains

$$C^n_{\mathrm{HH}}(A_N/R_N, A_N) = \mathrm{Hom}_{R_N}(\bar{A}_N^{\otimes_{R_N} n}, A_N)$$

with differential

$$\begin{aligned}(bF)(a_1, \dots, a_{n+1}) = {} & a_1 F(a_2, \dots, a_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i F(a_1, \dots, a_i a_{i+1}, \dots, a_{n+1}) \\ & + (-1)^{n+1} F(a_1, \dots, a_n) a_{n+1}.\end{aligned}$$

Let d_{CE} act on $C^n(\mathfrak{l}, S) = \mathrm{Hom}_{\mathbb{C}}(\Lambda^n \mathfrak{l}, S)$ by the adjoint Chevalley differential.

Theorem 67.1 (Relative Rees comparison)**.** *Antisymmetrization defines a quasi-isomorphism of R_N-complexes*

$$\mathrm{Alt}_N : C^{\bullet}_{\mathrm{HH}}(A_N/R_N, A_N) \longrightarrow \big(C^{\bullet}(\mathfrak{l}, S) \otimes R_N, \hbar d_{\mathrm{CE}}\big),$$

given by

$$(\mathrm{Alt}_N F)(x_1, \dots, x_n) = \mathrm{sym}_N^{-1} \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) F(x_{\sigma(1)}, \dots, x_{\sigma(n)}).$$

These maps commute with coefficient reduction in N.

Proof. Write $A = A_N$, $R = R_N$, and $A^e = A \otimes_R A^{\mathrm{op}}$. In homological degree n, set

$$K_n = A \otimes_R \Lambda_R^n \mathfrak{l}_N \otimes_R A.$$

Its augmentation is multiplication. For $X = x_1 \wedge \cdots \wedge x_n$, let $X_{\widehat{i}}$ and $X_{\widehat{ij}}$ omit the indicated entries without reordering. Define

$$\begin{aligned} \partial(a \otimes X \otimes b) = {} & \sum_i (-1)^{i+1} \big(a x_i \otimes X_{\widehat{i}} \otimes b - a \otimes X_{\widehat{i}} \otimes x_i b \big) \\ & + \hbar \sum_{i<j} (-1)^{i+j} a \otimes [x_i, x_j] \wedge X_{\widehat{ij}} \otimes b. \end{aligned}$$

The normalized relative bimodule bar has terms $\mathcal{B}_n = A \otimes_R \bar{A}^{\otimes_R n} \otimes_R A$. Define

$$\iota_n(a \otimes X \otimes b) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a [x_{\sigma(1)} | \cdots | x_{\sigma(n)}] b.$$

The outer bar faces give the first sum in ∂. Pair adjacent transpositions in the internal faces and use $xy - yx = \hbar[x, y]$ to obtain the second sum. For two inputs this reads

$$\begin{aligned} b\big(1[x|y]1 - 1[y|x]1\big) = {} & x[y]1 - y[x]1 + 1[x]y - 1[y]x \\ & - \hbar\, 1[[x, y]]1. \end{aligned}$$

Thus $b\iota = \iota\partial$. PBW makes $\mathfrak{l}_N$ a free summand of $\bar{A}$. Since every factorial is invertible in R, ι is injective. The identity $b^2 = 0$ then gives $\partial^2 = 0$.

Give K_n filtration degree equal to its outer PBW degrees plus n. The associated graded is the diagonal Koszul complex of $\operatorname{Sym}_R(\mathfrak{l}_N)$. Its differential uses differences of left and right polynomial generators. A linear change of variables makes these independent polynomial variables. They form a regular sequence even when R has nilpotents. For an augmented cycle, lift a primitive of its highest-degree symbol and subtract its differential. The filtration degree decreases, so this procedure terminates. Hence $K_\bullet \to A$ is a free bimodule resolution. The relative bar is also free over A^e, with its usual unit-insertion contraction. The map ι lifts the identity of A. Projectivity constructs its chain homotopy inverse degree by degree. Applying $\operatorname{Hom}_{A^e}(-, A)$ gives a cochain homotopy equivalence.

The induced differential on $\operatorname{Hom}_R(\Lambda_R^n \mathfrak{l}_N, A)$ is

$$\begin{aligned} (\delta_N \phi)(x_1, \dots, x_{n+1}) = {} & \sum_i (-1)^{i+1} [x_i, \phi(x_1, \dots, \widehat{x_i}, \dots, x_{n+1})]_A \\ & + \hbar \sum_{i<j} (-1)^{i+j} \phi([x_i, x_j], x_1, \dots, \widehat{x_i}, \dots, \widehat{x_j}, \dots, x_{n+1}). \end{aligned}$$

Commuting x past the factors of a symmetrized monomial gives

$$[x, \operatorname{sym}_N(f)]_A = \hbar \operatorname{sym}_N(\operatorname{ad}_x f).$$

Both sums therefore become $\hbar$ times the ordinary Chevalley differential. Precomposition with ι gives Alt_N. All formulas commute with coefficient reduction. □

67.2 The formal coefficient limit

Put $R = \mathbb{C}[\![\hbar]\!]$ and $A = \varprojlim_N A_N$. PBW identifies its underlying module with $S[\![\hbar]\!]$. Each coefficient of $\hbar$ is a polynomial. Define

$$A^{\widehat{\otimes}_R n} = \varprojlim_N A_N^{\otimes_{R_N} n}, \qquad C^\bullet_{\mathrm{HH,cont}}(A/R, A) = \varprojlim_N C^\bullet_{\mathrm{HH}}(A_N/R_N, A_N).$$

Equivalently, these are continuous R-multilinear normalized cochains on the completed relative tensors.

Corollary 67.2 (Formal relative Rees comparison)**.** *The compatible antisymmetrizations induce a quasi-isomorphism*

$$C^{\bullet}_{\mathrm{HH},\mathrm{cont}}(A/R, A) \longrightarrow \big(C^{\bullet}(\mathfrak{l}, S)[\![\hbar]\!], \hbar d_{\mathrm{CE}}\big).$$

Proof. Coefficient reduction is surjective in each cochain degree. For Hochschild cochains, lift outputs on the PBW input basis. The Chevalley exterior input has finite rank, so its outputs also lift. The comparison cones are acyclic and their transition maps are degreewise surjective. The long exact sequence shows that every transition kernel is acyclic. Choose a primitive of a compatible cycle at level zero. Lift it one level and correct its differential by a primitive in the transition kernel. Iteration gives compatible primitives at every order. The inverse-limit cone is therefore acyclic. □

The coefficient ring is part of this result. Absolute $\mathbb{C}$-linear cochains also detect variations of the coefficient parameter. For $\mathfrak{l} = 0$, the continuous absolute derivation $\partial_{\hbar}$ on R has no relative counterpart. For nonabelian $\mathfrak{l}$, the associated graded for the $\hbar$ filtration has zero differential, whereas d_{CE} need not vanish.

Chapter 68

The Kontsevich–Duflo map

68.1 Linear Poisson formality

For $\mathfrak{l} = \mathfrak{h}_{\leq d}$, put $A_d = U_\hbar(\mathfrak{h}_{\leq d})$ and

$$\mathcal{P}_d = \big(C^*(\mathfrak{h}_{\leq d}, \operatorname{Sym} \mathfrak{h}_{\leq d})[\![\hbar]\!], \hbar d_{\mathrm{CE}}\big).$$

This is the polynomial polyvector complex on $\mathfrak{h}^*_{\leq d}$ with differential $[\hbar\pi_d, -]$ for the linear Poisson tensor π_d. The relative comparison gives $C^\bullet_{\mathrm{HH,cont}}(A_d/R, A_d) \to \mathcal{P}_d$ as a map of complexes. Higher operations require a formality map with specified operadic structure.

Fix Kontsevich's formality morphism $\mathcal{U}$ on the affine space $\mathfrak{h}^*_{\leq d}$. Its twist by $\hbar\pi_d$ defines a star product $\star_d$ and an L_∞ quasi-isomorphism on the degree-shifted complexes. Let $T_d : (\operatorname{Sym} \mathfrak{h}_{\leq d}[\![\hbar]\!], \star_d) \to A_d$ be the algebra identification fixing the linear generators. This identification includes the equivalence between the Kontsevich and PBW products. PBW symmetrization alone identifies the underlying modules. The first Taylor coefficient of the twisted comparison is

$$\Phi_d(v) = (T_d)_* \sum_{r \geq 0} \frac{1}{r!} \mathcal{U}_{r+1}(\underbrace{\hbar\pi_d, \ldots, \hbar\pi_d}_{r}, v).$$

It has source $\mathcal{P}_d$ and target $C^\bullet_{\mathrm{HH,cont}}(A_d/R, A_d)$. At each order in $\hbar$, polynomial inputs have polynomial outputs. Its reduction modulo $\hbar$ is the Hochschild–Kostant–Rosenberg map, with the normalized antisymmetrization convention. That map is a quasi-isomorphism for the polynomial algebra. The complete $\hbar$ filtration then proves that Φ_d is a quasi-isomorphism. The higher Taylor coefficients supply the L_∞ identities after shifting by 1. These statements use the affine formality construction [103, Sections 6 and 8.3].

On invariant polynomials, the standard Duflo normalization gives

$$\mathrm{Duf}_\hbar = \mathrm{sym}_\hbar \circ j^{1/2}(\hbar\partial), \qquad j^{1/2}(x) = \det_{\mathfrak{h}_{\leq d}}{}^{1/2}\left(\frac{\sinh(\mathrm{ad}_x/2)}{\mathrm{ad}_x/2}\right).$$

It is an algebra isomorphism from $(\operatorname{Sym} \mathfrak{h}_{\leq d})^{\mathfrak{h}_{\leq d}}[\![\hbar]\!]$ to $Z(A_d)$, by the Duflo theorem [103, Section 8.3.4]. The square root has constant term one. This assertion concerns invariant degree-zero cohomology. It does not identify antisymmetrization with a homotopy-E_2 map.

68.2 The boundary action

The brace operad acts on $(C^\bullet_{\mathrm{HH,cont}}(A_d/R, A_d), A_d)$. Its closed color is the relative derived centre. A homotopy-E_2 refinement means a morphism or zigzag

$$\widetilde{\Phi}_d : \mathcal{P}_d \simeq C^\bullet_{\mathrm{HH,cont}}(A_d/R, A_d)$$

in complete R-linear homotopy-E_2 algebras, with the chosen Gerstenhaber structure on $\mathcal{P}_d$. A compatible Swiss-cheese refinement also specifies the mixed operations on the pair and their coherence homotopies.

Corollary 68.1 (Open–closed structure with a refinement)**.** *Given these refinements, the pair* $(\mathcal{P}_d, A_d)$ *has the transported homotopy Swiss-cheese action. Its open product is the enveloping product. The specified closed and mixed Taylor coefficients determine its closed bracket and boundary action.*

Proof. Transport the two-colored operations through the supplied equivalence of complete operadic algebras. The coherence homotopies are part of this equivalence. □

Chapter 69

The formal Hamiltonian limit

69.1 Weight completion and comparison data

The Hamiltonian Lie algebra and its boundary algebra have quotient systems

$$\mathfrak{h} = \varprojlim_{d} \mathfrak{h}_{\le d}, \qquad \widehat{U}_\hbar(\mathfrak{h}) = \varprojlim_{d} U_\hbar(\mathfrak{h}_{\le d}).$$

A Lie quotient does not by itself define a covariant map between Chevalley cochains with adjoint coefficients. Nor does an algebra quotient define a map between all Hochschild cochains with values in the algebra itself. A cochain must descend through the input quotient and the output quotient. Consequently the coefficient limit of Corollary 67.2 is distinct from this jet limit.

A *compatible jet comparison* consists of the following data. Choose complete R-linear cochain models P_d, H_d with transition maps, levelwise comparisons $P_d \simeq \mathcal{P}_d$ and $H_d \simeq C^\bullet_{\mathrm{HH,cont}}(A_d/R, A_d)$, and compatible quasi-isomorphisms $P_d \to H_d$. Require degreewise surjective transitions, or an explicit derived inverse-limit construction with the same exactness property. Specify identifications

$$\varprojlim_{d} P_d \simeq \mathcal{P}_\infty, \qquad \varprojlim_{d} H_d \simeq C^\bullet_{\mathrm{HH,cont}}(\widehat{U}_\hbar(\mathfrak{h})/R, \widehat{U}_\hbar(\mathfrak{h})),$$

where $\mathcal{P}_\infty$ is the proposed continuous Hamiltonian polyvector complex with differential $[\hbar\pi, -]$. The topology on the right combines the stated Hamiltonian weight completion with the $\hbar$ completion. The identifications must use this same topology. For higher operations, also supply compatible homotopy-E_2 and Swiss-cheese refinements whose operations converge in that topology.

Theorem 69.1 (Continuous comparison from compatible jets)**.** *A compatible jet comparison induces a continuous quasi-isomorphism*

$$\Phi_\infty : \mathcal{P}_\infty \xrightarrow{\sim} C^\bullet_{\mathrm{HH,cont}}(\widehat{U}_\hbar(\mathfrak{h})/R, \widehat{U}_\hbar(\mathfrak{h})).$$

Compatible homotopy-E_2 and Swiss-cheese refinements give the corresponding complete open–closed operations.

Proof. For surjective transitions, the cones of the levelwise maps are acyclic inverse systems with surjective maps in each degree. The primitive-lifting argument of Corollary 67.2 proves exactness of their limit. For a derived inverse limit, use its invariance under levelwise equivalences. Apply the supplied identifications of the two limits. Compatibility and convergence of the operadic maps give the final assertion. □

The finite PBW comparison constructs the relative centre at each fixed jet. A continuous field-theory comparison additionally requires the jet datum and a map from the specified field observables to $\mathcal{P}_\infty$. Universal graph formulas alone do not provide the cochain transition maps.

Principle 69.2 (Universal brane principle). The boundary algebra $\widehat{U}_\hbar(\mathfrak{h})$ is the universal formal Hamiltonian brane. Finite matrix branes are representation-theoretic images of this algebra. Its complete relative derived centre is the algebraic target for bulk observables. A field-theory identification requires the continuous comparison data stated above.

Book XIV

The finite-jet laboratory

Chapter 70

The first three Hamiltonian jets

70.1 Quadratic Hamiltonians

Set

$$e = \frac{1}{2}z_1^2, \qquad f = -\frac{1}{2}z_2^2, \qquad h = -z_1 z_2.$$

A direct differentiation gives

$$\{h, e\} = 2e, \qquad \{h, f\} = -2f, \qquad \{e, f\} = h.$$

Thus $\mathfrak{h}_{\le 2} = R_2$ is $\mathfrak{sl}_2$ with its standard Chevalley basis. The finite cotangent theory for $d = 2$ is the linear BV theory of $T^*[-1]\mathfrak{sl}_2$ and its boundary algebra is $U_\hbar(\mathfrak{sl}_2)$.

70.2 Cubic Hamiltonians

Let

$$v_j = z_1^{3-j} z_2^{j}, \qquad 0 \le j \le 3.$$

Then

$$\{h, v_j\} = (3 - 2j)v_j, \qquad \{e, v_j\} = j v_{j-1}, \qquad \{f, v_j\} = (3 - j)v_{j+1},$$

with the boundary terms understood as zero. Hence R_3 is the irreducible highest-weight-three module.

Proposition 70.1 (The cubic jet algebra)**.** *The degree-three jet algebra is the semidirect product*

$$\mathfrak{h}_{\le 3} \cong \mathfrak{sl}_2 \ltimes \mathrm{Sym}^3(\mathbb{C}^2),$$

where the second summand is abelian.

Proof. The bracket of two cubic polynomials has degree four and vanishes in the degree-three quotient. The bracket with a quadratic polynomial is the displayed $\mathfrak{sl}_2$ action. □

70.3 The first nonlinear layer

Write $w_k = z_1^{4-k} z_2^{k}$, $0 \le k \le 4$. In $\mathfrak{h}_{\le 4}$ the cubic modes satisfy

$$\{v_i, v_j\} = 3(j - i)w_{i+j-1}, \tag{70.1}$$

whenever the index lies between 0 and 4; the coefficient makes every remaining case vanish.

Calculation 70.2 (Two brackets)**.**

$$\{z_1^3, z_2^3\} = 9 z_1^2 z_2^2 = 9 w_2, \qquad \{z_1^2 z_2, z_1 z_2^2\} = 3 z_1^2 z_2^2 = 3 w_2.$$

The two independent cubic pairs reach the same quartic channel with amplitudes in the ratio $3 : 1$.

Proof. Differentiate each monomial in the two symplectic coordinates and substitute in the Poisson formula. The first pair gives $3z_1^2 \cdot 3z_2^2$. The second gives $2z_1z_2 \cdot 2z_1z_2 - z_1^2 \cdot z_2^2 = 3z_1^2z_2^2$. □

Theorem 70.3 (The quartic jet presentation)**.** *The Lie algebra $\mathfrak{h}_{\le 4}$ is generated by $e, f, h, v_0, v_1, v_2, v_3$. Its relations are the $\mathfrak{sl}_2$ relations, the irreducible action on the v_j, formula* (70.1)*, and the induced $\mathfrak{sl}_2$ action on R_4. The quartic subspace is abelian in the quotient.*

Proof. Quadratic and cubic generators produce every quartic monomial through (70.1). Brackets involving a quartic and a cubic have degree five, and brackets of two quartics have degree six, so both vanish after projection. Jacobi follows from the polynomial Poisson bracket before projection. □

Chapter 71

Moyal quantization in coordinates

71.1 The bidifferential formula

For polynomials on $\mathbb{C}^2$, the Moyal product is

$$f \star g = \sum_{n \geq 0} \frac{1}{n!} \left(\frac{\hbar}{2}\right)^n \sum_{r=0}^{n} (-1)^r \binom{n}{r} (\partial_1^{n-r} \partial_2^r f)(\partial_1^r \partial_2^{n-r} g). \tag{71.1}$$

Every polynomial pair gives a finite sum.

Theorem 71.1 (Quadratic exactness)**.** *For polynomials f, g of degree at most two,*

$$[f, g]_\star = \hbar\{f, g\}.$$

The quadratic generators therefore define a Lie representation of the Rees-scaled $\mathfrak{sl}_2$. Their associative algebra has an additional Casimir relation.

Proof. The commutator retains the odd powers of the Poisson bidifferential. Derivatives of order three vanish on quadratics, so the first-order term is the entire commutator. □

Proposition 71.2 (The oscillator quotient)**.** *Let $R_N = \mathbb{C}[\hbar]/(\hbar^{N+1})$ and let $W_N = R_N[q, p]$ have the Moyal product for $\{q, p\} = 1$. Let U_N have generators E, F, H with relations*

$$[H, E] = 2\hbar E, \qquad [H, F] = -2\hbar F, \qquad [E, F] = \hbar H.$$

Set $\Omega = EF + FE + H^2/2$. The assignments

$$E \longmapsto e = q^2/2, \qquad F \longmapsto f = -p^2/2, \qquad H \longmapsto h = -qp$$

induce an algebra isomorphism

$$U_N/(\Omega + 3\hbar^2/8) \xrightarrow{\sim} W_N^{(q,p)\mapsto(-q,-p)}.$$

These compatible isomorphisms also hold after $\hbar$-adic completion. If linear Hamiltonians are retained, the generated associative algebra is the full Weyl algebra.

Proof. Quadratic exactness proves the three commutator relations. The associative products are

$$\begin{aligned} e \star f &= -q^2p^2/4 - \hbar qp/2 - \hbar^2/8, \\ f \star e &= -q^2p^2/4 + \hbar qp/2 - \hbar^2/8, \\ h \star h &= q^2p^2 - \hbar^2/4. \end{aligned}$$

Thus Ω maps to $-3\hbar^2/8$. Its commutator with each of E, F, H vanishes by the defining relations. Give these generators PBW degree one. Give an even monomial $q^a p^b$ degree $(a + b)/2$. The leading symbol of $\Omega + 3\hbar^2/8$ is

$C = 2EF + H^2/2$. It is a non-zero-divisor in $R_N[E, F, H]$, since $2C$ is monic in H. Centrality and this property give

$$\operatorname{gr}\big(U_N/(\Omega + 3\hbar^2/8)\big) = R_N[E, F, H]/(C).$$

Indeed, every element of the two-sided ideal is a multiple of the central generator. Its leading symbol is the product of the two leading symbols. The quotient has basis $E^a H^b F^c$ with $a, c \geq 0$ and $b = 0, 1$. Their leading symbols map to unit multiples of the distinct monomials $q^{2a+b} p^{2c+b}$, precisely the even monomials. Hence the associated-graded map is an isomorphism. Lifting highest-degree terms proves surjectivity and comparing them proves injectivity. Every element at fixed N has finite polynomial degree. The formulas commute with reduction in N, so their inverse limit gives the formal isomorphism. Finally q, p generate W_N and topologically generate its formal limit. □

The primitive coproduct on U_N does not descend to this quotient. Modulo $\hbar$, it sends the classical relation to

$$\Delta(C) = C \otimes 1 + 1 \otimes C + 2E \otimes F + 2F \otimes E + H \otimes H.$$

The cross-term is nonzero in the tensor square of $\mathbb{C}[E, F, H]/(C)$. For example, evaluation at $(E, F, H) = (1, 0, 0)$ in the first factor and $(0, 1, 0)$ in the second gives 2. Thus the oscillator quotient is an associative algebra, with no coproduct inherited from these primitive generators.

Calculation 71.3 (Cubic correction)**.** For $f = z_1^3$ and $g = z_2^3$,

$$f \star g - g \star f = 9\hbar z_1^2 z_2^2 + \frac{3}{2}\hbar^3.$$

The first term quantizes the quartic Hamiltonian channel. The constant is the first wheel-free ordering correction visible beyond the quadratic Lie algebra.

Proof. In (71.1), the odd orders are $n = 1$ and $n = 3$. The first contributes $9\hbar z_1^2 z_2^2$. The third contributes $2(\hbar/2)^3(3!) = 3\hbar^3/2$. □

71.2 Weyl generators

With the convention $\{q, p\} = 1$, the symbols satisfy $[q, p]_\star = \hbar$. Their Weyl-ordered differential operators are

$$\widehat{q} = q, \qquad \widehat{p} = -\hbar\partial_q, \qquad [\widehat{q}, \widehat{p}] = \hbar.$$

The abstract Weyl algebra is defined by this commutator relation. Weyl ordering transports its multiplication to the Moyal product. The differential-operator representation realizes these relations over $\mathbb{C}[\hbar]$ and its formal completion. The finite quotient argument above uses the abstract R_N-algebra, so it does not require a faithful differential-operator representation at nilpotent $\hbar$.

Chapter 72

Chevalley coordinates and the Poisson differential

Choose a basis e_a of $\mathfrak{h}_{\leq d}$ with structure constants $[e_a, e_b] = f^c{}_{ab} e_c$. Let c^a be the odd Chevalley coordinate and let x_a be the even coordinate on $\mathfrak{h}^*_{\leq d}$. Then

$$d_{\mathrm{CE}} c^c = -\frac{1}{2} f^c{}_{ab} c^a c^b, \qquad d_{\mathrm{CE}} x_a = f^c{}_{ba} c^b x_c. \tag{72.1}$$

The first equation encodes the Lie bracket. The second is the adjoint action on $\mathrm{Sym}(\mathfrak{h}_{\leq d})$, viewed as functions on the dual.

Proposition 72.1 (Hamiltonian interpretation)**.** *For a polynomial $F(x)$,*

$$d_{\mathrm{CE}} F = c^a \{x_a, F\}_{\mathrm{lin}}.$$

Consequently $d_{\mathrm{CE}}^2 = 0$ is the Jacobi identity for the linear Poisson tensor.

Proof. The linear Poisson bracket is $\{x_a, x_b\} = f^c{}_{ab} x_c$. The Leibniz rule gives (72.1) on every polynomial. The square is one half of the Schouten bracket of the Poisson tensor with itself. □

72.1 The $\mathfrak{sl}_2$ invariant

For the quadratic jet, identify the coordinate functions with e, f, h. The invariant

$$q = 2ef + \frac{1}{2} h^2$$

is d_{CE}-closed. Its formal Duflo image is $\Omega + \hbar^2/2$, where $\Omega = ef + fe + h^2/2$ in the Rees enveloping algebra. This is the degree-zero invariant comparison. Under the oscillator quotient of Proposition 71.2, its image is $\hbar^2/8$.

Chapter 73

Residue duality and the formal current algebra

73.1 The continuous dual

Let

$$\rho_{ab} = z_1^{-a-1} z_2^{-b-1}\, \mathrm{d}z_1 \mathrm{d}z_2, \qquad a, b \geq 0.$$

The Grothendieck residue pairing is

$$\mathrm{Res}_0\Big(z_1^i z_2^j \rho_{ab} \Big) = \delta_{ia}\delta_{jb}.$$

The continuous dual of $\mathbb{C}[\![z_1, z_2]\!]$ is the direct sum of the ρ_{ab}. The continuous dual of the pointed algebra $\mathfrak{h} = \mathfrak{m}^2$ is the span of the modes with $a + b \geq 2$. The two modes of total degree one pair with the translation sector of $\mathfrak{h}^{\mathrm{aff}}$.

Proposition 73.1 (Coadjoint current formula)**.** *For $f = z_1^i z_2^j$ and ρ_{ab},*

$$\mathrm{ad}^*_f(\rho_{ab})(g) = -\rho_{ab}(\{f, g\}).$$

In the residue basis this is a finite linear combination of modes of bidegree

$$(a - i + 1, b - j + 1).$$

Proof. The definition is the coadjoint action. The Poisson bracket lowers the total degree of $f\,g$ by two, so residue conservation fixes the displayed bidegree. The Poisson formula contains two derivative terms. □

The shifted cotangent current algebra

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[-1]$$

is therefore a complete topological Lie algebra with a continuous invariant pairing. Its finite jet quotients are precisely the $\mathfrak{g}_d$ used above.

Chapter 74

All-arity open–closed operations

74.1 Hochschild braces

For Hochschild cochains $F \in C^{p}(A, A)$ and $G_i \in C^{q_i}(A, A)$, the brace

$$F\{G_1, \dots, G_r\}$$

is the signed sum over ordered insertions of the G_i into the inputs of F. The brace identities state that a nested insertion is the sum over all ways of distributing the inner insertions among the outer slots.

Theorem 74.1 (Brace realization with a refinement). *For $A = A_d$, supply the homotopy-E_2 and mixed refinements of Corollary 68.1. They identify the selected homotopy Gerstenhaber operations on $\mathcal{P}_d$ with the relative Hochschild centre operations. The chosen mixed refinement also identifies their action on the boundary algebra.*

Proof. The relative Hochschild complex has its brace operations. The supplied operadic equivalence preserves their coherent homotopies. An equivalence of complexes alone would not imply this conclusion. □

74.2 The first mixed identities

A relative Hochschild 1-cocycle D of an algebra concentrated in degree zero is an R-linear derivation:

$$D(ab) = D(a)b + aD(b).$$

For two such cocycles D, E, their Gerstenhaber bracket acts by

$$[D, E](a) = D(E(a)) - E(D(a)).$$

For a graded algebra and homogeneous derivations, these formulas become

$$D(ab) = D(a)b + (-1)^{|D||a|}aD(b),$$
$$[D, E](a) = D(E(a)) - (-1)^{|D||E|}E(D(a)).$$

Higher Hochschild cochains act through multilinear evaluations and braces. They need not act by derivations or carry a coproduct defining a diagonal action. A compatible mixed refinement transports these operations, including their homotopies.

Corollary 74.2 (Continuous open–closed structure with compatible jets). *Under the jet comparison and mixed refinement hypotheses of Theorem 69.1, the pair $(\mathcal{P}_\infty, \widehat{U}_\hbar(\mathfrak{h}))$ has the corresponding complete homotopy Swiss-cheese action. Its closed color is equivalent to the continuous relative derived centre.*

Proof. The supplied finite operations commute with the transition maps and converge in the specified completion. Their limit acts on the two limit objects. The comparison theorem identifies the closed complex with continuous relative Hochschild cochains. □

Chapter 75

Finite Chan–Paton ranks

75.1 Rank one

For $N = 1$, the commuting equation is automatic. The boundary coordinate algebra is $\mathbb{C}[x, y]$, and

$$J_1(f) = f(x, y).$$

The necklace bracket reduces to the ordinary Poisson bracket. Restriction to $f \in \mathfrak{m}^2$ gives the defining pointed-Hamiltonian representation. The two linear Hamiltonians act by translations and recover the affine center-of-mass extension.

75.2 Rank two and Cayley–Hamilton reduction

For a 2×2 matrix X,

$$X^2 - (\operatorname{Tr} X)X + (\det X)I = 0.$$

Hence every trace containing two consecutive powers of X reduces to traces with X-degree at most one after adjoining $\operatorname{Tr} X$ and $\det X$. The same statement holds for Y. On the commuting locus, simultaneous diagonalization identifies the invariant algebra with the symmetric polynomials in two ordered pairs.

Proposition 75.1 (First finite-rank trace relation)**.** *For commuting 2×2 matrices,*

$$p_{20} = p_{10}^2 - 2 \det X, \qquad p_{02} = p_{01}^2 - 2 \det Y,$$

and

$$p_{11}^2 - p_{20} p_{02} = -\big(x_1 y_2 - x_2 y_1\big)^2.$$

The stable generators remain independent until a relation forced by the rank is reached.

Proof. Write the joint eigenvalues as (x_1, y_1) and (x_2, y_2) and expand both sides. □

75.3 Stable passage

The transition from rank N to rank $N + 1$ adds a zero eigenvalue. Each fixed weighted degree stabilizes once N exceeds that degree. The inverse limit is therefore the completed multisymmetric algebra. The stable trace is an exact quotient of the universal centre on the polynomial trace sector.

Book XV

Global holomorphic symplectic descent

Chapter 76

Fedosov descent

76.1 The Weyl algebroid

All Hochschild cochains and Morita centres in this chapter are continuous and relative to $R = \mathbb{C}[\![\hbar]\!]$. The symplectic globalization uses its own sheafwise formality and Fedosov resolution data, independently of the finite Hamiltonian comparison. Let (S, ω) be a holomorphic symplectic surface. The bundle of formal Weyl algebras has fibre the completed Weyl algebra of a formal symplectic plane. A holomorphic symplectic connection and a Fedosov one-form define a flat connection

$$D_F = \nabla + \frac{1}{\hbar}[\gamma, -]$$

whose flat sections form a deformation-quantization algebroid $\mathcal{W}_{S,\hbar}$.

Theorem 76.1 (Globalized formality on a holomorphic symplectic surface)**.** *Fix a stable formality morphism and a compatible homotopy-E_2 refinement. There is a sheafwise homotopy-E_2 quasi-isomorphism*

$$\Phi_S : (\mathrm{PV}_S[\![\hbar]\!], [\hbar\pi, -]) \xrightarrow{\sim} C^{\bullet}_{\mathrm{HH}}(\mathcal{W}_{S,\hbar}, \mathcal{W}_{S,\hbar}),$$

where $\pi = \omega^{-1}$. A change of holomorphic symplectic connection or Fedosov one-form gives a gauge-equivalent resolution and a homotopic quasi-isomorphism. Holomorphic symplectomorphisms act equivariantly on the resulting homotopy class.

Proof. Fedosov resolutions replace both sheaves by acyclic sheaves of formal fibrewise polyvectors and polydifferential operators. The fixed covariant formality morphism acts fibrewise and intertwines the Fedosov differentials. A change of connection or Fedosov one-form is a gauge transformation in the pronilpotent resolution and therefore gives a homotopy. □

Corollary 76.2 (Global mixed-HT centre)**.** *The deformation-quantization category* $\mathrm{Perf}(\mathcal{W}_{S,\hbar})$ *has Morita centre*

$$Z_{\mathrm{Mor}}\big(\mathrm{Perf}(\mathcal{W}_{S,\hbar})\big) \simeq \mathbb{R}\Gamma(S, \mathrm{PV}_S[\![\hbar]\!], [\hbar\pi, -]).$$

For a K3 surface this computes the symplectic Poisson centre in the stated sheafwise model. A Hamiltonian BF identification additionally requires a comparison from its Chevalley observables to this symplectic polyvector complex.

Proof. The Morita centre of perfect modules over an algebroid is the derived global section complex of its Hochschild cochains. Apply the globalized formality equivalence and then derived global sections. A K3 surface is holomorphic symplectic, so the same sheafwise calculation applies whenever the chosen globalization data are available. □

Chapter 77

Factorization and descent

77.1 The algebraic mixed-HT realization

The sheaf

$$\mathcal{P}_S = (\mathrm{PV}_S[\![\hbar]\!], [\hbar\pi, -])$$

is a sheaf of complete homotopy-E_2 algebras. Dunn additivity gives its locally constant factorization realization in two oriented topological directions. The holomorphic sheaf variable retains the dependence on S.

Theorem 77.1 (Algebraic factorization realization)**.** *The homotopy-E_2 sheaf $\mathcal{P}_S$ determines a complete mixed holomorphic–topological factorization algebra*

$$\mathcal{A}^{\mathrm{alg}}_{S,\mathrm{bulk}} \quad \textit{on} \quad \mathbb{R}^2_{\mathrm{top}} \times S.$$

Its universal boundary category is $\mathrm{Perf}(\mathcal{W}_{S,\hbar})$*, and the bulk–boundary morphism is a local equivalence*

$$\mathcal{A}^{\mathrm{alg}}_{S,\mathrm{bulk}} \xrightarrow{\sim} Z^{\mathrm{der}}_{\mathrm{Mor}}\big(\mathrm{Perf}(\mathcal{W}_{S,\hbar})\big).$$

On a formal Darboux chart this map is the symplectic polyvector-to-Weyl Hochschild comparison. Comparing it with the pointed Hamiltonian centre additionally requires a map between those two polyvector complexes.

Proof. The factorization envelope of a complete homotopy-E_2 algebra is locally constant on oriented two-disks. Sheafwise application to $\mathcal{P}_S$ gives the first assertion. The Fedosov algebroid quantizes the symplectic Poisson tensor on S. Globalized formality identifies its Hochschild cochains with $\mathcal{P}_S$, and Morita invariance identifies Hochschild cochains with the centre of the perfect-module category. The construction is local on S and therefore respects factorization descent. □

77.2 BV realization

The cyclic Lie sheaf of formal Hamiltonians has the shifted-cotangent action

$$S_{\mathrm{BF}}(A, B) = \int_{\mathbb{R}^2 \times S} \Big\langle B, (\mathrm{d} + \bar{\partial})A + \frac{1}{2}[A, A] \Big\rangle.$$

A comparison of its classical observables with $\mathcal{A}^{\mathrm{alg}}_{S,\mathrm{bulk}}$ must specify a map from Hamiltonian Chevalley cochains to the symplectic polyvector complex. The finite Lie algebra identification alone does not supply that map. A compatible gauge fixing, extension of graph weights to compactified configurations, and solution of the quantum master equation produce a perturbative factorization realization of the same algebraic bulk–boundary map. The finite jet zero modes carry the exact quantum measure constructed above.

Book XVI

Matrix branes and trace geometry

Chapter 78

The derived commuting stack

Let $X, Y \in \mathfrak{gl}_N$ and let $\psi \in \mathfrak{gl}_N[1]$. The Koszul differential

$$QX = QY = 0, \qquad Q\psi = [X, Y]$$

resolves the commuting equation. The simultaneous conjugation action gives the boundary cdga

$$A_N = C^*_{\mathrm{CE}}\big(\mathfrak{gl}_N, \mathrm{Sym}(\mathfrak{gl}_N \oplus \mathfrak{gl}_N \oplus \mathfrak{gl}_N[1])\big).$$

Proposition 78.1 (Representation of the universal brane)**.** *Substitution of matrices defines a Lie morphism*

$$\rho_N : \mathfrak{h} \longrightarrow \mathrm{Der}(A_N), \qquad f \longmapsto \delta_f,$$

which integrates to an algebra morphism

$$\widehat{U_\hbar}(\mathfrak{h}) \longrightarrow \mathrm{End}(A_N)[\![\hbar]\!], \qquad f \longmapsto \hbar\, \delta_f.$$

Thus the N-brane stack is a representation of the universal boundary algebra.

Proof. For a polynomial Hamiltonian f, cyclic differentiation gives the matrix vector field

$$\delta_f(X) = -\partial^{\mathrm{cyc}}_Y f(X, Y), \qquad \delta_f(Y) = \partial^{\mathrm{cyc}}_X f(X, Y).$$

The cyclic Euler identity implies $\delta_f([X, Y]) = 0$. Hence δ_f commutes with the Koszul differential $Q\psi = [X, Y]$. The same formulas are equivariant for simultaneous conjugation, so δ_f descends to a derivation of the Chevalley–Eilenberg quotient A_N. The trace-moment-map calculation gives

$$[\delta_f, \delta_g] = \delta_{\{f,g\}},$$

first for polynomial Hamiltonians and then continuously for the formal completion. This proves that ρ_N is a continuous Lie morphism. The assignment $f \mapsto \hbar\delta_f$ satisfies

$$[\hbar\delta_f, \hbar\delta_g] = \hbar^2 \delta_{\{f,g\}} = \hbar\,(\hbar\delta_{\{f,g\}}),$$

which is the Rees relation. The universal property of the completed Rees enveloping algebra gives the displayed morphism. It is the universal boundary action evaluated in the N-dimensional Chan–Paton representation. □

Chapter 79

Necklace Hamiltonians

Let

$$F = \mathbb{C}\langle x, y\rangle, \qquad F_{\mathrm{cyc}} = F/[F, F].$$

For a cyclic word f, define cyclic derivatives by cutting the word at every occurrence of the selected letter. Put

$$\delta_f(x) = -\partial_y^{\mathrm{cyc}} f, \qquad \delta_f(y) = \partial_x^{\mathrm{cyc}} f.$$

Theorem 79.1 (Necklace Lie algebra)**.** *The bracket*

$$\{f, g\}_{\mathrm{neck}} = \left[\partial_x^{\mathrm{cyc}} f\ \partial_y^{\mathrm{cyc}} g - \partial_y^{\mathrm{cyc}} f\ \partial_x^{\mathrm{cyc}} g\right]_{\mathrm{cyc}}$$

makes $F_{\mathrm{cyc}}/\mathbb{C}$ a Lie algebra. The assignment $f \mapsto \delta_f$ is a Lie morphism. It preserves the moment map:

$$\delta_f([x, y]) = 0.$$

Proof. The cyclic Euler identity is

$$[x, \partial_x^{\mathrm{cyc}} f] + [y, \partial_y^{\mathrm{cyc}} f] = 0.$$

It gives the moment-map statement. The canonical double Poisson bracket on the free algebra descends to the displayed necklace bracket; its Jacobi identity gives the first assertion. Evaluation on generators gives $[\delta_f, \delta_g] = \delta_{\{f,g\}_{\mathrm{neck}}}$. □

Theorem 79.2 (Trace moment map)**.** *For*

$$J_N(f) = \operatorname{Tr} f(X, Y),$$

one has

$$\{J_N(f), J_N(g)\}_{\mathcal{M}_N} = J_N\big(\{f, g\}_{\mathrm{neck}}\big),$$

where $\mathcal{M}_N = \mathfrak{gl}_N^{2}$ carries $\operatorname{Tr}(\mathrm{d}X \wedge \mathrm{d}Y)$.

Proof. The differential of a trace is the matrix evaluation of the cyclic derivative. Contracting the two differentials with the inverse trace symplectic form gives the necklace formula. □

Chapter 80

Stable traces

On the semisimple commuting locus write the joint eigenvalues as (x_i, y_i). Define

$$p_{ab}^{(N)} = \sum_{i=1}^{N} x_i^a \, y_i^b.$$

Theorem 80.1 (Stable multisymmetric Poisson algebra)**.** *The inverse stable trace algebra is the completed polynomial algebra*

$$\mathcal{P}_\infty = \mathbb{C}[\![\, p_{ab} \mid a, b \geq 0,\ a + b > 0 \,]\!].$$

Its Poisson bracket is

$$\{p_{ab}, p_{cd}\} = (ad - bc)\, p_{a+c-1,b+d-1}.$$

Proof. The stable fundamental theorem for multisymmetric functions gives polynomial generators p_{ab}. The bracket follows from $\{x_i, y_j\} = \delta_{ij}$ and the Leibniz rule. □

Remark 80.2. The stable trace algebra is a quotient of the universal closed algebra after representation and scalarization. The universal centre retains Hochschild classes, stabilizer directions, and higher central operations that vanish under the trace.

Chapter 81

The two-brane cone

Separate center-of-mass and relative coordinates. Put

$$u = x_1 - x_2, \qquad v = y_1 - y_2, \qquad E = \frac{1}{2}u^2, \quad F = -\frac{1}{2}v^2, \quad H = -uv.$$

Then

$$H^2 + 4EF = 0.$$

Proposition 81.1 (The A_1 Kirillov–Kostant cone)**.** *With* $\{u, v\} = 2$,

$$\{H, E\} = 4E, \qquad \{H, F\} = -4F, \qquad \{E, F\} = 2H.$$

After the evident rescaling this is the $\mathfrak{sl}_2$ *Lie–Poisson bracket on the nilpotent cone.*

Proof. The three identities follow by direct differentiation. The quadratic equation is the determinant-zero equation for the traceless matrix

$$\begin{pmatrix} H & 2E \\ 2F & -H \end{pmatrix}.$$

□

Book XVII

Wheel corrections and exact quantum structure

Chapter 82

The Duflo wheel

First set the coefficient parameter to 1 in the polynomial Rees algebra. For the resulting ordinary $\mathfrak{sl}_2$ enveloping algebra, take generators

$$[h,e]=2e, \qquad [h,f]=-2f, \qquad [e,f]=h.$$

The invariant polynomial and the quadratic Casimir are

$$q=2ef+\frac{1}{2}h^2, \qquad \Omega=ef+fe+\frac{1}{2}h^2.$$

Calculation 82.1 (The zero-point shift)**.** The Duflo map gives

$$\mathrm{Duf}(q)=\Omega+\frac{1}{2}.$$

On the irreducible highest-weight-n module,

$$\left(\Omega+\frac{1}{2}\right)\Big|_{V_n}=\frac{(n+1)^2}{2}\,\mathrm{id}\,.$$

Proof. Write $x=ue+vf+wh$. Direct adjoint matrices give $\mathrm{Tr}(\mathrm{ad}_x^2)=8uv+8w^2$. Thus the quadratic term of $j^{1/2}(\partial)$ is $(\partial_e\partial_f+\partial_h^2)/6$. Applied to $q=2ef+h^2/2$, it gives $(2+1)/6=1/2$. Higher differential terms annihilate this quadratic polynomial. Symmetrization sends q to Ω. On a highest-weight vector, ef acts by n, fe by zero, and $h^2/2$ by $n^2/2$. Centrality gives the eigenvalue on the entire irreducible module. □

Restoring the Rees parameter gives $\mathrm{Duf}_\hbar(q)=\Omega+\hbar^2/2$. The polynomial calculation permits specialization at 1; arbitrary formal power series do not admit that specialization. Higher wheels assemble into $j^{1/2}$ on higher-degree invariants.

Chapter 83

The local centre with comparison data

Assume a compatible jet comparison and a complete homotopy-E_2 refinement as in Theorem 69.1. Put

$$\mathcal{A} = \widehat{U}_\hbar(\mathfrak{h}), \qquad \mathcal{A}^{\mathrm{alg}}_{\mathrm{bulk}} = \mathrm{Fact}_{\mathsf{E}_2}(\mathcal{P}_\infty).$$

The factorization envelope uses the two oriented topological directions. The formal holomorphic coefficient remains at the Darboux point. Let C_∂ be the category of perfect complete $\mathcal{A}$-modules in the chosen continuous R-linear Morita theory. Assume its centre is computed by $C^\bullet_{\mathrm{HH},\mathrm{cont}}(\mathcal{A}/R, \mathcal{A})$.

Theorem 83.1 (Local derived-centre comparison)**.** *These data give an equivalence of complete homotopy-E_2 factorization algebras*

$$\mathcal{A}^{\mathrm{alg}}_{\mathrm{bulk}} \xrightarrow{\sim} Z^{\mathrm{der}}_{\mathrm{Mor},R}(C_\partial).$$

At the formal insertion point the map is the selected refinement of Φ_∞. Every continuous boundary module has the induced derived centre action.

Proof. The compatible jet theorem gives the complete homotopy-E_2 comparison. The factorization envelope preserves this equivalence. The assumed continuous Morita model identifies its target with the centre. A centre element is a derived natural endomorphism of the identity functor, so evaluation gives the action on each boundary module. □

Corollary 83.2 (Global K3 comparison with descent data)**.** *Let S be a K3 surface with holomorphic symplectic form. Suppose the local comparisons extend to a sheafwise equivalence with the relative Hochschild centre of* $\mathrm{Perf}(\mathcal{W}_{S,\hbar})$*. Suppose further that the selected cyclic and mixed refinements extend and satisfy descent in the same completed category. Then derived global sections give the corresponding global centre, cyclic, and open–closed comparisons.*

Proof. Derived global sections preserve equivalences of sheaves. Apply them to the supplied sheafwise maps and their coherence homotopies. □

Chapter 84

Cyclic classes, stable traces, and Lie homology

A trace of the unit records the representation dimension. For $B = \mathbb{C}$, one has $\mathrm{Tr}_N(1) = N$, so ordinary traces cannot define a rank-compatible map on all of $HC_0(B)$. A scalar-reduced trace removes this dependence after choosing an augmentation.

Let B be a unital associative $\mathbb{C}$-algebra. The affine representation scheme $\mathrm{Rep}_N B$ represents unital algebra maps $B \to \mathrm{Mat}_N(D)$ for commutative $\mathbb{C}$-algebras D. Put

$$\mathcal{R}_N = \mathcal{O}(\mathrm{Rep}_N B)^{GL_N}, \qquad \mathfrak{c}_B = HC_0(B) = B/[B,B],$$

where $[B,B]$ denotes the linear span of commutators. The ordinary trace is the linear map

$$\mathrm{Tr}_N : \mathfrak{c}_B \longrightarrow \mathcal{R}_N, \qquad \mathrm{Tr}_N([b])(\rho) = \mathrm{Tr}(\rho(b)).$$

Cyclicity makes it independent of the representative, and conjugation invariance puts its image in $\mathcal{R}_N$.

For Lie and Poisson compatibility, specify a Lie bracket on $\mathfrak{c}_B$, a Poisson bracket on each $\mathcal{R}_N$, and the identities

$$\{\mathrm{Tr}_N\,\beta, \mathrm{Tr}_N\,\gamma\}_N = \mathrm{Tr}_N[\beta,\gamma]_{\mathrm{cyc}} \qquad (\beta,\gamma \in \mathfrak{c}_B). \tag{84.1}$$

A Hamiltonian comparison also specifies a Lie map $j : \mathfrak{h} \to \mathfrak{c}_B$ for this bracket. These are additional data on B. The preceding necklace bracket and trace moment map supply the bracket and trace data for $B = \mathbb{C}\langle x, y\rangle$ with its canonical symplectic bracket. The map j remains an additional comparison. An arbitrary unital algebra does not carry these structures by definition.

For rank stabilization, choose an augmentation $\epsilon : B \to \mathbb{C}$. It defines the rank-stabilization morphism

$$s_N : \mathrm{Rep}_N B \longrightarrow \mathrm{Rep}_{N+1} B, \qquad \rho \longmapsto \rho \oplus \epsilon.$$

Restriction gives $s_N^* : \mathcal{R}_{N+1} \to \mathcal{R}_N$. Define the reduced cyclic space and trace by

$$\bar{\mathfrak{c}}_B = B/(\mathbb{C}1 + [B,B]), \qquad t_N(\bar{b})(\rho) = \mathrm{Tr}(\rho(b)) - N\epsilon(b).$$

Let $\mathfrak{m}_N \subset \mathcal{R}_N$ be the ideal of functions vanishing at the representation $b \mapsto \epsilon(b) I_N$. The completed invariant rings and stable target are

$$\widehat{\mathcal{R}}_N = \varprojlim_{r \ge 1} \mathcal{R}_N/\mathfrak{m}_N^r, \qquad \mathcal{R}_\infty^\wedge = \varprojlim_N \widehat{\mathcal{R}}_N.$$

Use the inverse-limit topologies. Write J for the augmentation ideal in $\mathrm{Sym}_{\mathbb{C}} \bar{\mathfrak{c}}_B$ and put

$$\widehat{\mathrm{Sym}}\,\bar{\mathfrak{c}}_B = \varprojlim_{r \ge 1} \mathrm{Sym}_{\mathbb{C}} \bar{\mathfrak{c}}_B / J^r.$$

The stable matrix Lie algebra is $\mathfrak{gl}_\infty B = \varinjlim_N \mathfrak{gl}_N B$ under block-zero Lie embeddings. For its homology, use trivial coefficients $\mathbb{C}$. Let $V_B = \bigoplus_{n \ge 1} HC_{n-1}(B)$, with $HC_{n-1}(B)$ placed in homological degree n. The notation $\mathrm{Sym}_{\mathrm{gr}} V_B$ denotes the free graded-commutative algebra, with the Koszul sign rule.

Theorem 84.1 (Cyclic traces and stable Lie homology). *Assume the compatible jet comparison and its homotopy Gerstenhaber refinement. For the augmented algebra and completion above, there are maps*

$$\mathcal{P}_\infty \xrightarrow{\Phi_\infty} C^*_{\mathrm{HH},\mathrm{cont}}(\mathcal{A}/R,\mathcal{A}), \qquad \mathrm{Tr}_N : \mathfrak{c}_B \longrightarrow \mathcal{R}_N,$$

$$\widehat{\mathrm{Sym}}\,\bar{\mathfrak{c}}_B \longrightarrow \mathcal{R}^\wedge_\infty, \qquad \mathrm{Sym}_{\mathrm{gr}} V_B \xrightarrow{\sim} H^{\mathrm{CE}}_*(\mathfrak{gl}_\infty B;\mathbb{C}).$$

The first preserves the selected homotopy Gerstenhaber structure. The second is linear and, under the bracket data (84.1)*, is a Lie map. Under those data, its composite with j preserves the Hamiltonian bracket. The third is a continuous homomorphism of commutative algebras, and the fourth is an isomorphism of graded Hopf algebras. These maps are natural for morphisms preserving the stated data.*

Proof. The first arrow is the chosen comparison of Theorem 69.1. Its operadic refinement gives the asserted Gerstenhaber compatibility. The trace formula above constructs the second arrow. Equation (84.1) supplies its Lie compatibility, and the Lie property of j gives the composite assertion.

The augmentation annihilates commutators. Both terms of t_N vanish on a commutator, and their values on a scalar multiple of the unit cancel. Thus t_N is well defined on $\bar{\mathfrak{c}}_B$. At the trivial representation its value is zero, so its image lies in $\mathfrak{m}_N$. Moreover,

$$\begin{aligned}(s_N^* t_{N+1})(\bar{b})(\rho) &= \mathrm{Tr}(\rho(b)) + \epsilon(b) - (N+1)\epsilon(b)\\ &= t_N(\bar{b})(\rho).\end{aligned}$$

The point $N\epsilon$ maps to $(N+1)\epsilon$, so $s_N^*(\mathfrak{m}^r_{N+1}) \subset \mathfrak{m}^r_N$. Hence restriction defines the transition maps between the completed rings. The symmetric-algebra extension of t_N sends J^r into $\mathfrak{m}^r_N$. It therefore induces a ring homomorphism on each corresponding quotient. Taking the inverse limit in r gives a continuous map to $\widehat{\mathcal{R}}_N$. The displayed stabilization identity makes these maps compatible in N. Their inverse limit is the third arrow.

The diagonal Lie map $\mathfrak{gl}_\infty B \to \mathfrak{gl}_\infty B \oplus \mathfrak{gl}_\infty B$ induces the homology coproduct. Block sum in the opposite direction induces the homology product, using the Künneth isomorphism over $\mathbb{C}$. The Loday–Quillen–Tsygan theorem identifies primitive homology in degree n with $HC_{n-1}(B)$.[1] It identifies the full stable homology with the free graded-commutative Hopf algebra on these primitives, giving the fourth arrow. The trace formulas, the augmentation-preserving rank maps, and stable Lie homology are natural. Naturality of the first arrow requires preservation of the selected jet comparison. □

Scalar reduction need not preserve the supplied Poisson bracket. For example, for $B = \mathbb{C}[x, y]$ with $\{x, y\} = 1$ and $\epsilon(x) = \epsilon(y) = 0$, traces on N points satisfy

$$\{\mathrm{Tr}_N\, x, \mathrm{Tr}_N\, y\} = N, \qquad t_N(\bar{1}) = 0.$$

A reduced Poisson comparison consequently requires additional compatibility, such as vanishing of ϵ on the selected cyclic brackets, or a restriction to a Hamiltonian subalgebra with that property. Only multiplicativity and continuity are asserted for the third arrow. For $B = \mathbb{C}$, the reduced cyclic space is zero and that arrow is the identity on $\mathbb{C}$.

[1] J.-L. Loday and D. Quillen, *Cyclic homology and the Lie algebra homology of matrices*, Commentarii Mathematici Helvetici **59** (1984), 565–591, Theorem 6.2, pp. 584–586.

Chapter 85

Synthesis

At each fixed Hamiltonian jet, the Rees enveloping algebra has relative Hochschild complex equivalent to the Chevalley complex with differential $\hbar d_{\mathrm{CE}}$. PBW antisymmetrization proves this over every R_N and in the formal coefficient limit. The Duflo theorem gives the separate multiplicative comparison on invariant functions. Quadratic Moyal Hamiltonians generate the oscillator quotient of the enveloping algebra. Their associative Casimir relation prevents the primitive coproduct from descending.

The Hamiltonian jet limit requires the compatible cochain systems of Theorem 69.1. With its operadic refinement and continuous Morita model, the centre acts on every supplied continuous boundary module M through

$$\mathcal{P}_\infty \xrightarrow{\Phi_\infty} C^\bullet_{\mathrm{HH,cont}}(\mathcal{A}/R, \mathcal{A}) \longrightarrow \mathrm{RHom}_{\mathcal{A}}(M, M).$$

The last arrow evaluates a derived natural endomorphism at M. A matrix-brane realization must construct such a module and compare its centre action with the selected trace functions. A commuting square with scalar representation invariants requires these additional comparison maps. Likewise, Fedosov descent concerns symplectic polyvectors and a Weyl algebroid. Its relation to Hamiltonian Lie cochains requires the specified map between their carriers.

Book XVIII

Technical appendices to the preceding book

Chapter 86

Signs in the shifted cotangent complex

Let e_a have degree zero and let $e^a[-1]$ have degree -1. The Chevalley generators dual to them have degrees 1 and 0. Thus

$$C^*_{\mathrm{CE}}(T^*[-1]\mathfrak{l}) = \mathrm{Sym}\big(\mathfrak{l}^\vee[-1] \oplus \mathfrak{l}\big).$$

The degree-zero polynomial factor is $\mathrm{Sym}\,\mathfrak{l} = \mathcal{O}(\mathfrak{l}^*)$. The differential on a polynomial p is

$$(d_{\mathrm{CE}}p)(x) = \mathrm{ad}_x\, p,$$

which is the linear Poisson differential.

Chapter 87

A direct Jacobi check for Hamiltonian jets

For monomials

$$f = z_1^a z_2^b, \qquad g = z_1^c z_2^d,$$

one has

$$\{f, g\} = (ad - bc) z_1^{a+c-1} z_2^{b+d-1}.$$

Substitution of three monomials gives a scalar multiple of

$$\det((a,b),(c,d))\ \det((a+c-1, b+d-1),(e,f)) + \text{cyclic}.$$

Writing the three coefficients in terms of the 2×2 determinants of the exponent vectors shows pairwise cancellation. The coordinate-free proof is $[\pi, \pi]_{\mathrm{Sch}} = 0$ for $\pi = \partial_{z_1} \wedge \partial_{z_2}$.

Chapter 88

Bibliographic note

The finite affine formality map and the separate symplectic globalization use Kontsevich formality and its Fedosov resolution [22, 48, 103]. The relative Hochschild model is the Cartan–Eilenberg comparison, with both Chevalley terms scaled by $\hbar$. Its finite-base resolution proof is Theorem 67.1. Jet limits and operadic refinements require the additional data stated above. The all-arity centre action is the Deligne–Tamarkin brace action [104, 117]. The global deformation-quantization category is the holomorphic Weyl algebroid. The matrix trace and necklace formulas are the noncommutative symplectic moment-map construction.

Book entrance: Calabi–Yau Quantum Groups

An extension of geometric objects defines a correspondence. Its action on cohomology depends on the coefficient theory and the allowed pushforward. Tensor products and exchange require a coproduct and compatible intertwiners. A trace on a specified representation then produces an amplitude. The reconstruction problem is to retain these operations through the comparisons between Calabi–Yau categories, boundary fields, Hall algebras, and quantum groups.

Book XIX

Categorical and field-theoretic foundations

Chapter 89

Calabi–Yau categories

89.1 Smoothness, properness, and the Calabi–Yau class

Let C be a small stable $\Bbbk$-linear category. Smoothness means that the diagonal bimodule is perfect. Properness means that the morphism complexes are perfect over $\Bbbk$. A d-Calabi–Yau structure is a nondegenerate negative-cyclic class

$$\theta_{\mathrm{CY}} \in HC^{-}_{d}(C)$$

whose underlying Hochschild trace identifies the diagonal bimodule with its dual shifted by d.

For $E \in C$, the tangent complex of the derived moduli stack of objects is

$$T_E \mathcal{M}_C \simeq \mathrm{RHom}_C(E, E)[1].$$

The Calabi–Yau trace pairs two tangent vectors by composition and trace.

Theorem 89.1 (The two universal categorical outputs)**.** *Let C be a smooth proper d-Calabi–Yau category.*

(i) *The Hochschild cochains $CC^*(C)$ carry a framed E_2-algebra structure.*

(ii) *Whenever the moduli stack of objects is representable in the PTVV setting, $\mathcal{M}_C$ carries a $(2-d)$-shifted symplectic form.*

Proof. The cyclic Deligne theorem upgrades the brace E_2 structure on Hochschild cochains by the circle action supplied by the cyclic trace [16]. The shifted form at E is

$$\mathrm{RHom}(E,E)[1]^{\otimes 2} \xrightarrow{\circ} \mathrm{RHom}(E,E)[2] \xrightarrow{\mathrm{Tr}} \Bbbk[2-d].$$

Nondegeneracy is the Calabi–Yau duality isomorphism, and closedness is the negative-cyclic lift [15, 125]. □

The integer d controls the shift of the moduli form. The Hochschild operation remains framed E_2. These structures meet through deformation theory while retaining their separate operadic degrees.

Chapter 90

Local BV theories and factorization observables

90.1 Cyclic elliptic local Lie algebras

Let X be a complex manifold. A classical local field theory is specified by a sheaf $\mathcal{L}$ of elliptic L_∞ algebras, a local invariant pairing, and an action functional S satisfying the classical master equation. For an open set $U \subset X$, the classical observables are

$$\mathrm{Obs}^{\mathrm{cl}}(U) = C^*_{\mathrm{CE,cont}}(\mathcal{L}(U)).$$

Here $\mathcal{L}(U)$ consists of unrestricted smooth fields with its nuclear smooth-section topology. Continuous cochains use the completed symmetric powers of its continuous dual, with the completed tensor products of the elliptic-observable construction [30, 31]. The dual consists of compactly supported distributions. For $U \subset V$, restriction of fields $\mathcal{L}(V) \to \mathcal{L}(U)$ induces the covariant map on observables. For pairwise disjoint $U_i \subset V$, restrict to $\bigoplus_i \mathcal{L}(U_i)$ and multiply the pulled-back cochains. These are the prefactorization maps.

Theorem 90.1 (Factorization observables)**.** *An elliptic classical BV theory produces a factorization algebra* $\mathrm{Obs}^{\mathrm{cl}}$*. A renormalized quantization satisfying the quantum master equation produces a factorization algebra* $\mathrm{Obs}^{\mathrm{q}}$ *over* $\Bbbk[[\hbar]]$.

Proof. Restriction is a morphism of local L_∞ algebras, so its contravariant continuous Chevalley map commutes with the differential. For disjoint opens, the completed cochains of the finite direct sum are the completed tensor product of the cochains. This constructs the displayed maps and their composition identities. The elliptic descent theorem in this completed coefficient category supplies Weiss descent [30, 31]. In the quantum theory, renormalized time-ordered products define the structure maps, and the quantum master equation makes their differential square-zero and compatible with nested ultraviolet contractions [30, 31]. □

90.2 Boundary and defect restriction

Let $Y = C \times \mathbb{R}_{\geq 0}$, with C a complex curve, and let a boundary condition be given at $C \times \{0\}$. Boundary observables are holomorphic along C and ordered along the inward normal line. A compact boundary object b in the boundary factorization category gives

$$A_b = \mathrm{RHom}(b, b).$$

Theorem 90.2 (Ordered boundary algebra)**.** *When boundary observables are holomorphic on C and locally constant along the oriented normal direction, A_b is an ordered chiral algebra*

$$A_b \in \mathrm{Alg}_{\mathsf{E}_1}\left(\mathrm{Fact}_D(C), \otimes^!\right).$$

If b is a local compact generator and the Morita equivalences obey factorization descent, the boundary category is equivalent to A_b-modules in factorization coefficients.

Proof. Composition of boundary endomorphisms gives the internal E_1 multiplication. Holomorphic collision gives the factorization coefficient on C. The componentwise locality identity makes multiplication compatible with every disjoint-support factorization map. Local compact-generator Morita theory descends over the Ran diagrams. These are the recognition and Morita theorems of Volume I. □

Chapter 91

Three curve-level algebraic outputs

Let C be a smooth curve and let $\mathcal{D}_C$ be the fixed Ran category. The same geometric source can produce three algebraic outputs, and their distinction is functorial.

Definition 91.1 (Curve-level faces)**.** A transverse face is

$$A^{\perp} \in \mathrm{Alg}_{\mathsf{E}_1}(\mathrm{Fact}_D(C), \otimes^!).$$

An associative chiral face is

$$A^{\mathrm{ch}}_{\mathsf{Ass}} \in \mathrm{Alg}_{\mathsf{E}_1}(\mathcal{D}_C, \star^{\mathrm{ch}}).$$

A chiral Lie face is

$$\mathfrak{g}^{\mathrm{ch}} \in \mathrm{Alg}_{\mathsf{Lie}}(\mathcal{D}_C, \star^{\mathrm{ch}}).$$

Their bars are $\mathrm{Bar}^{\perp}$, $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}$, and $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}$.

Theorem 91.2 (Source separation)**.** *A boundary normal direction determines the transverse face. A chiral-convolution multiplication determines the associative chiral face. A Lie bracket for chiral convolution determines the Lie face. The commutator functor sends the associative chiral face to the Lie face. A chiral enveloping algebra $U^{\mathrm{ch}}(\mathfrak{g})$ satisfies*

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(U^{\mathrm{ch}}(\mathfrak{g})) \simeq \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(\mathfrak{g})$$

on the nilcomplete PBW locus. A mixed source joins the transverse and associative chiral faces through an aligned distributive law.

Proof. Each statement is the corresponding operadic restriction or adjunction in the named monoidal category. The PBW equivalence is the enveloping lane of chiral Koszul duality. The mixed statement is the distributive-law theorem of Volume I. □

This theorem replaces every dimension-based dispatch of an E_n level. The geometry of the source selects the available directions; the operadic level records the operations actually constructed.

Chapter 92

Holomorphic Chern–Simons theory

Let X be a complex threefold with holomorphic volume form Ω_X, and let $\mathfrak{a}$ be a quadratic Lie algebra. The BV fields are

$$\mathcal{E}_X = \Omega^{0,*}(X, \mathfrak{a})[1],$$

with degree -1 pairing

$$\omega(A, B) = \int_X \Omega_X \wedge \langle A, B \rangle.$$

The action is

$$S_{\mathrm{hCS}}(A) = \int_X \Omega_X \wedge \left(\frac{1}{2} \langle A, \bar{\partial} A \rangle + \frac{1}{6} \langle A, [A, A] \rangle \right).$$

Theorem 92.1 (Classical hCS factorization algebra)**.** *The action S_{hCS} satisfies the classical master equation. Its classical observables form a holomorphic factorization algebra on X.*

Proof. The Hamiltonian vector field is the Maurer–Cartan differential $\bar{\partial} A + \frac{1}{2}[A, A]$. Its square is the Dolbeault square, the derivation identity, and the Jacobi identity. The elliptic Dolbeault complex then gives factorization observables by the preceding theorem. □

A perturbative quantization is controlled by the local BV deformation complex. In complex dimension three, its one-loop gauge anomaly occupies the degree-four invariant-polynomial summand; mixed and gravitational components enter when the background fields carry the corresponding couplings. A quantization is a trivialization of the complete local anomaly class, compatible with restriction and factorization [33].

Chapter 93

Pushforward to a curve

Let $p : X \to C$ be a holomorphic map to a complex curve. For a factorization algebra $\mathbb{F}$ on X, define

$$(p_*\mathbb{F})(U) = \mathbb{F}(p^{-1}U).$$

Theorem 93.1 (Factorization pushforward)**.** *The assignment $p_*\mathbb{F}$ is a factorization algebra on C. Suppose in addition that p is proper and locally trivial in a topology compatible with the factorization theory, that $\mathbb{F}$ is locally constant in the fiber directions, and that product-chart excision and base change hold. Then on every trivializing open $U \subset C$ with fiber F_p there is a natural equivalence*

$$(p_*\mathbb{F})(U) \simeq \int_{F_p} \mathbb{F}|_{F_p \times U},$$

compatible on overlaps. If the resulting object is holomorphic and regular holonomic along C, Riemann–Hilbert algebraization gives a factorization D-module on C.

Proof. Disjoint opens in C have disjoint inverse images, so the factorization products push forward. Weiss descent is inherited from inverse images of Weiss covers. On a product chart, local constancy in the fiber and factorization excision identify the value with fiberwise factorization homology. The base-change hypothesis glues these local equivalences. Regular-holonomic Riemann–Hilbert transports the holomorphic factorization object to the algebraic D-module setting. □

Thus a curve shadow is attached to an actual map and an actual field theory. Different fibrations and boundary conditions produce different ordered chiral algebras from the same categorical source.

Chapter 94

The derived centre and the physical bulk

For an ordered boundary algebra A, define

$$Z_{\mathrm{ord}}(A) = \mathrm{RHom}_{A \otimes A^{\mathrm{op}}}(A, A).$$

It is a canonical E_2 algebra. A bulk theory acting on the boundary gives

$$\beta : \mathrm{Obs}_{\mathrm{bulk}}|_{\partial} \longrightarrow Z_{\mathrm{ord}}(A).$$

Definition 94.1 (Open–closed completeness)**.** A boundary condition is open–closed complete on a protected sector when β is an equivalence on that sector.

The definition turns the phrase ‘‘bulk equals centre’’ into a comparison theorem. The centre exists algebraically for every ordered boundary algebra; the physical bulk is identified with it by the open–closed map.

Book XX

Hall and BPS algebras

Chapter 95

Extension correspondences

Let $\mathcal{M} = \coprod_{\gamma \in \Gamma} \mathcal{M}_\gamma$ be a moduli stack in an exact or derived category. Let $\mathcal{M}^{\mathrm{ext}}_{\alpha,\beta}$ classify extensions

$$0 \longrightarrow E_\beta \longrightarrow E \longrightarrow E_\alpha \longrightarrow 0.$$

The correspondence is

$$\mathcal{M}_\alpha \times \mathcal{M}_\beta \xleftarrow{p} \mathcal{M}^{\mathrm{ext}}_{\alpha,\beta} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

Theorem 95.1 (Hall associativity)**.** *Assume a coefficient theory in which p^* and q_* are defined and satisfy base change for the two-step filtration stack. Then*

$$a * b = q_* p^* (a \boxtimes b)$$

defines an associative product.

Proof. The two parenthesizations are the two projections from the stack of filtrations $0 \subset E_1 \subset E_2 \subset E_3$. Base change identifies both composites with pullback to this stack followed by pushforward to E_3. □

For critical CoHAs, the coefficient theory includes vanishing cycles and a coherent orientation. For symplectic CoHAs, it includes the selected equivariant Borel–Moore theory. Completion is chosen so that each output degree receives a convergent sum.

Chapter 96

Orientation data and the threefold Hall product

For compact Calabi–Yau threefolds, natural orientation data on moduli of coherent sheaves and perfect complexes is constructed by Joyce and Upmeier [90]. A general cohomological Hall algebra for a 3-Calabi–Yau category uses strong orientation data, which carries the coherent multiplicativity required by the attractor and extension correspondences [97].

Theorem 96.1 (Threefold Hall algebra from strong orientation)**.** *Let C be a 3-Calabi–Yau category satisfying the geometric hypotheses of Kinjo–Park–Safronov, and assume strong orientation data. Then its cohomological Donaldson–Thomas complexes carry an associative cohomological Hall product. The product is the parabolic-induction correspondence supplied by the attractor Lagrangian correspondence.*

Proof. Kinjo–Park–Safronov prove the functoriality of the Donaldson–Thomas perverse sheaf under the attractor Lagrangian correspondence and use strong orientation to define the parabolic-induction morphism. Their base-change and associativity theorem gives the CoHA [97]. □

Definition 96.2 (Reduced $K3 \times E$ Hall datum)**.** A reduced $K3 \times E$ Hall datum is a strong-oriented threefold Hall datum together with descent along the elliptic translation action, a reduced deformation theory, and a separated completion in physical charge. A full quantum-group datum additionally includes a coproduct, Cartan extension, negative half, and nondegenerate continuous Hopf pairing.

Chapter 97

The geometric E-equivariant Hall source

Let S be a projective K3 surface, let E be an elliptic curve, and put

$$X = S \times E.$$

Write $\mathrm{Coh}_{\le 1}(X)$ for coherent sheaves whose support has dimension at most one. This is an extension-closed abelian category. Its numerical charge group is denoted Γ_X, and the degree of the projected one-cycle on E is the additive function

$$m : \Gamma_X \longrightarrow \mathbb{Z}_{\ge 0}.$$

For $\gamma \in \Gamma_X$, let $\mathcal{M}_\gamma$ be the moduli stack of objects of charge γ. Translation on the elliptic factor gives an action of E on every $\mathcal{M}_\gamma$ and on every extension stack.

Oberdieck and Shen construct the E-equivariant Grothendieck ring of stacks and its Hall convolution. In the present notation the completed algebra is

$$\mathcal{H}_X^{E,\mathrm{mot}} = \widehat{\bigoplus}_{\gamma \in \Gamma_X} K_0^E(\mathrm{St}/\mathcal{M}_\gamma),$$

where the completion is taken in effective curve class and Euler characteristic.

Theorem 97.1 (Geometric reduced Hall algebra)**.** *The extension correspondence in* $\mathrm{Coh}_{\le 1}(X)$ *defines an associative E-equivariant motivic Hall product on* $\mathcal{H}_X^{E,\mathrm{mot}}$*. The Oberdieck–Shen reduced integration morphism*

$$I_X^E : \mathcal{H}_X^{E,\mathrm{mot}} \longrightarrow \widehat{\mathbb{Q}[\Gamma_X]}[\epsilon]/(\epsilon^2)$$

is multiplicative on the regular subalgebra on which their integration theorem is defined. The coefficient of ϵ is the reduced contribution of the positive-dimensional E-orbits.

Proof. The stack of short exact sequences is E-equivariant because translation preserves exactness and numerical charge. The stack of two-step filtrations identifies the two composites of three Hall products, so equivariant base change proves associativity. The integration statement is the E-equivariant motivic Hall theorem of Oberdieck–Shen [123]. Their stabilizer stratification separates fixed contributions from moving elliptic orbits and packages the latter in the square-zero parameter ϵ. □

Chapter 98

Local and wrapped geometry inside the Hall algebra

Define the completed degree-zero and positive-degree pieces

$$\mathcal{H}_X^{\mathrm{loc}} = \widehat{\bigoplus}_{m(\gamma)=0} K_0^E(\mathrm{St}/\mathcal{M}_\gamma), \qquad \mathcal{H}_X^{\mathrm{wr}} = \widehat{\bigoplus}_{m(\gamma)>0} K_0^E(\mathrm{St}/\mathcal{M}_\gamma).$$

A pure one-dimensional support of degree zero projects to a finite subset of E. A support of positive degree projects onto E. The first sector is point-local on the elliptic curve; the second is wrapped.

Theorem 98.1 (Geometric local–wrapped decomposition)**.** *Hall multiplication respects elliptic degree:*

$$\mathcal{H}_{m_1} * \mathcal{H}_{m_2} \subseteq \mathcal{H}_{m_1+m_2}.$$

Consequently $\mathcal{H}_X^{\mathrm{loc}}$ is a subalgebra, $\mathcal{H}_X^{\mathrm{wr}}$ is a closed two-sided ideal, and

$$\mathcal{H}_X^{E,\mathrm{mot}} \cong \mathcal{H}_X^{\mathrm{loc}} \ltimes \mathcal{H}_X^{\mathrm{wr}}$$

as a complete vector space with multiplication determined by the Hall correspondences. The reduced integration morphism preserves this decomposition.

Proof. The cycle of the middle term of a short exact sequence is the sum of the cycles of its subobject and quotient. Projection to E therefore adds degrees. Degree zero is closed under addition, and positive degree absorbs addition by a nonnegative degree. The same grading is present on the target group algebra of I_X^E. □

This theorem supplies the geometric version of the local–wrapped splitting constructed from the Borcherds root algebra in the companion volume. The two constructions have the same additive elliptic degree and become comparable after a primitive root framing is chosen.

Chapter 99

Cohomological refinement and the Gram character

A cohomological refinement uses the Donaldson–Thomas sheaf on the derived moduli stack. The datum required for the convolution is the strong orientation coherence of Kinjo–Park–Safronov. A strong orientation enhancement of the natural Joyce–Upmeier orientation therefore produces

$$\mathcal{H}_X^{E,\mathrm{coh}} = \widehat{\bigoplus}_{\gamma\in\Gamma_X} H_*^{\mathrm{BM},E}(\mathcal{M}_\gamma,\Phi_\gamma),$$

with its cohomological Hall product.

Theorem 99.1 (Strong-oriented cohomological lift)**.** *For every strong orientation enhancement on the E-equivariant derived moduli problem, the attractor correspondence constructs an associative cohomological Hall algebra $\mathcal{H}_X^{E,\mathrm{coh}}$. Elliptic degree again gives a subalgebra in degree zero and a closed ideal in positive degree. Forgetting the cohomological coefficient system recovers the corresponding motivic Hall correspondences.*

Proof. The parabolic-induction theorem of Kinjo–Park–Safronov gives the cohomological product from the strong-oriented attractor Lagrangian correspondence [97]. Numerical charge and elliptic degree are constant on connected components of every correspondence, so the proof of the preceding decomposition applies verbatim. □

For the dyonic charge lattice

$$\Gamma_{\mathrm{phys}} = \widetilde{H}(S,\mathbb{Z})\oplus\widetilde{H}(S,\mathbb{Z}),$$

write

$$\Pi(Q,P) = \left(\frac{Q^2}{2}, Q\cdot P, \frac{P^2}{2}\right).$$

To compare sheaf charges with this lattice, choose an additive map

$$c : \Gamma_X \longrightarrow \Gamma_{\mathrm{phys}}.$$

Put $B_\Pi(a,b) = \Pi(a+b)-\Pi(a)-\Pi(b)$ and give $\widetilde{\Gamma}_{\mathrm{phys}} = \Gamma_{\mathrm{phys}}\times\mathbb{Z}^3$ the group law

$$(a,u)(b,v) = (a+b, u+v+B_\Pi(a,b)).$$

Then $j_\Pi(a) = (a,\Pi(a))$ is a homomorphism. Assume $j_\Pi c$ preserves the selected charge completions. On the regular subalgebra $\mathcal{H}_{\mathrm{reg}}$ of the reduced integration theorem, define

$$I^E_{X,\Pi,c} = (j_\Pi c)_* I^E_X : \mathcal{H}_{\mathrm{reg}} \longrightarrow \widehat{\mathbb{Q}[\widetilde{\Gamma}_{\mathrm{phys}}]}[\epsilon]/(\epsilon^2).$$

Write $I^E_{X,\Pi}$ for this map when c has been fixed.

Proposition 99.2 (Polarized reduced character)**.** *The map $I^{E}_{X,\Pi,c}$ is multiplicative. Its coefficients are the reduced invariants supplied by I^{E}_{X}, regraded by $j_{\Pi}c$. If $g \in 1 + F^{1}H$ is group-like in a complete filtered Hall bialgebra and the logarithm converges, then* $\log g$ *is primitive.*

Proof. Expanding $\Pi(a+b+d)$ gives

$$B_{\Pi}(a,b) + B_{\Pi}(a+b,d) = B_{\Pi}(b,d) + B_{\Pi}(a,b+d).$$

This proves associativity of the group law. The definition gives $j_{\Pi}(a+b) = j_{\Pi}(a)\, j_{\Pi}(b)$. Composing its group-algebra map with reduced integration proves multiplicativity on $\mathcal{H}_{\mathrm{reg}}$. For the last assertion, the two factors in $\Delta g = (g \widehat{\otimes} 1)(1 \widehat{\otimes} g)$ commute. Taking the convergent logarithm gives $\Delta \log g = \log g \widehat{\otimes} 1 + 1 \widehat{\otimes} \log g$. □

The primitive reduced Gromov–Witten theorem supplies the separate character $-\chi_{10}^{-1}$ in classes primitive in the K3 factor [122]. Comparing this character with the Hall character requires the same charge map, sector, and normalization. A primitive root framing must also preserve the extension bracket and parity of the chosen BPS states.

Chapter 100

BPS primitives

A connected graded bialgebra has primitive Lie algebra

$$\operatorname{Prim}(H) = \{x : \Delta x = x \otimes 1 + 1 \otimes x\}.$$

In cohomological Donaldson–Thomas theory, the BPS sheaf isolates the primitive part of the Hall algebra.

Theorem 100.1 (Two-Calabi–Yau BPS algebra)**.** *For the geometric two-Calabi–Yau categories covered by the BPS-algebra theorem, the BPS algebra is the positive enveloping algebra of a generalized Kac–Moody Lie algebra. This scope includes semistable sheaves on K3 and other symplectic surfaces under the stated good-moduli hypotheses.*

Proof. The theorem is the Davison–Hennecart–Schlegel Mejia construction [39]. The simple BPS sheaves give the simple-root spaces, extension correspondences give the bracket, and the integrality/PBW theorem identifies the Hall algebra with the positive enveloping algebra. □

This result supplies a global K3 generalized Kac–Moody algebra at the two-Calabi–Yau Hall level. Yangian structures arise from additional equivariant correspondences and coproducts in the surface and quiver models described below.

Chapter 101

The integral Hall–Borcherds amalgam

Let $\mathfrak{p}_X^+$ be a complete charge-graded BPS Lie superalgebra supplied by a geometric Hall source, and let $\mathfrak{n}_{\Delta_s,+}$ be the positive Igusa Borcherds Lie superalgebra. The polarized charge extension places both gradings over one abelian Cartan algebra $\mathfrak{h}_\Pi$. Form the two complete Borels

$$\mathfrak{b}_X = \mathfrak{h}_\Pi \ltimes \mathfrak{p}_X^+, \qquad \mathfrak{b}_\Delta = \mathfrak{h}_\Pi \ltimes \mathfrak{n}_{\Delta_s,+}.$$

The Cartan acts through the physical charge on the first Borel and through Gram degree on the second.

Definition 101.1 (Integral protected Lie algebra)**.** The integral Hall–Borcherds algebra is the completed amalgamated coproduct

$$\mathfrak{b}_{X\boxtimes\Delta} = \mathfrak{b}_X \widehat{\amalg}_{\mathfrak{h}_\Pi} \mathfrak{b}_\Delta$$

in complete charge-filtered dg Lie superalgebras.

Theorem 101.2 (Hall–Borcherds amalgamation)**.** *The completed amalgam exists and represents pairs of continuous Lie morphisms from $\mathfrak{b}_X$ and $\mathfrak{b}_\Delta$ that agree on $\mathfrak{h}_\Pi$. The Cartan projections of the two Borels induce canonical retractions*

$$r_X : \mathfrak{b}_{X\boxtimes\Delta} \longrightarrow \mathfrak{b}_X, \qquad r_\Delta : \mathfrak{b}_{X\boxtimes\Delta} \longrightarrow \mathfrak{b}_\Delta.$$

Its completed enveloping algebra is the amalgamated free product

$$\widehat{U}(\mathfrak{b}_{X\boxtimes\Delta}) \cong \widehat{U}(\mathfrak{b}_X) \widehat{*}_{\widehat{U}(\mathfrak{h}_\Pi)} \widehat{U}(\mathfrak{b}_\Delta).$$

The geometric Hall operations survive under r_X, while the Igusa bracket, PBW character, and automorphic determinant survive under r_Δ.

Proof. Construct the ordinary Lie coproduct from the free product of the two Borels and divide by the ideal identifying their two Cartan copies. Complete by total positive charge and proper root height. Every fixed filtered degree uses finitely many Lie words, so the completion is separated and has the stated mapping property. The projection $\mathfrak{b}_X \to \mathfrak{h}_\Pi \to \mathfrak{b}_\Delta$ together with the identity on $\mathfrak{b}_\Delta$ gives r_Δ by universality; the second retraction is symmetric. The enveloping functor preserves coproducts of Lie algebras, and PBW identifies the completed algebraic filtration with the displayed completed free product. □

Definition 101.3 (Comparison ideal)**.** A primitive comparison $\varphi : \mathfrak{n}_{\Delta_s,+} \to \mathfrak{p}_X^+$ preserving Cartan degree defines the closed ideal

$$\mathcal{J}_\varphi = \overline{\langle i_\Delta(x) - i_X(\varphi(x)) \mid x \in \mathfrak{n}_{\Delta_s,+} \rangle} \subset \mathfrak{b}_{X\boxtimes\Delta}.$$

The diagonal realization is

$$\mathfrak{b}_{X\simeq\Delta,\varphi} = \mathfrak{b}_{X\boxtimes\Delta} / \mathcal{J}_\varphi.$$

Proposition 101.4 (Graph quotient of a primitive comparison)**.** *Let $\varphi : \mathfrak{b}_\Delta \to \mathfrak{b}_X$ be a continuous Lie morphism restricting to the identity on $\mathfrak{h}_\Pi$. Use a completion in which the amalgam and closed quotient have their stated universal properties. Then*

$$\mathfrak{b}_{X \boxtimes \Delta} / \mathcal{J}_\varphi \cong \mathfrak{b}_X.$$

The two factor maps become id *and φ. If φ is an isomorphism, the quotient is also isomorphic to $\mathfrak{b}_\Delta$.*

Proof. The maps id and φ agree on $\mathfrak{h}_\Pi$. They give $q_\varphi : \mathfrak{b}_{X \boxtimes \Delta} \to \mathfrak{b}_X$, which kills the generators of $\mathcal{J}_\varphi$. The inclusion of $\mathfrak{b}_X$ gives a map s in the other direction. The composite $q_\varphi s$ is the identity. The other composite is the identity on the geometric factor. On the Borcherds factor, the defining relation identifies x with $s\varphi(x)$, so it is also the identity there. The two factor images topologically generate the quotient. Continuity proves that the maps are inverse on the completion. □

The graph quotient implements a specified primitive comparison. Constructing that comparison requires geometric simple states whose extension brackets satisfy the Borcherds relations. The quotient formula alone produces neither those states nor their orientation data.

Chapter 102

Coproducts, pairings, and doubles

A positive Hall algebra becomes a quantum group after additional structure is supplied. Let H^{+} be a complete graded bialgebra, let H^{0} be a Cartan algebra, and let H^{-} be the opposite or dual half. Let

$$\langle -, - \rangle : H^{+} \mathbin{\widehat{\otimes}} H^{-} \longrightarrow \Bbbk$$

be a nondegenerate Hopf pairing.

Definition 102.1 (Completed Hall double)**.** The completed Drinfeld double is the topological algebra

$$\widehat{D}(H^{+}) = H^{-} \mathbin{\widehat{\otimes}} H^{0} \mathbin{\widehat{\otimes}} H^{+}$$

with internal products in the three factors and cross-relations determined by the Hopf pairing. Its representation category is monoidal by the coproduct and braided by the universal R-matrix whenever the canonical tensor converges.

Theorem 102.2 (Double construction)**.** *Let H^{+} and H^{-} be paired complete graded bialgebras, let H^{0} be a Cartan bialgebra acting by the stated weight grading, and let*

$$\langle -, - \rangle : H^{+} \mathbin{\widehat{\otimes}} H^{-} \longrightarrow \Bbbk$$

be a continuous nondegenerate Hopf pairing. Assume the triangular multiplication map from $H^{-} \mathbin{\widehat{\otimes}} H^{0} \mathbin{\widehat{\otimes}} H^{+}$ is separated and complete. Then the pairing cross-relations determine a unique completed double with this triangular decomposition. On every continuous module category on which the canonical tensors converge, the completed canonical element gives the universal R-matrix.

Proof. The Hopf pairing defines the cross-relations between the paired halves and makes the three canonical embeddings bialgebra morphisms. The Hopf identities identify the two reductions of every triple product, proving associativity. Separated triangular decomposition gives uniqueness. Finite-degree canonical tensors form a compatible inverse system; convergence on the selected module category yields the quasitriangular element [50, 58]. □

Theorem 102.3 (Quantum Hall–Borcherds amalgam)**.** *Let H_{X} and H_{Δ} be complete Hopf algebras containing the same complete Cartan Hopf algebra H^{0}, with continuous Hopf retractions onto H^{0}. Their completed amalgamated free product*

$$H_{X \boxtimes \Delta} = H_{X} \mathbin{\widehat{*}_{H^{0}}} H_{\Delta}$$

has a unique complete Hopf structure extending the two coproducts, counits, and antipodes. It represents pairs of continuous Hopf morphisms agreeing on H^{0} and has canonical Hopf retractions to both factors. A mixed universal R-matrix is equivalent to a continuous cross-pairing satisfying the two hexagon identities; the intrinsic R-matrices of the two retracts remain defined independently.

Proof. The algebraic amalgamated free product exists because complete filtered algebras are obtained degreewise from finite quotients. The two coproducts agree on H^{0}, so the universal property extends them to a coproduct on the free product. Coassociativity is checked on the two generating Hopf subalgebras. The same argument extends the counit and antipode. The Hopf retractions are induced by the factor retractions onto H^{0}. A quasitriangular structure extending both factors requires the additional commutation data between the two generating halves; writing this data as a cross-pairing turns the quasitriangular and Yang–Baxter equations into the two hexagon identities. □

The deconcatenation coproduct on a bar complex and the representation-theoretic coproduct on a quantum group occupy different structures. A geometric comparison between them is a theorem about a selected model.

Chapter 103

Ordered chiral bialgebras

A tensor product of representations requires compatible actions on threefold products and the unit. Retain the factorization coefficient category $(\mathrm{Fact}_D(C), \otimes^!)$ on the smooth complex curve C. Fix a category $\widehat{\mathcal{V}}_C$ of complete filtered objects with specified presentations $F \simeq \varprojlim_r F_r$, where $F_r \in \mathrm{Fact}_D(C)$. Use derived limits in the fixed Ran enhancement and define

$$F \mathbin{\widehat{\otimes}^!} G = \varprojlim_{r,s} (F_r \otimes^! G_s).$$

Assume these limits remain factorizing and give $\widehat{\mathcal{V}}_C$ a symmetric monoidal structure with unit $\mathbf{1}$. Its associativity, symmetry, and unit equivalences must preserve the presentations and the Ran structure. Morphisms are continuous maps compatible with these presentations. These are hypotheses on the chosen completion.

Definition 103.1 (Ordered chiral bialgebra)**.** An ordered chiral bialgebra is a counital E_1-coalgebra object in complete ordered chiral algebras:

$$A \in \mathrm{CoAlg}_{\mathsf{E}_1}\Big(\mathrm{Alg}_{\mathsf{E}_1}(\widehat{\mathcal{V}}_C, \mathbin{\widehat{\otimes}^!})\Big).$$

It has continuous multiplication m and unit η, and continuous morphisms of unital ordered chiral algebras

$$\Delta : A \longrightarrow A \mathbin{\widehat{\otimes}^!} A, \qquad \epsilon : A \longrightarrow \mathbf{1}.$$

The counit is the chosen augmentation. Write τ for the coefficient symmetry, including Koszul signs. Suppressing ambient associativity and unit equivalences, compatibility reads

$$\Delta m = (m \mathbin{\widehat{\otimes}^!} m)(\mathrm{id} \mathbin{\widehat{\otimes}^!} \tau \mathbin{\widehat{\otimes}^!} \mathrm{id})(\Delta \mathbin{\widehat{\otimes}^!} \Delta),$$
$$\Delta\eta = \eta \mathbin{\widehat{\otimes}^!} \eta, \qquad \epsilon m = \epsilon \mathbin{\widehat{\otimes}^!} \epsilon, \qquad \epsilon\eta = \mathrm{id}_{\mathbf{1}}.$$

Coassociativity and the counit identities read

$$(\Delta \mathbin{\widehat{\otimes}^!} \mathrm{id})\Delta = (\mathrm{id} \mathbin{\widehat{\otimes}^!} \Delta)\Delta,$$
$$(\epsilon \mathbin{\widehat{\otimes}^!} \mathrm{id})\Delta = \mathrm{id}_A = (\mathrm{id} \mathbin{\widehat{\otimes}^!} \epsilon)\Delta.$$

For a homotopy-coherent presentation, these identities include the higher coherences of the displayed coalgebra object. Every operation and coherence preserves diagonal transitions and disjoint-locus factorization maps. An antipode is a continuous convolution inverse $S : A \to A$ of id_A:

$$m(S \mathbin{\widehat{\otimes}^!} \mathrm{id})\Delta = \eta\epsilon = m(\mathrm{id} \mathbin{\widehat{\otimes}^!} S)\Delta.$$

A bialgebra with an antipode is a Hopf object in this coefficient category.

A continuous left A-module is an object $M \in \widehat{\mathcal{V}}_C$ with a unital E_1 action $\rho_M : A \mathbin{\widehat{\otimes}^!} M \to M$. Its action and coherences are morphisms in these factorization coefficients. Write $\mathrm{Mod}_A(\widehat{\mathcal{V}}_C)$ for their category.

Theorem 103.2 (Representation fusion)**.** *An ordered chiral bialgebra makes* $\mathrm{Mod}_A(\widehat{\mathcal{V}}_C)$ *monoidal. Its product is* $M \mathbin{\widehat{\otimes}^!} N$ *with action*

$$\begin{aligned} A \mathbin{\widehat{\otimes}^!} (M \mathbin{\widehat{\otimes}^!} N) &\xrightarrow{\Delta \mathbin{\widehat{\otimes}^!} \mathrm{id}} (A \mathbin{\widehat{\otimes}^!} A) \mathbin{\widehat{\otimes}^!} (M \mathbin{\widehat{\otimes}^!} N) \\ &\xrightarrow{\mathrm{id} \mathbin{\widehat{\otimes}^!} \tau_{A,M} \mathbin{\widehat{\otimes}^!} \mathrm{id}} (A \mathbin{\widehat{\otimes}^!} M) \mathbin{\widehat{\otimes}^!} (A \mathbin{\widehat{\otimes}^!} N) \\ &\xrightarrow{\rho_M \mathbin{\widehat{\otimes}^!} \rho_N} M \mathbin{\widehat{\otimes}^!} N. \end{aligned}$$

The unit is $\mathbf{1}$ *with action induced by* ϵ*. The ambient associator and unit equivalences are module equivalences.*

Suppose a continuous universal R*-matrix is also given. It is an invertible morphism* $R : \mathbf{1} \to A \mathbin{\widehat{\otimes}^!} A$ *for the tensor-product algebra, satisfying*

$$R\Delta(a) = \Delta^{\mathrm{op}}(a) R, \qquad \Delta^{\mathrm{op}} = \tau_{A,A} \Delta,$$
$$(\Delta \mathbin{\widehat{\otimes}^!} \mathrm{id})(R) = R_{13} R_{23}, \qquad (\mathrm{id} \mathbin{\widehat{\otimes}^!} \Delta)(R) = R_{13} R_{12},$$
$$(\epsilon \mathbin{\widehat{\otimes}^!} \mathrm{id})(R) = \eta = (\mathrm{id} \mathbin{\widehat{\otimes}^!} \epsilon)(R).$$

The first equation is an identity of morphisms from A *to its completed tensor square. Leg notation uses the unit and symmetry of* $\widehat{\mathcal{V}}_C$*. In a homotopy-coherent presentation, require compatible lifts of these identities. Then* $c_{M,N} = \tau_{M,N} \circ R_{M,N}$ *is a braiding, where* $R_{M,N}$ *acts through the external tensor-product module action.*

Proof. The symmetry combines the independent module actions into an $A \mathbin{\widehat{\otimes}^!} A$-action. Restriction along the unital algebra map Δ gives the displayed action. For three modules, the two actions use $(\Delta \mathbin{\widehat{\otimes}^!} \mathrm{id})\Delta$ and $(\mathrm{id} \mathbin{\widehat{\otimes}^!} \Delta)\Delta$. Coassociativity makes the ambient associator A-linear. The counit identities make the ambient unit equivalences A-linear. The coalgebra coherences give the pentagon and triangle coherences. The completion hypotheses keep these maps continuous and factorizing. Quasitriangularity makes $c_{M,N}$ an intertwiner and invertibility makes it an equivalence. The two coproduct identities for R give the hexagons. Its counit identities give compatibility with the unit. □

The tensor in this construction is the completed transverse tensor $\otimes^!$ on factorization coefficients. A chiral-convolution or Hall coproduct requires its own construction and a comparison in the specified category. A spectral version additionally requires a difference coordinate, a Laurent-series completion, and compatible domains for iterated expansions. Its identities must hold in those domains before defining module actions.

Example 103.3 (A counital algebra map without coassociativity)**.** In ordinary complex vector spaces, let $A = \mathbb{C}[x]$ and put

$$\Delta(x) = x \otimes 1 + 1 \otimes x + x \otimes x^2, \qquad \epsilon(x) = 0.$$

Polynomial substitution extends these assignments to unital algebra maps. Writing $F(u, v) = u + v + uv^2$, the equations $F(0, v) = v$ and $F(u, 0) = u$ prove both counit identities. But

$$\begin{aligned} F(F(u, v), w) - F(u, F(v, w)) &= -uv^2w^4 - uv^2w^2 - 2uvw^3 - 2uvw \\ &= -uvw(1 + w^2)(vw + 2) \neq 0. \end{aligned}$$

On three regular modules, the two actions differ already on $1 \otimes 1 \otimes 1$. Thus the canonical tensor associator is not A-linear. This ordinary example shows why multiplicativity and the counit identities must be supplemented by coassociativity. It makes no assertion about a geometric realization in factorization coefficients.

Book XXI

Exact model cases

Chapter 104

The A_2 Hall algebra

For the quiver $1 \to 2$ over $\mathbb{F}_q$, the twisted Hall algebra has generators E_1, E_2 and $v^2 = q$. Counting subrepresentations gives

$$E_1^2 E_2 - (v + v^{-1}) E_1 E_2 E_1 + E_2 E_1^2 = 0.$$

For a representation of dimension vector $(2,1)$, the arrow has rank zero or one. The relevant flag counts are

rank	112	121	211
0	$q+1$	$q+1$	$q+1$
1	$q+1$	1	0.

After the Hall twists $v^{-1}, 1, v$, both rows give zero. This is the elementary geometric origin of the quantum Serre relation [73, 129].

Chapter 105

The tripled Jordan quiver and $\mathbb{C}^3$

Let Q have one vertex and loops x, y, z, with potential

$$W = \operatorname{Tr} x[y, z].$$

The Jacobi algebra is $\mathbb{C}[x, y, z]$. The critical CoHA uses the vanishing cycles of $\operatorname{Tr} W$ on the representation spaces. Its positive currents are generated by the one-vertex dimension sectors.

Theorem 105.1 (Affine-Yangian positive half)**.** *After the standard equivariant localization and completion, the critical CoHA of the tripled Jordan quiver is identified with the positive half of the affine Yangian of* $\mathfrak{gl}_1$*. The independently defined affine Yangian carries its standard Cartan and negative current halves; the positive-half comparison embeds the CoHA into that triangular presentation.*

Proof. The CoHA product becomes the shuffle product under localization. Dimensional reduction identifies the tripled-quiver critical theory with the equivariant surface CoHA, and Davison's affine-BPS theorem identifies its completed positive algebra with the strictly positive affine Yangian. The shuffle kernel satisfies the affine-Yangian current relations with equivariant parameters ϵ_1, ϵ_2, ϵ_3 and Calabi–Yau relation $\epsilon_1 + \epsilon_2 + \epsilon_3 = 0$. PBW and current comparison identify the positive algebra. The independently presented full affine Yangian is the triangular algebra generated by its negative currents, Cartan currents, and this positive half [38, 40, 130, 133]. □

A representation of the affine Yangian restricts along this positive-half map to an action of the critical CoHA. A vertex-algebra realization requires a state space, a vacuum, translation and fields satisfying the vertex identities. An identification with a $W_{1+\infty}$-type vacuum algebra therefore requires a separate representation theorem. The comparison with geometrically defined local fields must also preserve collision operations; its data are specified in Problem 5.5.

Chapter 106

Surface CoHAs and ADE Yangians

Let $S_\Gamma \to \mathbb{C}^2/\Gamma$ be the minimal resolution for a finite subgroup $\Gamma \subset \mathrm{SL}_2(\mathbb{C})$. The exceptional curves have ADE intersection matrix. Moduli of sheaves supported on these curves carry convolution correspondences.

Theorem 106.1 (Curve-supported surface CoHA)**.** *The completed positive CoHA of one-dimensional sheaves supported on the exceptional curves is identified with a nonstandard positive half of the affine Yangian of the corresponding ADE Lie algebra.*

Proof. The theorem is the curve-supported surface construction of Diaconescu–Porta–Sala–Schiffmann–Vasserot [45]. The simple curve sheaves give the simple generators, Hecke correspondences give the current operators, and the completed PBW comparison gives the affine Yangian presentation. □

This theorem is a local K3 model whenever an ADE configuration occurs on a K3 surface. The global K3 surface adds compact curve classes and monodromy beyond the local root lattice.

Chapter 107

Coulomb branches and shifted Yangians

For a three-dimensional $\mathcal{N} = 4$ gauge theory, the BFN Coulomb branch is defined by equivariant Borel–Moore homology of an affine-Grassmannian correspondence. Its convolution algebra quantizes the Coulomb branch [17]. In quiver gauge theories, shifted Yangians and their truncations provide presentations of these quantizations. Boundary conditions select the shift and truncation data.

The BFN construction supplies a second geometric source of Yangians. Its grading and coproduct arise from loop rotation and convolution, while the surface CoHA source arises from sheaf extensions. A comparison between the two is a model-specific equivalence.

Book XXII

K3 symmetries

Chapter 108

The Mukai lattice

For a K3 surface S, the Mukai lattice is

$$\widetilde{H}(S,\mathbb{Z}) = H^0(S,\mathbb{Z}) \oplus H^2(S,\mathbb{Z}) \oplus H^4(S,\mathbb{Z})$$

with pairing

$$((r,c,s),(r',c',s')) = c\cdot c' - rs' - r's.$$

It is even unimodular of signature $(4,20)$ and rank 24.

Chapter 109

The Nakajima Heisenberg algebra

Let

$$\mathbb{H}_S = \bigoplus_{n \geq 0} H^*(S^{[n]}).$$

For $m \in \mathbb{Z} \setminus \{0\}$ and $\alpha \in H^*(S)$, the Nakajima correspondence gives an operator $p_m(\alpha)$.

Theorem 109.1 (K3 Heisenberg action)**.** *The operators satisfy*

$$[p_m(\alpha), p_n(\beta)] = m\, \delta_{m+n,0} \int_S \alpha \cup \beta \ \mathrm{id}\,.$$

They generate the rank-twenty-four Heisenberg algebra acting irreducibly on $\mathbb{H}_S$ with vacuum $1 \in H^*(S^{[0]})$.

Proof. The commutator is the excess intersection of the two incidence correspondences. Opposite creation and annihilation correspondences meet along the diagonal with multiplicity m and intersection pairing $\int_S \alpha\beta$. Correspondences of the same sign commute. The vacuum construction gives the Fock representation [119]. □

This action is global, compact, and abelian at the Lie level. It is an exact K3 symmetry and supplies the rank-twenty-four Fock sector of any broader K3 algebra.

Chapter 110

The orthogonal LLV algebra

For a compact hyperkähler manifold, every Lefschetz class gives an $\mathfrak{sl}_2$ triple on cohomology. The Lie algebra generated by all such triples is the Looijenga–Lunts–Verbitsky algebra.

Theorem 110.1 (Orthogonal LLV symmetries)**.** *For a K3 surface S, the LLV algebra acting on $H^*(S)$ is*

$$\mathfrak{g}_{\mathrm{LLV}}(S) \cong \mathfrak{so}\big(\widetilde{H}(S), (-,-)_{\mathrm{Muk}}\big) \cong \mathfrak{so}(4, 20).$$

For an irreducible hyperkähler manifold M of $K3^{[n]}$ type with $n \geq 2$, one has $b_2(M) = 23$ and

$$\mathfrak{g}_{\mathrm{LLV}}(M) \cong \mathfrak{so}(4, 21).$$

Moduli spaces of stable objects on S carry their own LLV actions; Mukai and monodromy correspondences relate these actions to the K3 lattice.

Proof. The Looijenga–Lunts–Verbitsky theorem identifies the Lie algebra generated by all Lefschetz triples with the orthogonal algebra of the Mukai extension $H^2(M) \oplus U$. For S, the extension has signature $(4, 20)$. For $K3^{[n]}$ type with $n \geq 2$, the Beauville–Bogomolov lattice has signature $(3, 20)$, so adjoining the hyperbolic plane gives signature $(4, 21)$ [112, 136]. □

The LLV algebra is a classical cohomological symmetry. A Yangian deformation of the full orthogonal action consists of two further structures: a Drinfeld current presentation and a compatible spectral coproduct.

Chapter III

Local ADE Yangians

An ADE configuration of (-2)-curves on S gives a root sublattice of $H^2(S,\mathbb{Z})$. Sheaves supported on the configuration produce the surface CoHA of the preceding part and hence a completed positive affine ADE Yangian. Nakajima quiver varieties supply compatible Yangian actions on their equivariant cohomology.

Thus K3 geometry contains local nonabelian Yangian sectors indexed by ADE enhancements of the Picard lattice. Picard monodromy transports the ADE root sublattices over the corresponding Noether–Lefschetz loci. A family of sheaf moduli and Hecke correspondences compatible with that monodromy transports the Yangian actions themselves.

Chapter 112

The K3 symmetry trichotomy

The three exact structures are

Structure	Carrier	Operation
Mukai Heisenberg	$\bigoplus_n H^*(S^{[n]})$	creation and annihilation
LLV $\mathfrak{so}(4,20)$	$H^*(S)$; separately $\mathfrak{so}(4,21)$ on $K3^{[n]}$ type	Lefschetz Lie action
Local ADE Yangian	curve-supported equivariant moduli	Hecke and current correspondences

A global K3 quantum algebra is a system that contains these three actions with explicit comparison homomorphisms. The two-Calabi–Yau BPS theorem supplies a global generalized Kac–Moody positive algebra. A full Yangian or double is specified by a compatible coproduct, spectral parameter, negative half, Cartan part, pairing, and completion.

Chapter 113

K3 correspondences on fixed coefficients

Fix rational cohomology and a collection $\mathscr{M}_S$ of smooth projective moduli spaces on a projective K3 surface S. Include the Hilbert schemes $S^{[n]}$ and close the collection under finite products. For $M, N \in \mathscr{M}_S$, define the graded kernel space

$$K(M,N) = H^*(M \times N, \mathbb{Q})[2\dim_{\mathbb{C}} M].$$

A kernel of degree r acts from $H^j(M)$ to $H^{j+r}(N)$ by

$$T_\alpha(v) = (p_N)_*(p_M^* v \cup \alpha).$$

The shift in $K(M,N)$ accounts for integration over M. All pushforwards in this construction are proper.

Proposition 113.1 (Kernel composition)**.** *The product*

$$\beta \circ \alpha = (p_{13})_*(p_{12}^*\alpha \cup p_{23}^*\beta)$$

defines an associative graded category with diagonal identities. The kernel action identifies $K(M,N)$ *with* $\mathrm{Hom}^*(H^*(M), H^*(N))$.

Proof. The Künneth isomorphism and Poincaré duality identify $H^*(M \times N)[2\dim M]$ with $H^*(M)^\vee \otimes H^*(N)$. Under this identification, T_α is the associated linear map. Integration over the middle factor contracts its dual pair, so $T_{\beta\circ\alpha} = T_\beta T_\alpha$. Faithfulness then proves associativity and the diagonal identity. External products give tensor products with their Koszul signs. □

A geometric symmetry is specified by selected kernels in this category. Nakajima incidence kernels connect different Hilbert schemes. Lefschetz multiplication and its adjoint act on each individual cohomology space. Using every cohomology kernel would instead give all linear maps; the selected geometric kernels carry the representation-theoretic content.

Chapter 114

The algebra generated by geometric actions

Put $V_S = \bigoplus_{M \in \mathscr{M}_S} H^*(M, \mathbb{Q})$. Choose a set of homogeneous operators T_λ on V_S, each sending a vector with finite support to a vector with finite support. The Nakajima operators on the Hilbert-scheme summand satisfy this condition: each changes the point number by a fixed integer. Individual Lefschetz operators, extended by zero on other summands, also satisfy it. Let F be the free unital graded algebra on symbols t_λ.

Proposition 114.1 (Geometric action algebra)**.** *The map*

$$\rho : F \longrightarrow \mathrm{End}_{\mathbb{Q}}(V_S), \qquad t_\lambda \longmapsto T_\lambda$$

defines a faithful action of $A_S = F/\ker\rho$. For every algebra B, maps $A_S \to B$ are exactly maps $F \to B$ that kill $\ker\rho$*. Products and commutators of the selected kernels are their convolution products and commutators on the relevant moduli spaces.*

Proof. Finite words in the selected operators preserve finite support, so ρ is defined by composition. The quotient acts faithfully by definition of its kernel. The factorization property is the universal property of an algebra quotient. The convolution assertion follows from Proposition 113.1. □

An equivariant ADE construction uses a separate coefficient field, for example $\mathrm{Frac}\, H^*_T(\mathrm{pt}, \mathbb{Q})$, and its prescribed localization and proper pushforwards. Joining it to A_S requires maps between the actual state spaces that intertwine those operators and their coefficients. A completion of A_S requires a specified topology and continuity of every generator. The indefinite Mukai norm alone supplies neither condition. The desired common K3 quantum algebra must construct these local-to-global actions and their mixed convolution relations.

Chapter 115

Tensor reconstruction and geometric kernels

Let $\mathscr{C}$ be an essentially small rigid linear monoidal category and let $\omega : \mathscr{C} \to \mathrm{Vect}_{\mathrm{fd}}(\mathbb{Q})$ be a strong monoidal functor. A choice of geometric kernels specifies the morphisms of $\mathscr{C}$; it is part of the input. Define

$$H = \left(\bigoplus_{M \in \mathscr{C}} \omega(M)^* \otimes \omega(M) \right) \Big/ \left\langle [\lambda \circ \omega(f) \otimes v]_M - [\lambda \otimes \omega(f)v]_N \right\rangle,$$

where $f : M \to N$, $v \in \omega(M)$, and $\lambda \in \omega(N)^*$.

Proposition 115.1 (Matrix-coefficient coalgebra)**.** *For a basis (e_i) with dual basis (e^i), the formulas*

$$\Delta[\lambda \otimes v]_M = \sum_i [\lambda \otimes e_i]_M \otimes [e^i \otimes v]_M, \qquad \epsilon[\lambda \otimes v]_M = \lambda(v)$$

are basis-independent and make H a coalgebra. Tensor product gives its multiplication, and duals give its antipode. Each $\omega(M)$ is a comodule by

$$v \longmapsto \sum_i e_i \otimes [e^i \otimes v]_M.$$

Every selected kernel is a comodule morphism.

Proof. The identity tensor $\sum_i e_i \otimes e^i$ is basis-independent. Inserting it twice proves coassociativity; evaluation proves both counit identities. The defining relations express naturality under f and are preserved by these formulas. The strong monoidal constraints identify the coefficient of $M \otimes N$ with the product of coefficients of M and N. Their pentagon and unit identities give the bialgebra axioms. The evaluation and coevaluation identities for duals give the antipode identities. The same relations prove that $\omega(f)$ intertwines the coactions. □

The linear dual of H represents linear natural endomorphisms of ω. Tensor-compatible natural automorphisms are the corresponding multiplicative, unit-preserving families, not all elements of this dual. An equivalence with a comodule category requires the additional exactness and reconstruction hypotheses stated in the Tannaka theorem below. The action algebra A_S acts by selected kernels; the coend records symmetries commuting with those kernels.

Book XXIII

The exact $K3 \times E$ protected completion

Chapter 116

The protected input

The K3 elliptic genus is $2\phi_{0,1}$. The Borcherds half determines the weight-five denominator D_5 and its Borcherds Lie superalgebra $\mathfrak{g}_\Delta$. Fix its positive root spaces with their parity and proper height. The positive algebra and its root Fock space are defined from this Lie superalgebra, independently of a compact Hall realization.

For each finite zero-potential quiver window $Q_{\Delta,H}$, form the Ginzburg algebra $\Gamma_{\Delta,H}$. Restriction to full subquivers gives the projections of Proposition 122.1. Write Γ_Δ for this specified inverse system. A nondegenerate continuous Calabi–Yau structure on a completed limit is additional data. An ordinary Hall category $\mathcal{B}_\Delta$ must carry its chosen quiver coefficients and extension correspondences. Its identification with a root presentation requires the corresponding Hall theorem on that precise datum.

Definition 116.1 (Root-datum realizations)**.** A protected root-datum realization records

$$\mathfrak{P}(K3 \times E) = (\Gamma_\Delta, \mathcal{B}_\Delta, \mathcal{H}_{\mathrm{ip}}, \mathcal{F}^{\mathrm{hyb}}_{\Delta,E}, \widehat{U}_q(\mathfrak{g}_\Delta), \mathbb{A}_Z).$$

Here $\mathcal{H}_{\mathrm{ip}} = \bigoplus_{\alpha>0}(\mathfrak{g}_\Delta)_\alpha$ with its parity. The current-factorization, paired quantum-group, and Fredholm entries retain their own coefficient categories, completions, and defining hypotheses. Maps among these entries are specified realization maps, not consequences of using the same root labels.

The enveloping and Chevalley constructions are functorial under the primitive maps of Theorem 121.1. The root Fock trace is computed from the actual graded root spaces. A Fredholm realization adds a family of operators on a specified domain and the convergence and orientation data identifying its Pfaffian with the denominator. These are the analytic inputs to the square-root comparison below.

Definition 116.2 (Integral protected realization)**.** Let $\mathfrak{p}^+_X$ be the BPS primitive Lie algebra of a selected strong-oriented cohomological lift of the geometric E-equivariant Hall source. The integral protected realization is

$$\mathfrak{P}_{\mathrm{int}}(X, \Delta) = \left(\mathcal{H}^{E,\mathrm{mot}}_X, \mathfrak{P}(K3 \times E), \mathfrak{b}_{X\boxtimes\Delta}, r_X, r_\Delta, I^E_{X,\Pi}\right).$$

At the motivic level the first, second, and character components are defined. A strong orientation enhancement supplies $\mathfrak{p}^+_X$ and hence the amalgamated primitive component.

Proposition 116.3 (Primitive graph comparison)**.** *Suppose the two primitive Borels, their common Cartan action, and their completed amalgam are supplied as above. Their Cartan retractions induce retractions from the amalgam to each factor. For each specified primitive map φ preserving Cartan action, its graph quotient is canonically the geometric Borel.*

Proof. The identity on one Borel and the Cartan projection of the other agree on the common Cartan, so the amalgam gives the retraction. The quotient statement is Proposition 101.4. □

The motivic integration map and primitive comparison are different maps. Strong orientation gives the coefficient coherence of a cohomological Hall construction under its stated geometric hypotheses. Extracting a primitive Lie algebra additionally requires the appropriate BPS or coproduct construction.

Chapter 117

Cyclic open-string operations

Let A be a finite-dimensional, uncurved cyclic A_∞ algebra over $\mathbb{C}$, with a nondegenerate pairing of degree -3. Use the cohomological suspension $s : A \to A[1]$ of degree -1. The suspended operations $b_n : (A[1])^{\otimes n} \to A[1]$ have degree one. Our sign convention is the coderivation convention $b^2 = 0$ on the tensor coalgebra, with the Koszul rule for moving homogeneous inputs. Write ω for the resulting degree--1 symplectic form on $A[1]$. Fix the Hamiltonian convention $\iota_{X_F}\omega = dF$ and $[X_F, X_G] = X_{\{F,G\}}$.

For the universal degree-zero coordinate x on $A[1]$, put

$$S(x) = \sum_{n \geq 1} \frac{1}{n+1}\omega(x, b_n(x^{\otimes n})).$$

Cyclic invariance is read with the suspended Koszul signs. All functions are formal power series at the origin. An equivalent unsuspended convention writes this as $\frac{1}{2}\langle a, m_1 a\rangle + \sum_{n\geq 2} \frac{1}{n+1}\langle a, m_n(a^{\otimes n})\rangle$.

Proposition 117.1 (Cyclic master equation)**.** *The action satisfies* $\{S, S\} = 0$.

Proof. Cyclicity makes the $n+1$ derivatives of the arity-$n+1$ term equal, with the suspended Koszul signs. Thus $X_S(x) = \sum_{n\geq 1} b_n(x^{\otimes n})$. The coderivation identity implies $X_S^2 = 0$. The Hamiltonian identity gives $X_{\{S,S\}} = 2X_S^2 = 0$. Nondegeneracy of ω makes $\{S, S\}$ constant. Uncurvedness removes b_0, so S starts in order two and its bracket has no constant term. The constant is therefore zero. □

A finite-height Ginzburg realization requires such a cyclic model on the selected proper objects before this scalar action is applied. A Hall realization additionally requires a moduli problem with critical coefficient complexes. For an extension $A_0 \to F \to B_0$, put $L(A_0, B_0) = \det R\,\mathrm{Hom}(A_0, B_0)$. Three-Calabi–Yau duality gives

$$R\,\mathrm{Hom}(B_0, A_0) \simeq R\,\mathrm{Hom}(A_0, B_0)^\vee[-3].$$

Applying the determinant functor to the two derived Hom triangles gives

$$\det R\,\mathrm{Hom}(F, F) \cong \det R\,\mathrm{Hom}(A_0, A_0) \otimes \det R\,\mathrm{Hom}(B_0, B_0) \otimes L(A_0, B_0)^{\otimes 2}.$$

Indeed, the four subquotients are the two diagonal Hom complexes and the two cross complexes. Dualization and the odd shift cancel on their determinant lines, so the two cross determinants coincide. Orientation lines require square-root isomorphisms along this formula, compatible on triple flags and with any quotient descent. The cyclic master equation does not select these isomorphisms.

Chapter 118

An algebraic open–closed comparison

A finite-dimensional ordinary Lie algebra $\mathfrak{l}$ gives a precise coefficient model for an enveloping-algebra centre. Let $U = U(\mathfrak{l})$ over $\mathbb{C}$. The adjoint module U^{ad} has action $x \cdot u = xu - ux$. Its Chevalley cochains are

$$C^n(\mathfrak{l}, U^{\mathrm{ad}}) = \mathrm{Hom}_{\mathbb{C}}(\Lambda^n \mathfrak{l}, U),$$

with differential

$$\begin{aligned}(df)(x_1, \dots, x_{n+1}) = &\sum_i (-1)^{i+1} [x_i, f(x_1, \widehat{x_i}, \dots, x_{n+1})] \\ &+ \sum_{i<j} (-1)^{i+j} f([x_i, x_j], x_1, \widehat{x_i}, \widehat{x_j}, \dots, x_{n+1}).\end{aligned}$$

The second line places the bracket first and retains the order of the remaining arguments.

Theorem 118.1 (Hochschild–Chevalley comparison)**.** *Restriction followed by antisymmetrization defines a quasi-isomorphism*

$$C^{\bullet}_{\mathrm{HH}}(U, U) \longrightarrow C^{\bullet}(\mathfrak{l}, U^{\mathrm{ad}}), \qquad \phi \longmapsto \left[(x_1, \dots, x_n) \mapsto \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \phi(x_{\sigma(1)}, \dots, x_{\sigma(n)}) \right].$$

This is a comparison of ordinary cochain complexes.

Proof. The Chevalley bimodule resolution has terms $K_n = U \otimes \Lambda^n \mathfrak{l} \otimes U$. Its differential has the left-minus-right multiplication terms and the bracket terms with the signs dual to the displayed differential. Filter K_n by total PBW degree, counting each exterior generator as degree one. The multiplication terms preserve this degree and bracket terms lower it. The associated graded is the Koszul resolution of the diagonal in $\mathrm{Sym}(\mathfrak{l}) \otimes \mathrm{Sym}(\mathfrak{l})$, with regular sequence $x_i \otimes 1 - 1 \otimes x_i$. That Koszul complex is exact away from degree zero. The filtration is bounded below and exhaustive, and each element has finite filtration degree, so lifting a leading term proves exactness of $K_\bullet$ by induction. Its augmentation has image U. Thus $K_\bullet$ is a free bimodule resolution.

Antisymmetrization gives a map from $K_\bullet$ to the ordinary bar resolution, lifting the identity of U. Adjacent products in its bar differential pair to commutators; $xy - yx = [x, y]$ gives exactly the bracket terms of $K_\bullet$. The outer products give its left-minus-right terms. A comparison between projective resolutions lifting the identity is a homotopy equivalence. Applying $\mathrm{Hom}_{U \otimes U^{\mathrm{op}}}(-, U)$ gives the stated formula and quasi-isomorphism. □

This is the classical Cartan–Eilenberg comparison. For a finite Lie superalgebra, the same construction uses the super-exterior powers and permutation Koszul signs. Odd generators produce symmetric exterior directions, so the resolution can be unbounded; its algebraic direct sums and cochain products remain part of the convention. The PBW and super-Koszul proof applies with those conventions.

A physical mixed holomorphic–topological realization requires a map from its independently defined bulk observables to this cochain object. The theorem specifies the adjoint coefficients and the actual algebraic comparison. An E_2 refinement, a current-factorization realization, and compatibility with root-height transitions require maps preserving their additional operations. A quotient of Lie algebras alone does not provide a covariant map of Hochschild centres.

Chapter 119

The exact scalar and first-order observables

Set

$$\mathrm{Den}_{\Delta}(Z) = D_5(2Z), \qquad \jmath(Z) = Z/2, \qquad \jmath^{*}\,\mathrm{Den}_{\Delta} = D_5.$$

The one-particle root spaces give

$$\operatorname{sch}\mathrm{Fock}(\mathfrak{g}_{\Delta_5+}) = \prod_{\alpha>0}(1-e^{-\alpha})^{-\operatorname{sdim}\mathfrak{g}_{\Delta_5\alpha}}.$$

The oriented root operator gives

$$e^{-\rho}\,\mathrm{sPf}_{F}(\mathbb{A}_Z) = D_5(2Z).$$

The primitive reduced curve-counting theorem gives

$$\mathcal{Z}^{\mathrm{red}}_{K_3\times E} = -(\jmath^{*}\,\mathrm{Den}_{\Delta})^{-2} = -\chi_{10}^{-1}.$$

Theorem 119.1 (Protected square-root theorem)**.** *Assume the stated root-space Fredholm comparison with its orientation, Weyl shift, and convergence domain. Within that realization, the denominator section* Den_{Δ} *is simultaneously:*

(i) *the Weyl denominator of* $\mathfrak{g}_{\Delta_5}$*;*

(ii) *the Euler character of the bar–Chevalley complex;*

(iii) *the oriented Fredholm Pfaffian of* $\mathbb{A}_Z$*;*

(iv) *the PBW determinant of the parity-refined root superspace* $\mathcal{H}_{\mathrm{rp}}$*;*

(v) *the automorphic denominator whose half-coordinate pullback squares to the reduced scalar denominator.*

All five identifications preserve root degree and Weyl transport.

Proof. The first two are the denominator and bar–Chevalley theorems. The third is the Fredholm theorem. PBW for the Borcherds Lie superalgebra identifies the root superspace determinant, giving the fourth. The half-coordinate identity $\jmath^{*}\,\mathrm{Den}_{\Delta} = D_5$, the theta normalization, and the reduced curve-counting theorem give the fifth. □

Chapter 120

Physical meaning

The root-datum realizations retain several distinct constructions. The finite Ginzburg algebras and their projections form an inverse system. A completed three-Calabi–Yau sector requires a continuous nondegenerate Calabi–Yau structure and compatible cyclic models on the selected proper objects. The quiver Hall category retains its chosen coefficients and extension correspondences. A Hall theorem on that datum must identify its root presentation. The Borcherds root superspace defines the root-graded Fock space and its supercharacter. Its identification with physical one-particle states requires a parity-preserving graded comparison intertwining the primitive actions.

A realization on the elliptic curve must construct the current object with its local and wrapped supports, collision operations, and comparison maps. The quantum Borcherds algebra, with its specified pairing and completion, acts as a line-operator symmetry only through a representation preserving fusion and exchange. The root operator gives the Fredholm product on its specified convergence domain. A physical determinant interpretation requires an operator comparison and an orientation-preserving identification of the determinant lines and sections. The Fock character gives the scalar square. An operator trace additionally requires its state space and trace domain.

Simple-root states satisfying the defining relations give a primitive Lie map. The explicit enveloping and Chevalley extensions preserve products and brackets. A compact string realization must additionally construct its brane functor, Hall orientation maps, paired quantum coproduct, and trace-line comparison.

Chapter 121

Primitive comparisons and their algebraic extensions

Let $\mathfrak{n}_\Delta$ be the positive Borcherds Lie superalgebra. Choose homogeneous parity-preserving images of its simple generators in a charge-graded Lie superalgebra $\mathfrak{p}$. Assume these images satisfy every defining relation.

Theorem 121.1 (Extension of a primitive comparison)**.** *The chosen images determine a unique Lie-superalgebra map $f : \mathfrak{n}_\Delta \to \mathfrak{p}$. It induces*

$$U(f)(x_1 \cdots x_r) = f(x_1) \cdots f(x_r), \qquad C_*(f) = \mathrm{Sym}^c(f[1]).$$

These are respectively an enveloping-algebra map and a Chevalley chain-coalgebra map. Filtration-preserving maps extend to the specified inverse-limit completions. If f is surjective and both algebras have finite charge components and proper positive height, equality of their parity-refined PBW characters implies that f and $U(f)$ are isomorphisms.

Proof. The defining presentation gives f. Its tensor-algebra extension carries $xy - (-1)^{|x||y|}yx - [x, y]$ to the corresponding relation. It therefore descends to $U(f)$. The shifted symmetric-coalgebra map commutes with the Chevalley differential because f preserves brackets. Compatible maps on filtration quotients give maps on their inverse limits. Surjectivity of f gives surjectivity of $U(f)$. Character equality makes each finite charge component an isomorphism, separately in both parities. The PBW embeddings then imply injectivity of f. □

At level zero the current Lie conformal algebra is $\mathbb{C}[\partial] \otimes \mathfrak{n}_\Delta$ with $[x_\lambda y] = [x, y]$ on constant generators. The map $\partial^j x \mapsto \partial^j f(x)$ preserves this bracket and sesquilinearity. A nonzero central level additionally requires preservation of the specified invariant pairing. Hall convolution, Calabi–Yau structures, quantum coproducts, and determinant lines require their own comparison data.

Chapter 122

Finite quivers and Calabi–Yau completions

For a finite quiver Q with zero potential, let $\Gamma(Q, 0)$ be its path-length-completed Ginzburg algebra. Path products are read from left to right: if $a : i \to j$ and $b : j \to k$, then $ab : i \to k$. For each arrow $a : i \to j$, add $a^* : j \to i$ of degree -1. At each vertex add a loop t_i of degree -2. Original arrows have degree zero, and

$$da = da^* = 0, \qquad dt_i = e_i\Big(\sum_a [a, a^*]\Big)e_i.$$

Proposition 122.1 (Restriction to a full subquiver)**.** *If Q_0 is a full subquiver, retaining its generators and killing all other vertices, arrows, reverse arrows, and loops defines a continuous dg-algebra projection*

$$\Gamma(Q, 0) \longrightarrow \Gamma(Q_0, 0).$$

The inclusion of retained graded generators need not commute with the differential.

Proof. The projection respects source and target idempotents and extends multiplicatively. At a retained vertex, every term involving a discarded arrow vanishes. The other terms are precisely dt_i in the smaller algebra. All arrow differentials vanish, so the projection is a chain map. It preserves the path-length filtration and is continuous. For the final assertion, start with one vertex i and no arrows, where $dt_i = 0$. Add a vertex j and an arrow $a : i \to j$. In the larger algebra $dt_i = aa^*$, a nonzero length-two path. Thus the graded inclusion is not a dg map. □

Ginzburg completion supplies a three-Calabi–Yau algebra for each finite quiver in its stated setting. The displayed projections specify an inverse system for compatible zero-potential windows. A Calabi–Yau structure on a completed limit requires its own nondegeneracy and continuity argument. Comparisons with Hall categories and quantized Borcherds algebras must also preserve orientations, root relations, and the paired Hopf data.

Book XXIV

Quantum groups, exchange, and representation theory

Chapter 123

From positive half to full quantum group

Suppose the compact or local positive algebra H^+ is paired continuously with a negative half H^-, carries a completed coproduct, and admits Cartan data and a separated triangular completion. The completed double

$$\mathcal{U} = \widehat{D}(H^+)$$

has triangular decomposition $H^- \mathbin{\widehat{\otimes}} H^0 \mathbin{\widehat{\otimes}} H^+$. Its universal R-matrix acts on continuous tensor products of modules. The classical limit is a Lie bialgebra whose cobracket is the first-order term of the coproduct asymmetry.

For $\mathbb{C}^3$, H^+ is the positive affine Yangian. For local ADE surfaces, it is the completed positive affine ADE Yangian. For the Igusa algebra, the paired Borel construction of the preceding part supplies a height-complete formal quantum group; a compact Hall realization maps to it through primitive, coproduct, and pairing comparisons.

Chapter 124

Tannaka reconstruction

Let $\mathcal{L}$ be an essentially small rigid $\Bbbk$-linear abelian monoidal category with finite-dimensional morphism spaces, finite-length objects, and exact tensor product. Let

$$\omega : \mathcal{L} \to \mathrm{Vect}_{\mathrm{fd}}$$

be a faithful exact strong monoidal functor. The matrix-coefficient quotient

$$H = \int^{M \in \mathcal{L}} \omega(M)^* \otimes \omega(M)$$

exists in vector spaces. Tensor product makes it a bialgebra, and rigidity supplies its antipode. An ordinary braiding gives a coquasitriangular form. Parameter-dependent exchange requires its specified coefficient ring, domains of composition and compatible tensor constraints before a spectral reconstruction can be asserted.

Theorem 124.1 (Rigid Tannaka reconstruction)**.** *Under the preceding hypotheses, the comparison functor is a monoidal equivalence*

$$\mathcal{L} \simeq \mathrm{Comod}_{\mathrm{fd}}(H).$$

Braiding transports to the universal coquasitriangular form on H.

Proof. The exact faithful reconstruction theorem identifies $\mathcal{L}$ with the finite-dimensional comodules of its matrix-coefficient coalgebra. Tensor product and duality give the bialgebra and antipode structures, and the hexagons give the coquasitriangular identities. The coefficient formulas and the hypotheses are given in Theorem 11.4, following [55, Theorems 1.10.1 and 5.4.1]. □

The action algebra $\mathrm{End}(\omega)$ is the full dual of this coefficient coalgebra. Its coproduct may require a completed tensor product, as Example 11.5 shows. The coend instead carries an ordinary coproduct. Identifying it with a prescribed quantum-group presentation requires the actual matrix-coefficient map and its relations. The ordered boundary algebra gives composition, while a compatible coproduct and exchange give the separate tensor and braiding operations.

Chapter 125

KZ and rational exchange

For a simple Lie algebra $\mathfrak{g}$, the Knizhnik–Zamolodchikov connection is

$$\nabla = d - \frac{\hbar}{2\pi i} \sum_{i<j} \Omega_{ij}\, d\log(z_i - z_j).$$

Flatness follows from the infinitesimal braid relations. Drinfeld–Kohno identifies its monodromy tensor category with the corresponding Drinfeld–Jimbo category after a choice of associator.

Rational difference exchange instead gives the Yangian R-matrix

$$R(u) = 1 - \frac{\hbar P}{u}$$

in the fundamental representation. The Yang–Baxter equation is the confluence identity for three exchanges. The KZ and Yangian constructions share the geometry of configuration spaces while using different flat connections.

Chapter 126

The centre and the double

The Hochschild centre $Z_{\mathrm{ord}}(A)$ controls central operations on the ordered algebra in its specified coefficient category. To form a Drinfeld centre of modules, first specify a monoidal module category. For an ordered chiral bialgebra satisfying Definition 103.1, use

$$\mathcal{M}_A = \mathrm{Mod}_A(\widehat{\mathcal{V}}_C)$$

with the monoidal product of Theorem 103.2. An object of $Z(\mathcal{M}_A)$ is a module M together with natural equivalences

$$c_{M,N} : M \widehat{\otimes}^{!} N \xrightarrow{\sim} N \widehat{\otimes}^{!} M \qquad (N \in \mathcal{M}_A)$$

satisfying the half-braiding hexagon and unit identities, with their coherences. This categorical centre is braided. A comparison with the Hochschild centre must specify a realization functor, the source and target of the comparison, and the operations it preserves. Dualizability alone does not supply such a comparison. Derived loop-space descriptions apply in the geometric categories and under the hypotheses of their respective centre theorems [7].

The centre and the double answer complementary questions. The centre classifies central actions in the chosen category. The double reconstructs a quantum group from paired positive and negative halves. A field-theory comparison requires explicit realizations of the relevant line and bulk operations and a map compatible with both constructions.

Book XXV

A compact six-dimensional protected source

Chapter 127

Holomorphic cotangent BF theory on $K3 \times E$

Let S be a K3 surface with holomorphic symplectic form σ_S, let E be an elliptic curve with holomorphic form $\mathrm{d}z$, and put

$$X = S \times E, \qquad \Omega_X = \sigma_S \wedge \mathrm{d}z.$$

For a finite proper-height quotient $\mathfrak{n}_{\Delta_5,+}{}^{(H)}$ of the positive Igusa Borcherds algebra, take fields

$$A \in \Omega^{0,\bullet}(X, \mathfrak{n}_{\Delta_5,+}{}^{(H)})[1], \qquad B \in \Omega^{0,\bullet}(X, (\mathfrak{n}_{\Delta_5,+}{}^{(H)})^\vee)[1].$$

The invariant evaluation pairing and Ω_X define the BV action

$$S_H(A,B) = \int_X \Omega_X \wedge \langle B, \bar\partial A + \frac{1}{2}[A,A]\rangle. \tag{127.1}$$

This is the holomorphic cotangent theory of the dg Lie algebra of $\mathfrak{n}_{\Delta_5,+}{}^{(H)}$-valued Dolbeault forms.

Theorem 127.1 (Classical compact-background source)**.** *The action* (127.1) *satisfies the classical master equation and defines a holomorphic factorization algebra on the compact Calabi–Yau threefold X.*

Proof. The shifted cotangent pairing is invariant. The classical master equation is the Jacobi identity of $\mathfrak{n}_{\Delta_5,+}{}^{(H)}$ together with $\bar\partial^2 = 0$ and the derivation property of $\bar\partial$. Local observables of an elliptic cyclic L_∞ algebra form a factorization algebra by the standard BV construction. □

Chapter 128

Tree exactness and the quantum master equation

Every interaction vertex in (127.1) has one B-leg and two A-legs. The propagator pairs an A-leg with a B-leg.

Lemma 128.1 (Loop bound)**.** *A connected Feynman graph of the cotangent theory has loop number at most one.*

Proof. A graph with V interaction vertices has exactly V internal or external B-half-edges. Every internal edge consumes one of them, hence $E \leq V$. The loop number is $E - V + 1 \leq 1$. □

The positive algebra is graded by strictly positive root height. For homogeneous $x \in \mathfrak{n}_{\Delta_s,+}{}^{(H)}$, the operator ad_x raises height. Its coadjoint operator lowers dual height. A product of adjoint operators labelled by positive roots has total nonzero height shift and therefore has zero trace.

Proposition 128.2 (Vanishing of one-loop Lie weights)**.** *For every finite positive-height quotient, the Lie weight of each one-loop graph in the cotangent theory vanishes.*

Proof. A one-loop graph has $E = V$. Every B-leg is consumed by an internal edge, so its external legs are A-legs. Its Lie weight is a cyclic trace of adjoint operators, or the dual trace with the conventionally equivalent signs. Each external label has positive root height. The product therefore strictly raises height on the finite-dimensional adjoint representation and has zero trace. The same argument applies to the dual representation. □

This vanishing removes the one-loop Lie coefficients in a perturbative construction with the stated propagator and vertex types. A renormalized quantum factorization algebra still requires admissible analytic kernels, local extensions, the quantum master equation, and compatible height transitions on the chosen observable spaces. These are the remaining quantum realization data.

Chapter 129

Classical reduction along K3

Choose a Kähler metric on S and normalize a closed harmonic $(0,2)$-form η by $\int_S \sigma_S \wedge \eta = 1$. Let $K = \mathbb{C}[\eta]/(\eta^2)$ with $|\eta| = 2$ and zero differential. There is a unital dg-algebra map

$$i : K \longrightarrow \Omega^{0,*}(S), \qquad \eta \longmapsto \eta.$$

Its image is closed under multiplication because $(0,4)$-forms vanish. The Dolbeault cohomology of a K3 surface is one-dimensional in degrees zero and two and zero in degree one, so i is a quasi-isomorphism. It preserves the trace defined by integration against σ_S.

Let $\mathfrak{l}$ be a finite-dimensional Lie algebra, and write $\mathfrak{d} = \mathfrak{l} \ltimes \mathfrak{l}^*$ for its cotangent Lie algebra, with invariant evaluation pairing. For an open $U \subset E$, tensor with $\Omega^{0,*}(U)$, using the completed smooth tensor product in the S direction. The field comparison is

$$K \otimes \Omega^{0,*}(U) \otimes \mathfrak{d} \xrightarrow{i \otimes \mathrm{id}} \Omega^{0,*}(S \times U) \otimes \mathfrak{d}.$$

The bracket is multiplication of forms followed by the Lie bracket, with Koszul signs.

Proposition 129.1 (Classical field comparison)**.** *The displayed map is a trace-preserving dg Lie quasi-isomorphism, compatible with restriction in U. Its induced minimal operations in the K3 factor have no higher products.*

Proof. Multiplicativity of i proves bracket compatibility. Hodge projection p and $h = \bar\partial^* G$ on the compact surface satisfy $pi = \mathrm{id}$ and $\bar\partial h + h\bar\partial = \mathrm{id} - i\,p$. Tensoring these continuous operators with forms on U gives the same contraction, with the tensor-product signs. Thus $i \otimes \mathrm{id}$ is a quasi-isomorphism. Restriction acts only on the U factor and commutes with it. The normalized trace on K is the restriction of the Dolbeault trace. Choose the contraction with $hi = 0$. In every higher transferred tree, the first product of inputs from $i(K)$ remains in $i(K)$ and the next internal homotopy kills it. Thus every higher transferred product vanishes. □

The coefficient algebra K retains both the degree-zero and degree-two K3 modes. The proposition compares classical field complexes and their brackets. Constructing an equivalence of quantum observable factorization algebras requires compatible quantization, support conventions, and the induced maps on observable completions. The ordinary Hochschild comparison of Theorem 118.1 is a separate algebraic calculation, not that quantum field map.

Chapter 130

Protected states and the Igusa trace

Let

$$\mathcal{H}_{\mathrm{ip}} = \widehat{\bigoplus}_{\alpha>0} \mathfrak{g}_{\Delta,\alpha}$$

with its Borcherds parity. The multiparticle state space is the completed super-Fock space

$$\mathcal{H}_{\mathrm{Fock}} = \widehat{\mathrm{Sym}}(\mathcal{H}_{\mathrm{ip},\bar{0}}) \otimes \widehat{\Lambda}(\mathcal{H}_{\mathrm{ip},\bar{1}}).$$

For Z in the absolute-convergence chamber, let e^{-H_Z} act by $e^{-\alpha(Z)}$ on the root space of degree α.

Theorem 130.1 (Protected Hamiltonian trace)**.** *The supertrace is*

$$\mathrm{str}_{\mathcal{H}_{\mathrm{Fock}}}(e^{-H_Z}) = \prod_{\alpha>0} (1 - e^{-\alpha(Z)})^{-\dim \mathfrak{g}_{\Delta,\alpha}}.$$

After multiplication by the inverse Weyl line, its reciprocal is $\mathrm{Den}_\Delta(Z) = D_5(2Z)$*. After the half-coordinate pullback and doubling, the disconnected protected scalar is* $-(\jmath^* \mathrm{Den}_\Delta)^{-2} = -\chi_{10}^{-1}$ *in the Oberdieck–Pixton normalization.*

Proof. An even oscillator contributes $(1 - e^{-\alpha})^{-1}$ and an odd oscillator contributes $1 - e^{-\alpha}$. Multiplication over the root basis gives the product. The denominator identity and the reduced curve-counting theorem supply the final two equalities. □

Chapter 131

The compact-source comparison

The compact threefold provides the classical cotangent field theory and the dg Lie comparison of Proposition 129.1. The root datum separately gives the Borcherds algebra, its enveloping algebra, and the root-graded Fock space. The latter has the trace computed above. Proposition 128.2 supplies the finite one-loop Lie cancellation, while Theorem 118.1 computes the ordinary algebraic centre model with adjoint coefficients.

Problem 131.1 (Geometric compact comparison)**.** Construct objects E_i representing the simple branes and maps on their derived morphism complexes that preserve units, composition, and higher composition homotopies. Realize these maps on extension stacks with the reduced critical coefficients and orientation isomorphisms on triple flags. Construct compatible tensor and unit constraints on the resulting boundary functor. Compare the chosen quantum bulk action with the independently defined boundary centre, including support, collisions, and completion. Finally construct a trace and a map of its value line to the automorphic line in the same charge normalization.

A primitive map satisfying the root relations gives the enveloping and Chevalley maps of Theorem 121.1. A strong monoidal equivalence transports categorical centres. Neither construction determines the brane functor or its cyclic trace. These are the additional maps required for compact geometric reconstruction.

Book XXVI

Physics of the comparison maps

Chapter 132

Holomorphic–topological boundary theories

A three-dimensional holomorphic–topological theory on $C \times \mathbb{R}$ has local observables that are holomorphic along C and locally constant along $\mathbb{R}$. A boundary produces an ordered chiral algebra. A second topological direction permits exchange and gives an E_2 enhancement or a meromorphic braided line category. The bulk acts through the derived centre.

The finite cotangent BF model gives the exact local calculation:

$$\mathrm{Bar}(U(\mathfrak{l})) \simeq C_*(\mathfrak{l}).$$

For $\mathfrak{l} = \mathfrak{n}_{\Delta_5,+}{}^{(H)}$, this calculation realizes every finite-height Igusa denominator sector.

Chapter 133

Six-dimensional hCS on $K3 \times E$

Choose a holomorphic volume form $\Omega_S \wedge dz_E$. The hCS theory has classical fields

$$\Omega^{0,*}(S \times E, \mathfrak{a})[1]$$

and action S_{hCS}. Its factorization observables are native to the threefold. Pushforward along p_E gives a factorization algebra on E whose coefficient complex contains the Dolbeault modes of S.

A protected reduction to a chiral algebra consists of a deformation retract of the K3 Dolbeault complex, compatibility of the transferred L_∞ operations with the trace, a boundary or defect selecting ordered fusion, and a quantization trivializing the local anomaly. These data produce the ordered chiral algebra to which the Hall/BPS comparison may be applied.

The Hodge equality $\chi(\mathcal{O}_{S\times E}) = 0$ controls one compact supertrace. The gauge and gravitational anomalies are classes in the local BV deformation complex and are evaluated by their own characteristic forms.

Chapter 134

Omega backgrounds

Let a torus T act holomorphically on the local geometry and preserve the selected supercharge. For a generator V of the action, the equivariant differential is

$$Q_\Omega = Q + \iota_V, \qquad Q_\Omega^2 = \mathcal{L}_V.$$

After localization, fixed loci produce deformation parameters and often a spectral R-matrix. Toric threefolds and ADE surface resolutions carry such actions. A deformation family equipped with monodromy-compatible moduli correspondences and wall-crossing equivalences transports the localized algebra from algebraic or equivariant K3 fibers to the selected generic fiber.

Chapter 135

Anomalies as typed classes

The principal anomaly classes are

construction	deformation complex
hCS quantization	local BV cochains
critical CoHA orientation	Picard/gerbe theory of the critical stack
Hall coproduct	correspondence and localization complex
projective conformal blocks	Atiyah algebra of the determinant line
automorphic multiplier	character group of the modular cover.

A comparison theorem is a morphism between the corresponding complexes together with equality of the transported classes. This formulation turns numerical coincidences into geometric identities precisely when an intertwining map exists.

Book XXVII

Automorphic characters and scalar shadows

Chapter 136

Borcherds products as characters

Let $\mathfrak{n}_+$ be a locally finite positive Lie superalgebra. The Chevalley Euler character is

$$\chi C_*(\mathfrak{n}_+) = \prod_{\alpha>0} (1 - e^{-\alpha})^{\operatorname{sdim} \mathfrak{g}_\alpha}.$$

A Borcherds product becomes the Weyl-shifted character of $C_*(\mathfrak{n}_+)$. This construction produces a character of an existing algebra. Hall geometry supplies a second algebra, and a primitive comparison theorem identifies the two.

Chapter 137

The universal Borcherds weight formula

For a weight-zero weak Jacobi form

$$\varphi(\tau, z) = \sum c(n, \ell) q^n r^\ell,$$

the Borcherds lift has weight

$$\mathrm{wt}(\mathrm{Borch}(\varphi)) = \frac{c(0,0)}{2}.$$

For the K3 half-index, $c(0,0) = 10$ and the weight is 5. This formula lives entirely on the automorphic layer. Hodge Euler characteristics, Heisenberg ranks, central charges, and BV anomalies are evaluated by their own traces.

Chapter 138

The Gritsenko–Cléry atlas

The eight diagonal-divisor forms are

$$\Delta_{5},\ \Delta_{2},\ \Delta_{1},\ \Delta_{1/2},\ \nabla_{3},\ \nabla_{2},\ \nabla_{3/2},\ Q_{1}$$

with weights

$$5,\ 2,\ 1,\ \frac{1}{2},\ 3,\ 2,\ \frac{3}{2},\ 1.$$

Each form is an automorphic product on its own paramodular or Hecke congruence group with its own character or multiplier [28]. A geometric realization is specified row by row by a charge lattice, Hall problem, trace, and primitive algebra comparison.

Chapter 139

Scalarization hierarchy

The constructions descend through four levels:

$$\text{factorization or Hall object} \longrightarrow \text{primitive Lie algebra} \longrightarrow \text{PBW character} \longrightarrow \text{scalar partition function}.$$

Each arrow retains the quotient of structure displayed at its target. The first object contains supports, extension maps, and brackets. The second retains the infinitesimal algebra. The third retains signed graded dimensions. The fourth adds a trace and normalization. Upward reconstruction is supplied by the corresponding geometric operations and comparison morphisms.

Book XXVIII

The unified theorem system

Chapter 140

The foundational theorem

Definition 140.1 (Complete Calabi–Yau quantum-group datum)**.** A complete comparison datum specifies independent categorical, field, boundary, Hall, Hopf, and trace objects together with:

(i) PTVV-representable moduli for the smooth proper Calabi–Yau category;

(ii) a renormalized BV quantization satisfying the quantum master equation and a local compact factorization generator b;

(iii) a boundary reduction or proper locally trivial map to a curve satisfying factorization pushforward, together with the supplied curve-level faces and aligned distributive law;

(iv) a strong-oriented Hall coefficient theory whose pull–push maps satisfy base change and whose positive algebra is separated and complete;

(v) paired negative and Cartan halves, a completed coproduct, a continuous nondegenerate Hopf pairing, and a separated triangular completion on which the canonical tensors converge;

(vi) compatible open–closed and Hall–boundary morphisms;

(vii) a parity-refined primitive comparison for every automorphic identification.

Theorem 140.2 (Calabi–Yau quantum-group assembly)**.** *Let a complete datum consist of:*

(i) *a Calabi–Yau category and a representable moduli-of-objects problem;*

(ii) *a quantized cyclic elliptic BV theory and a boundary or defect generator;*

(iii) *a Hall extension correspondence with coefficient functor, base-change theorem, strong orientation data, and separated charge completion;*

(iv) *paired positive and negative bialgebras, Cartan data, a nondegenerate continuous Hopf pairing, and a completed double;*

(v) *every curve-level distributive, PBW, and analytic comparison used below.*

Then the datum determines functorially:

(i) *framed* E_2 *Hochschild cochains and the shifted-symplectic moduli structure;*

(ii) *the three curve-level algebraic faces and their three bars;*

(iii) *the positive Hall/BPS algebra and its primitive Lie algebra;*

(iv) *the completed quantum double and braided representation category;*

(v) *every supplied comparison among categorical, field-theoretic, Hall, current, quantum, and automorphic realizations;*

(vi) *the Weyl-shifted PBW/Chevalley character of any parity-refined primitive Lie-superalgebra identification.*

Proof. Cyclic Deligne and shifted symplectic geometry give the first output [15, 16, 125]. Boundary factorization gives the curve faces, and Volume I gives their bars and centres. The Hall correspondence and its base-change theorem give the associative product; strong orientation supplies the coefficient coherence required in the 3-Calabi–Yau setting [97]. The paired bialgebra data give the double and universal R-matrix. The comparison maps commute by hypothesis. PBW and Chevalley antisymmetrization give the final character. Each construction is natural under a morphism preserving the displayed data. □

The local quiver, equivariant surface, K3 correspondence, and Igusa Borcherds constructions have the carriers stated above. A geometric $K3 \times E$ comparison requires the primitive map and determinant-line map of Problem 131.1.

Chapter 141

Landscape of established realizations

Geometry	Positive algebra	Full or centred object	Character
A_2 quiver	$U_v^+(\mathfrak{sl}_3)$	$U_v(\mathfrak{sl}_3)$	PBW root product
$\mathbb{C}^3$	$Y^+(\widehat{\mathfrak{gl}}_1)$	$Y(\widehat{\mathfrak{gl}}_1)$	plane-partition and Fock traces
$\widetilde{\mathbb{C}^2/\Gamma}$	$Y^+(\widehat{\mathfrak{g}}_\Gamma)$	paired completion when the negative half and coproduct are supplied	surface Fock traces
$K3$	$U(\mathfrak{g}^+_{\mathrm{BPS}})$	Heisenberg and LLV actions	K3 Hall characters
$K3 \times E$	Borcherds root envelope; geometric reduced character	$U_q(\mathfrak{g}_{\Delta_5})$ and the root-current model and classical BF theory	$-\chi_{10}{}^{-1}$

The A_2 and $\mathbb{C}^3$ rows provide complete standard quantum-group models. The ADE row provides an established completed positive half together with its geometric current actions; its paired full double is formed from the negative half, Cartan data, coproduct, continuous Hopf pairing, and triangular completion stated in the table. The K3 row provides global two-Calabi–Yau BPS and cohomological symmetries. The final row records independently constructed root, character, and classical field objects. Their geometric Hall realization requires the comparison in Problem 131.1.

Chapter 142

Constructed comparisons and their geometric extensions

The Hall flag argument constructs associativity once its coefficient and orientation maps are defined. The tensor-action formula constructs representation fusion from a coassociative counital coproduct. The matrix-coefficient formulas construct the coend of a supplied fibre functor. The primitive presentation gives enveloping and Chevalley maps. The classical cyclic and K3 field calculations retain their actual differentials, traces, and brackets.

The quantum field realization requires admissible quantization and observable completions. The compact Hall realization requires the brane and orientation maps of Problem 131.1. The Fredholm and automorphic comparisons require their analytic domains and value-line isomorphisms. These constructions preserve the full geometric reconstruction problem beyond its finite algebraic and classical calculations.

Chapter 143

Actions and comparison maps

For a specified boundary algebra A_b, a coherent bulk action is a map

$$\beta : \mathrm{Obs}_{\mathrm{bulk}}|_{\partial} \longrightarrow Z_{\mathrm{ord}}(A_b).$$

The target is the derived endomorphism object of the diagonal bimodule in the chosen factorization coefficient category. A Hall construction instead starts with

$$\mathcal{M}_{\alpha} \times \mathcal{M}_{\beta} \xleftarrow{p} \mathcal{E}_{\alpha,\beta} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

Critical coefficients, orientations, and admissible pull–push maps give its product. Framed moduli supply a representation after their action correspondences and triple-flag identities have been constructed. A paired Hopf structure gives $D(H^{+}, H^{\circ}, H^{-})$. Its coproduct defines tensor products of continuous modules, and a convergent universal R-matrix gives exchange.

A strong monoidal equivalence between this module category and a boundary category transports categorical centres. It does not identify that categorical centre with the object $Z_{\mathrm{ord}}(A_b)$ without a realization theorem. A trace on a specified representation then gives an amplitude. Its comparison with an automorphic section requires a map of value lines with the same normalization.

Chapter 144

Comparison data for Calabi–Yau constructions

Comparisons among Calabi–Yau categories, holomorphic field theory, chiral algebras, Hall algebras, quantum groups, and automorphic products require the following structures:

(i) for a smooth proper d-Calabi–Yau category with its nondegenerate negative-cyclic class, the categorical outputs are framed E_2 Hochschild cochains and, when the moduli stack of objects is representable in the PTVV setting, $(2-d)$-shifted symplectic moduli; the shift d does not change the Hochschild operadic level to E_n;

(ii) a geometric BV theory and boundary category produce factorization observables and transverse ordered chiral algebras; associative chiral convolution is recorded as a separate supplied structure;

(iii) pushforward to a curve uses an actual map $p : X \to C$; formal cycle-specialization notation alone specifies no pushforward;

(iv) extension stacks, coefficient sheaves, orientations, and completions define the Hall algebra;

(v) coproduct, Cartan, Hopf pairing, and topology define the double;

(vi) K3 Heisenberg, LLV orthogonal symmetry, local ADE Yangians, and global BPS generalized Kac–Moody algebras occupy distinct strata joined by explicit comparison maps where constructed;

(vii) the graph of the $K3 \times E$ Gram map becomes multiplicative in the polarized central extension;

(viii) point-local and wrapped sectors form a hybrid factorization problem;

(ix) the Igusa form is the Weyl-shifted PBW/Chevalley character of $\mathfrak{g}_{\Delta_5}$, and compact Hall geometry maps to it through the primitive comparison;

(x) anomaly identities are transported through explicit maps between deformation complexes.

The categorical, geometric, Hall, and automorphic operations thus have separate carriers and comparison maps.

Book XXIX

Technical appendices to the preceding book

Chapter 145

Typed data table

Object	Minimal data	Primary operation
CY category	nondegenerate negative-cyclic trace	Hochschild and shifted-symplectic structures
Factorization algebra	local observables and Weiss descent	disjoint-open product
Transversely ordered chiral algebra	factorization coefficient with an internal E_1 structure for $\otimes^{!}$	ordered fusion
Associative chiral algebra	an E_1 structure for chiral convolution $\star^{\mathrm{ch}}$	nonlocal chiral composition
CoHA	extension correspondence and coefficient theory	pull–push product
Bialgebra	compatible product and coassociative counital coproduct	fusion of representations
Double	positive and negative halves, Cartan data, Hopf pairing	universal R-matrix
Borcherds product	Jacobi or vector-valued modular input	graded character

Chapter 146

The three K3 brackets

The Heisenberg bracket is the excess intersection of two Nakajima incidence kernels. The LLV bracket is the commutator of diagonal Lefschetz kernels and their adjoints. The local Yangian bracket is the convolution commutator of ADE Hecke kernels with spectral equivariance. In $\mathrm{Corr}_{K3}(S)$ all three are compositions of geometric kernels. Their mixed commutators are the corresponding triple-intersection classes. The action algebra and Tannaka coend retain these classes while preserving the three geometric constructions.

Chapter 147

Exact arithmetic checks

The companion verification calculation establishes:

(i) the low Fourier rows of $\phi_{0,1}$ through q^3 by exact theta-series division;

(ii) the vector-valued cocycle identity for the Gram polarization;

(iii) the rational Yang–Baxter equation on $(\mathbb{C}^2)^{\otimes 3}$;

(iv) the A_2 Hall flag counts at ranks zero and one;

(v) the type-II Gram matrix and reflection identities.

Each calculation uses integer or rational arithmetic and appears as a worked formula in the relevant chapter.

Book entrance: The Igusa Square Root

The Jacobi seed, Borcherds denominator, root superalgebra, local–wrapped currents, quantum group, Hall model, and Fredholm realization.

Book XXX

The automorphic determinant

Chapter 148

The protected Jacobi input

148.1 The K3 elliptic genus

Let S be a K3 surface. The right-moving supersymmetric index of a unitary $\mathcal{N}=(4,4)$ sigma model is the elliptic genus

$$Z_{K3}(\tau,z)=2\phi_{0,1}(\tau,z).$$

The normalized weak Jacobi form has weight zero, index one, and Fourier expansion

$$\phi_{0,1}(\tau,z)=\sum_{n\geq 0}\sum_{\ell\in\mathbb{Z}} f(n,\ell)\,q^{n}r^{\ell}.$$

In Eichler–Zagier normalization,

$$\phi_{0,1}(\tau,z)=4\sum_{a=2}^{4}\frac{\vartheta_a(\tau,z)^2}{\vartheta_a(\tau,0)^2}\in J^{\mathrm{w}}_{0,1}$$

[53]. The coefficient depends on the discriminant $4n-\ell^2$ and on ℓ modulo 2.

Theorem 148.1 (Exact low Fourier coefficients)**.** *The first four q-rows are*

$$\begin{aligned}\phi_{0,1}=\;&r^{-1}+10+r\\&+q\left(10r^{-2}-64r^{-1}+108-64r+10r^{2}\right)\\&+q^{2}\left(r^{-3}+108r^{-2}-513r^{-1}+808-513r+108r^{2}+r^{3}\right)\\&+q^{3}\left(-64r^{-3}+808r^{-2}-2752r^{-1}+4016\right.\\&\qquad\left.-2752r+808r^{2}-64r^{3}\right)+O(q^{4}).\end{aligned}$$

In particular,

$$f(0,0)=10,\qquad \frac{f(0,0)}{2}=5.$$

Proof. Insert the theta-series expansions into the preceding quotient formula. Work with $t=q^{1/8}$ and $x=r^{1/2}$; each quotient is an exact formal power series in t with Laurent-polynomial coefficients in x. The fractional powers cancel in the sum. Division through t^{24} gives the displayed rows. The arithmetic calculation is reproduced in Appendix 179. □

148.2 The virtual half-index

For every (n,ℓ) choose a finite-dimensional super vector space $U_{n,\ell}$ satisfying

$$\operatorname{sdim} U_{n,\ell}=f(n,\ell).$$

The integer $f(n,\ell)$ is its superdimension. In the ordinary Grothendieck group of finite-dimensional super vector spaces, the even and odd dimensions remain separate. The signed integer is their image under superdimension. The determinant depends only on that image.

Lemma 148.2 (Fock character and Berezinian). *Let $V = \bigoplus_\gamma V_\gamma$ be a locally finite graded super vector space. Then*

$$\operatorname{sch}\big(\operatorname{Sym}(V_{\bar 0}) \otimes \bigwedge(V_{\bar 1})\big) = \prod_\gamma (1 - x^\gamma)^{-\operatorname{sdim} V_\gamma},$$

and

$$\operatorname{sdet}_V(1 - x) = \prod_\gamma (1 - x^\gamma)^{\operatorname{sdim} V_\gamma}.$$

Proof. An even one-dimensional summand contributes the geometric series $(1 - x^\gamma)^{-1}$. An odd one-dimensional summand contributes $1 - x^\gamma$. Multiplication over a homogeneous basis proves both identities. □

Chapter 149

The Borcherds product

149.1 The effective chamber

Define

$$\Gamma_{\mathrm{eff}} = \{(n,\ell,m) \in \mathbb{Z}^3 : m > 0,\ n \geq 0\} \cup \{(n,\ell,0) : n > 0\} \cup \{(0,\ell,0) : \ell < 0\}.$$

The ordering is lexicographic in $(m, n, -\ell)$ and is precisely the ordering selected by the cusp chamber. For $\gamma = (n,\ell,m)$ put

$$V_\gamma = U_{nm,\ell}, \qquad \operatorname{sdim} V_\gamma = f(nm,\ell).$$

Definition 149.1 (Monic virtual determinant)**.** The monic determinant of the half-index is

$$D_5(Z) = q^{1/2} r^{1/2} s^{1/2} \operatorname{sdet}_V(1 - x) = q^{1/2} r^{1/2} s^{1/2} \prod_{\Gamma_{\mathrm{eff}}} (1 - q^n r^\ell s^m)^{f(nm,\ell)}.$$

Theorem 149.2 (Borcherds–Gritsenko–Nikulin product)**.** *The preceding product converges normally on the chosen cusp chamber and extends to a holomorphic genus-two modular form of weight five with a character of order two. It is the monic Igusa form D_5. The theta-normalized form is*

$$\Delta_5^\theta = 2^6 D_5.$$

Proof. The singular theta lift of the vector-valued form corresponding to $\phi_{0,1}$ produces the product, divisor, weight, and automorphic character. The weight is $f(0,0)/2 = 5$. The genus-two identification with the Igusa form and the product formula are the Gritsenko–Nikulin theorems [10, 11, 74, 76]. The product of the ten even theta constants has coefficient 2^6 at $q^{1/2} r^{1/2} s^{1/2}$, fixing the scalar. □

149.2 The theta constant normalization

Let $\mathfrak{E}$ be the ten even genus-two characteristics. Use the Gritsenko–Nikulin phase convention (17.17): the linear phase is $b^{\mathsf{T}} m$, without a shift by $a/2$. Define

$$\Delta_5^\theta(Z) = \prod_{\mathfrak{m} \in \mathfrak{E}} \vartheta[\mathfrak{m}](Z, 0).$$

The cusp leading terms of the ten factors contribute six factors of 2 and total exponent $(1/2, 1/2, 1/2)$. Thus

$$[q^{1/2} r^{1/2} s^{1/2}] \Delta_5^\theta = 64.$$

This coefficient selects a canonical scalar normalization of the automorphic section.

Proposition 149.3 (Ten-factor leading-term calculation). *Group the ten even characteristics by the first half-characteristic $a \in (\mathbb{Z}/2)^2$. Their leading contributions in the chosen cusp chamber are*

a	#	*scalar*	*monomial*
$(0,0)$	4	1	1
$(1,0)$	2	2	$q^{1/8}$
$(0,1)$	2	2	$s^{1/8}$
$(1,1)$	2	2	$q^{1/8}r^{1/4}s^{1/8}$.

Multiplication gives

$$1^4\,2^2\,2^2\,2^2\,q^{1/2}r^{1/2}s^{1/2} = 64q^{1/2}r^{1/2}s^{1/2}.$$

Proof. Insert each characteristic into the genus-two theta series and retain the unique terms of least order in the lexicographic cusp valuation $(m, n, -\ell)$. For $a = (0,0)$ the zero lattice vector contributes one. For each nonzero a, the two shortest lattice representatives are exchanged by sign and contribute the scalar two. Their quadratic exponents are the entries in the final column. Summing the exponents and multiplying the six factors of two gives the formula. □

Proposition 149.4 (Boundary factor of the Borcherds product). *The $m = 0$ part of the monic product is*

$$\prod_{n>0,\ell\in\mathbb{Z}} (1-q^n r^\ell)^{f(0,\ell)} \prod_{\ell<0} (1-r^\ell)^{f(0,\ell)} = (1-r^{-1}) \prod_{n\geq 1} (1-q^n r^{-1})(1-q^n)^{10}(1-q^n r).$$

Consequently the first Fourier–Jacobi coefficient is

$$q^{1/2}r^{1/2}(1-r^{-1}) \prod_{n\geq 1} (1-q^n r^{-1})(1-q^n)^{10}(1-q^n r) = \eta(\tau)^9 \vartheta_{11}(\tau, z),$$

where the odd theta function is fixed by the exact convention

$$\vartheta_{11}(\tau, z) = -q^{1/8}r^{-1/2} \prod_{n\geq 1} (1-q^{n-1}r)(1-q^n r^{-1})(1-q^n).$$

Proof. The q^0 row of $\phi_{0,1}$ has exactly three nonzero coefficients: $f(0,-1) = 1$, $f(0,0) = 10$, and $f(0,1) = 1$. Substitution into the $m = 0$ factors gives the first formula. The Jacobi triple product and $\eta(\tau) = q^{1/24} \prod_{n\geq 1}(1-q^n)$ give the second, including its leading sign and monomial. □

Corollary 149.5 (The scalar square). *The monic weight-ten form is*

$$\chi_{10} = D_5{}^2 = 2^{-12}\Delta_5^{\theta 2}.$$

Its automorphic character is trivial.

Proof. The identity $\Delta_5^\theta = 2^6 D_5$ gives the scalar factor. The character of D_5 has order two, so its square is trivial. □

Chapter 150

Divisor, character, and the meaning of the square root

The diagonal Humbert divisor is

$$H_1 = \{Z \in \mathbb{H}_2 : z = 0\}/\mathrm{Sp}_4(\mathbb{Z}).$$

Theorem 150.1 (Diagonal divisor). *The divisor of D_5 is the $\mathrm{Sp}_4(\mathbb{Z})$-orbit of H_1, with multiplicity one. The form transforms by the Maass character ν_5 of order two, and $\chi_{10} = D_5^2$ is scalar-valued.*

Proof. This is the defining diagonal-divisor property of the Igusa square root and the first row of the Cléry–Gritsenko classification [28, 85, 86]. □

Real roots of the denominator algebra define Weyl reflections. Imaginary roots contribute root spaces and product exponents. This division of labor is intrinsic to a Borcherds algebra: the Weyl group is generated by real simple roots, while the imaginary simple roots enter the correction term of the denominator identity.

150.1 Automorphic section, orientation line, and Pfaffian operator

An automorphic square root is a section of a line bundle with character. An orientation is a square root of a determinant line. A Pfaffian operator is a skew family whose regularized Pfaffian is a section of that orientation line. These form a sequence of successive refinements:

$$\text{automorphic section} \longrightarrow \text{determinant-line identification}$$
$$\longrightarrow \text{oriented square-root line} \longrightarrow \text{Fredholm family and Pfaffian section.}$$

Proposition 150.2 (Logical separation of the four levels). *The product theorem constructs the first object in the sequence. A determinant-line comparison constructs the second. A coherent square root of that line constructs the third. A first-order operator requires a symbol, domain, ellipticity or Fredholm condition, and regularization, and thereby produces the fourth refinement.*

Proof. Each stage adds a specified geometric structure. A determinant realization adds an isomorphism from a geometric determinant line to the automorphic line. An oriented realization adds a coherent square-root line and its parity refinement. An operator realization adds a principal symbol, a Fredholm domain, a spectrum, and a regularization whose Pfaffian is the resulting section. □

The title names the monic automorphic section D_5, characterized by the exact identity $D_5^2 = \chi_{10}$. Part IV formulates an analytic Pfaffian realization as a further geometric refinement.

Book **XXXI**

The denominator algebra

Chapter 151

The Lorentzian root lattice

Let

$$L = \mathbb{Z} f_2 \oplus \mathbb{Z} f_3 \oplus \mathbb{Z} f_{-2}$$

with bilinear form

$$(f_2, f_{-2}) = -1, \qquad (f_3, f_3) = 2,$$

and all remaining basis pairings zero. Its signature is $(2, 1)$. Define the type-II root lattice

$$L_{II} = 2\mathbb{Z} f_2 \oplus \mathbb{Z} f_3 \oplus 2\mathbb{Z} f_{-2} \cong U(4) \oplus \langle 2 \rangle.$$

The Gram-index map is

$$\alpha : \Gamma_{\mathrm{ind}} = \mathbb{Z}^3 \longrightarrow L_{II}, \qquad \alpha(n, \ell, m) = 2n f_2 - \ell f_3 + 2m f_{-2}.$$

Then

$$(\alpha(n,\ell,m), \alpha(n',\ell',m')) = -2\big(2nm' + 2n'm - \ell\ell'\big),$$

and

$$(\alpha(n,\ell,m), \alpha(n,\ell,m)) = -2(4nm - \ell^2).$$

151.1 Real simple roots and Weyl vector

Put

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3.$$

Proposition 151.1 (Type-II Cartan matrix)**.** *The Gram matrix of the real simple roots is*

$$\big((\delta_i, \delta_j)\big)_{i,j=1}^{3} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

The reflections

$$s_i(x) = x - (x, \delta_i)\delta_i$$

are integral involutions preserving L_{II} and its form.

Proof. Substitution into the basis pairing gives the matrix. Since $(\delta_i, \delta_i) = 2$, the reflection formula has integral coefficients. Direct expansion gives $s_i^2 = 1$ and $(s_i x, s_i y) = (x, y)$. □

The Weyl vector is

$$\rho = f_2 - \frac{1}{2} f_3 + f_{-2}, \qquad (\rho, \delta_i) = -1.$$

In the Gram-index variables, $e^{-\rho} = q^{1/2} r^{1/2} s^{1/2}$.

Chapter 152

The Gritsenko–Nikulin Lie superalgebra

The Gritsenko–Nikulin construction specifies a generalized Kac–Moody Lie superalgebra $\mathfrak{g}_{\Delta_5}$, with real simple roots δ_1, δ_2, δ_3 and the imaginary correction attached to $\phi_{0,1}$. Its generators and relations supply the bracket whose denominator is compared with the Borcherds product.

Theorem 152.1 (Root supermultiplicities and denominator)**.** *On the active positive support,*

$$\operatorname{smult} \mathfrak{g}_{\Delta_5\,\alpha(n,\ell,m)} = f(nm,\ell).$$

In the Lorentzian coordinate convention of Gritsenko–Nikulin,

$$\operatorname{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z) = 2^{-6}\Delta_5^{\theta}(2Z).$$

Proof. The real roots and Weyl chamber are the type-II case of the Gritsenko–Nikulin construction. The imaginary simple-root correction is chosen so that its Weyl–Kac–Borcherds product has exponent $f(nm,\ell)$. The resulting product is the doubled-variable monic Igusa form [74–76]. □

The theorem gives a Lie superalgebra with a bracket, Cartan, Weyl group, root parity, and denominator. The scalar product alone records the signed dimensions of the root spaces; the Gritsenko–Nikulin construction supplies the algebraic operations.

Chapter 153

The ordered bar as the denominator complex

Let $\mathfrak{n}_{\Delta_5,+}$ be the positive nilpotent subalgebra of $\mathfrak{g}_{\Delta_5}$. Fix a proper integral height h on the positive root cone and define

$$F^{>H}\mathfrak{n}_{\Delta_5,+} = \bigoplus_{h(\alpha)>H} \mathfrak{g}_{\Delta_5\,\alpha}, \qquad {\mathfrak{n}_{\Delta_5,+}}^{(H)} = \mathfrak{n}_{\Delta_5,+}/F^{>H}\mathfrak{n}_{\Delta_5,+}.$$

Height additivity makes $F^{>H}\mathfrak{n}_{\Delta_5,+}$ an ideal. Properness and local finiteness make ${\mathfrak{n}_{\Delta_5,+}}^{(H)}$ finite-dimensional and nilpotent.

Theorem 153.1 (Height-complete Bar–Chevalley comparison)**.** *For every H, antisymmetrization gives a natural quasi-isomorphism of conilpotent dg coalgebras*

$$\mathrm{Bar}\big(U({\mathfrak{n}_{\Delta_5,+}}^{(H)})\big) \simeq C_*({\mathfrak{n}_{\Delta_5,+}}^{(H)}).$$

Complete by the total height of each monomial, including repeated generators. Antisymmetrization induces a quasi-isomorphism in this separated completion:

$$\mathrm{Bar}_h^{\wedge}\big(U(\mathfrak{n}_{\Delta_5,+})\big) := \prod_{r\ge 0} \mathrm{Bar}\big(U(\mathfrak{n}_{\Delta_5,+})\big)_r \simeq C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta_5,+}) := \prod_{r\ge 0} C_*(\mathfrak{n}_{\Delta_5,+})_r.$$

Its root-graded Euler supercharacter is

$$\chi_{\mathrm{root}}\, C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta_5,+}) = \prod_{\alpha>0} (1-e^{-\alpha})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta_5\,\alpha}}.$$

Proof. For each H, PBW identifies the associated-graded comparison with the Koszul resolution of the free supercommutative algebra. In the full positive algebra, a monomial of total height r has at most r factors, drawn from finitely many finite-dimensional root spaces. Thus this filtration is finite in each total height. The comparison is a quasi-isomorphism there, and products of complexes of vector spaces are exact. This proves the completed assertion, as in Theorem 17.4. An even root vector contributes $1-e^{-\alpha}$ to the Euler supercharacter of Chevalley chains; an odd vector contributes its reciprocal. Multiplication gives the product. □

The inverse limit of enveloping algebras of Lie-height quotients is a different topology. For one even generator of height one it is $\mathbb{C}[x]$, whereas total-height completion is $\mathbb{C}[[x]]$; see (17.16).

Let $\mathbb{T}_{L_{II}}$ be the formal root torus and write Z_{root} for its logarithmic coordinate, normalized by

$$e^{-\alpha(n,\ell,m)} = Q^n R^\ell S^m, \qquad e^{-\rho} = Q^{1/2} R^{1/2} S^{1/2}.$$

Corollary 153.2 (Bar denominator identity)**.** *On the formal root torus,*

$$e^{-\rho}\chi_{\mathrm{root}} C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta_5,+}) = D_5(Z_{\mathrm{root}}).$$

The Gritsenko–Nikulin embedding of the root torus into the Siegel period domain sends Z_{root} to $2Z$ [74,76]. Hence

$$e^{-\rho}\,\chi_{\mathrm{root}}\,C^{!}_{*,h}(\mathfrak{n}_{\Delta_5,+}) = \mathrm{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z)$$

in Siegel period coordinates.

Proof. Multiply the Euler product in the preceding theorem by the Weyl monomial $e^{-\rho}$. The denominator theorem identifies the result on the root torus, and the stated embedding gives the Siegel-coordinate formula. □

The denominator is therefore the Weyl-shifted Euler supercharacter of the height-complete interval bar of the positive enveloping algebra.

Chapter 154

Finite-height currents

Retain the finite-height quotients ${\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)}$ of the preceding chapter. Define the current Lie conformal superalgebra

$$\mathrm{Cur}({\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)}) = \Bbbk[\partial] \otimes {\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)}, \qquad [a_\lambda b] = [a, b],$$

and its universal chiral enveloping algebra

$$V_H = U^{\mathrm{ch}}\Big(\mathrm{Cur}_E({\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)})\Big).$$

Theorem 154.1 (Height-complete current envelope)**.** *Let $V = U^{\mathrm{ch}}(\mathrm{Cur}_E \mathfrak{n}_{\Delta_{\mathrm{s}},+})$ and let V_r denote its total root-height r component. The completed current envelope is the pro-object*

$$\widehat{V}_{\Delta_{\mathrm{s}}} = \prod_{r\geq 0} V_r = \varprojlim_{H} V \Big/ \bigoplus_{r>H} V_r.$$

Every V_H is PBW-admissible as an associative chiral algebra, the current generators are primitive for the standard current coproduct, and

$$\mathrm{sch}\, \widehat{V}_{\Delta_{\mathrm{s}}} = \prod_{j\geq 1} \prod_{\alpha>0} (1 - t^j e^{-\alpha})^{-\operatorname{sdim} \mathfrak{g}_{\Delta_{\mathrm{s}}\alpha}}.$$

Moreover,

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(V_H) \simeq \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}\Big(\mathrm{Cur}_E({\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)})\Big),$$

and these equivalences pass to the separated height completion.

Proof. PBW orders the negative modes $a(-j)$, $j \geq 1$. Even modes contribute symmetric powers and odd modes exterior powers. A fixed total root height bounds the number of factors; fixing conformal energy as well bounds their mode indices. This gives the product and coefficientwise finiteness. The current coproduct sends $a(z)$ to $a(z) \otimes 1 + 1 \otimes a(z)$. Chiral antisymmetrization identifies the associative-chiral bar of the chiral enveloping algebra with the chiral Chevalley coalgebra in the stated PBW setting. The total-height quotients define a pro-vertex algebra. Uniform locality for arbitrary infinite-height sums requires an additional condition on a completed vertex-algebra realization. □

154.1 Classical local theory and exact quantum zero modes

For every H, the finite-dimensional nilpotent Lie superalgebra ${\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)}$ defines a cotangent BF theory on $E \times \mathbb{R}$. Its local dg Lie algebra is

$$\mathcal{L}_H = \Omega^{0,*}(E) \mathbin{\widehat{\otimes}} \Omega^*(\mathbb{R}) \otimes {\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)}$$

and the classical action is

$$S_H(A, B) = \int_{E\times\mathbb{R}} \Omega_E \wedge \Big\langle B, (\bar\partial + \mathrm{d}_{\mathbb{R}})A + \frac{1}{2}[A, A]\Big\rangle.$$

Theorem 154.2 (Finite-height BF realization). *The action S_H satisfies the classical master equation and defines a classical holomorphic–topological factorization algebra. Harmonic reduction along a compact interval produces the finite-dimensional cotangent BV theory of ${\mathfrak{n}_{\Delta_s,+}}^{(H)}$. The positive-root height grading makes every even adjoint endomorphism strictly height-increasing and every odd adjoint endomorphism odd; hence the modular supercharacter vanishes and the reduced theory satisfies the quantum master equation.*

Choose the standard complementary Lagrangian boundary polarizations, fixing the ${\mathfrak{n}_{\Delta_s,+}}^{(H)}[1]$ coordinate at one endpoint and its cotangent coordinate at the other. The algebraic boundary quantization, the interval amplitude between the two augmentation states, and the holomorphic boundary current envelope are then

$$U({\mathfrak{n}_{\Delta_s,+}}^{(H)}), \qquad \mathbf{1} \otimes^{\mathbb{L}}_{U({\mathfrak{n}_{\Delta_s,+}}^{(H)})} \mathbf{1} \simeq C_*({\mathfrak{n}_{\Delta_s,+}}^{(H)}), \qquad V_H.$$

The height quotient maps preserve the polarizations and yield the inverse systems

$$\widehat{U}_h(\mathfrak{n}_{\Delta_s,+}), \qquad C^{\wedge}_{*,h}(\mathfrak{n}_{\Delta_s,+}), \qquad \widehat{V}_{\Delta_s}.$$

Proof. The classical master equation is the Jacobi identity paired with invariance of the cotangent pairing. Elliptic descent gives the classical factorization algebra. Harmonic reduction along the interval gives a finite-dimensional BV space. If x is even of positive root height, ad_x raises height and is nilpotent on the finite quotient, so its restrictions to the even and odd summands both have trace zero; if x is odd, ad_x is odd and has supertrace zero. Thus the modular character vanishes and the finite BV quantum master equation follows.

The chosen boundary polarization quantizes the linear Kirillov–Kostant boundary bracket to $U({\mathfrak{n}_{\Delta_s,+}}^{(H)})$. Gluing the two augmentation states gives the two-sided derived bar, and the PBW antisymmetrization identifies it with $C_*({\mathfrak{n}_{\Delta_s,+}}^{(H)})$. Holomorphic descendants generate $U^{\mathrm{ch}}(\mathrm{Cur}_E\, {\mathfrak{n}_{\Delta_s,+}}^{(H)}) = V_H$. All maps preserve root height, so they pass coefficientwise to the separated inverse limits. □

Book XXXII

The compact scalar character

Chapter 155

Reduced curve counting on $K3 \times E$

Let S be a nonsingular projective K3 surface and E an elliptic curve. For a primitive class $\beta_h \in H_2(S, \mathbb{Z})$ with $\beta_h^2 = 2h - 2$, let $N_{g,h,d}$ be the reduced disconnected Gromov–Witten invariant of class (β_h, d) with the standard incidence insertion. Define

$$\mathcal{Z}(u, q, s) = \sum_{g,h,d} N_{g,h,d} u^{2g-2} q^{h-1} s^{d-1}.$$

Theorem 155.1 (Primitive Igusa cusp-form theorem)**.** *Under $r = e^{iu}$, the partition function is the Laurent expansion*

$$\mathcal{Z}(u, q, s) = -\frac{1}{\chi_{10}(r, q, s)}.$$

Equivalently, in theta normalization,

$$\mathcal{Z} = -2^{12} \Delta_5^{\theta^{-2}}.$$

Proof. Oberdieck and Pixton identify the primitive reduced partition function with the Laurent expansion of $-\chi_{10}^{-1}$ [122]; their normalization is the monic Borcherds product χ_{10}. The second formula is the identity $\chi_{10} = 2^{-12} \Delta_5^{\theta^2}$. □

The theorem identifies the numerical generating series in its stated primitive sector. The Borcherds root algebra supplies a separate algebraic character to compare with that series.

Corollary 155.2 (Exact character comparison)**.** *Let $\jmath : \mathbb{H}_2 \to \mathbb{H}_2$ be the half-coordinate map $\jmath(Z) = Z/2$, and let*

$$\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+}) = \mathrm{Sym}(\mathfrak{n}_{\Delta_5,+\bar{0}}) \otimes \bigwedge(\mathfrak{n}_{\Delta_5,+\bar{1}}).$$

Then

$$\mathcal{Z}^{\mathrm{red}}_{S \times E} = -\jmath^* \Big(e^{2\rho} \operatorname{sch}\big(\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+})^{\otimes 2}\big) \Big) = -\chi_{10}^{-1}.$$

Proof. Put

$$P = \prod_{\alpha > 0} (1 - e^{-\alpha})^{\operatorname{sdim} \mathfrak{g}_{\Delta_5 \alpha}}.$$

The denominator theorem gives $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = e^{-\rho} P = D_5(2Z)$, while PBW gives $\operatorname{sch} \mathrm{Fock}(\mathfrak{n}_{\Delta_5,+}) = P^{-1}$. Hence

$$e^{2\rho} \operatorname{sch}\big(\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+})^{\otimes 2}\big) = \mathrm{den}(\mathfrak{g}_{\Delta_5})^{-2}.$$

Pullback by $\jmath$ sends $\mathrm{den}(\mathfrak{g}_{\Delta_5})$ to $D_5(Z)$, whose square is χ_{10}. The reduced incidence convention supplies the overall minus sign in the Oberdieck–Pixton theorem. □

The equality compares the completed root-Fock supercharacter with the numerical reduced generating series. Its proof uses the denominator identity and the half-coordinate pullback. A map from compact geometric states to this Fock space, preserving operators and their trace, is an additional comparison.

Chapter 156

Extensions over the elliptic classifying stack

Work on locally noetherian complex schemes with the fppf topology. Let BE be the stack of right E-torsors. For a left E-stack M, an object of $[M/E](T)$ is a right torsor $P \to T$ and an equivariant map $P \to M$. Forgetting the map gives $[M/E] \to BE$. All products of stacks below are two-fibre products.

Proposition 156.1 (The common translation torsor)**.** *There is a natural equivalence*

$$[(M \times N)/E_{\mathrm{diag}}] \simeq [M/E] \times_{BE} [N/E].$$

The equivalences for any finite number of factors commute with regrouping and insertion of the terminal E-stack.

Proof. An object on the right is (P, m), (Q, n) together with a torsor isomorphism $\alpha : P \to Q$. Send it to $(P, m, n\alpha)$. Conversely, use a common torsor twice and its identity isomorphism. A pair of torsor maps a, b on the right satisfies $b\alpha = \alpha' a$, so b is determined by a. The same construction identifies the maps and their descent isomorphisms. It commutes with base change. For several factors, every regrouping gives one torsor with the same list of equivariant maps, proving all stated coherences. □

For $M = N = E$ with translation, the diagonal quotient is E, with coordinate $y - x$. Both separate quotients are points. Their fibre product over BE retains the relative translation, while their ordinary product loses it.

Let $\mathfrak{M}$ be the stack of coherent sheaves on $X = S \times E$, with families flat over the parameter scheme, and put $\mathfrak{Q} = [\mathfrak{M}/E]$. An object of $\mathfrak{Q}(T)$ is a torsor $P \to T$ and a flat sheaf on $S \times P$. Fix a polarization and write $\mathfrak{Q}_\alpha$ for a Hilbert-polynomial component. Translation preserves these polynomials. Let $\mathfrak{H}_{\alpha,\beta}$ classify short exact sequences $0 \to A \to F \to B \to 0$ on a common $S \times P$, with all terms flat over T and polynomials α, $\alpha + \beta$, β. Its span over BE is

$$\mathfrak{Q}_\alpha \times_{BE} \mathfrak{Q}_\beta \xleftarrow{p} \mathfrak{H}_{\alpha,\beta} \xrightarrow{q} \mathfrak{Q}_{\alpha+\beta}.$$

The category of spans of fppf stacks over BE has composition by fibre product and tensor product $\times_{BE}$. Its unit object is the identity map of BE.

Theorem 156.2 (Geometric Hall correspondence)**.** *For fixed α, β, q is representable by algebraic spaces and proper. The spans form a coherently associative unital Hall object in the category of spans over BE. Its unit is the zero-sheaf span $BE \leftarrow BE \to \mathfrak{Q}$.*

Proof. Over a fixed (P, F), the fibre of q classifies quotients $F \twoheadrightarrow B$ of polynomial β. Its kernel is flat, because F and B are flat. An automorphism fixing the quotient and F is the identity. Trivializing P fppf-locally identifies this fibre with the fixed-polynomial Quot scheme, projective over the parameter scheme by Grothendieck's theorem [77, Theorem 3.2]. The quotient functor descends to an algebraic space, and properness descends with it.

Both composites of three Hall spans classify $0 \subset A \subset D \subset F$ on the same torsor. One bracketing concatenates inclusions. The other starts with $A \subset F$ and a subobject $B \subset F/A$, and sets $D = F \times_{F/A} B$. These constructions are inverse on objects and isomorphisms. They preserve flatness and commute with base change. For longer flags all regroupings identify the same subobjects and quotients. Their comparison maps agree, proving the pentagon and higher coherences. Finally $\mathfrak{Q}_0 = BE$. The sequences with zero subobject or zero quotient give the unit and its coherences. □

Properness here concerns the full coherent-sheaf stack on a fixed pair of Hilbert-polynomial components. A stability restriction can replace its fibre by an open part of a Quot scheme, which need not be proper. A bounded charge window also needs closure under the extensions and flags used in its operations. A reduced critical-coefficient realization must supply pullback, pushforward, and orientation comparisons on these same flags. Its unit has stabilizer E. A coefficient theory for this quotient must include that nonaffine stabilizer, or work relative to BE, on the unit and every flag.

Chapter 157

Additive physical charge and quadratic automorphic degree

Let

$$\Lambda = \widetilde{H}(S, \mathbb{Z})$$

be the Mukai lattice of signature $(4, 20)$. The electric–magnetic charge lattice is

$$\Gamma_{\mathrm{phys}} = \Lambda \oplus \Lambda, \qquad c = (Q, P).$$

Its addition is the physical superselection law. Define the Gram map

$$\Pi(Q, P) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right) \in \Gamma_{\mathrm{ind}}.$$

Proposition 157.1 (Polarization identity)**.** *For $c_i = (Q_i, P_i)$,*

$$\Pi(c_1 + c_2) = \Pi(c_1) + \Pi(c_2) + B(c_1, c_2),$$

where

$$B(c_1, c_2) = \big(Q_1 \cdot Q_2, Q_1 \cdot P_2 + Q_2 \cdot P_1, P_1 \cdot P_2 \big).$$

The cross-term is a symmetric vector-valued two-cocycle:

$$B(c_1, c_2) + B(c_1 + c_2, c_3) = B(c_2, c_3) + B(c_1, c_2 + c_3).$$

Proof. Expand the three quadratic expressions. The cocycle identity is bilinearity of B. □

157.1 The polarized central extension

Define the abelian central extension

$$0 \longrightarrow \Gamma_{\mathrm{ind}} \longrightarrow \widetilde{\Gamma}_\Pi \longrightarrow \Gamma_{\mathrm{phys}} \longrightarrow 0, \qquad \widetilde{\Gamma}_\Pi = \Gamma_{\mathrm{phys}} \times \Gamma_{\mathrm{ind}},$$

with multiplication

$$(c, \lambda) \cdot (d, \mu) = \big(c + d, \lambda + \mu + B(c, d) \big).$$

The cocycle identity makes this multiplication associative. The polarization identity makes the graph map

$$j_\Pi : \Gamma_{\mathrm{phys}} \longrightarrow \widetilde{\Gamma}_\Pi, \qquad c \longmapsto (c, \Pi(c))$$

a homomorphism.

Let $A = \bigoplus_{c \in \Gamma_{\mathrm{phys}}} A_c$ be a charge-graded associative algebra, and let $\Bbbk[\Gamma_{\mathrm{ind}}]$ have monomials u^λ. Define the polarized central-extension algebra

$$\mathcal{R}_\Pi(A) = A \otimes \Bbbk[\Gamma_{\mathrm{ind}}]$$

with product

$$(a_c u^\lambda) \star (b_d u^\mu) = a_c b_d \, u^{\lambda+\mu+B(c,d)}.$$

Give $a_c u^\lambda$ degree $(c, \lambda) \in \widetilde{\Gamma}_\Pi$.

Theorem 157.2 (Graph-multiplicative polarization)**.** *The product $\star$ is associative and $\widetilde{\Gamma}_\Pi$-graded. The map*

$$\iota_\Pi : A \longrightarrow \mathcal{R}_\Pi(A), \qquad a_c \longmapsto a_c u^{\Pi(c)}$$

is an algebra homomorphism whose homogeneous degree is the graph element $j_\Pi(c)$.

Proof. The two parenthesizations differ by

$$B(c,d) + B(c+d,e) - B(d,e) - B(c,d+e),$$

which vanishes by the cocycle identity. The polarization formula gives

$$\iota_\Pi(a_c) \star \iota_\Pi(b_d) = a_c b_d \, u^{\Pi(c+d)}.$$

Thus the graph of the quadratic Gram map is multiplicative inside the central extension. □

The same construction applies to Lie superalgebras and their enveloping algebras. It retains additive physical charge, Gram degree, and the interaction cross-term in one grading group.

Chapter 158

Point-local and wrapped sectors

Let $p_E : S \times E \to E$ be projection.

Lemma 158.1 (Positive elliptic degree is wrapped)**.** *Let $C \subset S \times E$ be a proper one-dimensional subscheme whose pushforward $(p_E)_*[C]$ has positive degree. Then $p_E(C) = E$.*

Proof. Properness makes $p_E(C)$ closed. Positive pushforward degree implies that at least one irreducible component of C dominates a one-dimensional closed subset of E. Every one-dimensional closed subset of the irreducible curve E is E itself, so the image of that component, and hence the image of C, is all of E. □

Projection-finite objects define point-supported sectors on $\mathrm{Ran}(E)$. Positive elliptic degree defines wrapped sectors supported over the entire elliptic curve.

Definition 158.2 (The local–wrapped factorization operad)**.** Let Ch_E denote the chosen chiral/factorization operad on E and let Ass denote the ordered associative operad. Define the colored operad Hyb_E with colors

$$\mathsf{l}, \qquad \mathsf{w}_m \quad (m \geq 1)$$

by the following operation objects.

(i) Operations with local output are

$$\mathsf{Hyb}_E(\mathsf{l}^{\otimes k}; \mathsf{l}) = \mathsf{Ch}_E(k).$$

(ii) For $r \geq 1$ and $m = m_1 + \cdots + m_r$, operations with wrapped output are

$$\mathsf{Hyb}_E(\mathsf{l}^{\otimes k}, \mathsf{w}_{m_1}, \ldots, \mathsf{w}_{m_r}; \mathsf{w}_m) = \mathsf{Ch}_E(k) \otimes \mathsf{Ass}(r).$$

All other operation objects are initial.

Operadic composition uses chiral insertion and disjoint union in the first factor and ordered substitution in the second. The equality

$$(\gamma_1 \circ \gamma_2) \otimes (a_1 \circ a_2) = (\gamma_1 \otimes a_1) \circ (\gamma_2 \otimes a_2)$$

is the mixed interchange law. The arities $(k, r) = (k, 0), (0, 2), (k, 1)$ recover, respectively, local factorization, wrapped multiplication, and the action of local insertions on a wrapped sector.

Definition 158.3 (Hybrid local–wrapped algebra)**.** A hybrid local–wrapped algebra over E is an algebra over Hyb_E. It consists of a factorization algebra $\mathbb{F}_{\mathrm{loc}}$, a height- or charge-complete family $W = \prod_{m>0} W_m$, ordered products

$$\mu_{m_1,\ldots,m_r} : W_{m_1} \otimes \cdots \otimes W_{m_r} \longrightarrow W_{m_1+\cdots+m_r},$$

and factorization-compatible actions of the local operation objects $\mathsf{Ch}_E(k)$ on every wrapped product. The operad axioms state associativity, local factorization, and compatibility of the two operations.

A compact Hall theory of $S \times E$ produces a correspondence-valued Hyb_E-algebra when disjoint-support direct sums, wrapped extensions, and the mixed extension correspondences satisfy the same base-change identities.

Book XXXIII

The Borcherds algebra and its quantum group

Chapter 159

The local–wrapped root algebra

The positive cone is graded by $\gamma = (n, \ell, m)$, with elliptic degree m. Set

$$\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}} = \bigoplus_{\substack{\alpha>0\\ m(\alpha)=0}} \mathfrak{g}_{\Delta_5\,\alpha}, \qquad \mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}} = \bigoplus_{\substack{\alpha>0\\ m(\alpha)>0}} \mathfrak{g}_{\Delta_5\,\alpha}.$$

Proposition 159.1 (Semidirect local–wrapped decomposition)**.** *The space $\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}$ is a Lie subsuperalgebra, $\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}$ is a Lie ideal, and*

$$\mathfrak{n}_{\Delta_5,+} = \mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}} \ltimes \mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}.$$

The proper height filtration preserves both summands.

Proof. The bracket adds root degrees. The sum of two roots of elliptic degree zero again has degree zero, and addition of a nonnegative degree to a positive degree remains positive. Height is additive on every nonzero bracket. □

Define the complete enveloping Hopf algebra and current envelope

$$\mathbb{U}_{\Delta_5} = \widehat{U}_h(\mathfrak{n}_{\Delta_5,+}), \qquad \mathbb{V}_{\Delta_5,E} := \widehat{V}_{\Delta_5} = \prod_{r\geq 0} U^{\mathrm{ch}}(\mathrm{Cur}_E\, \mathfrak{n}_{\Delta_5,+})_r.$$

Here $\widehat{U}_h(\mathfrak{n}_{\Delta_5,+}) = \prod_{r\geq 0} U(\mathfrak{n}_{\Delta_5,+})_r$, and all completed tensor products sum the input heights to the output height. The adjoint action of $\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}$ on $\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}$ extends by derivations to $\widehat{U}_h(\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}})$. Let $\#$ denote the corresponding completed smash product.

Definition 159.2 (Current-generated local–wrapped operad)**.** The colored operad $\mathsf{Hyb}^{\mathrm{cur}}_{E,\Delta}$ is generated by the chiral current operations of $\mathrm{Cur}_E(\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}})$, the ordered multiplications of $\widehat{U}_h(\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}})$, and the adjoint derivation action of the former on the latter, modulo the chiral Jacobi, associative, derivation, and semidirect-product relations.

Theorem 159.3 (Local–wrapped Borcherds decomposition)**.** *Let $\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}$ be the sum of positive root spaces of elliptic degree $m = 0$ and let $\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}$ be the sum of those with $m > 0$. Then*

$$\mathfrak{n}_{\Delta_5,+} = \mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}} \ltimes \mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}},$$

where $\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}$ is a Lie subsuperalgebra and $\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}$ is a Lie ideal. For the proper-height completion,

$$\widehat{U}_h(\mathfrak{n}_{\Delta_5,+}) \cong \widehat{U}_h(\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}) \# \widehat{U}_h(\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}),$$

and

$$\widehat{C}_{*,h}(\mathfrak{n}_{\Delta_5,+}) \simeq \mathrm{Tot}\, C_*(\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}; \widehat{C}_{*,h}(\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}})).$$

The primitive Lie superalgebra of the completed enveloping algebra is the height completion $\widehat{\mathfrak{n}_{\Delta_5,+}}_h = \varprojlim_H \mathfrak{n}_{\Delta_5,+}{}^{(H)}$, and the Weyl-shifted Euler supercharacter of the completed Chevalley complex is $D_5(2Z)$.

Proof. Root addition adds elliptic degree. Thus the $m = 0$ summand is closed under the bracket, and every bracket with an $m > 0$ root again has $m > 0$. PBW orders wrapped generators before local generators and gives the completed smash product. The Chevalley complex of a semidirect product is the Hochschild–Serre total complex. Primitive elements of a completed enveloping algebra are recovered height by height. The denominator theorem supplies the last character identity. □

Remark 159.4. The theorem constructs a Borcherds root algebra with local and wrapped colors. The word Hall is reserved for an algebra obtained from geometric extension correspondences.

Chapter 160

The formal Igusa quantum group

Let $\mathfrak{b}_+$ be the positive Borcherds Borel half. Form the restricted graded dual $\mathfrak{b}_- := \mathfrak{b}_+^{\vee,\mathrm{gr}}$ with bracket and cobracket transported by the invariant pairing, identify the two Cartan copies along their common radical quotient, and complete by a proper root height. The completed Drinfeld double of this paired Borel datum is denoted ${\mathfrak{g}_{\Delta_5}}^{\mathrm{pair}}$. Its positive and negative halves, diagonal Cartan, invariant form, and canonical tensor form a height-admissible Manin triple and hence a quasitriangular Lie superbialgebra.

Definition 160.1 (Height-admissible Lie superbialgebra)**.** A root-graded Lie superbialgebra is height-admissible when every root space is finite-dimensional, every bounded-height region contains finitely many roots, and the bracket, cobracket, invariant tensor, and all iterated operations are coefficientwise finite in the height completion.

Proposition 160.2 (Height admissibility)**.** *The paired Gritsenko–Nikulin Borcherds Lie superbialgebra* ${\mathfrak{g}_{\Delta_5}}^{\mathrm{pair}}$ *is height-admissible.*

Proof. The Borcherds root spaces are finite-dimensional and the chosen height is proper. The bracket and cobracket preserve total root degree. In a bounded total height, finitely many decompositions of a root occur. The canonical tensor is therefore a coefficientwise finite sum in every height quotient. □

Theorem 160.3 (Quantized Borcherds superalgebra)**.** *Choose the symmetrizable Borcherds–Cartan superdatum and multiplicity index set that present* $\mathfrak{g}_{\Delta_5}$*. Hong's construction gives a quantized enveloping Hopf superalgebra*

$$U_q(\mathfrak{g}_{\Delta_5})$$

over $\mathbb{Q}(q)$*, generated by its Cartan elements and the real and imaginary simple-root generators, with triangular decomposition*

$$U_q(\mathfrak{g}_{\Delta_5}) \cong U_q^-(\mathfrak{g}_{\Delta_5}) \otimes U_q^0(\mathfrak{g}_{\Delta_5}) \otimes U_q^+(\mathfrak{g}_{\Delta_5}).$$

Its PBW root spaces have the same graded ranks as the classical Borcherds superalgebra. The canonical pairing of the two Borel halves yields a universal R-matrix in the root-height completion of

$$U_q^+(\mathfrak{g}_{\Delta_5}) \mathbin{\widehat{\otimes}} U_q^-(\mathfrak{g}_{\Delta_5}).$$

Consequently the quantization preserves the root-multiplicity data that enter $D_5(2Z)$*.*

Proof. The Gritsenko–Nikulin algebra is a symmetrizable Borcherds–Kac–Moody Lie superalgebra. Apply the generators-and-relations construction of the quantized Borcherds superalgebra. The nondegenerate bilinear form, triangular decomposition, centre, and universal R-matrix are established by Hong [82]. The PBW theorem identifies the associated graded root spaces with their classical counterparts. Since the denominator exponents are the classical signed root multiplicities, they are unchanged by the flat root-graded deformation. □

Remark 160.4. The quantum group is attached to the Borcherds generalized Cartan superdatum. A geometric Hall quantum group is identified with it through a Hall coproduct, pairing, completion, and primitive comparison.

Chapter 161

A holomorphic–topological BF realization

For every height bound H, let ${\mathfrak{n}_{\Delta_5,+}}^{(H)}$ be the finite-dimensional nilpotent quotient. On $E \times \mathbb{R}$ form the shifted cotangent theory of

$$\mathcal{L}_H = \Omega^{0,*}(E) \mathbin{\widehat{\otimes}} \Omega^*(\mathbb{R}) \otimes {\mathfrak{n}_{\Delta_5,+}}^{(H)}$$

with action

$$S_H(A, B) = \int_{E \times \mathbb{R}} \Omega_E \wedge \left\langle B, (\bar{\partial} + \mathrm{d}_{\mathbb{R}})A + \frac{1}{2}[A, A] \right\rangle.$$

Theorem 161.1 (Classical local and exact zero-mode realization)**.** *For every H:*

(i) *S_H satisfies the classical master equation and defines a classical holomorphic–topological factorization algebra;*

(ii) *harmonic reduction along a compact interval gives the finite-dimensional cotangent BV theory of ${\mathfrak{n}_{\Delta_5,+}}^{(H)}$, whose positive-root grading makes its modular supercharacter zero and hence gives the quantum master equation;*

(iii) *after choosing the complementary Lagrangian boundary polarizations, the boundary algebra is $U({\mathfrak{n}_{\Delta_5,+}}^{(H)})$, the interval amplitude is $\mathbf{1} \otimes^{\mathbb{L}}_{U({\mathfrak{n}_{\Delta_5,+}}^{(H)})} \mathbf{1} \simeq C_*({\mathfrak{n}_{\Delta_5,+}}^{(H)})$, and the holomorphic boundary current envelope is V_H.*

The transition maps yield a pro-object with boundary algebra $\widehat{U}_h(\mathfrak{n}_{\Delta_5,+})$, interval complex $C^{\wedge}_{*,h}(\mathfrak{n}_{\Delta_5,+})$, and current envelope $\widehat{V}_{\Delta_5}$.

Proof. The classical master equation is the Jacobi identity and invariance of the cotangent pairing. The classical observables satisfy factorization descent. Harmonic reduction is finite-dimensional. An even adjoint operator raises positive-root height and is nilpotent on the finite quotient; an odd adjoint operator is odd. Both have supertrace zero, so the modular term in the finite BV Laplacian vanishes. The chosen polarization quantizes the boundary Lie–Poisson bracket to the enveloping algebra; the two augmentation states give the two-sided bar, identified with Chevalley chains by PBW. Holomorphic descendants give the current algebra. All structure maps preserve height. □

The theorem gives the precise field-theoretic realization used in this volume: a classical local theory together with an exact quantum protected zero-mode sector.

Chapter 162

The determinant and Fredholm square-root operators

Let

$$\mathcal{V}_H = \bigoplus_{\substack{\alpha>0\\ h(\alpha)\le H}} \mathfrak{g}_{\Delta_5\alpha}$$

and define

$$\mathscr{D}_H|_{\mathfrak{g}_{\Delta_5\alpha}} = (1 - e^{-\alpha})\,\mathrm{id}\,.$$

Theorem 162.1 (Universal determinant section)**.** *The finite-height Berezinian is*

$$\mathrm{Ber}(\mathscr{D}_H) = \prod_{\substack{\alpha>0\\ h(\alpha)\le H}} (1 - e^{-\alpha})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta_5\alpha}}.$$

The coefficientwise limit satisfies

$$e^{-\rho}\,\mathrm{Ber}(\mathscr{D}_\infty) = D_5(2Z).$$

After pullback by $\jmath(Z) = Z/2$,

$$\jmath^*(e^{-\rho}\,\mathrm{Ber}(\mathscr{D}_\infty)) = D_5(Z), \qquad D_5(Z)^2 = \chi_{10}(Z).$$

Proof. The Berezinian of a diagonal endomorphism is the product of its eigenvalues with exponent equal to the superdimension of the eigenspace. Proper height makes every coefficient stationary. The denominator theorem supplies the Weyl line and the root-to-Siegel coordinate map. □

162.1 Trace-class realization on the root Hilbert superspace

Choose homogeneous orthonormal bases of the finite-dimensional root spaces and set

$$\mathcal{H}_\Delta = \widehat{\bigoplus}_{\alpha>0} \mathfrak{g}_{\Delta_5\alpha}$$

with its induced $\mathbb{Z}/2$ grading. Let $\mathfrak{C}_{\mathrm{abs}}$ be the subchamber on which

$$\sum_{\alpha>0} \dim \mathfrak{g}_{\Delta_5\alpha} \left|e^{-\alpha(Z)}\right| < \infty.$$

For $Z \in \mathfrak{C}_{\mathrm{abs}}$, define the trace-class diagonal operator

$$K_Z|_{\mathfrak{g}_{\Delta_5\alpha}} = e^{-\alpha(Z)}\,\mathrm{id}, \qquad D_Z = 1 - K_Z.$$

On each parity summand form the skew doubling

$$\mathbb{A}_{Z,\bar\epsilon} = \begin{pmatrix} 0 & D_{Z,\bar\epsilon} \\ -D_{Z,\bar\epsilon}^{\mathsf{t}} & 0 \end{pmatrix}.$$

Orient every finite-height truncation by the fixed root order and normalize its Pfaffian to 1 at $K_Z = 0$. The oriented Fredholm super-Pfaffian is the stabilized ratio

$$\mathrm{sPf}_F(\mathbb{A}_Z) = \frac{\mathrm{Pf}_F(\mathbb{A}_{Z,\bar{0}})}{\mathrm{Pf}_F(\mathbb{A}_{Z,\bar{1}})}.$$

Theorem 162.2 (Polarized Fredholm realization)**.** *Choose the protected parity refinement and the root-space Hilbert norms above. For every Z in the absolute-convergence chamber, the diagonal operator $K_Z|_{\mathfrak{g}_{\Delta_5\alpha}} = e^{-\alpha(Z)}\,\mathrm{id}$ is trace class on the even and odd root Hilbert spaces. For $D_Z = 1 - K_Z$,*

$$\mathrm{sdet}_F(D_Z) = \frac{\det_F(D_Z|_{\bar{0}})}{\det_F(D_Z|_{\bar{1}})} = \prod_{\alpha>0}(1 - e^{-\alpha(Z)})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta_5\alpha}}.$$

After choosing the standard polarization of $\mathcal{H}_\Delta \oplus \mathcal{H}_\Delta^*$ and its orientation, the paritywise skew block operators $\mathbb{A}_{Z,\bar{0}}$ and $\mathbb{A}_{Z,\bar{1}}$ have an oriented Fredholm super-Pfaffian

$$\mathrm{sPf}_F(\mathbb{A}_Z) = \frac{\mathrm{Pf}_F(\mathbb{A}_{Z,\bar{0}})}{\mathrm{Pf}_F(\mathbb{A}_{Z,\bar{1}})} = \mathrm{sdet}_F(D_Z).$$

Hence

$$e^{-\rho}\,\mathrm{sdet}_F(D_Z) = D_5(2Z).$$

Proof. The defining summability condition for $\mathfrak{C}_{\mathrm{abs}}$ is precisely the trace norm of the diagonal operator, summed in both parities. The Fredholm determinant of $1 - K_Z$ is the convergent product over eigenvalues; taking the even/odd ratio gives the superdeterminant. On each finite root-height truncation the Pfaffian of the oriented skew doubling equals the determinant of D_Z. Trace-norm convergence of the truncations passes this identity to the Fredholm limit. The final equality is the Gritsenko–Nikulin denominator formula. □

Remark 162.3. The polarization and orientation in this theorem belong to the root Hilbert space. A compact moduli-space orientation is a separate line bundle and is compared with this one by a specified intertwiner.

This constructs the square-root operator on the root Hilbert superspace where its stated summability condition holds. The nonemptiness estimate below concerns the minimal virtual superspace, whose dimensions are $|\mathrm{sdim}\,\mathfrak{g}_{\Delta_5\alpha}|$. Applying that estimate to the full root superspace requires a bound on both parity dimensions. A compact geometric realization requires an operator intertwiner and a compatible map of determinant lines.

Chapter 163

What the automorphic character determines

Theorem 163.1 (Denominator data and reconstruction data)**.** *The Borcherds product determines:*

(i) *the signed root multiplicity* $\operatorname{sdim} \mathfrak{g}_{\Delta_{\mathrm{s}}\alpha}$ *in every active root degree;*

(ii) *the Weyl vector and denominator character;*

(iii) *the divisor and automorphic multiplier of the section.*

A Lie bracket is supplied by the Borcherds–Kac–Moody presentation, parity dimensions by a parity-refined root datum, Hall multiplication by extension correspondences, and an oriented square root by a square-root line with monodromy. These reconstruction data refine the scalar product independently.

Proof. The first three items are read from the exponent, prefactor, and automorphy law. Independence is witnessed by explicit pairs: two brackets on the same graded vector space have the same PBW character; an even–odd pair has signed dimension zero; distinct Hall correspondences may have the same Euler product; and twisting a square-root line by a sign local system preserves its square. Hence each refined object is reconstructed from the additional datum named in the statement. □

Recognition Theorem 163.2 (Complete primitive recognition)**.** *Let* $\mathfrak{p}$ *be a positive graded Lie superalgebra and let* $\Upsilon : \mathfrak{n}_{\Delta_{\mathrm{s}},+} \twoheadrightarrow \mathfrak{p}$ *be a graded Lie map. The map is an isomorphism if either:*

(i) *the parity-refined Hilbert series of* $U(\mathfrak{n}_{\Delta_{\mathrm{s}},+})$ *and* $U(\mathfrak{p})$ *agree; or*

(ii) *each primitive degree has prescribed pure parity and the signed PBW characters agree.*

The induced height completions, Chevalley complexes, and current envelopes are then isomorphic.

Proof. Choose the first height containing a homogeneous kernel vector. All decomposable PBW terms in that height are determined by lower heights and agree. The primitive parity dimension therefore drops, contradicting the first hypothesis. In the second case the logarithm of the PBW product recovers signed primitive dimensions inductively, and prescribed parity recovers ordinary dimensions. Naturality of completion, Chevalley chains, and current envelopes gives the final assertion. □

Book XXXIV

The universal Borcherds Hall and three-Calabi–Yau models

Chapter 164

Finite simple-root windows

Let I_Δ be the multiset of simple roots of the Gritsenko–Nikulin Borcherds superalgebra. A repeated imaginary simple root is written once for each copy. The parity map is

$$p : I_\Delta \longrightarrow \mathbb{Z}/2.$$

Choose a proper height function

$$h : I_\Delta \longrightarrow \mathbb{Z}_{>0}$$

whose sublevel sets are finite. Put

$$I_H = \{ i \in I_\Delta : h(i) \le H \}.$$

The restriction of the invariant form gives a finite symmetric Borcherds–Cartan matrix

$$\begin{aligned} &A_H = (a_{ij})_{i,j \in I_H}, \\ &a_{ii} = 2 \qquad\qquad \text{for real } i, \\ &a_{ii} \le 0 \qquad\qquad \text{for imaginary } i, \\ &a_{ij} \le 0 \qquad\qquad \text{for } i \ne j. \end{aligned}$$

The evenness of the root lattice makes every imaginary diagonal entry even.

Construction 164.1 (The Igusa Borcherds quiver)**.** Order I_H. For $i < j$ place $-a_{ij}$ arrows from i to j. At an imaginary vertex i place

$$g_i = 1 - \frac{a_{ii}}{2}$$

loops. A real vertex has loop number zero. Denote the resulting quiver by $Q_{\Delta,H}$.

Proposition 164.2 (Euler form)**.** *The symmetrized Euler form of $Q_{\Delta,H}$ is A_H:*

$$\langle e_i, e_j \rangle + \langle e_j, e_i \rangle = a_{ij}.$$

Proof. For $i \ne j$, the total number of arrows between the two vertices is $-a_{ij}$, so the symmetrized Euler entry is a_{ij}. At i the value is $2 - 2g_i = a_{ii}$. □

The inclusions $I_H \subset I_{H+1}$ identify $Q_{\Delta,H}$ with a full subquiver of $Q_{\Delta,H+1}$. Representations supported on the old vertices form an exact extension-closed subcategory. This gives compatible Hall embeddings.

Chapter 165

The real-root seed window

The three real simple roots are

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3,$$

with Gram matrix

$$A_{\mathrm{re}} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

Choose the order $\delta_1 < \delta_2 < \delta_3$. The real-root quiver Q_{re} has two arrows from vertex i to vertex j for each $i < j$.

Proposition 165.1 (Real-root Hall seed)**.** *The symmetrized Euler form of Q_{re} is A_{re}. The Hall composition algebra generated by its three simple representations is the positive quantum Kac–Moody algebra with Cartan matrix A_{re}.*

Proof. The diagonal Euler entries are 2. The total number of arrows between distinct vertices is two, hence the symmetrized off-diagonal entries are -2. The Ringel–Green Hall theorem identifies the composition algebra with the positive quantum algebra defined by that symmetric Cartan matrix. □

The three vertices already contain the Weyl group and the reflective divisor. Imaginary vertices add the automorphic correction. The finite-height tower therefore begins with a completely explicit hyperbolic Hall algebra and enlarges it by one finite collection of imaginary simple roots at each stage.

165.1 The first Ginzburg differential

Let $a_{ij}^{(1)}, a_{ij}^{(2)} : i \to j$ be the two arrows for $i < j$. Add reverse arrows $(a_{ij}^{(r)})^* : j \to i$ of degree -1 and loops t_i of degree -2. The canonical three-Calabi–Yau completion has

$$d a_{ij}^{(r)} = 0, \qquad d(a_{ij}^{(r)})^* = 0,$$

and

$$d t_i = \sum_{a : s(a) = i} a a^* - \sum_{a : t(a) = i} a^* a. \tag{165.1}$$

The identity $d^2 = 0$ follows immediately. Formula (165.1) is the derived moment-map equation for the doubled real-root quiver.

Calculation 165.2 (Vertex one)**.** At vertex 1,

$$d t_1 = \sum_{r=1}^{2} \Big(a_{12}^{(r)} (a_{12}^{(r)})^* + a_{13}^{(r)} (a_{13}^{(r)})^* \Big).$$

At vertex 3,

$$dt_3 = -\sum_{r=1}^{2}\Big((a_{13}^{(r)})^* a_{13}^{(r)} + (a_{23}^{(r)})^* a_{23}^{(r)}\Big).$$

The sum over vertices is $\sum_a (aa^* - a^*a)$. It vanishes in the cyclic quotient and after matrix trace, but generally not in the path algebra.

Proof. The outgoing arrows at vertex 1 have targets 2 and 3. The incoming arrows at vertex 3 have sources 1 and 2. The displayed formulas follow from these incidence lists. Each arrow contributes one commutator to the sum over vertices. The trace of a commutator vanishes, giving the asserted trace identity. □

Chapter 166

Hall realization of the positive algebra

Let $\mathcal{A}_H = \mathrm{Rep}_{\mathbb{F}_q}(Q_{\Delta,H})$. Its twisted Hall product is

$$[M] * [N] = v^{\langle \dim M, \dim N\rangle} \sum_{[L]} |\{N' \subset L : N' \cong N,\ L/N' \cong M\}|[L], \qquad v^2 = q.$$

The class $[0]$ is the unit: the required subobject is unique on either side. Associativity follows by counting flags $N_1 \subset N_2 \subset L$ in two orders. The twist factors agree because the Euler form is bilinear:

$$\langle \alpha, \beta\rangle + \langle \alpha + \beta, \gamma\rangle = \langle \beta, \gamma\rangle + \langle \alpha, \beta + \gamma\rangle.$$

Thus this normalization defines the unital twisted Hall algebra on isomorphism classes.

Theorem 166.1 (Finite-window composition algebra)**.** *At each vertex i of $Q_{\Delta,H}$, choose the one-dimensional representation S_i supported at i, with every loop acting by zero. Use the generic composition algebra generated by these classes across finite fields, with the twisted subobject-count multiplication above. After extending coefficients from $\mathbb{Z}[v, v^{-1}]$ to $\mathbb{C}(v)$, this is the positive quantum generalized Kac–Moody algebra for the even Borcherds–Cartan matrix A_H. The parameter in Kang–Schiffmann is $v_{\mathrm{KS}} = v^{-1}$. Their specified semisimple perverse-sheaf construction on nilpotent representation varieties gives the corresponding canonical basis.*

Proof. This is the selected-simple composition construction of Kang–Schiffmann, Section 2.3 and Theorem 2.2, with one selected zero-loop simple at each vertex. Their geometric realization uses its split graded Grothendieck group over $\mathbb{Z}[v_{\mathrm{KS}}, v_{\mathrm{KS}}^{-1}]$ (Theorem 4.1), with the stated inversion of parameter [91]. The theorem does not use every simple object of the full representation category. A vertex with c loops has q^c one-dimensional simple representations over $\mathbb{F}_q$, whereas only the chosen zero-loop representation is the generator here. □

A semi-derived Hall realization of a full quantum group additionally specifies two-periodic complexes, the invertible acyclic classes and Cartan reduction [3, 113]. Its coproduct, pairing and completion are separate from the positive composition algebra used here.

166.1 Parity and the quantized superdatum

The Gritsenko–Nikulin denominator algebra carries a parity map on its simple-root multiset. It is additive on the free lattice of labelled simple generators. Projection to the Lorentzian degree can identify contributions of different parity. Both parity dimensions must therefore be retained in each projected root space. Hong's construction applied to this Borcherds superdatum gives the positive Borel

$$U_q^+(\mathfrak{g}_{\Delta_{\mathrm{s}}}{}^{(H)})$$

and the full quantized Borcherds superalgebra $U_q(\mathfrak{g}_{\Delta_{\mathrm{s}}}{}^{(H)})$.

Theorem 166.2 (Ordinary and super presentations)**.** *The ordinary quiver Hall presentation and Hong's quantum superalgebra use the same specified Cartan matrix and labelled simple generators. Their relation ideals and parity conventions are different inputs. Equality of their Cartan gradings alone does not identify their PBW characters.*

Proof. The two presentations are those of the preceding Hall construction and Hong's superalgebra construction. The difference already occurs for a single isotropic generator. The enveloping algebra of a one-dimensional even abelian Lie algebra is $\mathbb{C}[x]$. Declaring its generator odd only as a graded vector space leaves a nonzero vector x^2 in degree two. The enveloping algebra of a one-dimensional odd abelian Lie superalgebra is $\bigwedge(\xi)$, since its defining relation is $2\xi^2 = [\xi, \xi] = 0$. Its degree-two component is zero. Thus parity assignment to an ordinary PBW basis does not produce the super relation ideal or its character. A comparison must respect these relations degree by degree. □

Chapter 167

The three-Calabi–Yau completion

Let $kQ_{\Delta,H}$ be the path algebra. Its algebraic three-Calabi–Yau completion is

$$\Gamma_{\Delta,H} = \Pi_3(kQ_{\Delta,H}).$$

Write J for the ideal generated by all arrows in its graded quiver. Its completed version is the separate topological dg algebra $\widehat{\Gamma}_{\Delta,H} = \varprojlim_n \Gamma_{\Delta,H}/J^n$. The algebraic Calabi–Yau theorem is used for $\Gamma_{\Delta,H}$; a topological Calabi–Yau assertion for $\widehat{\Gamma}_{\Delta,H}$ requires its completed bimodule category. This is the Ginzburg model with zero potential: a degree-minus-one reverse arrow for every arrow and a degree-minus-two loop at each vertex. The loop differential is the signed sum of incident arrow–reverse-arrow products.

Theorem 167.1 (Borcherds–Ginzburg projections)**.** *For each finite quiver $Q_{\Delta,H}$, its undeformed three-Calabi–Yau completion $\Gamma_{\Delta,H}$ is the Ginzburg dg algebra with zero potential. If $H \le H'$ and $Q_{\Delta,H}$ is the full subquiver on a subset of vertices, there is a surjective dg algebra map*

$$\pi_{H'H} : \Gamma_{\Delta,H'} \longrightarrow \Gamma_{\Delta,H}.$$

It kills the added vertices, every arrow incident to them, the corresponding reverse arrows, and the added degree-minus-two loops. These maps compose strictly and extend continuously to the completed algebras.

Proof. The finite Calabi–Yau assertion is the Ginzburg model of Keller's completion theorem [68, 93]. We concatenate paths from left to right, so aa^* is based at the source of a. Original and reverse arrows have zero differential. For the degree-minus-two loop t_i, the differential is

$$dt_i = \sum_{s(a)=i} aa^* - \sum_{t(a)=i} a^*a.$$

The projection deletes exactly the terms involving an added vertex, leaving the same sum for the full subquiver. It therefore commutes with the differential. It preserves path length and extends to the completed path algebra. Successive projections kill the union of their omitted vertices and arrows, proving strict composition. □

The evident inclusion of graded quivers does not give the reverse dg map. A single old vertex has loop differential zero. Adding an arrow $a : i \to j$ changes that differential to aa^* at i. The map fixing the old loop therefore fails to commute with the differential.

Write $\Gamma_\Delta = (\Gamma_{\Delta,H}, \pi_{H'H})_H$ for this pro-system of dg algebras. It is separate from the ordinary Hall embeddings obtained by extension by zero on representations of a full subquiver. A direct system of Calabi–Yau dg categories compatible with those Hall embeddings requires additional dg functors. It does not follow from the inclusion of the displayed Ginzburg generators.

The quantum group retains its separately specified root-height completion. Its transition maps must preserve the Cartan generators, coproduct, and pairing. Quotienting the positive nilpotent Lie algebra by height and truncating a full quantum group are different operations.

Chapter 168

Hall states and protected superstates

The representation varieties of $Q_{\Delta,H}$ carry the perverse-sheaf complexes used in the composition-algebra construction. The full equivariant derived category is a larger carrier. Already for one real vertex and dimension one, its quotient is $B\mathbb{G}_m$, with $H^*(B\mathbb{G}_m;\mathbb{Q}) = \mathbb{Q}[c_1]$, where $|c_1| = 2$. Its total equivariant cohomology is infinite-dimensional. Finite root multiplicities therefore refer to the specified composition algebra, not to total equivariant cohomology of this stack.

Definition 168.1 (Igusa Hall category)**.** For every dimension vector $\mathbf{d}$, let

$$\mathfrak{R}_{H,\mathbf{d}} = [\mathrm{Rep}_{\mathbf{d}}(Q_{\Delta,H})/G_{\mathbf{d}}].$$

The finite-height Hall category is the direct sum of the equivariant constructible derived categories on the $\mathfrak{R}_{H,\mathbf{d}}$, equipped with Hall convolution along the stack of short exact sequences. Denote it by $\mathcal{B}_{\Delta,H}$ and set

$$\mathcal{B}_{\Delta} = \varinjlim_{H} \mathcal{B}_{\Delta,H}.$$

Independently, the protected one-particle superspace is the root-graded superspace

$$\mathcal{H}_{\mathrm{ip},H} = \bigoplus_{\alpha>0} \mathfrak{g}_{\Delta_5,\alpha}^{(H)},$$

with the parity of the Borcherds superalgebra. Its multiparticle completion is the PBW/Fock superspace of $U(\mathfrak{n}_{\Delta_5,+}^{(H)})$.

Theorem 168.2 (Hall and protected-state comparison)**.** *The composition subalgebra in the finite-window Hall theorem is the ordinary positive quantum generalized Kac–Moody algebra. This statement does not identify the Grothendieck group of the full category $\mathcal{B}_{\Delta,H}$ with that composition algebra. The protected one-particle superspace has signed root multiplicities*

$$\operatorname{sdim} \mathcal{H}_{\mathrm{ip},H,\alpha} = \operatorname{sdim} \mathfrak{g}_{\Delta_5,\alpha}^{(H)}.$$

Its height-complete PBW character is

$$\operatorname{sch}\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+}) = \prod_{\alpha>0} (1-e^{-\alpha})^{-\operatorname{sdim}\mathfrak{g}_{\Delta_5,\alpha}},$$

and its Weyl-shifted inverse is $D_5(2Z)$. The composition canonical basis and the protected root presentation use the specified simple-root indexing and Cartan grading; the parity refinement belongs to the Borcherds superdatum.

Proof. The composition-algebra assertion is the finite-window theorem. The protected assertion is PBW for the Borcherds Lie superalgebra. Proper height passes both constructions to the limit. The Gritsenko–Nikulin denominator theorem gives the final product identity. □

Chapter 169

Extension determinants and critical orientations

Let C be a three-Calabi–Yau dg category over $\mathbb{C}$, with its chosen trace. For families of objects A, B over a base T, let $C(A,B)$ denote the relative derived Hom complex on T. Suppose these complexes are perfect, and that the trace gives base-change-compatible duality

$$C(B,A) \simeq C(A,B)^{\vee}[-3].$$

Put $L(A,B) = \det C(A,B)$ and $D(A) = L(A,A)$. The determinant convention is $\det C = \bigotimes_i (\det C^i)^{(-1)^i}$, with symmetry sign specified by $\chi(C) \bmod 2$.

Theorem 169.1 (Extension determinant)**.** *For an exact triangle $A \to F \to B \to A[1]$, there is a base-change-compatible isomorphism*

$$D(F) \simeq D(A) \otimes D(B) \otimes L(A,B)^{\otimes 2}.$$

The determinant comparisons obtained by refining a flag agree, with the stated determinant symmetry signs.

Proof. Apply derived Hom to the triangle in each variable. Determinant additivity gives

$$D(F) \simeq L(B,A) \otimes D(A) \otimes D(B) \otimes L(A,B).$$

Dualization and the odd shift each invert the determinant line. Thus Calabi–Yau duality identifies $L(B,A)$ with $L(A,B)$. Reorder by determinant symmetry to obtain the formula. For a full flag, all iterated decompositions identify with the same ordered tensor product of $C(A_i, A_j)$, for its successive quotients A_i. Determinant additivity for refined filtrations proves coherence. Each operation commutes with base change. □

A square-root orientation consists of lines o_A and isomorphisms $o_A^{\otimes 2} \simeq D(A)$. Extension compatibility requires maps

$$o_A \otimes o_B \otimes L(A,B) \longrightarrow o_F$$

whose squares are the determinant comparisons. Choosing coordinates on a smooth critical chart trivializes its ambient canonical line. Hall convolution requires transition maps compatible with the cross line and the zero object. On three-step flags, the two orientation maps can differ by an element of μ_2, even when their squares agree. That sign must be trivialized coherently.

Example 169.2 (A nontrivial cross line)**.** Let X be a smooth projective Calabi–Yau threefold, with a chosen volume form, and let $x \in X$. On $X \times T$, where $T = \mathbb{P}^1$, take

$$A = \mathcal{O}_X \boxtimes \mathcal{O}_T(1), \qquad B = \mathcal{O}_x \boxtimes \mathcal{O}_T.$$

Then $C(A,B) = \mathcal{O}_T(-1)[0]$, and relative Serre duality gives $C(B,A) = \mathcal{O}_T(1)[-3]$. Thus $L(A,B) = L(B,A) = \mathcal{O}_T(-1)$. The self-Hom complexes of A and B have constant determinant lines, since the factors $\mathcal{O}_T(1)$ cancel in the first self-Hom. Nevertheless

$$D(A \oplus B) = D(A) \otimes D(B) \otimes \mathcal{O}_T(-2).$$

The orientation cross factor is $\mathcal{O}_T(-1)$. A sign change cannot replace this line of nonzero degree.

Corollary 169.3 (Oriented critical coefficients)**.** *Suppose the chosen critical charts have square-root orientation maps on overlaps and extension flags, with unit and flag coherences. Suppose their vanishing-cycle comparison maps preserve these data. Then the local vanishing-cycle complexes descend to the specified oriented coefficient object. Their Hall convolution is associative whenever its pullback and pushforward maps are defined and satisfy base change on the flag diagrams.*

Proof. The overlap cocycle identifies the local coefficient complexes by descent. On a three-step flag, orientation coherence identifies the two Thom–Sebastiani coefficient maps. Proper or compact-support base change, on the stipulated domain, identifies the two composites of convolution. The zero flag gives the unit equation. □

For the quiver charts, the remaining construction is an equivariant system of square roots and coefficient maps on the actual representation and extension stacks, compatible with changes of height. For compact $S \times E$, the reduced complexes and the common elliptic torsor must also be retained. A coordinate volume on an affine chart specifies neither descent system by itself.

Chapter 170

The colored elliptic Ran space

The elliptic degree is a global charge carried by the whole configuration. Define

$$\mathrm{Ran}^{\mathrm{wr}}(E) = \coprod_{m \geq 0} \mathrm{Ran}(E)_m$$

with monoidal operation

$$(I, m) \sqcup (J, n) = (I \sqcup J, m + n)$$

on disjoint supports. The component $m = 0$ is point-local. The components $m > 0$ are wrapped.

Let

$$\mathfrak{n}_{\Delta_s,+} = \bigoplus_{m \geq 0} \mathfrak{n}_{\Delta_s,+}[m]$$

be the elliptic-degree grading. The root bracket obeys

$$[\mathfrak{n}_{\Delta_s,+}[m], \mathfrak{n}_{\Delta_s,+}[n]] \subset \mathfrak{n}_{\Delta_s,+}[m + n].$$

Construction 170.1 (Hybrid factorization algebra)**.** Form the chiral Chevalley factorization coalgebra

$$\mathcal{F}^{\mathrm{hyb}}_{\Delta,E} = C^{\mathrm{ch}}_*\Big(\mathrm{Cur}_E(\mathfrak{n}_{\Delta_s,+})\Big).$$

Retain the elliptic-degree grading on every Chevalley word and place its degree-m summand on $\mathrm{Ran}(E)_m$. Disjoint union adds the degree labels.

Theorem 170.2 (Local–wrapped factorization)**.** *$\mathcal{F}^{\mathrm{hyb}}_{\Delta,E}$ is a graded factorization coalgebra on $\mathrm{Ran}^{\mathrm{wr}}(E)$. Its degree-zero color is the current theory of $\mathfrak{n}_{\Delta_s,+}[0]$. Every degree-$m$ color is a module for the degree-zero color. The factorization product of degrees m and n lands in degree $m + n$. Its chiral primitives are $\mathrm{Cur}_E(\mathfrak{n}_{\Delta_s,+})$, and constant modes recover $\mathfrak{n}_{\Delta_s,+}$.*

Proof. The chiral Chevalley construction is a factorization coalgebra for every chiral Lie algebra. The bracket preserves elliptic-degree addition, so its differential and coproduct preserve the graded Ran placement. The primitive theorem for the Chevalley coalgebra recovers the current Lie algebra. □

This construction retains the nonnegative root-degree label on a point-supported current theory. It does not replace a sheaf projecting onto the whole elliptic curve by a point-supported sheaf. The compact wrapped comparison must map its common-torsor extension correspondences to these graded current operations, preserving support and charge. The bracket on the present carrier is the Borcherds bracket.

Chapter 171

Physical charge and Gram degree

Let Γ_{phys} be an additive physical charge lattice and let

$$\Pi(c) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right)$$

be the Gram map. Its polarization is

$$B(c,d) = \Pi(c+d) - \Pi(c) - \Pi(d).$$

Define the central extension

$$\widetilde{\Gamma}_{\Pi} = \Gamma_{\mathrm{phys}} \times \Gamma_{\mathrm{ind}}$$

with multiplication

$$(c,\lambda)(d,\mu) = (c+d, \lambda+\mu+B(c,d)).$$

Theorem 171.1 (Graph homomorphism)**.** *The map*

$$j_{\Pi} : \Gamma_{\mathrm{phys}} \longrightarrow \widetilde{\Gamma}_{\Pi}, \qquad c \longmapsto (c, \Pi(c))$$

is a group homomorphism. For every Γ_{phys}-graded algebra A, the Rees lift

$$a_c \longmapsto a_c u^{\Pi(c)}$$

turns multiplication into an additive $\widetilde{\Gamma}_{\Pi}$-grading with cocycle $u^{-B(c,d)}$.

Proof. The defining multiplication gives

$$j_{\Pi}(c)\, j_{\Pi}(d) = (c+d, \Pi(c)+\Pi(d)+B(c,d)) = j_{\Pi}(c+d).$$

The algebra identity follows immediately. □

The universal Igusa category is graded directly by the Lorentzian root lattice. A geometric $K3 \times E$ theory enters through a charge functor to $\widetilde{\Gamma}_{\Pi}$.

Chapter 172

The universal protected field theory

For each H take cotangent BF theory with gauge Lie superalgebra ${\mathfrak{n}_{\Delta_s,+}}^{(H)}$ on $\mathbb{R}^2_{\mathrm{top}} \times E$. Its fields are

$$\Omega^\bullet(\mathbb{R}^2) \mathbin{\widehat{\otimes}} \Omega^{0,\bullet}(E) \mathbin{\widehat{\otimes}} \left({\mathfrak{n}_{\Delta_s,+}}^{(H)}[1] \oplus ({\mathfrak{n}_{\Delta_s,+}}^{(H)})^\vee[2] \right).$$

The action is

$$S_H = \int_{\mathbb{R}^2 \times E} \langle B, (\mathrm{d} + \bar{\partial}) A + \frac{1}{2}[A, A] \rangle.$$

Theorem 172.1 (Classical local theory and exact quantum harmonic sector)**.** *For every finite height H the shifted-cotangent action defines a classical mixed holomorphic–topological factorization algebra on $\mathbb{R}^2_{\mathrm{top}} \times E$. Harmonic reduction on a compact product cell gives the finite-dimensional cotangent BV theory of ${\mathfrak{n}_{\Delta_s,+}}^{(H)}$, and this reduced theory satisfies the quantum master equation. Complementary Lagrangian boundary polarizations give the boundary enveloping algebra $U({\mathfrak{n}_{\Delta_s,+}}^{(H)})$, the two-sided bar complex $C_*({\mathfrak{n}_{\Delta_s,+}}^{(H)})$, and the current envelope on E. The proper-height inverse system is coefficientwise finite and defines a complete classical local theory together with an exact quantum harmonic sector.*

Proof. The invariant cotangent pairing and the Jacobi identity give the classical master equation and factorization descent. Harmonic reduction leaves a finite positive nilpotent Lie superalgebra. Every even adjoint endomorphism raises proper height and is strictly upper triangular; every odd adjoint endomorphism has vanishing supertrace. The reduced BV divergence therefore vanishes. Canonical quantization in the selected polarization gives the enveloping algebra, and the augmentation boundary states give the two-sided bar, identified with Chevalley chains by PBW. Each map preserves height, so the inverse system is defined coefficientwise. □

Chapter 173

Absolute convergence of the virtual root operator

For each positive root α, let V_α be the minimal parity realization of the signed multiplicity: it is purely even of dimension $\operatorname{sdim} \mathfrak{g}_{\Delta,\alpha}$ when this integer is nonnegative and purely odd of dimension $-\operatorname{sdim} \mathfrak{g}_{\Delta,\alpha}$ when it is negative. Set

$$\mathcal{V}_\Delta = \widehat{\bigoplus}_{\alpha>0} V_\alpha .$$

This virtual root superspace is the canonical linear carrier of the denominator character; the Lie bracket remains carried by $\mathfrak{g}_{\Delta,\mathrm{s}}$.

Let $h(\alpha)$ be the proper height. For the simple-root height of (17.14), the theta-product bound of Lemma 17.8 gives $C_0 = 12$ and $C_1 = 2\pi\sqrt{2/3}$ in

$$\dim V_\alpha = \left|\operatorname{sdim} \mathfrak{g}_{\Delta,\alpha}\right| \le C_0 \exp(C_1 h(\alpha)) \tag{173.1}$$

for every positive root. The number of Gram triples of height H is polynomial in H.

Theorem 173.1 (Nonempty trace-class chamber)**.** *Let Z_0 lie in the interior of the positive Weyl chamber. For sufficiently large $t > 0$,*

$$\sum_{\alpha>0} \dim V_\alpha \left| e^{-\alpha(tZ_0)} \right| < \infty .$$

Hence the absolute-convergence chamber contains a conical tail in every interior direction.

Proof. Interior positivity gives a constant $c(Z_0) > 0$ with $\Re\alpha(Z_0) \ge c(Z_0)h(\alpha)$. Combine this with (173.1). The contribution of roots of height H is bounded by a polynomial in H times

$$\exp\bigl((C_1 - t c(Z_0))H\bigr).$$

The series converges when $t c(Z_0) > C_1$. □

Corollary 173.2 (Trace-class root Hamiltonian)**.** *On the conical tail, the diagonal operator*

$$K_Z\big|_{V_\alpha} = e^{-\alpha(Z)}\,\mathrm{id}$$

is trace class on both parity summands. The Fredholm product is holomorphic and equals the absolutely convergent Borcherds product.

Proof. The trace norm is the summable series of the preceding theorem, separated by parity. Standard trace-class determinant theory gives holomorphy, and diagonalization gives the product over root eigenvalues. □

Chapter 174

The virtual root Fredholm operator

Let

$$\mathcal{H}_{\Delta}^{\mathrm{vir}} = \widehat{\bigoplus}_{\alpha>0} V_{\alpha}$$

with an orthonormal basis in each minimal parity realization. This is the virtual root Hilbert space used in the preceding absolute-convergence theorem. In the absolute-convergence chamber define

$$K_Z\big|_{V_\alpha} = e^{-\alpha(Z)}\,\mathrm{id}, \qquad D_Z = 1 - K_Z.$$

Theorem 174.1 (Oriented Fredholm square root)**.** *The operator K_Z is trace class on the even and odd summands of the virtual root Hilbert space. Its Fredholm Berezinian is*

$$\mathrm{sdet}_F(D_Z) = \prod_{\alpha>0} (1 - e^{-\alpha(Z)})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta_5\alpha}}.$$

Choose an order of the positive roots, first by height and then lexicographically. This choice orients every finite-height root space. The skew doubling

$$\mathbb{A}_Z = \begin{pmatrix} 0 & D_Z \\ -D_Z^{\mathrm{t}} & 0 \end{pmatrix}$$

has an oriented Fredholm super-Pfaffian satisfying

$$\mathrm{sPf}_F(\mathbb{A}_Z) = \mathrm{sdet}_F(D_Z).$$

Consequently

$$e^{-\rho}\,\mathrm{sPf}_F(\mathbb{A}_Z) = D_5(2Z).$$

Proof. Absolute convergence of the Borcherds product is the trace-norm summability condition. The determinant formula follows from the spectrum. At finite height the Pfaffian of the oriented skew doubling is the determinant. Trace-norm convergence passes the identity to the limit. The denominator theorem supplies the Weyl shift. □

174.1 Automorphic orientation

Let $\mathcal{L}$ be the determinant of the Hodge bundle. The Weyl-normalized product is a section of $\mathcal{L}^5 \otimes \nu_5$, as in (17.19). In the chosen cusp frame it is represented by the Fredholm product. Scalar analytic continuation gives its weight-five automorphy law and quadratic character. To glue the operator families themselves requires transition intertwiners on their Hilbert spaces and determinant lines.

Corollary 174.2 (The full automorphic line)**.** *The square of the automorphic line is $\mathcal{L}^{10}$. Its flat character factor is ν_5 of order two. The root denominator is anti-invariant under each real simple reflection.*

Proof. Squaring the weight-five transformation law removes ν_5 and gives the weight-ten Hodge line. Anti-invariance is the sign in the Weyl denominator identity. These facts retain the Hodge factor as well as the character; they do not identify a geometric operator orientation with the flat character alone. □

Chapter 175

The universal Igusa protected effective theory

Definition 175.1 (Universal Igusa completion)**.** The protected completion selected by the K3 half-index is the quintuple

$$\mathfrak{K}_{\Delta} = \left(\Gamma_{\Delta}, \mathcal{B}_{\Delta}, \mathcal{H}_{\mathrm{ip}}, \mathcal{F}^{\mathrm{hyb}}_{\Delta,E}, \mathbb{A}_{Z}\right).$$

Its components are the pro-system of finite three-Calabi–Yau completions, the ordinary quiver Hall category, the Borcherds one-particle Lie superalgebra, the local–wrapped current factorization algebra, and the bounded virtual-root Fredholm family.

Theorem 175.2 (Igusa protected completion)**.** *The quintuple $\mathfrak{K}_{\Delta}$ satisfies:*

(i) *its one-particle Lie superalgebra is $\mathfrak{g}_{\Delta_5}$;*

(ii) *its ordinary Hall presentation and Hong's quantum supergroup retain the stated Cartan grading; comparison of their relation ideals and parity-refined PBW spaces requires an additional map;*

(iii) *its chiral primitive algebra on E is the local–wrapped current algebra of $\mathfrak{g}_{\Delta_5}$;*

(iv) *its raw multiparticle supercharacter is P^{-1}, where $e^{-\rho} P = \mathrm{Den}_{\Delta}$; equivalently,*

$$e^{\rho}\,\mathrm{sch}\,\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+}) = \mathrm{Den}_{\Delta}^{-1};$$

(v) *the half-coordinate pullback of the doubled Weyl-normalized character is $-{\chi_{10}}^{-1}$ in the Oberdieck–Pixton normalization;*

(vi) *its oriented Fredholm product, with the Weyl factor, is Den_{Δ}; after the stated coordinate pullback its automorphic line is $\mathcal{L}^5 \otimes \nu_5$.*

Proof. The first three statements are the Borcherds root, quiver Hall, Hong quantum-supergroup, Ginzburg, and colored-Ran theorems. PBW gives the fourth. The identity $\chi_{10} = {D_5}^2$ and the established reduced scalar theorem give the fifth. The Fredholm theorem gives the sixth. □

The tuple retains algebraic constructions associated with the Igusa presentation and their scalar characters. A geometric type-IIA or topological-string realization must construct compatible functors from its oriented extension category to these targets. Primitive states and extension amplitudes are inputs to those functors. Their action on morphisms, convolution, orientations, and chiral operations must also be specified and proved compatible.

Chapter 176

The primitive comparison and its extensions

Let $\mathfrak{p}$ be a positive-root-graded Lie superalgebra with finite-dimensional homogeneous pieces and a proper positive height. A primitive realization of the Igusa presentation is a parity-preserving homogeneous map from its labelled simple-root spaces to $\mathfrak{p}$, satisfying the defining relations.

Theorem 176.1 (Universal presentation map)**.** *Let $\mathfrak{p}$ be a positive-root-graded Lie superalgebra over $\mathbb{C}$, with finite-dimensional homogeneous pieces. Suppose a homogeneous parity-preserving map from every simple-root generating space of $\mathfrak{n}_{\Delta_5,+}$ to $\mathfrak{p}$ satisfies the defining Borcherds–Serre relations. It extends uniquely to a graded Lie-superalgebra map*

$$f : \mathfrak{n}_{\Delta_5,+} \longrightarrow \mathfrak{p}.$$

It induces continuous maps on height completions, enveloping algebras, Chevalley chains, and level-zero currents. The map f is an isomorphism if and only if it is surjective and the root-graded PBW series agree in each parity.

Proof. Extend the chosen maps to the free Lie superalgebra on the simple-root spaces. The relation hypothesis makes the defining ideal lie in its kernel, giving f. The generators prove uniqueness. On enveloping algebras the map sends $x_1 \cdots x_j$ to $f(x_1) \cdots f(x_j)$. On Chevalley chains it is the graded symmetric map on the shifted generators. It commutes with the Chevalley differential because f preserves brackets. The current map is $\partial^j x \mapsto \partial^j f(x)$, preserving $[x_\lambda y] = [x, y]$. Root degree preservation gives compatible maps at each height and hence on the stated limits.

If f is surjective, $U(f)$ is surjective. Proper positive height makes each root-graded PBW component finite-dimensional. Equality of its even and odd dimensions makes $U(f)$ bijective in every component. PBW embeds each Lie superalgebra into its enveloping algebra, so f is injective. The converse follows by applying the constructions to its inverse. □

A geometric comparison also requires functors on the actual Hall and critical categories, a compatible coproduct and Hopf pairing, and maps of determinant lines and operator domains. These categories and lines are not the domain of the primitive Lie map. The additional maps must be constructed and checked against that map. They are not consequences of its universal property.

Chapter 177

The eight diagonal-divisor forms

Cléry and Gritsenko classify eight genus-two modular forms with a simple diagonal divisor [28]. Their cusp contributions and multipliers are specified in Theorem 17.7. Each product defines a minimal virtual graded superspace and a formal Fock determinant. A trace-class realization additionally requires summability of the absolute multiplicities. A separately supplied reflective denominator presentation provides its bracket and Weyl action; examples are studied in [70–72, 74–76]. Half-integral weights retain their multiplier covers.

form	group	weight	character order
Δ_5	$\mathrm{Sp}_4(\mathbb{Z})$	5	2
Δ_2	Γ_2	2	4
Δ_1	Γ_3	1	6
$\Delta_{1/2}$	Γ_4	1/2	8
∇_3	$\Gamma_0^{(2)}(2)$	3	2
∇_2	$\Gamma_0^{(2)}(3)$	2	2
$\nabla_{3/2}$	$\Gamma_0^{(2)}(4)$	3/2	4
Q_1	$\Gamma_2(2)$	1	4

Theorem 177.1 (Formal product realizations and summability)**.** *Each locally finite, positive-height exponent system defines a graded virtual superspace, its super-Fock character and its formal determinant. If the absolute multiplicities satisfy an exponential height bound, and the number of degrees at each height is polynomially bounded, an interior conical tail gives a trace-class diagonal realization of the determinant. A reflective Borcherds presentation is additional algebraic input. Its ordinary quiver Hall algebra, Calabi–Yau completion and parity-refined quantum supergroup retain their respective definitions and comparison requirements.*

Proof. Local finiteness and positive height make every formal coefficient a finite sum. PBW for the corresponding abelian superspace gives the super-Fock character and the inverse determinant. Under the growth hypotheses, the trace norm is bounded by a polynomial times a decaying exponential, exactly as in Proposition 17.9. A bracket is not determined by this character. The further algebraic constructions apply only when their presentation, parity and coefficient data have been specified; equality with an ordinary Hall presentation requires its comparison map. □

Chapter 178

Algebraic and physical closure

Theorem 178.1 (Universal Igusa realization theorem)**.** *The constructions have the following inputs and outputs. An arrow labelled by a construction records its dependence on that input:*

$$\phi_{0,1} \xmapsto{\text{Borch}} D_5, \qquad \phi_{0,1} \xmapsto{\text{root datum}} \mathfrak{g}_{\Delta_5}, \qquad \mathcal{A}_\Delta \xmapsto{\text{quiver}} Q_{\Delta,H} \xmapsto{\text{Hall}} \mathcal{B}_\Delta,$$

$$kQ_{\Delta,H} \xmapsto{\Pi_3} \Gamma_{\Delta,H}, \qquad \mathfrak{g}_{\Delta_5} \xmapsto{\mathrm{Cur}_E} \mathcal{F}^{\mathrm{hyb}}_{\Delta,E}, \qquad \mathfrak{g}_{\Delta_5} \xmapsto{\text{Hong}} \widehat{U}_q(\mathfrak{g}_{\Delta_5}),$$

$$\mathfrak{g}_{\Delta_5} \xmapsto{\text{root spectrum}} \mathbb{A}_Z, \qquad e^{-\rho}\,\mathrm{sPf}_F(\mathbb{A}_Z) = \mathrm{Den}_\Delta(Z) = D_5(2Z),$$

$$\mathcal{Z}^{\mathrm{red}}_{K3\times E} = -(j^*\,\mathrm{Den}_\Delta)^{-2} = -\chi_{10}{}^{-1}.$$

The system contains parity-refined one-particle spaces, an ordinary Hall extension product, the Borcherds root bracket, a local–wrapped collision law, a quantum supergroup, an automorphic line, a bounded Fredholm determinant, and the reduced scalar square.

Proof. The Borcherds lift, Gritsenko–Nikulin presentation, and primitive reduced curve-counting theorem give the scalar identities. The enveloping and Chevalley maps are induced by the presentation. The quiver Hall algebra and finite Calabi–Yau completion have their respective quiver and path-algebra inputs. Their common grading specifies the degrees a comparison must preserve. A map between those constructions requires the additional functors, coefficient maps, and pairings described above. □

Book XXXV

Technical appendices to the preceding book

Chapter 179

Exact theta-series calculation

Set $t = q^{1/8}$ and $x = r^{1/2}$. Then

$$\vartheta_2(\tau, z) = \sum_{n \in \mathbb{Z}} t^{4n^2+4n+1} x^{2n+1},$$
$$\vartheta_3(\tau, z) = \sum_{n \in \mathbb{Z}} t^{4n^2} x^{2n},$$
$$\vartheta_4(\tau, z) = \sum_{n \in \mathbb{Z}} (-1)^n t^{4n^2} x^{2n}.$$

Squaring numerator and denominator, dividing in $\mathbb{Q}[x^{\pm 1}][[t]]$, and summing the three quotients gives the coefficient table in Chapter 1. The calculation uses integer convolution and formal-series inversion; fractional powers cancel before conversion from (t, x) to (q, r).

Chapter 180

The root reflection matrices

In the basis (f_2, f_3, f_{-2}), the form is

$$G = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 0 \end{pmatrix}.$$

The three reflection matrices are

$$S_1 = \begin{pmatrix} 1 & 4 & 4 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{pmatrix}, \quad S_2 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & -1 & 0 \\ 4 & 4 & 1 \end{pmatrix}, \quad S_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Direct multiplication gives

$$S_i^2 = I, \qquad S_i^{\mathsf{T}} G S_i = G.$$

Chapter 181

Automorphic, homological, and Hall comparisons

The half-index determinant, theta normalization, Borcherds denominator, reduced scalar theorem, charge Gram map, and geometric Hall comparison datum belong to the following constructions:

(i) the automorphic square root is completed entirely within the Borcherds product theorem;

(ii) the BKM bracket comes from the Gritsenko–Nikulin algebra;

(iii) the bar–Chevalley comparison gives the denominator a homological source;

(iv) the current and BF constructions give an exact first-order algebraic model;

(v) orientation data refine determinant lines, while an operator Pfaffian requires an actual Fredholm family;

(vi) Weyl reflections are attached to real roots, and imaginary roots enter through multiplicities;

(vii) the compact Hall comparison uses additive Mukai charge, the polarized central extension, and hybrid local–wrapped sectors.

Coherent boundary actions and BV fields

Boundary operations determine a deformation problem once their interface and coherence maps are supplied. A local realization gives a shifted cotangent BV theory. Comparing it with a physical bulk requires a map of fields, actions, and coherent boundary operations.

Book XXXVI

The full mixed boundary datum

Chapter 182

Three products and one interface

Holomorphic collisions and transverse composition act on different coefficient objects. A common boundary theory must specify their interaction on an interface. This interaction is part of the input to the following constructions.

Let X be a smooth complex curve and let $\mathcal{D}_X = D\text{-}\operatorname{Mod}(\operatorname{Ran} X)$ in a fixed six-functor enhancement. Write $\otimes^!$ for the pointwise product and $\star^{\mathrm{ch}}$ for chiral convolution.

Definition 182.1 (Complete mixed boundary datum)**.** A complete mixed boundary datum on X is a tuple

$$\mathbb{A} = (L, A^{\mathrm{ch}}, A^{\perp}, K, \iota, \lambda)$$

with the following components.

(i) L is a complete chiral Lie algebra.

(ii) A^{ch} is a complete associative algebra for $\star^{\mathrm{ch}}$.

(iii) $A^{\perp}$ is a complete E_1 algebra in $(\operatorname{Fact}_D(X), \otimes^!)$.

(iv) K is an $(A^{\mathrm{ch}}, A^{\perp})$ interface bimodule.

(v) $\iota : L \to (A^{\mathrm{ch}})_{\mathsf{Lie}}$ is a chiral Lie morphism.

(vi) λ is the aligned mixed distributive law relating the two module actions on K.

Every structure map is continuous for the chosen weight filtrations. The tuple includes the module actions, the mixed coherence homotopies, and their identities in the chosen enhancement. When a reduced bar is used, the relevant augmentations are supplied as additional data. Existence of this tuple for a physical boundary theory is a separate construction.

The three native bars are

$$\operatorname{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(L), \qquad \operatorname{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A^{\mathrm{ch}}), \qquad \operatorname{Bar}^{\perp}(A^{\perp}).$$

The interface has a fourth bar. Its degree-r terms are alternating words in the two associative colours with one marked interface letter. Its differential contracts adjacent compatible letters.

Construction 182.2 (Interface bar)**.** Let $\overline{A}^{\mathrm{ch}}$ and $\overline{A}^{\perp}$ be the augmentation ideals. Set

$$\operatorname{Bar}^{\mathrm{mix}}(\mathbb{A}; K) = \bigoplus_{p,q \geq 0} (s\overline{A}^{\mathrm{ch}})^{\otimes p} \otimes sK \otimes (s\overline{A}^{\perp})^{\otimes q}.$$

The differential is the sum of the internal differential, the two bar differentials, the left action on K, the right action on K, and the mixed terms supplied by λ.

Theorem 182.3 (Four-colour differential)**.** *The interface differential squares to zero. Its codimension-two terms are exactly the associative relations in the two colours, the two module relations, and the mixed distributive relation.*

Proof. Filter by total word length. Every component of the square contracts two adjacent edges. Disjoint contractions cancel by the determinant-line sign. Nested contractions in one colour cancel by associativity. A contraction meeting the interface cancels by the corresponding module axiom. The remaining square has one edge of each colour and is the aligned distributive square. The datum λ identifies its two composites. □

Chapter 183

Compatible modules and the mixed centre

An operation on a boundary theory must act on boundary-changing morphisms as well as on objects. Natural transformations retain this compatibility, and derived natural transformations retain its higher homotopies.

A compatible module is a triple $(M^{\mathrm{ch}}, M^{\perp}, \kappa)$ with an A^{ch} action, an $A^{\perp}$ action, and an interface comparison over K. Let $\mathrm{Mod}_{\mathbb{A}}^{\mathrm{mix}}$ be their stable category.

Definition 183.1 (Mixed derived centre)**.** The mixed derived centre is

$$Z_{\mathrm{mix}}(\mathbb{A}) = \mathrm{End}\Big(\mathrm{id}_{\mathrm{Mod}_{\mathbb{A}}^{\mathrm{mix}}}\Big).$$

It is computed by the endomorphisms of the diagonal compatible bimodule.

Theorem 183.2 (Universal interior action)**.** *$Z_{\mathrm{mix}}(\mathbb{A})$ carries an internal E_2 algebra structure. Every action already defined as a coherent natural endomorphism of the identity determines a point of this centre. The space of its factorizations through the representing endomorphism object is contractible. Morita-equivalent complete mixed data have equivalent centres.*

Proof. Endomorphisms of the identity functor form the higher centre of a module category. Composition and insertion of natural transformations give the brace operations and hence the E_2 structure. A supplied coherent natural action defines an endomorphism of the identity. A physical bulk insertion determines such an action only after its action on modules, morphisms, and higher compositions has been constructed. The factorization is the universal property of the endomorphism object. Morita equivalence identifies the module categories and their identity functors. □

Chapter 184

Relative induction from a chiral algebra

Let V be a chiral algebra with conformal vacuum. Its algebraic collar $\mathrm{Coll}(V)$ is the free transverse E_1 extension generated by one boundary copy of V, modulo the chiral relations already present in V. Weight completion gives $\widehat{\mathrm{Coll}}(V)$.

Theorem 184.1 (Universal collar)**.** *For every complete transversely ordered chiral algebra B and every chiral morphism $V \to B$, there is a unique continuous E_1 morphism*

$$\widehat{\mathrm{Coll}}(V) \longrightarrow B$$

extending the chiral map. The construction is left adjoint to restriction from transverse ordered algebras to chiral coefficients.

Proof. The algebraic collar is the relative operadic induction from the chiral operad to the mixed operad. Relative induction is left adjoint to restriction. Completion preserves the adjunction on positive complete objects because every map is determined weight by weight. □

When relative induction supplies a regular interface and a continuous aligned distributive law, these data define a mixed envelope with $L = V_{\mathsf{Lie}}$, $A^{\mathrm{ch}} = U^{\mathrm{ch}}(L)$, and $A^{\perp} = \widehat{\mathrm{Coll}}(V)$. Constructing that interface law in the selected geometric coefficient category is part of the envelope construction.

Book XXXVII

The deformation algebra and its BV envelope

Chapter 185

The centre-extended deformation algebra

Varying a boundary product changes its compatibility with the interface and the central action. The deformation complex must therefore include all these operations. A local field realization is further data needed to turn their deformation equations into a BV action.

Fix a cofibrant coloured operad $\mathcal{P}_{\mathrm{mix}}$ presenting the structures of a complete mixed datum and its central action. Let $\mathcal{P}_{\mathrm{mix}}^{!}$ be its bar cooperad. The convolution complex

$$\mathrm{Def}_{\mathbb{A}}^{\mathrm{full}} = \mathrm{Hom}_{\mathbb{S}}\left(\mathcal{P}_{\mathrm{mix}}^{!}, \mathrm{End}_{\mathbb{A}}\right)$$

contains the chiral operation, both associative products, the interface, and the centre action.

Theorem 185.1 (Complete deformation L_∞ algebra)**.** *The convolution complex has a canonical complete L_∞ structure. Its Maurer–Cartan points are complete mixed boundary structures together with a coherent central action. Twisting by the point defined by $\mathbb{A}$ gives the tangent algebra of deformations of the full datum.*

Proof. The infinitesimal decomposition maps of the bar cooperad insert collections of operations into one another. Composition in the endomorphism operad turns these decompositions into the convolution brackets. The cooperad identities imply the L_∞ identities. A Maurer–Cartan element is precisely an operad morphism from a cofibrant resolution of $\mathcal{P}_{\mathrm{mix}}$ to $\mathrm{End}_{\mathbb{A}}$. Twisting linearizes this mapping space at the selected structure. □

185.1 Field equations and Bianchi identities

Let α be a degree-one field in the localized deformation algebra. Its curvature is

$$\mathcal{F}(\alpha) = \sum_{n \geq 1} \frac{1}{n!} \ell_n(\alpha^{\otimes n}).$$

The linearization at α is

$$d_\alpha x = \sum_{n \geq 0} \frac{1}{n!} \ell_{n+1}(\alpha^{\otimes n}, x).$$

Theorem 185.2 (Universal Bianchi identity)**.** *For every α,*

$$d_\alpha \mathcal{F}(\alpha) = 0.$$

At a Maurer–Cartan point, $d_\alpha^2 = 0$.

Proof. Both identities are the exponential form of the L_∞ Jacobi relations. Expand the coderivation square on the cofree cocommutative coalgebra and evaluate it on e^{α} and $e^{\alpha} x$. □

Chapter 186

Shifted cotangent transgression

Let $\mathfrak{l}$ be a complete elliptic L_∞ algebra on a three-manifold M. Its shifted cotangent theory has fields

$$\mathcal{E} = T^*[-1]\mathfrak{l}[1] = \mathfrak{l}[1] \oplus \mathfrak{l}^\vee[-2].$$

The canonical action is

$$S(\alpha, \beta) = \langle \beta, \mathcal{F}(\alpha) \rangle.$$

Theorem 186.1 (Universal BV envelope)**.** *The action satisfies the classical master equation. Its Euler–Lagrange equations are*

$$\mathcal{F}(\alpha) = 0, \qquad d_\alpha^\vee \beta = 0.$$

The tangent complex at a solution is the cotangent complex of the Maurer–Cartan moduli problem.

Proof. The Hamiltonian vector field of S is the cotangent lift of the homological vector field defining the L_∞ structure. The square of that vector field is zero, hence $\{S, S\} = 0$. Variation in β gives the Maurer–Cartan equation. Variation in α gives the dual linearized equation. □

Definition 186.2 (Local realization and chiral BV envelope)**.** A local realization of the complete mixed datum consists of an elliptic local L_∞ algebra $\mathfrak{l}_{\mathbb{A}}^{\mathrm{full}}$ on $M = X^{\mathrm{an}} \times \mathbb{R}$, a specified continuous filtered L_∞ quasi-isomorphism

$$\Gamma(M, \mathfrak{l}_{\mathbb{A}}^{\mathrm{full}}) \longrightarrow \mathrm{Def}_{\mathbb{A}}^{\mathrm{full}},$$

and the local pairing needed for cotangent transgression. Here Γ denotes the chosen complex of global sections with its declared topology. The comparison includes compatibility with the specified boundary operations. The dual fields use the density-valued continuous dual, with compact support or boundary conditions that make the integrated pairing and integration by parts defined.

For a supplied local realization, define

$$\mathbf{BVEnv}_3^{\mathrm{full}}(\mathbb{A}) = T^*[-1]\,\mathrm{MC}(\mathfrak{l}_{\mathbb{A}}^{\mathrm{full}}).$$

Its action is the cotangent action displayed above. Constructing a local realization from an abstract mixed datum remains an additional existence problem.

Chapter 187

Intrinsic algebraic quantization

Let $\mathfrak{X}$ be a finite perfect derived formal moduli problem with a homological orientation μ: the corresponding Berezinian is invariant under the homological vector field. Functions on its shifted cotangent are completed polyvectors:

$$O(T^*[-1]\mathfrak{X}) = \widehat{\mathrm{PV}}(\mathfrak{X}) = \prod_{p \geq 0} \Gamma(\mathfrak{X}, \wedge^p T_{\mathfrak{X}})[p].$$

The homological orientation defines the divergence operator

$$\Delta_\mu : \widehat{\mathrm{PV}}(\mathfrak{X}) \longrightarrow \widehat{\mathrm{PV}}(\mathfrak{X})[1].$$

Theorem 187.1 (Oriented BV complex)**.** *The operator Δ_μ has order two and satisfies*

$$\Delta_\mu^2 = 0, \qquad [d_{\mathfrak{X}}, \Delta_\mu] = 0,$$

$$\Delta_\mu(fg) = \Delta_\mu(f)g + (-1)^{|f|} f \Delta_\mu(g) + (-1)^{|f|}\{f, g\}.$$

Consequently

$$\mathbf{QHdg}(\mathfrak{X}, \mu) = \left(\widehat{\mathrm{PV}}(\mathfrak{X})[\![\hbar]\!], d_{\mathfrak{X}} + \hbar\Delta_\mu\right)$$

is a square-zero quantum BV complex.

Proof. In oriented Darboux coordinates, Δ_μ is the sum of paired second derivatives. Their graded commutation gives $\Delta_\mu^2 = 0$. Invariance of the Berezinian under the homological vector field gives $[d_{\mathfrak{X}}, \Delta_\mu] = 0$. Expanding the operator on a product gives the bracket as the cross term. □

The orientation identifies polyvectors with differential forms by contraction. Under this identification,

$$\iota_\mu \Delta_\mu = d_{\mathrm{dR}}\, \iota_\mu.$$

Hence the BV complex is the Hodge-completed derived de Rham complex with differential

$$d_{\mathfrak{X}} + \hbar d_{\mathrm{dR}}.$$

The Rees parameter records Hodge degree.

Definition 187.2 (Intrinsic quantum complex of a mixed datum)**.** For a complete mixed datum $\mathbb{A}$ equipped with compatible homological orientations on its finite perfect charts, define $\mathbf{QHdg}_3^{\mathrm{full}}(\mathbb{A})$ by applying the preceding BV complex to finite perfect charts of

$$\mathrm{MC}(\mathbf{l}_{\mathbb{A}}^{\mathrm{full}})$$

and taking the separated inverse limit over the chart and weight filtrations.

Theorem 187.3 (Quantum master equation)**.** *A degree-zero action $S = S_0 + \hbar S_1 + \cdots$ defines a quantum state $e^{S/\hbar}\mu$ precisely when*

$$dS + \frac{1}{2}\{S, S\} + \hbar\Delta_\mu S = 0.$$

The equation is equivalent to

$$(d + \hbar\Delta_\mu)e^{S/\hbar} = 0.$$

Proof. Apply the second-order product formula to the exponential and collect powers of $\hbar$. □

Chapter 188

Perturbative factorization realization

The algebraic quantum differential does not specify a propagator or a space of singular kernels. Fix a local realization, elliptic gauge-fixing data, support and boundary conditions, and a renormalization prescription. A heat-kernel propagator is defined away from the diagonal. Extending its graph integrals to collisions requires counterterms and compatibility with the boundary conditions.

Definition 188.1 (Factorization realization data)**.** A factorization realization consists of renormalized local observable complexes, continuous disjoint-insertion maps, and coherent homotopies for their compositions. It includes extensions of graph weights to the required collision strata, their Stokes identities, and a quantum master equation in the permitted local functional complex. Weiss descent is a further required property of these actual complexes. A boundary realization also includes a comparison with the supplied mixed datum, preserving its two products, Lie operation, interface, and mixed law.

Under these hypotheses the disjoint-insertion maps and composition homotopies define the stated factorization object. The boundary comparison identifies its boundary operations with those of $\mathbb{A}$. The construction problem is to produce these data from the physical fields and action. A graphwise identity supplies the corresponding composition equation, whereas descent and boundary identification require their own maps and proofs.

Book XXXVIII

Boundary duality, exchange, and reduction

Chapter 189

The Koszul–Morita square

A change from modules to comodules must transport the interface as well as each algebra. The mixed bar supplies the comparison object. Exchange then asks for motion in an additional topological direction, while BRST reduction asks whether the differential respects every transported operation.

Let P be a dualizable boundary state for $A^{\perp}$. Its boundary dual is

$$A^{!}_{P} = \mathrm{RHom}_{A^{\perp}}(P, P).$$

The ordinary bar construction gives explicit twisted tensor functors. Their extension to the mixed datum must also transport the interface and its geometric operations.

Let k be a field and let $A_{\mathrm{alg}} = \bigoplus_{w \geq 0} A_w$ be an augmented associative algebra in cohomological degree zero. Assume $A_0 = k$, augmentation projects onto A_0, and multiplication adds weights. Set $A = \prod_{w \geq 0} A_w$ and $C = \prod_{w \geq 0} B(A_{\mathrm{alg}})_w$, with degreewise products and the weight topology. The bar suspension lowers cohomological degree by one, with $b_2[a|b] = [ab]$. The degree-one twisting map $\tau : C \to A$ sends $[a]$ to a and vanishes on the empty word and longer words.

A module or comodule has the form $\prod_{w \geq w_0} M_w$, with differential preserving weight, for some integer w_0. Actions and coactions preserve weight. Comodule coactions are counital and coassociative, and use the tensor product

$$(C \widehat{\otimes} M)_w = \bigoplus_{u+v=w} B(A_{\mathrm{alg}})_u \otimes M_v.$$

The same convention defines $A \widehat{\otimes} M$. These sums are finite in each weight. Repeated reduced coactions vanish in weight w after more than $w - w_0$ steps. Invert quasi-isomorphisms in each weight to obtain $D_{\mathrm{wt}}(A)$ and $D_{\mathrm{wt}}(C)$.

Proposition 189.1 (Ordinary twisted tensor equivalence)**.** *The functors*

$$F(M) = C \widehat{\otimes}_{\tau} M, \qquad G(N) = A \widehat{\otimes}_{\tau} N$$

induce inverse equivalences $D_{\mathrm{wt}}(A) \simeq D_{\mathrm{wt}}(C)$. Writing $\Delta c = \sum c_{(1)} \otimes c_{(2)}$ and $\rho n = \sum n_{(-1)} \otimes n_{(0)}$, their differentials are

$$d_F(c \otimes m) = d_C c \otimes m + (-1)^{|c|} c \otimes d_M m + \sum (-1)^{|c_{(1)}|} c_{(1)} \otimes \tau(c_{(2)}) m,$$
$$d_G(a \otimes n) = a \otimes d_N n - \sum a\tau(n_{(-1)}) \otimes n_{(0)}.$$

The counit and unit are

$$\epsilon_M(a \otimes c \otimes m) = a\epsilon_C(c)m, \qquad \eta_N(n) = \sum n_{(-1)} \otimes 1 \otimes n_{(0)}.$$

Proof. The twisting equation makes both differentials square to zero. The action, coaction, and counit identities show that the displayed maps are chain maps and satisfy the triangle identities. For $GF(M)$, the extra degeneracy

of the two-sided bar inserts $1[\bar{a}_0|a_1|\cdots|a_r]m$ into $a_0[a_1|\cdots|a_r]m$, where $\bar{a}_0 = a_0 - \epsilon_A(a_0)1$. With simplicial boundary $\sum_i (-1)^i d_i$, it contracts the augmented bar. Multiplying length-r words by $(-1)^r$ gives the displayed differential convention. This contraction preserves weight, so the counit is a weightwise quasi-isomorphism. For $FG(N)$, filter each weight by the weight of the final N-factor. The reduced coaction lowers that weight. The associated graded unit is the contracted right-sided bar $k \otimes N_v \to B(k, A_{\mathrm{alg}}, A_{\mathrm{alg}}) \otimes N_v$. The filtration is finite in each total weight, so the unit is also a quasi-isomorphism. The same finite filtrations show that F, G preserve weightwise quasi-isomorphisms, since tensoring complexes over k preserves them. Thus the unit and counit give the asserted inverse equivalences. □

For a right module R and a left module L with the same lower-weight bounds, the two-sided bar has underlying weight-completed complex

$$\prod_n \bigoplus_{u+v_1+\cdots+v_r+w=n} R_u \otimes s\bar{A}_{v_1} \otimes \cdots \otimes s\bar{A}_{v_r} \otimes L_w, \qquad v_i > 0,$$

where the sum also runs over $r \geq 0$ and $\bar{A} = \ker(A_{\mathrm{alg}} \to k)$. Internal differentials and the two endpoint actions and adjacent products form its bar differential. Its realization computes $R \otimes^{\mathbb{L}}_A L$ in these weight-graded categories. Comparing it with a derived cotensor product requires a two-sided twisting comparison on the chosen comodule category and completion. Weight completion here is not identified with derived augmentation-adic completion.

Proposition 189.2 (Centre transport under a mixed equivalence)**.** *Suppose an equivalence between the compatible mixed module categories is supplied, with coherent compatibility with all four bar objects and the interface operations. It identifies the derived endomorphism objects of their identity functors.*

Proof. Conjugate a natural endomorphism by the equivalence and a quasi-inverse. The unit and counit homotopies give inverse maps on derived endomorphism objects and preserve composition and its coherent operations. □

Constructing such a mixed equivalence requires the restriction, collision, interface, and distributive-law comparisons for the actual geometric coefficients. Its unit and counit must respect those maps and the chosen inverse limits. A diagonal bimodule comparison must produce an actual compatible bicomodule before tensor products can be identified with geometric cotensor products. Transport of mixed Hochschild chains and traces further requires their cyclic comparison and the trace domains used in their definition. These are additional mixed Koszul–Morita constructions. The ordinary weight-graded equivalence supplies its stated one-colour functors, without identifying the other mixed carriers or their physical traces.

Chapter 190

Planar exchange and Tannaka reconstruction

A second topological direction gives paths exchanging parallel defects. Their homotopy classes form braid groups. A rigid meromorphic line category $\mathscr{L}$ with a faithful fibre functor ω has coend

$$\mathcal{O}(\mathbb{G}) = \int^{M \in \mathscr{L}} \omega(M)^{*} \otimes \omega(M).$$

Theorem 190.1 (Quantum symmetry from exchange)**.** *The coend is a coquasitriangular bialgebra. Its continuous dual acts on every line object. The meromorphic braiding gives the universal R-matrix on completed matrix coefficients. A spectral rational braiding yields an RTT presentation.*

Proof. Tensor product in $\mathscr{L}$ gives the coproduct on matrix coefficients. Evaluation gives the counit. Rigidity gives the antipode on the Hopf envelope. The braiding defines the coquasitriangular pairing, and the hexagon identities give the Yang–Baxter equation. Writing this naturality on a generating fibre produces the RTT relation. □

Chapter 191

BRST reduction

Let Q be a square-zero derivation of a mixed datum. Extend Q factorwise to every bar word. Since Q is a derivation for each structure map, it anticommutes with every bar differential.

Theorem 191.1 (BRST–bar bicomplex)**.** *The total differential*

$$D = Q + b_{\mathrm{ch}} + b_{\perp} + b_{\mathrm{mix}}$$

satisfies $D^{2} = 0$. *A bounded-below complete filtration gives a convergent spectral sequence*

$$E_{1} = H_{Q}\big(\mathrm{Bar}^{\mathrm{mix}}(\mathbb{A}; K)\big) \Longrightarrow H_{D}\big(\mathrm{Bar}^{\mathrm{mix}}(\mathbb{A}; K)\big).$$

When Q-cohomology is concentrated and tensor-exact, the target is the mixed bar of the reduced datum.

Proof. The square vanishes by the derivation identity and the four-colour bar theorem. The filtration by BRST degree gives the spectral sequence. Concentration and tensor exactness identify the E_{1} differential with the bar differential of the reduced operations. □

Book XXXIX

Conformal covariance and Einstein completion

Chapter 192

Projective connections and the Virasoro tangent complex

A conformal boundary algebra transforms under changes of coordinates. Projective connections record its central term. Three-dimensional gravity supplies independently defined connection and coframe fields whose boundary transformation law can be compared with this algebraic structure.

Let $\mathcal{P}_c(X)$ be the formal moduli problem of projective connections with central parameter c. At a projective connection p, its tangent complex is locally

$$\Omega^{0,\bullet}(X, T_X) \xrightarrow{\mathcal{D}_{p,c}} \Omega^{0,\bullet}(X, K_X^{\otimes 2}),$$

where

$$\mathcal{D}_{p,c}(\epsilon) = \epsilon\, \partial p + 2(\partial \epsilon)\, p + \frac{c}{12} \partial^3 \epsilon.$$

Theorem 192.1 (Projective covariance)**.** *A conformal map* $\mathrm{Vir}_c \to A$ *defines a formal family of mixed structures over* $\mathcal{P}_c(X)$*. Its tangent algebra is the semidirect product of the Virasoro projective complex with the deformation algebra relative to a fixed projective connection.*

Proof. The conformal vector acts on fields by infinitesimal coordinate change. The central term is the affine cocycle in the transformation law of a projective connection. The action therefore gives a map from the projective tangent algebra to derivations of the mixed datum. The tangent algebra of the associated family is the semidirect product. □

Chapter 193

Three-dimensional gravity as a flat Cartan theory

Let M be an oriented three-manifold, let $\Lambda = -\ell^{-2}$, and let $\mathfrak{g}_\Lambda$ have basis J_a, P_a with

$$[J_a, J_b] = \epsilon_{ab}{}^{c} J_c, \qquad [J_a, P_b] = \epsilon_{ab}{}^{c} P_c, \qquad [P_a, P_b] = \ell^{-2} \epsilon_{ab}{}^{c} J_c.$$

The invariant pairing is $\langle J_a, P_b \rangle = \eta_{ab}$ and vanishes on the two diagonal summands. A Cartan connection is

$$\mathcal{A} = \omega^a J_a + e^a P_a.$$

Its curvature is

$$F_{\mathcal{A}} = \left(R^a + \frac{1}{2\ell^2} \epsilon^a{}_{bc} e^b \wedge e^c \right) J_a + T^a P_a.$$

Theorem 193.1 (Cartan–Einstein equivalence)**.** *On the locus where e is invertible, $F_{\mathcal{A}} = 0$ is equivalent to*

$$T^a = 0, \qquad G_{\mu\nu} + \Lambda g_{\mu\nu} = 0, \qquad g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu.$$

The Chern–Simons action for $\mathcal{A}$ equals the Einstein–Hilbert action with cosmological constant, up to its boundary term and normalization.

Proof. The P component of the flatness equation is the torsion equation. Invertibility of e gives the Levi–Civita spin connection. The J component states that the curvature two-form has constant sectional curvature $-\ell^{-2}$. In three dimensions the Riemann tensor is determined by the Ricci tensor, so this is the Einstein equation. Expanding the Chern–Simons three-form with the off-diagonal invariant pairing gives $e \wedge R + \ell^{-2} e^3$ plus an exact form [1, 138]. □

Chapter 194

Four equivalent linear complexes

Fix a constant-curvature solution (e, ω). There are four deformation complexes.

(i) The flat adjoint complex $\Omega^\bullet(M, \mathfrak{g}_\Lambda)$ with differential $d_{\mathcal{A}}$.

(ii) The Einstein–Cartan complex on $(\delta e, \delta\omega)$ with linearized torsion and curvature.

(iii) The metric Einstein complex on $h_{\mu\nu}$ with linearized Einstein operator.

(iv) The Calabi complex resolving local Killing fields by the Killing, linearized curvature, and Bianchi operators.

Theorem 194.1 (Chain equivalence of gravitational presentations)**.** *An invertible torsion-free dreibein gives chain maps among the four complexes. These maps are quasi-isomorphisms on every geodesically convex open set. Their cohomology describes infinitesimal isometries, physical linearized fields, and Bianchi identities in the four languages.*

Proof. The dreibein identifies internal translations with tangent vectors and internal Lorentz rotations with skew endomorphisms. Splitting $d_{\mathcal{A}}$ in this basis gives the linearized torsion and curvature operators. Eliminating $\delta\omega$ by the linearized torsion equation gives the metric Einstein operator. The symbol sequence is the Calabi symbol sequence and is exact on constant-curvature space. The Poincaré lemma for the flat adjoint connection and exactness of the symbol homotopies give local quasi-isomorphisms. □

Chapter 195

Brown–Henneaux and the projective boundary

Write the AdS connection as

$$A^{\pm} = \omega \pm \ell^{-1} e.$$

The Einstein action is the difference of two $\mathfrak{sl}_2$ Chern–Simons actions with level

$$k = \frac{\ell}{4G}.$$

Theorem 195.1 (Brown–Henneaux central charge)**.** *The classical asymptotic Poisson algebra of each chiral Chern–Simons sector has Virasoro central charge*

$$c = 6k = \frac{3\ell}{2G}.$$

The projective connection carried by the boundary stress tensor is the projective variable of the preceding conformal family.

Proof. Hamiltonian reduction of the boundary $\mathfrak{sl}_2$ current Poisson algebra gives the classical Virasoro bracket. Its central term is $6k$. Substitution of the gravitational level gives the second equality. The coadjoint transformation of the boundary stress tensor is the projective-connection transformation law [19]. □

Chapter 196

Einstein completion of a conformal pair

Let A_L and A_R be conformal mixed boundary data with equal Brown–Henneaux central charge. Their projective families map to the same $\mathcal{P}_c(X)$. Define their relative field algebra by the homotopy fibre product over the projective tangent algebra.

Definition 196.1 (Candidate Einstein BV completion)**.** Suppose the relative algebra and a continuous L_∞ action of the Cartan algebra have been supplied, together with their local realization. The associated cotangent candidate is

$$\mathbf{EinBV}_3(A_L, A_R) = T^*[-1]\,\mathrm{MC}\big(\mathfrak{g}_{\mathrm{grav}} \ltimes_{\mathrm{Ham}} \mathfrak{m}_{A_L,A_R}\big).$$

Here $\mathfrak{g}_{\mathrm{grav}}$ is the Cartan deformation algebra and $\mathfrak{m}_{A_L,A_R}$ is the relative mixed deformation algebra after projective matching.

Theorem 196.2 (Cotangent action for a supplied coupled algebra)**.** *Suppose projective matching and a continuous L_∞ action have constructed the displayed complete semidirect product. Suppose also that its local realization and integrated dual pairing are defined. Its shifted cotangent has action*

$$S(\alpha, \beta) = \langle \beta, \mathcal{F}(\alpha) \rangle$$

and satisfies the classical master equation. Its Bianchi identity is the L_∞ identity of the coupled algebra.

Proof. The supplied semidirect product has a square-zero homological vector field. Its cotangent lift is Hamiltonian for the displayed action and remains square-zero. This proves the classical master equation and the Bianchi identity by the preceding cotangent construction. □

The gravitational summand of this cotangent action is the cotangent theory of Cartan deformations. It introduces independent dual fields. To identify it with Einstein BV theory requires a comparison preserving the BV pairing and the action, including boundary terms, on the invertible-coframe sector. The flatness equations alone do not provide that comparison. A physical coupled completion must also construct the matter action and its equivariant moment map. The intended Einstein reconstruction therefore retains these field and action comparisons as existence problems.

Chapter 197

Higher-spin completion

Replace $\mathfrak{sl}_2$ by $\mathfrak{sl}_N$ and choose its principal $\mathfrak{sl}_2$ embedding. The adjoint representation decomposes as

$$\mathfrak{sl}_N \cong \bigoplus_{s=2}^{N} V_{2s-2}, \qquad \sum_{s=2}^{N} (2s-1) = N^2 - 1.$$

Theorem 197.1 (Principal higher-spin spectrum)**.** *The principal embedding gives one massless gauge field of each integer spin* $2, 3, \dots, N$*. Classical Drinfeld–Sokolov reduction of the boundary affine Poisson algebra gives the principal* W_N *algebra. A pair of principal* W_N *boundary data therefore has an* $\mathfrak{sl}_N \oplus \mathfrak{sl}_N$ *Einstein–Cartan completion with the same spin spectrum.*

Proof. The irreducible $\mathfrak{sl}_2$ module V_{2s-2} corresponds to a frame-like field of spin s. The dimension identity exhausts the adjoint. Principal Drinfeld–Sokolov reduction produces one generating current in each conformal weight $2, \dots, N$. The boundary currents and bulk gauge fields therefore match representation by representation [23, 78]. □

Book XL

Exact gravitational complexes and quantum tests

Chapter 198

The four complexes in coordinates

The Cartan flatness equations and the metric Einstein equations describe the same classical solutions on the invertible-coframe locus. Their infinitesimal complexes make the comparison explicit. Quantum determinants additionally depend on gauge fixing, boundary conditions, and determinant-line data.

198.1 Flat adjoint complex

At a flat Cartan connection $\mathcal{A}$, the first complex is

$$\Omega^0(M,\mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^1(M,\mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^2(M,\mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^3(M,\mathfrak{g}_\Lambda).$$

Degree zero is the gauge parameter, degree one is the fluctuation, degree two is the linearized curvature, and degree three is the Bianchi identity.

198.2 Einstein–Cartan complex

Write a gauge parameter as (ξ^a,λ^a) and a fluctuation as $(u^a,v^a)=(\delta e^a,\delta\omega^a)$. The first differential is

$$\begin{aligned} u^a &= D_\omega \xi^a + \epsilon^a{}_{bc} e^b \lambda^c, \\ v^a &= D_\omega \lambda^a + \ell^{-2}\epsilon^a{}_{bc} e^b \xi^c. \end{aligned}$$

The second differential is

$$\begin{aligned} \delta T^a &= D_\omega u^a + \epsilon^a{}_{bc} v^b \wedge e^c, \\ \delta F^a &= D_\omega v^a + \ell^{-2}\epsilon^a{}_{bc} e^b \wedge u^c. \end{aligned}$$

The third differential is the linearized pair of Cartan Bianchi identities.

Proposition 198.1 (Chain identity)**.** *The composite of the first two Einstein–Cartan differentials vanishes on a constant-curvature torsion-free background.*

Proof. Apply D_ω^2 to ξ and λ. Replace the background curvature by $-\ell^{-2}e\wedge e$ and use $D_\omega e=0$. The curvature terms cancel the explicit ℓ^{-2} terms. The remaining expressions are Jacobi identities for $\epsilon_{ab}{}^c$. □

198.3 Metric complex

The dreibein identifies the translation parameter with a vector field. Modulo Lorentz rotations,

$$h_{\mu\nu} = 2\eta_{ab} e^a_{(\mu} u^b_{\nu)}.$$

The metric differential begins with the Killing operator

$$K(\zeta)_{\mu\nu} = \nabla_\mu \zeta_\nu + \nabla_\nu \zeta_\mu.$$

The linearized cosmological Einstein operator is

$$(E_\Lambda h)_{\mu\nu} = \frac{1}{2}\big(- \nabla^2 h_{\mu\nu} - \nabla_\mu\nabla_\nu h + \nabla_\mu\nabla^\rho h_{\rho\nu} + \nabla_\nu\nabla^\rho h_{\rho\mu} \\ - g_{\mu\nu}(\nabla^\rho\nabla^\sigma h_{\rho\sigma} - \nabla^2 h)\big) - \Lambda(h_{\mu\nu} - g_{\mu\nu}h).$$

It satisfies $E_\Lambda K = 0$ and $\nabla^\mu (E_\Lambda h)_{\mu\nu} = 0$.

Theorem 198.2 (Explicit metric reduction)**.** *Solving the linearized torsion equation for $v = \delta\omega$ and substituting into the linearized curvature equation gives $E_\Lambda h = 0$. The kernel of the map from (u, v) to h is the infinitesimal Lorentz action.*

Proof. The torsion equation is algebraic in the antisymmetric part of the covariant derivative of u and determines v. The symmetric part of u is $h/2$. Substitution into $\delta F = 0$ yields the standard variation of the Ricci tensor and the displayed Einstein operator. An antisymmetric $u_{\mu\nu}$ is exactly a Lorentz rotation of the dreibein. □

198.4 Calabi complex

On a constant-curvature manifold, the Calabi complex begins

$$\Gamma(TM) \xrightarrow{K} \Gamma(S^2 T^* M) \xrightarrow{R'} \Gamma(\mathcal{R}) \xrightarrow{B'} \Gamma(\mathcal{B}),$$

where R' is the linearized Riemann tensor and B' is the linearized algebraic and differential Bianchi operator. In three dimensions the Riemann tensor is equivalent to the Einstein tensor, so the metric and Calabi complexes are locally quasi-isomorphic.

Chapter 199

Boundary BFV theory and the Virasoro moment map

The variation of Chern–Simons theory on a manifold with boundary gives the boundary symplectic form

$$\Omega_\partial = \frac{k}{4\pi}\int_{\partial M} \langle \delta A \wedge \delta A\rangle.$$

A chiral polarization selects $A_{\bar z}$ as coordinate. The residual moment map is the flatness constraint.

Theorem 199.1 (Boundary current algebra)**.** *Canonical quantization of the boundary symplectic form gives the affine current algebra at level k. Drinfeld–Sokolov reduction of the principal $\mathfrak{sl}_2$ constraint gives the Virasoro moment map*

$$T(z) = \frac{1}{2(k+h^\vee)} :J^a J_a:(z) + \textit{improvement},$$

with the classical Brown–Henneaux limit $c = 6k$.

Proof. The inverse of Ω_∂ is the current Poisson bracket. Quantization gives the affine central extension. The BRST complex of the principal constraint eliminates the gauge directions and retains the improved quadratic current. Its classical central term is $6k$ in the gravitational normalization. □

Chapter 200

Matter stress tensor and the Einstein moment map

Let $\mathfrak{m}$ be the relative field algebra of a conformal pair. Projective covariance gives a Hamiltonian action of vector fields and therefore a moment map

$$\mu_{\mathrm{matt}} : \mathfrak{m} \longrightarrow \Omega^{0,\bullet}(X, K_X^{\otimes 2})[-1].$$

Its classical component is the stress tensor.

Theorem 200.1 (Conservation from the master equation)**.** *The coupled BV action*

$$S_{\mathrm{tot}} = S_{\mathrm{grav}} + S_{\mathrm{matt}} + \langle h, \mu_{\mathrm{matt}} \rangle$$

satisfies the classical master equation precisely when the matter moment map is equivariant. The component of that equation linear in the diffeomorphism ghost is

$$\nabla^{\mu} T_{\mu\nu} = 0.$$

Proof. The master bracket of the coupling with itself is the defect of equivariance. The bracket with the gravitational action is the infinitesimal diffeomorphism of the moment map. Their cancellation is the Hamiltonian equivariance identity. Pairing the identity with the diffeomorphism ghost and integrating by parts gives stress-tensor conservation. □

Chapter 201

One-loop gravity and analytic torsion

Expand Chern–Simons theory around an acyclic flat connection $\mathcal{A}$. Gauge fixing gives the Laplacians of the flat adjoint de Rham complex.

Theorem 201.1 (One-loop determinant)**.** *The absolute value of the one-loop partition function is the square root of the Ray–Singer torsion of the flat adjoint bundle:*

$$|Z_{\text{1-loop}}(\mathcal{A})| = \tau_{\mathrm{RS}}(M, \operatorname{ad}\mathcal{A})^{1/2}.$$

The phase is the eta-invariant contribution fixed by the framing convention.

Proof. Bosonic and ghost Gaussian integrals give alternating determinants of the Laplacians in the flat de Rham complex. Their product is the analytic torsion. The odd signature operator supplies the determinant phase. This is the standard stationary-phase calculation for Chern–Simons theory [127, 138]. □

The same determinant complex is obtained from the Einstein–Cartan, metric, and Calabi presentations through their chain equivalences. One-loop equality is therefore a consequence of quasi-isomorphism together with compatible determinant-line orientations.

Chapter 202

BTZ, Cardy, and the area law

For a unitary modular boundary theory with left and right central charges c_L, c_R, the asymptotic entropy is

$$S_{\mathrm{Cardy}} = 2\pi\sqrt{\frac{c_L}{6}\left(L_0 - \frac{c_L}{24}\right)} + 2\pi\sqrt{\frac{c_R}{6}\left(\bar{L}_0 - \frac{c_R}{24}\right)}.$$

For the BTZ black hole,

$$L_0 - \frac{c}{24} = \frac{(r_+ + r_-)^2}{16G\ell}, \qquad \bar{L}_0 - \frac{c}{24} = \frac{(r_+ - r_-)^2}{16G\ell}, \qquad c = \frac{3\ell}{2G}.$$

Theorem 202.1 (Cardy–Bekenstein equality)**.** *Substitution gives*

$$S_{\mathrm{Cardy}} = \frac{2\pi r_+}{4G}.$$

This is the Bekenstein–Hawking entropy of the BTZ horizon.

Proof. Each square root becomes $\pi(r_+ \pm r_-)/(4G)$. Their sum is $2\pi r_+/(4G)$. The modular asymptotic is Cardy's theorem, and the geometric charges are the Chern–Simons holonomies of the BTZ solution [5, 25]. □

Chapter 203

Higher-spin holonomy and boundary charges

In $\mathfrak{sl}_N$ Chern–Simons gravity, a smooth Euclidean saddle is selected by the conjugacy class of the thermal holonomy. The invariant polynomials

$$\operatorname{Tr}(\mathrm{Hol}^{s}), \qquad 2 \le s \le N,$$

fix the spin-s charges. Drinfeld–Sokolov reduction identifies the same charges with zero modes of the W_N currents.

Theorem 203.1 (Bulk–boundary charge equality)**.** *For a principal $\mathfrak{sl}_N$ connection in Drinfeld–Sokolov gauge, the invariant polynomials of the spatial holonomy equal the classical W_N zero-mode Hamiltonians. Their Poisson brackets are the classical W_N brackets.*

Proof. Drinfeld–Sokolov gauge writes the connection as the principal raising operator plus lowest-weight components. The coefficients of the characteristic polynomial are the invariant polynomials and are polynomial functions of those components. Hamiltonian reduction gives the W_N Poisson structure on the same functions. □

Book XLI

Anomalies, Hall sources, and automorphic completion

Chapter 204

Four native anomaly classes

A local anomaly, a determinant-line curvature, and an arity-zero bar term belong to different complexes. A cancellation in one complex becomes a geometric cancellation only through a comparison preserving the relevant class and its primitive. Hall and automorphic constructions require their coefficient and orientation maps for the same reason.

The construction carries four curvature classes.

(i) A local BV anomaly lies in the degree-one local deformation complex.

(ii) A projective anomaly is the curvature of the determinant line over parameter space.

(iii) A framing anomaly is the transgression of a characteristic class to a three-manifold.

(iv) A curved Koszul term is the arity-zero component of a curved coalgebra.

Theorem 204.1 (Curvature-killing extension)**.** *Let κ be a closed central degree-two class in the deformation complex of a theory. Adjoining a degree-one generator u with $du = \kappa$ produces a central extension in which the lifted curvature vanishes. Tensoring with the inverse invertible field theory gives the geometric realization of the same cancellation.*

Proof. The extended differential squares to zero because $d\kappa = 0$ and κ is central. The new Maurer–Cartan equation contains the compensating field u. An invertible theory with curvature $-\kappa$ adds the opposite class under tensor product. □

Chapter 205

Hall and Calabi–Yau sources

Let C be a Calabi–Yau category with a Hall moduli problem. An oriented extension correspondence gives a Hall algebra H^+. A factorization pushforward to the boundary curve gives a chiral-convolution algebra. Relative induction supplies the remaining colours.

Proposition 205.1 (Transport under a supplied mixed Morita equivalence)**.** *Suppose a Hall construction supplies a complete mixed boundary datum and a Morita equivalence of its compatible module category with that of $\mathbb{A}$. The equivalence identifies their derived centres. A comparison of local deformation algebras, BV pairings, and actions is additional data for a BV comparison. Quantum line and amplitude comparisons additionally require the monoidal and cyclic structures used to define them.*

Proof. Conjugating natural transformations by the equivalence and its inverse identifies the endomorphisms of the two identity functors. The unit and counit homotopies provide the inverse comparison and preserve composition. This proves the centre comparison. It does not construct the additional local, symplectic, or cyclic maps. □

An algebra map alone does not induce the required forward centre map. For example, the homomorphism $\mathbb{C}[x] \to M_2(\mathbb{C})$ with $x \mapsto \operatorname{diag}(0,1)$ sends the central element x to a noncentral matrix. Its commutator with the matrix unit E_{12} is $-E_{12}$. Thus even the degree-zero condition fails for a centre map extending the given algebra map. Hall multiplication maps must be supplemented by an actual coherent module comparison before centre transport can be applied.

For $X = K3 \times E$, suppose a geometric Hall source supplies the complete charge-graded BPS Lie superalgebra and Cartan action required in Volume IV. Its semidirect product with the common Cartan defines $\mathfrak{b}_X$. Volume IV then constructs the completed Lie amalgam with the Igusa Borcherds Borel $\mathfrak{b}_\Delta$. Applying the completed enveloping functor gives the cocommutative Hopf algebra

$$H_{X\boxtimes\Delta} = \widehat{U}(\mathfrak{b}_X)\widehat{*}_{\widehat{U}(\mathfrak{h}_\Pi)}\widehat{U}(\mathfrak{b}_\Delta).$$

The two Hopf retractions preserve the geometric and automorphic sectors separately. A q-deformed mixed Hopf algebra is supplied by the quantum amalgam theorem when a geometric Hall coproduct and pairing are present.

205.1 Mixed realization of the amalgam

The Hopf retractions above are maps of their stated algebraic objects. A mixed realization must construct chiral and transverse operations, an interface, and a compatible distributive law. To realize the retractions, it must then supply maps preserving those operations and the chosen completions.

A BV retraction requires further maps between the local deformation algebras, together with compatible pairings and actions. The counterexample to forward centre functoriality prevents deducing these maps from the Hopf retractions alone. The intended geometric–automorphic theory asks for this mixed realization and for a primitive comparison quotient on its actual boundary operators. Its interaction

ideal is the kernel of that constructed quotient, rather than an ideal determined by equality of scalar characters.

Chapter 206

Igusa root states and the mixed-boundary comparison

The Gritsenko–Nikulin presentation defines the root Lie superalgebra $\mathfrak{g}_{\Delta_5}$ and its positive subalgebra $\mathfrak{n}_+$. Let $d_\alpha = \operatorname{sdim}(\mathfrak{g}_{\Delta_5})_\alpha$, and fix the positive cone, its proper height, and Weyl vector ρ. Put

$$P = \prod_{\alpha>0}(1-e^{-\alpha})^{d_\alpha}, \qquad \mathrm{Den}_\Delta = e^{-\rho}P = D_5(2Z), \qquad \jmath(Z) = Z/2.$$

The products are first interpreted in the positive-height completion.

Theorem 206.1 (Root character and reduced scalar)**.** *The root Fock space* $\operatorname{Sym}(\mathfrak{n}_+)$*, formed with super symmetry, has supercharacter* P^{-1}*. Consequently*

$$-\jmath^*\big(e^{2\rho}\operatorname{sch}(\operatorname{Sym}(\mathfrak{n}_+)^{\otimes 2})\big) = -(\jmath^*\,\mathrm{Den}_\Delta)^{-2} = -\chi_{10}^{-1}.$$

For a nonsingular projective K3 surface S and an elliptic curve E, this scalar also has an enumerative interpretation. Let $N_{g,h,d}$ be the reduced disconnected Gromov–Witten invariant in class (β_h, d), where β_h is nonzero and primitive with square $2h-2$. Use the incidence insertion $\beta_h^\vee \otimes [\text{point}]$, with $\langle \beta_h, \beta_h^\vee\rangle = 1$. Then the series

$$\mathcal{Z}(u,q,s) = \sum_{g,h,d\ge 0} N_{g,h,d}\,u^{2g-2}q^{h-1}s^{d-1}$$

is the Laurent expansion of $-\chi_{10}(r,q,s)^{-1}$ under $r = e^{iu}$, by [122, Theorem 1].

Proof. An even root vector contributes $(1-e^{-\alpha})^{-1}$ to the supercharacter of its symmetric algebra. An odd root vector contributes $1-e^{-\alpha}$, because its symmetric algebra in super vector spaces is an exterior algebra. Multiplying the two contributions gives P^{-1}. Proper height makes every coefficient calculation finite. The Weyl factor gives $e^{2\rho}P^{-2} = \mathrm{Den}_\Delta^{-2}$. Pullback by $\jmath$ gives $D_5(Z)^{-2} = \chi_{10}(Z)^{-1}$. Oberdieck–Pixton, Theorem 1, identifies the separately defined reduced numerical generating series with its negative [122]. This last theorem is an enumerative input; the character calculation supplies the root side. □

A primitive realization starts with parity-preserving root classes in a charge-graded Lie superalgebra $\mathfrak{p}$ satisfying the defining relations. Require the classes to preserve charge. The free Lie presentation gives an even charge-preserving map $f : \mathfrak{n}_+ \to \mathfrak{p}$.

Applying f to tensor words constructs $U(f)$. With trivial coefficients, use Chevalley chains $C_*^{\mathrm{CE}}(\mathfrak{a}) = \operatorname{Sym}(\mathfrak{a}[1])$, where the homological suspension raises degree by one and the differential contracts pairs by the bracket. Applying the suspension of f to every factor gives a chain map because f preserves brackets.

For the level-zero Lie conformal current, define $\mathrm{Cur}(\mathfrak{a}) = \mathbb{C}[\partial]\otimes\mathfrak{a}$, with ∂ even. Its generator bracket is $[x_\lambda y] = [x,y]$. Extend it by $[\partial x_\lambda y] = -\lambda[x_\lambda y]$ and $[x_\lambda \partial y] = (\partial+\lambda)[x_\lambda y]$. The formula $\partial^j x \mapsto \partial^j f(x)$ defines $\mathrm{Cur}(f)$. Charge preservation makes these maps compatible with the specified proper-height quotients and their inverse limits.

A complete mixed boundary realization requires further geometric maps. The compact source consists of sheaves and extension flags on a common elliptic torsor. Its critical realization must supply reduced coefficient complexes, orientation cross-line maps, unit and flag coherences, and the admissible pull–push operations. A chiral comparison must preserve actual support and moving-insertion operations. A quantum-double comparison must preserve coproducts, pairings and the domain of exchange operators. The finite Calabi–Yau quiver completions require their specified dg maps; ordinary full-subquiver inclusions do not in general give those maps.

The mixed datum to which a BV envelope is applied must therefore include these operations and compatible maps between them. Identifying the primitive algebra of that envelope requires an actual map of its deformation complex to the root complex. A determinant comparison requires a skew perfect family, a square-root line and section, and an isomorphism to the automorphic line preserving transport and normalization. An operator-trace interpretation further requires its state space and trace domain. Constructing this datum and these comparisons is the Igusa mixed-boundary problem. The root character identity specifies the scalar that their final evaluation must recover.

Chapter 207

The carrier lattice

The complete theory uses the carrier set

$$\mathfrak{C} = \{\mathrm{diag}, \mathrm{ch}, \bot, \mathrm{int}, \mathrm{ex}, \mathrm{cyc}, \mathrm{mod}, \mathrm{BV}, \mathrm{Ein}\}.$$

For a subset $S \subset \mathfrak{C}$, let $\mathrm{Str}_S(\mathbb{A})$ be the homotopy limit of all mapping spaces attached to incidence cells supported in S.

Theorem 207.1 (Carrier descent). *When S and T form a relation-excisive pair, the square*

$$\begin{array}{ccc} \mathrm{Str}_{S\cup T}(\mathbb{A}) & \longrightarrow & \mathrm{Str}_S(\mathbb{A}) \\ \downarrow & & \downarrow \\ \mathrm{Str}_T(\mathbb{A}) & \longrightarrow & \mathrm{Str}_{S\cap T}(\mathbb{A}) \end{array}$$

is a homotopy pullback. Thus a global structure is assembled from its carrier restrictions together with their common interface.

Proof. Relation-excision identifies the incidence category of $S \cup T$ with the pushout of the incidence categories for S and T over their intersection. Homotopy limits send this pushout to the displayed pullback. □

Definition 207.2 (Relation-complete carrier cover). Let $U \subseteq \mathfrak{C}$ and let $\mathcal{U} = \{S_i\}_{i\in I}$ be a finite family with $U = \bigcup_i S_i$. The family is *relation-complete* when the canonical functor from the Cech homotopy colimit

$$\operatorname*{hocolim}_{\varnothing \neq J \subseteq I} \mathcal{I}_{\bigcap_{j\in J} S_j} \longrightarrow \mathcal{I}_U$$

is an equivalence. Equivalently, every incidence cell and every relation among its faces is contained in one member of the cover, with all overlaps recorded by the intersections.

Theorem 207.3 (Power-set coherence). *The assignment*

$$S \longmapsto \mathrm{Str}_S(\mathbb{A})$$

is a contravariant functor on the full power set of $\mathfrak{C}$. For every relation-complete carrier cover $\mathcal{U} = \{S_i\}_{i\in I}$ of U, the restriction map is an equivalence

$$\mathrm{Str}_U(\mathbb{A}) \xrightarrow{\sim} \operatorname*{holim}_{\varnothing \neq J \subseteq I} \mathrm{Str}_{\bigcap_{j\in J} S_j}(\mathbb{A}).$$

Thus every subtheory, every intersection of subtheories, and the complete theory are values of one descent object.

Proof. An inclusion $S \subseteq T$ induces the full inclusion $\mathcal{I}_S \subseteq \mathcal{I}_T$ and therefore the restriction map on homotopy limits. Composition of inclusions gives functoriality. A relation-complete cover presents $\mathcal{I}_U$ as the displayed homotopy colimit of its intersection diagram. Mapping that colimit into the diagram of structure spaces converts it into the stated homotopy limit. The resulting equivalence is natural under refinement of the carrier cover. □

Definition 207.4 (Exact-support deformation complex)**.** For $S \subseteq \mathfrak{C}$, let $\mathrm{Def}_{\leq S}$ be the subcomplex generated by incidence cells supported in S, and set

$$\mathrm{Def}_{=S} = \mathrm{cofib}\Big(\operatorname*{hocolim}_{T \subsetneq S} \mathrm{Def}_{\leq T} \longrightarrow \mathrm{Def}_{\leq S}\Big).$$

Thus $\mathrm{Def}_{=S}$ retains the deformation cells whose minimal carrier support is exactly S.

Theorem 207.5 (Carrier-support spectral sequence)**.** *Let* $\mathrm{Def}_{\mathbb{A}}^{\mathrm{full}}$ *be the cellular convolution complex of the complete incidence diagram. Filter it by the cardinality of minimal carrier support. The finite filtration gives a convergent spectral sequence*

$$E_1^{p,q} = \bigoplus_{\substack{S \subseteq \mathfrak{C} \\ |S| = p}} H^{p+q}(\mathrm{Def}_{=S}) \Longrightarrow H^{p+q}\Big(\mathrm{Def}_{\mathbb{A}}^{\mathrm{full}}\Big).$$

A class on the S-summand is a deformation or curvature class whose first complete geometric carrier is S. The first differential is the signed incidence map between exact-support quotients.

Proof. The support-cardinality filtration is finite because $\mathfrak{C}$ is finite. Its pth associated-graded quotient is the direct sum of the exact-support complexes $\mathrm{Def}_{=S}$ with $|S| = p$. The spectral sequence of a finite filtered complex gives the displayed first page and converges strongly to the cohomology of the total convolution complex. The cellular determinant orientation gives the incidence signs. □

Chapter 208

Sixteen structural laws

The theory is governed by the following laws.

(i) A geometric carrier determines the type of its operation.

(ii) The diagonal stores the complete principal-part filtration.

(iii) The interval stores ordered fusion.

(iv) The plane stores exchange.

(v) The circle stores trace.

(vi) The stable curve stores modular sewing.

(vii) The transverse, associative-chiral, and chiral-Lie bars are distinct.

(viii) Chiral PBW compares the last two on enveloping objects.

(ix) The interface bar records the mixed distributive law.

(x) The derived centre is the universal interior action.

(xi) The deformation algebra is the tangent algebra of the centre-extended datum.

(xii) Maurer–Cartan curvature is the field equation.

(xiii) The L_∞ identity is the Bianchi identity.

(xiv) Shifted cotangent is the universal classical BV envelope.

(xv) Hodge-completed derived de Rham theory is the intrinsic algebraic quantum complex of the oriented formal moduli problem.

(xvi) A gravitational realization requires projective matching, a coupled local field algebra, and a comparison preserving the physical BV action and pairing.

Chapter 209

Coherent actions and gravitational reconstruction

A supplied mixed boundary datum defines compatible modules and their derived natural endomorphisms. A specified coloured operad organizes deformations of that datum together with its central action. These are constructions relative to the full operations and their coherence maps. The centre alone does not recover the datum from which the deformation algebra is formed.

A supplied local realization of the deformation algebra admits shifted cotangent transgression. With compatible invariant orientations, the finite perfect charts also carry the algebraic BV differential. The physical bulk is defined independently by its fields, action, gauge symmetry, boundary conditions, and quantization. Reconstruction requires a coherent action map from its observables to an independently defined geometric centre, followed by an equivalence theorem on the stated sector.

For gravity, the invertible coframe gives the classical Cartan–Einstein correspondence. The cotangent construction supplies a formal comparison theory with extra dual fields. Its identification with gravitational BV fields requires an action-preserving and pairing-preserving map. Quantum boundary currents, a continuum measure, global holonomy sectors, allowed fillings, and compatible states remain necessary for the quantum and global comparison.

Hall and Calabi–Yau geometry can supply boundary operations when their correspondences, coefficients, orientations, and comparison maps are constructed. Exchange additionally uses a monoidal category and a fibre functor. Cyclic pairings and sewing determine amplitudes only on their stated domains. These comparisons are the mathematical steps by which local boundary operations could reconstruct physical and geometric theories.

Book **XLII**

Technical appendices to the preceding book

Chapter 210

The Chern–Simons normalization

The gravitational charge calculation uses the action normalization. The mixed bar calculation uses the determinant orientation of its edges. The following formulas specify these conventions.

With invariant pairing $\langle J_a, P_b\rangle = \eta_{ab}$, the Chern–Simons action is

$$S_{\mathrm{CS}}[\mathcal{A}] = \frac{k}{4\pi}\int_M \langle \mathcal{A}\wedge \mathrm{d}\mathcal{A} + \frac{2}{3}\mathcal{A}\wedge\mathcal{A}\wedge\mathcal{A}\rangle.$$

For $A^{\pm} = \omega \pm \ell^{-1}e$ one finds

$$S_{\mathrm{CS}}[A^{+}] - S_{\mathrm{CS}}[A^{-}] = \frac{k}{2\pi\ell}\int_M \epsilon_{abc}e^{a}\wedge\left(R^{bc} + \frac{1}{3\ell^{2}}e^{b}\wedge e^{c}\right) + \int_{\partial M}(\cdots).$$

The equality with the Einstein–Hilbert normalization gives $k = \ell/(4G)$.

Chapter 211

The principal embedding count

The exponents of $\mathfrak{sl}_N$ are $1, 2, \ldots, N-1$. Under the principal $\mathfrak{sl}_2$ embedding,

$$\mathfrak{sl}_N = \bigoplus_{m=1}^{N-1} V_{2m}.$$

The dimensions satisfy

$$\sum_{m=1}^{N-1} (2m+1) = (N-1)(N+1) = N^2 - 1.$$

Writing $s = m+1$ gives one field at every spin $s = 2, \ldots, N$.

Chapter 212

Low-arity mixed signs

Let $x|k|y$ be a suspended interface word. The two length-lowering contractions are

$$b_L[x|k|y] = (-1)^{|x|}(xk)|y, \qquad b_R[x|k|y] = (-1)^{|x|+|k|}x|(ky).$$

Applying the remaining contraction gives the two composites $(xk)y$ and $x(ky)$ with opposite determinant-line signs. Their sum vanishes by the interface associativity relation. A transverse and a chiral contraction acquire the product orientation

$$\det(E_{\mathrm{ch}} \oplus E_{\perp}) = \det(E_{\mathrm{ch}}) \otimes \det(E_{\perp}),$$

so reversing their order contributes the Koszul sign and yields anticommutation.

Book XLIII

Geometric reconstruction

Chapter 213

What the comparison maps preserve

Extensions supply Hall multiplication through their flag correspondences. The cyclic pairing records duality on the underlying category. A coproduct combines representations, and a half-braiding specifies central exchange with every object. These operations have distinct carriers. A reconstruction must construct the maps between them before comparing traces or characters.

For a specified primitive Lie map, Theorem 121.1 constructs its enveloping and Chevalley extensions explicitly. Surjectivity and equality of finite parity-refined PBW components then prove an algebraic isomorphism. For a specified strong monoidal equivalence, Theorem 19.1 constructs the transported half-braiding and proves a braided equivalence of categorical centres. Neither statement constructs a geometric functor from a scalar character.

Chapter 214

K3 actions and their common carriers

The Mukai lattice has rank twenty-four and symmetric pairing of signature (4, 20). The Nakajima operators act on the Hilbert-scheme cohomology tower. Lefschetz operators act on each selected hyperkähler moduli space. The local ADE Yangian actions use their equivariant surface coefficients. These state spaces and coefficient fields are part of the respective constructions.

On smooth projective moduli spaces, Proposition 113.1 identifies rational cohomology kernels with graded linear operators and convolution with composition. A chosen collection of geometric operators therefore gives the faithful action algebra of Proposition 114.1. Its defining ideal consists of the actual relations of those operators. The construction neither inserts a spectral coproduct nor identifies local equivariant actions with global ones. Those comparisons require intertwining maps on their state spaces.

The coend of Proposition 115.1 instead constructs matrix coefficients for a specified tensor category and fibre functor. The selected kernels become comodule morphisms. This separates the algebra generated by geometric operations from tensor symmetries commuting with them. A Yangian realization must construct its current generators, relations, and tensor action on these geometric carriers.

Chapter 215

The compact Hall and automorphic comparison

For $X = S \times E$, the E-equivariant motivic Hall construction retains extensions of sheaves and the reduced integration map on its regular subalgebra. A chosen charge homomorphism gives the polarized character of Proposition 99.2. The Borcherds algebra and its root Fock character are independently defined.

A primitive root comparison must preserve brackets, parity, and charge. Its graph quotient is computed in Proposition 101.4. The quotient is the geometric Borel for every specified comparison; it is the Borcherds Borel as well only when that comparison is an isomorphism. The quotient construction does not provide the comparison map.

A geometric realization also needs the critical coefficient complexes and extension-compatible square roots of their determinant lines. The cross-line $\det R\,\mathrm{Hom}(A, B)$ enters the orientation formula for an extension of B by A. Compatibility on triple flags makes this data usable in Hall convolution. The brane functor, paired quantum coproduct, and trace-line comparison remain the maps specified in Problem 131.1.

Chapter 216

Classical fields, centres, and amplitudes

The cyclic A_∞ action satisfies the classical master equation by Proposition 117.1. The K3 field reduction of Proposition 129.1 retains both the degree-zero and degree-two Dolbeault modes. The finite one-loop Lie weights vanish by their positive-height trace calculation. Analytic quantization and compatible observable completions are separate steps in a quantum field comparison.

For an ordinary enveloping algebra, Theorem 118.1 computes the Hochschild centre complex using Chevalley cochains with adjoint coefficients. Its explicit antisymmetrization map identifies the two ordinary cochain models. A physical bulk comparison must construct its map into this centre or its chosen geometric refinement, preserving the actual local operations. A state and its pairing must then be compared separately. Thus the algebraic calculations specify concrete parts of reconstruction and the geometric maps still required to recover amplitudes.

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